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Quantum Mechanics

Non-relativistic Theory

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CHAPTER I

Fundamental Concepts of Quantum Mechanics

§ 1. The Uncertainty Principle

When classical mechanics and electrodynamics are applied in an attempt to explain atomic phenomena, they give results sharply at variance with experiment. This is already seen most clearly in the contradiction produced by applying ordinary electrodynamics to a model of the atom in which electrons move around the nucleus in classical orbits. Since any accelerated charge radiates, electrons in such orbits would be obliged to emit electromagnetic waves without interruption. Through radiation they would lose energy and eventually fall into the nucleus. Classical electrodynamics would therefore render the atom unstable—a conclusion entirely at odds with reality.

So profound a conflict between theory and experiment shows that a theory applicable to atomic phenomena—phenomena involving particles of very small mass in very small regions of space—requires a fundamental alteration of the basic classical ideas and laws.

A convenient point of departure for identifying these changes is the experimentally observed phenomenon known as electron diffraction[1]. When a uniform beam of electrons passes through a crystal, the transmitted beam displays a succession of intensity maxima and minima entirely analogous to the diffraction pattern observed for electromagnetic waves. Under certain conditions, therefore, the behavior of material particles—electrons—shows features characteristic of wave processes.

The depth of this phenomenon's conflict with ordinary ideas of motion is best shown by the following thought experiment, an idealization of electron diffraction by a crystal. Imagine a barrier impermeable to electrons, with two slits cut in it. If we observe an electron beam[2] passing through one slit while the other is closed, we obtain some intensity distribution on a solid screen behind the slit. Opening the second slit and closing the first gives another distribution. With both slits open at once, ordinary reasoning would lead us to expect simply the superposition of those two distributions: each electron, moving on its own trajectory, passes through one slit without affecting electrons passing through the other. Electron diffraction shows instead that the actual result is a diffraction pattern which, because of interference, is by no means the sum of the patterns from the separate slits. Plainly, this result cannot be reconciled in any way with the idea that electrons move along trajectories.

The mechanics governing atomic phenomena—the so-called *quantum* or *wave mechanics*—must therefore rest on concepts of motion fundamentally different from those of classical mechanics. Quantum mechanics has no concept of a particle trajectory.

[1] In fact, electron diffraction was discovered after quantum mechanics had been created. Our presentation, however, does not follow the historical sequence in which the theory developed; it is arranged instead to show as clearly as possible how the fundamental principles of quantum mechanics are connected with observed phenomena.

[2] The beam is assumed to be so dilute that interactions among its particles play no part.

This fact constitutes the content of the so-called *uncertainty principle*, a fundamental principle of quantum mechanics, due to *Heisenberg* (*W. Heisenberg*, 1927)[3]. Since it rejects the customary ideas of classical mechanics, the uncertainty principle may be said to have negative content. By itself it is naturally quite insufficient as the basis for constructing a new mechanics of particles. Such a theory must of course rest on certain positive propositions, which will be considered below (§ 2). Before those propositions can be formulated, however, the character of the problems posed to quantum mechanics must first be made clear.

For this purpose, let us first consider the special relation between quantum and classical mechanics.

Ordinarily, a more general theory can be formulated as a logically closed system independently of the less general theory that arises as its limiting case. Relativistic mechanics, for example, can be constructed from its own fundamental principles without any reference to Newtonian mechanics. The fundamental propositions of quantum mechanics, on the other hand, cannot in principle be formulated without invoking classical mechanics.

The absence of a definite trajectory for an electron[4] also deprives the electron, considered by itself, of any other dynamical characteristics[5]. It is therefore clear that no logically closed mechanics could be constructed for a system consisting solely of quantum objects. A quantitative description of electron motion requires the presence of physical objects which obey classical mechanics to sufficient accuracy. When an electron interacts with a "classical object," the state of that object is, in general, altered. Both the character and the magnitude of this alteration are governed by the electron's state and can therefore serve as its quantitative characteristic.

In this connection, the "classical object" is usually called an "apparatus," and its interaction with the electron is spoken of as a "measurement." It must be emphasized, however, that this does not mean a process of "measurement" involving a physicist as observer. In quantum mechanics, measurement means any interaction between a classical and a quantum object, occurring apart from and independently of any observer. The recognition of the profound role of the measurement concept in quantum mechanics is due to *Bohr* (*N. Bohr*).

We have defined an apparatus as a physical object which obeys classical mechanics to sufficient accuracy; a body of sufficiently great mass is one example. It should not be supposed, however, that an apparatus must be macroscopic. Under suitable conditions an object known to be microscopic can also play the part of an apparatus, since the meaning of "with sufficient accuracy" depends on the specific problem under consideration. In a Wilson chamber, for instance, one tracks the electron by its mist trail, whose thickness far exceeds atomic dimensions; at such a level of accuracy in locating the trajectory, the

[3]It is noteworthy that the complete mathematical apparatus of quantum mechanics had already been constructed by *W. Heisenberg* and *E. Schrödinger* in 1925–1926—prior to formulation of the uncertainty principle, which lays bare the physical content of that apparatus.

[4]In this and the next section, for brevity, we speak of an electron while meaning any quantum object in general: a particle or system of particles which obeys quantum but not classical mechanics.

[5]Here we mean quantities characterizing the electron's motion, not those characterizing the electron as a particle (charge and mass), which are parameters.

electron is a fully classical object.

Quantum mechanics thus occupies a very distinctive place among physical theories: it contains classical mechanics as its limiting case and, at the same time, requires that limiting case for its own foundation.

We can now formulate the problem addressed by quantum mechanics. A typical problem is to predict the outcome of a repeated measurement from the known outcomes of earlier measurements. We shall also see below that, compared with classical mechanics, quantum mechanics generally restricts the set of values that various physical quantities (energy, for example) may assume, that is, the values that may be found by measuring the quantity in question. The formalism of quantum mechanics must make it possible to determine these allowed values.

In quantum mechanics the measurement process has an essential peculiarity: it always acts upon the electron being measured, and for a given measurement accuracy this action cannot in principle be made arbitrarily weak. The more accurate the measurement, the stronger its action; only in measurements of very low accuracy can the action on the measured object be weak. This property of measurements is logically connected with the fact that the dynamical characteristics of an electron arise only through the measurement itself. It is clear that, if the action of the measurement process on the object could be made arbitrarily weak, the quantity being measured would have a definite value by itself, irrespective of the measurement.

Of the various types of measurement, the principal one is the measurement of an electron's coordinates. Within the domain of applicability of quantum mechanics, the electron's coordinates can always be measured[6] to any accuracy.

Suppose that successive measurements of the coordinates of an electron are made at specified time intervals Δt. As a rule, these results will fail to follow any smooth curve. Quite the opposite: the greater the measurement precision, the more abrupt and erratic the succession of results becomes, consistent with the absence of a path concept for the electron. A more or less smooth path is obtained only when the electron's coordinates are measured with a low degree of accuracy, for example, by the condensation of droplets of vapour in a Wilson chamber.

If instead, while keeping the measurement accuracy unchanged, the intervals Δt between measurements are reduced, adjacent measurements will, of course, give nearby coordinate values. Yet although the results of a series of successive measurements will lie in a small region of space, within that region they will be distributed quite irregularly and will by no means fit any smooth curve. In particular, as Δt tends to zero, the results of nearby measurements do not tend at all to lie on one straight line.

This last observation makes plain that a classical particle velocity does not exist in quantum mechanics: the ratio of the coordinate change between two time instants to the separation Δt tends to no well-defined limit. As we shall see later, however, quantum mechanics does admit a meaningful definition of a particle's velocity at a given instant, one that reduces to the classical velocity upon passing to classical mechanics.

In classical mechanics a particle possesses, at each instant, definite coordinates and

[6] We emphasize once again that by a "measurement made" we mean the interaction of the electron with a classical "apparatus"; this in no way presupposes the presence of an outside observer.

velocity; in quantum mechanics the situation is entirely different. If a measurement has given an electron definite coordinates, it then has no definite velocity at all. Conversely, an electron having a definite velocity cannot have a definite position in space. Indeed, the simultaneous existence of coordinates and velocity at any instant would imply the presence of a definite velocity, which the electron does not have. Thus, in quantum mechanics the electron's coordinates and velocity are quantities that cannot be measured exactly at the same time, that is, cannot simultaneously have definite values. One may say that the electron's coordinates and velocity are quantities which do not exist simultaneously. A quantitative relation governing the possibility of measuring coordinates and velocity inexactly at the same instant will be derived below. In classical mechanics, the state of a physical system is completely described by specifying all its coordinates and velocities at a given instant; from these initial data the equations of motion fully determine the system's behaviour at all future instants. In quantum mechanics such a description is impossible in principle, since the coordinates and their corresponding velocities do not exist simultaneously. Hence the state of a quantum system is described by fewer quantities than in classical mechanics, that is, its description is less detailed than a classical one.

From this follows a very important consequence regarding the character of predictions in quantum mechanics. Whereas a classical description is sufficient to predict with complete precision the future motion of a mechanical system, the less detailed quantum-mechanical description manifestly cannot be adequate for this purpose. Consequently, even when an electron is in a state described as completely as possible, its behaviour at subsequent instants is in principle not unique. Quantum mechanics therefore cannot make strictly definite predictions about the electron's future behaviour. For a specified initial state of an electron, a subsequent measurement may give different results. The task of quantum mechanics is only to determine the probability that one or another result will be obtained in that measurement. In some cases, of course, the probability of a particular measurement result may equal unity, that is, become a certainty, so that the result of the measurement is unique.

Every measurement process in quantum mechanics falls into one of two categories. The first, which includes the majority of measurements, comprises those that yield no determinate result with certainty for any state of the system. The second comprises measurements such that for every possible result there exists a state in which that result is obtained with certainty. It is precisely these latter measurements, which may be called *predictable*, that have the principal role in quantum mechanics. The quantitative characteristics of a state determined by such measurements are what quantum mechanics calls physical quantities. If in some state a measurement gives a unique result with certainty, we shall say that the corresponding physical quantity has a definite value in that state. Below we shall everywhere understand the expression "physical quantity" in precisely the sense stated here.

We shall repeatedly find below that by no means every collection of physical quantities in quantum mechanics can be measured simultaneously, that is, can have definite values simultaneously (we have already discussed one example: the velocity and coordinates of an electron).

Sets of physical quantities with the following property have an important role in quantum mechanics: the quantities can be measured simultaneously and, when they simultaneously have definite values, no other physical quantity (unless it is a function of them) can have a definite value in that state. We shall refer to such sets of physical quantities as *complete sets*.

Every description of an electron's state arises from some measurement. We now state what a complete description of a state means in quantum mechanics. Completely described states arise from the simultaneous measurement of a complete set of physical quantities. From the results of such a measurement one can, in particular, determine the probabilities of the results of any subsequent measurement, independently of everything that happened to the electron before the first measurement.

Below, everywhere except in § 14, by states of a quantum system we shall mean states described in precisely this complete manner.

§ 2. The superposition principle

The radical change in physical ideas about motion that quantum mechanics requires in comparison with classical mechanics naturally calls for an equally radical change in the theory's mathematical formalism. In this connection the first question that arises concerns the means of describing a state in quantum mechanics.

Let q stand for the totality of coordinates of a quantum system, and dq for the product of their differentials (it is called a volume element of the system's *configuration space*); for one particle, dq coincides with the volume element dV of ordinary space.

The mathematical formalism of quantum mechanics rests on the assertion that a system's state may be described by a certain (generally complex) function of the coordinates $\Psi(q)$, and that the squared modulus of this function determines the probability distribution of the coordinate values: $|\Psi|^2\, dq$ is the probability that a measurement made on the system will find coordinate values in the element dq of configuration space. The function Ψ is called the system's *wave function*[7].

Knowledge of the wave function makes it possible in principle to calculate the probabilities of the various outcomes of any measurement in general (not necessarily a coordinate measurement). Every one of these probabilities is given by an expression bilinear in Ψ and Ψ^*. In full generality, such an expression has the form

$$\iint \Psi(q)\Psi^*(q')\varphi(q,q')\, dq\, dq', \tag{2.1}$$

where the function $\varphi(q,q')$ depends on the kind and outcome of the measurement, and the integrations extend over the entire configuration space. The probability $\Psi\Psi^*$ itself for the different coordinate values is also an expression of this type[8].

As time passes, the system's state, and with it the wave function, generally changes. In this sense the wave function may also be regarded as a function of time. If the wave function is known at some initial instant, then, by the very meaning of a complete description of a state, it is thereby determined in principle at every future instant. The

[7]It was first introduced into quantum mechanics by *Schrödinger* (*E. Schrödinger*, 1926).

[8]It is obtained from (2.1) by taking $\varphi(q,q') = \delta(q-q_0)\,\delta(q'-q_0)$, where δ denotes the so-called δ-function defined below, in § 5; q_0 denotes the coordinate value whose probability is sought.

actual time dependence of the wave function is specified by equations to be derived below.

By definition, the sum of the probabilities of all possible values of the system's coordinates must equal unity. The integral of $|\Psi|^2$ over the whole configuration space must therefore equal unity:

$$\int |\Psi|^2 \, dq = 1. \tag{2.2}$$

This equality constitutes what is known as the *normalization condition* on wave functions. When the integral of $|\Psi|^2$ is convergent, by choosing a suitable constant coefficient, the function Ψ can always be normalized, as it is said. We shall see below, however, that the integral of $|\Psi|^2$ may diverge, in which case Ψ cannot be normalized by condition (2.2). Here $|\Psi|^2$ naturally does not yield the absolute coordinate probabilities; the ratio of $|\Psi|^2$ between two distinct points of configuration space, however, does determine the relative probability of the coordinate values.

Since all quantities with direct physical meaning that are calculated using the wave function have the form (2.1), in which Ψ occurs multiplied by Ψ^*, it is clear that a normalized wave function is defined only up to a constant *phase factor* of the form $e^{i\alpha}$, where α is any real number. This ambiguity is fundamental and cannot be removed; it is, however, immaterial, since it has no effect on any physical result.

The positive content of quantum mechanics is founded on a number of assertions about the properties of the wave function, as follows.

Suppose that in the state with wave function $\Psi_1(q)$ a certain measurement leads with certainty to a definite outcome, outcome 1, while in the state $\Psi_2(q)$ it leads to outcome 2. It is then postulated that every linear combination of Ψ_1 and Ψ_2, that is, every function of the form $c_1\Psi_1 + c_2\Psi_2$ (c_1, c_2 being constants), describes a state in which the same measurement yields either outcome 1 or outcome 2. Moreover, if the time dependence of the states is known, being given in one case by the function $\Psi_1(q,t)$ and in the other by $\Psi_2(q,t)$, then any linear combination of them also gives a possible time dependence of a state.

These assertions constitute the so-called *superposition principle for states*, the fundamental positive principle of quantum mechanics. It follows from this principle, in particular, that every equation satisfied by wave functions must be linear in Ψ.

Consider a system made up of two parts, and suppose its state is specified in such a way that each part is described completely[9]. One can then assert that the coordinate probabilities for q_1 in the first part are independent of those for the coordinates q_2 of the second, and therefore that the probability distribution for the whole system must equal the product of the probability distributions for its parts. It follows that the system's wave function $\Psi_{12}(q_1, q_2)$ may be written as a product of the wave functions $\Psi_1(q_1)$ and $\Psi_2(q_2)$ of its parts:

$$\Psi_{12}(q_1, q_2) = \Psi_1(q_1)\Psi_2(q_2). \tag{2.3}$$

If the two parts do not interact with each other, this relation between the wave functions

[9]This, of course, also gives a complete description of the state of the system as a whole. We emphasize, however, that the converse statement is by no means valid: in general, a complete description of the state of the system as a whole does not yet describe the states of its separate parts completely (see also § 14).

of the system and of its parts is preserved at future instants as well:

$$\Psi_{12}(q_1, q_2, t) = \Psi_1(q_1, t)\Psi_2(q_2, t). \tag{2.4}$$

§ 3. Operators

Consider some physical quantity f that characterizes the state of a quantum system. Strictly speaking, in what follows we ought to speak not of a single quantity but of an entire complete set of them at once. However, none of the reasoning is thereby essentially altered, and for the sake of brevity and simplicity we speak below of only a single physical quantity.

Those values which a given physical quantity may assume are termed its *eigenvalues* in quantum mechanics, and their aggregate is referred to as the eigenvalue *spectrum* of that quantity. In classical mechanics, quantities range, in general, over a continuous series of values. In quantum mechanics too there exist physical quantities (for example, coordinates) whose eigenvalues fill a continuous range; in such cases one speaks of a *continuous spectrum* of eigenvalues. Alongside these quantities, however, quantum mechanics also possesses quantities whose eigenvalues constitute a discrete set; one then speaks of a *discrete spectrum*.

For simplicity we shall first assume that the quantity f under consideration has a discrete spectrum; the case of a continuous spectrum is considered in § 5. We denote the eigenvalues of f by f_n, where the index n runs through the values $0, 1, 2, 3, \ldots$ We denote the wave function of the system in the state in which f has the value f_n by Ψ_n. The wave functions Ψ_n bear the name *eigenfunctions* of the physical quantity f. Every such function is taken to be normalized:

$$\int |\Psi_n|^2 \, dq = 1. \tag{3.1}$$

If the system is in some arbitrary state with wave function Ψ, then a measurement of f performed on it will yield as a result one of the eigenvalues f_n. By the superposition principle, the wave function Ψ must be a linear combination of those eigenfunctions Ψ_n that correspond to the values f_n which can be found with non-zero probability in a measurement performed on the system in the state under consideration. In the general case of an arbitrary state, Ψ may accordingly be written as the series

$$\Psi = \sum_n a_n \Psi_n, \tag{3.2}$$

where the summation is over all n, and a_n are certain constant coefficients.

Thus we arrive at the conclusion that every wave function can be, as one says, expanded in the eigenfunctions of any physical quantity. A system of functions in terms of which such an expansion can be carried out is spoken of as a *complete system of functions*.

The expansion (3.2) makes it possible to determine the probabilities of finding (by means of measurements) one or another value f_n of f for a system in the state with wave function Ψ. Indeed, according to what was said in the preceding section, these probabilities must be determined by certain expressions bilinear in Ψ and Ψ^* and must therefore be bilinear in a_n and a_n^*. Moreover, positivity of these expressions is obviously required. As for the probability associated with a given f_n: it must become unity for a system in the state $\Psi = \Psi_n$ and must vanish whenever expansion (3.2) of Ψ lacks a term containing the corresponding Ψ_n. The only essentially positive quantity satisfying this condition is the squared modulus of the coefficient a_n. Thus we arrive at the result that the squared modulus $|a_n|^2$ of every expansion coefficient in (3.2) gives the probability of the respective value f_n of f in the state with wave function Ψ. The sum of the probabilities of all possible values f_n must equal unity; in other words, the relation

$$\sum_n |a_n|^2 = 1. \tag{3.3}$$

must hold.

If Ψ were not normalized, then the relation (3.3) would also not hold. The sum $\sum_n |a_n|^2$ would then have to be determined by some expression bilinear in Ψ and Ψ^* and reducing to unity for normalized Ψ. The only such expression is the integral $\int \Psi\Psi^*\, dq$. Thus the equality

$$\sum_n a_n a_n^* = \int \Psi\Psi^*\, dq. \tag{3.4}$$

must hold.

On the other hand, multiplying the expansion $\Psi^* = \sum_n a_n^* \Psi_n^*$ of the complex conjugate function Ψ^* by Ψ and integrating, we obtain

$$\int \Psi\Psi^*\, dq = \sum_n a_n^* \int \Psi_n^* \Psi\, dq.$$

Comparing this with (3.4), we have

$$\sum_n a_n a_n^* = \sum_n a_n^* \int \Psi_n^* \Psi\, dq,$$

from which we find the following formula determining the coefficients a_n of the expansion of Ψ in the eigenfunctions Ψ_n:

$$a_n = \int \Psi\Psi_n^*\, dq. \tag{3.5}$$

Substituting (3.2) here, we obtain

$$a_n = \sum_m a_m \int \Psi_m \Psi_n^*\, dq,$$

whence it is seen that the eigenfunctions must satisfy the conditions

$$\int \Psi_m \Psi_n^*\, dq = \delta_{nm}, \tag{3.6}$$

where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$. The fact that the integrals of the products $\Psi_m \Psi_n^*$ with $m \neq n$ vanish is spoken of as the mutual *orthogonality* of the functions Ψ_n. The collection of eigenfunctions Ψ_n thus constitutes a complete system of normalized and mutually orthogonal (or, as one says for short, *orthonormal*) functions.

We introduce the concept of the *mean value* $\bar{f}$ of a quantity f in a given state. Following the standard definition, $\bar{f}$ is the sum of every eigenvalue f_n of the quantity in question, weighted by the respective probability $|a_n|^2$:

$$\bar{f} = \sum_n f_n |a_n|^2. \tag{3.7}$$

We write $\bar{f}$ in the form of an expression that involves not the expansion coefficients of Ψ but the function itself. Since (3.7) contains the products $a_n a_n^*$, it is clear that the desired expression must be bilinear in Ψ and Ψ^*. We introduce a certain mathematical *operator*, which we denote by $\hat{f}$[10], and define as follows. Let $(\hat{f}\Psi)$ denote the result of the action of the operator $\hat{f}$ on the function Ψ. We define $\hat{f}$ by requiring that the integral of $(\hat{f}\Psi)$ multiplied by the conjugate Ψ^* yield $\bar{f}$:

$$\bar{f} = \int \Psi^* (\hat{f}\Psi)\, dq. \tag{3.8}$$

It is easy to see that in the general case the operator $\hat{f}$ is some linear integral operator.[11] Indeed, using the expression (3.5) for a_n, the definition (3.7) of the mean value can be cast as

$$\bar{f} = \sum_n f_n a_n a_n^* = \int \Psi^* \left(\sum_n a_n f_n \Psi_n \right) dq.$$

Comparing with (3.8), we see that the result of the action of the operator $\hat{f}$ on the function Ψ has the form

$$(\hat{f}\Psi) = \sum_n a_n f_n \Psi_n. \tag{3.9}$$

Substituting here the expression (3.5) for a_n, we find that $\hat{f}$ is an integral operator of the form

$$(\hat{f}\Psi) = \int K(q, q')\Psi(q')\, dq', \tag{3.10}$$

where the function $K(q, q')$ (the so-called kernel of the operator) is

$$K(q, q') = \sum_n f_n \Psi_n^*(q')\Psi_n(q). \tag{3.11}$$

Thus, to each physical quantity in quantum mechanics there is assigned a definite linear operator.

[10] We shall everywhere denote operators by letters with a hat.

[11] A linear operator is one possessing the properties: $\hat{f}(\Psi_1 + \Psi_2) = \hat{f}\Psi_1 + \hat{f}\Psi_2$, $\hat{f}(a\Psi) = a\hat{f}\Psi$, where Ψ_1, Ψ_2 are arbitrary functions and a is an arbitrary constant.

From (3.9) it is seen that if Ψ is one of the eigenfunctions Ψ_n (so that all a_n except one are zero), then upon the action of the operator $\hat{f}$ on it, this function is simply multiplied by the eigenvalue f_n[12]:

$$\hat{f}\Psi_n = f_n\Psi_n. \tag{3.12}$$

The eigenfunctions of a physical quantity f are therefore solutions of the equation

$$\hat{f}\Psi = f\Psi,$$

in this equation f denotes a constant; the eigenvalues are those particular values of f that yield solutions obeying the necessary conditions. As will be seen later, the explicit form of operators for the various physical quantities can be established on direct physical grounds, whereupon this operator property enables one to obtain eigenfunctions and eigenvalues by solving the equations $\hat{f}\Psi = f\Psi$.

For a real physical quantity, both the eigenvalues themselves and their mean values in any state are real. This fact places a definite constraint on the properties of the associated operators. Setting the expression (3.8) equal to its own complex conjugate gives

$$\int \Psi^*(\hat{f}\Psi)\,dq = \int \Psi(\hat{f}^*\Psi^*)\,dq, \tag{3.13}$$

where $\hat{f}^*$ denotes the operator complex conjugate to $\hat{f}$[13]. For an arbitrary linear operator such a relation does not, in general, hold, so that it represents a certain restriction imposed on the possible form of the operators $\hat{f}$. For an arbitrary operator $\hat{f}$ one can define, as one says, the *transposed* operator $\tilde{\hat{f}}$, defined so that

$$\int \Phi(\hat{f}\Psi)\,dq = \int \Psi(\tilde{\hat{f}}\Phi)\,dq, \tag{3.14}$$

where Ψ, Φ are two different functions. Choosing as Φ the conjugate function Ψ^*, comparison with (3.13) reveals that

$$\tilde{\hat{f}} = \hat{f}^*. \tag{3.15}$$

An operator that satisfies this condition is called *Hermitian*[14]. Operators representing real physical quantities within the mathematical apparatus of quantum mechanics must therefore be Hermitian.

One may also formally consider complex physical quantities, i.e. quantities possessing complex eigenvalues. Suppose f is one such quantity; one may then form the complex-conjugate quantity f^*, whose eigenvalues are the complex conjugates of the eigenvalues of f. The operator corresponding to the quantity f^* will be denoted by $\hat{f}^+$. It is called the *adjoint* operator to $\hat{f}$ and must, in general, be distinguished from the complex-conjugate

[12]Below, wherever it cannot lead to misunderstanding, we shall omit the parentheses in the expression $(\hat{f}\Psi)$, the operator being understood to act on the expression written after it.

[13]By definition, if for the operator $\hat{f}$ we have $\hat{f}\psi = \varphi$, then the complex-conjugate operator $\hat{f}^*$ is the operator for which $\hat{f}^*\psi^* = \varphi^*$.

[14]For a linear integral operator of the form (3.10), Hermiticity requires that the kernel of the operator satisfy $K(q, q') = K^*(q', q)$.

operator $\hat{f}^*$. Indeed, by definition of the operator $\hat{f}^+$, the mean value of f^* in some state Ψ is

$$\overline{f^*} = \int \Psi^* \hat{f}^+ \Psi \, dq.$$

On the other hand, we have

$$(\bar{f})^* = \left[\int \Psi^* \hat{f} \Psi \, dq \right]^* = \int \Psi \hat{f}^* \Psi^* \, dq = \int \Psi^* \tilde{\hat{f}}^* \Psi \, dq.$$

Equating the two expressions, we find that

$$\hat{f}^+ = \tilde{\hat{f}}^*, \tag{3.16}$$

from which it is clear that $\hat{f}^+$ does not, in general, coincide with $\hat{f}^*$. The condition (3.15) can now be written in the form

$$\hat{f} = \hat{f}^+, \tag{3.17}$$

i.e. the operator of a real physical quantity coincides with its adjoint (Hermitian operators are also called *self-adjoint*).

Let us show how one can directly prove the mutual orthogonality of the eigenfunctions of a Hermitian operator corresponding to different eigenvalues. Let f_n, f_m be two different eigenvalues of the real quantity f, and Ψ_n, Ψ_m the corresponding eigenfunctions:

$$\hat{f} \Psi_n = f_n \Psi_n, \qquad \hat{f} \Psi_m = f_m \Psi_m.$$

Multiplying both sides of the first of these equalities by Ψ_m^*, and the complex conjugate of the second equality by Ψ_n, and subtracting these products termwise from each other, we obtain

$$\Psi_m^* \hat{f} \Psi_n - \Psi_n \hat{f}^* \Psi_m^* = (f_n - f_m) \Psi_n \Psi_m^*.$$

Integrate both sides of this equality over dq. Since $\hat{f}^* = \tilde{\hat{f}}$, by virtue of (3.14) the integral of the left-hand side of the equality vanishes, so that we obtain

$$(f_n - f_m) \int \Psi_n \Psi_m^* \, dq = 0,$$

whence, in view of $f_n \neq f_m$, the required orthogonality property of the functions Ψ_n and Ψ_m follows.

We have been speaking here all along of only a single physical quantity f, whereas we ought to speak, as was noted at the beginning of the section, of a complete system of simultaneously measurable physical quantities. Then we would find that to each of these quantities f, g, ... there corresponds its own operator $\hat{f}$, $\hat{g}$, ... The eigenfunctions Ψ_n correspond to states in which all the quantities under consideration have definite values, i.e. correspond to definite sets of eigenvalues f_n, g_n, ..., and are joint solutions of the system of equations

$$\hat{f} \Psi = f \Psi, \qquad \hat{g} \Psi = g \Psi, \ldots$$

§ 4. Addition and Multiplication of Operators

If $\hat{f}$ and $\hat{g}$ are the operators corresponding to two physical quantities f and g, then $\hat{f} + \hat{g}$ corresponds to their sum $f + g$. In quantum mechanics, however, the meaning of adding different physical quantities differs essentially according as the quantities are or are not measurable simultaneously. If f and g are simultaneously measurable, the operators $\hat{f}$ and $\hat{g}$ have common eigenfunctions. These are also eigenfunctions of $\hat{f} + \hat{g}$, whose eigenvalues are the sums $f_n + g_n$.

If f and g cannot have definite values simultaneously, their sum $f + g$ has a more restricted meaning. All that can be stated is that its mean value in an arbitrary state equals the sum of the separate mean values of the two terms:

$$\overline{f + g} = \bar{f} + \bar{g}. \tag{4.1}$$

The eigenvalues and eigenfunctions of $\hat{f} + \hat{g}$, on the other hand, will in general have no relation to those of f and g. Clearly, when $\hat{f}$ and $\hat{g}$ are Hermitian, so is $\hat{f} + \hat{g}$; its eigenvalues are therefore real and are the eigenvalues of the new quantity $f + g$ defined in this way.

We note the following theorem. Let f_0 and g_0 be the least eigenvalues of f and g, and let $(f + g)_0$ denote the corresponding quantity for $f + g$. Then

$$(f + g)_0 \geq f_0 + g_0. \tag{4.2}$$

The equality sign applies if f and g are simultaneously measurable. The proof follows from the evident fact that a quantity's mean value is always at least its least eigenvalue. In the state in which $(f + g)$ has the value $(f + g)_0$, we have $\overline{(f + g)} = (f + g)_0$; on the other hand, $\overline{(f + g)} = \bar{f} + \bar{g} \geq f_0 + g_0$, which gives (4.2).

Suppose again that f and g are simultaneously measurable. Besides their sum, we may introduce their product as a quantity whose eigenvalues are the products of the eigenvalues of f and g. The corresponding operator is readily seen to act by applying one operator to a function and then the other. Mathematically it is written as the product of $\hat{f}$ and $\hat{g}$. Indeed, if Ψ_n are common eigenfunctions of $\hat{f}$ and $\hat{g}$, then

$$\hat{f}\hat{g}\Psi_n = \hat{f}(\hat{g}\Psi_n) = \hat{f}g_n\Psi_n = g_n\hat{f}\Psi_n = g_n f_n \Psi_n$$

(the symbol $\hat{f}\hat{g}$ denotes the operator which acts on a function Ψ first by applying $\hat{g}$ to Ψ, and then by applying $\hat{f}$ to the function $\hat{g}\Psi$). We could equally well have used $\hat{g}\hat{f}$, whose factors occur in the opposite order. Plainly, the actions of these two operators on the functions Ψ_n give the same result. Since every wave function Ψ is expandable in the functions Ψ_n, it follows that $\hat{f}\hat{g}$ and $\hat{g}\hat{f}$ yield the same result when acting on an arbitrary function. This fact may be written symbolically as $\hat{f}\hat{g} = \hat{g}\hat{f}$, or

$$\hat{f}\hat{g} - \hat{g}\hat{f} = 0. \tag{4.3}$$

Two such operators $\hat{f}$ and $\hat{g}$ are said to *commute* with one another. We have therefore reached an important result: if two quantities f and g can have definite values simultaneously, their operators commute.

The converse theorem can also be proved (see § 11): when $\hat{f}$ and $\hat{g}$ commute, all their eigenfunctions can be chosen in common, which physically means that the corresponding physical quantities are simultaneously measurable. Thus commutation of the operators is a necessary and sufficient condition for simultaneous measurability of physical quantities.

Raising an operator to a power is a special case of multiplying operators. It follows from what has been said that the eigenvalues of $\hat{f}^p$ (where p is an integer) are the eigenvalues of $\hat{f}$ raised to the same pth power. More generally, any function of an operator, $\varphi(\hat{f})$, may be defined as the operator whose eigenvalues are the same function $\varphi(f)$ of the eigenvalues of $\hat{f}$. When $\varphi(f)$ has a Taylor-series expansion, this expansion reduces the action of $\varphi(\hat{f})$ to the actions of the various powers $\hat{f}^p$.

In particular, $\hat{f}^{-1}$ is called the operator *inverse* to $\hat{f}$. Clearly, successive application of $\hat{f}$ and $\hat{f}^{-1}$ to an arbitrary function leaves it unchanged; that is, $\hat{f}\hat{f}^{-1} = \hat{f}^{-1}\hat{f} = 1$. When f and g are not simultaneously measurable, their product has no such direct meaning. This is already apparent because $\hat{f}\hat{g}$ is then not Hermitian and consequently cannot correspond to a real physical quantity. Indeed, by the definition of the transposed operator,

$$\int \Psi \hat{f}\hat{g}\Phi \, dq = \int \Psi \hat{f}(\hat{g}\Phi)\, dq = \int (\hat{g}\Phi)(\tilde{f}\Psi)\, dq.$$

Here $\tilde{f}$ acts only on Ψ, while $\hat{g}$ acts on Φ; the integrand is therefore simply the product of the two functions $\hat{g}\Phi$ and $\tilde{f}\Psi$. Applying the definition of the transposed operator once more gives

$$\int \Psi \hat{f}\hat{g}\Phi \, dq = \int (\tilde{f}\Psi)(\hat{g}\Phi)\, dq = \int \Phi\tilde{g}\,\tilde{f}\Psi\, dq.$$

We have thus obtained an integral in which Ψ and Φ have exchanged places relative to the original one. In other words, $\tilde{g}\,\tilde{f}$ is the operator transposed to $\hat{f}\hat{g}$, so we may write

$$\widetilde{fg} = \tilde{g}\,\tilde{f}, \tag{4.4}$$

that is, the transpose of the product $\hat{f}\hat{g}$ is the product of the transposed factors written in reverse order. Taking the complex conjugate of both sides of (4.4), we find

$$(\hat{f}\hat{g})^+ = \hat{g}^+\hat{f}^+. \tag{4.5}$$

If both $\hat{f}$ and $\hat{g}$ are Hermitian, then $(\hat{f}\hat{g})^+ = \hat{g}\hat{f}$. Hence $\hat{f}\hat{g}$ is Hermitian only when the factors $\hat{f}$ and $\hat{g}$ commute.

From the products $\hat{f}\hat{g}$ and $\hat{g}\hat{f}$ of two noncommuting Hermitian operators one can nevertheless form another Hermitian operator, their *symmetrized product*

$$\frac{1}{2}(\hat{f}\hat{g} + \hat{g}\hat{f}). \tag{4.6}$$

It is also readily verified that the difference $\hat{f}\hat{g} - \hat{g}\hat{f}$ is an "anti-Hermitian" operator (that is, its transpose is equal to the negative of its complex conjugate). Multiplication by i makes it Hermitian; hence

$$i(\hat{f}\hat{g} - \hat{g}\hat{f}) \tag{4.7}$$

is likewise a Hermitian operator.

For brevity, later on we shall sometimes use the notation

$$\{\hat{f}, \hat{g}\} = \hat{f}\hat{g} - \hat{g}\hat{f} \tag{4.8}$$

for the so-called *commutator* of the operators. It is easy to verify the relation

$$\{\hat{f}\hat{g}, \hat{h}\} = \{\hat{f}, \hat{h}\}\hat{g} + \hat{f}\{\hat{g}, \hat{h}\}. \tag{4.9}$$

Notice that $\{\hat{f}, \hat{h}\} = 0$ and $\{\hat{g}, \hat{h}\} = 0$ do not in general imply that $\hat{f}$ and $\hat{g}$ commute with each other.

§ 5. Continuous spectrum

All the relations obtained in § 3, § 4 for the properties of discrete-spectrum eigenfunctions extend without difficulty to a continuous spectrum of eigenvalues.

Let f be a physical quantity possessing a continuous spectrum. We shall denote its eigenvalues simply by the same letter f without an index; the corresponding eigenfunctions will be written Ψ_f. In the same way that an arbitrary wave function Ψ admits the series expansion (3.2) in eigenfunctions of a discrete-spectrum quantity, it can also be expanded — this time as an integral — over the complete system of eigenfunctions of a continuous-spectrum quantity. Such an expansion takes the form

$$\Psi(q) = \int a_f \Psi_f(q)\, df, \tag{5.1}$$

where the integration is taken over the whole domain of values that the quantity f can assume.

More complicated than in the discrete-spectrum case is the question of normalizing the continuous-spectrum eigenfunctions. As we shall see below, the requirement that the integral of the squared modulus equal unity cannot be satisfied here. Instead, let us set ourselves the aim of normalizing the functions Ψ_f in such a way that $|a_f|^2\, df$ represents the probability for the physical quantity under consideration, in the state described by the wave function Ψ, to have a value in the given interval between f and $f + df$. Since the sum of the probabilities of all possible values of f must be equal to unity, we have

$$\int |a_f|^2\, df = 1 \tag{5.2}$$

(analogously to relation (3.3) for the discrete spectrum).

Proceeding exactly as we did in deriving formula (3.5), and using the same arguments, we write, on one hand,

$$\int \Psi\Psi^*\, dq = \int |a_f|^2\, df,$$

and, on other hand,

$$\int \Psi\Psi^*\, dq = \iint a_f^* \Psi_f^* \Psi\, df\, dq.$$

By comparing the two expressions we find the formula that determines the expansion coefficients,

$$a_f = \int \Psi(q)\Psi_f^*(q)\, dq, \tag{5.3}$$

exactly analogous to (3.5).

To derive the normalization condition we now substitute (5.1) into (5.3):

$$a_f = \int a_{f'} \left(\int \Psi_{f'} \Psi_f^* \, dq \right) df'.$$

This relation must hold for arbitrary a_f and must therefore be satisfied identically. For this to be so it is first of all necessary that the coefficient of $a_{f'}$ under the integral sign (i.e. the integral $\int \Psi_{f'} \Psi_f^* \, dq$) vanish for all $f' \neq f$. For $f' = f$ this coefficient must become infinite (otherwise the integral over df' would simply be equal to zero). Thus the integral $\int \Psi_{f'} \Psi_f^* \, dq$ is a function of the difference $f - f'$ which vanishes for non-zero values of the argument and becomes infinite when the argument is zero. We denote this function by $\delta(f' - f)$:

$$\int \Psi_{f'} \Psi_f^* \, dq = \delta(f' - f). \tag{5.4}$$

The manner in which the function $\delta(f' - f)$ becomes infinite at $f' - f = 0$ is determined by the requirement that there should be

$$\int \delta(f' - f) a_{f'} \, df' = a_f.$$

Clearly, for this to hold it is necessary that

$$\int \delta(f' - f) \, df' = 1.$$

A function defined in this way is called the *delta-function*[15]. Let us write out once more the formulas that define it. We have

$$\delta(x) = 0 \quad \text{at } x \neq 0, \qquad \delta(0) = \infty, \tag{5.5}$$

and moreover in such a way that

$$\int_{-\infty}^{+\infty} \delta(x) \, dx = 1. \tag{5.6}$$

As the limits of integration one may write any other pair between which the point $x = 0$ lies. If $f(x)$ is a function continuous at $x = 0$, then

$$\int_{-\infty}^{+\infty} \delta(x) f(x) \, dx = f(0). \tag{5.7}$$

In more general form this formula can be written as

$$\int \delta(x - a) f(x) \, dx = f(a), \tag{5.8}$$

where the region of integration includes the point $x = a$, and $f(x)$ is continuous at $x = a$. It is also obvious that the delta-function is even, i.e.

$$\delta(-x) = \delta(x). \tag{5.9}$$

[15]The delta-function was introduced into theoretical physics by *Dirac* (*P. A. M. Dirac*).

Finally, writing

$$\int_{-\infty}^{\infty} \delta(\alpha x)\, dx = \int_{-\infty}^{\infty} \delta(y)\frac{dy}{|\alpha|} = \frac{1}{|\alpha|},$$

we arrive at the conclusion that

$$\delta(\alpha x) = \frac{1}{|\alpha|}\delta(x), \tag{5.10}$$

where α is any constant. Formula (5.4) expresses the normalization rule for the eigenfunctions of the continuous spectrum; it replaces condition (3.6) of the discrete spectrum. We see that the functions Ψ_f and $\Psi_{f'}$ with $f \neq f'$ remain orthogonal to each other. The integrals of the squares $|\Psi_f|^2$ of the functions of the continuous spectrum, however, diverge.

The functions $\Psi_f(q)$ satisfy one further relation, similar to (5.4). To derive it we substitute (5.3) into (5.1), which gives

$$\Psi(q) = \int \Psi(q')\left(\int \Psi_f^*(q')\Psi_f(q)\, df\right) dq',$$

whence we immediately conclude that one must have

$$\int \Psi_f^*(q')\Psi_f(q)\, df = \delta(q - q'). \tag{5.11}$$

An analogous relation can, of course, be derived for the discrete spectrum, where it takes the form

$$\sum_n \Psi_n^*(q')\Psi_n(q) = \delta(q' - q). \tag{5.12}$$

On comparing the pair of formulas (5.1) and (5.4) with the pair (5.3) and (5.11), we see that, on the one hand, the functions $\Psi_f(q)$ effect the expansion of the function $\Psi(q)$ with the expansion coefficients a_f, while, on the other hand, formula (5.3) can be regarded as a completely analogous expansion of the function $a_f \equiv a(f)$ in the functions $\Psi_f^*(q)$, with $\Psi(q)$ playing the role of the expansion coefficients. The function $a(f)$, like $\Psi(q)$, determines the state of the system completely; it is spoken of as the wave function in the *f-representation* (and the function $\Psi(q)$ as the wave function in the *q-representation*). Just as $|\Psi(q)|^2$ determines the probability for the system to have coordinates in a given interval dq, so $|a(f)|^2$ determines the probability of values of the quantity f in a given interval df. The functions $\Psi_f(q)$, for their part, are, on the one hand, the eigenfunctions of the quantity f in the q-representation, while their complex conjugates $\Psi_f^*(q)$ serve as eigenfunctions of the coordinate q in the f-representation.

Suppose $\varphi(f)$ is a function of the quantity f, with φ and f standing in one-to-one correspondence. Every function $\Psi_f(q)$ may then equally be viewed as an eigenfunction of the quantity φ. In doing so, however, the normalization of these functions must be changed. Indeed, the eigen- functions $\Psi_\varphi(q)$ of the quantity φ must be normalized by the condition

$$\int \Psi_{\varphi(f')}\Psi_{\varphi(f)}^*\, dq = \delta[\varphi(f') - \varphi(f)],$$

whereas the functions Ψ_f are normalized by condition (5.4). The argument of the delta-function vanishes at $f' = f$. For f' close to f we have

$$\varphi(f') - \varphi(f) = \frac{d\varphi(f)}{df}(f' - f).$$

Making use of expression (5.10), one can write[16]

$$\delta[\varphi(f') - \varphi(f)] = \frac{1}{|d\varphi(f)/df|}\delta(f' - f). \tag{5.13}$$

Comparison of (5.13) with (5.4) now shows that the functions Ψ_φ and Ψ_f are related to each other by the relation

$$\Psi_{\varphi(f)} = \frac{1}{\sqrt{|d\varphi(f)/df|}}\Psi_f. \tag{5.14}$$

There exist physical quantities which, over some domain of their values, possess a discrete spectrum, and over another possess a continuous one. For the eigenfunctions of such a quantity, of course, all the same relations that were derived in this and the preceding sections continue to hold. It need only be noted that the complete system of functions is formed by the collection of the eigenfunctions of both spectra together. Therefore the expansion of an arbitrary wave function in the eigenfunctions of such a quantity has the form

$$\Psi(q) = \sum_n a_n \Psi_n(q) + \int a_f \Psi_f(q)\, df, \tag{5.15}$$

where the summation runs over the discrete spectrum, and the integration over the entire continuous spectrum.

An example of a quantity possessing a continuous spectrum is the coordinate q itself. One readily sees that the associated operator reduces to multiplication by q. Indeed, since the square $|\Psi(q)|^2$ governs the probability of different coordinate values, the mean value of the coordinate is

$$\bar{q} = \int q|\Psi|^2\, dq = \int \Psi^* q \Psi\, dq.$$

Comparing this expression with the definition of operators according to (3.8), we see that[17]

$$\hat{q} = q. \tag{5.16}$$

The eigenfunctions of this operator must be determined, according to the general rule, by the equation $q\Psi_{q_0} = q_0\Psi_{q_0}$, where q_0 is used temporarily to denote concrete values of

[16] In general, if $\varphi(x)$ is a single-valued function (its inverse, however, may be many-valued), then the formula

$$\delta[\varphi(x)] = \sum_i \frac{1}{|\varphi'(\alpha_i)|}\delta(x - \alpha_i), \tag{5.13a}$$

holds, where α_i are the roots of the equation $\varphi(x) = 0$.

[17] In what follows we shall agree, for simplicity of notation, to write operators that reduce to multiplication by some quantity everywhere simply as that quantity itself.

the coordinate, in distinction from the variable q. Since this equality can be satisfied either when $\Psi_{q_0} = 0$ or when $q = q_0$, it is clear that the normalized eigenfunctions are[18]

$$\Psi_{q_0} = \delta(q - q_0). \tag{5.17}$$

§ 6. The Limiting Transition

Classical mechanics is contained in quantum mechanics as a limiting case. We must determine how this limiting transition takes place.

In quantum mechanics a wave function describes the electron, governing the various possible values of its coordinate; thus far we know only that this function solves a certain linear partial differential equation. Classical mechanics, by contrast, treats the electron as a material particle moving along a trajectory fixed completely by the equations of motion. A relation analogous in a certain sense to that between quantum and classical mechanics occurs in electrodynamics between wave optics and geometrical optics. Wave optics describes electromagnetic waves by electric- and magnetic-field vectors satisfying a particular system of linear differential equations (Maxwell's equations). Geometrical optics, however, describes the propagation of light along definite paths—rays. This analogy leads one to conclude that the quantum-to-classical limiting transition is analogous to that from wave optics to geometrical optics.

We now recall how this latter transition is carried out mathematically (see II, § 53). Let u be any component of the field in an electromagnetic wave. It may be written $u = ae^{i\varphi}$, with real amplitude a and phase φ (the phase is called the eikonal in geometrical optics). The geometrical-optics limit corresponds to short wavelengths; mathematically, this means that φ changes greatly over short distances. In particular, its absolute magnitude may be regarded as large.

Correspondingly, assume that the classical limiting case in quantum mechanics is represented by wave functions of the form $\Psi = ae^{i\varphi}$, where a varies slowly and φ takes large values. In mechanics, as is known, a particle trajectory can be found from a variational principle according to which the so-called action S of the mechanical system must be a minimum (the principle of least action). In geometrical optics the ray path is instead fixed by Fermat's principle: the "optical path length" of the ray, that is, the difference between its phases at the end and at the beginning of the path, must be a minimum.

This analogy allows us to conclude that, in the classical limiting case, the phase φ of the wave function must be proportional to the mechanical action S of the physical system under consideration: $S = \text{const} \cdot \varphi$. The proportionality factor is called *Planck's constant* and is denoted by $\hbar$[19]. Since φ is dimensionless, it has the dimensions of action, and its value is

$$\hbar = 1{,}055 \cdot 10^{-27}\ \text{erg} \cdot \text{s}.$$

[18]The expansion coefficients of an arbitrary function Ψ in these eigenfunctions are $a_{q_0} = \int \Psi(q)\,\delta(q - q_0)\,dq = \Psi(q_0)$. The probability of values of the coordinate in a given interval dq_0 is $|a_{q_0}|^2\,dq_0 = |\Psi(q_0)|^2\,dq_0$, as it should be.

[19]It was introduced into physics by *Planck* (*M. Planck*, 1900). The constant $\hbar$ used throughout this book is, strictly speaking, Planck's constant h divided by 2π (Dirac's notation).

Thus the wave function of an "almost classical" (or, as it is called, *quasiclassical*) physical system has the form

$$\Psi = ae^{iS/\hbar}. \tag{6.1}$$

Planck's constant has a fundamental part in every quantum phenomenon. Its relative magnitude (in comparison with other quantities having the same dimensions) determines the "degree of quantumness" of a given physical system. Passage from quantum to classical mechanics corresponds to a large phase and can formally be described by taking the limit $\hbar \to 0$ (just as the passage from wave to geometrical optics corresponds to the zero-wavelength limit, $\lambda \to 0$).

We have established the limiting form of the wave function, but must still ask how it is related to classical motion along a trajectory. In general, motion described by a wave function does not at all turn into motion along one fixed trajectory. Its connection with classical motion is this: if the wave function, and hence the probability distribution of the coordinates, is specified at an initial instant, the distribution thereafter "moves" as required by the laws of classical mechanics (see the end of § 17 for details).

To obtain motion along a definite trajectory one must begin with a wave function of a special kind, appreciably different from zero only within a very small region of space (a so-called *wave packet*); the dimensions of this region may be made to tend to zero together with $\hbar$. One can then state that, in the quasiclassical case, the wave packet moves through space along the particle's classical trajectory.

Finally, in the limit, quantum-mechanical operators must reduce simply to multiplication by their corresponding physical quantities.

§ 7. The Wave Function and Measurements

Let us return to the measurement process, whose properties were discussed qualitatively in § 1, and show how those properties are connected with the mathematical apparatus of quantum mechanics.

Consider a system made up of two parts: a classical apparatus and an electron (treated as a quantum object). Measurement consists in bringing these two parts into interaction. The apparatus consequently passes from its initial state into some other state, and from this change we draw conclusions about the state of the electron. States of the apparatus are distinguished by the values of a physical quantity characterizing it (or quantities)—the apparatus "reading." Denote this quantity provisionally by g, and its eigenvalues by g_n. Because the apparatus is classical, these values generally range over a continuum; solely to simplify the notation in the formulas below, however, we shall suppose that the spectrum is discrete. The states of the apparatus are described by quasiclassical wave functions, which we denote by $\Phi_n(\xi)$; the index n corresponds to the reading g_n, and ξ stands collectively for its coordinates. That the apparatus is classical manifests itself in the circumstance that at each instant one can assert with certainty that it occupies one of the known states Φ_n, with some definite value of g. Such a statement would of course be false for a quantum system.

Let $\Phi_0(\xi)$ be the wave function of the apparatus in its initial state (before measurement), and let $\Psi(q)$ be an arbitrary normalized initial wave function of the electron (q denotes its coordinates). These functions describe apparatus and electron independently, so the initial wave function of the complete system is the product

$$\Psi(q)\Phi_0(\xi). \tag{7.1}$$

The apparatus and electron then interact. In principle, the equations of quantum mechanics allow the time variation of the system's wave function to be followed. After the measurement it will plainly no longer be a product of functions of ξ and q. Expanding it in the eigenfunctions Φ_n of the apparatus (which form a complete system of functions), we obtain a sum of the form

$$\sum_n A_n(q)\Phi_n(\xi), \tag{7.2}$$

where the $A_n(q)$ are certain functions of q.

At this point the "classical nature" of the apparatus enters, together with the double role of classical mechanics as both a limiting case and a basis of quantum mechanics. As already noted, because the apparatus is classical, the quantity g (its "reading") has a definite value at every instant. It follows that after measurement the actual state of apparatus plus electron is described not by the whole sum (7.2), but by only the one term corresponding to the apparatus reading g_n:

$$A_n(q)\Phi_n(\xi). \tag{7.3}$$

Consequently, $A_n(q)$ is proportional to the electron's wave function after measurement. It is not itself that wave function, as is already evident from the fact that $A_n(q)$ is not normalized. It contains both information about the properties of the state produced and the probability, determined by the system's initial state, that the apparatus will give its nth reading.

By the linearity of the equations of quantum mechanics, the relation between $A_n(q)$ and the initial electron wave function $\Psi(q)$ is in general given by a linear integral operator,

$$A_n(q) = \int K_n(q, q')\Psi(q')\, dq', \tag{7.4}$$

whose kernel $K_n(q, q')$ characterizes the measurement process in question.

We suppose that the measurement under consideration yields a complete description of the electron's state. In other words (see § 1), in the state produced, the probabilities for all quantities must be independent of the electron's preceding state (before measurement). Mathematically, this means that the form of the functions $A_n(q)$ must be fixed by the measurement process itself and must not depend on the initial electron wave function $\Psi(q)$.

The A_n must therefore have the form

$$A_n(q) = a_n\varphi_n(q), \tag{7.5}$$

where the φ_n are certain functions that we shall take to be normalized, while only the constants a_n depend on the initial state $\Psi(q)$. In the integral relation (7.4), this corresponds to a kernel $K_n(q, q')$ which factors into functions of q and q' separately:

$$K_n(q, q') = \varphi_n(q)\Psi_n^*(q'). \tag{7.6}$$

The linear dependence of the constants a_n on $\Psi(q)$ is then expressed by formulas of the form

$$a_n = \int \Psi(q)\Psi_n^*(q)\,dq, \tag{7.7}$$

where the $\Psi_n(q)$ are certain functions fixed by the measurement process.

The functions $\varphi_n(q)$ are the normalized wave functions of the electron after measurement. We thus see how the theory's mathematical formalism represents the possibility of producing, by a measurement, an electron state described by a definite wave function. When a measurement is made on an electron having a specified wave function $\Psi(q)$, the constants a_n have a simple physical meaning: by the general rules, $|a_n|^2$ is the probability that the measurement gives the nth result. The probabilities of all results sum to unity:

$$\sum_n |a_n|^2 = 1. \tag{7.8}$$

For an arbitrary normalized function $\Psi(q)$, the validity of (7.7) and (7.8) is equivalent (compare § 3) to the statement that $\Psi(q)$ can be expanded in the functions $\Psi_n(q)$. Hence the functions $\Psi_n(q)$ form a complete normalized and mutually orthogonal set.

If the electron's initial wave function coincides with one of the functions $\Psi_n(q)$, the corresponding constant a_n is plainly unity and all the others vanish. Put differently, if the measurement is performed on an electron in the state $\Psi_n(q)$, it will with certainty give a definite (nth) result.

Taken together, these properties of the functions $\Psi_n(q)$ demonstrate that they are eigenfunctions of a certain physical quantity that characterizes the electron (call it f), and the measurement being considered may be called a measurement of that quantity.

It is essential that, in general, the functions $\Psi_n(q)$ do not coincide with the functions $\varphi_n(q)$ (indeed, the latter need not even be mutually orthogonal or form an eigenfunction system of any operator). This fact expresses first of all the non-repeatability of measurement results in quantum mechanics. If the electron was in the state $\Psi_n(q)$, a measurement of f made upon it will reveal the value f_n with certainty. After the measurement, however, the electron is in a different state $\varphi_n(q)$, in which f need no longer have any definite value. Thus, if a second measurement were made immediately after the first, the value found for f would not coincide with the result of the first measurement[20]. To predict (in the sense of calculating the probability of) the result of a repeated measurement when the first result is known, one takes from the first measurement the wave function $\varphi_n(q)$ of the state it produced, and from the second the wave function $\Psi_m(q)$ of the state whose probability is sought. More explicitly, the equations of quantum mechanics determine a wave function $\varphi_n(q,t)$ which equals $\varphi_n(q)$ at the time of the first measurement. If the second measurement is made at time t, the probability of its mth result is the squared modulus of the integral $\int \varphi_n(q,t)\Psi_m^*(q)\,dq$.

We see that measurement in quantum mechanics is "two-faced": its roles with respect to past and future do not coincide. Toward the past it "verifies" the probabilities of the

[20]There is, however, an important exception to the non-repeatability of measurements: coordinate is the one quantity whose measurement is repeatable. Two measurements of an electron's coordinate made at a sufficiently short interval must give nearby values; otherwise the electron would have infinite velocity. The mathematical origin of this exception lies in the commutativity of the coordinate with the electron–apparatus interaction energy operator, which (in the nonrelativistic theory) depends solely on coordinates.

various possible results predicted from the state created by the preceding measurement. Toward the future it creates a new state (see also § 44). A profound irreversibility is therefore inherent in the measurement process itself.

This irreversibility has important consequences of principle. As will be seen later (see the end of § 18), the fundamental equations of quantum mechanics by themselves are symmetric under reversal of the sign of time; in this respect quantum mechanics does not differ from classical mechanics. The irreversibility of measurement, however, makes the two directions of time physically inequivalent in quantum phenomena, and thus produces a distinction between future and past.

CHAPTER II

Energy and Momentum

§ 8. The Hamiltonian

In quantum mechanics the wave function Ψ fully specifies the state of a physical system. This means that prescribing Ψ at some instant not only describes all the properties of the system at that instant, but also determines its behavior at every later instant—of course, only to the degree of completeness that quantum mechanics permits at all. Mathematically, this means that at each instant the value of the time derivative $\partial\Psi/\partial t$ must be determined by the value of Ψ itself at that same instant, and, by the superposition principle, this dependence must be linear. In the most general form we may write

$$i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi, \tag{8.1}$$

where $\hat{H}$ is a certain linear operator; the factor $i\hbar$ has been introduced here for a reason that will become clear below.

Since the integral $\int \Psi^*\Psi\, dq$ is constant in time, we have

$$\frac{d}{dt}\int |\Psi|^2\, dq = \int \frac{\partial\Psi^*}{\partial t}\Psi\, dq + \int \Psi^*\frac{\partial\Psi}{\partial t}\, dq = 0.$$

Substituting (8.1) here and applying the definition of the transposed operator in the first integral, we obtain (after suppressing the common factor $i/\hbar$)

$$\begin{aligned}\int \Psi\hat{H}^*\Psi^*\, dq - \int \Psi^*\hat{H}\Psi\, dq &= \int \Psi^*\tilde{H}^*\Psi\, dq - \int \Psi^*\hat{H}\Psi\, dq\\ &= \int \Psi^*(\tilde{H}^* - \hat{H})\Psi\, dq = 0.\end{aligned}$$

Because this equality must hold for an arbitrary function Ψ, it follows that identically $\hat{H}^+ = \hat{H}$; the operator $\hat{H}$ is therefore Hermitian. Let us determine the physical quantity to which it corresponds. We use the limiting expression (6.1) for the wave function and write

$$\frac{\partial\Psi}{\partial t} = \frac{i}{\hbar}\frac{\partial S}{\partial t}\Psi$$

(the slowly varying amplitude a need not be differentiated). Comparing this equality with definition (8.1), we see that in the limiting case $\hat{H}$ reduces to multiplication by $-\partial S/\partial t$. Thus this last quantity is the physical quantity into which the Hermitian operator $\hat{H}$ passes.

But $-\partial S/\partial t$ is precisely the Hamilton function H of the mechanical system. Hence $\hat{H}$ is the operator corresponding in quantum mechanics to the Hamilton function. This operator is termed the *Hamiltonian operator*, or simply the *Hamiltonian* of the system.

Once the form of the Hamiltonian is known, equation (8.1) fixes the wave functions of the given physical system. This fundamental equation of quantum mechanics is called the *wave equation*.

§ 9. Time Differentiation of Operators

In quantum mechanics, the concept of the time derivative of a physical quantity cannot be defined with the sense that it carries in classical mechanics. Indeed, the definition of the derivative in classical mechanics is connected with considering the values of the quantity at two close but different instants. But in quantum mechanics a quantity that has a definite value at some instant has no definite value at all at subsequent instants; this was discussed in more detail in § 1.

For this reason the concept of the time derivative has to be given a different definition in quantum mechanics. A natural course is to define the derivative $\dot{f}$ of a quantity f as that quantity whose mean value equals the time derivative of the mean value $\bar{f}$. Thus, by definition, we have

$$\bar{\dot{f}} = \dot{\bar{f}}. \tag{9.1}$$

Taking this definition as the starting point, it is not hard to derive an expression for the quantum-mechanical operator $\hat{\dot{f}}$ that corresponds to the quantity $\dot{f}$:

$$\begin{aligned}\bar{\dot{f}} = \dot{\bar{f}} = \frac{d}{dt}\int \Psi^* \hat{f}\Psi\, dq &= \\ &= \int \Psi^* \frac{\partial \hat{f}}{\partial t}\Psi\, dq + \int \frac{\partial \Psi^*}{\partial t}\hat{f}\Psi\, dq + \int \Psi^* \hat{f}\frac{\partial \Psi}{\partial t}\, dq.\end{aligned}$$

In this expression $\partial\hat{f}/\partial t$ stands for the operator that results from differentiating the operator $\hat{f}$ with respect to time, on which the latter may depend parametrically. Substituting for the derivatives $\partial\Psi/\partial t$, $\partial\Psi^*/\partial t$ their expressions according to (8.1), we obtain

$$\bar{\dot{f}} = \int \Psi^* \frac{\partial \hat{f}}{\partial t}\Psi\, dq + \frac{i}{\hbar}\int (\hat{H}^*\Psi^*)\hat{f}\Psi\, dq - \frac{i}{\hbar}\int \Psi^* \hat{f}(\hat{H}\Psi)\, dq.$$

Since the operator $\hat{H}$ is Hermitian,

$$\int (\hat{H}^*\Psi^*)(\hat{f}\Psi)\, dq = \int \Psi^* \hat{H}\hat{f}\Psi\, dq;$$

thus, we have

$$\bar{\dot{f}} = \int \Psi^* \left(\frac{\partial \hat{f}}{\partial t} + \frac{i}{\hbar}\hat{H}\hat{f} - \frac{i}{\hbar}\hat{f}\hat{H}\right)\Psi\, dq.$$

Since, on the other hand, by the definition of mean values we must have $\bar{\dot{f}} = \int \Psi^* \hat{\dot{f}}\Psi\, dq$, it is seen that the expression in parentheses under the integral is the required operator $\hat{\dot{f}}$[1]:

$$\hat{\dot{f}} = \frac{\partial \hat{f}}{\partial t} + \frac{i}{\hbar}(\hat{H}\hat{f} - \hat{f}\hat{H}). \tag{9.2}$$

[1] In classical mechanics, for the total time derivative of a quantity f that is a function of the generalized

If the operator $\hat{f}$ does not depend explicitly on time, then, up to a factor, $\hat{\dot{f}}$ reduces to the commutator of the operator $\hat{f}$ with the Hamiltonian.

The physical quantities whose operators have no explicit dependence on time and which, moreover, commute with the Hamiltonian—so that $\hat{\dot{f}} = 0$—form a very important category. Such quantities are called *conserved*. For these, $\bar{\dot{f}} = \dot{\bar{f}} = 0$, i.e. $\bar{f} = \text{const}$. Put differently, the quantity's mean value stays constant in time. One can assert, moreover, that if the quantity f has a definite value in a given state (i.e. if the wave function is an eigenfunction of the operator $\hat{f}$), then at later instants as well it will have a definite value—and the very same one.

§ 10. Stationary states

The Hamiltonian of a closed system (and also of a system in a constant, but not a variable, external field) cannot contain time explicitly. This follows because all instants of time are equivalent for such a physical system. Since, on the other hand, every operator commutes with itself, we conclude that the Hamilton function is conserved for systems not situated in a variable external field. A conserved Hamilton function is called the energy. The meaning of energy conservation in quantum mechanics is that if the energy has a definite value in a given state, this value remains constant in time.

States possessing definite energy values are called the *stationary states* of the system. They are described by wave functions Ψ_n that are eigenfunctions of the Hamiltonian operator, satisfying $\hat{H}\Psi_n = E_n\Psi_n$, where E_n are the energy eigenvalues. Accordingly, the wave equation (8.1) for the function Ψ_n,

$$i\hbar\frac{\partial\Psi_n}{\partial t} = \hat{H}\Psi_n = E_n\Psi_n,$$

can be integrated directly with respect to time, giving

$$\Psi_n = \exp\left(-\frac{i}{\hbar}E_n t\right)\psi_n(q), \tag{10.1}$$

coordinates q_i and momenta p_i of the system, we have

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \sum_i \left(\frac{\partial f}{\partial q_i}\dot{q}_i + \frac{\partial f}{\partial p_i}\dot{p}_i\right).$$

Substituting, according to Hamilton's equations, $\dot{q}_i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial q_i$, we obtain

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + [H, f], \qquad [H, f] \equiv \sum_i \left(\frac{\partial f}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial H}{\partial q_i}\right);$$

$[H, f]$ is the so-called *Poisson bracket* for the quantities f and H (see I, § 42). Comparing with expression (9.2), we see that in the transition to the classical limit the operator $i(\hat{H}\hat{f} - \hat{f}\hat{H})$ becomes zero to the first approximation, as it should, and in the next approximation (in $\hbar$) becomes the quantity $\hbar[H, f]$.

This result is also valid for any two quantities f and g: in the limit the operator $i(\hat{f}\hat{g} - \hat{g}\hat{f})$ becomes the quantity $\hbar[f, g]$, where $[f, g]$ is the Poisson bracket

$$[f, g] \equiv \sum_i \left(\frac{\partial g}{\partial q_i}\frac{\partial f}{\partial p_i} - \frac{\partial g}{\partial p_i}\frac{\partial f}{\partial q_i}\right).$$

This is a consequence of the fact that we can always, formally, conceive of a system whose Hamiltonian coincides with $\hat{g}$.

where ψ_n is a function of the coordinates alone. This determines the time dependence of the wave functions of stationary states.

We shall use the lower-case letter ψ for stationary-state wave functions without the time factor. These functions and the energy eigenvalues themselves are determined by

$$\hat{H}\psi = E\psi. \tag{10.2}$$

Among all stationary states, the one possessing the lowest energy is termed the *normal*, or *ground*, state of the system.

The expansion of any wave function Ψ over the stationary-state wave functions has the form

$$\Psi = \sum_n a_n \exp\left(-\frac{i}{\hbar}E_n(t)\right)\psi_n(q). \tag{10.3}$$

As usual, the squares $|a_n|^2$ give the probabilities of the various energy values of the system.

The coordinate probability distribution in a stationary state is determined by $|\Psi_n|^2 = |\psi_n|^2$ and is therefore independent of time. The same is true of the mean value $\bar{f} = \int \Psi_n^* \hat{f} \Psi_n, dq = \int \psi_n^* \hat{f} \psi_n, dq$ of any physical quantity f whose operator does not depend explicitly on time.

As stated above, the operator of every conserved quantity commutes with the Hamiltonian. Thus every conserved physical quantity can be measured simultaneously with the energy.

Among the stationary states there may be states corresponding to one and the same energy eigenvalue (or *energy level*) but differing in the values of other physical quantities. A level to which several distinct stationary states correspond is said to be *degenerate*. Physically, degeneracy is possible because the energy does not, in general, constitute by itself a complete set of physical quantities. The energy levels of a system are, in general, degenerate if there are two conserved quantities f and g whose operators do not commute. Let ψ be the wave function of a stationary state in which f, as well as the energy, has a definite value. Then $\hat{g}\psi$ cannot coincide with ψ up to a constant factor; otherwise g too would have a definite value, which is impossible because f and g cannot be measured simultaneously. On the other hand, $\hat{g}\psi$ is a Hamiltonian eigenfunction with the same energy E as ψ:

$$\hat{H}(\hat{g}\psi) = \hat{g}\hat{H}\psi = E(\hat{g}\psi).$$

Thus more than one eigenfunction corresponds to E, and the level is degenerate.

Any linear combination of wave functions belonging to the same degenerate energy level is also an eigenfunction with that energy. The choice of eigenfunctions for a degenerate eigenvalue is therefore not unique. Arbitrarily chosen eigenfunctions of a degenerate level are not in general mutually orthogonal, but suitable linear combinations can always be chosen to form a mutually orthogonal and normalized set[2].

These statements apply not only to energy eigenfunctions but to the eigenfunctions of any operator. Only functions belonging to different eigenvalues are automatically

[2]This can be done in infinitely many ways: a linear transformation of n functions involves n^2 independent coefficients, but the normalization and orthogonality conditions number only $n(n+1)/2$, which is fewer than n^2.

orthogonal; functions belonging to the same degenerate eigenvalue are not in general orthogonal.

If $\hat{H} = \hat{H}_1 + \hat{H}_2$, where one part contains only coordinates q_1 and the other only q_2, the eigenfunctions of $\hat{H}$ can be written as products of eigenfunctions of $\hat{H}_1$ and $\hat{H}_2$, and the energy eigenvalues are sums of the eigenvalues of these operators. The energy spectrum may be discrete or continuous. A stationary state in the discrete spectrum always corresponds to *finite motion*: neither the system nor any part of it recedes to infinity. Indeed, $\int |\Psi|^2, dq$ is finite over all space for a discrete-spectrum eigenfunction. Thus $|\Psi|^2$ decreases sufficiently rapidly and vanishes at infinity. Equivalently, the probability of infinite coordinate values is zero; the system performs finite motion, or is in a *bound* state.

For continuous-spectrum wave functions, $\int |\Psi|^2, dq$ diverges. $|\Psi|^2$ then determines coordinate probabilities only up to proportionality. The divergence is always associated with $|\Psi|^2$ failing to vanish at infinity, or vanishing too slowly. The integral over the region outside any arbitrarily large but finite closed surface therefore still diverges: the system, or some part of it, is at infinity. A superposition of continuous-spectrum stationary states may have a convergent integral and be localized in a finite region, but this region spreads without bound with time, and ultimately the system recedes to infinity.

Indeed, an arbitrary continuous-spectrum superposition is

$$\Psi = \int a_E \exp\left(-\frac{i}{\hbar}Et\right) \psi_E(q)\, dE,$$

and

$$|\Psi|^2 = \iint a_E a_{E'}^* \exp\left(\frac{i}{\hbar}(E' - E)t\right) \psi_E(q)\psi_{E'}^*(q)\, dE\, dE'.$$

If this expression is averaged over an interval T and then T is allowed to tend to infinity, the oscillating factors average to zero, and with them the entire integral. Thus the time-averaged probability of finding the system at any specified point of configuration space vanishes; this is possible only if the motion occupies all infinite space[3].

Thus stationary states of the continuous spectrum correspond to infinite motion of the system.

§ 11. Matrices

Assume for convenience that the system in question possesses a discrete energy spectrum (every relation derived below extends directly to the continuous-spectrum case as well). Let $\Psi = \sum_n a_n \Psi_n$ denote the expansion of an arbitrary wave function over the stationary-state wave functions Ψ_n. If this expansion is substituted into the definition (3.8) of the

[3]For a superposition of discrete-spectrum functions one would instead have

$$\overline{|\Psi|^2} = \overline{\sum_{n,m} a_n a_m^* \exp\left\{\frac{i}{\hbar}(E_m - E_n)t\right\} \psi_n \psi_m^*} = \sum_n |a_n \psi_n(q)|^2,$$

so the probability density remains finite under time averaging.

mean value of some quantity f, we obtain

$$\bar{f} = \sum_n \sum_m a_n^* a_m f_{nm}(t), \tag{11.1}$$

where $f_{nm}(t)$ denote the integrals

$$f_{nm}(t) = \int \Psi_n^* \hat{f} \Psi_m \, dq. \tag{11.2}$$

The set of quantities $f_{nm}(t)$ for all possible n, m is called the *matrix* of the quantity f, and each $f_{nm}(t)$ is spoken of as a *matrix element*, corresponding to the transition from state m to state n[4].

The time dependence of the matrix elements $f_{nm}(t)$ is determined (if the operator $\hat{f}$ does not contain t explicitly) by the time dependence of the functions Ψ_n. Substituting for them the expressions (10.1), we find that

$$f_{nm}(t) = f_{nm} e^{i\omega_{nm}t}, \tag{11.3}$$

where

$$\omega_{nm} = \frac{E_n - E_m}{\hbar} \tag{11.4}$$

is the transition frequency between states n and m, and the quantities

$$f_{nm} = \int \psi_n^* \hat{f} \psi_m \, dq \tag{11.5}$$

constitute the time-independent matrix of the quantity f, which is what one ordinarily uses[5].

One obtains the matrix elements of the derivative $\dot{f}$ by differentiating those of the quantity f with respect to time; this is an immediate consequence of the relation

$$\bar{\dot{f}} = \dot{\bar{f}} = \sum_n \sum_m a_n^* a_m \dot{f}_{nm}(t). \tag{11.6}$$

In view of (11.3) we thus have for the matrix elements of $\dot{f}$:

$$\dot{f}_{nm}(t) = i\omega_{nm} f_{nm}(t) \tag{11.7}$$

or (cancelling the time factor $\exp(i\omega_{nm}t)$ on both sides) for the time-independent matrix elements:

$$(\dot{f})_{nm} = i\omega_{nm} f_{nm} = \frac{i}{\hbar}(E_n - E_m) f_{nm}. \tag{11.8}$$

For simplicity of notation in the formulas, we derive below all relations for the time-independent matrix elements; exactly the same relations hold also for matrices that depend on time.

[4]The matrix representation of physical quantities was introduced by *Heisenberg* (*W. Heisenberg*) in 1925, even before Schrödinger's discovery of the wave equation. "Matrix mechanics" was subsequently developed by *Born, Heisenberg and Jordan* (*M. Born, P. Jordan*).

[5]In connection with the indeterminacy of the phase factor in the normalized wave functions (see § 2), the matrix elements f_{nm} (and $f_{nm}(t)$) are likewise defined only to within factors of the form $\exp[i(\alpha_m - \alpha_n)]$. Here too this indeterminacy does not affect the physical results.

For the matrix elements of the complex conjugate quantity f^* of f, taking into account the definition of the adjoint operator, we obtain

$$(f^*)_{nm} = \int \psi_n^* \hat{f}^+ \psi_m \, dq = \int \psi_n^* \tilde{\hat{f}}^* \psi_m \, dq = \int \psi_m \hat{f}^* \psi_n^* \, dq,$$

i.e.

$$(f^*)_{nm} = (f_{mn})^*. \tag{11.9}$$

For real physical quantities, which are usually all that we consider, we therefore have

$$f_{nm} = f_{mn}^*. \tag{11.10}$$

Such matrices, like the operators corresponding to them, are called *Hermitian*.

Matrix elements with $n = m$ are called *diagonal*. These elements do not depend on time at all, and from (11.10) it is clear that they are real. The element f_{nn} is the mean value of f in the state Ψ_n.

The rule of *matrix multiplication* is readily derived. As a preliminary step, observe that the formula

$$\hat{f}\psi_n = \sum_m f_{mn}\psi_m. \tag{11.11}$$

holds. This is nothing other than the expansion of the function $\hat{f}\psi_n$ in the functions ψ_m with coefficients determined according to the general rule (3.5). Bearing this formula in mind, we write

$$\hat{f}\hat{g}\psi_n = \hat{f}(\hat{g}\psi_n) = \sum_k g_{kn}\hat{f}\psi_k = \sum_{k,m} g_{kn} f_{mk}\psi_m.$$

Since, on the other hand, we must have $\hat{f}\hat{g}\psi_n = \sum_m (fg)_{mn}\psi_m$, we obtain

$$(fg)_{mn} = \sum_k f_{mk} g_{kn}. \tag{11.12}$$

This rule matches the standard mathematical prescription for multiplying matrices: one multiplies the rows of the first factor by the columns of the second.

Specifying a matrix is equivalent to specifying the operator itself. In particular, it makes it possible in principle to determine the eigenvalues of a given physical quantity and the eigenfunctions corresponding to them.

Let us consider the values of all quantities at some definite instant of time and expand an arbitrary wave function Ψ in the eigenfunctions of the Hamiltonian, i.e. in the time-independent wave functions ψ_m of the stationary states:

$$\Psi = \sum_m c_m\psi_m, \tag{11.13}$$

where the expansion coefficients are denoted by c_m. Let us substitute this expansion into the equation $\hat{f}\Psi = f\Psi$, which determines the eigenvalues and eigenfunctions of the quantity f. We have

$$\sum_m c_m \hat{f}\psi_m = f \sum_m c_m\psi_m.$$

Multiply this equation on both sides by ψ_n^* and integrate over dq. Each of the integrals $\int \psi_n^* \hat{f} \psi_m \, dq$ on the left-hand side of the equality is the corresponding matrix element f_{nm}. On the right-hand side, all integrals $\int \psi_n^* \psi_m \, dq$ with $m \neq n$ vanish by virtue of the orthogonality of the functions ψ_m, and $\int \psi_n^* \psi_n \, dq = 1$ by virtue of their normalization[6]:

$$\sum_m f_{nm} c_m = f c_n, \tag{11.14}$$

or $\sum_m (f_{nm} - f\delta_{nm}) c_m = 0$, where $\delta_{nm} = 1$ for $n = m$ and $\delta_{nm} = 0$ for $n \neq m$.

Thus we have obtained a system of homogeneous linear algebraic equations (with unknowns c_m). As is well known, such a system has nonzero solutions only if the determinant formed from the coefficients in the equations vanishes, i.e. under the condition

$$|f_{nm} - f\delta_{nm}| = 0. \tag{11.15}$$

The roots of this equation (where f plays the role of the unknown) give the possible values of the quantity f. The set of quantities c_m satisfying equations (11.14) upon setting f equal to one of these values yields the corresponding eigenfunction. Suppose that in the definition (11.5) for the matrix elements of f we adopt, as the functions ψ_n, the eigenfunctions of this very quantity; owing to the equation $\hat{f}\psi_n = f_n \psi_n$ we obtain

$$f_{nm} = \int \psi_n^* \hat{f} \psi_m \, dq = f_m \int \psi_n^* \psi_m \, dq.$$

Since the functions ψ_m are orthogonal and normalized, it follows that $f_{nm} = 0$ for $n \neq m$ and $f_{mm} = f_m$. Thus only diagonal matrix elements turn out to be nonzero, each of them being equal to the corresponding eigenvalue of the quantity f; a matrix in which only these elements are nonzero is said to be reduced to *diagonal form*. In particular, in the ordinary representation—where the stationary-state wave functions serve as the ψ_n—the energy matrix is diagonal (along with the matrices of all other physical quantities that have definite values in the stationary states). More broadly, when the matrix of a quantity f is constructed using the eigenfunctions of an operator g, one speaks of it as the matrix of f in the representation in which g is diagonal. Everywhere where it is not otherwise specified, by the matrix of a physical quantity we shall henceforth understand the matrix in the ordinary representation, in which the energy is diagonal. All that has been said above concerning the time dependence of matrices refers, of course, only to this ordinary representation[7].

The matrix representation of operators makes it possible to prove the theorem cited in § 4: if two operators commute, they possess a common complete system of eigenfunctions.

[6]In accordance with the general rule (§ 5), the set of coefficients c_n of the expansion (11.13) can be regarded as the wave function in the "energy representation" (the variable here being the index n, which enumerates the energy eigenvalues). The matrix f_{nm} then plays the role of the operator $\hat{f}$ in this representation, whose action on the wave function is given by the expression on the left-hand side of equation (11.14). The formula $\bar{f} = \sum_n \sum_m c_n^* (f_{nm} c_m)$ then corresponds to the general formula for the mean value of a quantity expressed through its operator and the wave function of the state in question.

[7]Keeping in mind that the energy matrix is diagonal, one readily verifies that the equality (11.8) is the operator relation (9.2) written in matrix form.

Let $\hat{f}$ and $\hat{g}$ be two such operators. From $\hat{f}\hat{g} = \hat{g}\hat{f}$ and the rule of matrix multiplication (11.12) it follows that

$$\sum_k f_{mk} g_{kn} = \sum_k g_{mk} f_{kn}.$$

Taking as the system of functions ψ_n, in terms of which the matrix elements are computed, the eigenfunctions of the operator $\hat{f}$, we have $f_{mk} = 0$ for $m \neq k$, so that the written equality reduces to $f_m g_{mn} = g_{mn} f_n$, or $g_{mn}(f_m - f_n) = 0$. If all the eigenvalues f_n of the quantity f are distinct, then for all $m \neq n$ we have $f_m - f_n \neq 0$, so that we must have $g_{mn} = 0$. Thus the matrix g_{mn} likewise turns out to be diagonal, i.e. the functions ψ_n are eigenfunctions also of the physical quantity g. If, however, among the values f_n there are equal ones, then the matrix elements g_{mn} corresponding to each such group of functions ψ_n will, in general, be nonzero. However, linear combinations of the functions ψ_n corresponding to one eigenvalue of the quantity f are likewise its eigenfunctions; one can always choose these combinations in such a way as to make the corresponding nondiagonal matrix elements g_{mn} vanish, and thus in this case too we obtain a system of functions that are eigenfunctions simultaneously for the operators $\hat{f}$ and $\hat{g}$.

Let us note a useful formula

$$\left(\frac{\partial H}{\partial \lambda}\right)_{nn} = \frac{\partial E_n}{\partial \lambda}, \tag{11.16}$$

where λ — a parameter on which the Hamiltonian $\hat{H}$ depends (and with it the energy eigenvalues E_n). Indeed, differentiating the equation $(\hat{H} - E_n)\psi_n = 0$ with respect to λ and then multiplying it from the left by ψ_n^*, we obtain

$$\psi_n^*(\hat{H} - E_n)\frac{\partial \psi_n}{\partial \lambda} = \psi_n^*\left(\frac{\partial E_n}{\partial \lambda} - \frac{\partial \hat{H}}{\partial \lambda}\right)\psi_n.$$

On integrating over dq, the left-hand side of this equality vanishes, since

$$\int \psi_n^*(\hat{H} - E_n)\frac{\partial \psi_n}{\partial \lambda}\, dq = \int \frac{\partial \psi_n}{\partial \lambda}(\hat{H} - E_n)^*\psi_n^*\, dq$$

by virtue of the Hermiticity of the operator $\hat{H}$. The right-hand side gives the desired equality.

In the modern literature, a system of notation (introduced by *Dirac*) is widely used, in which matrix elements are written as[8]

$$\langle n|f|m\rangle. \tag{11.17}$$

This symbol is composed of the notation for the quantity f and the symbols $|m\rangle$ and $|n\rangle$, denoting respectively the initial and final states. If there is a complete set of state functions $|n_1\rangle, |n_2\rangle, \ldots$, then the expansion coefficients of the wave function of the state $|m\rangle$ are denoted by $\langle n_i|m\rangle$:

$$\langle n_i|m\rangle = \int \psi_{n_i}^* \psi_m\, dq. \tag{11.18}$$

[8]We will use both ways of denoting matrix elements in this book. The notation (11.17) is especially convenient when each index has to be written as a combination of several letters.

§ 12. Transformation of matrices

The matrix elements of one and the same physical quantity may be defined with respect to different sets of wave functions. These may be, for example, wave functions of stationary states described by different sets of physical quantities, or wave functions of stationary states of the same system placed in different external fields. The question therefore arises of transforming matrices from one representation to another.

Let $\psi_n(q)$ and $\psi'_n(q)$ $(n = 1, 2, \ldots)$ be two complete systems of orthonormal functions. They are related by a linear transformation

$$\psi'_n = \sum_m S_{mn}\psi_m, \tag{12.1}$$

which is simply the expansion of the functions ψ'_n in the complete system ψ_m. This transformation can be written in operator form as

$$\psi'_n = \hat{S}\psi_n. \tag{12.2}$$

The operator $\hat{S}$ must satisfy a certain condition if the functions ψ'_n are to be orthonormal whenever the functions ψ_n are. Indeed, substituting (12.2) into $\int \psi'^*_m\psi'_n\,dq = \delta_{mn}$ and using the definition (3.14) of the transposed operator, we obtain

$$\int (\hat{S}\psi_m)^*\hat{S}\psi_n\,dq = \int \psi^*_m\tilde{S}^*\hat{S}\psi_n\,dq = \delta_{mn}.$$

For this equality to hold for every m, n, we must have $\tilde{S}^*\hat{S} = 1$, or

$$\tilde{S}^* \equiv \hat{S}^+ = \hat{S}^{-1}, \tag{12.3}$$

i.e. the inverse operator coincides with the adjoint. Operators having this property are called *unitary*. By virtue of this property the transformation $\psi_n = \hat{S}^{-1}\psi'_n$, inverse to (12.1), is given by

$$\psi_n = \sum_m S^*_{nm}\psi'_m. \tag{12.4}$$

Writing $\hat{S}^+\hat{S} = 1$ or $\hat{S}\hat{S}^+ = 1$ in matrix form gives the unitarity conditions

$$\sum_l S^*_{lm}S_{ln} = \delta_{mn} \tag{12.5}$$

or

$$\sum_l S^*_{ml}S_{nl} = \delta_{mn}. \tag{12.6}$$

Now consider a physical quantity f and write its matrix elements in the "new" representation, i.e. with respect to the functions ψ'_n. They are given by

$$\int \psi'^*_m\hat{f}\psi'_n\,dq = \int (\tilde{S}^*\psi^*_m)(\hat{f}\hat{S}\psi_n)\,dq = \int \psi^*_m\hat{S}^{-1}\hat{f}\hat{S}\psi_n\,dq.$$

It follows that the matrix of $\hat{f}$ in the new representation coincides with the matrix of the operator

$$\hat{f}' = \hat{S}^{-1}\hat{f}\hat{S} \tag{12.7}$$

in the old representation[9]. The *trace* of a matrix is defined as the sum of its diagonal elements and is written Tr f[10]:

$$\operatorname{Tr} f = \sum_n f_{nn}. \tag{12.8}$$

First note that the trace of a product of two matrices does not depend on the order of the factors:

$$\operatorname{Tr}(fg) = \operatorname{Tr}(gf). \tag{12.9}$$

Indeed, by the rule for matrix multiplication,

$$\operatorname{Tr}(fg) = \sum_n \sum_k f_{nk} g_{kn} = \sum_k \sum_n g_{kn} f_{nk} = \operatorname{Tr}(gf).$$

Similarly, the trace of a product of several matrices is unchanged by a cyclic permutation of the factors; thus

$$\operatorname{Tr}(fgh) = \operatorname{Tr}(hfg) = \operatorname{Tr}(ghf). \tag{12.10}$$

The most important property of the trace is its independence of the system of functions relative to which the matrix elements are defined. Indeed,

$$\operatorname{Tr} f' = \operatorname{Tr}\left(S^{-1} f S\right) = \operatorname{Tr}\left(S S^{-1} f\right) = \operatorname{Tr} f. \tag{12.11}$$

Also, a unitary transformation leaves invariant the sum of the squared moduli of the transformed functions. From (12.6),

$$\sum_i |\psi_i'|^2 = \sum_{k,l,i} S_{ki} S_{li}^* \psi_k \psi_l^* = \sum_{k,l} \delta_{kl} \psi_k \psi_l^* = \sum_k |\psi_k|^2. \tag{12.12}$$

Every unitary operator can be represented as

$$\hat{S} = e^{i\hat{R}}, \tag{12.13}$$

where $\hat{R}$ is Hermitian; indeed, $\hat{R}^+ = \hat{R}$ gives $\hat{S}^+ = e^{-i\hat{R}^+} = e^{-i\hat{R}} = \hat{S}^{-1}$.

We also note the expansion

$$\hat{f}' = \hat{S}^{-1} \hat{f} \hat{S} = \hat{f} + \{\hat{f}, i\hat{R}\} + \frac{1}{2}\{\{\hat{f}, i\hat{R}\}, i\hat{R}\} + \ldots, \tag{12.14}$$

which is readily verified directly by expanding the factors $\exp\left(\pm i\hat{R}\right)$ in powers of $\hat{R}$. This expansion can be useful when $\hat{R}$ is proportional to a small parameter, so that (12.14) becomes an expansion in powers of that parameter.

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[9] If $\{\hat{f}, \hat{g}\} = -i\hbar\hat{c}$ is the commutation rule for two operators, then after transformation (12.7) we obtain $\{\hat{f}', \hat{g}'\} = -i\hbar\hat{c}'$, so the rule is unchanged. It was noted in the note on p. 46 that c is the quantum analogue of the classical Poisson bracket $[f, g]$. Poisson brackets in classical mechanics, however, are invariant under canonical variable transformations (see Vol. I, § 45). One may therefore say that unitary transformations in quantum mechanics serve as the counterpart of canonical transformations in classical mechanics.

[10] The source's notation comes from the German *Spur*, trace. The notation Tr, from the English *trace*, is also used. Consideration of a matrix trace of course assumes convergence of the sum over n.

§ 13. Heisenberg representation of operators

In the mathematical apparatus of quantum mechanics as here presented, the operators corresponding to the various physical quantities act on functions of the coordinates and as such ordinarily contain no explicit dependence on time. The dependence of the mean values of physical quantities on time arises only through the time dependence of the wave function of the state, in accordance with the formula

$$\bar{f}(t) = \int \Psi^*(q,t)\hat{f}\Psi(q,t)\,dq. \tag{13.1}$$

The apparatus of quantum mechanics can, however, also be formulated in a somewhat different, equivalent form, in which the dependence on time is transferred from the wave functions to the operators. Although in this book we shall not make use of such a representation of operators—the so-called Heisenberg representation, as distinct from the Schrödinger representation—we shall formulate it here, with later applications in relativistic theory in view.

Let us introduce the unitary operator (cf. (12.13))

$$\hat{S} = \exp\left(-\frac{i}{\hbar}\hat{H}t\right), \tag{13.2}$$

where $\hat{H}$ is the Hamiltonian of the system. By definition its eigenfunctions coincide with the eigenfunctions of the operator $\hat{H}$, i.e. with the wave functions $\psi_n(q)$ of the stationary states, and moreover

$$\hat{S}\psi_n(q) = \exp\left(-\frac{i}{\hbar}E_n t\right)\psi_n(q) = \Psi_n(q,t). \tag{13.3}$$

It follows that expansion (10.3) of any wave function Ψ over the stationary-state wave functions can be recast in operator form as

$$\Psi(q,t) = \hat{S}\Psi(q,0), \tag{13.4}$$

i.e. the action of the operator $\hat{S}$ carries the wave function of the system at some initial instant of time over into the wave function at an arbitrary instant of time. On introducing, in accordance with (12.7), the time-dependent operator

$$\hat{f}(t) = \hat{S}^{-1}\hat{f}\hat{S}, \tag{13.5}$$

we obtain

$$\bar{f}(t) = \int \Psi^*(q,0)\hat{f}(t)\Psi(q,0)\,dq, \tag{13.6}$$

i.e. we bring the formula for the mean value of the quantity f (which serves as the definition of the operators) into a form in which the dependence on time is wholly transferred to the operator.

It is obvious that the matrix elements of the operator (13.5), with respect to the wave functions of stationary states, coincide with the time-dependent matrix elements $f_{nm}(t)$ defined by formula (11.3).

Finally, on differentiating expression (13.5) with respect to time (assuming here that the operators $\hat{f}$ and $\hat{H}$ themselves contain no t), we obtain the equation

$$\frac{\partial \hat{f}(t)}{\partial t} = \frac{i}{\hbar}\left[\hat{H}\hat{f}(t) - \hat{f}(t)\hat{H}\right], \tag{13.7}$$

which is analogous to formula (9.2) but has a somewhat different meaning: expression (9.2) constitutes the definition of the operator $\hat{\dot{f}}$ corresponding to the physical quantity $\dot{f}$, whereas the left-hand side of equation (13.7) contains the time derivative of the operator of the quantity f itself.

§ 14. The density matrix

Describing a system by its wave function represents the fullest description that quantum mechanics allows, in the sense indicated at the end of § 1.

States not admitting such a description arise when the system considered is part of a larger closed system. Suppose the closed system as a whole is in a state described by $\Psi(q, x)$, where x denotes the coordinates of the system considered and q the remaining coordinates of the closed system. In general this function does not factor into functions of x alone and q alone, so that the system has no wave function of its own[11].

Let f be a physical quantity pertaining to our system. Its operator acts on the x coordinates only, not on q. Its mean value is

$$\bar{f} = \iint \Psi^*(q, x)\hat{f}\Psi(q, x)\, dq\, dx. \tag{14.1}$$

Introduce the function

$$\rho(x, x') = \int \Psi(q, x)\Psi^*(q, x')\, dq, \tag{14.2}$$

where integration is only over q; it is called the system's *density matrix*. From (14.2),

$$\rho^*(x, x') = \rho(x', x). \tag{14.3}$$

The "diagonal elements"

$$\rho(x, x) = \int |\Psi(q, x)|^2\, dq$$

determine the coordinate probability distribution.

In terms of the density matrix,

$$\bar{f} = \int [\hat{f}\rho(x, x')]_{x'=x}\, dx. \tag{14.4}$$

Here $\hat{f}$ acts only on the variables x in $\rho(x, x')$; after its action has been evaluated, one sets $x' = x$. Thus the density matrix determines the mean of every quantity characterizing

[11] For $\Psi(q, x)$ to factor in this way at a given instant, the measurement that produced the state must describe completely and separately the system considered and the remainder of the closed system. For the factorization to persist, these two parts must also not interact (see § 2). Neither condition is assumed here.

the system and hence also the probabilities of its possible values. A state having no wave function can therefore be described by a density matrix. The density matrix contains none of the coordinates q that do not belong to the system, although it naturally depends in substance on the state of the closed system as a whole.

The density-matrix description is the most general form of quantum-mechanical description of systems. The wave-function description is the special case $\rho(x, x') = \Psi(x)\Psi^*(x')$. There is an important distinction. For a state possessing a wave function (a *pure* state), there is always a complete system of measurements whose results can be predicted with certainty (mathematically, Ψ is an eigenfunction of some operator). For states possessing only a density matrix (*mixed* states), no complete system of measurements has uniquely predictable results.

Suppose the system is closed, or became closed at some instant. We derive an equation governing the time dependence of its density matrix, analogous to the wave equation. The desired linear differential equation for $\rho(x, x', t)$ must also hold in the special case

$$\rho(x, x', t) = \Psi(x, t)\Psi^*(x', t).$$

Differentiating and using (8.1),

$$\begin{aligned} i\hbar\frac{\partial\rho}{\partial t} &= i\hbar\Psi^*(x', t)\frac{\partial\Psi(x, t)}{\partial t} + i\hbar\Psi(x, t)\frac{\partial\Psi^*(x', t)}{\partial t} \\ &= \Psi^*(x', t)\hat{H}\Psi(x, t) - \Psi(x, t)\hat{H}'^*\Psi^*(x', t). \end{aligned}$$

Here $\hat{H}$ acts on functions of x, while $\hat{H}'$ is the same operator acting on functions of x'. Bringing the other factors under the respective operator signs gives

$$i\hbar\frac{\partial\rho(x, x', t)}{\partial t} = (\hat{H} - \hat{H}'^*)\rho(x, x', t). \tag{14.5}$$

Let $\Psi_n(x, t)$ be the stationary-state wave functions. Expanding the density matrix in these functions gives the double series

$$\rho(x, x', t) = \sum_m \sum_n a_{mn}\psi_n^*(x')\psi_m(x)\exp\left[\frac{i}{\hbar}(E_n - E_m)t\right]. \tag{14.6}$$

This plays the same role for the density matrix as (10.3) does for the wave function. In place of the single set a_n there is the double set a_{mn}, which, like the density matrix, is Hermitian:

$$a_{mn} = a_{nm}^*. \tag{14.7}$$

Substitution in (14.4) gives

$$\bar{f} = \sum_m \sum_n a_{mn} \int \Psi_n^*(x, t)\hat{f}\Psi_m(x, t)\, dx,$$

or

$$\bar{f} = \sum_m \sum_n a_{mn} f_{nm}(t) = \sum_m \sum_n a_{mn} f_{nm} \exp\left[\frac{i}{\hbar}(E_n - E_m)t\right]. \tag{14.8}$$

This is analogous to (11.1)[12].

The a_{mn} must satisfy inequalities. Since the diagonal elements $\rho(x,x)$ are probabilities, they must be positive. Hence the quadratic form

$$\sum_n \sum_m a_{mn}\xi_m\xi_n^*$$

must be positive definite for arbitrary complex ξ_n. The standard conditions for quadratic forms therefore apply; in particular,

$$a_{nn} > 0, \tag{14.9}$$

and for every a_{mm}, a_{nn}, a_{mn},

$$a_{mm}a_{nn} \geq |a_{mn}|^2. \tag{14.10}$$

The pure case, in which the density matrix is a product of functions, corresponds to

$$a_{mn} = a_m a_n^*. \tag{14.11}$$

A simple criterion distinguishes a pure from a mixed state. In the pure case,

$$(a^2)_{mn} = \sum_k a_{mk}a_{kn} = \sum_k a_m a_k^* a_k a_n^* = a_m a_n^* \sum_k |a_k|^2 = a_m a_n^*,$$

and therefore

$$(a^2)_{mn} = a_{mn}. \tag{14.12}$$

Thus the square of the density matrix coincides with the matrix itself.

§ 15. Momentum

Consider a closed system of particles in no external field. Since all positions of the system as a whole are equivalent, its Hamiltonian is unchanged by an arbitrary parallel displacement. It is enough to require this for every infinitesimal displacement; it then holds for finite displacements as well.

Under an infinitesimal displacement $\delta\mathbf{r}$, every radius vector $\mathbf{r}_a$ acquires the same increment. Thus $\mathbf{r}_a \to \mathbf{r}_a + \delta\mathbf{r}$, and

$$\psi(\mathbf{r}_1 + \delta\mathbf{r}, \mathbf{r}_2 + \delta\mathbf{r}, \ldots) = \left(1 + \delta\mathbf{r}\cdot\sum_a \boldsymbol{\nabla}_a\right)\psi(\mathbf{r}_1, \mathbf{r}_2, \ldots).$$

Here a labels the particles and $\boldsymbol{\nabla}_a$ differentiates with respect to $\mathbf{r}_a$. The expression in parentheses is the operator of an infinitesimal displacement.

If a transformation does not change the Hamiltonian, transforming $\hat{H}\psi$ gives the same result as first transforming ψ and then applying $\hat{H}$. If $\hat{O}$ produces the transformation, then $\hat{O}(\hat{H}\psi) = \hat{H}(\hat{O}\psi)$, so $\hat{O}\hat{H} - \hat{H}\hat{O} = 0$: $\hat{H}$ commutes with $\hat{O}$.

[12] The quantities a_{mn} constitute the density matrix in the energy representation. Description of states by such a matrix was introduced independently by Landau and Bloch (*F. Bloch*) in 1927.

Here $\hat{O}$ is the infinitesimal-displacement operator. The unit operator commutes with every operator, and the constant $\delta\mathbf{r}$ may be taken outside $\hat{H}$; hence

$$\left(\sum_a \boldsymbol{\nabla}_a\right)\hat{H} - \hat{H}\left(\sum_a \boldsymbol{\nabla}_a\right) = 0. \tag{15.1}$$

An explicitly time-independent operator commuting with the Hamiltonian corresponds to a conserved quantity. The quantity whose conservation follows from the homogeneity of space is momentum (cf. Vol. I, § 7). Thus (15.1) expresses momentum conservation; $\sum_a \boldsymbol{\nabla}_a$ corresponds, up to a constant factor, to the total momentum, and each term to the momentum of one particle.

The factor relating the momentum operator to $\boldsymbol{\nabla}$ follows from the classical limit and equals $-i\hbar$:

$$\hat{\mathbf{p}} = -i\hbar\boldsymbol{\nabla}, \tag{15.2}$$

or

$$\hat{p}_x = -i\hbar\frac{\partial}{\partial x}, \qquad \hat{p}_y = -i\hbar\frac{\partial}{\partial y}, \qquad \hat{p}_z = -i\hbar\frac{\partial}{\partial z}.$$

Indeed, using (6.1),

$$\hat{\mathbf{p}}\Psi = -i\hbar\boldsymbol{\nabla}\Psi = \Psi\boldsymbol{\nabla}S.$$

Thus classically the operator acts by multiplication by $\boldsymbol{\nabla}S$, which is the classical momentum (see Vol. I, § 43).

The operator (15.2) is Hermitian, as required. For functions $\psi(x)$ and $\varphi(x)$ vanishing at infinity, integration by parts gives

$$\int \varphi\hat{p}_x\psi\, dx = -i\hbar\int \varphi\frac{\partial\psi}{\partial x}\, dx = i\hbar\int \psi\frac{\partial\varphi}{\partial x}\, dx = \int \psi\hat{p}_x^*\varphi\, dx,$$

which is the Hermiticity condition.

Since differentiations with respect to different variables commute,

$$\hat{p}_x\hat{p}_y - \hat{p}_y\hat{p}_x = 0, \quad \hat{p}_x\hat{p}_z - \hat{p}_z\hat{p}_x = 0, \quad \hat{p}_y\hat{p}_z - \hat{p}_z\hat{p}_y = 0. \tag{15.3}$$

Thus all three momentum components may have definite values simultaneously.

Their eigenfunctions and eigenvalues satisfy

$$-i\hbar\boldsymbol{\nabla}\psi = \mathbf{p}\psi, \tag{15.4}$$

with solutions

$$\psi = \text{const}\cdot e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \tag{15.5}$$

The three components completely determine the wave function and form one possible complete set. Their eigenvalues form a continuous spectrum from $-\infty$ to ∞.

By (5.4), $\int \psi_{\mathbf{p}'}^*\psi_{\mathbf{p}}, dV$ over all space should equal $\delta(\mathbf{p}'-\mathbf{p})$[13]. For later applications it is more natural to normalize to the delta function of the momentum difference divided by $2\pi\hbar$:

$$\int \psi_{\mathbf{p}'}^*\psi_{\mathbf{p}}, dV = \delta\left(\frac{\mathbf{p}'-\mathbf{p}}{2\pi\hbar}\right),$$

[13]For a vector argument, $\delta(\mathbf{a}) = \delta(a_x)\delta(a_y)\delta(a_z)$.

or equivalently

$$\int \psi^*_{\mathbf{p}'}\psi_{\mathbf{p}}, dV = (2\pi\hbar)^3\delta(\mathbf{p}' - \mathbf{p}). \tag{15.6}$$

Each one-dimensional factor obeys, for example, $\delta[(p'_x - p_x)/(2\pi\hbar)] = 2\pi\hbar\delta(p'_x - p_x)$. Integration uses[14]

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\alpha\xi}\, d\xi = \delta(\alpha). \tag{15.7}$$

It follows that const = 1 in (15.5)[15]:

$$\psi_{\mathbf{p}} = e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}. \tag{15.8}$$

Expansion of an arbitrary $\psi(\mathbf{r})$ in momentum eigenfunctions is its Fourier integral:

$$\psi(\mathbf{r}) = \int a(\mathbf{p})\psi_{\mathbf{p}}(\mathbf{r})\frac{d^3p}{(2\pi\hbar)^3} = \int a(\mathbf{p})e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}\frac{d^3p}{(2\pi\hbar)^3}. \tag{15.9}$$

Here $d^3p = dp_x dp_y dp_z$, and

$$a(\mathbf{p}) = \int \psi(\mathbf{r})\psi^*_{\mathbf{p}}(\mathbf{r})\, dV = \int \psi(\mathbf{r})e^{-i\mathbf{p}\cdot\mathbf{r}/\hbar}\, dV. \tag{15.10}$$

The function $a(\mathbf{p})$ is the momentum-representation wave function; $|a(\mathbf{p})|^2 d^3p/(2\pi\hbar)^3$ is the probability that the momentum lies in d^3p. Just as $\hat{\mathbf{p}}$ represents momentum in coordinate space, introduce $\hat{\mathbf{r}}$ for the coordinates in momentum space, defined by

$$\bar{\mathbf{r}} = \int a^*(\mathbf{p})\hat{\mathbf{r}}a(\mathbf{p})\frac{d^3p}{(2\pi\hbar)^3}. \tag{15.11}$$

But also $\bar{\mathbf{r}} = \int \psi^*\mathbf{r}\psi, dV$. Substituting (15.9) and integrating by parts,

$$\mathbf{r}\psi(\mathbf{r}) = \int i\hbar e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}\frac{\partial a(\mathbf{p})}{\partial \mathbf{p}}\frac{d^3p}{(2\pi\hbar)^3},$$

and hence

$$\bar{\mathbf{r}} = \int i\hbar a^*(\mathbf{p})\frac{\partial a(\mathbf{p})}{\partial \mathbf{p}}\frac{d^3p}{(2\pi\hbar)^3}.$$

Comparison with (15.11) gives

$$\hat{\mathbf{r}} = i\hbar\frac{\partial}{\partial \mathbf{p}}. \tag{15.12}$$

In this representation momentum is simply multiplication by $\mathbf{p}$.

[14]The formula means that its left-hand side has the defining property (5.8) of the delta function. Substitution in (5.8) gives the Fourier formula

$$f(a) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x)e^{i\xi(x-a)}\, dx\, \frac{d\xi}{2\pi}.$$

[15]With this normalization $|\psi|^2 = 1$: the function is normalized to "one particle per unit volume." This agreement is not accidental; cf. the note on p. 221.

Finally, express the operator of translation through an arbitrary finite distance $\mathbf{a}$ in terms of $\hat{\mathbf{p}}$. By definition, $\hat{T}_{\mathbf{a}}\psi(\mathbf{r}) = \psi(\mathbf{r}+\mathbf{a})$. Expanding $\psi(\mathbf{r}+\mathbf{a})$ in a Taylor series gives

$$\psi(\mathbf{r}+\mathbf{a}) = \psi(\mathbf{r}) + \mathbf{a}\cdot\frac{\partial\psi(\mathbf{r})}{\partial\mathbf{r}} + \dots$$

or, on introducing $\hat{\mathbf{p}} = -i\hbar\boldsymbol{\nabla}$,

$$\psi(\mathbf{r}+\mathbf{a}) = \left[1 + \frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}} + \frac{1}{2}\left(\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right)^2 + \dots\right]\psi(\mathbf{r}).$$

Thus

$$\hat{T}_{\mathbf{a}} = \exp\left(\frac{i}{\hbar}\mathbf{a}\cdot\hat{\mathbf{p}}\right). \qquad (15.13)$$

This is the required *finite-displacement operator*.

§ 16. Uncertainty Relations

Let us derive the commutation rules for the momentum and coordinate operators. Successive differentiation with respect to one of the variables x, y, z and multiplication by another of them gives a result independent of the order of these operations. Hence

$$\hat{p}_x y - y\hat{p}_x = 0, \qquad \hat{p}_x z - z\hat{p}_x = 0, \qquad (16.1)$$

and similarly for $\hat{p}_y$ and $\hat{p}_z$.

To derive the commutation rule for $\hat{p}_x$ and x, write

$$(\hat{p}_x x - x\hat{p}_x)\psi = -i\hbar\frac{\partial}{\partial x}(x\psi) + i\hbar x\frac{\partial\psi}{\partial x} = -i\hbar\psi.$$

We see that the action of $\hat{p}_x x - x\hat{p}_x$ reduces to multiplication of the function by $-i\hbar$; the same is of course true for the commutation of $\hat{p}_y$ with y and of $\hat{p}_z$ with z. Thus[16]

$$\hat{p}_x x - x\hat{p}_x = -i\hbar, \qquad \hat{p}_y y - y\hat{p}_y = -i\hbar, \qquad \hat{p}_z z - z\hat{p}_z = -i\hbar. \qquad (16.2)$$

Relations (16.1) and (16.2) can be written together as

$$\hat{p}_i x_k - x_k\hat{p}_i = -i\hbar\delta_{ik}, \qquad i, k = x, y, z. \qquad (16.3)$$

Before examining the physical meaning of these relations and their consequences, we give two formulas useful below. If $f(\mathbf{r})$ is a function of the coordinates, then

$$\hat{\mathbf{p}}f(\mathbf{r}) - f(\mathbf{r})\hat{\mathbf{p}} = -i\hbar\nabla f. \qquad (16.4)$$

Indeed,

$$(\hat{\mathbf{p}}f - f\hat{\mathbf{p}})\psi = -i\hbar[\nabla(f\psi) - f\nabla\psi] = -i\hbar\psi\nabla f.$$

[16] These relations, discovered in matrix form by *Heisenberg* in 1925, served as the starting point in the creation of quantum mechanics.

An analogous relation holds for the commutator with a function of the momentum operator:

$$f(\mathbf{p})\hat{\mathbf{r}} - \hat{\mathbf{r}}f(\mathbf{p}) = -i\hbar\frac{\partial f}{\partial \mathbf{p}}. \tag{16.5}$$

It may be derived in the same way as (16.4) by working in the momentum representation and using (15.12) for the coordinate operators.

Relations (16.1) and (16.2) show that a particle coordinate along one axis can have a definite value simultaneously with the momentum components along the other two axes; a coordinate and the momentum component along that same axis do not exist simultaneously. In particular, a particle cannot occupy a definite point in space while also having a definite momentum $\mathbf{p}$.

Let the particle occupy a finite spatial region whose dimensions along the three axes are of order Δx, Δy, Δz. Let its mean momentum be $\mathbf{p}_0$. Mathematically, this means that the wave function has the form $\psi = u(\mathbf{r})e^{i\mathbf{p}_0\cdot\mathbf{r}/\hbar}$, where $u(\mathbf{r})$ is appreciably different from zero only within the specified region.

Expand ψ in eigenfunctions of the momentum operator, that is, as a Fourier integral. The coefficients $a(\mathbf{p})$ in this expansion are given by (15.10) as integrals of functions of the form $u(\mathbf{r})e^{i(\mathbf{p}_0-\mathbf{p})\cdot\mathbf{r}/\hbar}$. For such an integral to be appreciably different from zero, the periods of the oscillating factor $e^{i(\mathbf{p}_0-\mathbf{p})\cdot\mathbf{r}/\hbar}$ must not be small compared with the dimensions Δx, Δy, and Δz of the region where $u(\mathbf{r})$ differs from zero. Thus $a(\mathbf{p})$ is appreciably different from zero only for values of $\mathbf{p}$ satisfying $(p_{0x} - p_x)\Delta x/\hbar \lesssim 1, \ldots$ Since $|a(\mathbf{p})|^2$ determines the probabilities of the various momentum values, the intervals of p_x, p_y, p_z in which $a(\mathbf{p})$ is nonzero are precisely the intervals in which the particle's momentum components may lie in the state considered. Denoting these intervals by Δp_x, Δp_y, and Δp_z, we therefore have

$$\Delta p_x \Delta x \sim \hbar, \qquad \Delta p_y \Delta y \sim \hbar, \qquad \Delta p_z \Delta z \sim \hbar. \tag{16.6}$$

These *uncertainty relations* were established by *Heisenberg* in 1927.

We see that the more accurately a particle coordinate is known (the smaller Δx is), the greater is the uncertainty Δp_x in the momentum component along the same axis, and conversely. In particular, if a particle occupies a strictly definite point of space ($\Delta x = \Delta y = \Delta z = 0$), then $\Delta p_x = \Delta p_y = \Delta p_z = \infty$. All momentum values are then equally probable. Conversely, if the particle has a strictly definite momentum $\mathbf{p}$, all its positions in space are equally probable (this also follows directly from the wave function (15.8), whose squared modulus is entirely independent of the coordinates).

If the coordinate and momentum uncertainties are characterized by their root-mean-square fluctuations,

$$\delta x = \sqrt{\overline{(x-\bar{x})^2}}, \qquad \delta p_x = \sqrt{\overline{(p_x-\bar{p}_x)^2}},$$

an exact bound can be given for the least possible value of their product (*H. Weyl*).

Consider the one-dimensional case, a packet with wave function $\psi(x)$ depending on only one coordinate; for simplicity, suppose that the mean values of x and p_x in this state are zero. Begin with the evident inequality

$$\int_{-\infty}^{\infty} \left| \alpha x \psi + \frac{d\psi}{dx} \right|^2 dx \geq 0,$$

where α is an arbitrary real constant. In evaluating this integral, note that

$$\int x^2 |\psi|^2 \, dx = (\delta x)^2,$$

$$\int \left(x \frac{d\psi^*}{dx} \psi + x\psi^* \frac{d\psi}{dx} \right) dx = \int x \frac{d|\psi|^2}{dx} \, dx = -\int |\psi|^2 \, dx = -1,$$

$$\int \frac{d\psi^*}{dx} \frac{d\psi}{dx} \, dx = -\int \psi^* \frac{d^2\psi}{dx^2} \, dx = \frac{1}{\hbar^2} \int \psi^* \hat{p}_x^2 \psi \, dx = \frac{1}{\hbar^2} (\delta p_x)^2,$$

and obtain

$$\alpha^2 (\delta x)^2 - \alpha + \frac{(\delta p_x)^2}{\hbar^2} \geq 0.$$

For this quadratic trinomial in α to be nonnegative for every value of α, its discriminant must be negative. It follows that

$$\delta x \, \delta p_x \geq \hbar/2. \tag{16.7}$$

The least possible value of the product is $\hbar/2$.

This value is attained by wave packets described by functions of the form

$$\psi = \frac{1}{(2\pi)^{1/4} \sqrt{\delta x}} \exp\left(\frac{i}{\hbar} p_0 x - \frac{x^2}{4(\delta x)^2} \right), \tag{16.8}$$

where p_0 and δx are constants. The probabilities of the various coordinate values in such a state are

$$|\psi|^2 = \frac{1}{\sqrt{2\pi}\, \delta x} \exp\left(-\frac{x^2}{2(\delta x)^2} \right),$$

that is, they have a Gaussian distribution about the origin (mean value $\bar{x} = 0$), with root-mean-square fluctuation δx. The wave function in the momentum representation is

$$a(p_x) = \int_{-\infty}^{\infty} \psi(x) \exp\left(-i \frac{p_x x}{\hbar} \right) dx.$$

Evaluation of the integral gives an expression of the form

$$a(p_x) = \text{const} \cdot \exp\left(-\frac{(\delta x)^2 (p_x - p_0)^2}{\hbar^2} \right).$$

The probability distribution of the momentum values, $|a(p_x)|^2$, is also Gaussian, about the mean $\bar{p}_x = p_0$, and has root-mean-square fluctuation $\delta p_x = \hbar/(2\delta x)$. Thus the product $\delta p_x \delta x$ is exactly $\hbar/2$.

Finally, let us derive one more useful relation. Let f and g be two physical quantities whose operators obey the commutation rule

$$\hat{f}\hat{g} - \hat{g}\hat{f} = -i\hbar\hat{c}, \tag{16.9}$$

where $\hat{c}$ is the operator of a physical quantity c. The factor $\hbar$ has been introduced on the right because in the classical limit ($\hbar \to 0$) all operators of physical quantities reduce to multiplication by those quantities and commute with one another. Thus, in the "quasiclassical" case, the right-hand side of (16.9) may be taken as zero in the first approximation. In the next approximation, $\hat{c}$ may be replaced by the operator of simple multiplication by c. We then have

$$\hat{f}\hat{g} - \hat{g}\hat{f} = -i\hbar c.$$

This equation is exactly analogous to $\hat{p}_x x - x\hat{p}_x = -i\hbar$, except that the quantity $\hbar c$ occurs in place of the constant $\hbar$[17]. By analogy with $\Delta x \Delta p_x \sim \hbar$, we therefore conclude that in the quasiclassical case the quantities f and g obey the uncertainty relation

$$\Delta f \Delta g \sim \hbar c. \tag{16.10}$$

In particular, if one of the quantities is the energy ($f \equiv H$) and the operator of the other (g) has no explicit time dependence, then by (9.2) $c = \dot{g}$, and the quasiclassical uncertainty relation is

$$\Delta E \Delta g \sim \hbar \dot{g}. \tag{16.11}$$

[17]The classical quantity c is the Poisson bracket of f and g (see the note on p. 46).

CHAPTER III

The Schrödinger Equation

§ 17. The Schrödinger equation

The Hamiltonian determines the form of a physical system's wave equation and consequently acquires fundamental significance in the entire mathematical apparatus of quantum mechanics.

For a free particle, the Hamiltonian is already established by the general requirements associated with the homogeneity and isotropy of space and with the Galilean relativity principle. In classical mechanics these requirements imply a quadratic dependence of the particle energy on its momentum: $E = p^2/2m$, where the constant m is called the mass of the particle (see I, § 4). In quantum mechanics the same requirements lead to the same relation for the eigenvalues of energy and momentum, quantities which for a free particle are conserved and can be measured simultaneously.

For the relation $E = p^2/2m$ to hold for every eigenvalue of energy and momentum, however, it must also hold for their operators:

$$\hat{H} = \frac{1}{2m}\left(\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2\right). \tag{17.1}$$

Substitution of (15.2) here gives the Hamiltonian of a freely moving particle in the form

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta, \tag{17.2}$$

where $\Delta = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is the Laplace operator. The Hamiltonian of a system of noninteracting particles is equal to the sum of the Hamiltonians of the individual particles:

$$\hat{H} = -\frac{\hbar^2}{2}\sum_a \frac{\Delta_a}{m_a}, \tag{17.3}$$

where the index a numbers the particles; Δ_a is the Laplace operator in which differentiation is performed with respect to the coordinates of the ath particle. In classical (nonrelativistic) mechanics, particle interaction is described by an additive term in the Hamilton function, the potential energy of interaction $U(\mathbf{r}_1, \mathbf{r}_2, \ldots)$, which is a function of the particle coordinates. Particle interaction in quantum mechanics is likewise described by adding the same function to the Hamiltonian of the system[1]:

$$\hat{H} = -\frac{\hbar^2}{2}\sum_a \frac{\Delta_a}{m_a} + U(\mathbf{r}_1, \mathbf{r}_2, \ldots); \tag{17.4}$$

[1]This assertion is not, of course, a logical consequence of the fundamental principles of quantum mechanics and must be regarded as a consequence of experimental data.

the first term may be regarded as the kinetic-energy operator and the second as the potential-energy operator. In particular, the Hamiltonian for one particle in an external field is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + U(x, y, z) = -\frac{\hbar^2}{2m}\Delta + U(x, y, z), \tag{17.5}$$

where $U(x, y, z)$ is the potential energy of the particle in the external field.

Substitution of expressions (17.2)–(17.5) in the general equation (8.1) yields the wave equations for the corresponding systems. Here we write out the wave equation for a particle in an external field:

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\Delta\Psi + U(x, y, z)\Psi. \tag{17.6}$$

Equation (10.2), determining the stationary states, assumes the form

$$\frac{\hbar^2}{2m}\Delta\psi + [E - U(x, y, z)]\psi = 0. \tag{17.7}$$

Equations (17.6) and (17.7) were established by *Schrödinger* in 1926 and are known as the *Schrödinger equations*.

For a free particle equation (17.7) becomes

$$\frac{\hbar^2}{2m}\Delta\psi + E\psi = 0. \tag{17.8}$$

This equation admits solutions that remain finite over all space for any positive value of the energy E. In states with definite directions of motion these solutions are the eigenfunctions of the momentum operator, with $E = p^2/2m$. The full (time-dependent) wave functions of these stationary states have the form

$$\Psi = \text{const} \cdot \exp\left[-\frac{i}{\hbar}(Et - \mathbf{p}\cdot\mathbf{r})\right]. \tag{17.9}$$

Each of these functions, a *plane wave*, describes a state in which the particle possesses definite energy E and momentum $\mathbf{p}$. The frequency of this wave is $E/\hbar$, and its wave vector is $\mathbf{k} = \mathbf{p}/\hbar$; the associated wavelength $\lambda = 2\pi\hbar/p$ is termed the *de Broglie wavelength of the particle*[2].

The energy spectrum of a freely moving particle is therefore continuous, extending from zero to $+\infty$. Each of these eigenvalues, except only $E = 0$, is degenerate, and the degeneracy is of infinite multiplicity. Indeed, every nonzero value of E corresponds to an infinite set of eigenfunctions (17.9), differing in the directions of the vector $\mathbf{p}$ while its absolute magnitude remains the same.

Let us trace how the limiting transition to classical mechanics occurs in the Schrödinger equation, considering for simplicity just one particle in an external field. Substitution in the Schrödinger equation (17.6) of the limiting expression (6.1), $\Psi = ae^{iS/\hbar}$, for the wave function gives, after the differentiations are performed,

$$a\frac{\partial S}{\partial t} - i\hbar\frac{\partial a}{\partial t} + \frac{a}{2m}(\nabla S)^2 - \frac{i\hbar}{2m}a\Delta S - \frac{i\hbar}{m}\nabla S\cdot\nabla a - \frac{\hbar^2}{2m}\Delta a + Ua = 0.$$

[2] The concept of a wave associated with a particle was first introduced by *de Broglie* (*L. de Broglie*) in 1924.

This equation contains purely real and purely imaginary terms (recall that S and a are real); equating each group separately to zero gives two equations:

$$\frac{\partial S}{\partial t} + \frac{1}{2m}(\nabla S)^2 + U - \frac{\hbar^2}{2ma}\Delta a = 0,$$

$$\frac{\partial a}{\partial t} + \frac{a}{2m}\Delta S + \frac{1}{m}\nabla S \cdot \nabla a = 0.$$

Neglecting the term containing $\hbar^2$ in the first of these equations, we obtain

$$\frac{\partial S}{\partial t} + \frac{1}{2m}(\nabla S)^2 + U = 0, \tag{17.10}$$

that is, as expected, the classical Hamilton–Jacobi equation for the action S of the particle. We note in passing that, as $\hbar \to 0$ classical mechanics is accurate up to and including quantities of first order in $\hbar$ (rather than zeroth order).

After multiplication by $2a$, the second equation obtained above can be rewritten as

$$\frac{\partial a^2}{\partial t} + \nabla \cdot \left(a^2 \frac{\nabla S}{m}\right) = 0. \tag{17.11}$$

This equation has a transparent physical meaning: a^2 is the probability density for finding the particle at one or another place in space ($|\Psi|^2 = a^2$); $\nabla S/m = \mathbf{p}/m$ is the classical velocity $\mathbf{v}$ of the particle. Equation (17.11) is therefore nothing other than the continuity equation, showing that the probability density is "transported" according to the laws of classical mechanics, with the classical velocity $\mathbf{v}$ at every point.

Problem

Find the transformation law of the wave function under a Galilean transformation.

Solution. Perform the transformation on the wave function for the free motion of a particle (a plane wave). Since every function Ψ can be expanded in plane waves, this will also determine the transformation law for an arbitrary wave function.

The plane waves in the reference frames K and K' (where K' moves with velocity $\mathbf{V}$ relative to K) are

$$\Psi(\mathbf{r}, t) = \text{const} \cdot \exp[i(\mathbf{p} \cdot \mathbf{r} - Et)/\hbar],$$

$$\Psi'(\mathbf{r}', t) = \text{const} \cdot \exp[i(\mathbf{p}' \cdot \mathbf{r}' - E't)/\hbar],$$

where $\mathbf{r} = \mathbf{r}' + \mathbf{V}t$, while the particle's momenta and energies in the two frames are related by

$$\mathbf{p} = \mathbf{p}' + m\mathbf{V}, \qquad E = E' + \mathbf{V} \cdot \mathbf{p}' + \frac{mV^2}{2}$$

(see I, § 8). Substitution of these expressions in Ψ gives

$$\begin{aligned} \Psi(\mathbf{r}, t) &= \Psi'(\mathbf{r}', t) \exp\left[\frac{i}{\hbar}\left(m\mathbf{V} \cdot \mathbf{r}' + \frac{mV^2}{2}t\right)\right] \\ &= \Psi'(\mathbf{r} - \mathbf{V}t, t) \exp\left[\frac{i}{\hbar}\left(m\mathbf{V} \cdot \mathbf{r} - \frac{mV^2}{2}t\right)\right]. \end{aligned} \tag{1}$$

Written this way, the formula no longer contains quantities characterizing the free motion of the particle; it therefore furnishes the sought-after general rule governing how the wave function of an arbitrary state of the particle transforms. For a system of particles, the exponent of the exponential in (1) must contain a summation over all the particles.

§ 18. Fundamental properties of the Schrödinger equation

The conditions to be satisfied by solutions of the Schrödinger equation are quite general in nature. The wave function, to begin with, must be single-valued and continuous throughout all of space. The continuity requirement remains in force even when the field $U(x, y, z)$ itself has surfaces of discontinuity. On any such surface, the wave function together with its derivatives must still be continuous. The latter are not continuous, however, if the potential energy U becomes infinite beyond some surface. A particle cannot enter at all a region of space where $U = \infty$, that is, $\psi = 0$ must hold everywhere in that region. Continuity of ψ requires it to vanish on the boundary of this region; in this case, however, the derivatives of ψ generally undergo a jump.

If the field $U(x, y, z)$ nowhere becomes infinite, the wave function must also be finite throughout space. The same condition must be observed when U becomes infinite at some point, provided it does not do so too rapidly, namely as $1/r^s$ with $s < 2$ (see also § 35).

Let $U_{\min}$ be the minimum value of the function $U(x, y, z)$. Since the particle's Hamiltonian is the sum of two terms, the kinetic- and potential-energy operators $\hat{T}$ and U, the mean energy in an arbitrary state is the sum $\bar{E} = \bar{T} + \bar{U}$. But all eigenvalues of the operator $\hat{T}$ (which coincides with the Hamiltonian of a free particle) are positive, and therefore the mean value $\bar{T} > 0$ as well. Taking account also of the evident inequality $\bar{U} > U_{\min}$, we find that $\bar{E} > U_{\min}$. Since this inequality holds for every state, it plainly holds for all energy eigenvalues as well:

$$E_n > U_{\min}. \tag{18.1}$$

Suppose a particle moves through a force field that goes to zero at large distances; following standard convention, we fix $U(x, y, z)$ so that it too tends to zero at infinity. One readily verifies that the eigenvalue spectrum for negative energies is then discrete: every state with $E < 0$ in a field that vanishes at infinity is a bound state. Indeed, in stationary states of the continuous spectrum, which correspond to infinite motion, the particle is at infinity (see § 10). At sufficiently large distances the presence of the field may be neglected and the particle's motion may be regarded as free; in free motion, however, the energy can only be positive.

Conversely, the positive eigenvalues form a continuous spectrum and correspond to infinite motion; for $E > 0$ the Schrödinger equation generally does not have (in the field under consideration) solutions for which the integral $\int |\psi|^2 \, dV$ converges[3].

We draw attention to the fact that, in quantum mechanics, a particle in finite motion may also be present in regions of space where $E < U$. Although the probability $|\psi|^2$

[3] From a purely mathematical standpoint, however, it must be qualified that, for certain forms of the function $U(x, y, z)$ (having no physical significance), a discrete set of values may drop out of the continuous spectrum.

of finding the particle tends rapidly to zero deeper inside such a region, it nevertheless differs from zero at every finite distance. This represents a fundamental departure from classical mechanics, where a particle is altogether unable to penetrate a region with $U > E$. In classical mechanics, penetration into this region is impossible because, for $E < U$, the kinetic energy would become negative, meaning that the velocity would be imaginary. In quantum mechanics, too, the kinetic-energy eigenvalues are positive; nevertheless, no contradiction arises here, because if the measurement process localizes the particle at some definite point of space, that same process disturbs the particle's state in such a way that it ceases altogether to have any definite kinetic energy.

If $U(x, y, z) > 0$ throughout space (with $U \to 0$ at infinity), inequality (18.1) gives $E_n > 0$. On the other hand, since the spectrum must be continuous for $E > 0$, it follows that in the case under consideration the discrete spectrum is absent altogether, that is, only infinite motion of the particle is possible.

Suppose that at some point, chosen as the coordinate origin, U tends to $-\infty$ according to the law

$$U \approx -\alpha/r^s, \qquad \alpha > 0. \tag{18.2}$$

Consider a wave function that is finite in a certain small region of radius r_0 around the origin and zero outside it. The uncertainty in the particle's coordinate values in such a wave packet is of order r_0; the uncertainty in the momentum value is therefore $\sim \hbar/r_0$. The mean kinetic energy in this state is of order $\hbar^2/mr_0^2$, and the mean potential energy is of order $-\alpha/r_0^s$. Suppose first that $s > 2$. Then, for sufficiently small r_0, the sum $\hbar^2/mr_0^2 - \alpha/r_0^s$ takes negative values of arbitrarily large absolute magnitude. But if the mean energy can take such values, this means in any event that there exist negative energy eigenvalues of arbitrarily large absolute magnitude. Energy levels with large $|E|$ correspond to motion of the particle in a very small region of space around the origin. The "normal" state will correspond to the particle being located at the origin itself, that is, the particle will "fall" into the point $r = 0$.

If $s < 2$, on the other hand, the energy cannot take negative values of arbitrarily large absolute magnitude. The diagonal spectrum starts from a certain finite negative value, and the particle does not fall to the center. We note that in classical mechanics a particle can in principle fall to the center in any attractive field (that is, for any positive s). The case $s = 2$ will be considered separately in § 35.

Next, let us investigate the character of the energy spectrum as a function of the behaviour of the field at large distances. Suppose that as $r \to \infty$ the potential energy, while negative, tends to zero according to the power law (18.2) (in this formula r is now large). Consider a wave packet "filling" a spherical shell of large radius r_0 and thickness $\Delta r \ll r_0$. Then again the order of magnitude of the kinetic energy is $\hbar^2/m(\Delta r)^2$, and that of the potential energy is $-\alpha/r_0^s$. Let r_0 increase while Δr is increased at the same time (so that Δr grows in proportion to r_0). If $s < 2$, then for sufficiently large r_0 the sum $\hbar^2/m(\Delta r)^2 - \alpha/r_0^s$ becomes negative. It follows that stationary states with negative energy exist in which the particle may be found with appreciable probability at large distances from the origin. But this means that negative energy levels of arbitrarily small absolute magnitude exist (it must be remembered that wave functions decay rapidly in the region of space where $U > E$). Thus, in the case under consideration, the discrete spectrum contains an infinite set of levels which crowd together towards the level $E = 0$.

If, however, the field falls off at infinity as $-1/r^s$ with $s > 2$, there are no negative levels of arbitrarily small absolute magnitude. The discrete spectrum ends at a level with nonzero absolute magnitude, so that the total number of levels is finite.

The Schrödinger equation for the stationary-state wave functions ψ, together with the conditions placed on its solutions, is real. Its solutions can therefore always be chosen real[4]. The eigenfunctions of nondegenerate energy values automatically prove to be real up to an immaterial phase factor. Indeed, ψ^* satisfies the same equation as ψ and is therefore also an eigenfunction for the same value of the energy; hence, if that value is nondegenerate, ψ and ψ^* must be essentially identical, that is, they may differ at most by a constant factor whose modulus equals unity. Wave functions that correspond to one and the same degenerate level need not be real, but a set of real functions can always be obtained by a suitable choice of their linear combinations.

The full (time-dependent) wave functions Ψ, however, are determined by an equation whose coefficients contain i. This equation nevertheless retains its form if t is replaced by $-t$ and complex conjugation is performed at the same time[5]. The functions Ψ can therefore always be chosen so that Ψ and Ψ^* differ only in the sign of time.

As is known, the equations of classical mechanics are unchanged under *time reversal*, i.e. upon reversing the sign of time. In quantum mechanics, as we see, symmetry with respect to the two directions of time manifests itself in the invariance of the wave equation under reversal of the sign of t accompanied by the replacement of Ψ with Ψ^*. It must be remembered, however, that here this symmetry applies only to the equations and not to the concept of measurement itself, which has a fundamental role in quantum mechanics (as discussed in detail in § 7).

§ 19. Flux Density

In classical mechanics a particle's velocity $\mathbf{v}$ is related to its momentum by $\mathbf{p} = m\mathbf{v}$. As expected, quantum mechanics has the same relation between the corresponding operators. This follows at once by calculating $\hat{\mathbf{v}} = \hat{\dot{\mathbf{r}}}$ from the general rule (9.2) for differentiating operators with respect to time:

$$\hat{\mathbf{v}} = \frac{i}{\hbar}\left(\hat{H}\mathbf{r} - \mathbf{r}\hat{H}\right).$$

Using (17.5) for $\hat{H}$ together with (16.5), we obtain

$$\hat{\mathbf{v}} = \hat{\mathbf{p}}/m. \tag{19.1}$$

The same relations plainly hold between the eigenvalues of velocity and momentum, and between their mean values in any state.

Like momentum, the velocity of a particle cannot have a definite value simultaneously with its coordinates. Velocity multiplied by an infinitesimal time element dt, however,

[4] This is not true for systems in a magnetic field.

[5] It is assumed that the potential energy U has no explicit dependence on time: the system is either closed or is in a constant (nonmagnetic) field.

determines the particle's displacement during dt. Thus the nonexistence of velocity simultaneously with coordinates means that if a particle occupies a definite point of space at one instant, it no longer has a definite position at the next, infinitesimally close instant.

We record a useful formula for the operator $\hat{\dot{f}}$, the time derivative of a quantity $f(\mathbf{r})$ which is a function of the particle's radius vector. Since f commutes with $U(\mathbf{r})$, we find

$$\hat{\dot{f}} = \frac{i}{\hbar}(\hat{H}f - f\hat{H}) = \frac{i}{2m\hbar}(\hat{\mathbf{p}}^2 f - f\hat{\mathbf{p}}^2),$$

while (16.4) gives

$$\hat{\mathbf{p}}^2 f = \hat{\mathbf{p}}\,(f\hat{\mathbf{p}} - i\hbar\nabla f)\,, \qquad f\hat{\mathbf{p}}^2 = (\hat{\mathbf{p}}f + i\hbar\nabla f)\,\hat{\mathbf{p}}.$$

Hence the desired expression is

$$\hat{\dot{f}} = \frac{1}{2m}\,(\hat{\mathbf{p}}\,\nabla f + \nabla f\cdot\hat{\mathbf{p}})\,. \tag{19.2}$$

Next, let us find the acceleration operator. We have

$$\hat{\dot{\mathbf{v}}} = \frac{i}{\hbar}(\hat{H}\hat{\mathbf{v}} - \hat{\mathbf{v}}\hat{H}) = \frac{i}{m\hbar}(\hat{H}\hat{\mathbf{p}} - \hat{\mathbf{p}}\hat{H}) = \frac{i}{m\hbar}(U\hat{\mathbf{p}} - \hat{\mathbf{p}}U).$$

Using (16.4), we find

$$m\hat{\dot{\mathbf{v}}} = -\nabla U. \tag{19.3}$$

In form, this operator equation coincides exactly with the equation of motion (Newton's equation) of classical mechanics.

The integral $\int_V |\Psi|^2\,dV$ over a finite volume V is the probability of finding the particle in that volume. Let us calculate its time derivative:

$$\frac{d}{dt}\int_V |\Psi|^2\,dV = \int_V \left(\Psi\frac{\partial\Psi^*}{\partial t} + \Psi^*\frac{\partial\Psi}{\partial t}\right)dV = \frac{i}{\hbar}\int_V \left(\Psi\hat{H}^*\Psi^* - \Psi^*\hat{H}\Psi\right)dV.$$

Substitution of

$$\hat{H} = \hat{H}^* = -\frac{\hbar^2}{2m}\Delta + U(x,y,z)$$

and use of the identity

$$\Psi\Delta\Psi^* - \Psi^*\Delta\Psi = \nabla\cdot(\Psi\nabla\Psi^* - \Psi^*\nabla\Psi)$$

give

$$\frac{d}{dt}\int_V |\Psi|^2\,dV = -\int_V \nabla\cdot\mathbf{j}\,dV,$$

where $\mathbf{j}$ denotes the vector[6]

$$\mathbf{j} = \frac{i\hbar}{2m}(\Psi\nabla\Psi^* - \Psi^*\nabla\Psi) = \frac{1}{2m}(\Psi\hat{\mathbf{p}}^*\Psi^* + \Psi^*\hat{\mathbf{p}}\Psi). \tag{19.4}$$

[6] If ψ is written as $|\psi|e^{i\alpha}$, then

$$\mathbf{j} = \frac{\hbar}{m}|\psi|^2\nabla\alpha. \tag{19.4a}$$

By Gauss's theorem, the integral of $\nabla \cdot \mathbf{j}$ can be transformed into an integral over the closed surface bounding V:

$$\frac{d}{dt}\int_V |\Psi|^2\, dV = -\oint \mathbf{j}\, d\mathbf{f}. \tag{19.5}$$

It follows that $\mathbf{j}$ may be called the *probability-flux-density vector*, or simply the *flux density*. Its surface integral is the probability that the particle crosses that surface per unit time. The vector $\mathbf{j}$ and the probability density $|\Psi|^2$ obey

$$\frac{\partial |\Psi|^2}{\partial t} + \nabla \cdot \mathbf{j} = 0, \tag{19.6}$$

an equation analogous to the classical continuity equation.

The free-motion wave function—the plane wave (17.9)—can be normalized so as to describe a particle flux of unit density (a flux in which, on average, one particle per unit time passes through a unit area of its cross-section). The function is

$$\Psi = \frac{1}{\sqrt{v}} \exp\left[-\frac{i}{\hbar}(Et - \mathbf{p}\cdot\mathbf{r})\right], \tag{19.7}$$

where v is the particle's speed. Indeed, substituting this function into (19.4) gives $\mathbf{j} = \mathbf{p}/mv$, the unit vector in the direction of motion. It is useful to show how the mutual orthogonality of wave functions of states with different energies follows directly from the Schrödinger equation. Let ψ_m and ψ_n be two such functions; they satisfy

$$-\frac{\hbar^2}{2m}\Delta\psi_m + U\psi_m = E_m\psi_m,$$

$$-\frac{\hbar^2}{2m}\Delta\psi_n^* + U\psi_n^* = E_n\psi_n^*.$$

Multiply the first equation by ψ_n^* and the second by ψ_m, and subtract them term by term. This gives

$$(E_m - E_n)\psi_m\psi_n^* = \frac{\hbar^2}{2m}(\psi_m\Delta\psi_n^* - \psi_n^*\Delta\psi_m) = \frac{\hbar^2}{2m}\nabla\cdot(\psi_m\nabla\psi_n^* - \psi_n^*\nabla\psi_m).$$

Integrating both sides over all space, the right-hand side vanishes after application of Gauss's theorem, and we obtain

$$(E_m - E_n)\int \psi_m\psi_n^*\, dV = 0.$$

Since $E_m \neq E_n$ by assumption, the required orthogonality relation follows:

$$\int \psi_m\psi_n^*\, dV = 0.$$

§ 20. The Variational Principle

The Schrödinger equation in its general form, $\hat{H}\psi = E\psi$, can be obtained from the variational principle

$$\delta \int \psi^*(\hat{H} - E)\psi \, dq = 0. \tag{20.1}$$

Since ψ is complex, variations with respect to ψ and ψ^* may be made independently. Variation with respect to ψ^* gives

$$\int \delta\psi^*(\hat{H} - E)\psi \, dq = 0,$$

and the arbitrariness of $\delta\psi^*$ then yields the required equation $\hat{H}\psi = E\psi$. Variation with respect to ψ gives nothing new. Indeed, varying ψ and using the Hermitian character of $\hat{H}$, we have

$$\int \psi^*(\hat{H} - E)\delta\psi \, dq = \int \delta\psi(\hat{H}^* - E)\psi^* \, dq = 0,$$

which yields the complex-conjugate equation $\hat{H}^*\psi^* = E\psi^*$.

The variational principle (20.1) requires an unconstrained extremum of the integral. It may be put in another form by treating E as a Lagrange multiplier in the problem of finding a constrained extremum of

$$\delta \int \psi^* \hat{H}\psi \, dq = 0 \tag{20.2}$$

subject to the supplementary condition

$$\int \psi\psi^* \, dq = 1. \tag{20.3}$$

The minimum value of the integral (20.2), subject to (20.3), is the first energy eigenvalue, that is, the energy E_0 of the normal state. The function ψ which realizes this minimum is correspondingly the normal-state wave function ψ_0[7]. The wave functions ψ_n $(n > 0)$ of the succeeding stationary states correspond only to extrema, not to true minima of the integral.

To obtain from the minimum condition for (20.2) the wave function ψ_1 and energy E_1 of the state immediately above the normal state, the trial functions ψ must be restricted not only by the normalization condition (20.3), but also by orthogonality to the normal-state wave function ψ_0: $\int \psi\psi_0 \, dq = 0$. In general, suppose that the wave functions $\psi_0, \psi_1, \ldots, \psi_{n-1}$ of the first n states are known (the states being ordered by increasing energy). The wave function of the next state minimizes (20.2) under the supplementary conditions

$$\int \psi^2 \, dq = 1, \qquad \int \psi\psi_m \, dq = 0, \qquad m = 0, 1, 2, \ldots, n-1. \tag{20.4}$$

[7]For the remainder of this section we take the wave functions ψ to be real, as they can always be chosen when no magnetic field is present.

We give here several general theorems which may be proved from the variational principle[8].

The normal-state wave function ψ_0 does not vanish (or, as one says, has no nodes) at any finite values of the coordinates[9]. In other words, it has the same sign throughout space. It follows that the wave functions ψ_n $(n > 0)$ of the other stationary states, being orthogonal to ψ_0, necessarily have nodal points (if ψ_n also had a constant sign, the integral $\int \psi_0 \psi_n \, dq$ could not vanish).

The absence of nodes in ψ_0 also implies that the normal energy level cannot be degenerate. Suppose otherwise, and let ψ_0 and ψ_0' be two different eigenfunctions belonging to the level E_0. Every linear combination $c\psi_0 + c'\psi_0'$ would also be an eigenfunction; but by a suitable choice of c and c', this function can always be made to vanish at any prescribed point of space, producing an eigenfunction with a node.

If motion is confined to a bounded region of space, then $\psi = 0$ on its boundary (see § 18). To determine the energy levels one must use the variational principle to find the minimum of (20.2) with this boundary condition. In this case the theorem that the normal-state wave function has no nodes states that ψ_0 vanishes nowhere inside the region.

Notice that increasing the size of the region of motion lowers every energy level E_n. This follows directly because enlarging the region enlarges the class of trial functions available for minimizing the integral, so its minimum value can only decrease. For discrete-spectrum states of a system of particles, the expression

$$\int \psi \hat{H} \psi \, dq = \int \left[-\sum_a \frac{\hbar^2}{2m_a} \psi \Delta_a \psi + U\psi^2 \right] dq$$

can be transformed into a form more convenient for actual variation. In the first term of the integrand, write

$$\psi \Delta_a \psi = \nabla_a \cdot (\psi \nabla_a \psi) - (\nabla_a \psi)^2.$$

The integral of $\nabla_a \cdot (\psi \nabla_a \psi)$ over dV_a becomes an integral over a closed surface at infinity. Since discrete-spectrum wave functions vanish sufficiently rapidly at infinity, this integral disappears. Thus

$$\int \psi \hat{H} \psi \, dq = \int \left[\sum_a \frac{\hbar^2}{2m_a} (\nabla_a \psi)^2 + U\psi^2 \right] dq. \tag{20.5}$$

§ 21. General Properties of One-Dimensional Motion

When a particle's potential energy depends on just one coordinate x, the wave function may be sought as a product of a function of y, z and a function of x alone. The

[8]Proofs of the theorems about zeros of eigenfunctions (see also the next section) may be found in: *M. A. Lavrent'ev and L. A. Lyusternik*, Course of the Calculus of Variations, Moscow: Gostekhizdat, 1950, Ch. IX; *R. Courant and D. Hilbert*, Methods of Mathematical Physics, Moscow: Gostekhizdat, 1951, Vol. I, Ch. VI.

[9]This theorem (and the consequences drawn from it below) is not valid in general for wave functions of systems containing several identical particles; see the end of § 63.

first is determined by the free-motion Schrödinger equation, and the second by the one-dimensional Schrödinger equation

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - U(x)]\psi = 0. \tag{21.1}$$

The problem of motion in a field with potential energy $U(x, y, z) = U_1(x) + U_2(y) + U_3(z)$, separated into a sum of functions each depending on only one coordinate, plainly reduces to the same kind of one-dimensional equations. In §§ 22–24 we shall consider several specific examples of such "one-dimensional" motion. First we establish some of its general properties.

We begin by showing that every discrete-spectrum energy level in a one-dimensional problem is nondegenerate. Suppose the contrary, and let ψ_1 and ψ_2 be two different eigenfunctions belonging to the same energy value. Since both satisfy the same equation (21.1),

$$\frac{\psi_1''}{\psi_1} = \frac{2m}{\hbar^2}(U - E) = \frac{\psi_2''}{\psi_2},$$

or $\psi_2\psi_1'' - \psi_1\psi_2'' = 0$ (a prime denotes differentiation with respect to x). Integration gives

$$\psi_1\psi_2' - \psi_2\psi_1' = \text{const}\,. \tag{21.2}$$

Since $\psi_1 = \psi_2 = 0$ at infinity, the constant must be zero, and hence

$$\psi_1\psi_2' - \psi_2\psi_1' = 0,$$

or $\psi_1'/\psi_1 = \psi_2'/\psi_2$. A second integration gives $\psi_1 = \text{const}\cdot\psi_2$; the two functions are therefore essentially identical.

For the discrete-spectrum wave functions $\psi_n(x)$, the following result, known as the *oscillation theorem*, can be stated: the function $\psi_n(x)$ corresponding to the $(n+1)$th eigenvalue E_n in increasing order vanishes n times at finite x[10].

Suppose that $U(x)$ tends to finite limits as $x \to \pm\infty$ (it need not in any way be monotonic). Take $U(+\infty)$ as the zero of energy, so that $U(+\infty) = 0$, and denote $U(-\infty)$ by U_0, with $U_0 > 0$. The discrete spectrum lies in the range of energies for which the particle cannot escape to infinity. The energy must therefore be less than both limiting values and, in particular, negative:

$$E < 0. \tag{21.3}$$

Of course $E > U_{\min}$ must also hold in every case, so $U(x)$ must possess at least one minimum with $U_{\min} < 0$.

Now consider the positive energies below U_0:

$$0 < E < U_0. \tag{21.4}$$

Here the spectrum is continuous and the particle motion in the corresponding stationary states is infinite, with the particle escaping toward $x = +\infty$. Every energy eigenvalue in

[10]If the particle can occupy only a bounded interval of the x axis, the zeros of $\psi_n(x)$ inside that interval are to be counted.

this part of the spectrum is likewise nondegenerate. To see this, it suffices to observe that the proof given above for the discrete spectrum requires ψ_1 and ψ_2 to vanish at only one of the two infinities (here they vanish as $x \to -\infty$).

At sufficiently large positive x, $U(x)$ can be neglected in the Schrödinger equation (21.1):

$$\psi'' + \frac{2m}{\hbar^2}E\psi = 0.$$

This equation has real solutions in the form of a standing plane wave,

$$\psi = a\cos(kx + \delta), \tag{21.5}$$

where a and δ are constants, and the "wave vector" is $k = p/\hbar = \sqrt{2mE}/\hbar$. This formula gives the asymptotic form, as $x \to +\infty$, of the wave functions for the nondegenerate energy levels in the continuous-spectrum interval (21.4). At large negative x, the Schrödinger equation becomes

$$\psi'' - \frac{2m}{\hbar^2}(U_0 - E)\psi = 0.$$

The solution which does not become infinite as $x \to -\infty$ is

$$\psi = be^{\kappa x}, \qquad \kappa = \frac{1}{\hbar}\sqrt{2m(U_0 - E)}. \tag{21.6}$$

This is the asymptotic wave function as $x \to -\infty$. The wave function thus decays exponentially into the region in which $E < U_0$.

Finally, when

$$E > U_0, \tag{21.7}$$

the spectrum is continuous and the motion is infinite in both directions. Every level in this part of the spectrum is doubly degenerate. The reason is that the corresponding wave functions are determined by the second-order equation (21.1), and both independent solutions meet the required conditions at infinity (whereas, for example, in the preceding case one solution became infinite as $x \to -\infty$ and had to be rejected). As $x \to +\infty$, the asymptotic wave function is

$$\psi = a_1 e^{ikx} + a_2 e^{-ikx}, \tag{21.8}$$

and similarly as $x \to -\infty$. The term containing e^{ikx} corresponds to a particle moving to the right, and that containing e^{-ikx} to a particle moving to the left.

Suppose $U(x)$ is even, $U(-x) = U(x)$. The Schrödinger equation (21.1) is then unchanged on reversing the sign of the coordinate. Hence, if $\psi(x)$ is a solution, $\psi(-x)$ is also a solution and differs from $\psi(x)$ only by a constant factor: $\psi(-x) = c\psi(x)$. Reversing x once more gives $\psi(x) = c^2\psi(x)$, so $c = \pm 1$. Thus, for a potential energy symmetric about $x = 0$, stationary-state wave functions may be either even, $\psi(-x) = \psi(x)$, or odd, $\psi(-x) = -\psi(x)$[11]. In particular, the ground-state wave function is even: it cannot have a node, whereas an odd function necessarily vanishes at $x = 0$, since $\psi(0) = -\psi(0) = 0$.

[11]This argument assumes that the stationary state is nondegenerate, that is, that its motion is not infinite in both directions. Otherwise the two wave functions belonging to the given energy level may transform into one another when the sign of x is reversed. Even then, the stationary-state wave functions are not necessarily even or odd, but definite parity can always be achieved by forming suitable linear combinations of the two solutions.

There is a simple method for normalizing one-dimensional-motion wave functions in the continuous spectrum, which determines the normalization coefficient directly from the asymptotic wave function at large $|x|$.

Consider the wave function for motion infinite in one direction ($x \to +\infty$). The normalization integral diverges as $x \to \infty$ (at $x \to -\infty$ the function decays exponentially, so the integral converges rapidly). To determine the normalization constant, one may therefore replace ψ by its asymptotic value for large positive x and take any finite value of x, say zero, as the lower limit of integration; this is equivalent to discarding a finite contribution beside a divergent one. We now show that a wave function normalized by

$$\int \psi_p^* \psi_{p'}\, dx = \delta\left(\frac{p-p'}{2\pi\hbar}\right) = 2\pi\hbar\delta(p-p') \tag{21.9}$$

(where p is the particle momentum at infinity) must have the asymptotic form (21.5) with $a = 2$:

$$\psi_p \approx 2\cos(kx+\delta) = e^{i(kx+\delta)} + e^{-i(kx+\delta)}. \tag{21.10}$$

Since we are not seeking to verify mutual orthogonality of functions corresponding to different p, when substituting (21.10) into the normalization integral we take p and p' arbitrarily close and may put $\delta = \delta'$ (in general, δ is a function of p). We then retain in the integrand only the terms which diverge when $p = p'$; equivalently, we drop terms containing factors $e^{\pm i(k+k')x}$. We thus obtain

$$\int \psi_p^* \psi_{p'}\, dx = \int_0^\infty e^{i(k'-k)x}\, dx + \int_0^\infty e^{-i(k'-k)x}\, dx = \int_{-\infty}^\infty e^{i(k'-k)x}\, dx,$$

which, by (15.7), agrees with (21.9). According to (5.14), conversion to normalization by an energy delta function is effected by multiplying ψ_p by

$$\left(\frac{d(p/2\pi\hbar)}{dE}\right)^{1/2} = \frac{1}{\sqrt{2\pi\hbar v}},$$

where v is the particle's speed at infinity. Thus

$$\psi_E = \frac{1}{\sqrt{2\pi\hbar v}}\psi_p = \frac{1}{\sqrt{2\pi\hbar v}}\left(e^{i(kx+\delta)} + e^{-i(kx+\delta)}\right). \tag{21.11}$$

Notice that each of the two traveling waves into which the standing wave (21.11) separates has flux density $1/(2\pi\hbar)$.

We may therefore state the following rule for normalizing, to an energy delta function, the wave function of motion infinite in one direction: write its asymptotic expression as the sum of two plane waves traveling in opposite directions, and choose the normalization coefficient so that the flux density in the wave traveling toward the origin (or away from it) equals $1/(2\pi\hbar)$.

The analogous argument gives the corresponding rule for wave functions of motion infinite in both directions. Such a wave function is normalized to an energy delta function when the sum of the fluxes in the waves traveling toward the origin from the positive and negative sides of the x axis equals $1/(2\pi\hbar)$.

§ 22. The Potential Well

As a simple example of one-dimensional motion, consider motion in a rectangular *potential well*, that is, in a field with the function $U(x)$ shown in Fig. 1: $U(x) = 0$ for $0 < x < a$, and $U(x) = U_0$ for $x < 0$ and $x > a$. It is evident in advance that the spectrum is discrete when $E < U_0$, whereas for $E > U_0$ there is a continuous spectrum of doubly degenerate levels.

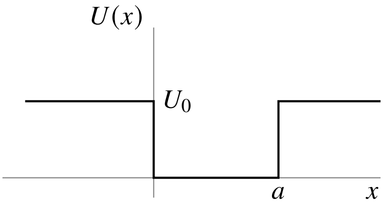

Fig. 1

In the region $0 < x < a$, the Schrödinger equation is

$$\psi'' + \frac{2m}{\hbar^2} E\psi = 0, \tag{22.1}$$

where a prime denotes differentiation with respect to x; outside the well it is

$$\psi'' + \frac{2m}{\hbar^2}(E - U_0)\psi = 0. \tag{22.2}$$

At $x = 0, a$, the solutions of these equations must pass continuously into one another, with continuous first derivative; at $x = \pm\infty$, the solution of (22.2) must remain finite (and for the discrete spectrum, $E < U_0$, it must vanish).

For $E < U_0$, the solution of (22.2) that vanishes at infinity is $\psi = \text{const} \cdot e^{\mp\kappa x}$, where

$$\kappa = \frac{1}{\hbar}\sqrt{2m(U_0 - E)} \tag{22.3}$$

(the minus and plus signs in the exponent refer respectively to the regions $x > a$ and $x < 0$). The probability $|\psi|^2$ of finding the particle decays exponentially into the region where $E < U(x)$. Instead of requiring ψ and ψ' to be continuous at the boundary of the potential well, it is convenient to require continuity of ψ and of the logarithmic derivative ψ'/ψ. With (22.3), the boundary condition becomes

$$|\psi'|/\psi = \mp\kappa. \tag{22.4}$$

Determining the energy levels for a well of arbitrary depth U_0 is left aside here (see Problem 2); we analyze in full only the limiting case of infinitely high walls ($U_0 \to \infty$).

When $U_0 = \infty$, motion occurs only on the interval bounded by $x = 0, a$ and, as stated in § 18, the boundary condition at these points is

$$\psi = 0. \tag{22.5}$$

(This condition also follows readily from the general condition (22.4). Indeed, when $U_0 \to \infty$, we also have $\kappa \to \infty$, and hence $\psi'/\psi \to \infty$; since ψ' cannot become infinite, it follows that $\psi = 0$.) Seek a solution of (22.1) inside the well in the form

$$\psi = c\sin(kx + \delta), \qquad k = \frac{\sqrt{2mE}}{\hbar}. \tag{22.6}$$

The condition $\psi = 0$ at $x = 0$ gives $\delta = 0$. The same condition at $x = a$ then gives $\sin ka = 0$, whence $ka = n\pi$ (n takes the positive integral values beginning with unity[12]), or

$$E_n = \frac{\pi^2\hbar^2}{2ma^2}n^2, \qquad n = 1, 2, 3, \ldots \tag{22.7}$$

The energy levels of a particle inside the potential well are thus determined. After normalization, the stationary-state wave functions are

$$\psi_n = \sqrt{\frac{2}{a}}\sin\left(\frac{\pi n}{a}x\right). \tag{22.8}$$

From these results one can write down directly the energy levels of a particle in a rectangular "potential box," that is, for three-dimensional motion in a field whose potential energy is $U = 0$ for $0 < x < a$, $0 < y < b$, $0 < z < c$, and $U = \infty$ outside this region. Namely, the levels are the sums

$$E_{n_1n_2n_3} = \frac{\pi^2\hbar^2}{2m}\left(\frac{n_1^2}{a^2} + \frac{n_2^2}{b^2} + \frac{n_3^2}{c^2}\right), \qquad n_1, n_2, n_3 = 1, 2, 3, \ldots, \tag{22.9}$$

and the corresponding wave functions are the products

$$\psi_{n_1n_2n_3} = \sqrt{\frac{8}{abc}}\sin\left(\frac{\pi n_1}{a}x\right)\sin\left(\frac{\pi n_2}{b}y\right)\sin\left(\frac{\pi n_3}{c}z\right). \tag{22.10}$$

Notice that, according to (22.7) or (22.9), the energy of the ground state is of order $E_0 \sim \hbar^2/ml^2$, where l denotes the linear dimensions of the region in which the particle moves. This result agrees with the uncertainty relations: for a coordinate uncertainty of order l, the momentum uncertainty, and with it the order of the momentum itself, is $\sim \hbar/l$; the corresponding energy is $\sim (\hbar/l)^2/m$.

Problem

1. Determine the probability distribution of the possible momentum values for the normal state of a particle in an infinitely deep rectangular potential well.

Solution. The coefficients $a(p)$ in the expansion of the function ψ_1 (22.8) in momentum eigenfunctions are

$$a(p) = \int \psi_p^*\psi_1\,dx = \sqrt{\frac{2}{a}}\int_0^a \sin\left(\frac{\pi}{a}x\right)\exp\left(-i\frac{px}{\hbar}\right)dx.$$

[12] For $n = 0$, one would obtain identically $\psi = 0$.

Evaluating the integral and taking the square of its modulus gives the required probability distribution

$$|a(p)|^2 \frac{dp}{2\pi\hbar} = \frac{4\pi\hbar^3 a}{\left(p^2a^2 - \pi^2\hbar^2\right)^2} \cos^2\left(\frac{pa}{2\hbar}\right) dp.$$

Problem

2. Determine the energy levels for the potential well shown in Fig. 2.

Solution. The discrete part of the spectrum is the range $E < U_1$, which is the range we consider. In the region $x < 0$, the wave function is

$$\psi = c_1 e^{\kappa_1 x}, \qquad \kappa_1 = \frac{1}{\hbar}\sqrt{2m(U_1 - E)},$$

and in the region $x > a$ it is

$$\psi = c_2 e^{-\kappa_2 x}, \qquad \kappa_2 = \frac{1}{\hbar}\sqrt{2m(U_2 - E)}.$$

Inside the well ($0 < x < a$), seek ψ in the form

$$\psi = c\sin(kx + \delta), \qquad k = \sqrt{2mE}/\hbar.$$

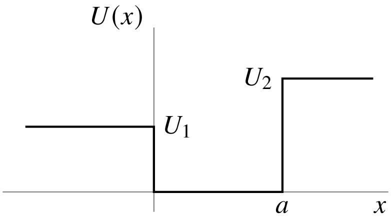

Fig. 2

Continuity of ψ'/ψ at the boundaries of the well gives

$$k\cot\delta = \kappa_1 = \sqrt{\frac{2m}{\hbar^2}U_1 - k^2},$$

$$k\cot(ak + \delta) = -\kappa_2 = -\sqrt{\frac{2m}{\hbar^2}U_2 - k^2},$$

or

$$\sin\delta = \frac{k\hbar}{\sqrt{2mU_1}}, \qquad \sin(ka + \delta) = -\frac{k\hbar}{\sqrt{2mU_2}}.$$

Eliminating δ, we obtain the transcendental equation

$$ka = n\pi - \arcsin\frac{k\hbar}{\sqrt{2mU_1}} - \arcsin\frac{k\hbar}{\sqrt{2mU_2}} \tag{1}$$

(where $n = 1, 2, 3, \ldots$, and the values of arcsin are taken between 0 and $\pi/2$), whose roots determine the energy levels $E = k^2\hbar^2/2m$. In general there is one root for each n; the values of n number the levels in increasing order.

Since the argument of arcsin cannot exceed unity, it is clear that the values of k can lie only in the interval from zero to $\sqrt{2mU_1}/\hbar$. The left-hand side of (1) is a monotonically increasing function of k, while the right-hand side is monotonically decreasing. Consequently, for a root of (1) to exist, its right-hand side at $k = \sqrt{2mU_1}/\hbar$ must be less than its left-hand side. In particular, the inequality

$$a\frac{\sqrt{2mU_1}}{\hbar} \geq \frac{\pi}{2} - \arcsin\sqrt{\frac{U_1}{U_2}}, \tag{2}$$

obtained for $n = 1$, is the condition for the well to contain at least one energy level. Thus, for given $U_1 \neq U_2$, there always exist sufficiently small values of the width a for which the well has no discrete energy level at all. When $U_1 = U_2$, condition (2) is plainly always satisfied.

When $U_1 = U_2 \equiv U_0$ (a symmetric well), equation (1) reduces to

$$\arcsin\frac{\hbar k}{\sqrt{2mU_0}} = \frac{n\pi - ka}{2}. \tag{3}$$

Introducing the variable $\xi = ka/2$, for odd n we obtain

$$\cos\xi = \pm\gamma\xi, \qquad \gamma = \frac{\hbar}{a}\sqrt{\frac{2}{mU_0}}, \tag{4}$$

where the roots for which $\tan\xi > 0$ must be selected. For even n we obtain

$$\sin\xi = \pm\gamma\xi, \tag{5}$$

where the roots for which $\tan\xi < 0$ must be selected. The energy levels are found from the roots of these two equations as $E = 2\xi^2\hbar^2/ma^2$; for $\gamma \neq 0$ only finitely many levels exist.

Consider as a special case a shallow well, in which $U_0 \ll \hbar^2/ma^2$, we have $\gamma \gg 1$, and equation (5) has no roots at all. Equation (4), however, has one root (with the upper sign on its right-hand side), equal to $\xi \approx (1/\gamma)(1 - 1/2\gamma^2)$. The well therefore contains only one energy level,

$$E_0 \approx U_0 - \frac{ma^2}{2\hbar^2}U_0^2,$$

lying close to its "top."

Problem

3. Determine the pressure exerted on the walls of a rectangular "potential box" by a particle inside it.

Solution. On a wall normal to the x axis, the force is given by the mean of the derivative $-\partial H/\partial a$ of the particle's Hamilton function with respect to the length of the box along the x axis; division of this force by the wall area bc gives the pressure. By (11.16), the required mean value is found by differentiating the energy eigenvalue (22.9). The resulting pressure is

$$p^{(x)} = \frac{\pi^2\hbar^2}{ma^3bc}n_1^2.$$

§ 23. The Linear Oscillator

Consider a particle executing small one-dimensional oscillations (the so-called *linear oscillator*). Its potential energy is $m\omega^2x^2/2$, where in classical mechanics ω is the natural frequency of the oscillations. The oscillator Hamiltonian is consequently

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2x^2}{2}. \tag{23.1}$$

Since the potential energy tends to infinity at $x = \pm\infty$, the particle can execute only finite motion. Accordingly, the entire energy spectrum of the oscillator is discrete.

Let us determine the oscillator's energy levels by the matrix method[13]. We begin with the equations of motion in the form (19.3), which in this case give

$$\hat{\ddot{x}} + \omega^2 x = 0. \tag{23.2}$$

In matrix form this equation is

$$(\ddot{x})_{mn} + \omega^2 x_{mn} = 0.$$

For the matrix elements of the acceleration, (11.8) gives $(\ddot{x})_{mn} = i\omega_{mn}(\dot{x})_{mn} = -\omega_{mn}^2 x_{mn}$. Hence

$$(\omega_{mn}^2 - \omega^2)x_{mn} = 0.$$

Thus all the matrix elements x_{mn} vanish except those for which $\omega_{mn} = \pm\omega$. Number all stationary states so that the frequencies $\pm\omega$ correspond to the transitions $n \to n \mp 1$, that is, $\omega_{n,n\mp1} = \pm\omega$. The only nonzero matrix elements are then $x_{n,n\mp1}$.

We shall suppose that the wave functions ψ_n have been chosen real. Since x is a real quantity, every matrix element x_{mn} is then real as well. The Hermiticity condition (11.10) now makes the matrix x_{mn} symmetric:

$$x_{mn} = x_{nm}.$$

To calculate the nonzero coordinate matrix elements, use the commutation rule

$$\hat{\dot{x}}\,\hat{x} - \hat{x}\,\hat{\dot{x}} = -i\frac{\hbar}{m},$$

written in matrix form as

$$(\dot{x}x)_{mn} - (x\dot{x})_{mn} = -\frac{i\hbar}{m}\delta_{mn}.$$

Using the matrix-multiplication rule (11.12), for $m = n$ this gives

$$i\sum_l(\omega_{nl}x_{nl}x_{ln} - x_{nl}\omega_{ln}x_{ln}) = 2i\sum_l \omega_{nl}x_{nl}^2 = -\frac{i\hbar}{m}.$$

Only the terms with $l = n \pm 1$ are nonzero in this sum, and therefore

$$(x_{n+1,n})^2 - (x_{n,n-1})^2 = \frac{\hbar}{2m\omega}. \tag{23.3}$$

[13]This was done by Heisenberg (1925), before Schrödinger's discovery of the wave equation.

This equality shows that the quantities $(x_{n+1,n})^2$ form an arithmetic progression, unbounded above but necessarily bounded below, since it can contain only positive terms. Thus far we have established only the relative ordering of the state labels n, not their absolute values; we may therefore choose arbitrarily the value of n assigned to the oscillator's first—the normal—state. Set it equal to zero. Correspondingly, $x_{0,-1}$ must be regarded as identically zero, and successive application of (23.3) for $n = 0, 1, \ldots$ gives

$$(x_{n,n-1})^2 = \frac{n\hbar}{2m\omega}.$$

The final expression for the nonzero coordinate matrix elements is therefore[14]

$$x_{n,n-1} = x_{n-1,n} = \sqrt{\frac{n\hbar}{2m\omega}}. \tag{23.4}$$

The matrix of $\hat{H}$ is diagonal, and its matrix elements H_{nn} are the required oscillator energy eigenvalues E_n. To calculate them, write

$$\begin{aligned} H_{nn} = E_n &= \frac{m}{2}\left[(\dot{x}^2)_{nn} + \omega^2 (x^2)_{nn}\right] \\ &= \frac{m}{2}\left[\sum_l i\omega_{nl}x_{nl}i\omega_{ln}x_{ln} + \omega^2 \sum_l x_{nl}x_{ln}\right] = \frac{m}{2}\sum_l (\omega^2 + \omega_{nl}^2)x_{ln}^2. \end{aligned}$$

Only the terms $l = n \pm 1$ are nonzero in the sum. Substitution of (23.4) gives

$$E_n = (n + 1/2)\hbar\omega, \qquad n = 0, 1, 2, \ldots \tag{23.5}$$

Thus the oscillator energy levels are separated by equal intervals $\hbar\omega$. The energy of the normal state ($n = 0$) is $\hbar\omega/2$; we emphasize that it is nonzero.

The result (23.5) may also be obtained by solving the Schrödinger equation. For the oscillator this equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{m\omega^2 x^2}{2}\right)\psi = 0. \tag{23.6}$$

It is convenient here to replace x by a dimensionless variable ξ, defined by

$$\xi = \sqrt{\frac{m\omega}{\hbar}}x. \tag{23.7}$$

We then obtain

$$\psi'' + \left(\frac{2E}{\hbar\omega} - \xi^2\right)\psi = 0. \tag{23.8}$$

(A prime here denotes differentiation with respect to ξ.)

For large ξ, the term $2E/\hbar\omega$ may be omitted in comparison with ξ^2; the equation $\psi'' = \xi^2\psi$ has the asymptotic integrals $\psi = e^{\pm\xi^2/2}$. (Indeed, differentiation of this

[14] We choose the undetermined phases α_n (see the note on p. 52) so that every matrix element in (23.4) has a plus sign before the root. A phase convention of this kind can always be arranged when a matrix has nonzero entries solely for transitions connecting states whose labels are adjacent.

function gives $\psi'' = \xi^2\psi$ when lower-order terms in ξ are neglected.) Since the wave function ψ must remain finite at $\xi = \pm\infty$, the minus sign must be chosen in the exponent. It is therefore natural to make the substitution

$$\psi = e^{-\xi^2/2}\chi(\xi) \tag{23.9}$$

in (23.8). For $\chi(\xi)$ we obtain the equation (introducing the notation $2E/\hbar\omega - 1 = 2n$; since we know in advance that $E > 0$, we have $n > -1/2$)

$$\chi'' - 2\xi\chi' + 2n\chi = 0, \tag{23.10}$$

where χ must be finite for every finite ξ and, at $\xi = \pm\infty$, may become infinite no faster than a finite power of ξ (so that ψ tends to zero).

Such solutions of (23.10) exist only for integer positive values of n, including zero (see § a of the mathematical supplements); this gives the energy eigenvalues (23.5), already known to us. For the different integral values of n, the corresponding solutions of (23.10) are

$$\chi = \text{const} \cdot H_n(\xi),$$

where $H_n(\xi)$ are the so-called Hermite polynomials, polynomials of degree n in ξ defined by

$$H_n(\xi) = (-1)^n e^{\xi^2} \frac{d^n e^{-\xi^2}}{d\xi^n}. \tag{23.11}$$

Choosing the constant so that ψ_n obeys the normalization condition

$$\int_{-\infty}^{+\infty} \psi_n^2(x)\, dx = 1,$$

we obtain (see (a.7))

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} \exp\left(-\frac{m\omega}{2\hbar}x^2\right) H_n\left(x\sqrt{\frac{m\omega}{\hbar}}\right). \tag{23.12}$$

Thus, the normal-state wave function is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right). \tag{23.13}$$

As expected, it has no zeros at finite x.

Evaluation of the integrals $\int_{-\infty}^{+\infty} \psi_n\psi_m\xi\, d\xi$ determines the coordinate matrix elements; the calculation naturally gives the same values (23.4).

We conclude by demonstrating how the wave functions ψ_n can be computed via the matrix method. In the matrices of the operators $\hat{\dot{x}} \pm i\omega\hat{x}$, the only nonzero elements are

$$(\dot{x} - i\omega x)_{n-1,n} = -(\dot{x} + i\omega x)_{n,n-1} = -i\sqrt{\frac{2\omega\hbar n}{m}}. \tag{23.14}$$

Using the general formula (11.11) and recalling that $\psi_{-1} \equiv 0$, we obtain

$$(\hat{\dot{x}} - i\omega x)\psi_0 = 0.$$

Substitution of $\hat{\dot{x}} = -i(\hbar/m)d/dx$ then gives

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar}x\psi_0,$$

whose normalized solution is (23.13). Further, since

$$(\hat{\dot{x}} + i\omega\hat{x})\psi_{n-1} = (\dot{x} + i\omega x)_{n,n-1}\psi_n = i\sqrt{\frac{2\omega\hbar n}{m}}\psi_n,$$

we obtain the recurrence formula

$$\begin{aligned}\psi_n &= \sqrt{\frac{m}{2\omega\hbar n}}\left(-\frac{\hbar}{m}\frac{d}{dx} + \omega x\right)\psi_{n-1} = \frac{1}{\sqrt{2n}}\left(-\frac{d}{d\xi} + \xi\right)\psi_{n-1}\\ &= -\frac{1}{\sqrt{2n}}\exp\frac{\xi^2}{2}\frac{d}{d\xi}\left(\exp\left(-\frac{\xi^2}{2}\right)\psi_{n-1}\right),\end{aligned}$$

whose n-fold application to (23.13) gives (23.12) for the normalized functions ψ_n.

Problem

1. Determine the probability distribution of the possible momentum values for the oscillator.

Solution. Rather than expanding a stationary-state wave function in momentum eigenfunctions, for the oscillator it is simpler to begin directly with the Schrödinger equation in the momentum representation. Substituting the coordinate operator (15.12), $\hat{x} = i\hbar d/dp$, into (23.1), we obtain the Hamiltonian in the momentum representation

$$\hat{H} = \frac{p^2}{2m} - \frac{m\omega^2\hbar^2}{2}\frac{d^2}{dp^2}.$$

The corresponding Schrödinger equation $\hat{H}a(p) = Ea(p)$ for the momentum-representation wave function $a(p)$ is

$$\frac{d^2a(p)}{dp^2} + \frac{2}{m\omega^2\hbar^2}\left(E - \frac{p^2}{2m}\right)a(p) = 0.$$

This equation has exactly the same form as (23.6), so its solutions can be written down directly by analogy with (23.12). The required probability distribution is therefore

$$|a_n(p)|^2\frac{dp}{2\pi\hbar} = \frac{1}{2^n n!\sqrt{\pi m\omega\hbar}}\exp\left(-\frac{p^2}{m\omega\hbar}\right)H_n^2\left(\frac{p}{\sqrt{m\omega\hbar}}\right)dp.$$

Problem

2. Use the uncertainty relation (16.7) to determine a lower bound on the possible energy values of the oscillator.

Solution. Noting that $\overline{x^2} = \bar{x}^2 + (\delta x)^2$ and $\overline{p^2} = \bar{p}^2 + (\delta p)^2$, and using (16.7), for the oscillator's mean energy we have

$$\bar{E} = \frac{m\omega^2}{2}\overline{x^2} + \frac{\overline{p^2}}{2m} \geq \frac{m\omega^2}{2}(\delta x)^2 + \frac{1}{2m}(\delta p)^2 \geq \frac{m\omega^2\hbar^2}{8(\delta p)^2} + \frac{(\delta p)^2}{2m}.$$

Finding the minimum of this expression as a function of δp, we obtain the lower bound for the mean values, and hence for all possible energy values: $E \geq \hbar\omega/2$.

Problem

3. Find the wave functions of linear-oscillator states that minimize the uncertainty relation, that is, states in which the root-mean-square fluctuations of coordinate and momentum in the wave packet are related by $\delta p\, \delta x = \hbar/2$ (E. Schrödinger, 1926)[15].

Solution. The required wave functions must have the form

$$\Psi(x,t) = \frac{1}{(2\pi)^{1/4}(\delta x)^{1/2}} \exp\left\{\frac{i\bar{p}x}{\hbar} - \frac{(x-\bar{x})^2}{4(\delta x)^2} - i\varphi(t)\right\}. \tag{1}$$

At every instant their coordinate dependence corresponds to (16.8), where $\bar{x} = \bar{x}(t)$ and $\bar{p} = \bar{p}(t) = m\dot{\bar{x}}(t)$ are the mean coordinate and momentum. By (19.3), for the linear oscillator ($U = m\omega^2 x^2/2$) we have $\dot{p} = -m\omega^2 x$, and hence for the mean values $\dot{\bar{p}} = -m\omega^2\bar{x}$, or

$$\ddot{\bar{x}} + \omega^2\bar{x} = 0, \tag{2}$$

that is, $\bar{x}(t)$ satisfies the classical equation of motion. The constant factor in (1) is fixed by the normalization condition $\int_{-\infty}^{\infty} |\Psi|^2 dx = 1$; besides this factor, Ψ may contain a phase factor with a time-dependent phase $\varphi(t)$. To find the unknown δx and the function $\varphi(t)$, we substitute (1) into the wave equation

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + \frac{m\omega^2 x^2}{2}\Psi = i\hbar\frac{\partial\Psi}{\partial t}.$$

Taking account of (2), the substitution gives

$$\left(\frac{x^2}{2} - x\bar{x}\right)\left(\frac{m^2\omega^2}{\hbar^2} - \frac{1}{4(\delta x)^4}\right) + \left[\frac{m^2\dot{\bar{x}}^2}{2\hbar^2} - \frac{\bar{x}^2}{8(\delta x)^4} + \frac{1}{4(\delta x)^2} - \frac{m}{\hbar}\dot{\varphi}(t)\right] = 0.$$

It follows that $(\delta x)^2 = \hbar/(2m\omega)$ and then

$$\dot{\varphi} = \frac{m}{2\hbar}(\dot{\bar{x}}^2 - \omega^2\bar{x}^2) + \frac{\omega}{2}, \qquad \varphi = \frac{1}{2\hbar}\bar{p}\bar{x} + \frac{\omega}{2}t.$$

[15] These states are called *coherent*.

Thus, finally,

$$\Psi(x,t) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left\{\frac{i\bar{p}x}{\hbar} - \frac{m\omega(x-\bar{x})^2}{2\hbar}\right\} \exp\left\{-\frac{i\omega t}{2} - \frac{i\bar{p}\bar{x}}{2\hbar}\right\}. \tag{3}$$

For $\bar{x} = 0$, $\bar{p} = 0$, this function becomes $\psi_0(x)e^{-i\omega t/2}$, the wave function of the oscillator ground state.

The oscillator's mean energy in a coherent state is

$$\bar{E} = \frac{\overline{p^2}}{2m} + \frac{m\omega^2\overline{x^2}}{2} = \frac{\bar{p}^2}{2m} + \frac{m\omega^2\bar{x}^2}{2} + \frac{\hbar\omega}{2} = \hbar\omega\left(\bar{n} + \frac{1}{2}\right); \tag{4}$$

the quantity $\bar{n}$ introduced here is the mean "number of quanta" $\hbar\omega$ in the state. We see that a coherent state is specified completely by giving a dependence $\bar{x}(t)$ that satisfies the classical equation (2). Its general form can be written

$$\frac{m\omega\bar{x} + i\bar{p}}{\sqrt{2m\hbar\omega}} = ae^{-i\omega t}, \qquad |a|^2 = \bar{n}. \tag{5}$$

The function (3) may be expanded in the wave functions of the oscillator's stationary states:

$$\Psi = \sum_{n=0}^{\infty} a_n\Psi_n, \qquad \Psi_n(x,t) = \psi_n(x)\exp\left\{-i\left(n+\frac{1}{2}\right)\omega t\right\}.$$

The coefficients of this expansion are[16]

$$a_n = \int_{-\infty}^{\infty} \Psi_n^*\Psi\, dx. \tag{6}$$

It follows that the probability for the oscillator to be in its nth state is

$$w_n = |a_n|^2 = e^{-\bar{n}}\frac{\bar{n}^n}{n!}, \tag{7}$$

that is, it is given by the familiar Poisson distribution.

Problem

4. Determine the energy levels of a particle moving in a field with the potential energy shown in Fig. 3 (Ph. Morse, 1929).

Solution. The spectrum of positive energy eigenvalues is continuous (and the levels are nondegenerate), while the spectrum of negative values is discrete.

The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - Ae^{-2\alpha x} + 2Ae^{-\alpha x})\psi = 0.$$

Introduce the new variable

$$\xi = \frac{2\sqrt{2mA}}{\alpha\hbar}e^{-\alpha x}$$

[16] Compare the calculation in Problem 1 of § 41.

(which ranges from zero to $+\infty$), and the notation (we consider the discrete spectrum, so $E < 0$)

$$s = \frac{\sqrt{-2mE}}{\alpha\hbar}, \qquad n = \frac{\sqrt{2mA}}{\alpha\hbar} - \left(s + \frac{1}{2}\right). \tag{1}$$

The Schrödinger equation then becomes

$$\psi'' + \frac{1}{\xi}\psi' + \left(-\frac{1}{4} + \frac{n + s + 1/2}{\xi} - \frac{s^2}{\xi^2}\right)\psi = 0.$$

As $\xi \to \infty$, ψ behaves asymptotically as $e^{\pm\xi/2}$, while as $\xi \to 0$ it is proportional to $\xi^{\pm s}$. Finiteness requires the solution behaving as $e^{-\xi/2}$ at $\xi \to \infty$ and as ξ^s at $\xi \to 0$.

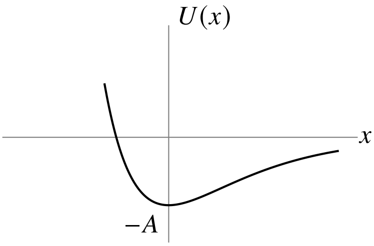

Fig. 3

Make the substitution

$$\psi = e^{-\xi/2}\xi^s w(\xi),$$

which gives for w the equation

$$\xi w'' + (2s + 1 - \xi)w' + nw = 0, \tag{2}$$

to be solved subject to the conditions that w is finite at $\xi = 0$, and that as $\xi \to \infty$ it becomes infinite no faster than a finite power of ξ. Equation (2) is the confluent hypergeometric equation (see § d of the mathematical supplements),

$$w = F(-n, 2s + 1, \xi).$$

The solution satisfying the required condition is obtained for a nonnegative integer n (and F then reduces to a polynomial). From the definitions (1), the energy levels are consequently

$$-E_n = A\left[1 - \frac{\alpha\hbar}{\sqrt{2mA}}\left(n + \frac{1}{2}\right)\right]^2,$$

where n takes integer positive values beginning at zero and ending at the greatest value for which

$$\frac{\sqrt{2mA}}{\alpha\hbar} > n + \frac{1}{2}$$

(so that the parameter s is positive, in agreement with its definition). Thus the discrete spectrum contains a finite sequence of levels. If

$$\frac{\sqrt{2mA}}{\alpha\hbar} < \frac{1}{2},$$

there is no discrete spectrum at all.

Problem

5. The same problem for $U = -U_0/\cosh^2 \alpha x$ (Fig. 4).

Solution. The positive-energy spectrum is continuous and the negative-energy spectrum is discrete; we consider the latter. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E + \frac{U_0}{\cosh^2 \alpha x}\right)\psi = 0.$$

Make the change of variable $\xi = \tanh \alpha x$ and introduce the notation

$$\varepsilon = \frac{\sqrt{-2mE}}{\hbar\alpha}, \qquad \frac{2mU_0}{\alpha^2\hbar^2} = s(s+1), \qquad s = \frac{1}{2}\left(-1 + \sqrt{1 + \frac{8mU_0}{\alpha^2\hbar^2}}\right).$$

We then obtain

$$\frac{d}{d\xi}\left[(1-\xi^2)\frac{d\psi}{d\xi}\right] + \left[s(s+1) - \frac{\varepsilon^2}{1-\xi^2}\right]\psi = 0.$$

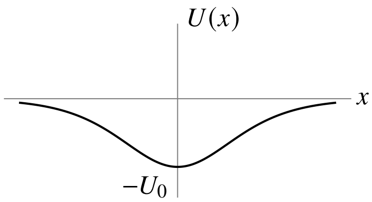

Fig. 4

This is the equation of the generalized Legendre functions. Reduce it to hypergeometric form by the substitution

$$\psi = (1-\xi^2)^{\varepsilon/2} w(\xi)$$

and the temporary change of variable $\frac{1}{2}(1-\xi) = u$:

$$u(1-u)w'' + (\varepsilon+1)(1-2u)w' - (\varepsilon - s)(\varepsilon + s + 1)w = 0.$$

The solution finite at $\xi = 1$ (that is, at $x = \infty$) is

$$\psi = (1-\xi^2)^{\varepsilon/2} F[\varepsilon - s, \varepsilon + s + 1, \varepsilon + 1, (1-\xi)/2].$$

For ψ to remain finite also at $\xi = -1$ (that is, at $x = -\infty$), we must have $\varepsilon - s = -n$, where $n = 0, 1, 2, \ldots$ (then F is a polynomial of degree n, finite at $\xi = -1$).

Thus the energy levels are fixed by $s - \varepsilon = n$, which gives

$$E_n = -\frac{\hbar^2\alpha^2}{8m}\left[-(1+2n) + \sqrt{1 + \frac{8mU_0}{\alpha^2\hbar^2}}\right]^2.$$

There is a finite number of levels, determined by the condition $\varepsilon > 0$, that is, $n < s$.

§ 24. Motion in a Uniform Field

Consider the motion of a particle in a uniform external field. Choose the direction of the field as the x axis, and let F be the force exerted by the field on the particle; in an electric field of strength E, this force is $F = eE$, where e is the particle's charge.

The potential energy of the particle in a uniform field has the form $U = -Fx + \text{const}$. Choosing the constant so that $U = 0$ at $x = 0$, we have $U = -Fx$. The Schrödinger equation for the problem is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E + Fx)\psi = 0. \tag{24.1}$$

Since U tends to $+\infty$ as $x \to -\infty$ and to $-\infty$ as $x \to \infty$, it is evident in advance that the energy levels form a continuous spectrum filling the entire interval from $-\infty$ to $+\infty$. All these eigenvalues are nondegenerate and correspond to motion that is finite on the side $x = -\infty$ and infinite in the direction $x \to +\infty$.

In place of the coordinate x, introduce the dimensionless variable

$$\xi = \left(x + \frac{E}{F}\right)\left(\frac{2mF}{\hbar^2}\right)^{1/3}. \tag{24.2}$$

Equation (24.1) then becomes

$$\psi'' + \xi\psi = 0. \tag{24.3}$$

This equation contains no energy parameter at all. Consequently, once a solution satisfying the required finiteness conditions has been found, it provides an eigenfunction for every value of the energy.

The solution of (24.3) that is finite for all x is (see § b of the mathematical supplements)

$$\psi(\xi) = A\Phi(-\xi), \tag{24.4}$$

where

$$\Phi(\xi) = \frac{1}{\sqrt{\pi}}\int_0^\infty \cos\left(\frac{u^3}{3} + u\xi\right) du$$

is the so-called Airy function, and A is a normalization factor to be determined below.

As $\xi \to -\infty$, the function $\psi(\xi)$ tends exponentially to zero. The asymptotic expression determining $\psi(\xi)$ for negative ξ of large absolute magnitude is (see (b.4))

$$\psi(\xi) \approx \frac{A}{2|\xi|^{1/4}} \exp\left(-\frac{2}{3}|\xi|^{3/2}\right). \tag{24.5}$$

For large positive ξ, the asymptotic expression for $\psi(\xi)$ is instead (see (b.5)[17]):

$$\psi(\xi) = \frac{A}{\xi^{1/4}} \sin\left(\frac{2}{3}\xi^{3/2} + \frac{\pi}{4}\right). \tag{24.6}$$

[17] We note in anticipation that the asymptotic expressions (24.5) and (24.6) correspond precisely to the quasiclassical forms of the wave function in classically forbidden and allowed regions, respectively (§ 47).

Following the general rule (5.4) for normalizing eigenfunctions of a continuous spectrum, put the functions (24.4) into a form normalized to a delta function of the energy:

$$\int_{-\infty}^{+\infty} \psi(\xi)\psi(\xi')\,dx = \delta(E' - E). \tag{24.7}$$

In § 21 a simple method was given for determining the normalization coefficient from the asymptotic form of the wave functions. Following that method, write (24.6) as a sum of two traveling waves:

$$\psi(\xi) \approx \frac{A}{2\xi^{1/4}}\left\{\exp\left[i\left(\frac{2}{3}\xi^{3/2} - \frac{\pi}{4}\right)\right] + \exp\left[-i\left(\frac{2}{3}\xi^{3/2} - \frac{\pi}{4}\right)\right]\right\}.$$

The flux density calculated for each of these two terms is

$$v\left(\frac{A}{2\xi^{1/4}}\right)^2 = \sqrt{\frac{2}{m}(E + Fx)}\left(\frac{A}{2\xi^{1/4}}\right)^2 = A^2\frac{(2\hbar F)^{1/3}}{4m^{2/3}}.$$

Equating it to $1/(2\pi\hbar)$, we find

$$A = \frac{(2m)^{1/3}}{\pi^{1/2}F^{1/6}\hbar^{2/3}}. \tag{24.8}$$

Problem

Determine the momentum-representation wave functions for a particle in a uniform field.

Solution. The Hamiltonian in the momentum representation is

$$\widehat{H} = \frac{p^2}{2m} - i\hbar F\frac{d}{dp},$$

so the Schrödinger equation for the wave function $a(p)$ is

$$-i\hbar F\frac{da}{dp} + \left(\frac{p^2}{2m} - E\right)a = 0.$$

Solving this equation gives the required functions

$$a_E(p) = \frac{1}{\sqrt{2\pi\hbar F}}\exp\left\{\frac{i}{\hbar F}\left(Ep - \frac{p^3}{6m}\right)\right\}.$$

These functions are normalized by the condition

$$\int_{-\infty}^{+\infty} a_E^*(p)a_{E'}(p)\,dp = \delta(E' - E).$$

§ 25. Transmission Coefficient

We consider particle motion in a field of the type depicted in Fig. 5: $U(x)$ increases monotonically from one constant limiting value ($U = 0$ as $x \to -\infty$) to another ($U = U_0$ as $x \to +\infty$). According to classical mechanics, a particle with energy $E < U_0$ moving in such a field from left to right arrives at the *potential wall*, reflects off it, and proceeds in the reverse direction; for $E > U_0$, however, the particle keeps traveling in its original direction with a reduced velocity. Quantum mechanics introduces a new phenomenon: even when $E > U_0$, the particle may be reflected from the potential wall. In principle, the reflection probability is to be calculated as follows.

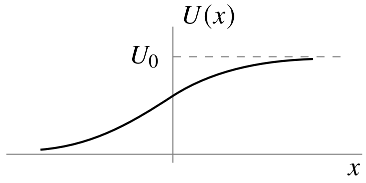

Figure 5

Let the particle move from left to right. At large positive values of x, the wave function must describe a particle that has passed "over the wall" and is moving in the positive direction of the x axis; that is, it must have the asymptotic form

$$\text{as } x \to \infty: \qquad \psi \approx A e^{ik_2 x}, \qquad k_2 = \frac{1}{\hbar}\sqrt{2m(E - U_0)} \tag{25.1}$$

(A is a constant). Having found a solution of the Schrödinger equation that satisfies this limiting condition, we calculate its asymptotic expression as $x \to -\infty$. This is a linear combination of two solutions of the free-motion equation and therefore has the form

$$\text{as } x \to -\infty: \qquad \psi \approx e^{ik_1 x} + B e^{-ik_1 x}, \qquad k_1 = \frac{1}{\hbar}\sqrt{2mE}. \tag{25.2}$$

The first term describes the particle impinging on the wall (ψ is taken normalized so that this term has unit coefficient); the second term depicts the particle reflected from the wall. The flux density is proportional to k_1 in the incident wave, to $k_1|B|^2$ in the reflected wave, and to $k_2|A|^2$ in the transmitted wave. Define the particle's *transmission coefficient* D as the ratio of the flux density in the transmitted wave to that in the incident wave:

$$D = \frac{k_2}{k_1}|A|^2. \tag{25.3}$$

Similarly, the *reflection coefficient* R may be defined as the ratio of the reflected flux density to the incident flux density; evidently $R = 1 - D$:

$$R = |B|^2 = 1 - \frac{k_2}{k_1}|A|^2 \tag{25.4}$$

(this relation between A and B follows automatically from the constancy of the flux along the x axis).

If a particle moves from left to right with energy $E < U_0$, then k_2 is purely imaginary and the wave function decreases exponentially as $x \to +\infty$. The reflected flux equals the incident flux; that is, the particle is totally reflected from the potential wall. We stress, however, that the probability for the particle to be located in the region $E < U$ remains nonzero even in this case, though it decreases rapidly with growing x.

For the general case of an arbitrary stationary state (energy $E > U_0$), the asymptotic form of the wave function both as $x \to -\infty$ and as $x \to +\infty$ is a sum of two waves propagating in the two directions along the x axis:

$$\begin{aligned} \psi &= A_1 e^{ik_1x} + B_1 e^{-ik_1x} \quad \text{as } x \to -\infty, \\ \psi &= A_2 e^{ik_2x} + B_2 e^{-ik_2x} \quad \text{as } x \to +\infty. \end{aligned} \tag{25.5}$$

Since both expressions are asymptotic forms of one and the same solution of a linear differential equation, the coefficients A_1, B_1 and A_2, B_2 are linearly related. Let $A_2 = \alpha A_1 + \beta B_1$, where α and β are constants (in general complex) that depend on the particular form of the field $U(x)$. The analogous relation for B_2 may then be written by using the fact that the Schrödinger equation is real. By this fact, if ψ is a solution of the given Schrödinger equation, the complex-conjugate function ψ^* is also a solution of the same equation.

The asymptotic form

$$\begin{aligned} \psi^* &= A_1^* e^{-ik_1x} + B_1^* e^{ik_1x} \quad \text{as } x \to -\infty, \\ \psi^* &= A_2^* e^{-ik_2x} + B_2^* e^{ik_2x} \quad \text{as } x \to +\infty \end{aligned}$$

differs from (25.5) only in the notation of the constant coefficients; hence $B_2^* = \alpha B_1^* + \beta A_1^*$, or $B_2 = \alpha^* B_1 + \beta^* A_1$. Thus the coefficients in (25.5) are related by equations of the form

$$A_2 = \alpha A_1 + \beta B_1, \qquad B_2 = \beta^* A_1 + \alpha^* B_1. \tag{25.6}$$

The condition that the flux be constant along the x axis gives the following relation for the coefficients in (25.5):

$$k_1\big(|A_1|^2 - |B_1|^2\big) = k_2\big(|A_2|^2 - |B_2|^2\big).$$

On expressing A_2 and B_2 here in terms of A_1 and B_1 according to (25.6), we obtain

$$|\alpha|^2 - |\beta|^2 = \frac{k_1}{k_2}. \tag{25.7}$$

The relations (25.6) can be used to show that the reflection coefficients are identical (at a given energy $E > U_0$) for particles propagating in the positive or negative direction along the x axis. Indeed, the first case is obtained by setting $B_2 = 0$ in the functions (25.5); then $B_1/A_1 = -\beta^*/\alpha^*$. In the second case we put $A_1 = 0$, whereupon $A_2/B_2 = \beta/\alpha^*$. The corresponding reflection coefficients are

$$R_1 = \left|\frac{B_1}{A_1}\right|^2 = \left|\frac{\beta^*}{\alpha^*}\right|^2, \qquad R_2 = \left|\frac{A_2}{B_2}\right|^2 = \left|\frac{\beta}{\alpha^*}\right|^2,$$

and it is clear that $R_1 = R_2$.

The quantities $B_1/A_1 = -\beta^*/\alpha^*$ and $A_2/B_2 = \beta/\alpha^*$ are naturally called the *reflection amplitudes* for motion in the positive and negative directions, respectively. These amplitudes have equal moduli but may differ by a phase factor.

Problem

1. For a particle with energy $E > U_0$, calculate the coefficient of reflection from a rectangular potential wall (Fig. 6).

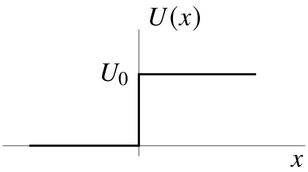

Figure 6

Solution. Throughout the region $x > 0$ the wave function has the form (25.1), and in the region $x < 0$ it has the form (25.2). The constants A and B are determined by the conditions that ψ and $d\psi/dx$ be continuous at $x = 0$:

$$1 + B = A, \qquad k_1(1 - B) = k_2 A,$$

whence

$$A = \frac{2k_1}{k_1 + k_2}, \qquad B = \frac{k_1 - k_2}{k_1 + k_2}.$$

The reflection coefficient is (25.4) [18]

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2 = \left(\frac{p_1 - p_2}{p_1 + p_2}\right)^2.$$

At $E = U_0$ ($k_2 = 0$), R becomes unity; as $E \to \infty$, it tends to zero as $R = (U_0/4E)^2$.

Problem

2. Determine the transmission coefficient of a particle through a rectangular potential barrier (Fig. 7).

Solution. Let $E > U_0$, and let the incident particle move from left to right. The wave function then has the following forms in the different regions:

$$\begin{aligned} &\psi = e^{ik_1x} + Ae^{-ik_1x} && \text{for } x < 0, \\ &\psi = Be^{ik_2x} + B'e^{-ik_2x} && \text{for } 0 < x < a, \\ &\psi = Ce^{ik_1x} && \text{for } x > a. \end{aligned}$$

(On the side $x > a$ there must be only the transmitted wave, propagating in the positive direction of the x axis.) The constants A, B, B', and C are determined by the conditions

[18] In the limiting case of classical mechanics the reflection coefficient must go to zero. However, the resulting expression does not contain a quantum constant at all. This seeming contradiction is resolved as follows. The classical limit corresponds to the situation where the particle's de Broglie wavelength $\lambda \sim \hbar/p$ is small in comparison with the characteristic dimensions of the problem, that is, with the distances over which the field $U(x)$ changes appreciably. In the schematic example under consideration this distance is zero (at $x = 0$), so that the limiting transition cannot be made.

that ψ and $d\psi/dx$ be continuous at the points $x = 0, a$. The transmission coefficient is $D = k_1|C|^2/k_1 = |C|^2$. Calculation gives

$$D = \frac{4k_1^2k_2^2}{(k_1^2 - k_2^2)^2 \sin^2 ak_2 + 4k_1^2k_2^2}.$$

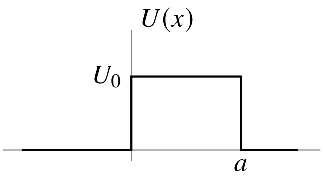

Figure 7

When $E < U_0$, k_2 is purely imaginary; the corresponding expression for D is obtained by replacing k_2 by $i\kappa_2$, where $\hbar\kappa_2 = \sqrt{2m(U_0 - E)}$:

$$D = \frac{4k_1^2\kappa_2^2}{(k_1^2 + \kappa_2^2)^2 \sinh^2(a\kappa_2) + 4k_1^2\kappa_2^2}.$$

Problem

3. Determine the reflection coefficient of a particle from the potential wall defined by

$$U(x) = U_0/(1 + e^{-\alpha x})$$

(see Fig. 5); the particle energy is $E > U_0$.

Solution. The Schrödinger equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\left(E - \frac{U_0}{1 + e^{-\alpha x}}\right)\psi = 0.$$

We must find the solution that, as $x \to +\infty$, has the form

$$\psi = \text{const} \cdot e^{ik_2x}.$$

Introduce the new variable

$$\xi = -e^{-\alpha x}$$

(which ranges from $-\infty$ to 0), and seek a solution in the form

$$\psi = \xi^{-ik_2/\alpha}w(\xi),$$

where $w(\xi)$ tends to a constant as $\xi \to 0$ (that is, as $x \to \infty$). For $w(\xi)$ we obtain an equation of hypergeometric type,

$$\xi(1 - \xi)w'' + \left(1 - \frac{2i}{\alpha}k_2\right)(1 - \xi)w' + \frac{1}{\alpha^2}(k_2^2 - k_1^2)w = 0,$$

whose solution is the hypergeometric function

$$w = F\left[-\frac{i}{\alpha}(k_1 - k_2), -\frac{i}{\alpha}(k_1 + k_2), -\frac{2i}{\alpha}k_2 + 1, \xi\right]$$

(a constant factor is omitted). As $\xi \to 0$, this function tends to 1 and thus satisfies the stated condition.

The asymptotic form of ψ as $\xi \to -\infty$ (that is, as $x \to -\infty$) is [19]

$$\begin{aligned}\psi &\approx \xi^{-ik_2/\alpha}\left[C_1(-\xi)^{i(k_2-k_1)/\alpha} + C_2(-\xi)^{i(k_1+k_2)/\alpha}\right]\\ &= (-1)^{ik_2/\alpha}\left[C_1 e^{ik_1 x} + C_2 e^{-ik_1 x}\right],\end{aligned}$$

where

$$\begin{aligned}C_1 &= \frac{\Gamma(-(2i/\alpha)k_1)\Gamma(-(2i/\alpha)k_2 + 1)}{\Gamma(-(i/\alpha)(k_1 + k_2))\Gamma(-(i/\alpha)(k_1 + k_2) + 1)},\\ C_2 &= \frac{\Gamma((2i/\alpha)k_1)\Gamma(-(2i/\alpha)k_2 + 1)}{\Gamma((i/\alpha)(k_1 - k_2))\Gamma((i/\alpha)(k_1 - k_2) + 1)}.\end{aligned}$$

The required reflection coefficient is $R = |C_2/C_1|^2$; calculation using the familiar formula

$$\Gamma(x)\Gamma(1 - x) = \frac{\pi}{\sin \pi x}$$

gives

$$R = \left\{\frac{\sinh[(\pi/\alpha)(k_1 - k_2)]}{\sinh[(\pi/\alpha)(k_1 + k_2)]}\right\}^2.$$

At $E = U_0$ ($k_2 = 0$), R becomes unity, while as $E \to \infty$ it tends to zero according to

$$R = \left(\frac{\pi U_0}{\alpha\hbar}\right)^2 \frac{2m}{E} \exp\left(-\frac{4\pi}{\alpha\hbar}\sqrt{2mE}\right).$$

In the limiting transition to classical mechanics, R vanishes, as it should.

Problem

4. Determine the transmission coefficient of a particle through the potential barrier defined by

$$U(x) = \frac{U_0}{\cosh^2(\alpha x)}$$

(Fig. 8); the particle energy is $E < U_0$.

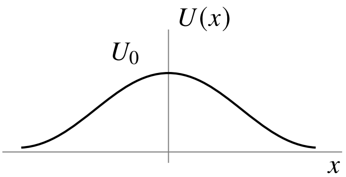

Figure 8

[19]See formula (e.6), in each of whose two terms only the first term of the expansion is to be retained; that is, the hypergeometric functions of $1/z$ are to be replaced by unity.

Solution. The Schrödinger equation for this problem is obtained from the example considered in Problem 5 of § 23 by changing the sign of U_0, with the energy E now taken to be positive.

The same method gives the solution

$$\psi = (1-\xi^2)^{-ik/2\alpha} F\left(-\frac{ik}{\alpha} - s, -\frac{ik}{\alpha} + s + 1, -\frac{ik}{\alpha} + 1, \frac{1-\xi}{2}\right), \tag{1}$$

where

$$\xi = \tanh(\alpha x), \qquad k = \frac{1}{\hbar}\sqrt{2mE}, \qquad s = \frac{1}{2}\left(-1 + \sqrt{1 - \frac{8mU_0}{\alpha^2\hbar^2}}\right).$$

This solution already satisfies the condition that as $x \to \infty$ (that is, as $\xi \to 1$, $(1-\xi) \approx 2e^{-2\alpha x}$) the wave function contain only the transmitted wave ($\sim e^{ikx}$). The asymptotic expression for the wave function as $x \to -\infty$ ($\xi \to -1$) is obtained by transforming the hypergeometric function via formula (e.7):

$$\psi \sim e^{-ikx}\frac{\Gamma(ik/\alpha)\Gamma(1-ik/\alpha)}{\Gamma(-s)\Gamma(1+s)} + e^{ikx}\frac{\Gamma(-ik/\alpha)\Gamma(1-ik/\alpha)}{\Gamma(-ik/\alpha - s)\Gamma(-ik/\alpha + s + 1)}. \tag{2}$$

Taking the squared modulus of the ratio of the coefficients in this function, we obtain the following expression for the transmission coefficient $D = 1 - R$:

$$D = \frac{\sinh^2(\pi k/\alpha)}{\sinh^2(\pi k/\alpha) + \cos^2\left(\frac{\pi}{2}\sqrt{1 - \frac{8mU_0}{\hbar^2\alpha^2}}\right)} \quad \text{when} \quad \frac{8mU_0}{\hbar^2\alpha^2} < 1,$$

$$D = \frac{\sinh^2(\pi k/\alpha)}{\sinh^2(\pi k/\alpha) + \cosh^2\left(\frac{\pi}{2}\sqrt{\frac{8mU_0}{\hbar^2\alpha^2} - 1}\right)} \quad \text{when} \quad \frac{8mU_0}{\hbar^2\alpha^2} > 1.$$

The first of these formulas also applies when $U_0 < 0$, that is, when the particle passes not over a potential barrier but over a potential well. It is noteworthy that then $D = 1$ if $1 + (8m|U_0|/\hbar^2\alpha^2) = (2n+1)^2$; thus, for certain values of the well depth $|U_0|$, particles passing over it are not reflected. This is already clear from expression (2), in which the term containing e^{-ikx} is entirely absent when s is a positive integer.

Problem

5. Determine how the transmission coefficient tends to zero as $E \to 0$, assuming that the potential energy $U(x)$ decreases rapidly at distances $|x| \gg a$, where a is the characteristic size of the interaction region.

Solution. In the range of distances $k|x| \ll 1$, the energy E may be neglected in the Schrödinger equation. If also $|x| \gg a$, the potential energy may be neglected as well, and the equation reduces to

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = 0,$$

whose solution we write as

$$\psi = a_1 + b_1 x, \quad x < 0, \qquad \psi = a_2 + b_2 x, \quad x > 0. \tag{1}$$

Solving the equation at distances $|x| \sim a$ determines the relation among a_1, b_1 and a_2, b_2. This relation is linear and has the form

$$a_1 = \rho a_2 + \mu b_2, \qquad b_1 = \nu a_2 + \tau b_2. \tag{2}$$

The coefficients ρ, μ, ν, and τ are real and independent of the energy, since the energy no longer enters the equation [20]. Solution (1) must agree with the first two terms in the expansions of (25.1) and (25.2) in powers of x, whence

$$a_1 = 1 + B, \qquad b_1 = ik(1 - B), \qquad a_2 = A, \qquad b_2 = ikA.$$

Substituting these expressions in (2) and solving the equations for A, we find, at small k, $A \approx 2ik/\nu$, whence

$$D \approx \frac{4k^2}{\nu^2} \sim E.$$

Thus the transmission coefficient tends to zero in proportion to the particle energy. This general relation is, of course, satisfied in the examples considered in Problems 2 and 4.

[20] By constancy of the flux, they satisfy $\rho\tau - \mu\nu = 1$.

CHAPTER IV

Angular Momentum

§ 26. Angular momentum

In deriving momentum conservation in § 15, we used the homogeneity of space for a closed system of particles. Besides being homogeneous, space is also isotropic: every direction in it is equivalent. The Hamiltonian of a closed system must therefore remain unchanged when the system as a whole is rotated through an arbitrary angle about an arbitrary axis. It is enough to impose this requirement for an arbitrary infinitesimal rotation.

Let $\delta\boldsymbol{\varphi}$ be the vector of an infinitesimal rotation. Its magnitude is the angle of rotation $\delta\varphi$, and its direction is the axis about which the rotation is made. Under such a rotation, the changes $\delta\mathbf{r}_a$ of the particles' radius vectors $\mathbf{r}_a$ are

$$\delta\mathbf{r}_a = \delta\boldsymbol{\varphi} \times \mathbf{r}_a.$$

An arbitrary function $\psi(\mathbf{r}_1, \mathbf{r}_2, \dots)$ is transformed into

$$\begin{aligned}\psi(\mathbf{r}_1 + \delta\mathbf{r}_1, \mathbf{r}_2 + \delta\mathbf{r}_2, \dots) &= \psi(\mathbf{r}_1, \mathbf{r}_2, \dots) + \sum_a \delta\mathbf{r}_a \boldsymbol{\nabla}_a \psi = \\ &= \psi(\mathbf{r}_1, \mathbf{r}_2, \dots) + \sum_a (\delta\boldsymbol{\varphi} \times \mathbf{r}_a) \cdot \boldsymbol{\nabla}_a \psi = \\ &= \left(1 + \delta\boldsymbol{\varphi} \cdot \sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)\right) \psi(\mathbf{r}_1, \mathbf{r}_2, \dots).\end{aligned}$$

The expression

$$1 + \delta\boldsymbol{\varphi} \cdot \sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)$$

is the operator of an infinitesimal rotation. The fact that such a rotation does not change the system's Hamiltonian is expressed (cf. § 15) by the commutation of the rotation operator with $\hat{H}$. Since $\delta\boldsymbol{\varphi}$ is a constant vector, this condition reduces to

$$\left(\sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)\right) \hat{H} - \hat{H} \left(\sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)\right) = 0, \tag{26.1}$$

which expresses a conservation law.

The quantity conserved in a closed system as a consequence of the isotropy of space is the system's *angular momentum* (cf. Vol. I, § 9). Thus, apart from a constant factor, the operator $\sum_a (\mathbf{r}_a \times \boldsymbol{\nabla}_a)$ must correspond to the total angular momentum of the system's motion, while each term $(\mathbf{r}_a \times \boldsymbol{\nabla}_a)$ must correspond to the angular momentum of one particle.

The proportionality factor must be chosen as $-i\hbar$. The resulting particle angular-momentum operator $-i\hbar(\mathbf{r}\times\boldsymbol{\nabla}) = \mathbf{r}\times\hat{\mathbf{p}}$ then agrees exactly with the classical expression $\mathbf{r}\times\mathbf{p}$. Henceforth we shall always measure angular momentum in units of $\hbar$. We denote the operator of the angular momentum of one particle, defined in this way, by $\hat{\mathbf{l}}$, and that of the entire system by $\hat{\mathbf{L}}$. Thus the particle angular-momentum operator is

$$\hbar\hat{\mathbf{l}} = \mathbf{r}\times\hat{\mathbf{p}} = -i\hbar(\mathbf{r}\times\boldsymbol{\nabla}) \tag{26.2}$$

or, in components,

$$\hbar\hat{l}_x = y\hat{p}_z - z\hat{p}_y, \qquad \hbar\hat{l}_y = z\hat{p}_x - x\hat{p}_z, \qquad \hbar\hat{l}_z = x\hat{p}_y - y\hat{p}_x.$$

For a system in an external field, angular momentum is not generally conserved. It may nevertheless be conserved when the field has a suitable symmetry. In a centrally symmetric field, for example, all spatial directions from the center are equivalent, so angular momentum about that center is conserved. When the field has axial symmetry, the same reasoning shows that what is conserved is the angular-momentum component directed along the symmetry axis. All these conservation laws of classical mechanics remain valid in quantum mechanics.

If a system's angular momentum is not conserved, it has no definite value in a stationary state. In such cases the mean angular momentum in a given stationary state can sometimes be of interest. Its mean value is readily seen to vanish in every nondegenerate stationary state. Indeed, the energy is unchanged when the sign of time is reversed; and since only one stationary state corresponds to the given energy level, replacing t by $-t$ must leave the state of the system unchanged . The mean values of all quantities, including angular momentum, must therefore also remain unchanged. But angular momentum reverses sign under time reversal, which would give $\bar{\mathbf{L}} = -\bar{\mathbf{L}}$; hence $\bar{\mathbf{L}} = 0$. A direct calculation confirms this: the integral $\int \psi^*\hat{\mathbf{L}}\psi\, dq$, which by definition gives the mean $\bar{\mathbf{L}}$, can be evaluated using the fact that nondegenerate wave functions are real (see the end of § 18). One then finds that

$$\bar{\mathbf{L}} = -i\hbar\int \psi^*\left(\sum_a(\mathbf{r}_a\times\boldsymbol{\nabla}_a)\right)\psi\, dq$$

is purely imaginary. Since $\bar{\mathbf{L}}$ must of course be real, it follows that $\bar{\mathbf{L}} = 0$.

We now determine the commutation rules between the angular-momentum operators and the coordinate and momentum operators. From (16.2) we readily obtain

$$\begin{aligned}
\{\hat{l}_x, x\} &= 0, \quad \{\hat{l}_x, y\} = iz, \quad \{\hat{l}_x, z\} = -iy,\\
\{\hat{l}_y, y\} &= 0, \quad \{\hat{l}_y, z\} = ix, \quad \{\hat{l}_y, x\} = -iz,\\
\{\hat{l}_z, z\} &= 0, \quad \{\hat{l}_z, x\} = iy, \quad \{\hat{l}_z, y\} = -ix.
\end{aligned} \tag{26.3}$$

For example,

$$\hat{l}_x y - y\hat{l}_x = \frac{1}{\hbar}(y\hat{p}_z - z\hat{p}_y)y - y(y\hat{p}_z - z\hat{p}_y)\frac{1}{\hbar} = -\frac{z}{\hbar}\{\hat{p}_y, y\} = iz.$$

All the relations (26.3) may be written in tensor form as

$$\{\hat{l}_i, x_k\} = i e_{ikl} x_l, \tag{26.4}$$

where e_{ikl} is the antisymmetric unit tensor of third rank[1] , with summation understood over each pair of repeated "dummy" indices. The angular-momentum and momentum operators satisfy analogous commutation relations:

$$\{\hat{l}_i, \hat{p}_k\} = i e_{ikl} \hat{p}_l. \tag{26.5}$$

These formulas make it easy to obtain the commutation rules among the operators for the components of angular momentum. We have

$$\begin{aligned}\hbar(\hat{l}_x \hat{l}_y - \hat{l}_y \hat{l}_x) &= \hat{l}_x (z\hat{p}_x - x\hat{p}_z) - (z\hat{p}_x - x\hat{p}_z)\hat{l}_x = \\ &= (\hat{l}_x z - z\hat{l}_x)\hat{p}_x - x(\hat{l}_x \hat{p}_z - \hat{p}_z \hat{l}_x) = -iy\hat{p}_x + ix\hat{p}_y = i\hbar \hat{l}_z.\end{aligned}$$

Thus

$$\{\hat{l}_y, \hat{l}_z\} = i\hat{l}_x, \qquad \{\hat{l}_z, \hat{l}_x\} = i\hat{l}_y, \qquad \{\hat{l}_x, \hat{l}_y\} = i\hat{l}_z \tag{26.6}$$

or

$$\{\hat{l}_i, \hat{l}_k\} = i e_{ikl} \hat{l}_l. \tag{26.7}$$

Exactly the same relations hold for the operators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ of the system's total angular momentum. Indeed, since the angular-momentum operators of different particles commute with one another, we have, for example,

$$\sum_a \hat{l}_{ay} \sum_a \hat{l}_{az} - \sum_a \hat{l}_{az} \sum_a \hat{l}_{ay} = \sum_a (\hat{l}_{ay}\hat{l}_{az} - \hat{l}_{az}\hat{l}_{ay}) = i \sum_a \hat{l}_{ax}.$$

Therefore

$$\{\hat{L}_y, \hat{L}_z\} = i\hat{L}_x, \qquad \{\hat{L}_z, \hat{L}_x\} = i\hat{L}_y, \qquad \{\hat{L}_x, \hat{L}_y\} = i\hat{L}_z. \tag{26.8}$$

Relations (26.8) show that the three components of angular momentum cannot have definite values simultaneously (except when all three components vanish at once; see below). In this respect angular momentum differs essentially from momentum, whose three components can be measured simultaneously.

From $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ we form the operator for the square of the magnitude of the angular-momentum vector:

$$\hat{\mathbf{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2. \tag{26.9}$$

[1]The antisymmetric unit tensor of third rank e_{ikl} (also called the unit axial tensor) is defined to be antisymmetric in all three indices, with $e_{123} = 1$. Evidently, of its 27 components, only the 6 for which i, k, l form a permutation of the numbers 1, 2, 3 are nonzero. A component is +1 if the permutation i, k, l is obtained from 1, 2, 3 by an even number of pairwise interchanges (transpositions), and is −1 for an odd number. Clearly,

$$e_{ikl}e_{ikm} = 2\delta_{lm}, \qquad e_{ikl}e_{ikl} = 6.$$

The components of the vector $\mathbf{C} = \mathbf{A} \times \mathbf{B}$, the vector product of $\mathbf{A}$ and $\mathbf{B}$, can be expressed through e_{ikl} as

$$C_i = e_{ikl} A_k B_l.$$

This operator commutes with each of $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$:

$$\{\hat{\mathbf{L}}^2, \hat{L}_x\} = 0, \qquad \{\hat{\mathbf{L}}^2, \hat{L}_y\} = 0, \qquad \{\hat{\mathbf{L}}^2, \hat{L}_z\} = 0. \tag{26.10}$$

Indeed, using (26.8), we find, for example,

$$\begin{aligned}
\{\hat{L}_x^2, \hat{L}_z\} &= \hat{L}_x\{\hat{L}_x, \hat{L}_z\} + \{\hat{L}_x, \hat{L}_z\}\hat{L}_x = -i(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x),\\
\{\hat{L}_y^2, \hat{L}_z\} &= i(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x),\\
\{\hat{L}_z^2, \hat{L}_z\} &= 0.
\end{aligned}$$

Adding these equalities gives the last relation in (26.10).

Physically, (26.10) means that the square of the angular momentum (that is, its magnitude) can have a definite value simultaneously with one of its components.

Instead of $\hat{L}_x$ and $\hat{L}_y$, it is often convenient to use their complex combinations

$$\hat{L}_+ = \hat{L}_x + i\hat{L}_y, \qquad \hat{L}_- = \hat{L}_x - i\hat{L}_y. \tag{26.11}$$

Direct calculation from (26.8) readily gives the following commutation rules for these combinations:

$$\{\hat{L}_+, \hat{L}_-\} = 2\hat{L}_z, \qquad \{\hat{L}_z, \hat{L}_+\} = \hat{L}_+, \qquad \{\hat{L}_z, \hat{L}_-\} = -\hat{L}_-. \tag{26.12}$$

It is also easy to verify that

$$\hat{\mathbf{L}}^2 = \hat{L}_+\hat{L}_- + \hat{L}_z^2 - \hat{L}_z = \hat{L}_-\hat{L}_+ + \hat{L}_z^2 + \hat{L}_z. \tag{26.13}$$

Finally, we give the frequently used expressions for the angular-momentum operator of one particle in spherical coordinates. Introducing these coordinates through the usual relations

$$x = r\sin\theta\cos\varphi, \qquad y = r\sin\theta\sin\varphi, \qquad z = r\cos\theta,$$

a straightforward calculation gives

$$\hat{l}_z = -i\frac{\partial}{\partial\varphi}, \tag{26.14}$$

$$\hat{l}_\pm = e^{\pm i\varphi}\left(\pm\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\varphi}\right). \tag{26.15}$$

Substitution in (26.13) gives the square of the particle's angular-momentum operator in the form

$$\hat{\mathbf{l}}^2 = -\left[\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} + \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)\right]. \tag{26.16}$$

Notice that, apart from a factor, this is the angular part of the Laplace operator.

§ 27. Eigenvalues of angular momentum

In finding the eigenvalues of a particle's angular-momentum projection onto a chosen direction, one may orient the polar axis along that direction and employ the spherical-coordinate form of the operator. Using (26.14), the equation $\hat{l}_z\psi = l_z\psi$ takes the form

$$-i\frac{\partial\psi}{\partial\varphi} = l_z\psi. \tag{27.1}$$

Its solution is

$$\psi = f(r,\theta)e^{il_z\varphi},$$

where $f(r,\theta)$ is an arbitrary function of r and θ. For ψ to be single-valued, it must be periodic in φ, with period 2π. Hence we find[2]

$$l_z = m, \qquad m = 0,\ \pm1,\ \pm2,\dots \tag{27.2}$$

Thus the eigenvalues l_z are the positive and negative integers, including zero. We denote the factor depending on φ that is characteristic of the eigenfunctions of $\hat{l}_z$ by

$$\Phi_m(\varphi) = \frac{1}{\sqrt{2\pi}}e^{im\varphi}. \tag{27.3}$$

These functions are normalized so that

$$\int_0^{2\pi}\Phi_m^*(\varphi)\Phi_{m'}(\varphi)\,d\varphi = \delta_{mm'}. \tag{27.4}$$

The eigenvalues of the z component of the total angular momentum of a system are evidently likewise positive and negative integers:

$$L_z = M, \qquad M = 0,\ \pm1,\ \pm2,\dots \tag{27.5}$$

(this follows because $\hat{L}_z$ is the sum of the mutually commuting operators $\hat{l}_z$ of the individual particles).

Since the direction of the z axis has not been distinguished in advance, the same result clearly holds for L_x, L_y, and, in general, for the component of angular momentum along any direction: each can take only integer values. At first sight this result may seem paradoxical, particularly if it is applied to two infinitely close directions. One must bear in mind, however, that the only common eigenfunction of the operators $\hat{L}_x$, $\hat{L}_y$, and $\hat{L}_z$ corresponds to the simultaneous values

$$L_x = L_y = L_z = 0;$$

in this case the angular-momentum vector, and therefore its projection on every direction, is zero. If at least one of the eigenvalues L_x, L_y, L_z is nonzero, the corresponding operators have no common eigenfunctions. Put differently, no state exists in which definite nonzero values are possessed simultaneously by two or three angular-momentum

[2] The customary notation of the eigenvalues of the angular-momentum projection by the letter m—the same letter that also denotes the particle mass—cannot in practice lead to confusion.

components referred to distinct directions; one can therefore only discuss the integer nature of a single component at a time.

Stationary states of a system that differ only in the value of M have the same energy. This follows already from general considerations, since the direction of the z axis has not been distinguished beforehand. Thus the energy levels of a system with conserved nonzero angular momentum are in any event degenerate[3].

We now turn to finding the eigenvalues of the square of angular momentum and show how they can be obtained solely from the commutation rules (26.8). Let ψ_M denote the wave functions of stationary states having the same value of $\mathbf{L}^2$, belonging to one degenerate energy level, and differing in the value of M[4].

First, since the two directions of the z axis are physically equivalent, every possible positive value $M = |M|$ has a corresponding negative value $M = -|M|$. Let L (a nonnegative integer) denote the largest possible value of $|M|$ for a given $\mathbf{L}^2$. The very existence of such an upper bound follows from the fact that the difference $\hat{\mathbf{L}}^2 - \hat{L}_z^2 = \hat{L}_x^2 + \hat{L}_y^2$ is the operator of the necessarily nonnegative physical quantity $L_x^2 + L_y^2$, and hence its eigenvalues cannot be negative.

Applying the operator $\hat{L}_z \hat{L}_\pm$ to an eigenfunction ψ_M of $\hat{L}_z$, and using the commutation rules (26.12), we obtain

$$\hat{L}_z \hat{L}_\pm \psi_M = (M \pm 1)\hat{L}_\pm \psi_M. \tag{27.6}$$

It follows that $\hat{L}_\pm \psi_M$ is, apart from a normalization constant, an eigenfunction corresponding to the value $M \pm 1$ of L_z:

$$\psi_{M+1} = \text{const} \cdot \hat{L}_+ \psi_M, \qquad \psi_{M-1} = \text{const} \cdot \hat{L}_- \psi_M. \tag{27.7}$$

If $M = L$ is put in the first of these equalities, then we must have identically

$$\hat{L}_+ \psi_L = 0, \tag{27.8}$$

since, by definition, there are no states with $M > L$. Applying $\hat{L}_-$ to this equality and using (26.13), we obtain

$$\hat{L}_- \hat{L}_+ \psi_L = \left(\hat{\mathbf{L}}^2 - \hat{L}_z^2 - \hat{L}_z\right)\psi_L = 0.$$

But since the ψ_M are common eigenfunctions of $\hat{\mathbf{L}}^2$ and $\hat{L}_z$,

$$\hat{\mathbf{L}}^2 \psi_L = \mathbf{L}^2 \psi_L, \qquad \hat{L}_z^2 \psi_L = L^2 \psi_L, \qquad \hat{L}_z \psi_L = L \psi_L,$$

[3] This circumstance is a special case of the general theorem stated in § 10 on the degeneracy of levels in the presence of at least two conserved quantities whose operators do not commute. Here those quantities are the components of angular momentum.

[4] Here it is understood that there is no additional degeneracy by which the same energy corresponds to different values of the square of angular momentum. This is true for the discrete spectrum (apart from the so-called "accidental degeneracy" in the Coulomb field; see § 36), and is, generally speaking, not true for energy levels in the continuous spectrum. Even when there is additional degeneracy, however, the eigenfunctions can always be chosen to represent states with definite values of $\mathbf{L}^2$; from them one can then select states with the same values of E and $\mathbf{L}^2$. The mathematical basis for this is that simultaneous diagonalization of the matrices of mutually commuting operators is always achievable. In analogous cases below, for brevity, we shall speak as if there were no additional degeneracy, with the understanding that, in view of what has just been said, the results obtained do not in fact depend on this assumption.

so the resulting equation gives

$$\mathbf{L}^2 = L(L+1). \tag{27.9}$$

Formula (27.9) determines the required eigenvalues of the square of angular momentum; L takes all nonnegative integer values. For a given value of L, the angular-momentum component $L_z = M$ can take the values

$$M = L, L-1, \ldots, -L, \tag{27.10}$$

that is, $2L+1$ distinct values in all. An energy level corresponding to angular momentum L is therefore $(2L+1)$-fold degenerate; this is usually called directional degeneracy of angular momentum. A state with zero angular momentum, $L = 0$ (so that all three of its components vanish), is nondegenerate. We note that the wave function of such a state is spherically symmetric. This is already clear from the fact that its change under any infinitesimal rotation, given by $\hat{\mathbf{L}}\psi$, vanishes in this case. For brevity we shall often use the customary phrase "angular momentum L" of the system, meaning angular momentum whose square equals $L(L+1)$; the z component of angular momentum is usually called simply its "projection."

We shall denote the angular momentum of one particle by the lowercase letter l; thus for it we write (27.9) as

$$\mathbf{l}^2 = l(l+1). \tag{27.11}$$

Let us calculate the matrix elements of L_x and L_y in the representation in which, together with the energy, $\mathbf{L}^2$ and L_z are diagonal (*M. Born, W. Heisenberg, P. Jordan*, 1926). We note that, since $\hat{L}_x$ and $\hat{L}_y$ commute with the Hamiltonian, their matrices are diagonal with respect to energy; that is, every matrix element for a transition between states of different energies (and different angular momenta L) is zero. It is therefore sufficient to consider matrix elements for transitions within the group of states with different values of M corresponding to a single degenerate energy level.

From (27.7) one sees that the matrix of $\hat{L}_+$ has nonzero elements only for transitions $M-1 \to M$, and that of $\hat{L}_-$ only for $M \to M-1$. With this in mind, equating the diagonal matrix elements of the two sides of (26.13) gives[5]

$$L(L+1) = \langle M|L_+|M-1\rangle\langle M-1|L_-|M\rangle + M^2 - M.$$

Since the operators $\hat{L}_x$ and $\hat{L}_y$ are Hermitian,

$$\langle M-1|L_-|M\rangle = \langle M|L_+|M-1\rangle^*,$$

we rewrite this equality as

$$\left|\langle M|L_+|M-1\rangle\right|^2 = L(L+1) - M(M-1) = (L-M+1)(L+M),$$

whence[6]

$$\langle M|L_+|M-1\rangle = \langle M-1|L_-|M\rangle = \sqrt{(L+M)(L-M+1)}. \tag{27.12}$$

[5]In the notation for matrix elements, for brevity, we omit all indices with respect to which they are diagonal (including the index L).

[6]The choice of sign in this formula is consistent with the choice of phase factors in the angular-momentum eigenfunctions.

For the nonzero matrix elements of L_x and L_y themselves, this gives

$$\begin{aligned}\langle M|L_x|M-1\rangle &= \langle M-1|L_x|M\rangle = \frac{1}{2}\sqrt{(L+M)(L-M+1)},\\ \langle M|L_y|M-1\rangle &= -\langle M-1|L_y|M\rangle = -\frac{i}{2}\sqrt{(L+M)(L-M+1)}.\end{aligned} \tag{27.13}$$

We draw attention to the absence of diagonal elements in the matrices of L_x and L_y. Since a diagonal matrix element gives the mean value of a quantity in the corresponding state, this means that in states with definite values of L_z the mean values are $\bar{L}_x = \bar{L}_y = 0$. Thus, if the projection of angular momentum on some direction in space has a definite value, the whole vector $\bar{\mathbf{L}}$ lies along that same direction.

§ 28. Angular-Momentum Eigenfunctions

Specifying the values of l and m does not determine a particle's wave function completely. Indeed, the expressions for the operators of these quantities in spherical coordinates contain only the angles θ and φ, so their eigenfunctions may include an arbitrary factor depending on r. Here we shall consider only the angular part of the wave function characteristic of angular-momentum eigenfunctions. Denote it by $Y_{lm}(\theta, \varphi)$ and impose the normalization condition

$$\int |Y_{lm}|^2\, do = 1$$

($do = \sin\theta\, d\theta\, d\varphi$ is the element of solid angle).

The calculation below shows that the problem of finding the common eigenfunctions of $\hat{\mathbf{l}}^2$ and $\hat{l}_z$ permits separation of the variables θ and φ. These functions may therefore be sought in the form

$$Y_{lm} = \Phi_m(\varphi)\Theta_{lm}(\theta), \tag{28.1}$$

where $\Phi_m(\varphi)$ are the eigenfunctions of $\hat{l}_z$ defined by (27.3). Since the functions Φ_m have already been normalized by (27.4), the functions Θ_{lm} must satisfy

$$\int_0^\pi |\Theta_{lm}|^2 \sin\theta\, d\theta = 1. \tag{28.2}$$

Functions Y_{lm} with different values of l or m are automatically mutually orthogonal:

$$\int_0^{2\pi}\int_0^\pi Y^*_{l'm'} Y_{lm} \sin\theta\, d\theta\, d\varphi = \delta_{ll'}\delta_{mm'}, \tag{28.3}$$

because they are eigenfunctions of angular-momentum operators belonging to different eigenvalues. The functions $\Phi_m(\varphi)$ are also separately orthogonal (see (27.4)), since they are eigenfunctions of $\hat{l}_z$ belonging to its different eigenvalues m. The functions $\Theta_{lm}(\theta)$ by themselves, however, are not eigenfunctions of any angular-momentum operator; they are mutually orthogonal for different l, but not for different m.

The most direct way to calculate the required functions is to solve the eigenfunction problem for $\hat{\mathbf{l}}^2$ written in spherical coordinates (formula (26.16)). The equation $\hat{\mathbf{l}}^2\psi = l^2\psi$ reads

$$\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2} + l(l+1)\psi = 0.$$

Substitution of ψ in the form (28.1) gives the following equation for Θ_{lm}:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta_{lm}}{d\theta}\right) - \frac{m^2}{\sin^2\theta}\Theta_{lm} + l(l+1)\Theta_{lm} = 0. \tag{28.4}$$

This equation is well known from the theory of spherical harmonics. It has solutions satisfying the finiteness and single-valuedness conditions for nonnegative integral values $l \geq |m|$, in agreement with the angular-momentum eigenvalues obtained above by the matrix method. The corresponding solutions are the associated Legendre polynomials $P_l^m(\cos\theta)$ (see § c of the mathematical supplements). Normalizing the solution by (28.2), we obtain[7]

$$\Theta_{lm} = (-1)^m i^l \sqrt{\frac{(2l+1)(l-m)!}{2(l+m)!}} P_l^m(\cos\theta). \tag{28.5}$$

Here $m \geq 0$ is assumed. For negative m, define Θ_{lm} by

$$\Theta_{l,-|m|} = (-1)^m \Theta_{l|m|}, \tag{28.6}$$

that is, for $m < 0$, Θ_{lm} is given by (28.5) with $|m|$ in place of m and with the factor $(-1)^m$ omitted. Thus, from the mathematical point of view, the angular-momentum eigenfunctions are spherical harmonics with a particular normalization. For later reference, we write their complete expression, incorporating all the definitions just given:

$$Y_{lm}(\theta,\varphi) = (-1)^{(m+|m|)/2} i^l \left[\frac{2l+1}{4\pi}\frac{(l-|m|)!}{(l+|m|)!}\right]^{1/2} P_l^{|m|}(\cos\theta)e^{im\varphi}. \tag{28.7}$$

In particular,

$$Y_{l0} = i^l \sqrt{\frac{2l+1}{4\pi}} P_l(\cos\theta). \tag{28.8}$$

Functions differing only in the sign of m are evidently related by

$$(-1)^{l-m} Y_{l,-m} = Y_{lm}^*. \tag{28.9}$$

For $l = 0$ (and consequently $m = 0$), the spherical harmonic reduces to a constant. In other words, the wave functions of particle states with zero angular momentum depend only on r and hence possess complete spherical symmetry, in accordance with the general statement made in § 27.

For a fixed m, the values of l beginning with $|m|$ enumerate, in ascending order, the successive eigenvalues of $\mathbf{l}^2$. The general theorem on zeros of eigenfunctions (§ 21) therefore implies that Θ_{lm} vanishes at $l - |m|$ different values of θ; in other words, its nodal lines are $l - |m|$ "circles of latitude" on the sphere. For the complete angular

[7] The normalization condition does not, of course, determine the choice of phase factor. The convention used in this book is the one most natural from the standpoint of the general theory of angular-momentum addition; it differs from the convention usually adopted by a factor i^l. The advantages of this choice will be apparent from the notes on pp. 278, 527, and 533.

functions, if real factors $\cos m\varphi$ or $\sin m\varphi$ are chosen instead of $e^{\pm i|m|\varphi}$[8], they also have $|m|$ "meridian circles" as nodal lines. The total number of nodal lines is therefore l.

Finally, let us show how the functions Θ_{lm} can be calculated by the matrix method. The procedure is analogous to the calculation of the oscillator wave functions in § 23. We start from (27.8), $\hat{l}_+ Y_{ll} = 0$. Using (26.15) for the operator $\hat{l}_+$ and substituting

$$Y_{ll} = \frac{1}{\sqrt{2\pi}} e^{il\varphi} \Theta_{ll}(\theta),$$

we obtain for Θ_{ll} the equation

$$\frac{d\Theta_{ll}}{d\theta} - l \cot\theta \cdot \Theta_{ll} = 0,$$

whence $\Theta_{ll} = \text{const} \cdot \sin^l \theta$. Determining the constant from the normalization condition gives

$$\Theta_{ll} = (-i)^l \sqrt{\frac{(2l+1)!}{2}} \frac{1}{2^l l!} \sin^l \theta. \tag{28.10}$$

Next, using (27.12), write

$$\hat{l}_- Y_{l,m+1} = (l_-)_{m,m+1} Y_{lm} = \sqrt{(l-m)(l+m+1)} Y_{lm}.$$

Repeated application of this formula gives

$$\sqrt{\frac{(l-m)!}{(l+m)!}} Y_{lm} = \frac{1}{\sqrt{(2l)!}} \hat{l}_-^{l-m} Y_{ll}.$$

The right-hand side is readily evaluated with the aid of (26.15) for $\hat{l}_-$, according to which

$$\hat{l}_- \left[f(\theta) e^{im\varphi} \right] = e^{i(m-1)\varphi} \sin^{1-m} \theta \frac{d}{d\cos\theta} \left(f \sin^m \theta \right).$$

Repeated application of this formula gives

$$\hat{l}_-^{l-m} e^{il\varphi} \Theta_{ll} = e^{im\varphi} \sin^{-m} \theta \frac{d^{l-m}}{(d\cos\theta)^{l-m}} \left(\sin^l \theta \cdot \Theta_{ll} \right).$$

Finally, using these relations together with (28.10) for Θ_{ll}, we obtain

$$\Theta_{lm}(\theta) = (-i)^l \sqrt{\frac{2l+1}{2} \frac{(l+m)!}{(l-m)!}} \frac{1}{2^l l! \sin^m \theta} \frac{d^{l-m}}{(d\cos\theta)^{l-m}} \sin^{2l} \theta, \tag{28.11}$$

which agrees with (28.5).

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[8] Each such function represents a state in which l_z has no definite value, but may take the values $\pm m$ with equal probability.

§ 29. Matrix elements of vectors

Return once more to a closed system of particles[9]. Let f stand for some scalar physical quantity of this system and $\hat{f}$ for its operator. Any scalar is invariant under a rotation of the coordinate system. Hence the scalar operator $\hat{f}$ is unchanged by a rotation; that is, it commutes with the rotation operator. We know, however, that the operator of an infinitesimal rotation coincides, apart from a constant factor, with the angular-momentum operator. Therefore

$$\{\hat{f}, \hat{\mathbf{L}}\} = 0. \tag{29.1}$$

It follows from the commutation of $\hat{f}$ with the angular-momentum operator that, for transitions between states with definite L and M, the matrix of f is diagonal in these indices. Moreover, specifying M determines only the orientation of the system relative to the coordinate axes, while the value of a scalar quantity does not depend on this orientation at all. Consequently, the matrix elements $\langle n'LM|f|nLM\rangle$ do not depend on M (the letter n provisionally denotes the collection of all quantum numbers, other than L and M, that specify the state of the system). A formal proof follows from the commutation of $\hat{f}$ and $\hat{L}_+$:

$$\hat{f}\hat{L}_+ - \hat{L}_+\hat{f} = 0. \tag{29.2}$$

Take the matrix element of this equality for the transition $n, L, M \to n', L, M+1$. Since the matrix of L_+ has only elements for transitions $n, L, M \to n, L, M+1$, we find

$$\begin{aligned}\langle n', L, M+1|f|n, L, M+1\rangle\langle n, L, M+1|L_+|n, L, M\rangle &= \\ = \langle n', L, M+1|L_+|n', L, M\rangle\langle n', L, M|f|n, L, M\rangle&,\end{aligned}$$

and, since the matrix elements of L_+ do not depend on the index n,

$$\langle n', L, M+1|f|n, L, M+1\rangle = \langle n', L, M|f|n, L, M\rangle, \tag{29.3}$$

which shows that all the elements $\langle n', L, M|f|n, L, M\rangle$ with different M (and identical remaining indices) are equal.

Applied to the Hamiltonian itself, this result gives the already known independence of the stationary-state energy from M, that is, the $(2L+1)$-fold degeneracy of the energy levels.

Next, let $\mathbf{A}$ be some vector physical quantity characterizing the closed system. Under a rotation of the coordinate system (in particular, an infinitesimal rotation, that is, under the action of the angular-momentum operator), the components of the vector transform into one another. The commutation of the operators $\hat{L}_i$ with the operators $\hat{A}_i$ must therefore again yield components of the same vector $\hat{A}_i$. Which components occur can be found by noting that, in the special case where $\mathbf{A}$ is the radius vector of a particle, the result must reduce to (26.4). We thus obtain the commutation rules

$$\{\hat{L}_i, \hat{A}_k\} = ie_{ikl}\hat{A}_l. \tag{29.4}$$

[9] Everything derived in this section applies equally to a particle in a centrally symmetric field (more broadly, to any situation in which the system's total angular momentum is conserved).

From these relations one can derive a number of results about the structure of the component matrices of the vector **A** (*M. Born, W. Heisenberg, P. Jordan*, 1926). First, they make it possible to find the *selection rules*, which determine the transitions for which matrix elements may be nonzero. We shall not, however, give the corresponding rather cumbersome calculations here, since it will be shown below (§ 107) that these rules are in fact direct consequences of the general transformation properties of vector quantities and can be obtained from them essentially without calculation. Here we merely state the rules without derivation.

Matrix elements of every component of a vector can be nonzero only for transitions in which the angular momentum L changes by no more than one:

$$L \to L, L \pm 1. \tag{29.5}$$

There is also an additional selection rule forbidding transitions between any two states with $L = 0$. This rule is an evident consequence of the complete spherical symmetry of states with zero angular momentum.

The selection rules for the angular-momentum projection M differ among the components of the vector. Specifically, matrix elements may be nonzero for transitions with the following changes of M:

$$\begin{aligned} &M \to M+1 && \text{for } A_+ = A_x + iA_y, \\ &M \to M-1 && \text{for } A_- = A_x - iA_y, \\ &M \to M && \text{for } A_z. \end{aligned} \tag{29.6}$$

It is also possible to determine in general form the dependence of the matrix elements of a vector on M. We again give these important and frequently used formulas without derivation, since they too are special cases of more general relations, applying to arbitrary tensor quantities, which will be derived in § 107. The nonzero matrix elements of A_z are given by

$$\begin{aligned} \langle n'LM|A_z|nLM\rangle &= \frac{M}{\sqrt{L(L+1)(2L+1)}}\langle n'L\|A\|nL\rangle, \\ \langle n'LM|A_z|n, L-1, M\rangle &= \sqrt{\frac{L^2-M^2}{L(2L-1)(2L+1)}}\langle n'L\|A\|n, L-1\rangle, \\ \langle n', L-1, M|A_z|nLM\rangle &= \sqrt{\frac{L^2-M^2}{L(2L-1)(2L+1)}}\langle n', L-1\|A\|nL\rangle. \end{aligned} \tag{29.7}$$

Here the symbol

$$\langle n'L'\|A\|nL\rangle$$

denotes a so-called *reduced matrix element*, a quantity independent of the quantum number M[10]. They are related by

$$\langle n'L'\|A\|nL\rangle = \langle nL\|A\|n'L'\rangle^*, \tag{29.8}$$

[10]The appearance in (29.7) and (29.9) of denominators depending on L accords with the general notation introduced in § 107. The usefulness of these denominators is seen, in particular, in the simple form taken by (29.12) for the matrix elements of the scalar product of two vectors.

One should regard the notation for the reduced matrix element as an indivisible entity (in contrast to what was said about the matrix element notation (11.17)).

which follows directly from the Hermiticity of $\hat{A}_z$.

The matrix elements of A_- and A_+ can be expressed through the same reduced elements. The nonzero matrix elements of A_- are

$$\begin{aligned}
\langle n', L, M-1|A_-|nLM\rangle &= \sqrt{\frac{(L-M+1)(L+M)}{L(L+1)(2L+1)}}\langle n'L\|A\|nL\rangle, \\
\langle n', L, M-1|A_-|n, L-1, M\rangle &= \sqrt{\frac{(L-M+1)(L-M)}{L(2L-1)(2L+1)}}\langle n'L\|A\|n, L-1\rangle, \\
\langle n', L-1, M-1|A_-|nLM\rangle &= -\sqrt{\frac{(L+M-1)(L+M)}{L(2L-1)(2L+1)}}\langle n', L-1\|A\|nL\rangle.
\end{aligned} \tag{29.9}$$

The matrix elements of A_+ need no separate formulas, since the reality of A_x and A_y gives

$$\langle n'L'M'|A_+|nLM\rangle = \langle nLM|A_-|n'L'M'\rangle^*. \tag{29.10}$$

Let us write out the formula that relates the matrix elements of the scalar $\mathbf{AB}$ to the reduced matrix elements of the two vectors $\mathbf{A}$ and $\mathbf{B}$. The calculation is conveniently made by writing the operator $\hat{\mathbf{A}}\hat{\mathbf{B}}$ as

$$\hat{\mathbf{A}}\hat{\mathbf{B}} = \frac{1}{2}(\hat{A}_+\hat{B}_- + \hat{A}_-\hat{B}_+) + \hat{A}_z\hat{B}_z. \tag{29.11}$$

The matrix of $\mathbf{AB}$, like that of any scalar, is diagonal in L and M. Calculation using (29.7)–(29.9) gives

$$\langle n'LM|\mathbf{AB}|nLM\rangle = \frac{1}{2L+1}\sum_{n'',L''}\langle n'L\|A\|n''L''\rangle\langle n''L''\|B\|nL\rangle, \tag{29.12}$$

where L'' takes the values $L, L\pm 1$.

For reference, we give the reduced matrix elements of the vector $\mathbf{L}$ itself. Comparing (29.9) and (27.12), we find

$$\begin{aligned}
\langle L\|L\|L\rangle &= \sqrt{L(L+1)(2L+1)}, \\
\langle L-1\|L\|L\rangle &= \langle L\|L\|L-1\rangle = 0.
\end{aligned} \tag{29.13}$$

A quantity frequently encountered in applications is the unit vector $\mathbf{n}$ along the radius vector of a particle. Let us find its reduced matrix elements. It is sufficient to calculate, for example, the matrix elements of $n_z = \cos\theta$ at zero angular-momentum projection, $m = 0$. We have

$$\langle l-1, 0|n_z|l0\rangle = \int_0^\pi \Theta^*_{l-1,0}\cos\theta\cdot\Theta_{l0}\sin\theta\, d\theta$$

with the functions Θ_{l0} from (28.11). Evaluation of the integral gives[11]

$$\langle l-1, 0|n_z|l0\rangle = \frac{il}{\sqrt{(2l-1)(2l+1)}}.$$

[11]The calculation is performed by integrating by parts $(l-1)$ times with respect to $d\cos\theta$. For the general formula for integrals of this kind, see (107.14).

The matrix elements for transitions $l \to l$ are zero (as they are for any polar vector belonging to a single particle; see below, (30.8)). Comparison with (29.7) now gives

$$\langle l-1\|n\|l\rangle = -\langle l\|n\|l-1\rangle = i\sqrt{l}, \qquad \langle l\|n\|l\rangle = 0. \tag{29.14}$$

Problem

Average the tensor $n_i n_k - (1/3)\delta_{ik}$ (where $\mathbf{n}$ is the unit vector along the particle's radius vector) over a state with a definite value of $|\mathbf{l}|$ but no preferred orientation (that is, l_z is unspecified).

Solution. The required mean value is an operator that can be expressed only in terms of $\hat{\mathbf{l}}$. We seek it in the form

$$\overline{n_i n_k} - \frac{1}{3}\delta_{ik} = a\left[\hat{l}_i\hat{l}_k + \hat{l}_k\hat{l}_i - \frac{2}{3}\delta_{ik}l(l+1)\right];$$

this is the most general form of a symmetric second-rank tensor with zero trace constructed from the components of $\hat{\mathbf{l}}$. To determine the constant a, multiply this equality on the left by $\hat{l}_i$ and on the right by $\hat{l}_k$, with summation over i and k. Since the vector $\mathbf{n}$ is perpendicular to the vector $\hbar\hat{\mathbf{l}} = \hat{\mathbf{r}}\times\hat{\mathbf{p}}$, we have $n_i\hat{l}_i = 0$. We replace the product $\hat{l}_i\hat{l}_i\hat{l}_k\hat{l}_k = (\hat{\mathbf{l}}^2)^2$ by its eigenvalue $l^2(l+1)^2$, and transform the product $\hat{l}_i\hat{l}_k\hat{l}_i\hat{l}_k$ using the commutation relations (26.7) as follows:

$$\begin{aligned}
\hat{l}_i\hat{l}_k\hat{l}_i\hat{l}_k &= \hat{l}_i\hat{l}_i\hat{l}_k\hat{l}_k - ie_{ikl}\hat{l}_i\hat{l}_l\hat{l}_k\\
&= (\hat{\mathbf{l}}^2)^2 - \frac{i}{2}e_{ikl}\hat{l}_i(\hat{l}_l\hat{l}_k - \hat{l}_k\hat{l}_l)\\
&= (\hat{\mathbf{l}}^2)^2 + \frac{1}{2}e_{ikl}e_{lkm}\hat{l}_i\hat{l}_m = (\hat{\mathbf{l}}^2)^2 - \hat{\mathbf{l}}^2\\
&= l^2(l+1)^2 - l(l+1)
\end{aligned}$$

(we have used $e_{ikl}e_{mkl} = 2\delta_{im}$). Straightforward simplification then gives

$$a = -\frac{1}{(2l-1)(2l+3)}.$$

§ 30. Parity of a state

Besides translations and rotations of the coordinate system (invariance under which expresses, respectively, the homogeneity and isotropy of space), there is one more operation that preserves the Hamiltonian of a closed system: the *inversion transformation*, which reverses the sign of every coordinate at once (equivalently, it flips the direction of all axes, turning a right-handed frame into a left-handed one and conversely). Invariance of the Hamiltonian under inversion is the mathematical expression of the mirror symmetry of space[12]. In classical mechanics, the invariance of the Hamilton function under

[12]Inversion likewise leaves invariant the Hamiltonian of particles moving in a centrally symmetric field (provided the coordinate origin coincides with the center of that field).

inversion gives no new conservation laws. In quantum mechanics, however, the situation is essentially different.

Introduce the inversion operator $\hat{P}$[13], whose action on a wave function $\psi(\mathbf{r})$ consists in reversing the signs of the coordinates:

$$\hat{P}\psi(\mathbf{r}) = \psi(-\mathbf{r}). \tag{30.1}$$

The eigenvalues P of this operator, defined by the equation

$$\hat{P}\psi(\mathbf{r}) = P\psi(\mathbf{r}), \tag{30.2}$$

are easily found. Two successive applications of the inversion operator give the identity: the arguments of the function are left entirely unchanged. In other words, $\hat{P}^2\psi = P^2\psi = \psi$, so that $P^2 = 1$, whence

$$P = \pm 1. \tag{30.3}$$

An eigenfunction of the inversion operator is therefore either left entirely unaltered or acquires an overall sign change. Wave functions (and the corresponding states) of the first type are called *even*, and those of the second type *odd*.

The invariance of the Hamiltonian under inversion (that is, the commutation of $\hat{H}$ and $\hat{P}$) therefore expresses the *law of conservation of parity*: if a state of a closed system has definite parity (if it is even or odd), that parity is conserved in time[14].

The angular-momentum operator is likewise invariant under inversion: inversion reverses the signs both of the coordinates and of the operators for differentiation with respect to them, and hence the operator (26.2) remains unchanged. Put differently, inversion and angular momentum commute, so that a definite parity can coexist with definite values of L and of the projection M. Moreover, all states that differ only in M have the same parity. This is already evident from the independence of the properties of a closed system from its orientation in space; formally, it can be proved from the commutation relation $\hat{L}_+\hat{P} - \hat{P}\hat{L}_+ = 0$ by the same method used to obtain (29.3) from (29.2). There are definite *parity selection rules* for the matrix elements of various physical quantities.

Consider scalar quantities first. Here one must distinguish *true scalars*, which remain entirely unchanged under inversion, from *pseudoscalars*, which change sign under inversion (the scalar product of an axial vector and a polar vector is a pseudoscalar). The operator $\hat{f}$ of a true scalar commutes with $\hat{P}$. It follows that if the matrix of P is diagonal, then f is likewise diagonal with respect to parity: matrix elements vanish except for the $g \to g$ and $u \to u$ transitions (here g and u label even and odd states). For the operator of a pseudoscalar quantity, on the other hand, $\hat{P}\hat{f} = -\hat{f}\hat{P}$; the operators $\hat{P}$ and $\hat{f}$ "anticommute." The matrix element of this equality for a transition $g \to g$ is

$$P_{gg}f_{gg} = -f_{gg}P_{gg},$$

and since $P_{gg} = 1$, we have $f_{gg} = 0$; in the same way one finds $f_{uu} = 0$. Thus the matrix of a pseudoscalar quantity has nonzero elements only for transitions in which the parity

[13] From the English word *parity*.

[14] To avoid misunderstanding, we recall that the discussion concerns nonrelativistic theory. Interactions exist in nature, and are treated in relativistic theory, that violate parity conservation.

changes. The selection rules for matrix elements of scalar quantities are therefore

$$\begin{aligned} &\text{true scalars:} \qquad g \to g, \quad u \to u, \\ &\text{pseudoscalars:} \qquad g \to u, \quad u \to g. \end{aligned} \tag{30.4}$$

These rules can also be obtained in another way, directly from the definition of matrix elements. Consider, for example, the integral $f_{ug} = \int \psi_u^* \hat{f} \psi_g \, dq$, where ψ_g is even and ψ_u is odd. On reversing the signs of all coordinates, the integrand changes sign if f is a true scalar. On the other hand, an integral over all space cannot change when the variables of integration are merely renamed. Hence $f_{ug} = -f_{ug}$, so that $f_{ug} = 0$.

The selection rules for vector quantities are obtained in the same way. One must remember that ordinary, polar vectors reverse sign under inversion, whereas axial vectors remain unchanged (an example is the angular-momentum vector, the cross product $\mathbf{r} \times \mathbf{p}$ of two polar vectors). Taking this into account, we find

$$\begin{aligned} &\text{polar vectors:} \qquad g \to u, \quad u \to g, \\ &\text{axial vectors:} \qquad g \to g, \quad u \to u. \end{aligned} \tag{30.5}$$

Let us determine the parity of a one-particle state with angular momentum l. Inversion ($x \to -x$, $y \to -y$, $z \to -z$) consists, in spherical coordinates, of the transformation

$$r \to r, \qquad \theta \to \pi - \theta, \qquad \varphi \to \varphi + \pi. \tag{30.6}$$

The angular dependence of the particle's wave function is given by the spherical function Y_{lm}, which, apart from a constant immaterial here, has the form $P_l^m(\cos\theta)e^{im\varphi}$. When φ is replaced by $\varphi + \pi$, the factor $e^{im\varphi}$ is multiplied by $(-1)^m$; when θ is replaced by $\pi - \theta$, $P_l^m(\cos\theta)$ becomes $P_l^m(-\cos\theta) = (-1)^{l-m} P_l^m(\cos\theta)$. Thus the entire function is multiplied by $(-1)^l$ (which is independent of m, in agreement with the statement above); hence the parity of a state with a given l is

$$P = (-1)^l. \tag{30.7}$$

Accordingly, even values of l yield even states and odd values yield odd ones.

For a vector physical quantity associated with a single particle, nonzero matrix elements can arise only in transitions with $l \to l, l \pm 1$ (§ 29). Combining this fact with (30.7) and with the preceding statements concerning parity changes in vector matrix elements, we conclude that matrix elements of vector quantities belonging to an individual particle are nonzero only for the transitions

$$\begin{aligned} &\text{polar vectors:} \qquad l \to l \pm 1, \\ &\text{axial vectors:} \qquad l \to l. \end{aligned} \tag{30.8}$$

§ 31. Addition of angular momenta

Consider a system consisting of two weakly interacting parts. When the interaction is entirely neglected, the conservation of angular momentum holds for each part independently, so that the angular momenta $\mathbf{L}_1$ and $\mathbf{L}_2$ of the two parts can be summed to give the total angular momentum $\mathbf{L}$ of the system. In the next approximation, once the weak interaction is included, the conservation laws for $\mathbf{L}_1$ and $\mathbf{L}_2$ cease to hold rigorously; nonetheless, the quantum numbers L_1 and L_2, which fix the squares of those angular momenta, remain "good" and can still serve for an approximate characterization of the system's state. In a pictorial, classical treatment of the angular momenta, one may say that in this approximation $\mathbf{L}_1$ and $\mathbf{L}_2$ precess about the direction of $\mathbf{L}$ while their magnitudes remain unchanged.

The consideration of such systems raises the question of the *rule for adding angular momenta*. What values of L are possible for given L_1 and L_2? The addition rule for the angular-momentum projections is evident: from $\hat{L}_z = \hat{L}_{1z} + \hat{L}_{2z}$ it follows that

$$M = M_1 + M_2. \tag{31.1}$$

There is no equally simple relation for the operators of the squares of the angular momenta, and we derive their "addition rule" as follows.

If the quantities $\mathbf{L}_1^2$, $\mathbf{L}_2^2$, L_{1z}, and L_{2z} are chosen as a complete set of physical quantities[15], each state is specified by the numbers L_1, L_2, M_1, and M_2. For given L_1 and L_2, the numbers M_1 and M_2 take, respectively, $2L_1 + 1$ and $2L_2 + 1$ values, so there are altogether $(2L_1 + 1)(2L_2 + 1)$ different states with the same L_1 and L_2. We denote the wave functions of the states in this description by $\varphi_{L_1 L_2 M_1 M_2}$.

Instead of the four quantities just stated, one can choose the four quantities $\mathbf{L}_1^2$, $\mathbf{L}_2^2$, $\mathbf{L}^2$, and L_z as a complete set. Each state is then characterized by the numbers L_1, L_2, L, and M (we denote the corresponding wave functions by $\psi_{L_1 L_2 L M}$). For given L_1 and L_2 there must, of course, still be $(2L_1 + 1)(2L_2 + 1)$ different states; that is, for fixed L_1 and L_2, the pair L, M can take $(2L_1 + 1)(2L_2 + 1)$ pairs of values. These values can be found by the following argument.

Adding the different allowed values of M_1 and M_2 gives the corresponding values of M:

$$\begin{array}{cc|l}
M_1 & M_2 & \quad M \\
\hline
L_1 & L_2 & L_1 + L_2 \\
\left.\begin{array}{c} L_1 \\ L_1 - 1 \end{array}\right. & \left.\begin{array}{c} L_2 - 1 \\ L_2 \end{array}\right\} & L_1 + L_2 - 1 \\
\left.\begin{array}{c} L_1 - 1 \\ L_1 \\ L_1 - 2 \end{array}\right. & \left.\begin{array}{c} L_2 - 1 \\ L_2 - 2 \\ L_2 \end{array}\right\} & L_1 + L_2 - 2 \\
\cdots & \cdots & \cdots
\end{array}$$

The largest possible value of M is evidently $M = L_1 + L_2$, and it corresponds to one state φ (one pair of values M_1, M_2). Consequently, the largest possible M in the states ψ, and hence the largest L, is also $L_1 + L_2$. Next, there are two states φ with $M = L_1 + L_2 - 1$. There must therefore also be two states ψ with this value of M; one is a state with

[15] Together with a number of other quantities which, along with the four stated here, make up a complete set. These remaining quantities play no part in the following argument; for conciseness we do not mention them at all and provisionally call the system of the four stated quantities complete.

$L = L_1 + L_2$ (and $M = L - 1$), and the other has $L = L_1 + L_2 - 1$ (with $M = L$). For $M = L_1 + L_2 - 2$ there are three different states φ. This means that, in addition to $L = L_1 + L_2$ and $L = L_1 + L_2 - 1$, the value $L = L_1 + L_2 - 2$ is also possible.

This argument can be continued in the same way as long as decreasing M by one increases by one the number of states with that value of M. It is easy to see that this continues until M reaches $|L_1 - L_2|$. As M is reduced further, the number of states ceases to increase and remains equal to $2L_2 + 1$ (if $L_2 \leq L_1$). Thus $|L_1 - L_2|$ is the smallest possible value of L.

We therefore obtain the result that, for given L_1 and L_2, L can take the values

$$L = L_1 + L_2, L_1 + L_2 - 1, \ldots, |L_1 - L_2|, \tag{31.2}$$

altogether $2L_2 + 1$ different values when $L_2 \leq L_1$. One readily checks that these indeed give $(2L_1 + 1)(2L_2 + 1)$ different pairs of values M, L. It is important to note that, apart from the $2L + 1$ different values of M for a given L, each possible value of L in (31.2) corresponds to just one state.

This result can be represented pictorially by the so-called *vector model*. Introduce two vectors $\mathbf{L}_1$ and $\mathbf{L}_2$ of lengths L_1 and L_2. The values of L are represented by the integer-valued lengths of vectors $\mathbf{L}$ obtained by vector addition of $\mathbf{L}_1$ and $\mathbf{L}_2$; the largest value, $L_1 + L_2$, is obtained when $\mathbf{L}_1$ and $\mathbf{L}_2$ are parallel, and the smallest, $|L_1 - L_2|$, when they are antiparallel.

In states with definite angular momenta L_1, L_2, and definite total angular momentum L, the scalar products $\mathbf{L}_1\mathbf{L}_2$, $\mathbf{L}\mathbf{L}_1$, and $\mathbf{L}\mathbf{L}_2$ also have definite values. These values are readily found. To calculate $\mathbf{L}_1\mathbf{L}_2$, write $\hat{\mathbf{L}} = \hat{\mathbf{L}}_1 + \hat{\mathbf{L}}_2$, or, on squaring and transposing terms,

$$2\hat{\mathbf{L}}_1\hat{\mathbf{L}}_2 = \hat{\mathbf{L}}^2 - \hat{\mathbf{L}}_1^2 - \hat{\mathbf{L}}_2^2.$$

Replacing the operators on the right by their eigenvalues gives the eigenvalue of the operator on the left:

$$\mathbf{L}_1\mathbf{L}_2 = \frac{1}{2}\{L(L+1) - L_1(L_1+1) - L_2(L_2+1)\}. \tag{31.3}$$

Similarly, we find

$$\mathbf{L}\mathbf{L}_1 = \frac{1}{2}\{L(L+1) + L_1(L_1+1) - L_2(L_2+1)\}. \tag{31.4}$$

Let us now establish the *rule for combining parities*. For a system that consists of two independent parts, the wave function Ψ equals the product of the individual wave functions Ψ_1 and Ψ_2. Hence, if the latter two have the same parity (both change sign, or both remain unchanged, when all coordinates change sign), the wave function of the whole system is even. If, on the other hand, Ψ_1 and Ψ_2 have opposite parities, Ψ is odd. These statements can be written as

$$P = P_1 P_2, \tag{31.5}$$

where P is the parity of the whole system, and P_1 and P_2 are the parities of its parts. This rule plainly generalizes directly to a system consisting of any number of noninteracting parts.

In particular, for a system of particles in a centrally symmetric field (with the interaction between the particles treated as weak), the parity of the state of the whole system is

$$P = (-1)^{l_1+l_2+\dots} \tag{31.6}$$

(see (30.7)). Note that the exponent involves the algebraic sum of the particle angular momenta, a quantity that, generally speaking, does not coincide with their "vector sum," i.e., with the angular momentum L of the system.

If a closed system decays into parts under the action of forces within the system itself, its total angular momentum and parity must be conserved. This may make the decay impossible even when it is energetically possible.

For example, consider an atom in an even state with angular momentum $L = 0$, whose energy would permit it to break up into a free electron and an ion occupying an odd state, also with $L = 0$. It is easy to show that this process is in fact impossible (one calls it *forbidden*). Indeed, by conservation of angular momentum the free electron would also have to possess zero angular momentum and would therefore be in an even state ($P = (-1)^0 = +1$). But the state ion+free electron would then be odd, whereas the initial state of the atom was even.

CHAPTER V

Motion in a Centrally Symmetric Field

§ 32. Motion in a central field

As in classical mechanics, the quantum-mechanical problem of two interacting particles can be reduced to a one-particle problem. For masses m_1, m_2 interacting through $U(r)$, where r is their separation,

$$\hat{H} = -\frac{\hbar^2}{2m_1}\Delta_1 - \frac{\hbar^2}{2m_2}\Delta_2 + U(r). \tag{32.1}$$

Here Δ_1, Δ_2 are the Laplace operators in the particle coordinates. Replace $\mathbf{r}_1, \mathbf{r}_2$ by

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1, \qquad \mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}. \tag{32.2}$$

Thus $\mathbf{r}$ is the relative radius vector and $\mathbf{R}$ the center-of-mass radius vector. Direct calculation gives

$$\hat{H} = -\frac{\hbar^2}{2(m_1 + m_2)}\Delta_R - \frac{\hbar^2}{2m}\Delta + U(r), \tag{32.3}$$

where $m = m_1 m_2/(m_1 + m_2)$ is the reduced mass. The Hamiltonian separates into two independent parts. Accordingly, $\psi(\mathbf{r}_1, \mathbf{r}_2) = \varphi(\mathbf{R})\psi(\mathbf{r})$: the first factor describes free motion of the center of mass, of mass $m_1 + m_2$, and the second describes relative motion as that of mass m in $U(r)$.

The Schrödinger equation in a central field is

$$\Delta\psi + \frac{2m}{\hbar^2}[E - U(r)]\psi = 0. \tag{32.4}$$

In spherical coordinates this becomes

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{1}{r^2}\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2\psi}{\partial\varphi^2}\right] + \frac{2m}{\hbar^2}[E - U(r)]\psi = 0. \tag{32.5}$$

Introducing the squared-angular-momentum operator (26.16),[1]

$$\frac{\hbar^2}{2m}\left[-\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\psi}{\partial r}\right) + \frac{\hat{\mathbf{l}}^2}{r^2}\psi\right] + U(r)\psi = E\psi. \tag{32.6}$$

[1] If the radial momentum operator is defined by

$$\hat{p}_r\psi = -i\hbar\frac{1}{r}\frac{\partial}{\partial r}(r\psi) = -i\hbar\left(\frac{\partial}{\partial r} + \frac{1}{r}\right)\psi,$$

then

$$\hat{H} = \frac{1}{2m}\left(\hat{p}_r^2 + \frac{\hbar^2\hat{\mathbf{l}}^2}{r^2}\right) + U(r),$$

formally the same as the classical Hamilton function in spherical coordinates.

Angular momentum is conserved in a central field. For stationary states with definite l, m, these values fix the angular dependence. Seek

$$\psi = R(r)Y_{lm}(\theta, \varphi), \tag{32.7}$$

where Y_{lm} are spherical functions. Since $\hat{\mathbf{l}}^2 Y_{lm} = l(l+1)Y_{lm}$, the radial function obeys

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - \frac{l(l+1)}{r^2}R + \frac{2m}{\hbar^2}[E - U(r)]R = 0. \tag{32.8}$$

This contains no $l_z = m$, consistently with the $(2l+1)$-fold directional degeneracy. Put

$$R(r) = \frac{\chi(r)}{r}. \tag{32.9}$$

Then

$$\frac{d^2\chi}{dr^2} + \left[\frac{2m}{\hbar^2}(E - U) - \frac{l(l+1)}{r^2}\right]\chi = 0. \tag{32.10}$$

If $U(r)$ is finite everywhere, ψ and hence R must remain finite at the origin, so

$$\chi(0) = 0. \tag{32.11}$$

This condition in fact also holds for fields diverging at the origin (§ 35).

Equation (32.10) is formally the one-dimensional Schrödinger equation with

$$U_l(r) = U(r) + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}, \tag{32.12}$$

the sum of $U(r)$ and the centrifugal energy

$$\frac{\hbar^2 l(l+1)}{2mr^2} = \frac{\hbar^2\hat{\mathbf{l}}^2}{2mr^2}.$$

Thus the central-field problem reduces to one-dimensional motion on a region bounded on one side by the condition at $r = 0$. The normalization is likewise one-dimensional:

$$\int_0^\infty |R|^2 r^2 dr = \int_0^\infty |\chi|^2 dr.$$

Energy levels for one-dimensional motion in such a region are nondegenerate (§ 21), so E completely determines the radial solution of (32.10). The angular part is fixed by l, m. Hence E, l, m completely determine the wave function: energy, squared angular momentum, and its projection form a complete set of physical quantities. The one-dimensional reduction permits use of the oscillation theorem (§ 21). At fixed l, order the discrete energy eigenvalues increasingly and number them $n_r = 0, 1, \ldots$. Then n_r is the number of finite-r nodes of the radial function, excluding $r = 0$, and is called the *radial quantum number*. The numbers l and m are also called the *azimuthal* and *magnetic quantum numbers*.

States with different l are conventionally denoted by

$$\begin{array}{c|cccccccc|c} l & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \ldots \\ & s & p & d & f & g & h & i & k & \ldots \end{array} \tag{32.13}$$

The normal state in a central field is always an s state: for $l \neq 0$ the angular wave function necessarily has nodes, whereas the normal-state wave function has none. The lowest possible eigenvalue at fixed l also rises with l, since angular momentum adds the positive term $\hbar^2 l(l+1)/(2mr^2)$ to the Hamiltonian.

To find the radial function near the origin, assume

$$\lim_{r\to 0} U(r)r^2 = 0. \tag{32.14}$$

Take the leading term as $R = \text{const}\cdot r^s$. From (32.8), after multiplication by r^2 and passage to $r \to 0$,

$$\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) - l(l+1)R = 0,$$

so $s(s+1) = l(l+1)$ and

$$s = l \quad \text{or} \quad s = -(l+1).$$

The latter solution is inadmissible, since it diverges at $r = 0$. Therefore

$$R_l \approx \text{const}\cdot r^l. \tag{32.15}$$

The probability of finding the particle between r and $r + dr$ is determined by $r^2|R|^2$ and is proportional to $r^{2(l+1)}$; it vanishes at the origin more rapidly for larger l.

§ 33. Spherical waves

The plane wave

$$\psi_{\mathbf{p}} = \text{const}\cdot\exp(i\mathbf{pr}/\hbar)$$

describes a free particle with definite momentum $\mathbf{p}$ and energy $E = p^2/2m$. We now consider stationary free-particle states having definite angular momentum and projection as well as energy. It is convenient to use the wave vector

$$k = \frac{p}{\hbar} = \frac{\sqrt{2mE}}{\hbar}. \tag{33.1}$$

The state with angular momentum l and projection m has

$$\psi_{klm} = R_{kl}(r)Y_{lm}(\theta,\varphi), \tag{33.2}$$

where

$$R''_{kl} + \frac{2}{r}R'_{kl} + \left[k^2 - \frac{l(l+1)}{r^2}\right]R_{kl} = 0. \tag{33.3}$$

The continuum wave functions satisfy

$$\int \psi^*_{k'l'm'}\psi_{klm}dV = \delta_{ll'}\delta_{mm'}\delta((k'-k)/2\pi).$$

Angular functions ensure orthogonality in l, m; the radial functions must obey

$$\int_0^\infty r^2 R_{k'l}R_{kl}dr = \delta((k'-k)/2\pi) = 2\pi\delta(k'-k). \tag{33.4}$$

For normalization on the energy scale instead, $\int r^2 R_{E'l} R_{El} dr = \delta(E'-E)$, (5.14) gives

$$R_{El} = R_{kl}\left(\frac{1}{2\pi}\frac{dk}{dE}\right)^{1/2} = \frac{1}{\hbar}\sqrt{\frac{m}{2\pi k}}R_{kl}. \tag{33.5}$$

For $l=0$, (33.3) reads $d^2(rR_{k0})/dr^2 + k^2 rR_{k0} = 0$. The finite solution normalized by (33.4) is

$$R_{k0} = 2\frac{\sin kr}{r}. \tag{33.6}$$

For $l \neq 0$, put

$$R_{kl} = r^l \chi_{kl}. \tag{33.7}$$

Then

$$\chi_{kl}'' + \frac{2(l+1)}{r}\chi_{kl}' + k^2\chi_{kl} = 0.$$

Differentiation gives

$$\chi_{kl}''' + \frac{2(l+1)}{r}\chi_{kl}'' + \left[k^2 - \frac{2(l+1)}{r^2}\right]\chi_{kl}' = 0.$$

Putting $\chi_{kl}' = r\chi_{k,l+1}$ yields the required equation for $\chi_{k,l+1}$, so

$$\chi_{k,l+1} = \frac{1}{r}\chi_{kl}'. \tag{33.8}$$

Hence

$$\chi_{kl} = \left(\frac{1}{r}\frac{d}{dr}\right)^l \chi_{k0},$$

where $\chi_{k0} = R_{k0}$ from (33.6), up to an arbitrary constant. Thus

$$R_{kl} = (-1)^l\, 2\frac{r^l}{k^l}\left(\frac{1}{r}\frac{d}{dr}\right)^l \frac{\sin kr}{r}. \tag{33.9}$$

The factor k^{-l} normalizes the function, and $(-1)^l$ is convenient. Equivalently,

$$R_{kl} = \sqrt{\frac{2\pi k}{r}}J_{l+1/2}(kr) = 2k\,j_l(kr), \tag{33.10}$$

where the spherical Bessel functions are

$$j_l(x) = \sqrt{\frac{\pi}{2x}}J_{l+1/2}(x). \tag{33.11}$$

[2]

[2]The first few are

$$j_0 = \frac{\sin x}{x}, \quad j_1 = \frac{\sin x}{x^2} - \frac{\cos x}{x}, \quad j_2 = \left(\frac{3}{x^3} - \frac{1}{x}\right)\sin x - \frac{3\cos x}{x^2}.$$

Another convention in the literature differs from (33.11) by a factor x.

At large r, the slowest-decaying term in (33.9) comes from differentiating the sine l times. Every operation $-d/dr$ shifts its argument by $-\pi/2$, giving

$$R_{kl} \approx \frac{2}{r}\sin\left(kr - \frac{\pi l}{2}\right). \tag{33.12}$$

Normalization may be performed from the asymptotic form (§ 21); comparison with (33.6) confirms the coefficient in (33.9). In the vicinity of the coordinate origin (small r), we expand $\sin kr$ in a series and retain only the term which, after differentiation, produces the lowest power of r[3]:

$$\left(\frac{1}{r}\frac{d}{dr}\right)^l \frac{\sin kr}{r} \approx \left(\frac{1}{r}\frac{d}{dr}\right)^l \frac{(-1)^l (kr)^{2l+1}}{r(2l+1)!} = \frac{(-1)^l k^{2l+1}}{(2l+1)!!}.$$

Therefore

$$R_{kl} \approx \frac{2k^{l+1}}{(2l+1)!!} r^l, \tag{33.13}$$

in agreement with (32.15).

Scattering theory also uses wave functions which are not finite in the usual sense, but describe particles emitted from or incident on the center. For $l = 0$, replace the standing wave (33.6) by an outgoing or incoming wave:

$$R^{\pm}_{k0} = \frac{A}{r} e^{\pm ikr}. \tag{33.14}$$

For arbitrary l,

$$R^{\pm}_{kl} = (-1)^l A \frac{r^l}{k^l}\left(\frac{1}{r}\frac{d}{dr}\right)^l \frac{e^{\pm ikr}}{r}. \tag{33.15}$$

In terms of Hankel functions,

$$R^{\pm}_{kl} = \pm iA\sqrt{\frac{k\pi}{2r}} H^{(1,2)}_{l+1/2}(kr), \tag{33.16}$$

of the first and second kind for + and − respectively. Their asymptotic forms are

$$R^{\pm}_{kl} \approx \frac{A}{r}\exp\left[\pm i\left(kr - \frac{\pi l}{2}\right)\right], \tag{33.17}$$

and near the origin

$$R^{\pm}_{kl} \approx A\frac{(2l-1)!!}{k^l} r^{-l-1}. \tag{33.18}$$

Normalize these functions to emission or absorption of one particle per unit time. Far away, each small part of a spherical wave is locally plane, with current density $j = v\psi\psi^*$ and $v = k\hbar/m$. The condition $\oint j\,df = 1$, or $\int jr^2 do = 1$, with do the solid-angle element, and the unchanged angular normalization require

$$A = \frac{1}{\sqrt{v}} = \sqrt{\frac{m}{k\hbar}}. \tag{33.19}$$

[3] The symbol !! denotes the product of all numbers of the same parity up to and including the specified number.

An asymptotic expression like (33.12) also holds for positive-energy motion in any field decreasing sufficiently rapidly with distance.[4] At large r, neglecting both the field and centrifugal energy leaves

$$\frac{1}{r}\frac{d^2(rR_{kl})}{dr^2} + k^2 R_{kl} = 0.$$

Its general solution is

$$R_{kl} \approx 2\frac{\sin(kr - l\pi/2 + \delta_l)}{r}, \tag{33.20}$$

where δ_l is the *phase shift*; the coefficient corresponds to $k/2\pi$ normalization.[5] The exact boundary condition of finiteness at the origin fixes δ_l. These phases depend on both l and k and are essential characteristics of continuum eigenfunctions.

Problem

1. Find the energy levels for $l = 0$ motion in the spherical square well $U(r) = -U_0$ for $r < a$, $U(r) = 0$ for $r > a$.

Solution. For $l = 0$ the wave functions depend only on r. Inside,

$$\frac{1}{r}\frac{d^2}{dr^2}(r\psi) + k^2\psi = 0, \qquad k = \frac{1}{\hbar}\sqrt{2m(U_0 - |E|)},$$

and the finite solution is $\psi = A\sin(kr)/r$. Outside,

$$\frac{1}{r}\frac{d^2}{dr^2}(r\psi) - \kappa^2\psi = 0, \qquad \kappa = \frac{1}{\hbar}\sqrt{2m|E|},$$

and the decaying solution is $\psi = A'e^{-\kappa r}/r$. Continuity of the logarithmic derivative of $r\psi$ at a gives

$$k\cot ka = -\kappa = -\sqrt{\frac{2mU_0}{\hbar^2} - k^2}, \tag{1}$$

or

$$\sin ka = \pm\sqrt{\frac{\hbar^2}{2ma^2U_0}}\, ka. \tag{2}$$

These equations implicitly determine the levels, using roots for which $\cot ka < 0$. The first $l = 0$ level is also the deepest level of all, the particle's normal state.

If the well is too shallow, there are no negative-energy levels. Graphically, the roots of $\pm\sin x = \alpha x$ are intersections of $y = \alpha x$ with $y = \pm\sin x$, retaining only portions with $\cot x < 0$, shown solid in Fig. 9.

[4] It will be shown in § 124 that the field must decrease faster than $1/r$.

[5] The term $-l\pi/2$ is included so that $\delta_l = 0$ in the absence of a field. Since the overall sign of a wave function is immaterial, δ_l is defined modulo π, not 2π, and can always be put between 0 and π.

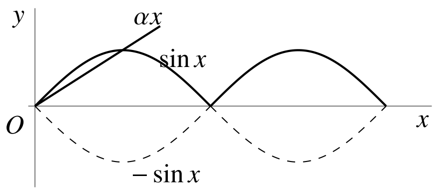

Fig. 9

For excessive α (small U_0) there are no intersections. The first appears at $\alpha = 2/\pi$, $x = \pi/2$. With $\alpha = \hbar/\sqrt{2ma^2U_0}$ and $x = ka$, the minimum depth is

$$U_{0\,\mathrm{min}} = \frac{\pi^2\hbar^2}{8ma^2}. \tag{3}$$

It increases as a decreases. At appearance, $ka = \pi/2$ and $E_1 = 0$. With further deepening, E_1 falls; for small $\Delta = U_0/U_{0\,\mathrm{min}} - 1$,

$$-E_1 = (\pi^2/16)U_{0\,\mathrm{min}}\Delta^2. \tag{4}$$

2. Find the order of levels with different l in a very deep spherical well, $U_0 \gg \hbar^2/ma^2$ (*W. Elsasser*, 1933).

SOLUTION. As $U_0 \to \infty$, the boundary condition is $\psi = 0$ at the wall (§ 22). Using (33.10) inside gives

$$J_{l+1/2}(ka) = 0.$$

Its roots determine the levels above the well bottom, $U_0 - |E| = \hbar^2k^2/2m$. Starting from the ground state, their order is

$$1s,\ 1p,\ 1d,\ 2s,\ 1f,\ 2p,\ 1g,\ 2d,\ 1h,\ 3s,\ 2f,\ldots$$

The preceding numeral orders levels having the same l.[6]

3. Find the order in which levels with different l appear as the well depth U_0 increases.

SOLUTION. At first appearance a new level has $E = 0$. Outside, the solution vanishing at infinity is $R_l = \mathrm{const}\cdot r^{-(l+1)}$. Continuity of R_l, R_l' implies

$$(r^{l+1}R_l)' = 0 \qquad \text{at } r = a.$$

This is equivalent[7] to $R_{l-1} = 0$, hence

$$J_{l-1/2}\left(\frac{a}{\hbar}\sqrt{2mU_0}\right) = 0;$$

for $l = 0$, replace $J_{l-1/2}$ by cos. The order of appearance is

$$1s,\ 1p,\ 1d,\ 2s,\ 1f,\ 2p,\ 1g,\ 2d,\ 3s,\ 1h,\ 2f,\ldots$$

[6] This notation is used for particle levels in nuclei (§ 118).

[7] From (33.7), (33.8), $(r^{-l}R_l)' \sim r^{-l}R_{l+1}$. Since (33.3) is invariant under $l \mapsto -l-1$, also $(r^{l+1}R_{-l-1})' \sim r^{l+1}R_{-l}$. Since R_{-l} and R_{l-1} obey the same equation, finally $(r^{l+1}R_l)' \sim r^{l+1}R_{l-1}$, as used here.

Differences from the deep-well ordering begin only for rather high levels.

4. Find the energy levels of the spatial oscillator $U = \frac{1}{2}m\omega^2 r^2$, their degeneracies, and the allowed orbital angular momenta.

SOLUTION. Separation in Cartesian coordinates gives three linear oscillators, hence

$$E = \hbar\omega(n_1 + n_2 + n_3 + 3/2) \equiv \hbar\omega(n + 3/2).$$

The degeneracy is the number of ways of writing n as a sum of three nonnegative integers,[8] namely

$$\frac{(n+1)(n+2)}{2}.$$

The stationary wave functions are

$$\psi_{n_1 n_2 n_3} = \text{const} \cdot e^{-\alpha^2 r^2/2} H_{n_1}(\alpha x) H_{n_2}(\alpha y) H_{n_3}(\alpha z), \tag{5}$$

where $\alpha = \sqrt{m\omega/\hbar}$. Since H_n gains $(-1)^n$ under a coordinate sign change, (5) has parity $(-1)^n$. Linear combinations at fixed $n_1 + n_2 + n_3 = n$ yield

$$\psi_{nlm} = \text{const} \cdot r^l e^{-\alpha^2 r^2/2} Y_{lm} F(-(n-l)/2, l + 3/2, \alpha^2 r^2), \tag{6}$$

where $|m| = 0, 1, \ldots, l$. The values of l are $0, 2, \ldots, n$ for even n and $1, 3, \ldots, n$ for odd n, as also follows by matching parities $(-1)^n$ and $(-1)^l$. Thus the level sequence is

$$(1s),\ (1p),\ (1d, 2s),\ (1f, 2p),\ (1g, 2d, 3s), \ldots,$$

with mutually degenerate states in parentheses.[9]

§ 34. Expansion of a plane wave

Consider a free particle moving with definite momentum $p = k\hbar$ along the positive z axis. Its wave function is

$$\psi = \text{const} \cdot e^{ikz}.$$

Expand it in the free-motion functions ψ_{klm} with definite angular momentum. Since $E = k^2\hbar^2/2m$ is definite, only functions with the same k enter. Since e^{ikz} is axially symmetric about z, only functions independent of φ, with $m = 0$, can occur. Thus

$$e^{ikz} = \sum_{l=0}^{\infty} a_l \psi_{kl0} = \sum_{l=0}^{\infty} a_l R_{kl} Y_{l0}.$$

Substitution of (28.8) and (33.9) gives

$$e^{ikz} = \sum_{l=0}^{\infty} C_l P_l(\cos\theta) \left(\frac{r}{k}\right)^l \left(\frac{1}{r}\frac{d}{dr}\right)^l \frac{\sin kr}{kr} \qquad (z = r\cos\theta).$$

[8] Equivalently, the number of ways to distribute n identical balls among three boxes.

[9] Note the degeneracy of levels with different l; see the note on p. 164.

Determine C_l by comparing coefficients of $(r\cos\theta)^n$ in the power-series expansions. On the right this term occurs only in the nth summand: for $l > n$ the radial expansion begins with higher powers of r, while for $n > l$, $P_l(\cos\theta)$ has lower powers of $\cos\theta$. The coefficient of $\cos^l\theta$ in $P_l(\cos\theta)$ is $(2l)!/2^l(l!)^2$ (see (c.1)). By (33.13), the relevant term is

$$(-1)^l C_l \frac{(2l)!(kr\cos\theta)^l}{2^l(l!)^2\, 1\cdot 3\ldots(2l+1)}.$$

The corresponding term in $\exp(ikr\cos\theta)$ is

$$\frac{(ikr\cos\theta)^l}{l!}.$$

Hence $C_l = (-i)^l(2l+1)$, and

$$e^{ikz} = \sum_{l=0}^{\infty}(-i)^l(2l+1)P_l(\cos\theta)\left(\frac{r}{k}\right)^l\left(\frac{1}{r}\frac{d}{dr}\right)^l\frac{\sin kr}{kr}. \tag{34.1}$$

At large distances,

$$e^{ikz} \approx \frac{1}{kr}\sum_{l=0}^{\infty} i^l(2l+1)P_l(\cos\theta)\sin\left(kr - \frac{l\pi}{2}\right). \tag{34.2}$$

In (34.1), z is along the wave vector $\mathbf{k}$. The expansion also has a coordinate-independent form. The addition theorem for spherical functions (c.11) expresses $P_l(\cos\theta)$ through spherical functions of the directions of $\mathbf{k}$ and $\mathbf{r}$:

$$e^{i\mathbf{kr}} = 4\pi\sum_{l=0}^{\infty}\sum_{m=-l}^{l} i^l j_l(kr)Y_{lm}^*\left(\frac{\mathbf{k}}{k}\right)Y_{lm}\left(\frac{\mathbf{r}}{r}\right). \tag{34.3}$$

Since $j_l(kr)$ depends only on kr, the symmetry in $\mathbf{k}$ and $\mathbf{r}$ is explicit; which spherical function is conjugated is immaterial.

Normalize e^{ikz} to unit probability-current density: one particle crosses unit area per unit time. The function is

$$\psi = \frac{1}{\sqrt{v}}e^{ikz} = \sqrt{\frac{m}{k\hbar}}e^{ikz} \tag{34.4}$$

(v is the speed; see (19.7)). Multiplying (34.1) by $\sqrt{m/k\hbar}$ and using $\psi^{\pm}_{klm} = R^{\pm}_{kl}Y_{lm}$ gives

$$\psi = \sum_{l=0}^{\infty}\sqrt{\pi(2l+1)}\,\frac{1}{ik\hbar}(\psi^+_{kl0} - \psi^-_{kl0}).$$

The squared coefficient of ψ^-_{kl0} (or ψ^+_{kl0}) gives the probability that a particle incident on the center (or leaving it) has angular momentum l. Since the incident flux density is unity, this probability has dimensions of area and can be interpreted as the target area in the xy plane for particles with angular momentum l. Denoting it by σ_l,

$$\sigma_l = \frac{\pi}{k^2}(2l+1). \tag{34.5}$$

For large l, summing over a range Δl, where $1 \ll \Delta l \ll l$, gives

$$\sum_{\Delta l} \sigma_l \approx \frac{\pi}{k^2} \cdot 2l\Delta l = 2\pi \frac{l\hbar^2}{p^2} \Delta l.$$

With the classical relation $\hbar l = \rho p$, where ρ is the impact parameter, this becomes

$$2\pi\rho\Delta\rho,$$

the classical result. This is not accidental: for large l the motion is semiclassical (§ 49).

PROBLEM

Expand a plane wave in wave functions with definite projection m of angular momentum on the y axis and definite momentum projection p_y on that axis.

SOLUTION. Introduce cylindrical coordinates y, ρ, φ with axis y. The functions have the form $Q_m(\rho)e^{im\varphi}e^{ip_y y/\hbar}$. Measuring φ from z,

$$e^{ikz} = e^{ik\rho\cos\varphi} = \sum_{m=-\infty}^{\infty} Q_m(\rho)e^{im\varphi}$$

(here $p_y = 0$), whence

$$Q_m(\rho) = \frac{1}{2\pi}\int_0^{2\pi} \exp[i(k\rho\cos\varphi - m\varphi)]\, d\varphi = i^m J_m(k\rho),$$

where J_m is a Bessel function. For $k\rho \gg 1$,

$$Q_m(\rho) \approx i^m \sqrt{\frac{2}{\pi k\rho}} \sin\left[k\rho - \frac{\pi}{2}\left(m - \frac{1}{2}\right)\right].$$

§ 35. Fall of a particle to the center

To clarify certain features of quantum-mechanical motion, it is useful to study a case which, although it has no immediate physical realization, involves a particle moving in a field whose potential energy becomes infinite at a point, the origin, according to $U(r) \approx -\beta/r^2$ with $\beta > 0$. The form of the field far from the origin will not matter. We saw in § 18 that this case lies precisely between those admitting ordinary stationary states and those in which the particle falls to the origin.

Near the origin the Schrödinger equation is

$$R'' + \frac{2}{r}R' + \frac{\gamma}{r^2}R = 0, \tag{35.1}$$

where $R(r)$ is the radial wave function and

$$\gamma = \frac{2m\beta}{\hbar^2} - l(l+1). \tag{35.2}$$

All terms of lower order in $1/r$ have been omitted. The energy E is assumed finite, so its term has likewise been dropped.

Put $R \sim r^s$. Then s satisfies

$$s(s+1) + \gamma = 0,$$

with roots

$$s_1 = -\frac{1}{2} + \sqrt{\frac{1}{4} - \gamma}, \qquad s_2 = -\frac{1}{2} - \sqrt{\frac{1}{4} - \gamma}. \tag{35.3}$$

For the subsequent analysis, isolate a small region of radius r_0 about the origin and replace $-\gamma/r^2$ inside it by the constant $-\gamma/r_0^2$. After determining the wave functions in this cut-off field, we shall examine the limit $r_0 \to 0$.

First suppose $\gamma < 1/4$. Then s_1 and s_2 are real negative numbers, with $s_1 > s_2$. For $r > r_0$, the general small-r solution is

$$R = Ar^{s_1} + Br^{s_2}, \tag{35.4}$$

where A and B are constants. For $r < r_0$, the solution finite at the origin of

$$R'' + \frac{2}{r}R' + \frac{\gamma}{r_0^2}R = 0$$

is

$$R = C\frac{\sin kr}{r}, \qquad k = \frac{\sqrt{\gamma}}{r_0}. \tag{35.5}$$

At $r = r_0$, both R and R' must be continuous. One condition may conveniently be written as continuity of the logarithmic derivative of rR, which gives

$$\frac{A(s_1+1)r_0^{s_1} + B(s_2+1)r_0^{s_2}}{Ar_0^{s_1+1} + Br_0^{s_2+1}} = k\cot kr_0,$$

or

$$\frac{A(s_1+1)r_0^{s_1} + B(s_2+1)r_0^{s_2}}{Ar_0^{s_1} + Br_0^{s_2}} = \sqrt{\gamma}\cot\sqrt{\gamma}.$$

Solving for B/A yields an expression of the form

$$\frac{B}{A} = \text{const}\cdot r_0^{s_1 - s_2}. \tag{35.6}$$

As $r_0 \to 0$, $B/A \to 0$ because $s_1 > s_2$. Of the two solutions of (35.1) that diverge at the origin, one must therefore choose the less rapidly divergent:

$$R = A\frac{1}{r^{|s_1|}}. \tag{35.7}$$

Now let $\gamma > 1/4$. The roots are complex:

$$s_1 = -\frac{1}{2} + i\sqrt{\gamma - \frac{1}{4}}, \qquad s_2 = s_1^*.$$

Repeating the argument again gives (35.6), which now becomes

$$\frac{B}{A} = \text{const} \cdot r_0^{i\sqrt{4\gamma-1}}. \tag{35.8}$$

This expression has no definite limit as $r_0 \to 0$, so a direct limiting procedure is impossible. Taking (35.8) into account, a real solution has the general form

$$R = \text{const} \cdot \frac{1}{\sqrt{r}} \cos\left(\sqrt{\gamma - \frac{1}{4}} \ln \frac{r}{r_0} + \text{const}\right). \tag{35.9}$$

The number of zeros of this function grows without bound as r_0 decreases. On the one hand, (35.9) applies at sufficiently small r for every finite particle energy E; on the other, the ground-state wave function must have no zeros. We therefore conclude that the ground state in this field corresponds to $E = -\infty$. In every discrete-spectrum state the particle is found mainly where $E > U$. Thus as $E \to -\infty$ it is confined to an infinitesimal region about the origin: the particle falls to the center.

The critical field U_{cr} at which fall to the center becomes possible corresponds to $\gamma = 1/4$. The smallest coefficient of $-1/r^2$ occurs for $l = 0$, and hence

$$U_{\text{cr}} = -\frac{\hbar^2}{8mr^2}. \tag{35.10}$$

Formula (35.3), with s_1, shows that the admissible Schrödinger solution near a point where $U \sim 1/r^2$ diverges no faster than $1/\sqrt{r}$ as $r \to 0$. If the field becomes infinite more slowly than $1/r^2$, then near the origin $U(r)$ may be neglected relative to the remaining terms. The free-motion solutions result, namely $\psi \sim r^l$ (§ 33). If instead the field diverges faster than $1/r^2$, as does $-1/r^s$ with $s > 2$, the wave function near the origin is proportional to $r^{s/4-1}$; see the problem in § 49. In every such case $r\psi$ vanishes at $r = 0$.

Next consider a field that at large distances decreases as $U \approx -\beta/r^2$ but is arbitrary at small distances. Suppose first that $\gamma < 1/4$. Only finitely many negative-energy levels can then exist[10]. Indeed, at $E = 0$ the large-distance Schrödinger equation is (35.1), with general solution (35.4). This function has no zeros for $r \neq 0$. All zeros of the required radial wave function therefore lie at finite distances from the origin, and their number is finite. In other words, the ordinal number of the level $E = 0$ that closes the discrete spectrum is finite.

For $\gamma > 1/4$, by contrast, the discrete spectrum contains infinitely many negative-energy levels. At $E = 0$ the large-distance wave function has the form (35.9), with infinitely many zeros, so its ordinal number is infinite. Finally, let $U = -\beta/r^2$ throughout space. If $\gamma > 1/4$, the particle falls to the center. If $\gamma < 1/4$, there are no negative-energy levels at all. The $E = 0$ wave function is then of the form (35.7) everywhere and has no zeros at finite distances; it therefore corresponds to the lowest level for the specified l.

§ 36. Motion in a Coulomb field (spherical coordinates)

An especially important case of motion in a central field is motion in a *Coulomb field*,

$$U = \pm\frac{\alpha}{r},$$

[10] At small r the field is assumed such that the particle does not fall to the center.

where $\alpha > 0$. We first consider attraction and write $U = -\alpha/r$. General considerations already show that the negative-energy spectrum is discrete, with infinitely many levels, whereas the positive-energy spectrum is continuous.

Equation (32.8) for the radial functions becomes

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} - \frac{l(l+1)}{r^2}R + \frac{2m}{\hbar^2}\left(E + \frac{\alpha}{r}\right)R = 0. \tag{36.1}$$

For the relative motion of two attracting particles, m denotes their reduced mass.

Calculations in a Coulomb field are conveniently made in special *Coulomb units*. We choose as the units of mass, length, and time

$$m, \qquad \frac{\hbar^2}{m\alpha}, \qquad \frac{\hbar^3}{m\alpha^2}.$$

All other units follow; the energy unit is $m\alpha^2/\hbar^2$. These units are used throughout this and the next section unless stated otherwise.[11] In these units (36.1) is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} - \frac{l(l+1)}{r^2}R + 2\left(E + \frac{1}{r}\right)R = 0. \tag{36.2}$$

Discrete spectrum. Introduce

$$n = \frac{1}{\sqrt{-2E}}, \qquad \rho = \frac{2r}{n}. \tag{36.3}$$

For negative energy n is real and positive. Equation (36.2) becomes

$$R'' + \frac{2}{\rho}R' + \left[-\frac{1}{4} + \frac{n}{\rho} - \frac{l(l+1)}{\rho^2}\right]R = 0, \tag{36.4}$$

where primes mean differentiation with respect to ρ.

At small ρ the admissible solution is proportional to ρ^l (see (32.15)). At large ρ, omission of the $1/\rho$ and $1/\rho^2$ terms gives $R'' = R/4$, hence $R = e^{\pm\rho/2}$. The solution vanishing at infinity therefore behaves as $e^{-\rho/2}$. Put

$$R = \rho^l e^{-\rho/2}\omega(\rho). \tag{36.5}$$

Then

$$\rho\omega'' + (2l + 2 - \rho)\omega' + (n - l - 1)\omega = 0. \tag{36.6}$$

The solution must grow no faster than a finite power at infinity and remain finite at the origin. The solution satisfying the latter condition is the confluent hypergeometric function

$$\omega = F(-n + l + 1, 2l + 2, \rho) \tag{36.7}$$

[11] For an electron of mass $m = 9.11 \cdot 10^{-28}$ g and $\alpha = e^2$, the Coulomb units coincide with the *atomic units*. The atomic length unit is $\hbar^2/me^2 = 0.529 \cdot 10^{-8}$ cm, the *Bohr radius*. The atomic energy unit is $me^4/\hbar^2 = 4.36 \cdot 10^{-11}$ erg = 27.21 eV; half of it is the *rydberg*, Ry. The atomic charge unit is $e = 4.80 \cdot 10^{-10}$ electrostatic units. Formally, formulas are converted to atomic units by putting $e = m = \hbar = 1$. For $\alpha = Ze^2$, Coulomb and atomic units differ.

(see mathematical appendix § d).[12] The condition at infinity holds only when $-n + l + 1$ is a nonpositive integer, so that (36.7) is a polynomial of degree $n - l - 1$; otherwise it grows as e^{ρ} (see (d.14)). Thus n is a positive integer and

$$n \geq l + 1. \tag{36.8}$$

From (36.3),

$$E = -\frac{1}{2n^2}, \qquad n = 1, 2, \ldots. \tag{36.9}$$

This determines the discrete energy levels. Infinitely many lie between the normal level $E_1 = -1/2$ and zero. Their spacings decrease with n, and they accumulate at $E = 0$, where the discrete and continuous spectra meet. In ordinary units,[13]

$$E = -\frac{m\alpha^2}{2\hbar^2 n^2}. \tag{36.10}$$

The integer n is the *principal quantum number*; the radial quantum number of § 32 is $n_r = n - l - 1$. At fixed n,

$$l = 0, 1, \ldots, n - 1. \tag{36.11}$$

Since the energy depends only on n, states with different l but the same n have equal energies. Each eigenvalue is therefore degenerate not only in the magnetic quantum number m, as in every central field, but also in l. This latter, *accidental* or *Coulomb*, degeneracy is peculiar to the Coulomb field. Since each l has $2l + 1$ values of m, the degeneracy of level n is

$$\sum_{l=0}^{n-1} (2l + 1) = n^2. \tag{36.12}$$

The stationary-state wave functions follow from (36.5) and (36.7). For integer parameters the confluent hypergeometric function is, apart from a factor, a generalized Laguerre polynomial (appendix § d), so

$$R_{nl} = \text{const} \cdot \rho^l e^{-\rho/2} L_{n+l}^{2l+1}(\rho).$$

[12] The second solution of (36.6) diverges as ρ^{-2l-1} when $\rho \to 0$.

[13] Formula (36.10) was first obtained by N. Bohr in 1913, before quantum mechanics. W. Pauli derived it in quantum mechanics by the matrix method in 1926, and Schrödinger derived it from the wave equation a few months later.

With $\int_0^\infty R_{nl}^2 r^2\,dr = 1$, the final form is[14]

$$\begin{aligned} R_{nl} &= -\frac{2}{n^2}\sqrt{\frac{(n-l-1)!}{[(n+l)!]^3}}\,e^{-r/n}(2r/n)^l L_{n+l}^{2l+1}(2r/n) \\ &= \frac{2}{n^{l+2}(2l+1)!}\sqrt{\frac{(n+l)!}{(n-l-1)!}}(2r)^l e^{-r/n} F(-n+l+1, 2l+2, 2r/n). \end{aligned} \tag{36.13}$$

(For the normalization integral see appendix § f, integral (f.6).[15]) Near the origin,

$$R_{nl} \approx r^l \frac{2^{l+1}}{n^{2+l}(2l+1)!}\sqrt{\frac{(n+l)!}{(n-l-1)!}}. \tag{36.14}$$

At large distances,

$$R_{nl} \approx (-1)^{n-l-1}\frac{2^n r^{n-1} e^{-r/n}}{n^{n+1}\sqrt{(n+l)!(n-l-1)!}}. \tag{36.15}$$

The normal-state function R_{10} decays over distances $r \sim 1$, or $r \sim \hbar^2/m\alpha$ in ordinary units.

Mean powers of r are $\overline{r^k} = \int_0^\infty r^{k+2} R_{nl}^2\,dr$. The general result follows from (f.7); some first values are

$$\begin{aligned} \bar r &= \tfrac{1}{2}[3n^2 - l(l+1)], \qquad \overline{r^2} = \tfrac{n^2}{2}[5n^2 + 1 - 3l(l+1)], \\ \overline{r^{-1}} &= \frac{1}{n^2}, \qquad \overline{r^{-2}} = \frac{1}{n^3(l+1/2)}. \end{aligned} \tag{36.16}$$

Continuous spectrum. Positive energies form a continuous spectrum from zero to infinity. Every eigenvalue is infinitely degenerate, with all integer $l \geq 0$ and all allowed m. Now n and ρ are purely imaginary:

$$n = -\frac{i}{\sqrt{2E}} = -\frac{i}{k}, \qquad \rho = 2ikr, \tag{36.17}$$

where $k = \sqrt{2E}$.[16] The continuum radial eigenfunctions are

$$R_{kl} = \frac{C_{kl}}{(2l+1)!}(2kr)^l e^{-ikr} F(i/k + l + 1, 2l+2, 2ikr), \tag{36.18}$$

[14] The first few functions are

$$R_{10} = 2e^{-r}, \quad R_{20} = \frac{1}{\sqrt{2}}e^{-r/2}(1 - r/2), \quad R_{21} = \frac{1}{2\sqrt{6}}e^{-r/2}r,$$
$$R_{30} = \frac{2}{3\sqrt{3}}e^{-r/3}(1 - 2r/3 + 2r^2/27),$$
$$R_{31} = \frac{8}{27\sqrt{6}}e^{-r/3}r(1 - r/6), \quad R_{32} = \frac{4}{81\sqrt{30}}e^{-r/3}r^2.$$

[15] It may also be evaluated by substituting (d.13) for the Laguerre polynomials and integrating by parts, as for the Legendre-polynomial integral (c.8).

[16] One could instead choose the complex conjugates $n = i/k$, $\rho = -2ikr$; the real functions R_{kl} are independent of this choice.

or, as a complex integral (appendix § d),

$$R_{kl} = C_{kl}(2kr)^l e^{-ikr} \frac{1}{2\pi i} \oint e^{\xi} \left(1 - \frac{2ikr}{\xi}\right)^{-i/k-l-1} \xi^{-2l-2} d\xi, \tag{36.19}$$

over the contour in Fig. 10.[17] With $\xi = 2ikr(t + 1/2)$ this becomes

$$R_{kl} = C_{kl} \frac{(-2kr)^{-l-1}}{2\pi} \oint e^{2ikrt} (t + \tfrac{1}{2})^{i/k-l-1} (t - \tfrac{1}{2})^{-i/k-l-1} dt. \tag{36.20}$$

The path encircles $t = \pm 1/2$ positively; this form directly shows that R_{kl} is real.

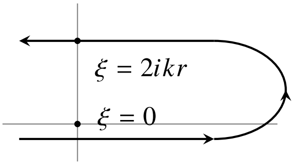

Fig. 10

Using (d.14), the asymptotic expansion is

$$R_{kl} = C_{kl} \frac{e^{-\pi/2k}}{kr} \operatorname{Re}\left\{ \frac{\exp[-i(kr - (\pi/2)(l+1) + (1/k)\ln 2kr)]}{\Gamma(l+1-i/k)} \times G(l+1+i/k, i/k - l, -2ikr) \right\}. \tag{36.21}$$

For normalization on the $k/2\pi$ scale, condition (33.4),

$$C_{kl} = 2ke^{\pi/2k} |\Gamma(l+1-i/k)|. \tag{36.22}$$

Indeed, at large r this gives

$$R_{kl} \approx \frac{2}{r} \sin\left(kr + k^{-1}\ln 2kr - \pi l/2 + \delta_l\right), \qquad \delta_l = \arg\Gamma(l+1-i/k). \tag{36.23}$$

This agrees with the general form (33.20), apart from the logarithmic phase; since $\ln r$ grows slowly compared with r, it does not affect the divergent normalization integral.

The gamma-function modulus in (36.22) is elementary. From $\Gamma(z+1) = z\Gamma(z)$ and $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$, one finds

$$\Gamma(l+1+i/k) = (l+i/k)\cdots(1+i/k)(i/k)\Gamma(i/k),$$
$$\Gamma(l+1-i/k) = (l-i/k)\cdots(1-i/k)\Gamma(1-i/k),$$

and

$$|\Gamma(l+1-i/k)| = \sqrt{\frac{\pi}{k}} \prod_{s=1}^{l} \sqrt{s^2 + k^{-2}} (\sinh(\pi/k))^{-1/2}.$$

[17]It may be replaced by any closed loop encircling $\xi = 0$ and $\xi = 2ikr$ positively. For integral l, the function $V(\xi) = \xi^{-n-l}(\xi - 2ikr)^{n-l}$ (appendix § d) returns to its original value around such a contour.

Thus

$$C_{kl} = \left[\frac{8\pi k}{1 - e^{-2\pi/k}}\right]^{1/2} \prod_{s=1}^{l} \sqrt{s^2 + \frac{1}{k^2}}, \tag{36.24}$$

with the product replaced by 1 at $l = 0$. The $E = 0$ radial function follows by taking $k \to 0$:

$$\begin{aligned} F(i/k + l + 1, 2l + 2, 2ikr) \to F(i/k, 2l + 2, 2ikr) \\ &= 1 - \frac{2r}{(2l+1)!} + \frac{(2r)^2}{(2l+2)(2l+3)2!} - \cdots \\ &= (2l+1)!(2r)^{-l-1/2} J_{2l+1}(\sqrt{8r}), \end{aligned}$$

where J_{2l+1} is a Bessel function, while $C_{kl} \approx \sqrt{8\pi}\, k^{-l+1/2}$. Hence

$$\left.\frac{R_{kl}}{\sqrt{k}}\right|_{k\to 0} = \sqrt{\frac{4\pi}{r}} J_{2l+1}(\sqrt{8r}). \tag{36.25}$$

At large r,[18]

$$\left.\frac{R_{kl}}{\sqrt{k}}\right|_{k\to 0} = \left(\frac{8}{r^3}\right)^{1/4} \sin\left(\sqrt{8r} - l\pi - \pi/4\right). \tag{36.26}$$

The factor $\sqrt{k}$ disappears on changing to energy-scale normalization, from R_{kl} to R_{El} by (33.5); R_{El} remains finite as $E \to 0$.

For the repulsive field $U = \alpha/r$, only a continuous positive-energy spectrum exists. Formally changing r to $-r$ in the attractive-field equation gives

$$\begin{aligned} R_{kl} &= \frac{C_{kl}}{(2l+1)!}(2kr)^l e^{ikr} F(i/k + l + 1, 2l + 2, -2ikr), \\ C_{kl} &= 2ke^{-\pi/2k}|\Gamma(l + 1 + i/k)| = \left(\frac{8\pi k}{e^{2\pi/k} - 1}\right)^{1/2} \prod_{s=1}^{l} \sqrt{s^2 + \frac{1}{k^2}}. \end{aligned} \tag{36.27}$$

At large r,

$$R_{kl} \approx \frac{2}{r} \sin\left(kr - k^{-1} \ln 2kr - l\pi/2 + \delta_l\right), \qquad \delta_l = \arg\Gamma(l + 1 + i/k). \tag{36.28}$$

Origin of the Coulomb degeneracy. Classical motion in an attractive Coulomb field has the special conserved quantity

$$\mathbf{A} = \frac{\mathbf{r}}{r} - \mathbf{p} \times \mathbf{l} = \text{const} \tag{36.29}$$

(see I, § 15). Its quantum operator is

$$\hat{\mathbf{A}} = \frac{\mathbf{r}}{r} - \frac{1}{2}(\hat{\mathbf{p}} \times \hat{\mathbf{l}} - \hat{\mathbf{l}} \times \hat{\mathbf{p}}), \tag{36.30}$$

which commutes with $\hat{H} = \hat{\mathbf{p}}^2/2 - 1/r$.

[18] This function corresponds to the semiclassical approximation (§ 49) in $(l + 1/2)^2 \ll r \ll k^{-2}$.

Direct calculation gives

$$[\hat{l}_i, \hat{A}_k] = ie_{ikl}\hat{A}_l, \qquad [\hat{A}_i, \hat{A}_k] = -2i\hat{H}e_{ikl}\hat{l}_l. \tag{36.31}$$

The mutual noncommutativity of the $\hat{A}_i$ means that A_x, A_y, A_z cannot all be definite. Each component, say $\hat{A}_z$, commutes with $\hat{l}_z$ but not with $\hat{\mathbf{l}}^2$. A new conserved quantity which cannot be measured together with the other conserved quantities causes the additional degeneracy (§ 10): this is the accidental degeneracy peculiar to the Coulomb field.

The same origin may be stated in terms of the Coulomb problem's enlarged quantum-mechanical symmetry (V. A. Fock, 1935). At fixed negative energy, replace $\hat{H}$ on the right of (36.31) by E and define $\hat{u}_i = \hat{A}_i/\sqrt{-2E}$. Then

$$[\hat{l}_i, \hat{u}_k] = ie_{ikl}\hat{u}_l, \qquad [\hat{u}_i, \hat{u}_k] = ie_{ikl}\hat{l}_l. \tag{36.32}$$

Together with $[\hat{l}_i, \hat{l}_k] = ie_{ikl}\hat{l}_l$, these are precisely the formal commutation rules for infinitesimal rotations in four-dimensional Euclidean space.[19] This is the symmetry of the Coulomb problem.[20]

The energy levels can again be derived from (36.32).[21] Introduce

$$\hat{\mathbf{j}}_1 = \tfrac{1}{2}(\hat{\mathbf{l}} + \hat{\mathbf{u}}), \qquad \hat{\mathbf{j}}_2 = \tfrac{1}{2}(\hat{\mathbf{l}} - \hat{\mathbf{u}}). \tag{36.33}$$

Then

$$[\hat{j}_{1i}, \hat{j}_{1k}] = ie_{ikl}\hat{j}_{1l}, \quad [\hat{j}_{2i}, \hat{j}_{2k}] = ie_{ikl}\hat{j}_{2l}, \quad [\hat{j}_{1i}, \hat{j}_{2k}] = 0. \tag{36.34}$$

These are the rules for two independent three-dimensional angular momenta. Thus the eigenvalues of $\hat{\mathbf{j}}_1^2$ and $\hat{\mathbf{j}}_2^2$ are $j_1(j_1+1)$ and $j_2(j_2+1)$, with $j_1, j_2 = 0, 1/2, 1, 3/2, \ldots$.[22] Directly from $\hat{\mathbf{u}}$ and $\hat{\mathbf{l}} = \mathbf{r} \times \hat{\mathbf{p}}$,

$$\hat{\mathbf{l}}\hat{\mathbf{u}} = \hat{\mathbf{u}}\hat{\mathbf{l}} = 0, \qquad \hat{\mathbf{l}}^2 + \hat{\mathbf{u}}^2 = -1 - \frac{1}{2E} \qquad \hat{\mathbf{l}}^2 + \hat{\mathbf{u}}^2 = -1 - \frac{1}{2E}.$$

Hence

$$\hat{\mathbf{j}}_1^2 = \hat{\mathbf{j}}_2^2 = -\tfrac{1}{4}(1 + 1/2E) = j(j+1),$$

so $E = -1/[2(2j+1)^2]$. Writing

$$2j + 1 = n, \qquad n = 1, 2, 3, \ldots, \tag{36.35}$$

gives $E = -1/(2n^2)$. The degeneracy is $(2j_1+1)(2j_2+1) = n^2$. Since $\hat{\mathbf{l}} = \hat{\mathbf{j}}_1 + \hat{\mathbf{j}}_2$, for $j_1 = j_2 = (n-1)/2$ the orbital momentum ranges from $l = 0$ to $2j = n - 1$.[23]

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[19] Here $\hat{l}_x, \hat{l}_y, \hat{l}_z$ generate rotations in the yz, xz, xy planes of Cartesian coordinates x, y, z, u, and $\hat{u}_x, \hat{u}_y, \hat{u}_z$ generate rotations in the xu, yu, zu planes.

[20] It appears explicitly in momentum-space wave functions; see V. A. Fock, Izv. AN SSSR, physical series, 1935, no. 2, p. 169; Zs. f. Physik 98 (1935), 145.

[21] This derivation is essentially Pauli's (1926).

[22] We anticipate § 54 in using the possible integral and half-integral values of j.

[23] Accidental degeneracy between different l also occurs in the central field $U = m\omega^2r^2/2$ (the spatial

Problem

1. Find the probability distribution of momentum values in the ground state of the hydrogen atom.

Solution. [24] The ground-state wave function is

$$\psi = R_{10}Y_{00} = \frac{1}{\sqrt{\pi}}e^{-r}.$$

In the **p**-representation,

$$a(\mathbf{p}) = \int \psi(\mathbf{r})e^{-i\mathbf{pr}}\,dV.$$

Using spherical coordinates with polar axis along **p** gives

$$a(\mathbf{p}) = \frac{8\sqrt{\pi}}{(1+p^2)^2},$$

and the probability density in momentum space is $|a(\mathbf{p})|^2/(2\pi)^3$.

2. Find the mean potential of the field produced by the nucleus and the electron in the hydrogen ground state.

Solution. The mean potential φ_e of the "electron cloud" is most simply found as the spherical solution of Poisson's equation with charge density $\rho = -|\psi|^2$:

$$\frac{1}{r}\frac{d^2}{dr^2}(r\varphi_e) = 4e^{-2r}.$$

Choosing the integration constants so that $\varphi_e(0)$ is finite and $\varphi_e(\infty) = 0$, and adding the nuclear potential, gives

$$\varphi = 1/r + \varphi_e(r) = (1/r + 1)e^{-2r}.$$

Thus $\varphi \approx 1/r$ for $r \ll 1$, while $\varphi \approx e^{-2r}$ for $r \gg 1$.

oscillator; Problem 4 of § 33), again because of an enlarged Hamiltonian symmetry. In $\hat{H} = \hat{\mathbf{p}}^2/2m + m\omega^2r^2/2$, both $\hat{p}_i$ and x_i enter as sums of squares. With

$$\hat{a}_i = \frac{m\omega x_i + i\hat{p}_i}{\sqrt{2m\hbar\omega}}, \qquad \hat{a}_i^+ = \frac{m\omega x_i - i\hat{p}_i}{\sqrt{2m\hbar\omega}},$$

one obtains $\hat{H} = \hbar\omega[\hat{\mathbf{a}}^+\hat{\mathbf{a}} + 3/2]$. This is invariant under arbitrary unitary transformations of $\hat{a}_i^+$ and $\hat{a}_j$, a group larger than the three-dimensional rotation group. The quantum peculiarity of the Coulomb and oscillator fields has its classical counterpart: particle trajectories are closed in these, and only these, fields.

[24] Atomic units are used in Problems 1 and 2.

3. Find the energy levels of a particle moving in the central field $U = A/r^2 - B/r$ (Fig. 11).

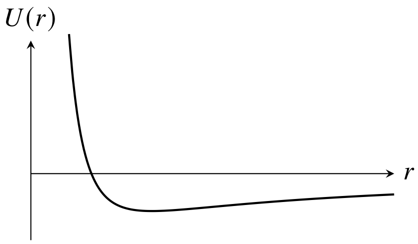


Fig. 11

Solution. Positive energies form a continuous spectrum and negative energies a discrete one; consider the latter. The radial equation is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \frac{2m}{\hbar^2}\left(E - \frac{\hbar^2 l(l+1)}{2mr^2} - \frac{A}{r^2} + \frac{B}{r}\right)R = 0. \tag{1}$$

Put $\rho = 2\sqrt{-mE}\,r/\hbar$ and define

$$\frac{2mA}{\hbar^2} + l(l+1) = s(s+1), \tag{2}$$

$$\frac{B}{\hbar}\sqrt{\frac{m}{-2E}} = n. \tag{3}$$

Then

$$R'' + \frac{2}{\rho}R' + (-\tfrac{1}{4} + n/\rho - s(s+1)/\rho^2)R = 0,$$

formally identical to (36.4). Hence

$$R = \rho^s e^{-\rho/2} F(-n + s + 1, 2s + 2, \rho),$$

where $n - s - 1 = p$ is a nonnegative integer and s is the positive root of (2). Formula (3) gives

$$-E_p = \frac{2B^2m}{\hbar^2}[2p + 1 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}]^{-2}.$$

4. Repeat the calculation for $U = A/r^2 + Br^2$ (Fig. 12).

Solution. There is only a discrete spectrum. The radial equation is

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \frac{2m}{\hbar^2}(E - \hbar^2 l(l+1)/(2mr^2) - A/r^2 - Br^2)R = 0.$$

Set

$$\xi = \frac{\sqrt{2mB}}{\hbar}r^2,$$

and define

$$l(l+1) + \frac{2mA}{\hbar^2} = 2s(2s+1), \qquad \sqrt{\frac{2m}{B}}\frac{E}{\hbar} = 4(n+s) + 3.$$

Then

$$\xi R'' + \frac{3}{2}R' + [n + s + \tfrac{3}{4} - \xi/4 - s(s+1/2)/\xi]R = 0.$$

The desired solution behaves as $e^{-\xi/2}$ at infinity and as ξ^s at the origin, where

$$s = \frac{1}{4}[-1 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}].$$

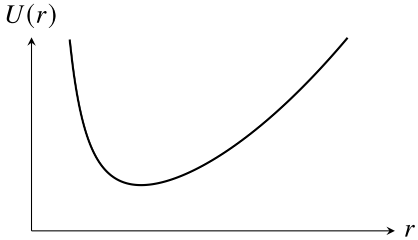


Fig. 12

Writing $R = e^{-\xi/2}\xi^s\omega$ gives

$$\xi\omega'' + (2s + 3/2 - \xi)\omega' + n\omega = 0,$$

so

$$\omega = F(-n, 2s + 3/2, \xi),$$

where n is a nonnegative integer. The equally spaced levels are

$$E_n = \hbar\sqrt{\frac{B}{2m}}[4n + 2 + \sqrt{(2l+1)^2 + 8mA/\hbar^2}], \qquad n = 0, 1, 2, \ldots .$$

§ 37. Motion in a Coulomb field (parabolic coordinates)

For motion in any centrally symmetric field, the Schrödinger equation can always be separated in spherical coordinates. In a Coulomb field, separation is also possible in the so-called parabolic coordinates. The solution of the Coulomb-field problem in these coordinates is useful in studying a number of problems in which a particular direction in space is singled out, for example by an external electric field in addition to the Coulomb field (§ 77). The parabolic coordinates ξ, η, and φ are defined by

$$x = \sqrt{\xi\eta}\cos\varphi, \qquad y = \sqrt{\xi\eta}\sin\varphi, \qquad z = \frac{1}{2}(\xi - \eta),$$
$$r = \sqrt{x^2 + y^2 + z^2} = \frac{1}{2}(\xi + \eta) \tag{37.1}$$

or, conversely,

$$\xi = r + z, \qquad \eta = r - z, \qquad \varphi = \arctan\frac{y}{x}; \tag{37.2}$$

ξ and η range from 0 to ∞, while φ ranges from 0 to 2π. The surfaces ξ = const and η = const are paraboloids of revolution whose axis is the z axis and whose focus is at the origin. This coordinate system is orthogonal. The line element is

$$dl^2 = \frac{\xi+\eta}{4\xi}\,d\xi^2 + \frac{\xi+\eta}{4\eta}\,d\eta^2 + \xi\eta\,d\varphi^2, \tag{37.3}$$

and the volume element is

$$dV = \frac{1}{4}(\xi+\eta)\,d\xi\,d\eta\,d\varphi. \tag{37.4}$$

Equation (37.3) gives the following expression for the Laplace operator:

$$\Delta = \frac{4}{\xi+\eta}\left[\frac{\partial}{\partial\xi}\left(\xi\frac{\partial}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left(\eta\frac{\partial}{\partial\eta}\right)\right] + \frac{1}{\xi\eta}\frac{\partial^2}{\partial\varphi^2}. \tag{37.5}$$

For a particle in the attractive Coulomb field

$$U = -\frac{1}{r} = -\frac{2}{\xi+\eta},$$

the Schrödinger equation becomes

$$\frac{4}{\xi+\eta}\left[\frac{\partial}{\partial\xi}\left(\xi\frac{\partial\psi}{\partial\xi}\right) + \frac{\partial}{\partial\eta}\left(\eta\frac{\partial\psi}{\partial\eta}\right)\right] + \frac{1}{\xi\eta}\frac{\partial^2\psi}{\partial\varphi^2} + 2\left(E + \frac{2}{\xi+\eta}\right)\psi = 0. \tag{37.6}$$

We seek the eigenfunctions ψ in the form

$$\psi = f_1(\xi) f_2(\eta) e^{im\varphi}, \tag{37.7}$$

where m is the magnetic quantum number. Substitute this expression into (37.6), after multiplying that equation by $(\xi+\eta)/4$, and separate the variables ξ and η. The resulting equations for f_1 and f_2 are

$$\begin{aligned} \frac{d}{d\xi}\left(\xi\frac{df_1}{d\xi}\right) + \left[\frac{E}{2}\xi - \frac{m^2}{4\xi} + \beta_1\right] f_1 = 0, \\ \frac{d}{d\eta}\left(\eta\frac{df_2}{d\eta}\right) + \left[\frac{E}{2}\eta - \frac{m^2}{4\eta} + \beta_2\right] f_2 = 0, \end{aligned} \tag{37.8}$$

where the "separation parameters" β_1 and β_2 are related by

$$\beta_1 + \beta_2 = 1. \tag{37.9}$$

Consider the discrete energy spectrum ($E < 0$). In place of E, ξ, and η, introduce the quantities

$$n = \frac{1}{\sqrt{-2E}}, \qquad \rho_1 = \xi\sqrt{-2E} = \frac{\xi}{n}, \qquad \rho_2 = \frac{\eta}{n}, \tag{37.10}$$

after which the equation for f_1 takes the form

$$\frac{d^2 f_1}{d\rho_1^2} + \frac{1}{\rho_1}\frac{df_1}{d\rho_1} + \left[-\frac{1}{4} + \frac{1}{\rho_1}\left(\frac{|m|+1}{2} + n_1\right) - \frac{m^2}{4\rho_1^2}\right] f_1 = 0, \tag{37.11}$$

with an identical equation for f_2; here we have also introduced the notation

$$n_1 = -\frac{|m|+1}{2} + n\beta_1, \qquad n_2 = -\frac{|m|+1}{2} + n\beta_2. \tag{37.12}$$

Proceeding as for equation (36.4), we find that at large ρ_1, f_1 behaves as $e^{-\rho_1/2}$, and at small ρ_1 as $\rho_1^{|m|/2}$. We therefore seek a solution of (37.11) in the form

$$f_1(\rho_1) = e^{-\rho_1/2}\rho_1^{|m|/2} w_1(\rho_1)$$

(and analogously for f_2), obtaining for w_1 the equation

$$\rho_1 w_1'' + (|m| + 1 - \rho_1)w_1' + n_1 w_1 = 0.$$

This is again the confluent hypergeometric equation. The solution satisfying the finiteness conditions is

$$w_1 = F(-n_1, |m| + 1, \rho_1),$$

and n_1 must be a nonnegative integer.

Thus, in parabolic coordinates every stationary state of the discrete spectrum is specified by three integers: the "parabolic quantum numbers" n_1 and n_2, and the magnetic quantum number m. From (37.9) and (37.12), the number n (the "principal quantum number") is

$$n = n_1 + n_2 + |m| + 1. \tag{37.13}$$

The energy levels are, of course, the same as in (36.9).

For a given n, the number $|m|$ can take the n values from 0 through $n-1$. With n and $|m|$ fixed, n_1 takes the $n-|m|$ values from 0 through $n-|m|-1$. For each prescribed $|m|$, functions with $m = \pm|m|$ may also be chosen. Hence the total number of distinct states for a given n is

$$2\sum_{m=1}^{n-1}(n-m) + (n-0) = n^2,$$

in agreement with the result obtained in § 36.

The discrete-spectrum wave functions $\psi_{n_1 n_2 m}$ must obey the normalization condition

$$\int |\psi_{n_1 n_2 m}|^2\, dV = \frac{1}{4}\int_0^\infty \int_0^\infty \int_0^{2\pi} |\psi_{n_1 n_2 m}|^2 (\xi + \eta)\, d\varphi\, d\xi\, d\eta = 1. \tag{37.14}$$

The normalized functions are

$$\psi_{n_1 n_2 m} = \frac{\sqrt{2}}{n^2} f_{n_1 m}\left(\frac{\xi}{n}\right) f_{n_2 m}\left(\frac{\eta}{n}\right)\frac{e^{im\varphi}}{\sqrt{2\pi}}, \tag{37.15}$$

$$f_{pm}(\rho) = \frac{1}{|m|!}\sqrt{\frac{(p+|m|)!}{p!}}F(-p, |m|+1, \rho)e^{-\rho/2}\rho^{|m|/2}. \tag{37.16}$$

Unlike the wave functions in spherical coordinates, those in parabolic coordinates are not symmetric about the plane $z = 0$. When $n_1 > n_2$, the particle is more likely to be found on the $z > 0$ side than on the $z < 0$ side; for $n_1 < n_2$, the reverse holds.

The continuous spectrum ($E > 0$) corresponds to a continuous spectrum of real values of the parameters β_1, β_2 in equations (37.8) (still, of course, related by (37.9)); we shall not write the corresponding wave functions here. Regarded as equations for the "eigenvalues" β_1 and β_2, equations (37.8) also possess, when $E > 0$, a spectrum of complex values. The associated wave functions will be given in § 135, where they will be used to solve the problem of scattering in a Coulomb field.

The existence of the stationary states $|n_1 n_2 m\rangle$ is tied to the additional conservation law (36.29). In these states, besides the energy, the quantities $l_z = m$ and A_z have definite values. Evaluation of the diagonal matrix elements of the operator $\hat{A}_z$ gives

$$A_z = \frac{n_1 - n_2}{n}. \tag{37.17}$$

Here $u_z = n_1 - n_2$, while the projections of the "angular momenta" $\mathbf{j}_1$ and $\mathbf{j}_2$ are

$$j_{1z} = \frac{m + n_1 - n_2}{2} \equiv \mu_1, \qquad j_{2z} = \frac{m - n_1 + n_2}{2} \equiv \mu_2. \tag{37.18}$$

These properties of the states $|n_1 n_2 m\rangle$ (or, equivalently, $|n\mu_1\mu_2\rangle$) make it easy to relate their wave functions to those of the states $|nlm\rangle$. Since $\mathbf{l} = \mathbf{j}_1 + \mathbf{j}_2$, a change between the two descriptions reduces to constructing wave functions for the addition of two angular momenta (considered below in § 106). In terms of the "angular momenta" $\mathbf{j}_1$ and $\mathbf{j}_2$, the states $|nlm\rangle$ and $|n_1 n_2 m\rangle$ are written as $|j_1 j_2 l m\rangle$ and $|j_1 j_2 \mu_1 \mu_2\rangle$, where, by (36.35) and (37.13),

$$j_1 = j_2 = \frac{n-1}{2} = \frac{n_1 + n_2 + |m|}{2}. \tag{37.19}$$

The general formulas (106.9)–(106.11) give

$$\begin{aligned} \psi_{nlm} &= \sum_{\mu_1 + \mu_2 = m} \langle lm|\mu_1\mu_2\rangle \psi_{n\mu_1\mu_2}, \\ \psi_{n\mu_1\mu_2} &= \sum_{l=0}^{n-1} \langle l, \mu_1 + \mu_2|\mu_1\mu_2\rangle \psi_{nlm} \end{aligned} \tag{37.20}$$

(*D. Park*, 1960).

CHAPTER VI

Perturbation Theory

§ 38. Time-independent perturbations

An exact solution of the Schrödinger equation can be found in only a comparatively small number of the simplest cases. Most problems in quantum mechanics lead to equations too complicated for exact solution. Frequently, however, a problem contains quantities of different orders of magnitude, including small quantities whose omission simplifies the problem enough to make an exact solution possible. The first step is then to solve this simplified problem exactly; the second is to calculate approximately the corrections due to the small terms that were omitted. The general method for calculating these corrections is called *perturbation theory*.

Suppose that the Hamiltonian of the physical system is

$$\hat{H} = \hat{H}_0 + \hat{V},$$

where $\hat{V}$ is a small correction, the *perturbation*, to the "unperturbed" operator $\hat{H}_0$. In § 38 and § 39 we consider perturbations $\hat{V}$ with no explicit time dependence; the same is assumed for $\hat{H}_0$. The conditions under which $\hat{V}$ may be regarded as small in comparison with $\hat{H}_0$ will be established below.

For the discrete spectrum, the problem may be stated as follows. The eigenfunctions $\psi^{(0)}$ and eigenvalues of $\hat{H}_0$ are assumed known; that is, the exact solutions of

$$\hat{H}_0\psi^{(0)} = E^{(0)}\psi^{(0)} \tag{38.1}$$

are known. Approximate solutions are required for

$$\hat{H}\psi = (\hat{H}_0 + \hat{V})\psi = E\psi, \tag{38.2}$$

namely approximate expressions for the eigenfunctions ψ_n and eigenvalues E_n of the perturbed operator $\hat{H}$.

Throughout this section all eigenvalues of $\hat{H}_0$ will be assumed nondegenerate. To simplify the derivation, we first suppose that the energy levels form a discrete spectrum only.

It is convenient to use matrix form from the outset. Expand the required function ψ in the functions $\psi_m^{(0)}$:

$$\psi = \sum_m c_m\psi_m^{(0)}. \tag{38.3}$$

Substitution in (38.2) gives

$$\sum_m c_m(E_m^{(0)} + \hat{V})\psi_m^{(0)} = \sum_m c_m E\psi_m^{(0)}.$$

Multiplying by $\psi_k^{(0)*}$ and integrating, we find

$$(E - E_k^{(0)})c_k = \sum_m V_{km} c_m. \tag{38.4}$$

Here V_{km} is the matrix of $\hat{V}$ in the unperturbed functions:

$$V_{km} = \int \psi_k^{(0)*} \hat{V} \psi_m^{(0)} \, dq. \tag{38.5}$$

Seek the coefficients and energy as series

$$E = E^{(0)} + E^{(1)} + E^{(2)} + \dots, \qquad c_m = c_m^{(0)} + c_m^{(1)} + c_m^{(2)} + \dots,$$

where $E^{(1)}, c_m^{(1)}$ have the same order of smallness as $\hat{V}$, $E^{(2)}, c_m^{(2)}$ are of second order, and so forth.

To calculate the corrections to the nth eigenvalue and eigenfunction, set $c_n^{(0)} = 1$ and $c_m^{(0)} = 0$ for $m \neq n$. For the first approximation, put $E = E_n^{(0)} + E_n^{(1)}$ and $c_k = c_k^{(0)} + c_k^{(1)}$ in (38.4), retaining first-order terms only. The equation with $k = n$ gives

$$E_n^{(1)} = V_{nn} = \int \psi_n^{(0)*} \hat{V} \psi_n^{(0)} \, dq. \tag{38.6}$$

Thus the first-order correction to $E_n^{(0)}$ is the mean value of the perturbation in the state $\psi_n^{(0)}$.

For $k \neq n$, equation (38.4) gives

$$c_k^{(1)} = \frac{V_{kn}}{E_n^{(0)} - E_k^{(0)}}, \qquad k \neq n. \tag{38.7}$$

The coefficient $c_n^{(1)}$ remains arbitrary and must be chosen so that $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$ is normalized through first order. This requires $c_n^{(1)} = 0$. Indeed,

$$\psi_n^{(1)} = \sum_m{}' \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} \psi_m^{(0)} \tag{38.8}$$

(the prime means that $m = n$ is omitted) is orthogonal to $\psi_n^{(0)}$; hence the integral of $|\psi_n^{(0)} + \psi_n^{(1)}|^2$ differs from unity only by a second-order quantity.

Formula (38.8) gives the first-order correction to the wave functions. It also shows the condition for applicability of the method:

$$|V_{mn}| \ll |E_n^{(0)} - E_m^{(0)}|. \tag{38.9}$$

Thus perturbation matrix elements must be small compared with the corresponding separations between unperturbed energy levels.

To find the second-order correction to $E_n^{(0)}$, substitute $E = E_n^{(0)} + E_n^{(1)} + E_n^{(2)}$ and $c_k = c_k^{(0)} + c_k^{(1)} + c_k^{(2)}$ in (38.4), and retain second-order terms. The equation with $k = n$ gives

$$E_n^{(2)} c_n^0 = \sum_m{}' V_{nm} c_m^{(1)},$$

and consequently

$$E_n^{(2)} = \sum_m{}' \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}. \tag{38.10}$$

Here (38.7) was used, together with Hermiticity, $V_{mn} = V_{nm}^*$. The second-order correction to the ground-state energy is always negative: if $E_n^{(0)}$ is the lowest eigenvalue, every term in (38.10) is negative. Higher approximations may be calculated in the same manner.

These results extend directly to an operator $\hat{H}_0$ that also has a continuous spectrum, while the perturbed state itself still belongs to the discrete spectrum. The appropriate continuous-spectrum integrals are simply added to the discrete sums. Denote continuous-spectrum states by an index ν ranging continuously; ν stands for the complete set of quantities needed to specify a state. If these states are degenerate, as they nearly always are, energy alone does not specify one.[1] For example, (38.8) becomes

$$\psi_n^{(1)} = \sum_m{}' \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}}\psi_m^{(0)} + \int \frac{V_{\nu n}}{E_n^{(0)} - E_\nu^{(0)}}\psi_\nu^{(0)}\,d\nu. \tag{38.11}$$

It is also useful to give the perturbed matrix elements of a physical quantity f, calculated through first order with $\psi_n = \psi_n^{(0)} + \psi_n^{(1)}$ and (38.8):

$$f_{nm} = f_{nm}^{(0)} + \sum_k{}' \frac{V_{nk} f_{km}^{(0)}}{E_n^{(0)} - E_k^{(0)}} + \sum_k{}' \frac{V_{km} f_{nk}^{(0)}}{E_m^{(0)} - E_k^{(0)}}. \tag{38.12}$$

In the first sum $k \neq n$, and in the second $k \neq m$.

Problem

1. Find the second-order correction to the eigenfunctions.

Solution. The coefficients $c_k^{(2)}$ for $k \neq n$ are obtained from (38.4) with $k \neq n$, written through second order; $c_n^{(2)}$ is chosen so that $\psi_n = \psi_n^{(0)} + \psi_n^{(1)} + \psi_n^{(2)}$ is normalized through second order. The result is

$$\begin{aligned}\psi_n^{(2)} = {} & \sum_m{}' \sum_k{}' \frac{V_{mk} V_{kn}}{\hbar^2 \omega_{nk} \omega_{nm}} \psi_m^{(0)} - \sum_m{}' \frac{V_{nn} V_{mn}}{\hbar^2 \omega_{nm}^2} \psi_m^{(0)} \\ & - \frac{\psi_n^{(0)}}{2} \sum_m{}' \frac{|V_{mn}|^2}{\hbar^2 \omega_{nm}^2},\end{aligned}$$

where

$$\omega_{nm} = \frac{1}{\hbar}(E_n^{(0)} - E_m^{(0)}).$$

2. Find the third-order correction to the energy eigenvalues.

[1] The wave functions $\psi_\nu^{(0)}$ must then be normalized to delta functions of the quantities ν.

SOLUTION. Retaining the third-order terms in (38.4) with $k = n$ gives

$$E_n^{(3)} = {\sum_k}' {\sum_m}' \frac{V_{nm}V_{mk}V_{kn}}{\hbar^2 \omega_{mn}\omega_{kn}} - V_{nn} {\sum_m}' \frac{|V_{nm}|^2}{\hbar^2 \omega_{mn}^2}.$$

3. Find the energy levels of the anharmonic linear oscillator with Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{m\omega^2 x^2}{2} + \alpha x^3 + \beta x^4.$$

SOLUTION. Matrix elements of x^3 and x^4 follow directly from the rule for matrix multiplication and the elements of x in (23.4). The nonzero elements of x^3 are

$$(x^3)_{n-3,n} = (x^3)_{n,n-3} = \left(\frac{\hbar}{m\omega}\right)^{3/2} \sqrt{\frac{n(n-1)(n-2)}{8}},$$

$$(x^3)_{n-1,n} = (x^3)_{n,n-1} = \left(\frac{\hbar}{m\omega}\right)^{3/2} \sqrt{\frac{9n^3}{8}}.$$

There are no diagonal elements in this matrix, so the term αx^3 gives no first-order correction when regarded as a perturbation of the harmonic oscillator. Its second-order correction has the same order as the first-order correction from βx^4. The diagonal elements of x^4 are

$$(x^4)_{n,n} = \left(\frac{\hbar}{m\omega}\right)^2 \frac{3}{4}(2n^2 + 2n + 1).$$

Equations (38.6) and (38.10) then give

$$\begin{aligned} E_n = \hbar\omega\left(n + \frac{1}{2}\right) &- \frac{15}{4}\frac{\alpha^2}{\hbar\omega}\left(\frac{\hbar}{m\omega}\right)^3 \left(n^2 + n + \frac{11}{30}\right) \\ &+ \frac{3}{2}\beta\left(\frac{\hbar}{m\omega}\right)^2 \left(n^2 + n + \frac{1}{2}\right). \end{aligned}$$

4. An infinite spherical potential well undergoes a small volume-preserving deformation into a slightly prolate or oblate ellipsoid of revolution with semiaxes $a = b$ and c. Find the splitting of its particle energy levels (*A. B. Migdal*, 1959).

SOLUTION. The boundary equation

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{c^2} = 1$$

becomes the sphere $x^2 + y^2 + z^2 = R^2$ under $x \to ax/R$, $y \to ay/R$, $z \to cz/R$. The particle Hamiltonian $\hat{H} = \hat{\mathbf{p}}^2/2M = -\hbar^2\Delta/2M$ (M is the mass; energy is measured from the bottom of the well) becomes $\hat{H} = \hat{H}_0 + \hat{V}$, with

$$\hat{H}_0 = -\frac{\hbar^2}{2M}\Delta, \qquad \hat{V} = -\frac{\hbar^2}{2M}\left[\left(\frac{R^2}{a^2} - 1\right)\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + \left(\frac{R^2}{c^2} - 1\right)\frac{\partial^2}{\partial z^2}\right].$$

The ellipsoidal-well problem is thereby reduced to the spherical-well problem. If the ellipsoid differs only slightly from a sphere of radius $R = (a^2c)^{1/3}$, $\hat{V}$ is a small perturbation. Introducing the "ellipticity" β, $|\beta| \ll 1$, by

$$a \approx R\left(1 - \frac{\beta}{3}\right), \qquad c \approx R\left(1 + \frac{2\beta}{3}\right),$$

we obtain

$$\hat{V} = \frac{\beta}{3M}(\hat{\mathbf{p}}^2 - 3\hat{p}_z^2).$$

In first order, the change from the spherical-well levels is

$$\Delta E_{nlm} = E_{nlm} - E_{nl}^{(0)} = \langle nlm|V|nlm\rangle.$$

Here l, m are the angular momentum and its projection on the ellipsoid axis, and n numbers the spherical-well levels for fixed l, which do not depend on m. The expression $\mathbf{p}^2 - 3p_z^2$ is the zz component of the traceless irreducible tensor $\delta_{ik}\mathbf{p}^2 - 3p_ip_k$. By (107.2) and (107.6), $\langle nlm|V|nlm\rangle$ is proportional to $(-1)^m \begin{pmatrix} l & 2 & l \\ -m & 0 & m \end{pmatrix}$, and hence

$$\langle nlm|V|nlm\rangle = \left(1 - \frac{3m^2}{l(l+1)}\right)\langle nl0|V|nl0\rangle.$$

(The table of $3j$ symbols is on p. 530.) We next write

$$\begin{aligned}\langle nl0|V|nl0\rangle &= \frac{2}{3}\beta E_{nl}^{(0)} + \beta\frac{\hbar^2}{M}\left\langle nl0\left|\frac{\partial^2}{\partial z^2}\right|nl0\right\rangle \\ &= \frac{2}{3}\beta E_{nl}^{(0)} - \frac{\beta\hbar^2}{M}\int\left|\frac{\partial\psi_{nl0}}{\partial z}\right|^2 r^2\,dr\,do.\end{aligned}$$

The Schrödinger equation $\hat{H}_0\psi_{nlm} = E_{nl}^{(0)}\psi_{nlm}$ was used in the first term, and the second was integrated by parts. Using (28.11) for Y_{l0}, the derivative of $\psi_{nl0} = R_{nl}(r)Y_{l0}(\theta, \varphi)$ is

$$\begin{aligned}\frac{\partial}{\partial z}\psi_{nl0} &= \left(\cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial\theta}\right)\psi_{nl0} \\ &= -\frac{i(l+1)}{[4(l+1)^2 - 1]^{1/2}}\left(R'_{nl} - \frac{l}{r}R_{nl}\right)Y_{l+1,0} \\ &\quad + \frac{il}{[4l^2 - 1]^{1/2}}\left(R'_{nl} + \frac{l+1}{r}R_{nl}\right)Y_{l-1,0}.\end{aligned}$$

The radial integrals are evaluated from

$$\int_0^\infty R_{nl}R'_{nl}r\,dr = -\frac{1}{2}\int_0^\infty R_{nl}^2\,dr,$$

$$\int_0^\infty R'^2_{nl}r^2\,dr = \frac{2M}{\hbar^2}E_{nl}^{(0)} - l(l+1)\int_0^\infty R_{nl}^2\,dr,$$

by integration by parts and the radial Schrödinger equation (33.3),

$$R''_{nl} + \frac{2}{r}R'_{nl} - \frac{l(l+1)}{r^2}R_{nl} = -\frac{2M}{\hbar^2}E^{(0)}_{nl}.$$

Terms containing integrals of R^2_{nl} cancel, leaving

$$\Delta E_{nlm} = 4\beta\frac{l(l+1)}{(2l-1)(2l+3)}\left[\frac{m^2}{l(l+1)} - \frac{1}{3}\right]E^{(0)}_{nl}.$$

Finally,

$$\frac{1}{2l+1}\sum_{m=-l}^{l} E_{nlm} = E^{(0)}_{nl},$$

so the centre of gravity of the multiplet is unchanged.

§ 39. The secular equation

We now turn to the case in which the unperturbed operator $\hat{H}_0$ has degenerate eigenvalues. Let $\psi^{(0)}_n$, $\psi^{(0)}_{n'}$, ... denote eigenfunctions belonging to the same energy eigenvalue $E^{(0)}_n$. As we know, the choice of these functions is not unique: they may be replaced by any s independent linear combinations of the same functions, where s is the degeneracy of the level $E^{(0)}_n$. This freedom disappears, however, when the wave functions are required to change only slightly under the applied small perturbation. For the moment, let $\psi^{(0)}_n$, $\psi^{(0)}_{n'}$, ... denote arbitrarily chosen unperturbed eigenfunctions. The correct zeroth-order functions are linear combinations of the form

$$c^{(0)}_n\psi^{(0)}_n + c^{(0)}_{n'}\psi^{(0)}_{n'} + \dots$$

The coefficients in these combinations and the first-order corrections to the eigenvalues are determined as follows.

Write equations (38.4) for $k = n, n', \dots$, substituting in first order $E = E^{(0)}_n + E^{(1)}$. For the c_k it is sufficient to retain their zeroth-order values $c_n = c^{(0)}_n$, $c_{n'} = c^{(0)}_{n'}$, ..., while $c_m = 0$ for $m \neq n, n', \dots$. We then obtain

$$E^{(1)}c^{(0)}_n = \sum_{n'} V_{nn'}c^{(0)}_{n'},$$

or

$$\sum_{n'}(V_{nn'} - E^{(1)}\delta_{nn'})c^{(0)}_{n'} = 0, \qquad (39.1)$$

where n and n' run over all values labeling states that belong to the given unperturbed eigenvalue $E^{(0)}_n$. This homogeneous system of linear equations for the $c^{(0)}_n$ has nonzero solutions if the determinant of its coefficients vanishes. Thus

$$|V_{nn'} - E^{(1)}\delta_{nn'}| = 0. \qquad (39.2)$$

This equation has degree s in $E^{(1)}$ and, in general, has s distinct real roots. These roots are the required first-order corrections to the eigenvalues. Equation (39.2) is called the *secular equation*[2]. The sum of its roots is the sum of the diagonal matrix elements V_{nn}, $V_{n'n'}, \ldots$; it is the coefficient of $E^{(1)s-1}$ in the equation.

Substituting the roots of (39.2) one by one into (39.1) and solving the resulting systems gives the coefficients $c_n^{(0)}$ and hence the correct zeroth-order eigenfunctions. As a result of the perturbation, the initially degenerate energy level is in general no longer degenerate, since the roots of (39.2) are in general different. The perturbation is said to lift the degeneracy. The lifting may be complete or partial; in the latter case the perturbed level remains degenerate, but with a lower multiplicity than before.

It may happen that, for some reason, all matrix elements for transitions within one group of mutually degenerate states $n, n', \ldots$ are especially small, or even vanish. It may then be useful, while retaining $V_{nn'}$ in first order, also to include in higher orders the matrix elements V_{nm}, with $m \neq n, n', \ldots$, for transitions to states of other energies. We shall do this by including V_{mn} through second order.

In (38.4) with $k = n$, set $E = E_n^{(0)} + E^{(1)}$ on the left-hand side; we retain the notation $E^{(1)}$ for the energy correction in the approximation now considered. Replace c_n by $c_n^{(0)}$. Since $c_m^{(0)} = 0$ for every $m \neq n, n', \ldots$, we have

$$E^{(1)} c_n^{(0)} = \sum_m V_{nm} c_m^{(1)} + \sum_{n'} V_{nn'} c_{n'}^{(0)}. \tag{39.3}$$

Equations (38.4) with $k = m \neq n, n', \ldots$, through first order, give

$$(E_n^{(0)} - E_m^{(0)}) c_m^{(1)} = \sum_{n'} V_{mn'} c_{n'}^{(0)},$$

and hence

$$c_m^{(1)} = \sum_{n'} \frac{V_{mn'}}{E_n^{(0)} - E_m^{(0)}} c_{n'}^{(0)}.$$

Substitution in (39.3) gives

$$E^{(1)} c_n^{(0)} = \sum_{n'} c_{n'}^{(0)} \left(V_{nn'} + \sum_m \frac{V_{nm} V_{mn'}}{E_n^{(0)} - E_m^{(0)}} \right).$$

This system now replaces (39.1). Its compatibility condition again gives a secular equation, differing from (39.2) by the replacement

$$V_{nn'} \longrightarrow V_{nn'} + \sum_m \frac{V_{nm} V_{mn'}}{E_n^{(0)} - E_m^{(0)}}. \tag{39.4}$$

[2]Also called the age equation (the French word *siècle* means an age); the name was borrowed from celestial mechanics.

Problem

1. Find the first-order corrections to the eigenvalue and the correct zeroth-order functions for a doubly degenerate level.

Solution. Equation (39.2) now has the form

$$\begin{vmatrix} V_{11} - E^{(1)} & V_{21} \\ V_{12} & V_{22} - E^{(1)} \end{vmatrix} = 0$$

(the indices 1 and 2 refer to two arbitrarily chosen unperturbed eigenfunctions $\psi_1^{(0)}$ and $\psi_2^{(0)}$ of the degenerate level). Solving it, we find

$$E^{(1)} = \frac{1}{2}[V_{11} + V_{22} \pm \hbar\omega^{(1)}], \tag{1}$$

where

$$\hbar\omega^{(1)} = \sqrt{(V_{11} - V_{22})^2 + 4|V_{12}|^2}$$

denotes the difference between the two values of the correction $E^{(1)}$. Solving (39.1) with these values of $E^{(1)}$, we obtain the coefficients in the normalized correct zeroth-order functions $\psi^{(0)} = c_1^{(0)}\psi_1^{(0)} + c_2^{(0)}\psi_2^{(0)}$:

$$\begin{aligned} c_1^{(0)} &= \left\{\frac{V_{12}}{2|V_{12}|}\left[1 \pm \frac{V_{11} - V_{22}}{\hbar\omega^{(1)}}\right]\right\}^{1/2}, \\ c_2^{(0)} &= \pm\left\{\frac{V_{21}}{2|V_{12}|}\left[1 \mp \frac{V_{11} - V_{22}}{\hbar\omega^{(1)}}\right]\right\}^{1/2}. \end{aligned} \tag{2}$$

Problem

2. Derive formulas for the first-order corrections to the eigenfunctions and the second-order corrections to the eigenvalues.

Solution. Let the functions $\psi_n^{(0)}$ be chosen as the correct zeroth-order functions. The matrix $V_{nn'}$ defined with these functions is evidently diagonal in the indices n, n' belonging to the same group of functions of a degenerate level. Its diagonal elements $V_{nn}, V_{n'n'}, \ldots$ equal the corresponding first-order corrections $E_n^{(1)}, E_{n'}^{(1)}, \ldots$.

Consider the perturbation of the eigenfunction $\psi_n^{(0)}$. In zeroth order, $E = E_n^{(0)}$, $c_n^{(0)} = 1$, and $c_m^{(0)} = 0$ for $m \neq n$. In first order, $E = E_n^{(0)} + V_{nn}$, $c_n = 1 + c_n^{(1)}$, and $c_m = c_m^{(1)}$. From the general system (38.4), take the equation with $k \neq n, n', \ldots$ and retain its first-order terms:

$$(E_n^{(0)} - E_k^{(0)})c_k^{(1)} = V_{kn}c_n^{(0)} = V_{kn},$$

whence

$$c_k^{(1)} = \frac{V_{kn}}{E_n^{(0)} - E_k^{(0)}} \quad \text{for} \quad k \neq n, n', \ldots \tag{1}$$

Next write the equation with $k = n'$, retaining its second-order terms:

$$E_n^{(1)} c_{n'}^{(1)} = V_{n'n'} c_{n'}^{(1)} + \sum_m{}' V_{n'm} c_m^{(1)}.$$

In the sum over m, the terms with $m = n, n', \ldots$ are omitted. Substituting $E_n^{(1)} = V_{nn}$ and (1) for $c_m^{(1)}$, we obtain, for $n' \neq n$,

$$c_{n'}^{(1)} = \frac{1}{V_{nn} - V_{n'n'}} \sum_m{}' \frac{V_{n'm} V_{mn}}{E_n^{(0)} - E_m^{(0)}}. \tag{2}$$

The coefficient $c_n^{(1)}$ is zero in this approximation. Formulas (1) and (2) determine the first-order correction $\psi_n^{(1)} = \sum_m c_m^{(1)} \psi_m^{(0)}$ to the eigenfunctions[3].

Finally, retaining the second-order terms in (38.4) with $k = n$, we obtain the second-order energy correction

$$E_n^{(2)} = \sum_m{}' \frac{V_{nm} V_{mn}}{E_n^{(0)} - E_m^{(0)}}, \tag{3}$$

which has the same formal appearance as (38.10).

Problem

3. At the initial time $t = 0$, a system is in a state $\psi_1^{(0)}$ belonging to a doubly degenerate level. Find the probability that at a later time t the system is in the other state $\psi_2^{(0)}$ of the same energy, when the transition is caused by a constant perturbation.

Solution. Construct the correct zeroth-order functions:

$$\psi = c_1 \psi_1 + c_2 \psi_2, \qquad \psi' = c_1' \psi_1 + c_2' \psi_2,$$

where c_1, c_2 and c_1', c_2' are the two pairs of coefficients given by (2) of Problem 1. Superscripts (0) have been omitted from all quantities for brevity. Conversely,

$$\psi_1 = \frac{c_2' \psi - c_2 \psi'}{c_1 c_2' - c_1' c_2}.$$

The functions ψ and ψ' belong to states with perturbed energies $E + E^{(1)}$ and $E + E^{(1)\prime}$, where $E^{(1)}$ and $E^{(1)\prime}$ are the two values of correction (1) in Problem 1. Introducing the time factors gives the time-dependent wave function

$$\Psi_1 = \frac{\exp[-(i/\hbar)Et]}{c_1 c_2' - c_1' c_2} \left[c_2' \psi \exp\left(-\frac{i}{\hbar} E^{(1)} t \right) - c_2 \psi' \exp\left(-\frac{i}{\hbar} E^{(1)\prime} t \right) \right].$$

[3]Notice that the requirement that the quantities (1) and (2) be small, and hence the condition for applicability of this perturbation method, still requires (38.9) only for transitions between states belonging to different energy levels. Transitions between states of the same degenerate level are accounted for by the secular equation in a certain sense exactly.

At $t = 0$, $\Psi_1 = \psi_1$. Expressing ψ and ψ' once more in terms of ψ_1 and ψ_2, we write Ψ_1 as a linear combination of ψ_1 and ψ_2 with time-dependent coefficients. The squared modulus of the coefficient of ψ_2 is the required transition probability w_{21}. Using (1) and (2) of Problem 1 gives

$$w_{21} = 2\frac{|V_{12}|^2}{(\hbar\omega^{(1)})^2}\left[1 - \cos\left(\omega^{(1)}t\right)\right].$$

Thus the probability oscillates periodically with frequency $\omega^{(1)}$. For times t small compared with the corresponding period, the expression in square brackets, and hence w_{21} itself, is proportional to t^2:

$$w_{21} = \frac{1}{\hbar^2}|V_{12}|^2 t^2.$$

This formula follows very simply by the method developed in the next section, using (40.4).

§ 40. Time-dependent perturbations

We now turn to perturbations with an explicit dependence on time. In this case there can be no question at all of corrections to the energy eigenvalues: when the Hamiltonian depends on time (as does the perturbed operator $\hat{H} = \hat{H}_0 + \hat{V}(t)$), energy is not conserved, and stationary states therefore do not exist. The problem is instead to calculate approximate wave functions from the wave functions of the unperturbed system's stationary states.

For this purpose we use the analogue of the familiar variation-of-constants method for solving linear differential equations (*P. A. M. Dirac*, 1926). Let $\Psi_k^{(0)}$ be the wave functions (including the time factor) of the unperturbed system's stationary states. An arbitrary solution of the unperturbed wave equation can then be written as $\Psi = \sum_k a_k \Psi_k^{(0)}$. We seek a solution of the perturbed equation

$$i\hbar\frac{\partial\Psi}{\partial t} = (\hat{H}_0 + \hat{V})\Psi \tag{40.1}$$

in the form

$$\Psi = \sum_k a_k(t)\Psi_k^{(0)}, \tag{40.2}$$

where the expansion coefficients are functions of time. Substitution of (40.2) into (40.1), with

$$i\hbar\frac{\partial\Psi_k^{(0)}}{\partial t} = \hat{H}_0\Psi_k^{(0)},$$

gives

$$i\hbar\sum_k \Psi_k^{(0)}\frac{da_k}{dt} = \sum_k a_k\hat{V}\Psi_k^{(0)}.$$

Multiplying both sides on the left by $\Psi_m^{(0)*}$ and integrating, we find

$$i\hbar\frac{da_m}{dt} = \sum_k V_{mk}(t)a_k, \tag{40.3}$$

where

$$V_{mk}(t) = \int \Psi_m^{(0)*} \hat{V} \Psi_k^{(0)}\, dq = V_{mk} e^{i\omega_{mk} t}, \qquad \omega_{mk} = \frac{E_m^{(0)} - E_k^{(0)}}{\hbar}$$

are the perturbation matrix elements including their time factor. If V itself depends explicitly on time, the quantities V_{mk} must of course also be regarded as functions of time.

Choose the wave function of the nth stationary state as the unperturbed wave function. The corresponding coefficients in (40.2) are $a_n^{(0)} = 1$ and $a_k^{(0)} = 0$ for $k \neq n$. To obtain the first approximation, write $a_k = a_k^{(0)} + a_k^{(1)}$ and put $a_k = a_k^{(0)}$ on the right-hand side of (40.3), which already contains the small quantities V_{mk}. Thus

$$i\hbar \frac{da_k^{(1)}}{dt} = V_{kn}(t). \tag{40.4}$$

To indicate the unperturbed function for which the correction is being calculated, introduce a second index on a_k and write

$$\Psi_n = \sum_k a_{kn}(t) \Psi_k^{(0)}.$$

The integrated form of (40.4) is accordingly

$$a_{kn}^{(1)} = -\frac{i}{\hbar} \int V_{kn}(t)\, dt = -\frac{i}{\hbar} \int V_{kn} e^{i\omega_{kn} t}\, dt. \tag{40.5}$$

This determines the wave functions in the first approximation.

Consider more closely the important case of a time-periodic perturbation

$$\hat{V} = \hat{F} e^{-i\omega t} + \hat{G} e^{i\omega t}, \tag{40.6}$$

where $\hat{F}$ and $\hat{G}$ do not depend on time. Hermiticity of V requires

$$\hat{F} e^{-i\omega t} + \hat{G} e^{i\omega t} = \hat{F}^{+} e^{i\omega t} + \hat{G}^{+} e^{-i\omega t},$$

whence $\hat{G} = \hat{F}^{+}$, or

$$G_{nm} = F_{mn}^{*}. \tag{40.7}$$

Using this relation, we have

$$V_{kn}(t) = V_{kn} e^{i\omega_{kn} t} = F_{kn} e^{i(\omega_{kn} - \omega)t} + F_{nk}^{*} e^{i(\omega_{kn} + \omega)t}. \tag{40.8}$$

Substitution in (40.5) and integration yield

$$a_{kn}^{(1)} = -\frac{F_{kn} e^{i(\omega_{kn} - \omega)t}}{\hbar(\omega_{kn} - \omega)} - \frac{F_{nk}^{*} e^{i(\omega_{kn} + \omega)t}}{\hbar(\omega_{kn} + \omega)}. \tag{40.9}$$

These expressions apply provided neither denominator vanishes[4]; that is, for every k (at fixed n),

$$E_k^{(0)} - E_n^{(0)} \neq \pm\hbar\omega. \tag{40.10}$$

[4]More precisely, neither may be so small that $a_{kn}^{(1)}$ ceases to be small in comparison with unity.

For a number of applications it is useful to know the matrix elements of an arbitrary quantity f, defined using the perturbed wave functions. To first order,

$$f_{nm}(t) = f_{nm}^{(0)}(t) + f_{nm}^{(1)}(t),$$

where

$$f_{nm}^{(0)}(t) = \int \Psi_n^{(0)*} \hat{f} \Psi_m^{(0)}\, dq = f_{nm}^{(0)} e^{i\omega_{nm}t},$$

$$f_{nm}^{(1)}(t) = \int \left(\Psi_n^{(0)*} \hat{f} \Psi_m^{(1)} + \Psi_n^{(1)*} \hat{f} \Psi_m^{(0)} \right) dq.$$

Inserting

$$\Psi_n^{(1)} = \sum_k a_{kn}^{(1)} \Psi_k^{(0)}$$

with the coefficients from (40.9), we obtain

$$\begin{aligned} f_{nm}^{(1)}(t) = -e^{i\omega_{nm}t} \sum_k \Bigg\{ &\left[\frac{f_{nk}^{(0)} F_{km}}{\hbar(\omega_{km} - \omega)} + \frac{f_{km}^{(0)} F_{nk}}{\hbar(\omega_{kn} + \omega)} \right] e^{-i\omega t} \\ &+ \left[\frac{f_{nk}^{(0)} F_{mk}^*}{\hbar(\omega_{km} + \omega)} + \frac{f_{km}^{(0)} F_{kn}^*}{\hbar(\omega_{kn} - \omega)} \right] e^{i\omega t} \Bigg\}. \end{aligned} \tag{40.11}$$

This formula applies if none of its terms becomes large, hence if all the frequencies ω_{kn} and ω_{km} remain sufficiently far from ω. At $\omega = 0$ it reduces to (38.12). The formulas above assume that the unperturbed energy levels form only a discrete spectrum. They extend immediately to the presence of a continuous spectrum as well (the states being perturbed still belonging to the discrete spectrum): one simply adds the corresponding integrals over the continuous spectrum to the sums over discrete levels. The denominators $\omega_{kn} \pm \omega$ in (40.9) and (40.11) must then remain nonzero as $E_k^{(0)}$ ranges through both the discrete and continuous spectra. If, as is usual, the continuous spectrum lies above all discrete levels, condition (40.10), for example, must be supplemented by

$$E_{\min}^{(0)} - E_n^{(0)} > \hbar\omega, \tag{40.12}$$

where $E_{\min}^{(0)}$ is the energy at the lower edge of the continuous spectrum.

Problem

1. Determine how the nth and mth solutions of the Schrödinger equation change under a periodic perturbation of the form (40.6), whose frequency ω satisfies $E_m^{(0)} - E_n^{(0)} = \hbar(\omega + \varepsilon)$, where ε is small.

Solution. The method developed above does not apply, since the coefficient $a_{mn}^{(1)}$ in (40.9) becomes large. Return to the exact equations (40.3), with $V_{mk}(t)$ given by (40.8). Evidently, the principal effect comes from terms on their right-hand sides whose time dependence is governed by the small frequency $\omega_{mn} - \omega$. Dropping all other terms gives

$$i\hbar \frac{da_m}{dt} = F_{mn} e^{i\varepsilon t} a_n, \qquad i\hbar \frac{da_n}{dt} = F_{mn}^* e^{-i\varepsilon t} a_m.$$

Put $a_n e^{i\varepsilon t} = b_n$. Then

$$i\hbar\dot{a}_m = F_{mn} b_n, \qquad i\hbar(\dot{b}_n - i\varepsilon b_n) = F^*_{mn} a_m,$$

and elimination of a_m gives

$$\ddot{b}_n - i\varepsilon\dot{b}_n + (1/\hbar^2)|F_{mn}|^2 b_n = 0.$$

Two independent solutions may be chosen as

$$a_n = Ae^{i\alpha_1 t}, \qquad a_m = -A\frac{\hbar\alpha_1}{F_{mn}} e^{i\alpha_2 t} \tag{1}$$

and

$$a_n = Be^{-i\alpha_2 t}, \qquad a_m = B\frac{\hbar\alpha_2}{F^*_{mn}} e^{-i\alpha_1 t}, \tag{2}$$

where A and B are constants (to be fixed by normalization), and

$$\alpha_1 = -\varepsilon/2 + \Omega, \quad \alpha_2 = \varepsilon/2 + \Omega, \quad \Omega = \sqrt{\varepsilon^2/4 + |\eta|^2}, \quad \eta = F_{mn}/\hbar.$$

Thus the perturbation transforms $\Psi_n^{(0)}$ and $\Psi_m^{(0)}$ into $a_n\Psi_n^{(0)} + a_m\Psi_m^{(0)}$, with a_n, a_m taken from (1) or (2).

Suppose that at $t = 0$ the system was in the state $\Psi_m^{(0)}$. At later times its state is the linear combination of the two solutions just found which reduces to $\Psi_m^{(0)}$ at $t = 0$:

$$\Psi = e^{i\varepsilon t/2}\left(\cos\Omega t - \frac{i\varepsilon}{2\Omega}\sin\Omega t\right)\Psi_m^{(0)} - \frac{i\eta^*}{\Omega} e^{-i\varepsilon t/2}\sin\Omega t\cdot\Psi_n^{(0)}. \tag{3}$$

The squared modulus of the coefficient of $\Psi_n^{(0)}$ is

$$\frac{|\eta|^2}{2\Omega^2}\left[1 - \cos(2\Omega t)\right]. \tag{4}$$

It is the probability that the system is in $\Psi_n^{(0)}$ at time t. This probability is periodic, with frequency 2Ω, and varies from zero to $|\eta|^2/\Omega^2$.

At exact resonance, $\varepsilon = 0$, (4) becomes

$$(1/2)\left[1 - \cos(2|\eta|t)\right].$$

It varies periodically between zero and unity: the system passes periodically from $\Psi_m^{(0)}$ to $\Psi_n^{(0)}$.

§41. Transitions produced by a perturbation acting for a finite time

Suppose that the perturbation $V(t)$ acts only during some finite time interval (or, alternatively, that $V(t)$ decreases sufficiently rapidly as $t \to \pm\infty$). Before the perturbation begins to act (or in the limit $t \to -\infty$), let the system be in the nth stationary state of the discrete spectrum. At any later instant its state is described by

$$\Psi = \sum_k a_{kn}\Psi_k^{(0)},$$

where, to first order,

$$\begin{aligned} a_{kn} &= a_{kn}^{(1)} = -\frac{i}{\hbar}\int_{-\infty}^{t} V_{kn} e^{i\omega_{kn}t}\,dt, \qquad k \neq n, \\ a_{nn} &= 1 + a_{nn}^{(1)} = 1 - \frac{i}{\hbar}\int_{-\infty}^{t} V_{nn}\,dt. \end{aligned} \tag{41.1}$$

The integration limits in (40.5) have been selected so that every $a_{kn}^{(1)}$ vanishes as $t \to -\infty$. Once the action of the perturbation has ended (or as $t \to \infty$), the coefficients a_{kn} acquire constant values $a_{kn}(\infty)$, and the system is in the state whose wave function is

$$\Psi = \sum_k a_{kn}(\infty)\Psi_k^{(0)}.$$

This function again satisfies the unperturbed wave equation, though it differs from the original function $\Psi_n^{(0)}$. By the general rules, the squared modulus of $a_{kn}(\infty)$ gives the probability that the system has energy $E_k^{(0)}$, that is, that it is in the kth stationary state.

Thus a perturbation can carry the system from its original stationary state into any other one. The probability of a transition from the initial (ith) stationary state to the final (fth) stationary state is[5]

$$w_{fi} = \frac{1}{\hbar^2}\left|\int_{-\infty}^{+\infty} V_{fi} e^{i\omega_{fi}t}\,dt\right|^2. \tag{41.2}$$

Now consider a perturbation which, once it has arisen, continues to act indefinitely (while remaining small throughout). In other words, $V(t)$ tends to zero as $t \to -\infty$ and to a finite nonzero limit as $t \to \infty$. Formula (41.2) cannot be used directly here, since its integral diverges. From the physical standpoint, however, this divergence is immaterial and is easily removed. Integration by parts gives

$$\begin{aligned} a_{fi} &= -\frac{i}{\hbar}\int_{-\infty}^{t} V_{fi} e^{i\omega_{fi}t}\,dt \\ &= -\left.\frac{V_{fi}e^{i\omega_{fi}t}}{\hbar\omega_{fi}}\right|_{-\infty}^{t} + \int_{-\infty}^{t} \frac{\partial V_{fi}}{\partial t}\frac{e^{i\omega_{fi}t}}{\hbar\omega_{fi}}\,dt. \end{aligned}$$

The first term vanishes at the lower limit; at the upper limit it formally coincides with the expansion coefficients in (38.8). (The extra periodic factor occurs simply because a_{fi} are expansion coefficients of the full wave function Ψ, whereas the c_{fi} of § 38 are expansion coefficients of the time-independent function ψ.) It is consequently clear that the limit of this term as $t \to \infty$ describes only the change of the original wave function $\Psi_i^{(0)}$ caused by the "constant part" $V(+\infty)$ of the perturbation, and therefore has nothing to do with transitions into other states. The transition probability is instead given by the square of the second term:

$$w_{fi} = \frac{1}{\hbar^2\omega_{fi}^2}\left|\int_{-\infty}^{+\infty} \frac{\partial V_{fi}}{\partial t} e^{i\omega_{fi}t}\,dt\right|^2. \tag{41.3}$$

[5] For uniformity, whenever transition probabilities are discussed below, we shall denote the initial and final states by the indices i and f, respectively. We shall also write the indices of transition probabilities specifically in the order fi, consistently with the convention for matrix-element indices.

The formulas obtained also apply when a transition proceeds from a discrete state to a continuous-spectrum state. The only difference is that one then speaks of the probability of transition from the specified (ith) state into states for which the quantities ν_f (see the end of § 38) lie between ν_f and $\nu_f + d\nu_f$. Thus, for example, (41.2) must be written as

$$dw_{if} = \frac{1}{\hbar^2}\left|\int_{-\infty}^{\infty} V_{fi}e^{i\omega_{fi}t}\,dt\right|^2 d\nu_f. \tag{41.4}$$

If $V(t)$ varies little over time intervals of order $1/\omega_{fi}$, the integral in (41.2) or (41.3) is very small. In the limit of arbitrarily slow variation of the applied perturbation, the probability of every transition that changes the energy (that is, with nonzero ω_{fi}) tends to zero. Hence, when an applied perturbation changes sufficiently slowly (*adiabatically*), a system initially in a particular nondegenerate stationary state continues to remain in that same state (see also § 53).

In the opposite limit, when a perturbation is switched on very rapidly—that is, *suddenly*—the derivatives $\partial V_{fi}/\partial t$ become infinite at the "switching instant." In the integral of $(\partial V_{fi}/\partial t)e^{i\omega_{fi}t}$, the comparatively slow factor $e^{i\omega_{fi}t}$ may then be taken outside the integral at its value at that instant. The remaining integration is immediate and gives

$$w_{fi} = \frac{|V_{fi}|^2}{\hbar^2\omega_{fi}^2}. \tag{41.5}$$

Transition probabilities for sudden perturbations can also be found when the perturbation is not small. Let the system initially be in a state described by an eigenfunction $\psi_i^{(0)}$ of the original Hamiltonian $\hat{H}_0$. If the Hamiltonian changes suddenly (in a time short compared with the periods $1/\omega_{fi}$ of transitions from the given state i to other states), the wave function has no time to change and remains what it was before the perturbation. It is no longer, however, an eigenfunction of the system's new Hamiltonian $\hat{H}$; the state $\psi_i^{(0)}$ is therefore not stationary. By the general rules of quantum mechanics, the probabilities w_{fi} for transition into any of the new stationary states are determined by expanding $\psi_i^{(0)}$ in the eigenfunctions ψ_f of $\hat{H}$:

$$w_{fi} = \left|\int \psi_i^{(0)}\psi_f^*\,dq\right|^2. \tag{41.6}$$

We now show how this general formula reduces to (41.5) when the change $\hat{V} = \hat{H} - \hat{H}_0$ of the Hamiltonian is small. Multiply the equations

$$\hat{H}_0\psi_i^{(0)} = E_i^{(0)}\psi_i^{(0)}, \qquad \hat{H}^*\psi_f^* = E_f\psi_f^*$$

by ψ_f^* and $\psi_i^{(0)}$, respectively, integrate over dq, and subtract the two results term by term. Using also the self-adjointness of $\hat{H}$, we obtain

$$(E_f - E_i^{(0)})\int \psi_f^*\psi_i^{(0)}\,dq = \int \psi_f^*\hat{V}\psi_i^{(0)}\,dq.$$

If $\hat{V}$ is small, then to first order E_f may be replaced by the nearby unperturbed level $E_f^{(0)}$, and ψ_f on the right-hand side by the corresponding function $\psi_f^{(0)}$. It follows that

$$\int \psi_f^* \psi_i^{(0)}\, dq = \frac{1}{\hbar\omega_{fi}} \int \psi_f^{(0)*} \hat{V} \psi_i^{(0)}\, dq,$$

so that (41.6) becomes (41.5).

Problem

1. A charged oscillator in its ground state is suddenly subjected to a uniform electric field. Determine the probabilities for the oscillator to pass into excited states under the action of this perturbation.

Solution. Potential energy of an oscillator in a uniform field exerting force F upon it is

$$U(x) = \frac{m\omega^2}{2} x^2 - Fx = \frac{m\omega^2}{2}(x - x_0)^2 + \text{const},$$

where $x_0 = F/m\omega^2$. Thus it again has a purely oscillator form, but with its equilibrium position displaced. Consequently, the stationary-state wave functions of the perturbed oscillator are $\psi_k(x - x_0)$, where $\psi_k(x)$ are the oscillator functions (23.12), while the initial wave function is $\psi_0(x)$ from (23.13). Using these functions and expression (23.11) for the Hermite polynomials, we find

$$\int_{-\infty}^{+\infty} \psi_0^{(0)} \psi_k\, dx = \frac{(-1)^k}{\sqrt{2^k \pi k!}} e^{-\xi_0^2/2} \int_{-\infty}^{\infty} e^{-\xi\xi_0} \frac{d^k}{d\xi^k} e^{-\xi^2 + 2\xi\xi_0}\, d\xi,$$

where $\xi_0 = x_0\sqrt{m\omega/\hbar}$. Integrating by parts k times reduces the integral here to

$$\xi_0^k \int_{-\infty}^{\infty} \exp\left(-\xi^2 + \xi\xi_0\right) d\xi = \xi_0^k \sqrt{\pi} \exp\frac{\xi_0^2}{4}.$$

Formula (41.6) therefore gives the required transition probability as

$$w_{k0} = \frac{\bar{k}^k}{k!} e^{-\bar{k}}, \qquad \bar{k} = \frac{\xi_0^2}{2} = \frac{F^2}{2m\hbar\omega^3}.$$

As a function of the integer k, this is a Poisson distribution with mean $\bar{k}$.

The perturbative regime corresponds to small F such that $\bar{k} \ll 1$. The excitation probabilities are then small and fall rapidly as k increases; the largest is $w_{10} \approx \bar{k}$.

Conversely, when F is large ($\bar{k} \gg 1$), excitation occurs with overwhelming probability: the chance that the oscillator remains in its normal state is $w_{00} = e^{-\bar{k}}$.

Problem

2. The nucleus of an atom in its normal state receives a sudden impulse and thereby acquires velocity **v**. Assume that the duration τ of the impulse is short both in comparison with the electronic periods and with a/v, where a is an atomic dimension. Determine the probability that this "shaking" excites the atom (*A. B. Migdal*, 1939).

Solution. Pass to the reference frame K' that moves with the nucleus after the impact. Because $\tau \ll a/v$, the nucleus may be treated as having hardly moved during the impact, and the electron coordinates in K' and in the original frame K therefore coincide immediately after the perturbation. The initial wave function in K' is

$$\psi_0' = \psi_0 \exp\left(-i\mathbf{q}\sum_a \mathbf{r}_a\right), \qquad \mathbf{q} = \frac{m\mathbf{v}}{\hbar},$$

where ψ_0 is the normal-state wave function for a stationary nucleus and the sum in the exponential extends over all Z electrons of the atom. According to (41.6), the probability sought for transition into the kth excited state is now

$$w_{k0} = \left|\left\langle k\left|\exp\left(-i\mathbf{q}\sum_a \mathbf{r}_a\right)\right|0\right\rangle\right|^2.$$

In particular, for $qa \ll 1$, expand the exponential under the integral. The integral of $\psi_k^*\psi_0$ vanishes by orthogonality, and hence

$$w_{k0} = \left|\left\langle k\left|\mathbf{q}\sum_a \mathbf{r}_a\right|0\right\rangle\right|^2.$$

Problem

3. Find the total probability of excitation and ionization of a hydrogen atom subjected to a sudden "shaking" (see the preceding problem).

Solution. The required probability can be evaluated as the difference

$$1 - w_{00} = 1 - \left|\int \psi_0^2 e^{-i\mathbf{q}\mathbf{r}}\, dV\right|^2,$$

where w_{00} is the probability that the atom remains in the ground state; $\psi_0 = (\pi a^3)^{-1/2} e^{-r/a}$ is the hydrogen ground-state wave function, and a is the Bohr radius. Evaluation of the integral gives

$$1 - w_{00} = 1 - \frac{1}{(1 + \frac{1}{4}q^2a^2)^4}.$$

In the limiting case $qa \ll 1$, this probability tends to zero as $1 - w_{00} \approx q^2a^2$; for $qa \gg 1$, it approaches unity as $1 - w_{00} \approx 1 - (2/qa)^8$.

Problem

4. Determine the probability that an electron is ejected from the K shell of an atom with large atomic number Z during nuclear β decay. Assume the β-particle velocity to be large relative to the velocity of a K electron (*A. B. Migdal*, 1941; *E. L. Feinberg*, 1939).

Solution. [6] Under the stated conditions, the time taken by the β particle to cross the K shell is short compared with the electron's orbital period; the change in nuclear charge may therefore be regarded as instantaneous. The perturbation is the change $V = 1/r$ in the nuclear field when its charge changes by an amount small compared with Z (namely, by 1). According to (41.5), the probability for either of the two K-shell electrons, whose energy is $E_0 = -Z^2/2$,[7] to enter a continuous-spectrum state of energy $E = k^2/2$ in the interval $dE = k\,dk$ is

$$dw = 2\frac{4|V_{0k}|^2}{(k^2+Z^2)^2}\,dk.$$

The integral defining V_{0k} receives its essential contribution from distances of order $1/Z$ near the nucleus. In this region a hydrogen-like expression may also be used for the continuous-spectrum wave function. The electron's final state must have $l = 0$, the same angular momentum as the initial state. From the function R_{10} and the function R_{k0} normalized on the $k/2\pi$ scale, obtained in § 36, together with formula (f.3) of the Mathematical Appendices, we find[8]

$$\left(\frac{1}{r}\right)_{0k} = \frac{4\sqrt{2\pi k}}{1-e^{-2\pi Z/k}}\,\frac{(1+ik/Z)^{iZ/k}(1-ik/Z)^{-iZ/k}}{1+k^2/Z^2},$$

and, since

$$|(1+i\alpha)^{i/\alpha}|^2 = \exp\left(-2\frac{\arctan\alpha}{\alpha}\right),$$

we finally obtain

$$dw = \frac{2^7}{Z^4(1+k^2/Z^2)^4}\,f\left(\frac{k}{Z}\right)k\,dk,$$

where

$$f(\alpha) = \frac{1}{1-e^{-2\pi/\alpha}}\exp\left(-4\frac{\arctan\alpha}{\alpha}\right).$$

The limiting values of $f(\alpha)$ are

$$f = e^{-4}\quad\text{for } \alpha \ll 1, \qquad f = \alpha/2\pi \quad\text{for } \alpha \gg 1.$$

The total probability of ionizing the K shell follows by integrating dw over all energies of the emitted electron. Numerical evaluation yields $w = 0.65Z^{-2}$.

[6] Atomic units are used in Problems 4 and 5.

[7] Here and below the K-electron states are treated as hydrogen-like (see § 74).

[8] It is convenient during the calculation to use Coulomb units, returning to atomic units in the final result.

Problem

5. Determine the probability that an electron is ejected from the K shell of a large-Z atom during nuclear α decay. The α particle moves slowly compared with a K electron, although its time of emergence from the nucleus is short compared with the electron's orbital period (*A. B. Migdal*, 1941; *J. Levinger*, 1953).

Solution. After the α particle has emerged, the perturbation acting upon the electron is adiabatic. The effect sought is therefore determined chiefly by times close to the nonadiabatic "switching instant," when the α particle, already outside the nucleus and moving freely, is still at distances small compared with the radius of the K orbit. The ionizing perturbation V is then the departure of the combined nuclear and α-particle field from the pure Coulomb field Z/r. For particles of atomic weights 4 and $A-4$, charges 2 and $Z-2$, and separation vt (where v is the relative speed of the nucleus and α particle), the dipole moment is

$$\frac{2(A-4)-(Z-2)4}{A}vt = \frac{2(A-2Z)}{A}\,vt.$$

Therefore, the dipole term of the field of the nucleus and the α-particle is given by[9]

$$V = \frac{2(A-2Z)}{A}\,vt\frac{z}{r^3},$$

where the z axis points along $\mathbf{v}$. The matrix element of this perturbation can be reduced to that of z. Taking the matrix element of the electron's equation of motion $\ddot{z} = -Zz/r^3$, we obtain

$$\left(\frac{z}{r^3}\right)_{0k} = \frac{(E-E_0)^2}{Z}z_{0k}.$$

By (41.2), the transition probability sought for either of the two K-shell electrons is

$$dw = 2\left|\int_0^\infty V_{0k}e^{i(E_0-E)t}\,dt\right|^2 dk = \frac{8(A-2Z)^2v^2}{A^2Z^2}|z_{0k}|^2\frac{dk}{2\pi}.$$

To evaluate the integral, an additional damping factor $e^{-\lambda t}$, $\lambda > 0$, is inserted in the integrand, and $\lambda \to 0$ is then taken in the result. For the matrix element of $z = r\cos\theta$, note that the initial orbital angular momentum is $l = 0$, so $\cos\theta$ has a nonzero matrix element only for transition to a state with $l = 1$. In that case

$$|(\cos\theta)_{01}|^2 = \frac{1}{3}, \qquad |z_{0k}|^2 = \frac{1}{3}|r_{0k}|^2.$$

Evaluation of r_{0k} using the radial functions R_{00} and R_{k1} then gives

$$dw = \frac{2^{11}(A-2Z)^2v^2}{3A^2Z^6(1+k^2/Z^2)^5}f\left(\frac{k}{Z}\right)k\,dk,$$

where f is the function defined in Problem 4.

[9] If the difference $A-2Z$ is small, it may also be necessary to include the next, quadrupole term.

§ 42. Transitions under a periodic perturbation

A different type of result is obtained for the probability of transition to continuous-spectrum states under a periodic perturbation. Suppose that at an initial time $t = 0$ the system is in the ith stationary state of the discrete spectrum. Let the frequency ω of the periodic perturbation satisfy

$$\hbar\omega > E_{\min} - E_i^{(0)}, \tag{42.1}$$

where $E_{\min}$ is the energy at which the continuous spectrum begins.

It follows at once from the results of § 40 that the principal role is played by continuous-spectrum states whose energies E_f lie close to the "resonant" energy $E_i^{(0)} + \hbar\omega$, that is, states for which $\omega_{fi} - \omega$ is small. For the same reason, only the first term in the perturbation matrix element (40.8), whose frequency $\omega_{fi} - \omega$ is close to zero, need be retained. Substituting this term in (40.5) and integrating gives

$$a_{fi} = -\frac{i}{\hbar}\int_0^t V_{fi}(t)\,dt = -F_{fi}\frac{\exp\left[i(\omega_{fi}-\omega)t\right]-1}{\hbar(\omega_{fi}-\omega)}. \tag{42.2}$$

The lower integration limit has been chosen so that $a_{fi} = 0$ at $t = 0$, in accordance with the initial condition.

The squared modulus of a_{fi} is

$$|a_{fi}|^2 = |F_{fi}|^2\frac{4\sin^2\dfrac{\omega_{fi}-\omega}{2}t}{\hbar^2(\omega_{fi}-\omega)^2}. \tag{42.3}$$

It is readily seen that, at large t, the function appearing here may be represented as proportional to t.

To see this, use the formula

$$\lim_{t\to\infty}\frac{\sin^2\alpha t}{\pi t\alpha^2} = \delta(\alpha). \tag{42.4}$$

For $\alpha \neq 0$ the displayed limit is zero, while for $\alpha = 0$,

$$\frac{\sin^2\alpha t}{t\alpha^2} = t,$$

so the limit is infinite. On integrating over $d\alpha$ from $-\infty$ to $+\infty$ and making the substitution $\alpha t = \xi$, we obtain

$$\frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{\sin^2\alpha t}{t\alpha^2}\,d\alpha = \frac{1}{\pi}\int_{-\infty}^{+\infty}\frac{\sin^2\xi}{\xi^2}\,d\xi = 1.$$

Thus the function on the left-hand side of (42.4) satisfies all the requirements defining the delta function.

Accordingly, for large t we may write

$$|a_{fi}|^2 = \frac{1}{\hbar^2}|F_{fi}|^2\pi t\,\delta\left(\frac{\omega_{fi}-\omega}{2}\right),$$

or, substituting $\hbar\omega_{fi} = E_f - E_i^{(0)}$ and using $\delta(ax) = \delta(x)/a$,

$$|a_{fi}|^2 = \frac{2\pi}{\hbar}|F_{fi}|^2\delta(E_f - E_i^{(0)} - \hbar\omega)t.$$

The quantity $|a_{fi}|^2 d\nu_f$ is the probability of transition from the initial state to states in a specified interval $d\nu_f$. At large t it is proportional to the interval of time elapsed since $t = 0$. The transition probability dw_{fi} per unit time is therefore[10]

$$dw_{fi} = \frac{2\pi}{\hbar}|F_{fi}|^2\delta(E_f - E_i^{(0)} - \hbar\omega)\, d\nu_f. \tag{42.5}$$

As expected, it is nonzero only for transitions to states with energy $E_f = E_i^{(0)} + \hbar\omega$. If the continuous-spectrum energy levels are nondegenerate, so that ν_f may be taken to mean the energy alone, the whole "interval" $d\nu_f$ reduces to a single state of energy $E = E_i^{(0)} + \hbar\omega$, and the probability of transition to this state is

$$w_{Ei} = \frac{2\pi}{\hbar}|F_{Ei}|^2. \tag{42.6}$$

It is also instructive to derive (42.5) in another way. Instead of switching the periodic perturbation on at the definite instant $t = 0$, let it grow slowly from $t = -\infty$ according to the exponential law $e^{\lambda t}$, where λ is positive and is ultimately allowed to tend to zero (*adiabatic switching*). The initial condition $a_{fi} = 0$ is then imposed at $t = -\infty$. The perturbation matrix element is now

$$V_{fi} = F_{fi}e^{i(\omega_{fi}-\omega)t+\lambda t},$$

and in place of (42.2) we write

$$a_{fi} = -\frac{i}{\hbar}\int_{-\infty}^{t} V_{fi}(t)\, dt = -F_{fi}\frac{\exp\left[i(\omega_{fi}-\omega)t + \lambda t\right]}{\hbar(\omega_{fi} - \omega - i\lambda)}. \tag{42.7}$$

Hence

$$|a_{fi}|^2 = \frac{1}{\hbar^2}|F_{fi}|^2\frac{e^{2\lambda t}}{(\omega_{fi}-\omega)^2 + \lambda^2}.$$

The transition probability per unit time is determined by

$$\frac{d}{dt}|a_{fi}|^2 = 2\lambda|a_{fi}|^2.$$

Now use the formula

$$\lim_{\lambda\to 0}\frac{\lambda}{\pi(\alpha^2 + \lambda^2)} = \delta(\alpha), \tag{42.8}$$

which is valid in the same sense as (42.4). Taking the limit $\lambda \to 0$ gives

$$\frac{d}{dt}|a_{fi}|^2 \longrightarrow \frac{2\pi}{\hbar^2}|F_{fi}|^2\delta(\omega_{fi} - \omega),$$

and we return once more to (42.5).

[10] It is readily verified that retaining the omitted second term in (40.8) would give additional expressions which, after division by t, tend to zero as $t \to +\infty$.

§ 43. Transitions within the continuous spectrum

One of the most important uses of perturbation theory is the calculation of transition probabilities within the continuous spectrum under a constant (time-independent) perturbation. As already noted, continuous-spectrum states are almost always degenerate. After choosing a definite set of unperturbed wave functions for a given energy level, the question can be posed as follows. The system is known to have occupied one of these states initially; we seek the probability that it passes into another state of the same energy. For transitions from an initial state i into states between ν_f and $\nu_f + d\nu_f$, (42.5), with $\omega = 0$ and a change of notation, gives directly

$$dw_{fi} = \frac{2\pi}{\hbar}|V_{fi}|^2\delta(E_f - E_i)\,d\nu_f. \tag{43.1}$$

As expected, this is nonzero only when $E_f = E_i$: a constant perturbation causes transitions solely between states of equal energy. For transitions from continuous-spectrum states, however, dw_{fi} cannot itself be treated as a transition probability. It does not even have the corresponding dimension, but has dimension 1/s. Expression (43.1) gives the number of transitions per unit time, and its dimension depends on the chosen normalization of the continuous-spectrum wave functions.[11]

We shall calculate the perturbed wave function which, before the perturbation begins to act, coincides with the original unperturbed function $\psi_i^{(0)}$. Following the procedure stated at the end of the preceding section, take the perturbation to be switched on adiabatically as $e^{\lambda t}$, with $\lambda \to 0$. Formula (42.7), after setting $\omega = 0$ and changing notation, gives

$$a_{fi}^{(1)} = V_{fi}\frac{\exp\left\{\dfrac{i}{\hbar}(E_f - E_i)t + \lambda t\right\}}{E_i - E_f + i\lambda}. \tag{43.2}$$

The perturbed wave function is

$$\Psi_i = \Psi_i^{(0)} + \int a_{fi}^{(1)}\Psi_f^{(0)}\,d\nu_f,$$

where the integration extends over the entire continuous spectrum.[12] Substitution of (43.2) yields

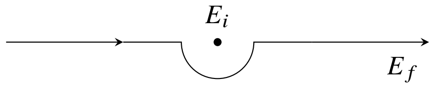
$$\Psi_i = \left[\psi_i^{(0)} + \int V_{fi}\psi_f^{(0)}\frac{d\nu_f}{E_i - E_f + i0}\right]\exp\left(-\frac{i}{\hbar}E_i t\right). \tag{43.3}$$

In the limit $\lambda \to 0$, the factor $e^{\lambda t}$ has been replaced by unity. The term $+i0$, meaning the limit of $i\lambda$ as the positive quantity λ tends to zero, specifies the integration prescription for E_f. Its differential occurs as a factor in $d\nu_f$, together with those of the other quantities characterizing continuous-spectrum states. Without $i\lambda$, the integrand in (43.3) would

[11] Phenomena covered by this theory include, for example, various collisions. In their initial and final states the system is an assemblage of free particles, while their mutual interaction serves as the perturbation. With a suitable wave- function normalization, (43.1) can then become a collision cross-section (see § 126).

[12] If a discrete spectrum is also present, the corresponding sum over its states must be added to the integral here and in the formulas below.

have a pole at $E_f = E_i$, and the integral would diverge near it. The term $i\lambda$ displaces this pole into the upper half-plane of the complex variable E_f. When $\lambda \to 0$, the pole returns to the real axis, but the integration path is now known to pass below it:



(43.4)

The time factor in (43.3) shows, as it should, that this function belongs to the same energy E_i as the original unperturbed function. Equivalently,

$$\psi_i = \psi_i^{(0)} + \int \frac{V_{fi}}{E_i - E_f + i0} \psi_f^{(0)} \, d\nu_f \tag{43.5}$$

satisfies the Schrödinger equation

$$(\hat{H}_0 + \hat{V})\psi_i = E_i \psi_i.$$

It is consequently natural that this expression agrees exactly with (38.8).[13]

The preceding calculation is the first approximation of perturbation theory. The second approximation is also readily obtained. One derives the next approximation for Ψ_i by the method of § 38, now knowing how the "divergent" integrals are to be taken. A direct calculation gives

$$\Psi_i = \left\{ \psi_i^{(0)} + \int \left[V_{fi} + \int \frac{V_{f\nu} V_{\nu i}}{E_i - E_\nu + i0} \, d\nu \right] \frac{\psi_f^{(0)} \, d\nu_f}{E_i - E_f + i0} \right\} \exp\left(-\frac{i}{\hbar} E_i t \right).$$

Comparison with (43.3) lets us write the corresponding probability formula (more precisely, the number of transitions) by direct analogy with (43.1):

$$dw_{fi} = \frac{2\pi}{\hbar} \left| V_{fi} + \int \frac{V_{f\nu} V_{\nu i}}{E_i - E_\nu + i0} \, d\nu \right|^2 \delta(E_i - E_f) \, d\nu_f. \tag{43.6}$$

The matrix element V_{fi} may vanish for the transition in question. The first-order effect is then absent altogether, and (43.6) becomes

$$dw_{fi} = \frac{2\pi}{\hbar} \left| \int \frac{V_{f\nu} V_{\nu i}}{E_i - E_\nu} \, d\nu \right|^2 \delta(E_f - E_i) \, d\nu_f. \tag{43.7}$$

In applications of this formula, $E_\nu = E_i$ is usually not a pole of the integrand. The prescription for integration over dE_ν is then immaterial, and the path may be taken directly along the real axis.

States ν for which $V_{f\nu}$ and $V_{\nu i}$ are nonzero are often called *intermediate* states for the transition $i \to f$. In a pictorial description, the transition appears to take place in two stages, $i \to \nu$ and $\nu \to f$; this description must not, of course, be understood literally. A

[13] The integration prescription in (43.5) can also be fixed by requiring the large-distance asymptotic form of ψ_i to contain only an outgoing wave, and no incoming wave (see § 136).

transition $i \to f$ may require not one but several successive intermediate states. Formula (43.7) extends directly to such cases. If two intermediate states are needed, then

$$dw_{fi} = \frac{2\pi}{\hbar} \left| \int \frac{V_{f\nu'} V_{\nu'\nu} V_{\nu i}}{(E_i - E_{\nu'})(E_i - E_\nu)}\, d\nu\, d\nu' \right|^2 \delta(E_f - E_i)\, d\nu_f. \tag{43.8}$$

Finally, the mathematical meaning of integrals taken along a path such as (43.4) is expressed by

$$\int \frac{f(x)\, dx}{x - a - i0} = \mathrm{P} \int \frac{f(x)\, dx}{x - a} + i\pi f(a), \tag{43.9}$$

where the interval of integration on the real axis contains $x = a$. Indeed, detouring around the pole $x = a$ on a semicircle of radius ρ, the full integral is the sum of the real-axis integrals from the lower limit to $a - \rho$ and from $a + \rho$ to the upper limit, plus $i\pi$ times the residue at the pole. As $\rho \to 0$, the real-axis pieces combine into the integral over the full interval understood as a principal value, as indicated by the marked integral sign, and (43.9) follows. The same formula is also written symbolically as

$$\frac{1}{x - a - i0} = P \frac{1}{x - a} + i\pi\delta(x - a); \tag{43.10}$$

the symbol P means that the principal value is to be taken when $f(x)/(x - a)$ is integrated.

§ 44. The uncertainty relation for energy

Consider a system made up of two weakly interacting parts. Suppose that at a certain instant these parts are known to have definite energies, denoted by E and ε. After an interval Δt, the energies are measured again, giving values E' and ε' which generally differ from E and ε. We can readily determine the order of magnitude of the most probable value of $E' + \varepsilon' - E - \varepsilon$ found by the measurement.

According to (42.3), with $\omega = 0$, the probability that a time-independent perturbation transfers the system during time t from energy E to energy E' is proportional to

$$\frac{\sin^2\left(\dfrac{E' - E}{2\hbar} t\right)}{(E' - E)^2}.$$

It follows that the most probable difference $E' - E$ is of order $\hbar/t$.

Applying this result to the present case, with the interaction between the two parts as the perturbation, gives

$$|E + \varepsilon - E' - \varepsilon'| \Delta t \sim \hbar. \tag{44.1}$$

Thus the shorter the interval Δt, the larger the observed energy change. Its order of magnitude, $\hbar/\Delta t$, is importantly independent of the strength of the perturbation.

The energy change specified by (44.1) will be observed however weak the interaction between the two parts may be. This is a purely quantum result with a profound physical meaning. It shows that in quantum mechanics two measurements can test energy conservation only to an accuracy of order $\hbar/\Delta t$, where Δt is the time between them.

Relation (44.1) is often called the uncertainty relation for energy. It must, however, be emphasized that its meaning is essentially different from that of the coordinate–momentum relation $\Delta p \Delta x \sim \hbar$. In the latter, Δp and Δx are the uncertainties in momentum and coordinate at one and the same instant; the relation says that these two quantities cannot have strictly definite values simultaneously.

The energies E and ε, by contrast, can be measured at each given instant with arbitrary precision. The quantity $(E + \varepsilon) - (E' + \varepsilon')$ in (44.1) is the difference between two exactly measured energy values at two different times, not an uncertainty in the energy at a particular instant.

If E is regarded as the energy of a system and ε as the energy of a "measuring apparatus," their interaction energy can be taken into account only to an accuracy $\hbar/\Delta t$. Denote the errors in the measurements of the corresponding quantities by $\Delta E, \Delta\varepsilon, \ldots$. In the favorable case where ε and ε' are known exactly ($\Delta\varepsilon = \Delta\varepsilon' = 0$),

$$\Delta(E - E') \sim \frac{\hbar}{\Delta t}. \tag{44.2}$$

This relation has important consequences for momentum measurement. Measuring the momentum of a particle (take an electron for definiteness) involves a collision with another, "measuring," particle whose momenta before and after the collision may be regarded as known exactly.[14] Applying momentum conservation to this collision gives three equations, the components of one vector equation, for six unknowns: the components of the electron momentum before and after the collision. More equations might be sought by arranging a sequence of collisions with measuring particles and applying momentum conservation to each. But the number of unknowns also grows, since it includes the electron momenta between collisions; for any number of collisions there are plainly three more unknowns than equations. Momentum measurement must therefore use energy conservation at each collision in addition to momentum conservation. As we have seen, the energy law can be applied only to an accuracy of order $\hbar/\Delta t$, with Δt the interval between the beginning and end of the process under consideration.

For simplicity, consider an idealized thought experiment in which the "measuring particle" is a perfectly reflecting plane mirror. Only the momentum component normal to the mirror then matters. Momentum and energy conservation give, for determining the particle momentum P,

$$p' + P' - p - P = 0, \tag{44.3}$$

$$|\varepsilon' + E' - \varepsilon - E| \sim \frac{\hbar}{\Delta t} \tag{44.4}$$

(P, E are the particle's momentum and energy; p, ε are those of the mirror; unprimed and primed quantities refer respectively to times before and after the collision). The quantities $p, p', \varepsilon, \varepsilon'$ of the "measuring particle" may be treated as known exactly, so their errors vanish. The equations above then give

$$\Delta P = \Delta P', \qquad |\Delta E' - \Delta E| \sim \frac{\hbar}{\Delta t}.$$

[14] For the present argument, how the energy of the "measuring" particle becomes known is immaterial.

But

$$\Delta E = \frac{\partial E}{\partial P}\Delta P = v\Delta P,$$

and similarly

$$\Delta E' = v'\Delta P' = v'\Delta P.$$

Hence

$$|(v'_x - v_x)\Delta P_x| \sim \frac{\hbar}{\Delta t}. \tag{44.5}$$

The suffix x has been attached to the velocities and momentum to stress that the relation applies separately to each component.

This is the required relation. It shows that measuring an electron's momentum to a specified accuracy ΔP is unavoidably accompanied by a change in its velocity, and therefore in the momentum itself.

The shorter the measurement process, the greater this change. The velocity change can be made arbitrarily small only as $\Delta t \to \infty$, but a momentum measurement extending over a long time can in general have meaning only for a free particle. Here the nonrepeatability of momentum measurements at short intervals, and the two-sided nature of measurement in quantum mechanics, are especially clear: one must distinguish the measured value of a quantity from the value produced by the measuring process.[15]

The argument at the beginning of this section, based on perturbation theory, can also be approached through the decay of a system under some perturbation. Let E_0 be an energy level calculated while entirely neglecting the possibility of decay. Denote by τ the *lifetime* of this state, the reciprocal of its decay probability per unit time. The same method gives

$$|E_0 - E - \varepsilon| \sim \hbar/\tau, \tag{44.6}$$

where E and ε are the energies of the two parts into which the system has decayed. Their sum $E + \varepsilon$ provides a measure of the system's energy before decay. The relation therefore shows that the energy of a decaying system in a *quasistationary* state can be defined only to an accuracy of order $\hbar/\tau$. This quantity is customarily called the level *width* Γ. Thus

$$\Gamma \sim \hbar/\tau. \tag{44.7}$$

§ 45. Potential energy treated as a perturbation

The case in which the particle's entire potential energy in an external field can be treated as the perturbation calls for separate consideration. The unperturbed Schrödinger equation is then the equation for free motion,

$$\Delta\psi^{(0)} + k^2\psi^{(0)} = 0, \qquad k = \frac{\sqrt{2mE}}{\hbar} = \frac{p}{\hbar} \tag{45.1}$$

[15] Relation (44.5), as well as the clarification of the physical meaning of the energy uncertainty relation, is due to N. Bohr (1928).

whose solutions are plane waves. Since the energy spectrum of free motion is continuous, this is a special instance of perturbation theory in a continuous spectrum. Here it is more convenient to obtain the solution directly, without using the general formulas.

The equation for the first-order correction $\psi^{(1)}$ to the wave function is

$$\Delta\psi^{(1)} + k^2\psi^{(1)} = \frac{2mU}{\hbar^2}\psi^{(0)} \tag{45.2}$$

(U is the potential energy). As is known from electrodynamics, its solution may be written in the form of "retarded potentials," namely[16]

$$\psi^{(1)}(x, y, z) = -\frac{m}{2\pi\hbar^2}\int \psi^{(0)}U(x', y', z')e^{ikr}\frac{dV'}{r}, \tag{45.3}$$

$$dV' = dx'\,dy'\,dz', \qquad r^2 = (x - x')^2 + (y - y')^2 + (z - z')^2.$$

Let us determine the conditions under which the field U may be regarded as a perturbation. Perturbation theory is applicable when $\psi^{(1)} \ll \psi^{(0)}$. Let a give the order of magnitude of the size of the region in which the field differs appreciably from zero. Suppose first that the particle's energy is so low that ak is at most of order unity. For an order-of-magnitude estimate the factor e^{ikr} in the integrand of (45.3) is then immaterial, and the whole integral is of order $\psi^{(0)}|U|a^2$. Thus $\psi^{(1)} \sim (ma^2|U|/\hbar^2)\psi^{(0)}$, and the condition becomes

$$|U| \ll \frac{\hbar^2}{ma^2} \quad \text{when} \quad ka \lesssim 1. \tag{45.4}$$

Notice that $\hbar^2/ma^2$ has a simple physical meaning: it gives the order of magnitude of the kinetic energy that a particle confined to a volume of linear dimensions a would have (for its momentum would be $\sim \hbar/a$ by the uncertainty relation).

As a particular example, consider a potential well so shallow that (45.4) is satisfied. Such a well has no negative energy levels (*R. Peierls*, 1929), as was already seen in the problem to § 33 for the special case of a spherically symmetric well. Indeed, at $E = 0$ the unperturbed wave function reduces to a constant, which may conventionally be set equal to unity: $\psi^{(0)} = 1$. Since $\psi^{(1)} \ll \psi^{(0)}$, the wave function for motion in the well, $\psi = 1 + \psi^{(1)}$, plainly vanishes nowhere. An eigenfunction without nodes belongs to the normal state, so $E = 0$ remains the lowest possible energy of the particle.

Thus, if a well is not sufficiently deep, only infinite motion of the particle is possible: the well cannot "capture" it. This result is specifically quantum-mechanical. In classical mechanics a particle can undergo finite motion in any potential well.

It must be stressed that everything just said applies only to a three-dimensional well. A one- or two-dimensional well (that is, one whose field depends on only one or two coordinates) always has negative energy levels (see the problems to this section). The reason is that in one and two dimensions the perturbation theory under consideration is wholly inapplicable at zero (or very small) energy E. [17]

[16]This is a particular integral of equation (45.2). Any solution of the equation with no right-hand side (that is, of the unperturbed equation (45.1)) may still be added to it.

[17]In two dimensions, $\psi^{(1)}$ is expressed (as follows from the theory of the two-dimensional wave equation) by

At high energies, where $ka \gg 1$, the factor e^{ikr} in the integrand is essential and greatly reduces the magnitude of the integral. In this case solution (45.3) can be put into another form. For its derivation, however, it is simpler to return directly to equation (45.2). Choose the direction of unperturbed motion as the x axis. The unperturbed wave function is then $\psi^{(0)} = e^{ikx}$ (the constant factor is conventionally set to unity). Seek a solution of

$$\Delta\psi^{(1)} + k^2\psi^{(1)} = \frac{2m}{\hbar^2}Ue^{ikx}$$

in the form $\psi^{(1)} = e^{ikx}f$. Since k is assumed large, in $\Delta\psi^{(1)}$ it suffices to retain only terms in which the factor e^{ikx} is differentiated at least once. The equation for f is then

$$2ik\frac{\partial f}{\partial x} = \frac{2mU}{\hbar^2},$$

and hence

$$\psi^{(1)} = e^{ikx}f = -\frac{im}{\hbar^2 k}e^{ikx}\int U\,dx. \tag{45.5}$$

The integral is of order $|\psi^{(1)}| \sim m|U|a/\hbar^2 k$, so in this case perturbation theory is applicable under the condition

$$|U| \ll \frac{\hbar^2}{ma^2}ka = \frac{\hbar v}{a}, \qquad ka \gg 1 \tag{45.6}$$

($v = k\hbar/m$ is the particle's velocity). This condition is weaker than (45.4). Consequently, if the field may be treated as a perturbation at low particle energies, it certainly may be so treated at high energies; the converse, generally speaking, does not hold. [18]

The applicability of the perturbation theory developed here to the Coulomb field requires separate consideration. In the field $U = \alpha/r$ there is no finite region of space outside which U is substantially smaller than inside it. The required condition follows by replacing the parameter a in (45.6) with the variable distance r; this gives

$$\frac{\alpha}{\hbar v} \ll 1. \tag{45.7}$$

Thus the Coulomb field may be treated as a perturbation at high particle energies. [19]

Finally, let us derive a formula giving an approximate wave function for a particle whose energy E is everywhere much greater than the potential energy U (no other conditions are required). To the first approximation, the coordinate dependence of the

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an integral analogous to (45.3), with $i\pi H_0^{(1)}(kr)dx'\,dy'$ in place of $e^{ikr}dx'\,dy'\,dz'/r$ ($H_0^{(1)}$ is a Hankel function), and $r = \sqrt{(x'-x)^2 + (y'-y)^2}$. As $k \to 0$, the Hankel function, and with it the entire integral, tends logarithmically to infinity.

Similarly, in one dimension the integrand defining $\psi^{(1)}$ contains $2\pi i e^{ikr}dx'/k$ (where $r = |x'-x|$), and as $k \to 0$, $\psi^{(1)}$ tends to infinity as $1/k$.

[18] In one dimension the condition for applicability of perturbation theory is the inequality (45.6) for all ka. The derivation of (45.4) given above for three dimensions cannot be carried out in one dimension because of the divergence, noted in the footnote on p. 204, of the function $\psi^{(1)}$ constructed in that manner.

[19] It should be borne in mind that the integral (45.5) with $U = \alpha/r$ diverges logarithmically at large $x/\sqrt{y^2+z^2}$. The Coulomb-field wave function obtained by perturbation theory is therefore inapplicable inside a narrow cone about the x axis.

wave function is the same as in free motion, whose direction we choose as the x axis. Accordingly, write $\psi = e^{ikx}F$, where F is a function of the coordinates that varies slowly in comparison with the factor e^{ikx} (although in general it cannot be said to be close to unity). Substitution in the Schrödinger equation gives

$$2ik\frac{\partial F}{\partial x} = \frac{2m}{\hbar^2}UF, \tag{45.8}$$

whence

$$\psi = e^{ikx}F = \text{const}\cdot e^{ikx}\exp\left(-\frac{i}{\hbar v}\int U\,dx\right). \tag{45.9}$$

This is the required expression. It must be remembered, however, that it is not valid at excessively large distances. In (45.8) the term ΔF, which contains second derivatives of F, was omitted. At large distances $\partial^2 F/\partial x^2$ tends to zero together with $\partial F/\partial x$. The derivatives with respect to the transverse coordinates y, z, on the other hand, do not tend to zero, and they may be neglected only if $x \ll ka^2$.

Problem

1. Determine the energy level in a shallow one-dimensional potential well; condition (45.4) is assumed to hold.

Solution. Make the assumption, confirmed by the result, that $|E| \ll |U|$. Then E may be neglected on the right-hand side of the Schrödinger equation within the well,

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}(U(x) - E)\psi,$$

and ψ may also be treated as a constant, which can be set equal to unity without loss of generality:

$$\frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}U.$$

Integrate this equality with respect to x between two points $\pm x_1$ such that $a \ll x_1 \ll 1/\kappa$, where a is the width of the well and $\kappa = \sqrt{2m|E|}/\hbar$. Since the integral of $U(x)$ converges, the integration on the right may be extended over the whole interval from $-\infty$ to $+\infty$:

$$\left.\frac{d\psi}{dx}\right|_{-x_1}^{x_1} = \frac{2m}{\hbar^2}\int_{-\infty}^{+\infty} U\,dx. \tag{1}$$

Far from the well the wave function has the form $\psi = e^{\mp\kappa x}$. Substitution in (1) gives

$$-2\kappa = \frac{2m}{\hbar^2}\int_{-\infty}^{+\infty} U\,dx \qquad \text{or} \qquad |E| = \frac{m}{2\hbar^2}\left(\int_{-\infty}^{+\infty} U\,dx\right)^2.$$

In accord with the assumption made, the magnitude of the level is thus a small quantity of higher (second) order than the depth of the well.

Problem

2. Determine the energy level in a shallow two-dimensional potential well $U(r)$ (r is the polar coordinate in the plane); the integral $\int_0^\infty rU\,dr$ is assumed to converge.

Solution. Proceeding as in the preceding problem, within the well we obtain

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{d\psi}{dr}\right) = \frac{2m}{\hbar^2}U.$$

Integrating with respect to r from 0 to r_1 (where $a \ll r_1 \ll 1/\kappa$), we find

$$\left.\frac{d\psi}{dr}\right|_{r=r_1} = \frac{2m}{\hbar^2 r_1}\int_0^\infty rU(r)\,dr. \qquad (1)$$

Far from the well, the two-dimensional free-motion equation

$$\frac{1}{r}\frac{d}{dr}\left(r\frac{d\psi}{dr}\right) + \frac{2m}{\hbar^2}E\psi = 0$$

has the solution (vanishing at infinity) $\psi = \text{const}\cdot H_0^{(1)}(i\kappa r)$. At small values of the argument, the leading term of this function is proportional to $\ln \kappa r$. Bearing this in mind, at $r \sim a$ equate the logarithmic derivatives of ψ evaluated inside the well (the right-hand side of (1)) and outside it. This gives

$$-\frac{1}{a\ln\kappa a} \approx \frac{2m}{\hbar^2 a}\int_0^\infty U(r)r\,dr,$$

and hence

$$|E| \sim \frac{\hbar^2}{ma^2}\exp\left[-\frac{\hbar^2}{m}\left|\int_0^\infty Ur\,dr\right|^{-1}\right].$$

Thus the energy level is exponentially small in comparison with the depth of the well.

CHAPTER VII

The Semiclassical Case

§ 46. The Wave Function in the Semiclassical Case

When the de Broglie wavelengths of the particles are small compared with the characteristic dimensions L that determine the conditions of the particular problem, the system behaves nearly classically. (This is analogous to the passage from wave optics to geometrical optics as the wavelength tends to zero.)

We shall now examine the properties of semiclassical systems in greater detail. In the Schrödinger equation

$$\sum_a \frac{\hbar^2}{2m_a}\Delta_a\psi + (E-U)\psi = 0$$

make the formal substitution

$$\psi = \exp\left(\frac{i}{\hbar}\sigma\right). \tag{46.1}$$

The resulting equation for σ is

$$\sum_a \frac{1}{2m_a}(\nabla_a\sigma)^2 - \sum_a \frac{i\hbar}{2m_a}\Delta_a\sigma = E - U. \tag{46.2}$$

Since the system is assumed to have almost classical properties, we seek σ as a series

$$\sigma = \sigma_0 + \frac{\hbar}{i}\sigma_1 + \left(\frac{\hbar}{i}\right)^2\sigma_2 + \cdots, \tag{46.3}$$

in powers of $\hbar$.

We begin with the simplest situation: one-dimensional motion of a single particle. Equation (46.2) then reduces to

$$\frac{1}{2m}\sigma'^2 - \frac{i\hbar}{2m}\sigma'' = E - U(x) \tag{46.4}$$

(a prime denotes differentiation with respect to x).

In the first approximation, put $\sigma = \sigma_0$ and omit the term containing $\hbar$ from the equation:

$$\frac{1}{2m}\sigma_0'^2 = E - U(x).$$

It follows that

$$\sigma_0 = \pm\int\sqrt{2m[E-U(x)]}\,dx.$$

The integrand is precisely the classical momentum $p(x)$ of the particle, expressed as a function of position. Defining $p(x)$ by taking the positive sign of the square root, we have

$$\sigma_0 = \pm \int p\,dx, \qquad p = \sqrt{2m(E-U)}, \tag{46.5}$$

as is to be expected from the limiting expression (6.1) for the wave function[1].

The neglect made in (46.4) is justified only when the second term on the left is small compared with the first. Thus $\hbar|\sigma''/\sigma'^2| \ll 1$, or

$$\left|\frac{d}{dx}\left(\frac{\hbar}{\sigma'}\right)\right| \ll 1.$$

To the first approximation, (46.5) gives $\sigma' = p$, and hence this condition may be written

$$\left|\frac{d\lambda\!\!\!^{-}}{dx}\right| \ll 1, \tag{46.6}$$

where $\lambda\!\!\!^{-} = \lambda/2\pi$, while $\lambda(x) = 2\pi\hbar/p(x)$ is the particle's de Broglie wavelength expressed as a function of x through the classical function $p(x)$. We have therefore obtained a quantitative criterion for the semiclassical approximation: over a distance of the order of the wavelength itself, the wavelength must vary only slightly. The approximation ceases to apply in regions where this condition is not fulfilled.

Condition (46.6) can also be put into another form. Since

$$\frac{dp}{dx} = \frac{d}{dx}\sqrt{2m(E-U)} = -\frac{m}{p}\frac{dU}{dx} = \frac{mF}{p},$$

where $F = -dU/dx$ is the classical force exerted on the particle by the external field, introducing this force gives

$$\frac{m\hbar|F|}{p^3} \ll 1. \tag{46.7}$$

This shows that the semiclassical approximation fails when the particle's momentum becomes too small. In particular, it necessarily fails near turning points, that is, near points at which classical mechanics would have the particle stop and then reverse its direction of motion. Such points are determined by $p(x) = 0$, or $E = U(x)$. As $p \to 0$, the de Broglie wavelength tends to infinity and plainly cannot be regarded as small.

It should be stressed, however, that condition (46.6) or (46.7) may by itself be insufficient to justify the semiclassical approximation. It was obtained by estimating the different terms of the differential equation (46.4), whose discarded term contains the highest derivative. What must actually be small are successive terms in the expansion of the solution of this equation, and smallness of the discarded term in the equation need not ensure this. For example, suppose that the solution for $\sigma(x)$ contains a term that grows with x in an almost linear fashion. The smallness of its second derivative does not prevent that term from accumulating a large value over a sufficiently long distance.

[1] As is well known, $\int p\,dx$ is the time-independent part of the action. The complete mechanical action of the particle is $S = -Et \pm \int p\,dx$. The term Et is absent from σ because we are considering a time-independent wave function ψ.

This situation generally arises when the field extends over distances large compared with the characteristic length L on which it changes appreciably (see the remark below concerning (46.11)). The semiclassical approximation then cannot be used to follow the wave function over large distances.

We proceed to calculate the next term of expansion (46.3). Terms of first order in $\hbar$ in (46.4) give

$$\sigma_0'\sigma_1' + \frac{1}{2}\sigma_0'' = 0, \qquad \sigma_1' = -\frac{\sigma_0''}{2\sigma_0'} = -\frac{p'}{2p}.$$

Integration yields

$$\sigma_1 = -\frac{1}{2}\ln p \tag{46.8}$$

(the integration constant is omitted).

Substitution into (46.1) and (46.3) gives the wave function

$$\psi = \frac{C}{\sqrt{p}}\exp\left(\frac{i}{\hbar}\int p\,dx\right) + \frac{C'}{\sqrt{p}}\exp\left(-\frac{i}{\hbar}\int p\,dx\right). \tag{46.9}$$

The factor $1/\sqrt{p}$ has a simple interpretation. The probability of finding the particle between x and $x + dx$ is determined by $|\psi|^2$ and is

therefore, in the main, proportional to $1/p$. This is just what one expects for a "semiclassical particle," since in classical motion the time spent by the particle in an interval dx is inversely proportional to its velocity (or momentum).

In classically forbidden regions, where $E < U(x)$, the function $p(x)$ is purely imaginary and the exponential arguments are real. The general form of the wave-equation solution in these regions is

$$\psi = \frac{C}{\sqrt{|p|}}\exp\left(-\frac{1}{\hbar}\int |p|\,dx\right) + \frac{C'}{\sqrt{|p|}}\exp\left(\frac{1}{\hbar}\int |p|\,dx\right). \tag{46.10}$$

One must bear in mind, however, that the accuracy of the semiclassical approximation does not permit exponentially small terms in the wave function to be retained against a background of exponentially large ones. In this sense, keeping both terms of (46.10) at once is generally inadmissible.

Higher-order terms in the wave function are ordinarily unnecessary. We shall nevertheless find one further term of expansion (46.3), in order to point out some features concerning the accuracy of the semiclassical approximation.

The terms of order $\hbar^2$ in (46.4) give

$$\sigma_0'\sigma_2' + \frac{1}{2}\sigma_1'^2 + \frac{1}{2}\sigma_1'' = 0,$$

and therefore, on substituting (46.5) and (46.8) for σ_0 and σ_1,

$$\sigma_2' = \frac{p''}{4p^2} - \frac{3p'^2}{8p^3}.$$

Integrating (the first term being integrated by parts) and introducing the force $F = pp'/m$, we obtain

$$\sigma_2 = \frac{mF}{4p^3} + \frac{m^2}{8}\int \frac{F^2}{p^5}\,dx.$$

In this approximation the wave function is

$$\psi = \exp[(i/\hbar)\sigma] = \exp[(i/\hbar)\sigma_0 + \sigma_1](1 - i\hbar\sigma_2),$$

or

$$\psi = \frac{\text{const}}{\sqrt{p}}\left[1 - \frac{im\hbar F}{4p^3} - \frac{im^2\hbar}{8}\int \frac{F^2}{p^5}\,dx\right]\exp\left(\frac{i}{\hbar}\int p\,dx\right). \tag{46.11}$$

The appearance of imaginary correction terms in the prefactor is equivalent to a real correction to the phase of the wave function (that is, an addition to the integral

$\hbar^{-1}\int p\,dx$ in its exponential). This correction is proportional to $\hbar$, and thus has order λ/L.

The second and third terms in the square brackets in (46.11) must be small compared with 1. For the former this requirement is the same as (46.7); for the latter, estimating the integral leads to (46.7) only if F^2 tends to zero sufficiently rapidly over distances $\sim L$.

§ 47. Boundary Conditions in the Semiclassical Case

Let $x = a$ be a turning point (so that $U(a) = E$), and suppose that $U > E$ for every $x > a$, making the region to the right of the turning point classically inaccessible. The wave function must decay farther into this region. At a sufficient distance from the turning point it has the form

$$\psi = \frac{C}{2\sqrt{|p|}}\exp\left(-\frac{1}{\hbar}\int_a^x |p|\,dx\right), \qquad x > a, \tag{47.1}$$

which corresponds to the first term of (46.10). To the left of the turning point, the wave function must instead be represented by a real combination of the two semiclassical solutions (46.9) of the Schrödinger equation:

$$\psi = \frac{C_1}{\sqrt{p}}\exp\left(\frac{i}{\hbar}\int_a^x p\,dx\right) + \frac{C_2}{\sqrt{p}}\exp\left(-\frac{i}{\hbar}\int_a^x p\,dx\right), \qquad x < a. \tag{47.2}$$

To determine the coefficients in this combination, one must follow the change of the wave function from positive to negative values of $x - a$, starting in the region where (47.1) applies. This path, however, passes through the neighborhood of the stopping point, where the semiclassical approximation fails and the exact solution of the Schrödinger equation must be considered. For small $|x - a|$,

$$E - U(x) \approx F_0(x - a), \qquad F_0 = -\left.\frac{dU}{dx}\right|_{x=a} < 0; \tag{47.3}$$

in other words, the problem in this region is that of motion in a constant field. Its exact Schrödinger-equation solution was found in § 24. The relation between the coefficients in (47.1) and (47.2) can therefore be obtained by comparison with asymptotic expressions (24.5) and (24.6) for that

exact solution on the two sides of the turning point. Notice here that (47.3) gives $p(x) = \sqrt{2mF_0(x - a)}$, and hence

$$\frac{1}{\hbar}\int_a^x p\,dx = \frac{2}{3\hbar}\sqrt{2mF_0}(x - a)^{3/2},$$

which coincides with the expression occurring in the argument of the exp or sin in (24.5) or (24.6). This reasoning relies essentially on a partial overlap between the domain where expansion (47.3) is valid and the semiclassical domain. If the motion is semiclassical throughout almost the whole field region, as assumed, there are values of $|x-a|$ small enough for (47.3) to apply but at the same time large enough to satisfy the semiclassical condition and permit the asymptotic forms (24.5) and (24.6)[2].

There is, however, another and methodologically more instructive procedure that avoids any need to use the exact solution. Formally regard $\psi(x)$ as a function of the complex variable x, and pass from positive to negative values of $x-a$ along a path lying entirely far from $x=a$, so that the semiclassical condition is formally satisfied all along it (A. Zwaan, 1929). We again consider values of $|x-a|$ for which expansion (47.3) is valid as well. The wave function (47.1) then takes the form

$$\psi(x) = \frac{C}{2(2m|F_0|)^{1/4}(x-a)^{1/4}} \exp\left\{-\frac{1}{\hbar}\sqrt{2m|F_0|}\int_a^x \sqrt{x-a}\,dx\right\}. \tag{47.4}$$

First follow the change of this function while passing around $x=a$ from right to left on a semicircle of radius ρ in the upper half of the complex x-plane. On this semicircle

$$x-a=\rho e^{i\varphi}, \qquad \int_a^x \sqrt{x-a}\,dx = \frac{2}{3}\rho^{3/2}\left(\cos\frac{3\varphi}{2}+i\sin\frac{3\varphi}{2}\right),$$

where the phase φ varies from 0 to π. The modulus of the exponential factor in (47.4) first increases (for $0<\varphi<2\pi/3$), and then decreases to 1. At the end of the passage the exponent is purely imaginary and equals

$$-\frac{i}{\hbar}\int_a^x \sqrt{2m|F_0|(a-x)}\,dx = -\frac{i}{\hbar}\int_a^x p(x)\,dx.$$

Meanwhile the prefactor in (47.4) changes under this continuation according to

$$(x-a)^{-1/4} \longrightarrow (a-x)^{-1/4}e^{-i\pi/4}.$$

Thus the whole function (47.4) becomes the second term in (47.2), with $C_2 = (1/2)Ce^{-i\pi/4}$.

There is a simple explanation for the fact that continuation through the upper half-plane determines only C_2 in (47.2). Follow the change of (47.2) along the same semicircle in the reverse direction, from left to right. At the start of the continuation, its first term rapidly becomes exponentially small relative to the second. But the semiclassical approximation cannot reveal exponentially small terms in ψ against the background of a large leading term. This is why the first term of (47.2) is “lost” in the continuation just performed.

To determine C_1, the continuation from right to left must be made along a semicircle in the lower half of the complex x-plane. In the same manner one finds that (47.4) then becomes the first term of (47.2), with $C_1 = (1/2)Ce^{i\pi/4}$.

[2]Indeed, expansion (47.3) is valid for $|x-a| \ll L$, where L is the characteristic distance over which the field $U(x)$ changes. The semiclassical condition (46.7), on the other hand, requires $|x-a|^{3/2} \gg \hbar/\sqrt{m|F_0|}$. These requirements are compatible because semiclassical motion far from the turning point (that is, at $|x-a| \sim L$) means that $L^{3/2} \gg \hbar/\sqrt{m|F_0|}$.

Consequently, the wave function (47.1) at $x > a$ corresponds at $x < a$ to

$$\psi = \frac{C}{\sqrt{p}} \cos\left(\frac{1}{\hbar}\int_a^x p\,dx + \frac{\pi}{4}\right).$$

The resulting connection rule may be written in a form independent of which side of the turning point contains the classically forbidden region:

$$\underset{U(x)>E}{\frac{2C}{\sqrt{|p|}} \exp\left\{-\frac{1}{\hbar}\int_a^x |p|\,dx\right\}} \longrightarrow \underset{U(x)<E}{\frac{C}{\sqrt{p}} \cos\left\{\frac{1}{\hbar}\int_a^x p\,dx - \frac{\pi}{4}\right\}} \tag{47.5}$$

(*H., A. Kramers*, 1926).

One point already plain from the derivation deserves renewed emphasis. The rule presupposes a specified boundary condition imposed on one side of the turning point. Consequently, it applies in only one specified direction. More precisely, rule (47.5) was obtained with the boundary condition $\psi \to 0$ deep within the classically forbidden region and must be used for passage from that region to the classically allowed region (as indicated by the arrow in (47.5))[3].

If the classically allowed region is bounded at $x = a$ by an infinitely high potential wall, the boundary condition there is $\psi = 0$ (see § 18). The semiclassical approximation is then valid all the way to the wall, and the wave function is

$$\begin{aligned} \psi &= \frac{C}{\sqrt{p}} \sin\left(\frac{1}{\hbar}\int_a^x p\,dx\right), && x < a, \\ \psi &= 0, && x > a. \end{aligned} \tag{47.6}$$

§ 48. The Bohr–Sommerfeld quantization rule

States belonging to the discrete energy spectrum become quasiclassical when the quantum number n, which gives the ordinal number of the state, is large. Indeed, this number specifies the number of nodes of the eigenfunction (see § 21). The separation of two successive nodes, however, has the same order of magnitude as the de Broglie wavelength. Hence, for large n this separation is small, and the wavelength is short in comparison with the dimensions of the region of motion.

Let us derive the condition that fixes the quantum energy levels in the quasiclassical case. Consider a particle executing bounded one-dimensional motion in a potential well. The classically accessible interval $b \le x \le a$ is bounded by two turning points[4].

[3] Passage in the reverse direction, however, ceases to be meaningful in the following sense: even a small change in the wave function on the right-hand side of (47.5) can produce an exponentially growing term in the function on the left-hand side.

[4] In classical mechanics a particle in this field would move periodically, with period (the time required to travel from b to a and back)

$$T = 2\int_b^a \frac{dx}{v}$$

(v is the particle velocity).

By rule (47.5), the boundary condition at $x = b$ gives, in the region to its right, the wave function

$$\psi = \frac{C}{\sqrt{p}} \cos\left(\frac{1}{\hbar}\int_b^x p\,dx - \frac{\pi}{4}\right). \tag{48.1}$$

Applying the same rule to the region on the left of $x = a$, we obtain another form of the same wave function:

$$\psi = \frac{C'}{\sqrt{p}} \cos\left(\frac{1}{\hbar}\int_x^a p\,dx - \frac{\pi}{4}\right).$$

For these expressions to agree throughout the interval, the sum of their phases, which is constant, must be an integral multiple of π:

$$\frac{1}{\hbar}\int_b^a p\,dx - \frac{\pi}{2} = n\pi$$

(with $C' = (-1)^n C$). It follows that

$$\frac{1}{2\pi\hbar}\oint p\,dx = n + \frac{1}{2}, \tag{48.2}$$

where $\oint p\,dx = 2\int_b^a p\,dx$ is evaluated over a complete period of the particle's classical motion. This is the condition determining the stationary states in the quasiclassical regime. It corresponds to the Bohr–Sommerfeld quantization rule of the old quantum theory.

The quantity $I = (1/2\pi)\oint p\,dx$ is called the adiabatic invariant (see I, § 49), and therefore (48.2) may be written

$$I(E) = \hbar(n + 1/2).$$

It was already noted in § 41 that, if the parameters change sufficiently slowly, or "adiabatically," a system remains in the same quantum state—in the present case, the state having a particular n. We now see that in the quasiclassical limit this statement becomes the classical theorem asserting the constancy of the adiabatic invariant under a slow change of parameters.

It is readily seen that the integer n equals the number of zeros of the wave function and is therefore the ordinal number of the stationary state. In fact, the phase in (48.1) increases from $-\pi/4$ at $x = b$ to $(n+1/4)\pi$ at $x = a$, so the cosine vanishes n times within this interval (outside the range $b \ll x \ll a$, the wave function decreases monotonically and has no zeros at finite distances)[5].

As explained above, n is large in the quasiclassical case. Nevertheless, retaining the term $1/2$ beside n in (48.2) is legitimate. Inclusion of the next correction terms in the phases of the wave functions would add to the right-hand side of (48.2) only terms of order λ/L, which are small compared with unity (see the remark at the end of § 46)[6] .

[5] Strictly, the zeros must be counted using the exact form of the wave function near the turning points. An analysis of this kind confirms the stated result.

[6] In certain instances, the exact formula for the energy levels $E(n)$ (viewed as a function of the quantum number n), obtained from the exact Schrödinger equation, retains the same form as $n \to \infty$. Examples are furnished by the energy levels in a Coulomb field and those of the harmonic oscillator. It is therefore natural that in these instances the quantization rule (48.2), which applies for large n, yields for $E(n)$ a formula identical to the exact one.

To normalize the wave function, it is sufficient to integrate $|\psi|^2$ only over $b \le x \le a$, since $\psi(x)$ decays exponentially outside this interval. The argument of the cosine in (48.1) varies rapidly, so to the required accuracy its square may be replaced by its mean value, $1/2$. Thus

$$\int |\psi|^2 dx \approx \frac{C^2}{2} \int_b^a \frac{dx}{p(x)} = \frac{\pi C^2}{2m\omega} = 1,$$

where $\omega = 2\pi/T$ is the frequency of the classical periodic motion. The normalized quasiclassical function is consequently

$$\psi = 2\sqrt{\frac{m\omega}{\pi p}} \cos\left(\frac{1}{\hbar}\int_b^x p\,dx - \frac{\pi}{4}\right). \tag{48.3}$$

It should be kept in mind that ω depends on the energy and, in general, differs from one level to another.

Relation (48.2) also admits another interpretation. The integral $\oint p\,dx$ is the area enclosed by the particle's closed

classical phase trajectory (that is, by the curve in the p, x plane—the particle's phase space). If this area is partitioned into cells, each of area $2\pi\hbar$, their total number is n. But n is the number of quantum states whose energies do not exceed the specified value (the value corresponding to the phase trajectory under consideration). Thus, in the quasiclassical case, one may associate with every quantum state a phase-space cell of area $2\pi\hbar$. Equivalently, the number of states assigned to a phase-space area element $\Delta p \Delta x$ is

$$\frac{\Delta p \Delta x}{2\pi\hbar}. \tag{48.4}$$

If the wave vector $k = p/\hbar$ is used in place of the momentum, this number takes the form $\Delta k \Delta x/2\pi$. As expected, it agrees with the known expression for the number of normal modes of a wave field (see II, § 52).

The general pattern in which levels are distributed through the energy spectrum can be deduced from the quantization rule (48.2). Let ΔE denote the separation of two adjacent levels, whose quantum numbers n thus differ by one. Since, for large n, ΔE is small in comparison with the level energy itself, (48.2) gives

$$\Delta E \oint \frac{\partial p}{\partial E}\,dx = 2\pi\hbar.$$

But $\partial E/\partial p = v$, and hence

$$\oint \frac{\partial p}{\partial E}\,dx = \oint \frac{dx}{v} = T.$$

We therefore find

$$\Delta E = \frac{2\pi\hbar}{T} = \hbar\omega. \tag{48.5}$$

Thus the spacing of adjacent levels is $\hbar\omega$. For a sequence of nearby levels whose differences in n are small compared with n itself, the corresponding frequencies ω may be regarded as approximately equal. It follows that, within any small portion of the quasiclassical part of the spectrum, the levels are equidistant, with the common interval $\hbar\omega$. This result is also foreseeable: in the quasiclassical regime, every frequency associated with a transition between energy levels must be an integer multiple of the classical frequency ω.

It is useful to trace the limiting behavior of the matrix elements of an arbitrary physical quantity f. We start from the requirement that the mean value of f in a quantum state must tend simply to its classical value, provided that, in the limit, the state itself describes motion of the particle along a definite trajectory. Such a state is represented by a wave packet (see § 6), formed by superposing stationary states with neighboring energies. Its wave function has the form

$$\psi = \sum_n a_n \psi_n,$$

where the coefficients a_n differ appreciably from zero only within an interval Δn of quantum-number values such that $1 \ll \Delta n \ll n$. The numbers n are assumed large, as required for the stationary states to be quasiclassical. By definition, the mean value of f is

$$\bar{f} = \int \psi^* f \psi \, dx = \sum_n \sum_m a_m^* a_n f_{mn} e^{i\omega_{mn}t},$$

or, replacing the summation over n, m by summations over n and the difference $s = m - n$, we obtain

$$\bar{f} = \sum_n \sum_s a_{n+s}^* a_n f_{n+s,n} e^{i\omega_s t},$$

where, in accordance with (48.5), $\omega_{mn} = \omega s$.

When evaluated with quasiclassical wave functions, the matrix elements f_{mn} decrease rapidly in magnitude as the difference $m - n$ grows, while varying slowly with n itself when $m - n$ is fixed. We may consequently write, to the same approximation,

$$\bar{f} = \sum_n \sum_s a_n^* a_n f_s e^{i\omega st} = \sum_n |a_n|^2 \sum_s f_s e^{i\omega st},$$

where $f_s = f_{n+s,n}$ and n denotes some mean value of the quantum number in the interval Δn. Since $\sum_n |a_n|^2 = 1$, it follows that

$$\bar{f} = \sum_s f_s e^{i\omega st}.$$

The resulting sum has the form of an ordinary Fourier series. Since $\bar{f}$ must approach the classical quantity $f(t)$ in the limit, we conclude that the matrix elements f_{mn} tend to the coefficients f_{m-n} in the Fourier-series expansion of the classical function $f(t)$.

Similarly, matrix elements for transitions between states in the continuous spectrum tend to the components in the Fourier-integral expansion of $f(t)$. Here the wave functions of the stationary states must be normalized to a delta function of the energy divided by $\hbar$.

All the preceding results extend directly to systems with many degrees of freedom undergoing finite motion for which the mechanical, or classical, problem permits complete separation of variables in the Hamilton–Jacobi method (the so-called conditionally periodic motion; see I, § 52). After the variables have been separated, the problem for each degree of freedom reduces to a one-dimensional one, and the corresponding quantization conditions are

$$\oint p_i\, dq_i = 2\pi\hbar(n_i + \gamma_i), \tag{48.6}$$

where the integral is taken over one period of the generalized coordinate q_i, while γ_i is a number of order unity determined by the form of the boundary conditions for that degree of freedom.[7]

For arbitrary multidimensional motion that is not conditionally periodic, a formulation of the quasiclassical quantization conditions requires a more subtle argument[8]. The notion of phase-space "cells," however, always applies in the same manner within the quasiclassical approximation. This is clear from the connection, noted above, between this notion and the number of normal modes of a wave field in a given spatial volume. In the general case of a system with s degrees of freedom, a phase-space volume element contains

$$\Delta N = \frac{\Delta q_1 \cdots \Delta q_s\, \Delta p_1 \cdots \Delta p_s}{(2\pi\hbar)^s} \tag{48.7}$$

quantum states[9].

Problem

1. Find, approximately, the number of discrete energy levels for a particle moving in a field $U(r)$ that satisfies the condition for the quasiclassical approximation.

Solution. The number of states "belonging" to the phase-space volume for momenta in the range $0 \le p \le p_{\max}$ and particle coordinates in the volume element dV is

$$\frac{(4\pi/3)p_{\max}^3\, dV}{(2\pi\hbar)^3}.$$

At a given r, the particle may have, in its classical motion, any momentum that obeys $E = (p^2/2m) + U(r) \le 0$. Substitution of $p_{\max} = \sqrt{-2mU(r)}$ gives the total number of discrete-spectrum states:

$$\frac{\sqrt{2}m^{3/2}}{3\pi^2\hbar^3} \int (-U)^{3/2}\, dV,$$

where the integration extends over the part of space in which $U < 0$. This integral diverges, and the number of levels is infinite, if U decreases at infinity as r^{-s} with $s < 2$, in agreement with the results of § 18.

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[7] For motion in a central field, for example,

$$\oint p_r\, dr = 2\pi\hbar(n_r + 1/2), \qquad \oint p_\theta\, d\theta = 2\pi\hbar(l - m + 1/2), \qquad \oint p_\varphi\, d\varphi = 2\pi\hbar m$$

(where $n_r = n - l - 1$ is the radial quantum number). The last equality follows simply because p_φ is the z component of the angular momentum and is equal to $\hbar m$.

[8] See *J., B. Keller* // Ann. Phys. 1958. V. 4. P. 180.

[9] In particular, $d^3p/(2\pi\hbar)^3$ for a single particle is the number of states associated with a momentum interval d^3p in a unit volume of space. This accounts for the agreement between the two interpretations of the plane-wave normalization (15.8) noted in the footnote on p. 68.

2. Find the corresponding number for a quasiclassical central field $U(r)$ (V. L. Pokrovskii).

Solution. In a centrally symmetric field, the number of states differs from the number of energy levels because the latter are degenerate with respect to the orientation of the angular momentum. The required number may be found by noting that the number of levels for a specified value of the angular momentum M is the same as the number of nondegenerate levels for one-dimensional motion with potential energy $U_{\text{eff}} = U(r) + M^2/(2mr^2)$. At a given r, the largest possible value of the momentum p_r for energies $E \leq 0$ is $p_{r\,\max} = \sqrt{-2mU_{\text{eff}}}$. Consequently, the number of states (that is, the number of levels) is

$$\frac{1}{2\pi\hbar}\oint dr\,dp_r = \frac{\sqrt{2m}}{\pi\hbar}\int\sqrt{-U - \frac{M^2}{2mr^2}}\,dr.$$

The required total number of discrete levels is obtained from this result by integration with the measure $dM/\hbar$ (which, in the semiclassical case, replaces summation over l), and is

$$\frac{\sqrt{2m}}{4\hbar^2}\int(-U)r\,dr.$$

§ 49. Semiclassical motion in a centrally symmetric field

For motion in a centrally symmetric field, the particle's wave function separates, as we know, into angular and radial parts. We first consider the angular part.

The dependence of the angular wave function on the angle φ (specified by the quantum number m) is so simple that there is no question of seeking approximate formulas for it. Its dependence on the polar angle θ, on the other hand, is semiclassical according to the general rule if the corresponding quantum number l is large (a more precise statement of this condition is given below).

Here we shall derive a semiclassical expression for the angular function only in the case of greatest practical importance, namely states with zero magnetic quantum number ($m = 0$)[10]. Apart from a constant factor, this function is the Legendre polynomial $P_l(\cos\theta)$ (see (28.8)) and satisfies the differential equation

$$\frac{d^2P_l}{d\theta^2} + \cot\theta\frac{dP_l}{d\theta} + l(l+1)P_l = 0. \tag{49.1}$$

The substitution

$$P_l = \frac{\chi}{\sqrt{\sin\theta}} \tag{49.2}$$

reduces it to the equation

$$\chi'' + \left[\left(l+\frac{1}{2}\right)^2 + \frac{1}{4\sin^2\theta}\right]\chi = 0, \tag{49.3}$$

[10]The opposite case, $m = l$, must in the limit correspond to motion in a classical orbit lying in the equatorial plane $\theta = \pi/2$. Indeed,

$$P_l^l(\cos\theta) = \text{const}\cdot\sin^l\theta;$$

as $l \to \infty$, this function (and with it $|\psi|^2$) tends to zero for all $\theta \neq \pi/2$.

which contains no first derivative and has the form of a one-dimensional Schrödinger equation.

In (49.3), the role of the "de Broglie wavelength" is played by $\lambda \sim 1/l$.

The requirement that the derivative $d\lambda/dx$ be small (condition (46.6)) gives the inequalities

$$\theta l \gg 1, \qquad (\pi - \theta)l \gg 1 \tag{49.4}$$

(the semiclassical conditions for the angular part of the wave function). For large l these conditions hold over almost the whole interval of values of θ, except only for angles very close to zero or to π.

When condition (49.4) holds, the second term in the square brackets in (49.3) may be neglected in comparison with the first:

$$\chi'' + \left(l + \frac{1}{2}\right)^2 \chi = 0.$$

The solution of this equation is

$$\chi = \sqrt{\sin\theta}\, P_l(\cos\theta) = A \sin\left[\left(l + \frac{1}{2}\right)\theta + \alpha\right] \tag{49.5}$$

(A and α are constants).

For angles $\theta \ll 1$, we may set $\cot\theta \approx 1/\theta$ in (49.1). Also replacing $l(l+1)$ approximately by $(l+1/2)^2$, we obtain

$$\frac{d^2 P_l}{d\theta^2} + \frac{1}{\theta}\frac{dP_l}{d\theta} + \left(l + \frac{1}{2}\right)^2 P_l = 0,$$

whose solution is the Bessel function of order zero,

$$P_l(\cos\theta) = J_0\left[\left(l + \frac{1}{2}\right)\theta\right], \qquad \theta \ll 1. \tag{49.6}$$

The constant factor has been set equal to unity because P_l must equal one at $\theta = 0$. The approximate expression (49.6) for P_l is valid for all angles $\theta \ll 1$. In particular, it may be used in the region $1/l \ll \theta \ll 1$, where it must agree with (49.5), which is valid whenever $\theta \gg 1/l$. When $\theta l \gg 1$, the Bessel function may be replaced by its asymptotic expression for large argument, giving

$$P_l \approx \sqrt{\frac{2}{\pi l \theta}} \sin\left[\left(l + \frac{1}{2}\right)\theta + \frac{\pi}{4}\right]$$

(in the coefficient, $1/2$ may be neglected in comparison with l). Comparison with (49.5) gives $A = \sqrt{2/\pi l}$ and $\alpha = \pi/4$. We thus arrive at the following expression for $P_l(\cos\theta)$, applicable in the semiclassical case[11]:

$$P_l(\cos\theta) = \sqrt{\frac{2}{\pi l \sin\theta}} \sin\left[\left(l + \frac{1}{2}\right)\theta + \frac{\pi}{4}\right]. \tag{49.7}$$

[11] Notice that it is precisely the replacement of $l(l+1)$ by $(l+1/2)^2$ that has yielded an expression which is multiplied by $(-1)^l$ when θ is replaced by $\pi - \theta$, as the function $P_l(\cos\theta)$ must be.

The normalized spherical function Y_{l0} therefore takes the form (compare (28.8))

$$Y_{l0} = \frac{1}{\pi\sqrt{\sin\theta}} \sin\left[\left(l + \frac{1}{2}\right)\theta + \frac{\pi}{4}\right]. \tag{49.8}$$

We now turn to the radial part of the wave function. It was shown in § 32 that the function $\chi(r) = rR(r)$ satisfies an equation identical with the one-dimensional Schrödinger equation having potential energy

$$U_l(r) = U(r) + \frac{\hbar^2}{2mr^2} l(l+1).$$

We may therefore apply the results of the preceding sections, with the function $U_l(r)$ understood as the potential energy.

The case $l = 0$ is the simplest. The centrifugal energy is absent, and if the field $U(r)$ satisfies the required condition (46.6), the radial wave function is semiclassical throughout space. At $r = 0$ we must have $\chi = 0$, and hence the semiclassical function $\chi(r)$ is determined in accordance with formulas (47.6).

If $l \neq 0$, the centrifugal energy must also satisfy condition (46.6). In the region of small r, where the centrifugal energy is of the order of the total energy, the wavelength is $\lambda = \hbar/p \sim r/l$, and condition (46.6) gives $l \gg 1$. Thus, when l is not large, the centrifugal energy violates the semiclassical condition in the region of small r. It is readily verified that the correct phase of the semiclassical wave function $\chi(r)$ is obtained by using the formulas for one-dimensional motion and replacing the coefficient $l(l+1)$ in the potential energy $U_l(r)$

by $(l + 1/2)^2$[12]:

$$U_l(r) = U(r) + \frac{\hbar^2}{2mr^2}\left(l + \frac{1}{2}\right)^2. \tag{49.9}$$

The applicability of the semiclassical approximation to a Coulomb field $U = \pm\alpha/r$ requires separate consideration. Of the entire region of motion, the most important part corresponds to distances r for which $|U| \sim |E|$, that is, $r \sim \alpha/|E|$. The semiclassical condition for motion in this region reduces to requiring that the wavelength $\lambda\!\!\!^{-} \sim \hbar/\sqrt{2m|E|}$ be small compared with the size $\alpha/|E|$ of the region. This gives

$$|E| \ll \frac{m\alpha^2}{\hbar^2}, \tag{49.10}$$

that is, the magnitude of the energy must be small compared with the energy of a particle in the first Bohr orbit. Condition (49.10) may also be written

$$\frac{\hbar v}{\alpha} \gg 1, \tag{49.11}$$

where $v \sim \sqrt{|E|/m}$ is the particle velocity. Notice that this condition is the reverse of condition (45.7) for the applicability of perturbation theory to the Coulomb field.

[12]Thus, in the simplest case of free motion ($U = 0$), the phase of the function calculated from (48.1), with U_l from (49.9), agrees at large r, as it should, with the phase of the function (33.12).

As for the region of small distances ($|U(r)| \gg E$), in a repulsive Coulomb field it is of no interest at all, since for $U > E$ the semiclassical wave functions decay exponentially. In an attractive field with small l, however, the particle may penetrate into a region where $|U| \gg |E|$, and the limits of applicability of the semiclassical approximation there must be determined. We use the general condition (46.7), putting

$$F = -\frac{dU}{dr} = \frac{\alpha}{r^2}, \qquad p \approx \sqrt{2m|U|} \sim \sqrt{\frac{m\alpha}{r}}.$$

We then find that the region of applicability of the semiclassical approximation is restricted to distances

$$r \gg \frac{\hbar^2}{m\alpha}, \tag{49.12}$$

that is, distances large compared with the "radius" of the first Bohr orbit.

Problem

Find the behavior of the wave function near the origin if, as $r \to 0$, the field tends to infinity as $\pm\alpha/r^s$, where $s > 2$.

Solution. For sufficiently small r, the wavelength is

$$\lambda\!\!\!^{-} \sim \frac{\hbar}{\sqrt{m|U|}} \sim \frac{\hbar r^{s/2}}{\sqrt{m\alpha}},$$

and therefore

$$\frac{d\lambda\!\!\!^{-}}{dr} \sim \frac{\hbar r^{s/2-1}}{\sqrt{m\alpha}} \ll 1;$$

thus the semiclassical condition is fulfilled. In an attractive field, $U \to -\infty$ as $r \to 0$. The region near the origin is then classically accessible, and the radial wave function $\chi \sim 1/\sqrt{p}$, whence

$$\psi \sim r^{s/4-1}.$$

In a repulsive field the region of small r is classically inaccessible. The wave function then tends exponentially to zero as $r \to 0$. Omitting the factor multiplying the exponential function, we have

$$\psi \sim \exp\left[-\frac{2\sqrt{m\alpha}}{(s-2)\hbar} r^{-(s/2-1)}\right].$$

§ 50. Passage through a potential barrier

Consider a particle moving in a field of the type shown in Fig. 13. The field contains a *potential barrier*: an interval in which the potential energy $U(x)$ is greater than the particle's total energy E. Classical mechanics does not allow the particle to penetrate such a barrier. In quantum mechanics, however, there is a nonzero probability that it will pass "through the barrier"; this phenomenon is also called the *tunnel effect*[13]. When

[13] We have already encountered examples of this kind in Problems 2 and 4 of § 25.

$U(x)$ satisfies the semiclassical conditions, a general expression can be obtained for the barrier transmission coefficient. These conditions imply, in particular, that the barrier is broad; consequently its semiclassical transmission coefficient is small.

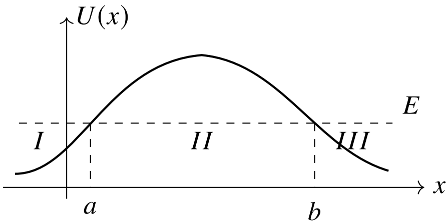

Fig. 13

We first solve an auxiliary problem so that it will not interrupt the calculation below. Suppose the semiclassical wave function to the right of the turning point $x = b$ (where $U(x) < E$) is the travelling wave

$$\psi = \frac{C}{\sqrt{p}} \exp\left(\frac{i}{\hbar}\int_b^x p\,dx + \frac{i\pi}{4}\right). \tag{50.1}$$

We seek the wave function of the same state in the region $x < b$. We shall use the same continuation in the complex x-plane as in § 47.

Writing

$$E - U(x) \approx F_0(x-b), \qquad F_0 > 0,$$

we put (50.1) in the form

$$\psi(x) = \frac{C}{(2mF_0)^{1/4}}\frac{1}{(x-b)^{1/4}} \exp\left\{\frac{i}{\hbar}(2mF_0)^{1/2}\int_b^x \sqrt{x-b}\,dx + \frac{i\pi}{4}\right\}.$$

Continue this expression from right to left round a semicircle in the upper half-plane:

$$x - b = \rho e^{i\varphi}, \qquad i\int_b^x \sqrt{x-b}\,dx = \frac{2}{3}\rho^{3/2}\left(-\sin\frac{3\varphi}{2} + i\cos\frac{3\varphi}{2}\right),$$

with the phase φ changing from 0 to π. During this continuation, $|\psi(x)|$ first decreases and then increases. At its end the function is

$$\psi(x) = \frac{C}{(2mF_0)^{1/4}}\frac{1}{(b-x)^{1/4}e^{i\pi/4}} \exp\left\{\frac{1}{\hbar}\int_x^b (2mF_0)^{1/2}\sqrt{b-x}\,dx + \frac{i\pi}{4}\right\}.$$

We have therefore obtained the connection rule[14]:

$$\frac{C}{\sqrt{p}}\exp\left\{\frac{i}{\hbar}\int_b^x p\,dx + \frac{i\pi}{4}\right\}\Bigg|_{x>b} \longrightarrow \frac{C}{\sqrt{|p|}}\exp\left\{\frac{1}{\hbar}\left|\int_b^x p\,dx\right|\right\}\Bigg|_{x<b}. \tag{50.2}$$

[14]If the continuation from right to left is instead made through the lower half-plane, $|\psi(x)|$ first grows and then falls. On the negative half-axis ($\varphi \to -\pi$) it becomes exponentially small, and retaining it against the exponentially large function (50.2) would be illegitimate. Over the part of the path on which $\psi(x)$ is exponentially large, the inaccuracy of the semiclassical approximation loses an exponentially small addition. As $\varphi \to -\pi$ this addition could have become an exponentially large term, which is therefore lost as well.

It is important that this rule presupposes a particular wave in the classically allowed region, namely one travelling to the right, and is to be used specifically for continuation from that region into the classically forbidden one.

We now return to the barrier transmission coefficient. Let a particle approach the barrier from region I, moving from left to right. Beyond the barrier, in region III, there is only the transmitted wave, which also travels to the right. Write its wave function as

$$\psi = \sqrt{\frac{D}{v}} \exp\left(\frac{i}{\hbar}\int_b^x p\,dx + \frac{i\pi}{4}\right), \tag{50.3}$$

where $v = p/m$ is the particle velocity and D is the current density in the wave. Rule (50.2) now gives the wave function within the barrier, in region II:

$$\psi = \sqrt{\frac{D}{|v|}} \exp\left(\frac{1}{\hbar}\left|\int_x^b p\,dx\right|\right) = \sqrt{\frac{D}{|v|}} \exp\left(\frac{1}{\hbar}\left|\int_a^b p\,dx\right| - \frac{1}{\hbar}\left|\int_a^x p\,dx\right|\right). \tag{50.4}$$

Finally, rule (47.5) yields, in region I before the barrier,

$$\psi = 2\sqrt{\frac{D}{v}} \exp\left(\frac{1}{\hbar}\int_a^b |p|\,dx\right) \cos\left(\frac{1}{\hbar}\int_x^a p\,dx - \frac{\pi}{4}\right).$$

If we set

$$D = \exp\left(-\frac{2}{\hbar}\int_a^b |p|\,dx\right), \tag{50.5}$$

this function becomes

$$\begin{aligned}\psi &= \frac{2}{\sqrt{v}} \cos\left(\frac{1}{\hbar}\int_a^x p\,dx + \frac{\pi}{4}\right) \\ &= \frac{1}{\sqrt{v}} \exp\left(\frac{i}{\hbar}\int_a^x p\,dx + \frac{i\pi}{4}\right) + \frac{1}{\sqrt{v}} \exp\left(-\frac{i}{\hbar}\int_a^x p\,dx - \frac{i\pi}{4}\right).\end{aligned}$$

The first term (which tends, as $x \to -\infty$, to the plane wave $\psi \sim e^{ipx/\hbar}$) is the wave incident on the barrier, and the second is the reflected wave. Our normalization assigns unit current density to the incident wave. Thus D, the current density of the transmitted wave, is the required barrier transmission coefficient. Formula (50.5) is applicable only when its exponential argument is large, so that D itself is small[15].

So far we have assumed that $U(x)$ obeys the semiclassical condition throughout the barrier, apart from the immediate neighborhoods of the turning points. In practice one often encounters barriers whose potential-energy curve rises or falls so sharply on one side that the semiclassical approximation fails there. The leading exponential factor in D remains the one in (50.5), but the pre-exponential factor, equal to unity in (50.5), changes. To calculate it one must in principle find the exact wave function in the non-semiclassical region and use its matching to fix the semiclassical wave function inside the barrier.

[15] The exponential smallness of D also explains why the incident and reflected amplitudes found in region I are equal: their exponentially small difference is lost within the semiclassical approximation.

Problem

1. Find the transmission coefficient for the potential barrier shown in Fig. 14: $U(x) = 0$ for $x < 0$, and $U(x) = U_0 - Fx$ for $x > 0$. Calculate only the exponential factor.

Solution. A direct calculation gives

$$D \sim \exp\left[-\frac{4\sqrt{2m}}{3\hbar F}(U_0 - E)^{3/2}\right].$$

2. Find the probability for a particle of zero angular momentum to escape from the spherically symmetric potential well $U(r) = -U_0$ for $r < r_0$, $U(r) = \alpha/r$ for $r > r_0$ (Fig. 15)[16].

Solution. A spherically symmetric problem reduces to a one-dimensional one, so the preceding formulas apply. Thus

$$w \sim \exp\left[-\frac{2}{\hbar}\int_{r_0}^{\alpha/E}\sqrt{2m\left(\frac{\alpha}{r} - E\right)}\,dr\right].$$

Evaluation of the integral gives finally

$$w \sim \exp\left\{-\frac{2\alpha}{\hbar}\sqrt{\frac{2m}{E}}\left[\arccos\sqrt{\frac{Er_0}{\alpha}} - \sqrt{\frac{Er_0}{\alpha}\left(1 - \frac{Er_0}{\alpha}\right)}\right]\right\}.$$

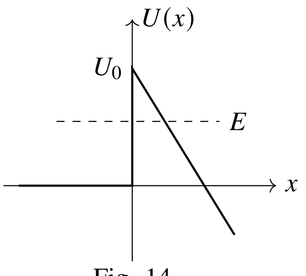

Fig. 14

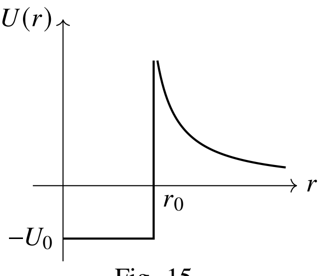

Fig. 15

In the limiting case $r_0 \to 0$, this expression becomes

$$w \sim \exp\left(-\frac{\pi\alpha}{\hbar}\sqrt{\frac{2m}{E}}\right) = \exp\left(-\frac{2\pi\alpha}{\hbar v}\right).$$

These formulas require a large exponent, that is, $\alpha/\hbar v \gg 1$. As expected, this is precisely condition (49.11) for semiclassical motion in a Coulomb field.

[16]This problem was first studied by G. F. Gamow (1928), and by *R.W.Gurney* and *E.U.Condon* (1929), in connection with the theory of radioactive α decay.

3. The field $U(x)$ consists of two symmetric potential wells (I and II in Fig. 16) separated by a barrier. If the particle could not penetrate the barrier, there would be levels corresponding to motion in either well alone, with the same energies for the two wells. Passage through the barrier splits every such level into a pair of nearby levels. Their states describe a particle moving in both wells at once. Find the splitting, assuming that $U(x)$ is semiclassical.

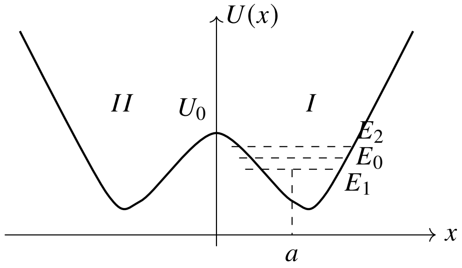

Fig. 16

Solution. To neglect barrier penetration, construct an approximate solution of the Schrödinger equation from the semiclassical function $\psi_0(x)$ describing motion with some energy E_0 in one well, say I. Thus it decays exponentially on both sides of that well. Normalize ψ_0 so that the integral of ψ_0^2 over well I is unity. Allowing for the small tunnelling probability splits E_0 into E_1 and E_2. The proper zeroth-order functions belonging to these levels are the symmetric and antisymmetric combinations of $\psi_0(x)$ and $\psi_0(-x)$:

$$\psi_1(x) = \frac{1}{\sqrt{2}}[\psi_0(x) + \psi_0(-x)], \qquad \psi_2(x) = \frac{1}{\sqrt{2}}[\psi_0(x) - \psi_0(-x)]. \tag{1}$$

In well I, $\psi_0(-x)$ is negligibly small compared with $\psi_0(x)$; in well II the reverse holds. Hence $\psi_0(x)\psi_0(-x)$ is negligible everywhere, and (1) is normalized so that the integrals of the squared functions over wells I and II are unity.

Write the Schrödinger equations

$$\psi_0'' + \frac{2m}{\hbar^2}(E_0 - U)\psi_0 = 0, \qquad \psi_1'' + \frac{2m}{\hbar^2}(E_1 - U)\psi_1 = 0.$$

Multiply the first by ψ_1 and the second by ψ_0, subtract them term by term, and integrate over dx from 0 to ∞. At $x = 0$ we have $\psi_1 = \sqrt{2}\psi_0$ and $\psi_1' = 0$, while

$$\int_0^\infty \psi_0\psi_1\,dx \approx \frac{1}{\sqrt{2}}\int_0^\infty \psi_0^2\,dx = \frac{1}{\sqrt{2}}.$$

It follows that

$$E_1 - E_0 = -\frac{\hbar^2}{m}\psi_0(0)\psi_0'(0).$$

The same calculation for $E_2 - E_0$ gives the opposite sign. Therefore

$$E_2 - E_1 = \frac{2\hbar^2}{m}\psi_0(0)\psi_0'(0).$$

Using (47.1), with the coefficient C from (48.3), gives

$$\psi_0(0) = \sqrt{\frac{\omega}{2\pi v_0}} \exp\left(-\frac{1}{\hbar}\int_0^a |p|\,dx\right), \qquad \psi_0'(0) = \frac{mv_0}{\hbar}\psi_0(0),$$

where $v_0 = \sqrt{2(U_0 - E_0)/m}$. We thus obtain

$$E_2 - E_1 = \frac{\omega\hbar}{\pi} \exp\left(-\frac{1}{\hbar}\int_{-a}^a |p|\,dx\right),$$

where a is the turning point associated with the energy E_0 (see Fig. 16).

4. Find the exact transmission coefficient D, without assuming it small, for the parabolic potential barrier $U(x) = -kx^2/2$ (*E.C.Kemble*, 1935)[17].

Solution. For arbitrary k and E, the motion is semiclassical at sufficiently large $|x|$, where

$$p = \sqrt{2m\left(E + \frac{1}{2}kx^2\right)} \approx x\sqrt{mk} + E\sqrt{\frac{m}{k}}\frac{1}{x},$$

and the asymptotic form of the Schrödinger-equation solution is

$$\psi = \text{const}\cdot\xi^{\pm i\varepsilon - 1/2}\exp\left(\pm i\xi^2/2\right),$$

where

$$\xi = x\left(\frac{mk}{\hbar^2}\right)^{1/4}, \qquad \varepsilon = \frac{E}{\hbar}\sqrt{\frac{m}{k}}.$$

We need the solution which, as $x \to +\infty$, contains only the wave transmitted through the barrier, travelling from left to right. Set

$$\psi = B\xi^{i\varepsilon - 1/2}\exp\left(i\xi^2/2\right) \qquad \text{as } x \to \infty, \tag{1}$$

$$\psi = (-\xi)^{-i\varepsilon - 1/2}\exp\left(-i\xi^2/2\right) + A(-\xi)^{i\varepsilon - 1/2}\exp\left(i\xi^2/2\right) \qquad \text{as } x \to -\infty. \tag{2}$$

The first term in (2) is the incident wave and the second the reflected wave; a wave propagates in the direction in which its phase increases. The relation between A and B follows from the validity of the asymptotic form of ψ throughout the sufficiently distant part of the complex ξ-plane. Follow the change in (1) along a semicircle of large radius ρ in the upper half-plane:

$$\xi = \rho e^{i\varphi}, \qquad i\xi^2 = \rho^2(-\sin 2\varphi + i\cos 2\varphi),$$

where φ runs from 0 to π. After this continuation, (1) becomes the second term of (2), with coefficient

$$A = B(e^{i\pi})^{i\varepsilon - 1/2} = -iBe^{-\pi\varepsilon}. \tag{3}$$

[17] The result also applies to transmission sufficiently near the top of any barrier $U(x)$ whose dependence on x is quadratic near its maximum.

On the portion $\pi/2 < \varphi < \pi$, where $|\exp(i\xi^2/2)|$ is exponentially large, an exponentially small quantity that should have produced the first term in (2) is lost[18].

With the normalization of the incident wave chosen in (2), conservation of particle number gives

$$|A|^2 + |B|^2 = 1. \tag{4}$$

Equations (3) and (4) yield the required transmission coefficient

$$D = |B|^2 = 1/(1 + e^{-2\pi\varepsilon}).$$

This result holds for every E. If the energy is negative and large in absolute value, $D \approx e^{-2\pi|\varepsilon|}$, in agreement with (50.5). For $E > 0$ the quantity

$$R = 1 - D = 1/(1 + e^{2\pi\varepsilon})$$

is the above-barrier reflection coefficient.

§ 51. Calculation of semiclassical matrix elements

Direct calculation of matrix elements of a physical quantity f using semiclassical wave functions presents serious difficulties. We assume that the energies of the states connected by the matrix element are not close, so the latter does not reduce to a Fourier component of f (§ 48). The difficulty arises because the exponential wave functions, with large imaginary exponents, make the integrand rapidly oscillating. Consider one-dimensional motion in $U(x)$ and, for simplicity, let the operator of f be just a function of x. Let ψ_1, ψ_2 be real wave functions for energies E_1, E_2, with $E_2 > E_1$ (Fig. 17). We must calculate

$$f_{12} = \int_{-\infty}^{+\infty} \psi_1 f \psi_2 \, dx. \tag{51.1}$$

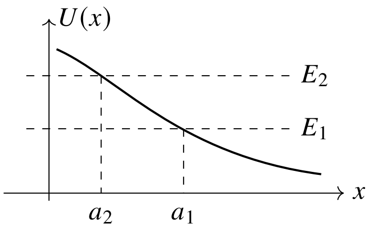

Fig. 17

By (47.5), on the two sides of $x = a_1$ and sufficiently far from it,

$$\psi_1 = \begin{cases} \dfrac{C_1}{\sqrt{|p_1|}} \exp\left(-\dfrac{1}{\hbar}\displaystyle\int_x^{a_1} |p_1|\, dx\right), & x < a_1, \\ \dfrac{2C_1}{\sqrt{p_1}} \cos\left(\dfrac{1}{\hbar}\displaystyle\int_{a_1}^{x} p_1\, dx - \dfrac{\pi}{4}\right), & x > a_1, \end{cases} \tag{51.2}$$

[18] Continuation through the lower half-plane would not determine A. On the portion $-\pi < \varphi < -\pi/2$ adjoining its left end, where ψ is given by (2), the term containing $\exp(i\xi^2/2)$ is exponentially small in comparison with the one containing $\exp(-i\xi^2/2)$.

and similarly for ψ_2, with 1 replaced by 2.

Substitution of these asymptotic expressions in (51.1) would give a wrong result. As shown below, the integral is exponentially small although its integrand is not. A relatively small change in the integrand may therefore change the order of the integral. This difficulty can be avoided as follows.

Write $\psi_2 = \psi_2^+ + \psi_2^-$ by resolving the cosine for $x > a_2$ into two exponentials. According to (50.2),

$$\psi_2^+ = \begin{cases} \dfrac{-iC_2}{2\sqrt{|p_2|}} \exp\left(\dfrac{1}{\hbar}\displaystyle\int_x^{a_2} |p_2|\, dx\right), & x < a_2, \\ \dfrac{C_2}{2\sqrt{p_2}} \exp\left(\dfrac{i}{\hbar}\displaystyle\int_{a_2}^{x} p_2\, dx - \dfrac{i\pi}{4}\right), & x > a_2, \end{cases} \tag{51.3}$$

and $\psi_2^- = (\psi_2^+)^*$.

Accordingly, (51.1) is the sum of two complex-conjugate integrals, $f_{12} = f_{12}^+ + f_{12}^-$. The integral

$$f_{12}^+ = \int_{-\infty}^{+\infty} \psi_1 f \psi_2^+\, dx$$

converges: although ψ_2^+ grows exponentially for $x < a_2$, ψ_1 decreases still faster for $x < a_1$, since $|p_1| > |p_2|$ throughout $x < a_2$.

Regard x as complex and displace the integration path into the upper half-plane. A positive imaginary increment of x produces a growing term in ψ_1 for $x > a_1$, but ψ_2^+ decreases faster because $p_2 > p_1$ there. The integrand therefore decreases.

The displaced path no longer crosses a_1, a_2, where the semiclassical approximation fails. Along the entire path we may use their asymptotic expressions in the upper half-plane:

$$\begin{aligned} \psi_1 &= \frac{C_1}{2[2m(U-E_1)]^{1/4}} \exp\left(\frac{1}{\hbar}\int_{a_1}^{x} \sqrt{2m(U-E_1)}\, dx\right), \\ \psi_2^+ &= \frac{-iC_2}{2[2m(U-E_2)]^{1/4}} \exp\left(-\frac{1}{\hbar}\int_{a_2}^{x} \sqrt{2m(U-E_2)}\, dx\right), \end{aligned} \tag{51.4}$$

where the roots are chosen positive on the real axis for $x < a_2$. Thus

$$\begin{aligned} f_{12}^+ = \frac{-iC_1C_2}{4\sqrt{2m}} \int \exp\bigg[&\frac{1}{\hbar}\int_{a_1}^{x} \sqrt{2m(U-E_1)}\, dx \\ &- \frac{1}{\hbar}\int_{a_2}^{x} \sqrt{2m(U-E_2)}\, dx\bigg] \frac{f(x)\, dx}{[(U-E_1)(U-E_2)]^{1/4}}. \end{aligned} \tag{51.5}$$

We seek to move the contour so as to reduce the exponential factor. Its exponent has an extremum only where $U(x) = \infty$; for $E_1 \neq E_2$ its derivative vanishes nowhere else. The only restriction on moving the contour upward is therefore the need to pass around the singular points of $U(x)$, which, by the general theory of linear differential equations, are also the singular points of $\psi(x)$.

The contour depends on the form of $U(x)$. If U has only one singular point x_0 in the upper half-plane, the path may be chosen as in Fig. 18. The immediate neighborhood of

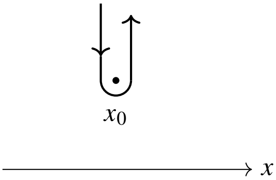

Fig. 18

the singularity gives the principal contribution, so $f_{12} = 2\,\mathrm{Re}\,f_{12}^{+}$ is mainly proportional to

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\,\mathrm{Im}\left[\int^{x_0}\sqrt{2m(E_2-U)}\,dx - \int^{x_0}\sqrt{2m(E_1-U)}\,dx\right]\right\}. \tag{51.6}$$

(L. D. Landau, 1932).[19] The lower limits may be any points in classically accessible regions and do not affect the imaginary parts. If $U(x)$ has several upper-half-plane singularities, x_0 in (51.6) is the one giving the exponent of smallest absolute magnitude.[20]

When E_1, E_2 are close, (51.6) simplifies and the matrix element reduces, by § 48, to a time Fourier component of the classical $f[x(t)]$. Put $E_{2,1} = E \pm \hbar\omega_{21}/2$ and expand in $\hbar\omega_{21}$:

$$f_{12} \sim \exp\left(-\omega_{21}\,\mathrm{Im}\int^{x_0}\frac{dx}{v}\right) = \exp(-\omega_{21}\,\mathrm{Im}\,\tau). \tag{51.6a}$$

Here

$$\tau = \int^{x_0}\frac{dx}{v}, \qquad v(x) = \sqrt{2(E-U(x))/m},$$

may be viewed as the *complex time* needed to reach x_0 in the complex x plane, and $v(x)$ as the corresponding "complex velocity." Formula (51.6a) indeed approximates the Fourier component when $\omega_{21}\,\mathrm{Im}\,\tau \gg 1$.

For motion in a central field the calculation is similar, but $U(r)$ is now the effective potential energy, the sum of potential and centrifugal terms, and differs for different l. For later use denote the two effective potentials generally by $U_1(r), U_2(r)$. The exponent in the integrand of (51.5) then has extrema not only at infinities of U_1 or U_2, but also at points satisfying

$$U_1(r) + U_2(r) = E_1 + E_2. \tag{51.7}$$

Thus in

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\,\mathrm{Im}\left[\int^{r_0}\sqrt{2m(E_2-U_2)}\,dr - \int^{r_0}\sqrt{2m(E_1-U_1)}\,dr\right]\right\} \tag{51.8}$$

the competing r_0 include both singular points and the roots of (51.7).

[19] Replacing the wave functions by their asymptotic expressions in deriving (51.5), (51.6) is legitimate because the order of the integral over the Fig. 18 contour is fixed by the order of its integrand; a relatively small change of the latter has no material effect.

[20] We assume that $f(x)$ itself has no singularities. Estimate (51.6) also assumes a prefactor of normal order. It may be anomalously small because of the particular problem. The simplest example is $f(x) = \text{const}$: the matrix element vanishes by orthogonality, which (51.6) does not reveal.

The central-field case also differs because the radial integral runs from 0, not $-\infty$, to $+\infty$:

$$f_{12} = \int_0^\infty \chi_1 f \chi_2 \, dr.$$

If the integrand is even, the range may formally be extended to the whole axis and nothing changes. This can occur when U_1, U_2 are even. Then χ_1, χ_2 are even or odd (see § 21),[21] and if $f(r)$ too is even or odd, their product may be even. If the integrand is not even, as necessarily happens if $U(r)$ is not even, the contour cannot be moved away from its initial point $r = 0$, and $r_0 = 0$ must also be included among the competing values in (51.8).

Problems

Problem

1. Calculate the semiclassical matrix elements, retaining only the exponential factor, in the field $U = U_0 e^{-\alpha x}$.

Solution. The only infinity of $U(x)$ is at $x \to -\infty$, so put $x_0 = -\infty$ in (51.6), extending the integrals to $+\infty$. Each separately diverges at $-\infty$; first integrate from $-x$ to $+\infty$ and then let $x \to \infty$. The result is

$$f_{12} \sim \exp\left[-\frac{\pi m}{\alpha \hbar}(v_2 - v_1)\right],$$

where $v_1 = \sqrt{2E_1/m}$ and $v_2 = \sqrt{2E_2/m}$ are the asymptotic free velocities as $x \to \infty$.

Problem

2. Do the same in the Coulomb field $U = \alpha/r$ for transitions between states with $l = 0$.

Solution. The only singular point of $U(r)$ is $r = 0$. The corresponding integral was evaluated in Problem 2 of § 50. Formula (51.8) gives

$$f_{12} \sim \exp\left[\frac{\pi\alpha}{\hbar}\left(\frac{1}{v_2} - \frac{1}{v_1}\right)\right].$$

Problem

3. Do the same for an anharmonic oscillator with

$$U(x) = \frac{m\omega^2}{2}x^2 + \beta x^4$$

under the condition

$$\hbar\omega \ll E_1, E_2 \ll \frac{m^2\omega^4}{\beta}. \tag{1}$$

[21] For even $U(r)$, the radial wave function $R(r)$ is even (odd) for even (odd) l, as follows from $R \sim r^l$ at small r.

SOLUTION. Extension of the argument to finite motion shows that (51.6) still applies. Choose $x_0 \to \pm\infty$; both points contribute at the same order. Thus

$$f_{12} \sim \exp\left\{-\frac{1}{\hbar}\left[\int_{a_2}^{-\infty} \sqrt{2m(U-E_2)}\,dx - \int_{a_1}^{-\infty} \sqrt{2m(U-E_1)}\,dx\right]\right\}.$$

Under condition (1), the main contribution comes from

$$\sqrt{\frac{E_1}{m\omega^2}}, \sqrt{\frac{E_2}{m\omega^2}} \ll |x| \ll \sqrt{\frac{m\omega^2}{\beta}}, \tag{2}$$

where

$$\frac{m\omega^2}{2}x^2 \gg E_1, E_2, \beta x^4.$$

Expanding the exponent in $E_{1,2}/U$, canceling the zeroth-order terms, and neglecting βx^4, we obtain

$$f_{12} \sim \exp\left(-\frac{E_2}{\hbar\omega}\int \frac{d|x|}{|x|} + \frac{E_1}{\hbar\omega}\int \frac{d|x|}{|x|}\right).$$

Cut off the logarithmic integrals at the boundaries of (2): above at $x \sim \sqrt{m\omega^2/\beta}$ and below at $x \sim a_{2,1} \sim \sqrt{E_{2,1}/(m\omega^2)}$. Then

$$f_{12} \sim \exp\left(-\frac{E_2}{2\hbar\omega}\ln\frac{m^2\omega^4}{\beta E_2} + \frac{E_1}{2\hbar\omega}\ln\frac{m^2\omega^4}{\beta E_1}\right).$$

Introducing the state numbers

$$n_1 \approx E_1/\hbar\omega, \qquad n_2 \approx E_2/\hbar\omega,$$

write the result as

$$f_{12} \sim \frac{n_2^{n_2/2}}{n_1^{n_1/2}}\left(\frac{\beta\hbar}{m\omega^3}\right)^{(n_2-n_1)/2}.$$

Since large x are essential to the solution, this result applies when $f(x)$ does not grow too rapidly at infinity. If $f(x)$ is a polynomial, its degree must be small compared with $n_2 - n_1$.

§ 52. Transition probability in the semiclassical case

Passage through a potential barrier exemplifies a process that cannot occur at all in classical mechanics. In the semiclassical case, the probability of such processes is exponentially small. The corresponding exponent can be found as follows.

Consider a transition of some system from one state to another. We solve the appropriate classical equations of motion and determine the transition "trajectory." Since the process is impossible in classical mechanics, this trajectory is complex. In general, the "transition point" q_0, at which the system formally passes from one state to the other, is complex as well; its position is fixed by the classical conservation laws. We next calculate the action $S_1(q_1, q_0) + S_2(q_0, q_2)$: in the first state the system moves from an arbitrary initial position q_1 to the “transition point” q_0, and in the second state it moves from q_0 to the final position q_2. The probability of the process is then

$$w \sim \exp\left\{-\frac{2}{\hbar}\,\mathrm{Im}[S_1(q_1, q_0) + S_2(q_0, q_2)]\right\}. \tag{52.1}$$

If several positions are possible for the “transition point,” the one for which the exponent in (52.1) has the smallest absolute value must be selected (this value must, of course, still be sufficiently large for (52.1) to be applicable at all)[22].

Formula (52.1) agrees with the rule obtained in the preceding section for calculating semiclassical matrix elements. It should be stressed, however, that it would be wrong to calculate the pre-exponential factor in the probability of a transition of this kind by squaring the corresponding matrix element.

The method of *complex classical trajectories* based on (52.1) is quite general and applies to transitions in systems having any number of degrees of freedom (L. D. Landau, 1932). When the transition point is real but lies in a classically forbidden region, formula (52.1), in the simplest case of one-dimensional motion, coincides with expression (50.5) for the probability of passage through a potential barrier.

Above-barrier reflection. Let us apply (52.1) to the one-dimensional problem of above-barrier reflection, in which the energy of the particle exceeds the height of the barrier. Here q_0 is to be understood as the complex coordinate x_0 of the “stopping point” where the particle reverses its direction of motion; thus, x_0 is a complex root of $U(x) = E$. We shall show that in this case the reflection coefficient can also be calculated more accurately, including its pre-exponential factor.

Once again (as in § 50), we have to relate the wave functions far on the right-hand side of the barrier (the transmitted wave) to those far on its left-hand side (the incident and reflected waves). We may do this without difficulty by proceeding as in ß 47 and 50 and regarding ψ as a function of the complex variable x.

Write the transmitted wave as

$$\psi_+ = \frac{1}{\sqrt{p}} \exp\left(\frac{i}{\hbar}\int_{x_1}^{x} p\,dx\right)$$

(where x_1 is any point on the real axis), and follow its change on going through the upper half-plane along a path C which passes around the turning point x_0 at a sufficient distance (Fig. 19). The final portion of this path must lie entirely in a region sufficiently far to the left that, along it, the error in the approximate semiclassical wave function of the incident wave is smaller than the small quantity ψ_- being sought. Going around x_0 changes the sign of the root $\sqrt{E - U(x)}$. Consequently, upon returning to the real axis, ψ_+ becomes a wave ψ_- traveling to the left, that is, the reflected wave[23]. Since the amplitudes of the incident and transmitted waves may be taken to coincide, the required reflection coefficient R is simply

[22] If the potential energy of the system itself has singular points, these points too must be included among the competing values of q_0.

[23] A path passing below x_0 (for example, simply along the real axis itself) instead transforms ψ_+ into the incident wave.

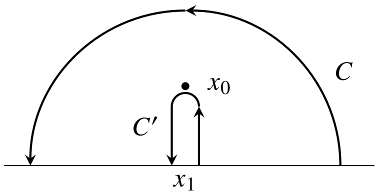

Fig. 19

the ratio of the squared moduli of ψ_- and ψ_+:

$$R = \left|\frac{\psi_-}{\psi_+}\right|^2 = \exp\left(-\frac{2}{\hbar}\,\mathrm{Im}\int_C p\,dx\right). \tag{52.2}$$

Once this formula has been obtained, the integration path in the exponent may be deformed in any manner. If it is changed into the path C' shown in Fig. 19, the integral reduces to twice the integral along the path from x_1 to x_0, giving

$$R = e^{-4\sigma(x_1,x_0)/\hbar}, \qquad \sigma(x_1,x_0) = \mathrm{Im}\int_{x_1}^{x_0} p(x)\,dx; \tag{52.3}$$

since $p(x)$ is real everywhere on the real axis, the choice of x_1 is immaterial[24]. Notice that the pre-exponential factor in (52.3) is equal to unity (V. L. Pokrovskii, S. K. Savvinykh, F. R. Ulinich, 1958)[25].

As stated above, among all possible values of x_0 one must choose the value for which the exponent in (52.3) has the smallest absolute magnitude, while that magnitude must still be sufficiently large compared with unity. (Only points x_0 for which $\sigma > 0$, that is, points in the upper half-plane, are of course to be considered.) It is also assumed that if the potential energy $U(x)$ itself has singular points in the upper half-plane, the values of the integral $\sigma(x_1, x_0)$ for these points are large[26]. Otherwise, such a point determines the exponent, but the pre-exponential factor will no longer be the one in (52.3). This last condition is inevitably violated as the energy E increases if $U(x)$ becomes infinite somewhere

in the upper half-plane. A stage is reached at which the point x_0 where $U = E$ is so close to the point x_∞ where $U = \infty$ that the two make comparable contributions to the reflection coefficient (the integral $\sigma(x_\infty, x_0) \sim 1$), and formula (52.3) ceases to apply. In

[24] In some cases, not only the amplitude relation but also the phase relation between the incident and reflected waves is of interest. These relations are characterized by the reflection amplitude, expressed in terms of the coefficients α and β introduced in § 25. The reasoning given above readily shows, in particular, that the reflection amplitude for a wave incident from the left is

$$\left(-\frac{\beta^*}{\alpha^*}\right) = -i\exp\left[\frac{2i}{\hbar}\left(\int_{x_1}^{x_0} p\,dx + p_1 x_1\right)\right], \qquad x_1 \to -\infty.$$

(The factor $(-i)$ is associated with the change in phase of the pre-exponential factor on going around the branch point; compare § 47.)

[25] The derivation of this result presented here is due to L. D. Landau (1961).

[26] The contour C in Fig. 19 must be taken below the singular points of $U(x)$.

the limiting regime where E is so large that the indicated integral is small relative to unity, perturbation theory becomes applicable (see Problem 2)[27].

Problems

Problem

1. In the quasiclassical approximation, to exponential accuracy, determine the probability of deuteron breakup in a collision with a heavy nucleus treated as a fixed centre of a Coulomb field (E. M. Lifshitz, 1939).

Solution. Collisions with zero orbital angular momentum make the largest contribution to the reaction probability. In the quasiclassical approximation these are "head-on" collisions, in which the motion of the particles reduces to one-dimensional motion.

Let E denote the energy of the deuteron, measured in units of ε, the binding energy of the proton and neutron within it; E_n and E_p denote the energies of the neutron and proton after their release (in the same units). We also introduce the dimensionless coordinate $q = r/(Ze^2/\varepsilon)$ (the nuclear charge is Ze), and write q_0 for its value (complex in general) at the "transition point," i.e. at the "instant of deuteron breakup." We represent E_n, E_p, and E in the form

$$E_n = v_n^2/2, \qquad E_p = v_p^2/2 + 1/q_0, \qquad E = v_d^2 + 1/q_0; \tag{1}$$

v_n, v_p, and v_d are the particle "velocities" at the instant of breakup, measured in units of $\sqrt{\varepsilon/m}$ (where m is the nucleon mass). Here v_n is real and equals the velocity of the released neutron, whereas v_p and v_d are complex. Conservation of energy and momentum at the transition point gives

$$E_p + E_n = E - 1, \qquad v_p + v_n = 2v_d, \tag{2}$$

whence

$$v_p = 2i + v_n, \qquad v_d = i + v_n, \qquad 1/q_0 = E + 1 - v_n^2 + 2iv_n.$$

Before the transition, the action of the system corresponds to motion of the deuteron in the nuclear field up to the breakup point; its imaginary part is

$$\begin{aligned} \operatorname{Im} S_1 &= Ze^2\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\int_\infty^{q_0}\sqrt{4\left(E-\frac{1}{q}\right)}\,dq = \\ &= Ze^2\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\left\{2q_0v_d - \frac{2}{\sqrt{E}}\cosh^{-1}\sqrt{q_0E}\right\}. \end{aligned} \tag{3}$$

After the transition, the action corresponds to motion of the neutron and proton away from the breakup point:

[27] The intermediate regime was considered by V. L. Pokrovskii and I. M. Khalatnikov (Zh. Eksp. Teor. Fiz. 1961. Vol. 40. P. 1713).

$$\begin{aligned}\operatorname{Im} S_2 &= Ze^2\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\left\{\int_{q_0}^{\infty} v_n\,dq + \int_{q_0}^{\infty}\sqrt{2\left(E_p - \frac{1}{q}\right)}\,dq\right\} = \\ &= Ze^2\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\left\{-v_n q_0 - v_p q_0 + \sqrt{\frac{2}{E_p}}\cosh^{-1}\sqrt{q_0 E_p}\right\}.\end{aligned} \tag{4}$$

According to (52.1), the probability of the process is

$$w \sim \exp\left\{-\frac{2Ze^2}{\hbar}\sqrt{\frac{m}{\varepsilon}}\operatorname{Im}\left[\sqrt{\frac{2}{E_p}}\cosh^{-1}\sqrt{q_0E_p} - \frac{2}{\sqrt{E}}\cosh^{-1}\sqrt{q_0 E}\right]\right\}. \tag{5}$$

In accordance with the origin of the first and second $\cosh^{-1}$ terms in the square brackets from expressions (4) and (3), the signs of their imaginary parts must agree with the signs of $\operatorname{Im} v_p$ and $\operatorname{Im} v_d$, respectively (the signs of the latter quantities in solving equations (2) were chosen so that $\operatorname{Im}(S_1 + S_2) > 0$ results).

Because w depends exponentially on E_n, the total breakup probability (including all values of E_n and $E_p = E - 1 - E_n$) is determined by the smallest absolute value of the exponent as a function of E_n. Analysis shows that this value is attained as $E_n \to 0$. Then $q_0 = 1/(E+1)$, and (5) gives

$$w \sim \exp\left\{-\frac{2Ze^2}{\hbar}\sqrt{\frac{m}{\varepsilon}}\left[\sqrt{\frac{2}{E-1}}\arccos\sqrt{\frac{E-1}{E+1}} - \frac{2}{\sqrt{E}}\arccos\sqrt{\frac{E}{E+1}}\right]\right\}.$$

The condition for this formula to apply is that the exponent be large compared with unity.

By calculating the imaginary part of the action $S = S_1 + S_2$ at nonzero values of E_n, one can find the energy distribution of the released particles. Near $E_n = 0$ we have[28]

$$\operatorname{Im} S(E_n) - \operatorname{Im} S(0) \approx E_n \left(\frac{d\operatorname{Im} S}{dE_n}\right)_{E_n=0}.$$

Evaluation of the derivative gives

$$\frac{dw}{dE_n} \sim \exp\left\{-\frac{2Ze^2}{\hbar}\sqrt{\frac{m}{\varepsilon}}E_n\left[\frac{3-E}{(E-1)(E+1)^2} + \frac{1}{\sqrt{2(E-1)^3}}\arccos\sqrt{\frac{E-1}{E+1}}\right]\right\}.$$

Problem

2. Determine the coefficient of over-barrier reflection at particle energies for which perturbation theory is applicable.

Solution. The result follows from formula (43.1), in which the initial and final wave functions are plane waves propagating in opposite directions and normalized, respectively, to unit flux density and to a delta function of momentum divided by $2\pi\hbar$.

[28] At $E_n = 0$ the function $\operatorname{Im} S(E_n)$ has a cusp, from which it increases in both directions—for positive and negative E_n (values $E_n < 0$ would correspond to capture of the neutron by the nucleus).

Here $d\nu = dp'/2\pi\hbar$, where p' is the momentum after reflection. Carrying out the integration over dp' in (43.1), with the delta function taken into account, gives

$$R = \frac{m^2}{\hbar^2 p^2}\left|\int_{-\infty}^{\infty} U(x) e^{2ipx/\hbar}\, dx\right|^2. \tag{1}$$

This formula is valid when the condition for applicability of perturbation theory is satisfied: $Ua/\hbar v \ll 1$, where a is the width of the barrier (see the note on p. 205), and at the same time $pa/\hbar \lesssim 1$. The latter condition ensures that the dependence $R(p)$ is not exponential; otherwise, the applicability of formula (1) requires further investigation.

Problem

3. Determine the coefficient of over-barrier reflection from a quasiclassical barrier when the function $U(x)$ has a kink at $x = x_0$.

Solution. If the function $U(x)$ has a singularity at some real x, the reflection coefficient is governed mainly by the field near that point. To calculate it, perturbation theory may be applied formally without requiring its condition of applicability to hold for every x; it is enough that the quasiclassical condition be satisfied. We then arrive at formula (1) of Problem 2, with the sole difference that the value of the function $p(x)$ at $x = x_0$ must appear there in place of the incident particle's momentum.

Choosing the kink as the point $x = 0$, we have near it

$$U = -F_1 x \quad \text{for } x > 0, \qquad U = -F_2 x \quad \text{for } x < 0,$$

with different values of F_1 and F_2. The integration over dx is performed by introducing the damping factor $e^{\pm\lambda x}$ into the integrand (after which $\lambda \to 0$ is taken). The result is

$$R = \frac{m^2\hbar^2}{16 p_0^6}(F_2 - F_1)^2,$$

where $p_0 = p(0)$.

§ 53. Transitions Caused by Adiabatic Perturbations

We noted already in § 41 that, in the limit of an arbitrarily slowly time-varying perturbation, the probability for a system to pass from one state to another tends to zero. We shall now examine this question quantitatively by calculating the transition probability produced by a slowly varying (adiabatic) perturbation (L. D. Landau, 1961).

Let the system's Hamiltonian be a slowly varying function of time, approaching definite limits as $t \to \pm\infty$. Let, further, $\psi_n(q,t)$ and $E_n(t)$ be the energy eigenfunctions and eigenvalues (with time serving as a parameter) obtained by solving the Schrödinger equation $\widehat{H}(t)\psi_n = E_n\psi_n$. Since the time variation of $\widehat{H}$ is adiabatic, the time dependences of E_n and ψ_n are likewise slow. Our problem is to determine the probability w_{21} that the system will be found in a given state ψ_2 as $t \to +\infty$, provided that it occupied state ψ_1 as $t \to -\infty$.

The slowness of the perturbation makes the "transition process" last a long time, so that the change in action over this interval (given by $-\int E(t)dt$) is large. In this sense the problem is semiclassical. The transition probability sought is governed essentially by those values $t = t_0$ for which

$$E_1(t_0) = E_2(t_0) \tag{53.1}$$

and which may be viewed as counterparts of the "transition instant" in classical mechanics (cf. § 52). Such a transition is, of course, classically impossible in reality; this appears in the complex character of the roots of equation (53.1). We must therefore investigate the solutions of the Schrödinger equation at complex values of the parameter t, near the point $t = t_0$ where two energy eigenvalues become equal.

As will be seen, the eigenfunctions ψ_1, ψ_2 vary strongly with t near this point. To determine this dependence, first introduce linear combinations of them, denoted by φ_1, φ_2, that satisfy

$$\int \varphi_1^2 dq = \int \varphi_2^2 dq = 0, \qquad \int \varphi_1\varphi_2 dq = 1. \tag{53.2}$$

This can always be achieved by a suitable choice of complex coefficients (functions of t). The functions φ_1, φ_2 are no longer singular at $t = t_0$.

We now seek the eigenfunctions as linear combinations

$$\psi = a_1\varphi_1 + a_2\varphi_2. \tag{53.3}$$

It should be borne in mind that, for complex values of the "time" t, the operator $\widehat{H}(t)$ depending on it (of the form (17.4)) still equals its transpose ($\widehat{H} = \widetilde{H}$), but is no longer Hermitian ($\widehat{H} \neq \widetilde{H}^*$), because the potential energy $U(t) \neq U(t)^*$.

Substitute (53.3) into the Schrödinger equation, multiply it on the left first by φ_1 and then by φ_2, and integrate over dq. With the notation

$$H_{ik}(t) = \int \varphi_i \widehat{H} \varphi_k, dq \tag{53.4}$$

and using $H_{12} = H_{21}$, which follows from the stated property of the Hamiltonian, we obtain the system

$$\begin{aligned} H_{11}a_1 + H_{12}a_2 &= Ea_2, \\ H_{12}a_1 + H_{22}a_2 &= Ea_1. \end{aligned} \tag{53.5}$$

Its solvability condition gives $(H_{12} - E)^2 = H_{11}H_{22}$, whose roots determine the energy eigenvalues:

$$E = H_{12} \pm \sqrt{H_{11}H_{22}}. \tag{53.6}$$

Equation (53.5) then gives

$$a_2/a_1 = \pm\sqrt{H_{11}/H_{22}}. \tag{53.7}$$

It follows from (53.6) that, for the two eigenvalues to coincide at $t = t_0$, either H_{11} or H_{22} must vanish there; suppose it is H_{11}. At a regular point a function generally vanishes in proportion to $t - t_0$. Hence

$$E(t) - E(t_0) = \pm \operatorname{const} \cdot \sqrt{t - t_0}, \tag{53.8}$$

so $E(t)$ has a branch point at $t = t_0$. At the same time $a_2 \sim \sqrt{t - t_0}$, and thus only one eigenfunction exists at $t = t_0$, coinciding with φ_1.

We now see that the problem posed is formally identical to the above-barrier reflection problem treated in § 52. Here the wave function $\Psi(t)$ is "semiclassical in time" (instead of semiclassical in the coordinate as in § 52). We must find the term $c_2\psi_2 \exp(-iE_2t/\hbar)$ in the wave function as $t \to +\infty$, given that $\Psi(t) = \psi_1 \exp(-iE_1t/\hbar)$ as $t \to -\infty$. This corresponds to finding the reflected wave at $x \to -\infty$ from the transmitted wave at $x \to +\infty$; the desired transition probability is $w_{21} = |c_2|^2$. The action $S = -\int E(t)dt$ is now an integral over time of a function having complex branch points, just as $p(x)$ had complex branch points in the integral $\int p, dx$. The present problem is therefore solved by continuing around a path in the complex t plane (from large negative to large positive values), in complete analogy with the continuation in the complex x plane used in § 52; we shall not repeat that reasoning.

Suppose that $E_2 > E_1$ on the real axis. The path must then pass through the upper half of the complex t plane (where the ratio $e^{-iE_2t/\hbar}/e^{-iE_1t/\hbar}$ increases). The result is the formula analogous to (52.2):

$$w_{21} = \exp\left(\frac{2}{\hbar}\,\mathrm{Im}\int_{C'} E(t)\,dt\right), \tag{53.9}$$

where the integral follows the contour shown in Fig. 19, but from left to right.

On the left branch of this contour $E = E_1$, while on its right branch $E = E_2$. Formula (53.9) may consequently be written as

$$w_{21} = \exp\left(-2\,\mathrm{Im}\int_{t}^{t_0} \omega_{21}(t)\,dt\right), \tag{53.10}$$

where $\omega_{21} = (E_2 - E_1)/\hbar$ and t is any point on the real t axis. For t_0 one must choose, among the roots of equation (53.1) in the upper half-plane, the one for which the absolute value of the exponent in (53.10) is smallest[29]. Moreover, direct transition from state 1 to state 2 may compete with "transition paths" passing through various intermediate states, whose probabilities are expressed by analogous formulas. Thus, for the "path" $1 \to 3 \to 2$, the integral in (53.10) is replaced by the sum

$$\int^{t_0^{(31)}} \omega_{31}(t)\,dt + \int^{t_0^{(23)}} \omega_{23}(t)\,dt,$$

whose upper limits are the "crossing points" of the terms $E_1(t)$, $E_3(t)$ and $E_2(t)$, $E_3(t)$, respectively. This result follows by using a contour enclosing both complex points[30].

[29] The competing values of t_0 must also include points at which $E(t)$ becomes infinite (although the pre-exponential factor in (53.10) would be different for such points).

[30] Intermediate states belonging to a continuous spectrum require separate treatment.

Problem

Determine the change in the adiabatic invariant of a classical oscillator governed by

$$\frac{d^2x}{dt^2} + \omega^2(t)x = 0 \tag{1}$$

when the frequency $\omega(t)$ varies slowly from the value ω_1 as $t \to -\infty$ to ω_2 as $t \to \infty$ (A. M. Dykhne, 1960).

Solution. Equation (1) is obtained from the Schrödinger equation by the changes of notation

$$\psi \to x, \qquad x \to t; \qquad p(x)/\hbar = k(x) \to \omega(t),$$

after which the problem is formally equivalent to reflection from a potential barrier, considered in § 25. The calculation of the change in the adiabatic invariant can thus be reduced to calculating the reflection amplitude.

Write the solution of (1) as $t \to \mp\infty$ in the form

$$x = A_1 e^{i\omega_1 t} + A_1^* e^{-i\omega_1 t}, \quad t \to -\infty,$$
$$x = A_2 e^{i\omega_2 t} + A_2^* e^{-i\omega_2 t}, \quad t \to \infty.$$

According to (25.6),

$$A_2 = \alpha A_1 + \beta A_1^*. \tag{2}$$

For an oscillator the adiabatic invariant is E/ω, so that

$$I_1 = m\omega_1\overline{x^2} = 2m\omega_1|A_1|^2, \qquad I_2 = 2m\omega_2|A_2|^2,$$

or, on substituting (2),

$$I_2 = 2m\omega_2[(|\alpha|^2 + |\beta|^2)|A_1|^2 + 2\,\mathrm{Re}(\alpha\beta^* A_1^2)].$$

Using relation (25.7), which in our notation reads $|\alpha|^2 = |\beta|^2 + \omega_1/\omega_2$, we find

$$I_2 - I_1 = 4m\omega_2[|\beta|^2|A_1|^2 + \mathrm{Re}(\alpha\beta^* A_1^2)]. \tag{3}$$

Slow variation of $\omega(t)$ in the case under consideration corresponds to the semiclassical situation of the preceding section in the barrier-reflection problem. In that situation β is exponentially small and $|\alpha|^2 \approx \omega_1/\omega_2$. (It is assumed that $\omega^2(t)$ has neither singularities nor zeros on the real t axis.) The method given in the preceding section for calculating the reflection amplitude yields the estimate

$$\Delta I = I_2 - I_1 \sim |\beta| \sim \exp\left(-2\,\mathrm{Im}\int_{t_1}^{t_0} \omega(t)\,dt\right),$$

where t_0 is the singular point in the upper half of the t plane that makes the largest contribution to ΔI. This formula agrees, for the harmonic oscillator considered here, with the results of § 51 (see Vol. I). When $\omega^2(t)$ has a simple zero in the upper half-plane, the formulas of the preceding section also determine the pre-exponential factor. (See the note on p. 241.)

Notice that the second—and principal—term in (3) depends on the initial phase of the oscillations. Averaging over this phase makes it vanish, so that

$$\overline{\Delta I} \approx 2RI_1,$$

where $R \approx (\omega_2/\omega_1)|\beta|^2$ is the "reflection coefficient."

CHAPTER VIII

Spin

§ 54. Spin

In classical mechanics as well as in quantum mechanics, conservation of angular momentum follows from the isotropy of space for a closed system. This already reveals the connection between angular momentum and rotational symmetry. In quantum mechanics, however, that connection becomes especially fundamental: it essentially supplies the principal content of the notion of angular momentum. This is all the more so because the classical definition of a particle's angular momentum as the product $\mathbf{r} \times \mathbf{p}$ loses its direct meaning here, since the position vector and momentum cannot be measured simultaneously.

As we saw in § 28, the values of l and m determine the angular dependence of a particle's wave function and, with it, all the rotational symmetry properties of that function. In their most general form these properties are specified by stating how wave functions transform when the coordinate system is rotated.

The wave function ψ_{LM} of a system of particles, with prescribed angular momentum L and projection M, is unchanged[1] only by rotations of the coordinate system about the z axis. After any rotation that alters the direction of this axis, the projection of the angular momentum on z no longer has a definite value. In the new coordinate axes the wave function must therefore become, in general, a superposition (linear combination) of the $2L + 1$ functions associated with the possible values of M at fixed L. Equivalently, rotations of the coordinate system transform the $2L + 1$ functions ψ_{LM} into one another[2]. The transformation law—that is, the superposition coefficients as functions of the angles through which the coordinate axes are rotated—is fixed completely by the value of L. Angular momentum thus takes on the meaning of a quantum number that classifies the states of a system by their transformation properties under rotations of the coordinate system. This aspect of angular momentum is especially important in quantum mechanics because it does not depend directly on an explicit angular dependence of the wave functions: their mutual transformation law can be formulated by itself, without referring to such a dependence.

Consider a composite particle (an atomic nucleus, for example) that is at rest as a whole and occupies a definite internal state. Besides having a definite internal energy, it has angular momentum of definite magnitude L, arising from the motion of the particles within it; this angular momentum can still have $2L + 1$ distinct orientations in space. In other words, when treating the motion of the composite particle as a whole, we must assign to it, in addition to its coordinates, one discrete variable: the projection of its

[1] Apart from an inessential phase factor.

[2] In mathematical terminology, these functions realize an *irreducible representation* of the rotation group. The number of functions transformed into one another is called the *dimension of the representation*; it is understood that no choice of other linear combinations of these functions can reduce this number.

internal angular momentum on some selected direction in space.

With the meaning of angular momentum just described, however, its origin ceases to be essential. We are therefore led naturally to the idea of an "intrinsic" angular momentum that must be assigned to a particle regardless of whether it is "composite" or "elementary."

Thus quantum mechanics requires an elementary particle to possess a certain "intrinsic" angular momentum unrelated to its motion through space. This property of elementary particles is specifically quantum mechanical (it vanishes in the limit $\hbar \to 0$) and admits no classical interpretation in principle.[3]

The intrinsic angular momentum of a particle is called its *spin*, in contrast to the angular momentum associated with the particle's motion in space, which is called *orbital angular momentum*.[4] The particle in question may be elementary, or it may be composite but behave as an elementary object within the class of phenomena under consideration (an atomic nucleus, for example). We shall denote the particle's spin, measured like orbital angular momentum in units of $\hbar$, by s.

For a particle with spin, its wave function must specify not only the probabilities of its possible positions in space but also the probabilities of the possible orientations of its spin. In other words, the wave function must depend not merely on three continuous variables—the particle's coordinates—but also on one discrete *spin variable*. This variable gives the value of the spin projection on a selected spatial direction (the z axis) and runs through a finite set of discrete values, which we shall subsequently denote by σ.

Let $\psi(x, y, z; \sigma)$ be such a wave function. In substance it is a set of several different functions of the coordinates, one for each value of σ; we shall call these functions the *spin components* of the wave function. The integral

$$\int |\psi(x, y, z; \sigma)|^2 \, dV$$

then gives the probability that the particle has the specified value of σ. The probability of finding the particle in the volume element dV with an arbitrary value of σ is

$$dV \sum_{\sigma} |\psi(x, y, z; \sigma)|^2.$$

When the quantum-mechanical spin operator is applied to a wave function, it acts specifically on the spin variable σ. In other words, it effects some transformation among the components of the wave function. Its explicit form will be found below. Even without that form, however, very general considerations suffice to show that $\hat{s}_x$, $\hat{s}_y$, and $\hat{s}_z$ obey the same commutation relations as the orbital-angular-momentum operators.

An angular-momentum operator is essentially identical with the operator of an infinitesimal rotation. In deriving the orbital-angular-momentum operator in § 26, we considered how a rotation acts on a function of the coordinates. That derivation is

[3] In particular, it would be entirely meaningless to picture the "intrinsic" angular momentum of an elementary particle as resulting from its rotation "about its own axis."

[4] The physical idea that the electron has an intrinsic angular momentum was proposed by Uhlenbeck and Goudsmit (G. Uhlenbeck, S. Goudsmit, 1925). Spin was introduced into quantum mechanics by Pauli (W. Pauli, 1927).

inapplicable to spin angular momentum, since the spin operator acts on the spin variable rather than on the coordinates. To obtain the required commutation relations, we must consequently consider an infinitesimal rotation in its general form, as a rotation of the coordinate system. Applying infinitesimal rotations in succession, first about the x and y axes and then about the same axes in reverse order, we readily find by direct computation that the difference between the outcomes of these two operations is equivalent to an infinitesimal rotation about the z axis (through an angle equal to the product of the rotation angles about the x and y axes). We shall omit these elementary calculations here. They lead once more to the standard commutation relations for the component operators of angular momentum; the same relations must therefore hold for the spin operators:

$$\{\hat{s}_x, \hat{s}_y\} = i\hat{s}_z \tag{54.1}$$

together with all their physical consequences.

The commutation relations (54.1) allow the possible values of the magnitude and components of spin to be determined. The entire derivation in § 27 (equations (27.7)–(27.9)) relied only on the commutation relations and is therefore fully applicable here, with s replacing L in those equations. It follows from equations (27.7) that the eigenvalues of a spin projection form a sequence whose adjacent terms differ by unity. We can no longer assert, however, that these values themselves must be integers, as was the case for the orbital-angular-momentum projection L_z. The argument at the beginning of § 27 does not apply here, since it rests on the expression (26.14) for the operator $\hat{l}_z$, which is specific to orbital angular momentum.

Furthermore, the sequence of eigenvalues of s_z is bounded above and below by values equal in magnitude and opposite in sign, which we denote by $\pm s$. The difference $2s$ between the largest and smallest values of s_z must be an integer or zero. Hence s may take the values $0, 1/2, 1, 3/2, \ldots$

Thus the eigenvalues of the square of the spin are

$$s^2 = s(s+1), \tag{54.2}$$

where s is either an integer (zero included) or a half-integer. For a given s, the spin component s_z can take the values $s, s-1, \ldots, -s$, a total of $2s+1$ values. Correspondingly, the wave function of a particle with spin s has $2s+1$ components.[5]

Experiment shows that most elementary particles—electrons, positrons, protons, neutrons, μ mesons, and all the hyperons (Λ, Σ, Ξ)—have spin $1/2$. There are also elementary particles—the π mesons and K mesons—with spin 0.

The total angular momentum of a particle is the sum of its orbital angular momentum $\mathbf{l}$ and its spin $\mathbf{s}$. Since their operators act on functions of entirely different variables, they of course commute with each other.

The eigenvalues of the total angular momentum

$$\mathbf{j} = \mathbf{l} + \mathbf{s} \tag{54.3}$$

[5] For each species of particle, s is a prescribed number. Consequently, in passing to the classical limit ($\hbar \to 0$), the spin angular momentum $\hbar s$ tends to zero. This argument is not meaningful for orbital angular momentum, because l can assume arbitrary values. The classical limit requires $\hbar$ to tend to zero and l simultaneously to infinity in such a way that the product $\hbar l$ remains finite.

are determined by the same "vector-model" rule as the sum of the orbital angular momenta of two distinct particles (§ 31). Namely, for prescribed l and s, the total angular momentum can have the values $l + s, l + s - 1, \dots, |l - s|$. Thus, for an electron (spin 1/2) with nonzero orbital angular momentum l, the total angular momentum can be $j = l \pm 1/2$; when $l = 0$, there is of course only the single value $j = 1/2$.

The total-angular-momentum operator $\mathbf{J}$ of a system of particles is the sum of the angular-momentum operators $\mathbf{j}$ for the individual particles, so that its values are again fixed by the vector-model rules. The angular momentum $\mathbf{J}$ can be written

$$\mathbf{J} = \sum_a \mathbf{j}_a = \mathbf{L} + \mathbf{S}, \qquad \mathbf{L} = \sum_a \mathbf{l}_a, \quad \mathbf{S} = \sum_a \mathbf{s}_a, \tag{54.4}$$

where $\mathbf{S}$ may be called the total spin and $\mathbf{L}$ the total orbital angular momentum of the system.

Notice that a half-integral (or integral) total spin of a system entails a half-integral (or integral) total angular momentum, since orbital angular momentum is always integral. In particular, a system composed of an even number of identical particles necessarily has integral total spin and hence integral total angular momentum as well.

The operators of a particle's total angular momentum $\mathbf{j}$ (or of the total angular momentum $\mathbf{J}$ of a system of particles) obey the same commutation rules as the operators of orbital angular momentum and spin. These are, in fact, the general commutation rules valid for every angular momentum. Equation (27.13), obtained from those rules for angular-momentum matrix elements, is likewise valid for any angular momentum, provided that its matrix elements are defined with respect to that same angular momentum's eigenfunctions. With the corresponding changes of notation, formulas (29.7)–(29.10) for the matrix elements of arbitrary vector quantities also remain valid.

Problem

A particle of spin 1/2 is in a state with the definite value $s_z = 1/2$. Find the probabilities of the possible values of the spin projection on an axis z' inclined at an angle θ to the z axis.

Solution.

The mean spin vector $\bar{\mathbf{s}}$ is evidently directed along the z axis and has magnitude 1/2. Projecting it onto the z' axis, we find that the mean spin along z' is $\bar{s}_{z'} = (1/2)\cos\theta$. On the other hand, $\bar{s}_{z'} = (1/2)(w_+ - w_-)$, where $w_\pm$ are the probabilities of the values $s_{z'} = \pm 1/2$. Using $w_+ + w_- = 1$ as well, we obtain

$$w_+ = \cos^2(\theta/2), \qquad w_- = \sin^2(\theta/2).$$

§ 55. The spin operator

In the remainder of this chapter we shall not be concerned with the coordinate dependence of wave functions. Thus, when discussing how $\psi(x, y, z; \sigma)$ behaves under a rotation of the coordinate system, we may suppose that the particle is at the origin. Its coordinates are then unchanged by the rotation, and the result describes specifically the dependence of ψ on the spin variable σ.

The variable σ differs from ordinary variables (coordinates) in being discrete. The most general linear operator acting on functions of a discrete variable may be written

$$(\hat{f}\psi)(\sigma) = \sum_{\sigma'} f_{\sigma\sigma'}\psi(\sigma'), \tag{55.1}$$

where $f_{\sigma\sigma'}$ are constants. The parentheses around $\hat{f}\psi$ emphasize that the spin argument following them belongs to the function produced by $\hat{f}$, not to the original function ψ. The quantities $f_{\sigma\sigma'}$ are readily seen to be the matrix elements of the operator as ordinarily defined by (11.5)[6]. Coordinate integration in (11.5) is now replaced by summation over the discrete variable, so that

$$f_{\sigma_2\sigma_1} = \sum_{\sigma} \psi^*_{\sigma_2}(\sigma)[\hat{f}\psi_{\sigma_1}(\sigma)]. \tag{55.2}$$

Here $\psi_{\sigma_1}(\sigma)$ and $\psi_{\sigma_2}(\sigma)$ are eigenfunctions of $\hat{s}_z$ belonging to $s_z = \sigma_1$ and $s_z = \sigma_2$. Each represents a state with a definite s_z, so only one wave-function component is nonzero[7]:

$$\psi_{\sigma_1}(\sigma) = \delta_{\sigma\sigma_1}, \qquad \psi_{\sigma_2}(\sigma) = \delta_{\sigma\sigma_2}. \tag{55.3}$$

By (55.1),

$$(\hat{f}\psi_{\sigma_1})(\sigma) = \sum_{\sigma'} f_{\sigma\sigma'}\psi_{\sigma_1}(\sigma') = \sum_{\sigma'} f_{\sigma\sigma'}\delta_{\sigma'\sigma_1} = f_{\sigma\sigma_1}.$$

Substitution of this result and $\psi_{\sigma_2}(\sigma)$ in (55.2) makes that equation an identity, proving the assertion.

Operators acting on functions of σ can therefore be represented by matrices of order $2s + 1$. In particular, this applies to the spin operator itself, whose action is, by (55.1),

$$(\hat{\mathbf{s}}\psi)(\sigma) = \sum_{\sigma'} \mathbf{s}_{\sigma\sigma'}\psi(\sigma'). \tag{55.4}$$

As stated at the end of § 54, the matrices $\hat{s}_x$, $\hat{s}_y$, and $\hat{s}_z$ are the matrices L_x, L_y, and L_z obtained in § 27, with L, M replaced by s, σ:

$$\begin{aligned} (s_x)_{\sigma,\sigma-1} &= (s_x)_{\sigma-1,\sigma} = \frac{1}{2}\sqrt{(s+\sigma)(s-\sigma+1)}, \\ (s_y)_{\sigma,\sigma-1} &= -(s_y)_{\sigma-1,\sigma} = -\frac{i}{2}\sqrt{(s+\sigma)(s-\sigma+1)}, \\ (s_z)_{\sigma\sigma} &= \sigma. \end{aligned} \tag{55.5}$$

[6] Notice that the indices on the matrix elements on the right of (55.1) occur in an order which is, in a certain sense, the reverse of the usual order in (11.11).

[7] More precisely one should write $\psi_{\sigma_1}(\sigma) = \psi(x, y, z)\,\delta_{\sigma_1\sigma}$; the coordinate factors irrelevant here have been omitted in (55.3).

We emphasize once again the need to distinguish the specified eigenvalue of s_z (σ_1 or σ_2) from the independent variable σ. This is precisely the reason for the difference between (11.11) and (55.1).

This defines the spin operator.

In the particularly important case $s = 1/2$, $\sigma = \pm 1/2$, these are two-rowed matrices. They are written

$$\hat{\mathbf{s}} = \frac{1}{2}\hat{\boldsymbol{\sigma}}, \tag{55.6}$$

where[8]

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{55.7}$$

The matrices (55.7) are called the *Pauli matrices*. As it should be for a matrix defined using eigenfunctions of s_z itself, $\hat{s}_z = \frac{1}{2}\hat{\sigma}_z$ is diagonal[9].

We record several characteristic properties of the Pauli matrices. Direct multiplication of (55.7) gives

$$\hat{\sigma}_y\hat{\sigma}_z = i\hat{\sigma}_x, \qquad \hat{\sigma}_z\hat{\sigma}_x = i\hat{\sigma}_y, \qquad \hat{\sigma}_x\hat{\sigma}_y = i\hat{\sigma}_z. \tag{55.8}$$

Combining these with the general commutation rules (54.1), we find

$$\hat{\sigma}_i\hat{\sigma}_k + \hat{\sigma}_k\hat{\sigma}_i = 0 \quad (i \neq k), \tag{55.9}$$

so the Pauli matrices anticommute. These relations immediately give the useful formulas

$$\hat{\boldsymbol{\sigma}}^2 = 3, \qquad (\hat{\boldsymbol{\sigma}}\mathbf{a})(\hat{\boldsymbol{\sigma}}\mathbf{b}) = \mathbf{ab} + i\hat{\boldsymbol{\sigma}}(\mathbf{a} \times \mathbf{b}), \tag{55.10}$$

where $\mathbf{a}$ and $\mathbf{b}$ are arbitrary vectors[10]. By these relations, any scalar polynomial in the matrices $\hat{\sigma}_i$ reduces to terms independent of $\hat{\boldsymbol{\sigma}}$ and terms linear in it. Hence every scalar function of $\hat{\boldsymbol{\sigma}}$ reduces to a linear function (see problem 1). Finally, the traces (sums of diagonal components) of the Pauli matrices and their products are

$$\operatorname{Tr}\hat{\sigma}_i = 0, \qquad \operatorname{Tr}(\hat{\sigma}_i\hat{\sigma}_k) = 2\delta_{ik}. \tag{55.11}$$

The following sections examine spin wave functions in detail, including their behavior under arbitrary coordinate rotations. We note here at once an important property: their behavior under rotations about the z axis.

Make an infinitesimal rotation $\delta\varphi$ about z. In terms of angular momentum, here spin, its operator is $1 + i\delta\varphi\hat{s}_z$. The functions $\psi(\sigma)$ therefore become $\psi(\sigma) + \delta\psi(\sigma)$, with

$$\delta\psi(\sigma) = i\delta\varphi\hat{s}_z\psi(\sigma) = i\sigma\psi(\sigma)\delta\varphi.$$

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[8] When matrices are written in the form (55.7), their rows and columns are indexed by the values of σ, with the row number corresponding to the first index of a matrix element and the column number to the second. Here these indices take the values $+1/2$ and $-1/2$. According to (55.4), applying the operator means multiplying the σth row of the matrix by the wave-function components arranged as the column $\psi = \begin{pmatrix} \psi(1/2) \\ \psi(-1/2) \end{pmatrix}$.

[9] Using the same letter for the spin projection and for the Pauli matrices cannot cause confusion: the Pauli matrices carry a hat.

[10] Terms on the right-hand sides of (55.8)–(55.10) that do not depend on $\hat{\boldsymbol{\sigma}}$ are, of course, to be understood as constants times the two-rowed unit matrix.

Writing this as $d\psi/d\varphi = i\sigma\psi(\sigma)$ and integrating, we find that a finite rotation through φ gives

$$\psi'(\sigma) = e^{i\sigma\varphi}\psi(\sigma). \tag{55.12}$$

For $\varphi = 2\pi$, the multiplier is $e^{2\pi i\sigma}$, the same for every σ and equal to $(-1)^{2s}$, since 2σ always has the same parity as $2s$. Thus a complete rotation of the coordinate system about z restores the wave functions of an integral-spin particle, but reverses the sign of those of a half-integral-spin particle.

Problem

1. Reduce an arbitrary function of the scalar $a + \mathbf{b}\hat{\boldsymbol{\sigma}}$, which is linear in the Pauli matrices, to a linear function.

Solution.

To find the coefficients in

$$f(a + \mathbf{b}\hat{\boldsymbol{\sigma}}) = A + B\mathbf{b}\hat{\boldsymbol{\sigma}},$$

choose z along $\mathbf{b}$. The eigenvalues of $a + \mathbf{b}\hat{\boldsymbol{\sigma}}$ are $a \pm b$, and those of the corresponding operator $f(a + \mathbf{b}\hat{\boldsymbol{\sigma}})$ are $f(a \pm b)$. Hence

$$A = \frac{1}{2}[f(a+b) + f(a-b)], \qquad B = \frac{1}{2b}[f(a+b) - f(a-b)].$$

2. Find the values of the scalar product $\mathbf{s}_1\mathbf{s}_2$ of the spin-$1/2$ angular momenta of two particles in states for which the total spin $\mathbf{S} = \mathbf{s}_1 + \mathbf{s}_2$ has a definite value, 0 or 1.

Solution.

The general addition formula (31.3) for any two angular momenta gives

$$\mathbf{s}_1\mathbf{s}_2 = 1/4 \quad \text{for } S = 1, \qquad \mathbf{s}_1\mathbf{s}_2 = -3/4 \quad \text{for } S = 0.$$

3. Which powers of the operator s_z for arbitrary spin s are independent?

Solution.

The operator

$$(\hat{s}_z - s)(\hat{s}_z - s + 1)\cdots(\hat{s}_z + s),$$

formed from the differences between $\hat{s}_z$ and every possible eigenvalue of s_z, gives zero on every wave function and is therefore itself zero. Consequently $(\hat{s}_z)^{2s+1}$ is expressible through lower powers of $\hat{s}_z$; only the powers from 1 through $2s$ are independent.

§ 56. Spinors

For zero spin the wave function has a single component, $\psi(0)$. The spin operator annihilates it: $\hat{\mathbf{s}}\psi = 0$. Since $\mathbf{s}$ is connected with the operator of infinitesimal rotations, this means that the wave function of a spin-zero particle is unchanged by rotations of the coordinate system; it is a scalar.

The wave function of a spin-1/2 particle has two components, $\psi(1/2)$ and $\psi(-1/2)$. For convenience in later generalizations, label these by upper indices 1 and 2. The two-component quantity

$$\psi = \begin{pmatrix}\psi^1\\ \psi^2\end{pmatrix} = \begin{pmatrix}\psi(1/2)\\ \psi(-1/2)\end{pmatrix} \tag{56.1}$$

is called a *spinor*.

Under an arbitrary rotation of the coordinate system, its components undergo the linear transformation

$$\psi^{1'} = a\psi^1 + b\psi^2, \qquad \psi^{2'} = c\psi^1 + d\psi^2. \tag{56.2}$$

This can be written

$$\psi^{\lambda'} = (\hat{U}\psi)^\lambda, \qquad \hat{U} = \begin{pmatrix}a & b\\ c & d\end{pmatrix}, \tag{56.3}$$

where $\hat{U}$ is the transformation matrix[11]. Its elements are generally complex functions of the rotation angles of the coordinate axes. Relations among them follow directly from the physical requirements on a spinor as a particle wave function.

Consider the bilinear form

$$\psi^1\varphi^2 - \psi^2\varphi^1, \tag{56.4}$$

where ψ and φ are two spinors. Direct calculation gives

$$\psi^{1'}\varphi^{2'} - \psi^{2'}\varphi^{1'} = (ad-bc)(\psi^1\varphi^2 - \psi^2\varphi^1),$$

so (56.4) transforms into itself under a coordinate rotation. A single function that transforms into itself can be regarded as belonging to spin zero; it must consequently be a scalar and remain altogether unchanged by a rotation. Hence

$$ad - bc = 1; \tag{56.5}$$

the determinant of the transformation matrix is unity[12].

Further relations follow from requiring the expression

$$\psi^1\psi^{1*} + \psi^2\psi^{2*}, \tag{56.6}$$

which gives the probability of finding the particle at a given point, to be a scalar. A transformation preserving the sum of squared moduli is unitary, so $\hat{U}^+ = \hat{U}^{-1}$ (see § 12). Subject to (56.5),

$$\hat{U}^{-1} = \begin{pmatrix}d & -b\\ -c & a\end{pmatrix}.$$

Equating this to the conjugate matrix

$$\hat{U}^+ = \begin{pmatrix}a^* & c^*\\ b^* & d^*\end{pmatrix}$$

[11]The notation $\hat{U}\psi$ means multiplication of the rows of $\hat{U}$ by the column ψ.

[12]A transformation of two quantities of this kind is called *binary*.

gives

$$a = d^*, \qquad b = -c^*. \tag{56.7}$$

By (56.5) and (56.7), the four complex quantities a, b, c, d contain only three independent real parameters, corresponding to the three angles that specify a rotation of a three-dimensional coordinate system.

Comparing the scalar expressions (56.4) and (56.6), we see that ψ^{1*}, ψ^{2*} must transform as $\psi^2, -\psi^1$. Relations (56.5) and (56.7) make this easy to verify[13].

Spinor algebra can be put into a form analogous to tensor algebra. Alongside the contravariant components ψ^1, ψ^2, with upper indices, introduce *covariant* components, with lower indices, by

$$\psi_1 = \psi^2, \qquad \psi_2 = -\psi^1. \tag{56.8}$$

The invariant combination (56.4) of two spinors then becomes the scalar product

$$\psi_\lambda \varphi^\lambda = \psi_1 \varphi^1 + \psi_2 \varphi^2. \tag{56.9}$$

Here and below, repeated dummy indices are summed as in tensor algebra. One rule specific to spinor algebra must be kept in mind. Since $\psi^\lambda \varphi_\lambda = \psi^1 \varphi_1 + \psi^2 \varphi_2 = -\psi^2 \varphi^1 - \psi^1 \varphi^2$, we have

$$\psi^\lambda \varphi_\lambda = -\psi_\lambda \varphi^\lambda. \tag{56.10}$$

It follows at once that the scalar product of any spinor with itself vanishes:

$$\psi_\lambda \psi^\lambda = 0. \tag{56.11}$$

As explained above, ψ_1, ψ_2 transform as ψ^{1*}, ψ^{2*}, that is,

$$\psi'_\lambda = (\hat{U}^* \psi)_\lambda. \tag{56.12}$$

The product $\hat{U}^* \psi$ may equally be written $\psi \tilde{\hat{U}}^*$ using the transpose of $\hat{U}^*$. Unitarity gives $\hat{U}^* = \widetilde{\hat{U}}^{-1}$, so that $\tilde{\psi}' = (\tilde{\psi} \hat{U}^{-1})$, or[14]

$$\psi'_\lambda = \psi_\mu (U^{-1})^\mu{}_\lambda. \tag{56.13}$$

Just as vectors lead to tensors in ordinary tensor algebra, one can introduce *spinors of higher rank*. A second-rank spinor is a four-component quantity $\psi^{\lambda\mu}$ whose components transform like the products $\psi^\lambda \varphi^\mu$ of components of two first-rank spinors. Alongside the contravariant components $\psi^{\lambda\mu}$ one may consider covariant $\psi_{\lambda\mu}$ and mixed $\psi_\lambda{}^\mu$ components, transforming respectively like $\psi_\lambda \varphi_\mu$ and $\psi_\lambda \varphi^\mu$. Spinors of any rank are defined analogously.

The passage from contravariant to covariant components and back can be written

$$\psi_\lambda = g_{\lambda\mu} \psi^\mu, \qquad \psi^\lambda = g^{\mu\lambda} \psi_\mu, \tag{56.14}$$

[13] This property is closely connected with time-reversal symmetry. Time reversal corresponds (see § 18) to replacing a wave function by its complex conjugate, while also reversing angular-momentum projections. Thus the functions complex conjugate to $\psi^1 = \psi(1/2)$ and $\psi^2 = \psi(-1/2)$ must have the same properties as the components belonging respectively to spin projections $-1/2$ and $1/2$.

[14] Notation of the form $\psi \hat{U}$, with ψ to the left, means multiplying the row of components (ψ_1, ψ_2) by the columns of $\hat{U}$.

where

$$g_{11} = g_{22} = 0, \qquad g_{12} = -g_{21} = 1 \tag{56.15}$$

is the *metric spinor* in a two-dimensional vector space[15]. Similarly, for example,

$$\psi_\lambda{}^\mu = g_{\lambda\nu}\psi^{\nu\mu}, \qquad \psi_{\lambda\mu} = g_{\lambda\nu}g_{\mu\rho}\psi^{\nu\rho},$$

so that $\psi_1{}^2 = -\psi^{11}$, $\psi_2{}^1 = -\psi^{21}$, $\psi_{11} = \psi^{22}$, and so forth.

The components $g_{\lambda\mu}$ themselves form the antisymmetric unit spinor of rank two. Their values are unchanged under coordinate transformations, and one readily verifies

$$g_{\lambda\nu}g^{\mu\nu} = \delta^\mu_\lambda, \tag{56.16}$$

where $\delta^1_1 = \delta^2_2 = 1$ and $\delta^2_1 = \delta^1_2 = 0$.

As in ordinary tensor algebra, spinor algebra has two basic operations: multiplication and contraction over a pair of indices. Multiplication of two spinors gives one of higher rank; for example, second- and third-rank spinors $\psi_{\lambda\mu}$ and $\varphi^{\nu\rho\sigma}$ give the fifth-rank spinor $\psi_{\lambda\mu}\varphi^{\nu\rho\sigma}$. Contraction over a pair of indices, meaning summation over equal values of one covariant and one contravariant index, lowers the rank by two. Thus contraction of $\psi_{\lambda\mu}{}^{\nu\rho\sigma}$ over μ and ν gives the third-rank spinor $\psi_{\lambda\mu}{}^{\mu\rho\sigma}$; contraction of $\varphi_\lambda{}^\mu$ gives the scalar $\varphi_\lambda{}^\lambda$. A rule analogous to (56.10) holds: interchanging the upper and lower positions of the contracted indices changes the sign, so $\varphi_\lambda{}^\lambda = -\varphi^\lambda{}_\lambda$. In particular, contracting two indices over which a spinor is symmetric gives zero. Thus for a symmetric second-rank spinor $\varphi_\lambda{}^\mu$, one has $\varphi_\lambda{}^\lambda = 0$.

A *symmetric spinor* of rank n is symmetric in all its indices. A nonsymmetric spinor can be symmetrized by summing the components obtained from all possible permutations of its indices. From the preceding result, no spinor of lower rank can be formed by contracting components of a symmetric spinor.

A spinor antisymmetric in all its indices can only have rank two. Indeed, each index has only two possible values; with three or more indices at least two must have equal values, and every component then vanishes identically. Every antisymmetric second-rank spinor is a scalar multiple of the unit spinor $g_{\lambda\mu}$. One consequence is

$$g_{\lambda\mu}\psi_\nu + g_{\mu\nu}\psi_\lambda + g_{\nu\lambda}\psi_\mu = 0, \tag{56.17}$$

where ψ_λ is arbitrary. The left-hand side is simply, as can be checked directly, an antisymmetric third-rank spinor and must therefore vanish.

The spinor formed by multiplying $\psi_{\lambda\mu}$ by itself and contracting one pair of indices is antisymmetric in the other pair, since

$$\psi_{\lambda\nu}\psi_\mu{}^\nu = -\psi_\lambda{}^\nu\psi_{\mu\nu}.$$

It must consequently reduce to $g_{\lambda\mu}$ times a scalar. Determining that scalar by requiring contraction over the second pair to give the proper result, we obtain

$$\psi_{\lambda\nu}\psi_\mu{}^\nu = -\frac{1}{2}\psi_{\rho\sigma}\psi^{\rho\sigma}g_{\lambda\mu}. \tag{56.18}$$

[15] The matrix (56.15) coincides with $i\hat{\sigma}_y$.

The components of the spinor $\varphi^{\lambda\mu\cdots}$, complex conjugate to the spinor $\psi_{\lambda\mu\ldots}$, transform as the components of the contravariant spinor $\varphi^{\lambda\mu\cdots}$, and conversely. Consequently, for any spinor the sum of the squared moduli of its components is an invariant.

§ 57. Wave functions of particles with arbitrary spin

Having developed the formal algebra of spinors of arbitrary rank, we can now turn to the immediate problem: the properties of wave functions for particles of arbitrary spin.

It is convenient to approach this question by considering a collection of n particles of spin $1/2$. The largest possible value of the z component of the system's total spin is $n/2$. It is obtained when $s_z = 1/2$ for every particle (all spins point in the same direction, along the z axis). In this case the total spin S of the system must likewise equal $n/2$.

Every component of the particle-system wave function $\psi(\sigma_1, \sigma_2, \ldots, \sigma_n)$ then vanishes except for $\psi(1/2, 1/2, \ldots, 1/2)$. Write the wave function as a product of n spinors $\psi^\lambda \varphi^\mu \ldots$, one for each particle. Each spinor has only its component with $\lambda, \mu, \ldots = 1$ different from zero. Consequently, the sole nonzero product is $\psi^1\varphi^1 \ldots$. The full set of these products, however, constitutes a spinor of rank n symmetric in all its indices. A transformation of coordinates (so that the spins no longer point along the z axis) yields a general spinor of rank n, which remains symmetric.

The spin properties of wave functions are, in essence, their behavior under rotations of the coordinates. They are therefore identical for a particle of spin s and for a system of $n = 2s$ spin-$1/2$ particles oriented so that the system has total spin s. It follows that the wave function of a spin-s particle is a symmetric spinor of rank $n = 2s$.

The number of independent components of a symmetric spinor of rank $2s$ is, as required, $2s + 1$. Indeed, the distinct components are precisely those whose indices contain $2s$ ones and no twos, then $2s - 1$ ones and one two, and so on, through no ones and $2s$ twos.

Mathematically, symmetric spinors classify the possible transformation types of quantities under rotations of the coordinates. Suppose that $2s + 1$ different quantities transform linearly among themselves, and that no choice of linear combinations can reduce their number. Their transformation law is then equivalent to that of the components of a symmetric spinor of rank $2s$. Any collection of functions, of whatever size, that transform linearly among themselves under coordinate rotations can be reduced, by a suitable linear transformation, to one or more symmetric spinors[16].

Thus an arbitrary spinor of rank n, $\psi^{\lambda\mu\nu\cdots}$, can be reduced to symmetric spinors of ranks n, $n-2$, $n-4$, ... In practice the reduction may be performed as follows. Symmetrizing $\psi^{\lambda\mu\nu\cdots}$ over all its indices produces a symmetric spinor of the same rank n. Next, contracting

the original spinor $\psi^{\lambda\mu\nu\cdots}$ over various pairs of indices gives spinors of rank $n-2$ such as $\psi^{\lambda}{}_{\lambda}{}^{\nu\cdots}$. Symmetrizing these in turn gives symmetric spinors of rank $n-2$. Contracting $\psi^{\lambda\mu\cdots}{}_{\lambda\mu}$ over two pairs and symmetrizing the resulting spinors gives symmetric spinors of rank $n-4$, and so forth.

It remains to connect the components of a symmetric spinor of rank $2s$ with the

[16]Equivalently, symmetric spinors realize the irreducible representations of the rotation group (see § 98).

$2s+1$ functions $\psi(\sigma)$, where $\sigma = s, s-1, \ldots, -s$. The component

$$\psi^{\overbrace{11\ldots1}^{s+\sigma}\overbrace{22\ldots2}^{s-\sigma}},$$

in which the index 1 occurs $s+\sigma$ times and the index 2 occurs $s-\sigma$ times, corresponds to a spin projection σ on the z axis. To verify this, replace the one particle of spin s once again by a system of $n = 2s$ spin-1/2 particles. The component just written corresponds to the product

$$\underbrace{\psi^1\varphi^1\ldots}_{s+\sigma}\underbrace{\chi^2\rho^2\ldots}_{s-\sigma};$$

this product represents a state in which $s+\sigma$ particles have spin projection $+1/2$ and $s-\sigma$ have projection $-1/2$. Their total projection is therefore $\frac{1}{2}(s+\sigma) - \frac{1}{2}(s-\sigma) = \sigma$. Finally, choose the proportionality factor between this spinor component and $\psi(\sigma)$ so that

$$\sum_{\sigma=-s}^{+s} |\psi(\sigma)|^2 = \sum_{\lambda,\mu,\ldots=1}^{2} |\psi^{\lambda\mu\ldots}|^2 \tag{57.1}$$

(This sum is a scalar, as it must be, since it gives the probability of finding the particle at the specified point in space.) In the sum on the right-hand side, a component having $(s+\sigma)$ indices equal to 1 occurs

$$\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}$$

times. It follows that the correspondence between the functions $\psi(\sigma)$ and the spinor components is given by

$$\psi(\sigma) = \left[\frac{(2s)!}{(s+\sigma)!(s-\sigma)!}\right]^{1/2} \psi^{\overbrace{11\ldots1}^{s+\sigma}\overbrace{22\ldots2}^{s-\sigma}}. \tag{57.2}$$

Relation (57.2) ensures not only condition (57.1), but also, as is readily checked, the more general condition

$$\psi^{\lambda\mu\ldots}\varphi_{\lambda\mu\ldots} = \sum_{\sigma}(-1)^{s-\sigma}\psi(\sigma)\varphi(-\sigma), \tag{57.3}$$

where $\psi^{\lambda\mu\ldots}$ and $\varphi_{\lambda\mu\ldots}$ are two different spinors of equal rank, while $\psi(\sigma)$ and $\varphi(\sigma)$ are the functions associated with them by (57.2). The factor $(-1)^{s-\sigma}$ appears because raising all the indices changes the sign of a component once for every index equal to 2.

Equations (55.5) specify the action of the spin operator on the wave functions $\psi(\sigma)$. We can readily determine how these operators act when the wave function is written as a spinor of rank $2s$. For spin 1/2, the functions $\psi(+1/2)$ and $\psi(-1/2)$ coincide with the spinor components ψ^1 and ψ^2. According to (55.6) and (55.7), the action of the spin operators on them gives

$$\begin{aligned}
&(\hat{s}_x\psi)^1 = \tfrac{1}{2}\psi^2, \quad (\hat{s}_y\psi)^1 = -\tfrac{i}{2}\psi^2, \quad (\hat{s}_z\psi)^1 = \tfrac{1}{2}\psi^1,\\
&(\hat{s}_x\psi)^2 = \tfrac{1}{2}\psi^1, \quad (\hat{s}_y\psi)^2 = \tfrac{i}{2}\psi^1, \quad (\hat{s}_z\psi)^2 = -\tfrac{1}{2}\psi^2.
\end{aligned} \tag{57.4}$$

To extend the construction to an arbitrary spin, consider once more a system of $2s$ particles of spin $1/2$, and express its wave function as a product of $2s$ spinors. The spin operator for this system is the sum of the individual particles' spin operators. Each term acts only on its corresponding spinor, with the result specified by formulas (57.4). Returning then to arbitrary symmetric spinors, that is, to the wave functions of a particle of spin s, gives

$$\begin{aligned}
(\hat{s}_x\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= \frac{s+\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma-1}\overbrace{22\ldots}^{s-\sigma+1}} + \frac{s-\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma+1}\overbrace{22\ldots}^{s-\sigma-1}},\\
(\hat{s}_y\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= -i\frac{s+\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma-1}\overbrace{22\ldots}^{s-\sigma+1}} + i\frac{s-\sigma}{2}\psi^{\overbrace{11\ldots}^{s+\sigma+1}\overbrace{22\ldots}^{s-\sigma-1}},\\
(\hat{s}_z\psi)^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}} &= \sigma\psi^{\overbrace{11\ldots}^{s+\sigma}\overbrace{22\ldots}^{s-\sigma}}.
\end{aligned} \tag{57.5}$$

Until now, spinors have been discussed as wave functions for the intrinsic angular momentum of elementary particles. Formally, however, the spin of a single particle is no different from the total angular momentum of any system regarded as a whole, when its internal structure is set aside.

Consequently, the transformation properties of spinors apply equally to the behaviour under spatial rotations of the wave functions ψ_{jm} for any particle (or particle system) having total angular momentum j, irrespective of whether that angular momentum is orbital or intrinsic. There must therefore be a definite correspondence between the transformation laws of the eigenfunctions ψ_{jm} under rotations of the coordinate system and those of the components of a symmetric spinor of rank $2j$.

In establishing this correspondence, two aspects of the dependence of the wave functions on the angular-momentum projection m, for fixed j, must be distinguished carefully. A wave function may be regarded as a probability amplitude for the possible values of m, or it may be an eigenfunction for a specified value of m.

Both aspects appeared at the beginning of § 55 in the eigenfunction $\delta_{\sigma\sigma_0}$ of s_z belonging to the value $s_z = \sigma_0$. Their mathematical difference is particularly transparent for a particle with spin $s = 1/2$. With respect to the variable σ, its spin function is a contravariant spinor of rank one and should therefore be written, in spinor notation, as ${\delta^{\lambda}}_{\sigma_0}$. With respect to σ_0, it is consequently a covariant spinor.

This observation is evidently quite general: formulas analogous to (57.2) establish a correspondence between the eigenfunctions ψ_{jm} and the components of a covariant symmetric spinor of rank $2j$[17]:

$$\psi_{jm} = \left[\frac{(2j)!}{(j+m)!(j-m)!}\right]^{1/2} \psi_{\underbrace{11\ldots}_{j+m}\underbrace{22\ldots}_{j-m}}. \tag{57.6}$$

For integral angular momentum j, the eigenfunctions are the spherical harmonics Y_{jm}. The case $j = 1$ is particularly important.

The three spherical functions Y_{1m} are

$$Y_{10} = i\sqrt{\frac{3}{4\pi}}\cos\theta = i\sqrt{\frac{3}{4\pi}}n_z, \qquad Y_{1,\pm 1} = \mp i\sqrt{\frac{3}{8\pi}}\sin\theta e^{\pm i\varphi} = \mp i\sqrt{\frac{3}{8\pi}}(n_x \pm i n_y)$$

where $\mathbf{n}$ is the unit vector in the radius-vector direction. Their transformation properties show that these three functions are equivalent to the components of a vector $\mathbf{a}$ under the correspondence

$$\psi_{10} = i a_z, \qquad \psi_{11} = -\frac{i}{\sqrt{2}}(a_x + i a_y), \qquad \psi_{1,-1} = \frac{i}{\sqrt{2}}(a_x - i a_y). \tag{57.7}$$

Comparing these expressions with (57.6) gives the following correspondence between the components of a symmetric spinor of rank two and those of a vector:

$$\psi_{11} = -\frac{i}{\sqrt{2}}(a_x + i a_y), \qquad \psi_{12} = \psi_{21} = \frac{i}{\sqrt{2}}a_z, \qquad \psi_{22} = \frac{i}{\sqrt{2}}(a_x - i a_y), \tag{57.8}$$

or

$$\psi^{11} = \frac{i}{\sqrt{2}}(a_x - i a_y), \qquad \psi^{12} = \psi^{21} = -\frac{i}{\sqrt{2}}a_z, \qquad \psi^{22} = -\frac{i}{\sqrt{2}}(a_x + i a_y). \tag{57.9}$$

Conversely,

$$a_z = i\sqrt{2}\psi^{12}, \qquad a_x = \frac{i}{\sqrt{2}}(\psi^{22} - \psi^{11}), \qquad a_y = \frac{1}{\sqrt{2}}(\psi^{11} + \psi^{22}). \tag{57.10}$$

With these definitions one readily verifies

$$\psi^{\lambda\mu}\varphi_{\lambda\mu} = \mathbf{a}\cdot\mathbf{b}, \tag{57.11}$$

where $\mathbf{a}$ and $\mathbf{b}$ are the vectors corresponding to the symmetric spinors $\psi^{\lambda\mu}$ and $\varphi_{\lambda\mu}$. One may also verify the spinor–vector correspondence[18]

$$\psi^{\lambda}{}_{\rho}\varphi^{\rho\mu} - \varphi^{\lambda}{}_{\rho}\psi^{\rho\mu} \longleftrightarrow i\sqrt{2}(\mathbf{a}\times\mathbf{b}). \tag{57.12}$$

[17] The same result can also be reached by a somewhat different argument. Expand the wave function ψ of a particle whose angular momentum is j in the eigenfunctions ψ_{jm}: $\psi = \sum_m a_m \psi_{jm}$. The coefficients a_m are then the probability amplitudes associated with the different values of m. In this sense, they correspond to the "components" $\psi(m)$ of the spin wave function, and this correspondence fixes their transformation law. On the other hand, the value of ψ at a specified point in space cannot depend on the coordinate system chosen; hence the sum $\sum a_m \psi_{jm}$ must be a scalar. Comparison with the scalar in (57.3) shows that the a_m must transform in the same way as $(-1)^{j-m}\psi_{j,-m}$.

[18] The mixed components of a symmetric spinor may be written as $\psi^{\lambda}{}_{\mu}$ without distinguishing $\psi^{\lambda\mu}$ from $\psi^{\mu\lambda}$.

Equations (57.10) can be written compactly in terms of the Pauli matrices:

$$a_i = \frac{i}{\sqrt{2}}(\hat{\sigma}_i)^{\lambda}{}_{\mu}\psi^{\mu}{}_{\lambda}. \tag{57.13}$$

The matrix indices on σ have been placed above and below in accordance with the positions of the spinor indices on $\psi^{\mu}{}_{\lambda}$. The origin of this formula becomes clear in the special case in which the rank-two spinor $\psi^{\mu}{}_{\lambda}$ reduces to the product of a rank-one spinor ψ^{μ} and its complex conjugate ψ^*_{λ}. The quantity

$$\frac{1}{2}\psi^*_{\lambda}\hat{\boldsymbol{\sigma}}^{\lambda}{}_{\mu}\psi^{\mu}$$

is then the mean spin for a particle with wave function ψ^{μ}, so its vector character is evident.

The correspondence (57.8) or (57.9) is a special case of a general rule. To every symmetric spinor of even rank $2j$, with integral j, one may associate a symmetric tensor of half that rank, j, whose contraction over every pair of indices vanishes; such a tensor will be called irreducible. This follows already from the equality of the numbers of independent components of these spinors and tensors, both being $2j+1$, as a direct count confirms[19]. The relation between the spinor and tensor components can be found from (57.8)–(57.10), by treating a spinor of the given rank as a product of rank-two spinors and the tensor as a product of vectors.

Problem

1. Rewrite the definition (57.4) of the spin-1/2 operator in terms of the spinor components of the vector **s**.

Solution.

Using (57.9), which relates the vector **s** to the spinor $s^{\lambda\mu}$, definition (57.4) becomes

$$s^{\lambda\mu}\psi^{\nu} = \frac{i}{2\sqrt{2}}(\psi^{\lambda}g^{\mu\nu} + \psi^{\mu}g^{\lambda\nu}).$$

2. Give the formulas defining the action of the spin operator on the vector wave function of a particle of spin 1.

Solution.

Equations (57.9) relate the components of the vector function $\boldsymbol{\psi}$ to the components of the spinor $\psi^{\lambda\mu}$, while the last of equations (57.5) gives

$$\hat{s}_z\psi_+ = -\psi_+, \qquad \hat{s}_z\psi_- = \psi_-, \qquad \hat{s}_z\psi_z = 0$$

where $\psi_{\pm} = \psi_x \pm i\psi_y$, or

$$\hat{s}_z\psi_x = -i\psi_y, \qquad \hat{s}_z\psi_y = i\psi_x, \qquad \hat{s}_z\psi_z = 0.$$

[19] In other words, the $2j+1$ components of an irreducible tensor of rank j (for integral j), the set of $2j+1$ spherical functions Y_{jm}, and the $2j+1$ components of a symmetric spinor of rank $2j$ all realize the same irreducible representation of the rotation group.

The remaining formulas follow by cyclic permutation of the indices x, y, z. Together they read

$$\hat{s}_i \psi_k = -i e_{ik\ell} \psi_\ell .$$

The complex vector $\boldsymbol{\psi}$ may be represented as $\boldsymbol{\psi} = e^{i\alpha}(\mathbf{u} + i\mathbf{v})$, where $\mathbf{u}$ and $\mathbf{v}$ are real vectors. By a suitable choice of the common phase α, these vectors may be made mutually perpendicular. The two vectors $\mathbf{u}$ and $\mathbf{v}$ specify a plane with the following property: the spin projection on the direction normal to it can take only the values ± 1.

§ 58. The finite-rotation operator

We return to spinor transformations and show how their coefficients can be expressed in terms of the angles through which the coordinate axes are rotated.

By the definition of the angular-momentum operator (here, spin), $1 + i\delta\varphi\,\mathbf{n}\hat{\mathbf{s}}$ is the operator for a rotation through $\delta\varphi$ about the direction of the unit vector $\mathbf{n}$. For a spin-$1/2$ wave function, a spinor of rank 1, one sets $\hat{\mathbf{s}} = \hat{\boldsymbol{\sigma}}/2$. A finite rotation through φ about the same direction is therefore represented by

$$\hat{U}_{\mathbf{n}} = \exp(i\varphi\,\mathbf{n}\hat{\boldsymbol{\sigma}}/2). \tag{58.1}$$

(Compare (15.13).) Like every function of the Pauli matrices (see Problem 1, § 55), this reduces to an expression linear in them:

$$\hat{U}_{\mathbf{n}} = \cos(\varphi/2) + i\mathbf{n}\hat{\boldsymbol{\sigma}} \sin(\varphi/2). \tag{58.2}$$

For a rotation about z,

$$\hat{U}_z(\varphi) = \cos\frac{\varphi}{2} + i\hat{\sigma}_z \sin\frac{\varphi}{2} = \begin{pmatrix} e^{i\varphi/2} & 0 \\ 0 & e^{-i\varphi/2} \end{pmatrix}. \tag{58.3}$$

Thus the spinor components transform as

$$\psi^{1'} = \psi^1 e^{i\varphi/2}, \qquad \psi^{2'} = \psi^2 e^{-i\varphi/2}.$$

In particular, a rotation through 2π changes the sign of the spinor components; every spinor of odd rank consequently has the same property (compare the end of § 55).

Similarly, rotations through φ about x or y have the matrices

$$\hat{U}_x(\varphi) = \begin{pmatrix} \cos(\varphi/2) & i\sin(\varphi/2) \\ i\sin(\varphi/2) & \cos(\varphi/2) \end{pmatrix}, \qquad \hat{U}_y(\varphi) = \begin{pmatrix} \cos(\varphi/2) & \sin(\varphi/2) \\ -\sin(\varphi/2) & \cos(\varphi/2) \end{pmatrix}. \tag{58.4}$$

For the special case of a rotation through π about y,

$$\psi^{1'} = \psi^2, \qquad \psi^{2'} = -\psi^1, \qquad \text{i.e.} \qquad \psi^{\lambda'} = \psi_\lambda. \tag{58.5}$$

We can now write the transformation matrix for an arbitrary rotation in terms of its Euler angles.

A rotation of the axes specified by Euler angles α, β, γ is performed in three stages: (1) rotation through α $(0 \leqslant \alpha \leqslant 2\pi)$ about z; (2) rotation through β $(0 \leqslant \beta \leqslant \pi)$ about

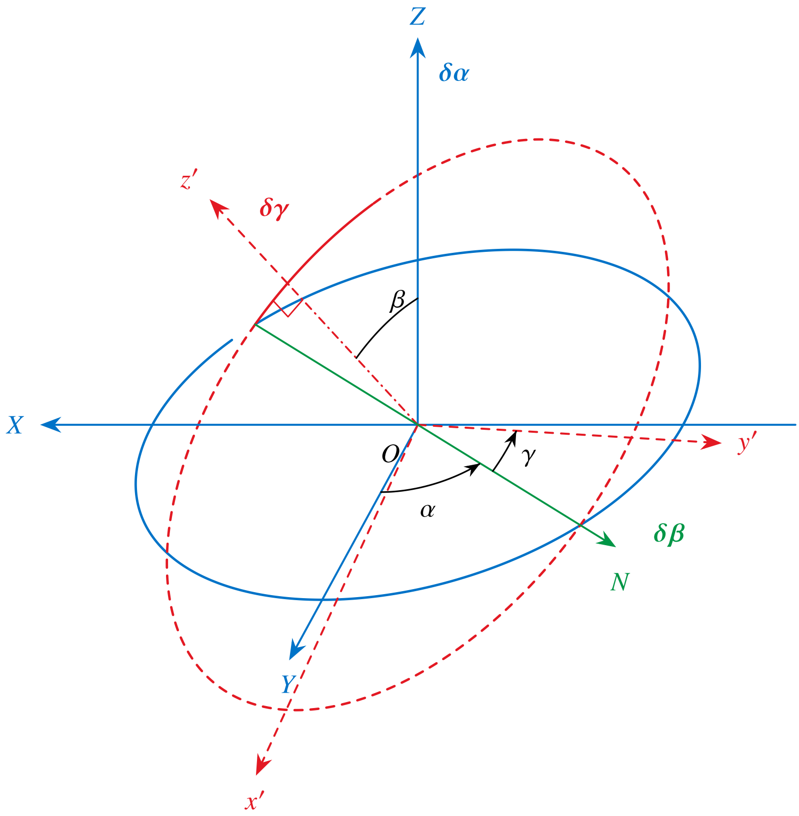

Fig. 20. Definition of the Euler angles.

the new position of y (ON in Fig. 20, the *line of nodes*); and (3) rotation through γ ($0 \leqslant \gamma \leqslant 2\pi$) about the final position z' of the z axis[20].

The angles α, β evidently coincide with the spherical angles φ, θ of the new z' axis relative to xyz: $\alpha = \varphi$, $\beta = \theta$.

For this sequence of rotations, the complete transformation matrix is

$$\hat{U}(\alpha, \beta, \gamma) = \hat{U}_z(\gamma)\hat{U}_y(\beta)\hat{U}_z(\alpha).$$

Direct multiplication gives

$$\hat{U}(\alpha, \beta, \gamma) = \begin{pmatrix} \cos(\beta/2)e^{i(\alpha+\gamma)/2} & \sin(\beta/2)e^{-i(\alpha-\gamma)/2} \\ -\sin(\beta/2)e^{i(\alpha-\gamma)/2} & \cos(\beta/2)e^{-i(\alpha+\gamma)/2} \end{pmatrix}. \tag{58.6}$$

By definition, higher-rank spinors transform as products of rank-1 spinor components.

[20]The systems xyz and $x'y'z'$ are, as always, right-handed; positive angles follow the direction of a right-handed screw advancing along the positive rotation axis.

This definition of the Euler angles, customary in quantum-mechanical applications, differs from that in § 35 (Volume I) because the second rotation is about y, not x. The angles α, β, γ are related to the angles φ, θ, ψ of Volume I (not to be confused with the spherical angles φ, θ) by $\psi = \gamma - \pi/2$.

In physical applications, however, the transformation laws of the corresponding wave functions ψ_{jm} are of greater interest than those of the spinors themselves.

Let ψ_{jm}, $m = j, j-1, \ldots, -j$, describe in the system xyz a state with specified angular momentum j, and let $\psi_{jm'}$ describe the same state relative to $x'y'z'$. Thus m is the value of J_z and m' that of $J_{z'}$. The two sets are related linearly; write

$$\psi_{jm} = \sum_{m'} D^{(j)}_{m'm} \psi_{jm'}. \tag{58.7}$$

The coefficients $D^{(j)}_{m'm}$ form, in the indices m', m, a matrix of order $2j+1$: the *finite-rotation matrix* $\hat{D}^{(j)}$. Its elements depend on the rotation angles α, β, γ of $x'y'z'$ relative to xyz.

The finite-rotation matrix can be constructed using the spinor representation of ψ_{jm}.

For $j = 1/2$, the two functions $\psi_{1/2,m}$ form a covariant spinor of rank 1. By (56.13), its transformation from $x'y'z'$ to xyz is effected by $\hat{U}$ in (58.6), so $\hat{D}^{(1/2)} = \hat{U}$[21]. Write its elements as

$$D^{(1/2)}_{m'm}(\alpha, \beta, \gamma) = e^{im'\gamma} d^{(1/2)}_{m'm}(\beta) e^{im\alpha},$$

where

$$\begin{array}{c|cc} d^{(1/2)}_{m'm}(\beta) & \frac{1}{2} & -\frac{1}{2} \\ \hline \frac{1}{2} & \cos(\beta/2) & \sin(\beta/2) \\ -\frac{1}{2} & -\sin(\beta/2) & \cos(\beta/2) \end{array} . \tag{58.8}$$

For arbitrary j, (57.6) relates ψ_{jm} to a symmetric covariant spinor of rank $2j$. Its transformation matrix is a product of $2j$ matrices $\hat{D}^{(1/2)}$, one acting on each spinor index. Multiplying and returning to ψ_{jm} gives

$$D^{(j)}_{m'm}(\alpha, \beta, \gamma) = e^{im'\gamma} d^{(j)}_{m'm}(\beta) e^{im\alpha}, \tag{58.9}$$

where[22]

$$\begin{aligned} d^{(j)}_{m'm}(\beta) = &\left[\frac{(j+m')!(j-m')!}{(j+m)!(j-m)!}\right]^{1/2} \left(\cos\frac{\beta}{2}\right)^{m'+m} \\ &\times \left(\sin\frac{\beta}{2}\right)^{m'-m} P^{(m'-m, m'+m)}_{j-m'}(\cos\beta), \end{aligned} \tag{58.10}$$

and

$$\begin{aligned} P^{(a,b)}_n(\cos\beta) = &\frac{(-1)^n}{2^n n!}(1-\cos\beta)^{-a}(1+\cos\beta)^{-b} \\ &\times \left(\frac{d}{d\cos\beta}\right)^n \left[(1-\cos\beta)^{a+n}(1+\cos\beta)^{b+n}\right] \end{aligned} \tag{58.11}$$

[21]The matrix indices in (58.7) are ordered precisely as required for multiplying the columns of $\hat{D}^{(j)}$ by the functions $\psi_{jm'}$ arranged in a row. Symbolically, (58.7) would read $\psi_{jm} = (\boldsymbol{\psi}_j \hat{D}^{(j)})_m$, consistently with (56.13).

[22]The calculation is given in A. R. Edmonds, *Angular Momentum in Quantum Mechanics*, Princeton, 1957 (see also the Russian translation of his article in *Deformation of Atomic Nuclei*, Moscow, 1958). The definition of $D^{(j)}_{m'm}$ in (58.9), (58.10) differs from Edmonds's book by interchanging α and γ (the more natural convention here), and from the article also by reversing the signs of all angles.

are the Jacobi polynomials[23]. Notice that

$$d^{(j)}_{m'm}(\beta) = 0 \quad \text{if } |m| > j \text{ or } |m'| > j. \tag{58.12}$$

The functions $d^{(j)}_{m'm}$ have several symmetry properties. Although these can be read from (58.11), (58.12), it is simpler to derive them directly from their definition as rotation coefficients.

The rotation matrix $\hat{D}^{(j)}$ is unitary. The inverse of the rotation (α, β, γ) is $(-\gamma, -\beta, -\alpha)$; since $\hat{d}$ is real, this gives

$$d^{(j)}_{m'm}(-\beta) = d^{(j)}_{mm'}(\beta). \tag{58.13}$$

We also have

$$d^{(j)}_{m'm}(\beta) = d^{(j)}_{-m,-m'}(\beta), \tag{58.14}$$

and

$$d^{(j)}_{m'm}(\pi) = (-1)^{j+m}\delta_{m',-m}, \qquad d^{(j)}_{m'm}(-\pi) = (-1)^{j-m}\delta_{m',-m}, \qquad d^{(j)}_{m'm}(0) = \delta_{m'm}. \tag{58.15}$$

For $j = 1/2$ these follow directly from (58.8), and for arbitrary j from the construction of the transformation matrix given above.

Perform a rotation through $\pi - \beta$ as successive rotations through π and $-\beta$:

$$d^{(j)}_{m'm}(\pi - \beta) = \sum_{m''} d^{(j)}_{m'm''}(\pi) d^{(j)}_{m''m}(-\beta) = (-1)^{j-m'} d^{(j)}_{-m',m}(-\beta),$$

or, using (58.13),

$$d^{(j)}_{m'm}(\pi - \beta) = (-1)^{j-m'} d^{(j)}_{m,-m'}(\beta). \tag{58.16}$$

Two rotations about the same axis give the same result in either order. Reversing the order of the rotations $-\beta$ and π and comparing with (58.16), we obtain

$$d^{(j)}_{m'm}(\beta) = (-1)^{m'-m} d^{(j)}_{-m',-m}(\beta). \tag{58.17}$$

Equations (58.17), (58.14), and (58.13) imply

$$d^{(j)}_{m'm}(\beta) = (-1)^{m'-m} d^{(j)}_{mm'}(\beta) = (-1)^{m'-m} d^{(j)}_{m'm}(-\beta). \tag{58.18}$$

These relations give corresponding symmetries of the complete functions $D^{(j)}_{m'm}$. In particular,

$$D^{(j)*}_{m'm}(\alpha, \beta, \gamma) = D^{(j)}_{m'm}(-\alpha, \beta, -\gamma) = (-1)^{m'-m} D^{(j)}_{-m',-m}(\alpha, \beta, \gamma). \tag{58.19}$$

Mathematically, the matrices $\hat{D}^{(j)}$ furnish unitary irreducible representations of the rotation group of dimension $2j + 1$ (see § 98). Hence the orthogonality and normalization relation is

$$\int D^{(j_1)*}_{m_1'm_1}(\alpha, \beta, \gamma) D^{(j_2)}_{m_2'm_2}(\alpha, \beta, \gamma) \frac{d\omega}{8\pi^2} = \frac{1}{2j_1 + 1} \delta_{j_1 j_2} \delta_{m_1 m_2} \delta_{m_1' m_2'}, \tag{58.20}$$

[23]For their relation to the hypergeometric series, see § e, formula (e.11).

where $d\omega = \sin\beta\, d\alpha\, d\beta\, d\gamma$.

Orthogonality in m and m' is supplied by the factor $\exp\{i(m\alpha+m'\gamma)\}$. Orthogonality in j belongs to the functions $d^{(j)}_{m'm}$:

$$\int_0^\pi d^{(j_1)}_{m'm}(\beta) d^{(j_2)}_{m'm}(\beta) \sin\beta\, d\beta = \frac{2}{2j_1+1}\delta_{j_1 j_2}. \tag{58.21}$$

For reference, we finally list the functions $d^{(j)}_{m'm}$ for several special values of their parameters. For $j = 1$,

$$\begin{array}{c|ccc}
d^{(1)}_{m'm}(\beta) & 1 & 0 & -1 \\
\hline
1 & \frac{1}{2}(1+\cos\beta) & \frac{1}{\sqrt{2}}\sin\beta & \frac{1}{2}(1-\cos\beta) \\
0 & -\frac{1}{\sqrt{2}}\sin\beta & \cos\beta & \frac{1}{\sqrt{2}}\sin\beta \\
-1 & \frac{1}{2}(1-\cos\beta) & -\frac{1}{\sqrt{2}}\sin\beta & \frac{1}{2}(1+\cos\beta)
\end{array} \tag{58.22}$$

For integral $j = l$ and $m' = 0$, (58.10), (58.11) give

$$d^{(l)}_{0m}(\beta) = \left[\frac{(l-m)!}{(l+m)!}\right]^{1/2} P^m_l(\cos\beta). \tag{58.23}$$

The origin of this formula is immediate from (58.7). Refer the functions $\psi_{jm'}$ on its right-hand side to the z' axis. For $j = l$,

$$\psi_{lm'}(\mathbf{n}_{z'}) = i^l \sqrt{\frac{2l+1}{4\pi}}\delta_{m'0}. \tag{58.24}$$

The function ψ_{jm} on the left is then the spherical harmonic $Y_{lm}(\beta, \alpha)$ of the spherical angles $\varphi = \alpha$, $\theta = \beta$ of z'. Substitution of (58.24) in (58.7) gives

$$Y_{lm}(\beta, \alpha) = i^l \sqrt{\frac{2l+1}{4\pi}} D^{(l)}_{0m}(\alpha, \beta, \gamma), \tag{58.25}$$

which is equivalent to (58.23).

Finally, when one index m or m' has its largest possible value,

$$d^{(j)}_{jm}(\beta) = (-1)^{j-m} d^{(j)}_{-j,m}(\beta) = \left[\frac{(2j)!}{(j+m)!(j-m)!}\right]^{1/2} \left(\cos\frac{\beta}{2}\right)^{j+m} \left(\sin\frac{\beta}{2}\right)^{j-m}. \tag{58.26}$$

§ 59. Partial polarization of particles

By a suitable choice of direction for the z axis, one component (say ψ^2) of any given spinor ψ^λ, the wave function of a spin-1/2 particle, can always be made zero. This is clear already because a spatial direction is specified by two quantities (angles), so the number of parameters at our disposal is precisely the number of quantities we wish to set equal to zero (the real and imaginary parts of the complex number ψ^2).

Physically, this means that if a particle of spin 1/2—for definiteness, let it be an electron—is in a state described by some spin wave function, a direction exists along which the particle's spin projection has the definite value $\sigma = 1/2$. The electron in such a state may be said to be *fully polarized*.

There are, however, electron states that may be termed *partially polarized*. Such states are specified not by wave functions, but solely by density matrices; in other words, they are mixed states with respect to spin (see § 14).

The electron's spin (or *polarization*) density matrix is a rank-two spinor $\rho^{\lambda}{}_{\mu}$, whose normalization is fixed by the condition

$$\rho^{\lambda}{}_{\lambda} = \rho^{1}{}_{1} + \rho^{2}{}_{2} = 1 \tag{59.1}$$

and satisfying the "Hermiticity" condition

$$(\rho^{\lambda}{}_{\mu})^{*} = \rho^{\mu}{}_{\lambda}. \tag{59.2}$$

For a pure spin state of an electron—that is, a completely polarized state—the spinor $\rho^{\lambda}{}_{\mu}$ reduces to a product of components of the wavefunction ψ^{λ}:

$$\rho^{\lambda}{}_{\mu} = \psi^{\lambda}\psi^{*}_{\mu}. \tag{59.3}$$

The diagonal components of the density matrix give the probabilities for the values $+1/2$ and $-1/2$ of the electron spin projection on the z axis. The mean value of this projection is therefore

$$\bar{s}_{z} = \frac{1}{2}(\rho^{1}{}_{1} - \rho^{2}{}_{2}),$$

or, using (59.1),

$$\rho^{1}{}_{1} = \frac{1}{2} + \bar{s}_{z}, \qquad \rho^{2}{}_{2} = \frac{1}{2} - \bar{s}_{z}. \tag{59.4}$$

In a pure state, the mean values of $s_{\pm} = s_x \pm i s_y$ are calculated as

$$\bar{s}_{\pm} = \psi^{*}_{\lambda}\hat{s}_{\pm}\psi^{\lambda}.$$

Since, according to (55.6) and (55.7), the operators $\hat{s}_{\pm}$ are represented by the matrices

$$\hat{s}_{+} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \hat{s}_{-} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

we find

$$\bar{s}_{+} = \psi^{*}_{1}\psi^{2}, \qquad \bar{s}_{-} = \psi^{*}_{2}\psi^{1}.$$

Accordingly, in a mixed state,

$$\rho^{1}{}_{2} = \bar{s}_{-}, \qquad \rho^{2}{}_{1} = \bar{s}_{+}. \tag{59.5}$$

With the Pauli matrices, equations (59.4) and (59.5) can be combined in the form

$$\rho^{\lambda}{}_{\mu} = \frac{1}{2}(\delta^{\lambda}{}_{\mu} + 2\hat{\boldsymbol{\sigma}}^{\lambda}{}_{\mu}\bar{\mathbf{s}}). \tag{59.6}$$

Thus every component of the electron polarization density matrix is expressed in terms of the mean values of the components of its spin vector. In other words, the real vector $\bar{\mathbf{s}}$ completely determines the polarization properties of a spin-1/2 particle. In the limiting case of full polarization, one component of this vector, with a suitable choice of axis directions, is 1/2, while the other two vanish. In the opposite case of an unpolarized state, all three components vanish. In the general case of arbitrary partial polarization and an arbitrary coordinate system, $0 \leqslant p \leqslant 1$, where

$$p = 2(\bar{s}_x^2 + \bar{s}_y^2 + \bar{s}_z^2)^{1/2}$$

is a quantity that may be called the electron's *degree of polarization*.

For a particle of arbitrary spin s, the density matrix is a spinor $\rho^{\lambda\mu\cdots}{}_{\rho\sigma\ldots}$ of rank $4s$, symmetric in its first $2s$ indices and in its last $2s$ indices, and subject to

$$\rho^{\lambda\mu\cdots}{}_{\lambda\mu\ldots} = 1, \tag{59.7}$$

$$(\rho^{\lambda\mu\ldots}{}_{\rho\sigma\ldots})^* = \rho^{\rho\sigma\cdots}{}_{\lambda\mu\ldots}. \tag{59.8}$$

To count the independent components of the density matrix, note that only $2s + 1$ essentially distinct combinations occur among the possible values of the indices $\lambda, \mu, \ldots$ (and likewise among $\rho, \sigma, \ldots$). The components of the spinor $\rho^{\lambda\mu\cdots}{}_{\rho\sigma\ldots}$ are also subject to the single relation (59.7). Their number is therefore $(2s + 1)^2 - 1 = 4s(s + 1)$. These components are complex, but relations (59.8) ensure that this does not enlarge the total number of independent quantities needed to characterize a particle's state of partial polarization; that number is thus $4s(s + 1)$[24]. By comparison, only $4s$ quantities describe a completely polarized particle: the wave function $\psi^{\lambda\mu\cdots}$ has $2s+1$ complex components, constrained by one normalization condition and containing one overall phase that is immaterial to the state.

Like every spinor of rank $4s$, the spinor $\rho^{\lambda\mu\cdots}{}_{\rho\sigma\ldots}$ is equivalent to a collection of irreducible tensors of ranks $4s$, $4s - 2$, ..., 0. In the present case there is just one tensor of each of these ranks, since the symmetry properties of $\rho^{\lambda\mu\cdots}{}_{\rho\sigma\ldots}$ allow each contraction to be carried out in only one way: over any one of the indices $\lambda, \mu, \ldots$ and any one of $\rho, \sigma, \ldots$. Moreover, the scalar (rank-zero tensor) is not present as an independent quantity at all, because condition (59.7) reduces it to unity.

§ 60. Time reversal and Kramers' theorem

In quantum mechanics, symmetry of motion under reversal of the sign of time means that if ψ is the wave function of a stationary state, the time-reversed wave function, denoted by ψ^{rev}, also describes a possible state of the same energy. At the end of § 18 it was stated that ψ^{rev} is the complex conjugate ψ^*. In this simple form the statement applies when particle spin is disregarded; spin requires a qualification.

Represent the wave function of a spin-s particle as a contravariant spinor $\psi^{\lambda\mu\cdots}$ of rank $2s$. Its complex conjugates $\psi^{\lambda\mu\ldots*}$ transform as the components of a covariant

[24] Specifying these quantities is equivalent to specifying the mean values of the components of the vector $\mathbf{s}$, together with all their powers and products of orders 2, 3, ..., $2s$ that cannot be reduced to still lower orders (see Problem 3 of § 55).

spinor. Time reversal therefore takes $\psi^{\lambda\mu\cdots}$ into a new wave function whose covariant components are

$$\psi^{\text{rev}}_{\lambda\mu\ldots} = \psi^{*}_{\lambda\mu\ldots}. \tag{60.1}$$

For a fixed set of index values, covariant and contravariant components correspond to opposite angular-momentum projections. Thus, in terms of $\psi_{s\sigma}$, time reversal interchanges $\psi_{s\sigma}$ and $\psi_{s,-\sigma}$, as expected because reversing time reverses angular momentum. Equation (60.1) fixes the precise relation:

$$\psi^{\text{rev}}_{s,-\sigma} = \psi^{*}_{s\sigma}(-1)^{s-\sigma}. \tag{60.2}$$

In other words, the replacement $\psi_{s\sigma} \longrightarrow \psi^{*}_{s\sigma}$ required by time reversal amounts to[25]

$$\psi_{s\sigma} \longrightarrow \psi_{s,-\sigma}(-1)^{s-\sigma}. \tag{60.3}$$

Applying the operation twice gives

$$\psi_{s\sigma} \longrightarrow \psi_{s,-\sigma}(-1)^{s-\sigma} \longrightarrow \psi_{s\sigma}(-1)^{s-\sigma}(-1)^{s+\sigma} = \psi_{s\sigma}(-1)^{2s}.$$

Thus two time reversals restore the wave function for integral spin, but change its sign for half-integral spin.

Consider an arbitrary system of interacting particles. When relativistic interactions are included, its orbital and spin angular momenta are generally not conserved separately; only the total J is conserved. With no external field, every energy level is $(2j + 1)$-fold degenerate. An external field generally removes this degeneracy. Can it remove it completely, leaving only simple levels? The answer is closely tied to time-reversal symmetry.

The equations of classical electrodynamics are invariant under reversal of time if the electric field is left unchanged and the magnetic field changes sign[26]. This fundamental property must persist in quantum mechanics. Time-reversal symmetry therefore holds not only for a closed system but also in any external electric field, provided no magnetic field is present.

The system's wave functions are spinors $\psi^{\lambda\mu\cdots}$ of rank $n = 2\sum s_a$, twice the sum of the spins of all particles; this sum need not equal the total spin S of the system. In an arbitrary electric field, a wave function and its time reverse must describe states of equal energy. A nondegenerate level requires these states to be identical, so their wave functions must agree up to a constant factor, both being expressed in the same covariant or contravariant form.

Write $\psi^{\text{rev}}_{\lambda\mu\ldots} = C\psi_{\lambda\mu\ldots}$ or, by (60.1),

$$\psi^{*}_{\lambda\mu\ldots} = C\psi_{\lambda\mu\ldots}, \tag{60.4}$$

where C is constant. Complex conjugation gives

$$\psi^{\lambda\mu\cdots} = C^{*}\psi^{\lambda\mu\ldots *}.$$

[25] Notice the agreement between the complex-conjugation rule for a spherical function, (28.9), and the general rule (60.3).

[26] See II, § 17; also the remark at the end of § 111.

Lower the indices on the left and correspondingly raise them on the right. That is, multiply both sides by $g_{\alpha\lambda} g_{\beta\mu} \ldots$ and sum over $\lambda, \mu, \ldots$; on the right use

$$g_{\alpha\lambda} g_{\beta\mu} \ldots = (-1)^n g_{\lambda\alpha} g_{\mu\beta} \ldots .$$

The result is

$$\psi_{\lambda\mu\ldots} = C^* (-1)^n \psi^*_{\lambda\mu\ldots}.$$

Substitution from (60.4) gives

$$\psi_{\lambda\mu\ldots} = (-1)^n C C^* \psi_{\lambda\mu\ldots}.$$

This identity requires $(-1)^n C C^* = 1$. Since $|C|^2$ is positive, this is possible only for even n, that is, integral $\sum s_a$. For odd n, or half-integral $\sum s_a$[27], condition (60.4) cannot hold.

An electric field can therefore remove degeneracy completely only for a system whose particle spins have an integral sum. For a half-integral sum, every level in an arbitrary electric field must be at least doubly degenerate; the two distinct states of equal energy correspond to complex-conjugate spinors[28] (H. A. Kramers, 1930).

One further mathematical remark is useful. A relation of the form (60.4) with real C is the condition that the spinor components can be associated with a set of real quantities; call it a "reality" condition for the spinor[29]. The impossibility of (60.4) for odd n says that no real quantity can correspond to an odd-rank spinor. For even n, by contrast, (60.4) can hold with real C. In particular, a symmetric rank-2 spinor corresponds to a real vector when (60.4) holds with $C = 1$:

$$\psi^{\lambda\mu *} = \psi_{\lambda\mu}$$

(as follows directly from (57.8), (57.9)). More generally, (60.4) with $C = 1$ is the reality condition for a symmetric spinor of any even rank.

[27] When $\sum s_a$ is integral (half-integral), all possible values of the total spin S are likewise integral (half-integral).

[28] If the electric field has high (cubic) symmetry, fourfold degeneracy may also occur; see § 99 and its problem.

[29] It is meaningless to call a spinor literally real, since complex-conjugate spinors have different transformation laws.

CHAPTER IX

Identity of Particles

§ 61. The indistinguishability principle for identical particles

In classical mechanics, identical particles (electrons, for example) retain their "individuality" despite having the same physical properties. One may imagine numbering the particles that make up a given physical system at some instant and thereafter following the motion of each along its own trajectory; the particles could then be identified at any later time.

In quantum mechanics the situation is entirely different. As has already been pointed out more than once, the uncertainty principle deprives the notion of an electron trajectory of all meaning. Even if an electron's position is known exactly at the present instant, at the next instant its coordinates have no definite values at all. We therefore gain nothing toward identifying electrons at later times by localizing and numbering them at some instant. If at another instant one electron is localized at a certain point in space, it is impossible to say which particular electron has reached that point. Quantum mechanics thus provides no possibility in principle of following each identical particle separately and thereby distinguishing one from another. One may say that identical particles lose their "individuality" completely in quantum mechanics. Here the sameness of their physical properties has a very profound consequence: the particles are wholly indistinguishable.

This *indistinguishability principle* for identical particles, as it is called, is fundamental to the quantum theory of systems made up of such particles. Consider first a system containing only two particles. Since they are identical, two states that differ merely by interchange of the particles must be physically equivalent in every respect. Consequently, under this interchange the system's wave function can change only by an

irrelevant phase factor. Let $\psi(\xi_1, \xi_2)$ be the wave function of the system, where ξ_1 and ξ_2 denote, by convention, the set consisting of the three coordinates and the spin projection of each particle. We must then have

$$\psi(\xi_1, \xi_2) = e^{i\alpha}\psi(\xi_2, \xi_1),$$

where α is a real constant. Interchanging the particles a second time restores the initial state, while multiplying ψ by $e^{2i\alpha}$. Hence $e^{2i\alpha} = 1$, so that $e^{i\alpha} = \pm 1$. Thus $\psi(\xi_1, \xi_2) = \pm\psi(\xi_2, \xi_1)$.

There are therefore only two possibilities: the wave function is either symmetric (that is, it remains entirely unchanged when the particles are interchanged) or antisymmetric (that is, its sign is reversed by the interchange). Clearly, the wave functions for all states of one and the same system must possess the same symmetry. Otherwise, the wave function of a state formed as a superposition of states with different symmetries would be neither symmetric nor antisymmetric.

This conclusion extends directly to systems containing any number of identical particles. The number of particles may be arbitrary. To see this, choose any pair of them.

Suppose, for example, that this pair has the property of being described by symmetric wave functions. The identity of the particles then requires every other pair of the same kind to have this property as well. Thus, in one possibility, the wave function remains wholly unchanged when any pair of particles is exchanged. It consequently remains unchanged under any permutation of the particles whatsoever. In the other possibility, it changes sign whenever a pair is exchanged. In the first case the wave function is called *symmetric*. In the second it is called *antisymmetric*. Which of these two properties applies depends on the kind of particle. Particles described by antisymmetric functions are said to obey *Fermi–Dirac statistics*. They are also called *fermions*. Particles described by symmetric functions are said to obey *Bose–Einstein statistics*. They are also called *bosons*[1].

The laws of relativistic quantum mechanics make it possible to prove (see IV, § 25) that the statistics obeyed by particles is uniquely tied to their spin: particles of half-integral spin are fermions, whereas those of integral spin are bosons.

The statistics of a composite particle are fixed by the parity of the number of elementary fermions among its constituents. Indeed, exchanging two identical composite particles amounts to simultaneously exchanging several pairs of identical elementary particles. An exchange of bosons leaves the wave function unchanged. An exchange of fermions reverses its sign. A composite containing an odd number of elementary fermions therefore obeys Fermi statistics. One containing an even number obeys Bose statistics. This conclusion agrees with the general rule stated above. An even number of half-integral-spin constituents gives the composite integral spin. An odd number gives it half-integral spin.

Thus atomic nuclei of odd atomic weight (that is, nuclei made up of an odd number of protons and neutrons) obey Fermi statistics, while nuclei of even atomic weight obey Bose statistics. For atoms, which contain electrons as well as a nucleus, the statistics is evidently determined by whether the sum of the atomic weight and atomic number is even or odd.

Consider a system of N identical particles whose mutual interaction may be neglected. Let $\psi_1, \psi_2, \ldots$ be the wave functions of the various stationary states available to each particle separately. The state of the entire system can be specified by listing the labels of the states occupied by the individual particles. We must determine how the wave function ψ of the full system is to be constructed from $\psi_1, \psi_2, \ldots$.

Let $p_1, p_2, \ldots, p_N$ label the states occupied by the individual particles. Two or more of these labels may coincide. For a system of bosons, the wave function is $\psi(\xi_1, \xi_2, \ldots, \xi_N)$. This function can be expressed as a sum of products. Each product has the form displayed below.

$$\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2)\cdots\psi_{p_N}(\xi_N) \tag{61.1}$$

[1]This terminology comes from the names of the statistics used to describe an ideal gas. In one case the gas consists of particles whose wave functions are antisymmetric. In the other it consists of particles whose wave functions are symmetric. In fact, the distinction here is not merely between two statistics. The mechanics themselves are also essentially different. Fermi statistics was proposed by Fermi (E. Fermi) for electrons in 1926. Its connection with quantum mechanics was established by Dirac in 1926. Bose statistics was proposed by Bose (S. Bose) for light quanta. Einstein generalized it in 1924.

over all possible permutations of the distinct indices $p_1, p_2, \ldots$; this sum plainly has the required symmetry. For example, for two particles occupying different states ($p_1 \neq p_2$),

$$\psi(\xi_1, \xi_2) = \frac{1}{\sqrt{2}} \left[\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2) + \psi_{p_1}(\xi_2)\psi_{p_2}(\xi_1)\right]. \tag{61.2}$$

The factor $1/\sqrt{2}$ provides normalization (all the functions $\psi_1, \psi_2, \ldots$ are mutually orthogonal and are assumed normalized). More generally, for a system containing an arbitrary number N of particles, the normalized wave function is

$$\psi_{N_1 N_2 \ldots} = \left(\frac{N_1! N_2! \cdots}{N!}\right)^{1/2} \sum \psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2) \cdots \psi_{p_N}(\xi_N), \tag{61.3}$$

where the sum runs over all permutations of distinct members among the indices $p_1, p_2, \ldots, p_N$, and N_i gives the number of these indices whose common value is i (with $\sum N_i = N$). On integrating $|\psi|^2$ over $d\xi_1 d\xi_2 \cdots d\xi_N$[2], every term vanishes except the squared moduli of the individual terms in the sum. Since the total number of terms in the sum (61.3) is plainly $N!/(N_1!N_2!\cdots)$, the normalization coefficient displayed there follows.

For a system of fermions, the wave function ψ is an antisymmetric combination of products (61.1). Thus, for two particles,

$$\psi(\xi_1, \xi_2) = \frac{1}{\sqrt{2}} \left[\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2) - \psi_{p_1}(\xi_2)\psi_{p_2}(\xi_1)\right]. \tag{61.4}$$

For the general case of N particles, the system's wave function is written as the determinant

$$\psi_{N_1 N_2 \ldots} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_{p_1}(\xi_1) & \psi_{p_1}(\xi_2) & \cdots & \psi_{p_1}(\xi_N) \\ \psi_{p_2}(\xi_1) & \psi_{p_2}(\xi_2) & \cdots & \psi_{p_2}(\xi_N) \\ \cdots & \cdots & \cdots & \cdots \\ \psi_{p_N}(\xi_1) & \psi_{p_N}(\xi_2) & \cdots & \psi_{p_N}(\xi_N) \end{vmatrix}. \tag{61.5}$$

Here an interchange of two particles corresponds to interchanging two columns of the determinant, which reverses its sign.

Expression (61.5) yields an important result. If any two of the labels $p_1, p_2, \ldots$ coincide, two rows of the determinant are identical and the determinant vanishes identically. It can be nonzero only when all the labels $p_1, p_2, \ldots$ are different. Thus, in a system of identical fermions, two or more particles cannot occupy the same state at the same time. This is the so-called *Pauli principle* (W. Pauli, 1925).

§ 62. Exchange interaction

The omission of particle spin from the Schrödinger equation in no way diminishes either the equation or the results obtained from it. Indeed, the electrical interaction between particles is independent of their spins[3].

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[2]Integration over $d\xi$ conventionally means here (and in §§ 64 and 65) integration over the coordinates together with summation over σ.

[3]This statement holds only in the nonrelativistic approximation. Once relativistic effects are included, the interaction of charged particles acquires a dependence on spin.

In mathematical terms, the Hamiltonian of a system of electrically interacting particles (with no magnetic field present) contains no spin operators. Consequently, when it acts on the wave function, it leaves the spin variables untouched. In fact, every component of the wave function therefore satisfies the Schrödinger equation. Equivalently, the wave function of a particle system can be expressed as the product

$$\psi(\xi_1, \xi_2) = \chi(\sigma_1, \sigma_2, \ldots)\varphi(\mathbf{r}_1, \mathbf{r}_2, \ldots),$$

where φ depends only on the particle coordinates and χ only on their spins. We shall call the former the *coordinate* or *orbital* wave function and the latter the *spin* wave function.

Thus, what the Schrödinger equation essentially determines is only the coordinate function φ; the function χ remains arbitrary. Whenever the spins themselves are of no interest, one may consequently use the Schrödinger equation with the coordinate function alone serving as the wave function, as was done in the preceding chapters.

Nevertheless, despite the stated independence of the electrical interaction from spin, the energy of a system displays a particular dependence on its total spin. Ultimately, this dependence arises from the indistinguishability principle for identical particles.

Consider a system containing just two identical particles. Solving the Schrödinger equation yields a sequence of energy levels, each associated with a definite coordinate wave function $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ that is either symmetric or antisymmetric.

This follows because the particles are identical: the Hamiltonian, and hence the Schrödinger equation, is invariant under their interchange. If an energy level is nondegenerate, interchanging $\mathbf{r}_1$ and $\mathbf{r}_2$ can

change $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ only by a constant factor. Repeating the interchange shows that this factor can only be ± 1[4]. Suppose first that the particles have zero spin. There is then no spin factor at all, and the wave function consists solely of $\varphi(\mathbf{r}_1, \mathbf{r}_2)$. This function must be symmetric, since particles of spin zero obey Bose statistics. Hence not every energy level produced by a formal solution of the Schrödinger equation can actually occur: levels whose φ is antisymmetric are impossible for the system under consideration.

Interchanging two identical particles is equivalent to inverting the coordinate system about an origin placed at the midpoint of the line joining them. Under inversion, on the other hand, φ must acquire the factor $(-1)^l$, where l is the orbital angular momentum of their relative motion (see § 30). Combining these observations, we conclude that two identical particles of zero spin can have only even orbital angular momentum.

Next let the system contain two particles of spin $1/2$, electrons for example. Its complete wave function—the product of $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ and $\chi(\sigma_1, \sigma_2)$—must be antisymmetric under interchange of the two particles. A symmetric coordinate function must therefore be paired with an antisymmetric spin function, and conversely. Write the spin function in spinor form, as a second-rank spinor $\chi^{\lambda\mu}$, with one index for the spin of each electron. Symmetry in the two spins corresponds to a symmetric spinor ($\chi^{\lambda\mu} = \chi^{\mu\lambda}$), whereas antisymmetry corresponds to an antisymmetric one ($\chi^{\lambda\mu} = -\chi^{\mu\lambda}$). A symmetric second-rank spinor, as we know, describes a system of total spin one; an antisymmetric spinor reduces to a scalar and thus corresponds to spin zero.

[4] In a degenerate level, linear combinations of its functions can always be chosen so as to satisfy this condition as well.

We have therefore obtained the following result. Energy levels associated with symmetric solutions $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ of the Schrödinger equation can occur

when the total spin of the system is zero, that is, when the two electron spins are "antiparallel" and add to zero. Energy values associated with antisymmetric $\varphi(\mathbf{r}_1, \mathbf{r}_2)$ require total spin one, so the electron spins must be "parallel." Thus the allowed energies of an electron system depend on its total spin. One may accordingly speak of a special interaction between the particles that produces this dependence. It is called the *exchange interaction*. This is an entirely quantum effect and disappears completely, together with spin itself, in the limiting transition to classical mechanics.

The two-electron system just considered has an important feature: every energy level admits one definite value of the total spin, either 0 or 1. As will be seen below (§ 63), this one-to-one assignment of spin values to energy levels persists for systems with any number of electrons. It does not, however, hold for systems composed of particles whose spin exceeds $1/2$.

Consider two particles of arbitrary spin s. Their spin wave function is a spinor of rank $4s$:

$$\chi^{\overbrace{\lambda\mu\ldots}^{2s}\overbrace{\rho\sigma\ldots}^{2s}},$$

one half of whose indices ($2s$ of them) describe the spin of one particle, while the other half describe that of the second. The spinor is symmetric in the indices within either group. Interchange of the particles interchanges all the indices $\lambda, \mu, \ldots$ in the first group with $\rho, \sigma, \ldots$ in the second. To obtain the spin function for a state of total spin S, one contracts this spinor over $2s - S$ pairs of indices (each pair containing one index of either group) and symmetrizes over the indices left over. The result is a symmetric spinor of rank $2S$.

But contraction over a pair of spinor indices, as we know, forms a combination antisymmetric in those indices. The spin wave function therefore acquires the factor $(-1)^{2s-S}$ when the particles are interchanged.

The complete wave function of two particles, meanwhile, must acquire $(-1)^{2s}$ under their interchange: $+1$ for integral s and -1 for half-integral s. It follows that the exchange symmetry of the coordinate wave function is governed by the factor $(-1)^S$, which depends on S alone.

We thus find that the coordinate wave function of two identical particles is symmetric for even total spin and antisymmetric for odd total spin.

Recalling the relation established above between particle interchange and coordinate inversion, we also conclude that for even (odd) spin S, the system can possess only even (odd) orbital angular momentum.

Here again there is a dependence between the possible energies of the system and its total spin, but it is no longer unique. Energy levels with symmetric (antisymmetric) coordinate functions can occur for every even (odd) value of S.

Let us count all the different states of two particles having even and odd S. The quantity S takes $2s + 1$ values, namely $2s, 2s - 1, \ldots, 0$. For each fixed S there are $2S + 1$ states distinguished by the z component of spin, giving $(2s + 1)^2$ states altogether. Let s be integral. Of the $2s + 1$ possible values of S, $s + 1$ are even and s are odd. The total number of states with even S is therefore

$$\sum_{S=0,2,\ldots,2s}(2S+1)=(2s+1)(s+1);$$

the remaining $s(2s+1)$ states have odd S. In the same way, for half-integral s one finds $s(2s+1)$ states with even S and $(s+1)(2s+1)$ with odd S.

Problem

1. Determine the exchange splitting of the energy levels of a two-electron system, treating the interaction between the electrons as a perturbation.

Solution. Let the particles, when their interaction is neglected, occupy states with orbital wave functions $\varphi_1(\mathbf{r}_1)$ and $\varphi_2(\mathbf{r}_2)$. The system states of total spin $S=0$ and $S=1$ correspond, respectively, to the symmetrized and antisymmetrized products

$$\varphi=\frac{1}{\sqrt{2}}[\varphi_1(\mathbf{r}_1)\varphi_2(\mathbf{r}_2)\pm\varphi_1(\mathbf{r}_2)\varphi_2(\mathbf{r}_1)].$$

In these states the mean values of the particle-interaction operator $U(\mathbf{r}_2-\mathbf{r}_1)$ are $A\pm J$, where

$$A=\iint U|\varphi_1(\mathbf{r}_1)|^2|\varphi_2(\mathbf{r}_2)|^2,dV_1dV_2,$$

$$J=\iint U\varphi_1(\mathbf{r}_1)\varphi_1^*(\mathbf{r}_2)\varphi_2(\mathbf{r}_2)\varphi_2^*(\mathbf{r}_1),dV_1dV_2.$$

The integral J is called the *exchange integral.* Dropping the additive constant A, which is not an exchange contribution, we obtain the level shifts $\Delta E_0=J$ and $\Delta E_1=-J$, with the subscript giving S. These quantities

can be represented as eigenvalues of the spin *exchange operator*[5]

$$\widehat{V}_{\text{ex}}=-\frac{1}{2}J(1+4\hat{\mathbf{s}}_1\hat{\mathbf{s}}_2) \tag{1}$$

(for the eigenvalues of the product $\mathbf{s}_1\mathbf{s}_2$, see problem 2 of § 55).

If, for example, the electrons belong to different atoms, the exchange integral decreases exponentially as the distance R between the atoms increases. The form of the integrand makes clear that the integral is determined by the "overlap" of the state wave functions $\varphi_1(\mathbf{r}_1)$ and $\varphi_2(\mathbf{r}_2)$. Using the asymptotic decay law for discrete-spectrum wave functions (cf. (21.6)), we find

$$J\sim\exp[-(\varkappa_1+\varkappa_2)R],\qquad \varkappa_1=\frac{1}{\hbar}\sqrt{2m|E_1|},\qquad \varkappa_2=\frac{1}{\hbar}\sqrt{2m|E_2|},$$

where E_1, E_2 are the electron energy levels in the two atoms.

2. Do the same for a system of three electrons.

[5]This operator was introduced by Dirac.

SOLUTION. Using formula (1) of problem 1, we write the pairwise exchange-interaction operator for three electrons as

$$\widehat{V}_{\text{ex}} = -\sum J_{ab}(1/2 + 2\hat{\mathbf{s}}_a\hat{\mathbf{s}}_b), \tag{1}$$

where the sum runs over the pairs 12, 13, and 23. From formulas (55.6), the matrix elements of $\hat{\mathbf{s}}_a\hat{\mathbf{s}}_b$ between states with different pairs σ_a, σ_b are

$$\langle 1/2, 1/2|\mathbf{s}_a\mathbf{s}_b|1/2, 1/2\rangle = 1/4, \quad \langle 1/2, -1/2|\mathbf{s}_a\mathbf{s}_b|1/2, -1/2\rangle = -1/4,$$

$$\langle 1/2, -1/2|\mathbf{s}_a\mathbf{s}_b| - 1/2, 1/2\rangle = 1/2.$$

We begin by finding the energy associated with the largest possible projection of the total spin, $M_S = \sigma_1 + \sigma_2 + \sigma_3$, namely $M_S = 3/2$. This also determines the energy of the state with total spin $S = 3/2$. Evaluation of the corresponding diagonal matrix element of (1) gives

$$\Delta E_{3/2} = -(J_{12} + J_{13} + J_{23}).$$

Now pass to the states with $M_S = 1/2$. This value can be realized in three ways, according to which of $\sigma_1, \sigma_2, \sigma_3$ equals $-1/2$, the other two being $1/2$. Direct treatment would therefore give a cubic secular equation. The calculation can be simplified at once, however, by observing that one root must be the already determined energy of the $S = 3/2$ state. The secular equation must consequently be divisible by $\Delta E - \Delta E_{3/2}$, so in this case there is no need to calculate the constant term of the cubic[6].

Calculating the leading terms, we obtain

$$(\Delta E)^3 + (J_{12}+J_{13}+J_{23})(\Delta E)^2 + [J_{12}J_{13}+J_{12}J_{23}+J_{13}J_{23}-(J_{12}^2+J_{13}^2+J_{23}^2)]\Delta E + \ldots = 0.$$

Division by $\Delta E + J_{12} + J_{13} + J_{23}$ then yields the two energy levels associated with states of spin $S = 1/2$:

$$\Delta E_{1/2} = \pm[J_{12}^2 + J_{13}^2 + J_{23}^2 - J_{12}J_{13} - J_{12}J_{23} - J_{13}J_{23}]^{1/2}.$$

Thus there are three energy levels in all, in agreement with the counting carried out in problem 1 of § 63.

3. From which states can the ^{8}Be nucleus decay into two α particles?

SOLUTION. Since an α particle has no spin, a system of two α particles can have only even orbital angular momentum (which is also its total angular momentum), and its states have even parity. The stated decay is therefore possible only from even-parity states of the ^{8}Be nucleus having even total angular momentum.

§ 63. Symmetry under permutations

For a system consisting of only two particles, we could assert that its coordinate wave functions for stationary states, $\varphi(\mathbf{r}_1, \mathbf{r}_2)$, must be either symmetric or antisymmetric. For a system containing an arbitrary number of particles, however, the solutions of

[6] This device is especially useful in analogous calculations for systems containing more particles.

the Schrödinger equation (the coordinate wave functions) need not be symmetric or antisymmetric under interchange of every pair of particles, unlike the complete wave functions, which include a spin factor. The reason is that interchanging only the coordinates of two particles does not yet amount to physically interchanging the particles themselves. Here the physical identity of the particles implies only that the system's Hamiltonian is invariant under particle permutations. Consequently, if one function solves the Schrödinger equation, then the functions obtained from it by the various permutations of its variables are solutions as well.

We first give some general remarks about permutations. A system of N particles admits $N!$ distinct permutations. Imagine that all particles have been numbered; every permutation can then be represented by a particular ordering of the numbers $1, 2, 3, \ldots$ Any such ordering can be produced from the natural ordering $1, 2, 3, \ldots$ by successively interchanging pairs of particles. A permutation is called *even* or *odd* according as it is effected by an even or odd number of pairwise interchanges. Let $\widehat{P}$ denote the permutation operators for the N particles, and introduce δ_P, equal to $+1$ when $\widehat{P}$ is an even permutation and to -1 when it is odd. If φ is symmetric in all the particles, then

$$\widehat{P}\varphi = \varphi,$$

whereas, if the function is antisymmetric in all the particles,

$$\widehat{P}\varphi = \delta_P \varphi.$$

From an arbitrary function $\varphi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$ one can form a symmetric function by the operation of *symmetrization*, written as

$$\varphi_{\text{sym}} = \text{const} \cdot \sum_P \widehat{P}\varphi, \tag{63.1}$$

where the sum extends over every possible permutation. The construction of an antisymmetric function (an operation also called *alternation*) can be written

$$\varphi_{\text{asym}} = \text{const} \cdot \sum_P \delta_P \widehat{P}\varphi. \tag{63.2}$$

We return to the behavior of the wave functions φ of a system of identical particles under permutations[7]. Mathematically, the symmetry of the system's Hamiltonian $\widehat{H}$ in all particles means that it commutes with every permutation operator $\widehat{P}$. These operators, however, do not commute with one another and hence cannot all be reduced simultaneously to diagonal form. Thus the wave functions φ cannot be chosen so that each is symmetric or antisymmetric under every separate pairwise interchange[8].

[7] In mathematical terms, the problem is to find the irreducible representations of the permutation group. Detailed accounts of the mathematical theory of permutation groups may be found in: H. Weyl, *Group Theory and Quantum Mechanics*; M. Hamermesh, *Group Theory and Applications to Problems in Physics*; I. G. Kaplan, *Symmetry in Many-Electron Systems*.

[8] Only a two-particle system has just one permutation operator, which can be reduced to diagonal form simultaneously with the Hamiltonian.

Our task is to determine the possible symmetry types, under permutations of the variables, of functions $\varphi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$ of N variables (or of sets of several such functions).

The symmetry must be incapable of further increase: every additional symmetrization or alternation applied to these functions must turn them either into linear combinations of the same functions or identically into zero.

We already know two operations that produce functions of maximal symmetry: symmetrization in all variables and alternation in all variables. They can be generalized as follows.

Partition the complete set of N variables $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N$ (or, equivalently, the indices $1, 2, 3, \ldots, N$) into several rows containing $N_1, N_2, \ldots$ elements (variables), where $N_1 + N_2 + \ldots = N$. Such a partition can be displayed pictorially by a diagram (a *Young diagram*) in which each of the numbers $N_1, N_2, \ldots$ is represented by a row of cells. Figure 21, for example, shows the partitions $6+4+4+3+3+1+1$ and $7+5+5+3+1+1$ for $N = 22$; one of the numbers $1, 2, 3, \ldots$ is to be placed in every square. If the rows are arranged in decreasing order of length (as in Fig. 21), the diagram consists not only of successive horizontal rows but also of vertical columns.

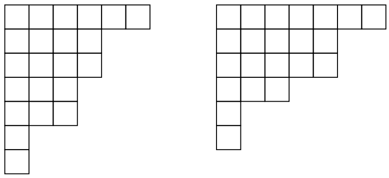


Fig. 21

Symmetrize an arbitrary function $\varphi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$ in the variables belonging to each row. Alternation may thereafter be performed only on variables from different rows; alternation in a pair of variables lying in the same row plainly gives zero identically.

Choose one variable from every row. Without loss of generality these variables may be regarded as occupying the first cells of their rows, since after symmetrization their ordering among the cells of a row is immaterial; now alternate in the chosen variables. Next strike out the first column and alternate in variables chosen one from each row of the shortened diagram; they may again be regarded as occupying the first cells of the shortened rows. Continuing in this manner, we obtain a function first symmetrized in the variables of every row and then alternated in those of every column. After alternation, of course, the function generally ceases to be symmetric in the variables of each row; symmetry remains only for variables in the cells of the first row that project beyond the other rows.

Assign the N variables in different ways to the rows of the Young diagram (their placement among the cells of any one row

is immaterial). This procedure gives a collection of functions that are carried into one another by an arbitrary permutation of the variables[9]. It must nevertheless be stressed that

[9]The order of symmetrization and alternation could instead be reversed: first alternate over the variables in each column, and then symmetrize over those in the rows. This produces nothing new, however, because the

these functions are not all linearly independent. In general, the number of independent functions is smaller than the number of possible assignments of the variables to the rows of the diagram; we shall not examine this point further here[10].

Thus every Young diagram specifies a particular symmetry type of functions under permutations. By constructing every Young diagram possible for a given N, we obtain all possible symmetry types. This reduces to partitioning the number N in every possible way into a sum of smaller terms, with N itself included among the allowed partitions. For $N = 4$, for example, the partitions are 4, 3 + 1, 2 + 2, 2 + 1 + 1, and 1 + 1 + 1 + 1.

A Young diagram can be associated with each energy level of the system; it specifies the permutation symmetry of the corresponding solutions of the Schrödinger equation. In general, a given energy value corresponds to several distinct functions that transform into one another under permutations. This "permutation degeneracy" is connected with the already noted noncommutativity of the operators $\widehat{P}$, each of which commutes with the Hamiltonian (cf. § 10, p. 48). It does not, however, signify any additional physical degeneracy of the energy levels. Multiplied by spin functions, all these distinct coordinate wave functions enter one particular combination—the complete wave function—which satisfies the condition of symmetry or antisymmetry according to the particles' spin.

For any given N, the various symmetry types always include two that are each represented by a single function. In one case the function is symmetric in every variable, and in the other it is antisymmetric (the Young diagram in the former case

is a single row of N boxes, while in the latter it is a single column).

We now pass to the spin wave functions $\chi(\sigma_1, \sigma_2, \ldots, \sigma_N)$. Their symmetry types under particle permutations are described by the same Young diagrams, with the particle spin projections playing the role of variables. The question arises: which diagram must correspond to the spin function when the diagram of the coordinate function is given? Suppose first that the particles have integral spin. The complete wave function ψ must then be symmetric in all particles. This requires the spin and coordinate functions to have symmetry described by the same Young diagram, while ψ is expressed as a particular bilinear combination of the two sets of functions. We shall not consider the construction of this combination here.

Suppose instead that the particles have half-integral spin. The complete wave function must then be antisymmetric in all particles. It can be shown that the Young diagrams of the coordinate and spin functions must be dual: each is obtained from the other by interchanging rows and columns (the two diagrams in Fig. 21 furnish an example).

We examine more closely the important case of particles with spin 1/2, such as electrons. Each spin variable $\sigma_1, \sigma_2, \ldots$ can assume only the two values $\pm 1/2$. A function antisymmetric in any two variables vanishes when those variables have equal values. It follows that χ can be alternated only in pairs of variables: upon alternation in three variables, at least two of them necessarily have equal values, and the result is identically zero.

For a system of electrons, therefore, the Young diagrams of the spin functions can

---

functions obtained by the two procedures are linear combinations of one another.

[10] A set of independent functions carried into one another forms a basis for an irreducible representation of the permutation group. The number of these functions is the dimension of the representation. For particles of spin 1/2, it is found in Problem 1 of this section.

contain columns only one or two cells long (that is, no more than two rows); in the Young diagrams for the coordinate functions, the analogous restriction applies to row lengths. Consequently, the number of possible permutation-symmetry types for N electrons equals the number of partitions of N into ones and twos. For even N it is $N/2+1$ (partitions containing $0, 1, \ldots, N/2$ twos), and for odd N it is $(N+1)/2$ (partitions containing $0, 1, \ldots, (N-1)/2$ twos). Figure 22 shows the possible Young diagrams, both coordinate and spin, for $N = 4$.

It is readily seen that each such symmetry type (each Young diagram) corresponds to a definite total spin S of the electron system. We shall represent

the spin functions in spinor form, as a spinor $\chi^{\lambda\mu\cdots}$ of rank N, whose indices—each associated with the spin of one particle—are the variables placed in the cells of the Young diagrams.

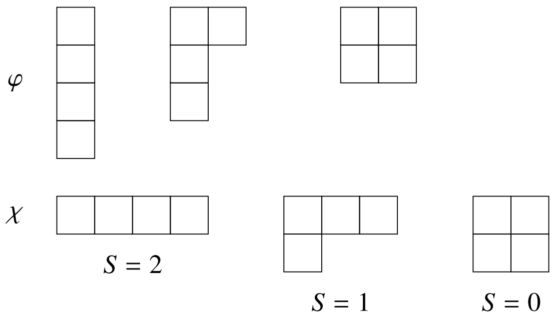


Fig. 22

Consider a spin Young diagram with two rows containing N_1 and N_2 boxes, respectively ($N_1 + N_2 = N$, $N_1 \geq N_2$). Each of the first N_2 columns has two boxes, so the spinor must be antisymmetric in each corresponding pair of indices. By contrast, it must be symmetric in the indices occupying the final $n = N_1 - N_2$ boxes of the first row. As we know, a rank-N spinor of this kind reduces to a symmetric spinor of rank n, and the latter has total spin $S = n/2$. In terms of the Young diagrams for the coordinate functions, therefore, a diagram having n one-box rows corresponds to a state whose total spin is $S = n/2$. For even N, the total spin may take the integer values from 0 through $N/2$; for odd N, it may take the half-integer values from $1/2$ through $N/2$, as required.

This one-to-one correspondence between Young diagrams and total spin is specific to systems of spin-$1/2$ particles; for a system of only two particles, we already encountered it in the preceding section. For N particles of spin s, the spin wave function is formed from a product of N symmetric spinors of rank $2s$, and is consequently a spinor of rank $2Ns$. When this spinor is symmetrized according to a particular Young diagram with N boxes, its independent components can usually be combined into several sets of linear combinations, each associated with a different value of the system's total spin S.

Just as the Young diagram for the spin functions of spin-$1/2$ particles cannot have columns containing more than two cells, so for particles of arbitrary spin s the column length must not exceed $2s+1$ cells.

If the number of particles N is an integral multiple of $2s+1$, the possible Young diagrams include a rectangular one in which every column contains $2s + 1$ cells. This diagram corresponds to one definite value of the total spin, $S = 0$. Hence any two spin Young diagrams that can be joined to form a rectangle

of height $2s + 1$ correspond to the same values of S[11]. This conclusion is simply a consequence of the fact that, when two angular momenta are added, their resultant can vanish only if the two angular momenta have equal magnitudes.

To conclude this section, return to the circumstance noted earlier (see the note on p. 86): for systems of several identical particles, one cannot assert that the wave function of the lowest-energy stationary state has no nodes. We can now refine that observation and explain its origin.

A nodeless wave function (here we mean the coordinate function) must necessarily be symmetric in all particles. If it were antisymmetric under interchange of any pair of particles, say 1 and 2, it would vanish at $\mathbf{r}_1 = \mathbf{r}_2$. Yet for a system of three or more electrons, a completely symmetric coordinate wave function is not allowed at all: the Young diagram of the coordinate function cannot contain a row longer than two cells. Thus, although the Schrödinger-equation solution belonging to the lowest eigenvalue has no nodes, as follows from the theorem of the calculus of variations, that solution may be physically inadmissible. The normal state of the system then corresponds to an eigenvalue other than the lowest one, and its wave function will in general have nodes. More generally, this situation occurs for particles of half-integral spin s in systems containing more than $2s + 1$ particles. In systems composed of bosons, by contrast, a completely symmetric coordinate wave function is always possible.

Problem

1. Determine how many energy levels have each possible value of the total spin S in a system of N particles of spin $1/2$ (F. Bloch, 1929).

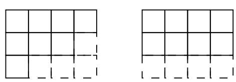
Solution. A prescribed value of the total-spin projection $M_S = \sum \sigma$ can be formed in

$$f(M_S) = \frac{N!}{(N/2 + M_S)!(N/2 - M_S)!}$$

different ways: assign $\sigma = 1/2$ to $N/2 + M_S$ particles and $\sigma = -1/2$ to all the others. Every energy level of total spin S supplies $2S + 1$ states whose projections are $M_S = S, S - 1, \ldots, -S$. It follows directly that the number of distinct levels having this value of S is

$$n(S) = f(S) - f(S + 1) = \frac{N!(2S + 1)}{(N/2 + S + 1)!(N/2 - S)!}.$$

[11] For example, when $s = 1$, the following pairs of diagrams have this property:



Complementary diagrams are drawn with solid and dashed lines.

The total number $n = \sum_S n(S)$ of distinct energy levels is

$$n = f(0) = \frac{N!}{[(N/2)!]^2}$$

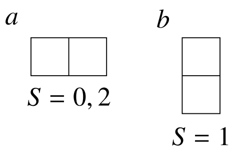
when N is even, and

$$n = f(1/2) = \frac{N!}{[(N+1)/2]![(N-1)/2]!}$$

when N is odd.

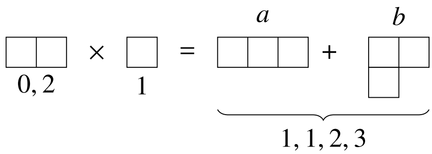
2. Find the total-spin values S that occur for the different symmetry types of the spin functions of a system consisting of two, three, or four particles, each of spin 1.

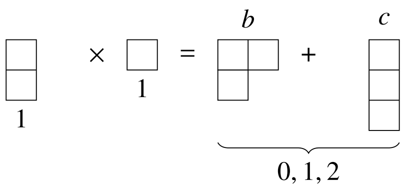
SOLUTION. For two particles, the correspondence follows by requiring the multiplier acquired by the spin function upon interchange of the particles to equal $(-1)^{2s-S}$ (see the end of § 62). Thus, for particles of spin $s = 1$, the correspondence is



(1)

The Young diagrams for three particles are obtained by adjoining one cell, in every possible way, to the diagrams in (1). Symbolically, this can be written

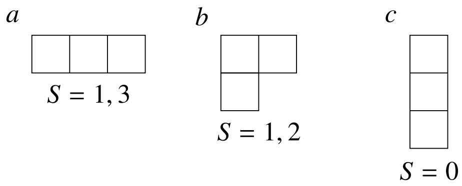




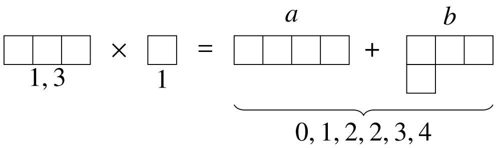
The values of S are written below the diagrams. The total spins for the three-particle system (the diagrams on the right) are obtained from the spins of the two-particle and one-particle systems (on the left) by the rule for addition of angular momenta.[12] To distribute

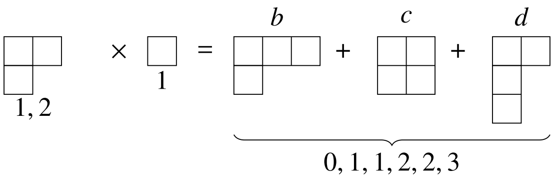
the resulting values of S among the individual diagrams on the right, observe first that diagram c (a column of three cells) corresponds to $S = 0$. Diagram b must therefore receive the remaining values 1 and 2 from the second equality; diagram a then receives the values 1 and 3 left by b in the first equality:

[12]The value 1 occurs twice below the diagrams on the right because it is produced once by adding angular momenta 0 and 1, and once by adding angular momenta 2 and 1.



The Young diagrams for four particles result from adding one cell to the diagrams in (2), subject to the restriction that no column may contain more than three cells:





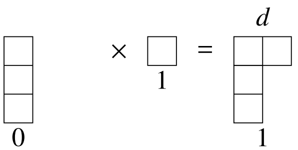
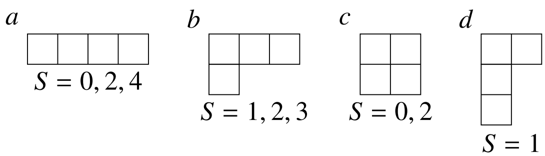


Diagram c combines with diagram $1a$ to make a rectangle whose columns contain three cells; hence it has the same values $S = 0, 2$ as $1a$. The values of S for diagram b are fixed by those remaining in the second equality, and those for diagram a are then fixed by the remainder in the first. The spin belonging to diagram d follows uniquely from the third equality:



§ 64. Second quantization. The case of Bose statistics

For the theory of systems formed by many identical particles, a special method known as *second quantization* is used extensively. This method is particularly indispensable in relativistic theory, where one has to treat systems in which even the number of particles is variable[13].

[13] Dirac developed the method of second quantization for photons in the theory of radiation (1927); Wigner and Jordan subsequently extended it to fermions (E. Wigner, P. Jordan, 1928).

Let $\psi_1(\xi), \psi_2(\xi), \ldots$ be some complete orthogonal normalized set of wave functions for the stationary states of one particle[14]. These may be the states of a particle in an arbitrarily selected external field, although plane waves are generally chosen: the wave functions of a free particle having specified momentum (and spin projection). To make the state spectrum discrete, the motion is then considered in a large but bounded region of space. For motion in a finite volume, the eigenvalues of the momentum components form a discrete sequence; the spacings of successive values are inversely proportional to the linear dimensions of the region and tend to zero as those dimensions increase.

In a system of free particles, the momentum of every particle is conserved separately. Consequently, the *occupation numbers* are conserved as well: the numbers $N_1, N_2, \ldots$ that state how many particles are in each of the states $\psi_1, \psi_2, \ldots$. In a system of interacting particles, the separate particle momenta are no longer conserved, and neither are the occupation numbers. For such a system one can speak only of the probability distribution among different values of the occupation numbers. Our aim is to construct a mathematical formalism in which these numbers themselves, rather than particle coordinates and spin projections, are the independent variables.

Dirac notation (see the end of § 11) is convenient in this formalism, with $N_1, N_2, \ldots$ chosen as the quantum numbers specifying a state. The states corresponding to the wave functions (61.3) and (61.5) will be denoted by $|N_1, N_2, \ldots\rangle$. Coordinate and spin variables then no longer appear explicitly.

With this choice of independent variables, the operators of the various physical quantities (including the system's Hamiltonian) must likewise be specified by how they act on functions of the occupation numbers. Such a specification can be obtained by starting from the usual matrix representation of operators. One must then consider

the operator matrix elements between wave functions for stationary states of the noninteracting-particle system. Since these states can be characterized by assigning definite values to the occupation numbers, the resulting matrix elements reveal how the operators act on those variables.

We first consider systems of particles obeying Bose statistics.

Let $\hat{f}_a^{(1)}$ be the operator of some quantity referring to one particle (particle a), so that it acts only on functions of the variables ξ_a. Introduce the operator symmetric in all particles

$$\hat{F}^{(1)} = \sum_a \hat{f}_a^{(1)} \tag{64.1}$$

(the sum runs over every particle), and determine its matrix elements with respect to the wave functions (61.3). It is readily seen first of all that these matrix elements can be nonzero only for transitions that leave all the numbers $N_1, N_2, \ldots$ unchanged (the diagonal elements), or for transitions in which one of these numbers rises by one while another falls by one. Indeed, each operator $\hat{f}_a^{(1)}$ acts on only one factor in the product $\psi_{p_1}(\xi_1)\psi_{p_2}(\xi_2)\cdots\psi_{p_N}(\xi_N)$, and hence its matrix element can differ from zero only when a single particle changes its state. This means that the number of particles in one state decreases by one, whereas the number in another state increases by one. The actual

[14]As in § 61, ξ denotes all the coordinates together with the particle's spin projection σ; integration over $d\xi$ will mean integration over the coordinates and summation over σ.

evaluation of these matrix elements is very simple; it is easier to carry it out than to follow a written account of it. We therefore give only the result. The off-diagonal elements are

$$\langle N_i, N_k - 1|\hat{F}^{(1)}|N_i - 1, N_k\rangle = f_{ik}^{(1)}\sqrt{N_i N_k}. \tag{64.2}$$

Only the indices in which the matrix element is not diagonal have been shown; the others are suppressed for brevity. Here $f_{ik}^{(1)}$ is the matrix element

$$f_{ik}^{(1)} = \int \psi_i^*(\xi)\hat{f}^{(1)}\psi_k(\xi)\, d\xi; \tag{64.3}$$

the operators $\hat{f}_a^{(1)}$ differ only in the designation of the variables on which they act, so the integrals (64.3) do not depend on a, and this index has been omitted. The diagonal matrix elements of $\hat{F}^{(1)}$ are the mean values

of $F^{(1)}$ in the states $\Psi_{N_1 N_2 \ldots}$. Their evaluation gives

$$\overline{F^{(1)}} = \sum_i f_{ii}^{(1)} N_i. \tag{64.4}$$

We now introduce the operators fundamental to the second-quantization method, $\hat{a}_i$. They act not on functions of coordinates but on functions of the occupation numbers. By definition, when $\hat{a}_i$ acts on the state $|N_1, N_2, \ldots\rangle$, it reduces the variable N_i by one and at the same time multiplies the function by $\sqrt{N_i}$[15]:

$$\hat{a}_i|N_1, N_2, \ldots, N_i, \ldots\rangle = \sqrt{N_i}|N_1, N_2, \ldots, N_i - 1, \ldots\rangle. \tag{64.5}$$

One may say that $\hat{a}_i$ lowers by one the number of particles in the ith state; it is accordingly called the particle *annihilation operator*. It can be represented by a matrix having just one nonzero element, namely

$$\langle N_i - 1|\hat{a}_i|N_i\rangle = \sqrt{N_i}. \tag{64.6}$$

The operator $\hat{a}_i^+$ adjoint to $\hat{a}_i$ is represented, by definition (see (11.9)), by the matrix with the single element

$$\langle N_i|\hat{a}_i^+|N_i - 1\rangle = \langle N_i - 1|\hat{a}_i|N_i\rangle^* = \sqrt{N_i}. \tag{64.7}$$

This means that its action on the function $|N_1, N_2, \ldots\rangle$ increases N_i by one:

$$\hat{a}_i^+|N_1, N_2, \ldots, N_i, \ldots\rangle = \sqrt{N_i + 1}|N_1, N_2, \ldots, N_i + 1, \ldots\rangle. \tag{64.8}$$

In other words, $\hat{a}_i^+$ increases by one the number of particles in the ith state; it is called the particle *creation operator*.

When the product $\hat{a}_i^+\hat{a}_i$ acts on a wave function, it can only multiply that function by a constant while leaving all the variables $N_1, N_2, \ldots$ unchanged: $\hat{a}_i$ lowers N_i by one, after which $\hat{a}_i^+$ restores its initial value. Direct multiplication of the matrices (64.6) and

[15]We have used the natural notation $\hat{a}|n\rangle$ for the result of applying the operator $\hat{a}$ to the wave function of the state $|n\rangle$.

(64.7) does indeed show that $\hat{a}_i^+ \hat{a}_i$ is represented by a diagonal matrix whose diagonal entries equal N_i:

$$\hat{a}_i^+ \hat{a}_i = N_i. \tag{64.9}$$

In the same way one finds

$$\hat{a}_i \hat{a}_i^+ = N_i + 1. \tag{64.10}$$

Subtracting these expressions gives the commutation rule for $\hat{a}_i$ and $\hat{a}_i^+$:

$$\hat{a}_i \hat{a}_i^+ - \hat{a}_i^+ \hat{a}_i = 1. \tag{64.11}$$

Operators with different indices i and k, which act on different variables (N_i and N_k), commute:

$$\hat{a}_i \hat{a}_k - \hat{a}_k \hat{a}_i = 0, \qquad \hat{a}_i \hat{a}_k^+ - \hat{a}_k^+ \hat{a}_i = 0, \qquad i \neq k. \tag{64.12}$$

From the properties of $\hat{a}_i$ and $\hat{a}_i^+$ just described, it follows at once that the operator

$$\hat{F}^{(1)} = \sum_{i,k} f_{ik}^{(1)} \hat{a}_i^+ \hat{a}_k \tag{64.13}$$

is identical with (64.1). Indeed, all matrix elements computed with (64.6) and (64.7) coincide with those in (64.2) and (64.4). This result is very important. The quantities $f_{ik}^{(1)}$ in (64.13) are merely numbers. We have thereby expressed an ordinary operator acting on functions of the coordinates as an operator acting on functions of the new variables, the occupation numbers N_i.

The result is readily generalized to operators of other kinds. Let

$$\hat{F}^{(2)} = \sum_{a>b} \hat{f}_{ab}^{(2)}, \tag{64.14}$$

where $\hat{f}_{ab}^{(2)}$ is the operator of a physical quantity referring to a pair of particles at once and therefore acts on functions of ξ_a and ξ_b. An analogous calculation shows that this operator can be expressed through $\hat{a}_i$ and $\hat{a}_i^+$ as

$$\hat{F}^{(2)} = \frac{1}{2} \sum_{i,k,l,m} \langle ik|f^{(2)}|lm\rangle \hat{a}_i^+ \hat{a}_k^+ \hat{a}_m \hat{a}_l, \tag{64.15}$$

$$\langle ik|f^{(2)}|lm\rangle = \iint \psi_i^*(\xi_1)\psi_k^*(\xi_2)\hat{f}^{(2)}\psi_l(\xi_1)\psi_m(\xi_2)\, d\xi_1 d\xi_2.$$

The generalization of these formulas to operators of any other kind that are symmetric in all the particles ($\hat{F}^{(3)} = \sum \hat{f}_{abc}^{(3)}$, and so forth) is evident.

These formulas also allow the Hamiltonian of the physical system under study, consisting of N interacting identical particles, to be expressed through $\hat{a}_i$ and $\hat{a}_i^+$. The Hamiltonian of such a system is, of course, symmetric in all the particles. In the nonrelativistic approximation[16], it is independent of their spins

and may be written in the general form

$$\hat{H} = \sum_a \hat{H}_a^{(1)} + \sum_{a>b} U^{(2)}(\mathbf{r}_a, \mathbf{r}_b) + \sum_{a>b>c} U^{(3)}(\mathbf{r}_a, \mathbf{r}_b, \mathbf{r}_c) + \cdots \tag{64.16}$$

[16] In the absence of a magnetic field.

Here $\hat{H}_a^{(1)}$ is the part of the Hamiltonian depending on the coordinates of only one particle, particle a:

$$\hat{H}_a^{(1)} = -\frac{\hbar^2}{2m}\Delta_a + U^{(1)}(\mathbf{r}_a), \tag{64.17}$$

where $U^{(1)}(\mathbf{r}_a)$ is the potential energy of one particle in the external field. The remaining terms in (64.16) describe the energy of interaction among the particles; terms depending respectively on the coordinates of two, three, and so on particles have been separated.

Writing the Hamiltonian in this form permits direct application of (64.13), (64.15), and the analogous formulas. Thus

$$\hat{H} = \sum_{i,k} H_{ik}^{(1)} \hat{a}_i^+ \hat{a}_k + \frac{1}{2} \sum_{i,k,l,m} \langle ik|U^{(2)}|lm\rangle \hat{a}_i^+ \hat{a}_k^+ \hat{a}_m \hat{a}_l + \cdots \tag{64.18}$$

This yields the required representation of the Hamiltonian as an operator acting on functions of the occupation numbers.

When the particles in a system do not interact, expression (64.18) retains only its first term:

$$\hat{H} = \sum_{i,k} H_{ik}^{(1)} \hat{a}_i^+ \hat{a}_k. \tag{64.19}$$

If the functions ψ_i are chosen to be eigenfunctions of the one-particle Hamiltonian $\hat{H}^{(1)}$, then the matrix $H_{ik}^{(1)}$ is diagonal, and its diagonal entries are the particle's energy eigenvalues ε_i. Consequently,

$$\hat{H} = \sum_i \varepsilon_i \hat{a}_i^+ \hat{a}_i;$$

replacing $\hat{a}_i^+ \hat{a}_i$ by its eigenvalues (64.9), we obtain the following expression for the energy levels of the system:

$$E = \sum_i \varepsilon_i N_i.$$

This is the trivial result that should indeed have emerged.

The formalism developed above can be put into a more compact form by introducing the so-called ψ-operators[17]

$$\hat{\psi}(\xi) = \sum_i \psi_i(\xi)\hat{a}_i, \qquad \hat{\psi}^+(\xi) = \sum_i \psi_i^*(\xi)\hat{a}_i^+, \tag{64.20}$$

where ξ is regarded as a parameter. It is clear from the properties of $\hat{a}_i$ and $\hat{a}_i^+$ stated above that $\hat{\psi}$ decreases the total number of particles in the system by one, whereas $\hat{\psi}^+$ increases it by one.

It is easy to see that $\hat{\psi}^+(\xi_0)$ creates a particle situated at the point ξ_0. Indeed, the action of $\hat{a}_i^+$ creates a particle in the state whose wave function is $\psi_i(\xi)$. It follows that applying $\psi_i^*(\xi_0)\hat{a}_i^+$ creates a particle in the state with wave function

$$\sum_i \psi_i^*(\xi)\psi_i(\xi_0) = \delta(\xi - \xi_0).$$

[17]Notice the analogy between (64.20) and the expansion $\psi = \sum a_i \psi_i$ of an arbitrary wave function in a complete set of functions. Here the expansion is, as it were, quantized once again, which accounts for the name of the entire method: second quantization.

Formula (5.12) has been used here; the result corresponds to a particle with specified coordinates and spin[18].

The commutation rules for the ψ-operators follow immediately from those for $\hat{a}_i$ and $\hat{a}_i^+$:

$$\hat{\psi}(\xi)\hat{\psi}(\xi') - \hat{\psi}(\xi')\hat{\psi}(\xi) = 0, \tag{64.21}$$

$$\hat{\psi}(\xi)\hat{\psi}^+(\xi') - \hat{\psi}^+(\xi')\hat{\psi}(\xi) = \sum_i \psi_i(\xi)\psi_i^*(\xi') = \delta(\xi - \xi'). \tag{64.22}$$

In terms of the ψ-operators, the second-quantized operator $\hat{F}^{(1)}$ takes the form

$$\hat{F}^{(1)} = \int \hat{\psi}^+(\xi)\hat{f}^{(1)}\hat{\psi}(\xi)\,d\xi \tag{64.23}$$

(the operator $\hat{f}^{(1)}$ is understood here to act in $\hat{\psi}(\xi)$ on functions of the parameters ξ). Substitution of $\hat{\psi}$ and $\hat{\psi}^+$ from (64.20), together with the definition (64.3), does indeed return us

to (64.13). Similarly, in place of (64.15) we have

$$\hat{F}^{(2)} = \frac{1}{2}\iint \hat{\psi}^+(\xi)\hat{\psi}^+(\xi')\hat{f}^{(2)}\hat{\psi}(\xi')\hat{\psi}(\xi)\,d\xi\,d\xi'. \tag{64.24}$$

In particular, the Hamiltonian of the system expressed through the ψ-operators is

$$\begin{aligned}\hat{H} = &\int \left\{-\frac{\hbar^2}{2m}\hat{\psi}^+(\xi)\Delta\hat{\psi}(\xi) + \hat{\psi}^+(\xi)U^{(1)}(\xi)\hat{\psi}(\xi)\right\} d\xi \\ &+ \frac{1}{2}\iint \hat{\psi}^+(\xi)\hat{\psi}^+(\xi')U^{(2)}(\xi,\xi')\hat{\psi}(\xi')\hat{\psi}(\xi)\,d\xi\,d\xi' + \cdots\end{aligned} \tag{64.25}$$

The operator $\hat{\psi}^+(\xi)\hat{\psi}(\xi)$ is formed from the ψ-operators in analogy with the product $\psi^*\psi$, which gives the probability density of a particle in a state whose wave function is ψ. It is called the particle *density operator*. The integral

$$\hat{N} = \int \hat{\psi}^+\hat{\psi}\,d\xi \tag{64.26}$$

serves in the second-quantization formalism as the operator for the total number of particles in the system. Indeed, on substituting the ψ-operators from (64.20) and using the normalization and mutual orthogonality of the wave functions, we find $\hat{N} = \sum_i \hat{a}_i^+\hat{a}_i$. Each term in this sum is the particle-number operator for the ith state; by (64.9), its eigenvalues are the occupation numbers N_i. Their sum is the total number of particles in the system[19].

Finally, if a system contains bosons of different kinds, separate operators $\hat{a}$ and $\hat{a}^+$ must be introduced for every kind of particle in the second-quantization method. Operators belonging to different kinds of particles evidently commute with one another.

[18] $\delta(\xi - \xi_0)$ conventionally denotes the product

$$\delta(x - x_0)\,\delta(y - y_0)\,\delta(z - z_0)\,\delta_{\sigma\sigma_0}.$$

[19] For systems having a prescribed number of particles, these statements (as well as the properties of the free-particle Hamiltonian (64.19)) appear trivial. In relativistic theory, however, their generalization leads to new results which are far from trivial (cf. Vol. IV, § 11).

§ 65. Second quantization: Fermi statistics

The essential basis of the method of second quantization remains unchanged for systems of identical fermions. The specific formulas for matrix elements of quantities and for the operators $\hat{a}_i$, however, naturally change.

The wave function $\psi_{N_1 N_2 \ldots}$ now has the form (61.5). Its antisymmetry first raises the question of choosing its sign. No such question arose with Bose statistics: because the wave function was symmetric, a sign once chosen was retained under every permutation of the particles.

To fix the sign of (61.5), adopt the following convention. Number all the states ψ_i once and for all in sequence. The rows of the determinant (61.5) are then always filled so that

$$p_1 < p_2 < p_3 < \cdots < p_N, \tag{65.1}$$

while its columns contain functions of the different variables in the order $\xi_1, \xi_2, \ldots, \xi_N$. No two of $p_1, p_2, \ldots$ can be equal, since the determinant would otherwise vanish. Equivalently, every occupation number N_i can have only the value 0 or 1.

Consider again an operator of the form (64.1), $\hat{F}^{(1)} = \sum_a \hat{f}_a^{(1)}$. For the same reasons as in § 64, its matrix elements can be nonzero only for transitions in which all occupation numbers remain unchanged, or in which one of them, N_i, decreases by one (from unity to zero) while another, N_k, increases by one (from zero to unity). For $i < k$ one readily finds

$$\langle 1_i, 0_k | \hat{F}^{(1)} | 0_i, 1_k \rangle = f_{ik}^{(1)} (-1)^{\Sigma(i+1,k-1)}. \tag{65.2}$$

Here $0_i, 1_i$ denote $N_i = 0, N_i = 1$, and $\sum(k,l)$ denotes the sum of the occupation numbers of all states from k through l.[20]

$$\sum(k,l) = \sum_{n=k}^{l} N_n.$$

The diagonal elements are still given by (64.4):

$$\overline{F^{(1)}} = \sum_i f_{ii}^{(1)} N_i. \tag{65.3}$$

For $\hat{F}^{(1)}$ to be representable in the form (64.13), the annihilation operators $\hat{a}_i$ must be defined as matrices

with elements

$$\langle 0_i | \hat{a}_i | 1_i \rangle = \langle 1_i | \hat{a}_i^+ | 0_i \rangle = (-1)^{\Sigma(1,i-1)}. \tag{65.4}$$

Multiplication of these matrices gives, for $k > i$,

$$\begin{aligned} \langle 1_i, 0_k | \hat{a}_i^+ \hat{a}_k | 0_i, 1_k \rangle &= \langle 1_i, 0_k | \hat{a}_i^+ | 0_i, 0_k \rangle \langle 0_i, 0_k | \hat{a}_k | 0_i, 1_k \rangle \\ &= (-1)^{\Sigma(1,i-1)} (-1)^{\Sigma(1,i-1) + \Sigma(i+1,k-1)}, \end{aligned}$$

or

$$\langle 1_i, 0_k | \hat{a}_i^+ \hat{a}_k | 0_i, 1_k \rangle = (-1)^{\Sigma(i+1,k-1)}. \tag{65.5}$$

[20] For $i > k$, the exponent in (65.2) must be $\sum(k+1, i-1)$. For $i = k \pm 1$, these sums are to be replaced by zero.

For $i = k$, the matrix $\hat{a}_i^+\hat{a}_i$ is diagonal, with element unity when $N_i = 1$ and zero when $N_i = 0$. Thus

$$\hat{a}_i^+\hat{a}_i = N_i. \tag{65.6}$$

Substitution of these expressions in (64.13) indeed gives (65.2), (65.3).

Multiplying $\hat{a}_i^+$ and $\hat{a}_k$ in the opposite order gives

$$\begin{aligned}\langle 1_i, 0_k|\hat{a}_k\hat{a}_i^+|0_i, 1_k\rangle &= \langle 1_i, 0_k|\hat{a}_k|1_i, 1_k\rangle\langle 1_i, 1_k|\hat{a}_i^+|0_i, 1_k\rangle \\ &= (-1)^{\sum(1,i-1)+\sum(i+1,k-1)+\sum(1,i-1)+1},\end{aligned}$$

or

$$\langle 1_i, 0_k|\hat{a}_k\hat{a}_i^+|0_i, 1_k\rangle = -(-1)^{\sum(i+1,k-1)}. \tag{65.7}$$

Comparison of (65.7) with (65.5) shows that these quantities have opposite signs, and hence

$$\hat{a}_i^+\hat{a}_k + \hat{a}_k\hat{a}_i^+ = 0, \qquad i \neq k.$$

For the diagonal matrix $\hat{a}_i\hat{a}_i^+$ we find

$$\hat{a}_i\hat{a}_i^+ = 1 - N_i. \tag{65.8}$$

Adding (65.6) gives

$$\hat{a}_i\hat{a}_i^+ + \hat{a}_i^+\hat{a}_i = 1.$$

Both relations can be written together as

$$\hat{a}_i\hat{a}_k^+ + \hat{a}_k^+\hat{a}_i = \delta_{ik}. \tag{65.9}$$

An analogous calculation for products of $\hat{a}_i$ and $\hat{a}_k$ yields

$$\hat{a}_i\hat{a}_k + \hat{a}_k\hat{a}_i = 0 \tag{65.10}$$

(in particular, $\hat{a}_i\hat{a}_i = 0$).

Thus operators $\hat{a}_i$ and $\hat{a}_k$ (or $\hat{a}_k^+$) with $i \neq k$ anticommute, whereas under Bose statistics they commuted. This distinction is natural. In the Bose case, $\hat{a}_i$ and $\hat{a}_k$ were completely independent: each $\hat{a}_i$ acted only on the single variable N_i, and its action did not depend on the other occupation numbers. Under Fermi statistics, by contrast, the action of $\hat{a}_i$ depends both on N_i itself and, as (65.4) shows, on the occupation numbers of all preceding states. The actions of different $\hat{a}_i$, $\hat{a}_k$ therefore cannot be regarded as independent.

Once the properties of $\hat{a}_i$, $\hat{a}_i^+$ have been defined, all the other formulas (64.13)–(64.18) remain fully valid. So do (64.23)–(64.25), which express operators of physical quantities through the field operators defined by (64.20). The commutation rules (64.21), (64.22), however, are replaced by the anticommutation relations

$$\hat{\psi}^+(\xi')\hat{\psi}(\xi) + \hat{\psi}(\xi)\hat{\psi}^+(\xi') = \delta(\xi - \xi'), \tag{65.11}$$

$$\hat{\psi}(\xi')\hat{\psi}(\xi) + \hat{\psi}(\xi)\hat{\psi}(\xi') = 0. \tag{65.12}$$

If a system contains different kinds of particles, separate second-quantized operators must be introduced for each kind, as mentioned at the end of the preceding section. Operators belonging to bosons commute with those belonging to fermions. Within nonrelativistic theory, operators belonging to different fermions may formally be taken either to commute or to anticommute; the method of second quantization gives the same results under either convention.

With a view to later use in relativistic theory, which permits transformations between different particles, the creation and annihilation operators of different fermions must nevertheless be taken to anticommute. This becomes clear if two different internal states of the same composite particle are regarded as different particles.

CHAPTER X

The Atom

§ 66. Atomic Energy Levels

In the nonrelativistic approximation, the stationary states of an atom are determined by the Schrödinger equation for a system of electrons moving in the Coulomb field of the nucleus and interacting electrically with one another. Electron-spin operators do not occur in this equation at all. For a system of particles in a centrally symmetric external field, both the total orbital angular momentum L and the parity of the state are conserved. Accordingly, every stationary state of an atom has a definite value of L and a definite parity. The coordinate-space wavefunctions of stationary states of identical particles also possess a definite permutation symmetry. Section 63 established that, for a system of electrons, every particular type of permutation symmetry—that is, every particular Young diagram—is associated with a particular value of the total spin of the system. The total electron spin S therefore provides a further label for each stationary atomic state.

For fixed S and L, an energy level is degenerate with respect to the possible spatial directions of the vectors $\mathbf{S}$ and $\mathbf{L}$. The degeneracy factors associated with their directions are $2S + 1$ and $2L + 1$, respectively. Hence the complete degeneracy of a level having specified L and S is $(2L + 1)(2S + 1)$.

The electromagnetic interaction between electrons does, however, contain relativistic effects that depend on their spins. Because of these effects, the atomic energy depends not only on the magnitudes of $\mathbf{L}$ and $\mathbf{S}$, but also on their relative orientation. Strictly speaking, once the relativistic interactions are included, neither the orbital angular momentum L nor the spin S of the atom is separately conserved. Only the combined angular momentum

$$\mathbf{J} = \mathbf{L} + \mathbf{S}$$

remains conserved. This is a universal exact law that follows from the isotropy of space for a closed system. Exact energy levels must consequently be labelled by the value J of the total angular momentum.

When the relativistic effects are comparatively small, as is often the case, they may instead be treated as a perturbation. Under this perturbation, the degenerate level with given L and S splits into a number of distinct, closely spaced levels having different values of the total angular momentum J.

To first order, these levels are obtained from the corresponding secular equation (Section 39). Their zeroth-order wavefunctions are certain linear combinations of the wavefunctions belonging to the original degenerate level with the specified values of L and S.

Within this approximation, the magnitudes of the orbital angular momentum and the spin may thus still be regarded as conserved, although their directions may not. The levels can therefore continue to be labelled by L and S as well.

Thus relativistic effects divide a level of specified L and S into levels with several different values of J. Such splitting is termed the level's *fine structure* (or *multiplet splitting*). The allowed values of J run from $L+S$ down to $|L-S|$. A level with given L and S therefore yields $2S+1$ distinct levels when $L > S$, or $2L+1$ when $L < S$. Each resulting level remains degenerate with respect to the directions of $\mathbf{J}$, with degeneracy $2J+1$. As a check, summing $2J+1$ over every allowed value of J gives $(2L+1)(2S+1)$, as required.

Atomic energy levels, also called atomic *spectral terms*, are customarily denoted by symbols analogous to those used for individual-particle states of definite angular momentum (Section 32). Different values of the total orbital angular momentum L are represented by capital Latin letters according to the correspondence

$$\begin{array}{cccccccccccc} L = & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & \dots \\ & S & P & D & F & G & H & I & K & L & M & N & \dots \end{array}$$

The number $2S+1$, called the *multiplicity* of the term, is placed as a superscript to the left of the symbol. One should remember, however, that this number equals the number of fine-structure components only when $L \geqslant S$[1]. The value of the total angular momentum J is placed as a subscript on the right. Thus ${}^2P_{1/2}$ and ${}^2P_{3/2}$ denote levels with $L=1$, $S=1/2$, and $J=1/2, 3/2$.

§ 67. Electron states in an atom

An atom containing more than one electron is a complicated system of mutually interacting electrons moving in the nuclear field. Strictly speaking, for such a system one can consider only states of the system as a whole. Nevertheless, it proves possible, to good accuracy, to introduce a state for each electron separately: a stationary state of its motion in an effective spherically symmetric field produced jointly by the nucleus and all the other electrons. In general, these fields differ from one electron to another within the atom. They must also all be determined simultaneously, since each depends on the states of every other electron. A field of this kind is called *self-consistent*.

Since the self-consistent field has central symmetry, every electron state has a definite value of its orbital angular momentum l. For a fixed l, the states of an individual electron are numbered in ascending order of energy by the *principal quantum number* n, whose possible values are $n = l+1, l+2, \dots$; this ordering convention is chosen to match the one used for the hydrogen atom. In a complex atom, however, the order in which levels of different l rise in energy generally differs from that in hydrogen. For hydrogen the energy is independent of l altogether, and consequently states with a larger n always have a higher energy. In a complex atom, by contrast, the level with, for example, $n=5$, $l=0$ lies below the level with $n=4$, $l=2$ (see § 73 for further details).

An individual-electron state with specified n and l is conventionally denoted by a symbol comprising a numeral for the principal quantum number and a letter for the value of l[2]. Thus $4d$ denotes the state $n=4$, $l=2$. A complete description of an atomic state

[1] For $2S+1 = 1, 2, 3, \dots$, the levels are called singlet, doublet, triplet, and so on, respectively.

[2] Another terminology refers to electrons with principal quantum numbers $n = 1, 2, 3, \dots$ as electrons of the K, L, M, ... shells, respectively (see § 74).

requires, in addition to the values of the total L, S, J, a listing of the states occupied by all its electrons. For example, the symbol $1s\,2p\,{}^3P_0$ designates a helium-atom state with $L = 1$, $S = 1$, $J = 0$, whose two electrons occupy

the $1s$ and $2p$ states. When several electrons have the same l and n, this is abbreviated by an exponent; thus $3p^2$ denotes two electrons in $3p$ states. The distribution of an atom's electrons among states with different l and n is called its *electron configuration*.

At fixed n and l, an electron may have different projections of its orbital angular momentum (m) and spin (σ) on the z axis. For a given l, the number m takes $2l + 1$ values, whereas σ is restricted to the two values $\pm 1/2$. There are consequently $2(2l + 1)$ different states with the same n, l; these states are said to be *equivalent*. By the Pauli principle, each can contain one electron. Hence no more than $2(2l + 1)$ electrons in an atom can simultaneously have identical n, l. A set of electrons occupying every state with specified n, l is called a *closed shell* of that type.

The difference between the energies of atomic levels having different L, S but the same electron configuration[3] results from the electrostatic interaction of the electrons. Usually these energy differences are comparatively small, being several times smaller than the separations between levels belonging to different configurations. The relative ordering of levels with the same configuration but different L, S is described by the following empirically established *Hund rule* (F. Hund, 1925):

The term of lowest energy is the one with the largest value of S permitted by the electron configuration and, for that S, the largest permitted value of L[4].

We now show how the atomic terms permitted for a specified electron configuration can be found. If the electrons are not equivalent, the possible values of L, S follow directly from the rule for angular-momentum addition. For instance, in the configuration $np, n'p$ (with different n, n'), the total

angular momentum L can take the values 2, 1, 0, while the total spin is $S = 0, 1$. Combining these values gives the terms ${}^{1/3}S$, ${}^{1/3}P$, ${}^{1/3}D$.

For equivalent electrons, however, the Pauli principle imposes restrictions. As an example, consider a configuration of three equivalent p electrons. For $l = 1$ (a p state), the orbital-angular-momentum projection m can be $1, 0, -1$, so there are six states with the following pairs m, σ:

$$
\begin{array}{llll}
\text{a)} & 1,,1/2, & \text{b)}\ 0,,1/2 & \text{c)}\ -1,,1/2, \\
\text{a}') & 1,,-1/2, & \text{b}')\ 0,,-1/2 & \text{c}')\ -1,,-1/2.
\end{array}
$$

The three electrons can be placed, one apiece, in any three of these states. The resulting atomic states have the following values of the projections $M_L = \sum m$, $M_S = \sum \sigma$ of the

[3] Here we disregard the fine structure of each multiplet level.

[4] The requirement that S be maximal can be justified as follows. Consider, for example, a two-electron system. It may have $S = 0$ or $S = 1$, and spin 1 corresponds to an antisymmetric coordinate wave function $\varphi(\mathbf{r}_1, \mathbf{r}_2)$. Such a function vanishes when $\mathbf{r}_1 = \mathbf{r}_2$; in other words, in the $S = 1$ state the probability that both electrons are close together is small. Their electrostatic repulsion is therefore comparatively weaker, and the energy is lower. Likewise, in a system of several electrons the greatest spin corresponds to the "most antisymmetric" coordinate wave function.

total orbital angular momentum and spin:

$$
\begin{array}{llll}
\mathrm{a}+\mathrm{a}'+\mathrm{b}) & 2,,1/2, & \mathrm{a}+\mathrm{a}'+\mathrm{c})\ 1,,1/2, & \mathrm{a}+\mathrm{b}+\mathrm{c})\ 0,,3/2, \\
 & & \mathrm{a}+\mathrm{b}+\mathrm{b}')\ 1,,1/2, & \mathrm{a}+\mathrm{b}+\mathrm{c}')\ 0,,1/2, \\
 & & \mathrm{a}+\mathrm{b}'+\mathrm{c})\ 0,,1/2, & \\
 & & \mathrm{a}'+\mathrm{b}+\mathrm{c})\ 0,,1/2. &
\end{array}
$$

(States with negative M_L, M_S need not be written out, since they add nothing new.) The presence of a state with $M_L = 2$, $M_S = 1/2$ shows that a ^2D term must occur; one state each with $(1, 1/2)$ and $(0, 1/2)$ must also belong to it. One state with $(1, 1/2)$ then remains, so there must be a ^2P term; one of the $(0, 1/2)$ states belongs to it as well. Finally, the remaining states $(0, 3/2)$ and $(0, 1/2)$ correspond to a ^4S term. Thus a configuration of three equivalent p electrons permits exactly one term of each of the types ^2D, ^2P, and ^4S.

Table 1 lists the possible terms for various configurations of equivalent p- and d-electrons. A number written beneath a term symbol gives the number of terms of that type in the configuration, whenever there is more than one. A configuration containing the largest possible number of equivalent electrons ($s^2, p^6, d^{10}, \ldots$) always has the term ^1S. Notice that the sets of terms are the same for two configurations when one contains as many electrons as the other lacks for a filled shell. This follows directly because a missing electron in a shell may be regarded as a *hole*, whose state is specified by the same quantum numbers as the state of the absent electron.

When Hund's rule is used to determine the normal term of an atom from its known electron configuration, only the unfilled shell need be considered, since the angular momenta of the electrons in filled shells cancel one another. Suppose, for example, that an atom has four d electrons outside its closed shells. The magnetic quantum number of a d electron can take five values: $0, \pm 1, \pm 2$. All four electrons can therefore have the same spin projection $\sigma = 1/2$, and the largest possible total spin is $S = 2$. We must then assign different values of m to the electrons so as to maximize $M_L = \sum m$; these are $2, 1, 0, -1$, giving $M_L = 2$. It follows that the greatest value of L compatible with $S = 2$ is likewise 2 (the ^5D term).

Table 1. All possible terms for configurations of equivalent electrons

p, p^5	^2P		
p^2, p^4	^1SD	^3P	
p^3,	^2PD	^4S	
d, d^9	^2D		
d^2, d^8	^1SDG	^3PF	
d^3, d^7	$^2P\underset{2}{D}FGH$	^4PF	
d^4, d^6	$^1\underset{2}{S}\underset{2}{D}F\underset{2}{G}I$	$^3\underset{2}{P}D\underset{2}{F}GH$	^5D
d^5,	$^2SP\underset{3}{D}\underset{2}{F}\underset{2}{G}HI$	$^4P\underset{2}{D}\underset{2}{F}G$	^6S

PROBLEM

Find the orbital wave functions for the possible states of a system of three equivalent p-electrons.

SOLUTION. In the ^4S state, the spin projections σ of all the electrons are equal, and hence their values of m are distinct. The wave function is given by a determinant of the form (61.5), constructed from the functions ψ_0, ψ_1, and ψ_{-1} (the subscript denotes m).

For the ^2D term, consider the state with the largest possible value $M_L = 2$. Two of the m projections must then equal 1 and the remaining one must equal 0. Let electrons 2 and 3 have $\sigma = +1/2$, and electron 1 have $\sigma = -1/2$ (in accordance with the total spin $S = 1/2$). The orbital wave function with the required symmetry is

$$\psi = \frac{1}{\sqrt{2}}\psi_1(1)[\psi_0(2)\psi_1(3) - \psi_0(3)\psi_1(2)]$$

(the numeral in a function's argument identifies the electron to which it refers).

For the ^2P term, take the state with $M_L = 1$ and the same electron-spin projections as above. This state can be formed with

two different sets of m values, so its orbital wave function is the linear combination

$$\psi = a\psi_{-111} + b\psi_{100},$$
$$\psi_{-111} = \psi_1(1)[\psi_{-1}(2)\psi_1(3) - \psi_{-1}(3)\psi_1(2)],$$
$$\psi_{100} = \psi_0(1)[\psi_1(2)\psi_0(3) - \psi_1(3)\psi_0(2)].$$

To determine the coefficients, use the relation

$$\hat{L}_+\psi = (\hat{l}_+^{(1)} + \hat{l}_+^{(2)} + \hat{l}_+^{(3)})\psi = 0,$$

which the wave function with $M_L = L$ must satisfy (see (27.8)). The matrix elements (27.12) give

$$\hat{l}_+\psi_1 = 0, \qquad \hat{l}_+\psi_{-1} = \sqrt{2}\psi_0, \qquad \hat{l}_+\psi_0 = \sqrt{2}\psi_1$$

and hence

$$\hat{L}_+\psi = \sqrt{2}(a-b)\psi_{011} = 0.$$

Thus $a - b = 0$; the normalization condition then gives $a = b = 1/2$.

Wave functions for states with $M_L < L$ are obtained from those found above by applying the operator $\hat{L}_-$.

§ 68. Hydrogen-like energy levels

Hydrogen, the simplest atom, is the only atom for which the Schrödinger equation admits an exact solution. The energy levels of hydrogen, and also of the one-electron ions He^+, Li^{++}, . . . , are given by the Bohr formula (36.10):

$$E = -\frac{mZ^2e^4}{2\hbar^2(1 + m/M)}\frac{1}{n^2}. \tag{68.1}$$

Here Ze is the nuclear charge, M the nuclear mass, and m the electron mass. Notice that the dependence on nuclear mass is very weak.

Equation (68.1) disregards every relativistic effect. In this approximation hydrogen has the special additional (*accidental*) degeneracy already discussed in § 36: at fixed principal quantum number n, the energy does not depend on the orbital angular momentum l.

Other atoms too have states with properties resembling those of hydrogen. These are highly excited states in which one electron has a large principal quantum number and therefore spends most of its time far from the nucleus. To some approximation, its motion may be treated as motion in the Coulomb field of the *ionic core*, whose effective charge is unity. The energy-level values obtained in this way are nevertheless too inaccurate, and require

a correction that allows for the departure of the short-range field from a pure Coulomb field. The form of this correction follows readily from the following argument.

States with large quantum numbers are semiclassical, so their energy levels may be found from the Bohr–Sommerfeld quantization rules (48.6). At distances from the nucleus small compared with the "orbital radius," the departure of the field from the Coulomb form may formally be represented by changing the boundary condition imposed on the wave function at $r = 0$. This changes the constant γ in the quantization condition for radial motion. Since the rest of that condition remains unaltered, the resulting energy-level formula must differ from the hydrogen formula by replacing the radial quantum number, or equivalently the principal quantum number n, with $n + \Delta_l$. Here Δ_l is a constant known as the *Rydberg correction*:

$$E = -\frac{me^4}{2\hbar^2}\frac{1}{(n+\Delta_l)^2}. \tag{68.2}$$

By its definition the Rydberg correction is independent of n, but it is of course a function of the excited electron's azimuthal quantum number l (written as the subscript on Δ), and also of the total atomic angular momenta L and S. At fixed L and S, Δ_l decreases rapidly as l increases. The larger l is, the less time the electron spends near the nucleus, and the more closely the energy levels must approach the hydrogen levels[5].

[5]As an illustration, consider the empirical Rydberg corrections for highly excited states of helium. The total spin of the helium atom may be $S = 0$ or 1, while in the states under consideration its total orbital angular momentum is the same as the angular momentum l of the excited electron (the second electron is in the $1s$ state). The Rydberg corrections are, for $S = 0$,

$$\Delta_0 = -0.140, \qquad \Delta_1 = +0.012, \qquad \Delta_2 = -0.0022,$$

and, for $S = 1$,

$$\Delta_0 = -0.296, \qquad \Delta_1 = -0.068, \qquad \Delta_2 = -0.0029.$$

PROBLEM

Find the asymptotic expression, at large distances from the ionic core, for the wave function of an electron in a hydrogen-like s state.

SOLUTION. At large distances, where $U = -1/r$ in atomic units, the required function $\psi(r)$ obeys the Schrödinger equation

$$\psi'' + \frac{2}{r}\psi' - \varkappa^2\psi + \frac{2}{r}\psi = 0,$$

where $\varkappa = \sqrt{2|E|}$. Seek a solution of the form $\psi = \text{const}\cdot r^{\nu} e^{-\varkappa r}$. On neglecting terms in the equation that decrease faster than ψ/r, one obtains

$$\psi = \text{const}\cdot r^{1/\varkappa - 1} e^{-\varkappa r}.$$

§ 69. The self-consistent field

For atoms containing more than one electron, the Schrödinger equation cannot be solved analytically. Approximate methods for calculating the energies and wave functions of stationary atomic states therefore become important. Foremost among them is the method usually called the self-consistent-field method. Its basic idea is to regard each electron in an atom as moving in a *self-consistent field* produced jointly by the nucleus and all the other electrons.

As an example, consider the helium atom and restrict attention to those of its terms in which both electrons occupy s states (with either equal or different n); the states of the atom as a whole are then S states. Let $\psi_1(r_1)$ and $\psi_2(r_2)$ be the electron wave functions. In s states they depend only on the respective distances r_1, r_2 from the nucleus. The wave function $\psi(r_1, r_2)$ of the complete atom is represented either by the symmetrized product

$$\psi = \psi_1(r_1)\psi_2(r_2) + \psi_1(r_2)\psi_2(r_1) \tag{69.1}$$

or by the antisymmetrized product

$$\psi = \psi_1(r_1)\psi_2(r_2) - \psi_1(r_2)\psi_2(r_1) \tag{69.2}$$

of the two functions, according as the total spin is $S = 0$ or $S = 1$[6]. We shall consider the latter case; ψ_1 and ψ_2 may then be taken to be mutually orthogonal[7].

Our objective is to determine a function of the form (69.2) that gives the best approximation to the true wave function of the atom. It is natural to base this construction

[6] The state of the helium atom with $S = 0$ is conventionally called a *parahelium* state, and one with $S = 1$ an *orthohelium* state.

[7] In general, the functions $\psi_1, \psi_2, \ldots$ for different one-electron states obtained by the self-consistent-field method are not mutually orthogonal, since they solve different equations rather than one common equation. In (69.2), however, ψ_2 can be replaced by $\psi_2' + \text{const}\cdot\psi_1$ without altering the wave function ψ of the whole atom. An appropriate choice of the constant always makes ψ_1 and ψ_2' mutually orthogonal.

on the variational principle, admitting only functions of the form (69.2) as competitors (the method described here was proposed by V. A. Fock in 1930).

The Schrödinger equation, as we know, follows from the variational principle

$$\iint \psi^* \hat{H} \psi \, dV_1 dV_2 = \min$$

subject to the additional condition

$$\iint |\psi|^2 dV_1 dV_2 = 1$$

(the integration is over the coordinates of both electrons in the helium atom). Variation gives

$$\iint \delta\psi^* (\hat{H} - E)\psi \, dV_1 dV_2 = 0, \tag{69.3}$$

from which the usual Schrödinger equation results when the variation of ψ is arbitrary. In the self-consistent-field method, by contrast, expression (69.2) is substituted for ψ in (69.3), and ψ_1 and ψ_2 are varied separately.

In other words, the integral is extremized among functions ψ having the form (69.2). This naturally yields an approximate energy eigenvalue and an approximate wave function, but the latter is the best among all functions representable in that form.

The Hamiltonian of the helium atom is[8]

$$\hat{H} = \hat{H}_1 + \hat{H}_2 + \frac{1}{r_{12}}, \qquad \hat{H}_1 = -\frac{1}{2}\Delta_1 - \frac{2}{r_1} \tag{69.4}$$

(r_{12} is the distance between the electrons). Substitute (69.2) in (69.3), carry out the variation, and set the coefficients of $\delta\psi_1$ and $\delta\psi_2$ in the integrand equal to zero. The resulting equations are

$$\begin{aligned} &[\tfrac{1}{2}\Delta + \tfrac{2}{r} + E - H_{22} - G_{22}(r)]\psi_1(r) + [H_{12} + G_{12}(r)]\psi_2(r) = 0,\\ &[\tfrac{1}{2}\Delta + \tfrac{2}{r} + E - H_{11} - G_{11}(r)]\psi_2(r) + [H_{12} + G_{12}(r)]\psi_1(r) = 0, \end{aligned} \tag{69.5}$$

where

$$G_{ab}(r_1) = \int \frac{\psi_a(r_2)\psi_b(r_2)}{r_{12}} dV_2, \quad H_{ab} = \int \psi_a \left(-\frac{1}{2}\Delta - \frac{2}{r}\right)\psi_b \, dV, \quad a, b = 1, 2. \tag{69.6}$$

These are the final equations furnished by the self-consistent-field method; naturally, they can be solved only numerically.[9]

The equations for more complicated cases must be derived in the same way. The atomic wave function to be placed in the variational integral is constructed as a linear combination of products of individual-electron wave functions. This combination must

[8] Atomic units are used throughout this section and its problems.

[9] Comparison of the energy levels calculated by the self-consistent-field method for light atoms with spectroscopic data indicates an accuracy of about 5% (and, in some cases, an even higher accuracy). For complex atoms, however, the error may be comparable with the separations between neighbouring levels and may consequently give an incorrect ordering of the levels.

be chosen so that, first, its permutation symmetry corresponds to the total spin S of the atomic state under consideration and, second, it corresponds to the specified total orbital angular momentum L of the atom[10].

By using a wave function with the required permutation symmetry in the variational principle, we thereby include the exchange interaction between the atomic electrons. Simpler equations, though with less accurate results, follow if both the exchange interaction and the dependence of the atomic energy on L for a fixed electron configuration are neglected (D. R. Hartree, 1928). Again taking helium as an example, the electron wave functions may then be written directly as ordinary Schrödinger equations,

$$[\tfrac{1}{2}\Delta_a + E_a - V_a(r_a)]\psi_a(r_a) = 0, \qquad a = 1, 2, \tag{69.7}$$

where V_a is the potential energy of one electron moving in the nuclear field and in the field of the distributed charge of the other electron:

$$V_1(r_1) = -\frac{2}{r_1} + \int \frac{1}{r_{12}}\psi_2^2(r_2)\, dV_2 \tag{69.8}$$

(and similarly for V_2). To find the energy E of the entire atom, observe that the electrostatic interaction of the two electrons is counted twice in $E_1 + E_2$, since it enters both the potential energy $V_1(r_1)$ of the first electron and $V_2(r_2)$ of the second. Hence E is obtained from $E_1 + E_2$ by subtracting the mean value of this interaction once, that is,

$$E = E_1 + E_2 - \iint \frac{1}{r_{12}}\psi_1^2(r_1)\psi_2^2(r_2)\, dV_1 dV_2. \tag{69.9}$$

The results of this simplified method can subsequently be refined by treating the exchange interaction and the dependence of the energy on L as perturbations.

Problems

Problem

1. Find an approximate value for the ground-state energy of the helium atom and of helium-like ions (a nucleus of charge Z with two electrons), treating the electron–electron interaction as a perturbation.

Solution. In the ion's ground state, both electrons occupy S states. The unperturbed energy is twice the ground-state energy of a hydrogen-like ion, since two electrons are present:

$$E^{(0)} = 2(-Z^2/2) = -Z^2.$$

The first-order correction is the mean electron–electron interaction energy evaluated in the state whose wave function is

$$\psi = \psi_1(r_1)\psi_2(r_2) = \frac{Z^2}{\pi} e^{-Z(r_1+r_2)} \tag{1}$$

[10]General procedures for constructing wave functions of an electron system in a central field are described in the book by I. G. Kaplan cited on p. 291.

(the product of two hydrogenic functions with $l = 0$). The integral

$$E^{(1)} = \iint \psi^2 \frac{1}{r_{12}}\, dV_1 dV_2$$

is most conveniently evaluated in the form

$$E^{(1)} = 2\int_0^\infty dV_2\, \rho_2 \frac{1}{r_2} \int_0^{r_2} \rho_1\, dV_1, \qquad dV_1 = 4\pi r_1^2 dr_1, \quad dV_2 = 4\pi r_2^2 dr_2$$

(the energy of the charge distribution $\rho_2 = |\psi_2|^2$ in the field of the spherically symmetric distribution $\rho_1 = |\psi_1|^2$; the integrand in the dV_2 integration is the energy of the charge $\rho_2(r_2)$ in the field of the sphere $r_1 < r_2$, while the factor 2 before the integral accounts for the configurations with $r_1 > r_2$). Hence $E^{(1)} = 5Z/8$, and finally

$$E = E^{(0)} + E^{(1)} = -Z^2 + \frac{5}{8}Z.$$

For the helium atom this gives $-E = 11/4 = 2.75$ (the actual ground-state energy of this atom is $-E = 2.90$ atomic units, or 78.9 eV).

Problem

2. Solve the same problem by the variational principle, approximating the wave function by the product of two hydrogenic functions with an effective nuclear charge.

Solution. Evaluate the integral

$$\iint \psi \hat{H} \psi\, dV_1 dV_2, \qquad \hat{H} = -\frac{1}{2}(\Delta_1 + \Delta_2) - \frac{Z}{r_1} - \frac{Z}{r_2} + \frac{1}{r_{12}},$$

using the function ψ from expression (1) of the preceding problem, with Z_{eff} in place of Z. The integral of ψ^2/r_{12} is evaluated in the same way as in Problem 1. The integral of $\psi\Delta_1\psi$ can be reduced to one involving ψ^2/r_1 by observing that the Schrödinger equation gives

$$\left(-\frac{1}{2}\Delta_1 - \frac{Z_{\text{eff}}}{r_1}\right)\psi_1 = -\frac{1}{2}Z_{\text{eff}}^2\psi_1.$$

The result is

$$\iint \psi \hat{H} \psi\, dV_1 dV_2 = Z_{\text{eff}}^2 - 2ZZ_{\text{eff}} + \frac{5}{8}Z_{\text{eff}}.$$

As a function of Z_{eff}, this expression has its minimum at $Z_{\text{eff}} = Z - 5/16$. The corresponding energy is

$$E = -\left(Z - \frac{5}{16}\right)^2.$$

For the helium atom, this gives $-E = 2.85$.

It should be noted that, with the value of Z_{eff} just found, wave function (1) is in fact the best not merely among functions of the form (1), but among all functions that depend only on the sum $r_1 + r_2$.

§ 70. The Thomas–Fermi equation

Numerical computation of the charge and field distributions in an atom by the self-consistent-field method is exceedingly laborious, especially for complex atoms. Yet it is precisely for complex atoms that another approximate method is available, whose merit lies in its simplicity; its results are, however, much less accurate than those of the self-consistent-field method.

This method (E. Fermi, L. Thomas, 1927) rests on the fact that, in complex atoms containing many electrons, most electrons have comparatively large principal quantum numbers. The quasiclassical approximation is applicable under these conditions. We can therefore apply the idea of "cells in phase space" (§ 48) to the states of the individual atomic electrons.

The phase-space volume corresponding to electrons whose momentum is less than p and whose positions lie in a physical-space volume element dV is $\frac{4}{3}\pi p^3 dV$.

This volume contains $4\pi p^3 dV/[3(2\pi)^3]$ cells[11], that is, possible states in which no more than

$$2\frac{4\pi p^3}{3(2\pi)^3}dV = \frac{p^3}{3\pi^2}dV$$

electrons (two electrons with mutually opposite spins in each cell). In the atom's normal state, the electrons present in every volume element dV must occupy, in phase space, the cells for momenta ranging from zero up to some maximum value p_0. The kinetic energy of the electrons will then be as small as possible at every point. If the number of electrons in the volume dV is written as n, dV (where n is the electron number density), the maximum electron momentum p_0 at each point is related to n by

$$\frac{p_0^3}{3\pi^2} = n.$$

Consequently, at a place where the electron density is n, the maximum kinetic energy of an electron is

$$\frac{p_0^2}{2} = \frac{1}{2}(3\pi^2 n)^{2/3}. \tag{70.1}$$

Let $\varphi(r)$ now denote the electrostatic potential, taken to vanish at infinity. The total energy of an electron is $p^2/2 - \varphi$. Plainly, every electron must have negative total energy; otherwise it would escape to infinity. Denote the maximum total electron energy at every point by $-\varphi_0$, where φ_0 is a positive constant. (Were this quantity not constant, electrons would pass from points with smaller φ_0 to points with larger φ_0.)

We may therefore write

$$\frac{p_0^2}{2} = \varphi - \varphi_0. \tag{70.2}$$

Equating (70.1) and (70.2), we obtain

$$n = [2(\varphi - \varphi_0)]^{3/2}\frac{1}{3\pi^2}, \tag{70.3}$$

which relates the electron density to the potential at every point of the atom.

[11]Atomic units are used throughout this section.

At $\varphi = \varphi_0$ the density n vanishes. Clearly, n must also be set equal to zero throughout the region where $\varphi < \varphi_0$, for there (70.2) would give a negative maximum kinetic energy.

Thus the equation $\varphi = \varphi_0$ defines the boundary of the atom. Outside a spherically symmetric charge distribution of zero total charge, however, there is no field. Hence $\varphi = 0$ at the boundary of a neutral atom, and the constant φ_0 must accordingly be set equal to zero for such an atom. For an ion, by contrast, φ_0 is nonzero.

We shall consider a neutral atom below and therefore put $\varphi_0 = 0$. Poisson's equation of electrostatics gives $\Delta\varphi = 4\pi n$; substitution of (70.3) yields the fundamental *Thomas–Fermi equation*

$$\Delta\varphi = \frac{8\sqrt{2}}{3\pi}\varphi^{3/2}. \tag{70.4}$$

The field distribution in the normal state of the atom is specified by the spherically symmetric solution of this equation subject to the following boundary conditions: as $r \to 0$, the field must approach the Coulomb field of the nucleus, so that $\varphi r \to Z$; as $r \to \infty$, $\varphi r \to 0$. Introduce a new variable x in place of r by

$$r = xbZ^{-1/3}, \qquad b = \frac{1}{2}\left(\frac{3\pi}{4}\right)^{2/3} = 0{,}885, \tag{70.5}$$

and a new unknown function χ in place of φ:

$$\varphi(r) = \frac{Z}{r}\chi\left(\frac{rZ^{1/3}}{b}\right) = \frac{Z^{4/3}}{b}\frac{\chi(x)}{x}\text{[12]}, \tag{70.6}$$

We then obtain

$$x^{1/2}\frac{d^2\chi}{dx^2} = \chi^{3/2} \tag{70.7}$$

with the boundary conditions $\chi = 1$ at $x = 0$ and $\chi = 0$ at $x = \infty$. This equation contains no parameters and hence defines a universal function $\chi(x)$. Table 2 gives this function as found by numerical integration of (70.7).

The function $\chi(x)$ decreases monotonically and reaches zero only at infinity[13]. In other words, the Thomas–Fermi model gives the atom no boundary: formally, it extends to infinity. At $x = 0$ the derivative has the value $\chi'(0) = -1.59$. Hence, as $x \to 0$, the function takes the form $\chi \approx 1 - 1.59x$, and the potential $\varphi(r)$ accordingly has the form:

$$\varphi(r) \approx Z/r - 1{,}80 \cdot Z^{4/3}. \tag{70.8}$$

The first term is the potential of the nuclear field, and the second is the potential produced by the electrons at the origin. Substitution of (70.6) into (70.3) gives the electron density in the form

$$n = Z^2 f\left(\frac{rZ^{1/3}}{b}\right), \qquad f(x) = \frac{32}{9\pi^3}\left(\frac{\chi}{x}\right)^{3/2}. \tag{70.9}$$

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[12] In ordinary units: $\varphi(r) = \dfrac{Ze}{r}\chi\left(\dfrac{rZ^{1/3}me^2}{0{,}885\hbar^2}\right)$.

[13] Equation (70.7) has the exact solution $\chi(x) = 144x^{-3}$, which tends to zero at infinity but does not obey the boundary condition at $x = 0$. It could be used as an asymptotic form of $\chi(x)$ for large x. This expression, however, gives values of reasonable accuracy only at very large x, while at large distances the Thomas–Fermi equation is itself no longer applicable (see below).

Table 2. Values of the function $\chi(x)$

x	$\chi(x)$	x	$\chi(x)$	x	$\chi(x)$
0,00	1,000	1,4	0,333	6	0,0594
0,02	0,972	1,6	0,298	7	0,0461
0,04	0,947	1,8	0,268	8	0,0366
0,06	0,924	2,0	0,243	9	0,0296
0,08	0,902	2,2	0,221	10	0,0243
0,10	0,882	2,4	0,202	11	0,0202
0,2	0,793	2,6	0,185	12	0,0171
0,3	0,721	2,8	0,170	13	0,0145
0,4	0,660	3,0	0,157	14	0,0125
0,5	0,607	3,2	0,145	15	0,0108
0,6	0,561	3,4	0,134	20	0,0058
0,7	0,521	3,6	0,125	25	0,0035
0,8	0,485	3,8	0,116	30	0,0023
0,9	0,453	4,0	0,108	40	0,0011
1,0	0,424	4,5	0,0919	50	0,00063
1,2	0,374	5,0	0,0788	60	0,00039

We see that in the Thomas–Fermi model the charge-density distributions in different atoms are similar, with the characteristic length scale set by $Z^{-1/3}$ (in ordinary units: $\hbar^2/(me^2Z^{1/3})$, i.e. the Bohr radius divided by $Z^{1/3}$). If distances are measured in atomic units, then, in particular, the distances at which the electron density is greatest will be the same for all Z. It may therefore be stated that most of the electrons in an atom of atomic number Z are located at distances from the nucleus of order $Z^{-1/3}$. Calculation shows that half the atom's total electronic charge lies inside a sphere of radius $1{,}33Z^{-1/3}$.

The same reasoning gives an average electron speed in the atom of order $Z^{2/3}$, with the speed estimated in order of magnitude as the square root of the energy.

The Thomas–Fermi equation fails both at excessively small and at excessively large distances from the nucleus. At small r, its domain of validity is bounded by inequality (49.12); at still smaller distances the quasiclassical approximation becomes invalid in the Coulomb field of the nucleus. Setting $\alpha = Z$ in (49.12), we find $1/Z$ as the lower limit on the distance. The quasiclassical approximation also becomes invalid in a complex atom at large r. Indeed, it is easy to see that at $r \sim 1$ the electron's de Broglie wavelength becomes of the same order as this distance itself, so the quasiclassical condition fails completely. This can be verified by estimating the terms in (70.2) and (70.4); the result is in fact evident beforehand, without calculation, since (70.4) contains no Z.

The validity of the Thomas–Fermi equation is therefore restricted to distances large compared with $1/Z$ and small compared with 1. In complex atoms, however, most of the electrons lie within this region.

The last fact means that the "outer boundary" of the atom in the Thomas–Fermi model lies at $r \sim 1$, so atomic dimensions do not depend on Z. The energy of the outer electrons, and hence the ionization potential of the atom, is likewise independent of Z[14].

The Thomas–Fermi method can be used to calculate the total ionization energy E, namely, the energy required to remove every electron from a neutral atom. To do this, one must

calculate the electrostatic energy of the Thomas–Fermi charge distribution in the atom. The required total energy equals half this electrostatic energy, since, in a system of particles interacting according to Coulomb's law, the mean kinetic energy is minus one half of the mean potential energy (by the virial theorem; see I, § 10).

The dependence of E on Z can be established in advance from simple considerations. The electrostatic energy of Z electrons in the field of a nucleus of charge Z, at a mean distance $Z^{-1/3}$ from it, is proportional to $Z \cdot Z/Z^{-1/3} = Z^{7/3}$. Numerical calculation gives $E = 20{,}8Z^{7/3}$ eV. The dependence on Z agrees well with experimental data, while the empirical coefficient is nearer 16.

We have already noted that nonzero positive values of the constant φ_0 correspond to ionized atoms. If χ is defined by $\varphi - \varphi_0 = Z\chi/r$, the same equation (70.7) is obtained for χ. We must now consider, however, solutions which vanish not at infinity, as in the neutral atom, but at finite values $x = x_0$; such a solution exists for every x_0. At $x = x_0$, the charge density vanishes together with χ, while the potential remains finite.

The value of x_0 is related to the degree of ionization as follows. By Gauss's theorem, the total charge within a sphere of radius r is

$$-r^2 \frac{\partial \varphi}{\partial r} = Z[\chi(x) - x\chi'(x)].$$

The ion's total charge z is obtained by putting $x = x_0$ here. Since $\chi(x_0) = 0$,

$$z = -Zx_0\chi'(x_0). \tag{70.10}$$

In Fig. 23 the heavy curve is $\chi = \chi(x)$ for a neutral atom, and the two curves below it are for ions with different degrees of ionization. Graphically, z/Z is represented by the length of the intercept made on the ordinate axis by the tangent to the curve at $x = x_0$.

Equation (70.7) also has solutions that vanish nowhere; these solutions diverge at infinity. They may be regarded as corresponding to negative values of the constant φ_0. Two such curves $\chi = \chi(x)$ are also shown in Fig. 23; they lie above the neutral-atom curve. At the point $x = x_1$ where

$$\chi(x_1) - x_1\chi'(x_1) = 0, \tag{70.11}$$

the total charge inside the sphere $x < x_1$ vanishes. Graphically, this is plainly the point at which the tangent to the curve passes through the origin. If the curve is terminated at this point, we may say that it defines $\chi(x)$ for a neutral atom whose charge density remains nonzero at its boundary. Physically, this corresponds to a kind of "compressed" atom confined within a prescribed finite volume[15].

[14]This model naturally does not reproduce the periodic dependence on Z of atomic dimensions and ionization potentials displayed in the periodic table. Empirical data also reveal a slight systematic increase of atomic dimensions and decrease of ionization potentials as Z increases.

[15]This treatment can be useful in studying the equation of state of matter at high compressions.

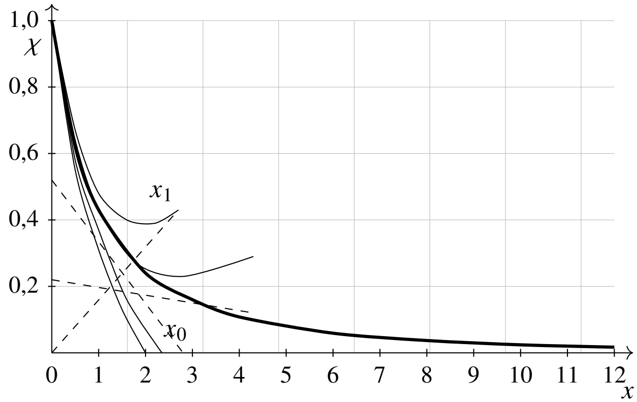

Fig. 23

The Thomas–Fermi equation does not allow for the exchange interaction between electrons. The effects associated with exchange are of the next order in $Z^{-2/3}$. Thus inclusion of the exchange interaction in the Thomas–Fermi method requires that all effects of this order be included at the same time[16].

PROBLEM

Find the relation between the energy of electrostatic interaction among the electrons and the energy of their interaction with the nucleus for a neutral atom in the Thomas–Fermi model.

SOLUTION. The potential φ_e of the field produced by the electrons is found by subtracting the nuclear-field potential Z/r from the total potential φ. The interaction energy among the electrons is therefore

$$U_{ee} = -\frac{1}{2}\int \varphi_e n\, dV = \frac{Z}{2}\int \frac{n}{r}\, dV - \frac{1}{2}\int \varphi n\, dV = \frac{Z}{2}\int \frac{n}{r}\, dV - \frac{(3\pi^2)^{2/3}}{4}\int n^{5/3} dV$$

(where φ has been expressed in terms of n by means of (70.3)). On the other hand, the electron–nucleus interaction energy U_{en} and the kinetic energy T are

$$U_{en} = -Z\int \frac{n}{r}\, dV, \qquad T = 2\int\int_0^{p_0} \frac{p^2}{2}\frac{4\pi p^2 dp\, dV}{(2\pi)^3} = \frac{3(3\pi^2)^{2/3}}{10}\int n^{5/3} dV.$$

Comparison of these expressions with the preceding equality gives

$$U_{ee} = -\frac{1}{2}U_{en} - \frac{5}{6}T.$$

At the same time, the virial theorem (see I, § 10) gives $2T = -U = -U_{en} - U_{ee}$ for a system of particles interacting according to Coulomb's law. Consequently,

$$U_{ee} = -\frac{1}{7}U_{en}.$$

[16]This was done by A. S. Kompaneets and E. S. Pavlovskii (ZhETF, 1956, vol. 31, p. 427) and D. A. Kirzhnits (ZhETF, 1957, vol. 32, p. 115).

§ 71. Wave functions of outer electrons near the nucleus

As we saw from the Thomas–Fermi model, the outer electrons in complex atoms (at large Z) lie predominantly at distances $r \sim 1$ from the nucleus[17]. Certain atomic properties, however, depend strongly on the electron density close to the nucleus (properties of this kind will arise in § 72 and 120). To estimate the order of magnitude of this density, let us follow how an atomic electron's wave function $\psi(r)$ changes as r passes from large distances ($r \sim 1$) to small ones.

At $r \sim 1$, the other electrons screen the nuclear field, and consequently the potential energy is $U(r) \sim 1/r \sim 1$. The energy of an electron level in this field is $E \sim 1$. At distances comparable in order of magnitude to the Bohr radius for a charge Z ($r \sim 1/Z$), by contrast, the nuclear field may be regarded as unscreened: $U = -Z/r$. In the intervening region, $1/Z \ll r \ll 1$, the magnitude $|U|$ of the potential energy is already large relative to the electron energy E, while

$$\frac{d}{dr}\frac{1}{p} \sim \frac{d}{dr}\frac{1}{\sqrt{|U|}} \ll 1$$

(p denotes the momentum). The electron's motion is therefore quasiclassical. The spherically symmetric quasiclassical wave function is

$$|\psi(r)| \sim \frac{1}{r\sqrt{p}} \sim \frac{1}{r|U|^{1/4}} \quad \text{for} \quad \frac{1}{Z} \ll r \ll 1. \tag{71.1}$$

The order of magnitude of its coefficient (~ 1) follows by matching it to the wave function at $r \sim 1$, where $\psi \sim 1$.

Using (71.1) as an order-of-magnitude expression at $r \sim 1/Z$, with $U = -Z/r$ inserted, gives the required value of the wave function near the nucleus[18]:

$$\psi\left(\frac{1}{Z}\right) \sim \sqrt{Z}. \tag{71.2}$$

In accord with the general properties of wave functions in a central field (§ 32), as the distance is reduced further, $\psi(r)$ either remains constant in order of magnitude (for an s electron) or begins to decrease (when $l \neq 0$).

The probability of finding the electron in the region $r \lesssim 1/Z$ is

$$w \sim |\psi|^2 r^3 \sim \frac{1}{Z^2}. \tag{71.3}$$

Naturally, formulas (71.2) and (71.3) specify only the systematic trend of these quantities as Z increases; they do not include the nonsystematic variations encountered from one element to the next.

[17] Atomic units are used throughout this section.

[18] To find the coefficient in this formula when the wave function in the region $r \sim 1$ is known, one would have to use expression (36.25) in the region $r \lesssim 1/Z$.

§ 72. Fine structure of atomic levels

A systematic derivation of the formulas for relativistic effects in the interaction of electrons belongs to another volume of this course (see Vol. IV, § 33, 83). Here we give only a general account of these effects as they apply to the study of atomic terms.

The relativistic terms in an atom's Hamiltonian fall into two classes: some are linear in the electron-spin operators, whereas others are quadratic in them. The former correspond, as it were, to an interaction between the orbital motion of the electrons and their spins; this is called the *spin–orbit interaction*. The latter describe the interaction among the electron spins (the *spin–spin interaction*). Both interactions have the same order, the second, in v/c, the ratio of an electron's speed to the speed of light. In practice, however, the spin–orbit interaction in heavy atoms is much greater than the spin–spin interaction. The reason is that the spin–orbit interaction grows rapidly with atomic number, while the spin–spin interaction is, to leading order, independent of Z altogether (see below).

The spin–orbit interaction operator has the form

$$\hat{V}_{sl} = \sum_a \hat{\mathbf{s}}_a \hat{\mathbf{A}}_a \tag{72.1}$$

(the sum extends over every electron of the atom), where $\hat{\mathbf{s}}_a$ are the electron-spin operators and $\hat{\mathbf{A}}_a$ are certain "orbital" operators, that is, operators acting on coordinate functions. In the self-consistent-field approximation, the operators $\hat{\mathbf{A}}_a$ are proportional to the electron orbital-angular-momentum operators $\hat{\mathbf{l}}_a$; hence $\hat{V}_{sl}$ may be written as

$$\hat{V}_{sl} = \sum_a \alpha_a \hat{\mathbf{s}}_a \hat{\mathbf{l}}_a. \tag{72.2}$$

The coefficients in this sum are expressed through the potential energy $U(r)$ of an electron in the self-consistent field as follows:

$$\alpha_a = \frac{\hbar^2}{2m^2c^2r_a}\frac{dU(r_a)}{dr_a}. \tag{72.3}$$

Since $|U(r)|$ decreases with distance from the nucleus, all the α_a are positive.

When the interaction (72.2) is treated as a perturbation, its contribution to the energy must be averaged over the unperturbed state. The principal contribution comes from distances close to the nucleus—distances of the order of the Bohr radius ($\sim \hbar^2/Zme^2$) for a nucleus of charge Ze. The nuclear field is practically unscreened in this region, and the potential energy is

$$U \sim -\frac{Ze^2}{r} \sim -\frac{Z^2me^4}{\hbar^2},$$

so that

$$\alpha \sim \frac{\hbar^2}{m^2c^2}\frac{U^2}{r} \sim Z^4\left(\frac{e^2}{\hbar c}\right)^2\frac{me^4}{\hbar^2}.$$

The mean value of α is obtained from this estimate by multiplying by the probability w of finding the electron near the nucleus. By (71.3), $w \sim Z^{-2}$; hence the final estimate for the spin–orbit interaction energy of an electron is

$$\alpha \sim \left(\frac{Ze^2}{\hbar c}\right)^2 \frac{me^4}{\hbar^2},$$

that is, it differs from the principal energy of an outer electron in the atom ($\sim me^4/\hbar^2$) only by the factor $(Ze^2/\hbar c)^2$. This factor increases rapidly with atomic number and becomes of order unity in heavy atoms.

The actual averaging of the perturbation operator (72.2) over the unperturbed states of the electron shell is carried out in two stages. First we average over the electronic state of the atom for specified values L and S of its total orbital angular momentum and total spin, but not over their directions. After this averaging, $\hat{V}_{sl}$ is still an operator; it must now, however, be expressible solely through operators for quantities that characterize the atom as a whole, rather than its individual electrons. These are the operators $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$.

Let the spin–orbit interaction operator averaged in this manner be denoted by $\hat{V}_{SL}$. Since it is linear in $\hat{\mathbf{S}}$, it has the form

$$\hat{V}_{SL} = A\hat{\mathbf{S}}\hat{\mathbf{L}}, \tag{72.4}$$

where A is a constant characteristic of the given unsplit term; thus it depends on S and L, but not on the atom's total angular momentum J[19].

The splitting energy of the degenerate level is now found by solving the secular equation formed from the matrix elements of the operator (72.4). Here, however, the appropriate zeroth-order functions in which the matrix of $\hat{V}_{SL}$ is diagonal are known beforehand: they are the wavefunctions of states having a definite value of the total angular momentum J. Averaging over such a state replaces the operator $\hat{\mathbf{L}}\hat{\mathbf{S}}$ by its eigenvalue, which, by (31.3), is

$$LS = \frac{1}{2}[J(J+1) - L(L+1) - S(S+1)].$$

All components of a multiplet have the same L and S, and only their relative positions are of interest. The splitting energy may therefore be written as

$$\frac{1}{2}AJ(J+1). \tag{72.5}$$

Consequently, the intervals between adjacent components, characterized by J and $J-1$, are

$$\Delta E_{J,J-1} = AJ. \tag{72.6}$$

[19]To clarify the meaning of this operation, recall that in quantum mechanics averaging means taking the corresponding diagonal matrix element. Partial averaging, on the other hand, consists in forming a collection of matrix elements that are diagonal only in some of all the quantum numbers specifying the system's state. Thus, in the present case, averaging the operator (72.2) means forming a matrix of the elements $(nM'_L M'_S|\hat{V}_{sl}|nM_L M_S)$ for every possible M_L, M'_L, M_S, and M'_S, while it is diagonal in all the remaining quantum numbers (whose collection is denoted by n). Correspondingly, the operators $\hat{\mathbf{S}}$ and $\hat{\mathbf{L}}$ are to be understood as the matrices $(M'_S|\mathbf{S}|M_S)$ and $(M'_L|\mathbf{L}|M_L)$, whose elements are given by (27.13). We shall repeatedly use this method of averaging in successive stages below.

This formula is known as the *Landé interval rule* (A. Landé, 1923).

The constant A may have either sign. If $A > 0$, the lowest component of a multiplet level is the one with the smallest possible J, namely $J = |L - S|$; such multiplets are called *normal*. If $A < 0$, the lowest level instead has $J = L + S$ (an *inverted multiplet*).

The sign of A is readily found for atomic ground states when the electron configuration contains just one incompletely filled shell. If this shell is no more than half full, Hund's rule (§ 67) says that all n electrons in it have parallel spins, so the total spin has its greatest possible value, $S = n/2$. Substituting $\mathbf{s}_a = \mathbf{S}/n$ into (72.2) and taking α_a (the same for every electron in one shell) outside the sum gives

$$\hat{V}_{SL} = \frac{\alpha}{2S}\hat{\mathbf{S}}\hat{\mathbf{L}},$$

and hence $A = \alpha/2S > 0$. If the shell is more than half full, we first add to (72.2), and subtract from it, an identical sum taken over the unoccupied places—the holes in the incomplete shell. Since a completely filled shell would have $\hat{V}_{sl} = 0$, the result represents the operator as $\hat{V}_{sl} = -\sum_a \alpha_a \hat{\mathbf{l}}_a \hat{\mathbf{s}}_a$, where the sum now extends only over the holes; moreover, the atom's total spin and orbital angular momentum are $\mathbf{S} = -\sum_a \mathbf{s}_a$ and $\mathbf{L} = -\sum_a \mathbf{l}_a$. Repeating the preceding argument therefore yields $A = -\alpha/2S$, so that $A < 0$.

The foregoing gives a simple rule for the value of J in the ground state of an atom having one incompletely filled shell. If this shell contains no more than half its maximum possible number of electrons, then $J = |L - S|$. If it is more than half full, then $J = L + S$.

As noted earlier, the spin–spin interaction, unlike the spin–orbit interaction, is for the most part independent of Z. This is already clear from its nature: it is a direct interaction of the electrons with one another and is unrelated to the field of the nucleus.

By analogy with (72.4), averaging the spin–spin interaction operator must give an expression quadratic in $\hat{\mathbf{S}}$. The expressions quadratic in $\hat{\mathbf{S}}$ are S^2 and $(\hat{\mathbf{L}}\hat{\mathbf{S}})^2$. The eigenvalues of the first do not depend on J, so it does not split the term. It may therefore be omitted, and we may write

$$\hat{V}_{ss} = B(\hat{\mathbf{S}}\hat{\mathbf{L}})^2, \tag{72.7}$$

where B is a constant. The eigenvalues of this operator contain terms independent of J, terms proportional to $J(J+1)$, and finally a term proportional to $J^2(J+1)^2$. The first group produces no splitting and is consequently of no interest; the second can be absorbed into (72.5), which is equivalent merely to changing the constant A. The last group gives the following contribution to the term energy:

$$\frac{1}{4}BJ^2(J+1)^2. \tag{72.8}$$

The construction of atomic levels described in §§ 66 and 67 is based on the picture in which the orbital angular momenta of the electrons combine to form the total orbital angular momentum L of the atom, and their spins combine to form its total spin S. As stated earlier, this treatment is possible only when the relativistic effects are small; more precisely, the fine-structure intervals must be small in comparison with the separations of levels having different L and S. This approximation is called the *Russell–Saunders case* (H. Russell, F. Saunders, 1925); it is also described as the LS type of coupling.

In fact, the range of validity of this approximation is limited. The levels of light atoms follow the LS scheme; with increasing atomic number, the relativistic interactions within the atom become stronger and the Russell–Saunders approximation ceases to apply[20]. It should also be observed that the approximation is inapplicable, in particular, to highly excited levels in which the atom has one electron in a state of large n, and this electron consequently lies mainly at great distances from the nucleus (§ 68). Its electrostatic interaction with the motion of the other electrons is relatively weak, while the relativistic interaction within the "atomic core" does not diminish.

At the opposite limiting extreme, the relativistic interaction is large relative to the electrostatic interaction (more precisely, relative to the part of the latter responsible for the energy dependence on L and S). Orbital angular momentum and spin cannot then be considered separately, since neither is conserved. Each electron is characterized by its own total angular momentum j, and these momenta combine to give the total angular momentum J of the atom. An atomic-level scheme of this kind is said to have jj coupling. In fact, this coupling type does not occur in pure form; the levels of very heavy atoms exhibit various forms of coupling intermediate between the LS and jj types[21].

A distinctive coupling type occurs in some highly excited states. The ionic core may then be in a Russell–Saunders state, that is, it may be characterized by values of L and S, while its coupling to the highly excited electron is of the jj type (again because the electrostatic interaction is weak for this electron).

The fine structure of the hydrogen atom's energy levels has certain distinctive features; its calculation will be given in another volume of this course (see IV, § 34). Here we shall merely note that, at a fixed principal quantum number n, the energy depends only on the electron's total angular momentum j. Consequently, the level degeneracy is not removed completely: for specified n and j, there are two corresponding states whose orbital angular momenta are $l = j \pm 1/2$ (except when j takes its largest value possible at the given n, namely $j = n - 1/2$). Thus the level with $n = 3$ splits into three levels: the states $s_{1/2}$ and $p_{1/2}$ correspond to one of them, $p_{3/2}$ and $d_{3/2}$ to another, and $d_{5/2}$ to the third.

§ 73. Mendeleev's periodic system of the elements

To explain the periodic variation of properties discovered by D. I. Mendeleev (1869) among elements arranged in order of increasing atomic number, one must examine the special features of the successive filling of atomic electron shells (N. Bohr, 1922).

In passing from one atom to the next, the charge increases by one unit and one electron is added to the shell. At first sight, one might expect the binding energies of successively added electrons to change monotonically as the atomic number rises. In fact, this is not so.

The normal state of the hydrogen atom contains only one electron, in the $1s$ state. In

[20] Although the quantitative formulas describing this type of coupling cease to apply, the classification of levels according to the same scheme may remain meaningful for heavier atoms as well, especially for the lowest states (including the ground state).

[21] For a fuller discussion of coupling types and their quantitative treatment, see E. Condon and G. Shortley, *The Theory of Atomic Spectra*, Moscow: IL, 1949.

the atom of the next element, helium, another electron is added in the same $1s$ state. The binding energy of each $1s$ electron in helium is, however, much greater than the binding energy of the electron in hydrogen. This follows naturally from the difference between the field experienced by an electron in H and the field encountered by an electron added to a He^+ ion: far from the nucleus the two fields are approximately the same, but near the nucleus the field of the He^+ ion, whose charge is $Z = 2$, is stronger than that of the hydrogen nucleus with $Z = 1$.

In lithium ($Z = 3$), the third electron enters the $2s$ state, since no more than two electrons can occupy the $1s$ states at the same time. For a specified Z, the $2s$ level lies above the $1s$ level; both fall as the nuclear charge increases. In going from $Z = 2$ to $Z = 3$, however, the first effect greatly outweighs the second, so that the binding energy of the third electron in Li is much smaller than that of the electrons in helium. Next, from Be ($Z = 4$) through Ne ($Z = 10$), one further $2s$ electron and then six $2p$ electrons are added in succession. Because the nuclear charge is increasing, the binding energies of the electrons added along this sequence rise on the whole. The next electron, added on passing to Na ($Z = 11$), occupies the $3s$ state; here the effect of entering a higher shell outweighs that of the increased nuclear charge, and the binding energy once more drops sharply.

This pattern of shell filling is characteristic of the entire sequence of elements. All electron states can be divided into groups that fill successively. While each group is being filled along the sequence, the binding energy generally rises, but it falls sharply when filling of the next group begins.

Figure 24 plots the ionization potentials of the elements known from spectroscopic data; they determine the binding energies of the electrons added in passing from each element to the next.

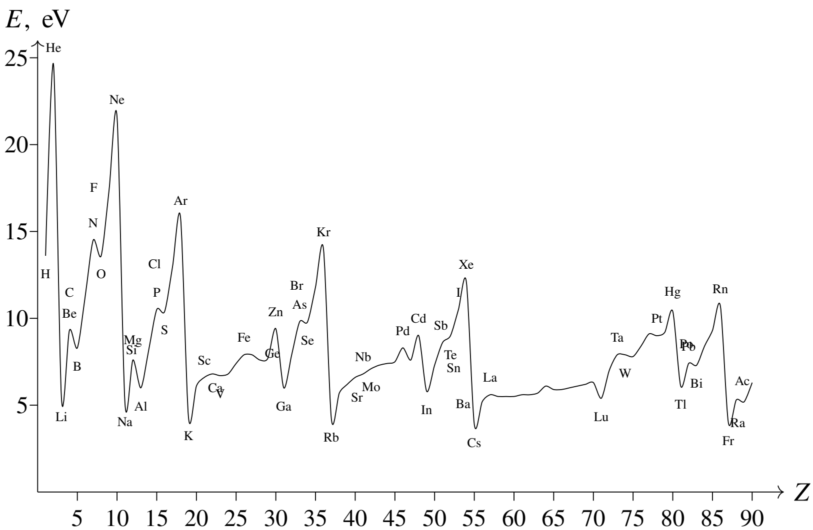

Figure 24

The states are divided among the successively filled groups as follows:

$$\begin{aligned} 1s &\ 2 \text{ electrons}, \\ 2s,\ 2p &\ 8 \text{ electrons}, \\ 3s,\ 3p &\ 8 \text{ electrons}, \\ 4s,\ 3d,\ 4p &\ 18 \text{ electrons}, \\ 5s,\ 4d,\ 5p &\ 18 \text{ electrons}, \\ 6s,\ 4f,\ 5d,\ 6p &\ 32 \text{ electrons}, \\ 7s,\ 6d,\ 5f &\ \ldots \end{aligned} \tag{73.1}$$

The first group is filled in H and He. Filling the second and third gives the first two short periods of the periodic system, each containing 8 elements. They are followed by two long periods of 18 elements each and by a long period that includes the rare-earth elements and contains 32 elements in all. The last group of states is not completely filled in the naturally occurring elements (or in the artificially produced transuranium elements).

To understand how the properties change while each group of states is being filled, the following feature of the d and f states, which distinguishes them from the s and p states, is important. For an electron in a heavy atom, the effective-potential-energy curves of the spherically symmetric field (the sum of the electrostatic and centrifugal fields) descend rapidly and almost vertically near the origin, have a deep minimum, and then begin to rise, approaching zero asymptotically. On their rising branches, the curves for the s and p states lie very close together. Thus an electron in either kind of state is located at approximately the same distance from the nucleus. The curves for the d states, and especially the f states, lie considerably farther to the left; the classically allowed region bounded by them ends much nearer the nucleus than it does for s and p states at the same total electron energy. In other words, an electron in a d or f state is found mainly much closer to the nucleus than one in an s or p state.

A number of atomic properties (including the chemical properties of the elements; see § 81) depend chiefly on the outer regions of the electron shells. The feature of the d and f states just described is therefore highly significant. For example, when the $4f$ states are being filled (in the rare-earth elements; see below), the added electrons lie much closer to the nucleus than the electrons in states filled earlier. Consequently,

these electrons have almost no effect on chemical properties, and all the rare-earth elements are consequently very similar chemically.

Elements whose d and f shells are filled (or which have no such shells at all) are called *main-group elements*; those in which these states are currently being filled are called *transition-group elements*. It is convenient to discuss the elements of these groups separately.

We begin with the main-group elements. Hydrogen and helium have the normal states

$${}_1\text{H}\ 1s\ {}^2S_{1/2}; \qquad {}_2\text{He}\ 1s^2\ {}^1S_0$$

(the subscript to the left of a chemical symbol always denotes the atomic number). The electron configurations of the remaining main-group elements are given in Table 3.

In each atom, the shells listed in the rightmost column on the same row, and those on every preceding row, are completely filled. The electron configuration in the shells

Table 3. Electron configurations of the main-group elements

	s	s^2	s^2p	s^2p^2	s^2p^3	s^2p^4	s^2p^5	s^2p^6	
$n = 2$	$_3$Li	$_4$Be	$_5$B	$_6$C	$_7$N	$_8$O	$_9$F	$_{10}$Ne	$1s^2$
3	$_{11}$Na	$_{12}$Mg	$_{13}$Al	$_{14}$Si	$_{15}$P	$_{16}$S	$_{17}$Cl	$_{18}$Ar	$2s^22p^6$
4	$_{19}$K	$_{20}$Ca							$3s^23p^6$
4	$_{29}$Cu	$_{30}$Zn	$_{31}$Ga	$_{32}$Ge	$_{33}$As	$_{34}$Se	$_{35}$Br	$_{36}$Kr	$3d^{10}$
5	$_{37}$Rb	$_{38}$Sr							$4s^24p^6$
5	$_{47}$Ag	$_{48}$Cd	$_{49}$In	$_{50}$Sn	$_{51}$Sb	$_{52}$Te	$_{53}$I	$_{54}$Xe	$4d^{10}$
6	$_{55}$Cs	$_{56}$Ba							$5s^25p^6$
6	$_{79}$Au	$_{80}$Hg	$_{81}$Tl	$_{82}$Pb	$_{83}$Bi	$_{84}$Po	$_{85}$At	$_{86}$Rn	$4f^{14}5d^{10}$
7	$_{87}$Fr	$_{88}$Ra							$6s^26p^6$
	$^2S_{1/2}$	$^1S_0$	$^2P_{1/2}$	$^3P_0$	$^4S_{3/2}$	$^3P_2$	$^2P_{3/2}$	$^1S_0$	

currently being filled is shown at the top; the principal quantum number of the electrons in these states is the number in the leftmost column of the same row. The normal states of the atom as a whole are shown at the bottom. Thus Al has the electron configuration $1s^22s^22p^63s^23p\ ^2P_{1/2}$.

The values of L and S in the normal state of an atom can be found (when its electron configuration is known) from

Hund's rule (§ 67), while J is determined by the rule given in § 72.

The noble-gas atoms (He, Ne, Ar, Kr, Xe, Rn) occupy a special place in the table: in each of them, the filling of one of the groups of states listed in (73.1) is completed. Their electron configurations are exceptionally stable (their ionization potentials are the largest in the corresponding series). The chemical inertness of these elements is also connected with this stability.

We see that the various states fill in a very regular manner along the sequence of main-group elements: for each principal quantum number n, the s states are filled first and the p states next. The electron configurations of ions of these elements are equally regular (as long as ionization does not affect electrons in d and f shells): each ion has the configuration of the preceding atom. Thus Mg^+ has the configuration of Na, and Mg^{++} that of Ne.

We now turn to the transition-group elements. The $3d$, $4d$, and $5d$ shells fill in groups of elements known respectively as the *iron*, *palladium*, and *platinum groups*.

Table 4 gives the electron configurations and atomic terms of these groups as established by experimental spectroscopic data. The table shows that the filling of the d shells is much less regular than that of the s and p shells in main-group atoms. Its characteristic feature is a "competition" between s and d states. Instead of the regular succession of configurations of the type d^ps^2 with increasing p, configurations of the form $d^{p+1}s$ or d^{p+2} are often more favourable. In the iron group, for example, Cr has the configuration $3d^54s$ rather than $3d^44s^2$; Ni, with eight d electrons, is followed directly by Cu with a completely filled d shell (and for that reason assigned above to the main groups). The terms of the ions show the same lack of regularity: their electron configurations usually do not coincide with those of the preceding atoms. For example, V^+ has the configuration $3d^4$ (not $3d^24s^2$, as in Ti), and Fe^+ has $3d^64s$ (rather than the

$3d^5 4s^2$ configuration of Mn). Notice that all ions occurring naturally in crystals and solutions contain only d electrons (and no s or p electrons) in their unfilled shells. Thus iron occurs in crystals or solutions only as Fe^{++} and Fe^{+++} ions, whose configurations are $3d^6$ and $3d^5$, respectively.

Table 4. Electron configurations of atoms in the iron, palladium, and platinum groups

			Iron group				
$_{21}$Sc	$_{22}$Ti	$_{23}$V	$_{24}$Cr	$_{25}$Mn	$_{26}$Fe	$_{27}$Co	$_{28}$Ni
$3d4s^2$	$3d^2 4s^2$	$3d^3 4s^2$	$3d^5 4s$	$3d^5 4s^2$	$3d^6 4s^2$	$3d^7 4s^2$	$3d^8 4s^2$
$^2D_{3/2}$	$^3F_2$	$^4F_{3/2}$	$^7S_3$	$^6S_{5/2}$	$^5D_4$	$^4F_{9/2}$	$^3F_4$
			Palladium group				
$_{39}$Y	$_{40}$Zr	$_{41}$Nb	$_{42}$Mo	$_{43}$Tc	$_{44}$Ru	$_{45}$Rh	$_{46}$Pd
$4d5s^2$	$4d^2 5s^2$	$4d^4 5s$	$4d^5 5s$	$4d^5 5s^2$	$4d^7 5s$	$4d^8 5s$	$4d^{10}$
$^2D_{3/2}$	$^3F_2$	$^6D_{1/2}$	$^7S_3$	$^6S_{5/2}$	$^5F_5$	$^4F_{9/2}$	$^1S_0$
			Platinum group				
$_{57}$La							
$5d6s^2$							
$^2D_{3/2}$							
$_{71}$Lu	$_{72}$Hf	$_{73}$Ta	$_{74}$W	$_{75}$Re	$_{76}$Os	$_{77}$Ir	$_{78}$Pt
$5d6s^2$	$5d^2 6s^2$	$5d^3 6s^2$	$5d^4 6s^2$	$5d^5 6s^2$	$5d^6 6s^2$	$5d^7 6s^2$	$5d^9 6s$
$^2D_{3/2}$	$^3F_2$	$^4F_{3/2}$	$^5D_0$	$^6S_{5/2}$	$^5D_4$	$^4F_{9/2}$	$^3D_3$

The situation is similar when the $4f$ shell is filled, as happens in the sequence of elements called the *rare earths* (Table 5)[22]. Filling the $4f$ shell is also not entirely regular, being characterized by competition among the $4f$, $5d$, and $6s$ states.

The last group of transition elements begins with actinium. In it the $6d$ and $5f$ shells are filled, in a manner analogous to the filling in the rare-earth sequence (Table 6).

Table 5. Electron configurations of rare-earth atoms

$_{58}$Ce	$_{59}$Pr	$_{60}$Nd	$_{61}$Pm	$_{62}$Sm	$_{63}$Eu	
$4f5d6s^2$	$4f^3 6s^2$	$4f^4 6s^2$	$4f^5 6s^2$	$4f^6 6s^2$	$4f^7 6s^2$	
$^1G_4$	$^4I_{9/2}$	$^5I_4$	$^6H_{5/2}$	$^7F_0$	$^8S_{7/2}$	
$_{64}$Gd	$_{65}$Tb	$_{66}$Dy	$_{67}$Ho	$_{68}$Er	$_{69}$Tm	$_{70}$Yb
$4f^7 5d6s^2$	$4f^9 6s^2$	$4f^{10} 6s^2$	$4f^{11} 6s^2$	$4f^{12} 6s^2$	$4f^{13} 6s^2$	$4f^{14} 6s^2$
$^9D_2$	$^6H_{15/2}$	$^5I_8$	$^4I_{15/2}$	$^3H_6$	$^2F_{7/2}$	$^1S_0$

Table 6. Electron configurations of atoms in the actinide group

$_{89}$Ac	$_{90}$Th	$_{91}$Pa	$_{92}$U	$_{93}$Np	$_{94}$Pu	$_{95}$Am	$_{96}$Cm
$6d7s^2$	$6d^2 7s^2$	$5f^2 6d7s^2$	$5f^3 6d7s^2$	$5f^4 6d7s^2$	$5f^6 7s^2$	$5f^7 7s^2$	$5f^7 6d7s^2$
$^2D_{3/2}$	$^3F_2$	$^4K_{11/2}$	$^5L_6$	$^6L_{11/2}$	$^7F_0$	$^8S_{7/2}$	$^9D_2$

We shall end this section by considering one noteworthy application of the Thomas–Fermi method. We have seen that the first p-shell electrons occur in the fifth element (B), that d-electrons appear at $Z = 21$ (Sc), and that f-electrons appear at $Z = 58$ (Ce). The Thomas–Fermi method predicts these values of Z in the following way.

In a many-electron atom, an electron of orbital angular momentum l moves with an

[22] Chemistry texts commonly include Lu among the rare-earth elements as well. This is incorrect, however, since its $4f$ shell is already full; Lu should be assigned to the platinum group, as has been done in Table 4.

"effective potential energy" equal to[23]

$$U_l(r) = -\varphi(r) + \frac{(l+1/2)^2}{2r^2}.$$

The first term is the potential energy in the electric field described by the Thomas–Fermi potential $\varphi(r)$. The second is the centrifugal energy, in which we write $(l+1/2)^2$ in place of

$l(l+1)$ because the motion is semiclassical. Since the total energy of an electron in an atom is negative, it is clear that if, for specified Z and l, $U_l(r) > 0$ for every r, the atom cannot contain any electrons at all with the value of l under consideration. If a particular l is fixed and Z is varied, then for sufficiently small Z one indeed finds $U_l(r) > 0$ everywhere. As Z is increased, a point is reached at which the curve $U_l = U_l(r)$ touches the abscissa axis; at larger Z there is a region in which $U_l(r) < 0$. Thus electrons with a given l first appear in the atom when the curve $U_l(r)$ is tangent to the abscissa axis, that is, when

$$U_l(r) = -\varphi(r) + \frac{(l+1/2)^2}{2r^2} = 0, \qquad U_l'(r) = -\varphi'(r) - \frac{(l+1/2)^2}{r^3} = 0.$$

Substitution of expression (70.6) for the potential gives

$$\begin{aligned} Z^{2/3}\frac{\chi(x)}{x} &= \left(\frac{2}{3\pi}\right)^{2/3}\frac{(l+1/2)^2}{2x^2}, \\ Z^{2/3}\frac{x\chi'(x)-\chi(x)}{x^2} &= -2\left(\frac{2}{3\pi}\right)^{2/3}\frac{(l+1/2)^2}{2x^3}. \end{aligned} \tag{73.2}$$

Dividing the second of these equations term by term by the first, we obtain the equation for x

$$\frac{\chi'(x)}{\chi(x)} = -\frac{1}{x},$$

after which Z is calculated from the first equation in (73.2). Numerical calculation gives

$$Z = 0{,}155(2l+1)^3.$$

This formula determines the values of Z at which electrons with a specified l first appear in the atom (with an error of about 10%). The exact integer values are obtained if the coefficient 0.155 is replaced by 0.17:

$$Z = 0{,}17(2l+1)^3. \tag{73.3}$$

For $l = 1, 2, 3$, rounding this formula's results to the nearest integers gives precisely the correct values 5, 21, and 58. For $l = 4$, formula (73.3) gives $Z = 124$; thus g electrons should first appear only in element 124.

[23] Atomic units are used, as in § 70.

§ 74. X-ray terms

An atom's inner electrons have such large binding energies that, when one of them passes into an unfilled outer shell (or is removed from the atom altogether), the resulting excited atom (or ion) is mechanically unstable against ionization accompanied by rearrangement of the electron shell and formation of a stable ion. Electron interactions within an atom are comparatively weak, however, so the probability of such a transition is still rather small and the excited state has a long lifetime τ. Its level "width" $\hbar/\tau$ (see § 44) is therefore small enough that the energies of an atom with an excited inner electron can meaningfully be treated as discrete energy levels of "quasistationary" atomic states. These levels are called *X-ray terms*[24].

X-ray terms are classified first by specifying the shell from which the electron has been removed, or, as one says, the shell in which a hole has formed. The particular destination of the electron has almost no effect on the atomic energy and is consequently immaterial.

The total angular momentum of all the electrons filling a shell is zero. Removing one electron gives the shell a certain angular momentum j. For the shell (n, l), j may take the values $l \pm 1/2$. The resulting levels could thus be denoted by $1s_{1/2}$, $2s_{1/2}$, $2p_{1/2}$, $2p_{3/2}$, ..., with the value of j written as a subscript on the symbol specifying the location of the hole. Special symbols are customary, however, with the following correspondence:

$$\begin{array}{ccccccccc} 1s_{1/2} & 2s_{1/2} & 2p_{1/2} & 2p_{3/2} & 3s_{1/2} & 3p_{1/2} & 3p_{3/2} & 3d_{3/2} & 3d_{5/2} \\ K & L_{\mathrm{I}} & L_{\mathrm{II}} & L_{\mathrm{III}} & M_{\mathrm{I}} & M_{\mathrm{II}} & M_{\mathrm{III}} & M_{\mathrm{IV}} & M_{\mathrm{V}} \end{array}$$

Levels with $n = 4, 5, 6$ are denoted analogously by the letters N, O, P.

Levels having the same n (and denoted by the same capital letter) lie close together and far from levels with other values of n. The reason is that inner electrons, being comparatively close to the nucleus, move in a field that is almost the unscreened nuclear field. Their states are therefore hydrogen-like, and to first approximation their energy is $-Z^2/2n^2$ (in atomic units), so it depends on n alone. When relativistic effects are taken into account, terms with different j separate from one another (compare the discussion in § 72 of the fine structure of the hydrogen levels). Thus, for example, L_{I} and L_{II} separate from L_{III}, while M_{I} and M_{II} separate from M_{III} and M_{IV}. Pairs of levels of this kind are called *relativistic doublets*. For a fixed j, the separation between terms with different l (for example, L_{I} from L_{II}, or M_{I} from M_{II}) results from the departure of the field acting on the inner electrons from the Coulomb field of the nucleus, that is, from allowing for the electron's interaction with the other electrons. Such doublets are called *screening doublets*. Near the nucleus, the potential due to all the other electrons supplies the leading correction to the electron's "hydrogen-like" energy; this contribution is proportional to $Z^{4/3}$ (see (70.8)). This correction, however, is independent of both n and l, and hence leaves the separations between levels unchanged. The inner-electron separations are $r \sim 1/Z$ (the Bohr radius in the field produced by charge Z), so the energy of the interaction in question is $\sim 1/r \sim Z$. With this correction included, the X-ray term energy can, to the same accuracy, be expressed as $-(Z - \delta)^2/2n^2$, where $\delta = \delta(n, l)$ is

[24]The name reflects the fact that transitions between these levels cause the atom to emit X-rays.

small relative to Z and may be taken as a measure of how strongly the nuclear charge is screened. Electronic shells can give rise not only to x-ray terms with one vacancy, but also to terms with two or three. For inner electrons the spin–orbit interaction is strong; consequently, the vacancies couple to one another according to the jj scheme.

The width of an x-ray term is fixed by the total probability of all possible rearrangements of the atom's electron shell that fill the vacancy in question. In heavy atoms, the chief contribution comes from passage of the vacancy from that shell into a higher one (that is, from the reverse electron transitions), with emission of an x-ray quantum. The probabilities of these "radiative" transitions, and hence the corresponding part of the level width, rise extremely rapidly with atomic number—as Z^4—but, at fixed Z, decrease in going from deeper levels to shallower ones. For lighter atoms (and at higher levels), a substantial, or even dominant, part is played by nonradiative transitions. In such a transition, the energy released when an electron from a higher level fills the hole is used to eject another inner electron from the atom (the so-called *Auger effect*); after this process the atom is left with two holes. To first order in $1/Z$, the probabilities of these processes and their contribution to the corresponding level width are independent of the atomic number (see the problem)[25].

Problem

Determine the asymptotic dependence of the Auger width of X-ray terms on the atomic number when the latter is sufficiently large.

Solution. For an Auger transition, the probability is proportional to the square of the matrix element given below.

$$M = \iint \psi_1'^* \psi_2'^* V \psi_1 \psi_2 \, dV_1 dV_2,$$

Here ψ_1, ψ_2 and ψ_1', ψ_2' are the initial and final wavefunctions of the two electrons taking part in the transition, while $V = e^2/r_{12}$ is their interaction energy. When Z is sufficiently large, the inner-electron wave functions can be treated as hydrogenic, while the shielding of the nuclear field by the remaining electrons can be ignored (the wave function of the electron undergoing ionization is also hydrogenic in the deep-atomic region relevant to the integral M). When every quantity in the calculation is expressed in Coulomb units (see § 36), the integral M contains just one quantity that depends on Z, namely $V = e^2/r_{12} = 1/Zr_{12}$; consequently, $M \sim 1/Z$. The transition probability, and hence the Auger width of the level ΔE, is proportional to Z^{-2}. Upon restoring ordinary units (the Coulomb unit of energy is $Z^2 me^4/\hbar^2$), we find that ΔE is independent of Z.

§ 75. Multipole Moments

In classical theory, the electrical properties of a system are characterized by its multipole moments of various orders, which are expressed in terms of the particles' charges and

[25] As an example, the Auger width of the K level is about 1 eV, while for higher levels it reaches values of roughly 10 eV.

coordinates. In quantum theory these quantities retain the same definitions, but the definitions must be understood as operator definitions.

The first multipole moment is the *dipole moment*, defined by the vector

$$\mathbf{d} = \sum e\mathbf{r}$$

(the sum extends over every particle in the system; for brevity, the label identifying a particle is suppressed). The matrix

of this operator—as with any polar vector (see § 30)—can have nonzero entries only for transitions between states of opposite parity. Consequently, all its diagonal entries are necessarily zero. Equivalently, in a stationary state the mean dipole moment of any particle system (an atom, for example) vanishes[26].

The same conclusion plainly applies to all 2^l-pole moments with odd l. The components of such a moment are odd polynomials of degree l in the coordinates; like the components of a polar vector, they reverse sign under coordinate inversion. They consequently obey the same parity selection rule.

The *quadrupole moment* of a system is defined as the symmetric tensor

$$\widehat{Q}_{ik} = \sum e(3x_i x_k - \delta_{ik}\mathbf{r}^2) \tag{75.1}$$

whose diagonal terms sum to zero. To determine the values of these quantities in a particular state of the system (say, an atom), the operator (75.1) must be averaged with the corresponding wave function. It is useful to carry out this averaging in two stages (cf. § 72).

Let $\widehat{Q}_{ik}$ denote the quadrupole-moment operator after averaging over electronic states with a prescribed value of the total angular momentum J, but without fixing its projection M_J.

An operator averaged in this way can depend only on operators for quantities that describe the atomic state as a whole. The "vector" $\widehat{\mathbf{J}}$ is the only such vector. The operator $\widehat{Q}_{ik}$ must therefore have the form

$$\widehat{Q}_{ik} = \frac{3Q}{2J(2J-1)}\left(\widehat{J}_i\widehat{J}_k + \widehat{J}_k\widehat{J}_i - \frac{2}{3}\widehat{\mathbf{J}}^2\delta_{ik}\right), \tag{75.2}$$

where the expression in parentheses has been constructed to be symmetric in i, k and to vanish when contracted over this pair of indices (the meaning of the coefficient Q is given below). Here the operators $\widehat{J}_i$ are to be understood as the familiar matrices (§ 27, 54) with respect to states having different M_J values. The operator $\widehat{\mathbf{J}}^2$ may, of course, simply be replaced by its eigenvalue $J(J+1)$.

[26]To prevent a possible misunderstanding, we stress that this statement concerns either a closed particle system or one placed in an external electric field having a center of symmetry. Thus, if the nuclei are regarded as "fixed," the general statement just made applies to the electron system of an atom, but not to that of a molecule.

It is also assumed that the energy level has no extra ("accidental") degeneracy beyond the degeneracy associated with the orientations of the total angular momentum. Otherwise one can form wave functions for stationary states that have no definite parity, and the corresponding diagonal entries of the dipole moment need not vanish.

Since definite values cannot be assigned simultaneously to all three components of the angular momentum $\mathbf{J}$, the same is true of the components of the tensor Q_{ik}. For the component Q_{zz}, we obtain

$$\widehat{Q}_{zz} = \frac{3Q}{J(2J-1)}\left(\widehat{J}_z^2 - \frac{1}{3}\widehat{\mathbf{J}}^2\right).$$

In a state in which $\mathbf{J}^2 = J(J+1)$ and $J_z = M_J$ have specified values, Q_{zz} likewise has a definite value:

$$Q_{zz} = \frac{3Q}{J(2J-1)}\left[M_J^2 - \frac{1}{3}J(J+1)\right]. \tag{75.3}$$

For $M_J = J$ (the angular momentum is directed "entirely" along the z axis), $Q_{zz} = Q$; this quantity is customarily called simply the quadrupole moment.

When $J = 0$, all matrix elements of the angular momentum vanish, and hence so do the operators (75.2). They also vanish identically for $J = 1/2$. This follows at once by directly multiplying the Pauli matrices (55.7), which represent the component matrices of every angular momentum equal to $1/2$.

This is not an accidental circumstance, but a special instance of a general rule: the tensor of a 2^l-pole moment (for even l) is nonzero only for system states whose total angular momentum satisfies

$$J \geqslant l/2. \tag{75.4}$$

The tensor of a 2^l-pole moment is an irreducible tensor of rank l (see II, § 41), and condition (75.4) follows from the general angular-momentum selection rules for matrix elements of such tensors: it is the condition under which diagonal matrix elements may be nonzero (§ 107). As noted above, the parity selection rule additionally requires l to be even.

One must also take into account that electric multipole moments are purely "orbital" quantities: their operators contain no spin operators. If the spin–orbit interaction can be neglected, so that L and S are conserved separately, the matrix elements of multipole moments therefore obey selection rules in L as well as in the quantum number J.

Problem

1. Find the relation between the atomic quadrupole-moment operators in states belonging to different fine-structure components of a level (that is, states with different J for prescribed L and S).

Solution. In states with fixed L and S, the quadrupole-moment operator, being a purely orbital quantity, depends only on $\widehat{\mathbf{L}}$. It is therefore given by the same formula (75.2), with $\widehat{\mathbf{J}}$ replaced by $\widehat{\mathbf{L}}$ (and with a different constant Q). A further average over a state with given J yields the operator (75.2):

$$\begin{aligned}\widehat{Q}_{ik} &= \frac{3Q_J}{2J(J-1)}\left[\widehat{J}_i\widehat{J}_k + \widehat{J}_k\widehat{J}_i - \frac{2}{3}J(J+1)\delta_{ik}\right] \\ &= \frac{3Q_L}{2L(L-1)}\overline{\left[\widehat{L}_i\widehat{L}_k + \widehat{L}_k\widehat{L}_i - \frac{2}{3}L(L+1)\delta_{ik}\right]}.\end{aligned} \tag{1}$$

We must determine the relation between Q_J and Q_L. Multiply equation (1) on the left by $\widehat{J}_i$ and on the right by $\widehat{J}_k$, summing over i and k, and then replace the resulting diagonal operators by their eigenvalues. In doing so,

$$\widehat{J}_i \widehat{L}_i \widehat{L}_k \widehat{J}_k = (\mathbf{JL})^2,$$

where, by (31.4),

$$2\mathbf{JL} = J(J+1) + L(L+1) - S(S+1).$$

The product $\widehat{J}_i \widehat{L}_k \widehat{L}_i \widehat{J}_k$, on the other hand, is transformed by means of

$$\{\widehat{L}_i, \widehat{L}_k\} = ie_{ikl}\widehat{L}_l, \qquad \{\widehat{J}_i, \widehat{L}_l\} = ie_{ilm}\widehat{L}_m,$$

in the same manner as in the problem following § 29, and gives

$$\widehat{J}_i \widehat{L}_k \widehat{L}_i \widehat{J}_k = (\mathbf{JL})^2 - (\mathbf{JL}).$$

Likewise,

$$\widehat{J}_i \widehat{J}_i \widehat{J}_k \widehat{J}_k = (\mathbf{J}^2)^2, \qquad \widehat{J}_i \widehat{J}_k \widehat{J}_i \widehat{J}_k = \mathbf{J}^2(\mathbf{J}^2 - 1).$$

Equation (1) consequently gives

$$Q_J = Q_L \frac{3(\mathbf{JL})(2\mathbf{JL} - 1) - 2J(J+1)L(L+1)}{(J+1)(2J+3)L(2L-1)}. \tag{2}$$

For $S = 1/2$, in particular, this formula becomes

$$\begin{aligned} Q_J &= Q_L, && \text{for } J = L + \frac{1}{2}, \\ Q_J &= Q_L \frac{(L-1)(2L+3)}{L(2L+1)}, && \text{for } J = L - \frac{1}{2}. \end{aligned} \tag{3}$$

Problem

2. Express the quadrupole moment of an electron (charge $-|e|$) having orbital angular momentum l in terms of the mean square of its distance from the center.

Solution. We have to average

$$Q_{zz} = -|e|\,\overline{r^2}(3\cos^2\theta - 1) = -|e|\,\overline{r^2}(3n_z^2 - 1)$$

over the state with angular momentum l and projection $m = l$. The mean value of the angular factor follows directly from the formula obtained in the problem after § 29, with $\widehat{l}_z$ replaced there by l. The result is

$$Q_l = |e|\,\overline{r^2}\frac{2l}{2l+3}. \tag{1}$$

As it should, this quantity has the sign opposite to the electron charge: a particle moving with its angular momentum directed along the z axis is found mainly near the plane $z = 0$, and therefore has $\cos^2\theta < 1/3$.

For an electron of prescribed $j = l \pm 1/2$, formulas (3) then give

$$Q_j = |e|\,\overline{r^2}\frac{2j-1}{2j+2}. \tag{2}$$

PROBLEM

3. Determine the ground-state quadrupole moment of an atom in which all ν electrons outside closed shells occupy equivalent states with orbital angular momentum l.

SOLUTION. Since the total quadrupole moment of the closed shells is zero, the atomic quadrupole-moment operator is the sum

$$\widehat{Q}_{ik} = \frac{3|e|\overline{r^2}}{(2l-1)(2l+3)} \sum \left[\widehat{l}_i\widehat{l}_k + \widehat{l}_k\widehat{l}_i - \frac{2}{3}l(l+1)\delta_{ik}\right],$$

taken over the ν outer electrons (formula (4) has been used here).

Suppose first that $\nu \leqslant 2l+1$, so that no more than half of the places in the shell are occupied. Hund's rule (§ 67) then makes the spins of all ν electrons parallel, and hence $S = \nu/2$. The atomic spin wave function is therefore symmetric, while the coordinate wave function must be antisymmetric in these electrons. It follows that the electrons must all have different m values. Thus the largest possible M_L, which is also L, equals

$$L = (M_L)_{\max} = \sum_{m=l-\nu+1}^{l} m = \frac{1}{2}\nu(2l-\nu+1).$$

The required Q_L is the eigenvalue of Q_{zz} at $M_L = L$. Consequently,

$$Q_L = \frac{6|e|\overline{r^2}}{(2l-1)(2l+3)} \sum_{m=l-\nu+1}^{l} \left[m^2 - \frac{l(l+1)}{3}\right],$$

and evaluation of the sum gives

$$Q_L = \frac{2l(2l-2\nu+1)}{(2l-1)(2l+3)}|e|\overline{r^2}. \tag{1}$$

The final conversion from Q_L to Q_J is made with formula (2).

For an atom whose outer shell is more than half filled, the preceding case is recovered by considering holes instead of electrons. The answer is therefore given by the same formula (6), with its overall sign reversed because a hole has charge $+|e|$; here ν is now to mean the number of unoccupied places in the shell, rather than the number of electrons.

§ 76. Atom in an electric field

Placing an atom in an external electric field changes its energy levels; this phenomenon is known as the *Stark effect*.

In a uniform external electric field the electrons move in an axially symmetric field, formed by the nucleus and the external field. Strictly, the total angular momentum is no longer conserved; only its projection M_J on the field direction remains conserved. States with different M_J have different energies, so the field removes the directional degeneracy. The removal is incomplete: states differing only in the sign of M_J still have equal energies. Indeed, the atom is symmetric under reflection in any plane through the symmetry axis, chosen below as z. States related by such a reflection must have equal energies, while reflection in a plane through an axis reverses the angular momentum about that axis.

Assume the field is weak enough that its additional energy is small compared with separations of neighbouring atomic levels, including fine-structure intervals. The shifts may then be calculated by perturbation theory (§ 38, 39). The perturbation is the energy of the electron system in the uniform field $\boldsymbol{\mathcal{E}}$:

$$\widehat{V} = -\mathbf{d}\boldsymbol{\mathcal{E}} = -\mathcal{E} d_z, \tag{76.1}$$

where $\mathbf{d}$ is the dipole moment. Zeroth-order levels are directionally degenerate, but this degeneracy is immaterial here, and perturbation theory may be applied as for nondegenerate levels. Only matrix elements of d_z for transitions with unchanged M_J can be nonzero (§ 29), so states with different M_J behave independently.

The first-order shift is given by diagonal elements of the perturbation, but diagonal dipole-moment elements vanish (§ 75). Hence the electric-field splitting is second order in the field.[27] Being quadratic in the field, the shift of level E_n has the form

$$\Delta E_n = -\frac{1}{2}\alpha_{ik}^{(n)}\mathcal{E}_i\mathcal{E}_k, \tag{76.2}$$

where $\alpha_{ik}^{(n)}$ is symmetric. With z along the field,

$$\Delta E_n = -\frac{1}{2}\alpha_{zz}^{(n)}\mathcal{E}^2. \tag{76.3}$$

The tensor $\alpha_{ik}^{(n)}$ is also the *polarizability* of the atom. In (11.16), take the parameters λ as the components $\mathcal{E}_i$ and put $\widehat{H} = \widehat{H}_0 - \mathcal{E}_i\widehat{d}_i$. Then the induced mean dipole moment is

$$\overline{d}_i^{(n)} = -\frac{\partial \Delta E_n}{\partial \mathcal{E}_i},$$

so that

$$\overline{d}_i^{(n)} = \alpha_{ik}^{(n)}\mathcal{E}_k. \tag{76.4}$$

By the general second-order formula (38.10),

$$\alpha_{ik}^{(n)} = -2\sum_m{}' \frac{(d_i)_{nm}(d_k)_{mn}}{E_n - E_m}. \tag{76.5}$$

The polarizability depends on the unperturbed state, including M_J. Its dependence on M_J follows generally by regarding the values $\alpha_{ik}^{(n)}$ as eigenvalues of

$$\widehat{\alpha}_{ik}^{(n)} = \alpha_n\delta_{ik} + \beta_n(\widehat{J}_i\widehat{J}_k + \widehat{J}_k\widehat{J}_i - \tfrac{2}{3}\delta_{ik}\widehat{\mathbf{J}}^2). \tag{76.6}$$

This is the general symmetric rank-two tensor depending on $\widehat{\mathbf{J}}$ (cf. § 75). Thus

$$\Delta E_n = -\frac{\mathcal{E}^2}{2}\{\alpha_n + 2\beta_n[M_J^2 - \tfrac{1}{3}J(J+1)]\}. \tag{76.7}$$

[27] Hydrogen is an exception: its Stark effect is linear in the field (see the next section). Other atoms in highly excited, and therefore hydrogen-like, states (§ 68) behave similarly in sufficiently strong fields.

Summing over all M_J makes the second term vanish; the first therefore shifts the centre of gravity of the split level. For $J = 1/2$ the level remains unsplit, consistently with Kramers' theorem (§ 60).

In a nonuniform external field that varies little across the atom, a splitting linear in the field may also arise from the atomic quadrupole moment. The quadrupole interaction operator has the classical form (see II, § 42)

$$\widehat{V} = \frac{1}{6}\frac{\partial^2\varphi}{\partial x_i \partial x_k}\widehat{Q}_{ik}, \tag{76.8}$$

where φ is the electric potential and the derivatives are evaluated at the atom.

Problem

1. Determine how the Stark splitting of the components of a multiplet level depends on J.

Solution. Reverse the order in which the perturbations are introduced: first consider Stark splitting without fine structure, then introduce spin–orbit interaction. Since spin does not interact with the external electric field, a level of orbital angular momentum L splits according to (76.2), with

$$\widehat{\alpha}_{ik} = a\delta_{ik} + b(\widehat{L}_i\widehat{L}_k + \widehat{L}_k\widehat{L}_i - \tfrac{2}{3}\delta_{ik}\widehat{\mathbf{L}}^2).$$

After spin–orbit interaction is introduced, average this operator over states of specified J but unspecified M_J, exactly as in problem 1 of § 75. Equations (76.6), (76.7) result, with

$$\alpha = a, \qquad \beta = b\frac{3(\mathbf{JL})[2(\mathbf{JL}) - 1] - 2J(J+1)L(L+1)}{J(J+1)(2J-1)(2J+3)}.$$

This determines the J dependence, though not the L, S dependence of a, b.

Problem

2. Determine the splitting of a doublet level ($S = 1/2$) in an arbitrary, not necessarily weak, electric field.

Solution. If the splitting is not small relative to the doublet interval, the electric-field perturbation and spin–orbit interaction must be included together:

$$\widehat{V} = A\widehat{\mathbf{S}}\widehat{\mathbf{L}} - \tfrac{1}{2}\mathcal{E}^2\{a + 2b[\widehat{L}_z^2 - \tfrac{1}{3}L(L+1)]\}.$$

Dropping terms common to all components,

$$\widehat{V} = \frac{A}{2}[\widehat{S}_+\widehat{L}_- + \widehat{S}_-\widehat{L}_+ + 2\widehat{S}_z\widehat{L}_z] - b\mathcal{E}^2\widehat{L}_z^2.$$

For fixed $M = M_J$, form the secular equation in the states $|M_L M_S\rangle = |M \mp 1/2, \pm 1/2\rangle$. Its elements are

$$\langle M - 1/2, 1/2|V|M - 1/2, 1/2\rangle = \frac{A}{2}(M - \tfrac{1}{2}) - b\mathcal{E}^2(M - \tfrac{1}{2})^2,$$

$$\langle M + 1/2, -1/2|V|M + 1/2, -1/2\rangle = -\frac{A}{2}(M + \tfrac{1}{2}) - b\mathcal{E}^2(M + \tfrac{1}{2})^2,$$

$$\langle M - 1/2, 1/2|V|M + 1/2, -1/2\rangle = \frac{A}{2}[(L + M + \tfrac{1}{2})(L - M + \tfrac{1}{2})]^{1/2}.$$

Hence (problem 1, § 39)

$$\Delta E = -b\mathcal{E}^2 M^2 \pm \sqrt{\frac{A^2}{4}(L+\tfrac{1}{2})^2 + b\mathcal{E}^2(b\mathcal{E}^2 + A)M^2}. \tag{1}$$

Terms common to the doublet have been omitted. Both signs apply for $|M| \leqslant L - 1/2$. For $|M| = L + 1/2$ only one state exists, and with the same additive constant

$$\Delta E = (\tfrac{A}{2} + b\mathcal{E}^2)(L+\tfrac{1}{2}) - b\mathcal{E}^2(L+\tfrac{1}{2})^2. \tag{2}$$

Problem

3. Determine the quadrupole splitting in an axially symmetric electric field.[28]

Solution. Here

$$\varphi_{xx} = \varphi_{yy} = a, \qquad \varphi_{zz} = -2a,$$

and the other second derivatives vanish. Equation (76.8) becomes

$$\frac{a}{6}(\widehat{Q}_{xx} + \widehat{Q}_{yy} - 2\widehat{Q}_{zz}) = \frac{Qa}{2J(2J-1)}(\widehat{\mathbf{J}}^2 - 3\widehat{J}_z^2),$$

and therefore

$$\Delta E = a\frac{Q}{2J(2J-1)}[J(J+1) - 3M_J^2].$$

Problem

4. Calculate the polarizability of hydrogen in its ground state.

Solution. Spherical symmetry makes $\alpha_{ik} = \alpha\delta_{ik}$, with

$$\alpha = -2e^2 \sum_k{}' \frac{|z_{0k}|^2}{E_0 - E_k}.$$

Introduce $\widehat{b}$ by $z = (m/\hbar)d\widehat{b}/dt$. Then $z_{0k} = (im/\hbar^2)(E_0 - E_k)b_{0k}$ and

$$\alpha = \frac{2ime^2}{\hbar^2}\sum_k z_{0k}b_{k0} = \frac{2ime^2}{\hbar^2}(zb)_{00}. \tag{1}$$

It suffices to know $\widehat{b}\psi_0$. From (9.2), writing $\widehat{b}\psi_0 = b(\mathbf{r})\psi_0$ and using $\widehat{H} = -\hbar^2\Delta/2m + U(r)$,

$$\tfrac{1}{2}\psi_0\Delta b + \nabla b\nabla\psi_0 = iz\psi_0.$$

With $b = \cos\theta f(r)$ this becomes

$$\frac{f''}{2} + \frac{f'}{r} - \frac{f}{r^2} + \frac{\psi_0'}{\psi_0}f' = ir. \tag{2}$$

The solution must keep $f\psi_0$ finite at zero and infinity. For hydrogen's ground state $\psi_0 = e^{-r/a_B}/\sqrt{\pi}$, $a_B = \hbar^2/me^2$, the solution is $f = -ira_B(a_B + r/2)$. Therefore[29]

$$\alpha = \frac{2i}{a_B}(rf\cos^2\theta)_{00} = \frac{2i}{3a_B}(rf)_{00} = \frac{9}{2}a_B^3.$$

[28] For an arbitrary field see problem 6, § 103.

[29] The next section obtains this result by another method.

PROBLEM

5. Calculate the polarizability of an electron in a bound s state in a potential well of range a, where $a\kappa \ll 1$, $\kappa = \sqrt{2m|E_0|}/\hbar$, and $|E_0|$ is the binding energy.

SOLUTION. The region inside the well may be neglected in $(z\widehat{b})_{00}$, and throughout space one may use

$$\psi_0 = \sqrt{\frac{\kappa}{2\pi}}\frac{e^{-\kappa r}}{r}.$$

Equation (2) becomes

$$\frac{f''}{2} - \kappa f' - \frac{f}{r^2} = ir,$$

whose boundary-condition solution is $f = -ir^2/2\kappa$. Equation (1) then gives

$$\alpha = \frac{me^2}{4\hbar^2\kappa^4}.$$

§ 77. The hydrogen atom in an electric field

Unlike the levels of other atoms, hydrogen levels placed in a uniform electric field split by an amount proportional to the first power of the field (the *linear Stark effect*). The reason is the accidental degeneracy of the hydrogen terms: at a fixed principal quantum number n, states having different l have equal energies. The dipole-moment matrix elements connecting these states are not zero; hence the secular equation produces a nonzero level displacement already in first order[30].

For the calculation it is useful to select the unperturbed wave functions so that, within every set of mutually degenerate states, the perturbation matrix is diagonal. This is achieved by quantizing the hydrogen atom in parabolic coordinates. Formulas (37.15) and (37.16) give the wave functions $\psi_{n_1n_2m}$ of its stationary states in these coordinates.

The perturbation operator, namely the electron energy in the field $\mathcal{E}$, is $\mathcal{E}z = \mathcal{E}(\xi-\eta)/2$ (the field points along the positive z axis, while the force on the electron points in the negative direction)[31]. We need matrix elements for transitions $n_1n_2m \to n_1'n_2'm'$ that leave the energy, and therefore n, unchanged. Among them, only the diagonal elements can be nonzero:

$$\begin{aligned}\int |\psi_{n_1n_2m}|^2\mathcal{E}z\,dV &= \frac{\mathcal{E}}{8}\int_0^\infty\int_0^\infty\int_0^{2\pi}(\xi^2-\eta^2)|\psi_{n_1n_2m}|^2, d\varphi, d\xi, d\eta \\ &= \frac{\mathcal{E}n}{4}\int_0^\infty\int_0^\infty f_{n_1m}^2(\rho_1)f_{n_2m}^2(\rho_2)(\rho_1^2-\rho_2^2), d\rho_1 d\rho_2.\end{aligned} \tag{77.1}$$

(Here we put $\xi = n\rho_1$, $\eta = n\rho_2$.) Diagonality in m is evident. Diagonality in n_1, n_2 follows from the mutual orthogonality of the functions f_{n_1m} with different n_1 and the same m (see below). The integrations over $d\rho_1$ and $d\rho_2$ in (77.1) separate; the resulting

[30] In the calculations below we disregard the fine structure of the hydrogen levels. Thus the field must be weak enough for perturbation theory to apply, yet strong enough that the Stark splitting greatly exceeds the fine structure. For the opposite case, see Problem 1 in Vol. IV, § 52.

[31] Atomic units are used throughout this section.

integrals are evaluated in § f of the Mathematical Appendices, integral (f.6). A short calculation then gives the first-order correction to the energy levels[32]

$$E^{(1)} = \frac{3}{2}\mathcal{E}n(n_1 - n_2) \tag{77.2}$$

or, in ordinary units,

$$E^{(1)} = \frac{3}{2}n(n_1 - n_2)|e|\mathcal{E}\frac{\hbar^2}{me^2}.$$

The two outermost components of the split level have $n_1 = n - 1$, $n_2 = 0$ and $n_1 = 0$, $n_2 = n - 1$. By (77.2), their separation is $3\mathcal{E}n(n-1)$; thus the total Stark splitting is approximately proportional to n^2. Its growth with the principal quantum number is natural: the farther the electron lies from the nucleus, the larger the atomic dipole moment.

The linear effect signifies that the unperturbed atom has a dipole moment whose mean value is

$$\overline{d}_z = -\frac{3}{2}n(n_1 - n_2). \tag{77.3}$$

This agrees with the fact that, in a state specified by parabolic quantum numbers, the atomic charge distribution is not symmetric about the plane $z = 0$ (see § 37). If $n_1 > n_2$, for example, the electron is found predominantly on the positive-z side; the atomic dipole moment therefore opposes the external field (the electron charge is negative!).

The preceding section showed that a uniform electric field cannot remove degeneracy completely: at least a twofold degeneracy remains between states whose angular-momentum projections on the field direction have opposite signs (here the projections are $\pm m$). Formula (77.2), however, shows that the linear Stark effect in hydrogen does not remove even this much degeneracy: for specified n and $n_1 - n_2$, the shift is wholly independent of m. Degeneracy is removed further by the second-order effect. Its calculation is especially relevant because the linear Stark effect vanishes altogether in states with $n_1 = n_2$.

Ordinary perturbation theory is inconvenient for calculating the quadratic effect, since it would require infinite sums of a complicated form. We shall instead use the following, slightly modified method.

The Schrödinger equation for hydrogen in a uniform electric field is

$$\frac{1}{2}\Delta\psi + \left(E + \frac{1}{r} - \mathcal{E}z\right)\psi = 0.$$

Like the equation at $\mathcal{E} = 0$, it separates in parabolic coordinates. The substitution (37.7), used in § 37, gives

$$\begin{aligned} &\frac{d}{d\xi}\left(\xi\frac{df_1}{d\xi}\right) + \left(\frac{E\xi}{2} - \frac{m^2}{4\xi} - \frac{\mathcal{E}\xi^2}{4}\right)f_1 = -\beta_1 f_1, \\ &\frac{d}{d\eta}\left(\eta\frac{df_2}{d\eta}\right) + \left(\frac{E\eta}{2} - \frac{m^2}{4\eta} + \frac{\mathcal{E}\eta^2}{4}\right)f_2 = -\beta_2 f_2, \qquad \beta_1 + \beta_2 = 1. \end{aligned} \tag{77.4}$$

[32] Schwarzschild and Epstein (K. Schwarzschild, P. Epstein, 1916) obtained this result from the old quantum theory; Pauli and Schrödinger obtained it by quantum mechanics in 1926.

In these equations regard E as a parameter fixed at some prescribed value, and β_1, β_2 as eigenvalues of the corresponding operators (which are readily verified to be self-adjoint). Solving the equations determines these quantities as functions of E and $\mathcal{E}$; the condition $\beta_1 + \beta_2 = 1$ then determines the energy as a function of the external field.

To solve (77.4) approximately, treat its field-dependent terms as a small perturbation. At zero order ($\mathcal{E} = 0$), the already known solutions are

$$f_1 = f_{n_1 m}(s\xi), \qquad f_2 = f_{n_2 m}(s\eta), \tag{77.5}$$

where the functions $f_{n_1 m}$ are those of (37.16), and the energy is replaced by the parameter

$$s = \sqrt{-2E}. \tag{77.6}$$

The corresponding values of β_1, β_2 are

$$\beta_1^{(0)} = \left(n_1 + \frac{|m|+1}{2}\right)s, \qquad \beta_2^{(0)} = \left(n_2 + \frac{|m|+1}{2}\right)s. \tag{77.7}$$

For fixed s, the functions f_1 belonging to different n_1 are mutually orthogonal; in (77.5) they are normalized by $\int_0^\infty f_1^2 d\xi = 1$, $\int_0^\infty f_2^2 d\eta = 1$.

The first-order corrections to β_1 and β_2 are the diagonal matrix elements of the perturbation:

$$\beta_1^{(1)} = \frac{\mathcal{E}}{4}\int_0^\infty \xi^2 f_1^2 d\xi, \qquad \beta_2^{(1)} = -\frac{\mathcal{E}}{4}\int_0^\infty \eta^2 f_2^2 d\eta.$$

Evaluation yields

$$\beta_1^{(1)} = \frac{\mathcal{E}}{4s^2}(6n_1^2 + 6n_1|m| + m^2 + 6n_1 + 3|m| + 2),$$

while $\beta_2^{(1)}$ follows by replacing n_1 with n_2 and reversing the sign.

In second order, the general formulas of perturbation theory give

$$\beta_1^{(2)} = -\frac{\mathcal{E}^2}{16}\sum_{n_1' \neq n_1} \frac{|(\xi^2)_{n_1 n_1'}|^2}{\beta_1^{(0)}(n_1) - \beta_1^{(0)}(n_1')}.$$

The integrals entering the matrix elements $(\xi^2)_{n_1 n_1'}$ are evaluated in § f of the Mathematical Appendices. The only nonzero elements are

$$(\xi^2)_{n_1, n_1-1} = (\xi^2)_{n_1-1, n_1} = -\frac{4}{s^2}(2n_1 + |m|)\sqrt{n_1(n_1+|m|)},$$
$$(\xi^2)_{n_1, n_1-2} = (\xi^2)_{n_1-2, n_1} = \frac{2}{s^2}\sqrt{n_1(n_1-1)(n_1+|m|)(n_1+|m|-1)}.$$

The differences in the denominators equal $\beta_1^{(0)}(n_1) - \beta_1^{(0)}(n_1') = s(n_1 - n_1')$. Calculation gives

$$\beta_1^{(2)} = -\frac{\mathcal{E}^2}{16s^5}(|m| + 2n_1 + 1)[4m^2 + 17(2|m|n_1 + 2n_1^2 + |m| + 2n_1) + 18]$$

(the expression for $\beta_2^{(2)}$ is obtained by replacing n_1 with n_2). Combining these results and substituting them in $\beta_1 + \beta_2 = 1$, we find

$$sn - \frac{\mathcal{E}^2 n}{16 s^5}[17n^2 + 5(n_1 - n_2)^2 - 9m^2 + 19] + \frac{3\mathcal{E}n}{2s^2}(n_1 - n_2) = 1.$$

Solving by successive approximations gives, through second order, the following energy $E = -s^2/2$:

$$E = -\frac{1}{2n^2} + \frac{3}{2}\mathcal{E}n(n_1 - n_2) - \frac{\mathcal{E}^2 n^4}{16}[17n^2 - 3(n_1 - n_2)^2 - 9m^2 + 19]. \tag{77.8}$$

The second term is the linear Stark effect already found, and the third is the required quadratic effect (C. Wentzel, I. Waller, P. Epstein, 1926). This third term is always negative: the quadratic effect always displaces the terms downward. Differentiating (77.8) with respect to the field gives the mean dipole moment; for $n_1 = n_2$ it is

$$\overline{d}_z = \frac{n^4}{8}(17n^2 - 9m^2 + 19)\mathcal{E}. \tag{77.9}$$

Thus the polarizability of hydrogen in its normal state ($n = 1, m = 0$) is $9/2$ (see also Problem 4 in § 76).

The absolute energy of the hydrogen terms decreases rapidly as n grows, whereas their Stark splitting increases. It is therefore useful to consider the Stark effect of highly excited levels in fields strong enough that the induced splitting is comparable with the energy of the level itself, so that perturbation theory no longer applies[33]. The semiclassical character of states with large n makes this possible.

With the substitution

$$f_1 = \chi_1/\sqrt{\xi}, \qquad f_2 = \chi_2/\sqrt{\eta} \tag{77.10}$$

equations (77.4) become

$$\begin{aligned} \frac{d^2\chi_1}{d\xi^2} + \left(\frac{E}{2} + \frac{\beta_1}{\xi} - \frac{m^2 - 1}{4\xi^2} - \frac{\mathcal{E}}{4}\xi\right)\chi_1 = 0, \\ \frac{d^2\chi_2}{d\eta^2} + \left(\frac{E}{2} + \frac{\beta_2}{\eta} - \frac{m^2 - 1}{4\eta^2} + \frac{\mathcal{E}}{4}\eta\right)\chi_2 = 0. \end{aligned} \tag{77.11}$$

Each has the form of a one-dimensional Schrödinger equation, with $E/4$ as the particle's total energy and, respectively, the potential energies

$$\begin{aligned} U_1(\xi) &= -\frac{\beta_1}{2\xi} + \frac{m^2 - 1}{8\xi^2} + \frac{\mathcal{E}}{8}\xi, \\ U_2(\eta) &= -\frac{\beta_2}{2\eta} + \frac{m^2 - 1}{8\eta^2} - \frac{\mathcal{E}}{8}\eta. \end{aligned} \tag{77.12}$$

[33]For high levels, perturbation theory requires the perturbation to be small only relative to the energy of the level itself (the electron binding energy), not relative to the spacing between levels. Indeed, in the semiclassical case represented by highly excited states, a perturbation is small if the force it produces is small compared with the forces acting in the unperturbed system; this is equivalent to the condition just stated.

Figures 25 and 26 show the approximate form of these functions (for $m > 1$). The Bohr–Sommerfeld quantization rule (48.2) gives

$$\int_{\xi_1}^{\xi_2} \sqrt{2[E/4 - U_1(\xi)]}\, d\xi = (n_1 + 1/2)\pi,$$
$$\int_{\eta_1}^{\eta_2} \sqrt{2[E/4 - U_2(\eta)]}\, d\eta = (n_2 + 1/2)\pi \tag{77.13}$$

(n_1, n_2 are integers)[34]. These equations specify implicitly how β_1, β_2 depend on E. Together with

$\beta_1 + \beta_2 = 1$, they consequently determine the energies of the levels shifted by the electric field. The integrals in (77.13) can be reduced to elliptic integrals, but the equations can be solved only numerically.

In strong fields the Stark effect is further complicated by another phenomenon: electric-field ionization of the atom (C. Lanczos, 1931). The potential energy $\mathcal{E}z$ of the electron in the external field becomes arbitrarily large and negative as $z \to -\infty$. When superposed on the electron's potential energy within the atom, it makes the region of possible motion for an electron with negative total energy E include not only the interior of the atom but also large distances from the nucleus toward the anode.

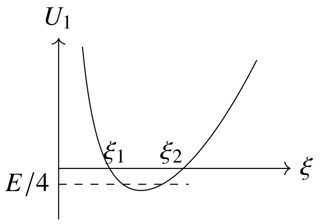

Figure 25

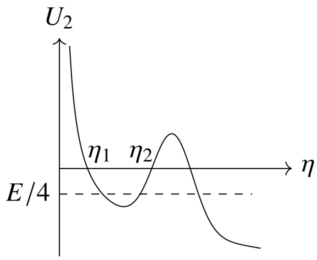

Figure 26

The two regions are separated by a potential barrier whose width decreases as the field increases. Quantum mechanics, however, always gives a nonzero probability for passage through a potential barrier. Here an electron's escape from the atomic region through the barrier is precisely ionization. In weak fields this probability is vanishingly small, but it rises exponentially with the field and becomes appreciable when the field is sufficiently strong[35].

[34] A detailed analysis shows that greater accuracy results if m^2 replaces $m^2 - 1$ in U_1, U_2. The integers n_1, n_2 then coincide with the parabolic quantum numbers.

[35] This phenomenon illustrates how a small perturbation can change the nature of the energy spectrum. Even a weak field $\mathcal{E}$ creates a potential barrier and makes the remote region in principle accessible to the electron. Strictly speaking, its motion thereby becomes infinite and the energy spectrum changes from discrete to continuous. Nevertheless, the formal perturbative solution retains a physical meaning: it specifies the energy levels of states that are, if not fully stationary, at least "nearly stationary." An atom initially prepared in such a state remains there for a long interval. At the same time, the entire perturbation series for the Stark splitting cannot converge in the strict sense; it is only asymptotic. Beyond some point in the series—a point increasingly remote for smaller perturbations—successive terms grow instead of decreasing.

Problem

1. Find the ionization probability per unit time for a hydrogen atom in its ground state in an electric field satisfying $\mathcal{E} \ll 1$ ($\mathcal{E} \ll m^2|e|^5/\hbar^4$ in ordinary units).

SOLUTION. [36] In parabolic coordinates the potential barrier lies "along the coordinate η" (Fig. 26); extraction of the electron toward $z \to -\infty$ corresponds to its passage into the large-η region. To determine the ionization probability, we must examine the wave function at large η and small ξ. (As will appear below, small values of ξ contribute to the integral giving the total probability current of the escaping electron.) Without the field, the ground-state wave function is

$$\psi = \frac{1}{\sqrt{\pi}} \exp\left(-\frac{\xi + \eta}{2}\right). \tag{1}$$

In the region of interest, the field may be taken not to alter the ξ dependence of ψ from (1). Its η dependence obeys

$$\frac{\partial^2 \chi}{\partial \eta^2} + \left(-\frac{1}{4} + \frac{1}{2\eta} + \frac{1}{4\eta^2} + \frac{\mathcal{E}\eta}{4}\right)\chi = 0, \tag{2}$$

where $\chi = \sqrt{\eta}\psi$. Choose a value η_0 inside the barrier such that $1 \ll \eta_0 \ll 1/\mathcal{E}$. For $\eta \gtrsim \eta_0$ the wave function is semiclassical. Matching it to (1) at $\eta = \eta_0$ gives, outside the barrier,

$$\chi = \left(\frac{\eta_0|p_0|}{\pi p}\right)^{1/2} \exp\left(-\frac{\xi + \eta_0}{2} + i\int_{\eta_0}^{\eta} p, d\eta + \frac{i\pi}{4}\right),$$

where

$$p(\eta) = \sqrt{-\frac{1}{4} + \frac{1}{2\eta} + \frac{1}{4\eta^2} + \frac{\mathcal{E}\eta}{4}}.$$

If η_1 denotes the root of $p(\eta) = 0$, then

$$|\chi|^2 = \frac{\eta_0|p_0|}{\pi p} \exp\left(-\xi - 2\int_{\eta_0}^{\eta_1} |p| d\eta - \eta_0\right). \tag{3}$$

For $\eta \gg 1$, put $|p_0| \approx 1/2$ and $p \approx \frac{1}{2}\sqrt{\mathcal{E}\eta - 1}$ in the prefactor; in the exponent, the next term in the expansion of $p(\eta)$ must also be retained. Integration then gives

$$|\chi|^2 = \frac{4}{\pi\mathcal{E}\sqrt{\mathcal{E}\eta - 1}} \exp\left(-\xi - \frac{2}{3\mathcal{E}}\right). \tag{4}$$

The total probability current across a plane perpendicular to the z axis, which is the required ionization probability w, is

$$w = \int_0^{\infty} |\psi|^2 v_z \, 2\pi\rho, d\rho.$$

[36] Atomic units are used in this problem.

At large η and small ξ we may put $d\rho = d\sqrt{\xi\eta} \approx \frac{1}{2}\sqrt{\eta/\xi}, d\xi$, while the electron velocity is $v_z \approx \sqrt{\mathcal{E}\eta - 1}$. Hence

$$w = \int_0^\infty |\chi|^2 \pi\sqrt{\mathcal{E}\eta - 1}, d\xi,$$

and finally

$$w = \frac{4}{\mathcal{E}} \exp\left(-\frac{2}{3\mathcal{E}}\right) \tag{5}$$

or, in ordinary units,

$$w = \frac{4m^3|e|^9}{\mathcal{E}\hbar^7} \exp\left(-\frac{2m^2|e|^5}{3\mathcal{E}\hbar^4}\right).$$

Problem

2. Find the probability that an electric field extracts an electron from a short-range-force potential well in which it occupies a bound s state. The field is weak in the sense $|e|\mathcal{E} \ll \hbar^2\varkappa^3/m$, where $\varkappa = \sqrt{2m|E|}/\hbar$, $|E|$ is the electron binding energy in the well, and m is its mass (Yu. N. Demkov, G. F. Drukarev, 1964).

Solution. As in Problem 1, large distances from the center ($\varkappa r \gg 1$) are relevant for a weak electric field. There the bound-state wave function of the electron in the well, with $\mathcal{E} = 0$, has the asymptotic form

$$\psi = \frac{A\sqrt{\varkappa}}{r} e^{-\varkappa r},$$

where A is a dimensionless constant depending on the particular form of the well[37]. In parabolic coordinates $r = (\xi + \eta)/2$, and for $\eta \gg \xi$ the wave function becomes

$$\psi \approx \frac{2A\sqrt{\varkappa}}{\eta} \exp\left[-\frac{\varkappa}{2}(\xi + \eta)\right]. \tag{6}$$

In the rest of this problem, mass, length, and time are measured in units of m, $1/\varkappa$, and $m/\hbar\varkappa^2$, respectively.

Function (6) factors into functions of ξ and η. In the electric field its dependence on ξ may be taken to remain that of (6). To determine its dependence on η, use the Schrödinger equation in parabolic coordinates. Separation again gives (77.11), now with $E = -1/2$, $m = 0$, and separation parameters satisfying $\beta_1 + \beta_2 = 0$. We must take $\beta_1 = 1/2$, and hence $\beta_2 = -1/2$; the η dependence of ψ therefore obeys

$$\frac{\partial^2\chi}{\partial\eta^2} + \left(-\frac{1}{4} - \frac{1}{2\eta} + \frac{1}{4\eta^2} + \frac{\mathcal{E}\eta}{4}\right)\chi = 0, \qquad \chi = \psi\sqrt{\eta}.$$

Solving this as equation (2) was solved, we obtain in place of (3)

$$|\chi|^2 = \frac{4A^2|p_0|}{\eta_0 p} \exp\left(-\xi - 2\int_{\eta_0}^{\eta_1} |p| d\eta - \eta_0\right),$$

[37] For example, if the well radius a is so small that $a\varkappa \ll 1$, then $A = (2\pi)^{-1/2}$; see § 133 for details.

with

$$p(\eta) = \sqrt{-\frac{1}{4} - \frac{1}{2\eta} + \frac{1}{4\eta^2} + \frac{\mathcal{E}\eta}{4}}.$$

Next, in place of (4), we find

$$|\chi|^2 = \frac{A^2\mathcal{E}}{\sqrt{\mathcal{E}\eta - 1}} \exp\left(-\xi - \frac{2}{3\mathcal{E}}\right),$$

and finally, instead of (5),

$$w = \pi A^2 \mathcal{E} \exp\left(-\frac{2}{3\mathcal{E}}\right)$$

or, in ordinary units,

$$w = \frac{\pi |e| \mathcal{E} A^2}{\hbar \varkappa} \exp\left(-\frac{2\hbar^2\varkappa^3}{3m|e|\mathcal{E}}\right).$$

Problem

3. To exponential accuracy, estimate the probability of extracting an electron from a potential well by a uniform alternating electric field $\mathcal{E} = \mathcal{E}_0 \cos \omega t$. The frequency and amplitude satisfy

$$\hbar\omega \ll |E|, \qquad |e|\mathcal{E}_0 \ll \hbar^2\varkappa^3/m,$$

where $\varkappa = \sqrt{2m|E|}/\hbar$ and $|E|$ is the binding energy in the well (L. V. Keldysh, 1964)[38].

Solution. Under these conditions the extraction probability w is exponentially small. To calculate only its exponential factor, without the prefactor, it suffices to regard the motion as one-dimensional along the field direction, chosen as the z axis.

It is convenient here to describe the electric field by a vector rather than a scalar potential: $A_z = A = -(c\mathcal{E}_0/\omega) \sin \omega t$. Outside the well the electron Hamiltonian is then

$$\widehat{H} = \frac{1}{2m}\left(-i\hbar\frac{\partial}{\partial z} + \frac{|e|\mathcal{E}_0}{\omega} \sin \omega t\right)^2$$

(see (111.3) below), and it contains no z. Introduce the dimensionless variables and parameters

$$\tau = \frac{\hbar\varkappa^2}{2m} t, \qquad \eta = 2\varkappa z, \qquad \Omega = \frac{2m\omega}{\hbar\varkappa^2} = \frac{\hbar\omega}{|E|}, \qquad F = \frac{|e|m\mathcal{E}_0}{\hbar^2\varkappa^3}.$$

The Schrödinger equation becomes

$$\frac{i}{4}\frac{\partial\Psi}{\partial\tau} = -\left(\frac{\partial}{\partial\eta} + \frac{iF}{\Omega} \sin \Omega\tau\right)^2 \Psi.$$

[38] One example is ionization of a singly charged negative ion by a strong light wave; here the potential well is produced by the interaction of the electron with a neutral atomic core. The condition $\hbar\omega \ll |E|$ permits the electromagnetic-wave field to be treated classically.

Its boundary condition requires the solution $\Psi(\eta, \tau)$, as $\eta \to 0$, to match the field-free wave function of the electron in the well, whose energy is $E = -|E|$:

$$\Psi \to e^{i\tau} \quad \text{as } \eta \to 0. \tag{7}$$

Since the problem is semiclassical, seek a solution to exponential accuracy as $\Psi = \exp iS$, where $S(\eta, \tau)$ is the classical action. The Hamiltonian is independent of η, so the generalized momentum $p_\eta = p$ is conserved along a classical trajectory. Thus

$$S = -\int_{\tau_0}^{\tau} H(p, \tau')d\tau' + \eta p + A, \qquad H(p, \tau) = 4\left(p + \frac{F}{\Omega}\sin\Omega\tau\right)^2, \tag{8}$$

where A, τ_0 are constants. Here p must mean the value whose trajectory reaches the specified point η at time τ; that is, p is a function of η, τ defined by the equation of motion

$$\eta = \int_{\tau_0}^{\tau} \frac{\partial H(p, \tau')}{\partial p} d\tau'. \tag{9}$$

To obtain a solution satisfying (7), regard A as a function of τ_0, and τ_0 as a function of coordinate and time fixed by

$$\frac{\partial S}{\partial \tau_0} = 0. \tag{10}$$

Plainly $A(\tau_0) = \tau_0$; equation (10) then becomes

$$H(p, \tau_0) + 1 = 0. \tag{11}$$

Equations (9) and (11) jointly determine $\tau_0(\eta, \tau)$ and $p(\eta, \tau)$, and therefore, after substitution in (8), the wave function $\Psi(\eta, \tau)$.

The required probability w is proportional to the current density along z. In the classically accessible region this density is $v_z|\Psi|^2$. That region begins at the point where $\operatorname{Im} S$ ceases to increase. There $(\partial \operatorname{Im} S/\partial\eta)_\tau = 0$; since $\partial S/\partial\eta = p$, we have $\operatorname{Im} p = 0$. Equations (9) and (11) then show that $\operatorname{Re} p = 0$ there as well. This condition fixes τ_0: setting $p = 0$ in (11) gives

$$\frac{4F^2}{\Omega^2}\sin^2\Omega\tau_0 = -1,$$

whence

$$\Omega\tau_0 = i \operatorname{Arsh}\gamma, \qquad \gamma = \frac{\Omega}{2F} = \frac{\sqrt{2m|E|}}{|e|\mathcal{E}_0}\omega.$$

(The imaginary "time" τ_0 expresses the classical impossibility of the process.) Finally,

$$w \sim \exp\left\{-2\operatorname{Im}\left[\int_{\tau}^{\tau_0} \frac{4F^2}{\Omega^2}\sin^2\Omega\tau'\, d\tau' + \tau_0\right]\right\}.$$

Evaluating the integral gives

$$w \sim \exp\left\{-\frac{2|E|}{\hbar\omega}f(\gamma)\right\}, \qquad f(\gamma) = \left(1 + \frac{1}{2\gamma^2}\right)\operatorname{Arsh}\gamma - \frac{\sqrt{1+\gamma^2}}{2\gamma}. \tag{12}$$

The limiting forms of $f(\gamma)$ are

$$f(\gamma) \approx \frac{2\gamma}{3} \quad \text{for } \gamma \ll 1, \qquad f(\gamma) \approx \ln 2\gamma - \frac{1}{2} \quad \text{for } \gamma \gg 1.$$

As $\gamma \to 0$, the limiting expression for w is the probability of extracting the particle from the well by a constant field.

Formula (12) applies when the exponent is large. In any event, this requires $\hbar\omega \ll |E|$.

CHAPTER XI

The Diatomic Molecule

§ 78. Electronic terms of a diatomic molecule

The great disparity between atomic nuclear masses and the electron mass is fundamental to molecular theory. Owing to this mass difference, nuclei in a molecule move slowly in comparison with the electrons. We may therefore study the electronic motion with the nuclei held fixed at prescribed separations. The energy levels U_n found for this system are called the molecule's *electronic terms*. For atoms, the energy levels were definite numbers; here, by contrast, the electronic terms are functions of parameters—the separations of the nuclei in the molecule. The quantity U_n also includes the electrostatic interaction energy of the nuclei. Thus U_n is, in substance, the total energy of the molecule for a specified configuration of fixed nuclei.

We begin the study of molecules with the simplest type, the *diatomic* molecule, for which the fullest theoretical investigation is possible. Its electronic terms depend on just one parameter: the internuclear distance r.

One of the principal ways of classifying atomic terms used the values of the total orbital angular momentum L. For molecules, however, the total orbital angular momentum of the electrons is not conserved in general, since the electric field of several nuclei has no central symmetry.

In a diatomic molecule the field does, nevertheless, possess axial symmetry about the line through both nuclei. The projection of the orbital angular momentum on this axis is consequently conserved, and molecular electronic terms can be classified by its values. The magnitude of this projection on the molecular axis is conventionally denoted by Λ; its possible values are $0, 1, 2, \ldots$ Terms with different Λ are designated by the capital Greek letters corresponding to the Latin symbols for atomic

terms having different L. Thus the cases $\Lambda = 0, 1, 2$ are called, respectively, Σ-, Π-, and Δ-terms; larger values of Λ ordinarily need not be considered.

Every electronic state of a molecule is further characterized by the total spin S of all its electrons. When S is nonzero, the different orientations of the total spin give a degeneracy of multiplicity $2S + 1$[1]. As for atoms, the number $2S + 1$ is called the term's *multiplicity* and is placed to the upper left of the term symbol. For example, $^3\Pi$ denotes a term with $\Lambda = 1$, $S = 1$.

Besides rotations through an arbitrary angle about the axis, the symmetry of the molecule permits reflection in any plane containing that axis. Such a reflection leaves the molecular energy unchanged, although the resulting state need not be wholly identical to the initial one. Under reflection in a plane through the molecular axis, the sign of the angular momentum about that axis changes (angular momentum is an axial vector). It follows that every electronic term with nonzero Λ is doubly degenerate: each

[1]We disregard here the fine structure arising from relativistic interactions (see below, § 83, 84).

energy belongs to two states that differ in the direction of the orbital-angular-momentum projection on the molecular axis. When $\Lambda = 0$, on the other hand, reflection does not alter the molecular state, so Σ-terms are nondegenerate. Reflection can only multiply the wave function of a Σ-term by a constant. Since two reflections in the same plane amount to the identity transformation, this constant is ± 1. We must therefore distinguish the Σ-terms whose wave function remains entirely unchanged under reflection from those whose wave function reverses sign. The former are denoted by Σ^+ and the latter by Σ^-.

For a molecule composed of two identical atoms, a further symmetry arises and supplies an additional characteristic of its electronic terms. A diatomic molecule with identical nuclei also has a center of symmetry at the midpoint of the line joining them (we choose this point as the coordinate origin)[2]. The Hamiltonian is therefore invariant under simultaneous reversal of the coordinates of all the molecular electrons, with the nuclear co-

ordinates left unchanged. The operator of this transformation[3] also commutes with the orbital-angular-momentum operator. Hence terms of definite Λ can be classified additionally by parity: the wave function of an *even* (g) state is unchanged when the electron coordinates reverse sign, whereas that of an *odd* (u) state changes sign. The parity indices u, g are conventionally written as subscripts on the term symbol: Π_u, Π_g, and so forth.

Empirically, the normal electronic state of the overwhelming majority of chemically stable diatomic molecules is known to be totally symmetric: the electronic wave function is invariant under every symmetry transformation of the molecule. In the great majority of cases, the total spin S also vanishes in the normal state. In other words, the molecular ground term is $^1\Sigma^+$, or $^1\Sigma_g^+$ when the molecule consists of identical atoms. Familiar exceptions to these rules are the molecules O_2 (normal term $^3\Sigma_g^-$) and NO (normal term $^2\Pi$).

Problem

Separate the variables in the Schrödinger equation for the electronic terms of the H_2^+ ion, using elliptic coordinates.

Solution. The Schrödinger equation for an electron in the field of two fixed protons is

$$\frac{\Delta\psi}{2} + \left(E + \frac{1}{r_1} + \frac{1}{r_2}\right)\psi = 0$$

(atomic units are used). Elliptic coordinates are defined by

$$\xi = \frac{r_1 + r_2}{R}, \qquad \eta = \frac{r_2 - r_1}{R}, \qquad 1 \leqslant \xi < \infty, \quad -1 \leqslant \eta \leqslant 1,$$

and the third coordinate is the angle φ of rotation about the axis through the two nuclei,

[2] It also has a symmetry plane perpendicular to the molecular axis and bisecting it. There is no need to treat this symmetry element separately, since the existence of such a plane follows automatically from the presence of a center and an axis of symmetry.

[3] This must not be confused with inversion of the coordinates of every particle in the system (compare § 86)!

whose separation is R (see I, § 48). In these coordinates the Laplacian is

$$\Delta = \frac{4}{R^2(\xi^2-\eta^2)}\left[\frac{\partial}{\partial\xi}\left[(\xi^2-1)\frac{\partial}{\partial\xi}\right]+\frac{\partial}{\partial\eta}\left[(1-\eta^2)\frac{\partial}{\partial\eta}\right]\right]+\frac{1}{R^2(\xi^2-1)(1-\eta^2)}\frac{\partial^2}{\partial\varphi^2}.$$

Setting

$$\psi = X(\xi)Y(\eta)e^{i\Lambda\varphi},$$

we obtain the following equations for X and Y:

$$\frac{d}{d\xi}\left[(\xi^2-1)\frac{dX}{d\xi}\right]+\left(\frac{ER^2\xi^2}{2}+2R\xi+A-\frac{\Lambda^2}{\xi^2-1}\right)X=0,$$

$$\frac{d}{d\eta}\left[(1-\eta^2)\frac{dY}{d\eta}\right]+\left(-\frac{ER^2\eta^2}{2}-A-\frac{\Lambda^2}{1-\eta^2}\right)Y=0,$$

where A is the separation parameter. Every electronic term $E(R)$ is characterized by three quantum numbers: Λ and two "elliptic quantum numbers" n_ξ, n_η, which specify the numbers of zeros of the functions $X(\xi)$ and $Y(\eta)$.

§ 79. Crossing of electronic terms

The electronic terms of a diatomic molecule, as functions of the internuclear distance r, can be displayed graphically by plotting the energy against r. The question whether curves representing different terms can cross is of considerable interest.

Let $U_1(r)$, $U_2(r)$ be two different electronic terms. If they cross at some point, then near that point the functions U_1, U_2 have nearly equal values. It is useful to formulate the question of whether such a crossing is possible as follows.

Consider a point r_0 at which $U_1(r)$, $U_2(r)$ have values that are very close but not equal (denote them by E_1 and E_2), and ask whether U_1 and U_2 can be made equal by shifting the point through a small amount δr. The energies E_1 and E_2 are eigenvalues of the Hamiltonian $\widehat{H}_0$ for the system of electrons in the field of nuclei separated by the distance r_0. If r is increased by δr, the Hamiltonian becomes $\widehat{H}_0+\widehat{V}$, where $\widehat{V}=\delta r\,\partial\widehat{H}_0/\partial r$ is a small correction; the values of U_1, U_2 at $r_0+\delta r$ may be regarded as eigenvalues of the new Hamiltonian. This viewpoint permits the values of the terms $U_1(r)$, $U_2(r)$ at $r_0+\delta r$ to be found by perturbation theory, with $\widehat{V}$ treated as a perturbation of $\widehat{H}_0$.

The usual perturbation-theory method is not applicable here, however, because the unperturbed energy eigenvalues E_1, E_2 are very close, and their difference is in general not large compared with the perturbation (condition (38.9) is not satisfied). Since the limit $E_2-E_1=0$ is the case of degenerate eigenvalues, it is natural to apply to nearby eigenvalues a method analogous to that developed in § 39.

Let ψ_1, ψ_2 be eigenfunctions of the unperturbed operator $\widehat{H}_0$ corresponding to the energies E_1, E_2. As the initial zeroth approximation, take linear combinations of ψ_1 and ψ_2, rather than the functions themselves, in the form

$$\psi = c_1\psi_1+c_2\psi_2. \tag{79.1}$$

Substitution into the perturbed equation

$$(\widehat{H}_0+\widehat{V})\psi = E\psi, \tag{79.2}$$

gives

$$c_1(E_1 + \widehat{V} - E)\psi_1 + c_2(E_2 + \widehat{V} - E)\psi_2 = 0.$$

Multiplying this equation successively from the left by ψ_1^* and ψ_2^* and integrating, we obtain the two algebraic equations

$$\left.\begin{aligned} c_1(E_1 + V_{11} - E) + c_2 V_{12} = 0, \\ c_1 V_{21} + c_2(E_2 + V_{22} - E) = 0. \end{aligned}\right\} \tag{79.3}$$

Since $\widehat{V}$ is Hermitian, the matrix elements V_{11} and V_{22} are real, while $V_{12} = V_{21}^*$. The compatibility condition for these equations is

$$\begin{vmatrix} E_1 + V_{11} - E & V_{12} \\ V_{21} & E_2 + V_{22} - E \end{vmatrix} = 0,$$

and hence

$$E = \frac{1}{2}(E_1 + E_2 + V_{11} + V_{22}) \pm \sqrt{\frac{1}{4}(E_1 - E_2 + V_{11} - V_{22})^2 + |V_{12}|^2}. \tag{79.4}$$

This formula determines the required first-approximation energy eigenvalues.

If the energies of the two terms at $r_0 + \delta r$ become equal (the terms cross), the two values of E given by (79.4) must coincide. The expression under the square root in (79.4) must therefore vanish. Since it is a sum of two squares, the conditions for a term-crossing point are the equations

$$E_1 - E_2 + V_{11} - V_{22} = 0, \qquad V_{12} = 0. \tag{79.5}$$

But there is only one adjustable parameter available to specify the perturbation $\widehat{V}$, namely the displacement δr. Thus the two equations (79.5) cannot in general be satisfied simultaneously (we suppose that ψ_1, ψ_2 have been chosen real, so that V_{12} is real as well).

It may happen, however, that the matrix element V_{12} vanishes identically; only one equation in (79.5) then remains, and it can be satisfied by an appropriate choice of δr. This occurs whenever the two terms under consideration have different symmetries. Here symmetry includes every possible kind: symmetry under rotations about the axis, reflections in planes, inversion, and permutations of the electrons. For a diatomic molecule, the terms may thus differ in Λ, parity, or multiplicity; Σ terms may also differ by being Σ^+ or Σ^-.

This statement holds because the perturbation operator (like the Hamiltonian itself) commutes with every molecular symmetry operator: the angular-momentum operator about the axis, the reflection and inversion operators, and the electron-permutation operators. The operator represents a scalar quantity and commutes with the angular-momentum and inversion operators. Consequently, its matrix elements can be nonzero only between states with the same angular momentum and parity. This result was established in §§ 29–30. The same proof, in essentially unchanged form, applies to the general case of an arbitrary symmetry operator. We shall not repeat it here, particularly since § 97 will also give a different general proof based on group theory.

We consequently reach the result that only terms of different symmetry can cross in a diatomic molecule; terms of the same symmetry cannot cross (E. Wigner, J. von Neumann, 1929). If some approximate calculation were to give two crossing terms of the same symmetry, the next approximation would separate them, as shown by the solid lines in Fig. 27.

We emphasize that this result is not confined to a diatomic molecule. It is in fact a general theorem of quantum mechanics, valid whenever the Hamiltonian contains a parameter and its eigenvalues are therefore functions of that parameter.

In the language of group theory (see § 96), the general condition that determines whether terms can cross is that they must belong to different irreducible representations of the symmetry group of the system's Hamiltonian[4].

In a polyatomic molecule, the electronic terms depend not on one parameter but on several: the distances between the different nuclei.

Let s be the number of independent internuclear distances; in an N-atom molecule ($N > 2$) with an arbitrary nuclear arrangement, this number is $s = 3N - 6$. Geometrically, each term $U_n(r_1, \ldots, r_s)$ is a surface in a space of $s + 1$ dimensions, and intersections of these surfaces may occur along manifolds of various dimensions, from 0 (intersection at a point) to $s - 1$.

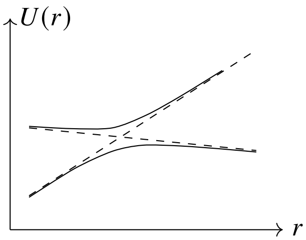

Fig. 27

The foregoing argument remains valid in full, except that the perturbation $\widehat{V}$ is now specified not by one parameter but by s parameters, the displacements $\delta r_1, \ldots, \delta r_s$. With as few as two parameters, however, the two equations (79.5) can in general be satisfied. We thus conclude that in polyatomic molecules any two terms can cross. If the terms have the same symmetry, their crossing is fixed by the two conditions (79.5), and hence the dimension of the manifold along which the crossing occurs is $s - 2$. If the terms have different symmetries, only one condition remains, and the crossing takes place along a manifold of $s - 1$ dimensions.

Thus, when $s = 2$, the terms are represented by surfaces in a three-dimensional

[4]The electronic terms of the H_2^+ ion appear to be an exception to this rule. These terms are specified by the angular-momentum projection Λ and two elliptic quantum numbers n_ξ, n_η (see the problem in § 78). Since all these numbers are associated with functions of different variables, there is in general nothing to prevent crossings between terms $E(R)$ that have the same Λ but different pairs n_ξ, n_η, even though their symmetry under rotations and reflections is the same. In fact, however, the separability of variables in the Schrödinger equation for this system means that its Hamiltonian possesses a higher symmetry than its geometrical properties alone imply; relative to this full symmetry group, states with different values of n_ξ, n_η belong to different types.

coordinate system. For terms of different symmetry these surfaces intersect along lines ($s - 1 = 1$), whereas for terms of the same symmetry they intersect at points ($s - 2 = 0$). The form of the surfaces near an intersection point in the latter case is readily found. The energy values near the points where the terms cross are given by (79.4). In this expression the matrix elements V_{11}, V_{22}, V_{12} are linear functions of the displacements δr_1, δr_2, and therefore also linear

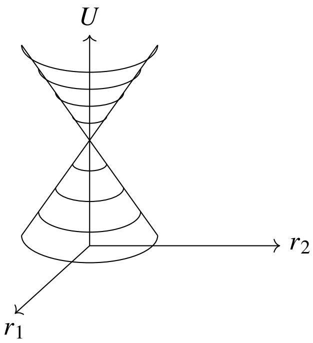

Fig. 28

functions of the distances r_1, r_2 themselves. As is known from analytic geometry, an equation of this form describes an elliptic cone. Thus near a crossing point the terms are represented by the surface of a two-napped elliptic cone of arbitrary orientation (Fig. 28).

§ 80. Relation between molecular and atomic terms

As the distance between the nuclei of a diatomic molecule is increased, the limiting result is a pair of isolated atoms (or ions). This raises the question of how an electronic term of the molecule corresponds to the states of the atoms obtained when they are separated (E. Wigner, E. Witmer, 1928). The correspondence is not unique: bringing together two atoms in specified states may produce a molecule in any of several electronic states.

First suppose that the molecule contains two unlike atoms. Let the isolated atoms occupy states with orbital angular momenta L_1, L_2 and spins S_1, S_2, where $L_1 \geqslant L_2$. Their projections on the line joining the nuclei take the values $M_1 = -L_1, -L_1+1, \ldots, L_1$ and $M_2 = -L_2, -L_2+1, \ldots, L_2$. As the atoms approach, the resulting angular momentum Λ is fixed by $\Lambda = |M_1 + M_2|$. Combining every possible pair of M_1 and M_2, we obtain the following frequencies for the distinct values of Λ:

$$\begin{array}{cl}
\Lambda = L_1 + L_2 & 2 \text{ times}, \\
L_1 + L_2 - 1 & 4 \text{ times}, \\
\dots\dots\dots & \dots\dots\dots\dots\dots \\
L_1 - L_2 & 2(2L_2+1) \text{ times}, \\
L_1 - L_2 - 1 & 2(2L_2+1) \text{ times}, \\
\dots\dots\dots & \dots\dots\dots\dots\dots \\
1 & 2(2L_2+1) \text{ times}, \\
0 & 2L_2+1 \text{ times}.
\end{array}$$

Terms with $\Lambda \neq 0$ are doubly degenerate, whereas those with $\Lambda = 0$ are nondegenerate. It follows that the possible results are

$$\begin{array}{rll}
1 \text{ term} & \text{with} & \Lambda = L_1 + L_2, \\
2 \text{ terms} & \text{with} & \Lambda = L_1 + L_2 - 1, \\
\dots\dots\dots & & \dots\dots\dots\dots \\
2L_2+1 \text{ terms} & \text{with} & \Lambda = L_1 - L_2, \\
2L_2+1 \text{ terms} & \text{with} & \Lambda = L_1 - L_2 - 1, \\
\dots\dots\dots & & \dots\dots\dots\dots \\
2L_2+1 \text{ terms} & \text{with} & \Lambda = 0;
\end{array} \tag{80.1}$$

altogether there are $(2L_2+1)(L_1+1)$ terms, whose values of Λ range from 0 to $L_1 + L_2$.

The spins S_1, S_2 of the two atoms combine, by the general rule for addition of angular momenta, into the total molecular spin and give the following

possible values of S:

$$S = S_1 + S_2, \quad S_1 + S_2 - 1, \dots, |S_1 - S_2|. \tag{80.2}$$

Combining each of these values with all the values of Λ in (80.1) gives the complete list of possible terms of the molecule formed.

For the Σ terms, their sign remains to be determined. This can readily be done by observing that, as $r \to \infty$, the molecular wavefunctions can be written as products (or sums of products) of the wavefunctions of the two atoms. The value $\Lambda = 0$ can arise either by adding nonzero projections $M_1 = -M_2$, or when $M_1 = M_2 = 0$. Denote the wavefunctions of the first and second atoms by $\psi^{(1)}_{M_1}$, $\psi^{(2)}_{M_2}$. For $M = |M_1| = |M_2| \neq 0$, form the symmetrized and antisymmetrized products

$$\psi^+ = \psi^{(1)}_M \psi^{(2)}_{-M} + \psi^{(1)}_{-M} \psi^{(2)}_M, \qquad \psi^- = \psi^{(1)}_M \psi^{(2)}_{-M} - \psi^{(1)}_{-M} \psi^{(2)}_M.$$

Reflection in a vertical plane (one containing the molecular axis) reverses the sign of the projection of angular momentum on the axis, so that $\psi^{(1)}_M$, $\psi^{(2)}_M$ pass respectively into $\psi^{(1)}_{-M}$, $\psi^{(2)}_{-M}$, and conversely. Under this reflection, ψ^+ remains unchanged, whereas ψ^- changes sign; hence the former corresponds to a Σ^+ term and the latter to a Σ^- term. Thus every value of M gives one Σ^+ term and one Σ^- term. Since M can take L_2 distinct values ($M = 1, \dots, L_2$), there are altogether L_2 terms of each type, Σ^+ and Σ^-.

If instead $M_1 = M_2 = 0$, the molecular wavefunction has the form $\psi = \psi^{(1)}_0 \psi^{(2)}_0$. To determine how $\psi^{(1)}_0$ behaves under reflection in a vertical plane, choose coordinates centered on the first atom, with the z axis along the molecular axis. Reflection in the vertical plane xz is equivalent to inversion through the coordinate origin followed by a rotation through 180° about the y axis. Inversion multiplies $\psi_0^{(1)}$ by P_1, where $P_1 = \pm 1$ is the parity of the given state of the first atom. The action on a wavefunction of an infinitesimal rotation, and hence of any finite rotation, is completely determined by the atom's total orbital angular momentum. It is therefore enough to consider the special case of a one-electron atom whose electron has orbital angular momentum l (and angular-momentum component $m = 0$ along z); replacing l by L in the result gives the desired answer for an arbitrary atom. Apart from a constant factor, the angular part of the electron wavefunction for $m = 0$ is $P_l(\cos\theta)$ (see (28.8)). A rotation through 180° about the y axis gives $x \to -x$, $y \to y$, $z \to -z$, or, in spherical coordinates, $r \to r$, $\theta \to \pi - \theta$, $\varphi \to \pi - \varphi$. Consequently $\cos\theta \to -\cos\theta$, and $P_l(\cos\theta)$ is multiplied by $(-1)^l$.

Thus reflection in a vertical plane multiplies $\psi_0^{(1)}$ by $(-1)^{L_1} P_1$. Similarly, $\psi_0^{(2)}$ is multiplied by $(-1)^{L_2} P_2$, so the net factor multiplying $\psi = \psi_0^{(1)}\psi_0^{(2)}$ is $(-1)^{L_1+L_2} P_1 P_2$. The term is Σ^+ or Σ^- according as this factor equals $+1$ or -1.

Collecting these results, of the total $(2L_2 + 1)$ Σ terms (for each possible multiplicity), $L_2 + 1$ are Σ^+ terms and L_2 are Σ^- terms if $(-1)^{L_1+L_2} P_1 P_2 = +1$; the numbers are interchanged if $(-1)^{L_1+L_2} P_1 P_2 = -1$.

We turn now to a molecule composed of identical atoms. The rules for combining the atomic spins and orbital angular momenta into the molecular values S and Λ remain the same as for a molecule made from different atoms. The new feature is that the terms may be even or odd. Two cases must be distinguished: the atoms may be in the same state or in different states.

If the atoms are in different states,[5] the total number of possible terms is doubled relative to the number for two different atoms. Indeed, inversion through the origin (placed at the midpoint of the molecular axis) interchanges the states of the two atoms. Symmetrizing or antisymmetrizing the molecular wavefunction with respect to this interchange gives two terms with the same Λ and S, one even and the other odd. Thus the total numbers of even and odd terms are equal.

If, however, the two atoms are in identical states, the total number of states remains the same as for a molecule with dissimilar atoms. With regard to the parity of these states, an analysis (not reproduced here because it is cumbersome)[6] gives the following results.

Let N_g, N_u denote the numbers of even and odd terms having specified values of Λ and S. Then

$$\begin{array}{ll} \text{if } \Lambda \text{ is odd,} & N_g = N_u, \\ \text{if } \Lambda \text{ is even and } S \text{ is even } (S = 0, 2, 4, \ldots), & N_g = N_u + 1, \\ \text{if } \Lambda \text{ is even and } S \text{ is odd } (S = 1, 3, \ldots), & N_u = N_g + 1. \end{array}$$

[5]This includes, in particular, the combination of a neutral atom with an ionized one.

[6]See E. Wigner, E. Witmer // Zs. Physik. 1928. V. 51 S. 850.

Finally, the Σ terms must further be divided into Σ^+ and Σ^-. The rule here is

$$\begin{aligned} &\text{if } S \text{ is even,} \quad N_g^+ = N_u^- + 1 = L + 1, \\ &\text{if } S \text{ is odd,} \quad N_u^+ = N_g^- + 1 = L + 1 \end{aligned}$$

(where $L_1 = L_2 \equiv L$). Every Σ^+ term has parity $(-1)^S$, and every Σ^- term has parity $(-1)^{S+1}$.

Besides the relation discussed above between molecular terms and the terms of the atoms obtained as $r \to \infty$, one may also ask how the molecular terms are related to the terms of the "united atom" that would result as $r \to 0$, when the two nuclei are brought to one point (for example, the relation between the terms of the H_2 molecule and the He atom). The following rules are obtained immediately. A term of the "united" atom with spin S, orbital angular momentum L, and parity P gives, when its constituent atoms are separated, molecular terms with spin S and axial angular-momentum projection $\Lambda = 0, 1, \ldots, L$, one term for each of these values of Λ. The parity of a molecular term is the same as the parity P of the atomic term (g for $P = +1$ and u for $P = -1$). The molecular term with $\Lambda = 0$ is a Σ^+ term if $(-1)^L P = +1$, and a Σ^- term if $(-1)^L P = -1$.

Problem

1. Determine the possible terms of the molecules H_2, N_2, O_2, and Cl_2 that can result when atoms in their ground states combine.

Solution. The rules stated in the text give the following possible terms:

$$\begin{array}{lll} \text{molecule } H_2 & (\text{atoms in } {}^2S \text{ states}): & {}^1\Sigma_g^+,\ {}^3\Sigma_u^+; \\ N_2 & (\text{atoms in } {}^4S \text{ states}): & {}^1\Sigma_g^+,\ {}^3\Sigma_u^+,\ {}^5\Sigma_g^+,\ {}^7\Sigma_u^+; \\ Cl_2 & (\text{atoms in } {}^2P \text{ states}): & 2{}^1\Sigma_g^+,\ {}^1\Sigma_u^-,\ {}^1\Pi_g,\ {}^1\Pi_u,\ {}^1\Delta_g, \\ & & 2{}^3\Sigma_u^+,\ {}^3\Sigma_g^-,\ {}^3\Pi_g,\ {}^3\Pi_u,\ {}^3\Delta_u; \\ O_2 & (\text{atoms in } {}^3P \text{ states}): & 2{}^1\Sigma_g^+,\ {}^1\Sigma_u^-,\ {}^1\Pi_g,\ {}^1\Pi_u,\ {}^1\Delta_g, \\ & & 2{}^3\Sigma_u^+,\ {}^3\Sigma_g^-,\ {}^3\Pi_u,\ {}^3\Pi_g,\ {}^3\Delta_u, \\ & & 2{}^5\Sigma_g^+,\ {}^5\Sigma_u^-,\ {}^5\Pi_g,\ {}^5\Pi_u,\ {}^5\Delta_g. \end{array}$$

(A numeral before a term symbol gives the number of terms of that type when the number exceeds one.)

Problem

2. Do the same for the molecules HCl and CO.

Solution. For unlike atoms, the parities of their states must also be taken into account. Formula (31.6) shows that the ground states of H, O, and C are even, while that of Cl is odd (for the electronic configurations of the atoms, see Table 3). The rules given in the text yield

$$\begin{array}{lll} \text{molecule HCl} & (\text{atoms in } {}^2S_g, {}^2P_u \text{ states}): & {}^{1,3}\Sigma^+,\ {}^{1,3}\Pi; \\ \text{molecule CO} & (\text{atoms in } {}^3P_g, {}^3P_g \text{ states}): & 2{}^{1,3,5}\Sigma^+,\ {}^{1,3,5}\Sigma^-, \\ & & 2{}^{1,3,5}\Pi,\ {}^{1,3,5}\Delta. \end{array}$$

§ 81. Valence

The capacity of atoms to join together into a molecule is described through the notion of *valence*. A definite valence is assigned to each atom, and when atoms combine, their valences must saturate one another: every valence bond of one atom must be matched by a valence bond of another. Thus, in the methane molecule CH_4, the four bonds of tetravalent carbon are saturated by bonds from four monovalent hydrogen atoms. To give valence a physical interpretation, we begin with the simplest case, in which two hydrogen atoms unite to form an H_2 molecule.

Consider two hydrogen atoms in their ground states (^2S). When brought near each other, they can form a system in either the $^1\Sigma_g^+$ or the $^3\Sigma_u^+$ molecular state. The singlet term has an antisymmetric spin wave function, while the triplet term has a symmetric one. For the coordinate wave function the symmetry is reversed: it is symmetric for the $^1\Sigma$ term and antisymmetric for the $^3\Sigma$ term. It is plain that only $^1\Sigma$ can be the ground term of the H_2 molecule. Indeed, an antisymmetric wave function $\psi(\mathbf{r}_1, \mathbf{r}_2)$ (where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the radius vectors of the two electrons) necessarily has nodes—it vanishes when $\mathbf{r}_1 = \mathbf{r}_2$—and therefore cannot describe the lowest state of the system.

Numerical calculation confirms that the $^1\Sigma$ electronic term has a pronounced minimum, corresponding to the formation of a stable H_2 molecule. In the $^3\Sigma$ state, by contrast, the energy $U(r)$ decreases monotonically as the distance between the nuclei increases; this describes mutual repulsion of the two H atoms[7] (Fig. 29).

Thus the total spin of a hydrogen molecule in its ground state is zero, $S = 0$. The molecules of virtually all chemically stable compounds of the main-group elements prove to have this property. Among inorganic molecules, the exceptions are the diatomic molecules O_2 (ground state $^3\Sigma$) and NO (ground state $^2\Pi$), as well as the triatomic molecules NO_2 and ClO_2 (total spin $S = 1/2$). The transition-group elements have special features; these will be considered below, after the valence properties of the main-group elements have been examined.

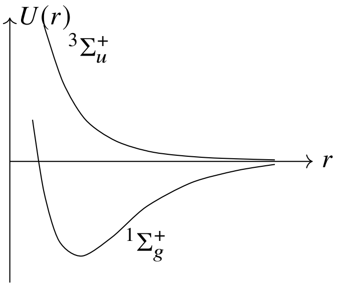

Fig. 29

The ability of atoms to unite is therefore connected with their spin (W. Heitler, H. London, 1927). Combination occurs in such a way that the atomic spins compensate

[7] Here we disregard the van der Waals attraction between the atoms (see § 89). Its existence means that the $U(r)$ curve for the $^3\Sigma$ term also has a minimum, situated at large distances. This minimum, however, is very shallow relative to that on the $^1\Sigma$ curve and would be invisible on the scale of Fig. 29.

one another. A convenient quantitative measure of an atom's ability to combine is the integer twice its spin. This number is the same as the atom's chemical valence. One must remember here that a given atom can have different valences according to the state in which it is found.

Let us apply this viewpoint to the main groups of the periodic system. In the normal state, first-group elements (the first column of Table 3, the alkali-metal group) have spin $S = 1/2$ and hence valence one. A state of higher spin could be produced only by exciting an electron out of a filled shell. Such states lie so high that an excited atom cannot form a stable molecule[8].

In the normal state, atoms of the second-group elements (the second column of Table 3, the alkaline-earth-metal group) have $S = 0$. These atoms therefore cannot enter chemical compounds while in their normal state. Fairly near the ground state, however, there is an excited state whose unfilled shell has configuration sp instead of s^2 and whose total spin is $S = 1$. In this state the atom has valence 2, which is the principal valence of second-group elements.

The normal-state configuration of third-group elements is s^2p, with spin $S = 1/2$. Excitation of an electron from the filled s shell, however, produces a nearby excited state of configuration sp^2 and spin $S = 3/2$. Elements of this group accordingly display both monovalent and trivalent behavior. The first elements of the group (B, Al) behave only as trivalent elements. The tendency to exhibit valence 1 becomes stronger with increasing atomic number, and Tl behaves to the same extent as a monovalent and as a trivalent element (for example, in TlCl and $TlCl_3$). The reason is that, for the early elements of the group, the energetic advantage from the greater bond energy in compounds of the trivalent element, relative to those of the monovalent element, outweighs the atomic excitation energy.

For fourth-group atoms the ground state has configuration s^2p^2 and spin 1, while a nearby excited state has configuration sp^3 and spin 2. The respective valences are 2 and 4. As in the third group, the first members of the fourth group (C, Si) chiefly exhibit the higher valence (CO, for example, is an exception), whereas the propensity for the lower valence increases with atomic number.

In atoms of the fifth group, the ground state has configuration s^2p^3 and spin $S = 3/2$, giving valence three. A higher-spin excited state can arise only when one of the electrons passes to a shell with the next principal quantum number. The nearest state of this kind has configuration sp^3s' and spin $S = 5/2$ (here s' conventionally denotes an electron s state whose principal quantum number is one greater than that of the s state). Although the excitation energy of this state is comparatively large, the excited atom can nevertheless enter a stable compound. Fifth-group elements consequently occur as both trivalent and pentavalent elements (nitrogen, for instance, is trivalent in NH_3 and pentavalent in HNO_3).

For sixth-group elements the ground-state configuration is s^2p^4 and the spin is 1, so the atom is divalent. Exciting one p electron gives the state s^2p^3s' with spin 2; excitation of one more s electron gives the state $sp^3s'p'$ with spin 3. In either excited state, the atom can become part of a stable molecule and displays valence 4 or 6, respectively. The

[8]For the elements Cu, Ag, and Au, see the end of this section.

first element of the sixth group (oxygen) has only valence 2, whereas later members also show the higher valences (thus sulfur is respectively divalent, tetravalent, and hexavalent in H_2S, SO_2, and SO_3).

In the seventh group (the halogens), atoms in the ground state (configuration s^2p^5, spin $S = 1/2$) are monovalent. They can, however, also form stable compounds in excited states of configurations s^2p^4s', $s^2p^3s'p'$, and $sp^3s'p'^2$, having spins 3/2, 5/2, and 7/2, respectively; these correspond to valences 3, 5, and 7. The first element of the group (F) is invariably monovalent, while later members also display the higher valences (chlorine, for example, is respectively mono-, tri-, penta-, and heptavalent in HCl, $HClO_2$, $HClO_3$, and $HClO_4$).

Finally, atoms of the noble-gas elements have completely filled shells in their ground state. Their spin is therefore $S = 0$, and their excitation energies are large. Accordingly, their valence is zero, and these elements are chemically inert.[9]

All these arguments call for the following general qualification. To say that an atom enters a molecule with the valence belonging to one of its excited states does not imply that separating the atoms to large distances must yield an excited atom. It means only that the molecular electron-density distribution near the nucleus of the atom in question resembles that in an isolated excited atom. Yet the limiting electron distribution reached as the internuclear distance grows may correspond to unexcited atoms.

When atoms form a molecule, their filled electron shells change little. The distribution of electron density in unfilled shells, on the other hand, can change substantially. In the most pronounced cases of a so-called *heteropolar bond*, all valence electrons pass from some atoms to others. The molecule may then be said to consist of ions whose charges, in units of e, are equal to their valences. The first-group elements are electropositive: in heteropolar compounds they give up electrons and form positive ions. Across the subsequent groups electropositivity gradually diminishes and gives way to electronegativity, which is strongest among the seventh-group elements. The same qualification made above for excited atoms in molecules must also be applied to heteropolarity. A heteropolar molecule need not necessarily yield two ions when its atoms are separated. Thus, separation of CsF would indeed give Cs^+ and F^- ions, but NaF tends instead to neutral Na and F atoms (because fluorine's electron affinity exceeds the ionization potential of cesium but is below that of sodium).

At the opposite limit is the so-called *homopolar bond*. The atoms in such a molecule remain neutral on average. Unlike heteropolar molecules, homopolar molecules possess no appreciable dipole moment. The distinction between heteropolar and homopolar bonding is entirely quantitative, and every intermediate case is possible.

We now turn to the transition-group elements. In the character of their valence

[9] A few of them nevertheless form stable compounds, for example with fluorine or oxygen. Such valences may arise when electrons are promoted out of the filled outer shell. The electrons then enter unoccupied f- (or d-) states lying relatively close in energy.

We should also note a distinctive attraction. It appears when a noble-gas atom interacts with an excited atom of the same element. Two identical atoms are then brought together while occupying different states. The number of possible states produced in this way is doubled (see § 80). Transfer of the excitation from one atom to the other takes the place of the exchange interaction that produces ordinary valence. The molecule He_2 provides an example. Bonding of the same kind occurs in molecular ions made up of two identical atoms. One example is H_2^+.

properties, the elements of the palladium and platinum groups differ little from main-group elements. The distinction is that their *d* electrons lie comparatively deep in the atom and therefore interact more weakly with other atoms in a molecule. As a result, "unsaturated" compounds, whose molecules have nonzero spin (in fact no greater than $1/2$), are encountered relatively often among compounds of these elements. Each element can show several valences, and here they may differ by one rather than only by two as in main-group elements (where a change of valence is tied to the excitation of an electron whose spin was compensated, thereby freeing the spins of a pair of electrons at once).

Rare-earth elements are characterized by an unfilled *f* shell. The *f* electrons lie much deeper than the *d* electrons and consequently take no part at all in valence. Thus only *s* and *p* electrons in unfilled shells determine the valence of rare-earth elements[10]. It should be borne in mind, however, that atomic excitation can transfer *f* electrons into *s* and *p* states and thereby increase the valence by one. Rare-earth elements therefore also show valences differing by one (in practice, they are all tri- and tetravalent).

The elements of the actinium group occupy a special position. Ac and Th contain no *f* electrons at all, and their *d* electrons participate in valence; their chemical properties therefore resemble those of the palladium and platinum groups rather than those of the rare earths. Although a U atom contains *f* electrons in its normal state, uranium likewise has no *f* electrons in its compounds. Finally, atoms of Np, Pu, Am, and Cm retain *f* electrons even in compounds, but for them too it is the *s* and *d* electrons that participate in valence. In this sense they are homologues of uranium. The greatest possible numbers of "unpaired" *s* and *d* electrons are 1 and 5, respectively; hence the maximum valence of actinium-group elements is six (whereas the maximum for rare-earth elements, with *s* and *p* electrons participating in valence, is $1 + 3 = 4$).

With respect to valence properties, the iron-group elements stand between the rare-earth elements and those of the palladium and platinum groups. In their atoms the *d* electrons lie relatively deep and, in many compounds, take no part at all in the valence bond. In these compounds, therefore, the iron-group elements behave like rare-earth elements. These include ionic compounds (for example, $FeCl_2$ and $FeCl_3$), in which the metal atom enters as a simple cation. As with the rare earths, iron-group elements can display many different valences in compounds of this type.

Another class of compounds formed by the iron-group elements consists of the so-called complex compounds. Their distinguishing feature is that the transition-element atom does not enter the molecule as a simple ion. Instead, it forms part of a composite, or complex, ion. One example is the ion MnO_4^- in $KMnO_4$. Another is the ion $Fe(CN)_6^{4-}$ in $K_4Fe(CN)_6$. In such complex ions the atoms

lie closer together than in simple ionic compounds, and their *d*-electrons take part in valence bonding. Accordingly, in complex compounds the iron-group elements behave like the elements of the palladium and platinum groups.

Finally, a qualification is needed concerning Cu, Ag, and Au, which were placed among the main groups in § 73: in a number of compounds they behave as transition elements. They can exhibit valence greater than one by transfer of electrons from the *d*

[10]The *d* electrons present in unfilled shells of the atoms of some rare-earth elements are immaterial, since in practice these atoms always enter compounds in excited states containing no *d* electrons.

shell to the energetically nearby p shell (for example, from $3d$ to $4p$ in Cu). In these compounds the atoms have an unfilled d shell and behave as transition elements (Cu like an iron-group element, and Ag and Au like palladium- and platinum-group elements).

Problem

Determine the electronic terms of the molecular ion H_2^+ formed by combining a hydrogen atom in its normal state with an H^+ ion, when the distance R between the nuclei is large compared with the Bohr radius (L. D. Landau, 1961; C. Herring, 1961)[11].

Solution. In its formulation this problem is analogous to Problem 3 of § 50. Instead of two one-dimensional potential wells, there are now two three-dimensional wells (around the two nuclei), with a common axial symmetry about the line joining the nuclei. The level $E_0 = -1/2$ (the ground level of the hydrogen atom)[12] splits into levels $U_g(R)$ and $U_u(R)$ (the ${}^2\Sigma_g^+$ and ${}^2\Sigma_u^+$ terms), corresponding to the electron wave functions

$$\psi_{g,u}(x, y, z) = \frac{1}{\sqrt{2}}[\psi_0(x, y, z) \pm \psi_0(-x, y, z)],$$

which are respectively symmetric and antisymmetric with respect to the plane $x = 0$ bisecting the distance between the nuclei (located at $x = \pm R/2$ on the x axis). Here $\psi_0(x, y, z)$ is the electron wave function in one of the potential wells. Proceeding exactly as in Problem 3 of § 50 gives

$$U_{g,u}(R) - E_0 = \mp \int \psi_0 \frac{\partial \psi_0}{\partial x}\, dy\, dz, \tag{1}$$

where the integration extends over the plane $x = 0$[13].

We seek the function ψ_0 (for motion, say, about nucleus 1 at $x = R/2$) in the form

$$\psi_0 = \frac{1}{\sqrt{\pi}} e^{-r_1} a, \tag{2}$$

where a varies slowly (for a hydrogen atom, a would be 1). The function ψ_0 must satisfy the Schrödinger equation

$$\frac{1}{2}\Delta\psi_0 + \left(-\frac{1}{2} - \frac{1}{R} + \frac{1}{r_1} + \frac{1}{r_2}\right)\psi_0 = 0 \tag{3}$$

(r_1 and r_2 are the electron's distances from nuclei 1 and 2). The electron's total energy appearing in this equation is the difference $E_0 - 1/R$, since E_0 itself also includes the energy $1/R$ of Coulomb repulsion between the nuclei.

[11] For the solution of the analogous problem for the H_2 molecule, see L. P. Gor'kov, L. P. Pitaevskii // Dokl. Akad. Nauk SSSR. 1963. Vol. 151. P. 822; C. Herring, M. Flicker // Phys. Rev. 1964. Vol. 134A. P. 362 (the computational error made in the first paper is corrected in the second).

[12] Atomic units are used in this problem.

[13] Thus the effect being sought is determined by the range of distances in which the electron interacts equally with the two nuclei.

Since ψ_0 decreases rapidly away from the x axis, only values of y, z small compared with R contribute appreciably to the integral (1). For $y, z \ll R$, substitution of (2) in (3) yields

$$\frac{\partial a}{\partial x} + \frac{R/2 + x}{R} a = 0$$

(second derivatives of the slowly varying function a have been neglected, and r_2 has been set equal to $R/2 + x$). The solution that tends to unity as $x \to R/2$ (that is, near nucleus 1) is

$$a = \frac{2R}{R + 2x} \exp\left(\frac{x}{R} - \frac{1}{2}\right).$$

Formula (1) now gives

$$U_{g,u} - E_0 = \mp\frac{4}{e}\int_{R/2}^{\infty} e^{-2r_1} 2\pi r_1, dr_1 = \mp 2Re^{-R-1}.$$

The splitting is therefore[14]

$$U_g - U_u = -4Re^{-R-1}. \tag{4}$$

At sufficiently large distances this exponentially decreasing expression becomes smaller than the second-order effect of the dipole interaction between the H atom and the H^+ ion. The polarizability of a hydrogen atom in its normal state is $9/2$ (see (77.9)), and the field of the H^+ ion is $\mathcal{E} = 1/R^2$; the corresponding interaction energy is consequently $-9/(4R^4)$. Including it gives

$$U_{g,u}(R) - E_0 = \mp 2Re^{-R-1} - \frac{9}{4R^4}. \tag{5}$$

The second term becomes comparable to the first only at $R = 10.8$. The term $U_u(R)$ also possesses a minimum. It lies at $R = 12.6$. Its value is $-5.8 \cdot 10^{-5}$ atomic units ($-1.6 \cdot 10^{-3}$ eV)[15].

§ 82. Vibrational and rotational structure of singlet terms of a diatomic molecule

As noted at the beginning of this chapter, the large disparity between the nuclear and electronic masses allows the determination of molecular energy levels to be divided into two parts. First one finds, as functions of the internuclear distance, the energy levels of the electrons for fixed nuclei (the electronic terms). One may then consider nuclear motion in a specified electronic state; the nuclei are thereby treated as particles interacting through $U_n(r)$, where U_n is the corresponding electronic term. Molecular motion consists of translation of the molecule as a whole and motion of the nuclei relative to their center of mass. The translational part is of no interest here, and the center of mass may be taken as fixed.

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[14] The analogous result for the H_2 molecule (see the papers cited above) is $U_g - U_u = -1{,}64R^{5/2}e^{-2R}$.

[15] This minimum, arising from van der Waals forces, is very shallow in comparison with the minimum of the term $U_g(R)$, which corresponds to the normal state of the stable ion H_2^+; that principal minimum lies at $R = 2.0$ and equals -0.60 atomic units (-16.3 eV).

For clarity, consider first electronic terms for which the total molecular spin S vanishes (singlet terms). The relative motion of two particles (the nuclei) interacting through $U(r)$ reduces to the motion of one particle of mass M (their reduced mass) in the central field $U(r)$. (Here $U(r)$ denotes the energy of the electronic term under consideration.) Motion in the central field, in turn, reduces to a one-dimensional problem with an effective energy equal to $U(r)$ plus the centrifugal energy.

Let $\mathbf{K}$ be the total angular momentum of the molecule, composed of the electronic orbital angular momentum $\mathbf{L}$ and the angular momentum of nuclear rotation. The operator of the nuclear centrifugal energy is then

$$B(r)(\mathbf{K}-\mathbf{L})^2,$$

where the notation

$$B(r) = \frac{\hbar^2}{2Mr^2}, \tag{82.1}$$

customary in the theory of diatomic molecules, has been introduced. Averaging this quantity over the electronic state (at fixed r) gives the centrifugal energy as a function of r, which must enter the effective potential energy $U_K(r)$. Hence

$$U_K(r) = U(r) + B(r)\overline{(\mathbf{K}-\mathbf{L})^2}, \tag{82.2}$$

where the bar denotes this averaging.

Carry out the average for a state with a definite square of the total angular momentum, $\mathbf{K}^2 = K(K+1)$ (K is an integer), and a definite projection of the electronic angular momentum on the molecular axis (the z axis), $L_z = \Lambda$. Expansion of (82.2) gives

$$U_K(r) = U(r) + B(r)K(K+1) - 2B(r)\overline{\mathbf{LK}} + B(r)\overline{\mathbf{L}^2}. \tag{82.3}$$

The last term depends only on the electronic state and contains no quantum number K; it can simply be absorbed into $U(r)$. We now show that this also applies to the penultimate term.

Recall that when the projection of an angular momentum on an axis has a definite value, the mean angular-momentum vector is directed along that same axis (see the remark at the end of § 27). Let $\mathbf{n}$ denote the unit vector along the z axis: $\overline{\mathbf{L}} = \Lambda\mathbf{n}$. In classical mechanics the rotational angular momentum of a two-particle system (the nuclei) is $\mathbf{r} \times \mathbf{p}$, where $\mathbf{r} = r\mathbf{n}$ is the radius vector between the particles and $\mathbf{p}$ is the momentum of their relative motion; this quantity is perpendicular to $\mathbf{n}$. The same statement holds in quantum mechanics for the operator of nuclear rotational angular momentum: $(\mathbf{K}-\mathbf{L})\mathbf{n} = 0$, or $\mathbf{Kn} = \mathbf{Ln}$. Equality of the operators entails equality of their eigenvalues, and since $\mathbf{nL} = L_z = \Lambda$, it follows that

$$\mathbf{Kn} = \Lambda. \tag{82.4}$$

Thus, in the penultimate term of (82.3), $\overline{\mathbf{LK}} = \mathbf{nK}\Lambda = \Lambda^2$, which is independent of K. After redefining $U(r)$, the effective potential energy finally becomes

$$U_K(r) = U(r) + B(r)K(K+1). \tag{82.5}$$

We also note that $K_z = \Lambda$ implies that, for a given Λ, the quantum number K can assume only the values

$$K \geqslant \Lambda. \tag{82.6}$$

Solving the one-dimensional Schrödinger equation with the potential energy (82.5) yields a sequence of energy levels. For each fixed K, let these levels be numbered in ascending order by $v = 0, 1, 2, \ldots$, with $v = 0$ corresponding to the lowest one. Nuclear motion therefore splits every electronic term into a set of levels characterized by the two quantum numbers K and v.

For a given electronic term, the number of these levels may be finite or infinite. Suppose the electronic state is such that, as $r \to \infty$, the molecule becomes two separate neutral atoms. Then $U(r)$, and with it $U_K(r)$, tends to the limiting constant $U(\infty)$ (the sum of the energies of the two isolated atoms) faster than $1/r^2$ (see § 89). Such a field has finitely many levels (see § 18), although their actual number in molecules is very large. Their distribution is as follows: for each fixed K there is a certain number of levels distinguished by v, and this number decreases as K increases, until a value of K is reached for which no levels remain at all.

If, however, the molecule separates into two ions as $r \to \infty$, then at large distances $U(r) - U(\infty)$ approaches the Coulomb attraction energy of the ions ($\sim 1/r$). Such a field has infinitely many levels, which crowd together on nearing the limiting value $U(\infty)$. We note that the first case applies to most molecules in the normal state; only a comparatively small number of molecules yield ions when their nuclei are drawn apart.

A complete general calculation of how the energy levels depend on the quantum numbers is not possible. It can be carried out only for levels with relatively low excitation, whose energies do not lie far above the ground level[16]. For such levels the quantum numbers K and v are small. Since molecular spectroscopy normally deals with levels of this kind, they are of particular interest.

In weakly excited states, nuclear motion can be described as small oscillations about equilibrium. Accordingly, expand $U(r)$ in powers of $\xi = r - r_e$, where r_e is the value of r at which $U(r)$ is minimal. Since $U'(r_e) = 0$, through second order $U(r) = U_e + M\omega_e^2\xi^2/2$, where $U_e = U(r_e)$ and ω_e is the oscillation frequency. In the second term of (82.5), the centrifugal energy, it is sufficient to put $r = r_e$, since this term already contains the small quantity $K(K+1)$. We thus have

$$U_K(r) = U_e + B_e K(K+1) + \frac{M\omega_e^2\xi^2}{2}, \tag{82.7}$$

where $B_e = \hbar^2/(2Mr_e^2) = \hbar^2/2I$ is the so-called *rotational constant* ($I = Mr_e^2$ is the molecule's moment of inertia).

The first two terms in (82.7) are constants, whereas the third represents a one-dimensional harmonic oscillator. The energy levels being sought may therefore be written down immediately as

$$E = U_e + B_e K(K+1) + \hbar\omega_e\left(v + \frac{1}{2}\right). \tag{82.8}$$

[16]All levels considered here originate from one specified electronic term.

Thus, in this approximation an energy level consists of three independent parts:

$$E = E^{el} + E^{r} + E^{v}. \tag{82.9}$$

The first term ($E^{el} = U_e$) is the electronic energy (including the energy of the Coulomb interaction between the nuclei at $r = r_e$). The second term

$$E^{r} = B_e K(K+1), \tag{82.10}$$

— the *energy of rotation* (or rotational energy) associated with the molecule's rotation[17]. Finally, the third term,

$$E^{v} = \hbar\omega_e\left(v + \frac{1}{2}\right), \tag{82.11}$$

is the energy of nuclear vibration within the molecule. In accordance with its definition, v numbers in ascending order the levels having a given K; it is called the *vibrational* quantum number.

For a fixed shape of the potential-energy curve $U(r)$, the frequency ω_e is inversely proportional to $\sqrt{M}$. The spacings ΔE^{v} between vibrational levels are consequently proportional to $1/\sqrt{M}$. The spacings ΔE^{r} between rotational levels contain the moment of inertia I in the denominator and are therefore proportional to $1/M$. By contrast, the spacings ΔE^{el} between electronic levels,

like the levels themselves, contain no M. Since m/M (m being the electron mass) is the small parameter in the theory of diatomic molecules, we find

$$\Delta E^{el} \gg \Delta E^{v} \gg \Delta E^{r}. \tag{82.12}$$

These inequalities express the distinctive distribution of molecular energy levels. Nuclear vibration resolves the electronic terms into levels lying comparatively close together. Molecular rotation, in turn, produces a still finer splitting of these levels[18].

In subsequent approximations the energy can no longer be separated into independent vibrational and rotational parts; rotation–vibration terms appear that contain both K and v. Successive approximations would give the levels E as a power-series expansion in the quantum numbers K and v.

We calculate here the approximation immediately following (82.8). For this purpose the expansion of $U(r)$ in powers of ξ must be continued through fourth-order terms

[17] The wave function describing the rotation of a diatomic molecule (with spin disregarded) has essentially the same form as the wave function of a symmetric top (§ 103). Unlike the top, a molecule's rotation is described by only two angles ($\alpha \equiv \varphi, \beta \equiv \theta$), which specify the direction of its axis. The rotational wave function differs from (103.8) in lacking the factor $e^{ik\gamma}/\sqrt{2\pi}$ and in the notation used for the quantum numbers. Since, by (82.4), Λ equals the projection of the total angular momentum $\mathbf{K}$ on the molecular axis (the ζ axis in § 103), the notation must be changed according to $J, M, k \to K, M, \Lambda$ (where now $M = K_z$). Thus, $\psi_{\text{rot}}(\varphi,\theta) = i^K\sqrt{(2K+1)/(4\pi)}D^{(K)}_{\Lambda M}(\varphi,\theta,0)$.

[18] As an example, values of U_e, $\hbar\omega_e$, and B_e (in electron-volts) for several molecules are

	H_2	N_2	O_2
$-U_e$	4,7	7,5	5,2
$\hbar\omega_e$	0,54	0,29	0,20
$10^3 B_e$	7,6	0,25	0,18

(compare the anharmonic-oscillator problem in § 38). Correspondingly, expand the centrifugal energy through terms in ξ^2. This gives

$$\begin{aligned} U_K(r) = U_e + \frac{M\omega_e^2}{2}\xi^2 + \frac{\hbar^2}{2Mr_e^2}K(K+1) - a\xi^3 + b\xi^4 \\ - \frac{\hbar^2}{Mr_e^3}K(K+1)\xi + \frac{3\hbar^2}{2Mr_e^4}K(K+1)\xi^2. \end{aligned} \tag{82.13}$$

Now calculate the correction to the eigenvalues (82.8), treating the last four terms in (82.13) as the perturbation operator. First-order perturbation theory suffices for the terms in ξ^2 and ξ^4, whereas second order must be calculated for those in ξ and ξ^3, since the diagonal matrix elements of ξ and ξ^3 vanish identically. All matrix elements needed for the calculation were found in § 23 and in problem 3 of § 38. The result is

conventionally written as

$$\begin{aligned} E = E^{el} + \hbar\omega_e\left(v + \frac{1}{2}\right) - x_e\hbar\omega_e\left(v + \frac{1}{2}\right)^2 \\ + B_v K(K+1) - D_e K^2(K+1)^2, \end{aligned} \tag{82.14}$$

where

$$B_v = B_e - \alpha_e\left(v + \frac{1}{2}\right) \equiv B_0 - \alpha_e v. \tag{82.15}$$

The constants x_e, B_e, α_e, and D_e are related to the constants entering (82.14) by

$$\begin{aligned} B_e = \frac{\hbar^2}{2I}, \qquad D_e = \frac{4B_e^3}{\hbar^2\omega_e^2}, \\ \alpha_e = \frac{6B_e^2}{\hbar\omega_e}\left(\frac{a\hbar}{M\omega_e^2}\sqrt{\frac{2}{MB_e}} - 1\right), \\ x_e = \frac{3}{2\hbar\omega_e}\left(\frac{\hbar}{M\omega_e}\right)^2\left(\frac{5}{2}\frac{a^2}{M\omega_e^2} - b\right). \end{aligned} \tag{82.16}$$

Terms independent of v and K have been included in E^{el}.

PROBLEM

Estimate the accuracy of the approximation that separates electronic and nuclear motion in a diatomic molecule.

SOLUTION. Write the complete molecular Hamiltonian as $\widehat{H} = \widehat{T}_r + \widehat{H}_{el}$, where $\widehat{T}_r = \widehat{\mathbf{p}}^2/2M$ is the kinetic-energy operator for the relative motion of the nuclei ($\widehat{\mathbf{p}} = -i\hbar\partial/\partial\mathbf{r}$; $\mathbf{r}$ is the vector separating the nuclei, and M their reduced mass). The Hamiltonian $\widehat{H}_{el}$ includes the electron kinetic-energy operators, the potential energy of their Coulomb interactions with one another and with the nuclei, and the Coulomb interaction energy of the nuclei[19]. Seek a solution of the Schrödinger equation

$$\widehat{H}\psi = (\widehat{T}_r + \widehat{H}_{el})\psi = E\psi \tag{1}$$

in the form

$$\psi = \sum_m \chi_m(\mathbf{r})\varphi_m(q, \mathbf{r}), \tag{2}$$

where the functions $\varphi_m(q, \mathbf{r})$ are orthonormal solutions of

$$\widehat{H}_{el}\varphi_m(q, \mathbf{r}) = U_m(r)\varphi_m(q, \mathbf{r}) \tag{3}$$

(q denotes the full set of electronic coordinates), and $U_m(r)$ are the eigenvalues of $\widehat{H}_{el}$ with r treated as a parameter. Substitute (2) into (1), multiply on the left by $\varphi_n^*(q, \mathbf{r})$, and integrate over dq. The result is

$$\left[\frac{\widehat{\mathbf{p}}^2}{2M} + V''_{nn} + U_n(r) - E\right]\chi_n(\mathbf{r}) = -{\sum_m}' (\widehat{V}'_{nm} + V''_{nm})\chi_m(\mathbf{r}), \tag{4}$$

where

$$\widehat{V}'_{nm} = \frac{1}{M}\mathbf{p}_{nm}\widehat{\mathbf{p}}, \qquad V''_{nm} = \frac{1}{2M}(\mathbf{p}^2)_{nm},$$

and $\mathbf{p}_{nm} = \int \varphi_n^* \widehat{\mathbf{p}}\varphi_m, dq$, while $(\mathbf{p}^2)_{nm}$ is likewise a matrix element with respect to the electronic wave functions; the diagonal element $\mathbf{p}_{nn}$ vanishes by symmetry.

The electronic functions φ_n vary appreciably only over distances of atomic order. Differentiating them with respect to $\mathbf{r}$ therefore introduces no large parameter M/m (m is the electron mass). Consequently, V''_{nn} is smaller than $U_n(r)$ by the ratio m/M and may be omitted. If the terms on the right-hand side of (4) are regarded as a small perturbation, then to zeroth order the functions $\chi_n(\mathbf{R})$ are solutions of

$$\left[\frac{\widehat{\mathbf{p}}^2}{2M} + U_n(r)\right]\chi_{nv} = E_{nv}\chi_{nv}, \tag{5}$$

which describes nuclear motion in the field $U_n(r)$ (v denotes the quantum numbers of this motion). Perturbation theory is applicable provided

$$|\langle nv'|\widehat{V}'_{nm} + V''_{nm}|mv\rangle| \ll |E_{nv'} - E_{mv}|.$$

On the right are energy differences belonging to different electronic terms; they are of zeroth order in the small parameter m/M. On the left are matrix elements over the nuclear wave functions. The term involving $\widehat{V}''_{nm}$ contains m/M and is certainly small. In the matrix element of $\widehat{V}'_{nm}$, the operator $\widehat{\mathbf{p}}$ acting on χ_{mv} multiplies it by a quantity of the order of the nuclear momentum. For small nuclear oscillations this momentum is $\sim \sqrt{M\hbar\omega_e}$; since at the same time ω_e is inversely proportional to $\sqrt{M}$, the matrix element $\langle nv'|\widehat{V}'_{nm}|mv\rangle$ is of order $(m/M)^{3/4}$.

[19] The Hamiltonian $\widehat{H}$ refers to a frame in which the center of mass of the whole molecule is at rest ($\mathbf{P}_n + \mathbf{P}_e = 0$, where $\mathbf{P}_n$ is the total momentum of the two nuclei and $\mathbf{P}_e$ the total electronic momentum). The term corresponding to the kinetic energy of the nuclear center of mass has nevertheless already been omitted: $\widehat{\mathbf{P}}_n^2/2(M_1 + M_2) = \mathbf{p}_e^2/2(M_1 + M_2)$. This term is certainly smaller than the electronic kinetic energy by the ratio m/M.

§ 83. Multiplet terms. Case a

We now turn to the classification of molecular levels for which the spin S is nonzero. At zeroth order, with relativistic effects omitted altogether, the energy of a molecule—just as for any particle system—does not change when the spin direction changes: the spin is "free," so every energy level is degenerate with respect to spin orientation. Relativistic effects remove this degeneracy, and the energy then depends on the magnitude of the spin projection on the molecular axis. We shall call the relativistic interactions in a molecule the *spin–axis interaction*. Here the main contribution, as for atoms, comes from coupling between the spins and the electrons' orbital motion[20].

The character and classification of the molecular levels depend essentially on the relative importance of the spin interaction with orbital motion, on the one hand, and molecular rotation, on the other. The importance of the latter is measured by the spacing of adjacent rotational levels. Accordingly, two limiting cases must be considered. In one, the spin–axis interaction energy is large compared with the differences between rotational levels; in the other, it is small. The first is customarily called case (or coupling type) a, and the second case b (F. Hund, 1933).

Case a occurs most often. The exceptions are the Σ terms, for which case b generally applies because their spin–axis interaction is weak (see below)[21]. For other terms, case b is sometimes found in the lightest molecules, where the spin–axis interaction is relatively weak while the rotational-level spacing is large (the moment of inertia is small).

Intermediate cases between a and b are, of course, also possible. It must also be kept in mind that, as the rotational quantum number changes, one and the same electronic state can pass continuously from case a to case b. This happens because the spacing of adjacent rotational levels grows with the rotational quantum number and, at sufficiently high values, can exceed the spin–axis interaction energy (case b), even when the lower rotational levels belong to case a.

In case a, the classification of levels differs little in principle from that of terms with zero spin. We first consider the electronic terms with the nuclei fixed, thus neglecting rotation completely; in addition to the projection Λ of the electrons' orbital angular momentum, we must now consider the projection of the total spin on the molecular axis, denoted by Σ[22], whose possible values are $S, S-1, \dots, -S$. We adopt the convention that Σ is positive when the direction of the spin projection agrees with that of the orbital angular momentum relative to the axis (recall that Λ denotes the absolute magnitude of the latter projection). The quantities Λ and Σ combine to give the total electronic angular-momentum projection on the molecular axis:

$$\Omega = \Lambda + \Sigma; \tag{83.1}$$

its possible values are $\Lambda + S, \Lambda + S - 1, \dots, \Lambda - S$. Consequently, an electronic term

[20] There is also, besides the spin–orbit and spin–spin couplings, a coupling of the spin and of the electrons' orbital motion to rotation of the molecule. This part of the interaction is very small, however; examining it can be of interest only for terms with spin $S = 1/2$ (see § 81).

[21] The ground electronic term of the O_2 molecule (the $^3\Sigma$ term) is a special case. Its coupling type lies between a and b (see Problem 3 in § 84).

[22] This must not be confused with the symbol for terms having $\Lambda = 0$!

with orbital angular momentum Λ gives rise to $2S + 1$ terms distinguished by their values of Ω (as with atomic terms, this separation is called the fine structure, or the multiplet splitting, of the electronic levels). The value of Ω is written as a subscript on the term symbol; for example, $\Lambda = 1$, $S = 1/2$ gives the terms $^2\Pi_{1/2}$ and $^2\Pi_{3/2}$.

For each of these terms, allowing for nuclear motion produces vibrational and rotational structures. The different rotational levels are characterized by values of the quantum number J, the molecule's total angular momentum, which includes the electrons' orbital and spin angular momenta and the rotational angular momentum of the nuclei[23]. This quantum number takes integer values beginning with $|\Omega|$:

$$J \geqslant |\Omega| \tag{83.2}$$

(the evident generalization of rule (82.6)).

We now derive quantitative expressions for the molecular levels in case a. We begin with the fine structure of the electronic term. The discussion of atomic-term fine structure in § 72 relied on formula (72.4), according to which the mean spin–orbit interaction is proportional to the projection of the atom's total spin on its orbital-angular-momentum vector. The exact analogue in a diatomic molecule is the spin–axis interaction: after averaging over the electronic state at a fixed internuclear separation r, it is proportional to the projection Σ of the molecule's total spin onto its axis, and the split electronic term can therefore be written as

$$U(r) + A(r)\Sigma,$$

where $U(r)$ is the energy of the original, unsplit term and $A(r)$ is some function of r; this function depends on the original term (in particular, on the value of Λ), but not on Σ. Because the quantum number normally used is Ω rather than Σ, it is more convenient to write $A\Omega$ in place of $A\Sigma$; the two expressions differ by $A\Lambda$, which may be included in $U(r)$. Thus the electronic term has the form

$$U(r) + A(r)\Omega. \tag{83.3}$$

Notice that the components of the split term are equally spaced: adjacent components, whose Ω values differ by unity, are separated by $A(r)$ independently of Ω.

General considerations readily show that A vanishes for Σ-terms. To see this, perform time reversal. The energy must remain unchanged under this operation, whereas the molecular state changes because the directions of the orbital and spin angular momenta relative to the axis are reversed. In the energy $A(r)\Sigma$, time reversal changes the sign of Σ; invariance of the energy then requires $A(r)$ to change sign as well. If $\Lambda \neq 0$, this gives no conclusion about the value of $A(r)$, since that function depends on the orbital angular momentum, which itself changes sign. For $\Lambda = 0$, however, $A(r)$ can in any event be asserted to remain unchanged and must consequently vanish identically. Thus, for Σ-terms, the spin–orbit interaction produces no splitting in first approximation; a splitting proportional to Σ^2 would arise only from this interaction in

[23] As is customary, the notation K is reserved for the molecule's total angular momentum when its spin is excluded. In case a, the quantum number K does not exist, since the angular momentum $\mathbf{K}$ is not conserved even approximately.

second approximation or from the spin–spin interaction in first approximation, and is therefore comparatively small. This is connected with the fact mentioned earlier that case b usually applies to Σ-terms.

Once the multiplet splitting has been determined, the molecule's rotation can be included as a perturbation, in complete analogy with the derivation at the beginning of the preceding section. The rotational angular momentum of the nuclei is obtained by subtracting the electrons' orbital angular momentum and spin from the total angular momentum. The centrifugal-energy operator therefore takes the form

$$B(r)(\hat{\mathbf{J}} - \hat{\mathbf{L}} - \hat{\mathbf{S}})^2.$$

Averaging this quantity over the electronic state and adding it to (83.3), we obtain the required effective potential energy $U_J(r)$:

$$\begin{aligned} U_J(r) &= U(r) + A(r)\Omega + B(r)\overline{(\mathbf{J} - \mathbf{L} - \mathbf{S})^2} \\ &= U(r) + A(r)\Omega + B(r)[\mathbf{J}^2 - 2\mathbf{J}(\overline{\mathbf{L} + \mathbf{S}}) + \overline{\mathbf{L}^2} + 2\overline{\mathbf{LS}} + \overline{\mathbf{S}^2}]. \end{aligned}$$

The eigenvalue of $\mathbf{J}^2$ is $J(J+1)$. Further, the same reasoning as in § 82 gives

$$\overline{\mathbf{L}} = \mathbf{n}\Lambda, \qquad \overline{\mathbf{S}} = \mathbf{n}\Sigma, \tag{83.4}$$

and we also have $(\hat{\mathbf{J}} - \hat{\mathbf{L}} - \hat{\mathbf{S}})\mathbf{n} = 0$; consequently, the eigenvalues satisfy

$$\mathbf{Jn} = (\mathbf{L} + \mathbf{S})\mathbf{n} = \Lambda + \Sigma = \Omega. \tag{83.5}$$

Inserting these values yields

$$U_J(r) = U(r) + A(r)\Omega + B(r)[J(J+1) - 2\Omega^2 + \overline{\mathbf{L}^2} + 2\overline{\mathbf{LS}} + \overline{\mathbf{S}^2}].$$

The average over the electronic state is evaluated with the zeroth-order wave functions[24]. At this order, however, the magnitude of the spin is conserved; consequently, $\mathbf{S}^2 = S(S+1)$. The wave function itself is a product of a spin function and a coordinate function; the moments $\mathbf{L}$ and $\mathbf{S}$ are therefore averaged independently, giving

$$\overline{\mathbf{LS}} = \Lambda\mathbf{n}\overline{\mathbf{S}} = \Lambda\Sigma.$$

Finally, the mean value of the squared orbital angular momentum $\mathbf{L}^2$ is independent of spin and is a function of r characteristic of the given (unsplit) electronic term. Every term that is a function of r but independent of J and Σ can be absorbed into $U(r)$, while a term proportional to Σ (or, equivalently, to Ω) can be incorporated into $A(r)\Omega$. The effective potential energy is thus

$$U_J(r) = U(r) + A(r)\Omega + B(r)[J(J+1) - 2\Omega^2]. \tag{83.6}$$

The molecular energy levels can now be found from this expression by the same procedure used in § 82 for formula (82.5). Expand $U(r)$ and $A(r)$ in powers of ξ,

[24]Zeroth order both with respect to the effect of molecular rotation and with respect to the spin–axis interaction.

retaining terms through second order in the expansion of $U(r)$ but only the zeroth-order terms in the second and third contributions; the resulting energy levels are

$$E = U_e + A_e\Omega + \hbar\omega_e\left(v + \frac{1}{2}\right) + B_e[J(J+1) - 2\Omega^2], \tag{83.7}$$

where $A_e = A(r_e)$ and B_e are constants characteristic of the given (unsplit) electronic term. Carrying the expansion further would produce additional terms of higher degree in the quantum numbers; we shall not write them out here.

§ 84. Multiplet terms. Case *b*

We now turn to case b. Here the effect of molecular rotation is stronger than the multiplet splitting. We must therefore first examine the rotational effect while neglecting the spin–axis interaction, and only afterward include the latter as a perturbation.

For a molecule with "free" spin, not only the total angular momentum **J** is conserved, but also the sum **K** of the electronic orbital angular momentum and the rotational angular momentum of the nuclei; it is related to **J** by

$$\mathbf{J} = \mathbf{K} + \mathbf{S}. \tag{84.1}$$

The quantum number K distinguishes the states of the rotating molecule with free spin that arise from a given electronic term. The effective potential energy $U_K(r)$ in a state with a specified K is plainly determined by the same formula (82.5) as for terms with $S = 0$:

$$U_K(r) = U(r) + B(r)K(K+1), \tag{84.2}$$

where K takes the values $\Lambda, \Lambda + 1, \ldots$

On including the spin–axis interaction, each term generally splits into $2S + 1$ terms (or into $2K + 1$ when $K < S$), distinguished by the values of the total angular momentum J[25]. By the general rule for addition of angular momenta, at a fixed K the number J ranges from $K + S$ to $|K - S|$:

$$|K - S| \leqslant J \leqslant K + S. \tag{84.3}$$

To calculate the splitting energy in first-order perturbation theory, the expectation value of the spin–axis interaction-energy operator must be found in a state of the zeroth approximation with respect to this interaction. In the present situation, this requires averaging both over the electronic state and over the molecular rotation (at a fixed r). The first average gives an operator of the form $A(r)\mathbf{n}\widehat{\mathbf{S}}$, proportional to the projection of the spin operator on the molecular axis.

We next average this operator over the molecular rotation, regarding the direction of the spin vector as arbitrary. Then

$$\overline{\mathbf{nS}} = \overline{\mathbf{n}}\mathbf{S}.$$

[25] In case b, the projection $\mathbf{nS}$ of the spin on the molecular axis has no definite value, so the quantum numbers Σ (and Ω) do not exist.

The average $\overline{\mathbf{n}}$ is a vector which, by symmetry, must be directed along the "vector" $\mathbf{K}$, the sole vector characterizing the molecular rotation. We may consequently write

$$\overline{\mathbf{n}} = \text{const} \cdot \mathbf{K}.$$

The proportionality coefficient is readily obtained by multiplying both sides by $\mathbf{K}$ and noting that the eigenvalues are $\mathbf{nK} = \Lambda$ (see (82.4)) and $\mathbf{K}^2 = K(K+1)$. Hence

$$\overline{\mathbf{nS}} = \frac{\Lambda}{K(K+1)}\mathbf{KS}.$$

Finally, by the general formula (31.3), the eigenvalue of the product $\mathbf{KS}$ is

$$\mathbf{KS} = \frac{1}{2}[J(J+1) - K(K+1) - S(S+1)]. \tag{84.4}$$

We thereby obtain the following expression for the required mean value of the spin–axis interaction energy:

$$\frac{A(r)\Lambda}{2K(K+1)}[J(J+1)-S(S+1)-K(K+1)] = \frac{A(r)\Lambda}{2K(K+1)}(J-S)(J+S+1) - \frac{1}{2}A(r)\Lambda.$$

This expression must be added to the energy (84.2). Since the term $\frac{1}{2}A(r)\Lambda$ does not depend on K or J, it can be absorbed into $U(r)$. Thus the effective potential energy finally takes the form

$$U_K(r) = U(r) + B(r)K(K+1) + A(r)\Lambda\frac{(J-S)(J+S+1)}{2K(K+1)}. \tag{84.5}$$

Expansion in powers of $\xi = r - r_e$ gives, in the usual way, the following expression for the molecular energy levels in case b:

$$E = U_e + \hbar\omega_e(v + 1/2) + B_e K(K+1) + A_e\Lambda\frac{(J-S)(J+S+1)}{2K(K+1)}. \tag{84.6}$$

As noted in the preceding section, for Σ terms the spin–orbit interaction produces no multiplet splitting in first order; to determine the fine structure one must include the spin–spin interaction, whose operator

is quadratic in the electron spins. Here we are concerned not with this operator itself, but with the result of averaging it over the electronic state of the molecule, just as was done for the spin–orbit interaction operator. Symmetry makes it clear that the required averaged operator must be proportional to the square of the projection of the molecule's total spin on its axis, and can therefore be written as

$$a(r)(\mathbf{Sn})^2, \tag{84.7}$$

where $a(r)$ is again a function of the distance r characteristic of the electronic term in question (symmetry also permits a term proportional to $\mathbf{S}^2$; it is of no interest, however, because the magnitude of the spin is simply constant). We shall not dwell here on deriving the cumbersome general formula for the splitting due to the operator (84.7); Problem 1 of this section gives the derivation for triplet Σ terms.

Doublet Σ terms form a special case. By Kramers' theorem (§ 60), a system of particles with total spin $S = 1/2$ necessarily retains a twofold degeneracy even when all internal relativistic interactions in the system are fully taken into account. Hence $^2\Sigma$ terms remain unsplit when both the spin–orbit and spin–spin interactions are included, to any order of approximation.

A splitting would arise here only upon including the relativistic interaction between the spin and the molecular rotation; this effect is very small. The averaged operator for this interaction must evidently have the form $\gamma\mathbf{KS}$, and its eigenvalues are given by (84.4) after setting $S = 1/2$ and $J = K \pm 1/2$. For $^2\Sigma$ terms we consequently obtain

$$E = U_e + \hbar\omega_e(v + 1/2) + B_e K(K + 1) \pm (\gamma/2)(K + 1/2) \tag{84.8}$$

(the constant $-\gamma/4$ has been included in U_e).

Problem

1. Determine the multiplet splitting of a $^3\Sigma$ term in case b (H. Kramers, 1929).

Solution. The multiplet splitting is governed by operator (84.7). To average it over the molecule's rotation, write the operator as $a_e n_i n_k S_i S_k$ and set $a_e = a(r_0)$. Since $\mathbf{S}$ is a conserved vector, the averaging acts only on the factor $n_i n_k$. The analogous formula derived in the problem to § 29 then gives

$$\overline{n_i n_k} = -\frac{\widehat{K}_i\widehat{K}_k + \widehat{K}_k\widehat{K}_i}{(2K-1)(2K+3)} + \cdots ;$$

terms proportional to δ_{ik} have not been displayed here, since their contribution to the energy would be independent of J and would therefore not produce the splitting of interest. Thus the splitting is determined by the operator

$$-a_e\frac{S_i S_k(\widehat{K}_i\widehat{K}_k + \widehat{K}_k\widehat{K}_i)}{(2K-1)(2K+3)}.$$

Since $\mathbf{S}$ commutes with $\mathbf{K}$,

$$S_i S_k \widehat{K}_i\widehat{K}_k = S_i\widehat{K}_i S_k\widehat{K}_k = (\mathbf{SK})^2,$$

where the eigenvalue of $\mathbf{SK}$ is given by (84.4). Further,

$$\begin{aligned} S_i S_k \widehat{K}_k\widehat{K}_i &= S_i S_k\widehat{K}_i\widehat{K}_k + iS_i S_k e_{kil}\widehat{K}_l \\ &= (\mathbf{SK})^2 - \frac{1}{2}(S_i S_k - S_k S_i)ie_{ikl}\widehat{K}_l \\ &= (\mathbf{SK})^2 + \frac{1}{2}e_{ikl}e_{ikm}S_m K_l = (\mathbf{SK})^2 + \mathbf{SK}. \end{aligned}$$

The three components E_K of the $^3\Sigma$ triplet ($S = 1$) correspond to $J = K, K \pm 1$. For the intervals between these components, we obtain

$$E_{K+1} - E_K = -a_e\frac{K+1}{2K+3}, \qquad E_{K-1} - E_K = -a_e\frac{K}{2K-1}.$$

2. Determine the energy of a doublet term (with $\Lambda \neq 0$) in cases intermediate between a and b (E. Hill, J. van Vleck, 1928).

SOLUTION. Since the rotational energy and the spin–axis interaction energy are assumed to be of the same order of magnitude, they must be treated together in perturbation theory. The perturbation operator is therefore[26]

$$\widehat{V} = B_e\widehat{\mathbf{K}}^2 + A_e\mathbf{n}\widehat{\mathbf{S}}.$$

As zeroth-approximation wave functions it is convenient to use the wave functions of states in which the angular momenta K and J have definite values (that is, the functions of case b). Since $S = 1/2$ for a doublet term, at a given J the quantum number K can take the values $K = J \pm 1/2$. To construct the secular equation, one must calculate the matrix elements $(nSKJ|V|nSK'J)$ (where n denotes the set of quantum numbers specifying the electronic term), with K, K' taking these values. The matrix of the operator $\mathbf{K}^2$ is diagonal, with diagonal elements $K(K+1)$. The matrix elements of $(\mathbf{nS})$ are calculated using the general formula (109.5), in which S, K, J take the roles of j_1, j_2, J; the required matrix elements of $\mathbf{n}$ are given by (87.4). The calculation yields the secular equation

$$\begin{vmatrix} B_e(J+1/2)(J+3/2) - \dfrac{A_e\Lambda}{2J+1} - E^{(1)} & \dfrac{A_e}{2J+1}\sqrt{(J+1/2)^2-\Lambda^2} \\ \dfrac{A_e}{2J+1}\sqrt{(J+1/2)^2-\Lambda^2} & B_e(J+1/2)(J-1/2) + \dfrac{A_e\Lambda}{2J+1} - E^{(1)} \end{vmatrix} = 0.$$

Solving this equation and adding $E^{(1)}$ to the unperturbed energy, we find

$$E = U_e + \hbar\omega_e(v+1/2) + B_eJ(J+1) \pm \sqrt{B_e^2(J+1/2)^2 - A_eB_e\Lambda + A_e^2/4}$$

(the constant $B_e/4$ has been included in U_e). Case a corresponds to $A_e \gg B_eJ$, while case b corresponds to the reverse inequality.

3. Determine the intervals between the components of a triplet $^3\Sigma$ level in the case intermediate between a and b.

SOLUTION. As in Problem 2, the rotational energy and the spin–spin interaction energy are treated together in perturbation theory. The perturbation operator is

$$\widehat{V} = B_e\widehat{\mathbf{K}}^2 + a_e(\mathbf{n}\widehat{\mathbf{S}})^2.$$

We use the wave functions of case b as zeroth-approximation wave functions. The matrix elements $(K|\mathbf{nS}|K')$ (all indices with respect to which the matrix is diagonal are suppressed) are again calculated from formulas (109.5) and (87.4), now with $\Lambda = 0$, $S = 1$. The nonzero elements are of the form

$$(J|\mathbf{nS}|J-1) = \sqrt{\frac{J+1}{2J+1}}, \qquad (J|\mathbf{nS}|J+1) = \sqrt{\frac{J}{2J+1}}.$$

[26] Averaging over the vibrations must precede averaging over the rotation. We have therefore replaced the functions $B(r)$ and $A(r)$ by the values B_e, A_e, retaining only the first terms of the expansion in ξ. The unperturbed energy levels are $E^{(0)} = U_e + \hbar\omega_e(v+1/2)$.

At a given J, the number K can take the values $K = J, J \pm 1$. For the matrix elements $(K|V|K')$ we find

$$\begin{aligned}
(J|V|J) &= B_e J(J+1) + a_e,\\
(J-1|V|J-1) &= B_e(J-1)J + a_e\frac{J+1}{2J+1},\\
(J+1|V|J+1) &= B_e(J+1)(J+2) + a_e\frac{J}{2J+1},\\
(J-1|V|J+1) = (J+1|V|J-1) &= a_e\frac{J(J+1)}{2J+1}.
\end{aligned}$$

The $K = J$ state is decoupled from the states with $K = J \pm 1$, and therefore gives the level $E_1 = (J|V|J)$. The other two levels, E_2 and E_3, follow from the quadratic secular equation for the two states $K = J - 1$ and $K = J + 1$, formed from the corresponding matrix elements. Since only the relative positions of the triplet components are of interest, subtract the same constant a_e from all three energies $E_{1,2,3}$. The result is

$$E_1 = B_e J(J+1), \qquad E_{2,3} = B_e(J^2+J+1) - \frac{a_e}{2} \pm \sqrt{B_e^2(2J+1)^2 - a_e B_e + \frac{a_e^2}{4}}.$$

In case b (small a_e), considering three levels with the same K and different J ($J = K, K\pm1$), we recover the formulas found in Problem 1.

§ 85. Multiplet terms. Cases c and d

Besides coupling cases a and b, and those intermediate between them, other types of coupling also exist. Their origin is as follows. The appearance of the quantum number Λ is due, in the final analysis, to the electric interaction between the two atoms in the molecule, which gives axial symmetry to the problem of determining the electronic terms (this interaction in a molecule is referred to as coupling of the orbital angular momentum to the axis). The magnitude of this interaction is measured by the separations between terms having different values of Λ. In all the preceding discussion, this interaction was tacitly assumed to be so strong that these separations are large both in comparison with the intervals of the multiplet splitting and with those of the rotational structure of the terms. There are, however, also converse situations in which the coupling of the orbital angular momentum to the axis is comparable with, or even small in comparison with, other effects. In such situations, of course, the projection of the orbital angular momentum on the axis cannot be regarded as conserved in any approximation, and the number Λ therefore loses its meaning.

If the coupling of the orbital angular momentum to the axis is weak compared with the spin–orbit coupling, this is called *case c*. It occurs in molecules containing a rare-earth atom. Such atoms have f electrons whose angular momenta are not compensated; their interaction with the molecular axis is weakened because the f electrons lie deep within the atom. Types of coupling intermediate between a and c occur in molecules composed of heavy atoms.

When the coupling of the orbital angular momentum to the axis is weak in comparison with the intervals of the rotational structure, it is called *case d*. This case occurs for high rotational levels (large J) of certain electronic terms of the lightest molecules (H_2, He_2). This class of terms involves a highly excited molecular electron; its coupling to the remaining electrons (the molecular "core," in the usual terminology) is so weak that its orbital angular momentum has no quantization along the molecular axis, whereas the core does have a definite axial angular-momentum projection Λ_{core}.

As the distance r between the nuclei increases, the interaction between the atoms weakens and ultimately becomes small in comparison with the spin–orbit interaction within the atoms. Consequently, when considering electronic terms at sufficiently large r, we shall be dealing with case c. This circumstance must be borne in mind when establishing the correspondence between the molecular electronic terms and the atomic states obtained as $r \to \infty$.

In § 80, we examined this correspondence with the spin–orbit interaction neglected. Once the fine structure of the terms is included, one must also determine how the values J_1 and J_2 of the total angular momenta of the isolated atoms correspond to the molecular quantum number Ω. We give only the results here, since the reasoning is entirely analogous to that used in § 80. If the molecule consists of unlike atoms, the possible values of Ω[27] obtained by combining atoms with angular momenta J_1 and J_2 ($J_1 \geqslant J_2$) are determined by the same scheme (80.1), with J_1, J_2 written in place of L_1, L_2 and $|\Omega|$ substituted for Λ.

The sole difference is that, when $J_1 + J_2$ is half-integral, the smallest value of $|\Omega|$ is $1/2$ rather than zero as shown in the scheme.

When $J_1 + J_2$ is integral, however, there are $2J_2 + 1$ terms with $\Omega = 0$; for these terms (as for the Σ terms when fine structure is neglected), the question of their sign arises.

If J_1 and J_2 are both half-integral, the number $2J_2 + 1$ is even, and there are equal numbers of terms that we denote conventionally by 0^+ and 0^-.

If J_1 and J_2 are both integral, then $J_2 + 1$ of the terms are 0^+ and J_2 are 0^- when $(-1)^{J_1+J_2}P_1P_2 = 1$, with the assignments reversed when $(-1)^{J_1+J_2}P_1P_2 = -1$.

When identical atoms occupying different states form the molecule, the resulting molecular states are the same as for unlike atoms, except that the total number of terms is doubled: every term occurs once with even parity and once with odd parity.

Finally, if the molecule consists of identical atoms in identical states (with angular momenta $J_1 = J_2 = J$), the total number of states remains the same as for unlike atoms, while their distribution according to parity is such that

$$\begin{array}{lll}
\text{if } J \text{ is integral,} & \Omega \text{ even:} & N_g = N_u + 1, \\
& \Omega \text{ odd:} & N_g = N_u, \\
\text{if } J \text{ is half-integral,} & \Omega \text{ even:} & N_u = N_g, \\
& \Omega \text{ odd:} & N_u = N_g + 1.
\end{array}$$

All the 0^+ terms are then even, and all the 0^- terms are odd.

As the nuclei approach one another, coupling of type c usually changes into coupling of type a[28]. The following interesting situation may then arise.

[27]When the two total atomic angular momenta $\mathbf{J}_1$ and $\mathbf{J}_2$ are added to form the resultant angular momentum $\boldsymbol{\Omega}$, the sign of Ω is plainly immaterial.

[28]A general correspondence between the classifications of terms in cases a and c cannot be established. It

As already stated, a term with $\Lambda = 0$ belongs to case *b*; from the standpoint of the case-*a* classification, this means that the multiplet levels with different values of Ω (and the same $\Lambda = 0$) correspond to the same energy. Such levels, however, can arise when atoms in different fine-structure states approach one another.

Thus, different pairs of fine-structure atomic states may correspond to one and the same molecular term. An analogous situation can arise for terms with $\Omega = 0$ that, as the nuclei approach one another, turn into a molecular term with $\Lambda \neq 0$ (and hence $\Sigma = -\Lambda$); these levels are doubly degenerate, since in case *a* the terms 0^+ and 0^- (which may originate from different pairs of atomic states) correspond to the same energy[29].

§ 86. Symmetry of molecular terms

In § 78 we have already examined certain symmetry properties of the terms of a diatomic molecule. These properties described the behavior of the wavefunctions under transformations that leave the nuclear coordinates unaffected. Thus, molecular symmetry under reflection in a plane containing its axis accounts for the distinction between the Σ^+ and Σ^- terms; symmetry under reversal of the signs of all electron coordinates (for a molecule made up of identical atoms)[30] leads to the division of the terms into even and odd ones.

These symmetry properties characterize the electronic terms and are the same for every rotational level belonging to one and the same electronic term.

Furthermore, the states of a molecule (like those of any system of particles in general—see § 30) are characterized by their behavior under inversion—the simultaneous reversal of the signs of the coordinates of all electrons and nuclei.

Accordingly, all molecular terms fall into two classes: *positive* terms, whose wavefunctions remain unchanged when the signs of the electron and nuclear coordinates are reversed, and *negative* terms, whose wavefunctions change sign under inversion.[31]

When $\Lambda \neq 0$, each term is doubly degenerate, corresponding to the two possible orientations of the angular momentum relative to the molecular axis. Under inversion the angular momentum itself does not change sign, whereas the direction of the molecular axis is reversed (the atoms exchange places!); consequently, the orientation of the angular momentum Λ relative to the molecule is reversed as well. The two wavefunctions belonging to the given energy level therefore transform into one another, and one can always form from them a linear combination invariant under inversion and another combination that changes sign under this transformation. Hence each term gives two states, one positive and the other negative. In fact, every term with $\Lambda \neq 0$ nevertheless undergoes splitting (see § 88), so these two states correspond to different energy values.

Σ terms require separate consideration in order to determine their sign. First of all, it is clear that spin has no bearing on the sign of a term: inversion acts only on the particle

requires a specific examination of the potential-energy curves, taking into account the rule that levels of the same symmetry do not cross (§ 79).

[29] Here we neglect the so-called Λ-doubling (see § 88).

[30] The coordinate origin is assumed to lie on the molecular axis, midway between the two nuclei.

[31] We follow the established terminology. It is unfortunate, since for an atom the behavior of terms under inversion is described by their parity, rather than by a sign.

For Σ terms, the sign discussed here must not be confused with the + and − signs written as superscripts!

coordinates and leaves the spin part of the wave function unchanged. Consequently, all components in the multiplet structure of any given term carry the same sign. In other words, the sign of a term depends only on K, and not on J[32].

The molecular wave function is the product of the electronic and nuclear wave functions. The result established in § 82 is that, in a Σ state, the nuclear motion is equivalent to that of a single particle with orbital angular momentum K moving in the spherically symmetric field $U(r)$. It follows that reversing the signs of the coordinates multiplies the nuclear wave function by $(-1)^K$ (see (30.7)). The electronic wave function characterizes the electronic term; to establish how it behaves under inversion, it must be considered in a coordinate system rigidly attached to the nuclei and rotating with them. Let xyz be a coordinate system fixed in space, and let $\xi\eta\zeta$ be a rotating coordinate system in which the molecule as a whole is at rest. Choose the directions of the $\xi\eta\zeta$ axes so that the ζ axis coincides with the molecular axis and points, say, from nucleus 1 to nucleus 2, while the positive directions of the $\xi\eta\zeta$ axes have the same mutual orientation as those of the xyz system (that is, if the xyz system is right-handed, the $\xi\eta\zeta$ system must also be right-handed). Inversion reverses the directions of the xyz axes, transforming a right-handed system into a left-handed one. At the same time, the $\xi\eta\zeta$ system must become left-handed as well. The ζ axis, however, is rigidly attached to the nuclei and retains its former direction; the direction of one of the axes ξ or η must therefore be reversed. Thus inversion in the fixed coordinate system is equivalent, in the moving system, to reflection in a plane containing the molecular axis. Under this reflection, however, the electronic wave function of a Σ^+ term remains unchanged, whereas that of a Σ^- term changes sign.

It follows that the sign of the rotational components of a Σ^+ term is fixed by the factor $(-1)^K$: every level with even K is positive, and every level with odd K is negative. For a Σ^- term, the sign of the rotational levels is fixed by $(-1)^{K+1}$; the levels with even K are all negative, while those with odd K are all positive.

If a molecule is composed of identical atoms[33], its Hamiltonian is also invariant under interchange of the coordinates of the two nuclei. A term is called symmetric with respect to the nuclei if its wave function is unchanged when the nuclei are interchanged, and antisymmetric if the wave function changes sign. Symmetry with respect to the nuclei is closely connected with the parity and the sign of the term. Interchanging the nuclear coordinates is equivalent to reversing the coordinates of all particles (electrons and nuclei), followed by reversal of the electronic coordinates alone. It follows that a term is symmetric with respect to the nuclei if it is even (odd) and at the same time positive (negative). If, on the other hand, the term is even (odd) and simultaneously negative (positive), it is antisymmetric with respect to the nuclei.

A general theorem established at the end of § 62 states that the coordinate wavefunction for a system of two identical particles is symmetric when the system's total spin is even and antisymmetric when it is odd. Applying this result to the two nuclei of a molecule composed of identical atoms, we find that the symmetry of the term is connected with the parity of the total spin I obtained by coupling the spins i of the two nuclei. The term is

[32]Recall that Σ terms usually belong to case b, so the quantum numbers K and J must be used.

[33]The two atoms must belong not merely to the same element, but also to the same isotope of that element.

symmetric for even I and antisymmetric for odd I[34]. In particular, if the nuclei have no spin ($i = 0$), then I is likewise zero; the molecule therefore has no antisymmetric terms at all. We thus see that nuclear spin has a substantial indirect effect on molecular terms, although its direct effect (the hyperfine structure of the terms) is utterly negligible.

Taking the nuclear spin into account introduces an additional degeneracy of the levels. In the same § 62, the numbers of states with even and odd values of I obtained by adding two spins i were calculated. Thus, for half-integral i, the number of states with even I is $i(2i + 1)$, while that with odd I is $(i + 1)(2i + 1)$. In view of the foregoing, we conclude that for half-integral i the ratio of the degeneracy multiplicities g_s, g_a[35] of the symmetric and antisymmetric terms is

$$\frac{g_s}{g_a} = \frac{i}{i+1}. \tag{86.1}$$

For integral i, we similarly find this ratio to be

$$\frac{g_s}{g_a} = \frac{i+1}{i}. \tag{86.2}$$

For a Σ^+ term, we found the signs of its rotational components to be set by $(-1)^K$. Thus, for example, the rotational components of the ${}^1\Sigma_g^+$ term are positive and hence symmetric for even K, whereas for odd K they are negative and consequently antisymmetric. In view of the results obtained above, we conclude that, for successive values of K, the nuclear statistical weights of the rotational components of the ${}^1\Sigma_g^+$ level alternate in the ratio (86.1) or (86.2). The same situation occurs for the ${}^1\Sigma_u^-$ levels, as well as for ${}^1\Sigma_g^-$ and ${}^1\Sigma_u^+$. In particular, for $i = 0$, the statistical weights vanish for the levels with even K of the ${}^1\Sigma_u^-$ and ${}^1\Sigma_g^-$ terms, and for the levels with odd K of the ${}^1\Sigma_g^+$ and ${}^1\Sigma_u^+$ terms. In other words, in the electronic states ${}^1\Sigma_u^-$ and ${}^1\Sigma_g^-$ there are no rotational states with even K, while in the states ${}^1\Sigma_g^+$ and ${}^1\Sigma_u^+$ there are no rotational states with odd K.

Because the interaction of the nuclear spins with the electrons is exceedingly weak, the probability of changing I is very small even in molecular collisions. Molecules that differ in the parity of I, and accordingly possess only symmetric or only antisymmetric terms, therefore behave in practice as distinct modifications of a substance. Examples are the so-called orthohydrogen and parahydrogen: in a molecule of the former, the spins $i = 1/2$ of the two nuclei are parallel ($I = 1$), whereas in one of the latter they are antiparallel ($I = 0$).

§ 87. Matrix elements for a diatomic molecule

This section gives several general formulas for matrix elements of physical quantities pertaining to a diatomic molecule. We first consider matrix elements for transitions between zero-spin states.

[34]In view of the relation between the parity, sign, and symmetry of terms, it follows that, for even total nuclear spin I, positive levels are even and negative levels are odd; for odd I, the reverse holds.

[35]In this connection, the degeneracy multiplicity of a level is often called its *statistical weight*. Equations (86.1) and (86.2) determine the ratios of the nuclear statistical weights of symmetric and antisymmetric levels.

Let **A** be a vector physical quantity that characterizes the molecule when its nuclei are fixed (for example, its electric or magnetic dipole moment). We first consider this quantity in the $\xi\eta\zeta$ coordinate system, which rotates with the molecule and whose ζ axis lies along the molecular axis. The molecule's angular momentum relative to this frame (that is, the electronic angular momentum **L**) is not conserved in its entirety, whereas its ζ component is conserved. The selection rules associated with the quantum number $L_\zeta = \Lambda$ therefore continue to apply (they coincide with the selection rules for M in § 29). Hence the vector can have nonzero matrix elements only of the following kinds:

$$\langle n'\Lambda|A_\zeta|n\Lambda\rangle, \quad \langle n'\Lambda|A_\xi + iA_\eta|n, \Lambda-1\rangle, \quad \langle n', \Lambda-1|A_\xi - iA_\eta|n\Lambda\rangle \tag{87.1}$$

(here n labels the electronic terms for a specified Λ).

When both terms are Σ terms, the selection rule associated with reflection symmetry in a plane containing the molecular axis must also be taken into account. Under such a reflection, the ζ component of an ordinary (polar) vector remains unchanged, whereas that of an axial vector reverses sign. It follows that, for a polar vector, A_ζ has nonzero matrix elements only in the transitions $\Sigma^+ \to \Sigma^+$ and $\Sigma^- \to \Sigma^-$, while for an axial vector this occurs for $\Sigma^+ \to \Sigma^-$. We do not discuss the components A_ξ, A_η, since transitions without a change in Λ are altogether impossible for them.

For a molecule made up of identical atoms, there is an additional selection rule involving parity. The components of a polar vector reverse sign under inversion. Its matrix elements can therefore be nonzero only for transitions between states of opposite parity (the reverse applies to an axial vector). In particular, every diagonal matrix element of a polar-vector component vanishes identically.

The relation between the matrix elements (87.1) and the matrix elements of the same vector in the space-fixed coordinate system xyz follows from the general formulas derived below (in § 110) for an arbitrary axially symmetric physical system.

On separating the dependence on the quantum number M_K that is common to any vector (the z projection of the total angular momentum of the molecule, **K**), one is left with the reduced matrix elements $\langle n'K'\Lambda'\|A\|nK\Lambda\rangle$. Their relation to the matrix elements (87.1) is specified by formula (110.7), with $k = k' = 1$ (as appropriate to a vector) and the corresponding relabeling of the quantum numbers (recall that, by (82.4), Λ equals the ζ component of the total angular momentum **K**). Using relation (107.1) between the components of a rank-one spherical tensor and the Cartesian components of a vector, together with the values of the $3j$ symbols in Table 9 (p. 530), we obtain the following formulas for matrix elements diagonal in Λ:

$$\begin{aligned} \langle n'K\Lambda\|A\|nK\Lambda\rangle &= \Lambda\sqrt{\frac{2K+1}{K(K+1)}}\langle n'\Lambda|A_\zeta|n\Lambda\rangle, \\ \langle n', K-1, \Lambda\|A\|nK\Lambda\rangle &= i\sqrt{\frac{K^2-\Lambda^2}{K}}\langle n'\Lambda|A_\zeta|n\Lambda\rangle \end{aligned} \tag{87.2}$$

and, for elements not diagonal in Λ,

$$\begin{aligned}
\langle n'K\Lambda\|A\|nK,\Lambda-1\rangle &= \left[\frac{(2K+1)(K+\Lambda)(K-\Lambda+1)}{4K(K+1)}\right]^{1/2} \\
&\quad\times\langle n'\Lambda|A_\xi+iA_\eta|n,\Lambda-1\rangle, \\
\langle n'K\Lambda\|A\|n,K-1,\Lambda-1\rangle &= i\left[\frac{(K+\Lambda)(K+\Lambda-1)}{4K}\right]^{1/2} \\
&\quad\times\langle n'\Lambda|A_\xi+iA_\eta|n,\Lambda-1\rangle, \\
\langle n',K-1,\Lambda\|A\|nK,\Lambda-1\rangle &= i\left[\frac{(K-\Lambda)(K-\Lambda+1)}{4K}\right]^{1/2} \\
&\quad\times\langle n'\Lambda|A_\xi+iA_\eta|n,\Lambda-1\rangle.
\end{aligned}\tag{87.3}$$

The other nonzero elements follow from those displayed above upon using the Hermiticity relations for the reduced matrix elements,

$$\langle nK\Lambda\|A\|n'K'\Lambda'\rangle = \langle n'K'\Lambda'\|A\|nK\Lambda\rangle^*,$$

and for the matrix elements in the $\xi\eta\zeta$ system,

$$\langle n\Lambda|A_\xi-iA_\eta|n'\Lambda'\rangle = \langle n'\Lambda'|A_\xi+iA_\eta|n\Lambda\rangle^*, \qquad \langle n\Lambda|A_\zeta|n'\Lambda'\rangle = \langle n'\Lambda'|A_\zeta|n\Lambda\rangle^*.$$

We record separately the formulas for matrix elements of the vector $\mathbf{A}=\mathbf{n}$, the unit vector along the molecular axis. In this case simply $A_\xi=A_\eta=0$, $A_\zeta=1$, so in the $\xi\eta\zeta$ system only the diagonal elements are nonzero: $\langle n\Lambda|A_\zeta|n\Lambda\rangle=1$. The reduced matrix elements are diagonal in every index except K; displaying only that index, we have

$$\langle K\|n\|K\rangle = \Lambda\sqrt{\frac{2K+1}{K(K+1)}}, \qquad \langle K-1\|n\|K\rangle = i\sqrt{\frac{K^2-\Lambda^2}{K}}. \tag{87.4}$$

(H. Hönl, F. London, 1925). For $\Lambda=0$, these formulas give

$$\langle K\|n\|K\rangle = 0, \qquad \langle K-1\|n\|K\rangle = i\sqrt{K},$$

which is precisely what is expected for the matrix elements of a unit vector in motion in a centrally symmetric field (see (29.14)).

We now indicate how the formulas obtained above are to be modified for transitions between states of nonzero spin. Here it is essential whether the states belong to case a or to case b.

If both states belong to case a, the changes in the formulas are essentially only notational. The quantum numbers K and M_K do not exist; in their place one has the total angular momentum J and its z projection M_J. In addition, the numbers S and $\Omega=\Lambda+\Sigma$ are introduced, so that the reduced matrix elements are written as

$$\langle n'J'S'\Omega'\Lambda'\|A\|nJS\Omega\Lambda\rangle.$$

Let $\mathbf{A}$ be any orbital vector (that is, one independent of spin). Its operator commutes with the spin operator $\widehat{\mathbf{S}}$, and its matrix is therefore diagonal in the quantum numbers S

and $S_\zeta = \Sigma$; hence the quantum number $\Omega = \Lambda + \Sigma$ changes together with Λ (that is, $\Omega' - \Omega = \Lambda' - \Lambda$). Equations (87.2)–(87.4) change only in that indices are added to the matrix elements and K, Λ must be replaced by J, Ω in all the other factors. For example, the first of equations (87.2) is replaced by

$$\langle n'JS\Omega\Lambda\|A\|nJS\Omega\Lambda\rangle = \Omega\sqrt{\frac{2J+1}{J(J+1)}}\langle n'\Omega\Lambda|A_\zeta|n\Omega\Lambda\rangle$$

(the diagonal index S has been omitted).

Now let $\mathbf{A} = \mathbf{S}$. Since the spin operator commutes both with the orbital angular momentum and with the Hamiltonian, its matrix is diagonal in n, Λ. It is not, however, diagonal in S and Σ (or Ω). For

the transitions $S, \Sigma \to S', \Sigma'$, the matrix elements of the components A_ξ, A_η, A_ζ are determined by (27.13), with S, Σ written in place of L, M. The transformation to the coordinate system xyz is then performed using (87.2), (87.3), with K, Λ replaced by J, Ω. In this manner, for example, we obtain

$$\begin{aligned}\langle J\Omega\|S\|J,\Omega-1\rangle &= \left[\frac{(2J+1)(J+\Omega)(J-\Omega+1)}{4J(J+1)}\right]^{1/2}\langle\Omega|S_\xi + iS_\eta|\Omega-1\rangle \\ &= \left[\frac{(2J+1)(J+\Omega)(J-\Omega+1)(S+\Sigma)(S-\Sigma+1)}{4J(J+1)}\right]^{1/2}.\end{aligned} \tag{87.5}$$

(the diagonal indices n, S, Λ have been omitted).

Next suppose that both states belong to case b, and that $\mathbf{A}$ is an orbital vector. The matrix elements are calculated in two stages. First we consider the rotating molecule without allowing for the coupling of $\mathbf{S}$ and $\mathbf{K}$; the matrix elements are diagonal in S and are given by the same formulas (87.2), (87.3). At the second stage, $\mathbf{K}$ is coupled with $\mathbf{S}$ to form the total angular momentum $\mathbf{J}$, and the transformation to the new matrix elements is made by means of the general formulas (109.3) (with K, S, J playing the parts of j_1, j_2, J in those formulas). Thus, for elements diagonal in J, K, Λ, we first find

$$\langle n'JK\Lambda\|A\|nJK\Lambda\rangle = (-1)^{K+J+S+1}(2J+1)\begin{Bmatrix} K & J & S \\ J & K & 1\end{Bmatrix}\langle n'K\Lambda\|A\|nK\Lambda\rangle$$

and then, using the value of the $6j$ symbol from Table 10 (p. 542) and the reduced matrix element from (87.2), we finally obtain

$$\langle n'JK\Lambda\|A\|nJK\Lambda\rangle = \Lambda\left[\frac{2J+1}{J(J+1)}\right]^{1/2}\frac{J(J+1)+K(K+1)-S(S+1)}{2K(K+1)}\langle n'\Lambda|A_\zeta|n\Lambda\rangle.$$

Matrix elements for transitions between states belonging respectively to cases a and b are calculated in the same manner; we shall not pursue that calculation here.

Problem

Determine the Stark splitting of the terms of a diatomic molecule having a permanent dipole moment, when the term belongs to case *a*.

Solution. The energy of a dipole **d** in an electric field $\boldsymbol{\mathcal{E}}$ is $-\mathbf{d}\boldsymbol{\mathcal{E}}$. By symmetry, the dipole moment of a diatomic molecule is plainly directed along its axis: $\mathbf{d} = d\mathbf{n}$ (where d is constant). Choosing the field direction as the z axis, we obtain the perturbation operator in the form $-dn_z\mathcal{E}$.

Evaluating the diagonal matrix elements of n_z from the formulas derived in the text, we find that in case a the splitting of the levels is given by[36]

$$\Delta E = -\mathcal{E} d M_J \frac{\Omega}{J(J+1)}.$$

Problem

The same problem for a term belonging to case b (with $\Lambda \neq 0$).

Solution. In the same manner we find

$$\Delta E = -\mathcal{E} d M_J \Lambda \frac{J(J+1) - S(S+1) + K(K+1)}{2K(K+1)J(J+1)}.$$

Problem

The same problem for a ${}^1\Sigma$ term.

Solution. When $\Lambda = 0$, the linear effect is absent, and second-order perturbation theory must be used. In carrying out the sum in the general formula (38.10), it is sufficient to retain only the terms corresponding to transitions between rotational components of the given electronic term (in the other terms, the energy differences in the denominators are large). We thus find

$$\Delta E = d^2\mathcal{E}^2 \left\{ \frac{|\langle KM_K|n_z|K-1, M_K\rangle|^2}{E_K - E_{K-1}} + \frac{|\langle KM_K|n_z|K+1, M_K\rangle|^2}{E_K - E_{K+1}} \right\},$$

where $E_K = BK(K+1)$. A straightforward calculation gives

$$\Delta E = \frac{d^2\mathcal{E}^2}{B} \frac{K(K+1) - 3M_K^2}{2K(K+1)(2K-1)(2K+3)}.$$

---

[36] This may appear to conflict with the general statement (see § 76) that there is no linear Stark effect. In fact, there is of course no conflict, because here the linear effect is associated with the twofold degeneracy of the levels with $\Omega \neq 0$; the formula obtained is consequently applicable provided that the energy of the Stark splitting is large compared with the energy of the so-called Λ-doubling (see § 88).

§ 88. Λ-doubling

The twofold degeneracy of terms with $\Lambda \neq 0$ (see § 78) is actually approximate. It holds only insofar as the influence of molecular rotation on the electronic state (and higher-order spin–orbit effects) is neglected, as throughout the preceding theory. Allowing for the interaction of the electronic state with rotation splits a term with $\Lambda \neq 0$ into two closely spaced levels. This phenomenon is called Λ-doubling (E. Hill, J. van Vleck, R. Kronig, 1928).

We again begin the quantitative treatment with singlet terms ($S = 0$). In § 82 the rotational-level energies were calculated to first order in perturbation theory by finding diagonal matrix elements (the mean value) of the operator

$$B(r)(\widehat{\mathbf{K}} - \widehat{\mathbf{L}})^2.$$

For the subsequent orders we must consider elements of this operator nondiagonal in Λ. The operators $\widehat{\mathbf{K}}^2$ and $\widehat{\mathbf{L}}^2$ are diagonal in Λ, so only $2B\widehat{\mathbf{K}}\widehat{\mathbf{L}}$ need be considered.

The matrix elements of $\widehat{\mathbf{K}}\widehat{\mathbf{L}}$ are conveniently calculated with (29.12), setting $\mathbf{A} = \mathbf{K}$, $\mathbf{B} = \mathbf{L}$; K, M_K play the roles of L, M, while n, Λ replaces n, where n denotes the set of quantum numbers (excluding Λ) that specifies the electronic term. Since the matrix of the conserved vector **K** is diagonal in n, Λ, while the matrix of **L** has nondiagonal elements only for transitions changing Λ by one (compare the discussion of an arbitrary vector **A** in § 87), formulas (87.3) give

$$\langle n'\Lambda K M_K|\mathbf{KL}|n, \Lambda - 1, K M_K\rangle = \frac{1}{2}\langle n'\Lambda|L_\xi + iL_\eta|n, \Lambda - 1\rangle\sqrt{(K + \Lambda)(K + 1 - \Lambda)}. \tag{88.1}$$

There are no matrix elements corresponding to a larger change in Λ.

The perturbing action of the elements for $\Lambda \to \Lambda - 1$ can produce an energy difference between the states with $\pm\Lambda$ only in order 2Λ of perturbation theory. The effect is consequently proportional to $B^{2\Lambda}$, that is, to $(m/M)^{2\Lambda}$ (M is the nuclear mass and m the electron mass). For $\Lambda > 1$ this quantity is too small to be of interest. Thus Λ-doubling is significant only for Π terms ($\Lambda = 1$), which are considered below.

For $\Lambda = 1$ we must go to second order. The corrections to the energy eigenvalues may be found from the general formula (38.10). The denominators of its summands contain energy differences of the form $E_{n,\Lambda,K} - E_{n',\Lambda-1,K}$. The terms involving K cancel in these differences, because at a fixed internuclear distance r the rotational energy has the same value $B(r)K(K + 1)$ for every term.

The dependence of the required splitting ΔE on K is therefore determined entirely by the squares of the matrix elements in the numerators. These include squares of the elements for transitions changing Λ from 1 to 0 and from 0 to −1; by (88.1) both have the same dependence on K. The splitting of a $^1\Pi$ term is consequently

$$\Delta E = \text{const}\ K(K + 1), \tag{88.2}$$

where, in order of magnitude, const $\sim B^2/\varepsilon$, and ε is of the order of the separations between neighboring electronic terms.

We turn to terms with nonzero spin ($^2\Pi$ and $^3\Pi$ terms; higher values of S scarcely occur). If the term belongs to case b, the multiplet splitting has no effect at all on the Λ-doubling of the rotational levels, which is still given by (88.2).

In case a, by contrast, spin has an essential influence. Each electronic term is characterized not only by Λ but also by Ω. Merely replacing Λ by $-\Lambda$ changes $\Omega = \Lambda + \Sigma$, and therefore gives an entirely different term. The mutually degenerate states are those with Λ, Ω and $-\Lambda, -\Omega$. This degeneracy can be removed not only by the interaction between orbital angular momentum and molecular rotation considered above, but also by the spin–orbit interaction. Conservation of the projection Ω of the total angular momentum on the molecular axis is an exact conservation law (for fixed nuclei), and hence cannot be violated by the spin–orbit interaction. The latter can, however, change Λ and Σ simultaneously, having matrix elements for the corresponding transitions, in such a way that Ω remains unchanged. Acting alone, or together with the orbit–rotation interaction (which changes Λ without changing Σ), this effect can produce Λ-doubling.

First consider ${}^2\Pi$ terms. For a ${}^2\Pi_{1/2}$ term ($\Lambda = 1, \Sigma = -1/2, \Omega = 1/2$), the splitting arises when both the spin–orbit and orbit–rotation interactions are included, each to first order. The former produces the transition $\Lambda = 1, \Sigma = -1/2 \to \Lambda = 0, \Sigma = 1/2$; the latter then takes the state $\Lambda = 0, \Sigma = 1/2$ into $\Lambda = -1, \Sigma = 1/2$, which differs from the initial state by reversal of the signs of Λ and Ω. The spin–orbit matrix elements do not depend on the rotational quantum number J, while the dependence of the orbit–rotation matrix elements is given by (88.1), with K, Λ under the radical replaced by J, Ω. Thus the Λ-doubling of the ${}^2\Pi_{1/2}$ term is

$$\Delta E_{1/2} = \text{const}\,(J + 1/2), \tag{88.3}$$

where const $\sim AB/\varepsilon$. For a ${}^2\Pi_{3/2}$ term, however, the splitting can arise only in higher orders, so in practice $\Delta E_{3/2} = 0$.

Finally consider ${}^3\Pi$ terms. For a ${}^3\Pi_0$ term ($\Lambda = 1, \Sigma = -1$), the splitting is produced by the spin–orbit interaction in second order (through the transitions $\Lambda = 1, \Sigma = -1 \to \Lambda = 0, \Sigma = 0 \to \Lambda = -1, \Sigma = 1$). Hence in this case the Λ-doubling is entirely independent of J:

$$\Delta E_0 = \text{const}, \tag{88.4}$$

where const $\sim A^2/\varepsilon$. For a ${}^3\Pi_1$ term, $\Sigma = 0$, so spin does not affect the splitting and we recover a formula of the form (88.2), with K replaced by J:

$$\Delta E_1 = \text{const}\ J(J + 1). \tag{88.5}$$

For a ${}^3\Pi_2$ term still higher orders are required, and one may take $\Delta E_2 = 0$.

Of the two levels produced by Λ-doubling, one is always positive and the other negative; this was already discussed in § 6. An analysis of the molecular wave functions determines the pattern in which positive and negative levels alternate. We give only the result of that analysis here.[37] If for some J the positive level lies below the negative one, their order is reversed in the doublet with $J + 1$: the positive level lies above the negative one, and so on. The ordering alternates with successive values of the total angular momentum (for case a terms; in case b the same statement holds for successive values of K).

[37] See the article by Wigner and Witmer cited on p. 376.

PROBLEM

Determine the Λ-splitting of a $^1\Delta$ term.

SOLUTION. Here the effect first appears in fourth-order perturbation theory. Its dependence on K is determined by products of four matrix elements (88.1), for transitions changing Λ as $2 \to 1$, $1 \to 0$, $0 \to -1$, $-1 \to -2$. This gives

$$\Delta E = \text{const}\,(K-1)K(K+1)(K+2),$$

where const $\sim B^4/\varepsilon^3$.

§ 89. Interaction between atoms at large distances

Consider two atoms whose separation is large compared with their dimensions, and determine their interaction energy. Equivalently, the problem is to find the form of the electronic terms at large internuclear distances.

To solve the problem, we use perturbation theory: the two isolated atoms form the unperturbed system, while the potential energy of their electric interaction serves as the perturbation operator. As is known (see Vol. II, §§ 41 and 42), the electric interaction of two charge systems separated by a large distance r can be expanded in powers of $1/r$; its successive terms describe the interactions of the two systems' total charges, dipole moments, quadrupole moments, and so on. For neutral atoms the total charges vanish. Here the expansion starts with the dipole–dipole interaction ($\sim 1/r^3$), followed by dipole–quadrupole terms ($\sim 1/r^4$), quadrupole–quadrupole (and dipole–octupole) terms ($\sim 1/r^5$), and so forth.

Suppose first that both atoms are in S states. It is readily seen that their interaction then has no effect in first-order perturbation theory. Indeed, to first order the interaction energy is the diagonal matrix element of the perturbation operator evaluated with the system's unperturbed wave functions (themselves products of the wave functions of the two atoms)[38]. In an S state, however, the diagonal matrix elements—that is, the expectation values of the atom's dipole, quadrupole, and higher moments—vanish as a direct consequence of the spherical symmetry of its charge-density distribution. Consequently, every term in the expansion of the perturbation operator in powers of $1/r$ gives zero in first-order perturbation theory[39].

In second order it is enough to retain in the perturbation operator the dipole interaction, since it decreases most slowly as r increases, namely the term

$$V = \frac{\mathbf{d}_1\mathbf{d}_2 - 3(\mathbf{d}_1\mathbf{n})(\mathbf{d}_2\mathbf{n})}{r^3} \tag{89.1}$$

[38] Exchange effects that decrease exponentially with distance are omitted here (cf. Problem 1 in § 62 and the problem in § 81).

[39] This does not, of course, mean that the expectation value of the atoms' interaction energy is exactly zero. It decreases exponentially with distance, and hence faster than any finite power of $1/r$; this accounts for the vanishing of every term in the expansion. The point is that expanding the interaction operator in multipole moments presupposes that the charges of the two atoms are separated from one another by the large distance r. The quantum-mechanical electron-density distribution, however, remains finite at large distances, although its values there are exponentially small.

(**n** is the unit vector directed from atom 1 to atom 2). Because the off-diagonal matrix elements of the dipole moment are generally nonzero, second-order perturbation theory produces a nonzero result which, being quadratic in V, is proportional to $1/r^6$. The second-order correction to the lowest eigenvalue is always negative (see § 38). Thus, for atoms in their ground states, the interaction energy has the form

$$U(r) = -\frac{\text{const}}{r^6}, \tag{89.2}$$

where const is positive[40] (F. London, 1928).

Thus, two atoms in ground S states, separated by a large distance, attract one another with a force $-dU/dr$ inversely proportional to the seventh power of their separation. Attractive forces acting between distant atoms are commonly called *van der Waals forces*. Such forces also produce a well in the potential-energy curves of electronic terms for atoms that do not form a stable molecule. These wells are, however, very shallow (their depths amount to only tenths or even hundredths of an electron-volt), and they occur at separations several times greater than the interatomic distances in stable molecules.

When only one atom is in an S state, the same result (89.2) is obtained for their interaction energy, because the disappearance of the dipole and higher moments of even one atom suffices to make the first-order contribution vanish. The constant in the numerator of (89.2) then depends not only on the states of both atoms but also on their relative orientation, that is, on the value Ω of the angular-momentum projection upon the axis joining the atoms.

The situation changes if both atoms possess nonzero orbital and total angular momenta. The mean dipole moment is zero in every atomic state (see § 75). By contrast, the mean values of the quadrupole moment (in states with $L \neq 0, J \neq 0, 1/2$) are different from zero. Consequently, the quadrupole–quadrupole term in the perturbation operator gives a nonzero result already in first order, and the atomic interaction energy decreases with the fifth rather than the sixth power of the distance:

$$U(r) = \frac{\text{const}}{r^5}. \tag{89.3}$$

The constant here may be either positive or negative; accordingly, either attraction or repulsion may occur. Here too, the constant is determined both by the individual atomic states and by the state of the system formed by the two atoms.

The interaction of two identical atoms occupying different states constitutes a special case. Here the unperturbed system (two isolated atoms) has an additional degeneracy because the two states can be interchanged between the atoms. Correspondingly, the first-order correction is determined by a secular equation containing both diagonal and off-diagonal matrix elements of the perturbation. For two atomic states of opposite parity whose values of L differ by 0 or ± 1, without both being zero (and with the same requirement imposed on J), the off-diagonal dipole-moment matrix elements associated with transitions between them are in general nonzero. The dipole term in the perturbation

[40] For example, its values (in atomic units) for pairs of atoms are: hydrogen, 6.5; helium, 1.5; argon, 68; krypton, 130.

operator therefore produces an effect already in first order. Thus, in this case the atomic interaction energy is proportional to $1/r^3$:

$$U(r) = \frac{\text{const}}{r^3}; \tag{89.4}$$

the constant may have either sign.

Usually, however, it is the interaction averaged over all possible orientations of the atomic angular momenta that is of interest (this formulation applies, for example, to the problem of atomic interactions in a gas). After this averaging, the mean values of every multipole moment vanish. Along with them, all first-order perturbation effects in the atomic interaction that are linear in these moments also vanish. Thus, after orientation averaging, the long-range forces between atoms obey law (89.2) in every case [41].

Let us also consider the related question of how a neutral atom interacts with an ion.

In first-order perturbation theory, this interaction is given by the expectation value of operator (76.8), the energy of a quadrupole in the ion's Coulomb field.

Because the ion's potential has the dependence $\varphi \sim 1/r$, the atom–ion interaction energy is proportional to $1/r^3$.

This effect occurs, however, only when the atom has a nonzero mean quadrupole moment.

Even in that case, it disappears upon averaging over every direction of the atomic angular momentum $\mathbf{J}$.

The next interaction in powers of $1/r$, which is always nonzero, arises in second-order perturbation theory from the dipole operator (76.1).

Since the ion's field strength varies as $1/r^2$, the energy of this interaction is proportional to $1/r^4$.

For an atom in an S-state, this energy is expressed through its polarizability α as

$$U = -\alpha e^2/2r^4. \tag{89.5}$$

If the atom is in its ground state, this energy is negative (as is any correction to the ground-state energy); an attractive force therefore acts between the atom and the ion[42].

[41] This law, obtained within nonrelativistic theory, holds only while retardation effects in the electromagnetic interaction remain negligible.

For this to be so, the separation r of the atoms must be small compared with c/ω_{0n}, where the ω_{0n} are transition frequencies between the ground state and the excited states of the atom. For the interaction of atoms with retardation taken into account, see Vol. IV, § 85.

[42] The same attraction occurs at large distances between an atom and an electron. This attraction enables atoms to form negative ions by attaching an electron, whose binding energy ranges from a fraction to several electron-volts. Not every atom, however, has this property. The reason is that, in a field which decreases at large distances as $1/r^4$ (or $1/r^3$), the number of levels corresponding to bound electron states is necessarily finite and, in particular cases, there may be no such levels at all.

PROBLEM

For two identical atoms in S-states, derive a formula that determines the van der Waals forces from the matrix elements of their dipole moments.

SOLUTION. The result follows by applying the general perturbation-theory formula (38.10) to operator (89.1).

The isotropy of atoms in an S state makes the result of the sum over all intermediate states clear beforehand: for each of the vectors $\mathbf{d}_1$ and $\mathbf{d}_2$, the squared matrix elements of its three components contribute equally, whereas terms formed from products of distinct components vanish.

The result is

$$U(r) = -\frac{6}{r^6} \sum_{n,n'} \frac{|\langle n|d_z|0\rangle|^2 |\langle n'|d_z|0\rangle|^2}{E_n + E_{n'} - E_0},$$

where E_0 and E_n are the unperturbed energies of the atom's ground and excited states.

Because the ground state is assumed to have $L = 0$, the matrix elements $(d_z)_{0n}$ can be nonzero only for transitions to P-states ($L = 1$).

Using formulas (29.7), we put $U(r)$ in its final form:

$$U(r) = -\frac{2}{3r^6} \sum_{n,n'} \frac{\langle n1\|d\|00\rangle^2 \langle n'1\|d\|00\rangle^2}{E_{n1} + E_{n'1} - 2E_{00}},$$

Here, in the nL indices of the energy levels and reduced matrix elements, the second symbol gives the value of L, while the first denotes the collection of all other quantum numbers specifying the energy level.

§ 90. Predissociation

The theory of diatomic molecules developed in this chapter rests chiefly on the assumption that a molecule's wave function can be separated into the product of an electronic wave function (in which the internuclear distance enters as a parameter) and a wave function describing the motion of the nuclei. Equivalently, certain small terms that couple the nuclear and electronic motions are omitted from the exact molecular Hamiltonian.

When these terms are included perturbatively, transitions arise between different electronic states. The transitions of greatest physical importance are those for which at least one of the states belongs to the continuous spectrum.

Figure 30 shows the potential-energy curves of two electronic terms (more precisely, the effective potential energy U_J for the specified rotational states of the molecule). The energy E' belongs to a particular vibrational level of a stable molecule in electronic state 2. In state 1, the same energy is in the continuous-spectrum region. Thus a transition from state 2 to state 1 causes the molecule to decay spontaneously; this phenomenon is called *predissociation*[43]. Consequently, the discrete-spectrum state represented by curve 2 actually has a finite lifetime. The discrete

energy level is therefore broadened: it acquires a certain width (see the end of § 44).

[43]Curve 1 need not have a minimum at all when it represents purely repulsive forces between the atoms.

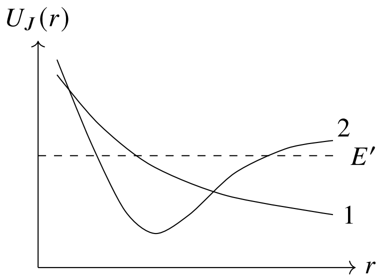

Fig. 30

If the total energy E lies above the dissociation threshold in both states, a transition from one state to the other represents what is known as a *collision of the second kind*. For example, in a $1 \to 2$ transition two atoms collide, enter excited states, and separate with reduced kinetic energy (as $r \to \infty$, curve 1 lies below curve 2; the difference $U_2(\infty) - U_1(\infty)$ is the atoms' excitation energy).

Because the nuclei are very massive, their motion is semiclassical. Determining the probability of the transitions under discussion is therefore a problem of the class considered in § 52. The general arguments given there show that the transition probability is controlled by a point at which the transition could occur classically[44]. Since the total energy of the two-atom system (the molecule) is conserved during such a transition, "classical feasibility" requires equality of the effective potential energies: $U_{J1}(r) = U_{J2}(r)$. The molecule's total angular momentum is conserved as well, so the centrifugal energies are identical in the two states. The stated condition consequently reduces to equality of the potential energies,

$$U_1(r) = U_2(r), \tag{90.1}$$

and contains no angular-momentum quantity whatever.

If equation (90.1) has no real roots in the classically accessible region (the region where $E > U_{J1}, U_{J2}$), then, according to § 52, the transition probability is exponentially small[45] . Transitions have an appreciable probability only when the potential-energy curves intersect within the classically accessible region (as shown in Fig. 30). In that event the exponent in formula (52.1) vanishes (so that this formula is, of course, inapplicable), and the transition probability is consequently given by a nonexponential expression (to be derived below). Condition (90.1) can

then be interpreted visually as follows. Equal potential (and total) energies also imply equal momenta. Thus, in place of (90.1), one may write

$$r_1 = r_2, \qquad p_1 = p_2, \tag{90.2}$$

[44]Alternatively, it may be the point $r = 0$, where the potential energy becomes infinite.

[45]A special situation arises when a transition involves a molecular term obtainable from two distinct pairs of atomic states (see the end of § 85), that is, when the potential-energy curve seems to divide into two branches toward increasing separations. In this situation the transition probability increases substantially; for an example, see A. I. Voronin, E. E. Nikitin // Optika i spektr. 1968. Vol. 25. P. 803.

where p is the momentum of the relative radial motion of the nuclei, while the subscripts 1 and 2 refer to the two electronic states. Thus, at the instant of transition, the separation and momentum of the nuclei may be said to remain unchanged (the so-called *Franck–Condon principle*). Physically, this follows because electronic velocities are large in comparison with nuclear velocities, and "during the electronic transition" the nuclei have no time to change their positions and velocities appreciably.

The selection rules for these transitions are readily established. First, there are two evident exact rules: neither the total angular momentum J nor the sign of the term (positive or negative; see § 86) may change in a transition. This follows directly because conservation of total angular momentum and invariance of the wave-function character under inversion of the coordinate system are exact laws for every closed system of particles.

There is also, to high accuracy, a rule forbidding transitions between states of different parity in a molecule made of identical atoms. Indeed, a state's parity is fixed uniquely by the nuclear spin and the term sign. Conservation of the term sign is exact, while nuclear spin is conserved to high accuracy because its interaction with the electrons is weak.

The existence of an intersection of the potential-energy curves requires terms of different symmetry (see § 79). Consider transitions that occur already in first-order perturbation theory. We first note that the Hamiltonian terms producing these transitions are precisely those responsible for the Λ-doubling of levels. Foremost among them are the terms describing spin–orbit interaction. Each is a product of two axial vectors, one of spin character and the other of coordinate character; these vectors, it should be stressed, are by no means simply $\widehat{\mathbf{S}}$ and $\widehat{\mathbf{L}}$. They consequently have nonzero matrix elements for transitions in which S and Λ change by $0, \pm 1$.

The case in which $\Delta S = \Delta\Lambda = 0$ simultaneously (with $\Lambda \neq 0$) must be excluded, since the symmetry of the term would then remain entirely unchanged in the transition. A transition between two Σ terms is possible only if their superscripts have opposite signs, one being Σ^+ and the other Σ^- (an axial vector has matrix elements only for transitions between Σ^+ and Σ^-; see § 87).

The Hamiltonian contains a term describing the coupling between molecular rotation and the orbital angular momentum. This term is proportional to $\widehat{\mathbf{J}} \cdot \widehat{\mathbf{L}}$, and its matrix elements are nonzero for transitions with $\Delta\Lambda = \pm 1$ and no change of spin. (For $\Delta\Lambda = 0$, matrix elements occur only for the ζ component of the vector, i.e. L_ζ; however, L_ζ is diagonal in the electronic states.)

Besides the terms just considered, a further perturbation arises because the nuclear kinetic-energy operator acts not only upon the nuclear wave function, but also upon the electronic function that depends parametrically on r. The corresponding Hamiltonian terms have the same symmetry as the unperturbed Hamiltonian. They can therefore produce transitions only between electronic terms of identical symmetry; the probability of these transitions is negligible because the terms do not intersect.

We now calculate the transition probability explicitly, taking a collision of the second kind for definiteness. By the general formula (43.1), the required probability is

$$w = \frac{4\pi^2}{\hbar^2} \left| \int \chi_{\mathrm{nuc},2} V(r) \chi_{\mathrm{nuc},1}\, dr \right|^2, \tag{90.3}$$

where $\chi_{\text{nuc}} = r\psi_{\text{nuc}}$ and $V(r)$ is the perturbing energy. The final wave function $\chi_{\text{nuc},2}$ must be normalized to a delta-function of the energy. With this normalization, the semiclassical function is

$$\chi_{\text{nuc},2} = \sqrt{\frac{2}{\pi\hbar v_2}} \cos\left(\frac{1}{\hbar}\int_{a_2}^{r} p_2, dr - \frac{\pi}{4}\right). \tag{90.4}$$

For the initial state we write

$$\chi_{\text{nuc},1} = \frac{2}{\sqrt{v_1}} \cos\left(\frac{1}{\hbar}\int_{a_1}^{r} p_1, dr - \frac{\pi}{4}\right). \tag{90.5}$$

It is normalized so that each of the two travelling waves carries unit current density. Substitution of these functions in (90.3) gives a dimensionless transition probability w, interpretable as the probability accumulated when the nuclei pass the point $r = r_0$ twice.

The matrix element contains a product of cosines, which we resolve into cosines of the sum and difference of their arguments. Near $r = r_0$ only the latter cosine is relevant, and hence

$$w = \frac{4}{\hbar^2}\left|\int \frac{V(r)}{\sqrt{v_1 v_2}} \cos\left(\frac{1}{\hbar}\int_{a_1}^{r} p_1 dr - \frac{1}{\hbar}\int_{a_2}^{r} p_2 dr\right) dr\right|^2.$$

The integral converges rapidly away from the intersection. Expanding the argument in powers of $\xi = r - r_0$ and using $p_1 = p_2$ at the intersection, we find

$$\int^{r} p_1 dr - \int^{r} p_2 dr \approx S_0 + \frac{1}{2}\left(\frac{dp_1}{dr} - \frac{dp_2}{dr}\right)_0 \xi^2.$$

Differentiation of $p_1^2/2\mu + U_1 = p_2^2/2\mu + U_2$ gives

$$v_1\frac{dp_1}{dr} - v_2\frac{dp_2}{dr} = F_1 - F_2,$$

and consequently

$$\int^{r} p_1 dr - \int^{r} p_2 dr \approx S_0 + \frac{F_1 - F_2}{2v}\xi^2.$$

Use of the familiar integral then yields

$$w = \frac{8\pi V^2}{\hbar v|F_2 - F_1|}\cos^2\left(\frac{S_0}{\hbar} + \frac{\pi}{4}\right). \tag{90.6}$$

The quantity $S_0/\hbar$ is large and varies rapidly with energy. On averaging, the squared cosine is replaced by its mean, giving

$$w = \frac{4\pi V^2}{\hbar v|F_2 - F_1|}. \tag{90.7}$$

(L. D. Landau, 1932). Every quantity on the right is evaluated at the intersection.

For predissociation, the decay probability per unit time is obtained by multiplying w by $\omega/2\pi$:

$$\frac{1}{\tau} = \frac{2\omega V^2}{\hbar v|F_2 - F_1|}. \tag{90.8}$$

In referring to an intersection of terms, we have meant the eigenvalues of the unperturbed Hamiltonian $\widehat{H}_0$, excluding the terms $\widehat{V}$ that produce transitions. Once those terms are included, the intersection disappears and the curves separate as shown in Fig. 31.

Let $U_{J1}(r)$ and $U_{J2}(r)$ be two eigenvalues of $\widehat{H}_0$. The method of § 79 gives

$$U_{b,a}(r) = \frac{1}{2}(U_{J1} + U_{J2}) \pm \frac{1}{2}\sqrt{(U_{J1} - U_{J2})^2 + 4V^2}. \tag{90.9}$$

The separation of the levels is

$$\Delta U = \sqrt{(U_{J1} - U_{J2})^2 + 4V^2}, \tag{90.10}$$

and its minimum value at $r = r_0$ is

$$(\Delta U)_{\min} = 2|V(r_0)|. \tag{90.11}$$

Near this point,

$$\Delta U = \sqrt{(F_2 - F_1)^2(r - r_0)^2 + 4V^2}. \tag{90.12}$$

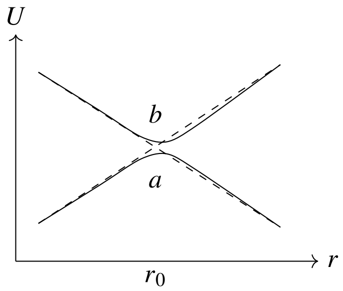

Fig. 31

These formulas require $(\Delta U)_{\min}$ to be small compared with the distance to the other terms. If condition (90.19) is not satisfied, ordinary perturbation theory cannot be used for the transition probability, and a more general treatment is needed.

In the neighborhood of the intersection, we treat the nuclear motion semiclassically, replace the velocity operator by the constant v, and put $\xi = r - r_0 = vt$. The problem reduces to

$$i\hbar\frac{\partial\Psi}{\partial t} = [\widehat{H}_0(t) + \widehat{V}(t)]\Psi. \tag{90.13}$$

Let ψ_a and ψ_b denote the electronic states associated with curves a and b. We seek a solution of the form

$$\Psi = a(t)\psi_a + b(t)\psi_b. \tag{90.14}$$

Under the boundary condition $a = 1, b = 0$ as $t \to -\infty$, the quantity $|b(\infty)|^2$ is the probability of transition from curve a to curve b in a single passage through the point. During a double passage, the transition can follow either $a \to b \to b$ or $a \to a \to b$; thus

$$w = 2|b(\infty)|^2[1 - |b(\infty)|^2]. \tag{90.15}$$

The value of $b(\infty)$ can be found by the method of § 53[46].

The curves $U_a(t)$ and $U_b(t)$ intersect at the imaginary points

$$t_0^{(\pm)} = \pm i\frac{2V}{|F_2 - F_1|v} = \pm i\tau_0. \tag{90.16}$$

Passing into the complex t plane and going around the point $t_0^{(+)}$, we use

$$\Delta U = \sqrt{(F_2 - F_1)^2 v^2 t^2 + 4V^2} \tag{90.17}$$

to obtain finally

$$w = 2\exp\left(-\frac{2\pi V^2}{\hbar v|F_2 - F_1|}\right)\left[1 - \exp\left(-\frac{2\pi V^2}{\hbar v|F_2 - F_1|}\right)\right]. \tag{90.18}$$

(C. Zener, 1932). The probability is small in either limiting case. When $V^2 \gg \hbar v|F_2 - F_1|$, it is

exponentially small (the adiabatic case); when

$$V^2 \ll \hbar v|F_2 - F_1| \tag{90.19}$$

formula (90.18) reduces to (90.7). Equation (90.17) shows that $\tau \sim |V|/(|F_2 - F_1|v)$ is the "nuclear transit time" past the intersection.

Finally, consider the phenomenon related to predissociation that is known as perturbations in the spectrum of diatomic molecules. If two discrete molecular levels E_1, E_2 are close, the possibility of a transition shifts them. According to (79.4),

$$E = \frac{E_1 + E_2}{2} \pm \sqrt{\left(\frac{E_1 - E_2}{2}\right)^2 + |V_{12}^{\text{nuc}}|^2}. \tag{90.20}$$

The levels move apart, by an amount that increases as $|E_1 - E_2|$ decreases. The matrix element V_{12}^{nuc} is calculated as above, except that the discrete-spectrum nuclear functions are normalized to unity. Comparison gives

$$|V_{12}^{\text{nuc}}|^2 = w\frac{\hbar\omega_1}{2\pi}\frac{\hbar\omega_2}{2\pi}. \tag{90.21}$$

[46] In § 53 the entire process was assumed to be adiabatic. Here adiabaticity may fail immediately around r_0; only adiabatic behavior at large $|t|$ and the possibility of retaining just two levels are essential.

Problem

1. Calculate the total cross section for second-kind collisions, as a function of the kinetic energy E of the colliding atoms, for transitions produced by spin–orbit interaction (L. D. Landau, 1932).

Solution. Since the nuclear motion is semiclassical, an impact parameter ρ may be introduced and we may set $d\sigma = 2\pi\rho, d\rho, w(\rho)$. For spin–orbit interaction, the matrix element $V(r)$ is independent of the angular momentum. The velocity at the intersection is

$$v = \sqrt{\frac{2}{\mu}\left(E - U - \frac{\rho^2 E}{r_0^2}\right)}.$$

Integration over ρ gives

$$\sigma = \frac{4\sqrt{2\mu}\pi^2 V^2 r_0^2}{\hbar|F_2 - F_1|}\frac{\sqrt{E-U}}{E}.$$

Problem

2. Do the same for transitions caused by the coupling of molecular rotation to orbital angular momentum (L. D. Landau, 1932).

Solution. The matrix element has the form $V(r) = MD/\mu r^2$, where $D(r)$ is the matrix element of the electronic orbital angular momentum. The same method gives

$$\sigma = \frac{16\sqrt{2}\pi^2 D^2 (E-U)^{3/2}}{3\hbar\mu r_0^2 |F_2 - F_1| E}.$$

Problem

3. Find the transition probability for energies E close to the potential energy U at the intersection.

Solution. For small $E - U$, formula (90.7) is inapplicable. Near the intersection, replace the curves by straight lines $U_1 = U - F_1\xi$, $U_2 = U - F_2\xi$, where $\xi = r - r_0$. The wave functions are those for one-dimensional motion in a uniform field. Calculation in the momentum representation gives

$$w = \frac{4\pi V^2 (2\mu)^{2/3}}{\hbar^{4/3}(F_1F_2)^{1/2}(F_2 - F_1)^{2/3}}\Phi^2\left[-\frac{E-U}{\hbar^{2/3}}\left(\frac{2\mu(F_2-F_1)^2}{F_1^2F_2^2}\right)^{1/3}\right], \tag{90.22}$$

where Φ is the Airy function (see § b of the Mathematical Supplements). For large $E - U$, this formula reduces to (90.7).

Problem

4. Find the charge-transfer probability in a distant, slow ($v \ll 1$) collision between a hydrogen atom and a proton (O. V. Firsov, 1951)[47].

Solution. Regard the system H + H$^+$ as a hydrogen molecular ion (see the problem in § 81). Charge transfer consists in moving the electron from the state ψ_1, localized at the first nucleus, to the state ψ_2 near the second. The stationary states are

$$\psi_{g,u} = \frac{1}{\sqrt{2}}(\psi_1 \pm \psi_2).$$

Their energies are $U_{g,u}(R)$. For the specified slow classical motion of the nuclei, the time dependence is supplied by the factors $\exp\left[-i\int U_{g,u}(t)dt\right]$. As $t \to \infty$, the charge-transfer probability is

$$w = \sin^2\eta, \qquad \eta = \frac{1}{2}\int_{-\infty}^{\infty}(U_u - U_g)dt.$$

At large impact parameters ρ, put $R = \sqrt{\rho^2 + v^2t^2}$ and use formula (4) of the problem in § 81:

$$\eta = \frac{4}{v}\int_{\rho}^{\infty}\frac{R^2 e^{-R-1}}{\sqrt{R^2 - \rho^2}}, dR.$$

For $\rho \gg 1$, putting $R = \rho(1 + x)$ gives

$$\eta = \frac{2\sqrt{2\pi}}{ev}\rho^{3/2}e^{-\rho}.$$

[47] Atomic units are used in this problem.

CHAPTER XII

Theory of Symmetry

§ 91. Symmetry transformations

For polyatomic molecules, just as for diatomic ones, the classification of terms is tied essentially to molecular symmetry. We shall therefore begin by examining the kinds of symmetry that a molecule may possess.

The symmetry of a body is specified by all displacements that bring the body into coincidence with itself; such displacements are called *symmetry transformations*. Every possible symmetry transformation can be expressed as a combination of one or more transformations belonging to three fundamental types. The three types, which are essentially distinct, are a *rotation* of the body through a definite angle about some axis, a *reflection* in a plane, and a *parallel translation* of the body through some distance. Evidently, only an unbounded medium (a crystal lattice) can possess symmetry of the last type. A body of finite dimensions, and hence a molecule in particular, can be symmetric only under rotations and reflections.

Suppose that rotation about an axis through an angle $2\pi/n$ makes a body coincide with itself. The axis is then called a *symmetry axis of order n*. Here n may be any integer $n = 2, 3, \ldots$; the value $n = 1$ describes rotation through 2π, equivalently through 0, and thus represents the identity transformation. We shall denote rotation through $2\pi/n$ about the chosen axis by C_n. Repeating this operation twice, three times, ... produces rotations through $2(2\pi/n), 3(2\pi/n), \ldots$, which likewise superpose the body on itself; these rotations may be written $C_n^2, C_n^3, \ldots$ Clearly, if p divides n, then

$$C_n^p = C_{n/p}. \tag{91.1}$$

In particular, after n rotations the initial position is restored, so the resulting transformation is the *identity*. It is customary to denote the latter by E, and hence

$$C_n^n = E. \tag{91.2}$$

If reflection in a certain plane causes a body to coincide with itself, that plane is called a *plane of symmetry*. We shall use σ for the operation of reflection in a plane. Two successive reflections in the same plane plainly give the identity transformation,

$$\sigma^2 = E. \tag{91.3}$$

Combining the two transformations—rotation and reflection—gives rise to the so-called *rotation–reflection axes*. A body has a rotation–reflection axis of order n when rotation about this axis through $2\pi/n$, followed by reflection in the plane perpendicular to it, brings the body into coincidence with itself (Fig. 32). It is readily seen that this supplies a genuinely new kind of symmetry only for even n. Indeed, when n is odd, repeating the rotation–reflection transformation n times is equivalent to a simple reflection in the plane normal to the axis: the total rotation angle is 2π, while an odd number of reflections in one and the same plane reduces to a single reflection. If the transformation is then repeated a further n times, one finds that the rotation–reflection axis amounts to the simultaneous presence of an independent symmetry axis of order n and a plane of symmetry perpendicular to it. For even n, by contrast, n repetitions of the rotation–reflection transformation return the body to its original position.

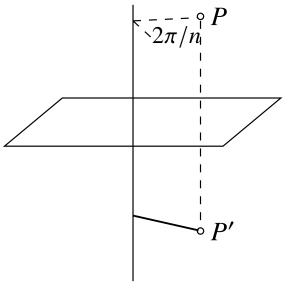

Figure 32

We denote a rotation–reflection transformation by S_n. If reflection in the plane normal to the given axis is denoted by σ_h, then, by definition,

$$S_n = C_n\sigma_h = \sigma_h C_n \tag{91.4}$$

(the outcome evidently does not depend on the order in which C_n and σ_h are performed).

The rotation–reflection axis of second order is an important special case. A rotation through π followed by reflection in the plane perpendicular to the rotation axis is readily recognized as an *inversion*. Under this transformation a point P of the body goes to another point P' on the continuation of the line joining P to the point O where the axis intersects the plane; moreover, $OP = OP'$.

A body symmetric under this transformation is said to have a center of symmetry (we shall denote inversion by I):

$$I \equiv S_2 = C_2\sigma_h. \tag{91.5}$$

It is also plain that $I\sigma_h = C_2$ and $IC_2 = \sigma_h$. Thus, a second-order axis, a plane of symmetry normal to it, and a center of symmetry at their intersection are mutually dependent: the presence of any two of these elements necessarily entails the third.

We now note several purely geometrical properties of rotations and reflections that are useful when studying the symmetry of bodies.

The product of two rotations about axes meeting at a point is a rotation about a third axis through that same point. Two reflections in intersecting planes are together equivalent to a rotation. Its axis is clearly the line in which the planes intersect, while its angle, as follows at once from a simple geometrical construction, is twice the angle between the two planes. Let $C(\varphi)$ denote rotation through φ about the axis, and let σ_v

and σ_v' denote reflections in two planes containing that axis.[1] The statement just made may then be written

$$\sigma_v \sigma_v' = C(2\varphi), \tag{91.6}$$

where φ is the angle between the two planes. The order of the reflections must be observed: $\sigma_v \sigma_v'$ gives a rotation directed from the plane σ_v' toward the plane σ_v, whereas interchanging the factors reverses the sense of rotation. Multiplication of (91.6) on the left by σ_v yields

$$\sigma_v' = \sigma_v C(2\varphi); \tag{91.7}$$

in other words, the product of a rotation and reflection in a plane containing the rotation axis is equivalent to reflection in another plane, inclined to the first at an angle equal to half the angle of rotation. In particular, this shows that a second-order symmetry axis and two mutually perpendicular symmetry planes through it are mutually dependent: if two are present, the third must be present as well.

We shall show that rotations through π about two axes that meet at an angle φ (the axes Oa and Ob in Fig. 33) have as their product a rotation through 2φ about the axis perpendicular to both (the line PP' in Fig. 33). The resulting transformation is known in advance to be a rotation. Under the first rotation, about Oa, the point P moves to P'; under the second, about Ob, it returns to its initial position. Consequently, the line PP' remains fixed and is therefore the rotation axis. To find the angle of rotation, observe that the first rotation leaves the axis Oa in place, whereas the second carries it into the position Oa', which makes an angle 2φ with Oa. The same argument shows that reversing the order of the transformations reverses the direction of rotation.

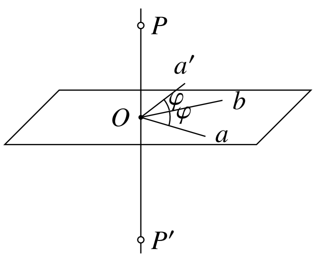

Figure 33

Although the result of two successive transformations generally depends on their order, there are several cases in which the order of operations is immaterial, so that the transformations commute. These cases are:

1) two rotations about the same axis;

2) reflections in two mutually perpendicular planes (their product is equivalent to a rotation through π about the line where the planes intersect);

3) rotations through π about two mutually perpendicular axes (their product is equivalent to a rotation through the same angle about the third perpendicular axis);

[1]The subscript v is customarily used for reflection in a plane containing the given axis (a "vertical" plane), whereas h denotes reflection in a plane perpendicular to the axis (a "horizontal" plane).

4) a rotation and reflection in a plane perpendicular to its axis;

5) any rotation (or reflection) together with inversion about a point on the rotation axis (or in the reflection plane); this follows from 1) and 4).

§ 92. Transformation groups

The collection of all symmetry transformations of a given body is called its group of symmetry transformations, or, more briefly, its *symmetry group*. Above, these transformations were described as geometrical motions of the body. In quantum-mechanical applications, however, it is more convenient to regard them as transformations of the coordinates under which the Hamiltonian of the system remains invariant. Clearly, if a rotation or reflection brings the system into coincidence with itself, the corresponding coordinate transformation leaves its Schrödinger equation unchanged. We shall therefore speak of the group of transformations under which the given Schrödinger equation is invariant[2].

Symmetry groups are conveniently studied by means of the general mathematical machinery known as *group theory*, whose foundations are presented below. We begin with groups containing only a finite number of distinct transformations (the so-called *finite groups*). Each transformation belonging to a group is called an *element of the group*.

Symmetry groups have the following evident properties. Every group contains the identity transformation E (called the *identity element of the group*). Group elements can be *multiplied* together; the product of two (or several) transformations means the result of applying them successively. Evidently, the product of any two elements of a group is an element of the same group. Element multiplication obeys the associative law $(AB)C = A(BC)$, where A, B, C are group elements. The commutative law does not hold in general, however: ordinarily $AB \neq BA$. For every group element A, the same group contains an *inverse element* A^{-1} (the inverse transformation), such that $AA^{-1} = E$. In some cases an element may coincide with its inverse; in particular, $E^{-1} = E$. It is evident that mutually inverse elements A and A^{-1} commute.

The inverse of the product AB of two elements is

$$(AB)^{-1} = B^{-1}A^{-1}$$

and the analogous statement holds for a product of more elements. This follows at once by carrying out the multiplication and using associativity.

[2]This viewpoint permits us to include not only the rotation and reflection groups considered here, but also other kinds of transformations that leave the Schrödinger equation unchanged. Among them are permutations of the coordinates of identical particles belonging to the system (a molecule or an atom). The collection of all permutations of identical particles that are possible in a given system is called its *permutation group*; such groups have already occurred in § 63. The general group properties developed below apply to permutation groups as well, but we shall not pursue a more detailed study of groups of this kind.

One comment is needed concerning the notation used in this chapter. Symmetry transformations are, in substance, operators of the same kind as those used throughout the book, and so their letters ought to carry hats. We omit the hats in deference to customary notation and because no ambiguity can result here. For the same reason the identity transformation is denoted by the usual symbol E, rather than by 1 as the notation of the other chapters would suggest. Finally, in this chapter the inversion operator is written I, in place of the symbol P used in § 30 and customary in present-day quantum- mechanics literature.

If all the elements of a group commute, the group is called *abelian*. The so-called *cyclic groups* form a special class of abelian groups. A cyclic group is one whose every element can be obtained by taking successive powers of a single element; thus it consists of the elements

$$A, A^2, A^3, \dots, A^n = E,$$

where n is an integer.

Let $\boldsymbol{G}$ be a group[3]. If a collection of its elements, denoted by $\boldsymbol{H}$, can be selected so that it too forms a group, then $\boldsymbol{H}$ is called a *subgroup* of $\boldsymbol{G}$. One and the same group element may belong to several different subgroups.

Take any element A of the group and form its successive powers. Since the total number of elements in the group is finite, the identity element must eventually be reached. If n is the smallest number for which $A^n = E$, then n is called the *order of the element* A, while the collection $A, A^2, \dots, A^n = E$ is the *period* of A. The period is denoted by $\{A\}$; it is itself a group, hence a subgroup of the original group, and specifically a cyclic subgroup.

To determine whether a given collection of group elements is a subgroup, it is enough to check that multiplication of any two of its elements gives an element contained in the same collection. Indeed, together with every element A, the collection will then contain all its powers, including A^{n-1} (where n is the order of the element), which serves as its inverse (since $A^{n-1}A = A^n = E$); it will plainly contain the identity as well.

The total number of elements in a group is called the *order* of the group. It is readily seen that the order of a subgroup divides the order of the whole group. To prove this, consider a subgroup $\boldsymbol{H}$ of a group $\boldsymbol{G}$, and let G_1 be an element of $\boldsymbol{G}$ that does not belong to $\boldsymbol{H}$. Multiplication of every element of $\boldsymbol{H}$ by G_1—on the right, for example—gives a set (or *coset*) of elements denoted by $\boldsymbol{H}G_1$. Every element of this coset plainly belongs to $\boldsymbol{G}$, but none belongs to $\boldsymbol{H}$. Indeed, if two elements H_a, H_b of $\boldsymbol{H}$ satisfied $H_aG_1 = H_b$, it would follow that $G_1 = H_a^{-1}H_b$; this would put G_1 in $\boldsymbol{H}$, contrary to our assumption. In the same way, if G_2 belongs to $\boldsymbol{G}$ but to neither $\boldsymbol{H}$ nor $\boldsymbol{H}G_1$, then none of the elements of $\boldsymbol{H}G_2$ belongs to either of those two sets. Continuing in this fashion eventually exhausts all elements of the finite group $\boldsymbol{G}$. Thus its elements are partitioned into sets (the *cosets* of $\boldsymbol{H}$ in $\boldsymbol{G}$)

$$\boldsymbol{H},\ \boldsymbol{H}G_1,\ \boldsymbol{H}G_2, \dots, \boldsymbol{H}G_m,$$

each containing h elements, where h is the order of the subgroup $\boldsymbol{H}$. Consequently the order of $\boldsymbol{G}$ is $g = hm$, which proves the assertion. The integer $m = g/h$ is called the *index* of $\boldsymbol{H}$ in $\boldsymbol{G}$.

If the order of a group is prime, the result just proved immediately shows that the group has no subgroups at all (apart from E and the group itself). The converse is also true: any group without subgroups must have prime order and must moreover be cyclic (otherwise it would contain elements whose periods would form subgroups).

We now introduce the important notion of *conjugate elements*. Two elements A and B are said to be conjugate to one another if

$$A = CBC^{-1},$$

[3]Groups will be denoted by bold italic letters.

where C is also an element of the group. Multiplying this equality by C on the right and by C^{-1} on the left gives the inverse relation $B = C^{-1}AC$. An essential property of conjugacy is transitivity: if A is conjugate to B, and B to C, then A and C are conjugate as well. In fact, from $B = P^{-1}AP$ and $C = Q^{-1}BQ$, with P, Q elements of the group, it follows that $C = (PQ)^{-1}A(PQ)$. One may therefore consider sets of elements that are mutually conjugate. Such sets are called *conjugacy classes*, or simply classes of the group. Any one of its elements A determines the entire class: once A is given, form the products GAG^{-1} as G ranges over all elements of the group; the same member of the class may, of course, be obtained more than once. In this way the whole group can be partitioned into classes, and each element evidently belongs to only one class. The identity element forms a class by itself, since $GEG^{-1} = E$ for every element G of the group. In an Abelian group the same is true of each element. All elements of such a group commute by definition, so every element is conjugate only to itself and hence constitutes its own class. It should be emphasized that a class other than E is by no means a subgroup: this already follows from the fact that it does not contain the identity element.

All elements in the same class have the same order. Indeed, if n is the order of A (so that $A^n = E$), then for the conjugate element $B = CAC^{-1}$ we likewise have $(CAC^{-1})^n = CA^nC^{-1} = E$.

Let $\boldsymbol{H}$ be a subgroup of $\boldsymbol{G}$, and let G_1 be an element of $\boldsymbol{G}$ outside $\boldsymbol{H}$. It is easily verified that the collection of elements $G_1\boldsymbol{H}G_1^{-1}$ satisfies all the requirements for a group and hence is also a subgroup of $\boldsymbol{G}$. The subgroups $\boldsymbol{H}$ and $G_1\boldsymbol{H}G_1^{-1}$ are called conjugate; each element of one is conjugate to an element of the other. Giving G_1 different values yields a number of conjugate subgroups, some of which may coincide. It can happen that every subgroup conjugate to $\boldsymbol{H}$ coincides with $\boldsymbol{H}$. In that case $\boldsymbol{H}$ is called a *normal divisor* (or an *invariant subgroup*) of $\boldsymbol{G}$. Thus, for example, every subgroup of an abelian group is plainly a normal divisor of that group.

Consider a group $\boldsymbol{A}$ with n elements $A, A', A'', \ldots$ and a group $\boldsymbol{B}$ with m elements $B, B', B'', \ldots$, and suppose that all the elements of $\boldsymbol{A}$ (apart from the identity E) are distinct from and commute with the elements of $\boldsymbol{B}$. Multiplying each element of $\boldsymbol{A}$ by every element of $\boldsymbol{B}$ produces a collection of nm elements that likewise forms a group. Indeed, for any two elements of the collection, $AB \cdot A'B' = AA' \cdot BB' = A''B''$, which is again an element of the same collection. The resulting group of order nm is denoted by $\boldsymbol{A} \times \boldsymbol{B}$ and is called the *direct product* of the groups $\boldsymbol{A}$ and $\boldsymbol{B}$.

Finally, we introduce the concept of group *isomorphism*. Two groups $\boldsymbol{A}$ and $\boldsymbol{B}$ of the same order are called isomorphic if a one-to-one correspondence can be established between their elements such that, if A corresponds to B and A' to B', then $A'' = AA'$ corresponds to $B'' = BB'$. Considered abstractly, two such groups plainly have identical properties, even though the concrete meanings of their elements differ.

§ 93. Point groups

For the transformations in the symmetry group of a finite body (a molecule in particular), there must be at least one point of the body that every transformation leaves fixed. Equivalently, all its symmetry axes and planes must share at least one point. Indeed, successive rotations about two nonintersecting axes, or reflections in nonintersecting planes, produce a translation of the body; such a displacement plainly cannot bring a finite body into coincidence with itself. Symmetry groups having this property are called *point groups*.

Before constructing the possible kinds of point groups, consider a simple geometric device for assigning their elements to classes. Let Oa be an axis, and let the group element A rotate through a certain angle about it. Suppose another transformation G of the same group, whether a rotation or a reflection, carries the axis Oa itself into the position Ob. We claim that $B = GAG^{-1}$ is then a rotation about Ob through the same angle as A about Oa. To see this, apply GAG^{-1} to the axis Ob. First G^{-1} takes Ob to Oa; the ensuing rotation A leaves that axis in place; and G finally returns it to its original position. Thus Ob remains fixed, so B is a rotation about that axis. Moreover, A and B lie in one class and consequently have the same order, which means that their rotation angles are equal.

It follows that two rotations through an equal angle belong to the same class whenever the group contains a transformation superposing one rotation axis on the other. The same reasoning shows that reflections in two different planes are in one class if some group transformation takes one plane into the other. Symmetry

axes or planes whose directions can be superposed in this way are termed *equivalent*.

When both rotations are about the same axis, a few further observations are needed. The inverse of C_n^k ($k = 1, 2, \dots, n-1$), a rotation about an n-fold symmetry axis, is $C_n^{-k} = C_n^{n-k}$. It is a rotation through $(n-k)(2\pi/n)$ in the original sense, or equivalently through $2k\pi/n$ in the opposite sense. If the group includes a rotation through π about a perpendicular axis, which reverses the direction of the axis under consideration, then the general result just proved places C_n^k and C_n^{-k} in the same class. Reflection σ_h in a plane perpendicular to the axis also reverses its direction, but a reflection reverses the sense of rotation as well. Hence the presence of σ_h does not make C_n^k and C_n^{-k} conjugate. By contrast, reflection σ_v in a plane containing the axis preserves the direction of that axis while reversing the sense of rotation. Thus $C_n^{-k} = \sigma_v C_n^k \sigma_v$, and such a symmetry plane makes C_n^k and C_n^{-k} members of one class. An axis for which equal rotations in opposite senses are conjugate will be called *two-sided*.

The following rule often simplifies the determination of the classes of a point group. Let $\boldsymbol{G}$ be a group without the inversion I, and let $\boldsymbol{C}_i$ consist of the two elements I and E. The direct product $\boldsymbol{G} \times \boldsymbol{C}_i$ has twice as many elements as $\boldsymbol{G}$: one half are the elements of $\boldsymbol{G}$ themselves, and the other half are obtained by multiplying them by I. Since I commutes with every point-group transformation, $\boldsymbol{G} \times \boldsymbol{C}_i$ likewise has twice the number of classes of $\boldsymbol{G}$. Each class A of $\boldsymbol{G}$ gives the two classes A and AI in the product group. In particular, inversion I always forms a class by itself.

We now list all possible point groups, beginning with the simplest and adjoining new symmetry elements in succession. Point groups will be denoted by bold Latin letters with suitable subscripts.

I. Group $\boldsymbol{C}_n$

The simplest symmetry type has only one n-fold symmetry axis. The group $\boldsymbol{C}_n$ consists of rotations about this axis and is plainly cyclic. Each of its n elements is a separate class. The group $\boldsymbol{C}_1$

contains only the identity transformation E and thus corresponds to complete absence of symmetry.

II. Group $\boldsymbol{S}_{2n}$

This is the group generated by a rotation–reflection axis of even order $2n$. It has $2n$ elements and is cyclic. In particular, $\boldsymbol{S}_2$ has only the elements E and I and is also written $\boldsymbol{C}_i$. Notice also that when the group order has the form $2n = 4p + 2$, inversion is one of its elements, since $(S_{4p+2})^{2p+1} = C_2\sigma_h = I$. Such a group can be expressed as the direct product $\boldsymbol{S}_{4p+2} = \boldsymbol{C}_{2p+1} \times \boldsymbol{C}_i$ and is also denoted by $\boldsymbol{C}_{2p+1,i}$.

III. Group $\boldsymbol{C}_{nh}$

This group results when a symmetry plane perpendicular to an n-fold symmetry axis is adjoined. The $2n$ elements of $\boldsymbol{C}_{nh}$ are the n rotations of $\boldsymbol{C}_n$ and the n rotation–reflection transformations $C_n^k\sigma_h$ $(k = 1, 2, 3, \ldots, n)$, including the reflection $C_n^n\sigma_h = \sigma_h$. Every pair of elements commutes, so the group is Abelian and has as many classes as elements. For even n $(n = 2p)$ it contains a center of symmetry, because $C_{2p}^p\sigma_h = C_2\sigma_h = I$. The simplest group $\boldsymbol{C}_{1h}$ has just E and σ_h and is also called $\boldsymbol{C}_s$.

IV. Group $\boldsymbol{C}_{nv}$

Adjoining to an n-fold symmetry axis a symmetry plane that contains it automatically supplies another $(n-1)$ planes, all intersecting along the axis at angles π/n. This follows immediately from the geometrical theorem (91.7).[4] The resulting group $\boldsymbol{C}_{nv}$ therefore has $2n$ elements: n rotations about the n-fold axis and n reflections σ_v in vertical planes. Figure 34 shows, as examples, the axes and symmetry planes of $\boldsymbol{C}_{3v}$ and $\boldsymbol{C}_{4v}$.

In finding the classes, first note that the symmetry planes containing the axis make this axis

two-sided. The actual division of the elements into classes differs for even and odd n.

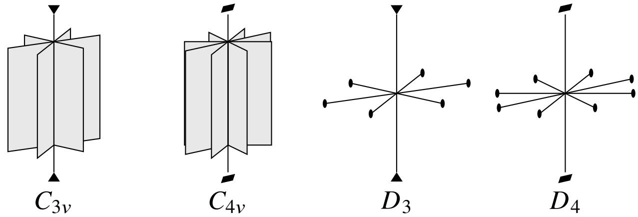

Figure 34

For odd n $(n = 2p+1)$, successive rotations C_{2p+1} carry any one plane into each of the other $2p$ planes. All the symmetry planes are therefore equivalent, and their reflections belong to a single class. Among the axial rotations there are $2p$ nonidentity operations; conjugate pairs form p two-element classes (C_{2p+1}^k and C_{2p+1}^{-k}, $k = 1, 2, \ldots, p$). The identity E supplies one further class, giving $p + 2$ classes altogether.

[4]A finite group cannot contain two symmetry planes meeting at an angle that is not a rational fraction of 2π. Two such planes would imply infinitely many further symmetry planes through the same line, generated by reflecting either plane in the other without limit. In other words, two planes of this kind at once imply full axial symmetry.

For even n ($n = 2p$), successive rotations C_{2p} superpose only alternate planes; adjacent planes cannot be brought into coincidence. There are consequently two sets of p equivalent planes, and hence two classes containing p reflections each. Of the axial rotations, $C_{2p}^{2p} = E$ and $C_{2p}^{p} = C_2$ each form a class alone. The remaining $2p - 2$ rotations occur in conjugate pairs, producing another $p - 1$ two-element classes. Thus $\boldsymbol{C}_{2p,v}$ has $p + 3$ classes in all.

V. Group $\boldsymbol{D}_n$

When a twofold axis perpendicular to an n-fold symmetry axis is adjoined, another $(n - 1)$ such axes appear. There are then n horizontal twofold axes meeting at angles π/n. The resulting group $\boldsymbol{D}_n$ has $2n$ elements: n rotations about the n-fold axis and n rotations through π about the horizontal axes. We shall denote the latter by U_2, reserving C_2 for rotation through π about the vertical

axis. Figure 34 also shows the axial systems of $\boldsymbol{D}_3$ and $\boldsymbol{D}_4$ as examples.

Exactly as in the preceding case, the n-fold axis is two-sided. For odd n all the horizontal twofold axes are equivalent, whereas for even n they split into two inequivalent sets. The $p + 3$ classes of $\boldsymbol{D}_{2p}$ are therefore E, two classes each containing p rotations U_2, the rotation C_2, and $(p - 1)$ classes each containing two rotations about the vertical axis. The group $\boldsymbol{D}_{2p+1}$ instead has $p + 2$ classes: E, one class of $2p + 1$ rotations U_2, and p two-element classes of rotations about the vertical axis.

An important special case is $\boldsymbol{D}_2$, whose axial system consists of three mutually perpendicular twofold axes. This group is also denoted by $\boldsymbol{V}$.

VI. Group $\boldsymbol{D}_{nh}$

Add to the axes of $\boldsymbol{D}_n$ a horizontal symmetry plane containing its n twofold axes. This automatically creates n vertical planes, each containing the vertical axis and one horizontal axis. The resulting group $\boldsymbol{D}_{nh}$ has $4n$ elements. Besides the $2n$ elements of $\boldsymbol{D}_n$, it includes n reflections σ_v and n rotation–reflection transformations $C_n^k \sigma_h$. The axes and planes of $\boldsymbol{D}_{3h}$ are shown in Fig. 35.

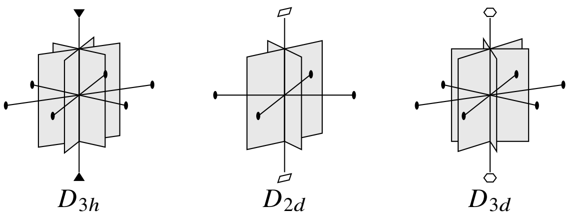

Figure 35

The reflection σ_h commutes with every other group element. Hence $\boldsymbol{D}_{nh}$ can be written as the direct product $\boldsymbol{D}_{nh} = \boldsymbol{D}_n \times \boldsymbol{C}_s$, where $\boldsymbol{C}_s$ is the two-element group consisting of E and σ_h. When n is even, inversion occurs among the elements and one may also write $\boldsymbol{D}_{2p,h} = \boldsymbol{D}_{2p} \times \boldsymbol{C}_i$.

It follows that the number of classes in $\boldsymbol{D}_{nh}$ is twice that in $\boldsymbol{D}_n$. Half of them coincide

with the classes of $\boldsymbol{D}_n$. These classes consist of rotations about axes. The others are obtained from them by multiplication by σ_h. All reflections σ_v in vertical planes belong to one class for odd n, but form two classes for even n. The rotation-reflections $\sigma_h C_n^k$ and $\sigma_h C_n^{-k}$ are conjugate in pairs.

VII. Group $\boldsymbol{D}_{nd}$

There is another way to adjoin symmetry planes to the axes of $\boldsymbol{D}_n$: place them vertically through the n-fold axis, midway between each pair of neighboring horizontal twofold axes. Once again, adding one such plane entails the presence of another $(n-1)$. The resulting system defines $\boldsymbol{D}_{nd}$. Figure 35 shows the systems for $\boldsymbol{D}_{2d}$ and $\boldsymbol{D}_{3d}$.

The group $\boldsymbol{D}_{nd}$ contains $4n$ elements. To the $2n$ elements of $\boldsymbol{D}_n$ are added n reflections in vertical planes, denoted by σ_d (the diagonal planes), and n transformations of the form $G = U_2\sigma_d$. To identify the latter, use (91.6) to write $U_2 = \sigma_h\sigma_v$, where σ_v is reflection in a vertical plane containing the particular twofold axis. Then $G = \sigma_h\sigma_v\sigma_d$; neither σ_v nor σ_h by itself is an element of the group. The planes σ_v and σ_d meet along the n-fold axis at an angle $(\pi/2n)(2k+1)$, with $k = 1, \ldots, (n-1)$, since adjacent planes here differ by $\pi/2n$. Equation (91.6) gives $\sigma_v\sigma_d = C_{2n}^{2k+1}$. Consequently $G = \sigma_h C_{2n}^{2k+1} = S_{2n}^{2k+1}$. These elements are rotation–reflection transformations about the vertical axis, which is therefore a rotation–reflection axis of order $2n$, not merely an n-fold symmetry axis.

Each diagonal plane reflects two adjacent horizontal twofold axes into one another. Hence all twofold axes in these groups are equivalent, whether n is even or odd. All the diagonal planes are likewise equivalent. The rotation–reflection transformations S_{2n}^{2k+1} and S_{2n}^{-2k-1} are conjugate in pairs.[5]

Applied to $\boldsymbol{D}_{2p,d}$, these observations give $2p+3$ classes: E; the rotation C_2 about the n-fold axis; $(p-1)$ classes of two conjugate rotations about that axis; a class of $2p$ rotations U_2; a class of $2p$ reflections σ_d; and p classes containing two rotation–reflection transformations each.

For odd n $(n = 2p+1)$, inversion is among the elements; one horizontal axis is then perpendicular to a vertical plane. Thus $\boldsymbol{D}_{2p+1,d} = \boldsymbol{D}_{2p+1} \times \boldsymbol{C}_i$, and its $2p+4$ classes follow directly from the $p+2$ classes of $\boldsymbol{D}_{2p+1}$.

VIII. Group $\boldsymbol{T}$ (tetrahedral group)

The axes of this group are the symmetry axes of a tetrahedron. They may be obtained by adding four inclined threefold axes to the axes of $\boldsymbol{V}$; rotations about them interchange the three twofold axes. A convenient picture places the three twofold axes through centers of opposite cube faces and the threefold axes along its body diagonals. Figure 36 shows one axis of each kind in the cube and tetrahedron.

The three twofold axes are mutually equivalent. The threefold axes are also equivalent, since rotations C_2 carry them into one another, but they are not two-sided. The 12 elements of $\boldsymbol{T}$ therefore divide into four classes: E, three rotations C_2, four rotations C_3, and four rotations C_3^2.

IX. Group $\boldsymbol{T}_d$

[5] Indeed,

$$\sigma_d S_{2n}^{2k+1}\sigma_d = \sigma_d\sigma_h C_{2n}^{2k+1}\sigma_d = \sigma_h\sigma_d C_{2n}^{2k+1}\sigma_d = \sigma_h C_{2n}^{-2k-1} = S_{2n}^{-2k-1}.$$

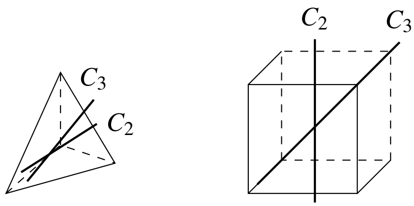

Figure 36

This group contains every symmetry transformation of a tetrahedron. Its axes and planes arise by adding to $\boldsymbol{T}$ symmetry planes, each passing through one twofold and two threefold axes. The twofold axes thereby become fourfold rotation–reflection axes, as in $\boldsymbol{D}_{2d}$. It is convenient to draw the three rotation–reflection axes through centers of opposite cube faces, the four threefold axes along body diagonals, and the six symmetry planes

through pairs of opposite edges. Figure 37 shows one axis and plane of each type.

Since the symmetry planes are vertical relative to the threefold axes, those axes are two-sided. All axes or planes of a given type are equivalent. The 24 elements consequently form five classes: E; eight rotations C_3 and C_3^2; six plane reflections; six rotation–reflection transformations S_4 and S_4^3; and three rotations $C_2 = S_4^2$.

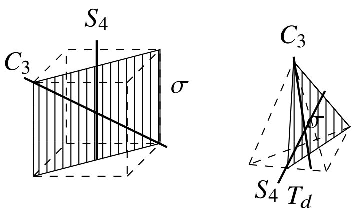

Figure 37

X. Group $\boldsymbol{T}_h$

This group is obtained from $\boldsymbol{T}$ by adding a center of symmetry: $\boldsymbol{T}_h = \boldsymbol{T} \times \boldsymbol{C}_i$. Three mutually perpendicular symmetry planes then appear, each passing through a pair of twofold axes, while the threefold axes become sixfold rotation–reflection axes. One such axis and one plane are shown in Fig. 38.

Its 24 elements are distributed among eight classes obtained directly from the classes of $\boldsymbol{T}$.

XI. Group $\boldsymbol{O}$ (octahedral group)

The axial system of this group is that of a cube: three fourfold axes pass through centers of opposite faces, four threefold axes through opposite vertices, and six twofold axes through midpoints of opposite edges (Fig. 39).

All axes of the same order are plainly equivalent, and every axis is two-sided. The 24 elements thus divide into five classes: E; eight rotations C_3 and C_3^2; six rotations C_4 and C_4^3; three rotations C_4^2; and six rotations C_2.

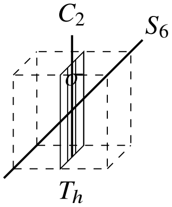

Figure 38

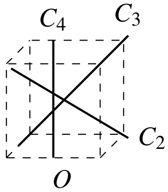

Figure 39

XII. Group $\boldsymbol{O}_h$

This is the group of all symmetry transformations of a cube.[6] It is produced by adding a center of symmetry to $\boldsymbol{O}$: $\boldsymbol{O}_h = \boldsymbol{O} \times \boldsymbol{C}_i$. The threefold axes of $\boldsymbol{O}$ thereby become sixfold rotation–reflection axes (the body diagonals of the cube). There also appear six symmetry planes through pairs of opposite edges and three planes parallel to the cube faces (Fig. 40). The group has 48 elements in ten classes, obtainable directly from those of $\boldsymbol{O}$. Five are the classes of $\boldsymbol{O}$ itself; the others are I; eight rotation–reflection transformations S_6 and S_6^5; six rotation–reflection transformations $C_4\sigma_h$ and $C_4^3\sigma_h$ about the fourfold axes; three reflections σ_h in planes horizontal relative to those axes; and six reflections σ_v in planes vertical relative to them.

XIII, XIV. Groups $\boldsymbol{Y}$, $\boldsymbol{Y}_h$ (icosahedral groups)

Only exceptionally do these groups occur in nature as molecular symmetry groups. We therefore state only that $\boldsymbol{Y}$ is the group of 60 rotations about the symmetry axes of an icosahedron (a regular 20-faced solid with triangular faces) or a pentagonal dodecahedron (a regular 12-faced solid with pentagonal faces). There are six fivefold, ten threefold, and fifteen twofold axes. Adding a center of symmetry gives $\boldsymbol{Y}_h = \boldsymbol{Y} \times \boldsymbol{C}_i$, the full symmetry group of these polyhedra.

This exhausts the possible types of point groups with finitely many elements. In addition, the continuous point groups, which have infinitely many elements, must be considered; this will be done in § 98.

§ 94. Group representations

Consider a symmetry group, and let ψ_1 be a single-valued function of the coordinates (in the configuration space of the physical system under consideration). A change of

[6] The groups $\boldsymbol{T}$, $\boldsymbol{T}_d$, $\boldsymbol{T}_h$, $\boldsymbol{O}$, and $\boldsymbol{O}_h$ are called *cubic groups*.

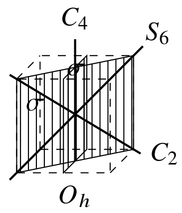

Figure 40

coordinates corresponding to an element G of the group takes this function into another function.

Apply in turn all g transformations of the group, where g is its order. In general, g different functions are then obtained from ψ_1. For particular choices of ψ_1, however, some of these functions may be linearly dependent. The result is a set of f ($f \leqslant g$) linearly independent functions $\psi_1, \psi_2, \ldots, \psi_f$. Under any symmetry transformation belonging to the group, these functions are transformed linearly among themselves. Thus, a transformation G takes each ψ_i ($i = 1, 2, \ldots, f$) into a linear combination of the form

$$\sum_{k=1}^{f} G_{ki}\psi_k,$$

where the constants G_{ki} depend on the transformation G. The collection of these constants is called the *transformation matrix*[7].

It is useful here to regard the group elements G as operators acting on the functions ψ_i. We may then write

$$\widehat{G}\psi_i = \sum_k G_{ki}\psi_k. \tag{94.1}$$

The functions ψ_i can always be chosen mutually orthogonal and normalized. The transformation matrix then has exactly the meaning of an operator matrix as defined in § 11:

$$G_{ik} = \int \psi_i^* \widehat{G}\psi_k \, dq. \tag{94.2}$$

The matrix corresponding to the product of two group elements G and H is obtained from their matrices by the ordinary matrix-multiplication rule (11.12):

$$(GH)_{ik} = \sum_l G_{il}H_{lk}. \tag{94.3}$$

The collection of the matrices for all elements of a group is called a *representation of the group*. The functions $\psi_1, \ldots, \psi_f$ by which these matrices are defined form the *basis of the representation*. Their number f is the *dimension of the representation*.

Consider the integrals $\int \psi_i^* \psi_k \, dq$. Since integration is over the entire space, their values plainly remain unchanged under every rotation or reflection of the coordinate

[7]Because the functions ψ_i are assumed to be single-valued, one definite matrix corresponds to every group element.

system. In other words, symmetry transformations preserve the orthonormality of the basis functions. It follows (see § 12) that the operators $\widehat{G}$ are unitary[8].

Consequently, the matrices representing the group elements are also unitary when the representation has an orthonormal basis.

Perform on $\psi_1, \ldots, \psi_f$ a linear unitary transformation

$$\psi_i' = \sum_k S_{ki}\psi_k. \tag{94.4}$$

This gives a new set $\psi_1', \ldots, \psi_f'$ which is likewise orthonormal (see § 12)[9]. If the functions ψ_i' are used as the basis, we obtain another representation of the same dimension. Representations related through a linear transformation of their basis functions are said to be *equivalent*; plainly, they are not essentially different.

The matrices of equivalent representations obey a simple relation. According to (12.7), the matrix of $\widehat{G}$ in the new representation equals, in the old representation, the matrix of the operator

$$\widehat{S}^{-1}\widehat{G}\widehat{S}. \tag{94.5}$$

The sum of the diagonal elements, that is, the trace, of the matrix representing a group element G is called its *character*; characters will be denoted by $\chi(G)$. An essential fact is that the matrices of equivalent representations have the same characters (see (12.11)). This gives special importance to describing a group representation through its characters: representations that differ essentially can thereby be distinguished at once from equivalent ones. Henceforth, only inequivalent representations will be described as different.

If S in (94.5) is taken to mean a group element relating conjugate elements G and G', we conclude

that, in any given representation of the group, matrices representing elements in the same class have equal characters.

The identity element E corresponds to the identity transformation. Its matrix in every representation is therefore diagonal, with all diagonal entries equal to one. Thus its character $\chi(E)$ is simply the dimension of the representation:

$$\chi(E) = f. \tag{94.6}$$

Consider a representation of dimension f. It may be possible, by an appropriate linear transformation (94.4), to divide the basis functions into sets containing $f_1, f_2, \ldots$ functions ($f_1 + f_2 + \cdots = f$), in such a way that every group element transforms the functions of each set only among themselves, without involving functions from the other sets. Such a representation is called *reducible*.

If no linear transformation of the basis can reduce the number of functions transformed among themselves, their representation is called *irreducible*. Every reducible representation can be decomposed into irreducible representations. This means that a

[8]What is essential in this argument is that the integrals either vanish (when $i \neq k$) or are certainly nonzero (when $i = k$), because the integrand $|\psi_i|^2$ is positive.

[9]Recall from (12.12) that, because the transformations are unitary, the sum of the squared moduli of the basis functions is invariant.

suitable linear transformation separates the basis functions into a number of sets, each of which transforms under the group elements according to an irreducible representation. Several distinct sets may transform according to the same irreducible representation; that irreducible representation is then said to occur in the reducible one the corresponding number of times.

Irreducible representations are an essential characteristic of a group and play the central part in every quantum-mechanical application of group theory. We shall state their principal properties[10].

It can be shown that the number of different irreducible representations equals the number r of classes in the group. We distinguish the characters of different irreducible representations by superscripts: the characters of the matrix for an element G in these representations are $\chi^{(1)}(G), \chi^{(2)}(G), \ldots, \chi^{(r)}(G)$.

The matrix elements of irreducible representations satisfy several orthogonality relations. First, for two different irreducible representations one has

$$\sum_G G_{ik}^{(\alpha)} G_{lm}^{(\beta)*} = 0, \tag{94.7}$$

where $\alpha \neq \beta$ labels the two irreducible representations and the sum extends over all group elements. For each individual irreducible representation,

$$\sum_G G_{ik}^{(\alpha)} G_{lm}^{(\alpha)*} = \frac{g}{f_\alpha} \delta_{il} \delta_{km}, \tag{94.8}$$

so only the sums of squared moduli of matrix elements are nonzero:

$$\sum_G |G_{ik}^{(\alpha)}|^2 = \frac{g}{f_\alpha}.$$

Relations (94.7) and (94.8) may be combined as

$$\sum_G G_{ik}^{(\alpha)} G_{lm}^{(\beta)*} = \frac{g}{f_\alpha} \delta_{\alpha\beta} \delta_{il} \delta_{km}. \tag{94.9}$$

In particular, this yields an important orthogonality relation for representation characters. Summing both sides of (94.9) over the index pairs i, k and l, m, we find

$$\sum_G \chi^{(\alpha)}(G) \chi^{(\beta)}(G)^* = g\delta_{\alpha\beta}. \tag{94.10}$$

For $\alpha = \beta$ this becomes

$$\sum_G |\chi^{(\alpha)}(G)|^2 = g.$$

Thus the sum of squared moduli of the characters of an irreducible representation equals the order of the group. This relation can also serve as a test of irreducibility: for a reducible representation the sum is necessarily greater than g (it equals ng if the representation contains n irreducible parts, all mutually different).

[10]Proofs of these properties may be found in any specialized course on group theory.

Equation (94.10) further shows that equality of the characters of two irreducible representations is sufficient, as well as necessary, for their equivalence.

Since characters belonging to elements of one class are equal, the sum in (94.10) actually has only r independent terms. It may therefore be rewritten

$$\sum_C g_C \chi^{(\alpha)}(C)\chi^{(\beta)}(C)^* = g\delta_{\alpha\beta}, \tag{94.11}$$

where the sum is over the r classes of the group, denoted provisionally by C, and g_C is the number of elements in class C.

Because the number of irreducible representations is the same as the number of classes, the quantities $f_{\alpha C} = \sqrt{g_C/g}\,\chi^{(\alpha)}(C)$ form a square matrix containing r^2 entries. The orthogonality relations in the first index, $\sum_C f_{\alpha C} f^*_{\beta C} = \delta_{\alpha\beta}$, then imply automatically the orthogonality relations in the second index, $\sum_\alpha f_{\alpha C} f^*_{\alpha C'} = \delta_{CC'}$. Hence, in addition to (94.11),

$$\sum_\alpha \chi^{(\alpha)}(C)\chi^{(\alpha)}(C')^* = \frac{g}{g_C}\delta_{CC'}. \tag{94.12}$$

Among the irreducible representations of every group there is always one trivial representation, realized by a single basis function invariant under all transformations of the group. This one-dimensional representation is called the *unit representation*; all its characters equal one. If one representation in the orthogonality relation (94.10) or (94.11) is the unit representation, then for the other we obtain

$$\sum_G \chi^{(\alpha)}(G) = \sum_C g_C \chi^{(\alpha)}(C) = 0, \tag{94.13}$$

that is, the sum of the characters of all group elements vanishes for every nonunit representation.

Relation (94.10) provides a very simple decomposition of any reducible representation into irreducible ones once the characters of both are known.

Let $\chi(G)$ be the characters of a reducible representation of dimension f, and let $a^{(1)}, a^{(2)}, \ldots, a^{(r)}$ give the numbers of times the corresponding irreducible representations occur in it, so that

$$\sum_{\beta=1}^{r} a^{(\beta)} f_\beta = f \tag{94.14}$$

(f_β are the dimensions of the irreducible representations). The characters $\chi(G)$ may then be written

$$\chi(G) = \sum_{\beta=1}^{r} a^{(\beta)}\chi^{(\beta)}(G). \tag{94.15}$$

Multiplying this equality by $\chi^{(\alpha)}(G)^*$ and summing over all G, relation (94.10) gives

$$a^{(\alpha)} = \frac{1}{g}\sum_G \chi(G)\chi^{(\alpha)}(G)^*. \tag{94.16}$$

Consider the $f = g$ dimensional representation realized by the g functions $\widehat{G}\phi$, where ϕ is a coordinate function of general form, so that the g functions $\widehat{G}\phi$ obtained from it are all linearly independent. Such a representation is called *regular*. Clearly, none of its matrices has any diagonal element except the matrix corresponding to the identity element. Hence $\chi(G) = 0$ for $G \neq E$, while $\chi(E) = g$. On decomposing this representation into irreducible ones, (94.16) gives $a^{(\alpha)} = (1/g)gf_\alpha = f_\alpha$. Thus every irreducible representation occurs in the regular representation a number of times equal to its dimension. Substitution in (94.14) gives

$$f_1^2 + f_2^2 + \cdots + f_r^2 = g; \tag{94.17}$$

Thus the squares of the dimensions of the group's irreducible representations add up to its order[11].

It follows in particular that every irreducible representation of an Abelian group (where $r = g$) is one-dimensional ($f_1 = f_2 = \cdots = f_r = 1$).

We also state, without proof, that the dimensions of a group's irreducible representations are divisors of its order.

The actual decomposition of the regular representation into irreducible parts is effected by the formula

$$\psi_i^{(\alpha)} = \frac{f_\alpha}{g} \sum_G G_{ik}^{(\alpha)*} \widehat{G}\phi. \tag{94.18}$$

It is readily verified that, for a fixed value of k, the functions $\psi_i^{(\alpha)}$ ($i = 1, 2, \ldots, f_\alpha$) defined by this formula transform among themselves according to

$$\widehat{G}\psi_i^{(\alpha)} = \sum_l G_{li}^{(\alpha)} \psi_l^{(\alpha)},$$

and hence form a basis of the αth irreducible representation. Giving k its different values produces f_α distinct sets of basis functions $\psi_i^{(\alpha)}$ for the same irreducible representation. This agrees with the fact that the irreducible representation occurs f_α times in the regular representation.

An arbitrary function ψ may be represented as a sum of functions transforming according to the irreducible representations of the group. The required formulas are

$$\psi = \sum_\alpha \sum_i \psi_i^{(\alpha)}, \qquad \psi_i^{(\alpha)} = \frac{f_\alpha}{g} \sum_G G_{ii}^{(\alpha)*} \widehat{G}\psi. \tag{94.19}$$

To prove them, substitute the second formula into the first and perform the summation over i. This gives

$$\psi = \frac{1}{g} \sum_\alpha f_\alpha \chi^{(\alpha)*}(G) \widehat{G}\psi. \tag{94.20}$$

The dimensions f_α are the characters $\chi^{(\alpha)}(E)$ of the identity element. Using the orthogonality relation (94.12), we find that the sum $\sum_\alpha f_\alpha \chi^{(\alpha)*}(G)$ is nonzero, and

[11] For point groups it is worth noting that, once r and g are specified, equation (94.17) can in fact be satisfied by a set of integers $f_1, \ldots, f_r$ in only one way.

equal to g, only when G is the identity element. The right-hand side of (94.20) therefore coincides identically with ψ.

Consider two different sets of functions $\psi_1^{(\alpha)}, \ldots, \psi_{f_\alpha}^{(\alpha)}$ and $\psi_1^{(\beta)}, \ldots, \psi_{f_\beta}^{(\beta)}$ realizing two irreducible representations of the group. The products $\psi_i^{(\alpha)}\psi_k^{(\beta)}$ form a set of $f_\alpha f_\beta$ new functions which can serve as the basis of a new representation of dimension $f_\alpha f_\beta$. This is called the *direct (or Kronecker) product* of the first two representations. It is irreducible only if at least one of f_α or f_β equals one. The characters of the direct product are plainly the products of the characters of its two component representations. Indeed, if

$$\widehat{G}\psi_i^{(\alpha)} = \sum_l G_{li}^{(\alpha)}\psi_l^{(\alpha)}, \qquad \widehat{G}\psi_k^{(\beta)} = \sum_m G_{mk}^{(\beta)}\psi_m^{(\beta)},$$

then

$$\widehat{G}\psi_i^{(\alpha)}\psi_k^{(\beta)} = \sum_{l,m} G_{li}^{(\alpha)}G_{mk}^{(\beta)}\psi_l^{(\alpha)}\psi_m^{(\beta)};$$

hence, denoting its characters by $(\chi^{(\alpha)} \times \chi^{(\beta)})(G)$, we obtain

$$(\chi^{(\alpha)} \times \chi^{(\beta)})(G) = \sum_{i,k} G_{ii}^{(\alpha)}G_{kk}^{(\beta)} = \sum_i G_{ii}^{(\alpha)} \sum_k G_{kk}^{(\beta)},$$

that is,

$$(\chi^{(\alpha)} \times \chi^{(\beta)})(G) = \chi^{(\alpha)}(G)\chi^{(\beta)}(G). \tag{94.21}$$

The two irreducible representations being multiplied may in particular coincide. In that event, there are two different sets $\psi_1, \ldots, \psi_f$ and $\phi_1, \ldots, \phi_f$ realizing the same representation, while its direct product with itself is realized by the f^2 functions $\psi_i\phi_k$ and has characters

$$(\chi \times \chi)(G) = [\chi(G)]^2.$$

This reducible representation can be decomposed at once into two representations of smaller dimension (although, in general, they too remain reducible). One is realized on the $f(f+1)/2$ functions $\psi_i\phi_k + \psi_k\phi_i$, and the other on the $f(f-1)/2$ functions $\psi_i\phi_k - \psi_k\phi_i$ $(i \neq k)$; evidently, within either set the functions transform only into one another. The first is called the *symmetric product* of the representation with itself (its characters are written as $[\chi^2](G)$), and the second the *antisymmetric product* (its characters are written as $\{\chi^2\}(G)$).

To determine the characters of the symmetric product, write

$$\begin{aligned}\widehat{G}(\psi_i\phi_k + \psi_k\phi_i) &= \sum_{l,m} G_{li}G_{mk}(\psi_l\phi_m + \psi_m\phi_l)\\ &= \frac{1}{2}\sum_{l,m}(G_{li}G_{mk} + G_{mi}G_{lk})(\psi_l\phi_m + \psi_m\phi_l).\end{aligned}$$

Its character is therefore

$$[\chi^2](G) = \frac{1}{2}\sum_{i,k}(G_{ii}G_{kk} + G_{ik}G_{ki}).$$

But

$$\sum_i G_{ii} = \chi(G), \qquad \sum_{i,k} G_{ik}G_{ki} = \chi(G^2);$$

and hence finally

$$[\chi^2](G) = \frac{1}{2}\{[\chi(G)]^2 + \chi(G^2)\}, \tag{94.22}$$

which expresses the characters of the symmetric product of a representation with itself through those of the original representation. In precisely the same manner, the characters of the antisymmetric product are found to be[12]

$$\{\chi^2\}(G) = \frac{1}{2}\{[\chi(G)]^2 - \chi(G^2)\}. \tag{94.23}$$

If the functions ψ_i and ϕ_i coincide, they can of course define only the symmetric product, realized by the squares ψ_i^2 and the products $\psi_i\psi_k$ $(i \neq k)$. Symmetric products of higher powers also arise in applications; their characters can be found in an analogous way.

We record a property of direct products that will be needed below. When the direct product of two distinct irreducible representations is resolved into irreducible parts, the identity representation occurs (and then only once) only if the two representations being multiplied are complex conjugates. For real representations, the identity representation occurs only in the direct product of an irreducible representation with itself (evidently, in its symmetric part). Indeed, to determine whether representation (94.21) contains the identity representation, one need only sum its characters over G, in accordance with (94.16), and divide the result by the group order g. The assertion then follows directly from the orthogonality relations (94.10).

We conclude with several remarks about the irreducible representations of a group that is the direct product of two other groups (this must not be confused with the direct product of two representations of one and the same group!). If the functions $\psi_i^{(\alpha)}$ realize an irreducible representation of group A, and the functions $\psi_k^{(\beta)}$ do the same for group B, then the products $\psi_k^{(\beta)}\psi_i^{(\alpha)}$ form a basis for an $f_\alpha f_\beta$-dimensional representation of the group $A \times B$, and this representation is irreducible. Its characters are obtained by multiplying the corresponding characters of the original representations (cf. the derivation of formula (94.21)); for an element $C = AB$ of the group $A \times B$, the corresponding character is

$$\chi(C) = \chi^{(\alpha)}(A)\chi^{(\beta)}(B). \tag{94.24}$$

Taking in this way all pairwise products of the irreducible representations of groups A and B gives every irreducible representation of $A \times B$.

§ 95. Irreducible representations of point groups

We now turn to determining explicitly the irreducible representations of point groups. The overwhelming majority of molecules possess only symmetry axes of orders two,

[12] For representations of dimension 2, it is useful to note that the characters $\{\chi^2\}(G)$ coincide with the determinants of the linear transformations G, as is readily checked by direct calculation.

three, four, and six. We shall therefore omit the icosahedral groups Y, Y_h; we shall consider the groups C_n, C_{nh}, C_{nv}, D_n, D_{nh} only for $n = 1, 2, 3, 4, 6$, and the groups S_{2n}, D_{nd} only for $n = 1, 2, 3$.

Table 7 gives the characters of the representations of these groups. Isomorphic groups possess the same representations. They are therefore placed together in a single table. In the first rows, the numerals preceding the symbols for group elements give the numbers of elements in the corresponding classes (see § 93). The first columns contain the adopted conventional labels for the representations. The letters A and B label one-dimensional representations. The letter E labels two-dimensional representations. The letter F labels three-dimensional representations. (The symbol E for a two-dimensional irreducible representation must not be confused with the symbol E for the identity element of the group!)[13] The basis functions of representations A are symmetric under rotations about the principal axis of order n. The basis functions of representations B are antisymmetric under these rotations. For functions having different symmetry under reflection σ_h, the labels differ in the number of primes, one or two. The subscripts g and u indicate symmetry with respect to inversion. Alongside the representation labels, the letters x, y, z indicate the representations according to which the coordinates themselves transform. The z axis is always chosen along the principal symmetry axis. The letters ε and ω denote:

$$\varepsilon = e^{2\pi i/3}, \qquad \omega = e^{2\pi i/6} = -\omega^4, \qquad \varepsilon + \varepsilon^2 = -1, \qquad \omega^2 - \omega = -1.$$

The action of the operator $\widehat{G}$ on a basis function ψ can only multiply that function by a scalar. This scalar must be a gth root of unity. It is therefore one of the values denoted by $\sqrt[g]{1}$. These are the only possible multipliers. Accordingly, the action has the following form.[14]

$$\widehat{G}\psi = e^{2\pi i k/g}\psi, \qquad k = 1, 2, \ldots, g.$$

The group C_{2h} (and the groups C_{2v} and D_2 isomorphic to it) is Abelian, so all its irreducible representations are likewise one-dimensional; their characters can only be ± 1 (since the square of every element is E).

Next consider the group C_{3v}. Relative to the group C_3, it has in addition the reflections σ_v in vertical planes (all belonging to one class). A function invariant under rotation about the axis (a basis function of the representation A of C_3) may be either symmetric or antisymmetric under the reflections σ_v. Functions that acquire the factors ε and ε^2 under the rotation C_3 (basis functions of the complex-conjugate representations E), however, are transformed into one another by reflection[15]. It follows that the group C_{3v} (and the group D_3 isomorphic to it) has two one-dimensional irreducible representations and one two-dimensional irreducible representation, with the characters listed in the

table. That these are indeed all the irreducible representations follows from $1^2 + 1^2 + 2^2 = 6$, which is the order of the group.

[13] The reason why two complex-conjugate one-dimensional representations are denoted as a single two-dimensional representation will become clear in § 96.

[14] For the point group C_n, one possible choice of functions ψ is $\psi = e^{ik\varphi}$. Here $k = 1, 2, \ldots, n$. The quantity φ is the angle of rotation about the axis, measured from a specified direction.

[15] These functions may be chosen, for example, as $\psi_1 = e^{i\varphi}$, $\psi_2 = e^{-i\varphi}$. Under reflection in a vertical plane, φ changes sign.

Table 7. Characters of irreducible representations of point groups

C_i	E	I
A_g	1	1
$A_u; x, y, z$	1	-1

C_2	E	C_2
$A; z$	1	1
$B; x, y$	1	-1

C_s	E	σ
$A'; x, y$	1	1
$A''; z$	1	-1

C_{2h}	E	C_2	σ_h	I
A_g	1	1	1	1
B_g	1	-1	-1	1
$A_u; z$	1	1	-1	-1
$B_u; x, y$	1	-1	1	-1

C_{2v}	E	C_2	σ_v	σ_v'
$A_1; z$	1	1	1	1
$B_2; y$	1	-1	-1	1
A_2	1	1	-1	-1
$B_1; x$	1	-1	1	-1

D_2	E	C_2	C_2'	C_2''
A	1	1	1	1
$B_3; x$	1	-1	-1	1
$B_1; z$	1	1	-1	-1
$B_2; y$	1	-1	1	-1

C_{3v}	E	$2C_3$	$3\sigma_v$
$A_1; z$	1	1	1
A_2	1	1	-1
$E; x, y$	2	-1	0

D_3	E	$2C_3$	$3U_2$
A_1	1	1	1
$A_2; z$	1	1	-1
$E; x, y$	2	-1	0

Table 7 (continued)

C_6	E	C_6	C_3	C_2	C_6^5	C_6^4
$A; z$	1	1	1	1	1	1
B	1	-1	1	-1	1	-1
E_1	1	ω	ω^2	-1	$-\omega$	$-\omega^2$
	1	$-\omega$	ω^2	-1	ω	$-\omega^2$
$E_2; x \pm iy$	1	ω^2	$-\omega$	1	ω^2	$-\omega$
	1	$-\omega^2$	$-\omega$	1	$-\omega^2$	ω

C_{4v}, D_4, D_{2d}	E	C_2	$2C_4$	$2U_2$	$2U_2'$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	1	-1	1	-1
B_2	1	1	-1	-1	1
$E; x, y$	2	-2	0	0	0

Table 7 (continued)

D_6, C_{6v}, D_{3h}	E	C_2	$2C_3$	$2C_6$	$3U_2$	$3U_2'$
A_1	1	1	1	1	1	1
A_2	1	1	1	1	-1	-1
B_1	1	-1	1	-1	1	-1
B_2	1	-1	1	-1	-1	1
E_2	2	2	-1	-1	0	0
$E_1; x, y$	2	-2	-1	1	0	0

T	E	$3C_2$	$4C_3$	$4C_3^2$
A	1	1	1	1
E	1	1	ε	ε^2
	1	1	ε^2	ε
$F; x, y, z$	3	-1	0	0

O, T_d	E	$8C_3$	$3C_2$	$6\sigma_d$	$6S_4$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
E	2	-1	2	0	0
$F_2; x, y, z$	3	0	-1	1	-1
F_1	3	0	-1	-1	1

Analogous reasoning gives the characters of representations of other groups of the same type. In particular, it gives those for C_{4v} and C_{6v}.

The group T is obtained from the group $D_2 \equiv V$ by adjoining rotations about four inclined axes of order three. A function invariant under the transformations of V (a basis function of the representation A) may acquire a factor 1, ε, or ε^2 under a rotation C_3. The basis functions of the three one-dimensional representations B_1, B_2, B_3 of V, on the other hand, are transformed into one another by rotations about the axes of order three (as is seen, for example, by taking the coordinates x, y, z themselves for these functions). We thus obtain three one-dimensional irreducible representations and one three-dimensional irreducible representation ($1^2 + 1^2 + 1^2 + 3^2 = 12$).

Finally, consider the isomorphic groups O and T_d. The group T_d is obtained from T by adjoining reflections σ_d in planes, each of which contains two axes of order three. A basis function of the identity representation A of T may be either symmetric or antisymmetric under these reflections (all of which belong to one class), giving two one-dimensional representations of T_d. Functions that acquire the factor ε or ε^2 under rotation about an axis of order three (a basis of the complex-conjugate representations E of T) are interchanged by reflection in a plane containing this axis, thereby giving one two-dimensional representation.

Lastly, of the three basis functions of the representation F of T, one is transformed into itself by reflection (and may either remain unchanged or change sign), while the other two are interchanged. Altogether, therefore, there are two one-dimensional, one two-dimensional, and two three-dimensional representations[16].

The representations of the remaining point groups of interest may be obtained directly from those already written down by observing that these groups are direct products of groups already considered with the group C_i (or C_s). Specifically,

$$\begin{gathered}
C_{3h} = C_3 \times C_s, \quad D_{2h} = D_2 \times C_i, \quad D_{3d} = D_3 \times C_i, \quad O_h = O \times C_i, \\
C_{4h} = C_4 \times C_i, \quad D_{4h} = D_4 \times C_i, \quad D_{6h} = D_6 \times C_i, \\
C_{6h} = C_6 \times C_i, \quad S_6 = C_3 \times C_i, \quad T_h = T \times C_i.
\end{gathered}$$

Each of these direct products has twice the original group's number of irreducible representations. Half are symmetric with respect to inversion. They are denoted by the index g. The other half are antisymmetric with respect to inversion. They are denoted by the index u.

The characters of these representations are obtained from those of the original group's representations by multiplication by ± 1 (in accordance with rule (94.24)). Thus, for the

[16]We mention that the icosahedral groups have irreducible representations of higher dimensions (four and five).

group D_{3d} we obtain the representations

D_{3d}	E	$2C_3$	$3U_2$	I	$2S_6$	$3\sigma_d$
A_{1g}	1	1	1	1	1	1
A_{2g}	1	1	−1	1	1	−1
E_g	2	−1	0	2	−1	0
A_{1u}	1	1	1	−1	−1	−1
A_{2u}	1	1	−1	−1	−1	1
E_u	2	−1	0	−2	1	0

§ 96. Irreducible representations and the classification of terms

Applications of group theory in quantum mechanics rest on the invariance of the Schrödinger equation for a physical system (an atom or molecule) under that system's symmetry transformations[17]. It follows directly that applying group elements to a function which solves the Schrödinger equation for some energy eigenvalue must produce further solutions of the same equation with that same energy. In other words, a symmetry transformation mixes among themselves the wave functions of the system's stationary states belonging to one energy level; these functions therefore realize a representation of the group. The essential point is that this representation is irreducible. Indeed, functions that necessarily mix with one another under symmetry transformations must in any event belong to the same energy level. By contrast, if the basis of a reducible representation were separated into several sets of functions that do not mix with one another, equality of the corresponding energy eigenvalues would be an exceedingly unlikely coincidence[18].

Thus each energy level of a system is associated with an irreducible representation of its symmetry group. The representation's dimension gives the degree of degeneracy of the level, namely the number of distinct states having that energy. Specifying the irreducible representation fixes all symmetry properties of the state—its response to the various symmetry transformations.

Irreducible representations of dimension greater than one occur only for groups containing elements that do not commute (Abelian groups possess only one-dimensional irreducible representations). It is worth recalling here that the connection between degeneracy and the existence of mutually noncommuting operators which nevertheless commute with the Hamiltonian was established earlier by arguments not involving group theory (see § 10).

An important qualification must be attached to all these statements. As was pointed out previously (see § 18), in the absence of a magnetic field, symmetry under reversal of the sign of time entails in quantum mechanics that complex-conjugate wave functions belong to the same energy eigenvalue. Consequently, when a set of functions and its complex-conjugate set realize two distinct (inequivalent) irreducible representations of

[17] Group-theoretical methods were first introduced into quantum mechanics by Wigner (E. P. Wigner, 1926).

[18] Unless special reasons exist for it. In this connection, recall the "accidental" degeneracy that arises when the system's Hamiltonian has a symmetry higher than the purely geometrical symmetry discussed in this chapter (cf. the end of § 36).

the group, these conjugate representations must be taken together as one "physically irreducible" representation of twice the dimension; this convention will be understood everywhere below. Examples occurred in the preceding section. The group C_3, for instance, has only one-dimensional representations, yet two of them are complex conjugates and physically correspond to doubly degenerate energy levels. (In a magnetic field there is no symmetry under reversal of the sign of time, and complex-conjugate representations therefore correspond to different energy levels[19]).

Suppose a physical system is acted on by a perturbation (the system is placed in an external field). To what extent can this perturbation split degenerate levels? The external field has a symmetry of its own[20]. If that symmetry is the same as, or higher than, the symmetry of the unperturbed system[21], then the perturbed Hamiltonian $\widehat{H} = \widehat{H}_0 + \widehat{V}$ has the same symmetry as the unperturbed operator $\widehat{H}_0$. No splitting of degenerate levels can then occur. If, however, the perturbation has lower symmetry than the unperturbed system, the symmetry of the Hamiltonian $\widehat{H}$ will be that of the perturbation $\widehat{V}$. The wave functions which formed an irreducible representation of the symmetry group of $\widehat{H}_0$ will still form a representation of the symmetry group of the perturbed operator $\widehat{H}$, but this representation may be reducible; the degenerate level then splits. The following example shows how the machinery of group theory gives a definite answer concerning the splitting of a particular level.

Let the unperturbed system have symmetry T_d. Consider a triply degenerate level corresponding to the irreducible representation F_2 of this group. Its characters are

	E	$8C_3$	$3C_2$	$6\sigma_d$	$6S_4$
F_2	3	0	−1	1	−1

Now let a perturbation of symmetry C_{3v} act on the system, with its threefold axis coinciding with one of the corresponding axes of T_d. The three wave functions of the degenerate level realize a representation of C_{3v}, a subgroup of T_d. The characters of this representation are simply those of the same elements in the original T_d representation:

E	$2C_3$	$3\sigma_v$
3	0	1

This representation, however, is reducible. From the characters of the irreducible representations of C_{3v}, it is readily decomposed into irreducible parts by the general rule (94.16). The result is the sum of the A_1 and E representations of C_{3v}. Hence the triply degenerate F_2 level separates into one nondegenerate A_1 level and one doubly degenerate E level. If instead the same system is subjected to a perturbation of symmetry C_{2v}, also

[19]Strictly, real characters (that is, equivalence of complex-conjugate representations) do not by themselves suffice to ensure that real basis functions can be chosen for a group representation. They do suffice for irreducible representations of point groups, although not for "double" point groups—see § 99.

[20]One example is provided by the energy levels of ionic d and f shells in a crystal lattice, weakly interacting with the surrounding atoms. Here the perturbation (external field) is the field exerted on the ion by the other atoms.

[21]If the symmetry group H is a subgroup of G, the symmetry associated with H is said to be lower than the higher symmetry of G. Plainly, for a sum of two expressions having symmetries G and H, respectively, the symmetry is the lower symmetry H.

a subgroup of T_d, the wave functions of the same F_2 level give a representation whose characters are

E	C_2	σ_v	σ'_v
3	−1	1	1

Decomposition into irreducible parts shows that it contains the representations A_1, B_1, and B_2. In this case, therefore, the level splits completely into three nondegenerate levels.

§ 97. Selection rules for matrix elements

Group theory serves not only to classify the terms of any symmetric physical system; it also supplies a straightforward way of determining selection rules for matrix elements of the various quantities that characterize the system.

The method rests on the following general theorem. Let $\psi_i^{(\alpha)}$ be one of the functions in a basis of an irreducible (nonidentity) representation of the symmetry group. Its integral over the entire space[22] then vanishes identically:

$$\int \psi_i^{(\alpha)}\, dq = 0. \tag{97.1}$$

To prove this, observe that an integral over all space is invariant under any change of the coordinate system and, in particular, under every symmetry transformation. Hence

$$\int \psi_i^{(\alpha)} dq = \int \widehat{G}\psi_i^{(\alpha)} dq = \int \sum_k G_{ki}^{(\alpha)} \psi_k^{(\alpha)} dq.$$

Now sum this equality over the elements of the group. On the left the integral is merely multiplied by the group order g, so that

$$g \int \psi_i^{(\alpha)} dq = \sum_k \int \psi_k^{(\alpha)} \sum_G G_{ki}^{(\alpha)} dq.$$

For any irreducible representation other than the identity representation, however, $\sum_G G_{ki}^{(\alpha)} = 0$ identically. This is the special case of the orthogonality relations (94.7) in which one of the two irreducible representations is the identity representation. The theorem follows.

If ψ is a basis function of some reducible representation, the integral $\int \psi\, dq$ can be nonzero only when that representation includes the identity representation. This assertion is an immediate consequence of the preceding theorem.

The matrix elements of a physical quantity f are the integrals

$$\langle \beta k | f | \alpha i \rangle = \int \psi_k^{(\beta)} \widehat{f} \psi_i^{(\alpha)} dq, \tag{97.2}$$

where α and β distinguish different energy levels of the system, while i and k enumerate the wave functions belonging to the same degenerate level[23]. Denote by $D^{(\alpha)}$ and $D^{(\beta)}$

[22] The configuration space of the physical system in question is meant here.

[23] After passing to "physically irreducible" representations, the basis functions may be chosen real. For this reason, (97.2) makes no distinction between the wave functions and their complex conjugates.

the irreducible representations of the system's symmetry group realized by $\psi_i^{(\alpha)}$ and $\psi_k^{(\beta)}$. Let D_f denote the representation of that group corresponding to the symmetry of f; it is determined by the tensor character of f. Thus, when f is a true scalar, its operator $\widehat{f}$ is invariant under every symmetry transformation, and D_f is the identity representation. The same conclusion holds for a pseudoscalar if the group contains symmetry axes only. If reflections also belong to the group, D_f is one-dimensional but is not the identity representation. For a vector quantity f, the representation D_f is realized by the three vector components that transform into one another; in general this representation is different for polar and axial vectors.

The products $\psi_k^{(\beta)}\widehat{f}\psi_i^{(\alpha)}$ realize the group representation given by the direct product $D^{(\beta)} \times D_f \times D^{(\alpha)}$. A matrix element can be nonzero when this representation contains the identity representation, or, equivalently, when $D^{(\beta)} \times D^{(\alpha)}$ contains D_f. In practice it is more convenient to resolve $D^{(\alpha)} \times D_f$ into irreducible parts. This immediately gives every type $D^{(\beta)}$ of state for which transitions from a state of type $D^{(\alpha)}$ can have nonzero matrix elements.

The simplest case is that of a scalar quantity, for which D_f is the identity representation. It follows at once that a matrix element can be nonzero only for a transition between states of the same type. Indeed, the direct product of two distinct irreducible representations $D^{(\alpha)} \times D^{(\beta)}$ does not contain the identity representation, whereas the direct product of an irreducible representation with itself always does. This is the most general statement of a theorem whose special cases have already arisen several times.

Matrix elements diagonal in energy call for separate treatment: they describe transitions between states belonging to one and the same term, in contrast to transitions between states belonging to two distinct terms of the same type. Here there is only one system of functions $\psi_1^{(\alpha)}, \psi_2^{(\alpha)}, \ldots$, rather than two different systems. The procedure for obtaining the selection rules now depends on how f behaves under time reversal.

Consider a state whose wave function has the form $\psi = \sum_i c_i\psi_i^{(\alpha)}$. In this state the mean value of f is

$$\bar{f} = \sum_{i,k} c_k^* c_i \langle \alpha k|f|\alpha i\rangle.$$

For the state with the complex-conjugate wave function $\psi^* = \sum_i c_i^*\psi_i^{(\alpha)}$, we have

$$\bar{f} = \sum_{i,k} c_k c_i^* \langle \alpha k|f|\alpha i\rangle = \sum_{i,k} c_i c_k^* \langle \alpha i|f|\alpha k\rangle.$$

When f is unchanged by time reversal, the two states share not only their energy level but also the same value $\bar{f}$. Since the coefficients c_i are arbitrary, one must therefore have:

$$\langle \alpha k|f|\alpha i\rangle = \langle \alpha i|f|\alpha k\rangle.$$

It is readily shown that the selection rules must then be found from the symmetric part $[D^{(\alpha)2}]$ rather than from the whole direct product $D^{(\alpha)} \times D^{(\alpha)}$. Nonzero matrix elements exist if $[D^{(\alpha)2}]$ contains D_f.[24]

[24] The product $[D^{(\alpha)2}]$ always includes the identity representation. Thus, for a scalar quantity, diagonal elements (as well as nondiagonal elements between states of the same type) can be nonzero.

If instead f changes sign under time reversal, replacing ψ by ψ^* must be accompanied by a reversal of the sign of $\bar{f}$. The same argument then yields

$$\langle \alpha k | f | \alpha i \rangle = -\langle \alpha i | f | \alpha k \rangle.$$

In this case the selection rules are fixed by the decomposition of the antisymmetric part of the direct product, $\{D^{(\alpha)2}\}$.

Problems

PROBLEM

1. Find the selection rules for the matrix elements of the electric dipole moment $\mathbf{d}$ and magnetic dipole moment $\boldsymbol{\mu}$ when the symmetry is O.

SOLUTION. The group O contains no reflections. Polar vectors ($\mathbf{d}$) and axial vectors ($\boldsymbol{\mu}$) therefore transform according to the same irreducible representation, F_1. The direct products of F_1 with the other representations of O decompose as

$$\begin{aligned} &F_1 \times A_1 = F_2, \qquad F_1 \times A_2 = F_1, \qquad F_1 \times E = F_1 + F_2, \\ &F_1 \times F_1 = A_1 + E + F_1 + F_2, \quad F_1 \times F_2 = A_2 + E + F_1 + F_2. \end{aligned} \tag{1}$$

Consequently, the nondiagonal (in energy) matrix elements can be nonzero for the transitions

$$F_1 \leftrightarrow A_1, E, F_1, F_2; \qquad F_2 \leftrightarrow A_2, E, F_1, F_2.$$

The symmetric and antisymmetric products of the irreducible representations of O are

$$\begin{aligned} &[A_1^2] = [A_2^2] = A_1, \qquad [E^2] = A_1 + E, \qquad [F_1^2] = [F_2^2] = A_1 + E + F_2, \\ &\{E^2\} = A_2, \qquad \{F_1^2\} = \{F_2^2\} = F_1. \end{aligned} \tag{2}$$

None of the symmetric products contains F_1. Hence the vector $\mathbf{d}$, which is invariant under time reversal, has no matrix elements diagonal in energy. The magnetic moment, on the other hand, reverses sign under time reversal and has diagonal matrix elements in states of type F_1 and F_2.

PROBLEM

2. Repeat the calculation for symmetry D_{3d}.

SOLUTION. The transformation laws of $\mathbf{d}$ and $\boldsymbol{\mu}$ differ in the group D_{3d}:

$$d_x, d_y \sim E_u, \quad d_z \sim A_{2u}, \qquad \mu_x, \mu_y \sim E_g, \quad \mu_z \sim A_{2g}$$

(here and in the problems below, the sign $\sim$ stands for the words "transforms according to the representation"). We have

$$\begin{aligned} &E_u \times A_{1g} = E_u \times A_{2g} = E_u, \qquad E_u \times A_{1u} = E_u \times A_{2u} = E_g, \\ &E_u \times E_u = A_{1g} + A_{2g} + E_g, \qquad E_u \times E_g = A_{1u} + A_{2u} + E_u. \end{aligned} \tag{3}$$

Thus the nondiagonal matrix elements of d_x, d_y can be nonzero for $E_u \leftrightarrow A_{1g}, A_{2g}, E_g$ and $E_g \leftrightarrow A_{1u}, A_{2u}, E_u$. The other selection rules are found in the same way:

$$\begin{aligned}\text{for } d_z: &\quad A_{1g} \leftrightarrow A_{2u};\ A_{2g} \leftrightarrow A_{1u};\ E_g \leftrightarrow E_u;\\ \text{for } \mu_x, \mu_y: &\quad E_g \leftrightarrow A_{1g}, A_{2g}, E_g;\ E_u \leftrightarrow A_{1u}, A_{2u}, E_u;\\ \text{for } \mu_z: &\quad A_{1g} \leftrightarrow A_{2g};\ A_{1u} \leftrightarrow A_{2u};\ E_g \leftrightarrow E_g;\ E_u \leftrightarrow E_u.\end{aligned}$$

The symmetric and antisymmetric products of the irreducible representations of D_{3d} are

$$\begin{aligned}&[A_{1g}^2] = [A_{1u}^2] = [A_{2g}^2] = [A_{2u}^2] = A_{1g},\\ &[E_g^2] = [E_u^2] = E_g + A_{1g},\\ &\{E_g^2\} = \{E_u^2\} = A_{2g}.\end{aligned} \tag{4}$$

It follows that every component of $\mathbf{d}$ has vanishing matrix elements diagonal in energy. For $\boldsymbol{\mu}$, diagonal matrix elements of μ_z exist for transitions between states belonging to a degenerate level of type E_g or E_u.

Problem

3. Find the selection rules for matrix elements of the electric quadrupole-moment tensor Q_{ik} for symmetry O.

Solution. With respect to the group O, the components of Q_{ik}, a symmetric tensor satisfying $\sum_i Q_{ii} = 0$, transform as follows:

$$Q_{xy},\ Q_{xz},\ Q_{yz} \sim F_2,$$

$$Q_{xx} + \epsilon Q_{yy} + \epsilon^2 Q_{zz}, \quad Q_{xx} + \epsilon^2 Q_{yy} + \epsilon Q_{zz} \sim E \quad (\epsilon = e^{2\pi i/3}).$$

Decomposing the direct products of F_2 and E with every representation of the group gives the following selection rules for nondiagonal matrix elements:

$$\begin{aligned}\text{for } Q_{xy}, Q_{xz}, Q_{yz}: &\quad F_1 \leftrightarrow A_2, E, F_1, F_2;\ F_2 \leftrightarrow A_1, E, F_1, F_2;\\ \text{for } Q_{xx}, Q_{yy}, Q_{zz}: &\quad E \leftrightarrow A_1, A_2, E;\ F_1 \leftrightarrow F_1, F_2;\ F_2 \leftrightarrow F_2.\end{aligned}$$

As follows from (2), diagonal matrix elements occur in the following states:

$$\begin{aligned}\text{for } Q_{xy}, Q_{xz}, Q_{yz}: &\quad F_1, F_2,\\ \text{for } Q_{xx}, Q_{yy}, Q_{zz}: &\quad E, F_1, F_2.\end{aligned}$$

Problem

4. Repeat the calculation for symmetry D_{3d}.

Solution. The components of Q_{ik} transform under D_{3d} according to

$$Q_{zz} \sim A_{1g}; \quad Q_{xx} - Q_{yy},\ Q_{xy} \sim E_g; \quad Q_{xz},\ Q_{yz} \sim E_g.$$

The component Q_{zz} behaves as a scalar. Decomposing the direct products of E_g with all the group representations, we obtain the selection rules for the nondiagonal matrix elements of the remaining components of Q_{ik}:

$$E_g \leftrightarrow A_{1g}, A_{2g}, E_g; \qquad E_u \leftrightarrow A_{1u}, A_{2u}, E_u.$$

As follows from (4), diagonal elements can be nonzero only for states of type E_g and E_u.

§ 98. Continuous groups

In addition to the finite point groups listed in § 93, there are also *continuous point groups*, whose number of elements is infinite. These are the groups associated with *axial* symmetry and with *spherical* symmetry.

The simplest axial-symmetry group is C_∞. It contains the rotations $C(\varphi)$ through any angle φ about the symmetry axis (and is called the *two-dimensional rotation group*). The groups C_n approach this group in the limit $n \to \infty$. In the same manner, the limiting forms of C_{nh}, C_{nv}, D_n, and D_{nh} are the continuous groups $C_{\infty h}$, $C_{\infty v}$, D_∞, and $D_{\infty h}$, respectively.

A molecule has axial symmetry only when all its atoms lie on a single straight line. If it is not symmetric with respect to its midpoint, its point group is $C_{\infty v}$; besides rotations about the axis, this group also includes reflections σ_v in every plane through the axis. If the molecule is symmetric with respect to its midpoint, its point group is $D_{\infty h} = C_{\infty v} \times C_i$. The groups C_∞, $C_{\infty h}$, and D_∞, on the other hand, cannot occur as molecular symmetry groups at all.

The group of complete spherical symmetry contains rotations through an arbitrary angle about every axis through the centre, together with reflections in every plane through that point. This group, which we shall denote by K_h, is the symmetry group of an isolated atom. It has as a subgroup the group K of all rotations in space (called the three-dimensional rotation group, or simply the *rotation group*). Adjoining a centre of symmetry to K gives the group K_h:

$$K_h = K \times C_i.$$

The elements of a continuous point group can be distinguished by one or several parameters ranging continuously over their possible values. For example, three Euler angles specifying a rotation of the coordinate system may serve as parameters of the rotation group.

The general properties of finite groups described in § 92, and the associated concepts (subgroups, conjugate elements, classes, and so forth), extend directly to continuous groups. Statements tied directly to the order of a group naturally cease to apply; one example is the assertion that the order of a subgroup divides the order of the group.

In $C_{\infty v}$ all symmetry planes are equivalent, and consequently all the reflections σ_v form a single class containing a continuous

set of elements. The symmetry axis is two-sided, so there is a continuous set of classes, each containing the two elements $C(\pm\varphi)$. The classes of $D_{\infty h}$ follow immediately from those of $C_{\infty v}$, since $D_{\infty h} = C_{\infty v} \times C_i$.

All axes in the rotation group K are equivalent and two-sided. Its classes therefore consist of rotations through an angle of a specified absolute magnitude $|\varphi|$ about an arbitrary axis. The classes of K_h follow directly from those of K.

The notion of representations, both reducible and irreducible, likewise extends directly to continuous groups. Every irreducible representation contains a continuous set of matrices, but the number of basis functions transformed into one another—the dimension of the representation—is finite. These functions can always be chosen so that the representation is unitary. A continuous group has infinitely many distinct irreducible representations, but they form a discrete sequence: they can be assigned successive numbers. The matrix elements and characters of these representations obey orthogonality relations generalizing the analogous relations for finite groups. In place of (94.9) we now have

$$\int G_{ik}^{(\alpha)} G_{lm}^{(\beta)*}\, d\tau_G = \frac{1}{f_\alpha}\delta_{\alpha\beta}\delta_{il}\delta_{km}\int d\tau_G, \tag{98.1}$$

and in place of (94.10),

$$\int \chi^{(\alpha)}(G)\chi^{(\beta)}(G)^*\, d\tau_G = \delta_{\alpha\beta}\int d\tau_G. \tag{98.2}$$

The integration in these formulas is the so-called invariant integration over the group. The integration element $d\tau_G$ is expressed in terms of the group parameters and their differentials in such a way that the action of any group transformation again produces the same integration element[25]. In the rotation group, for instance, one may take $d\tau_G = \sin\beta\, d\alpha\, d\beta\, d\gamma$, where α, β, and γ are the Euler angles specifying the rotation of the coordinate system (§ 58); then $\int d\tau_G = 8\pi^2$.

We have, in substance, already obtained the irreducible representations of the three-dimensional rotation group. We did so while finding the eigenvalues and eigenfunctions of the total angular momentum, although without employing the terminology of group theory. Apart from a constant factor, the operators for the components of angular momentum are the operators of infinitesimal rotations[26]; the angular-momentum eigenvalues therefore characterize the behaviour of wavefunctions under rotations in space. For angular momentum j there are $2j+1$ distinct eigenfunctions ψ_{jm}, distinguished by the values m of its projection and belonging to a single energy level of degeneracy $2j+1$. A rotation of the coordinate system mixes these functions among themselves, and they thus furnish an irreducible representation of the rotation group. In group-theoretic terms, therefore, the numbers j label the irreducible representations of the rotation group, with one representation of dimension $2j+1$ for every j. Since j assumes integral and half-integral values, the dimensions $2j+1$ run through all positive integers $1, 2, 3, \ldots$

The basis functions of these representations were already examined in substance in §§ 56 and 57 (and the representation matrices were found in § 58). A basis for the representation with a given j consists of the $2j+1$ independent components of a symmetric spinor of rank $2j$ (equivalently, the set of $2j+1$ functions ψ_{jm}).

[25] The assertions made here about the properties of irreducible representations of continuous groups hold only when the integrals (98.1) and (98.2) converge; in particular, the "group volume" $\int d\tau_G$ must be finite. This condition is satisfied by continuous point groups. It is not satisfied, for example, by the so-called Lorentz group, which will be encountered in the relativistic theory.

[26] In mathematical terminology these operators are called the *generators* of the rotation group.

The irreducible representations of the rotation group belonging to half-integral j have an essential special feature. On rotation through 2π, their basis functions—the components of a spinor of odd rank—change sign. But a 2π rotation is the identity element of the group. We must therefore conclude that representations with half-integral j are, as it is said, *double-valued*: in such a representation each group element (a rotation about some axis through an angle φ, $0 \leqslant \varphi \leqslant 2\pi$) corresponds not to one matrix but to two matrices whose characters have opposite signs[27].

As already noted, an isolated atom has the symmetry $K_h = K \times C_i$. From the standpoint of group theory, every atomic term is consequently associated with an irreducible representation of the rotation group K, which fixes the value of the atom's total angular

momentum J, and with an irreducible representation of C_i, which fixes the parity of the state[28].

Placing an atom in an external electric field splits its energy levels. The number of distinct levels thereby produced, and the symmetry of their states, can be found by the method described in § 96. To do this, the reducible $(2J + 1)$-dimensional representation of the symmetry group of the external field, furnished by the functions ψ_{JM}, must be decomposed into irreducible representations of that group. We therefore need the characters of the representation furnished by the functions ψ_{JM}. Since irreducible-representation characters have the same value for all elements of a class, it is enough to consider rotations about a single axis, chosen as the z axis. Under a rotation through φ about z, the wavefunctions ψ_{JM} are, as we know, multiplied by $e^{iM\varphi}$, where M is the projection of the angular momentum on that axis. The transformation matrix for the functions ψ_{JM} is consequently diagonal, and its character is

$$\chi^{(J)}(\varphi) = \sum_{M=-J}^{J} e^{iM\varphi} = \frac{e^{i(J+1)\varphi} - e^{-iJ\varphi}}{e^{i\varphi} - 1},$$

or[29]

$$\chi^{(J)}(\varphi) = \frac{\sin\Big((J + \tfrac{1}{2})\varphi\Big)}{\sin(\varphi/2)}. \tag{98.3}$$

[27] It should be noted that double-valued representations of a group are not representations in the strict sense, since they are realized by multivalued basis functions; see also § 99.

[28] The atomic Hamiltonian is also invariant under permutations of the electrons. In the non-relativistic approximation the spatial and spin wavefunctions separate, and one may speak of representations of the permutation group furnished by the spatial functions. Specifying an irreducible representation of the permutation group determines the total spin S of the atom (see § 63). Once relativistic interactions are included, however, a wavefunction cannot be separated into spatial and spin parts. Symmetry under simultaneous permutations of the coordinates and spins of the particles supplies no characteristic of a term, since the Pauli principle permits only total wavefunctions that are antisymmetric in all the electrons. Correspondingly, when relativistic interactions are included, spin is not, strictly speaking, conserved; only the total angular momentum J is conserved.

[29] To avoid misunderstanding, we emphasize that this formula uses a parametrization of group elements different from the Euler-angle parametrization: a transformation is specified by the direction of the rotation axis and by the angle φ about it. With this parametrization it can be shown, for example, that the integration in (98.2) is to be performed with the element $2(1 - \cos\varphi)\, d\varphi\, do$, where do is the element of solid angle for the direction of the rotation axis.

Under inversion I, on the other hand, all the functions ψ_{JM} with different M behave alike: they are multiplied by $+1$ or by -1 according as the atomic state is even or odd.

Consequently, the character is

$$\chi^{(J)}(I) = \pm(2J+1). \tag{98.4}$$

Finally, the characters for reflection in a plane σ and for a rotation-reflection through an angle φ are calculated by writing these symmetry transformations in the form

$$\sigma = IC_2, \qquad S(\varphi) = IC(\pi+\varphi).$$

We shall also consider the irreducible representations of the axial-symmetry group $C_{\infty v}$. This question was essentially answered already when we classified the electronic terms of a diatomic molecule, whose symmetry is precisely $C_{\infty v}$ when its two atoms are different. Two one-dimensional representations correspond to the terms 0^+ and 0^- (the terms with $\Omega = 0$): the identity representation A_1, and the representation A_2, in which the basis function is invariant under every rotation but changes sign under reflections in the planes σ_v. The doubly degenerate terms with $\Omega = 1, 2, \ldots$ correspond instead to two-dimensional representations, denoted by $E_1, E_2, \ldots$ Under a rotation through φ about the axis their basis functions are multiplied by $e^{\pm i\Omega\varphi}$, while reflection in a plane σ_v interchanges the two functions. The characters of all these representations are

$C_{\infty v}$	E	$2C(\varphi)$	$\infty\sigma_v$
A_1	1	1	1
A_2	1	1	-1
E_k	2	$2\cos k\varphi$	0

(98.5)

The irreducible representations of $D_{\infty h} = C_{\infty v} \times C_i$ follow directly from those of $C_{\infty v}$; they correspond to the classification of the terms of a diatomic molecule with identical nuclei.

If half-integral values are used for Ω, the functions $e^{\pm i\Omega\varphi}$ realize double-valued irreducible representations of $C_{\infty v}$, corresponding to molecular terms with half-integral spin[30].

§ 99. Double-valued representations of finite point groups

States of a system with half-integral spin (and therefore half-integral total angular momentum) correspond to double-valued representations of the point symmetry group of the system. This is a general property of spinors and hence applies to continuous and finite point groups alike. It is therefore necessary to determine the *double-valued* irreducible representations of finite point groups.

[30]Unlike the three-dimensional rotation group, here an appropriate choice of fractional values of Ω could produce not only single- and double-valued representations, but also triple-valued and still higher-valued ones. Physically possible eigenvalues of angular momentum, however, as the operator of an infinitesimal rotation in three dimensions, are determined by representations of the three-dimensional rotation group specifically. Thus triple-valued and higher-valued representations of the two-dimensional rotation group (and likewise of any finite symmetry group), although definable mathematically, have no physical meaning.

As already noted, double-valued representations are not, strictly speaking, true representations of a group at all. In particular, the relations discussed in § 94 do not apply to them. Thus, whenever those relations referred to all irreducible representations (for example, relation (94.17) for the sum of the squares of their dimensions), only true, single-valued representations were understood to be included.

The following formal device is convenient for finding double-valued representations (H. A. Bethe, 1929). We introduce, in a purely formal manner, a new group element Q: rotation through 2π about an arbitrary axis. It is regarded as distinct from the identity, but as becoming identical with E when applied twice, so that $Q^2 = E$. Accordingly, rotations C_n about n-fold symmetry axes give the identity only after $2n$ applications, rather than n:

$$C_n^n = Q, \qquad C_n^{2n} = E. \tag{99.1}$$

Inversion I, as an element commuting with every rotation, must still give E when applied twice. Two successive reflections in a plane, however, now give Q, not E:

$$\sigma^2 = Q, \qquad \sigma^4 = E \tag{99.2}$$

(this follows because a reflection can be written $\sigma = IC_2$). We thus obtain a collection of elements forming a fictitious point symmetry group whose order is twice that of the original group; such groups will be called *double point groups*. The double-valued representations of the actual point group are plainly single-valued, true representations of the corresponding double group, and can therefore be found by ordinary methods.

The number of classes in a double group is greater than in the original group (though not generally twice as great). The element Q commutes with all other group elements[31] and therefore always constitutes a class by itself. If a symmetry axis is two-sided, this means in the double group that C_n^k and $C_n^{2n-k} = QC_n^{n-k}$ are conjugate. Consequently, when twofold axes are present, the division of the elements into classes also depends on whether these axes are two-sided (this is immaterial in ordinary point groups, since C_2 coincides with the inverse rotation C_2^{-1}).

In the group T, for example, the twofold axes are equivalent and each is two-sided, while the threefold axes are equivalent but not two-sided. The 24 elements of the double group T'[32] are therefore divided among seven classes: E, Q, a class containing the three rotations C_2 and the three C_2Q, and the classes $4C_3$, $4C_3^2$, $4C_3Q$, $4C_3^2Q$.

All irreducible representations of a double point group include, first, representations identical with the single-valued representations of the ordinary group (where Q, like E, is represented by the unit matrix), and, second, the double-valued representations of the ordinary group, where Q is represented by the negative unit matrix. It is the latter that concern us here.

The double groups C_n' ($n = 1, 2, 3, 4, 6$) and S_4', like their corresponding ordinary groups, are cyclic[33]. All their irreducible representations are one-dimensional and can be found immediately, as explained in § 95.

[31] This is evident for rotations and inversion; for reflection in a plane it follows by representing the reflection as the product of inversion and a rotation.

[32] A prime on the symbol of an ordinary group will distinguish its double group.

[33] The groups $S_2' \equiv C_i'$ and $S_6' \equiv C_{3i}'$, which contain inversion I, are Abelian but not cyclic.

The irreducible representations of D'_n (or of the groups C'_{nv} isomorphic to them) can be found in the same way as for the corresponding ordinary groups. They are realized by functions $e^{\pm ik\varphi}$, where φ is the angle of rotation about the n-fold axis and k takes half-integral values (integral values correspond to the ordinary single-valued representations). Rotations about the horizontal twofold axes interchange these functions, while C_n multiplies them by $e^{\pm 2\pi ik/n}$.

The representations of the double cubic groups are somewhat harder to find. The 24 elements of T' are divided among seven

classes. There are consequently seven irreducible representations in all, four of which coincide with those of the ordinary group T. The sum of the squares of the dimensions of the remaining three must be 12, showing that all three are two-dimensional. Since C_2 and C_2Q belong to the same class, $\chi(C_2) = \chi(C_2Q) = -\chi(C_2)$, and hence $\chi(C_2) = 0$ in all three representations. Further, at least one of the three representations must be real, since complex representations can occur only in mutually conjugate pairs. Consider this representation and suppose that the matrix of C_3 has been reduced to diagonal form, with diagonal elements a_1, a_2. Since $C_3^3 = Q$, we have $a_1^3 = a_2^3 = -1$. For $\chi(C_3) = a_1 + a_2$ to be real, we must take $a_1 = e^{\pi i/3}$ and $a_2 = e^{-\pi i/3}$. We then find $\chi(C_3) = 1$ and $\chi(C_3^2) = a_1^2 + a_2^2 = -1$. One of the required representations has thus been found. Its direct products with the two mutually complex-conjugate one-dimensional representations of T give the other two.

Similar arguments, which we shall not give here, determine the representations of O'. Table 8 summarizes the characters of the representations of the double groups just listed (only the representations corresponding to double-valued representations of the ordinary groups are included). Double groups isomorphic to these have the same representations.

The remaining point groups are either isomorphic to those considered or are obtained by taking their direct products with C_i, so their representations require no separate calculation.

For the same reasons as with ordinary representations, two mutually complex-conjugate double-valued representations must be treated as one physically irreducible representation of twice the dimension. One-dimensional double-valued representations must be doubled even when their characters are real. The reason is that (see § 60) complex-conjugate wave functions are linearly independent in systems of half-integral spin. Thus, if a double-valued one-dimensional representation has real characters[34] and is realized by a function ψ, then although the complex-conjugate function ψ^* transforms according to an equivalent representation, ψ and ψ^* are nevertheless linearly independent. Since,

[34] Such representations occur for the groups C'_n with odd n; their characters are $\chi(C_n^k) = (-1)^k$.

Table 8. Double-valued representations of point groups

D'_2	E	Q	$C_2^{(x)}$	$C_2^{(y)}$	$C_2^{(z)}Q$				
			$C_2^{(x)}Q$	$C_2^{(y)}Q$	$C_2^{(z)}Q$				
E'	2	-2	0	0	0				
D'_3	E	Q	C_3	C_3^2	$3U_2$	$3U_2Q$			
			C_3^2Q	C_3Q					
E'_1	1	-1	-1	1	i	$-i$			
	1	-1	-1	1	$-i$	i			
E'_2	2	-2	1	-1	0	0			
D'_6	E	Q	C_2	C_3	C_3^2	C_6	C_6^5	$3U_2$	$3U'_2$
			C_2Q	C_3^2Q	C_3Q	C_6^5Q	C_6Q	$3U_2Q$	$3U'_2Q$
E'_1	2	-2	0	1	-1	$\sqrt{3}$	$-\sqrt{3}$	0	0
E'_2	2	-2	0	1	-1	$-\sqrt{3}$	$\sqrt{3}$	0	0
E'_3	2	-2	0	-2	2	0	0	0	0
D'_4	E	Q	C_2	C_4	C_4^3	$2U_2$	$2U'_2$		
			C_2Q	C_4^3Q	C_4Q	$2U_2Q$	$2U'_2Q$		
E'_1	2	-2	0	$\sqrt{2}$	$-\sqrt{2}$	0	0		
E'_2	2	-2	0	$-\sqrt{2}$	$\sqrt{2}$	0	0		
T'	E	Q	$4C_3$	$4C_3^2$	$4C_3Q$	$4C_3^2Q$	$3C_2$		
							$3C_2Q$		
E'	2	-2	1	-1	-1	1	0		
G'	2	-2	ϵ	$-\epsilon^2$	$-\epsilon$	ϵ^2	0		
	2	-2	ϵ^2	$-\epsilon$	$-\epsilon^2$	ϵ	0		
O'	E	Q	$4C_3$	$4C_3^2$	$3C_4^2$	$3C_4$	$3C_4^3$	$6C_2$	
			$4C_3^2Q$	$4C_3Q$	$3C_4^2Q$	$3C_4^3Q$	$3C_4Q$	$6C_2Q$	
E'_1	2	-2	1	-1	0	$\sqrt{2}$	$-\sqrt{2}$	0	
E'_2	2	-2	1	-1	0	$-\sqrt{2}$	$\sqrt{2}$	0	
G'	4	-4	-1	1	0	0	0	0	

on the other hand, complex-conjugate wave functions must belong to the same energy level, we see that such a representation must be doubled in physical applications.

Everything said in § 97 about finding selection rules for matrix elements of physical quantities f remains valid for states of a system with half-integral spin, with a change only for matrix elements diagonal in energy. Repeating the argument at the end of § 97, now taking formulas (60.2) and (60.3) into account, we find that if f is even or odd under time reversal, the selection rules are obtained by considering, respectively, the antisymmetric product $\{D^{(\alpha)2}\}$ or symmetric product $[D^{(\alpha)2}]$ of $D^{(\alpha)}$ with itself. This is the reverse of the rule stated in § 97 for systems with integral spin[35].

Problem

Determine how the levels of an atom (with specified values of the total angular momentum J) split when the atom is placed in a field of cubic symmetry O[36].

Solution. The wave functions of the atomic states with angular momentum J and different values of M_J realize a $(2J + 1)$-dimensional reducible representation of O, with characters given by (98.3). Decomposing this representation into irreducible parts (single-valued for integral J and double-valued for half-integral J) determines the required splitting (cf. § 96). The irreducible parts for the first several values of J are

$J = 0$	$1/2$	1	$3/2$	2	$5/2$	3
A_1	E_1'	F_1	G'	$E + F_2$	$E_2' + G'$	$A_2 + F_1 + F_2$

[35] For application of these rules, note that with double-valued representations the identity representation is contained not in the symmetric but in the antisymmetric product of a representation with itself. For a two-dimensional double-valued representation, $\{D^{(\alpha)2}\}$ is simply the identity representation.

[36] One example is an atom in a crystal lattice. Notice also that whether the symmetry group of the external field has a center of symmetry is immaterial here, since the behavior of the wave function under inversion (the evenness or oddness of the level) is unrelated to J.

CHAPTER XIII

Polyatomic Molecules

§ 100. Classification of Molecular Vibrations

For polyatomic molecules, group theory first of all resolves the question of classifying their electronic terms, that is, the energy levels at a specified arrangement of the nuclei. They are classified according to the irreducible representations of the point symmetry group possessed by the nuclear configuration under consideration. It must, however, be stressed that the classification obtained in this way pertains precisely to this particular arrangement of the nuclei, since a displacement of the nuclei generally destroys the symmetry of the configuration. Usually the arrangement in question is the one corresponding to the equilibrium positions of the nuclei. In that case the classification retains a certain meaning also for small vibrations of the nuclei, but naturally loses its meaning when the vibrations cannot be regarded as small.

For a diatomic molecule we did not encounter this question, since its axial symmetry is of course preserved under any displacement of the nuclei. The situation is similar for triatomic molecules. Three nuclei always lie in one plane, which is a symmetry plane of the molecule. Consequently, the electronic terms of a triatomic molecule can always be classified with respect to this plane (by whether the wave functions are symmetric or antisymmetric under reflection in the plane).

There is an empirical rule for the normal electronic terms of polyatomic molecules: in the overwhelming majority of molecules, the wave function of the normal electronic state is totally symmetric (this rule was already mentioned for diatomic molecules in § 78). In other words, it is invariant under every element of the molecular symmetry group, and thus belongs to the identity irreducible representation of the group.

Group-theoretical methods are especially important in the study of molecular vibrations (E. Wigner, 1930). A purely classical treatment of the vibrations

of a molecule as a system of a number of interacting particles (nuclei) must precede the quantum-mechanical study of this question.

As is known from mechanics (see I, § 23, 24), a system of N particles (not lying on one straight line) has $3N-6$ vibrational degrees of freedom; of the total $3N$ degrees of freedom, three correspond to translational and three to rotational motion of the system as a whole[1]. The energy of a system of particles executing small vibrations can be written as

$$E = \frac{1}{2}\sum_{i,k} m_{ik}\dot{u}_i\dot{u}_k + \frac{1}{2}\sum_{i,k} k_{ik}u_iu_k. \qquad (100.1)$$

Here m_{ik} and k_{ik} are constant coefficients, while u_i are components of the vectors giving the particles' displacements from their equilibrium positions (the indices i, k

[1] If all the particles lie on one straight line, the number of vibrational degrees of freedom is $3N-5$ (rotation then has only two corresponding coordinates, so that rotation of a linear molecule about its own axis has no meaning).

enumerate both vector components and particles). By a suitable linear transformation of the quantities u_i, the coordinates corresponding to translation and rotation of the system can be eliminated from (100.1), and the vibrational coordinates can be chosen so that both quadratic forms in (100.1) become sums of squares. Normalizing these coordinates so that all coefficients in the kinetic-energy expression become unity, we obtain the vibrational energy in the form

$$E = \frac{1}{2}\sum_{i,\alpha}\dot{Q}_{\alpha i}^2 + \frac{1}{2}\sum_{\alpha}\omega_\alpha^2\sum_i Q_{\alpha i}^2. \tag{100.2}$$

The vibrational coordinates $Q_{\alpha i}$ are called *normal coordinates*; ω_α are the frequencies of the corresponding independent vibrations. Several normal coordinates may have one and the same frequency (the frequency is then said to be *multiple*); the index α of a normal coordinate specifies the frequency, while $i = 1, 2, \ldots, f_\alpha$ enumerates the coordinates belonging to that same frequency (f_α is the multiplicity of the frequency).

Expression (100.2) for the molecular energy must be invariant under symmetry transformations. This means that, under every transformation belonging to the molecule's point symmetry group, the normal coordinates $Q_{\alpha i}$, $i = 1, 2, \ldots, f_\alpha$ (for each fixed α), transform linearly

among themselves in such a way that the sum of squares $\sum_i Q_{\alpha i}^2$ remains unchanged.

In other words, the normal coordinates belonging to each given natural frequency of the molecule realize an irreducible representation of its symmetry group; the frequency multiplicity determines the dimension of the representation. Irreducibility follows from the same considerations given in § 96 for solutions of the Schrödinger equation. Equality of frequencies corresponding to two different irreducible representations would be an extraordinarily unlikely coincidence. Here again a qualification is required: since physical normal coordinates are intrinsically real quantities, two complex-conjugate representations correspond to one natural frequency having twice the multiplicity.

These considerations make it possible to classify the natural vibrations of a molecule without solving the difficult problem of explicitly determining its normal coordinates. To do so, we must first find (by the method described below) the representation realized by all the vibrational coordinates together; we shall call it the complete *vibrational representation*. This representation is reducible, and by decomposing it into irreducible parts we thereby determine the multiplicities of the natural frequencies and the symmetry properties of the corresponding vibrations. The same irreducible representation may occur several times in the complete representation; this means that there are several different frequencies of the same multiplicity whose vibrations have the same symmetry.

To find the complete vibrational representation, we use the fact that the characters of a representation are invariant under a linear transformation of the basis functions. Hence, in calculating them, we may use as basis functions not the normal coordinates but simply the components u_i of the vectors giving the displacements of the nuclei from their equilibrium positions.

It is clear first of all that, when calculating the character of an element G of the point group, only those nuclei which remain fixed (more precisely, whose equilibrium positions remain fixed) under the given symmetry transformation need be considered. Indeed, if under the rotation or reflection G a nucleus 1 is moved to a new position previously occupied by another identical nucleus 2, the displacement of nucleus 1 is transformed under G into the displacement of nucleus 2. In other words, the rows of the matrix G_{ik} corresponding to this nucleus (that is, to its displacement u_i) will in any event contain no diagonal elements. The components of the displacement vector of a nucleus

whose equilibrium position is unaffected by the operation G, on the other hand, transform only among themselves and can therefore be considered independently of the displacement vectors of the other nuclei.

Consider first a rotation $C(\varphi)$ through an angle φ about a symmetry axis. Let u_x, u_y, u_z be the components of the displacement vector of a nucleus whose equilibrium position lies on the axis itself and is therefore unaffected by the rotation. Under the rotation these components transform like those of any ordinary (polar) vector, according to the formulas (the z axis coincides with the symmetry axis)

$$u'_x = u_x \cos\varphi + u_y \sin\varphi, \qquad u'_y = -u_x \sin\varphi + u_y \cos\varphi, \qquad u'_z = u_z.$$

The character, that is, the sum of the diagonal elements of the transformation matrix, is $1 + 2\cos\varphi$. If a total of N_C nuclei lie on this axis, the total character is

$$N_C(1 + 2\cos\varphi). \tag{100.3}$$

This character, however, corresponds to the transformation of all $3N$ displacements u_i; the part corresponding to the transformation of the translational displacement and the (small) rotation of the molecule as a whole must therefore be separated off. The translational displacement is specified by the displacement vector $\mathbf{U}$ of the molecular center of mass; its contribution to the character is consequently $1 + 2\cos\varphi$. The rotation of the molecule as a whole is specified by the rotation-angle vector $\delta\boldsymbol{\Omega}$[2].

The vector $\delta\boldsymbol{\Omega}$ is axial; under rotations of the coordinate system, however, an axial vector behaves just like a polar vector. Thus $\delta\boldsymbol{\Omega}$ also has the character $1 + 2\cos\varphi$. In all, therefore, we must subtract $2(1 + 2\cos\varphi)$ from (100.3). We finally obtain the character $\chi(C)$ of the rotation $C(\varphi)$ in the complete vibrational representation:

$$\chi(C) = (N_C - 2)(1 + 2\cos\varphi). \tag{100.4}$$

The character of the identity element E is evidently just the total number of vibrational degrees of freedom: $\chi(E) = 3N - 6$ (this also follows from (100.4) with $N_C = N$, $\varphi = 0$).

In the same way we calculate the character of the rotation–reflection transformation $S(\varphi)$ (a rotation through φ about the z axis followed by reflection in the xy plane). Under this transformation a vector changes according to

$$u'_x = u_x \cos\varphi + u_y \sin\varphi, \qquad u'_y = -u_x \sin\varphi + u_y \cos\varphi, \qquad u'_z = -u_z,$$

which has the character $(-1 + 2\cos\varphi)$. The character of the representation realized by all $3N$ displacements u_i is therefore

$$N_S(-1 + 2\cos\varphi), \tag{100.5}$$

[2] As is known, a small rotation angle can be regarded as a vector $\delta\boldsymbol{\Omega}$, whose magnitude equals the angle of rotation and whose direction is along the rotation axis in the sense fixed by the screw rule. A vector $\delta\boldsymbol{\Omega}$ defined in this way is evidently axial.

where N_S is the number of nuclei unaffected by the operation $S(\varphi)$ (this number can evidently be either zero or one). The displacement vector $\mathbf{L}$ of the center of mass has the character $(-1 + 2\cos\varphi)$. As for $\delta\mathbf{\Omega}$, being an axial vector it is unchanged by inversion of the coordinate system; on the other hand, the rotation–reflection transformation $S(\varphi)$ can be written

$$S(\varphi) = C(\varphi)\sigma_h = C(\varphi)C_2 I = C(\pi + \varphi)I,$$

that is, as a rotation through $\pi + \varphi$ followed by inversion. Consequently, the character of $S(\varphi)$ acting on $\delta\mathbf{\Omega}$ is equal to the character of $C(\pi + \varphi)$ acting on an ordinary vector, namely $1 + 2\cos(\pi + \varphi) = 1 - 2\cos\varphi$. The sum $(-1 + 2\cos\varphi) + (1 - 2\cos\varphi) = 0$, and we thus reach the result that expression (100.5) itself is the required character $\chi(S)$ of the rotation–reflection transformation $S(\varphi)$ in the complete vibrational representation:

$$\chi(S) = N_S(-1 + 2\cos\varphi). \tag{100.6}$$

In particular, the character of reflection in a plane ($\varphi = 0$) is $\chi(\sigma) = N_\sigma$, while the character of inversion ($\varphi = \pi$) is $\chi(I) = -3N_I$.

Once the characters χ of the complete vibrational representation have been found, it remains only to decompose it into irreducible representations. This is done from formula (94.16), using the character tables given in § 95 (see the problems to this section).

Group theory is not needed to classify the vibrations of a linear molecule. The total number of vibrational degrees of freedom is $3N - 5$. Among the vibrations one must distinguish those in which the atoms remain on one straight line from

those in which they do not[3]. The number of degrees of freedom for the motion of N particles along a line is N; one of them corresponds to translational motion of the molecule as a whole. Hence the number of normal coordinates for vibrations that leave the atoms on a straight line is $N - 1$; in general, these correspond to $N - 1$ different natural frequencies. The remaining $(3N - 5) - (N - 1) = 2N - 4$ normal coordinates belong to vibrations that destroy the linearity of the molecule; they correspond to $N - 2$ different double frequencies (each frequency has two normal coordinates, describing identical vibrations in two mutually perpendicular planes)[4].

Problems

Problem

1. Classify the normal vibrations of the NH_3 molecule (a regular pyramid with the N atom at its apex and the H atoms at the corners of its base; Fig. 41).

Solution. The molecular point symmetry group is C_{3v}. Rotations about the threefold axis leave only one atom (N) fixed, while reflections in the planes leave two atoms fixed (N and one H). From formulas (100.4), (100.6), the characters of the complete vibrational

[3] If the molecule is symmetric about its midpoint, there is one further characteristic of the vibrations; see problem 10 to this section.

[4] In the notation for the irreducible representations of the group $C_{\infty v}$ (§ 98), there are $N - 1$ vibrations of type A_1 and $N - 2$ vibrations of type E_1.

representation are

	E	$2C_3$	$3\sigma_v$
	6	0	2

Decomposing this representation into irreducible parts, we find that it contains the

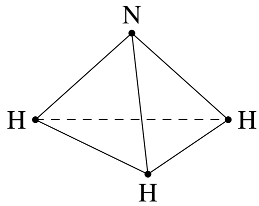

Figure 41

representation A_1 twice and E twice. There are therefore two simple frequencies corresponding to vibrations of type A_1, which preserve the full symmetry of the molecule (the so-called totally symmetric vibrations), and two double frequencies corresponding to normal coordinates that transform into one another according to the representation E.

PROBLEM

2. Do the same for the H_2O molecule (Fig. 42).

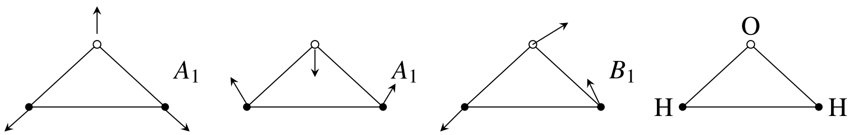

Figure 42

SOLUTION. The symmetry group is C_{2v}. The transformation C_2 leaves the O atom fixed, the transformation σ_v (reflection in the molecular plane) leaves all three atoms fixed, and the reflection σ'_v leaves only the O atom fixed. The characters of the complete vibrational representation are

	E	C_2	σ_v	σ'_v
	3	1	3	1

This representation decomposes into the irreducible representations $2A_1$, $1B_1$: there are two totally symmetric vibrations and one whose symmetry is specified by the representation B_1; all the frequencies are simple (Fig. 42 shows the corresponding normal vibrations).

Problem

3. Do the same for the CH_3Cl molecule (Fig. 43*a*).

Solution. The molecular symmetry group is C_{3v}. The same method shows that there are three totally symmetric vibrations A_1 and three double vibrations of type E.

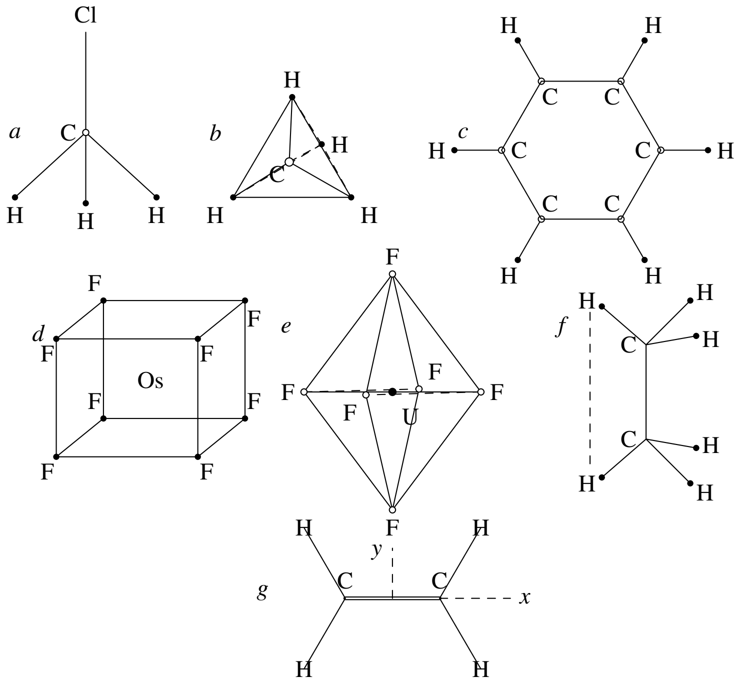
a
Cl
C
H
H
H
b
H
H
C
H
H
c
H
H
C
C
H
C
C
H
C
C
H
H
d
F
F
F
F
Os
F
F
F
F
e
F
F
F
F
F
U
F
f
H
H
C
H
C
H
H
H
g
H
H
y
C
C
x
H
H

Figure 43

PROBLEM

4. Do the same for the CH_4 molecule (the C atom is at the center and the H atoms are at the vertices of a tetrahedron; Fig. 43*b*).

SOLUTION. The molecular symmetry is T_d. The vibrations are $1A_1$, $1E$, $2F_2$.

PROBLEM

5. Do the same for the C_6H_6 molecule (Fig. 43*c*).

SOLUTION. The molecular symmetry is D_{6h}. The vibrations are

$$2A_{1g}, 1A_{2g}, 1A_{2u}, 1B_{1g}, 1B_{1u}, 1B_{2g}, 3B_{2u}, 1E_{1g}, 3E_{1u}, 4E_{2g}, 2E_{2u}.$$

PROBLEM

6. Do the same for the OsF_8 molecule (the Os atom is at the center and the F atoms are at the vertices of a cube; Fig. 43*d*).

SOLUTION. The molecular symmetry is O_h. The vibrations are

$$1A_{1g}, 1A_{2u}, 1E_g, 1E_u, 2F_{1u}, 2F_{2g}, 2F_{2u}.$$

PROBLEM

7. Do the same for the UF_6 molecule (the U atom is at the center and the F atoms are at the vertices of an octahedron; Fig. 43*e*).

SOLUTION. The molecular symmetry is O_h. The vibrations are

$$1A_{1g}, 1E_g, 2F_{1u}, 1F_{2g}, 1F_{2u}.$$

PROBLEM

8. Do the same for the C_2H_6 molecule (Fig. 43*f*).

SOLUTION. The molecular symmetry is D_{3d}. The vibrations are

$$3A_{1g}, 1A_{1u}, 2A_{2u}, 3E_g, 3E_u.$$

PROBLEM

9. Do the same for the C_2H_4 molecule (Fig. 43*g*; all the atoms lie in one plane).

SOLUTION. The molecular symmetry is D_{2h}. The vibrations are

$$3A_{1g}, 1A_{1u}, 2B_{1g}, 1B_{1u}, 2B_{3u}, 1B_{2g}, 2B_{2u}$$

(the coordinate axes are chosen as shown in the figure).

PROBLEM

10. Do the same for a linear molecule of N atoms symmetric about its midpoint.

SOLUTION. To the classification of the vibrations of a linear molecule considered in the text, one must add classification by their behavior under inversion in the center. The cases of even and odd N must be distinguished.

If N is even ($N = 2p$), there is no atom at the midpoint of the molecule. Give the p atoms in one half of the molecule independent displacements along the line, and the other p atoms equal and opposite displacements. We then find that p of the vibrations which leave the atoms on the line are symmetric about the center, while the remaining $(2p-1)-p = p-1$ vibrations of this type are antisymmetric about the center. Next, the p atoms have $2p$ degrees of freedom for motions in which the atoms are not constrained to the line. Giving symmetrically situated atoms equal and opposite displacements would produce $2p$ symmetric vibrations; from this number, however, two corresponding to rotation of the molecule must be subtracted. There are therefore $p-1$ double frequencies for vibrations that carry the atoms off the line and are symmetric about the center, and the same number ($(2p-2)-(p-1) = p-1$) that are antisymmetric. In the notation for the irreducible representations of the group $D_{\infty h}$ (see the end of § 98), there are p vibrations of type A_{1g} and $(p-1)$ vibrations of each of the types A_{1u}, E_{1g}, E_{1u}.

If N is odd ($N = 2p+1$), analogous reasoning shows that there are p vibrations of each of the types A_{1g}, A_{1u}, E_{1u}, and $(p-1)$ vibrations of type E_{1g}.

§ 101. Vibrational energy levels

In a quantum-mechanical treatment, the molecule's vibrational energies are given by the eigenvalues of the Hamiltonian. The Hamiltonian has the form

$$\widehat{H}^{(v)} = \frac{1}{2}\sum_{\alpha}\sum_{i=1}^{f_\alpha}\left(\widehat{P}_{\alpha i}^2 + \omega_\alpha^2 Q_{\alpha i}^2\right), \tag{101.1}$$

where $\widehat{P}_{\alpha i} = -i\hbar\,\partial/\partial Q_{\alpha i}$ are the momentum operators corresponding to the normal coordinates $Q_{\alpha i}$. Since this Hamiltonian separates into a sum of independent terms (the expressions in parentheses), the energy levels are given by the sums

$$E^{(v)} = \hbar\sum_{\alpha}\omega_\alpha\sum_{i}\left(v_{\alpha i} + \frac{1}{2}\right) = \sum_{\alpha}\hbar\omega_\alpha\left(v_\alpha + \frac{f_\alpha}{2}\right), \tag{101.2}$$

where $v_\alpha = \sum_i v_{\alpha i}$ and f_α is the multiplicity of the frequency ω_α. The wave

functions, in turn, are products of the corresponding wave functions of linear harmonic oscillators:

$$\psi = \prod_{\alpha}\psi_\alpha, \tag{101.3}$$

where

$$\psi_\alpha = \text{const}\cdot\exp\left(-\frac{1}{2}c_\alpha^2\sum_i Q_{\alpha i}^2\right)\prod_i H_{v_{\alpha i}}(c_\alpha Q_{\alpha i}); \tag{101.4}$$

H_v denotes the Hermite polynomial of degree v, and $c_\alpha = \sqrt{\omega_\alpha/\hbar}$.

If some of the frequencies ω_α are multiple, the vibrational energy levels are degenerate in general. The energy (101.2) depends only on the sum $v_\alpha = \sum_i v_{\alpha i}$. The degeneracy of a level is therefore the number of ways in which a given set of the numbers v_α can be formed from the numbers $v_{\alpha i}$. For a single number v_α, this number of ways is[5]

$$\frac{(v_\alpha + f_\alpha - 1)!}{v_\alpha!(f_\alpha - 1)!}.$$

Thus the total degeneracy is

$$\prod_\alpha \frac{(v_\alpha + f_\alpha - 1)!}{v_\alpha!(f_\alpha - 1)!}. \tag{101.5}$$

For twofold frequencies the factors in this product are $v_\alpha + 1$, and for threefold frequencies they are $(1/2)(v_\alpha + 1)(v_\alpha + 2)$.

It must be kept in mind that this degeneracy exists only while the vibrations are treated as purely harmonic. If terms of higher degree in the normal coordinates are included in the Hamiltonian (anharmonicity of the vibrations), the degeneracy is lifted in general, though not completely (see § 104 for further details).

The wave functions (101.3) belonging to the same degenerate vibrational term carry a certain representation (reducible in general) of the molecule's symmetry group. Functions belonging to different frequencies, however, transform independently of one another. Hence the representation carried by all the functions (101.3) is the product of the representations carried by the functions (101.4), so it is enough to consider only the latter.

The exponential factor in (101.4) is invariant under every symmetry transformation. In the Hermite polynomials, terms of any fixed degree transform only into one another (a symmetry transformation plainly does not alter the degree of each term). On the other hand, each Hermite polynomial is fully determined by its highest-degree term. Consequently, on writing

$$\prod_{i=1}^{f_\alpha} H_{v_{\alpha i}}(c_\alpha Q_{\alpha i}) = \text{const} \cdot Q_{\alpha 1}^{v_{\alpha 1}} Q_{\alpha 2}^{v_{\alpha 2}} \dots Q_{\alpha f_\alpha}^{v_{\alpha f_\alpha}} + \text{terms of lower degree},$$

it is sufficient to consider the highest-degree term alone.

Functions for which the sum $v_\alpha = \sum_i v_{\alpha i}$ has the same value belong to the same term. We thus obtain the representation carried by products of v_α quantities $Q_{\alpha i}$. This is precisely the symmetric product (see § 94) of v_α copies of the irreducible representation carried by the quantities $Q_{\alpha i}$ (L. Tisza, 1933).

For one-dimensional representations, finding the characters of their v-fold symmetric products is trivial[6]:

$$\chi_v(G) = [\chi(G)]^v.$$

For two- and three-dimensional representations, the following mathematical device is convenient[7]. The sum of the squares of the basis functions of an irreducible

[5] This is the number of ways to distribute v_α balls among f_α boxes.

[6] Here we use the notation $\chi_v(G)$ in place of the cumbersome $[\chi^v](G)$.

[7] Applied for this purpose by A. S. Kompaneyets (1940).

representation is invariant under all symmetry transformations. They may therefore be regarded formally as the components of a two- or three-dimensional vector, and the symmetry transformations as certain rotations (or reflections) acting on these vectors. We stress that, in general, these rotations and reflections have no connection with the actual symmetry transformations and, for every particular group element G, they also depend on the specific representation under consideration.

Consider two-dimensional representations in more detail. Let $\chi(G)$ be the character of some group element in a given two-dimensional representation, with $\chi(G) \neq 0$. Consider the matrix that transforms the components x and y of a two-dimensional vector under a rotation in the plane through an angle φ. The sum of its diagonal elements, namely its trace, is $2\cos\varphi$. By setting

$$2\cos\varphi = \chi(G), \tag{101.6}$$

we determine the angle of the rotation that formally corresponds to the element G in the given irreducible representation. The v-fold symmetric product of the representation has a basis consisting of the $v + 1$ quantities $x^v, x^{v-1}y, \ldots, y^v$. The characters of this representation are[8]

$$\chi_v(G) = \frac{\sin(v+1)\varphi}{\sin\varphi}. \tag{101.7}$$

The case $\chi(G) = 0$ requires separate consideration, since a zero character corresponds both to a rotation through $\pi/2$ and to a reflection. If $\chi(G^2) = -2$, the transformation is a rotation through $\pi/2$, and for $\chi_v(G)$ we obtain

$$\chi_v(G) = (-1)^{v/2}\frac{1+(-1)^v}{2}. \tag{101.8}$$

If instead $\chi(G^2) = 2$, then $\chi(G)$ must be treated as the character of a reflection (that is, the transformation $x \to x$, $y \to -y$); in that case

$$\chi_v(G) = \frac{1+(-1)^v}{2}. \tag{101.9}$$

Formulas for symmetric products of three-dimensional representations can be obtained in the same way. The rotation (or reflection) formally corresponding to a group element in the given representation is readily found with the aid of Table 7. It is the transformation corresponding to the given $\chi(G)$ in that one of the isomorphic groups in which the coordinates transform according to this representation. Thus, for the representation F_1 of the groups O and T_d, the transformation must be taken from the group O, whereas for the representation F_2 it must be taken from the group T_d. We shall not dwell here on the derivation of the corresponding formulas for the characters $\chi_v(G)$.

[8] For the calculation it is convenient to choose the basis functions in the form

$$(x+iy)^v, \quad (x+iy)^{v-1}(x-iy), \ldots, (x-iy)^v;$$

the rotation matrix is then diagonal, and the sum of its diagonal elements has the form

$$e^{iv\varphi} + e^{i(v-2)\varphi} + \ldots + e^{-iv\varphi}.$$

§ 102. Stability of symmetric molecular configurations

For a symmetric arrangement of the nuclei, a molecular electronic term can be degenerate if the symmetry group has irreducible representations of dimension greater than one.

We ask whether such a symmetric configuration can be a stable equilibrium configuration

of the molecule. We shall neglect the effect of spin (if there is any), which is generally negligible in polyatomic molecules. The degeneracy of electronic terms discussed below is therefore purely orbital and unrelated to spin.

For the configuration to be stable, the molecular energy, regarded as a function of the internuclear distances, must have a minimum at this arrangement of the nuclei. This means that the change of energy under a small displacement of the nuclei must contain no term linear in the displacements.

Let $\widehat{H}$ be the Hamiltonian of the molecule's electronic state, with the internuclear distances regarded as parameters. Denote by $\widehat{H}_0$ this Hamiltonian at the specified symmetric configuration. The normal vibrational coordinates $Q_{\alpha i}$ may be used as quantities describing small nuclear displacements. The expansion of $\widehat{H}$ in powers of $Q_{\alpha i}$ is

$$\widehat{H} = \widehat{H}_0 + \sum_{\alpha,i} V_{\alpha i} Q_{\alpha i} + \sum_{\alpha,\beta,i,k} W_{\alpha i,\beta k} Q_{\alpha i} Q_{\beta k} + \dots \tag{102.1}$$

The expansion coefficients $V, W, \dots$ are functions only of the electron coordinates. Under a symmetry transformation, the quantities $Q_{\alpha i}$ transform among themselves, and the sums in (102.1) thereby pass into other sums of the same form. We may therefore regard a symmetry operation formally as transforming the coefficients in these sums while leaving the $Q_{\alpha i}$ fixed. In particular, for each fixed α, the coefficients $V_{\alpha i}$ then transform according to the same representation of the symmetry group as the corresponding coordinates $Q_{\alpha i}$. This follows at once from the invariance of the Hamiltonian under every symmetry operation: the collection of terms of each fixed order in its expansion, and in particular the linear terms, must possess the same invariance[9].

Consider an electronic term E_0 that is degenerate at the symmetric configuration. A displacement of the nuclei that breaks the molecular symmetry will generally split the term. To first order in the nuclear displacements, the splitting is determined by the secular equation formed from matrix elements of the linear part of (102.1),

$$V_{\rho\sigma} = \sum_{\alpha,i} Q_{\alpha i} \int \psi_\rho V_{\alpha i} \psi_\sigma \, dq, \tag{102.2}$$

where ψ_ρ, ψ_σ are wave functions of electronic states belonging to this degenerate term (chosen to be real). Stability of the symmetric configuration requires the absence of splitting linear in Q: all roots of the secular equation must vanish identically, which means that the entire matrix $V_{\rho\sigma}$ must vanish. Naturally, only normal vibrations that

[9] Strictly, the quantities $V_{\alpha i}$ should transform according to the representation complex-conjugate to that of the $Q_{\alpha i}$. As noted earlier, however, if two complex-conjugate representations do not coincide, physically they must in any event be considered together as a single representation of twice the dimension. The qualification is therefore immaterial.

break the molecular symmetry are to be considered; the totally symmetric vibrations (corresponding to the unit representation of the group) must be omitted.

Since the $Q_{\alpha i}$ are arbitrary, the matrix elements (102.2) vanish only if all the integrals

$$\int \psi_\rho V_{\alpha i} \psi_\sigma \, dq \tag{102.3}$$

vanish.

Let $D^{(el)}$ be the irreducible representation according to which the electronic wave functions ψ_ρ transform, and let D_α denote the corresponding representation for the quantities $V_{\alpha i}$; as already stated, the representations D_α are those according to which the corresponding normal coordinates $Q_{\alpha i}$ transform. From the results of § 97, the integrals (102.3) can be nonzero if the product $[D^{(el)2}] \times D_\alpha$ contains the unit representation, or equivalently if $[D^{(el)2}]$ contains D_α. Otherwise every integral vanishes.

Thus the symmetric configuration is stable if $[D^{(el)2}]$ contains none of the irreducible representations D_α characterizing the molecular vibrations, apart from the unit representation. For a nondegenerate electronic state this condition is always satisfied, since the symmetric square of a one-dimensional representation is the unit representation.

As an example, consider a molecule of the CH_4 type, with one atom (C) at the center and four atoms (H) at the vertices of a tetrahedron. This configuration has symmetry T_d. Its degenerate electronic terms correspond to the representations E, F_1, F_2 of this group. The molecule has one normal vibration of type A_1 (a totally symmetric vibration), one double vibration E,

and two triple vibrations F_2 (see Problem 4 of § 100). The symmetric squares of E, F_1, F_2 are

$$[E^2] = A_1 + E, \qquad [F_1^2] = [F_2^2] = A_1 + E + F_2.$$

Each contains at least one of the representations E, F_2; hence the tetrahedral configuration under consideration is unstable for a degenerate electronic state.

This result is a general rule known as the *Jahn–Teller theorem* (H. A. Jahn, E. Teller, 1937): for a degenerate electronic state, every symmetric arrangement of nuclei is unstable, with the sole exception of an arrangement on one straight line. Owing to this instability, the nuclei move in such a way that the symmetry of their configuration is broken sufficiently to remove the term's degeneracy completely. In particular, the normal electronic term of a symmetric nonlinear molecule can only be nondegenerate[10].

As already mentioned, only linear molecules form an exception. This can easily be seen without group theory. A displacement by which a nucleus leaves the molecular axis is an ordinary vector with ξ- and η-components (the ζ-axis is directed along the molecular axis). We saw in § 87 that such vectors have matrix elements only for transitions in which the angular momentum Λ about the axis changes by one. A degenerate term of a linear molecule, however, corresponds to states with angular momenta Λ and $-\Lambda$ about the axis, where $\Lambda \geqslant 1$. A transition between them entails a change in angular momentum

[10]The physical idea that symmetry is destroyed in an electronic state whose degeneracy is due to that very symmetry was put forward by Landau (1934). Jahn and Teller (1937) proved the theorem by enumerating all possible types of symmetric nuclear arrangements in a molecule and examining each by the method given above.

of at least 2, and the matrix elements must therefore vanish. Thus a linear arrangement of nuclei in a molecule may be stable even for a degenerate electronic state.

A constructive general proof of the theorem rests on the following observation (E. Ruch, 1957).

Degeneracy of electronic states due to the symmetry of the nuclear arrangement can occur only for molecular point groups containing at least one rotation axis (C_n) or rotation–reflection axis (S_n) of order $n > 2$. Among the wave functions

of the mutually degenerate states (that is, among the basis functions of the corresponding representation $D^{(el)}$), there is then at least one whose electron density $\rho = |\psi|^2 = \psi^2$ is not invariant under rotations about this axis; the electric field created by the electrons is likewise not symmetric about the axis. At the same time, a nonlinear molecule contains equivalent nuclei away from the axis—nuclei carried into one another by the rotations C_n (or S_n). Equivalent nuclei thus lie at inequivalent points of the electric field. Yet equilibrium positions of charged particles in a field cannot possess an equivalence not required by the field's symmetry, except through an exceedingly improbable accident.

The full proof gives a concrete mathematical embodiment of this physical situation. We show how such a proof is constructed (E. Ruch, A. Schönhofer, 1965)[11].

In a nonlinear molecule, consider a nucleus (call it a) that lies outside the "center" of the molecule (that is, outside the fixed point of the transformations of its symmetry group) and is not on the principal symmetry axis, if one exists[12]. Let H be the set of symmetry operations of the molecule that leave nucleus a fixed. It is a subgroup of the full molecular symmetry group G and may be one of the point groups C_1, C_s, C_n, C_{nv}. The operations in G that are not in H carry nucleus a into other equivalent nuclei $a', a'', \ldots$; let s be the number of nuclei in this set. Clearly, the order of H is g/s, where g is the order of the full group G (that is, s is the index of H in G)[13].

Certainly $s \geqslant 3$, since the presumed existence of a multidimensional irreducible representation $D^{(el)}$ requires, as noted above, at least one symmetry axis of order higher than 2, while by assumption nucleus a does not lie on it.

When restricted from the group G to the lower-symmetry group H, the representation $D^{(el)}$ is generally reducible. Suppose that its decomposition into irreducible representations of H contains a one-dimensional representation; denote it by $d^{(el)}$. It is realized by an electronic wave function ψ, one of the basis functions of $D^{(el)}$. Because $d^{(el)}$ is one-dimensional, the square $\rho = \psi^2$ is invariant under all operations in H, and hence realizes the unit irreducible representation of this group.

The same unit representation of H can be realized by choosing as a basis one of the displacements Q_a of atom a—the displacement along the radius vector drawn from the molecular center to nucleus a.

Applying every operation of G to this displacement gives a basis for a certain representation of G, generally reducible; call it D_Q. Every transformation in G but not in H carries Q_a into a displacement of one of the other $s - 1$ equivalent nuclei $a', a'', \ldots$,

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[11] For details see E. Ruch, A. Schönhofer // Theoret. chim. acta (Berl.). 1965. Bd. 3. S. 291.

[12] In noncubic and nonicosahedral symmetry groups, the principal axis means an axis C_n or S_n of order $n > 2$.

[13] All elements of the group G can be partitioned into s cosets $H, G'H, G''H, \ldots$, where G', G'' are elements of the group carrying nucleus a into $a', a'', \ldots$

and displacements of different nuclei are plainly linearly independent. The dimension of D_Q is therefore s. Moreover, the displacements $Q_a, Q_{a'}, \dots$ forming the basis of D_Q cannot correspond either to a pure translation or to a pure rotation of the molecule as a whole: with three or more equivalent nuclei, no such motion can be composed from their radial displacements.

A representation of G can likewise be obtained by applying all its operations to the function $\rho = \psi^2$; call it D_ρ. The dimension of D_ρ may equal s, but it may also be smaller, since there is no prior reason for all s functions $\rho, G'\rho, G''\rho, \dots$ to be linearly independent. Nevertheless, one may assert that D_ρ, if it does not coincide with D_Q, is in any case contained wholly in it[14]. Furthermore, D_ρ is not the unit representation, since ψ^2 is certainly not invariant under the whole group G (only the sum of the squares of all basis functions of the multidimensional irreducible representation $D^{(el)}$ is invariant).

The properties of D_Q and D_ρ just established immediately give the desired result. Indeed, D_Q is part of the complete vibrational representation, while D_ρ is a part of $[D^{(el)2}]$ that does not contain the unit representation. Since D_ρ is contained in D_Q, it follows that $[D^{(el)2}]$ contains at least one nonunit vibrational representation D_α, as was to be proved.

The preceding argument still assumed, however, that the decomposition of $D^{(el)}$ into irreducible representations of the subgroup H contains a one-dimensional representation. This assumption holds in the overwhelming majority of cases. It is certainly true for $H = C_1, C_s, C_2, C_{2v}$, since all irreducible representations of these groups are one-dimensional. It is also certainly true for $H = C_n, C_{nv}$ with $n > 2$ when the dimension of $D^{(el)}$ is odd, because the groups C_n, C_{nv} have only one- and two-dimensional irreducible representations. Inspection of the character tables for irreducible representations of the point groups shows that the exceptions are two-dimensional representations of the cubic groups $G = O, T_d, O_h$ relative to the subgroups $H = C_3, C_{3v}$.

For definiteness, take $G = O$ and $H = C_3$; this choice affects only the notation for the representations. The two electronic functions ψ_1, ψ_2 realize the representation $D^{(el)} = E$ of O, and also the representation $d^{(el)} = E$ of C_3. The representation of C_3 realized by the products $\psi_1^2, \psi_2^2, \psi_1\psi_2$ is $[E^2] = A + E$. The three components of the vector of an arbitrary displacement $\mathbf{Q}_a$ of nucleus a realize the same representation of C_3. In this case the representation D_ρ of O is $D_\rho = [D^{(el)2}] = A_1 + E$; it does not contain the representation F_2 corresponding to a vector of translation or rotation of the molecule as a whole, and, besides the unit representation, it contains a nonunit one. Therefore the inclusion of D_ρ in D_Q (for the same reasons as above), which here has dimension $3s$, proves the instability of the molecule in this case as well[15].

In keeping with the qualification at the beginning of this section, the degeneracy of the electronic states was understood throughout the preceding discussion to have

[14] The general assertion is as follows. Suppose that the same representation (of dimension f) of a subgroup H is realized by different sets of basis functions, and that applying every transformation of G to one of these sets generates a representation of G of dimension sf, where s is the index of H in G. Then the representation of G generated in the same manner from any other such set either coincides with the first or is contained wholly in it. A rigorous proof is given in the paper cited on the preceding page.

[15] Four-dimensional representations of the icosahedral groups constitute one further exceptional case. It is treated analogously and leads to the same result.

a purely orbital origin. The Jahn–Teller theorem nevertheless remains valid when spin–orbit and spin–spin interactions are included, with the single difference that, in nonlinear molecules with half-integral spin, twofold Kramers degeneracy does not cause instability, in accordance with the general theorem proved in § 60. This case corresponds to two-dimensional double-valued irreducible representations of the double point groups. The absence of instability here can already be seen by the following formal argument. To determine the selection rules for the matrix elements (102.3) when $D^{(el)}$ is double-valued, one must consider not symmetric but antisymmetric products $\{D^{(el)2}\}$ (see § 99). For every two-dimensional double-valued irreducible representation, however, these products coincide with the unit representation, and therefore cannot contain representations corresponding to any molecular vibrations that are not totally symmetric.

§ 103. Quantization of rigid-rotor rotation

The study of the rotational levels of a polyatomic molecule is often complicated by the need to consider rotation together with vibration. As a preliminary problem, we shall consider the rotation of a molecule as a rigid body, that is, with its atoms "rigidly fixed" (a *rigid rotor*).

Let $\xi\eta\zeta$ be a coordinate system whose axes are directed along the three principal axes of inertia of the rotor and which rotates with it. The corresponding Hamiltonian is obtained by replacing the components J_ξ, J_η, J_ζ of its angular momentum in the classical expression for the energy by the corresponding operators:

$$\hat{H} = \frac{\hbar^2}{2}\left(\frac{\hat{J}_\xi^2}{I_A} + \frac{\hat{J}_\eta^2}{I_B} + \frac{\hat{J}_\zeta^2}{I_C}\right), \tag{103.1}$$

where I_A, I_B, I_C are the principal moments of inertia of the rotor.

The commutation rules for the operators $\hat{J}_\xi, \hat{J}_\eta, \hat{J}_\zeta$, the components of angular momentum in the rotating coordinate system, are not evident, since the usual derivation of the commutation rule concerns the components $\hat{J}_x, \hat{J}_y, \hat{J}_z$ in a fixed coordinate system. They are readily obtained, however, from the formula

$$(\hat{\mathbf{J}}\mathbf{a})(\hat{\mathbf{J}}\mathbf{b}) - (\hat{\mathbf{J}}\mathbf{b})(\hat{\mathbf{J}}\mathbf{a}) = -i\hat{\mathbf{J}}(\mathbf{a}\times\mathbf{b}), \tag{103.2}$$

where $\mathbf{a}, \mathbf{b}$ are any two vectors characterizing the given body (and commuting with one another). This formula is easily

verified by evaluating the left-hand side in the fixed coordinate system xyz, using the general commutation rules of the angular-momentum components with one another and with the components of an arbitrary vector.

Let $\mathbf{a}$ and $\mathbf{b}$ be unit vectors along the ξ and η axes. Then $\mathbf{a}\times\mathbf{b}$ is a unit vector along the ζ axis, and (103.2) gives

$$\hat{J}_\xi\hat{J}_\eta - \hat{J}_\eta\hat{J}_\xi = -i\hat{J}_\zeta. \tag{103.3}$$

The other two relations follow similarly.

Thus the commutation rules for the operators of the angular-momentum components in the rotating coordinate system differ from those in the fixed system only by the sign on the right-hand side[16]. It follows that all the results previously derived from the commutation rules for eigenvalues and matrix elements also hold for J_ξ, J_η, J_ζ, with the sole difference that every expression must be replaced by its complex conjugate. In particular, the eigenvalues of J_ζ (which in this section we denote by k, to distinguish them from the eigenvalues $J_z = M$) take the values $k = -J, \ldots, +J$, where J (an integer!) is the magnitude of the rotor angular momentum.

Spherical rotor. The energy eigenvalues of a rotating rotor are found most simply when all three principal moments of inertia are equal: $I_A = I_B = I_C = I$. For a molecule this occurs when it has the symmetry of one of the cubic point groups. The Hamiltonian (103.1) becomes

$$\hat{H} = \frac{\hbar^2}{2I}\hat{\mathbf{J}}^2,$$

and its eigenvalues are

$$E = \frac{\hbar^2}{2I}J(J+1). \tag{103.4}$$

Each of these energy levels is degenerate with respect to the $2J+1$ directions of the angular momentum relative to the rotor itself (that is, with respect to the values $J_\zeta = k$)[17].

Symmetric rotor. The energy levels are also readily calculated when only two moments of inertia of the rotor coincide: $I_A = I_B \neq I_C$. This occurs for molecules having a single symmetry axis of order greater than two. The Hamiltonian (103.1) takes the form

$$\hat{H} = \frac{\hbar^2}{2I_A}(\hat{J}_\xi^2 + \hat{J}_\eta^2) + \frac{\hbar^2}{2I_C}\hat{J}_\zeta^2 = \frac{\hbar^2}{2I_A}\hat{\mathbf{J}}^2 + \frac{\hbar^2}{2}\left(\frac{1}{I_C} - \frac{1}{I_A}\right)\hat{J}_\zeta^2. \tag{103.5}$$

It follows that in a state with definite J and k the energy is

$$E = \frac{\hbar^2}{2I_A}J(J+1) + \frac{\hbar^2}{2}\left(\frac{1}{I_C} - \frac{1}{I_A}\right)k^2, \tag{103.6}$$

which determines the energy levels of a symmetric rotor.

The degeneracy in k present for the spherical rotor is partially removed here. The energies coincide only for values of k differing in sign alone, corresponding to mutually opposite directions of the angular momentum relative to the rotor axis. Thus the levels of a symmetric rotor with $k \neq 0$ are doubly degenerate.

The stationary states of a symmetric rotor are therefore characterized by three quantum numbers: the angular momentum J and its projections on the rotor axis ($J_\zeta = k$) and on the space-fixed z axis ($J_z = M$); the rotor energy is independent of the last number. In this connection, the very fact that the magnitude of the angular momentum and its

[16]This circumstance expresses the fact that, as regards the action on the rotor wave function, a rotation of the xyz system is equivalent to the opposite rotation of the $\xi\eta\zeta$ system.

[17]Here and below we disregard the always present, physically inessential $(2J+1)$-fold degeneracy with respect to the directions of the angular momentum relative to the fixed coordinate system. When it is included, the total degeneracy of the energy levels of a spherical rotor is $(2J+1)^2$.

projections on a space-fixed axis and on an axis rigidly attached to the physical system are simultaneously measurable[18] follows because the operators $\hat{\mathbf{J}}^2$ and $\hat{J}_z$ commute not only with one another but also with $\hat{J}_\zeta = \hat{\mathbf{J}}\mathbf{n}$ ($\mathbf{n}$ is a unit vector along the ζ axis). This is readily checked by direct calculation, but is also evident beforehand. The angular-momentum operator reduces to the operator of an infinitesimal rotation, while the scalar product $\mathbf{Jn}$ of two vectors attached to the rotor is invariant under any rotation of the coordinate system.

The problem of determining the stationary-state wave functions of a symmetric rotor is consequently reduced to finding the common eigenfunctions of $\hat{\mathbf{J}}^2, \hat{J}_z, \hat{J}_\zeta$. This question, in turn, is mathematically closely connected with the transformation law of angular-momentum eigenfunctions under

finite rotations. Changing the notation for the quantum numbers, we write this law (58.7) as

$$\psi_{JM} = \sum_k D^{(J)}_{kM}(\alpha, \beta, \gamma)\psi_{Jk}. \tag{103.7}$$

We understand ψ_{JM} to be the wave function of a rotor state described relative to the fixed coordinate axes xyz, and ψ_{Jk} to be the wave functions of states described relative to the rotor-fixed axes $\xi\eta\zeta$. In coordinates rigidly attached to the physical system (the rotor), however, the quantities ψ_{Jk} have definite values independent of the orientation of the system in space; denote them by $\psi^{(0)}_{Jk}$. Formula (103.7) then gives the angular dependence of the functions ψ_{JM}. Suppose now that the state $|JM\rangle$ also has a definite value k of the angular-momentum projection on the ζ axis. This means that only one of all the quantities $\psi^{(0)}_{Jk}$, the one with the specified k, is nonzero. The sum in (103.7) then reduces to one term:

$$\psi_{JMk} = \psi^{(0)}_{Jk} D^{(J)}_{kM}(\alpha, \beta, \gamma).$$

We have thereby found the dependence of the wave functions of the states $|JMk\rangle$ on the Euler angles specifying the rotation of the rotor axes relative to the fixed axes. Normalizing the wave function by

$$\int |\psi_{JMk}|^2 \sin\beta \, d\alpha \, d\beta \, d\gamma = 1,$$

we obtain

$$\psi_{JMk} = i^J \sqrt{\frac{2J+1}{8\pi^2}} D^{(J)}_{kM}(\alpha, \beta, \gamma); \tag{103.8}$$

the phase factor is chosen so that for $k = 0$ the function (103.8) becomes the eigenfunction of a free integral angular momentum J (not attached in any way to the ζ axis) with projection M, that is, the usual spherical function (cf. (58.25)[19]).

Asymmetric rotor. When $I_A \neq I_B \neq I_C$, the energy levels cannot be calculated in general form. The degeneracy with respect to the directions of angular momentum

[18] These must not be confused with the projections (which are not simultaneously measurable) on two space-fixed axes!

[19] For a direct derivation of (103.8), without recourse to the theory of finite rotations, see Problem 1 of this section. For the calculation of matrix elements of various quantities with the wave functions (103.8), see § 110, 87 (the corresponding formulas differ from those for a diatomic molecule without spin only in the notation for the quantum numbers; cf. the note on p. 389).

relative to the rotor is completely removed, so that a given J corresponds to $2J + 1$ distinct nondegenerate levels. To calculate these levels (for a specified J), one must start from the Schrödinger equation written in matrix form (O. Klein, 1929). This is done as follows.

The wave functions ψ_{Jk} of rotor states with definite J and definite ζ-component of the angular momentum are the functions (103.8) found above (the index M of the z-component, on which the energy does not depend, will be omitted below for brevity); in these states the energy of the asymmetric rotor has no definite value. Conversely, in stationary states the projection J_ζ has no definite value, so definite values of k cannot be assigned to the energy levels. We seek the wave functions of these states as linear combinations

$$\psi_J = \sum_k c_k \psi_{Jk} \tag{103.9}$$

(all the functions are understood to have some one value of M, the same for every term). Substitution in the Schrödinger equation $\hat{H}\psi_J = E_J\psi_J$ gives the system

$$\sum_{k'} \big(\langle Jk|H|Jk'\rangle - E\delta_{kk'}\big)c_{k'} = 0, \tag{103.10}$$

and the solvability condition for this system gives the secular equation

$$|\langle Jk|H|Jk'\rangle - E\delta_{kk'}| = 0. \tag{103.11}$$

The roots of this equation determine the rotor energy levels; the system (103.10) then gives the linear combinations (103.9) that diagonalize the Hamiltonian, that is, the stationary-state wave functions of the rotor for specified J (and M). The calculation of matrix elements of any physical quantity with these wave functions is thus reduced to matrix elements for a symmetric rotor.

The operators $\hat{J}_\xi, \hat{J}_\eta$ have matrix elements only for transitions in which k changes by one, while $\hat{J}_\zeta$ has diagonal elements only (see formulas (27.13), with J, k in place of L, M). Consequently $\hat{J}_\xi^2, \hat{J}_\eta^2, \hat{J}_\zeta^2$, and hence $\hat{H}$, have

matrix elements only for transitions $k \to k, k \pm 2$. The absence of matrix elements for transitions between states with even and odd k causes the secular equation of degree $2J + 1$ to separate at once into two independent equations of degrees J and $J + 1$. One is formed from matrix elements for transitions among states with even k, and the other from those among states with odd k.

Each of these equations can in turn be reduced to two equations of lower degree. For this purpose one must use matrix elements defined not with the functions ψ_{Jk} but with the functions

$$\begin{aligned} \psi_{Jk}^+ &= \frac{1}{\sqrt{2}}(\psi_{Jk} + \psi_{J,-k}), \\ \psi_{Jk}^- &= \frac{1}{\sqrt{2}}(\psi_{Jk} - \psi_{J,-k}) \quad (k \neq 0), \\ \psi_{J0}^+ &= \psi_{J0}. \end{aligned} \tag{103.12}$$

Functions distinguished by the indices + and − have different symmetry (with respect to reflection in a plane through the ζ axis, which changes the sign of k), and hence matrix elements for transitions between them vanish. Secular equations may therefore be formed separately for the + and − states.

The Hamiltonian (103.1), together with the commutation rules (103.3), has a specific symmetry: it is invariant under simultaneous reversal of the signs of any two of the operators $\hat{J}_\xi, \hat{J}_\eta, \hat{J}_\zeta$. This symmetry formally corresponds to the group D_2. The levels of an asymmetric rotor may therefore be classified by the irreducible representations of this group. There are thus four types of nondegenerate levels, corresponding to the representations A, B_1, B_2, B_3 (see Table 7, p. 460).

It is easy to establish which asymmetric-rotor states belong to each type. To do so, we must determine the symmetry properties of the functions ψ_{Jk} and of the functions (103.12) formed from them. This could be done directly from (103.8). It is simpler, however, to start from the more familiar spherical functions, noting that, in their symmetry properties, wave functions of states with definite angular-momentum projection on the ζ axis coincide with angular-momentum eigenfunctions

$$\psi_{Jk} \sim Y^*_{Jk}(\theta, \varphi) \sim e^{-ik\varphi}\Theta_{Jk}(\theta), \tag{103.13}$$

where θ, φ are spherical angles in the $\xi\eta\zeta$ axes, and the sign $\sim$ here means "transforms as"; the complex conjugation in (103.13) is connected with the reversed sign on the right-hand sides of the commutation relations (103.3).

A rotation through π about the ζ axis (that is, the symmetry operation $C_2^{(\zeta)}$) multiplies the function (103.13) by $(-1)^k$:

$$C_2^{(\zeta)}: \qquad \psi_{Jk} \to (-1)^k \psi_{Jk}.$$

The operation $C_2^{(\eta)}$ may be regarded as a successive inversion and reflection in the $\xi\zeta$ plane; the first multiplies ψ_{Jk} by $(-1)^J$, while the second (reversal of the sign of φ) is equivalent to reversing the sign of k. Taking account of the definition of $\Theta_{J,-k}$ (28.6), we therefore obtain

$$C_2^{(\eta)}: \qquad \psi_{Jk} \to (-1)^{J+k} \psi_{J,-k}.$$

Finally, under $C_2^{(\xi)} = C_2^{(\eta)} C_2^{(\zeta)}$,

$$C_2^{(\xi)}: \qquad \psi_{Jk} \to (-1)^J \psi_{J,-k}.$$

Using these transformation laws, we find that the states corresponding to the functions (103.12) have the following symmetry types:

$$\begin{aligned}
&\psi^+_{Jk} \begin{cases} \text{even } J, \text{ even } k & -A, \\ \text{even } J, \text{ odd } k & -B_3, \\ \text{odd } J, \text{ even } k & -B_1, \\ \text{odd } J, \text{ odd } k & -B_2, \end{cases} \\
&\psi^-_{Jk} \begin{cases} \text{even } J, \text{ even } k & -B_1, \\ \text{even } J, \text{ odd } k & -B_2, \\ \text{odd } J, \text{ even } k & -A, \\ \text{odd } J, \text{ odd } k & -B_3. \end{cases}
\end{aligned} \tag{103.14}$$

A simple count gives the number of states of each type for a specified J. Specifically, the numbers of states of type A and of each type B_1, B_2, B_3 are

	A	B_1, B_2, B_3
Even J	$J/2+1$	$J/2$
Odd J	$(J-1)/2$	$(J+1)/2$

(103.15)

For an asymmetric rotor there are selection rules for matrix elements governing transitions among states of types A, B_1, B_2, B_3; these follow in the usual way from symmetry. Thus, for the components of a vector physical quantity $\mathbf{A}$, the selection rules are

$$\begin{aligned} &\text{for } A_\xi: \quad A \leftrightarrow B_3^{(\xi)}, \quad B_1^{(\zeta)} \leftrightarrow B_2^{(\eta)}, \\ &\text{for } A_\eta: \quad A \leftrightarrow B_2^{(\eta)}, \quad B_1^{(\zeta)} \leftrightarrow B_3^{(\xi)}, \\ &\text{for } A_\zeta: \quad A \leftrightarrow B_1^{(\zeta)}, \quad B_2^{(\eta)} \leftrightarrow B_3^{(\xi)} \end{aligned} \tag{103.16}$$

(for clarity, an index on the representation symbol indicates the axis about which a rotation has character +1 in that representation).

Problems

Problem

1. Find the wave functions of the states $|JMk\rangle$ of a symmetric rotor by direct calculation as common eigenfunctions of the operators $\hat{\mathbf{J}}^2, \hat{J}_z, \hat{J}_\zeta$ (F. Reiche, H. Rademacher, 1926).

Solution. . To obtain ψ_{JMk} as a function of the Euler angles α, β, γ, the operators of the angular-momentum projections on the fixed axes x, y, z must be expressed in terms of these angles. Since the operator of the projection on any axis is $-i\partial/\partial\varphi$, where φ is the angle of rotation about that axis, we may write

$$\hat{J}_x = -i\frac{\partial}{\partial\varphi_x}, \qquad \hat{J}_y = -i\frac{\partial}{\partial\varphi_y}, \qquad \hat{J}_z = -i\frac{\partial}{\partial\varphi_z},$$

where $\varphi_x, \varphi_y, \varphi_z$ are the angles of rotation about the corresponding axes. Derivatives with respect to these angles can be expressed in terms of derivatives with respect to α, β, γ by recalling that infinitesimal rotations add as vectors (directed along the axes of rotation). The directions of the vectors $\delta\alpha, \delta\beta, \delta\gamma$ of the infinitesimal rotations described by the Euler angles are shown in Fig. 20 (p. 270). Projecting them on the fixed axes xyz, we find the rotation angles about these axes in the form

$$\begin{aligned} \delta\varphi_x &= -\sin\alpha\,\delta\beta + \cos\alpha\sin\beta\,\delta\gamma, \\ \delta\varphi_y &= \cos\alpha\,\delta\beta + \sin\alpha\sin\beta\,\delta\gamma, \\ \delta\varphi_z &= \delta\alpha + \cos\beta\,\delta\gamma. \end{aligned}$$

Conversely,

$$\begin{aligned} \delta\alpha &= -\cot\beta\cos\alpha\,\delta\varphi_x - \cot\beta\sin\alpha\,\delta\varphi_y + \delta\varphi_z, \\ \delta\beta &= -\sin\alpha\,\delta\varphi_x + \cos\alpha\,\delta\varphi_y, \\ \delta\gamma &= \frac{\cos\alpha}{\sin\beta}\delta\varphi_x + \frac{\sin\alpha}{\sin\beta}\delta\varphi_y. \end{aligned}$$

These expressions give

$$\hat{J}_x = -i\left(-\cos\alpha\cot\beta\frac{\partial}{\partial\alpha} - \sin\alpha\frac{\partial}{\partial\beta} + \frac{\cos\alpha}{\sin\beta}\frac{\partial}{\partial\gamma}\right),$$
$$\hat{J}_y = -i\left(-\sin\alpha\cot\beta\frac{\partial}{\partial\alpha} + \cos\alpha\frac{\partial}{\partial\beta} + \frac{\sin\alpha}{\sin\beta}\frac{\partial}{\partial\gamma}\right),$$
$$\hat{J}_z = -i\frac{\partial}{\partial\alpha}.$$

When acting on ψ_{JMk}, the operators $\hat{J}_z = -i\partial/\partial\alpha$ and $\hat{J}_\zeta = -i\partial/\partial\gamma$ (γ is the angle of rotation about the ζ axis!) may be replaced by M and k (the corresponding dependence of the wave function on the Euler angles α and γ is given by the factor $\exp(i\alpha M + i\gamma k)$). We then have

$$\hat{J}_+ = \hat{J}_x + i\hat{J}_y = e^{i\alpha}\left(\frac{\partial}{\partial\beta} - M\cot\beta + \frac{k}{\sin\beta}\right),$$
$$\hat{J}_- = \hat{J}_x - i\hat{J}_y = e^{-i\alpha}\left(-\frac{\partial}{\partial\beta} - M\cot\beta + \frac{k}{\sin\beta}\right).$$

The rest of the derivation is exactly analogous to that given at the end of § 28. We begin with $\hat{J}_+\psi_{JJk} = 0$, which holds for the wave function with $M = J$. Hence

$$\left(\frac{\partial}{\partial\beta} - J\cot\beta + \frac{k}{\sin\beta}\right)\psi_{JJk} = 0.$$

The normalized solution of this equation is

$$\psi_{JJk} = i^J(-1)^{J-k}\left[\frac{(2J+1)!}{2(J+k)!(J-k)!}\right]^{1/2}\left(\cos\frac{\beta}{2}\right)^{J+k}\left(\sin\frac{\beta}{2}\right)^{J-k}\frac{e^{i(J\alpha+k\gamma)}}{2\pi}$$

(the normalization integral reduces to Euler's beta integral). This expression does indeed coincide, apart from a phase factor, with the function

$$\sqrt{\frac{2J+1}{8\pi^2}}D^{(J)}_{kJ}(\alpha,\beta,\gamma)$$

(cf. (58.26)); the phase factor has been chosen in accordance with the definition in (103.7).

The wave functions with $M < J$ are then calculated by repeated application to ψ_{JJk} of the formula

$$\hat{J}_-\psi_{J,M+1,k} = \sqrt{(J-M)(J+M+1)}\psi_{JMk}.$$

The final result coincides with (103.8), where the functions $D^{(J)}_{kM}$ are given by formulas (58.9)–(58.11), with the symmetry property (58.18) also taken into account.

Problem

2. Calculate the matrix elements $\langle Jk'|H|Jk\rangle$ for an asymmetric rotor.

Solution. . From formulas (27.13) we find

$$\langle k|J_\xi^2|k\rangle = \langle k|J_\eta^2|k\rangle = \frac{1}{2}[J(J+1)-k^2],$$

$$\begin{aligned}\langle k|J_\xi^2|k+2\rangle = \langle k+2|J_\xi^2|k\rangle = -\langle k|J_\eta^2|k+2\rangle = -\langle k+2|J_\eta^2|k\rangle \\ = \frac{1}{4}\sqrt{(J-k)(J-k-1)(J+k+1)(J+k+2)}\end{aligned}$$

(the diagonal indices J, J on the matrix elements are omitted throughout for brevity). Hence the required matrix elements
of the Hamiltonian are[20]

$$\begin{aligned}\langle k|H|k\rangle &= \frac{\hbar^2}{4}(a+b)[J(J+1)-k^2] + \frac{\hbar^2}{2}ck^2,\\ \langle k|H|k+2\rangle = \langle k+2|H|k\rangle &= \frac{\hbar^2}{8}(a-b)\sqrt{(J-k)(J-k-1)(J+k+1)(J+k+2)}.\end{aligned} \tag{1}$$

The matrix elements with respect to the functions (103.12) are expressed in terms of the elements (1) by

$$\begin{aligned}\langle k\pm|H|k\pm\rangle &= \langle k|H|k\rangle, \quad k \neq 1,\\ \langle 1\pm|H|1\pm\rangle &= \langle 1|H|1\rangle \pm \langle 1|H|-1\rangle,\\ \langle k\pm|H|k+2,\pm\rangle &= \langle k|H|k+2\rangle, \quad k \neq 0,\\ \langle 0+|H|2+\rangle &= \sqrt{2}\langle 0|H|2\rangle.\end{aligned} \tag{2}$$

Problem

3. Determine the energy levels of an asymmetric rotor for $J = 1$.

Solution. . The secular equation of third degree separates into three equations of first degree. One of them gives

$$E_1 = \langle 0+|H|0+\rangle = \frac{\hbar^2}{2}(a+b). \tag{3}$$

The other two energy levels can be written down at once, since it is evident beforehand that the three parameters a, b, c enter the problem symmetrically. Thus

$$E_2 = \frac{\hbar^2}{2}(a+c), \qquad E_3 = \frac{\hbar^2}{2}(b+c). \tag{4}$$

[20] In Problems 2–5, to simplify the notation in the formulas, we use $a = 1/I_A$, $b = 1/I_B$, $c = 1/I_C$.

The levels E_1, E_2, E_3 belong respectively to the symmetry types B_1, B_2, B_3[21]. The wave functions of these states are

$$\psi_1 = \psi^+_{10}, \qquad \psi_2 = \psi^-_{11}, \qquad \psi_3 = \psi^+_{11}.$$

Problem

4. Do the same for $J = 2$.

Solution. . The secular equation of fifth degree separates into three equations of first degree and one of second degree. One of the first-degree equations gives

$$E_1 = \langle 2-|H|2-\rangle = 2\hbar^2 c + \frac{\hbar^2}{2}(a+b) \tag{5}$$

(a level of type B_1). We infer that there must be two further levels (of types B_2 and B_3):

$$E_2 = 2\hbar^2 b + \frac{\hbar^2}{2}(a+c), \qquad E_3 = 2\hbar^2 a + \frac{\hbar^2}{2}(b+c).$$

The wave functions corresponding to these three levels are

$$\psi_1 = \psi^-_{22}, \qquad \psi_2 = \psi^-_{21}, \qquad \psi_3 = \psi^+_{21}.$$

The equation of second degree is

$$\begin{vmatrix} \langle 0+|H|0+\rangle - E & \langle 2+|H|0+\rangle \\ \langle 2+|H|0+\rangle & \langle 2+|H|2+\rangle - E \end{vmatrix} = 0. \tag{6}$$

Solving it, we obtain

$$E_{4,5} = \hbar^2(a+b+c) \pm \hbar^2[(a+b+c)^2 - 3(ab+bc+ac)]^{1/2}. \tag{7}$$

These levels are of type A. Their wave functions are linear combinations of ψ^+_{20} and ψ^+_{22}.

Problem

5. Do the same for $J = 3$.

Solution. . The secular equation of seventh degree separates into one equation of first degree and three of second degree. The first-degree equation gives

$$E_1 = \langle 2-|H|2-\rangle = 2\hbar^2(a+b+c) \tag{8}$$

(a level of type A). One of the second-degree equations is equation (6) of the preceding problem (with a different value of J). Its roots are

$$E_{2,3} = \frac{5\hbar^2}{2}(a+b) + \hbar^2 c \pm \hbar^2[4(a-b)^2 + c^2 + ab - ac - bc]^{1/2} \tag{9}$$

(levels of type B_1). The remaining levels are obtained from these by permuting the parameters a, b, c.

[21] This follows directly from symmetry considerations. For example, the energy E_1 is symmetric in the parameters a and b; this must be so for the energy of a state whose symmetry with respect to the ξ and η axes is the same (a state of type B_1).

PROBLEM

6. Determine the splitting of the levels of a system possessing a quadrupole moment in an arbitrary external electric field.

SOLUTION. . Choosing the principal axes of the tensor $\partial^2\varphi/\partial x_i \partial x_k$ as coordinate axes (cf. Problem 3 of § 76), we reduce the quadrupole part of the Hamiltonian of the system to the form

$$\hat{H} = A\hat{J}_x^2 + B\hat{J}_y^2 + C\hat{J}_z^2, \qquad A + B + C = 0.$$

Because this expression is completely analogous in form to the Hamiltonian (103.1), the stated problem is equivalent to finding the energy levels of an asymmetric rotor, except that the sum of the coefficients is now $A + B + C = 0$ and the angular momentum may also take half-integral values. For the latter, the calculations must be carried out anew by the same method; for integral J, the results of Problems 3–5 may be used. We finally obtain the following energy shifts ΔE for the first few values of J:

$$\begin{aligned} J = 1: &\qquad \Delta E = -A, -B, -C, \\ J = 3/2: &\qquad \Delta E = \pm\sqrt{(3/2)(A^2 + B^2 + C^2)}, \\ J = 2: &\qquad \Delta E = 3A, 3B, 3C, \ \pm\sqrt{6(A^2 + B^2 + C^2)}. \end{aligned}$$

For $J = 3/2$ the energy levels remain doubly degenerate, in accordance with Kramers' theorem (§ 60).

§ 104. Interaction between molecular vibration and rotation

Up to this point, rotation and vibration have been treated as independent motions of a molecule. In fact, when both occur simultaneously, a distinctive interaction arises between them (E. Teller, L. Tisza, O. Placzek, 1932–1933).

We begin with linear polyatomic molecules. A linear molecule can undergo two kinds of vibration (see the end of § 100): longitudinal vibrations with nondegenerate frequencies, and transverse vibrations with doubly degenerate frequencies. It is the latter that concern us here.

A molecule undergoing transverse vibrations generally has angular momentum. This follows at once from elementary mechanical considerations[22], but it can also be demonstrated by a quantum-mechanical treatment. Such a treatment also determines the possible values of this momentum in a specified vibrational state.

Suppose that one particular doubly degenerate frequency ω_α is excited in the molecule. The energy level with vibrational quantum number v_α has degeneracy $v_\alpha + 1$. It is represented by $v_\alpha + 1$ wave functions

$$\psi_{v_{\alpha1}v_{\alpha2}} = \text{const} \cdot \exp\left[-\frac{1}{2}c_\alpha^2(Q_{\alpha1}^2 + Q_{\alpha2}^2)\right] H_{v_{\alpha1}}(c_\alpha Q_{\alpha1}) H_{v_{\alpha2}}(c_\alpha Q_{\alpha2})$$

(where $v_{\alpha1} + v_{\alpha2} = v_\alpha$), or by any independent linear combinations of them. In every one of these functions, the polynomial multiplying the exponential has the same highest

[22]For example, two mutually perpendicular transverse vibrations whose phases differ by $\pi/2$ may be regarded as a pure rotation of the bent molecule about its longitudinal axis.

total degree in $Q_{\alpha 1}$ and $Q_{\alpha 2}$, namely v_α. Linear combinations of the functions $\psi_{v_{\alpha 1} v_{\alpha 2}}$ can evidently always be chosen as basis functions in the form

$$\begin{aligned}\psi_{v_\alpha l_\alpha} = \ &\text{const} \cdot \exp\left[-\frac{1}{2}c_\alpha^2(Q_{\alpha 1}^2 + Q_{\alpha 2}^2)\right] \times \\ &\times \left[(Q_{\alpha 1} + iQ_{\alpha 2})^{(v_\alpha + l_\alpha)/2}(Q_{\alpha 1} - iQ_{\alpha 2})^{(v_\alpha - l_\alpha)/2} + \dots\right].\end{aligned} \tag{104.1}$$

The square brackets contain a definite polynomial of which only the highest-degree term has been displayed; l_α is an integer having the $v_\alpha + 1$ possible values $l_\alpha = v_\alpha$, $v_\alpha - 2$, $v_\alpha - 4, \dots, -v_\alpha$.

The normal coordinates $Q_{\alpha 1}$ and $Q_{\alpha 2}$ of a transverse vibration describe two mutually perpendicular displacements from the molecular axis. Under a rotation through an angle φ about this axis, the highest-degree polynomial term, and hence the whole function $\psi_{v_\alpha l_\alpha}$, is multiplied by

$$\exp\left\{i\varphi\left(\frac{v_\alpha + l_\alpha}{2}\right) - i\varphi\left(\frac{v_\alpha - l_\alpha}{2}\right)\right\} = \exp(il_\alpha\varphi).$$

It follows that the function (104.1) describes a state whose angular momentum about the axis is l_α.

We thus obtain the following result. In a state in which a doubly degenerate frequency ω_α is excited with quantum number v_α, the molecule has an angular momentum about its own axis that takes the values

$$l_\alpha = v_\alpha, v_\alpha - 2, v_\alpha - 4, \dots, -v_\alpha. \tag{104.2}$$

This is called the molecule's *vibrational angular momentum.* If several transverse vibrations are excited at once, the total vibrational angular momentum is $\sum l_\alpha$. Together with the electronic orbital angular momentum, it gives the total angular momentum l of the molecule about its axis.

As in a diatomic molecule, the total molecular angular momentum J cannot be smaller than the angular momentum about the axis. Thus J assumes the values

$$J = |l|, |l| + 1, \dots$$

In other words, states with $J = 0, 1, \dots, |l| - 1$ do not exist.

For harmonic vibrations the energy depends only on the numbers v_α, not on l_α. Anharmonicity removes the degeneracy of the vibrational levels with respect to l_α, though only partially: the levels remain doubly degenerate, states related by simultaneous reversal of the signs of every l_α and of l having equal energy. In the approximation following the harmonic one, the energy acquires a term quadratic in the momenta l_α, of the form

$$\sum_{\alpha,\beta} g_{\alpha\beta} l_\alpha l_\beta$$

(the $g_{\alpha\beta}$ are constants). The remaining twofold degeneracy is removed by an effect analogous to Λ-doubling in diatomic molecules.

Before turning to nonlinear molecules, we must make a purely mechanical observation. For an arbitrary nonlinear system of particles, one must ask how vibrational motion can be separated from rotation at all, or, equivalently, what is meant by a "nonrotating system." At first sight, vanishing angular momentum might appear to provide the criterion for the absence of rotation:

$$\sum m(\mathbf{r} \times \mathbf{v}) = 0 \tag{104.3}$$

(the sum is over the particles of the system). The expression on the left, however, is not the total time derivative of any function of the coordinates. Consequently, this equality cannot be integrated with respect to time and recast as the vanishing of some coordinate function. Yet just such a formulation is needed if the concepts of "pure vibration" and "pure rotation" are to be defined meaningfully.

The condition to be adopted as the definition of no rotation is therefore

$$\sum m(\mathbf{r}_0 \times \mathbf{v}) = 0, \tag{104.4}$$

where $\mathbf{r}_0$ is the radius vector of the equilibrium position of a particle. Writing $\mathbf{r} = \mathbf{r}_0 + \mathbf{u}$, with $\mathbf{u}$ the displacement in a small vibration, gives $\mathbf{v} = \dot{\mathbf{r}} = \dot{\mathbf{u}}$. Integration of (104.4) with respect to time then yields

$$\sum m(\mathbf{r}_0 \times \mathbf{u}) = 0. \tag{104.5}$$

We shall regard molecular motion as the combination of a purely vibrational motion satisfying (104.5) and a rotation of the molecule as a whole[23]. On writing the angular momentum as

$$\sum m(\mathbf{r} \times \mathbf{v}) = \sum m(\mathbf{r}_0 \times \mathbf{v}) + \sum m(\mathbf{u} \times \mathbf{v}),$$

we see from the definition (104.4) of no rotation that the sum $\sum m(\mathbf{u} \times \mathbf{v})$ is to be understood as the vibrational angular momentum. It must be remembered, however, that this momentum is only one part of the total momentum of the system and is not itself conserved at all. Thus only a mean value of the vibrational angular momentum can be assigned to each vibrational state.

Molecules having no symmetry axis of order higher than two are of the asymmetric-top type. All vibrational frequencies of molecules of this kind are nondegenerate, since their symmetry groups have only one-dimensional irreducible representations. Hence every vibrational level is nondegenerate. In any nondegenerate state, however, the mean angular momentum vanishes (see § 26). Asymmetric-top molecules therefore have zero mean vibrational angular momentum in every state.

If the symmetry elements of a molecule include one axis of order higher than two, the molecule is of the symmetric-top type. Such a molecule has both nondegenerate and doubly degenerate vibrational frequencies. The mean vibrational angular momentum again vanishes for the former. A doubly degenerate frequency, on the other hand, corresponds to a nonzero mean projection of angular momentum on the molecular axis.

The energy of rotational motion of a symmetric-top molecule can readily be found with the vibrational angular momentum taken into account. Its operator differs from (103.5) by replacing the top's rotational angular momentum by the difference between the

[23]Translational motion is assumed to have been separated from the outset by choosing a coordinate system in which the molecular center of mass is at rest.

molecule's total conserved angular momentum $\mathbf{J}$ and its vibrational angular momentum $\mathbf{J}^{(v)}$:

$$\widehat{H}_{\text{rot}} = \frac{\hbar^2}{2I_A}(\widehat{\mathbf{J}} - \widehat{\mathbf{J}}^{(v)})^2 + \frac{\hbar^2}{2}\left(\frac{1}{I_C} - \frac{1}{I_A}\right)(\widehat{J}_\zeta - \widehat{J}_\zeta^{(v)})^2. \tag{104.6}$$

The energy sought is the mean value $\overline{H}_{\text{rot}}$. Terms in (104.6) containing squares of components of $\mathbf{J}$ give the purely rotational energy (103.6). Those containing squares of components of $\mathbf{J}^{(v)}$ give constants independent of the rotational quantum numbers and may be omitted. The terms containing products of components of $\mathbf{J}$ and $\mathbf{J}^{(v)}$ represent the effect of the interaction between molecular vibration and rotation that is of interest here. It is called the *Coriolis interaction*, in view of its correspondence to the Coriolis forces of classical mechanics. In averaging these terms, the mean values of the transverse (ξ, η) components of the vibrational angular momentum must be taken to vanish. The resulting mean Coriolis-interaction energy is

$$E_{\text{Cor}} = -\frac{\hbar^2}{I_C} k k_v, \tag{104.7}$$

where the integer k, as in § 103, is the projection of the total angular momentum on the molecular axis, while $k_v = \overline{J_\zeta^{(v)}}$ is the mean projection of the vibrational angular momentum characterizing the given vibrational state. Unlike k, k_v need not be an integer at all.

Finally, consider spherical-top molecules. These are molecules possessing the symmetry of one of the cubic groups. They have nondegenerate, doubly degenerate, and triply degenerate frequencies, corresponding to the occurrence of one-, two-, and three-dimensional irreducible representations among those of the cubic groups. As always, anharmonicity partially removes the degeneracy of the vibrational levels;

after these effects have been allowed for, only nondegenerate, doubly degenerate, and triply degenerate levels remain. We shall discuss precisely these levels after their splitting by anharmonicity.

For a spherical-top molecule, the mean vibrational angular momentum plainly vanishes not only in nondegenerate vibrational states but also in doubly degenerate ones. This follows directly from elementary symmetry considerations. Indeed, the vectors of the mean momenta in the two states belonging to one degenerate energy level would have to transform into each other under every symmetry transformation of the molecule. No cubic symmetry group, however, permits two directions that transform only into one another; only sets containing at least three directions are transformed among themselves.

The same argument shows that the mean vibrational angular momentum is nonzero in states belonging to triply degenerate vibrational levels. After averaging over the vibrational state, this momentum is represented by an operator whose matrix elements correspond to transitions among the three mutually degenerate states. In accordance with their number, this operator must have the form $\zeta\widehat{\mathbf{I}}$, where $\widehat{\mathbf{I}}$ is the operator of an angular momentum equal to unity (for which $2I + 1 = 3$), and ζ is a constant characteristic of the given vibrational level. After this averaging, the Hamiltonian of the rotational motion,

$$\widehat{H}_{\text{rot}} = \frac{\hbar^2}{2I}(\widehat{\mathbf{J}} - \widehat{\mathbf{J}}^{(v)})^2,$$

becomes the operator

$$\widehat{H}_{\text{rot}} = \frac{\hbar^2}{2I}\widehat{\mathbf{J}}^2 + \frac{\hbar^2}{2I}\overline{\widehat{\mathbf{J}}^{(v)2}} - \frac{\hbar^2}{I}\zeta\widehat{\mathbf{J}\mathbf{I}}. \tag{104.8}$$

The eigenvalues of the first term give the usual rotational energy (103.4), while the second term contributes an inessential constant independent of the rotational quantum number. The last term in (104.8) gives the required Coriolis splitting energy of the vibrational level. The eigenvalues of **JI** are obtained in the usual way. For a specified J, it can have three distinct values, corresponding to values $J+1$, $J-1$, and J of the vector $\mathbf{I}+\mathbf{J}$:

$$E_{\text{Cor}}^{(J+1)} = -\frac{\hbar^2}{I}\zeta J, \qquad E_{\text{Cor}}^{(J-1)} = \frac{\hbar^2}{I}\zeta(J+1), \qquad E_{\text{Cor}}^{(J)} = \frac{\hbar^2}{I}\zeta. \tag{104.9}$$

§ 105. Classification of molecular terms

The wave function of a molecule is the product of the electronic wave function, the wave function of the vibrational motion of the nuclei, and the rotational wave function. We have already discussed separately the classification and symmetry types of these functions. It remains to consider the classification of molecular terms as a whole, that is, the possible symmetry of the complete wave function.

Clearly, specifying the symmetry of all three factors under particular transformations also determines the symmetry of their product under those transformations. A complete characterization of the symmetry of a state must further specify the behavior of the complete wave function under simultaneous inversion of the coordinates of all particles (electrons and nuclei) in the molecule. A state is called *negative* or *positive* according as its wave function changes sign or remains unchanged under this transformation[24].

It must be borne in mind, however, that characterizing a state with respect to inversion is meaningful only for molecules without stereoisomers. Stereoisomerism means that inversion brings a molecule into a configuration that cannot be superposed on the original one by any rotation in space (the "right" and "left" modifications of a substance)[25]. Thus, in the presence of stereoisomerism, wave functions obtained from one another by inversion belong essentially to different molecules and there is no point in comparing them[26].

We saw in § 86 that in diatomic molecules nuclear spin has an important indirect influence on the scheme of molecular terms: it determines their degeneracies and, in some cases, altogether forbids levels of a particular symmetry. The same occurs in polyatomic molecules. Here, however, the analysis is considerably more complicated and requires group-theoretical methods in each particular case.

The idea of the method is as follows. In addition to the coordinate part

[24] As is customary, we use the same infelicitous terminology as for diatomic molecules (see § 86).

[25] For stereoisomerism to be possible, the molecule must possess no symmetry element involving reflection (an inversion center, a symmetry plane, or a rotation–reflection axis).

[26] Strictly speaking, quantum mechanics always gives a nonzero probability of transition from one modification to the other. This probability, however, which is associated with passage of the nuclei through a barrier, is extremely small.

(the only part considered so far), the complete wave function must also contain a spin factor, which is a function of the projections of the spins of all nuclei on some chosen direction in space. The projection σ of a nuclear spin takes $2i + 1$ values (i is the nuclear spin); on assigning all possible values to all $\sigma_1, \sigma_2, \ldots, \sigma_N$ (N is the number of atoms in the molecule), we obtain a total of $(2i_1 + 1)(2i_2 + 1) \ldots (2i_N + 1)$ different values of the spin factor. Under each symmetry transformation, certain nuclei (of the same kind) exchange places. If the spin values are imagined to "remain in place," the transformation is equivalent to permuting the spin values among the nuclei. The different spin factors are correspondingly transformed into one another and thus realize a certain representation (in general reducible) of the molecular symmetry group. By decomposing it into irreducible parts, we find the possible symmetry types of the spin wave function.

A general formula for the characters $\chi_{\mathrm{sp}}(G)$ of the representation realized by the spin factors is readily written down. It is enough to observe that only those spin factors for which the nuclei exchanging places have equal σ_a remain unchanged by the transformation; otherwise one spin factor passes into another and contributes nothing to the character. Since σ_a takes $2i_a + 1$ values, we find

$$\chi_{\mathrm{sp}}(G) = \prod (2i_a + 1), \tag{105.1}$$

where the product is over the groups of atoms that exchange places with one another under the given transformation G (one factor in the product for each group).

What interests us, however, is not so much the symmetry of the spin function as that of the coordinate wave function (symmetry under permutations of the nuclear coordinates while the electron coordinates remain fixed). These symmetries are directly related, since the complete wave function must remain unchanged or change sign when each pair of nuclei obeying Bose or Fermi statistics, respectively, is interchanged (in other words, it must be multiplied by $(-1)^{2i}$, where i is the spin of the nuclei being interchanged). Introducing the corresponding factor into the characters (105.1), we obtain the set of characters $\chi(G)$ of a representation containing all the irreducible representations according to which the coordinate wave functions transform:

$$\chi(G) = \prod (2i_a + 1)(-1)^{2i_a(n_a - 1)}. \tag{105.2}$$

Here n_a is the number of nuclei in each group that exchange places with one another under the given transformation. Decomposition of this representation into irreducible parts gives the possible symmetry types of the molecular coordinate wave functions, together with the degeneracies of the corresponding energy levels (here and below, degeneracy with respect to the different spin states of the nuclear system is meant)[27].

Each symmetry type of states is associated with definite values of the total spins of groups of equivalent nuclei in the molecule (that is, groups of nuclei interchanged by some symmetry transformations of the molecule). The correspondence is not one-to-one: in general, each symmetry type of states may occur with different values of the spins of groups of equivalent nuclei. This correspondence can likewise be established in each particular case by group-theoretical methods.

[27] In this connection the degeneracy of a level is often called its nuclear statistical weight (cf. the note on p. 407).

As an example, consider an asymmetric-rotor molecule, ethylene $^{12}C_2{}^1H_4$ (Fig. 43g, symmetry group D_{2h}). The superscript on a chemical symbol indicates the isotope to which the nucleus belongs; this specification is necessary because nuclei of different isotopes may have different spins. Here the spin of a ^{1}H nucleus is one half, while the ^{12}C nucleus has zero spin. We therefore need consider only the hydrogen atoms.

Choose the coordinate system as shown in Fig. 43g (the z axis is perpendicular to the molecular plane and the x axis is directed along its axis). Reflection in the plane $\sigma(xy)$ leaves all atoms in place, while the other reflections and rotations interchange the hydrogen atoms pairwise. Formula (105.2) gives the following characters of the representation:

E	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$	I	$C_2(z)$	$C_2(y)$	$C_2(x)$
16	16	4	4	4	4	4	4

Decomposing this representation into irreducible parts, we find that it contains the following irreducible representations of the group D_{2h}: $7A_g$, $3B_{1g}$, $3B_{2u}$, $3B_{3u}$. The number indicates the multiplicity with which the given irreducible representation occurs in the reducible one; these numbers are precisely the nuclear statistical weights of levels of the corresponding symmetry[28].

The classification obtained for the states of the ethylene molecule concerns the symmetry of the complete (coordinate) wave function, containing electronic, vibrational, and rotational parts. Usually, however, it is useful to approach these results from another direction. Namely, once the possible symmetries of the complete wave function are known, one can directly find which rotational levels are possible (and with which statistical weights) for a specified electronic and vibrational state.

Consider, for example, the rotational structure of the lowest vibrational level (with vibrations unexcited) of the normal electronic term, assuming the electronic wave function of the normal state to be totally symmetric (as is true in practice for nearly all polyatomic molecules). The symmetry of the complete wave function under rotations about symmetry axes then coincides with that of the rotational wave function. Comparison with the results above therefore shows that in ethylene the rotational levels of types A and B_1 (see § 103) are positive and have statistical weights 7 and 3, whereas levels of types B_2 and B_3 are negative and have statistical weight 3.

As in diatomic molecules (see the end of § 86), transitions between ethylene states of different nuclear symmetry practically do not occur, owing to the extreme weakness of the interaction of the nuclear spins with the electrons. Molecules in these states therefore behave as distinct modifications of the substance, so that ethylene $^{12}C_2{}^1H_4$ has four modifications with nuclear statistical weights 7, 3, 3, 3. Essential to this conclusion is that states of different symmetry belong to different energy levels (whose separations are large compared with the nuclear-spin interaction energy). It is therefore not valid for molecules having states of different nuclear symmetry that belong to the same degenerate energy level.

Consider the symmetric-rotor ammonia molecule $^{14}N^1H_3$ (Fig. 41, symmetry group C_{3v}). The spin of the ^{14}N nucleus is 1, and that of ^{1}H is one half. Formula (105.2) gives

[28]For the relation between the symmetry of states and the values of the total spin of the four H nuclei in the ethylene molecule, see Problem 1.

the characters of the representation of C_{3v} that interests us:

E	$2C_3$	$3\sigma_v$
24	6	−12

It contains the following irreducible representations of the group

C_{3v}: $12A_2, 6E$. Thus two types of levels are possible; their nuclear statistical weights are 12 and 6[29].

The rotational levels of a symmetric rotor are classified (for a given J) by the values of the quantum number k. As in the preceding example, consider the rotational structure of the normal electronic and vibrational states of the NH_3 molecule (that is, assume the electronic and vibrational wave functions to be totally symmetric). In determining the symmetry of the rotational wave function, one must remember that its behavior is meaningful only with respect to rotations about axes. We therefore replace the symmetry planes by second-order symmetry axes perpendicular to them (reflection in a plane is equivalent to rotation about such an axis followed by inversion). Thus, in place of C_{3v}, we must here consider the isomorphic point group D_3.

Under a rotation C_3 about the vertical third-order axis, the rotational wave functions with $k = \pm|k|$ are multiplied by $e^{\pm 2\pi i|k|/3}$; under a rotation U_2 about a horizontal second-order axis, they pass into one another, and hence realize a two-dimensional representation of D_3. When $|k|$ is not a multiple of three, this representation is irreducible: it is the representation E. The representation of C_{3v} corresponding to the complete wave function is obtained by multiplying the character $\chi(U_2)$ by +1 or −1 according as the term is positive or negative. But since $\chi(U_2) = 0$ in the representation E, in either case we again obtain the same representation E (now as a representation of C_{3v} rather than D_3). In view of the results above, we therefore conclude that when $|k|$ is not a multiple of three, both positive and negative levels with nuclear statistical weight 6 are possible (levels for which the complete coordinate wave function has symmetry type E). When $|k|$ is a nonzero multiple of three, the rotational functions realize a representation (of D_3) with characters

E	$2C_3$	$3U_2$
2	2	0

This representation is reducible and decomposes into A_1 and A_2. For the complete wave function to belong to the representation A_2 of C_{3v}, the rotational level A_1 must be negative and A_2 positive. Thus, for nonzero $|k|$ divisible by three, both positive and negative levels with nuclear statistical weight 12 are possible (levels of type A_2).

For the projection $k = 0$ there is just one rotational function, realizing a representation with characters[30]

E	$2C_3$	$3U_2$
1	1	$(-1)^J$

[29] Terms of symmetry A_2 correspond to total spin 3/2 of the hydrogen nuclei, and terms E to spin 1/2.

The occurrence of the two-dimensional representation E among the irreducible representations does not imply an additional degeneracy of the molecular energy levels. This is an instance of permutation degeneracy, discussed in § 63.

[30] Under a rotation through an angle π, an angular-momentum eigenfunction with magnitude J and zero projection is multiplied by $(-1)^J$.

For the complete wave function to have symmetry A_2, its behavior under inversion must consequently be determined by the factor $-(-1)^J$. Thus, at $k = 0$, levels with even (odd) J can only be negative (positive); in both cases the statistical weight is 12 (levels of type A_2).

Collecting these results, we obtain the following table of possible states at different values of the quantum number k for the normal electronic and vibrational term of the molecule $^{14}N^1H_3$ (+ and – denote positive and negative states):

	(+)	(−)
$\lvert k\rvert$ not divisible by three	$6E$	$6E$
$\lvert k\rvert$ divisible by three	$12A_2$	$12A_2$
$k = 0,\ J$ even	–	$12A_2$
$k = 0,\ J$ odd	$12A_2$	–

For given J and k, the energy levels of the NH_3 molecule are in general degenerate (see also the table for ND_3 in Problem 3). This degeneracy is partially removed by a distinctive effect associated with the flattened shape of the ammonia molecule and the small mass of the hydrogen atoms. A comparatively small vertical displacement of the atoms in this molecule can produce a transition between two configurations related by reflection in a plane parallel to the base of the pyramid (Fig. 44). These transitions split the levels,

separating the positive and negative levels (an effect analogous to the one-dimensional case considered in Problem 3 of § 50).

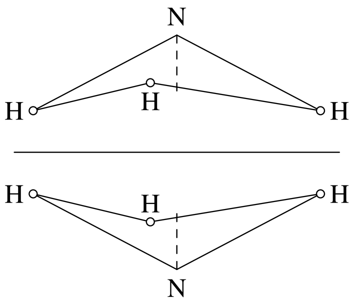

Figure 44

The magnitude of the splitting is proportional to the probability for the atoms to pass through the "potential barrier" separating the two molecular configurations. Although this probability is comparatively large in ammonia because of the properties just described, the splitting is nevertheless small ($1 \cdot 10^{-4}$ eV). An example of a spherical-rotor molecule is treated in Problem 5 of this section.

Problems

Problem

1. Establish the relation between the symmetry of states of the molecule $^{12}C_2{}^1H_4$ and the total spin of its hydrogen nuclei.

Solution. [31] The total spin of the four ^{1}H nuclei can have the values $I = 2, 1, 0$, and its projection M_I ranges from 2 to -2. Consider the representations realized by the spin factors belonging to each individual value of M_I, beginning with the maximum value.

For $M_I = 2$ there is only one spin factor, in which all nuclei have spin projection $+1/2$. For $M_I = 1$ there are four different spin factors, distinguished by which of the four nuclei is assigned spin projection $-1/2$. Finally, $M_I = 0$ is realized by six spin factors, according to which pair of nuclei is assigned spin projection $-1/2$. The characters of the corresponding three representations are

	E	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$	I	$C_2(z)$	$C_2(y)$	$C_2(x)$
$M_I = 2$	1	1	1	1	1	1	1	1
$M_I = 1$	4	4	0	0	0	0	0	0
$M_I = 0$	6	6	2	2	2	2	2	2

The first is the identity representation A_g. Since $M_I = 2$ can occur only for $I = 2$, it follows that spin $I = 2$ corresponds to a state of symmetry A_g.

The value $M_I = 1$ can occur for both $I = 1$ and $I = 2$. Accordingly subtracting the first representation from the second and decomposing the result into irreducible parts, we find that spin $I = 1$ corresponds to the states B_{1g}, B_{2u}, B_{3u}.

Finally, $M_I = 0$ can occur in all the cases in which $M_I = 1$ is possible, and also for $I = 0$. Accordingly subtracting the second representation from the third, we find two states A_g corresponding to spin $I = 0$.

Problem

2. Determine the symmetry types of the complete (coordinate) wave functions and the statistical weights of the corresponding levels for the molecules

$^{12}C_2{}^2H_4$, $^{13}C_2{}^1H_4$, $^{14}N_2{}^{16}O_4$ (all the molecules have the same shape); the spins are $i(^2H) = 1, i(^{13}C) = 1/2, i(^{14}N) = 1$.

Solution. By the same method as was applied in the text to $^{12}C_2{}^1H_4$, we find the following states (the coordinate axes are chosen as in the text):

Molecule	(+)	(−)
$^{12}C_2{}^2H_4$	$27A_g,\ 18B_{1g}$	$18B_{2u},\ 18B_{3u}$
$^{13}C_2{}^1H_4$	$16A_g,\ 12B_{1g}$	$12B_{2u},\ 24B_{2u}$
$^{14}N_2{}^{16}O_4$	$6A_g$	$3B_{3u}$

[31] For the method of solving such problems based on permutation-group theory, see the book by I. G. Kaplan cited on p. 291, Ch. VI, § 2.

Problem

3. Do the same for the molecule $^{14}N^2H_3$.

Solution. Just as for $^{14}N^1H_3$ in the text, we find the states $30A_1$, $3A_2$, $24E$. For the normal electronic and vibrational term, the following states are possible at different values of the quantum number k:

	(+)	(−)
$\lvert k\rvert$ not divisible by three	$24E$	$24E$
$\lvert k\rvert$ divisible by three	$30A_1, 3A_2$	$30A_1, 3A_2$
$k = 0$, J even	$30A_1$	$3A_2$
$k = 0$, J odd	$3A_2$	$30A_1$

Problem

4. Do the same for the molecule $^{12}C_2{}^1H_6$ (see Fig. 43f, symmetry D_{3d}).

Solution. The following types of states are possible: $7A_{1g}$, $1A_{1u}$, $3A_{2g}$, $13A_{2u}$, $9E_g$, $11E_u$.

For the normal electronic and vibrational term, the following states are obtained:

	(+)	(−)
$\lvert k\rvert$ not divisible by three	$9E_g$	$11E_u$
$\lvert k\rvert$ divisible by three	$7A_{1g}, 3A_{2g}$	$1A_{1u}, 13A_{2u}$
$k = 0$, J even	$7A_{1g}$	$1A_{1u}$
$k = 0$, J odd	$3A_{2g}$	$13A_{2u}$

Problem

5. Do the same for the methane molecule $^{12}C^1H_4$ (the H atoms are at the vertices and the C atom at the center of a tetrahedron).

Solution. The molecule is a spherical rotor and has symmetry T_d. By the same method, we find that states of the following types are possible: $5A_2$, $1E$, $3F_1$ (the corresponding total molecular spins are, respectively, 2, 0, 1). The rotational states of a spherical rotor are classified by the values of the total angular momentum J. The $(2J + 1)$ rotational functions belonging to a given J realize a $(2J + 1)$-dimensional representation of the group O, which is isomorphic to T_d and is obtained from it

by replacing every symmetry plane by a second-order axis perpendicular to it. The characters of this representation are determined by formula (98.3). Thus, for example, for $J = 3$ we obtain a representation with characters

E	$8C_3$	$6C_2$	$6C_4$	$3C_4^2$
7	1	−1	−1	−1

It contains the following irreducible representations of O: A_2, F_1, F_2. Considering again the rotational structure of the normal electronic and vibrational term, it follows that at

$J = 3$ states for which the complete wave function has symmetry A_2 can only be positive, whereas levels of the state F_1 may be either positive or negative. For the first few values of J, the following states are obtained in the same way (shown together with their nuclear statistical weights):

	$(+)$	$(-)$
$J = 0$	$-$	$5A_2$
$J = 1$	$3F_1$	$-$
$J = 2$	$1E$	$1E, 3F_1$
$J = 3$	$5A_2, 3F_1$	$3F_1$
$J = 4$	$1E, 3F_1$	$5A_2, 1E, 3F_1$

CHAPTER XIV

Addition of Angular Momenta

§ 106. $3j$ Symbols

The rule for adding angular momenta obtained in § 31 determines the possible values of the total angular momentum of a system composed of two particles (or more complex constituents) whose angular momenta are j_1 and j_2[1]. In fact, this rule is intimately connected with the way wave functions behave under rotations in space, and it follows directly from the properties of spinors.

Two symmetric spinors, of ranks $2j_1$ and $2j_2$, respectively, represent the wave functions of particles whose angular momenta are j_1 and j_2; the wave function of the combined system is their product

$$\psi^{(1)}_{\lambda\mu\ldots}\psi^{(2)}_{\rho\sigma\ldots}. \tag{106.1}$$

Symmetrization of this product over all its indices yields a symmetric spinor of rank $2(j_1 + j_2)$, which describes the state whose total angular momentum is $j_1 + j_2$. Next contract the product (106.1) over a single pair of indices, one belonging to $\psi^{(1)}$ and the other to $\psi^{(2)}$ (a contraction within either spinor would instead vanish); because each of the spinors $\psi^{(1)}$ and $\psi^{(2)}$ is symmetric, the particular indices chosen from $\lambda, \mu, \ldots$ and $\rho, \sigma, \ldots$ do not matter. Subsequent symmetrization gives a symmetric spinor of rank $2(j_1 + j_2 - 1)$, corresponding to a state with angular momentum $j_1 + j_2 - 1$[2].

Continuing in this way, we obtain the already familiar rule: j assumes, once each, all values from $j_1 + j_2$ down to $|j_1 - j_2|$.

Mathematically, this amounts to resolving the direct product $D^{(j_1)} \times D^{(j_2)}$ of two irreducible representations of the rotation group (of dimensions $2j_1 + 1$ and $2j_2 + 1$) into irreducible components. In this language, the angular-momentum addition rule is the decomposition

$$D^{(j_1)} \times D^{(j_2)} = D^{(j_1+j_2)} + D^{(j_1+j_2-1)} + \cdots + D^{(|j_1-j_2|)}.$$

To complete the solution of the angular-momentum addition problem, we must still determine how the wave function of a system with a specified total angular momentum is constructed from the wave functions of its two constituent particles.

[1] Strictly, throughout what follows we shall tacitly mean a system composed of constituents whose mutual interaction is weak enough for their angular momenta to be treated as conserved in the first approximation. The results presented below apply, of course, not only when the total angular momenta of two particles (or systems) are added, but also when the orbital angular momentum and the spin of a single system are added, provided that the spin–orbit coupling is sufficiently weak.

[2] For example, with $j_1 = j_2 = 1/2$, the wave function of the two-particle system is a spinor of rank two; when the total angular momentum is $j = 0$, this spinor is antisymmetric and hence reduces to a scalar. In general, a two-particle system has a wave function whose spinor rank is always $2(j_1 + j_2)$ and, in general, is not equal to $2j$, where j is the total angular momentum of the system. A spinor of this rank may nevertheless be equivalent to one of lower rank. More generally, j determines the symmetry of the system's spinor wave function: it is symmetric in $2j$ indices and antisymmetric in all the remaining indices.

Begin with the simplest case: two angular momenta adding to zero. Clearly, $j_1 = j_2$ and their projections obey $m_1 = -m_2$. Let ψ_{jm} denote normalized wave functions for the states of one particle with angular momentum j and projection m (in a nonspinor representation). The required system wave function ψ_0 is a sum of products of the two particles' wave functions with opposite values of m:

$$\psi_0 = \frac{1}{\sqrt{2j+1}} \sum_{m=-j}^{j} (-1)^{j-m} \psi^{(1)}_{jm} \psi^{(2)}_{j,-m}. \tag{106.2}$$

(j is the common value of j_1 and j_2). The factor preceding the sum follows from normalization. All the coefficients in the sum must have the same absolute value, since every possible value of the particles' angular-momentum projection m is equally probable. The pattern of alternating signs in (106.2) is readily obtained by writing the wave functions in spinor form. In spinor notation, the sum in (106.2) is the scalar (the system has zero total angular momentum!)

$$\psi^{(1)}_{\lambda\mu\ldots} \psi^{(2)\lambda\mu\ldots}, \tag{106.3}$$

formed from two spinors of rank $2j$. Once this is recognized, the signs in (106.2) follow immediately from formula (57.3).

It must nevertheless be kept in mind that, in general, only the relative signs of the terms in (106.2) are fixed; the overall sign may depend on the "order of addition" of the angular momenta. Indeed, lowering all the spinor indices (of which $j + m$ are ones and $j - m$ are twos) on $\psi^{(1)}$ and raising them on $\psi^{(2)}$ multiplies the scalar (106.3) by $(-1)^{2j}$; thus, for half-integral j, its sign changes.

Now consider a system of three particles, with angular momenta j_1, j_2, j_3 and projections m_1, m_2, m_3, whose total angular momentum is zero. This condition implies $m_1 + m_2 + m_3 = 0$, while j_1, j_2, j_3 must have values such that each can result from the vector addition of the other two. Geometrically, j_1, j_2, j_3 must therefore be the sides of a closed triangle; in other words, each is no smaller than the difference and no larger than the sum of the other two:

$$|j_1 - j_2| \le j_3 \le j_1 + j_2 \quad \text{and so on.}$$

It is clear that the algebraic sum $j_1 + j_2 + j_3$ is then an integer.

The wave function of this system has the form

$$\psi_0 = \sum_{m_1 m_2 m_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \psi^{(1)}_{j_1 m_1} \psi^{(2)}_{j_2 m_2} \psi^{(3)}_{j_3 m_3}, \tag{106.4}$$

where each m_i is summed from $-j_i$ to j_i. The coefficients in this formula are called Wigner $3j$ symbols. By definition, they are nonzero only when $m_1 + m_2 + m_3 = 0$.

Under a permutation of the indices 1, 2, 3, the wave function (106.4) can change only by an immaterial phase factor. In fact, the $3j$ symbols may be defined to be purely real (see below), in which case the only ambiguity in ψ_0 is its overall sign (just as for the function (106.2)). Thus, permuting the columns of a $3j$ symbol can either leave it unchanged or reverse its sign.

The most symmetric prescription for the coefficients in (106.4), and the one conventionally used to define the $3j$ symbols, is as follows. In spinor notation, ψ_0 is a scalar obtained from the product of three spinors $\psi^{(1)}_{\lambda\mu\ldots}$, $\psi^{(2)}_{\lambda\mu\ldots}$, $\psi^{(3)}_{\lambda\mu\ldots}$ by contracting all indices in pairs, each pair belonging to two different spinors. We agree that, in every pair associated with particles 1 and 2, the spinor index is written upstairs on $\psi^{(1)}$ and downstairs on $\psi^{(2)}$; in a pair associated with particles 2 and 3, it is upstairs on $\psi^{(2)}$ and downstairs on $\psi^{(3)}$; and in a pair associated with particles 3 and 1, it is upstairs on $\psi^{(3)}$ and downstairs on $\psi^{(1)}$. (There are, respectively, $j_1 + j_2 - j_3$, $j_2 + j_3 - j_1$, and $j_1 + j_3 - j_2$ pairs of these three "types," as is easily counted.) This prescription fixes the sign of ψ_0 uniquely.

With this definition, a cyclic permutation of the indices 1, 2, 3 evidently leaves ψ_0 unchanged. Hence a $3j$ symbol is invariant under a cyclic permutation of its columns. Interchanging any two indices, on the other hand, requires, as is readily seen, raising the lower and lowering the upper indices in all $j_1 + j_2 + j_3$ pairs. Thus ψ_0 is multiplied by $(-1)^{j_1+j_2+j_3}$; equivalently, the $3j$ symbols satisfy

$$\begin{pmatrix} j_2 & j_1 & j_3 \\ m_2 & m_1 & m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \quad \text{and so on,} \tag{106.5}$$

that is, interchanging two columns changes their sign when $j_1 + j_2 + j_3$ is odd.

Finally, it is easy to see that

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} = (-1)^{j_1+j_2+j_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \tag{106.6}$$

Indeed, reversing the signs of the z components of all angular momenta may be regarded as the result of a rotation through π about the y axis. This transformation is equivalent to raising every lower spinor index and lowering every upper one (see (58.5)).

Expression (106.4) can be used to obtain an important formula for the wave function ψ_{jm} of a two-particle system having specified values of j and m. Regard particles 1 and 2 together as a single system. Since the angular momentum j of this system and the angular momentum j_3 of particle 3 add to zero, we must have $j = j_3$ and $m = -m_3$. Formula (106.2) then gives

$$\psi_0 = \frac{1}{\sqrt{2j+1}} \sum_m (-1)^{j-m} \psi_{jm} \psi^{(3)}_{j,-m}. \tag{106.7}$$

This formula is to be compared with (106.4), in which $j, -m$ are written in place of j_3, m_3. Before doing so, however, one must account for the fact that the rule used to construct the sum in (106.7), according to (106.3), differs from the rule for constructing the sum (106.4). To bring (106.7) into the form (106.4), the upper and lower indices in the pairs associated with

particles 1 and 3 must be interchanged, as is easily verified; this supplies the extra factor $(-1)^{j_1-j_2+j_3}$. The result is[3]

$$\psi_{jm} = (-1)^{j_1-j_2+m}\sqrt{2j+1}\sum_{m_1m_2}\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m\end{pmatrix}\psi^{(1)}_{j_1m_1}\psi^{(2)}_{j_2m_2}, \tag{106.8}$$

where m_1 and m_2 are summed subject to $m_1 + m_2 = m$.

Formula (106.8) gives the required expansion of the system wave function in terms of the wave functions of the two particles with definite angular momenta j_1 and j_2. It may be written as

$$\psi_{jm} = \sum_{m_1m_2}(m_1m_2 \mid jm)\psi^{(1)}_{j_1m_1}\psi^{(2)}_{j_2m_2}, \qquad m_2 = m - m_1. \tag{106.9}$$

The coefficients

$$(m_1m_2 \mid jm) = (-1)^{j_1-j_2+m}\sqrt{2j+1}\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m\end{pmatrix} \tag{106.10}$$

form the matrix that transforms the complete orthonormal set of $(2j_1+1)(2j_2+1)$ wave functions for the states $|m_1m_2\rangle$ into the corresponding set of wave functions for the states $|jm\rangle$, with j_1 and j_2 fixed. They are called vector-coupling coefficients, or Clebsch–Gordan coefficients. The symbol $(m_1m_2 \mid jm)$ follows the general notation, introduced in (11.18), for coefficients that expand one system of functions in another. To shorten the notation, we have omitted the quantum numbers j_1, j_2, which are common to both systems of functions. When needed, they may be displayed explicitly: $(j_1m_1j_2m_2 \mid j_1j_2jm)$[4].

The transformation matrix (106.9) is unitary (see § 12). The coefficients of the inverse transformation

$$\psi^{(1)}_{j_1m_1}\psi^{(2)}_{j_2m_2} = \sum_{j=|j_2-j_1|}^{j_1+j_2}(jm \mid m_1m_2)\psi_{jm} \tag{106.11}$$

are therefore the complex conjugates of the coefficients in (106.9). We shall see below that these coefficients are real;

hence simply

$$(m_1m_2 \mid jm) = (jm \mid m_1m_2).$$

By the general rules of quantum mechanics, the squares of the expansion coefficients in (106.11) give the probabilities that the system has the various values of j, m (when j_1, m_1 and j_2, m_2 are specified).

The unitarity of the transformation (106.9) implies definite orthogonality relations for its coefficients. From (12.5) and (12.6),

$$\sum_{m_1m_2}(m_1m_2 \mid jm)(m_1m_2 \mid j'm') = \delta_{jj'}\delta_{mm'}, \tag{106.12}$$

$$\sum_{jm}(m_1m_2 \mid jm)(m_1'm_2' \mid jm) = \delta_{m_1m_1'}\delta_{m_2m_2'}. \tag{106.13}$$

[3]Under time reversal the wave functions transform, according to (60.2), as $\psi_{jm} \to (-1)^{j-m}\psi_{j,-m}$. It is readily checked that when the functions $\psi_{j_1m_1}$ and $\psi_{j_2m_2}$ on the right-hand side of (106.8) undergo this transformation, the function ψ_{jm} on the left-hand side transforms in the same way.

[4]The notation $C^{jm}_{j_1m_1j_2m_2}$ or $C^{j_1j_2j}_{m_1m_2m}$ is also used in the literature for the Clebsch–Gordan coefficients.

The explicit general expression for a $3j$ symbol is rather cumbersome. It may be written as[5]

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \left[\frac{(j_1+j_2-j_3)!(j_1-j_2+j_3)!(-j_1+j_2+j_3)!}{(j_1+j_2+j_3+1)!}\right]^{1/2}$$
$$\times [(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!(j_3+m_3)!(j_3-m_3)!]^{1/2}$$
$$\times \sum_z \frac{(-1)^{z+j_1-j_2-m_3}}{z!(j_1+j_2-j_3-z)!(j_1-m_1-z)!(j_2+m_2-z)!}$$
$$\times \frac{1}{(j_3-j_2+m_1+z)!(j_3-j_1-m_2+z)!}. \tag{106.14}$$

The sum extends over all integral values of z; since the factorial of a negative number is taken to be ∞, however, the number of terms is actually finite. The factor preceding the sum is manifestly symmetric in the indices 1, 2, 3; the symmetry of the sum itself becomes evident after a suitable relabeling of the summation variable z.

Besides the symmetry properties (106.5), (106.6), which follow simply from the definition of the $3j$ symbols, there are further symmetries whose derivation is more involved and will not be given here. They are conveniently stated by introducing a square (3×3) array of numbers related to the parameters of the $3j$ symbol as follows:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \longleftrightarrow \begin{bmatrix} j_2+j_3-j_1 & j_3+j_1-j_2 & j_1+j_2-j_3 \\ j_1-m_1 & j_2-m_2 & j_3-m_3 \\ j_1+m_1 & j_2+m_2 & j_3+m_3 \end{bmatrix}. \tag{106.15}$$

(The sum of the entries in every row and every column of this array is $j_1+j_2+j_3$.) Then: 1) interchanging any two columns of the array multiplies the $3j$ symbol by $(-1)^{j_1+j_2+j_3}$ (this is the same property as (106.5)); 2) the same statement holds for interchanging any two rows (for the two lower rows this is the same property as (106.6)); 3) the $3j$ symbol is unchanged when the rows of the array are replaced by its columns[6].

We list several simpler formulas for special cases. The value

$$\begin{pmatrix} j & j & 0 \\ m & -m & 0 \end{pmatrix} = \frac{(-1)^{j-m}}{\sqrt{2j+1}} \tag{106.16}$$

corresponds to formula (106.2). The formula

$$\begin{pmatrix} j_1 & j_2 & j_1+j_2 \\ m_1 & m_2 & -m_1-m_2 \end{pmatrix} = (-1)^{j_1-j_2+m_1+m_2}$$
$$\times \left[\frac{(2j_1)!(2j_2)!(j_1+j_2+m_1+m_2)!(j_1+j_2-m_1-m_2)!}{(j_1+j_2+1)!(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!}\right]^{1/2}, \tag{106.17}$$

[5] The expansion coefficients (106.9) were first calculated by Wigner (1931). The symmetry properties of these coefficients and the symmetric expression (106.14) for them were found by Racah (G. Racah, 1942). Perhaps the most direct calculation is to pass straight from the spinor representation of ψ_0 (normalized appropriately) to the sum (106.4), using the correspondence formula (57.6). Notice that the real coefficient in that formula automatically establishes that the $3j$ symbols are real. Another derivation is given in Edmonds's book (see the note on p. 272). The table of $3j$ symbols below is also taken from that book.

[6] See T. Regge // Nuovo Cimento. 1958. V. 10. P. 544; 1959. V. 11. P. 116. For the deeper mathematical aspects of the symmetry property (106.15), as well as of the property (108.3) of the $6j$ symbols given below, see the review article by Ya. A. Smorodinsky and L. A. Shelepin // UFN. 1972. V. 106. P. 3.

follows directly from (106.14). Deriving the formula

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ 0 & 0 & 0 \end{pmatrix} = (-1)^p \left[\frac{(j_1+j_2-j_3)!(j_1-j_2+j_3)!(-j_1+j_2+j_3)!}{(2p+1)!} \right]^{1/2} \times \frac{p!}{(p-j_1)!(p-j_2)!(p-j_3)!}, \tag{106.18}$$

where $2p = j_1 + j_2 + j_3$ is even, requires several additional calculations[7] (when $2p$ is odd, this $3j$ symbol vanishes by the symmetry property (106.6)).

For reference, Table 9 gives the values of the $3j$ symbols for $j_3 = 1/2, 1, 3/2, 2$. For each j_3, the smallest set of $3j$ symbols from which all the others can be obtained by means of (106.5), (106.6) is displayed.

Table 9. Formulas for $3j$ symbols

j_3	j_1	Value of $(-1)^{j-m}\begin{pmatrix} j_1 & j & j_3 \\ m_1 & -m-m_3 & m_3 \end{pmatrix}$
$\frac{1}{2}$	j	$m_3 = \frac{1}{2}:\quad -\left[\frac{j-m}{(2j+1)(2j+2)}\right]^{1/2}$
	$j+1$	$\left[\frac{j+m+1}{(2j+1)(2j+2)}\right]^{1/2}$
1	j	$m_3 = 0:\quad \frac{2m}{[2j(2j+1)(2j+2)]^{1/2}}$
	$j+1$	$\left[\frac{2(j+m+1)(j-m+1)}{(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	j	$m_3 = 1:\quad \left[\frac{2(j-m)(j+m+1)}{2j(2j+1)(2j+2)}\right]^{1/2}$
	$j+1$	$\left[\frac{(j-m)(j-m+1)}{(2j+1)(2j+2)(2j+3)}\right]^{1/2}$

Problem

Determine the angular dependence of the wave functions of a particle of spin 1/2 in states with specified orbital angular momentum l, total angular momentum j, and projection m.

Solution. . The problem is solved by the general formula (106.8), in which $\psi^{(1)}$ is to mean an eigenfunction of the orbital angular momentum (that is, a spherical harmonic Y_{lm_l}), while $\psi^{(2)}$ is the spin wave function $\chi(\sigma)$ (where $\sigma = \pm 1/2$):

$$\psi_{jm} = (-1)^{l+m-1/2}\sqrt{2j+1}\sum_{\sigma}\begin{pmatrix} l & \frac{1}{2} & j \\ m-\sigma & \sigma & -m \end{pmatrix} Y_{l,m-\sigma}\chi(\sigma).$$

[7]See the book by Edmonds cited above.

Table 9 (continued)

j_3	j_1	Value
$\frac{3}{2}$	$j+\frac{1}{2}$	$m_3=\frac{1}{2}:\quad -(j+3m+\frac{3}{2})\left[\frac{j-m+\frac{1}{2}}{2j(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	$j+\frac{3}{2}$	$\left[\frac{3(j-m+\frac{1}{2})(j-m+\frac{3}{2})(j+m+\frac{3}{2})}{(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
	$j+\frac{1}{2}$	$m_3=\frac{3}{2}:\quad -\left[\frac{3(j-m-\frac{1}{2})(j-m+\frac{1}{2})(j+m+\frac{3}{2})}{2j(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	$j+\frac{3}{2}$	$\left[\frac{(j-m-\frac{1}{2})(j-m+\frac{1}{2})(j-m+\frac{3}{2})}{(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
2	j	$m_3=0:\quad \frac{2[3m^2-j(j+1)]}{[(2j-1)2j(2j+1)(2j+2)(2j+3)]^{1/2}}$
	$j+1$	$-2m\left[\frac{6(j+m+1)(j-m+1)}{2j(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
	$j+2$	$\left[\frac{6(j+m+2)(j+m+1)(j-m+2)(j-m+1)}{(2j+1)(2j+2)(2j+3)(2j+4)(2j+5)}\right]^{1/2}$
	j	$m_3=1:\quad (1+2m)\left[\frac{6(j+m+1)(j-m)}{(2j-1)2j(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	$j+1$	$-2(j+2m+2)\left[\frac{(j-m+1)(j-m)}{2j(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
	$j+2$	$2\left[\frac{(j+m+2)(j-m+2)(j-m+1)(j-m)}{(2j+1)(2j+2)(2j+3)(2j+4)(2j+5)}\right]^{1/2}$
	j	$m_3=2:\quad \left[\frac{6(j-m-1)(j-m)(j+m+1)(j+m+2)}{(2j-1)2j(2j+1)(2j+2)(2j+3)}\right]^{1/2}$
	$j+1$	$-2\left[\frac{(j-m-1)(j-m)(j-m+1)(j+m+2)}{2j(2j+1)(2j+2)(2j+3)(2j+4)}\right]^{1/2}$
	$j+2$	$\left[\frac{(j-m-1)(j-m)(j-m+1)(j-m+2)}{(2j+1)(2j+2)(2j+3)(2j+4)(2j+5)}\right]^{1/2}$

Substitution of the $3j$ symbols gives

$$\psi_{l+1/2,m} = \sqrt{\frac{j+m}{2j}}\,\chi\left(\frac{1}{2}\right)Y_{l,m-1/2} + \sqrt{\frac{j-m}{2j}}\,\chi\left(-\frac{1}{2}\right)Y_{l,m+1/2},$$
$$\psi_{l-1/2,m} = -\sqrt{\frac{j-m+1}{2j+2}}\,\chi\left(\frac{1}{2}\right)Y_{l,m-1/2} + \sqrt{\frac{j+m+1}{2j+2}}\,\chi\left(-\frac{1}{2}\right)Y_{l,m+1/2}.$$

§ 107. Matrix elements of tensors

In § 29 we obtained formulas giving the dependence of the matrix elements of a vector physical quantity on the value of the angular-momentum projection. In fact, those formulas are a special instance of general relations that solve the same problem for an irreducible tensor (see p. 268) of arbitrary rank[8].

The set of $2k+1$ components of an irreducible tensor of rank k (where k is an integer) is, in its transformation properties, equivalent to the set of $2k+1$ spherical harmonics Y_{kq} ($q = -k, \dots, k$) (see the note on p. 268). Thus suitable linear combinations of the tensor components yield a set of quantities that transform under rotations in the same way as the functions Y_{kq}. We shall denote these quantities by f_{kq} and call their set a *spherical tensor* of rank k.

For example, a vector corresponds to $k = 1$, and the quantities f_{1q} are related to the vector components by

$$f_{10} = ia_z, \qquad f_{1,\pm1} = \mp\frac{i}{\sqrt{2}}(a_x \pm ia_y) \tag{107.1}$$

(cf. (57.7)). The analogous formulas for a tensor of rank two are

$$\begin{gathered} f_{20} = -\sqrt{3/2}\,a_{zz}, \qquad f_{2,\pm1} = \pm(a_{xz} \pm ia_{yz}), \\ f_{2,\pm2} = -\frac{1}{2}(a_{xx} - a_{yy} \pm 2ia_{xy}), \end{gathered} \tag{107.2}$$

with $a_{xx} + a_{yy} + a_{zz} = 0$[9].

Tensor products of two (or more) spherical tensors $f_{k_1q_1}$, $g_{k_2q_2}$ are formed according to the rules for adding angular momenta, with k_1 and k_2 formally playing the role of the "angular momenta" associated with the tensors. Thus two spherical tensors of ranks k_1 and k_2 can be combined into spherical tensors of ranks $K = k_1 + k_2, \dots, |k_1 - k_2|$ by the formulas

$$\begin{aligned} (f_{k_1}g_{k_2})_{KQ} &= \sum_{q_1q_2}(q_1q_2|KQ)f_{k_1q_1}g_{k_2q_2} = \\ &= (-1)^{k_1-k_2+Q}\sqrt{2K+1}\sum_{q_1q_2}\begin{pmatrix} k_1 & k_2 & K \\ q_1 & q_2 & -Q \end{pmatrix} f_{k_1q_1}g_{k_2q_2} \end{aligned} \tag{107.3}$$

[8]The treatment of the questions considered in § 107–109, as well as most of the results presented there, is due to Racah (1942/1943).

[9]It is understood that the quantities f_{kq} are complex only because of the change to spherical coordinates; the original Cartesian tensor components are real.

Compare (106.9). Let the two spherical tensors have the same rank k. Their scalar product is customarily defined, however, by

$$(f_k g_k)_{00} = \sum_q (-1)^{k-q} f_{kq} g_{k,-q}, \tag{107.4}$$

which differs by a factor $\sqrt{2k+1}$ from the definition obtained by putting $K = Q = 0$ in (107.3) (cf. (106.2))[10]. The same definition may also be written as

$$(f_k g_k)_{00} = \sum_q f_{kq} g_{kq}^*,$$

since complex conjugation of a spherical tensor obeys the rule

$$f_{kq}^* = (-1)^{k-q} f_{k,-q}$$

(cf. (28.9))[11].

Representing physical quantities as spherical tensors is particularly convenient when calculating their matrix elements, since the results of angular-momentum addition theory can then be used.

By the definition of matrix elements,

$$\widehat{f}_{kq} \psi_{njm} = \sum_{n'j'm'} \langle n'j'm' | f_{kq} | njm \rangle \psi_{n'j'm'}, \tag{107.5}$$

where ψ_{njm} are the wave functions of stationary states of the system, specified by its angular momentum j, its projection m, and the set n of the remaining quantum numbers. In their transformation properties, the functions on the two sides of (107.5) correspond to the functions on the respective sides of (106.11). The selection rules follow at once.

Matrix elements of the components f_{kq} of an irreducible tensor of rank k can be nonzero only for transitions $jm \to j'm'$ satisfying the angular-momentum "addition rule" $\mathbf{j}' = \mathbf{j} + \mathbf{k}$. The numbers j', j, and k must then satisfy the "triangle rule" (that is, they can be the lengths of the sides of a closed triangle), while $m' = m + q$. In particular, diagonal matrix elements can be nonzero only if $2j \geqslant k$.

The same analogy under transformations further shows that the coefficients in the sum (107.5) must be proportional to those in (106.11) (the *Wigner–Eckart theorem*). This fixes their dependence on m and m'. Accordingly, we write the matrix elements in the form

$$\langle n'j'm' | f_{kq} | njm \rangle = i^k (-1)^{j_{\max} - m'} \begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix} \langle n'j' \| f_k \| nj \rangle, \tag{107.6}$$

[10]If $\mathbf{A}$ and $\mathbf{B}$ are two vectors corresponding, by (107.1), to the spherical tensors f_{1q} and g_{1q}, then

$$(f_1 g_1)_{00} = \mathbf{AB}.$$

[11]We repeat the observation made above in connection with (106.8): with this rule, complex conjugation of the tensors of ranks k_1 and k_2 on the right-hand side of (107.3) produces the same conjugation of the rank-K tensor on its left-hand side.

where $j_{\max}$ is the larger of j and j'. The quantities $\langle n'j'\|f_k\|nj\rangle$, which do not depend on m, m', or q', are called *reduced matrix elements*. Formula (107.6) answers the question of how matrix elements depend on the angular-momentum projections. This dependence is governed entirely by symmetry with respect to the rotation group, whereas dependence on the other quantum numbers is determined by the physical nature of the quantities f_{kq} themselves[12].

The operators $\widehat{f}_{kq}$ are related by

$$\widehat{f}^{+}_{kq} = (-1)^{k-q}\widehat{f}_{k,-q}. \tag{107.7}$$

Their matrix elements therefore satisfy

$$\langle n'j'm'|f_{kq}|njm\rangle^* = (-1)^{k-q}\langle njm|f_{k,-q}|n'j'm'\rangle. \tag{107.8}$$

Substitution of (107.6) into this equality, followed by use of the $3j$-symbol properties (106.5) and (106.6), gives the "Hermiticity" relation for the reduced matrix elements[13]

$$\langle n'j'\|f_k\|nj\rangle = \langle nj\|f_k\|n'j'\rangle^*. \tag{107.9}$$

The matrix elements of the scalar (107.4) are diagonal in j and m. The rule of matrix multiplication gives

$$\begin{aligned}\langle n'jm|(f_k g_k)_{00}|njm\rangle &= \\ &= \sum_q (-1)^{k-q} \sum_{n''j''m''} \langle n'jm|f_{kq}|n''j''m''\rangle \\ &\qquad \times \langle n''j''m''|g_{k,-q}|njm\rangle.\end{aligned}$$

Substituting (107.6) here and summing over q and m'' with the aid of the $3j$-symbol orthogonality relation (106.12), we obtain

$$\langle n'jm|(f_k g_k)_{00}|njm\rangle = \frac{1}{2j+1}\sum_{n''j''}\langle n'j\|f_k\|n''j''\rangle\langle n''j''\|g_k\|nj\rangle. \tag{107.10}$$

In the same way one readily obtains the following formulas for sums of squared matrix elements:

$$\sum_{qm'}|\langle n'j'm'|f_{kq}|njm\rangle|^2 = \frac{1}{2j+1}|\langle n'j'\|f_k\|nj\rangle|^2, \tag{107.11}$$

$$\sum_{mm'}|\langle n'j'm'|f_{kq}|njm\rangle|^2 = \frac{1}{2k+1}|\langle n'j'\|f_k\|nj\rangle|^2. \tag{107.12}$$

In the first formula the sum is over q and m' for a specified m; in the second it is over m and m' for a specified q (with $m' = m + q$ in every case).

[12]Among other consequences, these results yield the selection rules stated in § 29 for vector matrix elements and the formulas (29.7)–(29.9) for them.

[13]The phase factor in definition (107.6) was chosen precisely so as to ensure this equality.

For reference, consider the case in which the quantities f_{kq} are the spherical harmonics Y_{kq} themselves, and give the expressions for their matrix elements in transitions between states of one particle with integral orbital angular momenta l_1 and l_2, that is, for the integrals

$$\langle l_1 m_1 | Y_{lm} | l_2 m_2 \rangle = \int Y^*_{l_1 m_1} Y_{lm} Y_{l_2 m_2} \, do. \tag{107.13}$$

Besides the selection rule corresponding to angular-momentum addition ($\mathbf{l} + \mathbf{l}_2 = \mathbf{l}_1$), these matrix elements obey the further rule that the sum $l + l_1 + l_2$ must be even. This follows from parity conservation: the product $(-1)^{l_1+l_2}$ of the parities of the two states must equal the parity $(-1)^l$ of the physical quantity under consideration (see § 30).

The matrix elements (107.13) are a special case of a more general integral that will be calculated in § 110 (see the note on p. 547). They are given by

$$\begin{aligned}\langle l_1 m_1 | Y_{lm} | l_2 m_2 \rangle = (-1)^{m_1} i^{-l_1+l_2+l} &\begin{pmatrix} l_1 & l & l_2 \\ -m_1 & m & m_2 \end{pmatrix} \begin{pmatrix} l_1 & l & l_2 \\ 0 & 0 & 0 \end{pmatrix} \\ &\times \left[\frac{(2l+1)(2l_1+1)(2l_2+1)}{4\pi} \right]^{1/2}. \end{aligned} \tag{107.14}$$

In particular, on setting $m_1 = m_2 = m = 0$, we find the integral of the product of three Legendre polynomials:

$$\int_{-1}^{1} P_l(\mu) P_{l_1}(\mu) P_{l_2}(\mu) \, d\mu = 2 \begin{pmatrix} l_1 & l & l_2 \\ 0 & 0 & 0 \end{pmatrix}^2. \tag{107.15}$$

§ 108. $6j$ symbols

In § 106 we defined the $3j$ symbols as the coefficients in the sum (106.4), which represents the wavefunction of a three-particle system with zero total angular momentum. From the standpoint of transformation properties under rotations, this sum is a scalar. It follows that the set of $3j$ symbols for specified values of j_1, j_2, j_3 (and all possible m_1, m_2, m_3) may be regarded as a collection of quantities which, under rotations, transform contragrediently to the products $\psi_{j_1 m_1} \psi_{j_2 m_2} \psi_{j_3 m_3}$, thereby making the entire sum invariant.

From this viewpoint, one may ask how to construct a scalar solely from $3j$-symbols. Such a scalar may depend only on the j values, and not on the

m values, which change under rotations. In other words, it must be expressible through sums over all the m values. Each such summation amounts to a "contraction" of the product of two $3j$-symbols according to the rule

$$\sum_m (-1)^{j-m} \begin{pmatrix} j & \cdots \\ m & \cdots \end{pmatrix} \begin{pmatrix} j & \cdots \\ -m & \cdots \end{pmatrix} \tag{108.1}$$

(compare the construction of the scalar (106.2)).

Since every "contraction" involves a pair of numbers m, a complete scalar must be constructed from a product of an even number of $3j$ symbols. By their orthogonality property, contracting a product of two $3j$ symbols gives the trivial result

$$\sum_{m_1 m_2 m_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} (-1)^{j_1+j_2+j_3-m_1-m_2-m_3}$$
$$= \sum_{m_1 m_2 m_3} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}^2 = 1$$

(here the equality $m_1 + m_2 + m_3 = 0$ and formulas (106.6) and (106.12) have been used). Hence the smallest number of factors required to obtain a nontrivial scalar is four. In each $3j$ symbol, the three numbers j geometrically form a closed triangle. Since every number j must occur in two $3j$ symbols during the contraction, it is clear that a scalar formed from a product of four $3j$ symbols contains six numbers j. Geometrically, they must be represented by the edge lengths of an irregular tetrahedron (Fig. 45), with one face corresponding to each $3j$ symbol. A definite convention for carrying out the contractions is adopted in defining the scalar in question; it is expressed by the formula

$$\begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix} = \sum_{\text{all } m} (-1)^{\sum_i (j_i - m_i)} \begin{pmatrix} j_1 & j_2 & j_3 \\ -m_1 & -m_2 & -m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_5 & j_6 \\ m_1 & -m_5 & m_6 \end{pmatrix}$$
$$\times \begin{pmatrix} j_4 & j_2 & j_6 \\ m_4 & m_2 & -m_6 \end{pmatrix} \begin{pmatrix} j_4 & j_5 & j_3 \\ -m_4 & m_5 & m_3 \end{pmatrix}. \tag{108.2}$$

The summation here is over all possible values of every number m; however, since the sum of the three m's in each $3j$ symbol must vanish, only

three of the six m's are actually independent. The quantities defined by (108.2) are called $6j$ symbols or Racah coefficients[14].

From definition (108.2), with account taken of the symmetry properties of the $3j$ symbols, it is readily verified that the $6j$ symbol is unchanged under any permutation of its three columns, and that the upper and lower numbers may be interchanged in any pair of columns. By virtue of these symmetry properties, the sequence of numbers $j_1 \ldots j_6$ in a $6j$ symbol can be represented in 24 equivalent forms[15].

The $6j$ symbols also have another, less evident symmetry property, which equates symbols having different sets of numbers j:

$$\begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix} = \begin{Bmatrix} j_1 & \frac{1}{2}(j_2+j_5+j_3-j_6) & \frac{1}{2}(j_3+j_6+j_2-j_5) \\ j_4 & \frac{1}{2}(j_2+j_5+j_6-j_3) & \frac{1}{2}(j_3+j_6+j_5-j_2) \end{Bmatrix}. \tag{108.3}$$

(*T. Regge*, 1959)[16].

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[14]The notation

$$W(j_1 j_2 j_4 j_5; j_3 j_6) = (-1)^{j_1+j_2+j_4+j_5} \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix}$$

is also used in the literature.

[15]If the tetrahedron in Fig. 45 is imagined to be regular, the 24 equivalent permutations of the numbers j can be obtained by applying the 24 symmetry transformations (rotations and reflections) of the tetrahedron.

[16]See the note on p. 529.

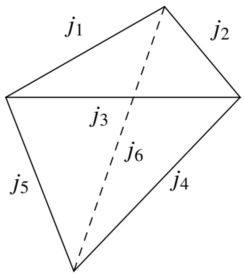

Figure 45

We give a useful relation between 6j and 3j symbols that follows from definition (108.2):

$$\sum_{m_4 m_5 m_6} (-1)^{j_4+j_5+j_6-m_4-m_5-m_6} \begin{pmatrix} j_1 & j_5 & j_6 \\ m_1 & -m_5 & m_6 \end{pmatrix} \begin{pmatrix} j_4 & j_2 & j_6 \\ m_4 & m_2 & -m_6 \end{pmatrix} \times \begin{pmatrix} j_4 & j_5 & j_3 \\ -m_4 & m_5 & m_3 \end{pmatrix} = \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix}. \tag{108.4}$$

The expression being summed on the left-hand side differs from the sum in (108.2) by the omission of one factor (a 3j-symbol). Thus the sum in (108.4) may be represented by a tetrahedron (Fig. 45) with one face removed; this is why the sum is not a scalar. In other words, in its transformation properties it corresponds to a single 3j-symbol—the one on the right-hand side of (108.4)—and must therefore be proportional to that symbol. The proportionality factor (the 6j-symbol on the right-hand side) is obtained at once by multiplying both sides by

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}$$

and summing over the remaining numbers m_1, m_2, m_3.

6j-symbols arise naturally in the following question concerning the addition of three angular momenta.

Let the three angular momenta j_1, j_2, j_3 combine to form a resultant angular momentum J. Specifying J (and its projection M), however, does not yet determine the state of the system uniquely: the state also depends on the order in which the angular momenta are added (or, as it is called, their coupling scheme).

Consider, for example, the following two coupling schemes: 1) first j_1 and j_2 are added to form the total angular momentum j_{12}, after which j_{12} and j_3 are added to give the final angular momentum J; 2) j_2 and j_3 are added to give j_{23}, and then j_{23} is combined with j_1 to give J. The first scheme corresponds to states in which, besides j_1, j_2, j_3, J, M, the quantity j_{12} has a definite value; let their wave functions be denoted by $\psi_{j_{12}JM}$ (the repeated indices $j_1 j_2 j_3$ being suppressed for brevity). Similarly, denote the wave functions for the second coupling scheme by $\psi_{j_{23}JM}$. In either case the "intermediate" angular momentum (j_{12} or j_{23}) can, in general, take more than one value. Thus, for fixed J, M, there are two distinct sets of states, distinguished respectively by the values of j_{12}

and j_{23}. By the general rules, the functions in these two sets are related by a certain unitary transformation

$$\psi_{j_{23}JM} = \sum_{j_{12}} \langle j_{12}|j_{23}\rangle \psi_{j_{12}JM}. \tag{108.5}$$

It is clear on physical grounds that the coefficients of this transformation do not depend on M: they cannot depend on the orientation of the entire system in space. They therefore depend only on the six angular momenta $j_1, j_2, j_3, j_{12}, j_{23}, J$, not on their projections, and so are scalar quantities (in the sense specified above). These coefficients can be calculated directly as follows.

Applying formula (106.9) twice, we find

$$\begin{aligned}
\psi_{j_{23}JM} &= \sum_{(m)} \langle m_1 m_{23}|JM\rangle \psi_{j_1 m_1} \psi_{j_{23} m_{23}} \\
&= \sum_{(m)} \langle m_1 m_{23}|JM\rangle \langle m_2 m_3|j_{23} m_{23}\rangle \psi_{j_1 m_1} \psi_{j_2 m_2} \psi_{j_3 m_3}, \\
\psi_{j_{12}JM} &= \sum_{(m)} \langle m_3 m_{12}|JM\rangle \langle m_1 m_2|j_{12} m_{12}\rangle \psi_{j_1 m_1} \psi_{j_2 m_2} \psi_{j_3 m_3}
\end{aligned}$$

(the sign (m) under the summation sign means that the summation is over all the numbers $m_1, m_2, \ldots$ occurring in the expression). Using the orthonormality of the functions ψ_{jm}, we now obtain

$$\begin{aligned}
\langle j_{12}|j_{23}\rangle &\equiv \int \psi^*_{j_{12}JM} \psi_{j_{23}JM}\, dq \\
&= \sum_{(m)} \langle m_3 m_{12}|JM\rangle \langle m_1 m_{23}|JM\rangle \langle m_1 m_2|j_{12} m_{12}\rangle \langle m_2 m_3|j_{23} m_{23}\rangle.
\end{aligned}$$

The sum on the right is taken at a specified value of M, but the result of the summation is in fact independent of M, for the reason given above. The summation may therefore be extended over the values of M as well, provided that the factor $1/(2J+1)$ is placed before the sum. Then, expressing the coefficients $\langle m_1 m_2|jm\rangle$ in terms of $3j$ symbols according to (106.10), we obtain

$$\langle j_{12}|j_{23}\rangle = (-1)^{j_1+j_2+j_3+J} \sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & J & j_{23} \end{Bmatrix}. \tag{108.6}$$

The connection of the $6j$ symbols with the transformation coefficients (108.5) makes it easy to derive several useful formulas for sums of products of $6j$ symbols.

First, by the unitarity of transformation (108.5) (and the reality of its coefficients), we have

$$\sum_{j} (2j+1)(2j''+1) \begin{Bmatrix} j_1 & j_2 & j' \\ j_3 & j_4 & j \end{Bmatrix} \begin{Bmatrix} j_3 & j_2 & j \\ j_1 & j_4 & j'' \end{Bmatrix} = \delta_{j'j''}. \tag{108.7}$$

Next consider three schemes for coupling three angular momenta, whose intermediate sums are respectively j_{12}, j_{23}, and j_{31}. By the rule for matrix multiplication, the corresponding transformation coefficients (108.6) are related by

$$\sum_{j_{23}} \langle j_{12}|j_{23}\rangle\langle j_{23}|j_{31}\rangle = \langle j_{12}|j_{31}\rangle.$$

Substitution of (108.6), followed by a change of index notation, gives

$$\sum_j (-1)^{j+j_3+j_6}(2j+1)\begin{Bmatrix} j_2 & j_4 & j_6 \\ j_1 & j_5 & j \end{Bmatrix}\begin{Bmatrix} j_4 & j_1 & j_6 \\ j_2 & j_5 & j_3 \end{Bmatrix} = \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix}. \tag{108.8}$$

Finally, consideration of different schemes for coupling four angular momenta yields[17] the following addition formula for products of three $6j$ symbols:

$$\begin{aligned} &\sum_j (-1)^{j+\sum_1^9 j_i}(2j+1)\begin{Bmatrix} j_4 & j_2 & j_6 \\ j_9 & j_8 & j \end{Bmatrix}\begin{Bmatrix} j_2 & j_1 & j_3 \\ j_7 & j & j_9 \end{Bmatrix}\begin{Bmatrix} j_4 & j_3 & j_5 \\ j_7 & j_8 & j \end{Bmatrix} \\ &\quad = \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix}\begin{Bmatrix} j_6 & j_1 & j_5 \\ j_7 & j_8 & j_9 \end{Bmatrix}. \end{aligned} \tag{108.9}$$

(*L. C. Biedenharn*, *J. P. Elliott*, 1953).

For reference, we give some explicit expressions for $6j$ symbols. In the general case a $6j$ symbol can be represented by the following sum:

$$\begin{aligned} \begin{Bmatrix} j_1 & j_2 & j_3 \\ j_4 & j_5 & j_6 \end{Bmatrix} &= \Delta(j_1j_2j_3)\Delta(j_1j_5j_6)\Delta(j_4j_2j_6)\Delta(j_4j_5j_3) \\ &\times \sum_z \frac{(-1)^z(z+1)!}{(z-j_1-j_2-j_3)!(z-j_1-j_5-j_6)!(z-j_4-j_2-j_6)!(z-j_4-j_5-j_3)!} \\ &\times (j_1+j_2+j_4+j_5-z)!(j_2+j_3+j_5+j_6-z)!(j_3+j_1+j_6+j_4-z)!, \end{aligned} \tag{108.10}$$

where

$$\Delta(abc) = \left[\frac{(a+b-c)!(a-b+c)!(-a+b+c)!}{(a+b+c+1)!}\right]^{1/2},$$

The sum runs over every positive integral value of z for which no factorial in the denominator has a negative argument. Table 10 gives formulas for the $6j$-symbols when one parameter is 0, 1/2, or 1.

We conclude with several remarks about higher-order scalars formed from $3j$ symbols.

After the $6j$ symbol, the next in complexity is the scalar formed by contracting a product of six $3j$ symbols. These $3j$ symbols contain 18 pairwise equal numbers j,

so that the resulting scalar depends on nine parameters j. It is conventionally called

[17] See the book by Edmonds cited above.

a $9j$ symbol and is defined as follows (*E. Wigner*, 1951)[18]:

$$\begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{Bmatrix} = \sum_{\text{all } m} \begin{pmatrix} j_{11} & j_{12} & j_{13} \\ m_{11} & m_{12} & m_{13} \end{pmatrix} \begin{pmatrix} j_{21} & j_{22} & j_{23} \\ m_{21} & m_{22} & m_{23} \end{pmatrix} \times \begin{pmatrix} j_{31} & j_{32} & j_{33} \\ m_{31} & m_{32} & m_{33} \end{pmatrix} \begin{pmatrix} j_{11} & j_{21} & j_{31} \\ m_{11} & m_{21} & m_{31} \end{pmatrix} \times \begin{pmatrix} j_{12} & j_{22} & j_{32} \\ m_{12} & m_{22} & m_{32} \end{pmatrix} \begin{pmatrix} j_{13} & j_{23} & j_{33} \\ m_{13} & m_{23} & m_{33} \end{pmatrix}. \tag{108.11}$$

This quantity can also be represented as a sum of products of three $6j$ symbols:

$$\begin{Bmatrix} j_{11} & j_{12} & j_{13} \\ j_{21} & j_{22} & j_{23} \\ j_{31} & j_{32} & j_{33} \end{Bmatrix} = \sum_{j} (-1)^{2j}(2j+1) \begin{Bmatrix} j_{11} & j_{21} & j_{31} \\ j_{32} & j_{33} & j \end{Bmatrix} \times \begin{Bmatrix} j_{12} & j_{22} & j_{32} \\ j_{21} & j & j_{23} \end{Bmatrix} \begin{Bmatrix} j_{13} & j_{23} & j_{33} \\ j & j_{11} & j_{12} \end{Bmatrix}. \tag{108.12}$$

The equivalence of (108.11) and (108.12) may be checked by inserting definition (108.2) into (108.12) and using the orthogonality properties of the $3j$-symbols.

The high symmetry of the $9j$-symbol follows directly from definition (108.11) and the symmetry properties of the $3j$-symbols. Interchanging any two rows or any two columns multiplies the $9j$-symbol by $(-1)^{\Sigma j}$. It is also unchanged by transposition, that is, by interchanging rows and columns.

Scalars of still higher order depend on still more parameters j. Their number must clearly be a multiple of three; these are the $3nj$-symbols. We shall not consider their properties here. We mention only that for every $n > 3$ there is more than one distinct type of $3nj$-symbol, and these types cannot be reduced to one another. Thus there are two distinct types of $12j$-symbols[19].

§ 109. Matrix elements in the coupling of angular momenta

Consider once more a system made up of two parts, which we shall call subsystems 1 and 2. Let $f^{(1)}_{kq}$ be a spherical tensor characterizing the first subsystem. By (107.6), its matrix elements between wave functions of that subsystem are

$$\langle n'_1 j'_1 m'_1 | f^{(1)}_{kq} | n_1 j_1 m_1 \rangle = i^k (-1)^{j_{1\,\max} - m'_1} \begin{pmatrix} j'_1 & k & j_1 \\ -m'_1 & q & m_1 \end{pmatrix} \times \langle n'_1 j'_1 \| f^{(1)}_k \| n_1 j_1 \rangle. \tag{109.1}$$

[18] According to the general contraction rule (108.1), the arguments m in the last three $3j$ symbols should be written with minus signs, and the factor $(-1)^{\Sigma(j-m)}$ should be inserted under the summation sign. Using property (106.6) of the $3j$ symbols, however, and noting that in this case, as is readily seen, the sum $\sum m$ over all nine numbers m is zero, we arrive at definition (108.11).

[19] For a fuller account of the theory of $9j$-symbols and of the properties of $3nj$-symbols, see the book by Edmonds cited on p. 272, and the books: A. P. Yutsis, I. B. Levinson, V. V. Vanagas. *Mathematical Apparatus of the Theory of Angular Momentum*. — Vilnius, 1960; D. A. Varshalovich, A. N. Moskalev, V. K. Khersonskii. *Quantum Theory of Angular Momentum*. — Moscow: Nauka, 1975.

Table 10. Formulas for $6j$ symbols

$$\begin{Bmatrix} a & b & c \\ 0 & c & b \end{Bmatrix} = \frac{(-1)^s}{\sqrt{(2b+1)(2c+1)}}, \qquad s = a+b+c$$

$$\begin{Bmatrix} a & b & c \\ \frac{1}{2} & c-\frac{1}{2} & b+\frac{1}{2} \end{Bmatrix} = (-1)^s \left[\frac{(s-2b)(s-2c+1)}{(2b+1)(2b+2)2c(2c+1)}\right]^{1/2}$$

$$\begin{Bmatrix} a & b & c \\ \frac{1}{2} & c-\frac{1}{2} & b-\frac{1}{2} \end{Bmatrix} = (-1)^s \left[\frac{(s+1)(s-2a)}{2b(2b+1)2c(2c+1)}\right]^{1/2}$$

$$\begin{Bmatrix} a & b & c \\ 1 & c-1 & b-1 \end{Bmatrix} = (-1)^s \left[\frac{s(s+1)(s-2a-1)(s-2a)}{(2b-1)2b(2b+1)(2c-1)2c(2c+1)}\right]^{1/2}$$

$$\begin{Bmatrix} a & b & c \\ 1 & c-1 & b \end{Bmatrix} = (-1)^s \left[\frac{2(s+1)(s-2a)(s-2b)(s-2c+1)}{2b(2b+1)(2b+2)(2c-1)2c(2c+1)}\right]^{1/2}$$

$$\begin{Bmatrix} a & b & c \\ 1 & c-1 & b+1 \end{Bmatrix} = (-1)^s \left[\frac{(s-2b-1)(s-2b)(s-2c+1)(s-2c+2)}{(2b+1)(2b+2)(2b+3)(2c-1)2c(2c+1)}\right]^{1/2}$$

$$\begin{Bmatrix} a & b & c \\ 1 & c & b \end{Bmatrix} = (-1)^{s+1} \frac{2[b(b+1)+c(c+1)-a(a+1)]}{[2b(2b+1)(2b+2)2c(2c+1)(2c+2)]^{1/2}}$$

We now need the matrix elements of these same quantities between wave functions of the complete system. We shall show how to express them through the reduced matrix elements already occurring in (109.1).

A state of the complete system is specified by the quantum numbers $j_1 j_2 J M n_1 n_2$, where J and M give the magnitude and projection of the total angular momentum. Since $f^{(1)}_{kq}$ belongs to subsystem 1, its operator commutes with the angular-momentum operator of subsystem 2. Its matrix is consequently diagonal in j_2, and likewise in the remaining quantum numbers n_2 of that subsystem. For brevity we shall suppress the indices j_2, n_2 and write the required matrix elements as

$$\langle n_1' j_1' J' M' | f^{(1)}_{kq} | n_1 j_1 J M \rangle.$$

Equation (107.6) fixes their dependence on M:

$$\langle n_1' j_1' J' M' | f^{(1)}_{kq} | n_1 j_1 J M \rangle = i^k (-1)^{J_{\max} - M'} \begin{pmatrix} J' & k & J \\ -M' & q & M \end{pmatrix} \times \langle n_1' j_1' J' \| f^{(1)}_k \| n_1 j_1 J \rangle. \tag{109.2}$$

To relate the reduced matrix elements on the right-hand sides of (109.1) and (109.2), use the definition of a matrix element to write

$$\begin{aligned} \langle n_1' j_1' J' M' | f^{(1)}_{kq} | n_1 j_1 J M \rangle &= \int \psi^*_{J'M'} \hat{f}^{(1)}_{kq} \psi_{JM} \, dq = \\ &= \sum_{m_1 m_1'} (-1)^{j_1' - j_2 + M' - M} \sqrt{(2J'+1)(2J+1)} \begin{pmatrix} j_1' & j_2 & J' \\ m_1' & m_2 & -M' \end{pmatrix} \\ &\quad \times \begin{pmatrix} j_1 & j_2 & J \\ m_1 & m_2 & -M \end{pmatrix} \langle n_1' j_1' m_1' | f^{(1)}_{kq} | n_1 j_1 m_1 \rangle. \end{aligned}$$

Insert (109.1) and (109.2) here and compare the resulting relation with (108.4). It then follows that the ratio of the reduced matrix elements in (109.1) and (109.2) is proportional to a particular $6j$ symbol. A detailed comparison of the two relations gives the final result

$$\begin{aligned} \langle n_1' j_1' J' \| f^{(1)}_k \| n_1 j_1 J \rangle &= (-1)^{j_{1\max} + j_2 + J_{\min} + k} \sqrt{(2J+1)(2J'+1)} \\ &\quad \times \begin{Bmatrix} j_1' & J' & j_2 \\ J & j_1 & k \end{Bmatrix} \langle n_1' j_1' \| f^{(1)}_k \| n_1 j_1 \rangle \end{aligned} \tag{109.3}$$

(here $j_{1\max}$ is the greater of j_1, j_1', while $J_{\min}$ is the lesser of J, J'). For a spherical tensor belonging to subsystem 2, the corresponding formula for the reduced matrix elements is

$$\begin{aligned} \langle n_2' j_2' J' \| f^{(2)}_k \| n_2 j_2 J \rangle &= (-1)^{j_1 + j_{2\min} + J_{\max} + k} \sqrt{(2J+1)(2J'+1)} \\ &\quad \times \begin{Bmatrix} j_2' & J' & j_1 \\ J & j_2 & k \end{Bmatrix} \langle n_2' j_2' \| f^{(2)}_k \| n_2 j_2 \rangle. \end{aligned} \tag{109.4}$$

Expressions (109.3) and (109.4) are not completely symmetric. The asymmetry lies in the exponent of -1. It results from the dependence of the wave-function phase on the order in which the angular momenta are coupled. This difference must be borne in mind in calculations of matrix elements. This is necessary when the elements are evaluated simultaneously for both subsystems.

We next derive a useful expression for the matrix elements, between wave functions of the complete system, of the scalar product [as defined in (107.4)] of two rank-k spherical tensors belonging to different subsystems and hence commuting with one another. By (107.10), these matrix elements can be written in terms of the reduced matrix elements of the two tensors, taken between wave functions of the complete system, in the form

$$\langle n_1' n_2' j_1' j_2' JM|(f_k^{(1)} f_k^{(2)})_{00}|n_1 n_2 j_1 j_2 JM\rangle = \\ = \frac{1}{2J+1}\sum_{J''}\langle n_1' j_1' J\|f_k^{(1)}\|n_1 j_1 J''\rangle\langle n_2' j_2' J''\|f_k^{(2)}\|n_2 j_2 J\rangle$$

(we have used here the fact that the matrix of a quantity belonging to either subsystem is diagonal in the quantum numbers of the other). Substitution of (109.3) and (109.4), followed by application of the summation formula (108.8), gives the required expression of the scalar-product matrix element in terms of the reduced matrix elements of the individual tensors between wave functions of the respective subsystems:

$$\langle n_1' n_2' j_1' j_2' JM|(f_k^{(1)} f_k^{(2)})_{00}|n_1 n_2 j_1 j_2 JM\rangle = \\ = (-1)^{j_{1\,\min}+j_{2\,\max}+J}\begin{Bmatrix} J & j_2' & j_1' \\ k & j_1 & j_2 \end{Bmatrix} \\ \times \langle n_1' j_1'\|f_k^{(1)}\|n_1 j_1\rangle\langle n_2' j_2'\|f_k^{(2)}\|n_2 j_2\rangle. \tag{109.5}$$

§ 110. Matrix elements in axially symmetric systems

The integral of a product of three D-functions is the starting point for calculating matrix elements of quantities that characterize systems of the symmetric-top type.

To derive the required expression, consider again expansion (106.11),

$$\psi_{j_1 m_1}\psi_{j_2 m_2} = \sum_j \langle jm|m_1 m_2\rangle \psi_{jm}, \qquad m = m_1 + m_2,$$

and subject both sides to a finite rotation of the coordinate system. Each ψ-function transforms by (58.7), and hence

$$\sum_{m_1' m_2'} D^{(j_1)}_{m_1' m_1} D^{(j_2)}_{m_2' m_2} \psi_{j_1 m_1'}\psi_{j_2 m_2'} = \sum_j \sum_{m'} \langle jm|m_1 m_2\rangle D^{(j)}_{m' m}\psi_{jm'}.$$

We now expand the functions $\psi_{jm'}$ on the right by means of (106.9). Comparing the coefficients of corresponding products $\psi_{j_1 m_1}$, $\psi_{j_2 m_2}$ then gives

$$D^{(j_1)}_{m_1' m_1}(\omega) D^{(j_2)}_{m_2' m_2}(\omega) = \sum_j \langle m_1' m_2'|jm'\rangle D^{(j)}_{m' m}(\omega)\langle m_1 m_2|jm\rangle \tag{110.1}$$

with $m = m_1 + m_2$ and $m' = m_1' + m_2'$; here ω collectively denotes the three Euler angles

α, β, and γ. In terms of $3j$-symbols, the same result reads

$$D^{(j_1)}_{m_1' m_1}(\omega) D^{(j_2)}_{m_2' m_2}(\omega) = \sum_j (2j+1) \begin{pmatrix} j_1 & j_2 & j \\ m_1' & m_2' & -m' \end{pmatrix} \times \\ \times \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix} D^{(j)*}_{-m',-m}(\omega), \tag{110.2}$$

where property (58.19) of the D-functions has also been used.

Multiply the two sides of (110.2) by $D^{(j)}_{-m',-m}(\omega)$ and integrate over $d\omega$, applying the orthogonality relation (58.20). The outcome is

$$\int D^{(j_1)}_{m_1' m_1}(\omega) D^{(j_2)}_{m_2' m_2}(\omega) D^{(j_3)}_{m_3' m_3}(\omega) \frac{d\omega}{8\pi^2} = \\ = \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1' & m_2' & m_3' \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix}. \tag{110.3}$$

An evident relabelling of the indices has been made here to display the symmetry more fully. Equation (110.3) is the desired formula[20].

Let $f_{kq'}$ be a spherical tensor of rank k describing the top in the body-fixed coordinate axes $x'y'z' \equiv \xi\eta\zeta$, with the ζ-axis directed along the top axis. An electric or magnetic multipole-moment tensor is one possible example. Denote by f_{kq} the components of this same tensor referred to the fixed coordinate axes xyz. The matrix of finite rotations relates the two sets of components according to

$$f_{kq} = \sum_{q'} D^{(k)}_{q'q}(\omega) f_{kq'}. \tag{110.4}$$

The wave functions for rotation of the system as a whole differ from D-functions only through their normalization:

$$\psi_{jm\mu} = i^j \sqrt{\frac{2j+1}{8\pi^2}} D^{(j)}_{\mu m}(\omega). \tag{110.5}$$

Here j is the system's total angular momentum, m is its projection on the fixed z-axis, and μ is its projection on the system axis. The phase factor is chosen so that, when j is integral and $\mu = 0$, (110.5) becomes an eigenfunction of free angular momentum (compare (103.8)). Evaluation, between these functions, of the matrix element of (110.4), with (110.3) and the representation (58.19) for the complex-conjugate D-function, yields

$$\langle j'\mu'm'|f_{kq}|j\mu m\rangle = i^{j-j'}(-1)^{\mu'-m'}\sqrt{(2j+1)(2j'+1)} \times \\ \times \begin{pmatrix} j' & k & j \\ -\mu' & q' & \mu \end{pmatrix} \begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix} \langle \mu'|f_{kq'}|\mu\rangle, \tag{110.6}$$

where $q' = \mu' - \mu$ and $q = m' - m$.

[20] For integral values $j_1 = l_1$, $j_2 = l_2$, $j_3 = l_3$ and for $m_1' = m_2' = m_3' = 0$, the functions $D^{(l)}_{0m}$ reduce, by (58.25), to spherical functions. Formula (110.3) then supplies the expression for the integral of three spherical functions (107.14).

Formula (110.6) answers the question posed. It fixes the dependence of the matrix elements on j, j' and on their projections m, m'. Their dependence on the quantum numbers μ, μ' necessarily remains unspecified: the possible values of these numbers are determined by the system's "internal" states, between which the "internal" matrix element $\langle \mu' | f_{kq'} | \mu \rangle$ is taken.

The m, m' dependence of the matrix elements in (110.6) is the one common to every system of specified total angular momentum. Separating it by introducing reduced matrix elements as in (107.6), we obtain

$$\begin{aligned}\langle j'\mu' \| f_k \| j\mu \rangle = i^{j-j'-k}(-1)^{j_{\max}-\mu'}\sqrt{(2j+1)(2j'+1)}\times \\ \times \begin{pmatrix} j' & k & j \\ -\mu' & q' & \mu \end{pmatrix} \langle \mu' | f_{kq'} | \mu \rangle.\end{aligned} \tag{110.7}$$

For fixed m, sum the squared modulus of (110.6) over every value of the final quantum number m' (and over $q = m' - m$). The result is independent of m and, by the general rule (107.11), is

$$\begin{aligned}\sum_{qm'} |\langle j'\mu'm' | f_{kq} | j\mu m \rangle|^2 = \frac{1}{2j+1} |\langle j'\mu' \| f_k \| j\mu \rangle|^2 = \\ = (2j'+1)\begin{pmatrix} j' & k & j \\ -\mu' & q' & \mu \end{pmatrix}^2 |\langle \mu' | f_{kq'} | \mu \rangle|^2.\end{aligned} \tag{110.8}$$

As they must be, the Hermiticity relations (107.9) for the reduced matrix elements (110.7) in the xyz coordinates agree with relations (107.8),

$$\langle \mu | f_{kq'} | \mu' \rangle = (-1)^{k-q'} \langle \mu' | f_{k,-q'} | \mu \rangle^*,$$

for the matrix elements in the $\xi\eta\zeta$ coordinates.

The rotation of an axially symmetric system such as a diatomic molecule (or an axial nucleus) is specified by only two angles, $\alpha \equiv \varphi$ and $\beta \equiv \theta$, which determine the direction of the system axis. Its rotational wave function differs from (110.5) by the omission of the factor $e^{i\mu\gamma}/\sqrt{2\pi}$ (compare the note on p. 389). This difference nevertheless leaves the matrix elements unchanged. Indeed, since the entire γ dependence of $D^{(j)}_{m'm}(\alpha, \beta, \gamma)$ is contained in the factor $e^{im'\gamma}$, relation (110.3) may be written

in the form

$$\begin{aligned}\delta_{m',0} \int D^{(j_1)}_{m'_1 m_1}(\alpha,\beta,0) D^{(j_2)}_{m'_2 m_2}(\alpha,\beta,0) D^{(j_3)}_{m'_3 m_3}(\alpha,\beta,0) \frac{\sin\alpha \, d\alpha \, d\beta}{4\pi} = \\ = \begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} \begin{pmatrix} j_1 & j_2 & j_3 \\ m'_1 & m'_2 & m'_3 \end{pmatrix},\end{aligned}$$

where $m' = m'_1 + m'_2 + m'_3$, and the value of the integral is unaltered. The selection rule for the projection of angular momentum on the system axis likewise retains its former form, $\mu' - \mu = q'$. It now follows from the orthogonality of the electronic wave functions, as a consequence of the molecule's symmetry about the ζ-axis. In (110.6) and (110.7), $\langle \mu' | f_{kq'} | \mu \rangle$ must in this case be understood as a matrix element between electronic states with the nuclei held fixed.

CHAPTER XV

Motion in a Magnetic Field

§ 111. The Schrödinger equation in a magnetic field

A particle with spin also possesses an "intrinsic" magnetic moment $\boldsymbol{\mu}$. Its quantum-mechanical operator is proportional to the spin operator $\hat{\mathbf{s}}$ and may therefore be written

$$\hat{\boldsymbol{\mu}} = \frac{\mu}{s}\hat{\mathbf{s}}, \tag{111.1}$$

where s is the magnitude of the particle's spin and μ is a constant characteristic of the particle. The eigenvalues of the projection of the magnetic moment are $\mu_z = \mu\sigma/s$. Thus the coefficient μ, which is usually referred to simply as the magnitude of the magnetic moment, is the largest possible value of μ_z and is attained for the spin projection $\sigma = s$.

The ratio $\mu/(\hbar s)$ gives the ratio of the particle's intrinsic magnetic moment to its intrinsic mechanical angular momentum when both are directed along the z axis. For ordinary orbital angular momentum this ratio is, as is known, $e/(2mc)$ (see II, § 44). The proportionality coefficient between the intrinsic magnetic moment and the spin of a particle is different. For the electron it is $-|e|/mc$, twice the ordinary value (this value follows theoretically from the relativistic Dirac wave equation; see IV, § 33). The intrinsic magnetic moment of the electron, whose spin is $1/2$, is consequently $-\mu_B$, where

$$\mu_B = \frac{|e|\hbar}{2mc} = 0.927 \cdot 10^{-20}\,\frac{\text{erg}}{\text{G}}. \tag{111.2}$$

This quantity is called the *Bohr magneton*.

The magnetic moments of heavy particles are conventionally measured in *nuclear magnetons*, defined as $e\hbar/(2m_p c)$, where m_p is the proton mass. Experiment gives the intrinsic magnetic moment of the proton as 2.79 nuclear magnetons, directed along the spin. The neutron's magnetic moment is directed opposite to its spin and has magnitude 1.91 nuclear magnetons.

Notice that the quantities $\boldsymbol{\mu}$ and $\mathbf{s}$ on the two sides of (111.1) have, as they must, the same vector character: both are axial vectors. An analogous equality for the electric dipole moment $\mathbf{d}$, namely $\mathbf{d} = \text{const} \cdot \mathbf{s}$, would be incompatible with symmetry under inversion of the coordinates, since inversion would change the relative sign of the two sides[1] .

In nonrelativistic quantum mechanics a magnetic field can be treated only as an external field. The magnetic interaction between particles is a relativistic effect, and a consistent relativistic theory is required to take it into account.

[1] It should be noted that this relation (and hence the existence of an electric moment in an elementary particle) would also be incompatible with time-reversal symmetry: reversing the sign of time leaves $\mathbf{d}$ unchanged but reverses the spin (as is clear, for example, from the definitions of these quantities for orbital motion: only coordinates enter the definition of $\mathbf{d}$, whereas the definition of the moment also contains the particle's velocity).

In classical theory the Hamilton function of a charged particle in an electromagnetic field is

$$H = \frac{1}{2m}\left(\mathbf{p} - \frac{e}{c}\mathbf{A}\right)^2 + e\varphi,$$

where φ and $\mathbf{A}$ are respectively the scalar and vector potentials of the field, while $\mathbf{p}$ is the generalized momentum of the particle (see II, § 16). If the particle has no spin, the transition to quantum mechanics is made in the usual way: the generalized momentum is replaced by the operator $\hat{\mathbf{p}} = -i\hbar\boldsymbol{\nabla}$, giving the Hamiltonian[2]

$$\hat{H} = \frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 + e\varphi. \tag{111.3}$$

If the particle has spin, however, this procedure is insufficient. Its intrinsic magnetic moment interacts directly with the magnetic field. This interaction is entirely absent from the classical Hamilton function, since spin itself, being a purely quantum effect, disappears in the classical limit. The correct Hamiltonian is obtained by adding to (111.3) the term $-\hat{\boldsymbol{\mu}}\mathbf{H}$, corresponding to the energy of a magnetic moment $\boldsymbol{\mu}$ in a field $\mathbf{H}$. Thus the Hamiltonian of a particle with spin is[3]

$$\hat{H} = \frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 - \hat{\boldsymbol{\mu}}\mathbf{H} + e\varphi. \tag{111.4}$$

When expanding the square $(\hat{\mathbf{p}} - (e/c)\mathbf{A})^2$, one must bear in mind that the operator $\hat{\mathbf{p}}$ does not in general commute with the vector $\mathbf{A}$, which is a function of the coordinates. We must therefore write

$$\hat{H} = \frac{1}{2m}\hat{\mathbf{p}}^2 - \frac{e}{2mc}(\hat{\mathbf{p}}\mathbf{A} + \mathbf{A}\hat{\mathbf{p}}) + \frac{e^2}{2mc^2}\mathbf{A}^2 - \frac{\mu}{s}\hat{\mathbf{s}}\mathbf{H} + e\varphi. \tag{111.5}$$

The commutation rule (16.4) for the momentum operator and an arbitrary function of the coordinates gives

$$\hat{\mathbf{p}}\mathbf{A} - \mathbf{A}\hat{\mathbf{p}} = -i\hbar\nabla\cdot\mathbf{A}. \tag{111.6}$$

Thus $\hat{\mathbf{p}}$ and $\mathbf{A}$ commute if $\nabla\cdot\mathbf{A} = 0$. This condition holds, in particular, for a uniform field when its vector potential is chosen as

$$\mathbf{A} = (1/2)(\mathbf{H}\times\mathbf{r}). \tag{111.7}$$

The equation $i\hbar\,\partial\Psi/\partial t = \hat{H}\Psi$, with Hamiltonian (111.4), is the generalization of the Schrödinger equation to the presence of a magnetic field. The wave functions acted on by the Hamiltonian in this equation are symmetric spinors of rank $2s$.

Wave functions of a particle in an electromagnetic field have an ambiguity associated with the ambiguity in the field potentials. As is known (see II, § 18), the latter are determined only up to a *gauge transformation*

$$\mathbf{A} \to \mathbf{A} + \boldsymbol{\nabla} f, \qquad \varphi \to \varphi - \frac{1}{c}\frac{\partial f}{\partial t}, \tag{111.8}$$

[2]Here we use the same letter $\mathbf{p}$ for the generalized momentum as for the ordinary momentum, instead of the $\mathbf{P}$ used in II, § 16, in order to emphasize that the same operator corresponds to it.

[3]Using the same letter for the magnetic field and the Hamiltonian cannot cause confusion: the symbol for the Hamiltonian carries a hat.

where f is an arbitrary function of the coordinates and time. Such a transformation does not affect the field strengths. It must therefore leave the solutions of the wave equation essentially unchanged as well; in particular, $|\Psi|^2$ must remain invariant. It is indeed readily verified that the original equation is recovered if, together with the replacement (111.8) in the Hamiltonian, the wave function is replaced according to

$$\Psi \to \Psi \exp\left(\frac{ie}{\hbar c} f\right). \tag{111.9}$$

This ambiguity of the wave function has no effect on any physically meaningful quantity whose definition does not explicitly contain the potentials.

In classical mechanics a particle's generalized momentum is related to its velocity by $m\mathbf{v} = \mathbf{p} - e\mathbf{A}/c$. To find the operator $\hat{\mathbf{v}}$ in quantum mechanics, one must form the commutator of the vector $\mathbf{r}$ with the Hamiltonian. A straightforward calculation gives

$$m\hat{\mathbf{v}} = \hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}, \tag{111.10}$$

which is exactly analogous to the classical relation. The operators of the velocity components obey the commutation rules

$$\left\{\hat{v}_x, \hat{v}_y\right\} = i\frac{e\hbar}{m^2 c} H_z, \qquad \left\{\hat{v}_y, \hat{v}_z\right\} = i\frac{e\hbar}{m^2 c} H_x, \qquad \left\{\hat{v}_z, \hat{v}_x\right\} = i\frac{e\hbar}{m^2 c} H_y, \tag{111.11}$$

These relations are readily verified by direct calculation. Thus, in a magnetic field the operators for the three velocity components of a (charged) particle do not commute. Consequently, the particle cannot simultaneously have definite velocity values along all three directions.

For motion in a magnetic field, time-reversal symmetry holds only if the sign of the field $\mathbf{H}$, and of the vector potential $\mathbf{A}$, is reversed as well. This means (see §§ 18 and 60) that the Schrödinger equation $\hat{H}\psi = E\psi$ must retain its form upon passage to the complex-conjugate quantities accompanied by reversal of the sign of $\mathbf{H}$. This is immediately evident for every term in the Hamiltonian (111.4) except $-\hat{\mathbf{s}}\mathbf{H}$. Under the specified transformation the term $-\hat{\mathbf{s}}\mathbf{H}\psi$ in the Schrödinger equation becomes $\hat{\mathbf{s}}^*\mathbf{H}\psi^*$ and appears at first sight to violate the required invariance, since the operator $\hat{\mathbf{s}}^*$ is not equal to $-\hat{\mathbf{s}}$.

One must take into account, however, that the wave function is actually a contravariant spinor $\psi^{\lambda\mu\cdots}$, which on complex conjugation becomes the covariant spinor $\psi^{\lambda\mu\cdots *}$ (see § 60). The spinor $\psi^*_{\lambda\mu\ldots}$, on the other hand, is contravariant. Using definitions (57.4) and (57.5) to find the components of $(\hat{\mathbf{s}}\mathbf{H}\psi)^*$ and expressing them in terms of $\psi^*_{\lambda\mu\ldots}$, one verifies that time reversal leads to a Schrödinger equation for the components $\psi^*_{\lambda\mu\ldots}$ of the same form as the original equation for the components $\psi^{\lambda\mu\cdots}$.

§ 112. Motion in a uniform magnetic field

For a particle in a constant uniform magnetic field, we shall determine the energy levels (L. D. Landau, 1930).

For a uniform field, it is convenient here to choose the vector potential not in the form (111.7), but as follows:

$$A_x = -Hy, \qquad A_y = A_z = 0. \tag{112.1}$$

(the z axis is chosen along the field). The Hamiltonian then takes the form

$$\hat{H} = \frac{1}{2m}\left(\hat{p}_x + \frac{eH}{c}y\right)^2 + \frac{\hat{p}_y^2}{2m} + \frac{\hat{p}_z^2}{2m} - \frac{\mu}{s}\hat{s}_z H. \tag{112.2}$$

First observe that the operator $\hat{s}_z$ commutes with the Hamiltonian (since the latter contains no operators for the other spin components). Thus the z projection of the spin is conserved, and $\hat{s}_z$ may therefore be replaced by its eigenvalue $s_z = \sigma$. The dependence of the wave function on spin is then immaterial, and ψ in the Schrödinger equation may be understood as an ordinary function of the coordinates. For this function we have

$$\frac{1}{2m}\left[\left(\hat{p}_x + \frac{eH}{c}y\right)^2 + \hat{p}_y^2 + \hat{p}_z^2\right]\psi - \frac{\mu}{s}\sigma H\psi = E\psi. \tag{112.3}$$

The coordinates x and z do not occur explicitly in the Hamiltonian of this equation. Hence the operators $\hat{p}_x$ and $\hat{p}_z$ (differentiation with respect to x and z) also commute with it; in other words, the x- and z-components of the generalized momentum are conserved. We therefore seek ψ in the form

$$\psi = \exp\left[\frac{i}{\hbar}(p_x x + p_z z)\right]\chi(y). \tag{112.4}$$

The eigenvalues p_x and p_z range over all values from $-\infty$ to ∞. Since $A_z = 0$, the z component of the generalized momentum is the same as the corresponding component of the ordinary momentum mv_z. The particle's velocity along the field can thus have any value; one may say that motion parallel to the field is "not quantized."

Substitution of (112.4) into (112.3) gives the following equation for $\chi(y)$:

$$\chi'' + \frac{2m}{\hbar^2}\left[\left(E + \frac{\mu\sigma}{s}H - \frac{p_z^2}{2m}\right) - \frac{m}{2}\omega_H^2(y - y_0)^2\right]\chi = 0, \tag{112.5}$$

where $y_0 = -cp_x/eH$ and

$$\omega_H = \frac{|e|H}{mc} \tag{112.6}$$

have been introduced.

Equation (112.5) has the same form as the Schrödinger equation (23.6) for a linear oscillator of frequency ω_H. It follows at once that the quantity in parentheses in (112.5), which plays the role of the oscillator energy, can take the values $(n + 1/2)\hbar\omega_H$, where $n = 0, 1, 2, \ldots$

We thus obtain the following energy levels for a particle in a uniform magnetic field:

$$E = \left(n + \frac{1}{2}\right)\hbar\omega_H + \frac{p_z^2}{2m} - \frac{\mu\sigma}{s}H. \tag{112.7}$$

The first term in this expression supplies the discrete energies associated with motion in the plane normal to the field; these are called the *Landau levels*. For an electron, $\mu/s = -|e|\hbar/mc$, and (112.7) becomes

$$E = \left(n + \frac{1}{2} + \sigma\right)\hbar\omega_H + \frac{p_z^2}{2m}. \tag{112.8}$$

The eigenfunctions $\chi_n(y)$ belonging to the energy levels (112.7) follow from (23.12) after the corresponding change of notation ($a_H = \sqrt{\hbar/m\omega_H}$):

$$\chi_n(y) = \frac{1}{\pi^{1/4}a_H^{1/2}\sqrt{2^n n!}}\exp\left(-\frac{(y-y_0)^2}{2a_H^2}\right)H_n\left(\frac{y-y_0}{a_H}\right). \tag{112.9}$$

In classical mechanics, a particle moves in a circle with a fixed center in the plane perpendicular to the field $\mathbf{H}$ (the xy plane). The quantity y_0, conserved in the quantum problem, corresponds to the classical y coordinate of the circle's center. The quantity $x_0 = (cp_y/eH) + x$ is conserved as well (its operator is readily verified to commute with the Hamiltonian (112.2)). It corresponds to the classical x coordinate of the circle's center[4].

The operators $\hat{x}_0$ and $\hat{y}_0$, however, do not commute with one another, so the coordinates x_0 and y_0 cannot simultaneously have definite values.

Since (112.7) does not involve p_x, whose values form a continuum, every energy level has a continuous degeneracy. This degeneracy becomes finite, however, when the motion in the xy plane is confined to a large but finite area $S = L_xL_y$. The number of distinct (now discrete) values of p_x in an interval Δp_x is $(L_x/2\pi\hbar)\Delta p_x$. All p_x for which the orbit centre lies within S are allowed (the orbit radius is neglected in comparison with the large length L_y). The conditions $0 < y_0 < L_y$ give $\Delta p_x = eHL_y/c$. Consequently, for fixed n and p_z, the number of states is $eHS/2\pi\hbar c$. If the region of motion is also bounded along the z axis, with length L_z, then an interval Δp_z contains $(L_z/2\pi\hbar)\Delta p_z$ possible values of p_z, and the number of states in that interval is

$$\frac{eHS}{2\pi\hbar c}\frac{L_z}{2\pi\hbar}\Delta p_z = \frac{eHV}{4\pi^2\hbar^2 c}\Delta p_z. \tag{112.10}$$

For an electron there is an additional degeneracy: the energy levels (112.8) are the same for states with quantum numbers n, $\sigma = 1/2$ and $n+1$, $\sigma = -1/2$.

Problems

[4] Indeed, for classical motion on a circle of radius cmv_t/eH (v_t is the projection of the velocity onto the xy plane; see Vol. II, § 21), we have

$$y_0 = -cp_x/eH = -(cm/eH)v_x + y.$$

It is evident from this expression that y is the coordinate of the circle's center. The other coordinate is

$$x_0 = (cm/eH)v_y + x = cp_y/eH + x.$$

Problem

1. Construct the electron wave functions for states in a uniform magnetic field with definite values of momentum and angular momentum along the field direction.

Solution. In cylindrical coordinates ρ, φ, z, with the z axis along the field, the vector potential in the gauge (111.7) has components $A_\varphi = H\rho/2$, $A_z = A_\rho = 0$, and the Schrödinger equation is[5]

$$-\frac{\hbar^2}{2M}\left[\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial\psi}{\partial\rho}\right)+\frac{\partial^2\psi}{\partial z^2}+\frac{1}{\rho^2}\frac{\partial^2\psi}{\partial\varphi^2}\right]-\frac{i\hbar\omega_H}{2}\frac{\partial\psi}{\partial\varphi}+\frac{M\omega_H^2}{8}\rho^2\psi = E\psi. \qquad (1)$$

We seek a solution in the form

$$\psi = \frac{e^{im\varphi}}{\sqrt{2\pi}}e^{ip_z z/\hbar}R(\rho)$$

and obtain for the radial function the equation

$$\frac{\hbar^2}{2M}\left(R''+\frac{1}{\rho}R'-\frac{m^2}{\rho^2}R\right)+\left[E-\frac{p_z^2}{2M}-\frac{M\omega_H^2}{8}\rho^2-\frac{\hbar\omega_H m}{2}\right]R = 0.$$

Introducing the new independent variable $\xi = (M\omega_H/2\hbar)\rho^2$, we rewrite this equation as

$$\xi R'' + R' + \left(-\frac{\xi}{4}+\beta-\frac{m^2}{4\xi}\right)R = 0, \qquad \beta = \frac{1}{\hbar\omega_H}\left(E-\frac{p_z^2}{2M}\right)-\frac{m}{2}.$$

As $\xi \to \infty$, the required function behaves as $e^{-\xi/2}$, while as $\xi \to 0$ it behaves as $\xi^{|m|/2}$. We therefore seek a solution of the form

$$R(\xi) = e^{-\xi/2}\xi^{|m|/2}w(\xi)$$

and obtain for $w(\xi)$ the confluent hypergeometric function equation:

$$w = F\left\{-\left(\beta-\frac{|m|+1}{2}\right), |m|+1, \xi\right\}.$$

For the wave function to be finite everywhere, $\beta - (|m|+1)/2$ must be a nonnegative integer n_ρ. The energy levels are then

$$E = \hbar\omega_H\left(n_\rho+\frac{|m|+m+1}{2}\right)+\frac{p_z^2}{2M},$$

which is equivalent to (112.7). The radial wave functions belonging to these levels are

$$R_{n_\rho m}(\rho) = \frac{1}{a_H^{1+|m|}}\left[\frac{(|m|+n_\rho)!}{2^{|m|}n_\rho!|m|!^2}\right]^{1/2}\exp\left(-\frac{\rho^2}{4a_H^2}\right)\rho^{|m|}F\left(-n_\rho, |m|+1, \frac{\rho^2}{2a_H^2}\right), \qquad (2)$$

where $a_H = \sqrt{\hbar/M\omega_H}$; the functions are normalized by $\int_0^\infty R^2\rho\, d\rho = 1$. Here the hypergeometric function is a generalized Laguerre polynomial.

[5] We write the electron charge as $e = -|e|$ and denote its mass here by M, to distinguish it from the angular momentum m. The term containing the particle's spin, which is inessential for this problem, is omitted.

PROBLEM

2. Determine the lowest energy level associated with a bound state of an electron in a shallow potential well $U(r)$ ($|U| \ll \hbar^2/ma^2$, where a is the range of the forces in the well) when a uniform magnetic field is also applied (Yu. A. Bychkov, 1960).

SOLUTION. The stated condition on the field $U(r)$ ensures that, in the absence of a magnetic field, perturbation theory is applicable to it; the well then has no bound states (see § 45). When a magnetic field is present as well, $U(r)$ can be treated as a perturbation only for motion in the plane transverse to **H**, whose (discrete) type of energy spectrum is not changed by adding U. The nature of the motion along **H**, however, changes: as will be seen, unbounded motion becomes bound, and the continuous spectrum becomes discrete. The well's field therefore cannot be treated by perturbation theory for this motion.

Accordingly, on separating variables in the Schrödinger equation (equation (1) of the preceding problem, supplemented by the term $U\psi$ on the left), we retain the radial functions $R(\rho)$ in the previous form (2); the lowest level corresponds to the quantum numbers $n_\rho = m = 0$. Substituting $\psi = R_{00}(\rho)\chi(z)$ into the Schrödinger equation, then multiplying it by $R_{00}(\rho)$ and integrating over $\rho\, d\rho$, we obtain for $\chi(z)$

$$-\frac{\hbar^2}{2m}\chi'' + \overline{U}(z)\chi = \varepsilon\chi, \tag{3}$$

where $\varepsilon = E - \hbar\omega_H/2$ and

$$\overline{U}(z) = \int_0^\infty U\left(\sqrt{z^2+\rho^2}\right) R_{00}^2(\rho)\rho\, d\rho$$

(m once again denotes the particle's mass). In form, this equation is the Schrödinger equation for one-dimensional motion in the potential well $\overline{U}(z)$, with ε as the energy of that motion. We may therefore use the result of Problem 1 in § 45, according to which the discrete energy level is

$$\varepsilon = -\frac{m}{2\hbar^2}\left[\int_{-\infty}^{\infty} \overline{U}(z)\, dz\right]^2 = -\frac{m}{2\hbar^2}\left[\int_{-\infty}^{\infty}\int_0^\infty U\left(\sqrt{z^2+\rho^2}\right) R_{00}^2(\rho)\rho\, d\rho\, dz\right]^2. \tag{4}$$

The wave function $R_{00}(\rho)$ decays over distances $\rho \sim a_H$. If the magnetic field is so weak that $a_H \gg a$, the integral over $d\rho$ is determined by the region $\rho \lesssim a$, in which one may set $R_{00}(\rho) \approx R_{00}(0) = 1/a_H$. Then

$$\varepsilon = -\frac{me^2H^2}{8\pi^2\hbar^4c^2}\left(\int U(r)\, dV\right)^2 \tag{5}$$

($dV = 2\pi\rho\, d\rho\, dz \to 4\pi r^2\, dr$). In the opposite case of a strong magnetic field, where $a_H \ll a$, the integral in (4) is determined by the region $\rho \lesssim a_H$, in which one may put $U(\sqrt{\rho^2+z^2}) \approx U(z)$. The integration over $d\rho$ then reduces to the normalization integral for R_{00} and equals 1, so that

$$\varepsilon = -\frac{2m}{\hbar^2}\left(\int_0^\infty U(z)\, dz\right)^2. \tag{6}$$

In both cases, an estimate of the integrals shows that $\varepsilon \ll \hbar\omega_H$.

Problem

3. Determine the energy levels of a hydrogen atom in a magnetic field so strong that $a_H \ll a_B$, where a_B is the Bohr radius (R. J. Elliott, R. Loudon, 1960).

Solution. Under the stated condition, $\hbar\omega_H \gg me^4/\hbar^2$, the effect of the nuclear Coulomb field on the electron's motion in the plane transverse to **H** may be treated as a small perturbation. We therefore return to the situation considered in Problem 2 and may use equation (3), with

$$\overline{U}(z) = -e^2 \int_0^\infty \frac{R_{00}^2(\rho)\rho}{\sqrt{\rho^2 + z^2}}\, d\rho. \tag{7}$$

By using the radial function R_{00} in this expression, we restrict the discussion below to the energy levels of the longitudinal motion that correspond to the lowest Landau level ($\hbar\omega_H/2$) of the transverse motion.

For the ground level, the conditions invoked in the solution of problem 1 in § 45 are satisfied: since $\chi_0(z)$ has no nodes, it does not vanish at $z = 0$. Formula (6), which follows from that solution, is therefore applicable. The ground-state wave function extends over distances $|z| \lesssim a_B$ and varies slowly across this region. The logarithmically divergent integral is consequently cut off at its upper end at distances $|z| \sim a_B$, and at its lower end at distances $|z| \sim a_H$ (there $|z| \sim \rho$, and replacing $\sqrt{\rho^2 + z^2}$ by $|z|$ in (7) is not permissible). The result is

$$\varepsilon_0 = -\frac{2me^4}{\hbar^2} \ln^2 \frac{a_B}{a_H} = -\frac{me^4}{2\hbar^2} \ln^2 \frac{\hbar^3 H}{m^2 c|e|^3}. \tag{8}$$

This formula is said to possess logarithmic accuracy: both the ratio a_B/a_H itself and its logarithm are assumed large, while the numerical factor in the logarithm's argument remains undetermined.

The excited states of the discrete spectrum are obtained by solving the Schrödinger equation (3) with $\overline{U}(z) \approx -e^2/z$ (which follows from (7) when $z \sim a_B \gg \rho$). Under the substitution $\chi = z\varphi(z)$, however, this equation takes the form

$$-\frac{\hbar^2}{2m}\frac{1}{z^2}\frac{d}{dz}\left(z^2 \frac{d\varphi}{dz}\right) - \frac{e^2}{z}\varphi = \varepsilon\varphi, \tag{9}$$

which has the same form as the equation for the radial wave functions of s states in the three-dimensional Coulomb problem. The required levels are therefore given by (36.10):

$$\varepsilon_n = -\frac{me^4}{2\hbar^2 n^2}, \tag{10}$$

where $n = 1, 2, 3, \ldots$ This expression likewise has only logarithmic accuracy: the next correction would be smaller than the leading term only by the factor $1/\ln(a_B/a_H)$.

Equation (9) determines the wave function only for $z > 0$; it can be continued into $z < 0$ either as $\chi(-z) = \chi(z)$ or as $\chi(-z) = -\chi(z)$. Accordingly, within the approximation considered, the levels (10) are twofold degenerate. This degeneracy is removed, however, in higher orders in a_H/a_B.

§ 113. The atom in a magnetic field

Consider an atom placed in a uniform magnetic field $\mathbf{H}$. Its Hamiltonian is

$$\hat{H} = \frac{1}{2m}\sum_a \left[\hat{\mathbf{p}}_a + \frac{|e|}{c}\mathbf{A}(\mathbf{r}_a)\right]^2 + U + \frac{|e|\hbar}{mc}\mathbf{H}\hat{\mathbf{S}}, \qquad (113.1)$$

where the sum extends over all electrons (the electron charge is written as $-|e|$); U is the energy of interaction of the electrons with the nucleus and with one another; $\hat{\mathbf{S}} = \sum_a \hat{\mathbf{s}}_a$ is the operator of the atom's total (electronic) spin.

If the vector potential of the field is chosen in the form (111.7), then, as already noted, the operator $\hat{\mathbf{p}}$ commutes with $\mathbf{A}$. Using this fact when expanding the square in (113.1), and denoting the Hamiltonian of the atom in the absence of a field by $\hat{H}_0$, we find

$$\hat{H} = \hat{H}_0 + \frac{|e|}{mc}\sum_a \mathbf{A}_a\hat{\mathbf{p}}_a + \frac{e^2}{2mc^2}\sum_a \mathbf{A}_a^2 + \frac{|e|\hbar}{mc}\mathbf{H}\hat{\mathbf{S}}.$$

On substituting $\mathbf{A}$ from (111.7), this becomes

$$\hat{H} = \hat{H}_0 + \frac{|e|}{2mc}\mathbf{H}\sum_a(\mathbf{r}_a \times \hat{\mathbf{p}}_a) + \frac{e^2}{8mc^2}\sum_a(\mathbf{H}\times\mathbf{r}_a)^2 + \frac{|e|\hbar}{mc}\mathbf{H}\hat{\mathbf{S}}.$$

But the vector product $\mathbf{r}_a \times \hat{\mathbf{p}}_a$ is the operator of the electron's orbital angular momentum, and summation over all the electrons gives the operator $\hbar\hat{\mathbf{L}}$ of the atom's total orbital angular momentum. Hence

$$\hat{H} = \hat{H}_0 + \mu_B(\hat{\mathbf{L}} + 2\hat{\mathbf{S}})\mathbf{H} + \frac{e^2}{8mc^2}\sum_a(\mathbf{H}\times\mathbf{r}_a)^2 \qquad (113.2)$$

(μ_B is the Bohr magneton). The operator

$$\hat{\boldsymbol{\mu}}_{\text{at}} = -\mu_B(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) \qquad (113.3)$$

may be regarded as the operator of the atom's "intrinsic" magnetic moment, which it possesses in the absence of a field.

An external magnetic field splits the atomic levels by removing their degeneracy with respect to the directions of the total angular momentum (the *Zeeman effect*). Let us determine the splitting energy for atomic levels characterized by definite values of the quantum numbers J, L, and S (that is, assuming LS coupling for the levels; see § 72).

We shall suppose that the magnetic field is so weak that $\mu_B H$ is small compared with the separations between the atom's energy levels, including the fine-structure intervals. The second and third terms in (113.2) may then be treated as a perturbation, the unperturbed levels being the individual components of the multiplets. In first approximation, the third term, which is quadratic in the field, may be neglected in comparison with the linear second term.

In this approximation the splitting energy ΔE is determined by the mean values of the perturbation in the (unperturbed) states differing in the value of the projection of the total angular momentum along the field. Taking this direction as the z axis, we have

$$\Delta E = \mu_B H(\overline{L}_z + 2\overline{S}_z) = \mu_B H(\overline{J}_z + \overline{S}_z). \qquad (113.4)$$

The mean value $\overline{J}_z$ is simply the specified eigenvalue $J_z = M_J$. The mean value $\overline{S}_z$ can be found as follows by means of averaging in stages (cf. § 72).

First average the operator $\hat{\mathbf{S}}$ over a state of the atom with specified S, L, and J, but not M_J. The operator $\overline{\mathbf{S}}$ obtained by this averaging can be "directed" only along $\hat{\mathbf{J}}$, the sole conserved "vector" characterizing the free atom. We may therefore write

$$\overline{\mathbf{S}} = \text{const} \cdot \mathbf{J}.$$

In this form, however, the equality has only a conditional meaning, since the three components of $\mathbf{J}$ cannot simultaneously have definite values. Its z projection does have a literal meaning:

$$\overline{S}_z = \text{const} \cdot J_z = \text{const} \cdot M_J,$$

as does the equality

$$\overline{\mathbf{SJ}} = \text{const} \cdot \mathbf{J}^2 = \text{const} \cdot J(J+1),$$

obtained by multiplying both sides by $\mathbf{J}$. Bringing the conserved vector $\mathbf{J}$ under the averaging sign, we write $\overline{\mathbf{S}}\mathbf{J} = \overline{\mathbf{SJ}}$. The mean value $\overline{\mathbf{SJ}}$ is equal to the eigenvalue

$$\mathbf{SJ} = \frac{1}{2}[J(J+1) - L(L+1) + S(S+1)],$$

which it has in a state with definite values of $\mathbf{L}^2$, $\mathbf{S}^2$, and $\mathbf{J}^2$ (cf. (31.4)). Determining the constant from the second equality and substituting it in the first, we thus obtain

$$\overline{S}_z = M_J \frac{\mathbf{JS}}{\mathbf{J}^2}. \tag{113.5}$$

Combining these results and substituting them in (113.4), we find the following final expression for the splitting energy:

$$\Delta E = \mu_B g M_J H, \tag{113.6}$$

where

$$g = 1 + \frac{J(J+1) - L(L+1) + S(S+1)}{2J(J+1)} \tag{113.7}$$

is called the *Landé factor*, or *gyromagnetic factor*. Notice that $g = 1$ when there is no spin ($S = 0$, and hence $J = L$), while $g = 2$ when $L = 0$ (and hence $J = S$)[6].

Formula (113.6) gives different energies for all $2J+1$ values $M_J = -J, -J+1, \ldots, J$. In other words, the magnetic field removes completely the degeneracy of the levels with respect to angular-momentum directions, unlike an electric field, which left the levels with $M_J = \pm|M_J|$ unsplit (§ 76)[7]. The linear splitting given by (113.6) is nevertheless absent if $g = 0$ (which is possible even for $J \neq 0$, for example in the state $^4D_{1/2}$).

[6] The splitting described by the general formulas (113.6), (113.7) is sometimes called the anomalous Zeeman effect. This infelicitous name arose historically because, before the discovery of electron spin, the effect described by (113.6) with $g = 1$ was regarded as normal.

[7] The reasoning used in this connection for the electric case in § 76 does not apply to a magnetic field. Indeed, $\mathbf{H}$ is an axial vector and therefore changes sign under reflection in a plane containing its direction. Consequently, states transformed into one another by this operation actually refer to an atom in different fields.

We saw in § 76 that the displacement of an atomic energy level in an electric field is related to its mean electric dipole moment. A corresponding relation exists in the magnetic case. In classical theory the potential energy of a system of charges is $-\boldsymbol{\mu}\mathbf{H}$, where $\boldsymbol{\mu}$ is the system's magnetic moment. In quantum theory this is replaced by the corresponding operator, so the system's Hamiltonian is

$$\hat{H} = \hat{H}_0 - \hat{\boldsymbol{\mu}}\mathbf{H} = \hat{H}_0 - \hat{\mu}_z H.$$

Applying formula (11.16), with the field H as the parameter λ, we find for the mean magnetic moment

$$\overline{\mu}_z = -\frac{\partial \Delta E}{\partial H}, \tag{113.8}$$

where ΔE is the displacement of the energy level of the atomic state in question. Substitution of (113.6) shows that an atom in a state having a definite value M_J of the angular-momentum projection along some direction z has a mean magnetic moment along that same direction:

$$\overline{\mu}_z = -\mu_B g M_J. \tag{113.9}$$

If the atom has neither spin nor orbital angular momentum ($S = L = 0$), the second term in (113.2) produces no level displacement in first or any higher approximation (since all matrix elements of $\mathbf{L}$ and $\mathbf{S}$ vanish). The whole effect in this case is therefore due to the third term in (113.2), and in first-order perturbation theory the level displacement is the mean value

$$\Delta E = \frac{e^2}{8mc^2}\sum_a \overline{(\mathbf{H}\times\mathbf{r}_a)^2}. \tag{113.10}$$

Writing $(\mathbf{H}\times\mathbf{r}_a)^2 = H^2 r_a^2 \sin^2\theta$, where θ is the angle between $\mathbf{r}_a$ and $\mathbf{H}$, and averaging over the directions of $\mathbf{r}_a$, we obtain $\overline{\sin^2\theta} = 1 - \overline{\cos^2\theta} = 2/3$ (the wave function of a state with $L = S = 0$ is spherically symmetric, so averaging over directions is performed independently of averaging over the distances r_a). Thus

$$\Delta E = \frac{e^2}{12mc^2} H^2 \sum_a \overline{r_a^2}. \tag{113.11}$$

The magnetic moment calculated from (113.8) is now proportional to the magnitude of the field (an atom with $L = S = 0$ naturally has no magnetic moment in the absence of a field). Writing it as χH, we may regard the coefficient χ as the magnetic susceptibility of the atom. For it we obtain the following *Langevin formula* (P. Langevin, 1905):

$$\chi = -\frac{e^2}{6mc^2}\sum_a \overline{r_a^2}. \tag{113.12}$$

This quantity is negative; that is, the atom is diamagnetic[8].

[8]We mention that the Thomas–Fermi model cannot be used to calculate the mean square distance of the electrons from the nucleus. Although the integral $\int nr^2\,dr$ with the Thomas–Fermi density $n(r)$ does converge, it converges too slowly, with the result that the values obtained differ greatly from empirical ones.

If $J = 0$ but $S = L \neq 0$, the level displacement linear in the field is likewise absent, but the quadratic second-order effect from the perturbation $-\hat{\boldsymbol{\mu}}_{\text{at}}\mathbf{H}$ exceeds the effect (113.11)[9]. This is because, according to the general formula (38.10), the second-order correction to an energy eigenvalue is determined by a sum of expressions whose denominators contain differences of unperturbed energy levels—in this case the fine-structure intervals of the level, which are small quantities. It was noted in § 38 that the second-order correction to a normal level is always negative. The magnetic moment in the normal state will therefore be positive; that is, an atom in the normal state with $J = 0$, $L = S \neq 0$ is paramagnetic (J. H. van Vleck, 1928).

In strong magnetic fields, when $\mu_B H$ is comparable to or greater than the fine-structure intervals, the level splitting departs from that predicted by (113.6), (113.7); this phenomenon is called the *Paschen–Back effect*.

The splitting energy is particularly easy to calculate when the Zeeman splitting is large compared with the fine-structure intervals but remains, of course, small compared with the separations between different multiplets (so that the third term in the Hamiltonian (113.2) may still be neglected in comparison with the second). In other words, the energy in the magnetic field greatly exceeds the spin–orbit interaction[10]. This interaction may therefore be neglected in first approximation. Then not only the projection of the total angular momentum but also the projections M_L and M_S of the orbital angular momentum and spin are conserved, and the splitting is given by

$$\Delta E = \mu_B H(M_L + 2M_S). \tag{113.13}$$

The multiplet splitting is superposed on the magnetic-field splitting. It is determined by the mean value of the operator $A\hat{\mathbf{L}}\hat{\mathbf{S}}$ (72.4) in a state with given M_L, M_S (we are considering the multiplet splitting associated with the spin–orbit interaction). For a specified value of one angular-momentum component, the mean values of the other two vanish. Hence $\overline{\mathbf{LS}} = M_L M_S$, and in the next approximation the level energies are given by

$$\Delta E = \mu_B H(M_L + 2M_S) + AM_L M_S. \tag{113.14}$$

The Zeeman splitting cannot be calculated in general for an arbitrary coupling scheme (not LS coupling). One can assert only that the splitting in a weak field is linear in the field and proportional to the projection M_J of the total angular momentum, that is, that it has the form

$$\Delta E = \mu_B g_{nJ} H M_J, \tag{113.15}$$

where g_{nJ} are coefficients characteristic of the term in question (the letter n denotes the set of quantum numbers, apart from J, that characterize the term). Although these coefficients cannot be calculated individually, it is possible to derive a useful formula for their sum over all possible states of the atom with the given electron configuration and total angular momentum.

[9] For $S = L \neq 0$, the off-diagonal matrix elements of L_z, S_z for the transitions $S, L, J \to S, L, J \pm 1$ are, in general, nonzero.

[10] For intermediate cases, in which the influence of the magnetic field is comparable to the spin–orbit interaction, the splitting cannot be calculated in general form (the case $S = 1/2$ is worked out in Problem 1).

By definition,

$$g_{nJ}M_J = \langle nJM_J|L_z + 2S_z|nJM_J\rangle.$$

The quantities $g_{SLJ}M_J$ (where g_{SLJ} is the Landé factor (113.7) corresponding to LS coupling) are the diagonal matrix elements

$$g_{SLJ}M_J = \langle SLJM_J|L_z + 2S_z|SLM_J\rangle,$$

calculated in another complete system of wave functions. The functions of either system are obtained from those of the other by a linear unitary transformation. Such a transformation, however, leaves the sum of the diagonal matrix elements unchanged (§ 12). It follows that

$$\sum_n g_{nJ}M_J = \sum_{SL} g_{SLJ}M_J,$$

or, since g_{nJ} and g_{SLJ} do not depend on M_J,

$$\sum_n g_{nJ} = \sum_{SL} g_{SLJ}. \tag{113.16}$$

The summation is over all states with the given value of J that are possible for the specified electron configuration. This is the required relation.

Problems

Problem

1. Determine the splitting of a term with $S = 1/2$ in the Paschen–Back effect.

Solution. The magnetic field and the spin–orbit interaction must be included simultaneously in perturbation theory; thus the perturbation operator has the form[11]

$$\hat{V} = A\hat{\mathbf{L}}\hat{\mathbf{S}} + \mu_B(\hat{L}_z + 2\hat{S}_z)H.$$

As the initial zeroth-approximation wave functions we choose functions corresponding to states with definite values of L, $S = 1/2$, M_L, and M_S (L is specified, $M_L = -L, \dots, L$; $M_S = \pm 1/2$). In the perturbed states only the sum $M \equiv M_J = M_L + M_S$ is conserved ($\hat{V}$ commutes with $\hat{J}_z$), so definite values of M can be assigned to the components of the split term.

The values $M = \pm(L + 1/2)$ can be realized in only one way: respectively with $|M_L M_S\rangle = |L, 1/2\rangle$ and $|-L, -1/2\rangle$. The energy corrections for states with these values of M are therefore simply the diagonal matrix elements $\langle M_L M_S|\hat{V}|M_L M_S\rangle$ at the indicated values of $|M_L M_S\rangle$. The remaining values of M can be realized in two ways, $|M - 1/2, 1/2\rangle$ and $|M + 1/2, -1/2\rangle$. Each M then corresponds to two different energy values, found from the secular equation formed from the matrix elements for transitions between these two states.

[11] We do not write in $\hat{V}$ a term proportional to $(\hat{\mathbf{L}}\hat{\mathbf{S}})^2$ (spin–spin interaction). It must be kept in mind, however, that for spin $S = 1/2$ the expression $(\hat{\mathbf{L}}\hat{\mathbf{S}})^2$, by virtue of the special properties of the Pauli matrices (see § 55), reduces to an expression in $\hat{\mathbf{L}}\hat{\mathbf{S}}$ and is therefore included in the formula displayed.

The matrix elements of **LS** are obtained by directly multiplying the matrices $\langle M_L|\mathbf{L}|M_L'\rangle$ and $\langle M_S|\mathbf{S}|M_S'\rangle$, and are

$$\langle M_L M_S|\mathbf{LS}|M_L M_S\rangle = M_L M_S,$$

$$\begin{aligned}\langle M+1/2, -1/2|\mathbf{LS}|M-1/2, 1/2\rangle &= \langle M-1/2, 1/2|\mathbf{LS}|M+1/2, -1/2\rangle \\ &= \frac{1}{2}\sqrt{(L+M+1/2)(L-M+1/2)}.\end{aligned}$$

In the absence of a magnetic field the term is a doublet whose components are separated by $\epsilon = A(L+1/2)$ (see (72.6)). Choose the lower of these levels as the energy origin. The final formulas for the levels in a magnetic field are then

$$E = \epsilon \pm \mu_B H(L+1) \quad \text{for } M = \pm(L+1/2),$$

$$E^{\pm} = \frac{\epsilon}{2} + \mu_B HM \pm \left[\frac{1}{4}(\epsilon^2 + \mu_B^2 H^2) + \frac{\mu_B HM\epsilon}{2L+1}\right]^{1/2}$$

for $M = L-1/2, \ldots, -(L-1/2)$.

For $\mu_B H/\epsilon \ll 1$, we obtain

$$E^{+} = \epsilon + \mu_B HM\frac{2(L+1)}{2L+1}, \qquad E^{-} = \mu_B HM\frac{2L}{2L+1},$$

in agreement with (113.6), (113.7), in which $S = 1/2$, $J = L \pm 1/2$ are to be substituted. For $\mu_B H/\epsilon \gg 1$,

$$E^{\pm} = \mu_B H(M \pm 1/2) + \frac{\epsilon}{2} \pm \frac{M\epsilon}{2L+1},$$

in agreement with (113.14).

Problem

2. Determine the Zeeman splitting of the terms of a diatomic molecule in case *a*.

Solution. The magnetic moment due to the motion of the nuclei is very small compared with the magnetic moment of the electrons. The magnetic-field perturbation for the molecule must therefore be written as for a system of electrons, that is, still in the form $\hat{V} = \mu_B \mathbf{H}(\hat{\mathbf{L}} + 2\hat{\mathbf{S}})$, where **L**, **S** are the electronic orbital and spin angular momenta.

Averaging the perturbation over the electronic state, we obtain in case *a*

$$\mu_B H n_z(\Lambda + 2\Sigma) = \mu_B H n_z(2\Omega - \Lambda).$$

The mean value of n_z over the molecular rotation is the diagonal matrix element

$$\langle JM|n_z|JM\rangle = \frac{\Omega M}{J(J+1)},$$

where $M \equiv M_J$ (the matrix element is calculated from the reduced matrix element given by (87.4), with $K, \Lambda \to J, \Omega$).

Thus the required splitting is

$$\Delta E = \mu_B HM\frac{\Omega(2\Omega - \Lambda)}{J(J+1)}.$$

PROBLEM

3. The same for case *b*.

SOLUTION. The diagonal matrix elements $\langle \Lambda KJ|V|\Lambda KJ\rangle$ that determine the required splitting could be calculated by the general rules set out in § 87. It is simpler, however, to carry out the calculation in a more direct form. Averaging the perturbation operator over the orbital and electronic states gives

$$\mu_B H(\Lambda n_z + 2\hat{S}_z)$$

(the spin operator is unaffected by this averaging). Next average over the molecular rotation (the mean value of n_z is found with (87.4)):

$$\mu_B H\left(\frac{\Lambda^2}{K(K+1)}\hat{K}_z + 2\hat{S}_z\right).$$

Finally, we average over the spin wave function. After the complete averaging, the mean vectors can be directed only along the sole conserved vector, the total angular momentum **J**. We therefore obtain (cf. (113.5))

$$\frac{\mu_B H}{J(J+1)}\left[\frac{\Lambda^2}{K(K+1)}(\mathbf{KJ}) + 2(\mathbf{SJ})\right] M$$

($M \equiv M_J$), or, finally,

$$\Delta E = \frac{\mu_B}{J(J+1)}\left\{\frac{\Lambda^2}{2K(K+1)}[J(J+1) + K(K+1) - S(S+1)] \right.$$
$$\left. + [J(J+1) - K(K+1) + S(S+1)]\right\} HM.$$

PROBLEM

4. A diamagnetic atom is in an external magnetic field. Determine the strength of the induced magnetic field at the center of the atom.

SOLUTION. For $S = L = 0$, the Hamiltonian contains no perturbation linear in the field at all, and consequently the atomic wave function has no first-order correction in the magnetic field. The change $\mathbf{j}'$ in the electron current in the atom induced by the external magnetic field is associated (in the same first approximation in H) only with the addition of the term $(|e|/mc)\mathbf{A}$ to the electron velocity operators. Hence[12]

$$\mathbf{j}' = -\rho\frac{e^2}{mc}\mathbf{A} = -\rho\frac{e^2}{2mc}(\mathbf{H}\times\mathbf{r}), \tag{1}$$

where ρ is the electron density in the atom. The strength of the magnetic field produced at the center of the atom by this additional current is

$$\mathbf{H}_{\text{ind}} = \frac{1}{c}\int \frac{\mathbf{r}\times\mathbf{j}'}{r^3}\,dV$$

[12]This expression corresponds to the Larmor precession of the atom's electron shell about the direction of the external magnetic field (see II, § 45).

(cf. (121.8) below). Substituting (1) here and averaging over the directions of $\mathbf{r}$ under the integral, we obtain

$$\mathbf{H}_{\text{ind}} = -\frac{e^2}{3mc^2}\mathbf{H}\int \frac{\rho}{r}\,dV = \frac{e}{3mc^2}\varphi_e(0)\mathbf{H}, \tag{2}$$

where $\varphi_e(0)$ is the potential of the field produced at its center by the atom's electron shell.

In the Thomas–Fermi model $\varphi_e(0) = -1{,}80Z^{4/3}me^3/\hbar^2$ (see (70.8)), and hence

$$\mathbf{H}_{\text{ind}} = -0{,}60\left(\frac{e^2}{\hbar c}\right)^2 Z^{4/3}\mathbf{H} = -3{,}2\cdot 10^{-5}Z^{4/3}\mathbf{H}.$$

§ 114. Spin in a time-dependent magnetic field

Consider an electrically neutral particle with a magnetic moment, placed in a uniform magnetic field which varies with time. The particle may be elementary (a neutron, for example) or composite (an atom). We suppose the field to be so weak that the particle's magnetic energy in it is small compared with the separations between its energy levels. The motion of the particle as a whole may then be considered for a specified internal state.

Let $\hat{\mathbf{s}}$ denote the operator of the particle's "intrinsic" angular momentum: its spin for an elementary particle, or the total angular momentum $\mathbf{J}$ for an atom. Write the magnetic-moment operator in the form (111.1). The Hamiltonian governing the motion of the neutral particle as a whole is

$$\widehat{H} = -\frac{\mu}{s}\hat{\mathbf{s}}\mathbf{H} \tag{114.1}$$

(only the spin-dependent part of the Hamiltonian has been displayed).

In a uniform field this operator contains no explicit coordinates[13]. The particle's wave function therefore separates into coordinate and spin factors. The former is merely the wave function for free motion; below we are concerned only with the spin factor. We shall show that the problem for a particle of arbitrary angular momentum s reduces to the simpler problem of the motion of a spin-$1/2$ particle (E. Majorana). For this purpose we use the device already employed in § 57. In place of one particle with spin s, we formally introduce a system of $2s$ spin-$1/2$ "particles." The operator $\hat{\mathbf{s}}$ is then represented by the sum $\sum_a \hat{\mathbf{s}}_a$ of their spin operators, while the wave function is represented by a product of $2s$ spinors of rank one. The Hamiltonian (114.1) accordingly separates into a sum of $2s$ independent Hamiltonians:

$$\widehat{H} = \sum_a \widehat{H}_a, \qquad \widehat{H}_a = -\frac{\mu}{s}\mathbf{H}\hat{\mathbf{s}}_a, \tag{114.2}$$

[13]The same reasoning also applies when a particle (including a charged one) moves through a nonuniform magnetic field and its motion can be treated semiclassically. The magnetic field encountered as the particle advances along its trajectory may then simply be regarded as a function of time, and the same equations may be used for the change of the spin wave function.

so that each of the $2s$ "particles" moves independently of all the others. Once this has been done, the products of components of the $2s$ rank-one spinors need only be replaced again by the components of an arbitrary symmetric spinor of rank $2s$.

Problems

Problem

1. Find the spin wave function of a neutral spin-$1/2$ particle in a uniform magnetic field whose direction is fixed but whose magnitude follows an arbitrary function $H(t)$.

Solution. The wave function is a spinor ψ^ν satisfying the wave equation

$$i\hbar\dot{\psi}^\nu = -2\mu\mathbf{H}\hat{\mathbf{s}}\psi^\nu. \tag{1}$$

Choose the field direction as the z axis. In spinor components, this equation then becomes

$$i\hbar\dot{\psi}^1 = -\mu H\psi^1, \qquad i\hbar\dot{\psi}^2 = \mu H\psi^2.$$

It follows that

$$\psi^1 = c_1 \exp\left(\frac{i\mu}{\hbar}\int H\,dt\right), \qquad \psi^2 = c_2 \exp\left(-\frac{i\mu}{\hbar}\int H\,dt\right).$$

The constants c_1, c_2 are fixed by the initial conditions and the normalization condition $|\psi^1|^2 + |\psi^2|^2 = 1$.

Problem

2. Solve the same problem for a uniform magnetic field of constant magnitude whose direction rotates uniformly (with angular velocity ω) about the z axis, maintaining an angle θ with it.

Solution. The components of the magnetic field are

$$H_x = H\sin\theta\cos\omega t, \qquad H_y = H\sin\theta\sin\omega t, \qquad H_z = H\cos\theta,$$

and (1) gives the system

$$\dot{\psi}^1 = i\omega_H(\cos\theta\cdot\psi^1 + \sin\theta\cdot e^{-i\omega t}\psi^2),$$
$$\dot{\psi}^2 = i\omega_H(\sin\theta\cdot e^{i\omega t}\psi^1 - \cos\theta\cdot\psi^2),$$

where $\omega_H = \mu H/\hbar$. The substitution

$$\psi^1 = e^{-i\omega t/2}\varphi^1, \qquad \psi^2 = e^{i\omega t/2}\varphi^2$$

reduces these equations to a system of linear equations with constant coefficients. Solving it gives

$$\psi^1 = e^{-i\omega t/2}\left(c_1 e^{i\Omega t/2} + c_2 e^{-i\Omega t/2}\right),$$

$$\psi^2 = 2\omega_H\sin\theta\, e^{i\omega t/2}\left[\frac{c_1}{\Omega + \omega + 2\omega_H\cos\theta}e^{i\Omega t/2} - \frac{c_2}{\Omega - \omega - 2\omega_H\cos\theta}e^{-i\Omega t/2}\right],$$

where

$$\Omega = \sqrt{(\omega + 2\omega_H\cos\theta)^2 + 4\omega_H^2\sin^2\theta}.$$

§ 115. Current density in a magnetic field

We derive the quantum-mechanical expression for the current density. The charged particle moves in a magnetic field. Our starting point is the formula[14]:

$$\delta H = -\frac{1}{c}\int \mathbf{j}\,\delta\mathbf{A}\,dV, \tag{115.1}$$

For charges distributed in space, equation (115.1) determines the change in the Hamilton function. The change in question results from varying the vector potential[15]. In quantum mechanics, it must be applied to the mean value of the Hamiltonian of a charged particle:

$$\overline{H} = \int \Psi^* \left[\frac{1}{2m}\left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 - \frac{\mu}{s}\mathbf{H}\hat{\mathbf{s}}\right]\Psi\,dV. \tag{115.2}$$

Varying this expression and recalling that $\delta\mathbf{H} = \nabla\times\delta\mathbf{A}$, we obtain

$$\begin{aligned}\delta\overline{H} = &\int \Psi^*\left[-\frac{e}{2mc}(\hat{\mathbf{p}}\,\delta\mathbf{A} + \delta\mathbf{A}\hat{\mathbf{p}}) + \frac{e^2}{mc^2}\mathbf{A}\,\delta\mathbf{A}\right]\Psi\,dV\\ &-\frac{\mu}{s}\int \nabla\times\delta\mathbf{A}\cdot\Psi^*\hat{\mathbf{s}}\Psi\,dV.\end{aligned} \tag{115.3}$$

We transform the term containing $\hat{\mathbf{p}}\,\delta\mathbf{A}$ by integration by parts:

$$\int \Psi^*\hat{\mathbf{p}}\,\delta\mathbf{A}\Psi\,dV = -i\hbar\int \Psi^*\nabla(\delta\mathbf{A}\Psi)\,dV = i\hbar\int \delta\mathbf{A}\Psi\nabla\Psi^*\,dV$$

(as usual, the integral over the surface at infinity vanishes). We also integrate the last term in (115.3) by parts, using the familiar vector-analysis identity

$$\mathbf{a}\nabla\times\mathbf{b} = -\nabla\cdot(\mathbf{a}\times\mathbf{b}) + \mathbf{b}\nabla\times\mathbf{a}.$$

The integral of the divergence term vanishes, leaving

$$\int \Psi^*\hat{\mathbf{s}}\Psi\nabla\times\delta\mathbf{A}\,dV = \int \delta\mathbf{A}\nabla\times(\Psi^*\hat{\mathbf{s}}\Psi)\,dV.$$

The final result is therefore

$$\begin{aligned}\delta\overline{H} = &-\frac{ie\hbar}{2mc}\int \delta\mathbf{A}(\Psi\nabla\Psi^* - \Psi^*\nabla\Psi)\,dV\\ &+\frac{e^2}{mc^2}\int \mathbf{A}\,\delta\mathbf{A}\Psi\Psi^*\,dV - \frac{\mu}{s}\int \delta\mathbf{A}\nabla\times(\Psi^*\hat{\mathbf{s}}\Psi)\,dV.\end{aligned}$$

[14]In this section, $\mathbf{j}$ denotes the electric-current density. It equals the particle-current density multiplied by the particle charge e.

[15]Consider a charge in a magnetic field. Its Lagrangian contains the term $(e/c)\mathbf{v}\mathbf{A}$. When the charge is represented as distributed in space, this term is instead written $(1/c)\int \mathbf{j}\mathbf{A}\,dV$. Consequently, variation of $\mathbf{A}$ changes the Lagrangian by

$$\delta L = \frac{1}{c}\int \mathbf{j}\,\delta\mathbf{A}\,dV.$$

An infinitesimal change in the Hamilton function is the negative of the change in the Lagrangian. See Vol. I, § 40.

Comparison with (115.1) gives the following expression for the current density:

$$\mathbf{j} = \frac{ie\hbar}{2m}[(\nabla\Psi^*)\Psi - \Psi^*\nabla\Psi] - \frac{e^2}{mc}\mathbf{A}\Psi^*\Psi + \frac{\mu}{s}c\nabla\times(\Psi^*\hat{\mathbf{s}}\Psi). \tag{115.4}$$

We emphasize that although the vector potential occurs explicitly in this expression, the result is nevertheless unambiguous, as it must be. This can be verified directly by observing that the gauge transformation of the vector potential must, in accordance with (111.8), be accompanied by the wave-function transformation (111.9).

It is also readily checked that the current (115.4), together with the charge density $\rho = e|\Psi|^2$, obeys the continuity equation, as required:

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\mathbf{j} = 0.$$

The final term of (115.4) is the contribution to the current density arising from the particle's magnetic moment. It has the form $c\nabla\times\mathbf{m}$, where

$$\mathbf{m} = \frac{\mu}{s}\Psi^*\hat{\mathbf{s}}\Psi = \Psi^*\hat{\boldsymbol{\mu}}\Psi \tag{115.5}$$

is the spatial density of magnetic moment.

Expression (115.4) is the mean value of the current density. It may be regarded as a diagonal matrix element of an operator, the current-density operator $\hat{\mathbf{j}}$. This operator has its simplest form in the second-quantization representation: one merely replaces Ψ and Ψ^* by the operators $\hat{\Psi}$ and $\hat{\Psi}^+$ (and, by the general rule, $\hat{\Psi}^+$ must stand to the left of $\hat{\Psi}$ in every term). The off-diagonal matrix elements of this operator can likewise be defined:

$$\mathbf{j}_{nm} = \frac{ie\hbar}{2m}[(\nabla\Psi_n^*)\Psi_m - \Psi_n^*\nabla\Psi_m] - \frac{e^2}{mc}\mathbf{A}\Psi_n^*\Psi_m + \frac{\mu}{s}c\nabla\times(\Psi_n^*\hat{\mathbf{s}}\Psi_m). \tag{115.6}$$

CHAPTER XVI

Structure of the Atomic Nucleus

§ 116. Isotopic invariance

At present there is still no complete theory of the so-called *nuclear forces*: the forces acting between nuclear particles (*nucleons*) and binding them together within an atomic nucleus. Consequently, for now the description of nuclear forces must rely on experiment to a much greater extent than would be necessary if a consistent theory were available.

The two kinds of particle belonging to the nucleon family differ first of all in their electrical properties: protons (p) carry positive charge, whereas neutrons (n) are electrically neutral. Both kinds nevertheless have the same spin $1/2$, and their masses are nearly equal (the proton and neutron masses are respectively 1836,1 and 1838,6 electron masses). This likeness is not accidental. In spite of their different electrical properties, the proton and neutron are particles very much alike, and this likeness is of fundamental importance.

It is found that, when the relatively weak electrical forces are set aside, the interaction between two protons closely resembles that between two neutrons. This property of nuclear forces is called *charge symmetry*[1].

To the accuracy with which this symmetry holds, one may say in particular that the two-proton (pp) and two-neutron (nn) systems possess states with the same properties. It is important here, of course, that protons and neutrons obey the same statistics (Fermi statistics). Hence only states having the same wave-function symmetry are possible for the pp and nn systems: the functions $\psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2)$ must be antisymmetric under simultaneous interchange of the particles' coordinates and spins.

Charge symmetry, however, is only one manifestation of a still deeper physical likeness between proton and neutron, known as *isotopic invariance*[2]. This deeper regularity establishes an analogy not only between the pp and nn systems (which are transformed into one another by exchanging all protons and neutrons), but also with the pn system, whose particles are different. A complete analogy is plainly impossible, since the allowed states of a pn system, being states of nonidentical particles, need not in any event be restricted to antisymmetric wave functions. Among the possible states of the pn system, however, there are states whose properties agree to high accuracy with those of a pair of identical nucleons[3]; naturally, these states are described by antisymmetric wave functions. The remaining pn states have symmetric wave functions and have no counterparts in the pp and nn systems.

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[1] One of its manifestations is the similarity in the properties (binding energy, energy spectrum, and so forth) of the so-called mirror nuclei, namely nuclei obtained from one another by replacing every proton by a neutron and conversely.

[2] This invariance is also called *isobaric* in the literature.

[3] This was established from an analysis of experimental data on the scattering of neutrons and protons by protons by *Breit, Condon, and Present* (G. Breit, E. U. Condon, R. D. Present, 1936).

Like charge symmetry, isotopic invariance applies only when electromagnetic interaction is neglected. A second reason for its approximate character is the small difference between neutron and proton masses: exact neutron–proton symmetry would plainly require their masses to coincide exactly[4].

A convenient formal apparatus can be introduced to describe isotopic invariance. Its natural route follows from observing that isotopic invariance amounts to the possibility of classifying the states of a nucleon system by the symmetry of its coordinate–spin wave functions ψ, without regard to which of the two kinds its nucleons belong. The required apparatus must therefore provide a new quantum number for characterizing the system's states, such that specifying this number uniquely fixes the symmetry of ψ. We have already encountered an analogous situation in studying a system of spin-$1/2$ particles. In fact, we saw (see § 63) that specifying the total spin S of such a system uniquely determines the symmetry of

its coordinate wave function φ, independently of which of the two possible values ($\pm 1/2$) are taken by the spin projections σ of the individual particles.

It is therefore natural, in a formal description of isotopic invariance, to regard the neutron and proton as two different *charge states* of one and the same particle (the nucleon). They are distinguished by the projection of a new vector $\boldsymbol{\tau}$, formally analogous to a spin-$1/2$ vector. This new quantity, customarily called *isotopic spin* (or simply *isospin*)[5], is a vector in an auxiliary "isotopic space" $\xi\eta\zeta$, which of course has nothing to do with physical space.

The projection of a nucleon's isotopic spin on the ζ axis can take only the two values $\tau_\zeta = \pm 1/2$. By convention, $+1/2$ is assigned to the proton and $-1/2$ to the neutron[6]. The isospins of several nucleons combine into the total isospin of the system according to the ordinary rules for adding spins. The ζ component of the system's total isospin is then the sum of the τ_ζ values of all its constituent particles. For a nucleus with proton number (that is, atomic number) Z, neutron number N, and mass number $A = Z + N$, we have

$$T_\zeta = \sum \tau_\zeta = \frac{Z - N}{2} = Z - \frac{A}{2}, \tag{116.1}$$

so, for a specified number of nucleons, T_ζ determines the total charge of the system. Thus T_ζ is strictly conserved, this being simply an expression of charge conservation.

The magnitude T of the system's total isotopic spin determines the symmetry of the "charge part" ω of its wave function, just as the total spin S determines the symmetry of the spin wave function. It thereby also fixes the symmetry of the coordinate–spin (that is, ordinary) wave function ψ, since the complete wave function of the nucleon system (the product $\psi\omega$) must have a definite symmetry. As for any fermions, it must be antisymmetric under simultaneous interchange of the coordinates, spins, and "charge variables" τ_ζ of the particles. Thus the definite symmetry of the wave functions ψ

for any nucleon system is represented in this scheme precisely by conservation of T.

Equivalently, isotopic invariance means that the system's properties are invariant

[4]It is reasonable to suppose that the neutron–proton mass difference is itself electromagnetic in origin.

[5]It was introduced by *Heisenberg* (1932) and applied to the description of isotopic invariance by *Cassen and Condon* (B. Cassen, E. U. Condon, 1936).

[6]The opposite convention is also used in the literature.

under arbitrary rotations in isotopic space. States differing only in the value of T_ζ, with T and all other quantum numbers fixed, have the same properties. Charge symmetry in particular—invariance under interchanging every neutron with a proton and conversely—is a special case of isotopic invariance. In this description it appears as invariance under simultaneous reversal of all τ_ζ, that is, under a rotation through 180° in isospace about an axis lying in the $\xi\eta$ plane.

The evident violation of isotopic invariance by the Coulomb interaction is also formally apparent in this scheme: the Coulomb interaction depends on charge, hence on the ζ components of isospin, which are not invariant under rotations in $\xi\eta\zeta$ space.

Consider, for example, a system of two nucleons. Its total isotopic spin can have the values $T = 1$ and $T = 0$. For $T = 1$, the possible projection values are $T_\zeta = 1, 0, -1$. According to (116.1), these correspond to charges 2, 1, and 0; thus a system with $T = 1$ can be realized as pp, pn, or nn. The charge part ω of the wave function is symmetric for $T = 1$ (just as a spin value $S = 1$ corresponds to a symmetric spin function; compare § 62). Consequently, $T = 1$ corresponds to states with antisymmetric ordinary wave functions ψ. For $T = 0$, only $T_\zeta = 0$ is possible and the corresponding ω is antisymmetric; this case therefore comprises the pn states with symmetric wave functions ψ.

Isotopic spin has an associated operator $\hat{\boldsymbol{\tau}}$ acting on the charge variable τ_ζ in the wave function in the same manner as the spin operator $\hat{\mathbf{s}}$ acts on the spin variable σ. Because the formal analogy is complete, the operators $\hat{\tau}_\xi$, $\hat{\tau}_\eta$, $\hat{\tau}_\zeta$ are represented by the same Pauli matrices (55.7) as $\hat{s}_x$, $\hat{s}_y$, $\hat{s}_z$.

We note several combinations of these operators that have a simple, direct meaning. The sum

$$\hat{\tau}_+ = \hat{\tau}_\xi + i\hat{\tau}_\eta = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

is an operator which turns a neutron wave function into a proton wave function

when applied to it, while giving zero on a proton function. Similarly, the operator

$$\hat{\tau}_- = \hat{\tau}_\xi - i\hat{\tau}_\eta = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

turns a proton into a neutron and annihilates a neutron. Finally, the operator

$$\frac{1}{2} + \hat{\tau}_\zeta = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

leaves a proton function unchanged and annihilates a neutron function; it may be called the nucleon charge operator (in units of e).

We shall also show how the operator $\hat{P}$ that interchanges two particles can be expressed through their isospin operators $\hat{\boldsymbol{\tau}}_1$, $\hat{\boldsymbol{\tau}}_2$. By definition, its action on the wave function $\psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2)$ of a two-particle system interchanges the coordinates and spins of the particles, that is, the variables $\mathbf{r}_1, \sigma_1$ and $\mathbf{r}_2, \sigma_2$. Its eigenvalues are ± 1, obtained when it acts on a symmetric or an antisymmetric function ψ:

$$\hat{P}\psi_{\text{sym}} = \psi_{\text{sym}}, \qquad \hat{P}\psi_{\text{antisym}} = -\psi_{\text{antisym}}. \tag{116.2}$$

We saw above that the functions ψ_{sym} and ψ_{antisym} correspond respectively to charge functions ω_T with total isospin $T = 0$ and $T = 1$. Hence, if $\hat{P}$ is to be represented in a form acting on charge variables, it must have the properties

$$\hat{P}\omega_0 = \omega_0, \qquad \hat{P}\omega_1 = -\omega_1. \tag{116.3}$$

These conditions are satisfied by the operator $1 - \hat{\mathbf{T}}^2$. This is seen immediately by noting that ω_T is an eigenfunction of $\hat{\mathbf{T}}^2$ with eigenvalue $T(T+1)$. Finally, putting $\hat{\mathbf{T}} = \hat{\boldsymbol{\tau}}_1 + \hat{\boldsymbol{\tau}}_2$ and using the fact that $\hat{\boldsymbol{\tau}}_1^2$ and $\hat{\boldsymbol{\tau}}_2^2$ have the same definite value $\tau(\tau+1) = 3/4$, we obtain the required final expression[7]

$$\hat{P} = 1 - \hat{\mathbf{T}}^2 = -\frac{1}{2} - 2\hat{\boldsymbol{\tau}}_1\hat{\boldsymbol{\tau}}_2. \tag{116.4}$$

There are definite isospin selection rules for matrix elements of various physical quantities of a nucleon system (L. A. Radicati, 1952). Let F be a quantity (of arbitrary tensor character) which is additive in the sense that its value for the system equals the sum of its values for the individual nucleons. Write its operator as

$$\hat{F} = \sum_p \hat{f}_p + \sum_n \hat{f}_n,$$

where the sums extend over all protons and all neutrons in the system. This expression may identically be rewritten as

$$\begin{aligned}\hat{F} &= \sum\left(\frac{1}{2} + \hat{\tau}_\zeta\right)\hat{f}_p + \sum\left(\frac{1}{2} - \hat{\tau}_\zeta\right)\hat{f}_n \\ &= \frac{1}{2}\sum(\hat{f}_p + \hat{f}_n) + \sum(\hat{f}_p - \hat{f}_n)\hat{\tau}_\zeta,\end{aligned} \tag{116.5}$$

where every sum in the last expression runs over all nucleons, both protons and neutrons. The first term in (116.5) is a scalar, while the second is the ζ component of a vector in isospace. They therefore obey the same isospin selection rules as those applying to scalars and vectors in ordinary space with respect to orbital angular momentum (see § 29). An isotopic scalar allows only transitions for which T does not change. The ζ component of an isotopic vector has matrix elements only for transitions with $\Delta T = 0, \pm 1$; in addition, $\Delta T = 0$ transitions between states with $T_\zeta = 0$ are forbidden, that is, for systems having equal numbers of neutrons and protons. The last rule follows because the matrix element of a transition with $\Delta T = 0$ is proportional to T_ζ; see (29.7).

For the nuclear dipole moment, for example, the quantities f_p are the products $e\mathbf{r}$, while $f_n = 0$. The first term in (116.5) is then

$$\frac{e}{2}\sum \mathbf{r} = \frac{e}{2m}\sum m\mathbf{r},$$

that is, it is proportional to the radius vector of the center of mass and can be made zero by a suitable choice of origin. In other words, the nuclear dipole moment reduces to the ζ component of an isotopic vector.

[7] An operator of this form, constructed from the ordinary spins of the particles, was encountered earlier in the problems to § 62.

§ 117. Nuclear forces

The *nuclear forces* specific to nucleons are distinguished above all by their short range: at distances of order 10^{-13} cm they fall off exponentially.

In the nonrelativistic limit, nuclear forces may be regarded as independent of the nucleon velocities and as possessing a potential (inside a nucleus, nucleon velocities are about one quarter of the speed of light; see below). The interaction potential energy U of two nucleons depends not only on their separation r, but also, and quite strongly, on their spins[8]. Only a systematic theory of nuclear forces could, of course, determine the exact dependence on r. The form of the spin dependence, however, follows already from simple arguments based on the properties of spin operators.

Only three vectors are available on which the interaction energy U can depend: the unit vector $\mathbf{n}$ directed along the radius vector between the two nucleons, and their spins $\mathbf{s}_1$ and $\mathbf{s}_2$. By the general properties of the spin-1/2 operator, every function of it reduces to a linear function (see § 55). We must also note that the product $\mathbf{ns}$ is a pseudoscalar rather than a true scalar (since $\mathbf{n}$ is a polar vector and $\mathbf{s}$ an axial one). It is therefore evident that only two independent scalar quantities, each linear in either spin, can be formed from $\mathbf{n}$, $\mathbf{s}_1$, and $\mathbf{s}_2$: $\mathbf{s}_1\mathbf{s}_2$ and $(\mathbf{ns}_1)(\mathbf{ns}_2)$[9].

Consequently, as regards its spin dependence, the interaction operator for two nucleons can be written as a sum of three independent terms,

$$\hat{U}_{\text{ord}} = U_1(r) + U_2(r)(\hat{\mathbf{s}}_1\hat{\mathbf{s}}_2) + U_3(r)\big[3(\hat{\mathbf{s}}_1\mathbf{n})(\hat{\mathbf{s}}_2\mathbf{n}) - \hat{\mathbf{s}}_1\hat{\mathbf{s}}_2\big], \tag{117.1}$$

one of which is spin-independent while the other two depend on the spins. The third term has been written in this form so that it vanishes upon averaging over the directions of $\mathbf{n}$; the forces represented by this term are usually called *tensor forces*.

We have attached the label "ordinary" to the interaction (117.1) in order to emphasize that this operator does not change the

charge state of the nucleons. An interaction in which a proton turns into a neutron, and conversely, is also possible. The operator of this "exchange" interaction differs in form from (117.1) through the presence of the particle permutation operator (116.4):

$$\hat{U}_{\text{ex}} = \big\{U_4(r) + U_5(r)(\hat{\mathbf{s}}_1\hat{\mathbf{s}}_2) + U_6(r)\big[3(\hat{\mathbf{s}}_1\mathbf{n})(\hat{\mathbf{s}}_2\mathbf{n}) - \hat{\mathbf{s}}_1\hat{\mathbf{s}}_2\big]\big\}\,\hat{P}. \tag{117.2}$$

The complete interaction operator is the sum

$$\hat{U} = \hat{U}_{\text{ord}} + \hat{U}_{\text{ex}}. \tag{117.3}$$

Thus the interaction of two nucleons is specified by six different functions of their separation. In general, all these terms have the same order of magnitude[10].

[8] This sharply distinguishes the interaction of nucleons from that of electrons: electron spin–spin interaction is purely relativistic in origin and is small in atoms.

[9] It is understood here that the nuclear forces are invariant under spatial inversion, and hence cannot contain pseudoscalar terms. There is at present no experimental evidence indicating otherwise.

[10] We also mention that an interaction depending on the nucleon velocities is described, to first order in the velocities, by an operator of the form

$$[\varphi_1(r) + \varphi_2(r)\hat{P}]\hat{\mathbf{L}}\hat{\mathbf{S}},$$

where $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is the orbital angular momentum of the relative nucleon motion ($\mathbf{p}$ is its momentum), and $\mathbf{S} = \mathbf{s}_1 + \mathbf{s}_2$; this operator contains two functions of r. Terms of the forms $\mathbf{pn}$ and $\mathbf{Sn}$ are excluded by invariance under spatial inversion and time reversal.

The spin operators occurring in (117.1) and (117.2) may be expressed in terms of the total-spin operator $\mathbf{S}$. Indeed, on squaring the equalities $\hat{\mathbf{S}} = \hat{\mathbf{s}}_1 + \hat{\mathbf{s}}_2$ and $\hat{\mathbf{S}}\mathbf{n} = \hat{\mathbf{s}}_1\mathbf{n} + \hat{\mathbf{s}}_2\mathbf{n}$, and using $\hat{\mathbf{s}}_1^2 = \hat{\mathbf{s}}_2^2 = 3/4$ and $(\hat{\mathbf{s}}_1\mathbf{n})^2 = (\hat{\mathbf{s}}_2\mathbf{n})^2 = 1/4$ (see (55.10)), we obtain

$$\hat{\mathbf{s}}_1\hat{\mathbf{s}}_2 = \frac{1}{2}\left(\hat{\mathbf{S}}^2 - \frac{3}{2}\right), \qquad (\hat{\mathbf{s}}_1\mathbf{n})(\hat{\mathbf{s}}_2\mathbf{n}) = \frac{1}{2}\left[(\hat{\mathbf{S}}\mathbf{n})^2 - \frac{1}{2}\right]. \tag{117.4}$$

The operator $\hat{\mathbf{S}}^2$ commutes with $\hat{\mathbf{S}}$; hence the interactions represented by the first two terms of (117.1) and (117.2) preserve the system's total-spin vector. The tensor interaction, by contrast, contains the operator $(\hat{\mathbf{S}}\mathbf{n})^2$, which commutes with the square $\hat{\mathbf{S}}^2$ but not with the vector $\hat{\mathbf{S}}$ itself. It follows that only the magnitude of the total spin remains conserved, not its direction.

For a system of two nucleons the total spin S may take the values 0 and 1. Its total isotopic spin T has the same two possible values. All states of the system therefore separate into four groups, distinguished by the pair of numbers S, T. For the states in each group

For $S = 0$, the corresponding interaction operator has the form $A(r)$. For $S = 1$, it has the form $A(r) + B(r)[(\hat{\mathbf{S}}\mathbf{n})^2 - 2/3]$. In these cases the general operator (117.3) reduces to the respective one (see Problem 1)[11].

For fixed S and T, the states of the system are classified by total angular momentum J and parity. As we know, $T = 0$ corresponds to states with symmetric wave functions ψ, whereas for $T = 1$ these functions are antisymmetric. On the other hand, S fixes the symmetry of the wave function in the spin variables: it is symmetric for $S = 1$ and antisymmetric for $S = 0$. Consequently, the pair S, T also fixes the symmetry in the spatial variables, and hence the parity of the state. Thus, a system with isotopic spin $T = 0$ can have only even triplet states ($S = 1$) or odd singlet states ($S = 0$); for $T = 1$, the states are odd triplets or even singlets.

Since the spin vector is not conserved, orbital angular momentum is not generally conserved either; only the sum $\mathbf{J} = \mathbf{L} + \mathbf{S}$ is conserved. Its magnitude L may nevertheless be conserved simply because fixed values of J, S, and parity (or of J, S, and T) can be compatible with just one definite value of L; recall that the parity of a two-particle system is $(-1)^L$. Thus an odd state with $S = 1$, $J = 1$ can have only $L = 1$, and is therefore a ${}^3P_1$ state. In other cases, two different values of L may correspond to the prescribed values of J, S, and parity, so that L is not conserved. For example, in an odd state with $S = 1$, $J = 2$, either $L = 1$ or $L = 3$ is possible; such a state is a superposition ${}^3P_2 + {}^3F_2$.

We thus arrive at the following possible states of a two-nucleon system (the sign $\pm$ indicates the parity):

[11]The sign of the observed quadrupole moment of the deuteron implies that, in the $T = 0$, $S = 1$ state, the coefficient $B(r)$ of the tensor forces is negative. Experimental data on the properties of the deuteron further show that the nucleon interaction in this state contains a strong attraction with a deep potential well. The presence of tensor forces makes it difficult to express this fact in terms of properties of the functions $A(r)$ and $B(r)$. Nucleon-scattering data show that attraction is also present for $T = 1$, $S = 0$. It is weaker and, in particular, does not result in a stable two-particle system.

$$
\begin{aligned}
\text{for } T = 1: \quad & {}^3P_0^-,\ {}^3P_1^-,\ ({}^3P_2 + {}^3F_2)^-,\ {}^3F_3^-, \ldots; \\
& {}^1S_0^+,\ {}^1D_2^+,\ {}^1G_4^+, \ldots, \\
\text{for } T = 0: \quad & ({}^3S_1 + {}^3D_1)^+,\ {}^3D_2^+, \\
& ({}^3D_3 + {}^3G_3)^+, \ldots;\ {}^1P_1^-,\ {}^1F_3^-, \ldots
\end{aligned}
$$

In general, nuclear forces are nonadditive. This means that the interaction in a system containing more than two nucleons does not reduce to the sum of the interactions between all particle pairs. It appears, however, that three-body and higher interactions play a relatively minor role compared with pair interactions. The properties of complex nuclei can therefore be treated to a considerable extent on the basis of pairwise interactions.

Experimental information on nuclei shows that, as the particle number A increases, a nucleon system begins to behave as macroscopic "nuclear matter," whose volume and energy grow in proportion to A (apart from effects due to the Coulomb interaction of the protons and to the presence of a free nuclear surface). The property of nuclear forces responsible for this behavior is called their *saturation*.

This saturation property imposes certain restrictions on the functions $U_1, \ldots, U_6$ that specify pairwise nucleon interactions. Suppose all the particles are confined within a region whose dimensions are of the order of the range of the nuclear forces; every pair then interacts. If some configuration of the nucleons, including their spin orientations, makes all these pair forces attractive, the potential energy of the system is negative and proportional to A^2. Its kinetic energy, by contrast, is positive and proportional to $A^{5/3}$, a lower power of A[12]. Under these conditions, a sufficiently large collection of nucleons would indeed contract into a small volume independent of A, and hence would not form nuclear matter. Saturation of the nuclear forces must therefore require the absence of configurations whose negative interaction energy is proportional to A^2 (see problem 2).

The fact that the volume of nuclear matter is proportional to the particle number is expressed by a relation of the form

$$R = r_0 A^{1/3}, \tag{117.5}$$

which connects the nuclear radius R with the number A of particles in the nucleus. Experimental data (from electron scattering by nuclei) give $r_0 \approx 1.1 \cdot 10^{-13}$ cm.

Let us determine the limiting nucleon momentum in nuclear matter (compare § 70). The phase-space volume corresponding to particles that occupy a unit volume of physical space and have momenta $p \leqslant p_0$ is $4\pi p_0^3/3$. Dividing this by $(2\pi\hbar)^3$ gives the number of "cells," each capable of holding two protons and two neutrons. Taking the numbers of protons and neutrons to be equal, we obtain

$$4\frac{4\pi}{3}\left(\frac{p_0}{2\pi\hbar}\right)^3 = \frac{A}{V},$$

[12] The density n of particles confined in the given volume is proportional to their number A, while the kinetic energy per particle is then proportional to $n^{2/3}$ (compare (70.1)). The total kinetic energy is therefore proportional to $AA^{2/3}$.

where V is the nuclear volume. Substitution of (117.5) then gives

$$p_0 = \left(\frac{3\pi^2 A}{2V}\right)^{1/3}\hbar = \frac{(9\pi)^{1/3}\hbar}{2r_0} = 1.4\cdot 10^{-14}\ \mathrm{g\cdot cm\cdot s^{-1}}.$$

The corresponding energy $p_0^2/2m_p$ (where m_p is the nucleon mass) is ~ 30 MeV, and the velocity is $p_0/m_p \approx c/4$.

Problems

PROBLEM

1. Find the interaction operators for two nucleons in states with definite values of S and T.

SOLUTION. The required operators $\hat{U}_{ST}$ follow from the general expressions (117.1)–(117.3), with relations (116.3) and (117.4) taken into account:

$$\begin{aligned}
\hat{U}_{00} &= U_1 - \frac{3}{4}U_2 + U_4 - \frac{3}{4}U_5,\\
\hat{U}_{01} &= U_1 - \frac{3}{4}U_2 - U_4 + \frac{3}{4}U_5,\\
\hat{U}_{10} &= U_1 + \frac{1}{4}U_2 + U_4 + \frac{1}{4}U_5 + \frac{1}{2}(U_3 + U_6)\left[3(\hat{\mathbf{S}}\mathbf{n})^2 - 2\right],\\
\hat{U}_{11} &= U_1 + \frac{1}{4}U_2 - U_4 - \frac{1}{4}U_5 + \frac{1}{2}(U_3 - U_6)\left[3(\hat{\mathbf{S}}\mathbf{n})^2 - 2\right].
\end{aligned}$$

PROBLEM

2. Find the saturation conditions for the nuclear forces, assuming that tensor forces are absent and that the ranges of all the remaining types of force are equal.

SOLUTION. Consider several limiting cases (between which all other possible cases lie) for the state of a system of A nucleons, and write the conditions under which the interaction energy of an "average" nucleon pair in this system is positive.

Let the total spin and isotopic spin of the nucleus have their largest possible values: $S_{\mathrm{nuc}} = T_{\mathrm{nuc}} = A/2$ (all particles in the system are protons with parallel spins). Every nucleon pair then has $S = T = 1$, giving the condition

$$U_{11} > 0. \tag{1}$$

Now let $T_{\mathrm{nuc}} = A/2$, $S_{\mathrm{nuc}} = 0$. Every nucleon pair then has $T = 1$, while the mean value of s_z for a single nucleon is zero. The latter means that a nucleon is equally likely to have $s_z = 1/2$ or $s_z = -1/2$; under these conditions the probabilities for a nucleon pair to be in states with $S = 0$ and $S = 1$ are respectively $1/2$ and $3/4$ (they are proportional to the number $2S + 1$ of possible S_z values). The condition that the mean pair energy be positive is therefore

$$\frac{1}{4}U_{01} + \frac{3}{4}U_{11} > 0. \tag{2}$$

Similarly, consideration of the state with $T_{\text{nuc}} = 0$, $S_{\text{nuc}} = A/2$ leads to

$$\frac{1}{4}U_{10} + \frac{3}{4}U_{11} > 0. \tag{3}$$

In the state with $T_{\text{nuc}} = S_{\text{nuc}} = 0$, the probability that a nucleon pair has $S = T = 1$ is $3/4 \cdot 3/4$, the probability that it has $T = 1$, $S = 0$ is $3/4 \cdot 1/4$, and so on. We thus find the condition

$$\frac{9}{16}U_{11} + \frac{3}{16}(U_{10} + U_{01}) + \frac{1}{16}U_{00} > 0. \tag{4}$$

Finally, let the system contain $A/2$ protons and $A/2$ neutrons, with the spins of all protons parallel to one another and antiparallel to the spins of all neutrons. A given nucleon is equally likely to be a p or an n, that is, to have $T_\zeta = 1/2$ or $T_\zeta = -1/2$; the probability that a nucleon pair has $T = 0$ is $1/4$. In this case one nucleon in the pair is a p and the other an n, and hence $S_z = 0$. This value of S_z may arise with equal probability from states having $S = 0$ or $S = 1$. Consequently, the probabilities for the pair to occupy a state with $T = 0$, $S = 0$ or with $T = 0$, $S = 1$ are each $1/4 \cdot 1/2 = 1/8$. The probability for the state $T = 1$, $S = 0$ is the same, while the remaining $5/8$ belongs to the state $T = S = 1$. Taking all this into account, we obtain

$$\frac{1}{8}(U_{00} + U_{01} + U_{10}) + \frac{5}{8}U_{11} > 0. \tag{5}$$

Inequalities (1)–(5) constitute the required system of saturation conditions for the nuclear forces.

§ 118. The shell model

Many nuclear properties admit a good description in terms of the *shell model*, whose basic picture resembles the description of an atom's electron shell. In this approach, each nucleon is treated as moving in the self-consistent field produced by all the remaining nucleons. Because nuclear forces have a short range, this field falls off rapidly beyond the volume enclosed by the nuclear "surface." The state of the entire nucleus is therefore specified by listing the states occupied by its individual nucleons.

The self-consistent field has spherical symmetry, with its center naturally at the nuclear center of mass. This gives rise, however, to a difficulty. In the self-consistent-field method, the system's wave function is constructed as a product of one-particle wave functions (or as a suitably symmetrized sum of such products). A function of this sort does not keep the center of mass at rest: although it yields zero for the mean velocity of the center of mass, it also assigns finite probabilities to nonzero values of that velocity[13].

The difficulty can be circumvented by first removing the motion of the center of mass whenever a physical quantity is evaluated with the self-consistent-field wave functions $\psi(\mathbf{r}_1, \ldots, \mathbf{r}_A)$. Let $f(\mathbf{r}_i, \mathbf{p}_i)$ denote any physical quantity, regarded as a function of the nucleon coordinates and momenta. To evaluate its matrix elements with the functions ψ,

[13]For electrons in an atom this difficulty does not arise at all, since the center of mass is automatically kept at rest by its coincidence with the position of the stationary heavy nucleus.

one leaves $\psi(\mathbf{r}_i)$ unchanged and instead replaces the arguments of f according to

$$\mathbf{r}_i \to \mathbf{r}_i - \mathbf{R}, \qquad \mathbf{p}_i \to \mathbf{p}_i - \frac{\mathbf{P}}{A}, \tag{118.1}$$

where $\mathbf{R}$ is the radius vector of the nuclear center of mass, A is the number of particles in the nucleus, and $\mathbf{P}$ is the momentum of the nucleus as a whole. The second replacement in (118.1) amounts to subtracting the center-of-mass velocity $\mathbf{V}$ from the nucleon velocities, $\mathbf{v}_i \to \mathbf{v}_i - \mathbf{V}$; here $\mathbf{P}$ and $\mathbf{V}$ are related by $\mathbf{P} = Am_p\mathbf{V}$ (S. Gartenhaus, C. Schwartz, 1957).

For example, the nuclear dipole-moment operator is $\hat{\mathbf{d}} = e\sum_p \mathbf{r}_p$, the sum extending over every proton in the nucleus. In self-consistent-field matrix elements, however, it must be replaced by $e\sum_p(\mathbf{r}_p - \mathbf{R})$. The coordinate of the nuclear center is

$$\mathbf{R} = \frac{1}{A}\left(\sum_p \mathbf{r}_p + \sum_n \mathbf{r}_n\right),$$

where the two sums run over all protons and all neutrons, respectively. Since the nucleus contains Z protons, the final replacement of the dipole-moment operator is

$$e\sum_p \mathbf{r}_p \to e\left(1 - \frac{Z}{A}\right)\sum_p \mathbf{r}_p - e\frac{Z}{A}\sum_n \mathbf{r}_n. \tag{118.2}$$

Thus the protons occur with an "effective charge" $e(1 - Z/A)$, while the neutrons have the "charge" $-eZ/A$. As (118.2) shows, the correction terms that enter the dipole moment have relative order of magnitude unity. By contrast, it is readily seen that the corrections to the magnetic moment and to the higher electric multipole moments are of relative order $\sim 1/A$.

In the nonrelativistic approximation, a nucleon's interaction with the self-consistent field is independent of its spin. Such a dependence could only be represented by a term proportional to $\hat{\mathbf{s}}\mathbf{n}$, where $\mathbf{n}$ is the unit vector along the nucleon's radius vector $\mathbf{r}$; this product is a pseudoscalar rather than a true scalar.

Velocity-dependent relativistic terms nevertheless introduce a spin dependence into the nucleon energy. The largest such term is linear in the velocity. The three vectors $\mathbf{s}$, $\mathbf{n}$, and $\mathbf{v}$ can be combined into the true scalar $(\mathbf{n}\times\mathbf{v})\mathbf{s}$. Consequently the operator for a nucleon's *spin–orbit coupling* in the nucleus has the form

$$\hat{V}_{sl} = -\varphi(r)(\mathbf{n}\times\hat{\mathbf{v}})\hat{\mathbf{s}}, \tag{118.3}$$

where $\varphi(r)$ is some function of r (compare also the note on p. 579). Since $m_p(\mathbf{r}\times\mathbf{v})$ is the particle's orbital angular momentum $\hbar\hat{\mathbf{l}}$, (118.3) may also be written

$$\hat{V}_{sl} = -f(r)\hat{\mathbf{l}}\hat{\mathbf{s}}, \tag{118.4}$$

with $f = \hbar\varphi/(rm_p)$. Notice that this interaction is of first order in v/c, whereas the spin–orbit coupling of an atomic electron is a second-order effect (§ 72). The reason for this distinction is that nuclear forces already depend on spin in the nonrelativistic approximation, whereas the nonrelativistic interaction between electrons (the Coulomb forces) has no spin dependence.

The energy of the spin–orbit interaction is concentrated chiefly near the nuclear surface; in other words, $f(r)$ decreases toward the nuclear interior. Indeed, an interaction of this form could not exist at all in unbounded nuclear matter. The homogeneity of such a system leaves no preferred direction along which the vector **n** could point.

The interaction (118.4) splits a nucleon level of orbital angular momentum l into two levels whose angular momenta are $j =$
$l \pm 1/2$. Since

$$\begin{aligned} \mathbf{ls} &= l/2 \qquad &\text{for } j = l + 1/2, \\ \mathbf{ls} &= -(l+1)/2 \quad &\text{for } j = l - 1/2, \end{aligned} \tag{118.5}$$

by (31.3), the magnitude of the splitting is

$$\Delta E = E_{l-1/2} - E_{l+1/2} = \overline{f(r)}(l + 1/2). \tag{118.6}$$

Experiment places the $j = l + 1/2$ level (parallel **l** and **s**) deeper than the $j = l - 1/2$ level. Hence $f(r) > 0$.

A nucleon's spin–orbit coupling within a nucleus is weak relative to its interaction with the self-consistent field. At the same time it is generally larger than the energy of direct interaction between two nucleons in the nucleus, because the latter decreases more rapidly as the atomic weight increases.

This hierarchy of interaction energies requires nuclear levels to be classified by the jj-coupling scheme. For each nucleon its spin and orbital angular momentum combine to form the total angular momentum $\mathbf{j} = \mathbf{l} + \mathbf{s}$. This is a definite quantity because the coupling between **l** and **s** is not destroyed by the direct interaction of the particles with one another (M. Göppert-Mayer, 1949; O. Haxel, J. H. Jensen, H. E. Suess, 1949)[14]. The individual nucleons have angular-momentum vectors **j**. These vectors are then added together. Their sum is the total nuclear angular momentum **J**. This quantity is usually called simply the *nuclear spin*. The name treats the nucleus as though it were an elementary particle. In this respect, nuclear levels and atomic levels are classified quite differently. Consider the electron shell of an atom. Its relativistic spin–orbit coupling is generally weak. One comparison is with direct electric interactions. These interactions are stronger. Exchange interactions are likewise stronger. Atomic levels are therefore usually classified by LS coupling.

Each nucleon state in a nucleus is characterized by its angular momentum j and parity. Although neither **l** nor **s** is separately conserved, the magnitude of the nucleon's orbital angular momentum is still definite. Indeed, a given j can arise either from a state with $l = j - 1/2$ or from one with $l = j + 1/2$. For fixed half-integral j, these two states have opposite parities $(-1)^l$; specifying j and the parity thus fixes the quantum number l as well.

Nucleon states having the same l and j are conventionally numbered, in order of increasing energy, by a "principal quantum number" n that takes integer values beginning

[14] Only in the lightest nuclei is the coupling closer to the LS type.

with 1[15]. The various states are denoted by $1s_{1/2}$, $1p_{1/2}$, $1p_{3/2}$, and so forth. The numeral preceding the letter is the principal quantum number; the letters s, p, d, ... indicate l in the usual way; and the subscript on the letter gives j. A state with specified n, l, and j can contain at most $2j + 1$ neutrons and the same number of protons.

For a specified configuration, the state of the nucleus as a whole is customarily written as a numeral giving J, with a + or − subscript to show the state's parity. In the shell model that parity is determined by whether the algebraic sum of the l values of all nucleons is even or odd.

Analysis of experimental information about nuclear properties reveals a number of regularities in the arrangement of nuclear levels.

First, nucleon-level energies rise as the orbital angular momentum l increases. This rule follows because a larger l increases the particle's centrifugal energy and hence lowers its binding energy.

Second, at fixed l, the level with $j = l + 1/2$ (corresponding to parallel $\mathbf{l}$ and $\mathbf{s}$) lies deeper than the level with $j = l - 1/2$. This rule was already mentioned above in connection with the properties of a nucleon's spin–orbit coupling inside a nucleus.

A further rule concerns nuclear isotopic spin. Recall that the projection T_ζ is fixed by the weight and number of the nucleus (see (116.1)). For a specified T_ζ, the isospin magnitude may assume any value satisfying $T \geqslant |T_\zeta|$. Ordinarily the nuclear ground state has the smallest allowed isospin, namely

$$T_{\text{ground}} = |T_\zeta| = (1/2)(N - Z). \tag{118.7}$$

This rule is tied to the character of the neutron–proton interaction: in an np system, the $T = 0$ state (the deuteron state) has a greater binding energy than the $T = 1$ state (see the note on p. 580).

Some rules can also be stated for the spins of nuclear ground states. They specify how the angular momenta $\mathbf{j}$ of the individual nucleons combine into the total nuclear spin. These rules express the tendency of protons and neutrons that occupy identical nuclear states to form pp and nn pairs whose angular momenta are opposite. The binding energy of such a pair is of order 1–2 MeV.

One consequence is that, if both the proton number and the neutron number are even (an *even–even nucleus*), all nucleon angular momenta cancel in pairs and the total nuclear angular momentum vanishes.

If either the number of protons or the number of neutrons is odd, and all the nucleons outside the filled shells occupy identical states, the total nuclear angular momentum usually equals that of a single nucleon. It is as though, after every possible proton and neutron pair had formed, just one nucleon with uncancelled angular momentum remained. The total angular momentum of any filled shell is automatically zero.

For *odd–odd nuclei*, in which both Z and N are odd, no sufficiently general rule determines the ground-state spin.

A detailed discussion of the particular sequence in which nuclear shells are filled would require extensive analysis of the available experimental data and lies outside the

[15]This differs from the convention for atomic electron levels, in which the allowed values of n begin with $l + 1$.

scope of this book. We shall give only a few further general observations.

In studying atoms, we found that their electron states can be divided into groups such that completing one group and beginning the next reduces an electron's binding energy. Nuclei exhibit an analogous pattern, with nucleon states distributed among the following groups:

$$\begin{array}{ll} 1s_{1/2} & \text{2 nucleons,} \\ 1p_{3/2},\ 1p_{1/2} & \text{6 nucleons,} \\ 1d_{5/2},\ 1d_{3/2},\ 2s_{1/2} & \text{12 nucleons,} \\ 1f_{7/2},\ 2p_{3/2},\ 1f_{5/2},\ 2p_{1/2},\ 1g_{9/2} & \text{30 nucleons,} \\ 2d_{5/2},\ 1g_{7/2},\ 1h_{11/2},\ 2d_{3/2},\ 3s_{1/2} & \text{32 nucleons,} \\ 2f_{7/2},\ 1h_{9/2},\ 1i_{13/2},\ 2f_{5/2},\ 3p_{3/2},\ 3p_{1/2} & \text{44 nucleons.} \end{array} \tag{118.8}$$

For each group the table gives the total number of available proton or neutron places. It follows that one of these groups is completed whenever the total proton number Z, or the total neutron number N, reaches one of the values

$$2,\ 8,\ 20,\ 50,\ 82,\ 126.$$

These are conventionally called *magic numbers*[16].

The so-called *doubly magic nuclei*, for which both Z and N are magic numbers, possess exceptional stability. Relative to neighboring nuclei, their affinity for one additional nucleon is anomalously small, while their first excited levels are anomalously high[17].

Within each group in (118.8), the states have been listed approximately in the sequence in which they are progressively filled along the series of nuclei. In reality, substantial irregularities occur in this filling. It must also be remembered that, in heavy nuclei far from magic numbers, the separations between different levels may be comparable with the "pairing energy." Under these conditions, the very notion of individual states for the two members of a pair largely loses its meaning.

We now make several remarks on calculating a nucleus's magnetic moment in the shell model. By the magnetic moment of a nucleus we naturally mean the moment averaged over the motion of the particles within it. The mean magnetic moment $\overline{\boldsymbol{\mu}}$ is evidently directed along the nuclear spin $\mathbf{J}$, since the latter is the nucleus's only preferred direction. Its operator is therefore

$$\hat{\overline{\boldsymbol{\mu}}} = \mu_0 g \hat{\mathbf{J}}, \tag{118.9}$$

Here μ_0 is the nuclear magneton. The symbol g denotes the gyromagnetic factor. Consider the projection of this moment. Its eigenvalue is $\overline{\mu}_z = \mu_0 g M_J$. Conventionally, μ denotes the nuclear magnetic moment. Compare this usage with (111.1). It is defined as the largest projection value. Thus $\mu = \mu_0 g J$. With this notation, one obtains:

$$\hat{\overline{\boldsymbol{\mu}}} = \mu(\hat{\mathbf{J}}/J). \tag{118.10}$$

[16] The $1f_{7/2}$ states, with their 8 places, are sometimes separated into a group of their own, reflecting the fact that 28 too has, to some extent, the properties of a magic number.

[17] Examples are ${}^4_2He_2$, ${}^{16}_8O_8$, ${}^{40}_{20}Ca_{20}$, and ${}^{208}_{82}Pb_{126}$. The ${}^4He$ nucleus cannot bind an additional nucleon at all. The number at the lower right is the value of N in the nucleus.

The nuclear magnetic moment is composed of the moments of nucleons outside filled shells, since those of nucleons in filled shells cancel one another. Every nucleon contributes a magnetic moment having two parts: a spin part and, for a proton, an orbital part. It is thus expressed as the sum $g_s\hat{\mathbf{s}} + g_l\hat{\mathbf{l}}$. (Here and below we omit the factor μ_0. We follow the usual convention. Magnetic moments are understood to be measured in units of the nuclear magneton.) The orbital and spin gyromagnetic factors are $g_l = 1$, $g_s = 5{,}585$ for a proton, and $g_l = 0$, $g_s = -3{,}826$ for a neutron.

After averaging over a nucleon's motion in the nucleus, its magnetic moment is proportional to $\mathbf{j}$. Writing it as $g_j\hat{\mathbf{j}}$, we have

$$g_j\hat{\mathbf{j}} = g_s\hat{\mathbf{s}} + g_l\hat{\mathbf{l}} = \frac{1}{2}(g_l + g_s)\hat{\mathbf{j}} + \frac{1}{2}(g_l - g_s)(\hat{\mathbf{l}} - \hat{\mathbf{s}}).$$

Multiplying both sides by $\hat{\mathbf{j}} = \hat{\mathbf{l}} + \hat{\mathbf{s}}$ and passing to the eigenvalues gives

$$g_j j(j+1) = \frac{1}{2}(g_l + g_s)j(j+1) + \frac{1}{2}(g_l - g_s)[l(l+1) - s(s+1)].$$

On setting $s = 1/2$ and $j = l \pm 1/2$, we obtain

$$g_j = g_l \pm \frac{g_s - g_l}{2l+1} \quad \text{for } j = l \pm 1/2. \tag{118.11}$$

With the gyromagnetic factors quoted above, the proton magnetic moment $\mu_p = g_j j$ is

$$\begin{aligned} \mu_p &= \left(1 - \frac{2{,}29}{j+1}\right) j \quad && \text{for } j = l - 1/2, \\ \mu_p &= j + 2{,}29 && \text{for } j = l + 1/2, \end{aligned} \tag{118.12}$$

and the neutron magnetic moment is

$$\begin{aligned} \mu_n &= \frac{1{,}91}{j+1} j \quad && \text{for } j = l - 1/2, \\ \mu_n &= -1{,}91 && \text{for } j = l + 1/2. \end{aligned} \tag{118.13}$$

(T. Schmidt, 1937).

If only one nucleon lies outside the filled shells, (118.12) or (118.13) directly supplies the nuclear magnetic moment. The addition of the magnetic moments is also elementary for two nucleons (see Problem 1). With more nucleons, the magnetic moment must be averaged using a system wave function properly constructed from the individual nucleon wave functions. The nucleon configuration together with the state of the whole nucleus determines this construction uniquely whenever the configuration admits only a single system state with the specified J and T (see, for example, Problem 3). Otherwise the nuclear state is a mixture of several independent states with the same J and T, and the coefficients in the linear combination that forms the nuclear wave function are generally unknown[18].

[18]In practice, however, the accuracy of the "one-particle" scheme for calculating nuclear magnetic moments is not high. The paired values in (118.12) and (118.13) serve more as upper and lower bounds than as exact moment values.

Finally, the presence of nucleon spin–orbit coupling in a nucleus gives the protons an additional magnetic moment beyond (118.9) (M. Göppert-Mayer, J. H. Jensen, 1952). The reason is that, when an interaction operator depends explicitly on a particle's velocity, an external field is introduced by the replacement $\hat{\mathbf{p}} \to \hat{\mathbf{p}} - (e/c)\mathbf{A}$. Making this replacement in (118.3), and using (111.7) for the vector potential, produces the following additional term in the proton Hamiltonian:

$$\varphi(r)\frac{e}{cm_p}(\mathbf{n}\times\mathbf{A})\hat{\mathbf{s}} = f(r)\frac{e}{2c\hbar}\{\mathbf{r}\times(\mathbf{H}\times\mathbf{r})\}\hat{\mathbf{s}} = f(r)\frac{e}{2c\hbar}\{\mathbf{r}\times(\hat{\mathbf{s}}\times\mathbf{r})\}\mathbf{H}.$$

This term is equivalent to an additional magnetic moment whose operator is

$$\hat{\boldsymbol{\mu}}_{\text{add}} = -\frac{e}{2c\hbar}f(r)\{\mathbf{r}\times(\hat{\mathbf{s}}\times\mathbf{r})\} = -\frac{e}{2c\hbar}r^2 f(r)\{\hat{\mathbf{s}} - (\hat{\mathbf{s}}\mathbf{n})\mathbf{n}\}. \tag{118.14}$$

Problem

1. Find the magnetic moment of a two-nucleon system, whose total mechanical angular momentum is $\mathbf{J} = \mathbf{j}_1 + \mathbf{j}_2$, expressing it in terms of the individual nucleon magnetic moments μ_1 and μ_2.

Solution. A derivation analogous to that of (118.11) gives

$$\frac{\mu}{J} = \frac{1}{2}\left(\frac{\mu_1}{j_1}+\frac{\mu_2}{j_2}\right) + \frac{1}{2}\left(\frac{\mu_1}{j_1}-\frac{\mu_2}{j_2}\right)\frac{(j_1-j_2)(j_1+j_2+1)}{J(J+1)}.$$

Problem

2. Determine the possible states of three nucleons with angular momenta $j = 3/2$ and identical principal quantum numbers.

Solution. We proceed as in § 67 when the possible states of equivalent electrons were found. Each nucleon may occupy one of eight states, characterized by the following pairs (m_j, τ_ζ):

$$\begin{gathered}(3/2, 1/2),\quad (1/2, 1/2),\quad (-1/2, 1/2),\quad (-3/2, 1/2),\\ (3/2, -1/2),\quad (1/2, -1/2),\quad (-1/2, -1/2),\quad (-3/2, -1/2).\end{gathered}$$

Combining three distinct states at a time, we obtain the following pairs (M_J, T_ζ) for the three-nucleon system:

$$(7/2, 1/2),\quad 2(5/2, 1/2),\quad (3/2, 3/2),\quad 4(3/2, 1/2),\quad (1/2, 3/2),\quad 5(1/2, 1/2).$$

The numeral preceding a pair gives the number of corresponding states; states with negative M_J and T_ζ need not be listed. These pairs correspond to system states with the following (J, T) values:

$$(7/2, 1/2),\quad (5/2, 1/2),\quad (3/2, 3/2),\quad (3/2, 1/2),\quad (1/2, 1/2).$$

Problem

3. Taking isotopic invariance into account, determine the magnetic moment of the ground state for a configuration consisting of two neutrons and one proton in $p_{3/2}$ states with the same n[19].

Solution. The ground state of this configuration has $J = 3/2$, while the rule stated in the text gives its isospin the smallest possible value, $T = |T_\zeta| = 1/2$.

We determine the system wave function belonging to the largest possible value $M_J = 3/2$. Subject to the Pauli-principle requirements for the two identical nucleons, this M_J value may be realized by the following triples of m_j values for the nucleons p, n, n, respectively:

$$(3/2, 3/2, -3/2), \quad (3/2, 1/2, -1/2), \quad (1/2, 3/2, -1/2), \quad (1/2, 3/2, 1/2).$$

The required wave function $\Psi^{JM_J}_{TT_\zeta}$ is therefore a linear combination of the form

$$\begin{aligned}\Psi^{3/2\;3/2}_{1/2\;-1/2} &= a[\psi^{3/2}_{1/2}\psi^{3/2}_{-1/2}\psi^{-3/2}_{-1/2}] + b[\psi^{3/2}_{1/2}\psi^{1/2}_{-1/2}\psi^{-1/2}_{-1/2}]\\ &\quad + c[\psi^{-3/2}_{-1/2}\psi^{1/2}_{1/2}\psi^{-1/2}_{-1/2}] + d[\psi^{3/2}_{-1/2}\psi^{1/2}_{-1/2}\psi^{-1/2}_{1/2}],\end{aligned} \tag{1}$$

where $[\ldots]$ denotes a normalized antisymmetrized product, that is, a determinant of the form (61.5), of the individual nucleon wave functions $\psi^{m_j}_{\tau_\zeta}$.

The function (1) must vanish under the action of each of the operators

$$\hat{T}_- = \sum_{i=1}^{3} \hat{\tau}_-^{(i)} \quad \text{and} \quad \hat{J}_+ = \sum_{i=1}^{3} \hat{j}_+^{(i)}$$

(compare the problem in § 67). The operators $\hat{\tau}_-^{(i)}$ turn the proton function of the ith nucleon into a neutron function (and annihilate a neutron function). It is therefore readily seen that $\hat{T}_-$ turns the first term in (1) into a determinant with two identical rows and hence into zero, while the determinants from the other three terms become identical. We thus obtain the condition $b + c + d = 0$. Next, for a single nucleon with $j = 3/2$ and the various m_j values, (27.12) gives

$$\hat{j}_+\psi^{3/2} = 0, \quad \hat{j}_+\psi^{1/2} = \sqrt{3}\psi^{3/2}, \quad \hat{j}_+\psi^{-1/2} = 2\psi^{1/2}, \quad \hat{j}_+\psi^{-3/2} = \sqrt{3}\psi^{-1/2}.$$

It then follows that applying $\hat{J}_+$ to (1) yields

$$\hat{J}_+\Psi^{3/2\;3/2}_{1/2\;-1/2} = \sqrt{3}(a + b - c)[\psi^{3/2}_{1/2}\psi^{3/2}_{-1/2}\psi^{-1/2}_{-1/2}] + 2(c - d)[\psi^{3/2}_{-1/2}\psi^{1/2}_{1/2}\psi^{1/2}_{-1/2}].$$

The sign changes in some terms result from interchanging rows of the determinant. Requiring this expression to vanish gives

$$a + b - c = 0, \qquad c - d = 0.$$

[19]The ^{7}Li nucleus has this configuration outside the filled shell $(1s_{1/2})^4$.

Together with normalization of (1), these relations yield

$$a = \frac{3}{\sqrt{15}}, \qquad b = -\frac{2}{\sqrt{15}}, \qquad c = d = \frac{1}{\sqrt{15}}.$$

The mean value of the proton (or neutron) magnetic-moment projection in a state with given m_j is $\mu_p m_j/j$ (or $\mu_n m_j/j$). The mean system moment calculated with (1) is consequently

$$\overline{\mu} = \overline{\mu}_z = \frac{9}{15}\mu_p + \frac{4}{15}\mu_p + \frac{1}{15}\left(\frac{1}{3}\mu_p + \frac{2}{3}\mu_n\right) + \frac{1}{15}\left(-\frac{1}{3}\mu_p + \frac{3}{4}\mu_n\right) = \frac{1}{15}(13\mu_p + \mu_n).$$

Equations (118.12) and (118.13) give, for a nucleon in a $p_{3/2}$ state, $\mu_n = -1{,}91$ and $\mu_p = 3{,}79$. The result is $\mu = 3{,}03$.

§ 119. Nonspherical nuclei

A system of particles moving in a spherically symmetric field cannot possess a rotational energy spectrum; indeed, in quantum mechanics the very notion of rotation has no meaning for such a system. This applies also to the shell model of the nucleus considered in the preceding section, with its spherically symmetric self-consistent field.

In quantum mechanics, a rigorous division of a system's energy into internal and rotational parts has no meaning in general. Such a division can only be approximate, and is possible when physical circumstances make it a good approximation to regard the system as a collection of particles moving in a prescribed field that lacks spherical symmetry. The rotational structure of the levels then arises when allowance is made for this field to rotate relative to a fixed coordinate system. We encountered such a case, for example, in molecules, whose electronic terms may be defined as the energy levels of the electron system moving in the prescribed field of fixed nuclei.

Experiment shows that most nuclei do not, in fact, have a rotational structure. For them, therefore, a spherically symmetric self-consistent field is a good approximation: apart from quantum fluctuations, these nuclei are spherical.

There is, however, a class of nuclei whose energy spectrum is of rotational type (this class includes nuclei in the approximate atomic-weight ranges $150 < A < 190$ and $A > 220$). This property means that the approximation of a spherically symmetric self-consistent field is wholly unsuitable for them. In principle, their self-consistent field must be sought without any prior assumption about its symmetries, so that the shape of the nucleus is itself determined "self-consistently." Experiment indicates that the appropriate model for nuclei in this class is a self-consistent field having an axis of symmetry and a symmetry plane perpendicular to it (that is, the symmetry of an ellipsoid of revolution). The conception of nonspherical nuclei was developed most fully in the work of *A. Bohr and B. R. Mottelson* (1952–1953).

We emphasize that these are two qualitatively different classes of nuclei. One manifestation of this distinction is that a nucleus is either spherical or nonspherical with a degree of nonsphericity that is by no means small.

Unfilled shells in a nucleus promote the appearance of nonsphericity; nucleon pairing also appears to play an important part in this phenomenon. Closed shells, by contrast, favor a spherical nucleus. A characteristic example is the doubly magic nucleus $^{208}_{82}\mathrm{Pb}$. Because its nucleon configuration is very strongly closed, this nucleus (and nuclei near it) is spherical, thereby producing the interruption in the sequence of nonspherical heavy nuclei.

The energy levels of a nonspherical nucleus are represented as the sum of two parts: the levels of the "stationary" nucleus and the energy of its rotation as a whole. In even–even nuclei, the intervals within this rotational structure are small compared with the separations between levels of the "stationary" nucleus.

The classification of the levels of a nonspherical nucleus is in many respects analogous to the classification of the levels of a diatomic molecule composed of identical atoms, since in the two cases the fields in which the particles (nucleons or electrons) move have the same symmetry. We may therefore use directly a number of the results obtained in Chapter XI[20].

Let us first consider the classification of states of the "stationary nucleus." In an axially symmetric field, only the projection of the angular momentum on the symmetry axis is conserved. Each nuclear state is consequently characterized first of all by the magnitude Ω of the projection of its total angular momentum[21]; this magnitude may be integral or half-integral. According to the behavior of the wave function when the signs of the coordinates of all nucleons relative to the nuclear center are reversed, the levels are divided into even (g) and odd (u) levels.

For $\Omega = 0$ a further distinction is made between positive and negative states, according to the behavior of the wave function under reflection in a plane through the nuclear axis (see § 78).

The ground states of nonspherical even–even nuclei are 0_g states (the numeral gives the value of Ω), corresponding to zero angular momentum and to the highest symmetry of the wave function. This is a consequence of the pairwise pairing of all the neutrons and all the protons. If, however, a nucleus contains an odd number of protons or neutrons, one may consider the state of the "odd" nucleon in the self-consistent field of the even–even nuclear "core."

The value of Ω is then determined by the projection of this nucleon's angular momentum. Similarly, in an odd–odd nucleus Ω is obtained from the angular-momentum projections of the odd neutron and odd proton ($\Omega = |\omega_p \pm \omega_n|$).

At the same time, it must be stressed that definite values cannot be assigned to the projections of the nucleon's orbital angular momentum and spin.

Although the spin–orbit coupling of a nucleon is small compared with the energy of its interaction with the self-consistent field of the core, it is not small compared with the spacings between neighboring energy levels of the nucleon in that field. Yet precisely this latter condition would be required for perturbation theory to apply and permit the nucleon's orbital angular momentum and spin to be treated separately to a

[20]We stress that the analogy concerns the classification of the levels of a diatomic molecule, not of a symmetric top. For particles moving in an axially symmetric field, rotation about the field axis has no meaning, just as rotation about any axis has no meaning for a system in a centrally symmetric field.

[21]By definition, $\Omega \geqslant 0$ (as with the positive quantum number Λ in diatomic molecules). Recall that negative values of Ω for diatomic molecules could occur only because Ω was defined as the sum $\Lambda + \Sigma$, while Σ could be positive or negative according to the relative directions of orbital angular momentum and spin.

good approximation[22].

We now turn to the rotational structure of the levels of a nonspherical nucleus. Its intervals are small compared with the spin–orbit interaction of the nucleons in the nucleus; this situation corresponds to case *a* in the theory of diatomic molecules (see § 83).

The total angular momentum $\mathbf{J}$ of the rotating nucleus is, of course, conserved. For a specified Ω, its magnitude J takes the sequence of values beginning with Ω:

$$J = \Omega, \quad \Omega + 1, \quad \Omega + 2, \ldots \tag{119.1}$$

(see (83.2)). There is an additional restriction on the possible values of J for nuclei with $\Omega = 0$: in the states 0_g^+ and 0_u^+, J takes only even values, whereas in 0_g^- and 0_u^- it takes odd values (see § 86). In particular, for the rotational levels of the ground term of an even–even nucleus (0_g^+), J takes the values $0, 2, 4, \ldots$

The rotational energy of the nucleus is

$$E_{\text{rot}} = \frac{\hbar^2}{2I} J(J+1), \tag{119.2}$$

where I is the moment of inertia of the nucleus about an axis perpendicular to its symmetry axis. This formula corresponds to the analogous expression in the theory of diatomic molecules (the term depending

on J in (83.6)). The lowest level corresponds to the least possible value of J, namely $J = \Omega$.

By (119.2), the rotational level structure obeys definite interval rules that, for a specified Ω, are independent of the other characteristics of the level. Thus the components of the rotational structure of the ground term of an even–even nucleus (with $J = 2, 4, 6, 8, \ldots$) lie above the lowest level ($J = 0$) by intervals whose ratios are $1 : 3$, $3 : 7 : 12 \ldots$

Formula (119.2), however, is insufficient for states with $\Omega = 1/2$, which may occur in nuclei containing an odd number of nucleons.

In this case the interaction of the odd nucleon with the centrifugal field of the rotating nucleus makes a contribution to the energy comparable with (119.2). Its dependence on J can be found as follows.

As is known from mechanics (see Volume I, § 39), the energy of a particle in a rotating coordinate system contains an additional term equal to the product of the angular velocity and the particle's angular momentum. The corresponding term in the nuclear Hamiltonian may be written as $2b\hat{\mathbf{K}}\hat{\boldsymbol{\sigma}}$, where b is a constant, $\hat{\mathbf{K}}$ is the rotational angular momentum of the nuclear core (the nucleus without the last nucleon), and $\hat{\boldsymbol{\sigma}}$ is the angular momentum of the nucleon. The latter is to be understood here in a purely formal sense (in reality, an angular-momentum vector of a nucleon in an axial nuclear field does not exist): it is an operator analogous to the spin-1/2 operator, producing transitions between states whose angular-momentum projections are $\pm 1/2$, in accordance with $\Omega = 1/2$[23].

[22] In spherical nuclei it was nevertheless possible to determine the value of l by using conservation of parity together with conservation of angular momentum.

[23] The special feature of the case $\Omega = 1/2$ is precisely the existence of matrix elements of the energy

Since $\mathbf{K} = \mathbf{J} - \boldsymbol{\sigma}$, the eigenvalues of this operator are

$$2b\mathbf{K}\boldsymbol{\sigma} = b\left[J(J+1) - K(K+1) - \frac{3}{4}\right].$$

Adding, for convenience, the J-independent constant $b/2$, we find this quantity to be $\pm b(J + 1/2)$ when $J = K \pm 1/2$.

This expression can be written $(-1)^{J-1/2}b(J + 1/2)$ on noting that the angular momentum K of the core, which is an even–even nucleus, is an even integer. Thus,

the final expression for the rotational energy of a nucleus with $\Omega = 1/2$ is

$$E_{\text{rot}} = \frac{\hbar^2}{2I}J(J+1) + (-1)^{J-1/2}b\left(J + \frac{1}{2}\right) \tag{119.3}$$

(A. Bohr and B. Mottelson, 1953). Notice that if b is positive and sufficiently large, the level with $J = 3/2$ can lie below that with $J = 1/2$. The normal ordering of rotational levels, in which the lowest level has the least possible J, may therefore be violated.

The moment of inertia of a nonspherical nucleus cannot be calculated as that of a rigid body of prescribed shape. Such a calculation would be possible only if the nucleons moving in the nuclear self-consistent field could be regarded as having no direct interactions with one another. In reality, pairing reduces the moment of inertia below the value for a rigid body.

The magnetic moment $\boldsymbol{\mu}$ of a nonspherical nucleus consists of the magnetic moment of the "stationary" nucleus and a moment associated with the rotation of the nucleus. After averaging over the motion of the nucleons in the nucleus, the first is directed along the nuclear axis. Denoting its magnitude by μ' and the unit vector along that axis by $\mathbf{n}$, we write it as $\mu'\mathbf{n}$. The magnetic moment due to rotation, after the same averaging, is directed along $\mathbf{J} - \Omega\mathbf{n}$, the total mechanical angular momentum of the nucleus less the angular momentum of the nucleons in the "stationary nucleus"[24].

Thus

$$\boldsymbol{\mu} = \mu'\mathbf{n} + g_r(\mathbf{J} - \Omega\mathbf{n}). \tag{119.4}$$

Here g_r is the gyromagnetic factor for nuclear rotation. Since only the protons contribute to the magnetic moment arising from rotation,

$$g_r = \frac{I_p}{I_p + I_n}, \tag{119.5}$$

where I_n and I_p are the neutron and proton parts of the nuclear moment of inertia (for a system consisting solely of protons one would simply have $g_r = 1$). In general, the ratio (119.5) does not equal the ratio Z/A of the number of protons to the total nuclear mass.

After averaging over the nuclear rotation, the magnetic moment is directed along the conserved vector $\mathbf{J}$:

$$\hat{\boldsymbol{\mu}} = \frac{\mu}{J}\hat{\mathbf{J}} = (\mu' - \Omega g_r)\overline{\mathbf{n}} + g_r\hat{\mathbf{J}}.$$

perturbation for transitions between states that differ only in the sign of the angular-momentum projection and therefore have the same energy. An energy shift consequently appears already in first-order perturbation theory. This phenomenon is analogous to the Λ-doubling of the levels of a diatomic molecule with $\Omega = 1/2$ (see § 88).

[24]This form can be used only when $\Omega \neq 1/2$ (see Problem 2).

As usual, we multiply both sides of this equality by $\hat{\mathbf{J}}$ and pass to the eigenvalues. In the nuclear ground state $\Omega = J$, and we obtain

$$\mu = (\mu' + g_r)\frac{J}{J+1}. \tag{119.6}$$

PROBLEM

1. Express the quadrupole moment Q of a rotating nucleus in terms of the quadrupole moment Q_0 relative to axes fixed in the nucleus (A. Bohr, 1951).

SOLUTION. The quadrupole-moment tensor operator of the rotating nucleus is expressed in terms of Q_0 as

$$Q_{ik} = \frac{3}{2}Q_0\left(n_i n_k - \frac{1}{3}\delta_{ik}\right);$$

this is a symmetric tensor of zero trace, constructed from the components of the unit vector $\mathbf{n}$ along the nuclear axis, with $Q_{zz} = Q_0$. Averaging over the rotational state of the nucleus proceeds as in the solution of the problem to § 29, except that $n_i J_i = \Omega$ rather than zero, and gives an expression of the form (75.2), with

$$Q = Q_0\frac{3\Omega^2 - J(J+1)}{(2J+3)(J+1)}.$$

For the nuclear ground state with $\Omega = J$, this becomes

$$Q = Q_0\frac{(2J-1)J}{(2J+3)(J+1)}.$$

As J increases, the ratio Q/Q_0 tends to unity, though rather slowly.

PROBLEM

2. Determine the magnetic moment in the ground state of a nucleus with $\Omega = 1/2$.

SOLUTION. In this case the magnetic-moment operator may be written, using the operator $\hat{\boldsymbol{\sigma}}$ introduced in the text, as

$$\hat{\boldsymbol{\mu}} = 2\mu'\hat{\boldsymbol{\sigma}} + g_r\hat{\mathbf{K}}, \qquad \hat{\mathbf{K}} = \hat{\mathbf{J}} - \hat{\boldsymbol{\sigma}}.$$

The remaining calculation is analogous to that carried out in the text. If the nuclear ground level has $J = 1/2$ (so that $K = J - 1/2 = 0$), the result is $\mu = \mu'$. If instead the ground state has $J = 3/2$ (so that $K = J + 1/2 = 2$), then $\mu = \frac{9}{5}g_r - \frac{3}{5}\mu'$.

PROBLEM

3. Determine the energies of the first few levels in the rotational structure of the ground state of an even–even nucleus having the symmetry of a triaxial ellipsoid.

SOLUTION. The ground state of an even–even nucleus corresponds to the most symmetric wave function of the "stationary" nucleus, namely a function with the symmetry belonging to the representation A_1 of the group D_2. For a specified J, there are therefore only $(J/2)+1$ distinct levels when J is even, or $(J-1)/2$ when J is odd. For $J=2$ they are given by formula (7) obtained in the problems to § 103, and for $J=3$ by formula (8).

§ 120. Isotope shift

Properties specific to the nucleus—its finite mass, dimensions, and spin— distinguish it from a fixed point centre of a Coulomb field and have a definite effect on the electronic energy levels of an atom.

One such effect is the so-called *isotope shift* of the levels: the change in a level's energy on passing from one isotope of an element to another. What is of physical interest is, of course, not the change in a single level but the change in the separation of two levels, which is observed as a spectral line. Consequently, it is not the energy of the entire electronic shell that must be considered, but only the part of it associated with the electron taking part in the spectral transition in question.

For light atoms the chief origin of the isotope shift is the finiteness of the nuclear mass. Allowance for the motion of the nucleus adds to the atomic Hamiltonian the term

$$\frac{1}{2M}\left(\sum_i \hat{\mathbf{p}}_i\right)^2,$$

where M is the mass of the nucleus and $\mathbf{p}_i$ are the electron momenta[25]. The isotope shift due to this effect is thus the mean value

$$\frac{1}{2}\left(\frac{1}{M_1}-\frac{1}{M_2}\right)\overline{\left(\sum_i \mathbf{p}_i\right)^2}, \tag{120.1}$$

evaluated with the wave function of the atomic state concerned; M_1 and M_2 are the nuclear masses of the two isotopes.

For heavy atoms the principal contribution to the isotope shift comes from the finite extent of the nucleus. In practice, this effect is appreciable only for levels of an outer electron in an s-state. Indeed, unlike the wave functions of states with $l \neq 0$, the s-state wave function does not vanish as $r \to 0$, so the probability of finding the electron within

[25] In the atom's centre-of-mass frame, the momenta of the nucleus and the electrons have zero sum: $\mathbf{p}_{\text{nuc}} + \sum_i \mathbf{p}_i = 0$. Their total kinetic energy is therefore

$$\frac{\mathbf{p}_{\text{nuc}}^2}{2M} + \frac{1}{2m}\sum_i \mathbf{p}_i^2 = \frac{1}{2M}\left(\sum_i \mathbf{p}_i\right)^2 + \frac{1}{2m}\sum_i \mathbf{p}_i^2.$$

the "volume of the nucleus" is comparatively large. We shall calculate the isotope shift in this case[26] .

Let $\varphi(r)$ denote the actual electrostatic potential of the nuclear field, as distinct from the Coulomb potential Ze/r of a point charge Ze. The resulting change in the electron's energy relative to its value in the pure Coulomb field Ze/r is

$$\Delta E = -e\int\left(\varphi - \frac{Ze}{r}\right)\psi^2(r)\,dV, \tag{120.2}$$

where $\psi(r)$ is the electron wave function (for an s-state it is real and spherically symmetric). Although the integral is formally over all space, the difference $\varphi - Ze/r$ in its integrand is nonzero only inside the nuclear volume.

On the other hand, the wave function of an s-state tends to a constant as $r \to 0$ (see § 32), and this limiting value is already reached, to sufficient accuracy, outside the nucleus. We may therefore take ψ^2 outside the integral, replacing $\psi(r)$ by its value at $r = 0$ as calculated for the Coulomb field of a point charge.

To transform the remaining integral, use the identity $\Delta r^2 = 6$ and rewrite (120.2) as

$$\Delta E = -\frac{1}{6}e\psi^2(0)\int\left(\varphi - \frac{Ze}{r}\right)\Delta r^2\,dV = -\frac{1}{6}e\psi^2(0)\int r^2\Delta\left(\varphi - \frac{Ze}{r}\right)dV.$$

In this transformation of the volume integral, the surface integral at infinity has been set equal to zero. Now $\Delta(1/r) = -4\pi\delta(\mathbf{r})$, while $r^2\delta(\mathbf{r}) = 0$ for all r. The electrostatic Poisson equation gives $\Delta\varphi = -4\pi\rho$, where here ρ is the density of electric charge in the nucleus. It follows finally that

$$\Delta E = \frac{2\pi}{3}\psi^2(0)Ze^2\overline{r^2}, \tag{120.3}$$

where

$$\overline{r^2} = \frac{1}{Ze}\int\rho r^2\,dV$$

is the nuclear proton mean-square radius. (For a uniform distribution of protons in the nucleus it would be $\overline{r^2} = 3R^2/5$, with R the geometrical nuclear radius.) The isotope shift of a level is determined by the difference between (120.3) for the two isotopes.

In § 71 the magnitude of $\psi(0)$ was estimated and found to depend on the atomic number (assumed large) as $\sqrt{Z}$. The splitting in (120.3) is therefore proportional to R^2Z^2.

§ 121. Hyperfine structure of atomic levels

Another atomic effect associated with specific properties of the nucleus is the splitting of atomic energy levels through the interaction of the electrons with the nuclear spin. This is the so-called *hyperfine structure* of the levels. Since this interaction is weak, the intervals of the resulting structure are very small, even relative to the fine-structure intervals.

[26]The calculation below does not include relativistic effects. The omitted effects concern the electron's motion near the nucleus. It is valid under the condition $Ze^2/(\hbar c) \ll 1$.

The hyperfine structure must therefore be considered separately for each fine-structure component.

In this section we denote the nuclear spin by **i** (in accordance with the convention of atomic spectroscopy), while retaining **J** for the total angular momentum of the atomic electron shell. Let the total angular momentum of the atom, including the nucleus, be $\mathbf{F} = \mathbf{J} + \mathbf{i}$. Each hyperfine-structure component is characterized by a definite value of this angular momentum. From the general rules for addition of angular momenta, the quantum number F takes the values

$$F = J + i, \quad J + i - 1, \ldots, |J - i|, \tag{121.1}$$

so that every level of given J splits into $2i + 1$ components if $i < J$, or into $2J + 1$ components if $i > J$.

Because the mean electron distances r in an atom are large compared with the nuclear radius R, the principal contributions to hyperfine splitting come from the interactions of the electrons with the lowest-order nuclear multipole moments. These are the magnetic dipole and electric quadrupole moments (the mean electric dipole moment vanishes; see § 75).

The nuclear magnetic moment is of order $\mu_{\text{nuc}} \sim eRv_{\text{nuc}}/c$, where v_{nuc} denotes the velocities of nucleons in the nucleus. Its interaction energy with the electron magnetic moment ($\mu_{\text{el}} \sim e\hbar/mc$) is of order

$$\frac{\mu_{\text{nuc}}\mu_{\text{el}}}{r^3} \sim \frac{e^2\hbar}{mc^2}\frac{Rv_{\text{nuc}}}{r^3}. \tag{121.2}$$

The nuclear quadrupole moment is $Q \sim eR^2$; the interaction energy of its field with the electron charge is of order

$$\frac{eQ}{r^3} \sim \frac{e^2R^2}{r^3}. \tag{121.3}$$

Comparison of (121.2) and (121.3) shows that the magnetic interaction, and hence the level splitting that it produces, exceeds the quadrupole interaction by a factor $(v_{\text{nuc}}/c)(\hbar/mcR) \sim 15$. Although v_{nuc}/c is comparatively small, $\hbar/mcR$ is large.

The operator for the magnetic interaction of the electrons with the nucleus has the form

$$\hat{V}_{iJ} = a\hat{\mathbf{i}}\hat{\mathbf{J}} \tag{121.4}$$

(in analogy with the electron spin–orbit interaction (72.4)). Its dependence on F, and consequently that of the splitting which it produces, is

$$\frac{a}{2}F(F+1) \tag{121.5}$$

(cf. (72.5)).

The quadrupole interaction operator is formed from the operator $\hat{Q}_{ik}$ of the nuclear quadrupole-moment tensor and the components of the electron angular momentum $\hat{\mathbf{J}}$. It is proportional to the scalar $\hat{Q}_{ik}\hat{J}_i\hat{J}_k$ constructed from these operators, and thus has the form

$$b\left[\hat{i}_i\hat{i}_k + \hat{i}_k\hat{i}_i - \frac{2}{3}i(i+1)\delta_{ik}\right]\hat{J}_i\hat{J}_k; \tag{121.6}$$

This result uses the fact that Q_{ik} can be expressed through the nuclear-spin operator by a formula of the form (75.2). Evaluating the eigenvalues of operator (121.6)—by exactly the same procedure as in Problem 1 of § 84—gives the following dependence of the quadrupole hyperfine level splitting on the quantum number F:

$$\frac{b}{2}F^2(F+1)^2 + \frac{b}{2}F(F+1)[1 - 2J(J+1) - 2i(i+1)]. \tag{121.7}$$

Magnetic hyperfine splitting is especially pronounced for levels associated with an outer electron in an s state, because such an electron has a comparatively high probability of being near the nucleus.

We calculate the hyperfine splitting for an atom having one outer s electron (E. Fermi, 1930). Its motion in the self-consistent field of the remaining electrons and the nucleus is described by a spherically symmetric wavefunction $\psi(r)$[27].

We seek the electron–nucleus interaction operator as the energy operator $-\hat{\boldsymbol{\mu}}\hat{\mathbf{H}}$ of the nuclear magnetic moment $\hat{\boldsymbol{\mu}} = \mu\hat{\mathbf{i}}/i$ in the magnetic field produced by the electron at the origin. The familiar formula of electrodynamics gives this field as

$$\hat{\mathbf{H}} = \frac{1}{c}\int \frac{\mathbf{n}\times\hat{\mathbf{j}}}{r^2}\,dV, \tag{121.8}$$

where $\hat{\mathbf{j}}$ is the current-density operator produced by the moving electron spin, while $\mathbf{r} = \mathbf{n}r$ is the radius vector from the centre to the volume element dV[28]. According to (115.4),

$$\hat{\mathbf{j}} = -2\mu_B c\,\nabla\times\psi^2\hat{\mathbf{s}} = -2\mu_B c\frac{d\psi^2(r)}{dr}\,\mathbf{n}\times\hat{\mathbf{s}}$$

(μ_B is the Bohr magneton). Writing $dV = r^2\,dr\,d\Omega$ and performing the integrations, we find

$$\hat{\mathbf{H}} = -2\mu_B\int_0^\infty \frac{d\psi^2}{dr}\,dr\int \mathbf{n}\times(\mathbf{n}\times\hat{\mathbf{s}})\,d\Omega = -2\mu_B\psi^2(0)\frac{8\pi}{3}\hat{\mathbf{s}}.$$

Thus the interaction operator is

$$\hat{V}_{is} = -\hat{\boldsymbol{\mu}}\hat{\mathbf{H}} = \frac{16\pi}{3i}\mu\mu_B\psi^2(0)\hat{\mathbf{i}}\hat{\mathbf{s}}. \tag{121.9}$$

If the total atomic angular momentum is $J = S = 1/2$, the hyperfine splitting produces a doublet, with $F = i \pm 1/2$. Equations (121.5) and (121.9) give the separation of its two levels as

$$E_{i+1/2} - E_{i-1/2} = \frac{8\pi}{3i}\mu\mu_B(2i+1)\psi^2(0). \tag{121.10}$$

Since $\psi(0)$ is proportional to $\sqrt{Z}$ (see § 71), this splitting increases in proportion to the atomic number.

[27] The calculation below assumes that $Ze^2/(\hbar c) \ll 1$ (cf. the note on p. 601).

[28] See II, § 43, formula (43.7). There the vector $\mathbf{R}$ points in the opposite direction, from dV to the centre (the point at which the field is observed).

Problem

1. Calculate the hyperfine splitting due to the magnetic interaction for an atom having, outside closed shells, one electron with orbital angular momentum l (E. Fermi, 1930).

Solution. Vector potential and magnetic-field intensity produced by the nuclear magnetic moment $\boldsymbol{\mu}$ are

$$\mathbf{A} = \frac{\boldsymbol{\mu} \times \mathbf{n}}{r^2}, \qquad \mathbf{H} = \frac{3\mathbf{n}(\boldsymbol{\mu}\mathbf{n}) - \boldsymbol{\mu}}{r^3}$$

$(\nabla \cdot \mathbf{A} = 0)$. These expressions give the interaction operator in the form

$$\frac{|e|}{mc}\mathbf{A}\hat{\mathbf{p}} + \frac{|e|\hbar}{mc}\hat{\mathbf{H}}\hat{\mathbf{s}} = \frac{2\mu_B}{r^3}\hat{\boldsymbol{\mu}}\left[\hat{\mathbf{l}} + 3(\hat{\mathbf{s}}\mathbf{n})\mathbf{n} - \hat{\mathbf{s}}\right].$$

After averaging over a state of specified j, the expression in square brackets is directed along $\mathbf{j}$. We may therefore write

$$\hat{V}_{ij} = 2\mu_B(\hat{\boldsymbol{\mu}}\hat{\mathbf{j}})[\hat{\mathbf{l}}\hat{\mathbf{j}} + 3(\hat{\mathbf{s}}\mathbf{n})(\mathbf{n}\hat{\mathbf{j}}) - \hat{\mathbf{s}}\hat{\mathbf{j}}]\frac{\overline{r^{-3}}}{j(j+1)}.$$

The mean $\overline{n_i n_k}$ was evaluated in the problem to § 29. Using that result and passing to eigenvalues, we obtain

$$\frac{2\mu_B\mu}{i}(\hat{\mathbf{i}}\hat{\mathbf{j}})\left[\hat{\mathbf{l}}\hat{\mathbf{j}} + \frac{2l(l+1)\hat{\mathbf{s}}\hat{\mathbf{j}} - 6(\hat{\mathbf{s}}\hat{\mathbf{l}})(\hat{\mathbf{j}}\hat{\mathbf{l}})}{(2l-1)(2l+3)}\right]\frac{\overline{r^{-3}}}{j(j+1)},$$

and a straightforward calculation finally gives

$$\frac{\mu_B\mu}{i}\frac{l(l+1)}{j(j+1)}F(F+1)\overline{r^{-3}},$$

where $\mathbf{F} = \mathbf{j} + \mathbf{i}$ and $j = l \pm 1/2$. The average of r^{-3} is taken over the radial part of the electron wavefunction.

Problem

2. Determine the Zeeman splitting of the hyperfine-structure components of an atomic level (S. A. Goudsmit, R. F. Bacher, 1930).

Solution. In formula (113.4), where the field is assumed so weak that the splitting which it produces is small compared with the hyperfine intervals, the averaging must now be carried out both over the electronic state and over the orientations of the nuclear spin. The first averaging gives $\Delta E = \mu_B g_J \overline{J_z} H$, with the same g_J as in (113.7). By analogy with (113.5), the second averaging gives

$$\overline{J_z} = \frac{(\mathbf{JF})}{F^2}M_F.$$

Consequently,

$$\Delta E = \mu_B g_F H M_F, \qquad g_F = g_J\frac{F(F+1) + J(J+1) - i(i+1)}{2F(F+1)}.$$

§ 122. Hyperfine structure of molecular levels

The origin of the hyperfine structure of molecular energy levels is analogous to that of atomic levels.

In the overwhelming majority of molecules the total electronic spin is zero. For such molecules, the principal source of hyperfine level splitting is the quadrupole interaction between the nuclei and the electrons. Naturally, only nuclei whose spin i differs from 0 and $1/2$ participate in this interaction, since otherwise the quadrupole moment vanishes.

Because the nuclei move comparatively slowly within a molecule, the operator of the quadrupole interaction is averaged over a molecular state in two stages. It must first be averaged over the electronic state with the nuclei held fixed, and then over the rotation of the molecule.

Consider first a diatomic molecule. After the first averaging stage, the interaction of each nucleus with the electrons is represented by an operator proportional to the scalar $\hat{Q}_{ik}n_i n_k$. This scalar is formed from the operator of the nuclear quadrupole-moment tensor and the unit vector $\mathbf{n}$ along the molecular axis, the sole quantity specifying the orientation of the molecule relative to the direction of the nuclear spin. Since $\hat{Q}_{ii} = 0$, this operator may be written as

$$b\hat{i}_i\hat{i}_k\left(n_i n_k - \frac{1}{3}\delta_{ik}\right); \tag{122.1}$$

for a specified value of the projection i_ζ of the nuclear spin on the molecular axis, its value is $b[i_\zeta^2 - \frac{1}{3}i(i+1)]$.

On averaging the operator (122.1) over molecular rotation, it is instead expressed in terms of the operator $\hat{\mathbf{K}}$ of the conserved rotational angular momentum. The product $n_i n_k$ is averaged by the formula derived in the problem to § 29 (with $\mathbf{K}$ in place of $\mathbf{l}$), and the result is

$$-\frac{b}{(2K-1)(2K+3)}\hat{i}_i\hat{i}_k\left[\hat{K}_i\hat{K}_k + \hat{K}_k\hat{K}_i - \frac{2}{3}\delta_{ik}K(K+1)\right]. \tag{122.2}$$

The eigenvalues of this operator are found in the same manner as described for the operator (121.6).

For a polyatomic molecule, (122.1) is replaced, in general, by an operator of the form

$$b_{ik}\hat{i}_i\hat{i}_k, \tag{122.3}$$

where b_{ik} is a traceless tensor that constitutes a particular characteristic of the molecule's electronic state. Once it has been averaged over the molecule's rotation, this tensor is expressed in terms of the total rotational angular momentum $\mathbf{J}$ by a formula of the form

$$\overline{b_{ik}} = b\left[\hat{J}_i\hat{J}_k + \hat{J}_k\hat{J}_i - \frac{2}{3}J(J+1)\delta_{ik}\right]. \tag{122.4}$$

In principle, the coefficient b can be written in terms of the components of the tensor b_{ik} referred to the molecule's principal axes of inertia ξ, η, and ζ. These axes are fixed in the molecule. Consequently, the components $b_{\xi\xi}, \ldots$ characterize the molecule without being altered by the averaging. To obtain this relation, consider the scalar $\overline{b_{ik}J_iJ_k}$. Its evaluation using (122.4) gives

$$\overline{b_{ik}J_iJ_k} = bJ(J+1)\left[\frac{4}{3}J(J+1) - 1\right] \tag{122.5}$$

(the calculation is analogous to the one carried out in the problem to § 29). On the other hand, expanding the tensor product along the ξ, η, ζ axes yields

$$\overline{b_{ik}J_iJ_k} = b_{\xi\xi}\overline{J_\xi^2} + b_{\eta\eta}\overline{J_\eta^2} + b_{\zeta\zeta}\overline{J_\zeta^2}. \tag{122.6}$$

Here it has been used that the mean values of the products $J_\xi J_\zeta, \ldots$ vanish[29]. The wave functions of the relevant rotational states of the top allow, in principle, the mean values of $J_\xi^2, \ldots$ to be calculated. For a symmetric top, in particular, the result has the simple form

$$\overline{J_\zeta^2} = k^2, \qquad \overline{J_\xi^2} = \overline{J_\eta^2} = \frac{1}{2}[J(J+1) - k^2].$$

When the nuclear spins are 1/2, the quadrupole interaction is absent. One of the principal sources of hyperfine splitting in this case is the direct magnetic interaction of the nuclear magnetic moments with each other. The interaction operator of two magnetic moments $\boldsymbol{\mu}_1 = \mu_1\hat{\mathbf{i}}_1/i_1$, $\boldsymbol{\mu}_2 = \mu_2\hat{\mathbf{i}}_2/i_2$ is

$$\frac{\mu_1\mu_2}{i_1i_2r^3}[\hat{\mathbf{i}}_1\hat{\mathbf{i}}_2 - 3(\hat{\mathbf{i}}_1\mathbf{n})(\hat{\mathbf{i}}_2\mathbf{n})].$$

To determine the splitting energy, this operator must be averaged over the molecular state in the manner described above.

In a molecule containing heavy atoms, the electronic shell also mediates an indirect coupling between the nuclear moments. Its contribution to the hyperfine splitting can be comparable to that of their direct coupling. Formally, the indirect contribution arises in second-order perturbation theory from the coupling of a nuclear spin to the electrons. The results of § 121 show directly that its magnitude relative to the direct nuclear-moment interaction is of order $(Ze^2/\hbar c)^2$. For large Z, this ratio is of order unity.

Finally, the interaction of a nuclear moment with molecular rotation also makes a definite contribution to the hyperfine splitting of molecular levels. As a moving system of charges, a rotating molecule produces a magnetic field. This field can be calculated from the standard formulas of electrodynamics for the given current density $\mathbf{j} = \rho(\boldsymbol{\Omega}\times\mathbf{r})$, where ρ is the charge density of the electrons and nuclei in the molecule at rest, and $\boldsymbol{\Omega}$ is its angular velocity of rotation. The level splitting is the energy of the nuclear magnetic moment in this field; the components of the molecule's angular velocity must here be expressed in terms of the components of its angular momentum (cf. § 103).

[29] Indeed, in a representation in which the matrix of one component of $\mathbf{J}$ (say J_ζ) is diagonal, the matrices of the products $J_\xi J_\zeta$, $J_\eta J_\zeta$ have nonzero elements only when the quantum number k changes by 1. The wavefunctions of the stationary states of an asymmetric top, however, contain functions ψ_{Jk} whose values of k differ by an even integer (see § 103).

CHAPTER XVII

Elastic Collisions

§ 123. General theory of scattering

In classical mechanics, the collision of two particles is specified completely by their velocities and by the impact parameter (the distance at which they would pass one another if there were no interaction). In quantum mechanics the question itself must be posed differently: for motion at specified velocities, a trajectory has no meaning, and consequently neither does an impact parameter. The task of the theory is then only to calculate the probability that, after the collision, the particles will be deflected—or, as one says, *scattered*—through a particular angle. Here we consider so-called elastic collisions, in which the particles undergo no transformations and, if they are composite, their internal states remain unchanged.

Like every two-body problem, an elastic collision reduces to the scattering of one particle, with the reduced mass, by the field $U(r)$ of a fixed force centre[1]. The reduction is made by passing to the frame in which the centre of mass of the two particles is at rest. Let the scattering angle in this frame be θ. It is connected by simple formulae with the deflection angles θ_1 and θ_2 of the two particles in the "laboratory" frame, in which one of the particles (the second) was at rest before the collision:

$$\tan\theta_1 = \frac{m_2\sin\theta}{m_1 + m_2\cos\theta}, \qquad \theta_2 = \frac{\pi-\theta}{2}, \tag{123.1}$$

where m_1 and m_2 are the particle masses (see I, § 17). In particular, when the masses are equal ($m_1 = m_2$), this gives simply

$$\theta_1 = \frac{\theta}{2}, \qquad \theta_2 = \frac{\pi-\theta}{2}; \tag{123.2}$$

the sum $\theta_1 + \theta_2 = \pi/2$, so that the particles fly apart at a right angle.

Throughout the remainder of this chapter, unless expressly stated otherwise, we use the centre-of-mass frame and denote the reduced mass of the colliding particles by m.

A free particle moving along the positive z-axis is represented by a plane wave, which we write as $\psi = e^{ikz}$; thus we choose a normalization for which the current density in the wave equals the particle velocity v. Far from the centre, the scattered particles are represented by an outgoing spherical wave $f(\theta)e^{ikr}/r$, where $f(\theta)$ is a function of the scattering angle θ (the angle between the z-axis and the direction of the scattered particle). This function is called the *scattering amplitude*. Hence the exact wave function solving the Schrödinger equation with potential energy $U(r)$ must have, at large distances, the asymptotic form

$$\psi \approx e^{ikz} + \frac{f(\theta)}{r}e^{ikr}. \tag{123.3}$$

[1]We neglect the spin–orbit interaction of the particles (if they possess spin). By assuming a centrally symmetric field, we also exclude from consideration processes such as electron scattering by molecules.

The probability per unit time for a scattered particle to cross the surface element $dS = r^2 d\Omega$ (where $d\Omega$ is the element of solid angle) is $vr^{-2}|f|^2 dS = v|f|^2 d\Omega$[2]. Its ratio to the current density of the incident wave is

$$d\sigma = |f(\theta)|^2 d\Omega. \tag{123.4}$$

This quantity has the dimensions of area and is called the *effective cross-section* (or simply the *cross-section*) for scattering into the solid-angle element $d\Omega$. On putting $d\Omega = 2\pi \sin\theta \, d\theta$, we obtain the cross-section

$$d\sigma = 2\pi \sin\theta |f(\theta)|^2 d\theta \tag{123.5}$$

for scattering into the angular interval from θ to $\theta + d\theta$.

The solution of the Schrödinger equation describing scattering in the central field $U(r)$ must plainly be axially symmetric about the z-axis, the direction of the incident particles. Any such solution may be represented as a

superposition of continuous-spectrum wave functions for particles moving in the given field with specified energy $\hbar^2 k^2/2m$, with orbital angular momenta of various magnitudes l and vanishing z-components (these functions do not depend on the azimuthal angle φ about the z-axis and are therefore axially symmetric). The required wave function thus has the form

$$\psi = \sum_{l=0}^{\infty} A_l P_l(\cos\theta) R_{kl}(r), \tag{123.6}$$

where the A_l are constants and the radial functions $R_{kl}(r)$ satisfy

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{dR_{kl}}{dr}\right) + \left[k^2 - \frac{l(l+1)}{r^2} - \frac{2m}{\hbar^2}U(r)\right] R_{kl} = 0. \tag{123.7}$$

The coefficients A_l must be chosen so that (123.6) has the asymptotic form (123.3) at large distances. We shall show that this requires

$$A_l = \frac{1}{2k}(2l+1) i^l \exp(i\delta_l), \tag{123.8}$$

where δ_l are the phase shifts of the functions R_{kl}. This will at the same time express the scattering amplitude in terms of these phases.

The asymptotic form of R_{kl} is given by (33.20):

$$\begin{aligned} R_{kl} &\approx \frac{2}{r}\sin\left(kr - \frac{l\pi}{2} + \delta_l\right) \\ &= \frac{1}{ir}\{(-i)^l \exp[i(kr + \delta_l)] - i^l \exp[-i(kr + \delta_l)]\}. \end{aligned}$$

[2]This argument tacitly assumes that the incident particle beam is bounded by a broad, but finite, aperture—broad so as to avoid diffraction effects—as in actual scattering experiments. There is consequently no interference between the two terms in (123.3): the square $|\psi|^2$ is evaluated at points where the incident wave is absent.

Substitution of this expression and (123.8) into (123.6) gives the asymptotic wave function

$$\psi \approx \frac{1}{2irk}\sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta)[(-1)^{l+1}e^{-ikr}+S_l e^{ikr}], \tag{123.9}$$

where we have introduced the notation

$$S_l = \exp(2i\delta_l). \tag{123.10}$$

On the other hand, after the same rearrangement, the expansion (34.2) of a plane wave reads

$$e^{ikz} \approx \frac{1}{2ikr}\sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta)[(-1)^{l+1}e^{-ikr}+e^{ikr}].$$

We see that all terms containing e^{-ikr} cancel, as they must, in the difference $\psi - e^{ikz}$. The coefficient of e^{ikr}/r in this difference, which is the scattering amplitude, is therefore

$$f(\theta) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)(S_l-1)P_l(\cos\theta). \tag{123.11}$$

This formula expresses the scattering amplitude in terms of the phases δ_l (H. Faxen, J. Holtsmark, 1927)[3].

Integrating $d\sigma$ over all angles gives the total scattering cross-section σ, which is the ratio of the total probability per unit time that a particle is scattered to the current density in the incident wave. Substituting (123.11) in

$$\sigma = 2\pi\int_0^{\pi}|f(\theta)|^2\sin\theta\, d\theta$$

and using the mutual orthogonality of Legendre polynomials with different l, together with

$$\int_0^{\pi}P_l^2(\cos\theta)\sin\theta\, d\theta = \frac{2}{2l+1},$$

we obtain the following expression for the total cross-section:

$$\sigma = \frac{4\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)\sin^2\delta_l. \tag{123.12}$$

Each term of this sum is the *partial cross-section* σ_l for the scattering of particles with the specified orbital angular

[3]Of fundamental interest is the inverse problem of recovering the scattering potential when the phases δ_l are presumed known. This problem was solved by *I. M. Gelfand, B. M. Levitan, and V. A. Marchenko*. In principle, to determine $U(r)$ it is sufficient to know $\delta_0(k)$ as a function of the wave vector over the whole range $0 \le k < \infty$, together with the coefficients a_n in the asymptotic expressions (as $r \to \infty$)

$$R_{n0} \approx a_n e^{-\kappa_n r}/r \quad (\kappa_n = \sqrt{2m|E_n|}/\hbar)$$

for the wave functions of states belonging to discrete (negative) energy levels E_n, if such levels exist. Determining $U(r)$ from these data reduces to solving a certain linear integral equation. A systematic account may be found in: V. de Alfaro and T. Regge, *Potential Scattering*. Moscow: Mir, 1966.

momentum l. The greatest possible value of this cross-section is

$$\sigma_{l\,\max} = \frac{4\pi}{k^2}(2l+1). \tag{123.13}$$

Comparison with (34.5) shows that the number of particles scattered with angular momentum l can be four times the number of such particles in the incident flux. This circumstance is a purely quantum effect associated with interference between scattered and unscattered particles.

We shall also find it convenient below to use the *partial scattering amplitudes* f_l, defined as the coefficients in the expansion

$$f(\theta) = \sum_{l=0}^{\infty}(2l+1)f_l P_l(\cos\theta). \tag{123.14}$$

According to (123.11), they are related to the phases δ_l by

$$f_l = \frac{1}{2ik}(S_l - 1) = \frac{1}{2ik}(e^{2i\delta_l} - 1), \tag{123.15}$$

and the partial cross-sections are

$$\sigma_l = 4\pi(2l+1)|f_l|^2. \tag{123.16}$$

Problem

Express the scattering amplitude in terms of phase shifts in two dimensions. The field is $U = U(\rho)$, where $\rho = \sqrt{x^2+z^2}$, and the particle flux is directed along the z-axis.

Solution. In two dimensions, far from the scatterer the wave function is a superposition of a plane wave and an outgoing cylindrical wave:

$$\psi = e^{ikz} + f(\varphi)\frac{e^{ik\rho}}{\sqrt{-i\rho}}. \tag{1}$$

Here φ is the angle between the z-axis and the scattering direction, and $f(\varphi)$ is the scattering amplitude, whose dimension in two dimensions is the square root of length. The factor $-i = \exp(-i\pi/2)$ beneath the radical has been introduced to simplify the subsequent formulae. The scattering cross-section per unit length along the y-axis is

$$d\sigma = |f|^2 d\varphi.$$

It has the dimension of length.

The wave function must be expanded in functions with a definite projection m of angular momentum on the y-axis, of the form $Q_m(\rho)e^{im\varphi}$. At large distances from the scatterer, the radial functions differ from the free-motion functions obtained in the problem to § 34 only by a phase shift:

$$Q_m(\rho) \approx i^m\sqrt{\frac{2}{\pi k\rho}}\sin\left[k\rho - \frac{\pi}{2}\left(m - \frac{1}{2}\right) + \delta_m\right],$$

with $\delta_m = \delta_{-m}$. Repeating the argument of this section and using the plane-wave expansion from the problem to § 34, we find that the function with asymptotic form (1) is given by the series

$$\psi = \sum_{m=-\infty}^{\infty} e^{i\delta_m} Q_m(\rho) e^{im\varphi},$$

and that the scattering amplitude is

$$f(\varphi) = \frac{1}{i\sqrt{2\pi k}} \sum_{m=-\infty}^{\infty} (e^{2i\delta_m} - 1) e^{im\varphi}. \tag{2}$$

Integration gives the total cross-section

$$\sigma = \int_0^{2\pi} |f|^2 d\varphi = \sum_{m=-\infty}^{\infty} \sigma_m, \qquad \text{where} \qquad \sigma_m = \sigma_{-m} = \frac{4}{k}\sin^2\delta_m.$$

It is readily verified that

$$\operatorname{Im} f(0) = \sqrt{\frac{k}{8\pi}}\,\sigma, \tag{3}$$

which is the optical theorem in two dimensions (see formula (125.9) below).

§ 124. Analysis of the general formula

The formulas obtained above apply, in principle, to scattering by any field $U(r)$ that tends to zero at infinity. Their analysis therefore reduces to examining the properties of the phases δ_l which occur in them.

To estimate the order of magnitude of δ_l at large l, we use the fact that the motion is quasiclassical for large l (see § 49). The phase of the wave function is consequently determined by the integral

$$\int_{r_0}^{r} \sqrt{k^2 - \frac{(l+1/2)^2}{r^2} - \frac{2mU(r)}{\hbar^2}}\, dr + \frac{\pi}{4},$$

where r_0 is a root of the expression under the radical (the region $r > r_0$ is classically accessible). From this we subtract the phase

$$\int_{r_0}^{r} \sqrt{k^2 - \frac{(l+1/2)^2}{r^2}}\, dr + \frac{\pi}{4}$$

of the free-motion wave function and then let $r \to \infty$; by definition, the result is δ_l. For large l, the value of r_0 is also large. Hence $U(r)$ is small throughout the integration region, and approximately

$$\delta_l = -\int_{r_0}^{\infty} \frac{mU(r)\,dr}{\hbar^2\sqrt{k^2 - (l+1/2)^2/r^2}}. \tag{124.1}$$

In order of magnitude, this integral (provided it converges) is

$$\delta_l \sim \frac{mU(r_0)r_0}{k\hbar^2}. \tag{124.2}$$

The corresponding magnitude of r_0 is $r_0 \sim l/k$.

If $U(r)$ approaches zero at infinity as r^{-n} with $n > 1$, the integral (124.1) converges and the phases δ_l are finite. For $n \leqslant 1$, on the contrary, the integral diverges, and the phases δ_l are infinite. This statement applies to every l: convergence of (124.1) depends on the large-r behavior of $U(r)$, while at large distances (where the field $U(r)$ is already weak) radial motion is quasiclassical for any l. The interpretation of formulas (123.11) and (123.12) when δ_l is infinite will be given below.

First consider convergence of the series (123.12), which represents the total scattering cross-section. At large l the phases satisfy $\delta_l \ll 1$, as follows from (124.1) when $U(r)$ decreases faster than $1/r$. We may therefore put $\sin^2 \delta_l \approx \delta_l^2$, so that the tail of the series (123.12) has the order of $\sum_{l \gg 1} l\delta_l^2$. The usual integral test shows that this series converges if $\int_0^\infty l\delta_l^2\, dl$ converges. Substitution of (124.2), followed by replacing l with kr_0, gives the integral

$$\int^\infty U^2(r_0) r_0^3\, dr_0.$$

When $U(r)$ falls off at infinity as r^{-n} with $n > 2$, this integral converges and the total cross-section is finite. If, however, the field $U(r)$ decreases as $1/r^2$ or more slowly, the total cross-section is infinite. Physically, this is because a slowly decreasing field gives a very large probability of scattering through small angles. Recall in this connection that, in classical mechanics, a particle passing at any finite impact parameter ρ, however large, still undergoes a deflection through some small

but nonzero angle whenever the field vanishes only as $r \to \infty$; the total scattering cross-section is therefore infinite for every law by which $U(r)$ decreases.[4] This argument does not apply in quantum mechanics, if only because scattering through a given angle can be discussed only when that angle is large compared with the uncertainty in the direction of motion. If the impact parameter is known to an accuracy $\Delta\rho$, it produces an uncertainty $\sim \hbar/\Delta\rho$ in the transverse momentum component, and thus an angular uncertainty $\sim \hbar/(mv\Delta\rho)$.

Because small-angle scattering is so important when $U(r)$ decreases slowly, one naturally asks whether the scattering amplitude $f(\theta)$ might diverge at $\theta = 0$ even when $U(r)$ falls off faster than $1/r^2$. On setting $\theta = 0$ in (123.11), the distant terms of the sum give an expression proportional to $\sum_{l \gg 1} l\delta_l$. The same reasoning as before reduces the condition for finiteness of the sum to convergence of

$$\int^\infty U(r_0) r_0^2\, dr_0,$$

which already diverges for $U(r) \sim r^{-n}$ when $n \leqslant 3$. Thus the scattering amplitude becomes infinite at $\theta = 0$ ($n \leqslant 1$) in fields which fall off as $1/r^3$ or more slowly.

Finally, consider the case in which the phase δ_l itself is infinite, as happens for $U(r) \sim r^{-n}$ with $n \leqslant 1$. The preceding results make it clear in advance that, with so slow a decrease of the field, both the total cross-section and the scattering amplitude at

[4]This appears as the divergence of the integral $\int 2\pi\rho\, d\rho$ which defines the total cross-section in classical mechanics.

$\theta = 0$ are infinite. It remains to determine $f(\theta)$ for $\theta \neq 0$. We first note the formula[5]

$$\sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta) = 4\delta(1-\cos\theta). \tag{124.3}$$

In other words, this sum vanishes for every $\theta \neq 0$. Consequently, for $\theta \neq 0$ the unity in square brackets in every term of (123.11) may be

omitted, leaving

$$f(\theta) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta)e^{2i\delta_l}. \tag{124.4}$$

Multiplication of the right-hand side by the constant factor $\exp(-2i\delta_0)$ does not affect the cross-section, which is determined by $|f(\theta)|^2$; it merely changes the phase of the complex function $f(\theta)$ by an inessential constant. On the other hand, in the difference $\delta_l - \delta_0$ between expressions (124.1), the divergent integral of $U(r)$ cancels and a finite quantity remains. Thus, in the case considered, the scattering amplitude may be calculated from

$$f(\theta) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta)e^{2i(\delta_l-\delta_0)}. \tag{124.5}$$

§ 125. The unitarity condition in scattering

The scattering amplitude in an arbitrary field (which need not be central) obeys certain relations that follow from general physical requirements.

At large distances, the asymptotic form of the wave function for elastic scattering in an arbitrary field is

$$\psi \approx e^{ikr\mathbf{n}\mathbf{n}'} + \frac{1}{r}f(\mathbf{n},\mathbf{n}')e^{ikr}. \tag{125.1}$$

This notation differs from (123.3) in that the scattering amplitude now depends on the directions of two unit vectors: one along the incident direction (**n**), and the other along the scattered direction (**n′**), rather than only on the angle between them.

Any linear combination of functions of the form (125.1), with different incident directions **n**, likewise represents a possible scattering process. Multiply the functions (125.1) by arbitrary coefficients $F(\mathbf{n})$ and integrate over all directions **n** (with solid-angle element $d\Omega$). Such a linear combination may then be written as

$$\int F(\mathbf{n})e^{ikr\mathbf{n}\mathbf{n}'}d\Omega + \frac{e^{ikr}}{r}\int F(\mathbf{n})f(\mathbf{n},\mathbf{n}')d\Omega. \tag{125.2}$$

Since r may be arbitrarily large, the factor $e^{ikr\mathbf{n}\mathbf{n}'}$ in the first integral oscillates rapidly as the direction of the variable vector **n** changes. The integral is therefore determined

[5] This formula is the expansion of the δ-function in Legendre polynomials and follows directly on multiplying both sides by $\sin\theta P_l(\cos\theta)$ and integrating with respect to $d\theta$. Here the integral $\int_0^\infty \delta(x)\,dx$ of the even function $\delta(x)$ is taken to be $1/2$.

mainly by neighborhoods of those values of $\mathbf{n}$ at which the exponent is extremal ($\mathbf{n} = \pm\mathbf{n}'$). Within either neighborhood, the factor $F(\mathbf{n}) \approx F(\pm\mathbf{n}')$ may be taken outside the integral; integration then gives[6]

$$\frac{2\pi i}{kr}F(-\mathbf{n}')e^{-ikr} - \frac{2\pi i}{kr}F(\mathbf{n}')e^{ikr} + \frac{e^{ikr}}{r}\int f(\mathbf{n},\mathbf{n}')F(\mathbf{n})d\Omega.$$

Dropping the common factor $2\pi i/k$, rewrite this expression in the compact operator form

$$\frac{e^{-ikr}}{r}F(-\mathbf{n}') - \frac{e^{ikr}}{r}\hat{S}F(\mathbf{n}'), \tag{125.3}$$

where

$$\hat{S} = 1 + 2ik\hat{f}, \tag{125.4}$$

and $\hat{f}$ is the integral operator

$$\hat{f}F(\mathbf{n}') = \frac{1}{4\pi}\int f(\mathbf{n},\mathbf{n}')F(\mathbf{n})d\Omega. \tag{125.5}$$

The operator $\hat{S}$ is called the scattering operator (or scattering *matrix*), or simply the S-matrix; it was introduced by *W. Heisenberg* in 1943.

The first term in (125.3) is a wave converging on the centre, whereas the second is a wave diverging from it. Conservation of particle number in elastic scattering means that the total particle currents in the converging and diverging waves are equal. Equivalently, these waves must have the same normalization. The scattering operator must consequently be unitary (see § 12), so that

$$\hat{S}\hat{S}^{+} = 1, \tag{125.6}$$

or, on substituting (125.4) and multiplying out,

$$\hat{f} - \hat{f}^{+} = 2ik\hat{f}\hat{f}^{+}. \tag{125.7}$$

Finally, using the definition (125.5), the *unitarity* condition for scattering takes the form

$$f(\mathbf{n},\mathbf{n}') - f^{*}(\mathbf{n}',\mathbf{n}) = \frac{ik}{2\pi}\int f(\mathbf{n},\mathbf{n}'')f^{*}(\mathbf{n}',\mathbf{n}'')d\Omega''. \tag{125.8}$$

When $\mathbf{n} = \mathbf{n}'$, the integral on the right is precisely the total scattering cross-section

$$\sigma = \int |f(\mathbf{n},\mathbf{n}'')|^2 d\Omega''.$$

The difference on the left then reduces to the imaginary part of the amplitude $f(\mathbf{n},\mathbf{n})$. We thus obtain the following general relation between the total elastic-scattering cross-section and the imaginary part of the forward-scattering amplitude:

$$\operatorname{Im} f(\mathbf{n},\mathbf{n}) = \frac{k}{4\pi}\sigma \tag{125.9}$$

[6]To evaluate the integral, deform the integration path in the complex μ plane, where $\mu = \cos\theta$ (θ is the angle between $\mathbf{n}$ and $\mathbf{n}'$), so that it bends into the upper half-plane while its endpoints $\mu = \pm 1$ remain fixed. The function $e^{ikr\mu}$ then decays rapidly on moving away from either endpoint.

(the so-called *optical theorem* for scattering).

Another general property of the scattering amplitude follows from the requirement of time-reversal symmetry. In quantum mechanics this symmetry means that, if ψ describes a possible state, the complex-conjugate function ψ^* also corresponds to a possible state (see § 18). Hence the wave function

$$-\frac{e^{-ikr}}{r}F^*(-\mathbf{n}') - \frac{e^{ikr}}{r}\hat{S}^*F^*(\mathbf{n}'),$$

which is the complex conjugate of (125.3), also describes a possible scattering process. Introduce a new arbitrary function by $-\hat{S}^*F^*(\mathbf{n}') = \Phi(-\mathbf{n}')$. From the unitarity of $\hat{S}$ we have

$$F^*(\mathbf{n}') = -\hat{S}^{*-1}\Phi(-\mathbf{n}') = -\tilde{\hat{S}}\Phi(-\mathbf{n}');$$

introducing the coordinate-inversion operator $\hat{P}$, which reverses the signs of $\mathbf{n}$ and $\mathbf{n}'$, we write

$$F^*(-\mathbf{n}') = -\hat{P}\tilde{\hat{S}}\hat{P}\Phi(\mathbf{n}').$$

The time-reversed wave function therefore becomes

$$\frac{e^{-ikr}}{r}\Phi(-\mathbf{n}') - \frac{e^{ikr}}{r}\hat{P}\tilde{\hat{S}}\hat{P}\Phi(\mathbf{n}').$$

In substance, it must coincide with the original wave function (125.3). Comparison shows that this requires

$$\hat{P}\tilde{\hat{S}}\hat{P} = \hat{S}, \tag{125.10}$$

after which the two functions differ only in the name given to the arbitrary function.

The corresponding relation for the scattering amplitude is obtained by passing from the operator equality (125.10) to its matrix form. Transposition interchanges the initial and final vectors $\mathbf{n}$ and $\mathbf{n}'$, while inversion reverses their signs. Consequently,

$$S(\mathbf{n}, \mathbf{n}') = S(-\mathbf{n}', -\mathbf{n}), \tag{125.11}$$

or equivalently,

$$f(\mathbf{n}, \mathbf{n}') = f(-\mathbf{n}', -\mathbf{n}). \tag{125.12}$$

This relation (the so-called *reciprocity theorem*) expresses the natural result that two scattering processes which are time reverses of one another have equal amplitudes. Time reversal interchanges the initial and final states and reverses the particles' directions of motion in both.

The general relations just obtained simplify for scattering in a central field. The amplitude $f(\mathbf{n}, \mathbf{n}')$ then depends only on the angle θ between $\mathbf{n}$ and $\mathbf{n}'$, so (125.12) becomes an identity. The unitarity condition (125.8), on the other hand, becomes

$$\operatorname{Im} f(\theta) = \frac{k}{4\pi}\int f(\gamma)f^*(\gamma')d\Omega'', \tag{125.13}$$

where γ and γ' are the angles made by $\mathbf{n}$ and $\mathbf{n}'$, respectively, with some direction $\mathbf{n}''$ in space. If $f(\theta)$ is expanded as in (123.14), the addition theorem for spherical functions (p. 10) applied to (125.13) gives the following relation for the partial amplitudes:

$$\operatorname{Im} f_l = k|f_l|^2. \tag{125.14}$$

The same formula follows directly from (123.15), according to which $|2ikf_l + 1|^2 = 1$. For scattering in a central field the optical theorem (125.9) also follows at once from (123.11) and (123.12).

Writing (125.14) as $\mathrm{Im}(1/f_l) = -k$, we see that f_l must have the form

$$f_l = \frac{1}{g_l - ik}, \tag{125.15}$$

where $g_l = g_l(k)$ is real; it is related to the phase δ_l by

$$g_l = k \cot \delta_l. \tag{125.16}$$

We shall use this representation of the amplitude repeatedly below.

For scattering in a central field, let us now trace the connection between the scattering operator introduced above and the quantities used in the theory of § 123.

Since orbital angular momentum is conserved in a central field, the scattering operator commutes with the angular-momentum operator. In other words, the S-matrix is diagonal in the l-representation. Because $\hat{S}$ is unitary, its eigenvalues must have unit modulus; they are therefore of the form $\exp(2i\delta_l)$, with real δ_l. It is readily seen that these quantities are the phase shifts of the wave functions, so the eigenvalues of the S-matrix coincide with the quantities S_l introduced in § 123, (123.10); correspondingly, the eigenvalues of $\hat{f} = (\hat{S} - 1)/(2ik)$ coincide with the partial amplitudes (123.15). Indeed, choose $P_l(\cos\theta)$ for $F(\mathbf{n})$ (then $F(-\mathbf{n}) = P_l(-\cos\theta) = (-1)^l P_l(\cos\theta)$). The wave function (125.3) must coincide with the solution of the Schrödinger equation represented by a single term of the sum in (123.9). This means precisely that

$$\hat{S}P_l(\cos\theta) = S_l P_l(\cos\theta).$$

For a plane wave incident along the z axis, the function $F(\mathbf{n})$ in (125.3) is the delta-function $F = 4\delta(1 - \cos\theta)$, where θ is the angle between $\mathbf{n}$ and the z axis. Here the delta-function is defined as stated in the note on p. 616, and the coefficient before it is chosen so that substitution in the right-hand side of (125.5) yields simply $f(\theta)$ (where now θ is the angle between $\mathbf{n}'$ and the z axis). Expanding the delta-function in the form (124.3),

$$F = 4\delta(1 - \cos\theta) = \sum_{l=0}^{\infty} (2l + 1) P_l(\cos\theta), \tag{125.17}$$

and applying $\hat{f}$ to it gives, as required, the scattering amplitude in the form (123.14).

We conclude with one further observation. Mathematically, the unitarity condition (125.8) shows that an arbitrarily prescribed function $f(\mathbf{n}, \mathbf{n}')$ cannot necessarily serve as the scattering amplitude for some field. In particular, an arbitrary $f(\theta)$ need not be the scattering amplitude for any central field. By (125.13), its real and imaginary parts must satisfy a definite relation. If $f(\theta) = |f|e^{i\alpha}$ is written with $|f|$ prescribed for all angles, (125.13) becomes an integral equation from which the unknown phase $\alpha(\theta)$ can in principle be found. In other words, knowledge of the scattering cross-section (the square $|f|^2$) at every angle permits, in principle, reconstruction of the amplitude as well.

This reconstruction is not wholly unique, however: it determines the amplitude only up to the replacement

$$f(\theta) \to -f^*(\theta), \tag{125.18}$$

which leaves (125.13) invariant and of course does not change the cross-section $|f|^2$ (the transformation (125.18) is equivalent to reversing the signs of all the phases δ_l in (123.11) simultaneously). The ambiguity is removed, however, if the scattering amplitude is considered as a function not only of angle but also of energy. We shall see below (§ 128, 129) that the analytic properties of the amplitude as a function of energy are not invariant under (125.18).

§ 126. Born's formula

The scattering cross-section can be calculated in general form in the very important case where the scattering field may be treated as a perturbation[7]. Section 45 showed that this is possible if at least one of the two conditions

$$|U| \ll \frac{\hbar^2}{ma^2} \tag{126.1}$$

or

$$|U| \ll \frac{\hbar v}{a} = \frac{\hbar^2 ka}{ma^2} \tag{126.2}$$

is satisfied. Here a is the range of the field $U(r)$, and U denotes its order of magnitude in the principal region where it is present. When the first

condition holds, the approximation applies at every velocity. The second condition shows that it applies in any event to sufficiently fast particles.

Following § 45, write the wave function as $\psi = \psi^{(0)} + \psi^{(1)}$, where $\psi^{(0)} = e^{i\mathbf{kr}}$ describes the incident particle with wave vector $\mathbf{k} = \mathbf{p}/\hbar$. Formula (45.3) gives

$$\psi^{(1)}(x, y, z) = -\frac{m}{2\pi\hbar^2} \int \frac{U(x', y', z')e^{i(\mathbf{kr}' + kR)}}{R}\, dV'. \tag{126.3}$$

Choose the scattering center as origin, introduce the radius vector $\mathbf{R}_0$ to the observation point of $\psi^{(1)}$, and let $\mathbf{n}'$ be the unit vector along $\mathbf{R}_0$. If the radius vector of the volume element dV' is $\mathbf{r}'$, then $\mathbf{R} = \mathbf{R}_0 - \mathbf{r}'$. Far from the center, $R_0 \gg r'$, and hence

$$R = |\mathbf{R}_0 - \mathbf{r}'| \approx R_0 - \mathbf{r}'\mathbf{n}'.$$

Insertion in (126.3) yields the asymptotic expression

$$\psi^{(1)} \approx -\frac{m}{2\pi\hbar^2} \frac{e^{ikR_0}}{R_0} \int U(\mathbf{r}')e^{i(\mathbf{k}-\mathbf{k}')\mathbf{r}'} dV',$$

where $\mathbf{k}' = k\mathbf{n}'$ is the wave vector after scattering. Comparison with the definition (123.3) gives the scattering amplitude

$$f = -\frac{m}{2\pi\hbar^2} \int U e^{-i\mathbf{qr}} dV, \tag{126.4}$$

[7] In the general theory developed in § 123, this approximation corresponds to all the phases δ_l being small; in addition, these phases must be calculable from the Schrödinger equation with the potential energy treated as a perturbation (see Problem 4).

after relabeling the integration variables and introducing

$$\mathbf{q} = \mathbf{k}' - \mathbf{k}, \tag{126.5}$$

whose magnitude is

$$q = 2k \sin\frac{\theta}{2}, \tag{126.6}$$

with θ the angle between $\mathbf{k}$ and $\mathbf{k}'$, that is, the scattering angle.

Finally, taking the squared modulus of the amplitude gives the cross-section into the solid-angle element do:

$$d\sigma = \frac{m^2}{4\pi^2\hbar^4}\left|\int U e^{-i\mathbf{qr}} dV\right|^2 do. \tag{126.7}$$

Thus, scattering that changes the momentum by $\hbar\mathbf{q}$ is governed by the squared modulus of the matching Fourier component of the field U. Formula (126.7) was first obtained by *Born* (M. Born, 1926); this approximation is often called the *Born approximation* in collision theory.

In this approximation the relation

$$f(\mathbf{k}, \mathbf{k}') = f^*(\mathbf{k}', \mathbf{k}) \tag{126.8}$$

holds between the amplitudes of the direct and reverse scattering processes in the literal sense: the initial and final momenta are interchanged, without reversal of their signs as under time reversal. Scattering therefore has an additional symmetry besides the reciprocity theorem (125.12). This symmetry is closely connected with the small scattering amplitudes of perturbation theory and follows directly from the unitarity condition (125.8) if its integral term, quadratic in f, is neglected[8].

Formula (126.7) can also be obtained in another way, though this leaves the phase of the scattering amplitude undetermined. Start from the general formula (43.1) for the transition probability between continuous-spectrum states,

$$dw_{fi} = \frac{2\pi}{\hbar}|U_{fi}|^2\delta(E_f - E_i)d\nu_f.$$

Here it is applied to the transition from an incident free-particle state of momentum $\mathbf{p}$ to a state of momentum $\mathbf{p}'$ scattered into the solid-angle element do'. Choose $d^3p'/(2\pi\hbar)^3$ as the state interval $d\nu_f$. With $E_f - E_i = (p'^2 - p^2)/2m$, we find

$$dw_{\mathbf{p}'\mathbf{p}} = \frac{4\pi m}{\hbar}|U_{\mathbf{p}'\mathbf{p}}|^2\delta(p'^2 - p^2)\frac{d^3p'}{(2\pi\hbar)^3}. \tag{126.9}$$

The incident and scattered wave functions are plane waves. Since the state interval was chosen as an element of $\mathbf{p}'/(2\pi\hbar)$ space, the final wave function must be normalized to a delta-function of $\mathbf{p}'/(2\pi\hbar)$:

$$\psi_{\mathbf{p}'} = e^{i\mathbf{p}'\mathbf{r}/\hbar}. \tag{126.10}$$

[8] It follows that this property already disappears in the second approximation of perturbation theory. We shall verify this directly in § 130 in connection with (130.13).

Normalize the initial wave function to unit current density:

$$\psi_{\mathbf{p}} = \sqrt{\frac{m}{p}}\, e^{i\mathbf{pr}/\hbar}. \tag{126.11}$$

Then (126.9) has the dimension of area and is the differential scattering cross-section.

The delta-function in (126.9) means that $p' = p$, as required for elastic scattering. It may be eliminated by using spherical coordinates in momentum space, replacing d^3p' by $p'^2dp'do' = (1/2)p'd(p'^2)do'$, and integrating over $d(p'^2)$. This amounts to replacing p' by p in the integrand and gives

$$d\sigma = \frac{4\pi^2}{\hbar^4}p^2\left|\int \psi^*_{\mathbf{p}'}U\psi_{\mathbf{p}}dV\right|^2 do'.$$

Substitution of (126.10) and (126.11) returns (126.7).

In the form (126.7), the formula applies to a field $U(x, y, z)$ depending on coordinates in any combination, not merely on r. For a central field $U(r)$, however, it can be transformed further.

In the integral

$$\int U(r)e^{-i\mathbf{qr}}dV$$

use spherical coordinates r, ϑ, φ, taking the polar axis along $\mathbf{q}$; ϑ distinguishes the polar angle from the scattering angle θ. Integration over ϑ and φ gives

$$\int_0^\infty\int_0^\pi\int_0^{2\pi} U(r)e^{iqr\cos\vartheta}r^2\sin\vartheta\, d\varphi\, d\vartheta\, dr = 4\pi\int_0^\infty U(r)\frac{\sin qr}{qr}r^2dr.$$

Thus the scattering amplitude in a central field is

$$f = -\frac{2m}{\hbar^2}\int_0^\infty U(r)\frac{\sin qr}{q}r\, dr. \tag{126.12}$$

At $\theta = 0$, and hence $q = 0$, this integral diverges if $U(r)$ decreases at infinity as $1/r^3$ or more slowly, in agreement with the general results of § 124.

Notice an interesting point. The momentum p and scattering angle θ enter (126.12) only through q. In the Born approximation the cross-section therefore depends on p and θ only in the combination $p\sin(\theta/2)$.

Returning to arbitrary fields $U(x, y, z)$, consider the limiting cases of small ($ka \ll 1$) and large ($ka \gg 1$) velocities.

At small velocities one may put $e^{-i\mathbf{qr}} \approx 1$ in (126.4), so that

$$f = -\frac{m}{2\pi\hbar^2}\int U\, dV, \tag{126.13}$$

and, for $U = U(r)$,

$$f = -\frac{2m}{\hbar^2}\int_0^\infty U(r)r^2dr. \tag{126.14}$$

The scattering is isotropic and independent of velocity, consistently with the general results of § 132.

In the opposite, high-velocity limit the scattering is strongly anisotropic and directed forward, within a narrow cone of aperture $\Delta\theta \sim 1/ka$. Outside this cone q is large, the factor $e^{-i\mathbf{qr}}$ oscillates rapidly, and its integral multiplied by the slowly varying function U is nearly zero.

The decrease of the cross-section at large q is not universal, but depends on the particular field. If $U(r)$ has a singularity at $r = 0$ or at some other real value of r, the neighborhood of that point dominates the integral in (126.12), and the decrease follows a power law. The same holds when $U(r)$ is nonsingular but not even: the neighborhood of $r = 0$ then gives the main contribution. If $U(r)$ is even, the integration may formally be extended to negative r, along the entire real axis. Provided $U(r)$ has no singularities on that axis, the contour can then be shifted into the complex plane until it is caught by the nearest complex singularity. The integral consequently decreases exponentially at large q.

It must be remembered, however, that the Born approximation is generally unsuitable for calculating this exponentially small quantity (see also § 131).

Although the differential cross-section inside the cone $\Delta\theta \sim 1/ka$ is essentially independent of velocity, the shrinking aperture makes the total cross-section decrease at high energies, if $\int d\sigma$ converges at all. It decreases with the solid angle cut out by the cone, proportionally to $(\Delta\theta)^2 \sim 1/k^2a^2$, and hence inversely with energy.

In many applications of collision theory, scattering is characterized by

$$\sigma_{\text{tr}} = \int (1 - \cos\theta) d\sigma, \tag{126.15}$$

often called the *transport cross-section*. Arguments like those above show that at high velocities it is inversely proportional to the square of the energy.

Problems

Problem

1. Find in the Born approximation the scattering cross-section of a spherical potential well: $U = -U_0$ for $r < a$, and $U = 0$ for $r > a$.

Solution. The integral in (126.12) gives

$$d\sigma = 4a^2 \left(\frac{mU_0a^2}{\hbar^2}\right)^2 \frac{(\sin qa - qa\cos qa)^2}{(qa)^6}\, do.$$

Integration over all angles, conveniently performed with $q = 2k\sin(\theta/2)$ and $do = 2\pi q\, dq/k^2$, gives

$$\sigma = \frac{2\pi}{k^2}\left(\frac{mU_0a^2}{\hbar^2}\right)^2 \left[1 - \frac{1}{(2ka)^2} + \frac{\sin 4ka}{(2ka)^3} - \frac{\sin^2 2ka}{(2ka)^4}\right].$$

In the limiting cases,

$$\sigma = \begin{cases} \dfrac{16\pi a^2}{9}\left(\dfrac{mU_0a^2}{\hbar^2}\right)^2, & ka \ll 1, \\ \dfrac{2\pi}{k^2}\left(\dfrac{mU_0a^2}{\hbar^2}\right)^2, & ka \gg 1. \end{cases}$$

Problem

2. The same for the field $U = U_0 e^{-r^2/a^2}$.

Solution. It is convenient to use (126.7), taking $\mathbf{q}$ along one coordinate axis. The result is

$$d\sigma = \frac{\pi a^2}{4}\left(\frac{mU_0a^2}{\hbar^2}\right)^2 e^{-q^2a^2/2}do,$$

and the total cross-section is

$$\sigma = \frac{\pi^2}{2k^2}\left(\frac{mU_0a^2}{\hbar^2}\right)^2 (1 - e^{-2k^2a^2}).$$

The applicability conditions are (126.1), (126.2), with U_0 for U. Moreover, the formula for $d\sigma$ is invalid when the magnitude of the exponent is large[9].

Problem

3. The same for $U = (\alpha/r)e^{-r/a}$.

Solution. Evaluation of (126.12) gives

$$d\sigma = 4a^2\left(\frac{\alpha m a}{\hbar^2}\right)^2 \frac{do}{(q^2a^2+1)^2}.$$

The total cross-section is

$$\sigma = 16\pi a^2\left(\frac{\alpha m a}{\hbar^2}\right)^2 \frac{1}{4k^2a^2+1}.$$

Using α/a for U in (126.1), (126.2), the applicability condition is $\alpha m a/\hbar^2 \ll 1$ or $\alpha/\hbar v \ll 1$.

[9]This failure of perturbation theory is readily seen by calculating the second-approximation amplitude (see (130.13) below): its pre-exponential factor is small compared with the first-approximation coefficient, but the magnitude of its negative exponent is only half as large.

Problem

4. Determine the phases δ_l for scattering in a central field in the case corresponding to the Born approximation.

Solution. For the radial wave function $\chi = rR$ in the field $U(r)$ and the free-motion function $\chi^{(0)}$, the equations are (see (32.10))

$$\chi'' + \left[k^2 - \frac{l(l+1)}{r^2} - \frac{2m}{\hbar^2}U\right]\chi = 0, \qquad \chi^{(0)''} + \left[k^2 - \frac{l(l+1)}{r^2}\right]\chi^{(0)} = 0.$$

Multiply the first by $\chi^{(0)}$ and the second by χ, subtract, and integrate over dr, using $\chi = 0$ at $r = 0$. This gives

$$\chi'(r)\chi^{(0)}(r) - \chi(r)\chi^{(0)'}(r) = \frac{2m}{\hbar^2}\int_0^r U\chi\chi^{(0)}\,dr.$$

Treating U as a perturbation, put $\chi \approx \chi^{(0)}$ on the right. At $r \to \infty$ use (33.12), (33.20) on the left and insert the exact expression (33.10) in the integral. The result is

$$\sin\delta_l \approx \delta_l = -\frac{\pi m}{\hbar^2}\int_0^\infty U(r)\left[J_{l+1/2}(kr)\right]^2 r\,dr.$$

The same formula could also be obtained by directly expanding the Born amplitude (126.4) in Legendre polynomials according to (123.11), for small δ_l.

Problem

5. In the Born approximation, find the total cross-section for fast particles ($ka \gg 1$) in the field $U = \alpha/(r^2 + a^2)^{n/2}$, where $n > 2$.

Solution. As will emerge, partial amplitudes with large l dominate. We may therefore use (123.11), replacing the sum over l by an integral. Since all $\delta_l \ll 1$ in the Born approximation,

$$\sigma \approx \frac{4\pi}{k^2}\int_0^\infty 2l\delta_l^2\,dl. \tag{1}$$

For large l, (124.1) gives

$$\delta_l = -\frac{\alpha m}{\hbar^2}\int_{l/k}^\infty \frac{dr}{(r^2 + a^2)^{n/2}(k^2 - l^2/r^2)^{1/2}}.$$

With $r^2 + a^2 = (a^2 + l^2/k^2)/t$, this reduces to Euler's integral and gives

$$\delta_l = -\frac{\alpha m}{2\hbar^2 k(a^2 + l^2/k^2)^{(n-1)/2}}\,\frac{\Gamma(1/2)\Gamma((n-1)/2)}{\Gamma(n/2)}. \tag{2}$$

The integral (1) is governed by $l \sim ak \gg 1$, justifying the assumption. Evaluation gives

$$\sigma = \frac{\pi^2}{n-2}\left[\frac{\Gamma((n-1)/2)}{\Gamma(n/2)}\right]^2\left(\frac{\alpha m}{k\hbar^2 a^{n-2}}\right)^2. \tag{3}$$

By (126.2), the condition for the Born approximation is $\alpha m/(\hbar^2 k a^{n-1}) \ll 1$. Notice the dependence $\sigma \sim k^{-2}$, as anticipated by the general statements in the text.

PROBLEM

6. Find the Born-approximation scattering amplitude in two dimensions, for $U = U(x, z)$ and a particle beam incident along the z axis.

SOLUTION. Using the note on p. 204 and the familiar asymptotic form of the Hankel function

$$H_0^{(1)}(u) \approx \sqrt{\frac{2}{\pi u}} e^{i(u-\pi/4)} \quad \text{as } u \to \infty, \tag{4}$$

we find, at large distance ρ_0 from the field axis (the y axis),

$$\psi^{(1)} \sim \frac{f(\varphi)}{\sqrt{\rho_0}} e^{ik\rho_0},$$

where

$$f(\varphi) = -\frac{m}{\hbar^2}\sqrt{\frac{i}{2\pi k}} \int U(\boldsymbol{\rho}) e^{-i\mathbf{q}\boldsymbol{\rho}} d^2\rho$$

is the scattering amplitude. Here $\boldsymbol{\rho} = (x, z)$ is the two-dimensional radius vector, $d^2\rho = dx\,dz$, and φ is the scattering angle in the xz plane. In two dimensions the amplitude has the dimension of the square root of length, while the scattering cross-section $d\sigma = |f|^2 d\varphi$ has the dimension of length.

§ 127. The semiclassical case

We shall trace how the classical limit emerges from the quantum-mechanical theory of scattering.

Excluding the zero scattering angle θ from consideration, we can write the scattering amplitude given by the exact theory in the form (124.4):

$$f(\theta) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta)e^{2i\delta_l}. \tag{127.1}$$

We know that semiclassical wave functions have large phases. It is therefore natural to anticipate that the limiting transition in scattering theory corresponds to large phases δ_l. The main contribution to the sum (127.1) comes from terms with large l. We may consequently replace $P_l(\cos\theta)$ by the asymptotic expression (49.7), written as

$$P_l(\cos\theta) \approx -\frac{i}{\sqrt{2\pi l \sin\theta}}\left\{\exp\left[i\left(l+\frac{1}{2}\right)\theta + \frac{i\pi}{4}\right] - \exp\left[-i\left(l+\frac{1}{2}\right)\theta - \frac{i\pi}{4}\right]\right\}.$$

Substitution in (127.1) gives

$$f(\theta) = \frac{1}{k}\sum_l \sqrt{\frac{l}{2\pi\sin\theta}}\left(\exp\left\{i\left[2\delta_l - \left(l+\frac{1}{2}\right)\theta - \frac{\pi}{4}\right]\right\} - \exp\left\{i\left[2\delta_l + \left(l+\frac{1}{2}\right)\theta + \frac{\pi}{4}\right]\right\}\right). \tag{127.2}$$

Regarded as functions of l, the exponential factors oscillate rapidly, since their phases are large. Most of the terms in (127.2) therefore cancel one another. The sum is determined chiefly by the range of l close to the value at which the exponent of one of the exponentials is stationary, that is, close to a root of

$$2\frac{d\delta_l}{dl} \pm \theta = 0. \tag{127.3}$$

This range contains many terms for which the exponential factors remain nearly constant, because their exponents vary slowly near the stationary point; these terms therefore do not cancel.

In the semiclassical case, the phases δ_l can be written (see § 124) as the limit, as $r \to \infty$, of the difference between the phase

$$\frac{\pi}{4} + \frac{1}{\hbar}\int_{r_0}^{r} \sqrt{2m[E - U(r)] - \frac{\hbar^2(l+1/2)^2}{r^2}}\, dr$$

of the semiclassical wave function in the field $U(r)$ and the free-motion wave-function phase $kr - \pi l/2$ (see § 33). Thus,

$$\delta_l = \int_{r_0}^{\infty} \left[\sqrt{\frac{2m}{\hbar^2}(E - U) - \frac{(l+1/2)^2}{r^2}} - k\right] dr + \frac{\pi}{2}\left(l + \frac{1}{2}\right) - kr_0. \tag{127.4}$$

This expression is to be substituted into equation (127.3). When differentiating the integral, one must bear in mind that its limit r_0 also depends on l. The resulting term $k\,dr_0/dl$, however, is cancelled by the derivative of the term $-kr_0$ in δ_l. The quantity $\hbar(l + 1/2)$ is the particle's angular momentum. In classical mechanics it may be written as $m\rho v$, where ρ is the *impact parameter* and v is the particle's velocity at infinity.

After making this substitution, equation (127.3) takes the final form

$$2\int_{r_0}^{\infty} \frac{\rho\, dr}{r^2\sqrt{1 - U/E - \rho^2/r^2}} = \pi \pm \theta. \tag{127.5}$$

In a repulsive field this equation has a root for ρ only with the minus sign before θ on the right-hand side, and in an attractive field only with the plus sign.

Equation (127.5) coincides exactly with the classical equation relating the scattering angle to the impact parameter (see I, § 18). It is also straightforward to verify that the cross-section takes its classical form.

To do so, expand the exponent in (127.2) in powers of $l' = l - l_0(\theta)$, where $l_0(\theta)$ is determined by (127.3)–(127.5). For definiteness, consider the first term in (127.2), and accordingly take the lower sign in (127.3), the repulsive case. From (127.3), observe that

$$\frac{d\delta_l}{dl} = \frac{1}{2}\frac{d\theta}{dl_0}.$$

We then have

$$i\left[2\delta_l - \left(l + \frac{1}{2}\right)\theta - \frac{\pi}{4}\right] \approx i\left[2\delta_{l_0} - \left(l_0 + \frac{1}{2}\right)\theta - \frac{\pi}{4}\right] + i\frac{1}{2}\frac{d\theta}{dl_0}l'^2.$$

We now replace the summation over l in (127.2) by integration over dl' near $l' = 0$. Treating l' as a complex variable, direct the integration path near that point along the direction of steepest descent of the exponential, at an angle $\pi/4$ or $-\pi/4$ to the real axis according to the sign of $d\theta/dl_0$. In other words, put $l' = \xi \exp(\pm i\pi/4)$ and integrate over real ξ. Since the integral converges rapidly, its limits may be extended from $-\infty$ to ∞:

$$\int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}\left|\frac{d\theta}{dl_0}\right|\xi^2\right) d\xi = \sqrt{2\pi\left|\frac{dl_0}{d\theta}\right|}.$$

The result is

$$f(\theta) = \frac{1}{k}\left(\frac{l_0}{\sin\theta}\left|\frac{dl_0}{d\theta}\right|\right)^{1/2} \exp\left\{i\left[2\delta_{l_0} - \left(l_0 + \frac{1}{2}\right)\theta - \frac{\pi}{4}\right]\right\}. \qquad (127.6)$$

It follows that

$$d\sigma = |f|^2\, 2\pi \sin\theta\, d\theta = 2\pi\frac{l_0}{k^2}\left|\frac{dl_0}{d\theta}\right| d\theta, \qquad (127.7)$$

and on introducing the impact parameter by $\rho = l_0/k$, we recover the classical formula $d\sigma = 2\pi\rho\, d\rho$.

Thus, for scattering through a prescribed angle θ to be classical, both the value of l satisfying (127.3) and the phase δ_l at that value must be large[10]. This condition has a simple meaning. To speak of classical scattering through an angle θ for a particle passing with impact parameter ρ, the quantum-mechanical uncertainties in both quantities must be relatively small: $\Delta\rho \ll \rho$ and $\Delta\theta \ll \theta$. The uncertainty of the scattering angle is of order $\Delta\theta \sim \Delta p/p$, where p is the particle momentum and Δp is the uncertainty in its transverse component. Since $\Delta p \sim \hbar/\Delta\rho \gg \hbar/\rho$,

we have $\Delta\theta \gg \hbar/p\rho$, and consequently, in any event,

$$\theta \gg \frac{\hbar}{\rho m v}. \qquad (127.8)$$

Replacing the angular momentum $m\rho v$ by $\hbar l$, we obtain $\theta l \gg 1$, which is the same as the condition $\delta_l \gg 1$ (because $\delta_l \sim l\theta$, as follows from (127.3)).

The classical deflection angle can be estimated as the ratio of the transverse momentum increment Δp acquired during the collision time $\tau \sim \rho/v$ to the initial momentum mv. The force on a particle in the field $U(r)$ at a distance ρ is $F = -dU(\rho)/d\rho$, and hence $\Delta p \sim F\rho/v$, so that $\theta \sim \rho F/mv^2$. Strictly, this estimate is valid only for $\theta \ll 1$, but as an order-of-magnitude estimate it may be extended to $\theta \sim 1$. Substitution in (127.8) gives the condition for semiclassical scattering:

$$|F|\rho^2 \gg \hbar v. \qquad (127.9)$$

This inequality must hold for all values of ρ for which $|U(\rho)| \lesssim E$ still holds.

If the field $U(r)$ falls off faster than $1/r$, condition (127.9) necessarily fails at sufficiently large ρ. Large ρ, however, correspond to small θ; hence scattering through

[10]The relation between θ and ρ given by (127.5) need not be single-valued; more than one value of ρ may then correspond to the same θ. In that event, $f(\theta)$ is a sum of expressions (127.6) with the corresponding values of l_0. At extrema of $\theta(\rho)$, the derivative $d\rho/d\theta$, and with it the classical differential cross-section $d\sigma/do$, becomes infinite; the classical approximation is, of course, inadequate near such an angle (see Problem 2).

sufficiently small angles cannot be classical. If instead the field falls off more slowly than $1/r$, small-angle scattering will be classical. Whether large-angle scattering is also classical in this case depends on how the field behaves at short distances.

For the Coulomb field $U = \alpha/r$, condition (127.9) holds when $\alpha \gg \hbar v$. This is the reverse of the condition under which the Coulomb field may be treated as a perturbation. We shall see, however, that for accidental reasons the quantum theory of scattering in a Coulomb field gives the same result as the classical theory in every case.

Problems

1. Find the total semiclassical scattering cross-section in a field whose form at sufficiently large distances is $U = \alpha/r^n$, with $n > 2$.

SOLUTION. Bearing in mind that the main contribution comes from phases δ_l with large l, calculate them from (124.1):

$$\delta_l = -\frac{m\alpha}{\hbar^2}\int_{l/k}^{\infty}\frac{dr}{r^n\sqrt{k^2 - l^2/r^2}} = -\frac{m\alpha k^{n-2}}{2\hbar^2 l^{n-1}}\frac{\Gamma(1/2)\Gamma((n-1)/2)}{\Gamma(n/2)}. \tag{1}$$

(for evaluation of the integral, cf. Problem 5 of § 126). Replacing the summation in (123.11) by integration, write

$$\sigma = \frac{4\pi}{k^2}\int_0^{\infty} 2l \sin^2\delta_l \, dl.$$

After the substitution $\delta_l = u$ and an integration by parts with respect to du, the integral reduces to a Γ-function. The result is

$$\sigma = \frac{2\pi}{n-1}\sin\left[\frac{\pi}{2}\left(\frac{n-3}{n-1}\right)\right]\Gamma\left(\frac{n-3}{n-1}\right)\left[\frac{\sqrt{\pi}\,\Gamma((n-1)/2)}{\Gamma(n/2)}\right]^{2/(n-1)}\left(\frac{m\alpha}{\hbar^2 k}\right)^{2/(n-1)}. \tag{2}$$

(for $n = 3$, resolving the indeterminate form gives $\sigma = 2\pi^2\alpha m/\hbar v$).

For this result to apply, it is necessary first of all that $l \gg 1$ when $\delta_l \sim 1$; this gives the inequality

$$\frac{m\alpha k^{n-2}}{\hbar^2} \gg 1.$$

Another condition follows from requiring the field $U(r)$ to have the form under consideration already at distances

$$r \sim l/k \sim (m\alpha/\hbar^2 k)^{1/(n-1)}$$

(where l is obtained from $\delta_l \sim 1$), which make the main contribution to the integral (1). If this form is reached only at distances $r \gg a$, where a is the characteristic size of the field, this gives the condition

$$\frac{m\alpha}{\hbar^2 k a^{n-1}} \gg 1,$$

which sets an upper limit on the permissible speeds. Recall that in the opposite case, at sufficiently high speeds and subject to $m\alpha/\hbar^2 k a^{n-1} \ll 1$, one has $\sigma \sim k^{-2}$ (cf. Problem 5 of § 126).

PROBLEM

2. Determine the distribution of scattering over angle near an extremum of the classical deflection angle $\theta(\rho)$, regarded as a function of the impact parameter $\rho = l/k$.

SOLUTION. An extremum of $\theta(l)$ at some $l = l_0$ means, by (127.3), that the phase near this point has the form

$$\delta_l \approx \delta_{l_0} + \frac{\theta_0}{2} l' + \frac{\alpha}{3} l'^3,$$

where $\theta_0 = \theta(l_0)$ and $l' = l - l_0$ (again choosing, for definiteness, the case of the lower sign in (127.3)); the constant α is negative or positive according as $\theta(l)$ has a maximum or a minimum. In place of (127.6), the scattering amplitude is

$$|f(\theta)| = \frac{1}{k}\left(\frac{l_0}{2\pi \sin\theta_0}\right)^{1/2} \left|\int_{-\infty}^{\infty} \exp\left[i\left(\frac{2\alpha}{3}l'^3 - \theta' l'\right)\right] dl'\right|,$$

Here $\theta' = \theta - \theta_0$. Expressing the integral in terms of the Airy function by (b.3), we finally obtain for the scattering cross section[11]

$$d\sigma = \frac{4\pi l_0}{\alpha^{2/3} k^2} \Phi^2\left(-\frac{\theta'}{\alpha^{1/3}}\right) d\theta.$$

§ 128. Analytic properties of the scattering amplitude

Several important properties of the scattering amplitude can be established by studying it as a function of the scattered particle's energy E, formally regarded as a complex variable.

Consider motion in a field $U(r)$ which tends sufficiently rapidly to zero at infinity; the required rate will be specified below. To simplify the argument, suppose initially that the particle's orbital angular momentum is $l = 0$. Write the asymptotic form

of the wave function, a solution of the Schrödinger equation with $l = 0$ at an arbitrary specified E, as

$$\chi = r\psi = A(E)\exp\left(-\frac{\sqrt{-2mE}}{\hbar} r\right) + B(E)\exp\left(\frac{\sqrt{-2mE}}{\hbar} r\right), \tag{128.1}$$

and regard E as complex. We define $\sqrt{-E}$ to be positive when E is real and negative. The wave function is assumed normalized by some fixed condition, say $\psi(0) = 1$.

On the left-hand part of the real axis ($E < 0$), the exponential factors in the two terms of (128.1) are real; one decreases and the other increases as $r \to \infty$. Since χ is real, $A(E)$ and $B(E)$ are real for $E < 0$. Consequently their values at any two points symmetric about the real axis are complex conjugates:

$$A(E^*) = A^*(E), \qquad B(E^*) = B^*(E). \tag{128.2}$$

[11] Scattering of this type occurs in the theory of the rainbow and is therefore called *rainbow scattering*.

Continuing from the negative to the positive real semiaxis through the upper half-plane gives, for $E > 0$,

$$\chi = A(E)e^{ikr} + B(E)e^{-ikr}, \qquad k = \frac{\sqrt{2mE}}{\hbar}. \tag{128.3}$$

Continuation through the lower half-plane would instead give

$$\chi = A^*(E)e^{-ikr} + B^*(E)e^{ikr}.$$

Since χ must be single-valued in E, it follows that

$$A(E) = B^*(E) \quad \text{for } E > 0. \tag{128.4}$$

This also follows directly from the reality of χ for $E > 0$. The coefficients $A(E)$ and $B(E)$ themselves are nevertheless multivalued because of the square root $\sqrt{-E}$ in (128.1). Remove this ambiguity by cutting the complex plane along the positive real semiaxis. The cut makes $\sqrt{-E}$, and hence $A(E)$ and $B(E)$, single-valued. Their values on the upper and lower edges are complex conjugates; in (128.3), A and B are evaluated on the upper edge.

The complex plane cut in this way will be called the *physical sheet* of the Riemann surface. With our definition,

$$\operatorname{Re}\sqrt{-E} > 0 \tag{128.5}$$

throughout this sheet. In particular, on the upper edge of the cut, $\sqrt{-E}$ becomes $-i\sqrt{E}$.

In (128.3), the factors e^{ikr} and e^{-ikr}, and therefore the two terms in χ, have the same order of magnitude, so this asymptotic expression is always valid. Everywhere else on the physical sheet, the first term in (128.1) decreases exponentially and the second increases as $r \to \infty$, by (128.5). The terms then have different orders, and (128.1) may cease to be a legitimate asymptotic form: the smaller term against the larger may exceed the permitted accuracy. For (128.1) to be valid, the ratio of the small term to the large one must not be below the relative order U/E neglected in the Schrödinger equation on passing to the asymptotic region. Thus $U(r)$ must decrease as $r \to \infty$ faster than

$$\exp\left(-\frac{2\sqrt{2m}}{\hbar} r \operatorname{Re}\sqrt{-E}\right). \tag{128.6}$$

If this holds for every $\operatorname{Re}\sqrt{-E} > 0$, that is, if $U(r)$ decreases faster than

$$e^{-cr} \tag{128.6a}$$

for every positive constant c, an asymptotic expression of the form (128.1) is valid throughout the physical sheet. As a solution of an equation with finite coefficients, it has no singularities in E. Hence $A(E)$ and $B(E)$ are regular throughout the physical sheet except at $E = 0$, the end of the cut and therefore their branch point.[12]

[12]Throughout this section we study the scattering amplitude on the physical sheet. Later, however, it will sometimes be necessary to consider the second, *unphysical* sheet of the Riemann surface (see § 134). On that sheet

$$\operatorname{Re}\sqrt{-E} < 0. \tag{128.5a}$$

The unphysical sheet is reached directly downward through the cut from the positive semiaxis.

Bound states in $U(r)$ have wave functions tending to zero as $r \to \infty$. The second term of (128.1) must therefore vanish, so the discrete energy levels are zeros of $B(E)$. Since the Schrödinger equation has only real eigenvalues, all zeros of $B(E)$ on the physical sheet are real and lie on the negative real semiaxis.

For $E > 0$, $A(E)$ and $B(E)$ are directly related to the scattering amplitude. Comparing (128.3) with (33.20), written as

$$\chi = \text{const} \cdot [e^{i(kr+\delta_0)} - e^{-i(kr+\delta_0)}], \tag{128.7}$$

gives

$$-\frac{A(E)}{B(E)} = e^{2i\delta_0(E)}. \tag{128.8}$$

By (123.15), the $l = 0$ scattering amplitude is

$$\hbar f_0 = \frac{1}{2ik}(e^{2i\delta_0} - 1) = \frac{\hbar}{2\sqrt{-2mE}}\left(\frac{A}{B} + 1\right), \tag{128.9}$$

where A and B are taken on the upper edge of the cut.

Considering the amplitude as a function of E over the physical sheet, we see that the discrete energy levels are its simple poles. If $U(r)$ satisfies (128.6a), the amplitude has no other singular points, by the preceding argument[13].

Let us calculate the residue at a pole corresponding to a discrete level $E = E_0 < 0$. The equations for χ and its energy derivative are

$$\chi'' + \frac{2m}{\hbar^2}(E - U)\chi = 0, \qquad \left(\frac{\partial\chi}{\partial E}\right)'' + \frac{2m}{\hbar^2}(E - U)\frac{\partial\chi}{\partial E} = -\frac{2m}{\hbar^2}\chi.$$

Multiply the first by $\partial\chi/\partial E$, the second by χ, subtract, and integrate over dr. One obtains

$$\chi'\frac{\partial\chi}{\partial E} - \chi\left(\frac{\partial\chi}{\partial E}\right)' = \frac{2m}{\hbar^2}\int_0^r \chi^2 dr. \tag{128.10}$$

Apply this at $E = E_0$ and $r \to \infty$. The integral on the right becomes unity if the bound-state wave function is normalized by $\int \chi^2 dr = 1$. In the left-hand side insert (128.1), noting that near $E = E_0$

$$A(E) \approx A(E_0) = A_0, \qquad B(E) \approx (E + |E_0|)\left.\frac{dB}{dE}\right|_{E=E_0} = \beta(E + |E_0|).$$

The result is $\beta = -1/(A_0\hbar\sqrt{2m|E_0|})$.

These expressions show that near $E = E_0$ the leading term of the scattering amplitude, which is the $l = 0$ amplitude, is

$$f = \frac{\hbar^2 A_0^2}{2m}\frac{1}{E + |E_0|}. \tag{128.11}$$

[13] Apart from $E = 0$, which is singular because of the stated singularity of $A(E)$ and $B(E)$. The scattering amplitude itself remains finite as $E \to 0$ (see § 132). For brevity, this qualification will not be repeated below.

Thus its residue at a discrete level is determined by the coefficient A_0 in the asymptotic expression

$$\chi = A_0 \exp\left(-\frac{\sqrt{2m|E_0|}}{\hbar} r\right) \tag{128.12}$$

for the normalized wave function of the corresponding stationary state.

Return now to the analytic properties when (128.6a) is not fulfilled. In such fields only the increasing term in (128.1) is a valid part of the asymptotic solution throughout the physical sheet. Accordingly, $B(E)$ can still be said to have no singularities.

Under these conditions $A(E)$ can be defined in the complex plane only by analytic continuation of the coefficient in the asymptotic expression for χ on the positive real semiaxis, where both terms are valid. Such a continuation generally gives different results according as it is made from the upper or lower edge of the cut.

To make the definition single-valued, define $A(E)$ in the upper and lower half-planes by analytic continuation from the upper and lower sides, respectively, of the positive semiaxis. In general the cut must then

be extended along the whole real axis. The function so defined still satisfies $A(E^*) = A^*(E)$, but need not be real on either the positive or negative real semiaxis. In principle it may also have singularities.

Nevertheless, there is a class of fields for which $A(E)$ has no singularities inside the physical sheet even though (128.6a) does not hold.

To see this, regard χ as a function of complex r at a fixed complex E. It is enough to consider E in the upper half-plane, since the values of $A(E)$ in the two half-planes are complex conjugates. For values of r such that Er^2 is real and positive, the two terms in (128.1) have the same order. This restores the situation for positive E and real r, in which both asymptotic terms are valid for any field tending to zero at infinity. Thus $A(E)$ cannot have singular points at those E for which $U(r) \to 0$ as $r \to \infty$ along the ray on which $Er^2 > 0$. As E ranges over the upper half-plane, this condition selects the lower right quadrant of the complex r-plane. We conclude that $A(E)$ has no singularities within the physical sheet also when[14]

$$U(r) \to 0, \quad \text{as } r \to \infty \text{ in the right half-plane} \tag{128.13}$$

(L. D. Landau, 1961).

Conditions (128.6a) and (128.13) cover a very broad class of fields. As a rule, therefore, the scattering amplitude has no singularities inside either half-plane. On the negative semiaxis itself, which belongs to the physical sheet when there is no cut there, it has poles at the bound-state energies; if a cut is present, other singularities may also occur there.

The latter situation occurs, in particular, for fields of the form

$$U = \text{const} \cdot r^n e^{-r/a} \tag{128.14}$$

for any n. On the segment $0 < -E < \hbar^2/(8ma^2)$ of the negative semiaxis, (128.6) holds; there can be no cut there, and the amplitude has only the poles of bound states. On

[14] Since $U(r)$ is real on the real axis, $U(r^*) = U^*(r)$. Thus fulfillment of (128.13) in the lower right quadrant automatically implies its fulfillment throughout the right half-plane.

the remainder of the negative semiaxis there may also be *extraneous* poles and other singularities (S. T. Ma, 1946). They arise because (128.14) ceases to tend to zero along the ray $Er^2 > 0$ as soon as E passes below the negative semiaxis, when that ray passes leftward beyond the imaginary axis in the complex r-plane.

Next consider the analytic properties as $|E| \to \infty$. When $E \to +\infty$ along the real axis, the Born approximation applies and the scattering amplitude tends to zero. The same situation holds when E tends to infinity in the complex plane along a line $\arg E = \text{const}$, if one considers complex r for which $Er^2 > 0$. If $U \to 0$ along $\arg r = -(1/2)\arg E$ and has no singular points on that line, the Born approximation is applicable and the amplitude again tends to zero. As $\arg E$ ranges from 0 to π, $\arg r$ ranges from 0 to $-\pi/2$.

It follows that the scattering amplitude tends to zero at infinity in every direction of the E plane if $U(r)$ has no singular points in the right half-plane of r and tends to zero at infinity there.

Although the discussion was phrased for $l = 0$, all these results also apply to partial scattering amplitudes with any nonzero angular momentum. The only difference in the derivation is that the exact free radial wave functions (33.16) must replace the factors $e^{\pm ikr}$ in the asymptotic expressions for χ[15].

For $l \neq 0$, formulas (128.9) and (128.11) require some modifications. Instead of (128.7), one has

$$\chi_l = rR_l = \text{const} \cdot \left\{ \exp\left[i\left(kr - \frac{l\pi}{2} + \delta_l \right) \right] - \exp\left[-i\left(kr - \frac{l\pi}{2} + \delta_l \right) \right] \right\} \qquad (128.15)$$

and the partial amplitude f_l, defined as in (123.15), is

$$f_l = \frac{\hbar}{2\sqrt{-2mE}} \left[(-1)^l \frac{A}{B} + 1 \right]. \qquad (128.16)$$

In place of (128.11), the leading term of the scattering amplitude near a level $E = E_0$ of angular momentum l is

$$f \approx (2l+1) f_l P_l(\cos\theta) = (-1)^{l+1} \frac{\hbar^2 A_0^2}{2m} \frac{1}{E + |E_0|} (2l+1) P_l(\cos\theta). \qquad (128.17)$$

§ 129. The dispersion relation

In the preceding section we investigated the analytic properties of partial scattering amplitudes at fixed l. We found that possible "extraneous" singularities and irregular behavior at infinity complicate those properties. Clearly, the full amplitude, regarded as a function of energy at a fixed scattering angle, has the same complications. There is, however, one exception: the amplitude for zero-angle scattering. We shall now show that its analytic properties are substantially simpler.

[15]The limiting form (33.17) of those functions may be used only for $E > 0$. Elsewhere in the E plane the two terms in χ have different orders; using the limiting expressions would generally introduce an error larger than that due to neglecting U in the Schrödinger equation.

Write the Schrödinger equation for the scattered particle's wave function as

$$\Delta\psi + k^2\psi = \frac{2mU}{\hbar^2}\psi. \tag{129.1}$$

We shall formally regard this as a wave equation with a right-hand side, that is, as the equation of retarded potentials familiar from electrodynamics.

As is known, the solution describing "radiation" in a direction $\mathbf{k}'$ at large distances R_0 from the center is (see II, § 66)

$$\psi_{\rm sc} = -\frac{1}{4\pi}\frac{e^{ikR_0}}{R_0}\int \frac{2mU}{\hbar^2}\psi e^{-i\mathbf{k}'\mathbf{r}}dV. \tag{129.2}$$

Here this expression is the scattered-particle wave function, and the coefficient of e^{ikR_0}/R_0 is the scattering amplitude $f(\theta, E)$. In particular, setting $\mathbf{k}' = \mathbf{k}$, where $\mathbf{k}$ is the wave vector of the incident particle, gives the amplitude for scattering through angle zero:

$$f(0, E) = -\frac{m}{2\pi\hbar^2}\int U\psi e^{-ikz}dV \tag{129.3}$$

(the z axis is directed along $\mathbf{k}$). Of course, this formula is only formal, since the integrand again contains the unknown wave function. It does, nevertheless, permit definite conclusions about the analytic properties of $f(0, E)$ as a function of E[16].

At large r, the function ψ under the integral consists of an incident wave and an outgoing wave. The latter is proportional to e^{ikr}, so its contribution to the integral contains $e^{ik(r-z)}$ in the integrand. On continuation into the complex plane from the upper bank of the cut along the positive real semiaxis, ik becomes $-\sqrt{-2mE}/\hbar$; throughout the physical sheet, Re $\sqrt{-E} > 0$. Since $r \geqslant z$, we have $\mathrm{Re}[ik(r-z)] < 0$, and the integral converges for every complex E. For the incident-wave part of ψ, which is proportional to e^{ikz}, the exponential factors cancel altogether, so that contribution also converges.

For any complex E, the function ψ in (129.3) is uniquely specified as the solution of the Schrödinger equation which, besides the plane wave, contains only a part that decays as $r \to \infty$. The entire convergent integral (129.2) is therefore uniquely specified as well, and its singularities can arise only when ψ becomes infinite. This occurs at the discrete energy levels[17].

It is also readily seen that $f(0, E)$ stays finite as $|E| \to \infty$. For large $|E|$, the term containing U in the Schrödinger equation (129.1) can be omitted, leaving only the plane wave in ψ: $\psi \sim e^{ikz}$. The integral (129.2) then becomes

$$f(0, \infty) = -\frac{m}{2\pi\hbar^2}\int U\,dV.$$

[16] It is understood that the field $U(r)$ decreases sufficiently rapidly as $r \to \infty$ for $f(0, E)$ to exist at all when $E > 0$ (see § 124).

[17] To avoid misunderstanding, we emphasize that ψ here is the complete wave function of the system, normalized by setting the coefficient of the plane wave in its asymptotic expression equal to one (compare (123.3)). In the preceding section, however, the parts ψ_l of the wave function corresponding to definite l were considered, with each ψ_l normalized by some arbitrary condition. If the complete function ψ is expanded in the functions ψ_l, their coefficients in ψ are proportional to $1/B_l$. Thus, for $l = 0$, the function (128.3)

must enter ψ in the form $$\frac{\text{const}}{r}\frac{1}{B}[(A+B)e^{ikr} - 2iB\sin kr].$$

Consequently, ψ becomes infinite at the zeros of $B_l(E)$, that is, at the discrete energy levels.

As it should, this agrees with the Born amplitude (126.4) for scattering through zero angle ($q = 0$); denote it by $f_B(0)$.

We conclude that the zero-angle scattering amplitude is regular everywhere on the physical sheet, infinity included, apart from the necessary poles on the negative real semiaxis at the discrete energy levels[18].

Consider the integral

$$\frac{1}{2\pi i}\int_C \frac{f(0, E') - f_B}{E' - E}\, dE', \tag{129.4}$$

taken over the contour shown in Fig. 46, which consists of a circle at infinity and a path around the cut along the positive semiaxis. The integral over the circle vanishes because $f(0, \infty) - f_B = 0$. Integration along the two banks of the cut gives

$$\frac{1}{\pi}\int_0^\infty \frac{\operatorname{Im} f(0, E')}{E' - E}\, dE'.$$

Here we used the convention adopted in § 128: for real positive E, the physical scattering amplitude is specified on the upper bank of the cut, and its value on the lower bank is the complex conjugate.

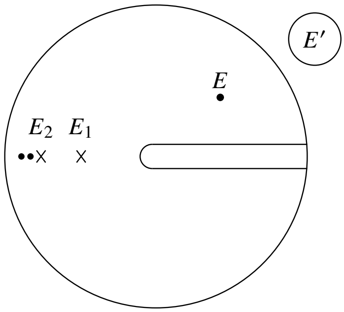


Figure 46

On the other hand, Cauchy's theorem says that (129.4) equals the sum of $f(0, E) - f_B$ and the residues R_n of the integrand at all poles $E' = E_n$ of $f(0, E')/(E' - E)$,

where the E_n are the discrete energy levels. Formula (128.17) gives these residues as

$$R_n = \frac{d_n}{E_n - E}, \qquad d_n = -(-1)^{l_n}(2l_n + 1)\frac{\hbar^2 A_{0n}^2}{2m} \tag{129.5}$$

(l_n is the angular momentum of the state whose energy is E_n).

It follows that

$$f(0, E) = f_B + \frac{1}{\pi}\int_0^\infty \frac{\operatorname{Im} f(0, E')}{E' - E}\, dE' + \sum_n \frac{d_n}{E - E_n}. \tag{129.6}$$

[18] The idea of this proof is due to L. D. Faddeev (1958).

This is the so-called *dispersion relation*. From the values of the imaginary part for $E > 0$, it determines $f(0, E)$ at every point of the physical sheet (D. Wong, 1957; N. N. Khuri, 1957).

As the point E approaches the upper bank of the cut, the real-axis integral in (129.6) must pass below the pole at $E' = E$. If the detour is made along an infinitesimal semicircle (Fig. 47), its contribution to the right-hand side of (129.6) is $i \operatorname{Im} f(0, E)$, while the remaining integral from zero to infinity is to be understood as a principal value. We thereby obtain

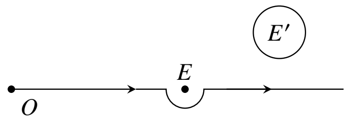


Figure 47

$$\operatorname{Re} f(0, E) = f_B + \frac{1}{\pi} \mathrm{P} \int_0^\infty \frac{\operatorname{Im} f(0, E')}{E' - E} dE' + \sum_n \frac{d_n}{E - E_n}, \tag{129.7}$$

which expresses the real part of the zero-angle scattering amplitude at $E > 0$ through its imaginary part. Recall that, by (125.9), the latter is directly related to the total scattering cross section.

§ 130. The scattering amplitude in the momentum representation

The scattering amplitude involves only the directions of the scattered particle's initial and final momenta. It is therefore natural that the same concept can

also be reached by formulating the scattering problem in the momentum representation, where the spatial distribution of the process as a whole is not considered at all. Let us show how this is done.

First transform the original Schrödinger equation

$$-\frac{\hbar^2}{2m} \Delta\psi(\mathbf{r}) + [U(\mathbf{r}) - E]\psi(\mathbf{r}) = 0 \tag{130.1}$$

to the momentum representation, passing from coordinate wave functions to their Fourier components

$$a(\mathbf{q}) = \int \psi(\mathbf{r}) e^{-i\mathbf{qr}} dV. \tag{130.2}$$

Conversely,

$$\psi(\mathbf{r}) = \int a(\mathbf{q}) e^{i\mathbf{qr}} \frac{d^3q}{(2\pi)^3}. \tag{130.3}$$

Multiply (130.1) by $e^{-i\mathbf{qr}}$ and integrate over dV. Two integrations by parts in the first term give

$$\int e^{-i\mathbf{qr}} \Delta\psi(\mathbf{r}) dV = \int \psi(\mathbf{r}) \Delta e^{-i\mathbf{qr}} dV = -q^2 a(\mathbf{q}).$$

In the second term, inserting (130.3), we find

$$\int U(\mathbf{r})\psi(\mathbf{r})e^{-i\mathbf{qr}}dV = \iint U(\mathbf{r})e^{-i\mathbf{qr}}a(\mathbf{q}')e^{i\mathbf{q}'\mathbf{r}}dV\frac{d^3q'}{(2\pi)^3}$$
$$= \int U(\mathbf{q}-\mathbf{q}')a(\mathbf{q}')\frac{d^3q'}{(2\pi)^3},$$

where $U(\mathbf{q})$ denotes the Fourier component of $U(\mathbf{r})$ [19]:

$$U(\mathbf{q}) = \int U(\mathbf{r})e^{-i\mathbf{qr}}dV.$$

Thus the Schrödinger equation in the momentum representation is

$$\left(\frac{\hbar^2q^2}{2m} - E\right)a(\mathbf{q}) + \int U(\mathbf{q}-\mathbf{q}')a(\mathbf{q}')\frac{d^3q'}{(2\pi)^3} = 0. \tag{130.4}$$

Notice that this is an integral equation, not a differential equation.

Write the wave function describing the scattering of particles with momentum $\hbar\mathbf{k}$ as

$$\psi_\mathbf{k}(\mathbf{r}) = e^{i\mathbf{kr}} + \chi_\mathbf{k}(\mathbf{r}), \tag{130.5}$$

where $\chi_\mathbf{k}(\mathbf{r})$ has asymptotically, as $r \to \infty$, the form of an outgoing spherical wave. Its Fourier component is

$$a_\mathbf{k}(\mathbf{q}) = (2\pi)^3\delta(\mathbf{q}-\mathbf{k}) + \chi_\mathbf{k}(\mathbf{q}), \tag{130.6}$$

and substitution in (130.4) gives the following equation for $\chi_\mathbf{k}(\mathbf{q})$[20]:

$$-\frac{\hbar^2}{2m}(k^2-q^2)\chi_\mathbf{k}(\mathbf{q}) = U(\mathbf{q}-\mathbf{k}) + \int U(\mathbf{q}-\mathbf{q}')\chi_\mathbf{k}(\mathbf{q}')\frac{d^3q'}{(2\pi)^3}. \tag{130.7}$$

It is useful to transform this equation by introducing a new unknown function in place of $\chi_\mathbf{k}(\mathbf{q})$, defined by

$$\chi_\mathbf{k}(\mathbf{q}) = -\frac{2m}{\hbar^2}\frac{F(\mathbf{k},\mathbf{q})}{q^2-k^2-i0}. \tag{130.8}$$

This removes the singularity at $q^2 = k^2$ from the coefficients in (130.7), which becomes

$$F(\mathbf{k},\mathbf{q}) = U(\mathbf{q}-\mathbf{k}) - \frac{2m}{\hbar^2}\int \frac{U(\mathbf{q}-\mathbf{q}')F(\mathbf{k},\mathbf{q}')}{q'^2-k^2-i0}\frac{d^3q'}{(2\pi)^3}. \tag{130.9}$$

The term $i0$, meaning the limit $i\varepsilon$ as $\varepsilon \to +0$, has been included in (130.8) to assign a definite meaning to the integral in (130.9): it specifies how the pole $q'^2 = k^2$ is bypassed (compare § 43). We shall show that this prescription gives the required asymptotic form of

$$\chi_\mathbf{k}(\mathbf{r}) = -\frac{2m}{\hbar^2}\int \frac{F(\mathbf{k},\mathbf{q})e^{i\mathbf{qr}}}{q^2-k^2-i0}\frac{d^3q}{(2\pi)^3}. \tag{130.10}$$

[19] For notational convenience, $\mathbf{q}$ is written as an argument of the Fourier component rather than as a subscript.

[20] By the properties of the delta-function, the product $(q^2-k^2)\delta(\mathbf{q}-\mathbf{k})$ gives zero when multiplied by an arbitrary function $f(\mathbf{q})$ nonsingular at $\mathbf{q} = \mathbf{k}$ and integrated over d^3q. In this sense, $(q^2-k^2)\delta(\mathbf{q}-\mathbf{k}) = 0$.

Write $d^3q = q^2 dq\, do_q$ and first integrate over do_q, the directions of $\mathbf{q}$ relative to $\mathbf{r}$. An integration of this kind was already carried out in transforming the first term of (125.2); at large r it gives

$$\chi_{\mathbf{k}}(\mathbf{r}) = -\frac{2m}{\hbar^2}\frac{2\pi i}{r}\int_0^\infty \frac{F(\mathbf{k}, q\mathbf{n}')e^{iqr} - F(\mathbf{k}, -q\mathbf{n}')e^{-iqr}}{q^2 - k^2 - i0} q\, dq \frac{1}{(2\pi)^3},$$

where $\mathbf{n}' = \mathbf{r}/r$, or

$$\chi_{\mathbf{k}}(\mathbf{r}) = -\frac{im}{2\pi^2\hbar^2 r}\int_{-\infty}^\infty \frac{F(\mathbf{k}, q\mathbf{n}')e^{iqr} q\, dq}{q^2 - k^2 - i0}.$$

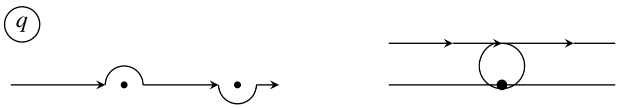
The integrand has poles at $q = k + i0$ and $q = -k - i0$, bypassed below and above, respectively, in integration through the complex q plane (Fig. 48a). Shift the contour slightly into the upper half-plane, replacing it by a straight line parallel to the real axis and a closed loop enclosing the pole $q = k$ (Fig. 48b). As $r \to \infty$, the integral along the straight line vanishes because the integrand contains $\exp(-r \operatorname{Im} q)$. The closed-loop integral is the residue at $q = k$ multiplied by $2\pi i$. We finally obtain

$$\chi_{\mathbf{k}}(\mathbf{r}) = -\frac{m}{2\pi\hbar^2}\frac{e^{ikr}}{r} F(k\mathbf{n}, k\mathbf{n}'), \tag{130.11}$$

where $\mathbf{n}$ is the unit vector along $\mathbf{k}$. Thus the required asymptotic form is obtained, and the scattering amplitude is

$$f(\mathbf{n}, \mathbf{n}') = -\frac{m}{2\pi\hbar^2} F(k\mathbf{n}, k\mathbf{n}'). \tag{130.12}$$

The scattering amplitude is therefore determined by the value at $q = k$ of the function $F(\mathbf{k}, \mathbf{q})$ satisfying (130.9).



(a) (b)

Figure 48

When perturbation theory applies, (130.9) is readily solved by successive iteration. In the first approximation, omitting the integral term altogether,

one has $F(\mathbf{k}, \mathbf{q}) = U(\mathbf{q} - \mathbf{k})$. In the next approximation, insert this first-order expression for F in the integral term. After a slight change of variables, (130.12) then gives

$$f(\mathbf{n}, \mathbf{n}') = -\frac{m}{2\pi\hbar^2}\left\{U(\mathbf{k}' - \mathbf{k}) + \frac{2m}{\hbar^2}\int \frac{U(\mathbf{k}' - \mathbf{k}'')U(\mathbf{k}'' - \mathbf{k})}{k^2 - k''^2 + i0}\frac{d^3k''}{(2\pi)^3}\right\}, \tag{130.13}$$

where $\mathbf{k} = k\mathbf{n}$ and $\mathbf{k}' = k\mathbf{n}'$. The first term is the first-Born-approximation formula (126.4); the second is the contribution of the second approximation to the scattering amplitude[21].

Formula (130.13) demonstrates the fact mentioned in § 126: already in the second approximation the scattering amplitude loses the symmetry (126.8). At first sight the integral term in (130.13) might also seem symmetric under interchange of the initial and final states. It is not, because complex conjugation changes the integration contour, namely the direction in which the pole is bypassed.

§ 131. Scattering at high energies

If the potential energy is not small in comparison with $\hbar^2/ma^2$ (where a is, as usual, the range of the field), a situation may arise in which the energy of the scattered particles is so high that

$$|U| \ll E \sim \frac{\hbar^2}{ma^2}(ka)^2, \tag{131.1}$$

while at the same time

$$|U| \gtrsim \frac{\hbar^2}{ma^2}ka = \frac{\hbar v}{a}; \tag{131.2}$$

it is understood, of course, that

$$ka \gg 1. \tag{131.3}$$

This is scattering of fast particles for which the Born approximation does not apply, since neither (126.1) nor (126.2) holds.

This case can be studied with the wave function (45.9),

$$\psi = e^{ikz}F(\mathbf{r}), \qquad F(\mathbf{r}) = \exp\left(-\frac{i}{\hbar v}\int_{-\infty}^{z} U\,dz\right), \tag{131.4}$$

whose applicability requires only $|U| \ll E$.

Section 45 noted that this expression is valid only for $z \ll ka^2$ and hence cannot be continued directly to distances where (123.3) is already valid. No such continuation is needed: to calculate the amplitude it is enough to know the wave function at $a \ll z \ll ka^2$. The integral in the exponent of $F(\mathbf{r})$ may then be extended to infinity:

$$\psi = e^{ikz}S(\boldsymbol{\rho}), \tag{131.5}$$

where

$$S(\boldsymbol{\rho}) = \exp[2i\delta(\boldsymbol{\rho})], \qquad \delta(\boldsymbol{\rho}) = -\frac{1}{2\hbar v}\int_{-\infty}^{\infty} U\,dz \tag{131.6}$$

and $\boldsymbol{\rho}$ is the radius vector in the xy plane.

Fast particles scatter mainly through small angles, which we shall consider. The momentum change $\hbar\mathbf{q}$ is then relatively small ($q \ll k$), so $\mathbf{q}$ may be regarded as

[21] This result can of course also be obtained without passing to the momentum representation. Comparison of (43.1) and (43.6) makes it evident that the second-approximation formula differs from the first by replacing $U(\mathbf{k}' - \mathbf{k})$ with the expression in braces in (130.13).

perpendicular to the incident wave vector $\mathbf{k}$, lying in the xy plane. The scattered wave is obtained by subtracting the incident wave e^{ikz}, the value of (131.4) at $z = -\infty$, from (131.5). The amplitude for scattering to $\mathbf{k}' = \mathbf{k} + \mathbf{q}$ is proportional to the corresponding Fourier component of the scattered wave[22]:

$$f \sim \int [S(\boldsymbol{\rho}) - 1] e^{-i\mathbf{q}\boldsymbol{\rho}} d^2\rho,$$

where $d^2\rho = dx\,dy$. The proportionality factor can be found by comparison with the limiting case of the Born approximation, as below.

There is another calculation which gives a definite expression directly. Use (129.2), inserting ψ from (131.4). Since (45.8) gives

$$\frac{2m}{\hbar^2} UF = 2ik \frac{\partial F}{\partial z},$$

the amplitude, the coefficient of e^{ikR_0}/R_0, is

$$f = \frac{ik}{2\pi} \int \frac{\partial F}{\partial z} e^{-i\mathbf{q}\boldsymbol{\rho}} dx\,dy\,dz = \frac{ik}{2\pi} \int [F(\infty) - F(-\infty)] e^{-i\mathbf{q}\boldsymbol{\rho}} dx\,dy.$$

Substitution of F finally gives[23]

$$f = \frac{ik}{2\pi} \int [S(\boldsymbol{\rho}) - 1] e^{-i\mathbf{q}\boldsymbol{\rho}} d^2\rho. \tag{131.7}$$

If the energy is so high that $\delta \sim |U|a/\hbar v \ll 1$, the Born approximation applies. Expanding $S - 1 \approx 2i\delta$, (131.7) gives, in agreement with (126.4),

$$f = -\frac{m}{2\pi\hbar^2} \int U e^{-i\mathbf{q}\boldsymbol{\rho}} d^2\rho\,dz.$$

The total cross-section follows from (131.7) by the optical theorem (125.9). The forward amplitude is the value at $q = 0$, whence

$$\sigma = 2 \int \mathrm{Re}(1 - S) d^2\rho = 4 \int \sin^2 \delta(\rho) d^2\rho. \tag{131.8}$$

The integrand may be regarded as the cross-section for particles with impact parameters in $d^2\rho$[24].

Formula (131.7) does not assume a central field. It is instructive to obtain it directly from the exact general formula (123.11) for a central field.

[22] This method of defining the scattering amplitude is analogous to that used for Fraunhofer diffraction (see Vol. II, § 61). It is precisely diffraction effects that invalidate (131.4) at $z \gtrsim ka^2$.

[23] In two dimensions, for a field $U(x, z)$, the scattering amplitude defined as in the problem to § 123 is

$$f = \sqrt{\frac{ik}{2\pi}} \int [S(x) - 1] e^{-iqx} dx. \tag{131.7a}$$

Here $|f|^2 d\varphi$ is the scattering cross-section per unit length along the y axis, and φ is the scattering angle in the xz plane; compare the Born amplitude in Problem 6 of § 126.

[24] Section 152 will generalize (131.7), (131.8) to scattering by a system of particles.

Under (131.1)–(131.3), partial amplitudes with large l dominate. The wave functions are therefore semiclassical, and (124.1) may be used for δ_l. Putting $r_0 \approx l/k$ and $r^2 = z^2 + l^2/k^2$ gives

$$\delta_l \approx -\frac{m}{\hbar^2}\int_{l/k}^{\infty}\frac{U(r)\,dr}{\sqrt{k^2 - l^2/r^2}} = -\frac{m}{\hbar^2 k}\int_0^{\infty} U\left(\sqrt{z^2 + l^2/k^2}\right)dz,$$

which is $\delta(\rho)$ from (131.6) at $\rho = l/k$[25].

For small angles ($\theta \ll 1$), the Legendre polynomials at large l have the form (49.6)

$$P_l(\cos\theta) \approx J_0(\theta l) = \frac{1}{2\pi}\int_0^{2\pi} e^{-i\theta l\cos\varphi}\,d\varphi.$$

Insert this in (123.11) and replace the sum over large l by an integral:

$$f = \frac{1}{k}\int\frac{d\varphi}{2\pi}\int f_l e^{-i\theta l\cos\varphi}\,l\,dl = \frac{k}{2\pi}\int f_l e^{-i\mathbf{q}\boldsymbol{\rho}}\,d^2\rho, \tag{131.9}$$

where $\mathbf{q}$ and $\boldsymbol{\rho}$ are two-dimensional vectors of magnitudes $q = k\theta$ and $\rho = l/k$. Inserting (123.15) with $\delta_l = \delta(l/k)$ returns (131.7).

For a central field, after integration over the polar angle φ in the xy plane, with $d^2\rho = \rho\,d\rho\,d\varphi$, (131.7) becomes

$$f = ik\int_0^{\infty}\{\exp[2i\delta(\rho)] - 1\}J_0(q\rho)\rho\,d\rho. \tag{131.10}$$

Section 126 noted that the Born approximation fails for high-energy scattering through large angles when the cross-section is exponentially small. The method developed here also fails under those conditions. Such cases are in fact semiclassical situations to which perturbation theory does not apply.

By the general rules of the semiclassical approximation (compare § 52, 53), the exponent governing the exponential decrease of the cross-section can be found by considering "complex trajectories" in the classically inaccessible region[26].

In the classical scattering problem, the relation between the deflection angle θ in $U(r)$ and the impact parameter ρ is

$$\frac{\pi}{2} \pm \frac{\theta}{2} = \int_{r_0}^{\infty}\frac{\rho\,dr}{r^2\sqrt{1 - \rho^2/r^2 - U/E}}, \tag{131.11}$$

[25] The semiclassical function $2\hbar\delta(\rho)$ is the change of action produced by U as the particle passes along its classical trajectory. For a fast particle the trajectory may be taken as straight, and $2\delta(\rho)$ is then the difference of the classical action integrals

$$\frac{1}{\hbar}\int_{-\infty}^{\infty}\sqrt{k^2 - 2mU/\hbar^2}\,dz - \frac{1}{\hbar}\int_{-\infty}^{\infty}k\,dz \approx -\frac{m}{\hbar^2 k}\int_{-\infty}^{\infty}U\,dz.$$

Thus $2\delta(\rho)$ plays a role analogous to the eikonal of geometrical optics, and this scattering approximation is often called the *eikonal approximation*. The amplitude itself is by no means reduced to its semiclassical expression, since in general the conditions $\theta l \gg 1$ and $\delta_l \gg 1$ do not hold.

[26] For the pre-exponential factor, see A. Z. Patashinskii, V. L. Pokrovskii, and I. M. Khalatnikov, ZhETF 45, 989 (1963).

where r_0 is the minimum distance from the center and is a root of

$$1 - \frac{\rho^2}{r^2} - \frac{U}{E} = 0 \tag{131.12}$$

(see (127.5)). We are interested in angles through which a classical particle could not be deflected[27]. They correspond to complex solutions $\rho(\theta)$ of (131.11), with complex r_0. From this function and the classical orbital angular momentum $mv\rho$, calculate

$$S(\theta) = mv \int \rho(\theta) d\theta. \tag{131.13}$$

Here v is the particle's velocity at infinity. The scattering amplitude is

$$f \sim \exp\left(\frac{iS}{\hbar}\right). \tag{131.14}$$

Equation (131.12) generally has more than one complex root. The r_0 used in (131.11) must be the one producing the smallest positive value of $\operatorname{Im} S$. In addition, if $U(r)$ has complex singular points, these too must be included among the competing values of r_0[28].

The region $r \sim r_0$ contributes most to (131.11). At high energies the term U/E under the radical may then be omitted, and integration gives

$$\rho = r_0 \cos\frac{\theta}{2}. \tag{131.15}$$

If r_0 is a singular point of $U(r)$, it depends only on the field, not on ρ or E. Calculating S from (131.13), the amplitude is then

$$f \sim \exp\left(-\frac{2mv}{\hbar} \operatorname{Im} r_0 \sin\frac{\theta}{2}\right). \tag{131.16}$$

If instead r_0 must be a root of (131.12), the exponent depends on the particular field. For

$$U = U_0 e^{-(r/a)^2},$$

which has no singular points at finite distances, the equation

$$-\frac{U}{E} = 1 - \frac{\rho^2}{r^2} \approx \sin^2\frac{\theta}{2}$$

gives

$$r_0 = a\sqrt{\ln\left(-\frac{E}{U_0}\sin^2\frac{\theta}{2}\right)}. \tag{131.17}$$

Because r_0 depends only weakly on θ, it may be treated as constant in (131.13); the amplitude is then (131.16) with r_0 from (131.17).

Problems

[27] This method is not restricted to large E; it applies to every case of exponentially small scattering.

[28] Recall from § 126 that if $U(r)$ has a singularity at real r, the cross-section does not in general decrease exponentially.

Problem

1. Find the total cross-section for scattering by a spherical square well of radius a and depth U_0, under (131.1): $U_0 \ll \hbar^2 k^2/m$.

Solution. We have

$$\int_{-\infty}^{\infty} U\,dz = -2U_0\sqrt{a^2-\rho^2}.$$

By (131.7), the forward amplitude ($q = 0$) is

$$\begin{aligned} f(0) &= -\frac{ik}{2\pi}\int_0^a \left[\exp\left(-\frac{2iU_0}{\hbar v}\sqrt{a^2-\rho^2}\right)-1\right]2\pi\rho\,d\rho \\ &= -ika^2\int_0^1 (e^{2i\nu x}-1)x\,dx = \frac{ka^2}{2i}\left[1-\frac{i}{\nu}-\frac{1}{2\nu^2}(e^{2i\nu}-1)\right], \end{aligned}$$

where $\nu = U_0 a/\hbar v$ is the "Born parameter." The optical theorem (125.9) then gives

$$\sigma = 2\pi a^2\left[1+\frac{1}{2\nu^2}-\frac{\sin 2\nu}{\nu}-\frac{\cos 2\nu}{2\nu^2}\right].$$

In the Born limit $\nu \ll 1$, this gives $\sigma = 2\pi a^2\nu^2$, in agreement with Problem 1 of § 126. In the opposite limit $\nu \gg 1$, $\sigma = 2\pi a^2$, twice the geometrical cross-section. This has a simple meaning. Every particle with $\rho < a$ is scattered and removed from the incident beam, so the well behaves as an "absorbing" sphere. By Babinet's principle (see Vol. II, end of § 61), the total cross-section is twice the "absorption" cross-section.

Problem

2. The same for $U = U_0\exp(-r^2/a^2)$.

Solution. Here

$$\int_{-\infty}^{\infty} U\,dz = a\sqrt{\pi}U_0 e^{-\rho^2/a^2}.$$

Insertion in (131.7), followed by the evident change of variable, gives

$$f(0) = -\frac{ika^2}{2}\int_0^{\nu\sqrt{\pi}}(e^{-iu}-1)\frac{du}{u},$$

where again $\nu = U_0 a/\hbar v$. Thus

$$\sigma = 2\pi a^2\int_0^{\nu\sqrt{\pi}}(1-\cos u)\frac{du}{u}.$$

For $\nu \ll 1$, the integrand becomes $u/2$ and $\sigma = \pi a^2\nu^2/2$, in agreement with Problem 2 of § 126 for $ka \gg 1$. For $\nu \gg 1$, write it as $(1-e^{-\lambda u}\cos u)/u$ and then let the small parameter λ tend to zero. Integration by parts gives

$$\int_0^{\nu\sqrt{\pi}}(1-\cos u)\frac{du}{u} \approx \ln\left(\nu\sqrt{\pi}\right)-\int_0^{\infty}\ln u\sin u\,du = \ln\left(\nu\sqrt{\pi}\right)+C,$$

where C is Euler's constant. Hence

$$\sigma = 2\pi a^2\ln\left(\nu\sqrt{\pi}e^C\right) \quad \text{for } \nu \gg 1.$$

Problem

3. Find the small-angle cross-section for electron scattering by a magnetic field confined to a cylindrical region of radius a (Y. Aharonov, D. Bohm, 1959).

Solution. Let the magnetic field point along the y axis, the axis of the cylindrical region, and take the incident direction as the z axis. The scattering is independent of y, so the problem below is two-dimensional in the xz plane.

Outside the cylinder $\mathbf{H} = 0$, but the vector potential is nonzero:

$$\mathbf{A} = \frac{\Phi}{2\pi}\nabla\varphi, \tag{1}$$

where φ is the polar angle in the xz plane and Φ the magnetic flux. Indeed, over a circle of radius $r > a$ in this plane,

$$\int \mathbf{H}\,dx\,dz = \oint \mathbf{A}\,d\mathbf{l} = \frac{\Phi}{2\pi}\int_0^{2\pi} d\varphi = \Phi.$$

The potential (1) changes the phase of the electron plane wave; by (111.9),

$$\psi = e^{ikz}\exp\left(-\frac{ie\Phi}{2\pi\hbar c}\varphi\right). \tag{2}$$

This expression fails in a narrow region along the positive z semiaxis, since particles passing through the field region have their motion disturbed. This explains the apparent multivaluedness of (2) on circling the origin, when φ increases by 2π. In reality a discontinuity of finite width lies near the positive z semiaxis, where (2) is inapplicable; on its two sides φ takes values differing by 2π, for example $\mp\pi$.

For scattering through small θ with small momentum transfer $q \approx k\theta$, where $qa \ll 1$ and $\theta \ll 1$, transverse distances $x \sim 1/q \gg a$ are important and the width of the cut may be neglected. In the region $z \gg |x|$, the x dependence of ψ on either side

of the z axis may also be neglected, giving[29]

$$\psi = e^{ikz}F(x), \qquad F(x) = \begin{cases}\exp(-ie\Phi/2\hbar c), & x > 0,\\ \exp(ie\Phi/2\hbar c), & x < 0.\end{cases} \tag{3}$$

The “two-dimensional” amplitude is calculated from (131.7a)[30]. For $q \neq 0$,

$$f = \sqrt{\frac{ik}{2\pi}}\left\{\left[\exp\left(-\frac{ie\Phi}{2\hbar c}\right) - 1\right]\int_0^{\infty} e^{-iqx}dx + \text{c.c.}\right\}.$$

Evaluate the integral by inserting $e^{-\lambda x}$ and then taking $\lambda \to 0$. The result is

$$f = \frac{i}{q}\sqrt{\frac{2ik}{\pi}}\sin\frac{e\Phi}{2\hbar c}.$$

[29] Formula (3), like (131.4), fails at excessively large z, when diffraction effects become important.

[30] For $q \neq 0$, this formula can also be obtained, as explained in the text, without using the Schrödinger equation in a potential field. We denote the scattering angle by θ to distinguish it from the polar angle φ.

Hence

$$d\sigma = |f|^2 d\theta = \frac{2}{\pi k\theta^2}\sin^2\frac{e\Phi}{2\hbar c}\,d\theta \tag{4}$$

and for $e\Phi/\hbar c \ll 1$,

$$d\sigma = \frac{e^2\Phi^2}{2\pi k\hbar^2 c^2\theta^2}d\theta,$$

the perturbative result.

We draw attention to the periodic dependence of cross section (4) on the magnetic-field strength, and also to the divergence of the total cross section (owing to $\theta \to 0$), even though the field is confined to a finite region of space; both are effects specific to quantum theory.

§ 132. Scattering of slow particles

Consider elastic scattering in the limiting case of low particle velocities. The wavelength is assumed large compared with the range a of $U(r)$ ($ka \ll 1$), and the energy small compared with the field within that range. We must determine the limiting dependence of the phases δ_l on small k.

For $r \lesssim a$, only the k^2 term may be neglected in the exact Schrödinger equation (123.7):

$$R_l'' + \frac{2}{r}R_l' - \frac{l(l+1)}{r^2}R_l = \frac{2m}{\hbar^2}U(r)R_l. \tag{132.1}$$

For $a \ll r \ll 1/k$, the $U(r)$ term may also be omitted:

$$R_l'' + \frac{2}{r}R_l' - \frac{l(l+1)}{r^2}R_l = 0. \tag{132.2}$$

Its general solution is

$$R_l = c_1 r^l + \frac{c_2}{r^{l+1}}. \tag{132.3}$$

The constants c_1, c_2, different for each l, can in principle be found only by solving (132.1) for the specified $U(r)$.

At $r \sim 1/k$, $U(r)$ may be omitted but k^2 may not:

$$R_l'' + \frac{2}{r}R_l' + \left[k^2 - \frac{l(l+1)}{r^2}\right]R_l = 0. \tag{132.4}$$

The free-motion solution is

$$R_l = c_1\frac{(-1)^l(2l+1)!!}{k^{2l+1}}r^l\left(\frac{1}{r}\frac{d}{dr}\right)^l\frac{\sin kr}{r} + c_2\frac{k^l}{(2l-1)!!}\left(\frac{1}{r}\frac{d}{dr}\right)^l\frac{\cos kr}{r}. \tag{132.5}$$

The coefficients make this reduce to (132.3) for $r \ll 1/k$, matching the solutions in the regions $kr \ll 1$ and $kr \sim 1$.

For $kr \gg 1$ it becomes

$$R_l \sim \frac{c_1(2l+1)!!}{rk^{l+1}}\sin\left(kr - \frac{\pi l}{2}\right) + \frac{c_2 k^l}{r(2l-1)!!}\cos\left(kr - \frac{\pi l}{2}\right),$$

or

$$R_l \sim \text{const} \cdot \frac{1}{r} \sin\left(kr - \frac{\pi l}{2} + \delta_l\right). \tag{132.6}$$

Here

$$\tan \delta_l = \frac{c_2}{c_1(2l-1)!!(2l+1)!!} k^{2l+1}; \tag{132.7}$$

all phases are small at small k. By (123.15), $f_l = (e^{2i\delta_l} - 1)/(2ik) \approx \delta_l/k$, and hence

$$f_l \sim k^{2l}. \tag{132.8}$$

All $l \neq 0$ amplitudes are therefore small compared with the $l = 0$ or s-wave amplitude. Neglecting them,

$$f(\theta) \approx f_0 = \frac{\delta_0}{k} = \frac{c_2}{c_1} = -a, \tag{132.9}$$

so $d\sigma = a^2 do$ and

$$\sigma = 4\pi a^2. \tag{132.10}$$

Slow-particle scattering is isotropic and energy-independent[31]. The constant a, positive or negative, is the *scattering length*.

These arguments assume sufficiently rapid decrease of $U(r)$. At large r, the second term of (132.3) is small. Retaining it is legitimate only if the terms $\sim c_2/r^{l+1}r^2$ kept in (132.2) exceed the omitted term $UR_l \sim Uc_1r^l$. Thus $U(r)$ must

decrease faster than $1/r^{2l+3}$ for (132.8) to hold. In particular, (132.9) and energy-independent isotropic scattering require decrease faster than $1/r^3$.

If U decreases exponentially, further terms in the power expansion of f_l can be characterized. Section 128 showed that $f_l(E)$ is real for real negative E[32]. The same is true of $g_l(E)$ in $f_l = 1/(g_l - ik)$, since ik is real for $E < 0$; by definition g_l is also real for $E > 0$. Hence g_l has an expansion in integral powers of E, or even powers of k. Consequently $f_l(k)$ expands in integral powers of ik: even powers have real coefficients and odd powers imaginary ones. By (132.8), f_l begins with k^{2l}, and g_l begins with k^{-2l}.

For a power-law tail $U \sim \beta r^{-n}$ with $n \leq 3$, the constant-amplitude result fails. For $n < 1$, at sufficiently low velocity and practically every impact parameter ρ,

$$\rho|U(\rho)| \gtrsim \hbar v, \tag{132.11}$$

so scattering is classical. For $1 < n < 2$, this holds over a substantial range of moderate ρ, giving classical scattering through appreciable angles, while another range satisfies

$$\rho|U(\rho)| \ll \hbar v, \tag{132.12}$$

the perturbative condition. For $n > 2$, at large distances

$$|U| \ll \frac{\hbar^2}{mr^2}, \tag{132.13}$$

[31] For electron scattering by atoms, the length compared with $1/k$ is the atomic radius, which in complex atoms reaches several Bohr radii. Its large value restricts constant cross-sections to energies of fractions of an electron volt. At higher energies the cross-section depends strongly on energy (the Ramsauer effect).

[32] At small E, (128.6) already holds for decrease as $e^{-r/a}$.

so the distant contribution is perturbative even if the nearer region is not[33]. Choose $r_0 \ll 1/k$ so that (132.13) holds for $r \gg r_0$. By (126.12), the contribution from this region is

$$-\frac{2m\beta}{\hbar^2 q}\int_{r_0}^{\infty}\frac{\sin qr}{r^{n-1}}dr = -\frac{2m\beta}{\hbar^2}q^{n-3}\int_{qr_0}^{\infty}\frac{\sin x}{x^{n-1}}dx. \tag{132.14}$$

For $2 < n < 3$ the lower limit is convergent and may be replaced by zero at $kr_0 \ll 1$, making this the dominant contribution:

$$f \sim q^{-(3-n)}, \qquad 2 < n < 3. \tag{132.15}$$

For $n = 3$ the lower-limit divergence is logarithmic and again dominant:

$$f \sim \ln\frac{\text{const}}{q}, \qquad n = 3. \tag{132.16}$$

For $n > 3$ the distant contribution vanishes as $k \to 0$ and (132.9) governs. Nevertheless (132.14) is anomalous: repeated integration by parts separates even powers of k, leaving a convergent term proportional to k^{n-3}, which is not generally even.
[34]

Problems

Problem

1. Find the slow-particle cross-section for a spherical square well of depth U_0 and radius a.

Solution. Assume $ka \ll 1$ and $k \ll \varkappa$, where $\varkappa = \sqrt{2mU_0}/\hbar$. Only δ_0 matters. With $l = 0$, (132.1) for $\chi = rR_0$ gives $\chi'' + \varkappa^2\chi = 0$ inside, whose regular solution is $\chi = A\sin\varkappa r$. Outside, $\chi = B\sin(kr + \delta_0)$. Continuity of χ'/χ at a gives

$$\varkappa\cot a\varkappa = k\cot(ka + \delta_0) \approx \frac{k}{ka + \delta_0},$$

and hence[35]

$$f = \frac{\tan\varkappa a - \varkappa a}{\varkappa}.$$

For $\varkappa a \ll 1$, it gives $\sigma = (4/9)\pi a^2(\varkappa a)^4$, agreeing with Born approximation.

[33] Low-velocity scattering never becomes semiclassical here, because (132.11) is incompatible with the simultaneous condition $|U(\rho)| \gtrsim E$.

[34] If $n = 2p + 1$ is an odd integer, $n - 3 = 2p - 2$ is even, but (132.14) still has an anomalous part proportional to $q^{2p-2}\ln q$.

[35] This formula fails when $\varkappa a$ is close to an odd multiple of $\pi/2$. Then the discrete spectrum has a negative-energy level near zero (see Problem 1 of § 33), and the formulas of the next section are required.

PROBLEM

2. The same for a spherical square potential barrier of height U_0.

SOLUTION. Replace U_0 by $-U_0$, and hence $\varkappa$ by $i\varkappa$:

$$f = \frac{\tanh \varkappa a - \varkappa a}{\varkappa}.$$

For $\varkappa a \gg 1$, $f = -a$ and $\sigma = 4\pi a^2$. This is scattering by an impenetrable sphere; classical mechanics would give one quarter of this value, πa^2.

PROBLEM

3. Find the low-energy cross-section in $U = \alpha/r^n$, where $\alpha > 0$, $n > 3$.

SOLUTION. For $l = 0$, (132.1) is

$$\chi'' - \gamma^2 \frac{\chi}{r^n} = 0, \qquad \gamma^2 = \frac{2m\alpha}{\hbar^2}.$$

With

$$r = \left(\frac{2\gamma}{(n-2)x}\right)^{2/(n-2)}, \qquad \chi = \sqrt{r}\,\varphi,$$

it becomes

$$\varphi'' + \frac{1}{x}\varphi' - \left[1 + \frac{1}{(n-2)^2 x^2}\right]\varphi = 0.$$

The solution vanishing at $r = 0$ is, up to a factor,

$$\chi = \sqrt{r}\, H^{(1)}_{1/(n-2)}\left(\frac{2i\gamma}{n-2} r^{-(n-2)/2}\right).$$

Using the small-argument formulas for $H^{(1)}_p$, at $\gamma \ll r \ll 1/k$ one finds $\chi = \text{const}(c_1 r + c_2)$ and

$$f = -\left(\frac{\gamma}{n-2}\right)^{2/(n-2)} \frac{\Gamma[(n-3)/(n-2)]}{\Gamma[(n-1)/(n-2)]}.$$

PROBLEM

4. Find the slow-particle amplitude for a field with tail $U = \beta r^{-n}$, $2 < n \le 3$.

SOLUTION. The leading term is (132.14) with lower limit zero:

$$f = -\frac{\pi m\beta}{\hbar^2 \Gamma(n-1)\cos(\pi n/2)} q^{n-3}, \qquad 2 < n < 3, \tag{1}$$

and for $n = 3$,

$$f = -\frac{2m\beta}{\hbar^2} \ln\frac{\text{const}}{q}. \tag{2}$$

Expansion of (1) in Legendre polynomials gives

$$f_l = -\frac{m\beta}{2\hbar^2} \frac{\Gamma[(n-1)/2]\Gamma[l-(n-3)/2]}{\Gamma(n/2)\Gamma[(n+1)/2+l]} k^{n-3}. \tag{3}$$

For $n > 3$, (1) gives the anomalous part. In partial amplitudes, (3) is always dominant when $2l > n-3$; then $f_l \sim k^{n-3}$ replaces (132.8).

Problem

5. Find the slow-particle amplitude in $U(r) = -U_0 e^{-r/a}$, $U_0 > 0$.

SOLUTION. Put

$$x = 2a\varkappa e^{-r/2a}, \qquad \varkappa = \sqrt{2mU_0}/\hbar.$$

Equation (132.1) for $\chi = rR_0$ becomes $\chi_{xx} + x^{-1}\chi_x + \chi = 0$, with solution $\chi = AJ_0(x) + BN_0(x)$. The condition $\chi(0) = 0$ at $r = 0$ gives

$$A/B = -N_0(2\varkappa a)/J_0(2\varkappa a).$$

For $a \ll r \ll 1/k$, $x \ll 1$ (with, of course, $a\varkappa \exp(-1/ak) \ll 1$), and

$$\chi \approx A + B\frac{2}{\pi}\ln\frac{\gamma x}{2} = A + B\frac{2}{\pi}\ln(a\varkappa\gamma) - \frac{B}{\pi a}r,$$

where $\gamma = e^C = 1{,}78\ldots$. Comparison with (132.3) gives

$$f = -a\left[\frac{\pi A}{B} + 2\ln(a\varkappa\gamma)\right] = a\left[\frac{\pi N_0(2\varkappa a)}{J_0(2\varkappa a)} - 2\ln(a\varkappa\gamma)\right].$$

For $\varkappa a \ll 1$, $f = 2a^3\varkappa^2$; for $\varkappa a \gg 1$, $f = -2a\ln(\varkappa a\gamma)$.

Problem

6. Find the low-energy amplitude in second-order perturbation theory (I. Ya. Pomeranchuk, 1948).

SOLUTION. As $k \to 0$, the integral in the second term of (130.13) is

$$-\int U(\mathbf{k}' - \mathbf{k}'')\frac{U(\mathbf{k}'' - \mathbf{k})}{k''^2}d^3k'' = -2\pi^2\iint\frac{U(\mathbf{r})U(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}dV\,dV',$$

using

$$\int e^{i\mathbf{k}(\mathbf{r}-\mathbf{r}')}\frac{4\pi}{k^2}\frac{d^3k}{(2\pi)^3} = \frac{1}{|\mathbf{r} - \mathbf{r}'|}.$$

Thus

$$f = -\frac{m}{2\pi\hbar^2}\int U\,dV + \left(\frac{m}{2\pi\hbar^2}\right)^2\iint\frac{U(\mathbf{r})U(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}dV\,dV'. \tag{1}$$

For a central field,

$$f = -\frac{2m}{\hbar^2}\int Ur^2dr + \frac{8m^2}{\hbar^4}\iint_{r'>r}U(r)U(r')r^2dr\,r'dr'.$$

The second term in (1) is always positive. Hence for repulsion the first Born approximation overestimates, and for attraction underestimates, the low-energy cross-section.

PROBLEM

7. Determine the energy dependence of the slow-particle scattering amplitude in two dimensions.

SOLUTION. At large distances the wave function is given by Problem 1 of § 124. As in three dimensions, the $m = 0$ state dominates, so f is angle-independent. Replacing $e^{ik\rho}/\sqrt{\rho}$ by the exact free solution with this asymptotic form gives, for all $\rho \gg a$,

$$\psi = e^{ikz} + \frac{f}{2}\sqrt{\pi k}\, H_0^{(1)}(k\rho). \tag{1}$$

At $\rho \ll 1/k$, use $H_0^{(1)}(x) \approx (2i/\pi)\ln[2i/(\gamma x)]$ to obtain

$$\psi \approx \left(1 + f\sqrt{\frac{k}{\pi}}\ln\frac{2i}{\gamma k}\right) - f\sqrt{\frac{k}{\pi}}\ln\rho. \tag{2}$$

This is the general free solution $C_1 + C_2\ln\rho$ for $1/k \gg \rho \gg a$. The real, energy-independent ratio fixed by the zero-energy equation at $\rho \sim a$ is written

$$C_1/C_2 = -\ln r_0, \tag{3}$$

where r_0 has dimensions of length. Comparison gives

$$f = -\left[\sqrt{\frac{k}{\pi}}\ln\frac{2i}{\gamma k r_0}\right]^{-1}. \tag{4}$$

Thus, unlike the three-dimensional case, the two-dimensional cross-section increases as the energy decreases. For scattering by an infinitely high cylindrical barrier of radius a, the constant r_0 equals a.

§ 133. Resonance scattering at low energies

The scattering of slow particles ($ka \ll 1$) in an attractive field requires special consideration when the discrete spectrum of negative energy levels contains an s-state whose energy is small compared with the magnitude of the field U within its range a. Denote this level by ε ($\varepsilon < 0$). The energy E of the scattered particle, itself small, is close to ε, or, as one says, nearly in resonance with it. As we shall see, this greatly increases the scattering cross-section.

The presence of a shallow level can be included in scattering theory by a formal method based on the following observations.

In the exact Schrödinger equation for $\chi = rR_0(r)$ (with $l = 0$),

$$\chi'' + \frac{2m}{\hbar^2}[E - U(r)]\chi = 0,$$

E may be neglected in comparison with U in the "inner" region of the field ($r \sim a$):

$$\chi'' - \frac{2m}{\hbar^2}U(r)\chi = 0, \qquad r \sim a. \tag{133.1}$$

Conversely, the field U may be neglected in the "outer" region ($r \gg a$):

$$\chi'' + \frac{2m}{\hbar^2} E\chi = 0, \qquad r \gg a. \tag{133.2}$$

The solution of (133.2) would have to be matched at some r_1, with $1/k \gg r_1 \gg a$, to the solution of (133.1) satisfying $\chi(0) = 0$. The matching condition is continuity of the ratio χ'/χ, which is independent of the overall normalization factor of the wave function.

Instead of considering the motion in the region $r \sim a$, however, impose on the outer solution a suitably chosen boundary condition on χ'/χ at small r. Since the outer solution varies slowly as $r \to 0$, this condition may formally be assigned to $r = 0$. Equation

(133.1) in the region $r \sim a$ does not contain E; the boundary condition that replaces it must therefore also be independent of the particle energy. In other words, it must have the form

$$\left.\frac{\chi'}{\chi}\right|_{r\to 0} = -\kappa, \tag{133.3}$$

where κ is a constant. Since κ is independent of E, the same condition (133.3) must apply to the solution of the Schrödinger equation at the small negative energy $E = -|\varepsilon|$, that is, to the wave function of the corresponding stationary state. At $E = -|\varepsilon|$, (133.2) gives

$$\chi = A_0 \exp\left(-\frac{\sqrt{2m|\varepsilon|}}{\hbar} r\right) \tag{133.4}$$

(A_0 is a constant), and substitution in (133.3) shows that κ is positive and equal to

$$\kappa = \frac{\sqrt{2m|\varepsilon|}}{\hbar}. \tag{133.5}$$

Now apply (133.3) to the free-motion wave function

$$\chi = \text{const} \cdot \sin(kr + \delta_0),$$

which is the exact general solution of (133.2) for $E > 0$. The required phase δ_0 is then given by

$$\cot\delta_0 = -\frac{\kappa}{k} = -\sqrt{\frac{|\varepsilon|}{E}}. \tag{133.6}$$

Since E is restricted here only by $ak \ll 1$ and need not be small compared with $|\varepsilon|$, the phase δ_0, and hence the s-wave scattering amplitude, need not be small.

The phases δ_l for $l > 0$, and the corresponding partial amplitudes, remain small as before. The total amplitude may therefore still be identified with the s-wave amplitude,

$$f = \frac{1}{2ik}(e^{2i\delta_0} - 1) = \frac{1}{k(\cot\delta_0 - i)}.$$

Substitution of (133.6) gives

$$f = -\frac{1}{\kappa + ik}. \tag{133.7}$$

The total scattering cross-section is consequently

$$\sigma = 4\pi|f|^2 = \frac{4\pi}{k^2 + \kappa^2} = \frac{2\pi\hbar^2/m}{E + |\varepsilon|}. \tag{133.8}$$

Thus the scattering remains isotropic, but its cross-section depends on the energy and, in the resonance region ($E \sim |\varepsilon|$), is large compared with the square a^2 of the range of the field (since $1/k \gg a$). We emphasize that the form of (133.8) is independent of the details of the interaction at short interparticle distances and is determined entirely by the value of the resonance level[36].

The formula obtained is somewhat more general than the assumption used in its derivation. Subject $U(r)$ to a small change; the constant κ in (133.3) will then change as well. By a suitable change of $U(r)$, κ can be made zero and then small and negative. The same formula (133.7) for the amplitude and (133.8) for the cross-section are thereby obtained. In the latter, however, $|\varepsilon| = \hbar^2\kappa^2/2m$ is now merely a constant characteristic of $U(r)$ and by no means an energy level in this field. In such cases the field is said to have a *virtual level*: although no level near zero actually exists, a small change in the field would suffice to produce one.

On analytic continuation of (133.7) into the complex E plane, ik becomes $-\sqrt{-2mE}/\hbar$ on the negative real semiaxis (see § 128), and the scattering amplitude is seen to have a pole at $E = -|\varepsilon|$, in agreement with the general results of § 128. Conversely, as expected, a virtual level corresponds to no singularity of the scattering amplitude on the physical sheet; the amplitude does have a pole at $E = -|\varepsilon|$ on the unphysical sheet (see the note on p. 637).

Formally, (133.7) corresponds to the case in which the first term in the expansion of $g_0(k)$ in (125.15),

$$f_0 = \frac{1}{g_0(k) - ik},$$

is negative and anomalously small. A more accurate formula includes the next term of the expansion:

$$f_0 = \frac{1}{-\kappa_0 + (1/2)r_0k^2 - ik} \tag{133.9}$$

(L. D. Landau, Ya. A. Smorodinsky, 1944). Recall that, for a sufficiently rapidly decreasing field, the functions $g_l(k)$ are expanded in even powers of k; see § 132. We have denoted $g_0(0)$ by $-\kappa_0$, reserving the symbol κ for (133.5), which is related to the energy level ε. From the foregoing, κ is the value of $-ik = \kappa$ that makes the denominator in (133.9) vanish, that is, a root of

$$\kappa = \kappa_0 + (1/2)r_0\kappa^2. \tag{133.10}$$

The correction $r_0k^2/2$ in the denominator of (133.9) is small compared with κ_0 by virtue of the assumed smallness of k, but it has a "normal" order of magnitude in itself:

[36] Equation (133.8) was first obtained by Wigner (E. Wigner, 1933); the idea of the derivation given here is due to Bethe and Peierls (H. A. Bethe, R. Peierls, 1935).

$r_0 \sim a$ (this coefficient is always positive; see Problem 1). It should be emphasized that including this term remains a legitimate refinement of the scattering-amplitude formula in which contributions from $l \neq 0$ have been neglected: it gives a relative correction of order ak to f, whereas the contribution from $l = 1$ scattering is of relative order $(ak)^3$.

As $k \to 0$, $f_0 \to -1/\kappa_0$; thus $1/\kappa_0$ is the scattering length a introduced in the preceding section. The coefficient r_0 in

$$g_0(k) = k \cot \delta_0 = -1/a + (1/2) r_0 k^2 \tag{133.11}$$

is called the *effective range of the interaction*[37].

From (133.9), the scattering cross-section is

$$\sigma = \frac{4\pi}{[\kappa_0 - (1/2) r_0 k^2]^2 + k^2}.$$

If the term of order k^4 in the denominator is neglected (although it is a legitimate term), this formula can be written, using

(133.10), as

$$\sigma = \frac{4\pi(1 + r_0\kappa)}{k^2 + \kappa_0^2} = \frac{2\pi\hbar^2}{m} \frac{1 + r_0\kappa}{E + |\varepsilon|}. \tag{133.12}$$

Return to (133.4), the wave function of the bound state in the "outer" region, and relate its normalization coefficient to the parameters introduced above. Evaluation of the residue of (133.9) at its pole $E = \varepsilon$ and comparison with (128.11) give

$$A_0^2 = \frac{2\kappa}{1 - r_0\kappa} \approx 2\kappa(1 + r_0\kappa). \tag{133.13}$$

The second term is a small correction to the first, since $\kappa r_0 \sim \kappa_0 a \ll 1$. Without this correction, $A_0^2 = 2\kappa$, and therefore

$$\chi = \sqrt{2\kappa}\, e^{-\kappa r}, \qquad \psi = \sqrt{\frac{\kappa}{2\pi}} \frac{e^{-\kappa r}}{r}, \tag{133.14}$$

which corresponds to normalizing as though (133.14) held throughout space.

We now briefly consider resonance in scattering with nonzero orbital angular momentum.

The expansion of $g_l(k)$ begins with a term proportional to k^{-2l}. Retaining its first two terms, write the partial scattering amplitude as

$$f_l = \frac{1}{-bk^{-2l}(\varepsilon - E) + ik}, \tag{133.15}$$

[37] We give the values of a and r_0 for the important case of two-nucleon interactions. For a neutron and proton with parallel spins (the isotopic state $T = 0$), $a = 5.4 \cdot 10^{-13}$ cm and $r_0 = 1.7 \cdot 10^{-13}$ cm; these values correspond to a true level of energy $|\varepsilon| = 2.23$ MeV, the deuteron ground state. For a neutron and proton with antiparallel spins (the isotopic state $T = 1$), $a = -24 \cdot 10^{-13}$ cm and $r_0 = 2.7 \cdot 10^{-13}$ cm; these values correspond to a virtual level $|\varepsilon| = 0.067$ MeV. By isotopic invariance, the latter values must also apply to two neutrons with antiparallel spins. The nn system in an s-state cannot have parallel spins at all, by the Pauli principle.

where b and ε are two constants, with $b > 0$ (see below). The resonance case corresponds to an anomalously small coefficient of k^{-2l}, that is, to an anomalously small ε. Despite the smallness of E, however, the term $b\varepsilon E^{-l}$ can still be large compared with k.

If $\varepsilon < 0$, the denominator in (133.15) has a real root $E \approx -|\varepsilon|$, so ε is a discrete energy level of angular momentum l[38]. Unlike resonance in s-wave scattering, however, the amplitude (133.15) nowhere becomes large compared with a; the resonant scattering amplitude with angular momentum $l+1$ is only of the same order as the nonresonant amplitude with angular momentum l.

If $\varepsilon > 0$, the amplitude (133.15) reaches order $1/k$ in the region $E \sim \varepsilon$, and is therefore large compared with a. The relative width of this region is small: $\Delta E/E \sim (ak)^{2l-1}$. Thus a sharply defined resonance occurs in this case. This pattern of resonance scattering results because a positive level with $l \neq 0$, though not a true discrete level, is a quasidiscrete level. The centrifugal potential barrier makes the probability that a low-energy particle escapes from this state to infinity small, so the state's "lifetime" is long (see § 134). This accounts for the difference between resonance scattering with $l \neq 0$ and resonance in an s-state, where there is no centrifugal barrier.

For $\varepsilon > 0$, the denominator in (133.15) vanishes at $E = E_0 - i\Gamma/2$, where

$$E_0 \approx \varepsilon, \qquad \Gamma = \frac{2\sqrt{2m}}{b\hbar}E^{l+1/2}. \tag{133.16}$$

This pole of the scattering amplitude lies on the "unphysical" sheet. The small quantity Γ is the width of the quasidiscrete level (see § 134).

Finally, we mention an interesting property of the phases δ_l which is readily established from the preceding results.

Regard $\delta_l(E)$ as continuous functions of the energy, without reducing them to the interval between 0 and π (compare the note on p. 144). We shall show that

$$\delta_l(0) - \delta_l(\infty) = n_l\pi, \tag{133.17}$$

where n_l is the number of discrete levels of angular momentum l in the attractive field $U(r)$ (N. Levinson, 1949).

First note that, in a field satisfying $|U| \ll \hbar^2/ma^2$, the Born approximation applies at every energy, so $\delta_l(E) \ll 1$ for all E. Here $\delta_l(\infty) = 0$, since the scattering amplitude tends to zero as $E \to \infty$, while $\delta_l(0) = 0$ in accordance with the general results of § 132. At the same time such a field has no discrete levels (see § 45), so $n_l = 0$. Now follow the change in $\delta_l(\Lambda) - \delta_l(\infty)$, where Λ is a specified small quantity, as the potential well $U(r)$ is gradually deepened. As it deepens, the first, second, and subsequent levels appear in succession at the upper edge of the well. Each time this happens,

$\delta_l(\Lambda)$ increases by π[39]. On reaching the specified $U(r)$ and then taking $\Lambda \to 0$, we obtain (133.17).

[38]When $\varepsilon < 0$ and E is close to $|\varepsilon|$, one has $f_l \approx (-1)^{l+1}|\varepsilon|^l/[b(E+|\varepsilon|)]$. Comparison with (128.17) shows that $b > 0$.

[39]In (133.6), this corresponds to a change of δ_0 from 0 to π when, at a specified small value of k, κ changes from a negative value $-\kappa \gg k$ to a positive value $\kappa \gg k$. For $l \neq 0$, the same follows from $k \cot \delta_l = -bE^{-l}(E-\varepsilon)$ when, at fixed $E = \Lambda$, ε changes from $\varepsilon \gg \Lambda$ to $-\varepsilon \gg \Lambda$.

Problems

PROBLEM

1. Express the effective range r_0 in terms of the bound-state wave function ($E = \varepsilon$) in the "inner" region $r \sim a$ (Ya. A. Smorodinsky, 1948).

SOLUTION. Let χ_0 be the wave function in the region $r \sim a$, normalized by $\chi_0 \to 1$ as $r \to \infty$. The square of the wave function throughout space can then be written

$$\chi^2 = A_0^2(e^{-2\kappa r} + \chi_0^2 - 1)$$

(this expression becomes $A_0^2 e^{-2\kappa r}$ when $\kappa r \gg 1$ and $A_0^2\chi_0^2$ when $\kappa r \ll 1$).

It must satisfy the normalization condition

$$\int_0^\infty \chi^2 dr = A_0^2\left[\frac{1}{2\kappa} - \int_0^\infty (1-\chi_0^2)dr\right] = 1,$$

and comparison with (133.13) gives

$$r_0 = 2\int_0^\infty (1-\chi_0^2)dr.$$

It follows from (133.1), with $U(r) < 0$, whose solution is χ_0, that $\chi_0(r) < \chi_0(\infty) = 1$. Hence $r_0 > 0$ always.

PROBLEM

2. Determine the change of the phases δ_l when the field $U(r)$ is varied.

SOLUTION. Varying $U(r)$ in the Schrödinger equation

$$\chi_l'' + \frac{2m}{\hbar^2}\left[E - \frac{\hbar^2 l(l+1)}{2mr^2} - U\right]\chi_l = 0$$

gives the corresponding equation for $\delta\chi_l$. Multiply the first equation by $\delta\chi_l$ and the second by χ_l, subtract them term by term, and integrate with respect to r. We find

$$(\chi_l'\delta\chi_l - \chi_l\delta\chi_l')|_{r\to\infty} = \frac{2m}{\hbar^2}\int_0^\infty \chi_l^2\delta U\, dr.$$

Substituting the asymptotic expressions

$$\chi_l = \sin(kr - l\pi/2 + \delta_l), \qquad \delta\chi_l = \delta(\delta_l)\cos(kr - l\pi/2 + \delta_l)$$

in the left-hand side gives

$$\delta(\delta_l) = -\frac{2m}{\hbar^2 k}\int_0^\infty \chi_l^2\delta U\, dr.$$

This formula permits definite conclusions about the signs of the phases δ_l regarded as continuous functions of energy. To remove the ambiguity in defining these functions, an additive constant that is a multiple of π, normalize them by $\delta_l(\infty) = 0$.

Starting from $U = 0$, where every $\delta_l = 0$, and gradually increasing $|U|$, we find that all $\delta_l < 0$ in a repulsive field ($U > 0$), whereas $\delta_l > 0$ in an attractive field ($U < 0$). In a repulsive field $\delta_l(0) = 0$, and the phases are therefore small at low energies; the scattering amplitude is consequently negative: $f \approx \delta_0/k < 0$. In an attractive field, the analogous conclusion that f is positive can be drawn only if there are no levels. Otherwise, at small E, the phases δ_l are close not to zero but to $n_l\pi$ (see (133.17)), and nothing can be concluded about the sign of f.

Problem

3. Find the scattering length a and effective range r_0 for a spherical square potential well of radius a and depth U_0 containing a single discrete energy level close to zero.

Solution. Proceed as in Problem 1 of § 132, except that inside the well the particle energy $E = \hbar^2 k^2/2m$ is not neglected in comparison with U_0. The equation determining δ_0 is

$$k\cot(\delta_0 + ak) = K\cot aK, \qquad K = \frac{1}{\hbar}\sqrt{2m(U_0 + E)}.$$

For the well to contain only one level near zero, one must have

$$U_0 = \frac{\pi^2\hbar^2}{8ma^2}(1+\Lambda)$$

with $\Lambda \ll 1$ (see Problem 1 of § 33). Expanding the preceding equation in powers of ka and Λ gives

$$k\cot\delta_0 \approx -(\pi^2/8a)\Lambda + ak^2/2,$$

whence $a = 1/\kappa_0 = 8a/\pi^2\Lambda$ and $r_0 = a$. As expected, κ_0 agrees with $\sqrt{2m|E_1|}/\hbar$, where E_1 is the energy of the level in the well (see Problem 1 of § 33).

Problem

4. Express the integral $\int_0^a \chi^2 dr$ of the square of the s-state wave function in terms of the phase $\delta_0(k)$ for a field $U(r)$ that differs from zero only inside a sphere of radius a (O. Lüders, 1955).

Solution. According to (128.10),

$$\int_0^a \chi^2 dr = \frac{1}{2k}\left[\chi'\frac{\partial\chi}{\partial k} - \chi\left(\frac{\partial\chi}{\partial k}\right)'\right]_{r=a},$$

where a prime denotes differentiation with respect to r (and the derivatives with respect to E in (128.10) have been replaced by derivatives with respect to $k = \sqrt{2mE}/\hbar$). Since the field already vanishes at $r = a$, the free-motion wave function $\chi = 2\sin(kr + \delta_0)$ may be used on the right-hand side (normalized as in
(33.20)). The result is

$$\int_0^a \chi^2 dr = 2\left(a + \frac{d\delta_0}{dk}\right) - \frac{1}{k}\sin[2(ka + \delta_0)] > 0.$$

Since the integral of χ^2 is necessarily positive, the expression on the right-hand side must also be positive[40].

[40]This inequality was obtained earlier by another method by Wigner (1955).

§ 134. Resonance at a quasidiscrete level

A system capable of decay has, strictly speaking, no discrete energy spectrum: the emitted particle escapes to infinity, so the motion is infinite and the spectrum continuous. The decay probability may nevertheless be very small, as for a particle enclosed by a high, broad potential barrier, or when decay requires a spin change produced by weak spin–orbit interaction.

Such systems admit *quasistationary* states in which particles remain "inside" for a long lifetime $\tau \sim 1/w$, leaving only after decay. Their spectrum is *quasidiscrete*: broadened levels of width $\Gamma \sim \hbar/\tau$, small compared with their separations.

Formally, replace the usual condition of finiteness at infinity by the condition of an outgoing spherical wave, representing a particle ultimately leaving the system. This complex boundary condition no longer requires real energy eigenvalues; instead they have the form

$$E = E_0 - i\Gamma/2, \tag{134.1}$$

with $E_0, \Gamma > 0$. The time factor is

$$e^{-iEt/\hbar} = e^{-iE_0 t/\hbar - \Gamma t/2\hbar},$$

so probabilities decrease as $e^{-\Gamma t/\hbar}$[41]. Thus

$$w = \Gamma/\hbar. \tag{134.2}$$

At large distance the outgoing wave contains $\exp\left[ir\sqrt{2m(E_0 - i\Gamma/2)}/\hbar\right]$ and grows exponentially, so its normalization integral diverges. This resolves the apparent conflict between temporal decay of $|\psi|^2$ and conservation of the normalization integral.

To describe energies near a quasidiscrete level, write as in § 128

$$R_l = \frac{1}{r}\left[A_l(E)e^{-\sqrt{-2mE}r/\hbar} + B_l(E)e^{\sqrt{-2mE}r/\hbar}\right]. \tag{134.3}$$

For real positive E,

$$R_l = \frac{1}{r}[A_l(E)e^{ikr} + B_l(E)e^{-ikr}], \qquad k = \frac{\sqrt{2mE}}{\hbar}, \tag{134.4}$$

with $A_l = B_l^*$, and B_l taken on the upper edge of the positive-axis cut. Absence of the incoming wave requires

$$B_l(E_0 - i\Gamma/2) = 0. \tag{134.5}$$

Thus quasidiscrete levels, like true discrete levels, are zeros of B_l, but they lie on the unphysical sheet. Reaching the point below the positive axis on the physical sheet by

[41] This shows why Γ must be positive. The outgoing-wave condition, or the equivalent pole-bypass rule of § 130, ensures this. For a constant perturbation V coupling a discrete level n to continuum states ν,

$$E_n^{(2)} = \int \frac{|V_{n\nu}|^2 d\nu}{E_n^{(0)} - E_\nu + i0}, \qquad \Gamma = -2\,\mathrm{Im}\,E_n^{(2)} = 2\pi \int |V_{n\nu}|^2 \delta(E_n^{(0)} - E_\nu)d\nu,$$

in agreement with (43.1).

circling $E = 0$ reverses $\sqrt{-E}$ and turns the outgoing wave into an incoming one; retaining its outgoing character requires continuation directly downward through the cut.

For real positive E near a narrow level, expand

$$B_l(E) = (E - E_0 + i\Gamma/2)b_l. \tag{134.6}$$

Then

$$R_l = \frac{1}{r}[(E - E_0 - i\Gamma/2)b_l^* e^{ikr} + (E - E_0 + i\Gamma/2)b_l e^{-ikr}], \tag{134.7}$$

and

$$\begin{aligned} e^{2i\delta_l} &= \frac{E - E_0 - i\Gamma/2}{E - E_0 + i\Gamma/2} e^{2i\delta_l^{(0)}} \\ &= \left[1 - \frac{i\Gamma}{E - E_0 + i\Gamma/2}\right] e^{2i\delta_l^{(0)}}, \end{aligned} \tag{134.8}$$

where

$$e^{2i\delta_l^{(0)}} = (-1)^{l+1} b_l^*/b_l. \tag{134.9}$$

Far from resonance $\delta_l = \delta_l^{(0)}$. Using $e^{2i\arctan\lambda} = (1 + i\lambda)/(1 - i\lambda)$,

$$\delta_l = \delta_l^{(0)} - \arctan\frac{\Gamma}{2(E - E_0)}, \tag{134.10}$$

so the phase changes by π across the resonance.

At $E = E_0 - i\Gamma/2$, $R_l = -i\Gamma b_l^* e^{ikr}/r$. If the probability inside is normalized to unity, its outgoing flux $v|i\Gamma b_l^*|^2$ equals (134.2), giving

$$|b_l|^2 = 1/(\hbar v\Gamma). \tag{134.11}$$

For elastic scattering near a quasidiscrete level of the compound system, insert (134.8) into the corresponding term of (123.11):

$$f(\theta) = f^{(0)}(\theta) - \frac{2l+1}{k}\frac{\Gamma/2}{E - E_0 + i\Gamma/2} e^{2i\delta_l^{(0)}} P_l(\cos\theta). \tag{134.12}$$

Here $f^{(0)}$ is the nonresonant *potential-scattering* amplitude; the second term is the *resonance-scattering* amplitude[42][43]. Its pole is on the unphysical sheet. Validity requires

$$|E - E_0| \ll D, \tag{134.13}$$

where D is the distance to neighboring quasidiscrete levels.

For slow particles only s scattering matters. If E_0 belongs to $l = 0$, $f^{(0)} = -\alpha$[44], and $e^{2i\delta_0^{(0)}}$ may be replaced by unity.

[42] For charged particles, use (135.11) for $\delta_l^{(0)}$.

[43] Formula (133.15) near a positive slow-particle level with $l \neq 0$ has precisely this resonance form, with E_0, Γ from (133.16) and $e^{2i\delta_l^{(0)}} \approx 1$.

[44] The field is assumed to decrease sufficiently rapidly. Section 145 applies these results to slow-neutron scattering by nuclei.

Hence

$$f(\theta) = -\alpha - \frac{\Gamma/2}{k(E - E_0 + i\Gamma/2)}. \tag{134.14}$$

Within $|E - E_0| \sim \Gamma$ the resonance term dominates, though farther away the two terms may be comparable.

This derivation assumes E_0 is not close to the branch point $E = 0$. For the first quasidiscrete level with $E_0 \ll D$, (134.6) may fail, as is already shown by (134.14), which lacks the constant $E \to 0$ limit required for s scattering.

Near zero, expand B_0 in powers of $\sqrt{-E}$, whose sign changes on passing between the edges of the cut. For real positive E, write

$$B_0(E) = (E - \varepsilon_0 + i\gamma\sqrt{E})b_0(E), \tag{134.15}$$

where ε_0, γ are real and b_0 has no nearby zeros [45]. The level condition is

$$E_0 - i\Gamma/2 - \varepsilon_0 + i\gamma\sqrt{E_0 - i\Gamma/2} = 0. \tag{134.16}$$

Positive E_0, Γ require positive ε_0, γ. When $\varepsilon_0 \gg \gamma^2$, one has $E_0 = \varepsilon_0$ and $\Gamma = 2\gamma\sqrt{\varepsilon_0}$.

Replacing E_0 by ε_0 and Γ by $2\gamma\sqrt{E}$ in the subsequent formulas gives

$$f = -\alpha - \frac{\hbar\gamma}{\sqrt{2m}(E - \varepsilon_0 + i\gamma\sqrt{E})}. \tag{134.17}$$

Here m is the reduced mass. The amplitude now has the required constant limit at $E \to 0$.

Expression (134.17) also covers a true discrete level near zero. If $|\varepsilon_0| \ll \gamma^2$ and $E \ll \gamma^2$, neglect E in the resonance denominator and also neglect α:

$$f = -[ik - \sqrt{2m}\,\varepsilon_0/(\hbar\gamma)]^{-1}.$$

This is (133.7), with $\varkappa = -\sqrt{2m}\,\varepsilon_0/(\hbar\gamma)$, and describes a resonance at $E = \varepsilon_0^2/\gamma^2$, a true or virtual discrete level according as $\varkappa$ is positive or negative.

§ 135. The Rutherford formula

Coulomb-field scattering is important in physical applications. It is also noteworthy because, in this instance, the quantum-mechanical collision problem admits a complete exact solution.

When a direction is distinguished—here, the direction of the incident particle—the Schrödinger equation for a Coulomb field is conveniently solved in the parabolic coordinates ξ, η, φ (see § 37). The problem of scattering by a central field

has axial symmetry, and the wave function ψ is consequently independent of the angle φ. Write a separated solution of the Schrödinger equation (37.6) as

$$\psi = f_1(\xi) f_2(\eta) \tag{135.1}$$

[45] It determines the potential-scattering phase. For slow particles, $b_0(E) = \text{const} \cdot i(1 + i\alpha k) + \cdots$.

(formula (37.7) with $m = 0$). Separation of the variables then gives equations (37.8) with $m = 0$[46]:

$$\frac{d}{d\xi}\left(\xi\frac{df_1}{d\xi}\right) + \left[\frac{k^2}{4}\xi - \beta_1\right] f_1 = 0,$$
$$\frac{d}{d\eta}\left(\eta\frac{df_2}{d\eta}\right) + \left[\frac{k^2}{4}\eta - \beta_2\right] f_2 = 0,$$
$$\beta_1 + \beta_2 = 1. \tag{135.2}$$

The scattered particle naturally has positive energy; we have put $E = k^2/2$. The signs in (135.2) refer to a repulsive field. An attractive field gives exactly the same final scattering cross-section.

We require a solution of the Schrödinger equation which, for negative z and large r, behaves as the plane wave

$$\psi \sim e^{ikz} \quad \text{for} \quad -\infty < z < 0, \quad r \to \infty.$$

This describes a particle incident in the positive z direction. It will follow below that a single separated solution (135.1), rather than a sum of solutions having different β_1 and β_2, can meet this condition.

In parabolic coordinates the condition reads

$$\psi \sim \exp\left[\frac{ik}{2}(\xi - \eta)\right] \quad \text{for} \quad \eta \to \infty \quad \text{at every } \xi.$$

It can hold only when

$$f_1(\xi) = e^{ik\xi/2}, \tag{135.3}$$

while $f_2(\eta)$ obeys

$$f_2(\eta) \sim e^{-ik\eta/2} \quad \text{for} \quad \eta \to \infty. \tag{135.4}$$

Substitution of (135.3) into the first equation (135.2) confirms that this function does solve it when $\beta_1 = ik/2$. With $\beta_2 = 1 - \beta_1$, the second equation (135.2) therefore becomes

$$\frac{d}{d\eta}\left(\eta\frac{df_2}{d\eta}\right) + \left(\frac{k^2}{4}\eta - 1 + \frac{ik}{2}\right) f_2 = 0.$$

Seek its solution in the form

$$f_2(\eta) = e^{-ik\eta/2} w(\eta), \tag{135.5}$$

where $w(\eta)$ tends to a constant limit as $\eta \to \infty$. The equation for $w(\eta)$ is

$$\eta w'' + (1 - ik\eta)w' - w = 0. \tag{135.6}$$

The change of variable $\eta_1 = ik\eta$ reduces this to the confluent hypergeometric equation with parameters $\alpha = -i/k$, $\gamma = 1$. Among its solutions we must choose the one which,

[46]Coulomb units are used throughout this section (see p. 154).

after multiplication by $f_1(\xi)$, contains only an outgoing (that is, scattered) spherical wave and no incoming one. That solution is

$$w = \text{const} \cdot F\left(-\frac{i}{k}, 1, ik\eta\right).$$

Collecting these results, we obtain the following exact solution of the Schrödinger equation for the scattering process:

$$\psi = \exp\left(-\frac{\pi}{2k}\right)\Gamma\left(1+\frac{i}{k}\right)\exp\left[\frac{ik}{2}(\xi-\eta)\right]F\left(-\frac{i}{k}, 1, ik\eta\right). \tag{135.7}$$

The normalization constant in ψ has been selected so that the incident plane wave has unit amplitude (as verified below).

To separate the incident and scattered waves in this function, we examine its large-distance form. The first two terms in asymptotic expansion (d.14) of the confluent hypergeometric function give, at large η,

$$\begin{aligned} F\left(-\frac{i}{k}, 1, ik\eta\right) &\approx \frac{(-ik\eta)^{i/k}}{\Gamma(1+i/k)}\left(1+\frac{1}{ik^3\eta}\right) \\ &\quad + \frac{(ik\eta)^{-i/k}e^{ik\eta}}{\Gamma(-i/k)ik\eta} \\ &= \frac{e^{\pi/2k}}{\Gamma(1+i/k)}\left(1+\frac{1}{ik^3\eta}\right)\exp\left(\frac{i}{k}\ln k\eta\right) \\ &\quad - \frac{(i/k)e^{\pi/2k}}{\Gamma(1-i/k)}\frac{e^{ik\eta}}{ik\eta}\exp\left(-\frac{i}{k}\ln k\eta\right). \end{aligned}$$

Insert this expression in (135.7) and change to spherical coordinates, $\xi - \eta = 2z$ and $\eta = r - z = r(1-\cos\theta)$. The resulting

asymptotic form of the wave function is

$$\begin{aligned} \psi &= \left[1+\frac{1}{ik^3 r(1-\cos\theta)}\right]\exp\left\{ikz+\frac{i}{k}\ln[kr(1-\cos\theta)]\right\} \\ &\quad + \frac{f(\theta)}{r}\exp\left\{ikr-\frac{i}{k}\ln(2kr)\right\}, \end{aligned} \tag{135.8}$$

where

$$f(\theta) = -\frac{1}{2k^2\sin^2(\theta/2)}\frac{\Gamma(1+i/k)}{\Gamma(1-i/k)}\exp\left(-\frac{2i}{k}\ln\sin\frac{\theta}{2}\right). \tag{135.9}$$

The first term of (135.8) is the incident wave. The Coulomb field falls off so slowly that the incident plane wave remains distorted even far from the center: the phase has a logarithmic contribution and the amplitude contains a term of order $1/r$[47]. The scattered

[47]This distortion is already intelligible from the classical picture. Consider a family of classical hyperbolic Coulomb trajectories with a common incident direction parallel to the z axis. At large distances from the scattering center ($z \to -\infty$), the equation of a surface normal to these trajectories tends, as may readily be shown, not to $z = \text{const}$ but to $z + k^{-2}\ln k(r-z) = \text{const}$. This is precisely the constant-phase surface of the incident wave in (135.8).

spherical wave, represented by the second term of (135.8), likewise has a logarithmic distortion in its phase. These departures from the usual asymptotic form (123.3) are nevertheless immaterial: their corrections to the flux density vanish as $r \to \infty$.

Hence the scattering cross-section $d\sigma = |f(\theta)|^2 d\Omega$ is

$$d\sigma = \frac{d\Omega}{4k^2 \sin^4(\theta/2)},$$

or, in ordinary units,

$$d\sigma = \left(\frac{\alpha}{2mv^2}\right)^2 \frac{d\Omega}{\sin^4(\theta/2)}. \tag{135.10}$$

Here $v = k\hbar/m$ is the particle velocity. This is the familiar *Rutherford formula* of classical mechanics. Thus quantum and classical mechanics predict the same result for scattering in a Coulomb field (N. Mott and W. Gordon, 1928). The Born formula (126.12), as expected, also leads to (135.10).

For reference, we give the scattering amplitude (135.9) as an expansion in spherical functions. Substitution in (124.5) of the phases from (36.28) yields it[48]:

$$\exp\left(2i\delta_l^{\mathrm{C}}\right) = \frac{\Gamma(l+1+i/k)}{\Gamma(l+1-i/k)}. \tag{135.11}$$

Consequently,

$$f(\theta) = \frac{1}{2ik}\sum_l (2l+1)\frac{\Gamma(l+1+i/k)}{\Gamma(l+1-i/k)} P_l(\cos\theta). \tag{135.12}$$

The signs in amplitude (135.9) correspond to repulsion in the Coulomb field. For attraction, (135.9) must be replaced by its complex conjugate. In that case $f(\theta)$ becomes infinite at the poles of $\Gamma(1-i/k)$, namely when the argument of the gamma function is zero or a negative integer (here $\operatorname{Im} k > 0$, and $r\psi$ decays at infinity). The corresponding energies are

$$\frac{k^2}{2} = -\frac{1}{2n^2}, \qquad n = 1, 2, 3, \ldots,$$

and coincide with the discrete energy levels of the attractive Coulomb field (compare § 128).

§ 136. A system of continuous-spectrum wave functions

In studying motion in a centrally symmetric field in Chapter V, we considered stationary states in which the particle has specified values not only of the energy but also of the orbital angular momentum l and its projection m. The wave functions of these states in the discrete spectrum (ψ_{nlm}) and continuous spectrum (ψ_{klm}, with energy $\hbar^2 k^2/2m$) together form a complete system in which the wave function of an arbitrary state may be expanded. Such a system is not, however, suited to the formulation of

[48]The quantity δ_l^{C} in that formula differs from the true Coulomb phase of the outgoing wave by a quantity common to all l.

problems in scattering theory. A different system is convenient here, one in which the continuous-spectrum wave functions are characterized by specified asymptotic behavior: at infinity there is a plane wave $\exp(i\mathbf{kr})$ together with an outgoing spherical wave. In these states the particle has a specified energy, but neither its angular momentum nor its projection is specified.

According to (123.6), (123.7), these wave functions, denoted here by $\psi_{\mathbf{k}}^{(+)}$, are

$$\psi_{\mathbf{k}}^{(+)} = \frac{1}{2k}\sum_{l=0}^{\infty} i^l(2l+1)e^{i\delta_l}R_{kl}(r)P_l\left(\frac{\mathbf{kr}}{kr}\right). \tag{136.1}$$

The argument of the Legendre polynomials has been written as $\cos\theta = \mathbf{kr}/kr$, so this expression is no longer tied to any particular choice of coordinate axes (as it was in (123.6), where the z axis coincided with the propagation direction of the plane wave). Giving $\mathbf{k}$ all possible values produces a set of wave functions which, as we shall now show, are mutually orthogonal and normalized by the usual continuous-spectrum rule

$$\int \psi_{\mathbf{k}'}^{(+)*}\psi_{\mathbf{k}}^{(+)}\,dV = (2\pi)^3\delta(\mathbf{k}'-\mathbf{k}). \tag{136.2}$$

For the proof[49], note that the product $\psi_{\mathbf{k}'}^{(+)*}\psi_{\mathbf{k}}^{(+)}$ is a double sum over l and l' of terms containing the products

$$P_l\left(\frac{\mathbf{kr}}{kr}\right)P_{l'}\left(\frac{\mathbf{k}'\mathbf{r}}{k'r}\right).$$

Integration over the directions of $\mathbf{r}$ is carried out by

$$\int P_l\left(\frac{\mathbf{kr}}{kr}\right)P_{l'}\left(\frac{\mathbf{k}'\mathbf{r}}{k'r}\right)d\Omega = \delta_{ll'}\frac{4\pi}{2l+1}P_l\left(\frac{\mathbf{kk}'}{kk'}\right) \tag{136.3}$$

(compare formula (c.12) in the Mathematical Supplements), after which there remains

$$\begin{aligned}\int \psi_{\mathbf{k}'}^{(+)*}\psi_{\mathbf{k}}^{(+)}\,dV = {}& \frac{\pi}{kk'}\sum_{l=0}^{\infty}(2l+1)\exp[i\delta_l(k)-i\delta_l(k')]P_l(\cos\gamma)\\ &\times\int_0^{\infty}R_{k'l}R_{kl}r^2dr,\end{aligned}$$

where γ is the angle between $\mathbf{k}$ and $\mathbf{k}'$. The radial functions R_{kl} are orthogonal and normalized according to

$$\int_0^{\infty}R_{k'l}R_{kl}r^2dr = 2\pi\delta(k'-k).$$

[49]Only the mutual orthogonality of the functions $\psi_{\mathbf{k}}^{(+)}$ essentially requires a separate proof. Their normalization could be established directly from their asymptotic form (compare § 21). In this sense, (136.2) is already evident from the fact that, as $r \to \infty$, the only nondecreasing term in these functions is $\psi_{\mathbf{k}}^{(+)} \approx e^{i\mathbf{kr}}$.

We may therefore put $k = k'$ in the coefficients before the integrals. Using also (124.3), we obtain

$$\int \psi_{\mathbf{k}'}^{(+)*}\psi_{\mathbf{k}}^{(+)}\,dV = \frac{2\pi^2}{k^2}\delta(k'-k)\sum_{l=0}^{\infty}(2l+1)P_l(\cos\gamma)$$
$$= \frac{8\pi^2}{k^2}\delta(k'-k)\delta(1-\cos\gamma).$$

The expression on the right is zero for every $\mathbf{k} \neq \mathbf{k}'$, while multiplication by $2\pi k^2 \sin\gamma\, d\gamma\, dk/(2\pi)^3$ and integration over all $\mathbf{k}$ space gives unity, proving (136.2).

Besides the system $\psi_{\mathbf{k}}^{(+)}$, one may introduce a system corresponding to states in which a plane wave and an incoming spherical wave are present at infinity. Denote these functions by $\psi_{\mathbf{k}}^{(-)}$; they are obtained from $\psi_{\mathbf{k}}^{(+)}$ by

$$\psi_{\mathbf{k}}^{(-)} = \psi_{-\mathbf{k}}^{(+)*}. \tag{136.4}$$

Indeed, complex conjugation changes the outgoing wave (e^{ikr}/r) into an incoming wave (e^{-ikr}/r), while the plane wave becomes $e^{-i\mathbf{kr}}$. To retain the former definition of $\mathbf{k}$, with plane wave $e^{i\mathbf{kr}}$, one must also replace $\mathbf{k}$ by $-\mathbf{k}$, as in (136.4). Noting that $P_l(-\cos\theta) = (-1)^l P_l(\cos\theta)$, (136.1) gives

$$\psi_{\mathbf{k}}^{(-)} = \frac{1}{2k}\sum_{l=0}^{\infty} i^l(2l+1)e^{-i\delta_l}R_{kl}(r)P_l\left(\frac{\mathbf{kr}}{kr}\right). \tag{136.5}$$

The Coulomb field is a particularly important case. Here $\psi_{\mathbf{k}}^{(+)}$ and $\psi_{\mathbf{k}}^{(-)}$ can be written in closed form directly from (135.7). Express the parabolic coordinates by

$$\frac{k}{2}(\xi-\eta) = kz = \mathbf{kr}, \qquad k\eta = k(r-z) = kr - \mathbf{kr}.$$

Thus, for a repulsive Coulomb field[50],

$$\psi_{\mathbf{k}}^{(+)} = e^{-\pi/2k}\Gamma\left(1+\frac{i}{k}\right)e^{i\mathbf{kr}}F\left(-\frac{i}{k}, 1, i(kr-\mathbf{kr})\right), \tag{136.6}$$

$$\psi_{\mathbf{k}}^{(-)} = e^{-\pi/2k}\Gamma\left(1-\frac{i}{k}\right)e^{i\mathbf{kr}}F\left(\frac{i}{k}, 1, -i(kr+\mathbf{kr})\right). \tag{136.7}$$

The wave functions for an attractive Coulomb field follow by simultaneously reversing the signs of k and r:

$$\psi_{\mathbf{k}}^{(+)} = e^{\pi/2k}\Gamma\left(1-\frac{i}{k}\right)e^{i\mathbf{kr}}F\left(\frac{i}{k}, 1, i(kr-\mathbf{kr})\right), \tag{136.8}$$

$$\psi_{\mathbf{k}}^{(-)} = e^{\pi/2k}\Gamma\left(1+\frac{i}{k}\right)e^{i\mathbf{kr}}F\left(-\frac{i}{k}, 1, -i(kr+\mathbf{kr})\right). \tag{136.9}$$

[50] Coulomb units are used.

The effect of the Coulomb field on the particle's motion near the origin may be characterized by the ratio of $|\psi_{\mathbf{k}}^{(+)}|^2$ or $|\psi_{\mathbf{k}}^{(-)}|^2$ at $r = 0$ to the squared modulus of the free-motion wave function $\psi_{\mathbf{k}} = e^{i\mathbf{kr}}$. Using

$$\Gamma\left(1+\frac{i}{k}\right)\Gamma\left(1-\frac{i}{k}\right) = \frac{i}{k}\Gamma\left(\frac{i}{k}\right)\Gamma\left(1-\frac{i}{k}\right) = \frac{\pi}{k\sinh(\pi/k)},$$

we readily find, for a repulsive field,

$$\frac{|\psi_{\mathbf{k}}^{(+)}(0)|^2}{|\psi_{\mathbf{k}}|^2} = \frac{|\psi_{\mathbf{k}}^{(-)}(0)|^2}{|\psi_{\mathbf{k}}|^2} = \frac{2\pi}{k(e^{2\pi/k}-1)} \tag{136.10}$$

and, for an attractive field,

$$\frac{|\psi_{\mathbf{k}}^{(+)}(0)|^2}{|\psi_{\mathbf{k}}|^2} = \frac{|\psi_{\mathbf{k}}^{(-)}(0)|^2}{|\psi_{\mathbf{k}}|^2} = \frac{2\pi}{k(1-e^{-2\pi/k})}. \tag{136.11}$$

The functions $\psi_{\mathbf{k}}^{(+)}$ and $\psi_{\mathbf{k}}^{(-)}$ play an essential role in applications of perturbation theory to the continuous spectrum. Suppose that a perturbation $\hat{V}$ causes a transition of a particle between continuous-spectrum states. The transition probability is determined by the matrix element

$$\int \psi_f^* \hat{V} \psi_i \, dV. \tag{136.12}$$

The question arises: which particular solutions of the wave equation should be chosen as the initial (ψ_i) and final (ψ_f)

wave functions in order to obtain the amplitude for a particle to pass from a state of momentum $\hbar\mathbf{k}$ to a state of momentum $\hbar\mathbf{k}'$ at infinity[51]? We shall show that one must choose

$$\psi_i = \psi_{\mathbf{k}}^{(+)}, \qquad \psi_f = \psi_{\mathbf{k}'}^{(-)} \tag{136.13}$$

(A. Sommerfeld, 1931).

This becomes clear by considering how the question would be treated by perturbation theory applied not only to $\hat{V}$ but also to the field $U(r)$ in which the particle moves. To zeroth order in U, the matrix element (136.12) is

$$V_{\mathbf{k}'\mathbf{k}} = \int e^{-i\mathbf{k}'\mathbf{r}} \hat{V} e^{i\mathbf{kr}} dV.$$

At higher orders in U, this integral is replaced by a series whose terms are integrals of the form

$$\int \frac{V_{\mathbf{k}'\mathbf{k}_1} U_{\mathbf{k}_1\mathbf{k}_2} \dots U_{\mathbf{k}_n\mathbf{k}}}{(E_{\mathbf{k}} - E_{\mathbf{k}_1} + i0)\dots(E_{\mathbf{k}} - E_{\mathbf{k}_n} + i0)} \, d^3k_1 \dots d^3k_n$$

(compare § 43, 130). The numerators contain matrix elements with respect to unperturbed plane waves, arranged in various sequences, and every pole is bypassed in the integrations

[51] One example of such a process is an electron which, on colliding with a stationary heavy nucleus, emits a photon and thereby changes its energy and direction of motion. The perturbation $\hat{V}$ is the interaction of the electron with the radiation field, while the Coulomb field of the nucleus is the field U for which $\psi_{\mathbf{k}}^{(+)}$ and $\psi_{\mathbf{k}}^{(-)}$ are defined (see IV, § 92, 96). Another example is the collision of an electron with an atom, accompanied by ionization of the latter (see Problem 4 in § 148).

according to the same specified rule. On the other hand, this series may be obtained as the matrix element (136.12), with ψ_i and ψ_f represented by perturbation series in the field U. The fact that the result must be a sum of integrals in which all poles are bypassed according to the same rule means that the poles in the series for ψ_i and ψ_f^* are bypassed by that rule as well. Solving the wave equation by perturbation theory with this bypass rule automatically produces a solution whose asymptotic form contains an outgoing wave together with the plane wave. In other words, the wave functions which at zeroth order in U were

$$\psi_i = e^{i\mathbf{kr}}, \qquad \psi_f^* = e^{-i\mathbf{k}'\mathbf{r}},$$

must be replaced by the exact solutions $\psi_{\mathbf{k}}^{(+)}$ and $\psi_{-\mathbf{k}'}^{(+)} = (\psi_{\mathbf{k}'}^{(-)})^*$, respectively. This proves the rule ((136.13)).

The choice of $\psi_{\mathbf{k}'}^{(-)}$ as the final wave function applies also to transitions from a discrete-spectrum state to a continuous-spectrum state (the question of how to choose ψ_i naturally does not arise in that case).

§ 137. Collisions of identical particles

The collision of two identical particles requires separate consideration. In quantum mechanics, particle identity produces a peculiar exchange interaction between them, which has a substantial effect on scattering (N. Mott, 1930)[52].

For a two-particle system, the orbital wavefunction must be symmetric or antisymmetric under interchange of the particles, according as their combined spin is even or odd (see § 62). Consequently, the scattering wavefunction obtained by solving the ordinary Schrödinger equation must likewise be symmetrized or antisymmetrized in the particle labels. Interchanging the particles amounts to reversing the radius vector that joins them. In the frame where the centre of mass is at rest, this leaves r unchanged while replacing the angle θ by $\pi - \theta$ (and hence sends $z = r\cos\theta$ to $-z$). Thus, in place of the asymptotic expression (123.3) for the wavefunction, we must write

$$\psi = e^{ikz} \pm e^{-ikz} + \frac{1}{r}e^{ikr}\left[f(\theta) \pm (\pi - \theta)\right]. \tag{137.1}$$

Because the particles are identical, one cannot say which is scattered and which is the scatterer. In the centre-of-mass frame there are two identical incident waves propagating towards each other, e^{ikz} and e^{-ikz}. The outgoing spherical wave in (137.1) accounts for the scattering of both particles, and the flux calculated from it gives the probability that either particle is scattered into a given solid-angle element do. The scattering cross-section is the ratio of this flux to the flux density in each incident

plane wave; as before, it is therefore the squared modulus of the coefficient of e^{ikr}/r in (137.1).

If the total spin of the colliding particles is even, the cross-section is

$$d\sigma_s = |f(\theta) + f(\pi - \theta)|^2 do, \tag{137.2}$$

[52]The direct spin–orbit interaction is still not considered here.

while for odd total spin it is

$$d\sigma_a = |f(\theta) - f(\pi - \theta)|^2 do. \tag{137.3}$$

The appearance of the interference term $f(\theta)f^*(\pi-\theta) + f^*(\theta)f(\pi-\theta)$ is characteristic of the exchange interaction. If the particles were distinguishable, as in ordinary classical mechanics, the probability for either of them to be scattered into do would simply be the sum of the probabilities for one to be deflected through θ and the oppositely moving one through $\pi - \theta$. The cross-section would then be

$$\{|f(\theta)|^2 + |f(\pi-\theta)|^2\} do.$$

In the low-velocity limit, the scattering amplitude, provided the interaction decreases sufficiently rapidly with distance, tends to an angle-independent constant (§ 132). Equation (137.3) then shows that $d\sigma_a$ vanishes: only particles with even total spin scatter from one another.

Equations (137.2) and (137.3) assume a definite value for the combined spin of the colliding particles. If the particles are not in definite spin states, the cross-section must instead be found by averaging over all spin states, taking them to be equally probable.

It was shown in § 62 that among the $(2s+1)^2$ spin states of two particles of spin s, $s(2s+1)$ states have even total spin and $(s+1)(2s+1)$ have odd total spin if s is half-integral, with the assignments reversed if s is integral. First let s be half-integral. The probabilities of even and odd S are then $s/(2s+1)$ and $(s+1)/(2s+1)$, respectively. Hence

$$d\sigma = \frac{s}{2s+1} d\sigma_s + \frac{s+1}{2s+1} d\sigma_a. \tag{137.4}$$

Substitution of (137.2) and (137.3) gives

$$d\sigma \left\{|f(\theta)|^2 + |f(\pi-\theta)|^2 - \frac{1}{2s+1}[f(\theta)f(\pi-\theta)^* + f(\theta)^* f(\pi-\theta)]\right\} do. \tag{137.5}$$

For integral s, similarly,

$$d\sigma = \left\{|f(\theta)|^2 + |f(\pi-\theta)|^2 + \frac{1}{2s+1}[f(\theta)f(\pi-\theta)^* + f(\theta)^* f(\pi-\theta)]\right\} do. \tag{137.6}$$

As an example, consider two electrons interacting through the Coulomb law $U = e^2/r$. Substitution of (135.9) in (137.5) with $s = 1/2$ gives, after a simple calculation in ordinary units,

$$\begin{aligned} d\sigma = \left(\frac{e^2}{m_0 v^2}\right)^2 \Bigg[& \frac{1}{\sin^4(\theta/2)} + \frac{1}{\cos^4(\theta/2)} \\ & - \frac{1}{\sin^2(\theta/2)\cos^2(\theta/2)} \cos\left(\frac{e^2}{\hbar v} \ln \tan^2 \frac{\theta}{2}\right)\Bigg] do \end{aligned} \tag{137.7}$$

(we have introduced the electron mass m_0 in place of the reduced mass $m = m_0/2$). This formula simplifies appreciably when the speed is so high that $e^2 \ll v\hbar$, precisely the condition for perturbation theory to apply to the Coulomb field. The cosine in the third term can then be replaced by unity, giving

$$d\sigma = \left(\frac{2e^2}{m_0 v^2}\right)^2 \frac{4 - 3\sin^2\theta}{\sin^4\theta}\, do. \tag{137.8}$$

The opposite limit, $e^2 \gg v\hbar$, corresponds to the classical limit (see the end of § 127). In (137.7) this limit is reached in a distinctive manner. For $e^2 \gg v\hbar$, the cosine in the third term oscillates rapidly. At any prescribed θ, (137.7) generally gives a cross-section appreciably different from Rutherford's. Averaging over even a small interval of θ, however, removes the oscillatory term and gives the classical formula.

All these formulas refer to the centre-of-mass frame. Passage to a frame in which one particle was at rest before the collision is accomplished,

according to (123.2), simply by replacing θ by 2ϑ. Thus (137.7) gives for electron collisions

$$d\sigma = \left(\frac{2e^2}{m_0 v^2}\right)^2 \left[\frac{1}{\sin^4\vartheta} + \frac{1}{\cos^4\vartheta} - \frac{1}{\sin^2\vartheta\cos^2\vartheta} \times \cos\left(\frac{e^2}{\hbar v}\ln\tan^2\vartheta\right)\right] \cos\vartheta\, do, \tag{137.9}$$

where do is the solid-angle element in the new frame. When θ is replaced by 2ϑ, do must be replaced by $4\cos\vartheta\, do$, since $\sin\theta\, d\theta\, d\varphi = 4\cos\vartheta\sin\vartheta\, d\vartheta\, d\varphi$.

Problem

Determine the scattering cross-section of two identical spin-1/2 particles having prescribed mean spins $\bar{\mathbf{s}}_1$ and $\bar{\mathbf{s}}_2$.

SOLUTION. The dependence of the cross-section on the particle polarizations must occur through a term proportional to the scalar $\bar{\mathbf{s}}_1\bar{\mathbf{s}}_2$. Seek $d\sigma$ in the form $a + b\bar{\mathbf{s}}_1\bar{\mathbf{s}}_2$. For unpolarized particles, $\bar{\mathbf{s}}_1 = \bar{\mathbf{s}}_2 = 0$, the second term is absent and (137.4) gives $d\sigma = a = (d\sigma_s + 3d\sigma_a)/4$. If both particles are fully polarized in the same direction, $\bar{\mathbf{s}}_1\bar{\mathbf{s}}_2 = 1/4$, the system is certainly in a state with $S = 1$; hence $d\sigma = a + b/4 = d\sigma_a$. Solving these two equations for a and b, we find

$$d\sigma = \frac{1}{4}(d\sigma_s + 3d\sigma_a) + (d\sigma_a - d\sigma_s)\bar{\mathbf{s}}_1\bar{\mathbf{s}}_2.$$

§ 138. Resonance scattering of charged particles

When charged nuclear particles are scattered (for example, protons by protons), the short-range nuclear forces are accompanied by the slowly decreasing Coulomb interaction. The theory of resonance scattering is then constructed by the same method as in § 133. The only difference is that, outside the range of the nuclear forces ($r \gg a$), the exact general solution of the Schrödinger equation in a Coulomb field must be used in place of the free-motion solution (133.2). The particle speed is still assumed small only to the extent that $ka \ll 1$; the relation between $1/k$ and the Coulomb unit of length $a_c = \hbar^2/(mZ_1Z_2e^2)$, where m is the reduced mass of the colliding particles, may be arbitrary[53].

For motion with $l = 0$ in a repulsive Coulomb field, the Schrödinger equation for the radial function $\chi = rR_0$ is

$$\chi'' + \left(k^2 - \frac{2}{r}\right)\chi = 0 \tag{138.1}$$

(Coulomb units are used here). In § 36 a solution of this equation was found subject to the requirement that χ/r remain finite at $r = 0$. Denote this solution here by F_0; it has the form (see (36.27), (36.28))

$$F_0 = Ae^{ikr}krF\left(\frac{i}{k} + 1, 2, -2ikr\right), \qquad A^2 = \frac{2\pi/k}{e^{2\pi/k} - 1}. \tag{138.2}$$

Its asymptotic expression at large distances is

$$F_0 \approx \sin\left[kr - \frac{1}{k}\ln(2kr) + \delta_0^{\mathrm{C}}\right], \qquad \delta_0^{\mathrm{C}} = \arg\Gamma\left(1 + \frac{i}{k}\right), \tag{138.3}$$

and the first terms of its expansion at small r ($kr \ll 1$, $r \ll 1$) are

$$F_0 = Akr(1 + r + \ldots). \tag{138.4}$$

With the changed boundary condition, however, the behavior of the function at the origin is immaterial, and we need the general solution of (138.1), a linear combination of its two independent integrals.

The parameters of the confluent hypergeometric function in (138.2), including the integral value $\gamma = 2$, put us precisely in the case mentioned at the end of § d of the Mathematical Supplements. Following the instructions given there, a second integral of (138.1) is obtained by replacing F in (138.2) by another linear combination of the two terms whose sum gives the confluent hypergeometric function according to (d.14). Choosing the difference of these terms gives the second independent solution of (138.1), denoted by G_0:[54]

$$G_0 = 2\,\mathrm{Im}\,\frac{Ae^{-ikr}kr}{\Gamma(1 + i/k)}(-2ikr)^{-1+i/k}G\left(1 - \frac{i}{k}, -\frac{i}{k}, -2ikr\right) \tag{138.5}$$

(the function F_0 is the real part of the expression occurring here). Its asymptotic form at large distances is

$$G_0 \approx \cos\left(kr - \frac{1}{k}\ln 2kr + \delta_0^{\mathrm{C}}\right), \tag{138.6}$$

and the first terms of the expansion at small r are

$$G_0 = \frac{1}{A}\{1 + 2r[\ln 2r + 2C - 1 + h(k)] + \ldots\}, \tag{138.7}$$

[53]The theory presented below was developed by L. D. Landau and Ya. A. Smorodinsky (1944).

[54]The functions F_0 and G_0, and the similarly defined functions F_l and G_l for $l \neq 0$, are called the *regular* and *irregular Coulomb functions*, respectively.

where $C = 0.577\ldots$ is Euler's constant and $h(k)$ denotes the function

$$h(k) = \operatorname{Re}\psi\left(-\frac{i}{k}\right) + \ln k \tag{138.8}$$

(where $\psi(z) = \Gamma'(z)/\Gamma(z)$ is the logarithmic derivative of the Γ-function)[55].

Write the general integral of (138.1) as

$$\chi = \text{const}\cdot(F_0 \cot\delta_0 + G_0), \tag{138.9}$$

where $\cot\delta_0$ is a constant. This notation is chosen so that the asymptotic form of the solution is

$$\chi \sim \sin\left[kr - \frac{1}{k}\ln(2kr) + \delta_0^{\mathrm{C}} + \delta_0\right]. \tag{138.10}$$

Thus δ_0 is the additional shift of the wave-function phase caused by the short-range forces. It must be related to the constant in the boundary condition $(\chi'/\chi)|_{r\to 0} = \text{const}$ which replaces consideration of the wave function within the range of the nuclear forces. Because the logarithmic derivative χ'/χ diverges as $\ln r$ when $r \to 0$, this condition must now be imposed not at zero but at an arbitrarily small, yet finite, value $r = \rho$. Calculating $\chi'(\rho)/\chi(\rho)$ from (138.4) and (138.7) and equating it to a constant gives

$$kA^2\cot\delta_0 + 2[\ln 2\rho + 2C + h(k)] = \text{const}.$$

The left-hand side contains the k-independent constants $2\ln 2\rho + 4C$; include them in the constant and then denote it by $-\kappa$. The final expression for $\cot\delta_0$, now in ordinary units, is

$$\cot\delta_0 = -\frac{1}{\pi}\left(e^{2\pi/ka_c} - 1\right)\left[h(ka_c) + \frac{\kappa a_c}{2}\right]. \tag{138.11}$$

In the limit $1/a_c \to 0$, corresponding to uncharged particles, (138.11) becomes $\cot\delta_0 = -\kappa/k$, in agreement with (133.6). Figure 49 shows the function $h(x)$[56].

[55] The expansion (138.7) follows from (138.5) by use of (d.17). The familiar relation

$$\psi(1+z) = \psi(z) + \frac{1}{z},$$

which follows readily from $\Gamma(z+1) = z\Gamma(z)$, and the values $\psi(1) = -C$, $\psi(2) = -C + 1$ have been used.

[56] The function $h(k)$ may be calculated from

$$h(k) = k^{-2}\sum_{n=1}^{\infty}\frac{1}{n(n^2 + k^{-2})} - C + \ln k,$$

which follows readily from

$$\psi(z) = -C - \frac{1}{z} + z\sum_{n=1}^{\infty}\frac{1}{n(n+z)}$$

(see E. Whittaker and G. Watson, *A Course of Modern Analysis*, Vol. II, § 12.16; Moscow: Fizmatgiz, 1963). The limiting expressions for $h(k)$ are

$$h(k) \approx \frac{k^2}{12} \quad \text{for } k \ll 1, \qquad h(k) = -C + \ln k + \frac{1.2}{k^2} \quad \text{for } k \gg 1$$

(the latter gives values of $h(k)$ with an error below 4% already for $k > 2.5$).

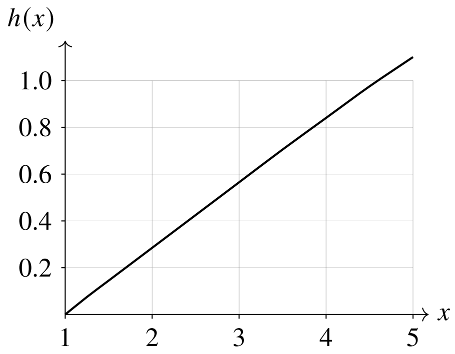


Figure 49. The function $h(x)$.

Thus, in the presence of the Coulomb interaction, the following quantity is "constant":

$$\frac{2\pi \cot \delta_0}{a(e^{2\pi/ka_c} - 1)} + \frac{2}{a_c} h(ka_c) = -\kappa. \tag{138.12}$$

We have put the word "constant" in quotation marks because κ is actually the first term in the expansion, in powers of the small quantity ka, of a certain function that depends on the properties of the short-range forces. As noted in § 133, a resonance at low energies corresponds to an anomalously small value of the constant κ. Greater accuracy therefore requires retaining the next term of the expansion ($\sim k^2$), whose coefficient has a "normal" order of magnitude; that is, $-\kappa$ in (138.12) must be replaced by[57]

$$-\kappa_0 + \frac{1}{2} r_0 k^2.$$

The presence of a resonance may be associated, as noted in § 133, with either an actual or a virtual discrete bound state of the system. One can show[58] that the sign of the constant κ remains the criterion for deciding whether the level is actual or virtual.

According to (138.10), the full phase shifts of the wave functions are $\delta_l^{\mathrm{C}} + \delta_l$. Therefore the scattering cross-section is

$$f(\theta) = \frac{1}{2ik} \sum_{l=0}^{\infty} (2l+1) \left[\exp\left(2i\delta_l^{\mathrm{C}} + 2i\delta_l\right) - 1\right] P_l(\cos\theta). \tag{138.13}$$

Write the difference in square brackets as

$$\begin{aligned} \exp\left(2i\delta_l^{\mathrm{C}} + 2i\delta_l\right) - 1 = {} & \left[\exp\left(2i\delta_l^{\mathrm{C}}\right) - 1\right] \\ & + \left[\exp\left(2i\delta_l^{\mathrm{C}}\right)\left(e^{2i\delta_l} - 1\right)\right]. \end{aligned} \tag{138.14}$$

[57] For proton–proton scattering, the constants $a = 1/\kappa_0$ and r_0 have the values $a = -7.8 \cdot 10^{-13}$, $r_0 = 2.8 \cdot 10^{-13}$ cm (the Coulomb unit of length is $2\hbar^2/m_p e^2 = 57.6 \cdot 10^{-13}$ cm). These values refer to a proton pair with antiparallel spins (when the spins are parallel, the Pauli principle prevents the two-proton system from occupying an s state at all).

[58] See L. D. Landau and Ya. A. Smorodinsky, ZhETF, vol. 14 (1944), p. 269.

The Coulomb phases δ_l^{C} make contributions of the same order to the scattering amplitude for every l. The phases δ_l associated with the short-range forces, however, are small for $l \neq 0$ at low energies. Thus, on substituting (138.14) in (138.13), retain the first bracket in every term of the sum; these terms sum to the Coulomb scattering amplitude (135.9),

$$f_{\mathrm{C}}(\theta) = -\frac{1}{2a_c k^2 \sin^2(\theta/2)} \exp\left(-\frac{2i}{ka_c} \ln \sin\frac{\theta}{2} + 2i\delta_0^{\mathrm{C}}\right). \tag{138.15}$$

Retain the second bracket in (138.14) only in the term with $l = 0$. The full scattering amplitude is therefore

$$f(\theta) = f_{\mathrm{C}}(\theta) + \frac{1}{2ik}(e^{2i\delta_0} - 1)\exp\left(2i\delta_0^{\mathrm{C}}\right). \tag{138.16}$$

The second term may be called the nuclear scattering amplitude. It must be emphasized, however, that this separation is conventional. Through the definition of δ_0 in (138.11), the Coulomb interaction substantially affects this term too, making it entirely different from what the same short-range forces would give for uncharged particles. In particular, as $ka_c \to 0$, δ_0, and hence the entire second term in (138.16), tends exponentially to zero, as $\exp(-2\pi/ka_c)$; nuclear scattering is then completely masked by Coulomb repulsion.

The two parts of the amplitude interfere in the scattering cross-section:

$$\begin{aligned} \frac{d\sigma}{d\Omega} = |f(\theta)|^2 = \left(\frac{Z_1 Z_2 e^2}{2mv^2}\right)^2 \Bigg[\frac{1}{\sin^4(\theta/2)} \\ -\frac{4ka_c}{\sin^2(\theta/2)} \sin\delta_0 \cos\left(\frac{2}{ka_c}\ln\sin\frac{\theta}{2} + \delta_0\right) + 4(ka_c)^2 \sin^2\delta_0\Bigg]. \end{aligned} \tag{138.17}$$

Here the colliding particles are assumed distinct. For identical particles, the scattering amplitude would first have to be symmetrized before its square is taken (compare § 137).

§ 139. Elastic collisions of fast electrons with atoms

Elastic collisions of fast electrons with atoms may be treated by the Born approximation when the incident electron moves much faster than the electrons in the atom.

Because the electron and the atom differ greatly in mass, the atom may be regarded as stationary during the collision. The reference frame in which the centre of inertia is at rest then coincides with the frame in which the atom is at rest. Thus, in formula (126.7), $\mathbf{p}$ and $\mathbf{p}'$ are the electron momenta before and after the collision, m is the electron mass, and the angle θ is the same as the electron deflection angle ϑ. The potential energy $U(r)$ occurring in (126.7), however, must be specified appropriately.

In § 126 we calculated the matrix element $U_{\mathbf{p}'\mathbf{p}}$ of the interaction energy with respect to the free-particle wave functions before and after the collision. For a collision with an atom, the wave functions describing the atom's internal state must also be included. In elastic scattering this state does not change. Consequently, $U_{\mathbf{p}'\mathbf{p}}$ must be defined as the matrix element with respect to the electron wave functions $\psi_{\mathbf{p}}$ and $\psi_{\mathbf{p}'}$, taken diagonally with respect to the atomic wave function. In other words, U in formula (126.7) is to be understood as the potential energy of interaction between the electron and the atom, averaged over the latter's wave function. It is equal to $e\varphi(r)$, where $\varphi(r)$ is the potential of the field produced at the point r by the mean charge distribution in the atom.

If $\rho(r)$ denotes the charge density in the atom, the potential φ obeys Poisson's equation

$$\Delta\varphi = -4\pi\rho(r).$$

The matrix element $U_{\mathbf{p}'\mathbf{p}}$ sought is essentially the Fourier component of U (and hence of φ) corresponding to the wave vector $\mathbf{q} = \mathbf{k}' - \mathbf{k}$. Applying Poisson's equation separately to each Fourier component gives

$$\Delta(\varphi_{\mathbf{q}} e^{i\mathbf{qr}}) = -q^2\varphi_{\mathbf{q}} e^{i\mathbf{qr}} = -4\pi\rho_{\mathbf{q}} e^{i\mathbf{qr}},$$

whence $\varphi_{\mathbf{q}} = 4\pi\rho_{\mathbf{q}}/q^2$, or

$$\int \varphi e^{-i\mathbf{qr}} dV = \frac{4\pi}{q^2}\int \rho e^{-i\mathbf{qr}} dV. \tag{139.1}$$

The charge density $\rho(r)$ consists of the electronic charges and the nuclear charge:

$$\rho = -en(r) + Ze\delta(r),$$

where $en(r)$ is the electron-charge density in the atom. Multiplication by $e^{-i\mathbf{qr}}$ followed by integration yields

$$\int \rho e^{-i\mathbf{qr}} dV = -e\int n e^{-i\mathbf{qr}} dV + Ze.$$

We therefore obtain the following expression for the integral of interest:

$$\int U e^{-i\mathbf{qr}} dV = \frac{4\pi e^2}{q^2}[Z - F(q)], \tag{139.2}$$

where

$$F(q) = \int n e^{-i\mathbf{qr}} dV \tag{139.3}$$

defines the quantity $F(q)$, called the *atomic form factor*. It depends both on the scattering angle and on the speed of the incident electron.

Finally, substitution of (139.2) into (126.7) gives the cross-section for elastic scattering of fast electrons by an atom in the form[59]

$$d\sigma = \frac{4m^2e^4}{\hbar^4 q^4}[Z - F(q)]^2 do, \qquad q = \frac{2mv}{\hbar}\sin\frac{\vartheta}{2}. \tag{139.4}$$

[59]We neglect exchange effects between the scattered fast electron and the atomic electrons; that is, we do not symmetrize the wave function of the system. The validity of this neglect is evident beforehand: in the "exchange integral", interference between the rapidly oscillating free-particle wave function and the atomic-electron wave function makes the associated contribution to the scattering amplitude small.

Consider first the limiting case $qa_0 \ll 1$, where a_0 gives the order of magnitude of the atomic dimensions. Small q correspond to small scattering angles,

$\vartheta \ll v_0/v$, where $v_0 \sim \hbar/ma_0$ is of the order of the speeds of the atomic electrons.

Expand $F(q)$ in powers of q. The zeroth-order term is $\int n\,dV$, the total number Z of electrons in the atom. The first-order term is proportional to $\int \mathbf{r}n(r)dV$, and therefore to the mean value of the atomic dipole moment; this value vanishes identically (see § 75). The expansion must consequently be continued through the second-order term, giving

$$Z - F(q) = \frac{q^2}{6}\int nr^2 dV;$$

on substituting this into (139.4), we find

$$d\sigma = \left|\frac{me^2}{3\hbar^2}\int nr^2 dV\right|^2 do. \tag{139.5}$$

Thus, in the small-angle region the cross section is independent of the scattering angle and is fixed by the mean square of the distances between the atomic electrons and the nucleus.

In the opposite limiting case of large q ($qa_0 \gg 1$, $\vartheta \gg v_0/v$), the factor $e^{-i\mathbf{qr}}$ in the integrand of (139.3) oscillates rapidly, so that the integral as a whole is nearly zero. Hence $F(q)$ may be neglected in comparison with Z, leaving

$$d\sigma = \left(\frac{Ze^2}{2mv^2}\right)^2 \frac{do}{\sin^4(\vartheta/2)}, \tag{139.6}$$

which is the Rutherford cross-section for scattering by the atomic nucleus.

Let us also calculate the transport cross-section

$$\sigma_{\text{tr}} = \int (1 - \cos\vartheta)d\sigma. \tag{139.7}$$

For angles $\vartheta \ll v_0/v$, equation (139.5) gives

$$d\sigma = \text{const}\cdot\sin\vartheta\,d\vartheta \approx \text{const}\cdot\vartheta\,d\vartheta,$$

where const is independent of ϑ. In this range the integrand of (139.7) is therefore proportional to $\vartheta^3 d\vartheta$, and the integral rapidly converges at its lower limit. In the range $1 \gg \vartheta \gg v_0/v$, on the other hand,

$$d\sigma \approx \text{const}\cdot(d\vartheta/\vartheta^3),$$

and the integrand is proportional to $d\vartheta/\vartheta$; the integral (139.7) is thus logarithmically divergent.

It follows that precisely this angular range supplies the main part of the integral, so that integration may be restricted to it. The lower limit must be of the order of v_0/v; write it as $e^2/(\gamma\hbar v)$, where γ is a dimensionless constant. The result is

$$\sigma_{\text{tr}} = 4\pi\left(\frac{Ze^2}{mv^2}\right)^2 \ln\frac{\gamma\hbar v}{e^2}. \tag{139.8}$$

An exact calculation of the constant γ requires consideration of scattering through angles $\vartheta > v_0/v$ and cannot be carried out in general. The value of σ_{tr} depends only weakly on this constant, since γ occurs inside a logarithm multiplied by the large quantity $\hbar v/e^2$.

For a numerical determination of the form factor of heavy atoms, the Thomas–Fermi distribution of the density $n(r)$ may be used. We have seen that, in the Thomas–Fermi model, $n(r)$ has the form

$$n(r) = Z^2 f(rZ^{1/3}/b)$$

(all quantities in this and the following formulas are measured in atomic units). It is readily seen that the integral (139.3), evaluated with this $n(r)$, contains q only in a particular combination with Z:

$$F(q) = Z\varphi(bqZ^{-1/3}). \tag{139.9}$$

Table 11 gives values of the function $\varphi(x)$ common to all atoms[60].

Table 11. Thomas–Fermi atomic factor

x	$\varphi(x)$	x	$\varphi(x)$	x	$\varphi(x)$
0	1.000	1.08	0.422	2.17	0.224
0.15	0.922	1.24	0.378	2.32	0.205
0.31	0.796	1.39	0.342	2.48	0.189
0.46	0.684	1.55	0.309	2.64	0.175
0.62	0.589	1.70	0.284	2.79	0.167
0.77	0.522	1.86	0.264	2.94	0.156
0.93	0.469	2.02	0.240		

With the atomic form factor (139.9), the cross-section (139.4) becomes

$$d\sigma = \frac{4Z^2}{q^4}[1-\varphi(bqZ^{-1/3})]^2 do = Z^{2/3}\Phi\left(Z^{-1/3}v\sin\frac{\vartheta}{2}\right)do, \tag{139.10}$$

The total cross-section can be obtained by integration. The small-ϑ range makes the chief contribution to the integral. We may therefore write

$$d\sigma \approx Z^{2/3}\Phi(Z^{-1/3}v\vartheta/2)\cdot 2\pi\vartheta\, d\vartheta,$$

and extend the integration over $d\vartheta$ to infinity:

$$\sigma = 2\pi Z^{2/3}\int_0^\infty \Phi\left(Z^{-1/3}\frac{v\vartheta}{2}\right)\vartheta\, d\vartheta = \frac{8\pi}{v^2}Z^{4/3}\int_0^\infty x\Phi(x)dx.$$

Thus σ has the form

$$\sigma = \text{const}\cdot\frac{Z^{4/3}}{v^2}. \tag{139.11}$$

It is similarly easy to verify that the constant γ in (139.8) is proportional to $Z^{-1/3}$.

[60]It must be kept in mind that this formula does not apply at small q, consistently with the fact that the integral of nr^2 cannot in practice be evaluated by the Thomas–Fermi method (see the note on p. 563). One should also remember that the Thomas–Fermi model does not reproduce the individual atomic properties that disturb their systematic variation with atomic number.

Problem

Calculate the cross-section for elastic scattering of fast electrons by a hydrogen atom in its ground state.

Solution. The ground-state wave function of the hydrogen atom is $\psi = e^{-r}/\sqrt{\pi}$ (in atomic units), and hence $n = e^{-2r}/\pi$. The angular integration in (139.3), carried out as in the derivation of (126.12), gives

$$F = \frac{4\pi}{q}\int_0^\infty n(r)\sin qr\,dr = \left(1 + \frac{q^2}{4}\right)^{-2}.$$

Substitution in (139.4) yields

$$d\sigma = \frac{4(8+q^2)^2}{(4+q^2)^4}do,$$

where $q = 2v\sin(\vartheta/2)$. To evaluate the total cross-section, write

$$do = 2\pi\sin\vartheta\,d\vartheta = \frac{2\pi}{v^2}q\,dq$$

and integrate with respect to q from 0 to $2v$. But v is assumed large and the integral converges, so its upper limit may be extended to infinity. We thus obtain

$$\sigma = \frac{7\pi}{3v^2}.$$

The transport cross section is evaluated as the integral

$$\sigma_{\mathrm{tr}} = \frac{1}{2v^2}\int q^2 d\sigma.$$

On changing the integration variable according to $4 + q^2 = u$, and replacing the upper limit by infinity everywhere except in the term du/u, we obtain

$$\sigma_{\mathrm{tr}} = \frac{4\pi}{v^2}\left(\ln v + \frac{1}{12}\right)$$

in agreement with (139.8).

§ 140. Scattering with spin–orbit interaction

So far we have considered only collisions of particles whose interaction is independent of their spins. Under these conditions the spins either have no effect on scattering at all or act indirectly through exchange effects (§ 137).

We now generalize the scattering theory developed in § 123 to the case in which the interaction depends essentially on the particle spins, as in collisions of nuclear particles.

Consider in detail the simplest case, where one colliding particle, taken for definiteness to be the incident-beam particle, has spin 1/2, and the other, the target particle, has spin zero.

For a specified half-integral total angular momentum j, the orbital angular momentum can have only the two values $l = j \pm 1/2$, corresponding to states of different parity. Conservation of j and parity therefore also implies conservation of the magnitude of the orbital angular momentum.

The operator $\hat{f}$ (see § 125) now acts on both the orbital and spin variables of the system's wave function. It must commute with the operator of the conserved quantity $\hat{\mathbf{l}}^2$. Its most general form is

$$\hat{f} = \hat{a} + \hat{b}\,\hat{\mathbf{l}} \cdot \hat{\mathbf{s}}, \tag{140.1}$$

where $\hat{a}$, $\hat{b}$ are orbital operators depending only on $\hat{\mathbf{l}}^2$.

The S-matrix, and hence the matrix of $\hat{f}$, is diagonal with respect to the wave functions of states having specified values of the conserved quantities l and j (and of the projection m of the total angular momentum); its diagonal elements are expressed in terms of the wave-function phases δ by (123.15). At fixed l, for total angular momentum $j = l + 1/2$ and $j = l - 1/2$, the eigenvalues of $\hat{\mathbf{l}} \cdot \hat{\mathbf{s}}$ are respectively $l/2$ and $-(l+1)/2$ (see (118.5)). Thus the diagonal matrix elements a_l and b_l of $\hat{a}$ and $\hat{b}$ obey

$$a_l + \frac{l}{2} b_l = \frac{1}{2ik}(e^{2i\delta_l^+} - 1), \qquad a_l - \frac{l+1}{2} b_l = \frac{1}{2ik}(e^{2i\delta_l^-} - 1), \tag{140.2}$$

where δ_l^+ and δ_l^- correspond to the states $j = l + 1/2$ and $j = l - 1/2$.

Our interest, however, is not in the diagonal elements of $\hat{f}$ themselves for states of specified l

and j, but in the scattering amplitude as a function of the directions of the incident and scattered waves. This amplitude is still an operator, but only in the spin variables; it is not diagonal in the spin projection σ. In the remainder of this section, $\hat{f}$ denotes this operator.

To find it, apply (140.1) to the function (125.17) corresponding to a plane wave incident along the z axis. Thus

$$\hat{f} = \sum_{l=0}^{\infty} (2l+1)(a_l + b_l \hat{\mathbf{l}} \cdot \hat{\mathbf{s}}) P_l(\cos\theta). \tag{140.3}$$

It remains to evaluate the action of $\hat{\mathbf{l}} \cdot \hat{\mathbf{s}}$ on $P_l(\cos\theta)$. This may be done by writing

$$\hat{\mathbf{l}} \cdot \hat{\mathbf{s}} = \frac{1}{2}(\hat{l}_+ \hat{s}_- + \hat{l}_- \hat{s}_+) + \hat{l}_z \hat{s}_z$$

(see (29.11)) and using (27.12) for the matrix elements of $l_\pm$; it is even simpler to use the operator expressions (26.14), (26.15) directly. A short calculation gives

$$\hat{\mathbf{l}} \cdot \hat{\mathbf{s}} P_l(\cos\theta) = i\mathbf{v} \cdot \hat{\mathbf{s}} P_l^1(\cos\theta),$$

where P_l^1 is an associated Legendre polynomial and $\mathbf{v}$ is a unit vector in the direction $\mathbf{n} \times \mathbf{n}'$, perpendicular to the scattering plane. Here $\mathbf{n}$ is the incident direction along the z axis and $\mathbf{n}'$ is the scattering direction specified by the spherical angles θ, φ.

Solving (140.2) for a_l, b_l and substituting in (140.3), we obtain

$$\hat{f} = A + 2B\mathbf{v} \cdot \hat{\mathbf{s}}, \tag{140.4}$$

$$A = \frac{1}{2ik}\sum_{l=0}^{\infty}[(l+1)(e^{2i\delta_l^+}-1)+l(e^{2i\delta_l^-}-1)]P_l(\cos\theta),$$
$$B = \frac{1}{2k}\sum_{l=1}^{\infty}(e^{2i\delta_l^+}-e^{2i\delta_l^-})P_l^1(\cos\theta). \tag{140.5}$$

The matrix elements of this operator are scattering amplitudes for specified values of the spin projection in the initial (σ) and final (σ') states. Consider a cross-section summed over every possible σ' and averaged with the probabilities of the different σ in the initial state, that is, in the

incident beam. It is calculated as

$$d\sigma = \overline{(\hat{f}^+\hat{f})_{\sigma\sigma}}d\Omega; \tag{140.6}$$

taking diagonal elements of $\hat{f}^+\hat{f}$ performs the sum over final states, while the bar denotes averaging over the initial state[61]. If all spin directions are equally probable in the initial state, this average reduces to taking the trace of the matrix divided by the number of possible spin projections:

$$d\sigma = \frac{1}{2}\,\mathrm{Tr}\Big(\hat{f}^+\hat{f}\Big)d\Omega. \tag{140.7}$$

On substituting (140.4) in (140.6), the mean square $(\mathbf{v}\cdot\mathbf{s})^2$ is $v^2s^2/3 = s(s+1)/3 = 1/4$. The result is

$$\frac{d\sigma}{d\Omega} = |A|^2 + |B|^2 + 2\,\mathrm{Re}(AB^*)\mathbf{v}\cdot\mathbf{P}, \tag{140.8}$$

where $\mathbf{P} = 2\bar{\mathbf{s}}$ is the initial beam polarization, defined as the ratio of the mean initial spin to its greatest possible value (1/2). Recall that for spin 1/2, the vector $\bar{\mathbf{s}}$ completely characterizes the spin state (§ 59).

Notice that polarization of the incident beam produces an azimuthal asymmetry of the scattering. Through the factor $\mathbf{v}\cdot\mathbf{P}$ in the last term, (140.8) depends not only on the polar angle θ but also on the azimuth φ of $\mathbf{n}'$ relative to $\mathbf{n}$ (provided the polarization is not perpendicular to $\mathbf{v}$, so that $\mathbf{v}\cdot\mathbf{P}\neq 0$).

The polarization of the scattered particles can be calculated from

$$\mathbf{P}' = 2\frac{\overline{(\hat{f}^+\mathbf{s}\hat{f})_{\sigma\sigma}}}{\overline{(\hat{f}^+\hat{f})_{\sigma\sigma}}}. \tag{140.9}$$

For example, if the initial state is unpolarized ($\mathbf{P} = 0$), a short calculation gives

$$\mathbf{P}' = \frac{2\,\mathrm{Re}(AB^*)\mathbf{v}}{|A|^2+|B|^2}. \tag{140.10}$$

[61] If the squared modulus $|f_{0n}|^2$ of a matrix element of an operator for a transition $0\to n$ is summed over the final states n, then

$$\sum_n |f_{0n}|^2 = \sum_n f_{0n}(f_{0n})^* = \sum_n f_{0n}(f^+)_{n0} = (ff^+)_{00}.$$

To avoid misunderstanding, we emphasize that the adjoint sign + in (140.6) and below refers to $\hat{f}$ as a spin operator and, in particular, does not imply interchange of $\mathbf{n}$ and $\mathbf{n}'$.

Thus scattering generally produces polarization perpendicular to the scattering plane. This effect is absent in the Born approximation, however: if all phases δ are small, then to first order in them A is real and B is purely imaginary, so $\mathrm{Re}(AB^*) = 0$.

That $\mathbf{P}'$ in (140.10) is directed along $\mathbf{v}$ is evident in advance: $\mathbf{P}'$ is an axial vector, and $\mathbf{v}$ is the only axial vector that can be formed from the available polar vectors $\mathbf{n}$ and $\mathbf{n}'$. The same property therefore also holds for the polarization produced when an unpolarized beam of spin-1/2 particles is scattered by an unpolarized target of nuclei with any spin, not only zero[62].

The formulation of the reciprocity theorem for scattering in the presence of spins must take account of the fact that time reversal changes the signs not only of momenta but also of angular momenta. Symmetry under time reversal must therefore be expressed as equality of the amplitudes of processes differing not only by interchange of initial and final states and reversal of the directions of motion, but also by reversal of the spin projections in both states. The signs of these amplitudes may nevertheless differ, since time reversal introduces into the spin wave function the factor $(-1)^{s-\sigma}$ according to (60.3). The reciprocity theorem must therefore be formulated as follows[63]:

$$\begin{aligned} &f(\sigma_1, \sigma_2, \mathbf{n}; \sigma_1', \sigma_2', \mathbf{n}') \\ &\quad = (-1)^{\Sigma(s-\sigma)} f(-\sigma_1', -\sigma_2', -\mathbf{n}'; -\sigma_1, -\sigma_2, -\mathbf{n}). \end{aligned} \tag{140.11}$$

Here $f(\sigma_1, \sigma_2, \mathbf{n}; \sigma_1', \sigma_2', \mathbf{n}')$ is the scattering amplitude for the spin projections of the colliding particles to change from σ_1, σ_2 to σ_1', σ_2'. The sum in the exponent is over both particles before and after the scattering.

In the Born approximation scattering has an additional symmetry: the probabilities are the same for processes differing by interchange of the initial and final states without reversal of the particle momenta and spin projections, as under time reversal (see § 126). Combining this property with the reciprocity theorem, we find that scattering is symmetric under reversal of all momenta and spin projections without their interchange. It follows that, in the Born approximation, polarization cannot be produced when any unpolarized beam is scattered by an unpolarized target. Indeed, under this transformation the polarization vector $\mathbf{P}$ changes sign, whereas the only vector $\mathbf{k} \times \mathbf{k}'$ along which $\mathbf{P}$ can be directed is unchanged. Thus the property noted above for spin-1/2 particles scattered by spin-zero particles is in fact general.

For arbitrary spins of the colliding particles, the general formulas for the angular distributions are very cumbersome, and we shall not derive them here. We shall only count the parameters that determine these distributions.

The case considered above, of particles with spins 1/2 and zero, is characterized in particular by there being only one state of the two-particle system for specified values of J and parity, apart from the immaterial orientation of the total angular momentum in space. Each such state contributes one real parameter, a phase δ, to the scattering

[62] Here we mean a target of nuclei whose spin directions are completely random. Recall that for $s > 1/2$ the mean spin vector does not fully determine the spin state; its vanishing does not by itself mean a complete absence of spin ordering.

[63] The derivation is analogous to that of (125.12). Spin factors must be introduced in the amplitudes of the incoming and outgoing waves in the wave function, and (125.10) is replaced by $\hat{K}^{-1}\hat{S}\hat{K} = \hat{S}$, where $\hat{K}$ not only performs inversion but also transforms the spin state according to (60.3).

amplitude. For other spins, there are generally several different states with the same total angular momentum J and parity; they differ in the total particle spin S and the orbital angular momentum l of their relative motion. Let the number of these states be n. It is readily seen that every such group contributes $n(n+1)/2$ independent real parameters to the scattering amplitude.

Indeed, for these states the S-matrix is a unitary symmetric matrix, by the reciprocity theorem, with $n \cdot n$ complex elements. The number of independent quantities is conveniently counted by writing $\hat{S} = \exp(i\hat{R})$. Unitarity is then automatic when $\hat{R}$ is an arbitrary Hermitian operator (see (12.13)). If S is symmetric, so is R; being Hermitian, it is then real. A real symmetric matrix

has $n(n+1)/2$ independent components.

For example, for two spin-$1/2$ particles $n = 2$. Indeed, at fixed J there are four states: two with $l = J$ and total spin $S = 0$ or 1, and two with $l = J \pm 1$, $S = 1$. Clearly two are even (l even) and two are odd (l odd).

The general scattering amplitude for spin-$1/2$ particles, as an operator in the spin variables of both particles, is readily written from the required invariance conditions: it must be a scalar invariant under time reversal. We have available the two axial spin vectors $\mathbf{s}_1$, $\mathbf{s}_2$ and the two ordinary polar vectors $\mathbf{n}$, $\mathbf{n}'$. Each of the operators $\mathbf{s}_1$, $\mathbf{s}_2$ must enter the amplitude linearly, since every function of a spin-$1/2$ operator reduces to a linear function. The most general operator satisfying these conditions can be written

$$\begin{aligned}\hat{f} = A &+ B(\mathbf{s}_1 \cdot \boldsymbol{\lambda})(\mathbf{s}_2 \cdot \boldsymbol{\lambda}) + C(\mathbf{s}_1 \cdot \boldsymbol{\mu})(\mathbf{s}_2 \cdot \boldsymbol{\mu}) + D(\mathbf{s}_1 \cdot \boldsymbol{\nu})(\mathbf{s}_2 \cdot \boldsymbol{\nu}) \\ &+ E(\mathbf{s}_1 + \mathbf{s}_2) \cdot \boldsymbol{\nu} + F(\mathbf{s}_1 - \mathbf{s}_2) \cdot \boldsymbol{\nu}.\end{aligned} \tag{140.12}$$

The coefficients A, B, ... are scalars depending only on $\mathbf{n} \cdot \mathbf{n}'$, that is, on the scattering angle θ and the energy. The vectors $\boldsymbol{\lambda}$, $\boldsymbol{\mu}$, $\boldsymbol{\nu}$ are three mutually perpendicular unit vectors directed respectively along $\mathbf{n} + \mathbf{n}'$, $\mathbf{n} - \mathbf{n}'$, and $\mathbf{n} \times \mathbf{n}'$. Time reversal corresponds to

$$\mathbf{s}_1 \to -\mathbf{s}_1, \quad \mathbf{s}_2 \to -\mathbf{s}_2, \quad \mathbf{n} \to -\mathbf{n}', \quad \mathbf{n}' \to -\mathbf{n}.$$

Then $\boldsymbol{\lambda} \to -\boldsymbol{\lambda}$, $\boldsymbol{\mu} \to \boldsymbol{\mu}$, and $\boldsymbol{\nu} \to -\boldsymbol{\nu}$, making the invariance of (140.12) evident.

For mutual scattering of nucleons, protons and neutrons, the last term in (140.12) is absent. This follows already from the fact that the nuclear forces between nucleons conserve the magnitude of the total spin S, whereas $\mathbf{s}_1 - \mathbf{s}_2$ does not commute with S^2. The remaining terms in (140.12) can be expressed through the total-spin operator $\mathbf{S}$ by (117.4), and therefore commute with S^2. For scattering of identical nucleons, pp or nn, the coefficients A, B, ... as functions of the scattering angle also satisfy symmetry relations following from the identity of the particles (see Problem 2).

Problems

PROBLEM

1. For scattering of spin-1/2 particles by spin-zero particles, determine the polarization after scattering when it was also nonzero before scattering.

SOLUTION. The calculation from (140.9) is conveniently performed in components, choosing the z axis along $\mathbf{v}$. The result is

$$\mathbf{P}' = \frac{(|A|^2 - |B|^2)\mathbf{P} + 2|B|^2\mathbf{v}(\mathbf{v}\cdot\mathbf{P}) + 2\,\mathrm{Im}(AB^*)\mathbf{v}\times\mathbf{P} + 2\mathbf{v}\,\mathrm{Re}(AB^*)}{|A|^2 + |B|^2 + 2\,\mathrm{Re}(AB^*)\mathbf{v}\cdot\mathbf{P}}.$$

PROBLEM

2. Find the symmetry conditions satisfied, as functions of the angle θ, by the coefficients in the scattering amplitude of two identical nucleons (R. Oehme, 1955).

SOLUTION. Regroup the terms in (140.12) so that each is nonzero only for singlet ($S = 0$) or triplet ($S = 1$) states of the nucleon system:

$$\begin{aligned} f = a(\mathbf{s}_1\cdot\mathbf{s}_2 - 1/4) + b(\mathbf{s}_1\cdot\mathbf{s}_2 + 3/4) + c[1/4 + (\mathbf{s}_1\cdot\boldsymbol{\nu})(\mathbf{s}_2\cdot\boldsymbol{\nu})] \\ + d[(\mathbf{s}_1\cdot\mathbf{n})(\mathbf{s}_2\cdot\mathbf{n}') + (\mathbf{s}_1\cdot\mathbf{n}')(\mathbf{s}_2\cdot\mathbf{n})] + e(\mathbf{s}_1 + \mathbf{s}_2)\cdot\boldsymbol{\nu}. \end{aligned} \tag{1}$$

Using (117.4), it is readily verified that the first term is nonzero only for $S = 0$, and the others only for $S = 1$. By particle identity, the scattering amplitude must be symmetric under interchange of particle coordinates for $S = 0$ and antisymmetric for $S = 1$. This transformation means $\theta \to \pi - \theta$, or equivalently reversal of one of $\mathbf{n}$, $\mathbf{n}'$ (compare § 137). The conditions are

$$\begin{aligned} a(\pi-\theta) = a(\theta), \quad b(\pi-\theta) = -b(\theta), \quad c(\pi-\theta) = -c(\theta), \\ d(\pi-\theta) = d(\theta), \qquad e(\pi-\theta) = e(\theta). \end{aligned} \tag{2}$$

By isotopic invariance, the scattering amplitude is the same for nn and pp scattering and for np scattering in the isotopic state $T = 1$. The np system can also be in a state with $T = 0$; its scattering amplitude is then characterized by different coefficients $a, b, \ldots$ in (1), which do not have the symmetry properties (2).

§ 141. Regge poles

Section 128 treated the scattering amplitude's analytic properties in the complex particle energy E, with orbital angular momentum l serving as a parameter restricted to real integral values. Further properties of methodological importance emerge when E is held real and l is instead allowed to vary continuously as a complex variable[64].

As in § 128, consider radial wave functions whose asymptotic form as $r \to \infty$ is

$$\chi_l = rR_l = A(l,E)\exp\left(-\frac{\sqrt{-2mE}}{\hbar}r\right) + B(l,E)\exp\left(\frac{\sqrt{-2mE}}{\hbar}r\right). \tag{141.1}$$

[64] These properties were first investigated by Regge (T. Regge, 1958).

These functions solve the Schrödinger equation (32.8), in which l is now regarded as a complex parameter; one of the two independent solutions is selected by the condition

$$R_l \approx \text{const} \cdot r^l \quad \text{as } r \to 0. \tag{141.2}$$

This condition restricts the allowed values of l. The general form of a solution of (32.8) at small r is

$$R_l \approx c_1 r^l + c_2 r^{-l-1}$$

(see the end of § 32). For the second solution to be distinguished unambiguously against the background of the first and excluded, r^{-l-1} must be larger than r^l as $r \to 0$. For complex l this gives $\operatorname{Re} l > \operatorname{Re}(-l-1)$, or

$$\operatorname{Re}\left(l + \frac{1}{2}\right) > 0. \tag{141.3}$$

We henceforth consider this half-plane of complex l, to the right of $l = -1/2$.

As a solution of a differential equation with coefficients analytic in l, $R(r; l, E)$ is analytic in this parameter and has no singularities in (141.3). This applies to (141.1), so $A(l, E)$ and $B(l, E)$ have no singularities in l. Retention of both terms of (141.1) as $r \to \infty$ is understood to be legitimate. For $E > 0$ this is always so; for $E < 0$ it holds if $U(r)$ satisfies (128.6) or (128.13). The asymptotic behaviour in r depends only on E, not on l, so complex l does not alter the conditions for reaching the asymptotic form.

Comparison with (128.15) gives

$$S(l, E) = \exp[2i\delta(l, E)] = e^{i\pi l}\frac{A(l, E)}{B(l, E)}, \tag{141.4}$$

also for complex l, although the phase shift δ is then not real.

For real l and $E > 0$, (128.4) gives $A(l, E) = B^*(l, E)$. Hence for complex l

$$A(l^*, E) = B^*(l, E) \quad \text{for } E > 0, \tag{141.5}$$

and $S(l, E)$ satisfies *complex unitarity*:

$$S^*(l, E) S(l^*, E) = 1. \tag{141.6}$$

Because $A(l, E)$ and $B(l, E)$ have no singularities as functions of l, the only singularities (poles) of $S(l, E)$, and hence of the partial scattering amplitude $f(l, E)$, occur at zeros of $B(l, E)$. Poles of the scattering amplitude in the complex l-plane are termed *Regge poles*. Their locations naturally depend on the value of the real parameter E. The functions $l = \alpha_i(E)$ that specify these locations are termed *Regge trajectories*: as E varies, the poles follow definite curves in the l-plane (below we shall omit the index i that labels the poles).

We first show that for $E < 0$ all $\alpha(E)$ are real. Consider

$$\chi'' + \left[\frac{2m}{\hbar^2}(E - U) - \frac{\alpha(\alpha + 1)}{r^2}\right]\chi = 0. \tag{141.7}$$

Multiplying by χ^*, integrating over dr, and integrating the first term by parts gives

$$-\int_0^\infty |\chi'|^2 dr + \frac{2m}{\hbar^2}\int_0^\infty (E-U)|\chi|^2 dr - \alpha(\alpha+1)\int_0^\infty \frac{|\chi|^2}{r^2} dr = 0.$$

For $B = 0$, the Regge-pole condition, the wave function decreases exponentially as $r \to \infty$, so all integrals converge. The first two terms and the integral in the last term are real. Therefore

$$\operatorname{Im}[\alpha(\alpha+1)] = \operatorname{Im}\left(\alpha+\frac{1}{2}\right)^2 = 2\operatorname{Re}\left(\alpha+\frac{1}{2}\right)\operatorname{Im}\alpha = 0.$$

Since only poles in (141.3) are considered, $\operatorname{Re}(\alpha + 1/2) > 0$, and

$$\operatorname{Im}\alpha(E) = 0 \quad \text{for } E < 0. \tag{141.8}$$

Differentiate (141.7) with respect to E, multiply the result by χ, multiply (141.7) by $\partial\chi/\partial E$, and subtract. This gives

$$\left[\chi'\frac{\partial\chi}{\partial E} - \chi\left(\frac{\partial\chi}{\partial E}\right)'\right]' - \frac{2m}{\hbar^2}\chi^2 + \frac{\chi^2}{r^2}\frac{d\alpha(\alpha+1)}{dE} = 0.$$

Integration over dr from 0 to ∞, using again the vanishing of χ at infinity, yields

$$\frac{d\alpha(\alpha+1)}{dE}\int_0^\infty \frac{\chi^2}{r^2} dr = \frac{2m}{\hbar^2}\int_0^\infty \chi^2 dr. \tag{141.9}$$

Since α and the wave function are real, both integrals are positive. Thus

$$\frac{d\alpha(\alpha+1)}{dE} = 2\left(\alpha+\frac{1}{2}\right)\frac{d\alpha}{dE} > 0,$$

and, since $\alpha + 1/2 > 0$,

$$\frac{d\alpha}{dE} > 0 \quad \text{for } E < 0.$$

Thus $\alpha(E)$ increases monotonically with E for $E < 0$.

Negative energies at which $\alpha(E)$ takes physical values, the integers $l = 0, 1, 2, \ldots$, are discrete energy levels. This gives a new classification of bound states by the Regge trajectories on which they lie.

For example, in an attractive Coulomb field the scattering-matrix elements are[65]

$$S_l = \frac{\Gamma(l+1-i/k)}{\Gamma(l+1+i/k)}. \tag{141.10}$$

Here k is in Coulomb units. The poles occur when the argument of $\Gamma(l + 1 - i/k)$ is a negative integer or zero. Since $k = i\sqrt{-2E}$ for $E < 0$,

$$\alpha(E) = -n_r - 1 + \frac{1}{\sqrt{-2E}}, \qquad E < 0, \tag{141.11}$$

[65] Compare (135.11), in which the signs before k must be reversed in passing from repulsion to attraction.

where $n_r = 0, 1, 2, \ldots$ labels the trajectories. Setting $\alpha(E) = l = 0, 1, 2, \ldots$ gives the Bohr formula

$$E = -\frac{1}{2(n_r + l + 1)^2}.$$

Here n_r is the radial quantum number specifying the number of radial nodes. Each prescribed n_r, and hence each trajectory, corresponds to infinitely many levels distinguished by orbital angular momentum.

We turn to $\alpha(E)$ for $E > 0$. Recall that $A(l, E)$ and $B(l, E)$, as functions of complex E, are defined on a plane cut along the positive real semiaxis. The roots $l = \alpha(E)$ of $B(l, E) = 0$ have the same cut. On its two edges they take complex-conjugate values, with $\operatorname{Im}\alpha > 0$ on the upper edge. We explain this by physical reasoning.

For complex l, the centrifugal energy and effective potential $U_l = U + l(l+1)/(2mr^2)$ are complex. Repeating the derivation of § 19 gives

$$\frac{\partial}{\partial t}|\psi|^2 + \operatorname{div}\mathbf{j} = 2|\psi|^2 \operatorname{Im} U_l.$$

For $l = \alpha$ and $\operatorname{Im}\alpha > 0$, one also has $\operatorname{Im} U_l > 0$; the positive right-hand side is as though particles were emitted inside the field. The asymptotic wave function, containing only the first term of (141.1) when $B = 0$, must accordingly be outgoing. This occurs on the upper edge of the cut; compare (128.1) and (128.3).

Since $\alpha(E)$ is complex for $E > 0$, it cannot take physical values $l = 0, 1,$

$2, \ldots$, but may approach them. We show that a resonance then appears in the corresponding partial amplitude.

Let l_0 be an integer near $\alpha(E)$, and let the positive real energy E_0 satisfy $\operatorname{Re}\alpha(E_0) = l_0$. Near it,

$$\alpha(E) \approx l_0 + i\eta + \beta(E - E_0), \tag{141.12}$$

where $\eta = \operatorname{Im}\alpha(E_0)$ is real. On the upper edge, $\eta > 0$, and proximity to l_0 means $\eta \ll 1$. The constant $\beta = d\alpha/dE$ at E_0 can also be treated as real and positive. Since α is nearly real, $\chi(r; \alpha, E)$ is nearly real. Neglecting higher orders in η and the imaginary part of χ, positivity of β follows from the positive integrals in (141.9)[66].

Because $l = \alpha(E)$ is a zero of $B(l, E)$, near α, E_0 this function is proportional to $\alpha - l$. Hence

$$B(l_0, E) \approx \text{const} \cdot [\beta(E - E_0) + i\eta]. \tag{141.13}$$

This has the form (134.6), with energy E_0 and quasidiscrete-level width $\Gamma = 2\eta/\beta > 0$. Thus proximity of a Regge trajectory

to integral l at $E > 0$ corresponds to quasistationary states. The same classification principle therefore applies as for stationary states: a trajectory may correspond to a family of discrete and quasidiscrete levels.

[66]To clarify these integrals, the asymptotic region $r \gg a$, where a is the field range and (141.1) applies, contributes only slightly when η is small. If $l = \alpha(E)$ is a zero of $B(l, E)$, then by (141.5) $l = \alpha^*$ is a zero of $A(l, E)$. Thus $A(\alpha, E)$, and hence $\chi(r; \alpha, E)$ for $r \gg a$, is of order $\eta^{1/2}$ (see (134.11)). On the upper edge of the cut the wave function contains e^{ikr}: $\chi(r; \alpha, E) = A(\alpha, E)e^{ikr}$. Taking E as $E + i\delta$, $\delta \to +0$, gives k a small positive imaginary part and ensures convergence of (141.9). Physically, the small contribution from $r \gg a$ reflects that E_0 is a quasistationary state; the particle reaches this region only through improbable decay. The main contribution comes from $r \sim a$, where the wave function is nearly real.

Complex l gives a useful integral representation of the complete scattering amplitude for $E > 0$, whose series is

$$f(\mu) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)[S(l,E)-1]P_l(\mu), \qquad \mu = \cos\theta. \tag{141.14}$$

First define $P_l(\mu)$ for complex as well as integral l. Let it be the solution of (c.2)

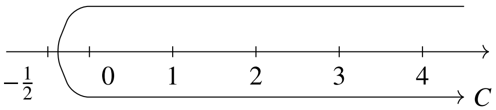
$$(1-\mu)^2 P_l''(\mu) - 2\mu P_l'(\mu) + l(l+1)P_l(\mu) = 0 \tag{141.15}$$

with $P_l(1) = 1$. As a function of l, it then has no finite singularities[67].

The series (141.14) equals

$$f(\mu) = \frac{1}{4k}\int_C \frac{2l+1}{\sin\pi l}[S(l,E)-1]P_l(-\mu)\,dl, \tag{141.16}$$

where C encircles clockwise all points $l = 0, 1, 2, \ldots$ on the real axis and closes at infinity:



All poles $l = \alpha_1, \alpha_2, \ldots$ of $S(l,E)$, which for $E > 0$ lie off the real axis, must remain outside C.

Indeed, (141.16) is $-2\pi i$ times the sum of residues of the integrand at $l = 0, 1, 2, \ldots$, the poles of $1/\sin\pi l$, whose residues are $(-1)^l/\pi$. Since $P_l(-\mu) = (-1)^l P_l(\mu)$ for integral l, this reduces (141.16) to (141.14)[68].

Problem

Show that the phase shifts for successive integral l satisfy

$$\delta_{l+1}(E) - \delta_l(E) < \pi/2.$$

Solution. Regard l as a continuous real variable and differentiate (32.10):

$$\left[\frac{\partial\chi}{\partial l}\right]'' + \left[\frac{2m}{\hbar^2}(E-U) - \frac{l(l+1)}{r^2}\right]\frac{\partial\chi}{\partial l} = \frac{2l+1}{r^2}\chi.$$

Multiply this equation by χ, the original equation by $\partial\chi/\partial l$, and subtract:

$$\left[\chi\frac{\partial\chi'}{\partial l} - \chi'\frac{\partial\chi}{\partial l}\right]' = (2l+1)\frac{\chi^2}{r^2}.$$

[67] Comparison with (e.2) expresses it through the hypergeometric function

$$P_l(\mu) = F\left(-l, l+1, 1; \frac{1-\mu}{2}\right).$$

[68] A fuller account of these topics in nonrelativistic theory is given in the book by de Alfaro and Regge cited in the note on p. 612.

Integrate from 0 to ∞. At $r = 0$ the bracket vanishes; at infinity use (33.20). Thus

$$4k\left(\frac{\pi}{2} - \frac{\partial \delta_l}{\partial l}\right) = (2l+1)\int_0^\infty \frac{\chi^2}{r^2}\,dr > 0,$$

so $\partial\delta_l/\partial l < \pi/2$. Integration from l to $l+1$ gives the required inequality. Combined with (133.17), it proves that the number n_l of discrete levels does not increase with l. At $E \to \infty$, where the Born approximation applies, $\delta_l(\infty) = 0$. Hence

$$n_{l+1} - n_l = -\frac{1}{\pi}[\delta_{l+1}(0) - \delta_l(0)] < \frac{1}{2}, \qquad n_{l+1} - n_l \le 0.$$

CHAPTER XVIII

Inelastic Collisions

§ 142. Elastic scattering in the presence of inelastic processes

Collisions accompanied by a change in the internal state of the colliding particles are called inelastic. Here these changes are understood in the broadest sense; even the type of particle may change. Thus atoms may be excited or ionized, and nuclei may be excited or disintegrated. When a collision, for example a nuclear reaction, may be accompanied by different physical processes, one speaks of different *reaction channels*.

The presence of inelastic channels also has a definite effect on the properties of elastic scattering.

In the general case of several reaction channels, the asymptotic wave function of the colliding-particle system is a sum with one term for each possible channel. Among these is a term describing the particles in their unchanged initial state, the *entrance channel*. It is the product of the wave functions of the particles' internal states and a function describing their relative motion in the center-of-mass system. This last function is of interest here; denote it by ψ and determine its asymptotic form.

In the entrance channel, ψ consists of an incident plane wave and an outgoing spherical wave corresponding to elastic scattering. It may also be represented as a sum of incoming and outgoing waves, as in § 123. The difference is that the asymptotic expression for the radial functions $R_l(r)$ cannot be taken as a standing wave. A standing wave is the sum of incoming and outgoing waves of equal amplitude. This accords with the physical meaning of purely elastic scattering, but when inelastic channels are present, the outgoing-wave amplitude must be smaller than the incoming-wave amplitude. Thus the asymptotic expression for ψ is given by (123.9),

$$\psi = \frac{1}{2ikr}\sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta)\left[(-1)^{l+1}e^{-ikr}+S_l e^{ikr}\right], \tag{142.1}$$

except that S_l is no longer defined by (123.10), but is a generally complex quantity with modulus less than unity. The elastic scattering amplitude is expressed through these quantities by (123.11):

$$f(\theta) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)(S_l-1)P_l(\cos\theta). \tag{142.2}$$

In place of (123.12), the total elastic scattering cross-section σ_e is

$$\sigma_e = \frac{\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)|1-S_l|^2. \tag{142.3}$$

The total inelastic scattering cross-section, also called the *reaction cross-section* σ_r, summed over every possible channel, can likewise be expressed in terms of S_l. For each

l, the outgoing-wave intensity is reduced relative to the incoming-wave intensity by the factor $|S_l|^2$. This reduction must be attributed entirely to inelastic scattering. Hence

$$\sigma_r = \frac{\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)(1-|S_l|^2), \tag{142.4}$$

and the total cross-section is

$$\sigma_t = \sigma_e + \sigma_r = \frac{2\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)(1-\operatorname{Re} S_l). \tag{142.5}$$

The partial elastic scattering amplitude with angular momentum l, defined by (123.15), is

$$f_l = \frac{S_l - 1}{2ik}, \tag{142.6}$$

and each term of the sums in (142.3), (142.4) is the partial elastic or inelastic cross-section for particles of angular momentum l:

$$\begin{aligned}\sigma_e^{(l)} &= \frac{\pi}{k^2}(2l+1)|1-S_l|^2,\\ \sigma_r^{(l)} &= \frac{\pi}{k^2}(2l+1)(1-|S_l|^2),\\ \sigma_t^{(l)} &= \frac{2\pi}{k^2}(2l+1)(1-\operatorname{Re} S_l).\end{aligned} \tag{142.7}$$

The value $S_l = 1$ means that there is no scattering at all for this l. The case $S_l = 0$ corresponds to complete "absorption" of particles with angular momentum l: the partial outgoing wave with this l is absent from (142.1). The elastic and inelastic cross-sections are then equal:

$$\sigma_e^{(l)} = \sigma_r^{(l)} = \frac{\pi}{k^2}(2l+1). \tag{142.8}$$

We also note that elastic scattering may occur without inelastic scattering, when $|S_l| = 1$, but the converse is impossible: inelastic scattering necessarily entails elastic scattering. For a specified $\sigma_r^{(l)}$, the partial elastic cross-section must lie in the interval

$$\sqrt{\sigma_0} - \sqrt{\sigma_0 - \sigma_r^{(l)}} \le \sqrt{\sigma_e^{(l)}} \le \sqrt{\sigma_0} + \sqrt{\sigma_0 - \sigma_r^{(l)}}, \tag{142.9}$$

where $\sigma_0 = (2l+1)\pi/k^2$.

Taking $f(\theta)$ from (142.2) at $\theta = 0$ and comparing it with (142.5) gives

$$\operatorname{Im} f(0) = \frac{k}{4\pi}\sigma_t, \tag{142.10}$$

which generalizes the optical theorem (125.9). Here $f(0)$ is still the elastic zero-angle scattering amplitude, but σ_t includes the inelastic part as well.

The imaginary parts of the partial amplitudes f_l are related to $\sigma_t^{(l)}$ by

$$\operatorname{Im} f_l = \frac{k}{4\pi}\frac{\sigma_t^{(l)}}{2l+1}, \tag{142.11}$$

which follows directly from (142.6), (142.7).

The fact that the coefficients S_l in the asymptotic wave function do not have unit modulus has no effect on the conclusions of § 128 concerning the singular points of the elastic scattering amplitude as a function of complex E; they remain valid in the presence of inelastic processes. The analytic properties do change in that the amplitude is now nonreal on the negative real semiaxis ($E < 0$), and its values on the upper and lower edges of the cut for $E > 0$ are not complex conjugates. More generally, its values at points in the upper and lower half-planes symmetric about the real axis are not complex conjugates either.

On passing from the upper edge of the cut to the lower by a full circuit around $E = 0$, the root $\sqrt{E}$ changes sign; hence the real quantity k at $E > 0$ changes sign. The incoming and outgoing waves in (142.1) exchange roles, so the new coefficient S_l is $1/S_l$, the reciprocal of its former value, which is by no means S_l^*. Denote the amplitudes on the upper and lower edges by $f_l(k)$ and $f_l(-k)$; only $f_l(k)$ is, of course, the physical amplitude. From (142.6),

$$f_l(k) = \frac{S_l - 1}{2ik}, \qquad f_l(-k) = -\frac{1/S_l - 1}{2ik}.$$

Eliminating S_l gives

$$f_l(k) - f_l(-k) = 2ik f_l(k) f_l(-k). \tag{142.12}$$

In the absence of inelastic processes, one would have $f(-k) = f^*(k)$, and (142.12) and (142.11) would coincide.

Rewriting (142.12) as

$$\frac{1}{f_l(k)} - \frac{1}{f_l(-k)} = -2ik,$$

shows that $1/f_l(k) + ik$ must be an even function of k. Denoting this function by $g_l(k^2)$, we have

$$f_l(k) = \frac{1}{g_l(k^2) - ik}. \tag{142.13}$$

The even function $g_l(k^2)$ is no longer real, however, as it was in (125.15)[1].

When a particle beam passes through a scattering medium containing many scattering centers, it gradually weakens as particles undergoing various collision processes leave the beam. This attenuation is determined entirely by the elastic zero-angle scattering amplitude and, under certain conditions (see below), can be described by the following formal method[2].

Let $f(0, E)$ be the zero-angle scattering amplitude for each particle of the medium. Assume that f is small compared with the mean distance $d \sim (V/N)^{1/3}$ between particles, so each may be treated as a separate scatterer. Introduce as an auxiliary quantity an

[1] This argument, and hence the conclusion that g_l is even, assumes that the interaction decreases sufficiently rapidly as $r \to \infty$ to ensure the absence of cuts in the left half-plane of E, thereby allowing a full circuit around $E = 0$.

[2] The ideas presented below apply, in particular, to the scattering by nuclei of fast neutrons with energies of order hundreds of MeV. Their wavelength is so short that the nucleus may be regarded as an inhomogeneous macroscopic medium.

effective field U_{eff} of a fixed center, defined so that its Born zero-angle scattering amplitude is precisely the true amplitude $f(0,E)$. This does not imply that the Born approximation applies to the calculation of $f(0,E)$ from the true particle interaction. By definition (see (126.4)),

$$\int U_{\text{eff}}\,dV = -\frac{2\pi\hbar^2}{m}f(0,E), \tag{142.14}$$

where m is the mass of the scattered particle. Since f is complex, so is the field thus defined. An estimate of both sides of (142.14) relates its range a to its magnitude:

$$a^3 U_{\text{eff}} \sim \hbar^2 f/m. \tag{142.15}$$

The definition (142.14) is, of course, not unique. Impose the additional condition that U_{eff} satisfy the applicability condition for perturbation theory:

$$|U_{\text{eff}}| \ll \hbar^2/ma^2 \tag{142.16}$$

(then $|f| \ll a$). The attenuation of the scattered beam can then be described as propagation of a plane wave through a homogeneous medium in which the particle has the constant potential energy

$$\overline{U}_{\text{eff}} = \frac{N}{V}\int U_{\text{eff}}\,dV = -\frac{N}{V}\frac{2\pi\hbar^2}{m}f(0,E), \tag{142.17}$$

obtained by averaging the effective fields of all N particles over the volume V of the medium. This is clear if one first considers scattering in a region of the medium which contains many scattering centers but in which the scattering effect is still small; such regions can be selected by virtue of (142.16). The attenuation on passing through this region is determined by the zero-angle amplitude, which in the Born approximation is in turn determined by the integral of the scattering field over the whole volume of the region. This means precisely that the scattering properties of interest are determined completely by the field (142.17) averaged over the volume.

The beam passing through the medium may therefore be described by a plane wave proportional to e^{ikz} with wave vector

$$k = \frac{1}{\hbar}\sqrt{2m(E-\overline{U}_{\text{eff}})}.$$

Introduce the incident-particle wave vector $k_0 = \sqrt{2mE}/\hbar$ and write $k = nk_0$. The quantity

$$n = \sqrt{1-\frac{\overline{U}_{\text{eff}}}{E}} = \sqrt{1+\frac{N}{V}\frac{2\pi\hbar^2}{mE}f(0,E)} \tag{142.18}$$

acts as the *refractive index* of the medium for the beam passing through it. It is generally complex, since the amplitude is complex, and its imaginary part determines the attenuation of the beam intensity. If $E \gg |\overline{U}_{\text{eff}}|$, then (142.18) gives, as it should,

$$\operatorname{Im} n = \frac{N}{V}\frac{\pi\hbar^2}{mE}\operatorname{Im} f(0,E) = \frac{N}{V}\frac{\sigma_t}{2k},$$

where σ_t is the total scattering cross-section; the optical theorem (142.10) has been used. This expression gives the evident attenuation law

$$|e^{ikz}|^2 \sim \exp\left(-\frac{N}{V}\sigma_t z\right).$$

Besides absorption, the complex refractive index (142.18), through its real part, determines the refraction law at entry into and exit from the scattering medium[3].

Problem

Neutrons are scattered by a heavy nucleus, and their wavelength is small compared with the nuclear radius a ($ka \gg 1$). Assume that every neutron incident with orbital angular momentum $l < ka \equiv l_0$, that is, with impact parameter $\rho = \hbar l/mv = l/k < a$, is absorbed by the nucleus, while for $l > l_0$ it does not interact with the nucleus at all. Determine the elastic scattering cross-section at small angles.

Solution. Under these conditions the neutron motion is mainly semiclassical, while the elastic scattering results from a small deflection entirely analogous to Fraunhofer diffraction of light by a black sphere. The required cross-section can therefore be written directly from the familiar solution of the diffraction problem[4]:

$$d\sigma_e = \pi a^2 \frac{J_1^2(ka\theta)}{\pi\theta^2}\, d\Omega.$$

The same result follows from (142.3). By assumption, $S_l = 0$ for $l < l_0$ and $S_l = 1$ for $l > l_0$. Hence the elastic scattering amplitude is

$$f(\theta) = -\frac{1}{2ik}\sum_{l=0}^{l_0}(2l+1)P_l(\cos\theta).$$

The main contribution comes from large l. Accordingly, replace $2l+1$ by $2l$, use the approximation (49.6) for $P_l(\cos\theta)$ at small θ, and replace summation by integration:

$$f(\theta) = \frac{i}{k}\int_0^{l_0} lJ_0(\theta l)\, dl = \frac{i}{k\theta} l_0 J_1(\theta l_0) = \frac{ia}{\theta} J_1(ka\theta),$$

[3] An interesting application of (142.17) is the shift of the higher levels of an alkali-metal atom immersed in a foreign gas. In a highly excited state the valence electron is at a mean distance r from the atomic center large compared with the dimensions a of both the ionic core and the foreign neutral atoms. The latter atoms within a sphere of radius of order r act as scattering centers for the valence electron and shift its energy level by (142.17). Since the de Broglie wavelength of the excited valence electron is also large compared with a, the amplitude is $f(0, E) \approx -a$, where a is the scattering length (compare (132.9)). The result is a constant level shift $2\pi\hbar^2 a\nu/m$, where m is the electron mass and ν is the number density of the foreign gas (E. Fermi, 1934).

[4] See II, § 61, Problem 3. Diffraction by a black sphere is equivalent to diffraction by a circular aperture in an opaque screen. The scattering cross-section is obtained by dividing the intensity of the diffracted waves by the incident current density.

as required[5]. The total elastic scattering cross-section is

$$\sigma_e = \pi a^2 \int_0^\infty \frac{J_1^2(ka\theta)}{\pi\theta^2} 2\pi\theta \, d\theta = \pi a^2$$

(the upper integration limit may be extended to infinity because the integral converges rapidly). As required under these conditions (compare (142.8)), it equals the absorption cross-section, which is simply the geometric cross-sectional area of the sphere. The total cross-section is $\sigma_t = 2\pi a^2$.

§ 143. Inelastic scattering of slow particles

The derivation in § 132 of the low-energy limiting law for elastic scattering extends readily to situations in which inelastic processes are also present.

As before, scattering with $l = 0$ gives the principal contribution at low energies. Recall from the results of § 132 that the corresponding element of the S-matrix was

$$S_0 = e^{2i\delta_0} \approx 1 + 2i\delta_0 = 1 - 2ik\alpha.$$

The properties of the wave function used in § 132 change in only one respect. Its boundary condition at infinity, the asymptotic form (142.1), is now complex rather than the real standing wave appropriate to purely elastic scattering. Consequently the constant $\alpha = -c_2/c_1$ is complex as well. The modulus $|S_0|$ is no longer unity; the condition $|S_0| < 1$ requires the imaginary part in $\alpha = \alpha' + i\alpha''$ to be negative, $\alpha'' < 0$.

Substitution of S_0 into (142.7) gives the elastic and inelastic scattering cross sections

$$\sigma_e = 4\pi|\alpha|^2, \tag{143.1}$$

$$\sigma_r = (4\pi/k)|\alpha''|. \tag{143.2}$$

Thus the elastic-scattering cross section remains independent of velocity. The cross section for inelastic processes, by contrast, is inversely proportional to the particle velocity—the so-called $1/v$ *law* (H. A. Bethe, 1935). Hence, as the velocity decreases, inelastic processes become increasingly important relative to elastic scattering[6].

The limiting laws (143.1) and (143.2) are, of course, only the leading terms in expansions of the cross sections in powers of k. Remarkably, the next term in each expansion introduces no new constants beyond the quantities already present in (143.1) and (143.2) (F. L. Shapiro, 1958). This follows from the evenness of the function $g_0(k^2)$ in the expression (142.13), $f_0(k) = 1/[g_0(k^2) - ik]$, for the partial scattering amplitude with $l = 0$. At small k, this function therefore has an expansion in even powers of k, so

[5] The diffraction scattering of fast charged particles by a "black" nucleus may be treated similarly. The boundary value l_0 must then be found by requiring the shortest distance between the nucleus and a particle moving on a classical trajectory in the Coulomb field to equal the nuclear radius. One must again put $S_l = 0$ for $l < l_0$, while for $l > l_0$, $S_l = e^{2i\delta_l}$, where δ_l are the Coulomb phases from (135.11). See A. I. Akhiezer and I. Ya. Pomeranchuk, *Some Problems of Nuclear Theory*, Moscow: Gostekhizdat, 1950, § 22; J. Physics USSR, 1945, Vol. 9, p. 471.

[6] The velocity dependence of partial reaction cross sections for nonzero orbital angular momenta l can be found in the same way. They are proportional to $\sigma_r^{(l)} \sim k^{2l-1}$. The elastic-scattering cross sections $\sigma_e^{(l)}$ remain proportional to k^{4l} and thus, for the same l, decrease as $k \to 0$ more rapidly than $\sigma_r^{(l)}$.

the term following $g_0 \approx -1/\alpha$ is of order k^2. Even when that term is neglected, two terms may still be retained in the expansion of $f_0(k)$: $f_0(k) \approx -\alpha(1 - ik\alpha)$. Correspondingly, the next terms can also be kept in the cross sections, giving

$$\sigma_e = 4\pi|\alpha|^2(1 - 2k|\alpha''|), \tag{143.3}$$

$$\sigma_r = (4\pi|\alpha''|/k)(1 - 2k|\alpha''|). \tag{143.4}$$

These results assume that the interaction falls off sufficiently rapidly at large distances. We saw in § 132 that the elastic-scattering amplitude approaches a constant limit as $k \to 0$ when the field $U(r)$ decreases faster than r^{-3}. The same condition is required for the corresponding law (143.1) to hold when inelastic channels are present[7]. The $1/v$ law for the reaction cross section, however, requires only the weaker condition that the field fall off faster than r^{-2}, as is clear from the following illustrative justification of this law. The probability that a reaction occurs in a collision is proportional to the squared modulus of the incident particle's wave function in the "reaction zone" (the region $r \sim a$). In physical terms, this statement means that, for example, a slow neutron colliding with a nucleus can produce a reaction only after it has "penetrated" into the nucleus. If the interaction falls off more rapidly than r^{-2}, then, in passing from large r down to $r \sim a$, it leaves the order of magnitude of the wavefunction

unchanged; equivalently, as $k \to 0$, the ratio $|\psi(a)/\psi(\infty)|^2$ approaches a finite limit (this follows because the term $U\psi$ in the Schrödinger equation is small relative to $\Delta\psi$). The reaction cross section is obtained by dividing $|\psi|^2$ by the flux density. Taking ψ to be a plane wave normalized to unit flux density gives $|\psi|^2 \sim 1/v$, which is the required result. When charged nuclear particles collide, a slowly decaying Coulomb field is present alongside the short-range nuclear forces. This field can substantially alter the magnitude of the incident wave in the reaction zone. To obtain the reaction cross section, one multiplies $1/v$ by the ratio of the squared moduli of the Coulomb and free wavefunctions (as $r \to 0$). This ratio follows from formulas (136.10) and (136.11). Hence, in Coulomb units, we find

$$\sigma_r = \frac{2\pi A}{k^2|e^{\pm 2\pi/k} - 1|}; \tag{143.5}$$

The plus sign in the exponent corresponds to repulsion, whereas the minus sign corresponds to attraction. The coefficient A is the constant in the $1/v$ law; when the speed is large compared with the Coulomb unit ($k \gg 1$), the Coulomb interaction is immaterial, and the law $\sigma_r = A/k$ is recovered. If, however, the velocity is small relative to the Coulomb unit ($k \ll 1$, i.e. in ordinary units $Z_1 Z_2 e^2/\hbar v \gg 1$, where $Z_1 e$ and $Z_2 e$ are the charges of the colliding particles), the Coulomb interaction has the dominant role in setting the magnitude of the wave function in the reaction region. For a collision of attracting particles, we then have

$$\sigma_r = 2\pi A/k^2, \tag{143.6}$$

whereas for a collision of particles that repel one another

[7]Formula (143.3), which includes the next term in the expansion in powers of k, requires U to decrease faster than r^{-4}.

$$\sigma_r = (2\pi A/k^2)e^{-2\pi/k}. \tag{143.7}$$

In the latter case, the cross section tends to zero as $k \to 0$. The exponential factor that distinguishes (143.7) from (143.6) is the probability of penetrating the Coulomb potential barrier; in ordinary units, it has the form $\exp(-2\pi Z_1 Z_2 e^2/\hbar v)$. It should be noted that the limiting law (143.6) governs not only the total cross section, but also every partial cross section of angular momentum l[8]. The reason is that every term in the expansion (136.1) of the functions $\psi_{\mathbf{k}}^{(+)}$ (which occur in the formulas used above

(136.10), (136.11)) contains an R_{kl} with the same limiting dependence on k. Indeed, in the limit $k \to 0$, the radial functions (for attraction) are given by (36.25), and near the center we have $R_{kl} \sim \sqrt{k}r^l$. The contribution from each individual l to the square of the wave function in the reaction region is $\sim a^{2l}/k$, so all such contributions have the same dependence on k, although each is suppressed by the small factor $(a/a_c)^{2l}$ ($a_c = \hbar^2/(mZ_1Z_2e^2)$ — the Coulomb unit of length).

§ 144. The scattering matrix in the presence of reactions

The cross-section σ_r considered in § 142, 143 was the total cross-section of all possible inelastic scattering channels. We shall now show how to construct the general theory of inelastic collisions in which each channel can be considered separately.

Suppose that a collision of two particles again produces two particles, whether the same or different. Number all reaction channels possible at the specified energy and attach the corresponding indices to the quantities belonging to them.

Let channel i be the entrance channel. In this channel the wave function of the relative motion in the center-of-mass system is the familiar sum of an incident plane wave and an elastically scattered outgoing wave:

$$\psi_i = \exp(ik_i z) + f_{ii}(\theta)\frac{\exp(ik_i r)}{r}. \tag{144.1}$$

The squared amplitude f_{ii} gives the elastic scattering cross-section in channel i:

$$d\sigma_{ii} = |f_{ii}|^2 d\Omega. \tag{144.2}$$

In the other channels, indexed by f, the wave functions of relative motion are outgoing waves. For a reason that will become clear below, it is convenient to write them as[9]

$$\psi_f = f_{fi}(\theta)\sqrt{m_f/m_i}\,\frac{\exp(ik_f r)}{r}, \tag{144.3}$$

where $\mathbf{k}_f$ is the wave vector of relative motion of the reaction products in channel f, θ is its angle with the z axis, and m_i, m_f are the reduced masses of the two initial and two final particles. The scattered current in the solid-angle element $d\Omega_f$ is obtained by

[8]The same statement applies to law (143.7).

[9]As before (compare the note on p. 187), the initial state is denoted by i and the final state by f. In the scattering amplitude, the final-state index is placed to the left of the initial-state index, in accordance with the order of matrix-element indices. For uniformity, the indices in cross-section symbols will be ordered in the same way.

multiplying $|\psi_f|^2$ by $v_f r^2 d\Omega_f$, and the cross-section of the corresponding reaction by dividing this current by the incident current density v_i:

$$d\sigma_{fi} = |f_{fi}|^2 \frac{p_f}{p_i} d\Omega_f, \tag{144.4}$$

where $p_i = m_i v_i$, $p_f = m_f v_f$.

In § 125 the scattering operator $\widehat{S}$ was introduced as the operator that transforms an incoming wave into an outgoing wave. With several channels, this operator has matrix elements for transitions between different channels. Elements diagonal in the channel indices correspond to elastic scattering, and off-diagonal elements to the various inelastic processes; all these elements remain operators in the other variables. They are defined as follows.

As in § 125, introduce operators $\widehat{f}_{ii}$, $\widehat{f}_{fi}$ associated with the amplitudes f_{ii}, f_{fi} by

$$\widehat{S}_{fi} = \delta_{fi} + 2i\sqrt{k_i k_f}\, \widehat{f}_{fi}. \tag{144.5}$$

This definition indeed gives an S-matrix that must satisfy unitarity. Write the entrance-channel wave function as a set of incoming and outgoing waves, as in § 125:

$$\begin{aligned} \psi_i &= F(-\mathbf{n}')\frac{\exp(-ik_i r)}{r\sqrt{v_i}} - (1 + 2ik_i\widehat{f}_{ii})F(\mathbf{n}')\frac{\exp(ik_i r)}{r\sqrt{v_i}} \\ &= F(-\mathbf{n}')\frac{\exp(-ik_i r)}{r\sqrt{v_i}} - \widehat{S}_{ii}F(\mathbf{n}')\frac{\exp(ik_i r)}{r\sqrt{v_i}}. \end{aligned} \tag{144.6}$$

For convenience, an extra factor $v_i^{-1/2}$ has been introduced here relative to (125.3). With the adopted amplitude notation, the wave function in channel f is

$$\psi_f = 2ik_i\sqrt{m_f/m_i}\, \widehat{f}_{fi}F(\mathbf{n}')\frac{\exp\big(ik_f r\big)}{r\sqrt{v_i}} = \widehat{S}_{fi}F(\mathbf{n}')\frac{\exp\big(ik_f r\big)}{r\sqrt{v_f}}. \tag{144.7}$$

The incoming-wave current must equal the sum of the outgoing-wave currents in all channels. This is the evident requirement that the sum of the probabilities of all

possible elastic and inelastic processes in a collision be unity. Because of the factors $\sqrt{v}$ in the spherical-wave denominators, the velocity drops out of their current densities. The condition therefore simply requires the normalization of the incoming wave to equal that of the complete set of outgoing waves. It remains the unitarity condition for the scattering operator, now understood also as a matrix in the channel indices. For the operators $\widehat{f}_{fi}$ this condition is

$$\widehat{f}_{fi} - \widehat{f}^{+}_{if} = 2i\sum_n k_n \widehat{f}_{fn}\widehat{f}^{+}_{in}, \tag{144.8}$$

analogous to (125.7). The index + denotes complex conjugation and transposition in every matrix index other than the channel number.

The S-matrix is diagonal in states of specified orbital angular momentum l; distinguish the corresponding matrix elements by a superscript (l). Applying $\widehat{f}_{ii}$ and $\widehat{f}_{fi}$ to (125.17) gives the elastic and inelastic amplitudes

$$f_{ii} = \frac{1}{2ik_i}\sum_{l=0}^{\infty}(2l+1)(S_{ii}^{(l)}-1)P_l(\cos\theta),$$
$$f_{fi} = \frac{1}{2i\sqrt{k_i k_f}}\sum_{l=0}^{\infty}(2l+1)S_{fi}^{(l)}P_l(\cos\theta). \tag{144.9}$$

The corresponding integral cross-sections are

$$\sigma_{ii} = \frac{\pi}{k_i^2}\sum_{l=0}^{\infty}(2l+1)|1-S_{ii}^{(l)}|^2, \qquad \sigma_{fi} = \frac{\pi}{k_i^2}\sum_{l=0}^{\infty}(2l+1)|S_{fi}^{(l)}|^2. \tag{144.10}$$

The first agrees with (142.3). The total reaction cross-section σ_r from entrance channel i is $\sigma_r = \sum_f' \sigma_{fi}$ over all $f \neq i$. By unitarity, $\sum_f' |S_{fi}|^2 = 1 - |S_{ii}|^2$, and (142.4) for σ_r is recovered.

Time-reversal symmetry of the scattering process, the reciprocity theorem, is expressed by

$$\widehat{S}_{fi} = \widehat{S}_{i^*f^*}, \tag{144.11}$$

or equivalently

$$\widehat{f}_{fi} = \widehat{f}_{i^*f^*}. \tag{144.12}$$

Here i^* and f^* denote states obtained from i and f by reversing the particle momenta and spin projections[10]; they are called the *time-reversed* states of i and f. Equations (144.11), (144.12) generalize (125.11), (125.12) for elastic scattering[11].

Equation (144.12) gives the following relation between reaction cross-sections:

$$\frac{d\sigma_{fi}}{p_f^2 d\Omega_f} = \frac{d\sigma_{i^*f^*}}{p_i^2 d\Omega_{i^*}}. \tag{144.13}$$

This expresses the *principle of detailed balance*.

As noted in § 126, when perturbation theory applies, its first approximation provides, besides the reciprocity theorem, an additional relation between the amplitudes of the direct and reverse processes in the literal sense, $i \to f$ and $f \to i$. This property, $f_{fi} = f_{if}^*$, holds in the same approximation for inelastic processes as well. The cross-sections then obey

$$\frac{d\sigma_{fi}}{p_f^2 d\Omega_f} = \frac{d\sigma_{if}}{p_i^2 d\Omega_i}. \tag{144.14}$$

The distinction between $i \to f$ and $i^* \to f^*$ disappears for integral cross-sections integrated over all directions of $\mathbf{p}_f$, summed over the spin directions of the final particles

[10] For composite particles such as atoms or atomic nuclei, "spin" here means the total intrinsic angular momentum, formed from both the spins and the orbital angular momenta of the internal motions of the constituents.

[11] We disregard here a factor -1 that can arise in collisions of particles with spin (compare (140.11)). This does not, of course, affect the cross-section relation (144.13).

s_{1f}, s_{2f}, and averaged over the direction of $\mathbf{p}_i$ and the spins s_{1i}, s_{2i} of the initial particles. Denote this cross-section by $\overline{\sigma}_{fi}$:

$$\overline{\sigma}_{fi} = \frac{1}{4\pi(2s_{1i}+1)(2s_{2i}+1)} \sum_{(m_s)} \int d\sigma_{fi}\, d\Omega_i.$$

The sum is over the spin projections of all particles. The factor before the sum and integral occurs because quantities belonging to the initial particles are averaged rather than summed. Write (144.13) as

$$p_i^2 d\sigma_{fi}\, d\Omega_{i^*} = p_f^2 d\sigma_{i^*f^*}\, d\Omega_f$$

and perform the stated operations. The required relation is

$$g_i p_i^2 \overline{\sigma}_{fi} = g_f p_f^2 \overline{\sigma}_{if}; \tag{144.15}$$

where

$$g_i = (2s_{1i}+1)(2s_{2i}+1), \qquad g_f = (2s_{1f}+1)(2s_{2f}+1), \tag{144.16}$$

are the numbers of possible spin orientations of the initial and final particle pairs. These numbers are called the *statistical weights* of states i and f.

Finally, note the following property of f_{fi}. We saw in the preceding section that, as $p_i \to 0$, the reaction cross-section varies as $\sigma_{fi} \sim 1/p_i$ when the interaction decreases sufficiently rapidly at large distances. According to (144.4), this means that $f_{fi} \to$ const as $p_i \to 0$. By the symmetry (144.12), f_{fi} also tends to a constant as $p_f \to 0$. We shall return to this property in § 147.

§ 145. The Breit–Wigner formulas

In § 134 quasistationary states were introduced as states having a finite but comparatively long lifetime. A broad class of such states occurs in nuclear reactions at moderate energies which proceed through the formation of a *compound nucleus*[12].

The physical picture is that the incident particle interacts with the nucleons and "merges" with the nucleus, forming a compound system in which its energy is distributed among many nucleons. Resonance energies correspond to quasidiscrete levels of this system. Their lifetimes are long compared with nuclear periods because, most of the time, the energy is shared among so many particles that none can overcome the attraction of the others and leave the nucleus. Only rarely is enough energy concentrated on one particle. The compound nucleus can decay in various ways, corresponding to the possible reaction channels[13].

This character of the collision shows that inelastic processes do not affect the potential part of the elastic amplitude, which is unrelated to compound- nucleus properties (see

[12] The concept of the compound nucleus was introduced by N. Bohr (1936).

[13] The competing processes also include radiative capture of the incident particle, in which the excited compound nucleus emits a γ quantum and passes to its ground state. This process too is "slow," because a radiative transition has comparatively low probability.

§ 134); they change only its resonance part. For the same reason, inelastic amplitudes involving a compound nucleus are purely resonant. The resonance denominator retains the form $E - E_0 + i\Gamma/2$, associated with the vanishing of the incoming-wave coefficient at $E = E_0 - i\Gamma/2$, and Γ still determines the total decay probability of the quasistationary state by any route.

These considerations, together with unitarity, determine the amplitudes. Number all decay channels symmetrically, without first choosing the entrance channel, using indices $a, b, c, \ldots$. Consider partial amplitudes for the angular momentum l of the quasistationary state[14]. Seek them in the form

$$f_{ab}^{(l)} = \frac{1}{2ik_a}(e^{2i\delta_a} - 1)\delta_{ab} - \frac{1}{2\sqrt{k_a k_b}} e^{i(\delta_a+\delta_b)} \frac{\Gamma M_{ab}}{E - E_0 + (1/2)i\Gamma}. \tag{145.1}$$

The index l on δ_a and M_{ab} is suppressed. The first term occurs only for $a = b$ and is the potential elastic amplitude in channel a; the δ_a are the phases $\delta_l^{(0)}$ of (134.12). The second term is resonant. Its coefficient is written in this form to simplify unitarity.

Since the fixed magnitude of orbital angular momentum is unchanged by time reversal, reciprocity here simply requires $f_{ab}^{(l)}$ to be symmetric in a, b. Thus $M_{ab} = M_{ba}$.

The unitarity conditions are

$$\operatorname{Im} f_{ab}^{(l)} = \sum_c k_c f_{ac}^{(l)} f_{bc}^{(l)*} \tag{145.2}$$

(compare (144.8)). Substitution of (145.1) gives

$$\frac{M_{ab}^*}{E - E_0 - (1/2)i\Gamma} - \frac{M_{ab}}{E - E_0 + (1/2)i\Gamma} = \frac{i\Gamma \sum_c M_{ac} M_{bc}^*}{(E - E_0)^2 + (1/4)\Gamma^2}.$$

For this to hold identically in E, first $M_{ab} = M_{ab}^*$, so the M_{ab} are real, and then

$$M_{ab} = \sum_c M_{ac} M_{bc}. \tag{145.3}$$

Thus the matrix M equals its square.

A real symmetric M can be diagonalized by an orthogonal transformation U. Denoting its eigenvalues by $M^{(\alpha)}$, write

$$\sum_{a,b} U_{\alpha a} U_{\beta b} M_{ab} = M^{(\alpha)} \delta_{\alpha\beta},$$

with

$$\sum_c U_{\alpha c} U_{\beta c} = \delta_{\alpha\beta}. \tag{145.4}$$

Conversely,

$$M_{ab} = \sum_\alpha U_{\alpha a} U_{\alpha b} M^{(\alpha)}. \tag{145.5}$$

[14] We first disregard the complicating influence of particle spins.

Equation (145.3) gives $M^{(\alpha)} = (M^{(\alpha)})^2$, so every eigenvalue is 0 or 1. If only $M^{(1)} = 1$ is nonzero, then

$$M_{ab} = U_{1a}U_{1b}. \tag{145.6}$$

If several eigenvalues are nonzero, M_{ab} is a sum of terms involving independent sets $U_{1a}, U_{2a}, \ldots$, constrained only by orthogonality. This would describe accidental degeneracy, with several compound-nucleus quasistationary states at the same quasidiscrete energy[15]. Excluding these uninteresting degenerate levels, the elements of M factor into quantities depending on one channel each.

Define

$$|U_{1a}| = \sqrt{\Gamma_a/\Gamma}.$$

Then (145.6) becomes

$$M_{ab} = \pm\sqrt{\Gamma_a\Gamma_b}/\Gamma, \tag{145.7}$$

where the sign depends on those of U_{1a} and U_{1b}. Since $\sum_c U_{1c}U_{1c} = 1$,

$$\sum_a \Gamma_a = \Gamma. \tag{145.8}$$

The Γ_a are the *partial widths* of the channels. Equations (145.1), (145.7), (145.8) give the required general amplitudes.

Now select one entrance channel[16]. Denote its partial width by Γ_e, the *elastic width*, and the widths of the reactions by $\Gamma_{r_1}, \Gamma_{r_2}, \ldots$. The total elastic amplitude is

$$f_e(\theta) = f^{(0)}(\theta) - \frac{2l+1}{2k}\frac{\Gamma_e}{E - E_0 + (1/2)i\Gamma}e^{2i\delta_l^{(0)}}P_l(\cos\theta), \tag{145.9}$$

where $\mathbf{k}$ is the incident wave vector and $f^{(0)}$ is the potential scattering amplitude. This differs from (134.12) by replacing Γ in the resonance numerator with the smaller Γ_e.

The inelastic amplitudes are purely resonant. Their differential and integral cross-sections are

$$d\sigma_{r_a} = \frac{(2l+1)^2}{4k^2}\frac{\Gamma_e\Gamma_{r_a}}{(E-E_0)^2 + (1/4)\Gamma^2}[P_l(\cos\theta)]^2 d\Omega, \tag{145.10}$$

$$\sigma_{r_a} = (2l+1)\frac{\pi}{k^2}\frac{\Gamma_e\Gamma_{r_a}}{(E-E_0)^2 + (1/4)\Gamma^2}. \tag{145.11}$$

The total inelastic cross-section is

$$\sigma_r = (2l+1)\frac{\pi}{k^2}\frac{\Gamma_e\Gamma_r}{(E-E_0)^2 + (1/4)\Gamma^2}, \tag{145.12}$$

where $\Gamma_r = \Gamma - \Gamma_e$ is the total inelastic width.

[15]This is especially clear if every $M^{(\alpha)} = 1$. Equations (145.4), (145.5) then give $M_{ab} = \delta_{ab}$, so there are no transitions between different channels. The case would represent several independent quasidiscrete states, each realized in elastic scattering in one channel.

[16]These formulas were first obtained by *Breit and Wigner* (O. Breit, E. Wigner, 1936).

Integrating over the resonance region, and extending $E - E_0$ from $-\infty$ to ∞ because σ_r falls rapidly, gives

$$\int \sigma_r \, dE = (2l+1)\frac{2\pi^2}{k^2}\frac{\Gamma_e \Gamma_r}{\Gamma}. \tag{145.13}$$

For slow neutrons only s-wave scattering matters, and the potential amplitude is the real constant $-\alpha$. Instead of (134.14),

$$f_e = -\alpha - \frac{\Gamma_e}{2k(E - E_0 + (1/2)i\Gamma)}. \tag{145.14}$$

The total elastic cross-section is

$$\sigma_e = 4\pi\alpha^2 + \frac{\pi}{k^2}\frac{\Gamma_e^2 + 4\alpha k \Gamma_e (E - E_0)}{(E - E_0)^2 + (1/4)\Gamma^2}. \tag{145.15}$$

The first term is the potential-scattering cross-section. Potential and resonance scattering interfere. Only very near the level, $E - E_0 \sim \Gamma$, may α be negligible (recall $|\alpha k| \ll 1$), giving

$$\sigma_e = \frac{\pi}{k^2}\frac{\Gamma_e^2}{(E - E_0)^2 + (1/4)\Gamma^2}. \tag{145.16}$$

The total elastic plus inelastic cross-section is then

$$\sigma_t = \sigma_e + \sigma_r = \frac{\pi}{k^2}\frac{\Gamma_e \Gamma}{(E - E_0)^2 + (1/4)\Gamma^2}. \tag{145.17}$$

When potential scattering is negligible,

$$\sigma_e = \sigma_t \frac{\Gamma_e}{\Gamma}, \qquad \sigma_{r_a} = \sigma_t \frac{\Gamma_{r_a}}{\Gamma}.$$

Here σ_t, the sum over every resonant process, is the cross-section for forming the compound nucleus. Each elastic or inelastic cross-section is σ_t times the relative probability for the corresponding decay, given by the partial width divided by the total width. This factorization of M_{ab} matches the two-stage physical picture: formation of a compound nucleus in a quasistationary state, followed by decay through one channel[17].

As noted in § 134, these formulas require only $|E - E_0| \ll D$, where D is the spacing of neighboring compound-nucleus levels with the same angular momentum. In this form, however, they do not permit the limit $E \to 0$ when zero lies in the resonance region. One must then replace E_0 by a related constant ε_0, and Γ_e by $\gamma_e\sqrt{E}$, while keeping Γ_r constant (H. A. Bethe, G. Placzek, 1937)[18].

The resulting inelastic cross-section (145.12) grows as $1/\sqrt{E}$ when $E \to 0$, in agreement with § 143.

[17]The calculations above referred to reactions $a + X = b + Y$, with two particles both before and after. This assumption is not fundamental, as is clear from the physical result. Formulas such as (145.11) for integral cross-sections also apply when more than one particle is emitted.

[18]For inelastic processes possible at low energies, such as radiative capture, $E = 0$ is not a threshold. A similar replacement for Γ_{r_a} would be required near the threshold of that reaction, below which it is impossible.

Spin generally gives cumbersome formulas. Consider only slow-neutron scattering, where $l = 0$. Coupling the target spin i to the neutron spin $s = 1/2$ gives compound-nucleus spin $j = i \pm 1/2$; assume $i \neq 0$. Each quasidiscrete level has one definite j. The reaction cross-section is therefore (145.12), with $l = 0$, multiplied by the probability $g(j)$ that the nucleus–neutron system has the required j.

Suppose neutron and target spins are randomly oriented. There are $2(2i + 1)$ orientations of the pair, and $2j + 1$ correspond to a specified j. Hence

$$g(j) = \frac{2j+1}{2(2i+1)}. \tag{145.18}$$

The elastic formula changes similarly. Both j values contribute to potential scattering, so multiply the second term of (145.15) by the $g(j)$ of the resonant level, and replace $4\pi\alpha^2$ by $\sum_j g(j)\, 4\pi\alpha^{(j)2}$.

Compound-nucleus formation also implies general properties of the angular distribution of reaction products. Each quasistationary state has a definite parity, which is inherited by the final system $(b + Y)$. Under inversion its wave function and reaction amplitude can only acquire a factor ± 1, so the cross-section is unchanged. In the center-of-mass system inversion sends

$\theta \to \pi - \theta$, $\varphi \to \pi + \varphi$. Thus the distribution is invariant under this replacement. After averaging over all spin directions, it depends only on θ and is symmetric under $\theta \to \pi - \theta$[19].

The great number of densely spaced compound-nucleus levels makes the detailed energy dependence of scattering cross-sections extremely complicated and obscures systematic changes between nuclei. It is therefore useful to average over energy intervals large compared with level spacings, suppressing the resonance detail. We also cease to distinguish types of inelastic process and divide scattering, in the sense described below, only into "elastic" and "inelastic" parts[20].

To explain the averaging, again disregard spin and consider $l = 0$. From (142.7),

$$\sigma_e = \frac{\pi}{k^2}|S-1|^2, \qquad \sigma_r = \frac{\pi}{k^2}(1-|S|^2), \qquad \sigma_t = \frac{\pi}{k^2}\,2(1-\operatorname{Re} S), \tag{145.19}$$

where the superscript (0) is suppressed. Since σ_t is linear in S, its energy average is

$$\overline{\sigma}_t = \frac{\pi}{k^2}\,2(1-\operatorname{Re}\overline{S}), \tag{145.20}$$

leaving the slowly varying k^{-2} outside the average. Define the "elastic" cross-section in the averaged picture as

$$\overline{\sigma}_e^{\text{opt}} = (\pi/k^2)|\overline{S}-1|^2, \tag{145.21}$$

which generally differs from $\overline{\sigma_e}$. Thus the outgoing-wave amplitude Se^{ikr}/r is averaged before defining elastic scattering. This preserves the form of an elastically scattered

[19]For spinless particles the differential reaction cross-section is simply proportional to $[P_l(\cos\theta)]^2$, making the symmetry evident.

[20]The averaging method leading to the optical model of nuclear scattering was proposed by *Weisskopf, Porter, and Feshbach* (V. F. Weisskopf, C. E. Porter, H. Feshbach, 1954).

wave packet and describes the "coherent" part. Elastic scattering through the long-lived compound nucleus is excluded, since its formation and decay erase the incident packet's specific character. Define the averaged "inelastic" cross-section by $\overline{\sigma}_a^{\mathrm{opt}} = \overline{\sigma}_t - \overline{\sigma}_e^{\mathrm{opt}}$:

$$\overline{\sigma}_a^{\mathrm{opt}} = (\pi/k^2)(1 - |\overline{S}|^2). \tag{145.22}$$

It includes both truly inelastic processes and elastic scattering through an intermediate compound nucleus.

This interpretation gives the correct limiting pictures and provides a reasonable interpolation. For well-resolved low-energy resonances ($\Gamma \ll D$), near each level

$$S = \left(1 - \frac{i\Gamma_e}{E - E_0 + (1/2)i\Gamma}\right)\exp\left(2i\delta^{(0)}\right).$$

Averaging gives

$$\overline{S} = (1 - \pi\overline{\Gamma}_e/D)\exp\left(2i\delta^{(0)}\right), \tag{145.23}$$

where $\overline{\Gamma}_e$ and D are the mean elastic width and mean level spacing in the interval, and slowly varying $\delta^{(0)}(E)$ is held constant. Hence

$$\overline{\sigma}_a^{\mathrm{opt}} = (\pi/k^2)(2\pi\overline{\Gamma}_e/D), \tag{145.24}$$

neglecting terms of order Γ/D[21]. This equals the mean of (145.17), the compound-nucleus formation cross-section.

As excitation energy increases, level spacings decrease and decay probabilities, hence total widths, increase, until levels overlap and the notion of quasidiscrete levels largely loses meaning. Irregularities in $S(E)$ are smoothed, so the exact and averaged functions become close, and (145.22) coincides with σ_r in (145.19). At high energies, decay back through the entrance channel is negligible beside the many other decay modes; all compound-nucleus processes may then be regarded as inelastic.

Thus the averaged scattering is again determined by one quantity, $\overline{S}$, now a smooth function of energy. In the *optical model*, the nucleus is approximated by a force field with a complex potential. Its imaginary part produces absorption in addition to elastic scattering. This absorption, with cross-section (145.22), is identified with "inelastic" scattering in the averaged picture.

§ 146. Final-state interaction in reactions

Particles created in a reaction interact with one another, and this may substantially alter their distributions in energy and angle. Naturally, the effect should be especially pronounced when the interacting particles have a low relative speed. This situation arises, for example, in nuclear reactions that eject two or more nucleons; here the effect is due to the nuclear forces acting between the free nucleons[22].

[21] Terms from the influence of other levels near a given level would be of the same order.

[22] The results given below were obtained by A. B. Migdal (1950) and, independently, by Watson (K. M. Watson, 1952).

Denote by $\mathbf{p}_0$ the momentum of the center of mass of the pair of emitted nucleons, and by $\mathbf{p}$ the momentum of their relative motion. Suppose that $p \ll p_0$, so that the relative energy $E = p^2/m$

(m is the nucleon mass) is small in comparison with the energy of the center-of-mass motion, $E_0 = p_0^2/4m$. At the same time, suppose that E_0 is large in comparison with the energy ε of a level (real or virtual) belonging to the two-nucleon system. In other words, only the relative motion of the nucleons is assumed to be "slow"; the nucleons themselves are "fast."

When the particles formed in the reaction are inside the "reaction region", the reaction probability is proportional to the squared modulus of their wave function. This region corresponds to separations of the order of the nuclear-force range a (compare the analogous argument in § 143 for the initial particles). Here we require only the dependence of the reaction probability on the characteristics of the relative motion of one nucleon pair. It is therefore sufficient to consider the wave function $\psi_{\mathbf{p}}(\mathbf{r})$ of this motion alone. Thus, the probability of producing a nucleon pair with relative momentum in the interval d^3p is

$$dw_{\mathbf{p}} = \text{const} \cdot |\psi_{\mathbf{p}}(a)|^2 d^3p. \tag{146.1}$$

As shown in § 136, to determine the probability that a system undergoing scattering makes a transition to a state with a specified direction of motion, one must use the functions $\psi_{\mathbf{p}}^{(-)}$ as the final-state wave functions; at infinity these contain, besides a plane wave, only an incoming wave, and they must be normalized to a delta-function in momentum. Furthermore, the functions $\psi_{\mathbf{p}}^{(-)}$ are obtained directly from the functions $\psi_{\mathbf{p}}^{(+)}$ by complex conjugation and reversal of the sign of $\mathbf{p}$; the latter contain outgoing spherical waves at infinity, that is, they correspond to the problem of two particles scattering from one another. On substitution into (146.1), this distinction is in general immaterial, so that in (146.1) $\psi_{\mathbf{p}}$ may be understood as the function $\psi_{\mathbf{p}}^{(+)}$; the problem is thereby reduced to the previously considered problem of resonant scattering of slow particles.

Although the actual form of the function $\psi_{\mathbf{p}}$ in the region $r \sim a$ is unknown, determining how the probability depends on the energy E requires only that this function be considered at distances $r \gtrsim 1/k \gg a$ (where $\mathbf{k} = \mathbf{p}/\hbar$; it is assumed that $ak \ll 1$) and then continued, as an order-of-magnitude estimate, to distances $r \sim a$[23]. In this procedure, The spherical wave (which contains the factor $1/r$) makes the dominant contribution to $\psi_{\mathbf{p}}$. This wave is a collection of partial waves with different values of l, whose amplitudes are the corresponding scattering amplitudes. To determine $|\psi_{\mathbf{p}}(a)|^2$, it is enough here to retain only the s wave, since at low energies the scattering amplitudes with $l \neq 0$ are relatively small. Thus, according to formula (133.7), we have

$$\psi_{\mathbf{p}} \sim \frac{1}{\kappa + ik} \frac{e^{ikr}}{r}, \tag{146.2}$$

[23]This procedure is permissible because, in the region $r \ll 1/k$, the energy E may be neglected in the Schrödinger equation that determines the function $\psi_{\mathbf{p}}$. Consequently, the dependence of ψ on E in this region is determined entirely by matching it to the function in the region $r \sim 1/k$.

where $\kappa = \sqrt{m|\varepsilon|}/\hbar$, while ε is the energy of a bound (or virtual) state of the two-nucleon system[24]. Substituting this expression into (146.1), we obtain

$$dw_{\mathbf{p}} = \text{const} \cdot \frac{d^3 p}{E + |\varepsilon|}. \tag{146.3}$$

Thus, the distribution over momentum directions (in the center-of-mass frame of the two nucleons) is isotropic.

The relative-motion energy distribution, on the other hand, is given by

$$dw_E = \text{const} \cdot \frac{\sqrt{E}\, dE}{E + |\varepsilon|}. \tag{146.4}$$

We thus see that the nucleon interaction makes the distribution develop a maximum at low E (for $E \sim |\varepsilon|$)[25].

In the laboratory frame, small values of the relative momentum ($p \ll p_0$) correspond to small angles θ between the momenta of the two nucleons. The maximum in the distribution over E therefore corresponds, in this frame, to an angular correlation between the nucleons' directions of emission, manifested by an enhanced probability of small values of θ.

Let $\mathbf{p}_1$ and $\mathbf{p}_2$ be the momenta of the nucleons in the laboratory frame. Then

$$\mathbf{p}_0 = \mathbf{p}_1 + \mathbf{p}_2, \qquad \mathbf{p} = \frac{1}{2}(\mathbf{p}_2 - \mathbf{p}_1)$$

(recall that the reduced mass of two identical particles is $m/2$). Taking the vector product of these two equalities, we obtain $\mathbf{p}_0 \times \mathbf{p} = \mathbf{p}_1 \times \mathbf{p}_2$; for $p \ll p_0$, this gives

$$p_0 p_\perp = p_1 p_2 \sin\theta \approx \frac{p_0^2}{4}\theta,$$

or $\theta = 4p_\perp/p_0$, where $p_\perp$ is the component of the vector $\mathbf{p}$ transverse to the direction of $\mathbf{p}_0$, and θ is the small angle between the directions of $\mathbf{p}_1$ and $\mathbf{p}_2$. Rewriting formula (146.3) in the form

$$dw_{\mathbf{p}} = \text{const} \cdot \frac{2\pi p_\perp\, dp_\perp\, dp_\parallel}{(1/m)(p_\perp^2 + p_\parallel^2) + |\varepsilon|}$$

and, on integrating over $dp_\parallel$, we find the probability distribution with respect to the angle θ. Since the integral converges rapidly, its limits may be extended from $-\infty$ to ∞, and the final result is

[24]Here we mean an np pair with parallel or antiparallel spins, or an nn pair with antiparallel spins. For a pp pair, however, the situation is complicated by Coulomb repulsion; this case must be treated using the theory presented in § 138.

[25]Strictly speaking, the constant coefficients in formulas (146.3) and (146.4) may also depend on E, through the other parts of the wave function of the complete system of reaction products. This dependence is weak, however: as a function of E, such a coefficient changes appreciably only over the full range of energies ($\sim E_0$) that a nucleon pair can acquire in this reaction. Hence, for the distribution in the region $E \ll E_0$, this dependence may be neglected in comparison with the strong dependence described by formula (146.4).

$$dw_\theta = \text{const} \cdot \frac{\theta\, d\theta}{\sqrt{\theta^2 + 4|\varepsilon|/E_0}}. \tag{146.5}$$

The angular distribution referred to the solid-angle element $do \approx 2\pi\theta\, d\theta$ has a maximum at $\theta \sim \sqrt{|\varepsilon|/E_0}$.

§ 147. Behavior of cross sections near a reaction threshold

If the sum of the internal energies of the reaction products exceeds that of the initial particles, the reaction has a *threshold*: it can occur only when the kinetic energy E of the colliding particles (in the center-of-mass system) exceeds a certain threshold value E_{th}. We shall examine the energy dependence of the reaction cross section near this threshold. We assume that the reaction again produces only two particles (a reaction of the type $A + B = A' + B'$).

Near threshold, the relative velocity v' of the particles produced is small. This reaction is inverse to one in which the velocity of the colliding particles is small. Its cross section as a function of v' can therefore be obtained directly, by detailed balance (144.13), from the known energy dependence of the reaction for which v' would be the velocity in the entrance channel (§ 143). For the broad class of reactions in which there is no Coulomb interaction between A' and B' (for example, nuclear reactions producing a slow neutron), the reaction cross section is consequently proportional to $v'^2(1/v')$, and hence[26]

$$\sigma_r \sim v'. \tag{147.1}$$

This also gives its dependence on the energy of the colliding particles: v', and therefore the reaction cross section, is proportional to the square root of $E - E_{\text{th}}$:

$$\sigma_r = A\sqrt{E - E_{\text{th}}}. \tag{147.2}$$

Scattering amplitudes in the different channels are related by the unitarity conditions. Because of this relation, the opening of a new channel produces characteristic features in the energy dependence of the cross sections of other processes as well, including elastic scattering (E. Wigner, 1948; A. I. Baz', 1957; G. Breit, 1957). To establish the origin and form of this effect, consider the simplest case, in which only elastic scattering is possible below the reaction threshold.

Close to threshold, A' and B' are produced in a state of orbital angular momentum $l = 0$ (this is exactly the condition corresponding to the law (147.2)). If the participating particles have no spin, orbital angular momentum is conserved, and the $A + B$ system is also in an s state. According to (142.7), the partial reaction cross section for $l = 0$ is related to the elastic-scattering element of the S matrix by

$$\sigma_r = \frac{\pi}{k^2}(1 - |S_0|^2), \tag{147.3}$$

[26] The constancy, noted at the end of § 144, of the limit approached by the amplitude f_{fi} as $p_f \to 0$ is precisely consistent with this result: the cross section (144.4) is proportional to p_f.

where k is the wave vector of the colliding particles. Equating (147.2) and (147.3), we find that above and near the threshold, to terms of order $\sqrt{E-E_{\text{th}}}$, the modulus $|S_0|$ is

$$|S_0| = 1 - \frac{k_{\text{th}}^2}{2\pi} A\sqrt{E-E_{\text{th}}}, \qquad E > E_{\text{th}}, \tag{147.4}$$

where $k_{\text{th}} = \sqrt{2mE_{\text{th}}}/\hbar$ (m is the reduced mass of A and B). Below threshold only elastic scattering occurs, so that

$$|S_0| = 1, \qquad E < E_{\text{th}}. \tag{147.5}$$

The scattering amplitude, and hence S_0, must nevertheless be analytic throughout the energy domain. To the same accuracy, a function having the values (147.4) and (147.5) above and below threshold is

$$S_0 = e^{2i\delta_0}\left(1 - \frac{k_{\text{th}}^2}{2\pi} A\sqrt{E-E_{\text{th}}}\right), \tag{147.6}$$

where δ_0 is constant. (For $E < E_{\text{th}}$, the root $\sqrt{E-E_{\text{th}}}$ becomes imaginary, and the modulus of the parenthesized expression differs from unity only by a higher-order small quantity.)

For every $l \neq 0$, however, inelastic scattering is absent, and therefore

$$S_l = e^{2i\delta_l}, \qquad l \neq 0, \tag{147.7}$$

where, in the threshold region, each phase δ_l is to be replaced by its value at $E = E_{\text{th}}$[27].

Substitution of these S_l in (142.2) gives the following scattering amplitude near the reaction threshold:

$$f(\theta, E) = f_{\text{th}}(\theta) - \frac{k_{\text{th}}}{4\pi} A\sqrt{E-E_{\text{th}}}\, e^{2i\delta_0}, \tag{147.8}$$

where $f_{\text{th}}(\theta)$ is the amplitude at $E = E_{\text{th}}$. It follows that the differential scattering cross section is

$$\frac{d\sigma}{d\Omega} = \begin{cases} |f_{\text{th}}(\theta)|^2 + \dfrac{k_{\text{th}}}{2\pi} A\sqrt{E-E_{\text{th}}}\,\mathrm{Im}\{f_{\text{th}}(\theta)e^{-2i\delta_0}\}, & E > E_{\text{th}}, \\ |f_{\text{th}}(\theta)|^2 - \dfrac{k_{\text{th}}}{2\pi} A\sqrt{E_{\text{th}}-E}\,\mathrm{Re}\{f_{\text{th}}(\theta)e^{-2i\delta_0}\}, & E < E_{\text{th}}. \end{cases}$$

Writing $f_{\text{th}} = |f_{\text{th}}|e^{i\alpha(\theta)}$, we obtain finally

$$\frac{d\sigma}{d\Omega} = |f_{\text{th}}(\theta)|^2 - \frac{k_{\text{th}}}{2\pi} A|f_{\text{th}}(\theta)|\sqrt{|E-E_{\text{th}}|}\begin{cases} \sin(2\delta_0-\alpha), & E > E_{\text{th}}, \\ \cos(2\delta_0-\alpha), & E < E_{\text{th}}. \end{cases} \tag{147.9}$$

According as the angle $2\delta_0 - \alpha$ lies in the first, second, third, or fourth quadrant, the energy dependence described by this formula has the form shown in Fig. 50a, b, c, or d. In every case there are two branches on opposite sides of a common vertical tangent.

[27] Since the functions $\delta_l(E)$ are real both for $E > E_{\text{th}}$ and for $E < E_{\text{th}}$, they are expandable in integral powers of $E - E_{\text{th}}$.

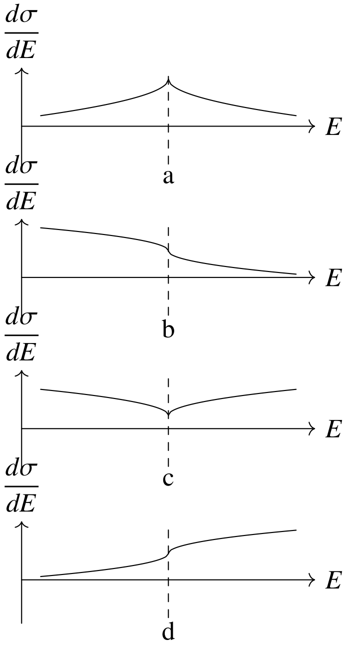


Figure 50

On integrating (147.9) over $d\Omega$, only the isotropic part of $f_{\text{th}}(\theta)$ contributes to the integrals of the second terms: this is the partial amplitude $(e^{2i\delta_0} - 1)/(2ik_{\text{th}})$ of elastic s-wave scattering. The total elastic-scattering cross section near threshold is therefore

$$\sigma = \sigma_{\text{th}} - 2A\sqrt{|E - E_{\text{th}}|}\begin{cases}\sin^2\delta_0, & E > E_{\text{th}},\\ \sin\delta_0\cos\delta_0, & E < E_{\text{th}}.\end{cases} \tag{147.10}$$

This dependence has form a or b in Fig. 50, according as $\sin\delta_0\cos\delta_0$ is positive or negative.

Thus, the existence of a reaction threshold produces a characteristic feature in the energy dependence of the elastic-scattering cross section. Particle spin naturally changes the quantitative formulas, but not the general character of the phenomenon[28]. If reactions other than elastic scattering are also possible below threshold, analogous features occur in their cross sections. At $E = E_{\text{th}}$ every such cross section has a singular point, near which it is a linear function of $\sqrt{|E - E_{\text{th}}|}$, with different slopes above and below threshold.

In nuclear reactions that emit a positively charged particle, the reaction products A' and B' repel each other by Coulomb forces. In this case, as $v' \to 0$ (that is, $E \to E_{\text{th}}$), the reaction cross section and all its energy derivatives tend exponentially to zero; consequently, the cross sections of other processes acquire no threshold feature.

Finally, consider a reaction producing two slow particles of opposite charge, between

[28]For nonzero spins the $A' + B'$ system in an s state may have nonzero total angular momentum, allowing several orbital states of the $A + B$ system.

which the Coulomb force is attractive. Detailed balance relates its cross section to the cross section (143.6) of the inverse reaction between two slow, mutually attracting particles. It follows that, as $v' \to 0$, the cross section tends to a constant limit:

$$\sigma_r = \text{const} \quad \text{as } v' \to 0, \tag{147.11}$$

so that beyond threshold the reaction begins immediately with a finite cross section.

We now determine the threshold feature of the elastic-scattering cross section in this case (A. I. Baz', 1959). Unlike the neutral-particle case considered above, this cannot be done directly from the known above-threshold law (147.11) by the same simple argument. The complication is that, in the near-threshold region below E_{th}, the $A' + B'$ system has bound states corresponding to the discrete energy levels in an attractive Coulomb field. Energetically, these states can be formed in collisions of A and B, but the possibility of elastic scattering makes them only quasistationary. Their existence must nevertheless give rise to resonance effects in subthreshold elastic scattering, analogous to Breit–Wigner resonances.

To solve the problem, consider the structure of the wave functions describing the collision. Since there are two channels, the Schrödinger equation of the interacting system has two independent solutions finite throughout configuration space; denote two arbitrarily chosen and normalized solutions by ψ_1 and ψ_2. Linear combinations of these functions can describe scattering with either channel as entrance channel. Denote by a and b the channels belonging to the particle pairs A, B and A', B', and let $\psi = a_1\psi_1 + a_2\psi_2$ correspond to entrance channel a; it describes elastic scattering of A and B together with the reaction $A + B \to A' + B'$. Near threshold the coefficients a_1, a_2 depend essentially on the small momentum k_b, whereas the arbitrarily chosen functions ψ_1, ψ_2 themselves have no singular behavior at $k_b = 0$.

At large distances, ψ must be the sum of two terms describing the motion of the particle pairs in channels a and b. Each term is the product of the particles' "internal" functions and the wave function for their relative motion[29]. In channel a the latter is $R_a^- - S_{aa}R_a^+$, and in channel b it is $-S_{ab}R_b^+$, where R^+ and R^- are the outgoing and incoming waves in the corresponding channels. At distances r_0 large compared with the range of the short-range forces and small compared with $1/k_b$, these functions and their derivatives must match values obtained from ψ in the "reaction zone." The matching conditions have the form

$$a_1\alpha_1 + a_2\alpha_2 = (R_a^- - S_{aa}R_a^+)|_{r_0}, \qquad a_1\beta_1 + a_2\beta_2 = -S_{ab}R_b^+|_{r_0},$$

where $\alpha_1, \alpha_2, \beta_1, \beta_2, \ldots$ are quantities calculated from ψ_1 and ψ_2; by the argument above, they may be treated near threshold as constants independent of k_b. Dividing corresponding members of the first and second pairs of these equations gives two linear equations for a_1/a_2 and S_{aa}. Their coefficients contain only one quantity with a "critical" dependence on k_b, namely the logarithmic derivative of the outgoing wave in channel b.

[29]The law (147.11) holds not only for the total cross section but also for partial cross sections of different angular momenta l (compare the end of § 143). The threshold feature considered below therefore occurs in every partial scattering cross section. Its character is already fully revealed for $l = 0$, which is the case treated below. To simplify notation, the subscript 0 is omitted from the corresponding partial amplitudes.

Define it by

$$\Lambda = \frac{1}{2\pi}\frac{(rR_b^+)'}{rR_b^+}\bigg|_{r=r_0}.$$

There is no need to solve these equations explicitly. It is enough to observe that S_{aa}, which determines the elastic-scattering amplitude, is a linear-fractional function of Λ. Below threshold Λ is real, since R_b^+ is real: it solves a real Schrödinger equation with the real boundary condition at infinity of decay as $e^{-\varkappa_b r}$, where $\varkappa_b = \sqrt{2m_b(E_{\text{th}} - E)}/\hbar$. At the same time $|S_{aa}| = 1$ below threshold. Hence the linear-fractional function $S_{aa}(\Lambda)$ must have the form

$$S_{aa} = \frac{1+\beta\Lambda}{1+\beta^*\Lambda}e^{2i\eta}, \tag{147.12}$$

where η is real and β is a complex constant.

Let us determine Λ as a function of k_b. Since the force between A' and B' is attractive and Coulombic, rR_b^+ is a Coulomb wave function asymptotically proportional to $e^{ik_b r}$ at infinity. In a repulsive Coulomb field this function is $G_0 + iF_0$, with G_0 and F_0 given by (138.4) and (138.7). An attractive field is obtained by changing simultaneously the signs of k and r[30]. Making this replacement and evaluating the logarithmic derivative (see § 138), we obtain[31]

$$\Lambda = \frac{i}{1-\exp(-2\pi/k_b)} - \frac{1}{\pi}\left\{\ln k_b + \frac{1}{2}\left[\psi\left(\frac{i}{k_b}\right) + \psi\left(-\frac{i}{k_b}\right)\right]\right\}. \tag{147.13}$$

Here k_b is assumed real, so this formula applies above threshold. As $k_b \to 0$, the first term in (147.13) tends to i, while the second tends to zero (see the note on p. 695). Thus above threshold

$$\Lambda = i, \qquad E > E_{\text{th}}. \tag{147.14}$$

The continuation below threshold is made by replacing k with $i\varkappa$. Equation (147.13) then gives, as $\varkappa \to 0$[32]

$$\Lambda = \cot\frac{\pi}{\varkappa_b}, \qquad E < E_{\text{th}}. \tag{147.15}$$

These formulas answer the question posed. The elastic-scattering cross section is

$$\sigma_e = \frac{\pi}{k_a^2}|S_{aa} - 1|^2.$$

Above threshold,

$$S_{aa} = \frac{1+i\beta}{1+i\beta^*}e^{2i\eta}. \tag{147.16}$$

[30]Coulomb units are used below. Formally, changing the signs of k and r corresponds to changing the sign of the Coulomb unit of length.

[31]For simplicity in the formulas that follow, a real constant independent of k_b, namely $-\ln 2r_0 - 2C$, has been omitted from the braces. This merely amounts to an inessential redefinition of the complex quantity β and the real quantity η in (147.12).

[32]The first term in (147.13) becomes $-(1/2)\cot(\pi/\varkappa_b) + i/2$, while the expression in braces becomes $(\pi/2)\cot(\pi/\varkappa_b) + i\pi/2$. Here one uses $\psi(x) - \psi(-x) = -\pi\cot\pi x - 1/x$, obtainable by logarithmic differentiation of $\Gamma(x)\Gamma(-x) = -\pi/(x\sin\pi x)$, together with the limiting form $\psi(x) \approx \ln x - 1/2x$ as $x \to \infty$.

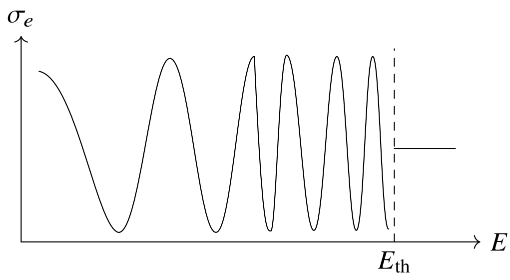


Figure 51

Like the reaction cross section, the scattering cross section is constant in this region. Notice that $|S_{aa}| \le 1$ requires $\operatorname{Im}\beta > 0$.

Below threshold we find

$$S_{aa} = e^{2i\eta}\,\frac{\beta - \tan(\pi/\varkappa_b)}{\beta^* - \tan(\pi/\varkappa_b)}. \tag{147.17}$$

This expression has infinitely many resonances accumulating toward $E = E_{\rm th}$. The resonance energies are the roots of $S_{aa} = -1$, that is,

$$\operatorname{Re}\left\{e^{i\eta}\left(\beta - \tan\frac{\pi}{\varkappa_b}\right)\right\} = 0;$$

they are shifted from the pure Coulomb levels, the roots of $\tan(\pi/\varkappa_b) = 0$, by the short-range forces. As E approaches threshold, the elastic-scattering cross section oscillates between zero and $4\pi/k_a^2$, as shown schematically in Fig. 51. The width of the entire subthreshold region displaying resonance structure is set by the energy of the first Coulomb level[33].

§ 148. Inelastic collisions of fast electrons with atoms

Inelastic collisions of fast electrons with atoms can be treated in the Born approximation, in the same manner as the elastic collisions in § 139[34]. As before, the condition for applicability of the Born approximation requires the speed of the incident electron to be large in comparison with the speeds of the atomic electrons. The energy lost in the collision, however, may have any value. If the electron loses a substantial part of its energy, the atom is ionized and the energy is transferred to one of its electrons. But of these two electrons we may always call the one having the greater speed after the collision the scattered electron. Thus, when the incident electron is fast, the scattered electron will be fast as well.

[33] Another interesting threshold case is electron-impact ionization of an atom when the electron energy only slightly exceeds the first ionization energy. Under these conditions the collision may be treated semiclassically, but the presence of three charged particles in the final state makes the problem very difficult. A general solution was given by Wannier (G. H. Wannier // Phys. Rev. 1953. V. 90. P. 817). The ionization probability of a neutral atom is proportional to $(E - I)^\alpha$, where $\alpha = (1/4)(\sqrt{9173} - 1) \approx 1.13$, and $E - I$ is the electron energy in excess of the ionization threshold.

[34] Most of the results presented in §§ 148–150 were obtained by H. A. Bethe (1930).

For collisions between an electron and an atom, the frame of reference in which their center of mass is at rest may be regarded,

as noted above, as coinciding with the frame in which the atom is at rest; in what follows, we shall use precisely this latter frame.

An inelastic collision changes the atom's internal state. The atom may pass from its ground state to an excited state belonging to either the discrete or the continuous spectrum; in the latter case, the atom is ionized. In deriving general formulas, these cases may be treated together.

We begin, as in § 126, with the general formula for a transition probability between states of a continuous spectrum, applying it to a system composed of the incident electron and the atom. Let $\mathbf{p}$ and $\mathbf{p}'$ be the momenta of the incident electron, and let E_0 and E_n be the atomic energies before and after the collision, respectively. In place of (126.9), the transition probability is

$$dw_n = \frac{2\pi}{\hbar}|\langle n, \mathbf{p}'|U|0, \mathbf{p}\rangle|^2 \delta\left(\frac{p'^2 - p^2}{2m} + E_n - E_0\right)\frac{d^3p'}{(2\pi\hbar)^3}, \tag{148.1}$$

where the matrix element is taken of the interaction energy between the incident electron and the atom,

$$U = \frac{Ze^2}{r} - \sum_{a=1}^{Z} \frac{e^2}{|\mathbf{r} - \mathbf{r}_a|}$$

($\mathbf{r}$ is the radius vector of the incident electron, $\mathbf{r}_a$ those of the atomic electrons, and the origin is chosen at the atomic nucleus; m is the electron mass).

The electron wave functions $\psi_{\mathbf{p}}$ and $\psi_{\mathbf{p}'}$ are given by the same formulas (126.10) and (126.11); dw is then the collision cross-section $d\sigma$. Denote the atomic wave functions in the initial and final states by ψ_0 and ψ_n. If the final atomic state belongs to the discrete spectrum, ψ_n, like ψ_0, is normalized in the usual way to unity. If the atom passes to a state in the continuous spectrum, the wave function is normalized to a δ-function in the parameters ν specifying these states (such parameters may be, for example, the atomic energy and the components of the momentum of the electron ejected from the atom during ionization). The resulting cross-sections determine the probability of a collision in which the atom passes to continuous-spectrum states whose parameter values lie between ν and $\nu + d\nu$.

Integration in (148.1) over the magnitude of p' gives

$$d\sigma_n = \frac{mp'}{4\pi^2\hbar^4}|\langle n\mathbf{p}'|U|0\mathbf{p}\rangle|^2 do',$$

where p' is determined by conservation of energy:

$$\frac{p^2 - p'^2}{2m} = E_n - E_0. \tag{148.2}$$

Substituting the electron wave functions (126.10) and (126.11) into the matrix element, we obtain

$$d\sigma_n = \frac{m^2}{4\pi^2\hbar^4}\frac{p'}{p}\left|\iint Ue^{-i\mathbf{q}\mathbf{r}}\psi_n^*\psi_0 \, d\tau \, dV\right|^2 do \tag{148.3}$$

($d\tau = dV_1 dV_2 \dots dV_Z$ is the configuration-space element for the Z atomic electrons, and the prime on do has been omitted)[35]. For $n = 0$ and $p = p'$, formula (148.3) becomes the elastic-scattering cross-section formula.

By orthogonality of ψ_n and ψ_0, the term in U containing the interaction Ze^2/r with the nucleus vanishes upon integration over $d\tau$. Thus, for inelastic collisions,

$$d\sigma_n = \frac{m^2}{4\pi^2\hbar^4}\frac{p'}{p}\left|\sum_a \iint \frac{e^2}{|\mathbf{r}-\mathbf{r}_a|} e^{-i\mathbf{q}\mathbf{r}}\psi_n^*\psi_0 \, d\tau \, dV\right|^2 do. \tag{148.4}$$

The integration over dV can be carried out in the same way as in § 139. The integral

$$\varphi_{\mathbf{q}}(\mathbf{r}_a) = \int \frac{e^{-i\mathbf{q}\mathbf{r}}}{|\mathbf{r}-\mathbf{r}_a|} dV$$

is formally the Fourier component of the potential produced at the point $\mathbf{r}$ by charges distributed in space with density $\rho = \delta(\mathbf{r}-\mathbf{r}_a)$. Formula (139.1) therefore gives

$$\varphi_{\mathbf{q}}(\mathbf{r}_a) = \frac{4\pi}{q^2} e^{-i\mathbf{q}\mathbf{r}_a}. \tag{148.5}$$

Substitution of this expression into (148.4) finally gives the following *general* expression for the cross-section of inelastic collisions:

$$d\sigma_n = \left(\frac{e^2 m}{\hbar^2}\right)^2 \frac{4k'}{kq^4}\left|\left\langle n\left|\sum_a e^{-i\mathbf{q}\mathbf{r}_a}\right|0\right\rangle\right|^2 do, \tag{148.6}$$

where the matrix element is evaluated with the atomic wave functions, and the momenta have been replaced by the wave vectors $\mathbf{k} = \mathbf{p}/\hbar$,

$\mathbf{k}' = \mathbf{p}'/\hbar$. The formula gives the probability that, in a collision, the atom is carried into its nth excited state while the electron is scattered into the solid-angle element do. The vector $-\hbar\mathbf{q}$ is the momentum transferred by the electron to the atom in the collision.

For calculations it is sometimes more convenient to specify the cross-section per element dq of the magnitude of $\mathbf{q}$, rather than per element of solid angle. The vector $\mathbf{q}$ is defined by $\mathbf{q} = \mathbf{k}' - \mathbf{k}$; its magnitude satisfies

$$q^2 = k^2 + k'^2 - 2kk'\cos\vartheta. \tag{148.7}$$

Hence, for prescribed k and k', that is, for a prescribed energy loss by the electron,

$$q\,dq = kk'\sin\vartheta\,d\vartheta = \frac{kk'}{2\pi}do. \tag{148.8}$$

Formula (148.6) can therefore be rewritten as

$$d\sigma_n = 8\pi\left(\frac{e^2}{\hbar v}\right)^2 \frac{dq}{q^3}\left|\left\langle n\left|\sum_a e^{-i\mathbf{q}\mathbf{r}_a}\right|0\right\rangle\right|^2. \tag{148.9}$$

[35] In this form, this is the general perturbation-theory formula, applicable not only to electron–atom collisions but to any inelastic collision of two particles. It determines the scattering cross-section in the coordinate frame in which the centre of mass of the particles is at rest (in that case, m is the reduced mass of the two particles).

The vector $\mathbf{q}$ plays an important part in the calculations below. We examine in more detail how it is related to the scattering angle ϑ and to the energy $E_n - E_0$ transferred in the collision. It will be seen below that the main contribution comes from collisions producing scattering through small angles ($\vartheta \ll 1$) and transferring an energy small in comparison with the energy $E = mv^2/2$ of the incident electron: $E_n - E_0 \ll E$. The difference $k - k'$ is then also small ($k - k' \ll k$), and consequently

$$E_n - E_0 = \frac{\hbar^2}{2m}(k^2 - k'^2) \approx \frac{\hbar^2}{m}k(k - k') = \hbar v(k - k').$$

Since ϑ is small, (148.7) gives $q^2 \approx (k - k')^2 + (k\vartheta)^2$, and hence

$$q = \sqrt{\left(\frac{E_n - E_0}{\hbar v}\right)^2 + (k\vartheta)^2}. \tag{148.10}$$

The minimum value of q is

$$q_{\min} = \frac{E_n - E_0}{\hbar v}. \tag{148.11}$$

At small angles, further regions can be distinguished according to the relation between the small quantities ϑ and v_0/v, where v_0 is of the order of an atomic-electron speed. For energy transfers of the order of the atomic-electron energy ε_0 ($E_n - E_0 \sim \varepsilon_0 \sim mv_0^2$), if $(v_0/v)^2 \ll \vartheta \ll 1$, then

$$q = k\vartheta = mv\vartheta/\hbar. \tag{148.12}$$

(the first term under the radical in (148.10) may be neglected in comparison with the second); consequently, in this angular region q is independent of the amount of energy transferred. When $\vartheta \ll 1$, q may be either large or small compared with $1/a_0$ (where a_0 is a quantity of the order of atomic dimensions). Under the same assumption concerning the amount of energy transferred, we have

$$qa_0 \sim 1 \quad \text{when} \quad \vartheta \sim v_0/v. \tag{148.13}$$

We now return to the general formula (148.9) and consider the case of small q ($qa_0 \ll 1$, i.e. $\vartheta \ll v_0/v$).

In this case the exponential factors may be expanded in powers of $\mathbf{q}$: $e^{-i\mathbf{q}\mathbf{r}_a} \approx 1 - i\mathbf{q}\mathbf{r}_a = 1 - iqx_a$ (the x axis is along the vector $\mathbf{q}$). On substituting this expansion in (148.9), the terms containing 1 vanish by virtue of the orthogonality of the wave functions ψ_0 and ψ_n, and we obtain

$$d\sigma_n = 8\pi\left(\frac{e}{\hbar v}\right)^2 \frac{dq}{q}|\langle n|d_x|0\rangle|^2 = \left(\frac{2e}{\hbar v}\right)^2 |\langle n|d_x|0\rangle|^2 \frac{d\Omega}{\vartheta^2}, \tag{148.14}$$

where $d_x = e\sum_a x_a$ is the component of the atomic dipole moment. We see that the scattering cross-section (at small q) is determined by the squared modulus of the dipole-moment matrix element for the transition corresponding to the change of the atomic state[36].

[36] Usually the quantity of physical interest is the cross-section $d\sigma_n$, summed over all orientations of the atomic moment in the final state and averaged over its orientations in the initial state. After this summation and averaging, the square $|\langle n|d_x|0\rangle|^2$ no longer depends on the direction of the x axis.

It may happen, however, that the dipole-moment matrix element for the given transition vanishes identically because of the selection rules (a forbidden transition). The expansion of $\exp(-i\mathbf{q}\mathbf{r}_a)$ must then be continued to the next term, and we obtain

$$d\sigma_n = 2\pi \left(\frac{e^2}{\hbar v}\right)^2 \left|\left\langle n \left| \sum_a x_a^2 \right| 0 \right\rangle\right|^2 q\,dq. \tag{148.15}$$

We now consider the opposite limiting case of large q ($qa_0 \gg 1$). Large q means that the momentum transferred to the atom is large compared with the initial intrinsic momenta of the atomic electrons. It is physically clear in advance that in this case the atomic electrons may be treated as free, and the collision with the atom as an elastic collision of the incident electron with atomic electrons initially at rest. This also follows from the general formula (148.9). At large q, the integrand in the matrix element contains rapidly oscillating factors $\exp(-i\mathbf{q}\mathbf{r}_a)$, and the integral can fail to be small only if ψ_n contains the same factor.

Such a function ψ_n corresponds to an ionized atom from which an electron has been ejected with momentum $-\hbar\mathbf{q} = \mathbf{p} - \mathbf{p}'$, fixed simply by momentum conservation, just as in a collision between two free electrons.

In a collision involving a large momentum transfer, the two electrons (the incident and atomic electrons) may both emerge with speeds of comparable magnitude. The exchange effects associated with the identity of the colliding particles, which were not taken into account in the general formula (148.9), then become important. The scattering cross-section for fast electrons with exchange included is given by (137.9); that formula applies in the frame in which one of the electrons was at rest before the collision. For fast electrons, the cosine in the last term of (137.9) may be replaced by unity. Multiplying also by the number Z of electrons in the atom, we obtain the cross-section for collision of an electron with an atom in the form

$$d\sigma = 4Z\left(\frac{e^2}{mv^2}\right)^2 \left(\frac{1}{\sin^4\vartheta} + \frac{1}{\cos^4\vartheta} - \frac{1}{\sin^2\vartheta\cos^2\vartheta}\right)\cos\vartheta\,d\Omega. \tag{148.16}$$

In this formula it is convenient to express the scattering angle in terms of the energy acquired by the electrons after the collision. As is known, when a particle of energy $E = mv^2/2$ collides with a particle of the same mass at rest, the energies of the particles after the collision are $\varepsilon = E\sin^2\vartheta$ and $E - \varepsilon = E\cos^2\vartheta$. To obtain the cross-section referred to the interval $d\varepsilon$, we express $d\Omega$ in terms of $d\varepsilon$ according to $\cos\vartheta\,d\Omega = 2\pi\cos\vartheta\sin\vartheta\,d\vartheta = (\pi/E)d\varepsilon$. Substitution in (148.16) gives the final formula

$$d\sigma_\varepsilon = \pi Z e^4 \left[\frac{1}{\varepsilon^2} + \frac{1}{(E-\varepsilon)^2} - \frac{1}{\varepsilon(E-\varepsilon)}\right]\frac{d\varepsilon}{E}. \tag{148.17}$$

If one of the energies ε or $E - \varepsilon$ is small compared with the other, only one of the three terms in this formula (the first or the second) is important. This corresponds to the fact that, when the two electron energies differ greatly, the exchange effect is unimportant and we must return to the usual Rutherford formula[37].

[37] For the collision of a positron with an atom, the exchange effect is altogether absent, and the Rutherford formula $d\sigma_\varepsilon = (\pi Z e^4/E)d\varepsilon/\varepsilon^2$ applies for all $q \gg 1/a_0$.

Integration of the differential cross-section over all angles (or, equivalently, over dq) gives the total cross-section σ_n for a collision that excites the given atomic state. The dependence of σ_n on the incident electron's speed is essentially connected with whether the atomic dipole-moment matrix element for the corresponding transition is present or absent. Suppose first that

this element is nonzero. At small q, the cross-section $d\sigma_n$ is then given by (148.14), and we see that the integral over dq diverges logarithmically as q decreases. In the region of large q, by contrast, the cross-section (for a specified energy transfer $E_n - E_0$) decreases exponentially as q increases, because of the already noted rapidly oscillating factor in the integrand of the matrix element in (148.9). Thus the region of small q gives the main contribution to the integral over dq, and we may restrict the integration to the range from the minimum value $q_{\min}$ in (148.11) to some value $\sim 1/a_0$. The result is

$$\sigma_n = 8\pi \left(\frac{e}{\hbar v}\right)^2 |\langle n|d_x|0\rangle|^2 \ln\left(\beta_n \frac{v\hbar}{e^2}\right), \tag{148.18}$$

where β_n is a dimensionless constant that cannot be calculated in general[38].

If, on the other hand, the dipole-moment matrix element vanishes for the given transition, the integral over dq converges rapidly both at small q (as is clear from (148.15)) and at large q. In this case the principal region for the integral is $q \sim 1/a_0$. No general quantitative formula can be obtained here, and we conclude that σ_n is inversely proportional to the square of the speed:

$$\sigma_n = \frac{\text{const}}{v^2}. \tag{148.19}$$

This follows directly from the general formula (148.9), according to which $d\sigma_n$ at $q \sim 1/a_0$ is proportional to v^{-2}.

Let $d\sigma_r$ be the cross-section for inelastic scattering into a given solid-angle element, irrespective of the state to which the atom passes. To find it, expression (148.9) must be summed over all $n \neq 0$, that is, over all atomic states (of both the discrete and continuous spectra) except the ground state. We exclude from consideration both large and extremely small angles and assume that $(v_0/v)^2 \ll \vartheta \ll 1$. Then, according to (148.12), q is independent of the energy transferred[39].

The last circumstance makes it easy to calculate the total cross-section for inelastic collisions, that is, the sum

$$\begin{aligned} d\sigma_r = \sum_{n\neq 0} d\sigma_n &= 8\pi \left(\frac{e^2}{\hbar v}\right)^2 \sum_{n\neq 0} \left|\left\langle n \left| \sum_a e^{-i\mathbf{q}\mathbf{r}_a} \right| 0 \right\rangle\right|^2 \frac{dq}{q^3} \\ &= \left(\frac{2e^2}{mv^2}\right)^2 \sum_{n\neq 0} \left|\left\langle n \left| \sum_a e^{-i\mathbf{q}\mathbf{r}_a} \right| 0 \right\rangle\right|^2 \frac{d\Omega}{\vartheta^4}. \end{aligned} \tag{148.20}$$

[38] We assume that $E_n - E_0$ is of the order of the atomic-electron energy ε_0. For large energy transfers ($E_n - E_0 \sim E \gg \varepsilon_0$), formulae (148.14), (148.18) are inapplicable in any event, since the dipole-moment matrix element becomes very small and one cannot retain only the first term of the expansion in q.

[39] The sum in (148.9) also extends over states with $E_n - E_0 \gg \varepsilon_0$, for which (148.12) does not hold. Transitions involving a large energy transfer have, however, a comparatively small cross-section, and these terms contribute little to the sum. The condition $\vartheta \ll 1$ permits exchange effects to be neglected.

To do this, note that for any quantity f, the rule for matrix multiplication gives

$$\sum_n |f_{0n}|^2 = \sum_n f_{0n}(f_{0n})^* = \sum_n f_{0n}(f^+)_{n0} = (ff^+)_{00}.$$

The summation here is over all n, including $n = 0$. Therefore

$$\sum_{n\neq 0} |f_{0n}|^2 = \sum_n |f_{0n}|^2 - |f_{00}|^2 = (ff^+)_{00} - |f_{00}|^2. \tag{148.21}$$

Applying this relation to $f = \sum_a e^{-i\mathbf{q}\mathbf{r}_a}$, we obtain

$$d\sigma_r = \left(\frac{2e^2}{mv^2}\right)^2 \left\{\left\langle\left|\sum_a e^{-i\mathbf{q}\mathbf{r}_a}\right|^2\right\rangle - \left|\left\langle\sum_a e^{-i\mathbf{q}\mathbf{r}_a}\right\rangle\right|^2\right\} \frac{d\Omega}{\vartheta^4}, \tag{148.22}$$

where $\langle\ldots\rangle$ denotes averaging over the ground state of the atom (that is, taking the diagonal matrix element 00). By definition, the mean value $\left\langle\sum e^{-i\mathbf{q}\mathbf{r}_a}\right\rangle$ is the atomic factor $F(q)$ of the atom in its ground state. In the first term in braces we may write

$$\left|\sum_{a=1}^{Z} e^{-i\mathbf{q}\mathbf{r}_a}\right|^2 = Z + \sum_{a\neq b} e^{i\mathbf{q}(\mathbf{r}_a-\mathbf{r}_b)}.$$

Thus we find the general formula

$$d\sigma_r = \left(\frac{2e^2}{mv^2}\right)^2 \left\{Z - F^2(q) + \left\langle\sum_{a\neq b} e^{i\mathbf{q}(\mathbf{r}_a-\mathbf{r}_b)}\right\rangle\right\} \frac{d\Omega}{\vartheta^4}. \tag{148.23}$$

This formula is greatly simplified at small q, when it may be expanded in powers of q ($v_0/v \ll qa_0 \ll 1$, corresponding to angles $(v_0/v)^2 \ll \vartheta \ll v_0/v$). Rather than carry out the expansion in formula (148.23), it is more convenient to start anew   repeat the summation over n, now using (148.14) for $d\sigma_n$. Summation by means of (148.21), with $f = d_x$, and the fact that $\langle d_x\rangle = 0$ give

$$d\sigma_r = \left(\frac{2e}{\hbar v}\right)^2 \langle d_x^2\rangle \frac{do}{\vartheta^2}. \tag{148.24}$$

It is instructive to compare this result with the small-angle elastic-scattering cross-section (139.5). The latter is independent of ϑ, whereas the inelastic cross-section into a solid-angle element do increases as $1/\vartheta^2$ when ϑ decreases.

For angles $v_0/v \ll \vartheta \ll 1$, and hence $qa_0 \gg 1$, the second and third terms within the braces in (148.23) are small. We then simply obtain

$$d\sigma_r = Z\left(\frac{2e^2}{mv^2}\right)^2 \frac{do}{\vartheta^4}, \tag{148.25}$$

which is Rutherford scattering by the Z atomic electrons, with exchange neglected. Recall that the differential elastic-scattering cross-section (139.6) is proportional to Z^2, rather than to Z.

Finally, angular integration gives the total cross-section σ_r for inelastic scattering through arbitrary angles and with arbitrary atomic excitation. Proceeding exactly as in the calculation of σ_n in (148.18), we find

$$\sigma_r = 8\pi \left(\frac{e}{\hbar v}\right)^2 \langle d_x^2\rangle \ln\left(\beta\frac{v\hbar}{e^2}\right). \tag{148.26}$$

Problems[40]

1. Determine the angular distribution, for $v^{-2} \ll \vartheta \ll 1$, of the inelastic scattering of fast electrons by a hydrogen atom in its ground state.

Solution. For hydrogen, the third term in the braces in (148.23) is absent, while the atomic factor $F(q)$ was calculated in the problem following § 139. Its substitution gives

$$d\sigma_r = \frac{4}{v^4\vartheta^4}\left[1-\left(1+\frac{v^2\vartheta^2}{4}\right)^{-4}\right] do.$$

2. Find the differential cross-section for collisions of electrons with a ground-state hydrogen atom that excite the nth discrete level, where n is the principal quantum number.

Solution. The matrix elements are conveniently evaluated in parabolic coordinates. Choose the z axis along $\mathbf{q}$; then $e^{i\mathbf{qr}} = e^{iqz} = e^{iq(\xi-\eta)/2}$. The ground-state wave function is $\psi_{000} = \pi^{-1/2}e^{-(\xi+\eta)/2}$. Matrix elements are nonzero only for transitions to states with $m = 0$. The wave functions of these states are

$$\psi_{n_1n_20} = \frac{1}{\sqrt{\pi n^2}}\exp\left(-\frac{\xi+\eta}{2n}\right) F\left(-n_1, 1, \frac{\xi}{n}\right) F\left(-n_2, 1, \frac{\eta}{n}\right)$$

with $n = n_1 + n_2 + 1$. The required matrix elements are represented by the integrals

$$\langle n_1n_20|e^{i\mathbf{qr}}|000\rangle = \int_0^\infty\int_0^\infty \exp\left(i\frac{q}{2}(\xi-\eta)\right)\psi_{000}\psi_{n_1n_20}\frac{\xi+\eta}{4}\,2\pi\,d\xi\,d\eta.$$

The integrations use the formulas collected in § f of the Mathematical Appendix. Their evaluation yields

$$|\langle n_1n_20|e^{i\mathbf{qr}}|000\rangle|^2 = 2^8n^6q^2\frac{[(n-1)^2+(qn)^2]^{n-3}}{[(n+1)^2+(qn)^2]^{n+3}}[(n_1-n_2)^2+(qn)^2].$$

All states having the same $n_1 + n_2 = n - 1$ have equal energies. Summing over all possible values of $n_1 - n_2$ at fixed n, and inserting the result into (148.9), gives the required cross-section:

$$d\sigma_n = \frac{2^{11}\pi}{v^2}n^7\left[\frac{n^2-1}{3}+(qn)^2\right]\frac{[(n-1)^2+(qn)^2]^{n-3}}{[(n+1)^2+(qn)^2]^{n+3}}\frac{dq}{q}.$$

3. Determine the total cross-section for excitation of the first excited state of the hydrogen atom.

[40] Atomic units are used throughout these problems.

SOLUTION. Integrate

$$d\sigma_2 = \frac{2^8\pi}{v^2}\frac{dq}{q(q^2+9/4)^5}$$

over q from $q_{\min} = (E_2 - E_1)/v = 3/8v$ to $q_{\max} = 2v$, retaining only terms of the highest order in v. Elementary integration gives[41]

$$\sigma_2 = \frac{2^{18}\pi}{3^{10}v^2}\left(\ln 4v - \frac{25}{24}\right) = \frac{4\pi}{v^2}\cdot 0.555\ln\frac{v^2}{0.50}.$$

4. Find the cross-section for ionization of a ground-state hydrogen atom when the secondary electron is emitted in a specified direction. The secondary electron's energy is small compared with the primary electron's energy, so exchange effects are immaterial (H. Massey, C. Mohr, 1933).

SOLUTION. The initial atomic wave function is $\psi_0 = \pi^{-1/2}e^{-r}$. In the final state the atom is ionized. Denote the wave vector of the emitted secondary electron by $\boldsymbol{\kappa}$, so that its energy is $\kappa^2/2$. This state is described by $\psi_{\boldsymbol{\kappa}}^{(-)}$ from (136.9), whose "outgoing" part at infinity consists only of a plane wave propagating in the $\boldsymbol{\kappa}$ direction. The function $\psi_{\boldsymbol{\kappa}}^{(-)}$ is normalized to a delta function in $\boldsymbol{\kappa}/2\pi$-space. The cross-section calculated from it is therefore referred to $d^3\kappa/(2\pi)^3$, or to $\kappa^2 d\kappa\, do_\kappa/(2\pi)^3$, where do_κ is the solid-angle element for the secondary electron's direction. Thus

$$d\sigma = \frac{4k'\kappa^2}{(2\pi)^3 kq^4}|\langle\boldsymbol{\kappa}|e^{-i\mathbf{qr}}|0\rangle|^2 do\, do_\kappa\, d\kappa$$

where do is the solid-angle element for the scattered electron, and

$$\langle\boldsymbol{\kappa}|e^{-i\mathbf{qr}}|0\rangle = \int \psi_{\boldsymbol{\kappa}}^{(-)*}e^{-i\mathbf{qr}}\psi_0\, dV = \frac{e^{-\pi/2\kappa}\Gamma(1-i/\kappa)}{\pi^{1/2}}I,$$

$$I = \left\{-\frac{\partial}{\partial\lambda}\int \exp(-i\mathbf{qr} - i\boldsymbol{\kappa}\mathbf{r} - \lambda r)F\left(\frac{i}{\kappa}, 1, i(\kappa r + \boldsymbol{\kappa}\mathbf{r})\right)\frac{dV}{r}\right\}_{\lambda=1}.$$

Carry out the integration in parabolic coordinates with the z axis along $\boldsymbol{\kappa}$. Let φ be measured from the plane $(\mathbf{q}, \boldsymbol{\kappa})$:

$$I = \left\{-\frac{1}{2}\frac{\partial}{\partial\lambda}\int_0^\infty\int_0^\infty\int_0^{2\pi}\exp\left[-\frac{i}{2}q(\xi-\eta)\cos\gamma + iq\sqrt{\xi\eta}\sin\gamma\cos\varphi\right.\right.$$
$$\left.\left.-\frac{\lambda}{2}(\xi+\eta) - \frac{i}{2}\kappa(\xi-\eta)\right]F(i/\kappa, 1, i\kappa\xi)\, d\varphi\, d\xi\, d\eta\right\}_{\lambda=1}.$$

Here γ is the angle between $\boldsymbol{\kappa}$ and $\mathbf{q}$. The integrations over $d\varphi\, d\eta$ are readily performed by substituting $\sqrt{\eta}\cos\varphi = u$ and $\sqrt{\eta}\sin\varphi = v$, after which

$$\frac{I}{2\pi} = \left\{\frac{\partial}{\partial\lambda}\int_0^\infty \exp\left[\frac{-q^2\sin^2\gamma + \lambda^2 + (\kappa + q\cos\gamma)^2}{2[i(\kappa+q\cos\gamma)-\lambda]}\xi\right]\frac{F(i/\kappa, 1, i\kappa\xi)\, d\xi}{i(\kappa+q\cos\gamma)-\lambda}\right\}_{\lambda=1}.$$

[41] The cross-section can likewise be calculated for arbitrary n. Numerical calculation also gives the total inelastic-scattering cross-section for hydrogen: $\sigma_r = (4\pi/v^2)\ln(v^2/0.160)$. Of this result, excitation of discrete-spectrum states and ionization account, respectively, for $\sigma_{\text{exc}} = (4\pi/v^2)\cdot 0.715\ln(v^2/0.45)$ and $\sigma_{\text{ion}} = (4\pi/v^2)\cdot 0.285\ln(v^2/0.012)$.

The remaining integral follows from (f.3) with $\gamma = 1$ and $n = 0$.

The subsequent calculations are lengthy but elementary. They lead to the following cross-section:

$$d\sigma = \frac{2^8 k'\kappa[q^2 + 2q\kappa\cos\gamma + (\kappa^2+1)\cos^2\gamma]}{\pi k q^2[q^2 + 2q\kappa\cos\gamma + 1 + \kappa^2]^4} \times \frac{\exp\left\{-\dfrac{2}{\kappa}\arctan\dfrac{2\kappa}{q^2-\kappa^2+1}\right\}}{[(q+\kappa)^2+1][(q-\kappa)^2+1](1-e^{-2\pi/\kappa})} do\, do_\kappa\, d\kappa.$$

Integration over all directions of emission of the secondary electron is elementary and gives, at the specified emitted-electron energy $\kappa^2/2$, the distribution of scattering directions

$$d\sigma = \frac{2^{10}k'\kappa[q^2 + (1/3)(1+\kappa^2)]\exp\{-\frac{2}{\kappa}\arctan[2\kappa/(q^2-\kappa^2+1)]\}}{kq^2[(q+\kappa)^2+1]^3[(q-\kappa)^2+1]^3(1-e^{-2\pi/\kappa})} do\, d\kappa.$$

For $q \gg 1$ this expression has a sharp maximum near $\kappa = q$. Close to the maximum,

$$d\sigma = \frac{2^5}{3\pi\kappa^4}\frac{d\kappa\, do}{[1+(q-\kappa)^2]^3}.$$

Integrating over $do = 2\pi q\, dq/k^2 \approx (2\pi\kappa/k^2)d(q-\kappa)$ gives $8\pi\, d\kappa/k^2\kappa^3$, in agreement, as required, with the first term of (148.17).

§ 149. Effective stopping

An important quantity in applications of collision theory is the mean energy lost by the colliding particle. It is convenient to characterize this loss by

$$d\varkappa = \sum_n (E_n - E_0)\, d\sigma_n, \tag{149.1}$$

which we shall call the (differential) *effective stopping*; the sum naturally extends over states of both the discrete and continuous spectra, and $d\varkappa$ refers to scattering into the given element of solid angle[42].

The general expression for the effective stopping of fast electrons is

$$d\varkappa = 8\pi\left(\frac{e^2}{\hbar v}\right)^2 \sum_n (E_n - E_0)\left|\left\langle n\left|\sum_a e^{-i\mathbf{q}\mathbf{r}_a}\right|0\right\rangle\right|^2 \frac{dq}{q^3} \tag{149.2}$$

($d\sigma_n$ is taken from (148.9)). As in the derivation of (148.23), let us omit the range of extremely small angles and again suppose that $(v_0/v)^2 \ll \vartheta \ll 1$. The value of q is then independent of the energy transferred, and the sum over n can be evaluated in general form.

[42] When an electron passes through a gas, scattering by different atoms occurs independently, and $N\, d\varkappa$ (N is the number of atoms per unit volume of the gas) is the energy lost by the electron per unit path length in collisions that deflect it into the given element of solid angle.

This evaluation uses a *sum rule*, derived as follows. The matrix elements of a quantity f (a function of the coordinates) and of its time derivative $\dot{f}$ obey

$$(\dot{f})_{0n} = -(i/\hbar)(E_n - E_0) f_{0n}. \tag{149.3}$$

Hence

$$\sum_n (E_n - E_0)|f_{0n}|^2 = \sum_n (E_n - E_0) f_{0n} (f_{0n})^* = \sum_n (E_n - E_0) f_{0n} (f^+)_{n0}$$
$$= i\hbar \sum_n (\dot{f})_{0n} (f^+)_{n0} = i\hbar (\dot{f} f^+)_{00}.$$

The wave functions of the atom's stationary states may be chosen real. The matrix elements of the coordinate function f then satisfy $f_{0n} = f_{n0}$, while (149.3) gives correspondingly $(\dot{f})_{0n} = -(\dot{f})_{n0}$. The sum in question may therefore also be written

$$-i\hbar \sum_n (f^+)_{0n} (\dot{f})_{n0} = -i\hbar (f^+ \dot{f})_{00}.$$

Taking half the sum of these two forms yields the required rule:

$$\sum_n (E_n - E_0)|f_{0n}|^2 = \frac{i\hbar}{2} (\dot{f} f^+ - f^+ \dot{f})_{00}. \tag{149.4}$$

Apply it to $f = \sum_a e^{-i\mathbf{q}\mathbf{r}_a}$. By (19.2), its time derivative is represented by the operator

$$\hat{\dot{f}} = -\frac{\hbar}{2m} \sum_a [e^{-i\mathbf{q}\mathbf{r}_a} (\mathbf{q}\boldsymbol{\nabla}_a) + (\mathbf{q}\boldsymbol{\nabla}_a) e^{-i\mathbf{q}\mathbf{r}_a}].$$

Direct calculation gives

$$\hat{\dot{f}} \hat{f}^+ - \hat{f}^+ \hat{\dot{f}} = -\frac{i\hbar}{m} q^2 Z.$$

Substitution in (149.4) gives

$$\sum_n \frac{2m}{\hbar^2 q^2} (E_n - E_0) \left| \left\langle n \left| \sum_a e^{-i\mathbf{q}\mathbf{r}_a} \right| 0 \right\rangle \right|^2 = Z, \tag{149.5}$$

which performs the needed summation[43].

Thus the differential effective stopping is

$$d\varkappa = 4\pi \frac{Ze^4}{mv^2} \frac{dq}{q} = \frac{2Ze^4}{mv^2} \frac{do}{\vartheta^2}. \tag{149.6}$$

Its domain of validity is $(v_0/v)^2 \ll \vartheta \ll 1$, or $v_0/v \ll a_0 q \ll v/v_0$.

Next define the total effective stopping $\varkappa(q_1)$ for all collisions in which the momentum transfer does not exceed a value q_1 satisfying $v_0/v \ll a_0 q_1 \ll v/v_0$:

$$\varkappa(q_1) = \sum_n \int_{q_{\min}}^{q_1} (E_n - E_0) d\sigma_n; \tag{149.7}$$

[43] Nowhere in deriving this relation did we use the fact that the state bearing the index 0 is the atom's ground state. The relation therefore holds for any initial state.

$q_{\min}$ is given by (148.11). The integral sign cannot be moved outside the sum because $q_{\min}$ depends on n.

Divide the integration range into two parts, from $q_{\min}$ to q_0 and from q_0 to q_1, where q_0 is chosen so that $v_0/v \ll q_0 a_0 \ll 1$. Throughout the first of these ranges, expression (148.14) may be used for $d\sigma_n$:

$$\varkappa(q_0) = 8\pi\left(\frac{e}{\hbar v}\right)^2 \sum_n |\langle n|d_x|0\rangle|^2 (E_n - E_0) \int_{q_{\min}}^{q_0} \frac{dq}{q},$$

and hence

$$\varkappa(q_0) = 8\pi\left(\frac{e}{\hbar v}\right)^2 \sum_n |\langle n|d_x|0\rangle|^2 (E_n - E_0) \ln \frac{q_0 \hbar v}{E_n - E_0}. \tag{149.8}$$

In the range from q_0 to q_1, on the other hand, the summation over n may be carried out first. This reduces $d\varkappa$ to (149.6), whose integration over dq gives

$$\varkappa(q_1) - \varkappa(q_0) = 4\pi \frac{Ze^4}{mv^2} \ln \frac{q_1}{q_0}. \tag{149.9}$$

To transform these results, use the sum rule obtained from (149.4) by setting

$$\hat{f} = \frac{d_x}{e} = \sum_a x_a, \qquad \hat{\dot{f}} = \frac{1}{m}\sum_a \hat{p}_{xa}.$$

Commuting $\hat{f}^+$ with $\hat{\dot{f}}$ gives ($\hat{f}^+$ here is identical to $\hat{f}$) $\hat{\dot{f}}\hat{f}^+ - \hat{f}^+\hat{\dot{f}} = -(i\hbar/m)Z$, so that[44]

$$\sum_n N_{0n} \equiv \sum_n \frac{2m}{(e\hbar)^2}(E_n - E_0)|\langle n|d_x|0\rangle|^2 = Z. \tag{149.10}$$

The quantities N_{0n} are called the *oscillator strengths* of the corresponding transitions. Introduce an average atomic energy I by

$$\ln I = \frac{\sum_n N_{0n} \ln(E_n - E_0)}{\sum_n N_{0n}} = \frac{1}{Z}\sum_n N_{0n} \ln(E_n - E_0). \tag{149.11}$$

Using (149.10), equation (149.8) may be written as $\varkappa(q_0) = 4\pi Z e^4 (mv^2)^{-1} \ln(q_0 \hbar v/I)$. Adding (149.9), we finally obtain

$$\varkappa(q_1) = \frac{4\pi Z e^4}{mv^2} \ln \frac{q_1 \hbar v}{I}. \tag{149.12}$$

Only one constant characteristic of the atom occurs in this formula[45].

[44] The same observation made in connection with (149.5) applies to this relation.

[45] For hydrogen, $I = 0{,}55 me^4/\hbar^2 = 14{,}9$ eV. For heavy atoms, good accuracy may be expected if the constant I is calculated by the Thomas–Fermi method. The dependence on Z of values of I found in this way is readily established. In the quasiclassical case, the differences between energy levels correspond to the natural frequencies of the particle system. The mean natural frequency of an atom is of order v_0/a_0; hence $I \sim \hbar v_0/a_0$. In the Thomas–Fermi model, atomic-electron velocities vary as $Z^{2/3}$, while the atomic dimensions vary as $Z^{-1/3}$. It follows that I must be proportional to Z: $I = \text{const}\cdot Z$. Experimental data give const ~ 10 eV.

Expressing q_1 through the scattering angle ϑ_1 by $q_1 = mv\vartheta_1/\hbar$, we obtain the effective stopping for scattering through all angles $\vartheta \leqslant \vartheta_1$:

$$\varkappa(\vartheta_1) = 4\pi \frac{Ze^4}{mv^2} \ln \frac{mv^2\vartheta_1}{I}. \tag{149.13}$$

If $q_1 a_0 \gg 1$ (that is, $\vartheta_1 \gg v_0/v$), then $\varkappa$ can be expressed as a function of the greatest energy that the incident electron can transfer to the atom. The preceding section showed that, for $qa_0 \gg 1$, the atom is ionized and practically all the momentum $\hbar\mathbf{q}$ and the energy are transferred to one atomic electron. Thus $\hbar q$ and ε are related as the momentum and energy of an electron: $\varepsilon = \hbar^2 q^2/2m$. Substitution of $q_1^2 = 2m\varepsilon_1/\hbar^2$ in (149.12) gives the effective stopping in collisions with energy transfer $\varepsilon \ll \varepsilon_1$:

$$\varkappa(\varepsilon_1) = \frac{2\pi Ze^4}{mv^2} \ln \frac{2m\varepsilon_1 v^2}{I^2}. \tag{149.14}$$

We conclude with the following observation. The discrete energy levels of an atom are associated chiefly with excitation of one (outer) electron; excitation of two electrons already usually requires enough energy to ionize the atom. Consequently, within the sum of oscillator strengths, transitions to discrete-spectrum states contribute only a quantity of order unity, whereas ionizing transitions contribute one of order Z. Collisions accompanied by ionization therefore provide the dominant stopping mechanism for heavy atoms.

Problem

Determine the total effective stopping of an electron by a hydrogen atom ($I = 0{,}55$ atomic units); at large energy transfers, the faster of the two colliding electrons is regarded as the primary electron.

Solution. When the primary and secondary electrons emerge from a collision with comparable energies, exchange must be taken into account. Therefore, for stopping with energy transfer from a value ε_1 ($I \ll \varepsilon_1 \ll v^2$) up to the maximum $\varepsilon_{\max} = E/2 = v^2/4$ (because of our definition of the primary electron), one must use cross section (148.17):

$$\begin{aligned}\varkappa(\varepsilon_{\max}) - \varkappa(\varepsilon_1) &= \frac{\pi}{E} \int_{\varepsilon_1}^{E/2} \varepsilon \left[\frac{1}{\varepsilon^2} + \frac{1}{(E-\varepsilon)^2} - \frac{1}{\varepsilon(E-\varepsilon)} \right] d\varepsilon \\ &= \frac{\pi}{E} \left(\ln \frac{E}{8\varepsilon_1} + 1 \right).\end{aligned}$$

Adding (149.14), we find, in atomic units,[46]

$$\varkappa = \frac{4\pi}{v^2} \ln \left(\frac{v^2}{2I} \sqrt{e/2} \right) = \frac{4\pi}{v^2} \ln \frac{v^2}{0{,}94}.$$

[46]For positron collisions with a hydrogen atom there is no exchange effect, and the total stopping follows simply by putting $\varepsilon_{\max} = E = v^2/2$ for ε_1 in (149.14): $\varkappa = (4\pi/v^2)\ln(v^2/0{,}55)$.

§ 150. Inelastic collisions of heavy particles with atoms

Expressed in terms of the particle speed, the condition for applicability of the Born approximation to collisions of heavy particles with atoms remains the same as for electrons: $v \gg v_0$.

This follows directly from the general condition (126.2) for applicability of perturbation theory, $Ua_0/\hbar v \ll 1$, upon observing that the particle mass does not enter it at all, while $Ua_0/\hbar$ is of the order of the atomic electron speed.

In the coordinate system in which the center of mass of the atom and particle is at rest, the cross section is determined by the general formula (148.3) (where m must now mean the reduced mass of the particle and atom). It is more convenient, however, to consider the collision in the coordinate system in which the scattering atom is at rest before the collision. For this purpose we start from (148.1). In the coordinate system in which the atom was at rest before the collision, the argument of the delta-function expressing energy conservation is

$$\frac{p'^2}{2M} - \frac{p^2}{2M} + \frac{(\mathbf{p}' - \mathbf{p})^2}{2M_a} + E_n - E_0, \tag{150.1}$$

where M is the mass of the incident particle and M_a the mass of the atom; the third term is the atom's recoil kinetic energy (which could be neglected completely in an electron collision).

When a fast heavy particle collides with an atom, the change in the particle's momentum is almost always small compared with its initial momentum. If this condition holds, the atom's recoil energy may be neglected in the argument of the delta-function. We then return exactly to formula (148.3), except that m must be replaced by the mass M of the incident particle (not by the reduced mass of the particle and atom). Since the momentum transfer is assumed small compared with the initial momentum, we put $p \approx p'$. Thus, for the cross section in the coordinate system in which the atom was at rest before the collision,

we obtain

$$d\sigma_n = \frac{M^2}{4\pi^2\hbar^4} \left| \iint Ue^{-i\mathbf{qr}} \psi_n^* \psi_0 \, d\tau \, dV \right|^2 d\Omega. \tag{150.2}$$

To allow for a particle whose charge differs from the electron charge, we shall write ze^2 in place of e^2, where ze is the charge of the incident particle. The general formula for inelastic scattering, expressed in the form (148.9), is

$$d\sigma_n = 8\pi \left(\frac{ze^2}{\hbar v}\right)^2 \left| \left\langle n \left| \sum_a e^{-i\mathbf{qr}_a} \right| 0 \right\rangle \right|^2 \frac{dq}{q^3}, \tag{150.3}$$

contains no particle mass. It follows that all formulas derived from it remain applicable to collisions of heavy particles as well, provided those formulas are expressed in terms of v and q.

It is readily seen how formulas expressed in terms of the scattering angle ϑ (the deflection angle of the heavy particle colliding with the atom) must be modified. First observe that in an inelastic collision of a heavy particle, ϑ is always small. Indeed, when the momentum transfer is large compared with atomic-electron momenta, an inelastic collision with an atom may be regarded as an elastic collision with free electrons; but when a heavy particle collides with a light one (an electron), the heavy particle is hardly deflected. In other words, the momentum transfer from the heavy particle to the atom is small compared with the particle's initial momentum (the exception is elastic scattering through large angles, which is, however, extremely improbable).

Consequently, throughout the angular range one may put

$$q = \frac{1}{\hbar}\sqrt{\left(\frac{E_n - E_0}{v}\right)^2 + (Mv\vartheta)^2}, \tag{150.4}$$

which in practice reduces to

$$q\hbar \approx Mv\vartheta \tag{150.5}$$

except at the very smallest angles. On the other hand, when considering electron collisions with an atom, we wrote (for small angles)

$$q = \frac{1}{\hbar}\sqrt{\left(\frac{E_n - E_0}{v}\right)^2 + (mv\vartheta)^2}.$$

Comparison of the two expressions shows that the formulas obtained for electron collisions with atoms,

when expressed in terms of the speed and deflection angle, are converted into formulas for heavy-particle collisions by making the replacement everywhere (including in the solid-angle element $d\Omega = 2\pi \sin\vartheta\, d\vartheta \approx 2\pi\vartheta\, d\vartheta$)

$$\vartheta \longrightarrow \frac{M}{m}\vartheta, \tag{150.6}$$

at the same incident-particle speed v. Qualitatively, this means that the entire small-angle scattering pattern is narrowed by the factor m/M (at a specified speed).

The rules obtained also apply to elastic scattering at small angles. Applying the transformation (150.6) to (139.4) with $\vartheta \ll 1$, we obtain the cross section

$$d\sigma_e = 8\pi\left(\frac{ze^2}{Mv^2}\right)^2\left[Z - F\left(\frac{Mv\vartheta}{\hbar}\right)\right]^2\frac{d\vartheta}{\vartheta^3}. \tag{150.7}$$

Elastic scattering of heavy particles through angles $\vartheta \sim 1$, on the other hand, reduces to Rutherford scattering by the atomic nucleus.

Inelastic scattering with ionization of the atom at large momentum transfer requires separate consideration. Unlike electron-impact ionization, there are of course no exchange effects here. A characteristic feature of heavy particles is that a large momentum transfer ($qa_0 \gg 1$) does not at all imply deflection through a large angle; ϑ always remains small. The ionization cross section for emission of an electron with energy between ε and $\varepsilon + d\varepsilon$ follows directly from (148.25), which we write as

$$d\sigma_r = 8\pi\left(\frac{ze^2}{\hbar v}\right)^2 Z\frac{dq}{q^3},$$

and in which we put $\hbar^2 q^2/2m = \varepsilon$ (the entire momentum $\hbar\mathbf{q}$ is transferred to one atomic electron). This gives

$$d\sigma_r = \frac{2\pi Z z^2 e^4}{mv^2}\frac{d\varepsilon}{\varepsilon^2}. \tag{150.8}$$

For collisions of heavy particles with atoms, integral effective cross sections and stopping are of particular interest.

The total inelastic-scattering cross section is determined by the previous formula (148.26). The total effective stopping is obtained by substituting in (149.12), for q_1, the greatest possible momentum transfer $q_{\max}$. The latter is readily expressed in terms of the particle speed as follows. Since $\hbar q_{\max}$ is still small compared with the particle's initial momentum Mv,

The corresponding change in its energy is related to the momentum change by $\Delta E = \mathbf{v}\cdot\hbar\mathbf{q}$.

On the other hand, at large momentum transfer essentially all this energy is transferred to one atomic electron, so that we may write

$$\varepsilon = \frac{\hbar^2 q^2}{2m} = \hbar\mathbf{v}\mathbf{q} \leqslant \hbar v q.$$

It follows that $\hbar q \leqslant 2mv$, that is,

$$\hbar q_{\max} = 2mv, \qquad \varepsilon_{\max} = 2mv^2. \tag{150.9}$$

Note that the greatest particle deflection angle in inelastic scattering is

$$\vartheta_{\max} = \frac{\hbar q_{\max}}{Mv} = \frac{2m}{M}.$$

Substitution of (150.9) in (149.12) gives the total effective stopping of a heavy particle:

$$\kappa = \frac{4\pi Z z^2 e^4}{mv^2}\ln\frac{2mv^2}{I}. \tag{150.10}$$

§ 151. Neutron scattering

In a number of physical problems in collision theory, one must determine how the intrinsic motion of the scattering centres affects the scattering process. Under certain conditions these problems can be treated by a special perturbative method developed by Fermi (1936), even though perturbation theory need not apply to scattering by an individual centre. One such question is the scattering of slow neutrons by a system of atoms—for example, by a molecule.

For definiteness, we shall discuss precisely this problem below.

Electrons scatter neutrons to a negligible extent, so that in practice all scattering occurs at the nuclei[47]. We assume that the scattering amplitude of an individual nucleus is small compared with the interatomic distances. The amplitude of the wave

[47] It is also assumed that the molecule has no magnetic moment. Otherwise there is an additional, specific scattering effect associated with the interaction between the magnetic moments of the molecule and the neutron.

scattered by each nucleus in the molecule is then already small at the positions of the other nuclei. Under these conditions, the molecular scattering amplitude reduces to the sum of the amplitudes for scattering by the individual nuclei.

In general, perturbation theory does not apply to a collision between a neutron and a nucleus: although nuclear forces have a short range, they are very large within that range. What matters, however, is that the scattering amplitude for a slow neutron (whose wavelength is large compared with the nuclear dimensions) is a constant independent of velocity. Let f_a be the amplitude for scattering by nucleus a; then $|f_a|^2 d\Omega$ is the differential cross-section for elastic neutron scattering by a free nucleus (in their centre-of-mass frame).

This constant amplitude may be obtained formally from perturbation theory if the interaction of the neutron with the nucleus is represented by the "point" potential energy

$$U(\mathbf{r}) = -\frac{2\pi\hbar^2}{M} f\delta(\mathbf{r}), \tag{151.1}$$

where M is the reduced mass of the neutron and nucleus. On substituting this expression in the Born formula (126.4), the delta function turns the integral into a constant independent of $\mathbf{q}$. The "field" $U(\mathbf{r})$ defined in this way is called a *pseudopotential*. We stress that its introduction is possible precisely because f is constant.

For a neutron of arbitrary energy, the scattering amplitude generally depends separately on the initial and final momenta $\mathbf{p}$ and $\mathbf{p}'$, and not merely on their difference $\mathbf{q}$; an amplitude calculated in the Born approximation, by contrast, can depend only on $\mathbf{q}$[48].

If the scattering nucleus undergoes a prescribed motion (molecular vibrations, for example), averaging over this motion "smears" the interaction (151.1) over a region whose dimensions are, in general, large compared with the scattering amplitude f. Such a smeared interaction satisfies condition (126.1) for applicability of the Born approximation.

We shall therefore describe the interaction of the neutron with the molecule by the pseudopotential

$$U(\mathbf{r}) = -2\pi\hbar^2 \sum_a \frac{f_a}{M_a}\delta(\mathbf{r} - \mathbf{R}_a), \tag{151.2}$$

where the sum runs over all nuclei in the molecule, $\mathbf{R}_a$ are their radius vectors, and $\mathbf{r}$ is the neutron radius vector. Substitution in the perturbation-theory formula (148.3), with the reduced mass M_m of the molecule and neutron in place of m, gives the following expression for the cross-section of neutron scattering by the molecule in their centre-of-mass frame:

$$d\sigma_n = M_\mathrm{m}^2 \frac{p'}{p} \left| \sum_a \frac{f_a}{M_a} \langle n | e^{-i\mathbf{q}\mathbf{R}_a} | 0 \rangle \right|^2 d\Omega. \tag{151.3}$$

[48] We also stress that, although the pseudopotential gives the correct scattering amplitude when perturbation theory is applied formally, this by no means implies that perturbation theory is actually applicable to such a field. On the contrary, for a potential well whose depth U_0 tends to infinity according to $U_0 a^3$ = const (where a, the well radius, tends to zero), conditions (126.1) and (126.2) are certainly not satisfied.

The matrix elements here are taken with the wave functions of stationary states of nuclear motion having energies E_0 and E_n, while p and p' are related by energy conservation:

$$\frac{p^2 - p'^2}{2M_\mathrm{m}} = E_n - E_0.$$

Formula (151.3) describes an inelastic collision accompanied by a specified change in the state of nuclear motion in the molecule (a transition $0 \longrightarrow n$). It solves the problem posed: from the assumed known amplitudes for neutron scattering by free nuclei, it determines the molecular scattering cross-section while allowing both for the intrinsic motion of the nuclei and for interference between scattering at different nuclei.

If the nuclei have nonzero spin, one must also allow for the dependence of the scattering amplitudes f_a on the total spin of the scattering nucleus and neutron. This can be done as follows.

The total spin of nucleus a and the neutron may take the two values $j_a = i_a \pm 1/2$, where i_a is the nuclear spin. Denote the corresponding scattering amplitudes by f_a^+ and f_a^-. We construct a spin operator whose eigenvalues at the two values of j_a are respectively f_a^+ and f_a^-. Such an operator is

$$\hat{f}_a = a_a + b_a \hat{\mathbf{s}}\hat{\mathbf{i}}_a, \tag{151.4}$$

where $\hat{\mathbf{i}}_a$ and $\hat{\mathbf{s}}$ are the nuclear and neutron spin operators, and

$$a_a = \frac{1}{2i_a + 1}\left[(i_a + 1) f_a^+ + i_a f_a^-\right], \qquad b_a = \frac{2}{2i_a + 1}(f_a^+ - f_a^-). \tag{151.5}$$

This follows at once by noting that, for a specified j, the eigenvalue of $\mathbf{si}$ is

$$\mathbf{si} = \frac{1}{2}\left[j(j+1) - i(i+1) - \frac{3}{4}\right].$$

The operators (151.4), rather than f_a, must be substituted in (151.3), with their matrix elements taken for the transition under consideration. If neither the incident neutrons nor the target nuclei are polarized, the scattering cross-section must be averaged accordingly.

Problems

Problem

1. Average formula (151.3), assuming completely random orientations of the neutron and nuclear spins. All nuclei in the molecule are distinct.

Solution. The averages over the neutron-spin and nuclear-spin directions are independent, and each spin averages to zero; hence $\overline{\mathbf{si}_a} = 0$. If the molecule contains no identical atoms, there is no exchange interaction between nuclear spins; because their direct interaction is negligible, the spin directions of different nuclei in the molecule may be regarded as independent. Products of the form $(\mathbf{si}_1)(\mathbf{si}_2)$ therefore also vanish on averaging. For the squares $(\mathbf{si})^2$, however,

$$\overline{(\mathbf{si})^2} = \frac{1}{3}\mathbf{s}^2\mathbf{i}^2 = \frac{s(s+1)i(i+1)}{3} = \frac{i(i+1)}{4}.$$

The averaged cross-section is consequently

$$d\sigma_n = M_{\mathrm{m}}^2 \frac{p'}{p} \left\{ \left| \sum_a \frac{a_a}{M_a} \langle n|e^{-i\mathbf{qR}_a}|0\rangle \right|^2 + \frac{1}{4} \sum_a \frac{i_a(i_a+1)}{M_a^2} b_a^2 \left|\langle n|e^{-i\mathbf{qR}_a}|0\rangle\right|^2 \right\} d\Omega.$$

Problem

2. Apply formula (151.3) to the scattering of slow neutrons by para- and orthohydrogen (J. Schwinger, E. Teller, 1937).

Solution. Before the matrix elements of the spin operators are evaluated, expression (151.3) for scattering by an H_2 molecule has the form

$$d\sigma_n = \frac{16p'}{9p} \left| a\langle n|e^{-i\mathbf{qr}/2} + e^{i\mathbf{qr}/2}|0\rangle + b\hat{\mathbf{s}}\langle n|\hat{\mathbf{i}}_1 e^{-i\mathbf{qr}/2} + \hat{\mathbf{i}}_2 e^{i\mathbf{qr}/2}|0\rangle \right|^2 d\Omega, \qquad (1)$$

$$a = \frac{1}{4}(3f^+ + f^-), \qquad b = f^+ - f^-$$

(the radius vectors of the two nuclei relative to the molecular centre of mass are $\pm\mathbf{r}/2$).

The rotational and vibrational states of the molecule are specified by the quantum numbers K, M_K, and v, whose collection is to be understood as n in (1). In the ground electronic state of H_2, even K can occur only for total nuclear spin $I = 0$ (parahydrogen), and odd K only for $I = 1$ (orthohydrogen); see § 86. Two cases must therefore be distinguished: (1) transitions between rotational states whose K values have the same parity, possible only without a change of I (ortho–ortho and para–para transitions); (2) transitions between states whose K values have opposite parity, possible only with a change of I (ortho–para and para–ortho transitions).

In the first case,

$$\langle n|e^{-i\mathbf{qr}/2}|0\rangle = \langle n|e^{i\mathbf{qr}/2}|0\rangle = \left\langle n \left| \cos\frac{\mathbf{qr}}{2} \right| 0 \right\rangle$$

(recall that the rotational wave function is multiplied by $(-1)^K$ when the sign of $\mathbf{r}$ is reversed). The spin operator in (1) then becomes $2a + b\hat{\mathbf{s}}\hat{\mathbf{I}}$, where $\hat{\mathbf{I}} = \hat{\mathbf{i}}_1 + \hat{\mathbf{i}}_2$. In accordance with the preceding argument, this operator is diagonal in I. The square $(2a + b\mathbf{sI})^2$ is averaged as in problem 1 and gives

$$4a^2 + \frac{b^2}{4} I(I+1).$$

It follows that

$$d\sigma_n = \frac{4p'}{9p} \left| \left\langle n \left| \cos\frac{\mathbf{qr}}{2} \right| 0 \right\rangle \right|^2 \left\{ (3f^+ + f^-)^2 + I(I+1)(f^+ - f^-)^2 \right\} d\Omega. \qquad (2)$$

In the second case,

$$\langle n|e^{i\mathbf{qr}/2}|0\rangle = -\langle n|e^{-i\mathbf{qr}/2}|0\rangle = i\left\langle n \left| \sin\frac{\mathbf{qr}}{2} \right| 0 \right\rangle,$$

and the spin operator in (1) reduces to $\hat{\mathbf{s}}(\hat{\mathbf{i}}_1 - \hat{\mathbf{i}}_2)$; it has only matrix elements off diagonal in I. The squared modulus of these elements, summed over all possible projections of the final total spin I', is calculated as the mean value (diagonal element) of the square $(\mathbf{s}, \mathbf{i}_1 - \mathbf{i}_2)^2$ (see the note on p. 704) and is

$$\overline{(\mathbf{s}, \mathbf{i}_1 - \mathbf{i}_2)^2} = \frac{1}{3} \cdot \frac{3}{4} \overline{(\mathbf{i}_1 - \mathbf{i}_2)^2} = \frac{1}{4}(2\mathbf{i}_1^2 + 2\mathbf{i}_2^2 - \mathbf{I}^2) = \frac{1}{4}[3 - I(I+1)].$$

Thus

$$d\sigma_n = (1)(3)\frac{4p'}{9p}\left|\left\langle n\left|\sin\frac{\mathbf{qr}}{2}\right|0\right\rangle\right|^2 (f^+ - f^-)^2 d\Omega, \tag{3}$$

where the factor (1) applies to ortho–para transitions and the factor (3) to para–ortho transitions.

If the neutrons are so slow that their wavelength is also large compared with the molecular dimensions, one may put $\cos(qr/2) = 1$ and $\sin(qr/2) = 0$ in the matrix elements in (2) and (3). All of them then vanish except the diagonal element 00; naturally, only elastic scattering is possible under these conditions. Its cross-section is

$$d\sigma_e = \frac{4}{9}\left[(3f^+ + f^-)^2 + I(I+1)(f^+ - f^-)^2\right] d\Omega.$$

Problem

3. Find the cross-section for neutron scattering by a bound proton treated as an isotropic spatial oscillator of frequency ω (E. Fermi, 1936).

Solution. Since the proton is regarded as oscillating about a point fixed in space, the meaning of the derivation of (151.3) requires us to set $M_\mathrm{m} = M$ and $M_a = M/2$ in that formula, where M is the proton mass. Then

$$d\sigma_n = \frac{p'}{p}\frac{\sigma_0}{\pi}\sum\left|\int e^{-i\mathbf{qr}}\psi_{000}(\mathbf{r})\psi_{n_1n_2n_3}(\mathbf{r})\,dV\right|^2 d\Omega,$$

where $\sigma_0 = 4\pi|f|^2$ is the scattering cross-section for a free proton, and $\psi_{n_1n_2n_3}$ are the eigenfunctions of the spatial oscillator corresponding to the energy levels $E_n = \hbar\omega(n + 3/2)$. The sum is over all n_1, n_2, n_3 with the specified sum $n_1 + n_2 + n_3 = n$. The functions $\psi_{n_1n_2n_3}$ are products of the wave functions of three linear oscillators (see problem 4 in § 33). The required integral therefore separates into a product of three integrals of the form

$$\int_{-\infty}^{\infty}\exp\left(-\frac{iq_x x}{2} - \frac{\alpha^2x^2}{2} - \frac{\alpha^2x^2}{2}\right)H_{n_1}(\alpha x)\,dx$$

with $\alpha = \sqrt{M\omega/\hbar}$. These integrals are evaluated by substituting $H_n(x)$ in the form (a.4) and integrating by parts n times. The result is

$$d\sigma_n = \frac{1}{\pi}\frac{v'}{v}\frac{\sigma_0}{2^n\alpha^{2n}}\sum\frac{q_x^{2n_1}q_y^{2n_2}q_z^{2n_3}}{n_1!n_2!n_3!}\exp\left(-\frac{q^2}{2\alpha^2}\right)d\Omega.$$

The sum is evaluated by the multinomial formula, giving finally

$$d\sigma_n = \frac{\sigma_0}{\pi n!}\sqrt{\frac{E'}{E}}\left(\frac{q^2}{2\alpha^2}\right)^n \exp\left(-\frac{q^2}{2\alpha^2}\right) d\Omega.$$

In particular, the elastic-scattering cross-section ($n = 0$, $E = E'$) is

$$d\sigma_e = \frac{\sigma_0}{\pi}\exp\left(-\frac{q^2}{2\alpha^2}\right) d\Omega, \qquad \sigma_e = \sigma_0\frac{\hbar\omega}{E}\left[1 - \exp\left(-\frac{4E}{\hbar\omega}\right)\right].$$

If $E/\hbar\omega \longrightarrow 0$, then $\sigma_e \longrightarrow 4\sigma_0$.

§ 152. Inelastic scattering at high energies

The eikonal approximation used in § 131 for the mutual scattering of two particles can be generalized so as to include processes, among them inelastic ones, occurring when a fast particle collides with a system of particles—a "target" (R. J. Glauber, 1958).

The basic assumptions remain the same in this generalization. The incident-particle energy E is taken to be so large that $E \gg |U|$ and $ka \gg 1$, where U is the energy of its interaction with the target particles and a is the range of that interaction. We consider scattering with a relatively small momentum transfer: the change $\hbar\mathbf{q}$ in the incident particle's momentum is small compared with its initial momentum $\hbar\mathbf{k}$, so that $q \ll k$. Here, however, this condition implies not only a small scattering angle but also a comparatively small energy transfer.

We shall also suppose that the incident particle's speed v is large compared with the speeds v_0 of particles within the target:

$$v \gg v_0. \tag{152.1}$$

For the scattering of charged particles by atoms, this condition is equivalent to the applicability of the Born approximation (compare §§ 148 and 150): $v \gg v_0$ automatically entails $|U|a/\hbar v \ll 1$. Thus the theory developed here is not needed in that case. The situation is different for nuclear targets, whose constituents are bound by nuclear rather than Coulomb forces. For definiteness, we shall below speak of a fast particle scattered by a nucleus[49].

Condition (152.1) permits the incident particle's motion to be considered at prescribed positions of the nucleons in the nucleus[50]. In other words, the wave function of the particle–target system can be written as

$$\psi(\mathbf{r}, \mathbf{R}_1, \mathbf{R}_2, \ldots) = \varphi(\mathbf{r}; \mathbf{R}_1, \mathbf{R}_2, \ldots)\Phi_i(\mathbf{R}_1, \mathbf{R}_2, \ldots). \tag{152.2}$$

Here $\Phi_i(\mathbf{R}_1, \mathbf{R}_2, \ldots)$ is the wave function of some internal state i of the nucleus ($\mathbf{R}_1, \mathbf{R}_2, \ldots$ are the position vectors of its nucleons). The factor $\varphi(\mathbf{r}; \mathbf{R}_1, \mathbf{R}_2, \ldots)$ is

[49] For nuclei of any appreciable mass, condition (152.1) requires relativistic values of v. In presenting the formalism of this section within nonrelativistic theory, we leave aside the question of its actual applicability to particular scattering processes.

[50] This approximation is analogous to the one underlying molecular theory, in which the electronic state is considered for a prescribed arrangement of the nuclei.

the wave function of the scattered particle ($\mathbf{r}$ is its position vector) for fixed values of $\mathbf{R}_1, \mathbf{R}_2, \ldots$, which serve as parameters in the Schrödinger equation

$$\left[-\frac{\hbar^2}{2m}\Delta + \sum_a U_a(\mathbf{r}-\mathbf{R}_a)\right]\varphi = \frac{\hbar^2 k^2}{2m}\varphi, \tag{152.3}$$

where $U_a(\mathbf{r}-\mathbf{R}_a)$ is the interaction energy of the particle with nucleon a, and $\hbar\mathbf{k}$ is its momentum at infinity[51].

If a solution of (152.3) is found with the asymptotic form

$$\varphi = e^{i\mathbf{kr}} + F(\mathbf{n}', \mathbf{n}; \mathbf{R}_1, \mathbf{R}_2, \ldots)\frac{e^{ikr}}{r}, \tag{152.4}$$

where $\mathbf{n}' = \mathbf{r}/r$ and $\mathbf{n} = \mathbf{k}/k$, then the wave function (152.2),

$$\psi = e^{i\mathbf{kr}}\Phi_i + F\Phi_i\frac{e^{ikr}}{r}, \tag{152.5}$$

describes scattering by a nucleus which, before the collision, is in its state i: in (152.5), the incident wave $e^{i\mathbf{kr}}$ occurs multiplied by Φ_i.

The second term in (152.5) represents the scattered wave. This expression can be used to determine the scattering amplitude only when the incident particle's energy changes by sufficiently little, that is, when the change in the nucleus's internal energy is small. Indeed, in treating the particle's motion in the constant field of nucleons "fixed in place," as in (152.3), we thereby neglect a possible change in the energy of this motion.

To extract the scattering amplitude for a definite change of the nucleus's internal state, ψ must be expanded in the form

$$\psi = e^{i\mathbf{kr}}\Phi_i + \sum_f f_{fi}(\mathbf{n}', \mathbf{n})\Phi_f\frac{e^{ikr}}{r}, \tag{152.6}$$

where the sum runs over the different states of the nucleus. The quantity $f_{fi}(\mathbf{n}', \mathbf{n})$ is then the desired amplitude for scattering accompanied by the specified nuclear transition $i \to f$, as a function of the scattering angle (the angle between $\mathbf{n}$ and $\mathbf{n}'$). Comparison of (152.6) and (152.5) gives

$$f_{fi}(\mathbf{n}', \mathbf{n}) = \int \Phi_f^* F \Phi_i \, d\tau, \tag{152.7}$$

where $d\tau = d^3R_1 d^3R_2 \ldots$ is the volume element in the nucleus's configuration space. We emphasize again that this formula applies only when the energy difference between states i and f is relatively small.

The solution (152.4) of (152.3) is itself obtained by the method described in § 131[52]. In analogy with (131.7), we have

$$F(\mathbf{n}', \mathbf{n}; \mathbf{R}_1, \mathbf{R}_2, \ldots) = \frac{k}{2\pi i}\int [S(\boldsymbol{\rho}, \mathbf{R}_1, \mathbf{R}_2, \ldots) - 1]e^{-i\mathbf{q}\boldsymbol{\rho}}\, d^2\rho, \tag{152.8}$$

[51] Equation (152.3) assumes that the particle's interaction with the nucleus reduces to the sum of its pairwise interactions with the individual nucleons.

[52] It was noted in § 131 that the initial expression (131.4) for the wave function applies only at distances $a \ll ka^2$. This fact did not affect the subsequent derivation in § 131. For scattering by a system of particles (a nucleus), however, it gives an additional restriction: expression (131.4) must be valid throughout the entire scattering system. Hence $R_0 \ll ka^2$, where R_0 is the nuclear radius and a is the range of the potentials U.

with the notation

$$
\begin{aligned}
S(\boldsymbol{\rho}, \mathbf{R}_1, \mathbf{R}_2, \dots) &= \exp[2i\delta(\boldsymbol{\rho}, \mathbf{R}_1, \mathbf{R}_2, \dots)],\\
\delta(\boldsymbol{\rho}, \mathbf{R}_1, \mathbf{R}_2, \dots) &= \sum_a \delta_a(\boldsymbol{\rho} - \mathbf{R}_{a\perp}),\\
\delta_a(\boldsymbol{\rho} - \mathbf{R}_{a\perp}) &= -\frac{1}{2\hbar v}\int_{-\infty}^{\infty} U_a(\mathbf{r} - \mathbf{R}_a)\,dz.
\end{aligned}
\tag{152.9}
$$

Recall that $\boldsymbol{\rho}$ is the projection of the position vector $\mathbf{r}$ on the xy plane perpendicular to $\mathbf{k}$ (and $\mathbf{R}_{a\perp}$ is the corresponding projection of $\mathbf{R}_a$). Further, $\hbar\mathbf{q} = \mathbf{p}' - \mathbf{p}$ is the change in the scattered particle's momentum, of which only the transverse components occur in (152.8). The functions δ_a determine the amplitudes for elastic scattering by the separate free nucleons according to

$$
f^{(a)} = \frac{k}{2\pi i}\int \{\exp[2i\delta_a(\boldsymbol{\rho})] - 1\} e^{-i\mathbf{q}\boldsymbol{\rho}}\,d^2\rho. \tag{152.10}
$$

For $i = f$, equations (152.7) and (152.8) give the amplitude for elastic scattering by the nucleus:

$$
f_{ii}(\mathbf{n}', \mathbf{n}) = \frac{k}{2\pi i}\int [\overline{S}(\boldsymbol{\rho}) - 1] e^{-i\mathbf{q}\boldsymbol{\rho}}\,d^2\rho, \tag{152.11}
$$

where the bar denotes averaging over the internal state of the nucleus:

$$
\overline{S}(\boldsymbol{\rho}) = \int S(\boldsymbol{\rho}, \mathbf{R}_1, \mathbf{R}_2, \dots)|\Phi_i(\mathbf{R}_1, \mathbf{R}_2, \dots)|^2\,d\tau. \tag{152.12}
$$

This is a generalization of the earlier formula (131.7).

Putting $\mathbf{n}' = \mathbf{n}$ in (152.11) and using the optical theorem (142.10), we obtain the total scattering cross-section

$$
\sigma_t = 2\int (1 - \operatorname{Re}\overline{S})\,d^2\rho. \tag{152.13}
$$

The integrated elastic-scattering cross-section σ_e is obtained by integrating $|f_{ii}|^2$ over the directions $\mathbf{n}'$. At small scattering angles θ, we have $q \simeq k\theta$ and the solid-angle element is $d\Omega \simeq d^2q/k^2$. Hence

$$
\sigma_e = \int |f_{ii}|^2 \frac{d^2q}{k^2}.
$$

Writing $f_{ii}f_{ii}^*$, with f_{ii} from (152.11), as a double integral over $d^2\rho\,d^2\rho'$, we integrate over d^2q by means of

$$
\int e^{-i\mathbf{q}(\boldsymbol{\rho}-\boldsymbol{\rho}')}\,d^2q = (2\pi)^2\delta(\boldsymbol{\rho} - \boldsymbol{\rho}'),
$$

after which the delta function is removed by integration. The result is

$$
\sigma_e = \int |\overline{S} - 1|^2\,d^2\rho. \tag{152.14}
$$

Finally, the total reaction cross-section is

$$\sigma_r = \sigma_t - \sigma_e = \int (1 - |\overline{S}|^2)\, d^2\rho. \tag{152.15}$$

Notice the correspondence between (152.13)–(152.15) and the general formulas (142.3)–(142.5). If in the latter we pass from summation over large l to integration over $d^2\rho$, with $\rho = l/k$, and replace S_l by the function $\overline{S}(\rho)$, we obtain (152.13)–(152.15).

Problem

1. Express the amplitude for elastic scattering of a fast particle by a deuteron in terms of the amplitudes for scattering by a proton and a neutron (R. J. Glauber, 1955).

Solution. According to (152.11), the amplitude for elastic scattering by a deuteron is

$$\begin{aligned} f^{(d)}(\mathbf{q}) = \frac{k}{2\pi i} \int |\psi_d(\mathbf{R})|^2 \Big\{ \exp\Big[2i\delta_n\Big(\boldsymbol{\rho} - \frac{\mathbf{R}_\perp}{2}\Big) \\ + 2i\delta_p\Big(\boldsymbol{\rho} + \frac{\mathbf{R}_\perp}{2}\Big)\Big] - 1\Big\} e^{-i\mathbf{q}\boldsymbol{\rho}}\, d^3R\, d^2\rho. \end{aligned} \tag{1}$$

Here $\psi_d(\mathbf{R})$ is the wave function of the relative motion of the neutron (n) and proton (p) in the deuteron; $\mathbf{R} = \mathbf{R}_n - \mathbf{R}_p$, while $\mathbf{R}_\perp$ is the projection of $\mathbf{R}$ on the plane perpendicular to the incident particle's wave vector $\mathbf{k}$. Write the difference in braces in (1) as

$$e^{2i\delta_n + 2i\delta_p} - 1 = (e^{2i\delta_n} - 1) + (e^{2i\delta_p} - 1) + (e^{2i\delta_n} - 1)(e^{2i\delta_p} - 1).$$

The integrals can then be transformed by using the definition (152.10) of the neutron and proton scattering amplitudes, $f^{(n)}$ and $f^{(p)}$, together with the inverse formulas.

We obtain

$$\begin{aligned} f^{(d)}(\mathbf{q}) = f^{(n)}(\mathbf{q})F(\mathbf{q}) + f^{(p)}(\mathbf{q})F(-\mathbf{q}) \\ - \frac{1}{2\pi i k} \int F(2\mathbf{q}') f^{(n)}\Big(\frac{\mathbf{q}}{2} + \mathbf{q}'\Big) f^{(p)}\Big(\frac{\mathbf{q}}{2} - \mathbf{q}'\Big)\, d^2q', \end{aligned} \tag{2}$$

where

$$F(\mathbf{q}) = \int |\psi_d(\mathbf{R})|^2 e^{-i\mathbf{q}\mathbf{R}/2}\, d^3R$$

is the deuteron form factor. Setting $\mathbf{q} = 0$ in (2), with $F(0) = 1$, and using the optical theorem (142.10), we find the total cross-section for scattering by the deuteron:

$$\sigma_t^{(d)} = \sigma_t^{(n)} + \sigma_t^{(p)} + \frac{2}{k^2} \operatorname{Re} \int F(2\mathbf{q}) f^{(n)}(\mathbf{q}) f^{(p)}(-\mathbf{q})\, d^2q. \tag{3}$$

2. Determine the cross-section for breakup of a fast deuteron into a free neutron and proton when it is scattered by a heavy absorbing nucleus. The nuclear radius R_0 is

large compared both with the deuteron's wavelength ($kR_0 \gg 1$, where $\hbar k$ is the deuteron momentum) and with the deuteron's radius (E. L. Feinberg, 1954; R. J. Glauber, 1955; A. I. Akhiezer and A. G. Sitenko, 1955).

SOLUTION. For an incident deuteron plane wave, a large ($kR_0 \gg 1$) absorbing nucleus acts as an opaque screen at which the wave diffracts. The wave function of the incident deuterons is $e^{i\mathbf{kr}}\psi_d(\mathbf{R})$, where $\psi_d(\mathbf{R})$ is the internal deuteron wave function.

Here $\mathbf{R} = \mathbf{R}_n - \mathbf{R}_p$ is the vector from the proton to the neutron in the deuteron, and $\mathbf{r} = (\mathbf{R}_n + \mathbf{R}_p)/2$ is the position vector of their centre of mass. The absorbing nucleus "cuts out" the part of this function for which the transverse coordinates of the neutron and proton, $\boldsymbol{\rho}_n$ and $\boldsymbol{\rho}_p$, fall within the nucleus's "shadow," namely inside the circle of radius R_0. In other words, the wave function becomes

$$\psi = e^{i\mathbf{kr}} S(\boldsymbol{\rho}_n, \boldsymbol{\rho}_p)\psi_d(\mathbf{R}),$$

where $S = 1$ for $\rho_n, \rho_p \geqslant R_0$, while $S = 0$ if either ρ_n or ρ_p is less than R_0[53]. Apart from the factor ψ_d, this function corresponds to the form (131.5) of the incident wave, in which diffraction-induced bending of rays is ignored. The factor S therefore has the same meaning as in §§ 131 and 152.

In analogy with (152.13) and (152.14), the total deuteron scattering cross-section σ_t, including all inelastic processes, and the elastic-scattering cross-section σ_e are

$$\sigma_t = 2\int (1 - \overline{S})\, d^2\rho, \qquad \sigma_e = \int (\overline{S} - 1)^2\, d^2\rho,$$

where $\boldsymbol{\rho} = (\boldsymbol{\rho}_n + \boldsymbol{\rho}_p)/2$ and the reality of S has been used; S is averaged over the deuteron ground state:

$$\overline{S}(\boldsymbol{\rho}) = \int S|\psi_d(\mathbf{R})|^2\, d^3R.$$

We use the approximate wave function

$$\psi_d(R) = \sqrt{\frac{\varkappa}{2\pi}}\frac{e^{-\varkappa R}}{R},$$

valid at distances R beyond the range of the nuclear forces acting between neutron and proton (compare (133.14)); here $\varkappa = \sqrt{m|E_d|}/\hbar$, $|E_d|$ is the deuteron binding energy, and m is the nucleon mass. By definition of S, the difference $1 - S$ is nonzero if one or both nucleons enter the circle of radius R_0 and are absorbed by the nucleus. Therefore

$$\sigma_{\text{cap}} = \int (1 - \overline{S})\, d^2\rho = \frac{1}{2}\sigma_t \tag{1}$$

is the cross-section for capture of one or both nucleons. On the other hand, $\sigma_t = \sigma_{\text{cap}} + \sigma_e + \sigma_{\text{br}}$, where σ_{br} is the "diffractive" deuteron-breakup cross-section sought. Thus

$$\sigma_{\text{br}} = \frac{1}{2}\sigma_t - \sigma_e = \int \overline{S}(1 - \overline{S})\, d^2\rho. \tag{2}$$

[53] The Coulomb interaction between the deuteron and the nucleus is neglected.

For $R_0\varkappa \gg 1$, only small distances, of order $1/\varkappa$, from the nuclear edge contribute materially to the integral (2). Integration along the edge then gives a factor $2\pi R_0$, while integration in the perpendicular direction may be performed as though the shadow region were bounded by a straight line. Taking this line as the y axis and directing the x axis outward from the shadow, we have

$$\sigma_{\text{br}} = 2\pi R_0 \int_0^{\infty} \overline{S}(x)[1 - \overline{S}(x)]\, dx.$$

Here the integral

$$\overline{S}(x) = \int_{-2x}^{2x} \int \int |\psi_d(\mathbf{R})|^2\, dX\, dY\, dZ, \qquad R = \sqrt{X^2 + Y^2 + Z^2},$$

is taken over the region $X_n, X_p \geqslant 0$ for a fixed value of $x = (X_n + X_p)/2$, or equivalently over $|X| = |X_n - X_p| \leqslant 2x$. It is transformed by passing to X, R, and the polar angle in the YZ plane, with $dY\, dZ \to 2\pi R\, dR$, and becomes

$$\overline{S}(x) = 1 - e^{-4\varkappa x} + 4\varkappa x \int_{4\varkappa x}^{\infty} \frac{e^{-\xi}}{\xi}\, d\xi. \tag{3}$$

The integral (2), with this function $\overline{S}(x)$, is evaluated by repeated integrations by parts, using

$$\int_0^{\infty} x^n \ln x\, e^{-ax}\, dx = \frac{n!}{a^{n+1}} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} - C - \ln a\right).$$

The result is

$$\sigma_{\text{br}} = \frac{\pi R_0}{3\varkappa} \left(\ln 2 - \frac{1}{4}\right).$$

Under the same condition $\varkappa R_0 \gg 1$, the capture cross-section is

$$\sigma_{\text{cap}} = \pi R_0^2 + \frac{\pi R_0}{4\varkappa}$$

(the integral (1) over $\rho > R_0$ is evaluated with (3), while the integral over $\rho < R_0$ gives πR_0^2). This cross-section includes both capture of the deuteron as a whole and capture of only one nucleon with release of the other (the stripping reaction). The cross-section for the latter is calculated as the impact area, averaged with $|\psi_d|^2$, corresponding to the entry of just one of the two nucleons into the shadow region, and is

$$\sigma_{\text{cap}\, n} = \sigma_{\text{cap}\, p} = \frac{\pi R_0}{4\varkappa}$$

(R. Serber, 1947).

Mathematical Supplements

§ a. Hermite Polynomials

The equation

$$y'' - 2xy' + 2ny = 0 \tag{a.1}$$

belongs to a class of equations soluble by the *Laplace method*[54].

In general, the method applies to linear equations of the form

$$\sum_{m=0}^{n} (a_m + b_m x) \frac{d^m y}{dx^m} = 0,$$

whose coefficients are at most linear in x. Its procedure is as follows. Form the polynomials

$$P(t) = \sum_{m=0}^{n} a_m t^m, \qquad Q(t) = \sum_{m=0}^{n} b_m t^m$$

and from them construct

$$Z(t) = \frac{1}{Q} \exp \int \frac{P}{Q}\, dt,$$

which is determined up to a constant factor. A solution of the equation can then be written as the complex integral

$$y = \int_C Z(t) e^{xt}\, dt.$$

The path C is to be chosen so that this integral is finite and nonzero, and the function

$$V = e^{xt} QZ$$

must return to its initial value after t has traversed the whole of C; the contour may be either closed or open. For equation (a.1),

$$P = t^2 + 2n; \qquad Q = -2t, \qquad Z = -\frac{1}{2t^{n+1}} e^{-t^2/4}, \qquad V = \frac{1}{t^n} e^{xt - t^2/4},$$

and hence its solution takes the form

$$y = \int \exp\left(xt - \frac{t^2}{4}\right) \frac{dt}{t^{n+1}}. \tag{a.2}$$

For physical applications it is sufficient to consider $n > -1/2$. At these values either C_1 or C_2 in Fig. 52 may serve as the integration path. The required conditions are satisfied because V vanishes at their endpoints, $t = +\infty$ or $t = -\infty$[55].

[54] See, for example, É. Goursat, *Course in Mathematical Analysis*, vol. II, Moscow: Gostekhizdat, 1933; V. I. Smirnov, *Course of Higher Mathematics*, vol. III, part 2, Moscow: Nauka, 1974.

[55] These paths cannot be used for negative integral n, since the integral (a.2) along either path would then vanish identically.

We shall determine the values of n for which (a.1) has a solution that is finite at every finite x and, as $x \to \pm\infty$, grows no faster than a finite power of x. Begin with nonintegral n. Integration in (a.2) along C_1 and C_2 gives two independent solutions. Transform the C_1 integral by introducing u through $t = 2(x - u)$. On omitting a constant factor, the result is

$$y = e^{x^2} \int_{C_1'} \frac{e^{-u^2}}{(u-x)^{n+1}}\, du, \tag{a.3}$$

where C_1' is the contour in the complex u plane shown in Fig. 53.

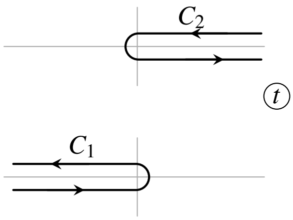


Figure 52

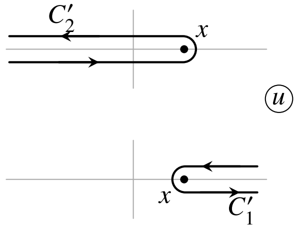


Figure 53

As $x \to +\infty$, the whole contour C_1' recedes to infinity, and the integral in (a.3) tends to zero as e^{-x^2}. For $x \to -\infty$, however, the integration path extends along the entire real axis and the integral in (a.3) does not decrease exponentially; the leading growth of $y(x)$ is therefore e^{x^2}. In the same way, one readily verifies that the integral (a.2) over C_2 diverges exponentially as $x \to \infty$.

When n is a nonnegative integer, the integrals over the straight parts of the path cancel pairwise. Both versions of (a.3), over C_1' and C_2', then reduce to an integral over a closed contour surrounding the point $u = x$. We thus obtain the solution

$$y(x) = e^{x^2} \oint \frac{e^{-u^2}}{(u-x)^{n+1}}\, du,$$

which has the required properties. By Cauchy's familiar formula for the derivatives of an analytic function,

$$f^{(n)}(x) = \frac{n!}{2\pi i} \oint \frac{f(t)}{(t-x)^{n+1}}\, dt,$$

this solution, apart from a constant factor, is the *Hermite polynomial*

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}. \tag{a.4}$$

Expanded and ordered by descending powers of x, H_n is

$$H_n(x) = (2x)^n - \frac{n(n-1)}{1}(2x)^{n-2} + \frac{n(n-1)(n-2)(n-3)}{1 \cdot 2}(2x)^{n-4} - \ldots \tag{a.5}$$

It contains only powers of x having the same parity as n. The first few Hermite polynomials are

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x, \\ H_4 = 16x^4 - 48x^2 + 12. \tag{a.6}$$

To evaluate the normalization integral, substitute the expression (a.4) for $e^{-x^2}H_n$ and integrate by parts n times:

$$\int_{-\infty}^{+\infty} e^{-x^2} H_n^2(x)\,dx = \int_{-\infty}^{+\infty} (-1)^n H_n(x) \frac{d^n}{dx^n} e^{-x^2}\,dx = \int_{-\infty}^{+\infty} e^{-x^2} \frac{d^n H_n}{dx^n}\,dx.$$

But $d^n H_n/dx^n$ is the constant $2^n n!$, and consequently

$$\int_{-\infty}^{+\infty} e^{-x^2} H_n^2(x)\,dx = 2^n n! \sqrt{\pi}. \tag{a.7}$$

§ b. The Airy function

The equation

$$y'' - xy = 0 \tag{b.1}$$

is likewise of Laplace type. Applying the general procedure, we form the functions

$$P = t^2, \qquad Q = -1, \qquad Z = -\exp\!\left(-t^3/3\right), \qquad V = \exp\!\left(xt - t^3/3\right),$$

and hence a solution can be written as

$$y(x) = \mathrm{const}\cdot \int_C \exp\!\left(xt - t^3/3\right) dt, \tag{b.2}$$

where the contour C must be selected so that V vanishes at both of its ends. These ends must therefore recede to infinity through the regions of the complex t-plane in which $\mathrm{Re}(t^3) > 0$ (the shaded regions in Fig. 54).

A solution finite for every x follows from choosing C as shown in the figure. The contour may be displaced arbitrarily, subject only to the requirement that its ends tend to infinity through the same two shaded sectors (I and III in Fig. 54). Notice that a contour passing, for example, through sectors III and II would instead give a solution that diverges as $x \to \infty$.

Moving C until it lies on the imaginary axis puts (b.2) into the form (with $t = iu$)

$$\Phi(x) = \frac{1}{\sqrt{\pi}} \int_0^{\infty} \cos\left(ux + \frac{u^3}{3}\right) du. \tag{b.3}$$

Here the constant in (b.2) has been set equal to $-i/(2\sqrt{\pi})$, and the function defined in this way has been denoted by $\Phi(x)$; it is called the *Airy function*[56].

For large x, an asymptotic expression for $\Phi(x)$ can be found by evaluating (b.2) by the saddle-point method. When $x > 0$, the exponent of the integrand is stationary at $t = \pm\sqrt{x}$, and its direction of steepest descent is parallel to the imaginary axis. Accordingly, for large positive x we expand the exponent in powers of

[56]We use the definition proposed by V. A. Fock (see G. D. Yakovleva, *Tables of Airy Functions*. Moscow: Nauka, 1969; $\Phi(x)$ is one of Fock's two functions and is denoted there by $V(x)$). Another definition of the Airy function is also used in the literature; it differs from (b.3) by a constant factor: $\mathrm{Ai}\,x = \Phi(x)/\sqrt{\pi}$, so that $\int \mathrm{Ai}\,x\,dx = 1$.

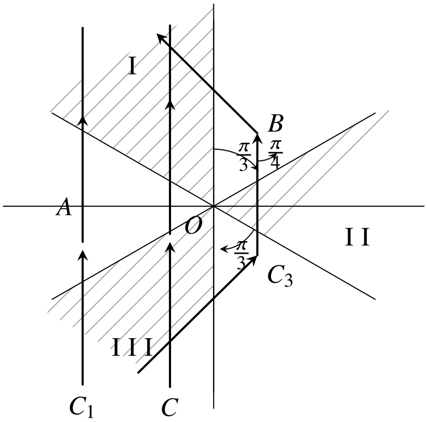


Figure 54

$t + \sqrt{x}$ and integrate along the line C_1 (Fig. 54), which is parallel to the imaginary axis (with $OA = \sqrt{x}$). Substitution of $t = -\sqrt{x} + iu$ gives

$$\Phi(x) \approx \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{+\infty} \exp\left(-\frac{2}{3}x^{3/2} - \sqrt{x}\,u^2\right) du,$$

and therefore

$$\Phi(x) \approx \frac{1}{2x^{1/4}} \exp\left(-\frac{2}{3}x^{3/2}\right). \tag{b.4}$$

Thus $\Phi(x)$ decreases exponentially at large positive x.

To obtain the asymptotic expression at large negative x, observe that for $x < 0$ the exponent is stationary at

$$t = i\sqrt{|x|} \qquad \text{and} \qquad t = -i\sqrt{|x|},$$

while at these two points the respective directions of steepest descent are the lines making angles $\mp\pi/4$ with the real axis. Taking the broken line C_3 as the integration contour (with $OB = \sqrt{|x|}$), straightforward transformations give

$$\Phi(x) = \frac{1}{|x|^{1/4}} \sin\left(\frac{2}{3}|x|^{3/2} + \frac{\pi}{4}\right). \tag{b.5}$$

Thus, for large negative x, $\Phi(x)$ is oscillatory. Its first (and largest) maximum may be noted: $\Phi(-1.02) = 0.95$.

The Airy function can be expressed through Bessel functions of order $1/3$. As is readily verified, equation (b.1) has the solution

$$\sqrt{x}\, Z_{1/3}\left(\frac{2}{3}x^{3/2}\right),$$

where $Z_{1/3}(x)$ denotes any solution of the Bessel equation of order $1/3$. The solution agreeing with (b.3) is

$$\Phi(x) = \frac{\sqrt{\pi x}}{3}\left[I_{-1/3}\left(\frac{2}{3}x^{3/2}\right) - I_{1/3}\left(\frac{2}{3}x^{3/2}\right)\right] \equiv \sqrt{\frac{x}{3\pi}}K_{1/3}\left(\frac{2}{3}x^{3/2}\right), \quad \text{for } x > 0,$$
$$\Phi(x) = \frac{\sqrt{\pi|x|}}{3}\left[J_{-1/3}\left(\frac{2}{3}|x|^{3/2}\right) + J_{1/3}\left(\frac{2}{3}|x|^{3/2}\right)\right], \quad \text{for } x < 0. \tag{b.6}$$

Here

$$I_\nu(x) = i^{-\nu}J_\nu(ix), \qquad K_\nu(x) = \frac{\pi}{2\sin\nu\pi}\left[I_{-\nu}(x) - I_\nu(x)\right].$$

The recurrence relations

$$K_{\nu-1}(x) - K_{\nu+1}(x) = -\frac{2\nu}{x}K_\nu(x),$$
$$2K'_\nu(x) = -K_{\nu-1}(x) - K_{\nu+1}(x)$$

then readily yield the following expression for the derivative of the Airy function:

$$\Phi'(x) = -\frac{x}{\sqrt{3\pi}}K_{2/3}\left(\frac{2}{3}x^{3/2}\right), \qquad x > 0. \tag{b.7}$$

At $x = 0$,

$$\Phi(0) = \frac{\sqrt{\pi}}{3^{2/3}\Gamma(2/3)} = 0.629,$$
$$\Phi'(0) = \frac{3^{1/6}\Gamma(2/3)}{2\sqrt{\pi}} = -0.459. \tag{b.8}$$

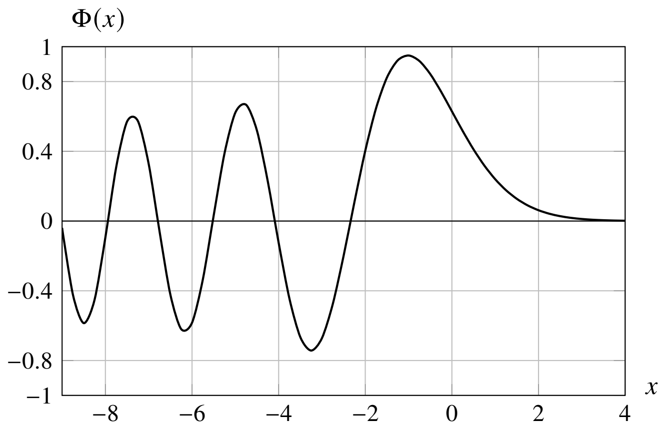


Figure 55

Figure 55 shows the graph of the Airy function.

§ c. Legendre polynomials[57]

The *Legendre polynomials* $P_l(\cos\theta)$ are defined by

$$P_l(\cos\theta) = \frac{1}{2^l l!}\frac{d^l}{(d\cos\theta)^l}(\cos^2\theta - 1)^l. \tag{c.1}$$

They satisfy the differential equation

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP_l}{d\theta}\right) + l(l+1)P_l = 0. \tag{c.2}$$

The associated Legendre polynomials are defined by

$$\begin{aligned} P_l^m(\cos\theta) &= \sin^m\theta\frac{d^m P_l(\cos\theta)}{(d\cos\theta)^m} \\ &= \frac{1}{2^l l!}\sin^m\theta\frac{d^{l+m}}{(d\cos\theta)^{l+m}}(\cos^2\theta-1)^l \end{aligned} \tag{c.3}$$

or, equivalently, by

$$P_l^m(\cos\theta) = (-1)^m\frac{(l+m)!}{(l-m)!2^l l!}\sin^{-m}\theta\frac{d^{l-m}}{(d\cos\theta)^{l-m}}(\cos^2\theta-1)^l, \tag{c.4}$$

where $m = 0, 1, \ldots, l$. The associated polynomials obey the equation

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dP_l^m}{d\theta}\right) + \left[l(l+1) - \frac{m^2}{\sin^2\theta}\right]P_l^m = 0. \tag{c.5}$$

The normalization integral of the Legendre polynomials, $\int_{-1}^{1}[P_l(\mu)]^2 d\mu$ ($\mu = \cos\theta$), is evaluated by substituting (c.1) into it and integrating by parts l times. It then becomes

$$\frac{(-1)^l}{2^{2l}(l!)^2}\int_{-1}^{1}(\mu^2-1)^l\frac{d^{2l}}{d\mu^{2l}}(\mu^2-1)^l\,d\mu = \frac{(2l)!}{2^{2l}(l!)^2}\int_{-1}^{1}(1-\mu^2)^l\,d\mu.$$

With the substitution $u = (1-\mu)/2$, this integral reduces to Euler's B integral and gives

$$\int_{-1}^{1}[P_l(\mu)]^2 d\mu = \frac{2}{2l+1}. \tag{c.6}$$

In the same way, one readily verifies that the functions $P_l(\mu)$ with different l are mutually orthogonal:

$$\int_{-1}^{1}P_l(\mu)P_{l'}(\mu)\,d\mu = 0, \qquad l \neq l'. \tag{c.7}$$

The normalization integral for the associated polynomials can be calculated similarly: write $[P_l^m(\mu)]^2$ as the product of expressions (c.3) and (c.4), and integrate by parts $l-m$ times. The result is

$$\int_{-1}^{1}[P_l^m(\mu)]^2 d\mu = \frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}. \tag{c.8}$$

[57] The mathematical literature contains many good accounts of the theory of spherical functions. Here, for reference, we give only some basic relations, without attempting any systematic presentation of the theory of these functions.

It is likewise easy to verify that functions P_l^m with different l (and the same m) are mutually orthogonal:

$$\int_{-1}^{1} P_l^m(\mu) P_{l'}^m(\mu)\, d\mu = 0, \qquad l \neq l'. \tag{c.9}$$

Integrals of products of three Legendre polynomials were considered in § 107.

The following *addition theorem* holds for the Legendre polynomials. Let γ be the angle between two directions specified by the spherical angles θ, φ and θ', φ': $\cos\gamma = \cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\varphi - \varphi')$. Then

$$\begin{aligned} P_l(\cos\gamma) &= P_l(\cos\theta) P_l(\cos\theta') \\ &\quad + \sum_{m=1}^{l} 2\frac{(l-m)!}{(l+m)!} P_l^m(\cos\theta) P_l^m(\cos\theta') \cos[m(\varphi - \varphi')]. \end{aligned} \tag{c.10}$$

In terms of the spherical functions (defined as in (28.7)), this theorem may also be written

$$P_l(\mathbf{n}\mathbf{n}') = \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_{lm}^*(\mathbf{n}') Y_{lm}(\mathbf{n}). \tag{c.11}$$

Here $\mathbf{n}$ and $\mathbf{n}'$ are two unit vectors, and $Y_{lm}(\mathbf{n})$ denotes the spherical function of the polar angle and azimuth of the direction $\mathbf{n}$ relative to a fixed coordinate system.

Multiply (c.10) by $P_{l'}(\cos\theta)$ and integrate it over $do = \sin\theta\, d\theta\, d\varphi$. Integration with respect to $d\varphi$ makes all terms on the right-hand side containing factors $\cos[m(\varphi - \varphi')]$ vanish. Taking account of (c.6) and (c.7), we obtain

$$\int P_l(\cos\gamma) P_{l'}(\cos\theta)\, do = \delta_{ll'} \frac{4\pi}{2l+1} P_l(\cos\theta').$$

This result can be written in the symmetric form

$$\int P_l(\mathbf{n}_1\mathbf{n}_2) P_{l'}(\mathbf{n}_1\mathbf{n}_3)\, do_1 = \delta_{ll'} \frac{4\pi}{2l+1} P_l(\mathbf{n}_2\mathbf{n}_3), \tag{c.12}$$

where $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$ are three unit vectors, and the integration is performed over the directions of one of them, $\mathbf{n}_1$. Finally, we list the first few normalized spherical functions Y_{lm}:

$$Y_{00} = \frac{1}{\sqrt{4\pi}}, \qquad Y_{10} = i\sqrt{\frac{3}{4\pi}}\cos\theta,$$

$$Y_{1,\pm 1} = \pm i\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{\pm i\varphi},$$

$$Y_{20} = \sqrt{\frac{5}{16\pi}}(1 - 3\cos^2\theta),$$

$$Y_{2,\pm 1} = \pm\sqrt{\frac{15}{8\pi}}\cos\theta\sin\theta\, e^{\pm i\varphi},$$

$$Y_{2,\pm 2} = -\sqrt{\frac{15}{32\pi}}\sin^2\theta\, e^{\pm 2i\varphi},$$

$$Y_{30} = -i\sqrt{\frac{7}{16\pi}}\cos\theta(5\cos^2\theta - 3),$$

$$Y_{3,\pm 1} = \pm i\sqrt{\frac{21}{64\pi}}\sin\theta(5\cos^2\theta - 1)e^{\pm i\varphi},$$

$$Y_{3,\pm 2} = -i\sqrt{\frac{105}{32\pi}}\cos\theta\sin^2\theta\, e^{\pm 2i\varphi},$$

$$Y_{3,\pm 3} = \pm i\sqrt{\frac{35}{64\pi}}\sin^3\theta\, e^{\pm 3i\varphi}.$$

§ d. The confluent hypergeometric function

The *confluent hypergeometric function* is defined by the series

$$F(\alpha,\gamma,z) = 1 + \frac{\alpha}{\gamma}\frac{z}{1!} + \frac{\alpha(\alpha+1)}{\gamma(\gamma+1)}\frac{z^2}{2!} + \dots, \tag{d.1}$$

which converges for every finite z. The parameter α is arbitrary, whereas γ is assumed to be neither zero nor a negative integer. When α is a negative integer (or zero), $F(\alpha,\gamma,z)$ becomes a polynomial of degree $|\alpha|$.

The function $F(\alpha,\gamma,z)$ obeys the equation

$$zu'' + (\gamma - z)u' - \alpha u = 0, \tag{d.2}$$

as is readily established by direct substitution.[58] On setting $u = z^{1-\gamma}u_1$, this equation is changed into another of the same form:

$$zu_1'' + (2 - \gamma - z)u_1' - (\alpha - \gamma + 1)u_1 = 0. \tag{d.3}$$

It follows that, for nonintegral γ, equation (d.2) also possesses the particular integral $z^{1-\gamma}F(\alpha - \gamma + 1, 2 - \gamma, z)$, which is linearly independent

[58] Equation (d.2) with a negative integral value of γ requires no separate treatment, since reduction to equation (d.3) converts it to the case of positive integral γ.

of (d.1). Thus the general solution of (d.2) is

$$u = c_1 F(\alpha, \gamma, z) + c_2 z^{1-\gamma} F(\alpha - \gamma + 1, 2 - \gamma, z). \tag{d.4}$$

Unlike the first term, the second has a singular point at $z = 0$.

Equation (d.2) is of Laplace type, so its solutions admit representations by contour integrals. Following the general procedure, we form the functions

$$P(t) = \gamma t - \alpha, \qquad Q(t) = t(t-1), \qquad Z(t) = t^{\alpha-1}(t-1)^{\gamma-\alpha-1},$$

and hence

$$u = \int e^{tz} t^{\alpha-1}(t-1)^{\gamma-\alpha-1}\, dt. \tag{d.5}$$

The integration path must be chosen so that, after it is traversed, the function $V(t) = e^{tz} t^{\alpha}(t-1)^{\gamma-\alpha}$ returns to its initial value. Applying the same procedure to equation (d.3) gives a contour integral of another form for u:

$$u = z^{1-\gamma} \int e^{tz} t^{\alpha-\gamma}(t-1)^{-\alpha}\, dt.$$

It is useful here to make the substitution $tz \to t$, which brings the integral to the form

$$u(z) = \int e^{t}(t-z)^{-\alpha} t^{\alpha-\gamma}\, dt, \tag{d.6}$$

with the corresponding function $V(t) = e^t t^{\alpha-\gamma+1}(t-z)^{1-\alpha}$.

In general, the integrand in (d.6) has two singular points, at $t = z$ and at $t = 0$. Choose an integration contour C that comes from infinity ($\operatorname{Re} t \to -\infty$), encircles both singular points in the positive sense, and then returns to infinity (Fig. 56). This contour fulfills the required conditions, since $V(t)$ vanishes at its ends. The integral (d.6) taken along C is nonsingular at $z = 0$; it must consequently agree, apart from a constant factor, with the nonsingular function $F(\alpha, \gamma, z)$.

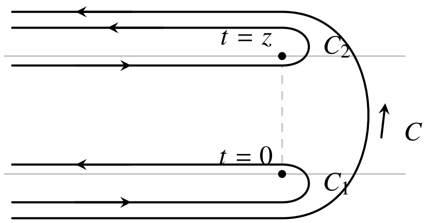


Figure 56

At $z = 0$ the two singular points of the integrand coalesce. A familiar formula from the theory of the Γ-function gives

$$\frac{1}{2\pi i} \int_C e^t t^{-\gamma}\, dt = \frac{1}{\Gamma(\gamma)}. \tag{d.7}$$

Since $F(\alpha, \gamma, 0) = 1$, it is evident that

$$F(\alpha, \gamma, z) = \frac{\Gamma(\gamma)}{2\pi i} \int_C e^t (t-z)^{-\alpha} t^{\alpha-\gamma}\, dt. \tag{d.8}$$

The integrand in (d.5) has singular points at $t = 0$ and $t = 1$. If $\mathrm{Re}(\gamma - \alpha) > 0$ and γ is not a positive integer, the path of integration may be taken to be a contour C' that leaves $t = 1$, encircles $t = 0$ in the positive sense, and returns to $t = 1$ (Fig. 57). For $\mathrm{Re}(\gamma - \alpha) > 0$, traversal of this contour returns the function $V(t)$ to its initial value, zero.[59] The integral so defined is likewise nonsingular at $z = 0$ and is related to $F(\alpha, \gamma, z)$ by

$$F(\alpha, \gamma, z) = -\frac{1}{2\pi i} \frac{\Gamma(1-\alpha)\Gamma(\gamma)}{\Gamma(\gamma-\alpha)} \oint_{C'} e^{tz} (-t)^{\alpha-1} (1-t)^{\gamma-\alpha-1}\, dt. \tag{d.9}$$

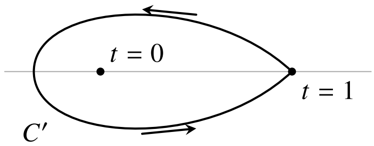


Figure 57

The following qualification concerning the integrals (d.8) and (d.9) is necessary. For nonintegral α and γ, their integrands are multiple-valued functions. At each point their values are understood to be selected by requiring every complex quantity raised to a power to have the argument of smallest absolute value.

We also note the useful relation

$$F(\alpha, \gamma, z) = e^z F(\gamma - \alpha, \gamma, -z), \tag{d.10}$$

which follows at once upon making the substitution $t \to t + z$ in (d.8).

It was already mentioned that when $\alpha = -n$, with n a positive integer, $F(\alpha, \gamma, z)$ reduces to a polynomial. A compact formula can be found for these polynomials. Making the substitution $t \to 1 - (t/z)$ in (d.9) and applying the Cauchy formula to

the resulting integral, we find

$$F(-n, \gamma, z) = \frac{z^{1-\gamma} e^z}{\gamma(\gamma+1)\cdots(\gamma+n-1)} \frac{d^n}{dz^n} \left(e^{-z} z^{\gamma+n-1} \right). \tag{d.11}$$

If, in addition, $\gamma = m$, where m is a positive integer, one also has

$$F(-n, m, z) = \frac{(-1)^{m-1} e^z z^{1-m-n}}{m(m+1)\cdots(m+n-1)} \frac{d^{m+n-1}}{dz^{m+n-1}} \left(e^{-z} z^n \right). \tag{d.12}$$

This formula results by applying the Cauchy formula to the integral obtained from (d.8) through the substitution $t \to z - t$.

[59] If γ is a positive integer, any contour encircling both $t = 0$ and $t = 1$ may be chosen as C'.

Up to a constant factor, the polynomials $F(-n, m, z)$ $(0 \leq m \leq n)$ are the *generalized Laguerre polynomials*:

$$\begin{aligned} L_n^m(z) &= (-1)^m \frac{(n!)^2}{m!(n-m)!} F(-(n-m), m+1, z) \\ &= \frac{n!}{(n-m)!} e^z \frac{d^m}{dz^m} (e^{-z} z^{n-m}) \\ &= (-1)^m \frac{n!}{(n-m)!} e^z z^{-m} \frac{d^{n-m}}{dz^{n-m}} (e^{-z} z^n) . \end{aligned} \tag{d.13}$$

For $m = 0$, the polynomials L_n^m are denoted by $L_n(z)$ and are called simply the *Laguerre polynomials*; according to (d.13),

$$L_n(z) = e^z \frac{d^n}{dz^n} (e^{-z} z^n) .$$

The integral representation (d.8) is useful in finding the asymptotic expansion of the confluent hypergeometric function for large z. Deform the contour until it becomes two contours C_1 and C_2 (see Fig. 56), which encircle $t = 0$ and $t = z$, respectively; the lower branch of C_2 and the upper branch of C_1 are to be regarded as joining at infinity. To obtain an expansion in inverse powers of z, take the factor $(-z)^{-\alpha}$ outside the integrand. In the integral along C_2, make the substitution $t \to t + z$; this changes C_2 into C_1. Formula (d.8) is thereby written as

$$\begin{aligned} F(\alpha, \gamma, z) = {} & \frac{\Gamma(\gamma)}{\Gamma(\gamma - \alpha)} (-z)^{-\alpha} G(\alpha, \alpha - \gamma + 1, -z) \\ & + \frac{\Gamma(\gamma)}{\Gamma(\alpha)} e^z z^{\alpha - \gamma} G(\gamma - \alpha, 1 - \alpha, z), \end{aligned} \tag{d.14}$$

where

$$G(\alpha, \beta, z) = \frac{\Gamma(1-\beta)}{2\pi i} \int_{C_1} \left(1 + \frac{t}{z}\right)^{-\alpha} t^{\beta - 1} e^t \, dt. \tag{d.15}$$

When powers are taken in (d.14), $-z$ and z must be assigned the arguments of smallest absolute value. Finally, expand $(1 + t/z)^{-\alpha}$ in powers of t/z in the integrand and use (d.7). The resulting asymptotic series for $G(\alpha, \beta, z)$ is

$$G(\alpha, \beta, z) = 1 + \frac{\alpha\beta}{1! z} + \frac{\alpha(\alpha+1)\beta(\beta+1)}{2! z^2} + \ldots \tag{d.16}$$

Equations (d.14) and (d.16) specify the asymptotic expansion of $F(\alpha, \gamma, z)$.

For positive integral γ, the second term in the general solution (d.4) of equation (d.2) either coincides with the first (when $\gamma = 1$) or becomes altogether meaningless (when $\gamma > 1$). In this case, the two terms in (d.14) may be chosen as a pair of linearly independent solutions, that is, the integrals (d.8) taken along C_1 and C_2. (These contours, like C, satisfy the required conditions, so their integrals are also solutions of (d.2).) Their asymptotic forms have already been given; it remains to determine their expansions in ascending powers of z. For this purpose, begin with (d.14) and the analogous equality for the function $z^{1-\gamma} F(\alpha - \gamma + 1, 2 - \gamma, z)$. From these two equalities, express

$G(\alpha, \alpha-\gamma+1, -z)$ in terms of $F(\alpha, \gamma, z)$ and $F(\alpha-\gamma+1, 2-\gamma, z)$, then put $\gamma = p+\varepsilon$ (p being a positive integer) and take the limit $\varepsilon \to 0$, resolving the indeterminate forms by l'Hopital's rule. A fairly lengthy calculation gives the expansion

$$\begin{aligned} G(\alpha, \alpha-p+1, -z) = -\frac{\sin\pi\alpha\,\Gamma(p-\alpha)}{\pi\Gamma(p)} z^{\alpha} \Bigg\{ &\ln z\, F(\alpha, p, z) \\ &+ \sum_{s=0}^{\infty} \frac{\Gamma(p)\Gamma(\alpha+s)[\psi(\alpha+s)-\psi(p+s)-\psi(s+1)]}{\Gamma(\alpha)\Gamma(s+p)\Gamma(s+1)} z^{s} \\ &+ \sum_{s=1}^{p-1} (-1)^{s+1} \frac{\Gamma(s)\Gamma(\alpha-s)\Gamma(p)}{\Gamma(\alpha)\Gamma(p-s)} z^{-s} \Bigg\}, \end{aligned} \tag{d.17}$$

where ψ denotes the logarithmic derivative of the Γ-function: $\psi(\alpha) = \Gamma'(\alpha)/\Gamma(\alpha)$.

§ e. The hypergeometric function

Within the circle $|z| < 1$, the *hypergeometric function* is defined by the series

$$F(\alpha, \beta, \gamma, z) = 1 + \frac{\alpha\beta}{\gamma}\frac{z}{1!} + \frac{\alpha(\alpha+1)\beta(\beta+1)}{\gamma(\gamma+1)}\frac{z^2}{2!} + \dots, \tag{e.1}$$

and for $|z| > 1$ it is obtained by analytic continuation of this series (see (e.6)). The hypergeometric function is one particular integral of the differential equation

$$z(1-z)u'' + [\gamma - (\alpha+\beta+1)z]u' - \alpha\beta u = 0. \tag{e.2}$$

The parameters α and β are arbitrary, while $\gamma \neq 0, -1, -2, \dots$. Evidently, $F(\alpha, \beta, \gamma, z)$ is symmetric in the parameters α and β[60].

A second independent solution of (e.2) is

$$z^{1-\gamma} F(\beta-\gamma+1, \alpha-\gamma+1, 2-\gamma, z);$$

it has a singular point at $z = 0$.

For reference, we give here a number of relations satisfied by the hypergeometric function.

For all z, provided $\mathrm{Re}(\gamma-\alpha) > 0$, the function $F(\alpha, \beta, \gamma, z)$ can be represented by the integral

$$\begin{aligned} F(\alpha, \beta, \gamma, z) = &-\frac{1}{2\pi i}\frac{\Gamma(1-\alpha)\Gamma(\gamma)}{\Gamma(\gamma-\alpha)} \\ &\times \oint_{C'} (-t)^{\alpha-1}(1-t)^{\gamma-\alpha-1}(1-tz)^{-\beta}\, dt, \end{aligned} \tag{e.3}$$

[60] The confluent hypergeometric function is obtained from $F(\alpha, \beta, \gamma, z)$ by the limiting transition $F(\alpha, \gamma, z) = \lim_{\beta\to\infty} F(\alpha, \beta, \gamma, z/\beta)$. The notations ${}_2F_1(\alpha, \beta, \gamma, z)$ and ${}_1F_1(\alpha, \gamma, z)$ are also used in the literature for the hypergeometric and confluent hypergeometric functions, respectively. The indices to the left and right of F give the numbers of parameters appearing in the numerators and denominators, respectively, of the terms of the series.

taken over the contour C' shown in Fig. 57. Direct substitution readily verifies that this integral does indeed satisfy (e.2); the constant factor has been chosen so that its value at $z = 0$ is unity.

The substitution

$$u = (1-z)^{\gamma-\alpha-\beta} u_1$$

in (e.2) gives an equation of the same form, with the parameters $\gamma - \alpha$, $\gamma - \beta$, γ in place of α, β, γ, respectively. It follows that

$$F(\alpha, \beta, \gamma, z) = (1-z)^{\gamma-\alpha-\beta} F(\gamma-\alpha, \gamma-\beta, \gamma, z) \tag{e.4}$$

(both sides obey the same equation and have equal values at $z = 0$).

Making the substitution $t \to t/(1 - z + zt)$ in the integral (e.3) gives the following relation between hypergeometric functions of the variables z and $z/(z-1)$:

$$F(\alpha, \beta, \gamma, z) = (1-z)^{-\alpha} F\left(\alpha, \gamma-\beta, \gamma, \frac{z}{z-1}\right). \tag{e.5}$$

The value of the multivalued expression $(1-z)^{-\alpha}$ in this formula (and of analogous expressions in all the formulas below) is fixed by requiring the complex quantity being raised to a power to be taken with the argument of smallest absolute value.

We next state, without proof, an important formula relating hypergeometric functions of the variables z and $1/z$:

$$\begin{aligned} F(\alpha, \beta, \gamma, z) = {} & \frac{\Gamma(\gamma)\Gamma(\beta-\alpha)}{\Gamma(\beta)\Gamma(\gamma-\alpha)} (-z)^{-\alpha} F\left(\alpha, \alpha+1-\gamma, \alpha+1-\beta, \frac{1}{z}\right) \\ & + \frac{\Gamma(\gamma)\Gamma(\alpha-\beta)}{\Gamma(\alpha)\Gamma(\gamma-\beta)} (-z)^{-\beta} F\left(\beta, \beta+1-\gamma, \beta+1-\alpha, \frac{1}{z}\right). \end{aligned} \tag{e.6}$$

This formula expresses $F(\alpha, \beta, \gamma, z)$ as a series convergent for $|z| > 1$; it therefore supplies the analytic continuation of the original series (e.1).

The formula

$$\begin{aligned} F(\alpha, \beta, \gamma, z) = {} & \frac{\Gamma(\gamma)\Gamma(\gamma-\alpha-\beta)}{\Gamma(\gamma-\alpha)\Gamma(\gamma-\beta)} F(\alpha, \beta, \alpha+\beta+1-\gamma, 1-z) \\ & + \frac{\Gamma(\gamma)\Gamma(\alpha+\beta-\gamma)}{\Gamma(\alpha)\Gamma(\beta)} (1-z)^{\gamma-\alpha-\beta} \\ & \times F(\gamma-\alpha, \gamma-\beta, \gamma+1-\alpha-\beta, 1-z) \end{aligned} \tag{e.7}$$

relates hypergeometric functions of z and $1-z$ (we give this relation also without proof). Combining (e.7) with (e.6), we obtain

the relations

$$\begin{aligned} F(\alpha, \beta, \gamma, z) = {} & \frac{\Gamma(\gamma)\Gamma(\beta-\alpha)}{\Gamma(\beta)\Gamma(\gamma-\alpha)} (1-z)^{-\alpha} F\left(\alpha, \gamma-\beta, \alpha+1-\beta, \frac{1}{1-z}\right) \\ & + \frac{\Gamma(\gamma)\Gamma(\alpha-\beta)}{\Gamma(\alpha)\Gamma(\gamma-\beta)} (1-z)^{-\beta} F\left(\beta, \gamma-\alpha, \beta+1-\alpha, \frac{1}{1-z}\right), \end{aligned} \tag{e.8}$$

$$
\begin{aligned}
F(\alpha,\beta,\gamma,z) = {} & \frac{\Gamma(\gamma)\Gamma(\gamma-\alpha-\beta)}{\Gamma(\gamma-\beta)\Gamma(\gamma-\alpha)} z^{-\alpha} F\left(\alpha, \alpha+1-\gamma, \alpha+\beta+1-\gamma, \frac{z-1}{z}\right) \\
& + \frac{\Gamma(\gamma)\Gamma(\alpha+\beta-\gamma)}{\Gamma(\alpha)\Gamma(\beta)} (1-z)^{\gamma-\alpha-\beta} z^{\beta-\gamma} \\
& \times F\left(1-\beta, \gamma-\beta, \gamma+1-\alpha-\beta, \frac{z-1}{z}\right).
\end{aligned} \tag{e.9}
$$

Each term in the sums on the right-hand sides of (e.6)–(e.9) is itself a solution of the hypergeometric equation.

If α (or β) is a negative integer (or zero), $\alpha = -n$, the hypergeometric function becomes a polynomial of degree n and may be written as

$$
F(-n,\beta,\gamma,z) = \frac{z^{1-\gamma}(1-z)^{\gamma+n-\beta}}{\gamma(\gamma+1)\cdots(\gamma+n-1)} \frac{d^n}{dz^n}\left[z^{\gamma+n-1}(1-z)^{\beta-\gamma}\right]. \tag{e.10}
$$

Apart from a constant factor, these polynomials are the *Jacobi polynomials*, defined by

$$
\begin{aligned}
P_n^{(a,b)}(z) &= \frac{(a+1)(a+2)\cdots(a+n)}{n!} F\left(-n, a+b+n+1, a+1, \frac{1-z}{2}\right) \\
&= \frac{(-1)^n}{2^n n!}(1-z)^{-a}(1+z)^{-b} \frac{d^n}{dz^n}\left[(1-z)^{a+n}(1+z)^{b+n}\right].
\end{aligned} \tag{e.11}
$$

For $a = b = 0$, the Jacobi polynomials coincide with the Legendre polynomials. For $n = 0$, $P_0^{(a,b)} = 1$.

§f. Evaluation of integrals containing confluent hypergeometric functions

Consider an integral of the form

$$
J_{\alpha\gamma}^{\nu} = \int_0^{\infty} e^{-\lambda z} z^{\nu} F(\alpha,\gamma,kz)\, dz. \tag{f.1}
$$

It is assumed to converge. This requires $\operatorname{Re}\nu > -1$ and $\operatorname{Re}\lambda > |\operatorname{Re} k|$; if α is a negative integer,

the second condition may instead be replaced by $\operatorname{Re}\lambda > 0$. Use the integral representation (d.9) for $F(\alpha,\gamma,kz)$ and carry out the integration with respect to z under the contour-integral sign. This gives

$$
\begin{aligned}
J_{\alpha\gamma}^{\nu} = {} & -\frac{1}{2\pi i}\frac{\Gamma(1-\alpha)\Gamma(\gamma)}{\Gamma(\gamma-\alpha)} \lambda^{-\nu-1}\Gamma(\nu+1) \\
& \times \oint_{C'} (-t)^{\alpha-1}(1-t)^{\gamma-\alpha-1}\left(1-\frac{k}{\lambda}t\right)^{-\nu-1} dt.
\end{aligned}
$$

Taking (e.3) into account, we finally find

$$
J_{\alpha\gamma}^{\nu} = \Gamma(\nu+1)\lambda^{-\nu-1} F\left(\alpha, \nu+1, \gamma, \frac{k}{\lambda}\right). \tag{f.2}
$$

In the cases where $F(\alpha, \nu+1, \gamma, k/\lambda)$ reduces to polynomials, the corresponding expressions for $J^{\nu}_{\alpha\gamma}$ are in elementary functions:

$$J^{\gamma+n-1}_{\alpha\gamma} = (-1)^n \Gamma(\gamma) \frac{d^n}{d\lambda^n} \left[\lambda^{\alpha-1}(\lambda-k)^{-\alpha}\right], \tag{f.3}$$

$$J^{\nu}_{-n,\gamma} = (-1)^n \frac{\Gamma(\nu+1)(\lambda-k)^{\gamma+n-\nu-1}}{\gamma(\gamma+1)\cdots(\gamma+n-1)} \frac{d^n}{d\lambda^n} \left[\lambda^{-\nu-1}(\lambda-k)^{\nu-\gamma+1}\right], \tag{f.4}$$

$$\begin{aligned} J^{n}_{\alpha m} = {} & \frac{(-1)^{m-n}}{k^{m-1}(1-\alpha)(2-\alpha)\cdots(m-1-\alpha)} \\ & \times \left\{ -(m-1)! \frac{d^n}{d\lambda^n} \left[\lambda^{\alpha-1}(\lambda-k)^{m-\alpha-1}\right] \right. \\ & + n!(m-n-1)\cdots(m-1)\lambda^{\alpha-n-1}(\lambda-k)^{-1+m-n-\alpha} \\ & \left. \times \frac{d^{m-n-2}}{d\lambda^{m-n-2}} \left[\lambda^{m-\alpha-1}(\lambda-k)^{\alpha-1}\right] \right\}, \end{aligned} \tag{f.5}$$

where m, n are integers and $0 \le n \le m-2$.

Next, evaluate the integral

$$J_\nu = \int_0^\infty e^{-kz} z^{\nu-1} [F(-n, \gamma, kz)]^2 \, dz \tag{f.6}$$

(n is a positive integer, $\operatorname{Re}\nu > 0$). We begin with the more general integral having $e^{-\lambda z}$ rather than e^{-kz} in its integrand. Write one of the functions $F(-n, \gamma, kz)$ as the integral (d.9), after which integration with respect to z by

formula (f.3) gives

$$\begin{aligned} & \int_0^\infty e^{-\lambda z} z^{\nu-1} [F(-n, \gamma, kz)]^2 \, dz \\ & = -\frac{1}{2\pi i}(-1)^n \frac{\Gamma(1+n)\Gamma^2(\gamma)\Gamma(\nu)}{\Gamma^2(\gamma+n)} \\ & \quad \times \oint_{C'} (\lambda - kt - k)^{\gamma+n-\nu}(-t)^{-n-1}(1-t)^{\gamma+n-1} \\ & \qquad \times \frac{d^n}{d\lambda^n} \left[(\lambda-kt)^{-\nu}(\lambda-kt-k)^{\nu-\gamma}\right] dt. \end{aligned}$$

The derivative of order n with respect to λ can plainly be replaced by an expression in terms of a derivative of the same order with respect to t. Having done this, put $\lambda = k$, thereby returning to the integral J_ν:

$$\begin{aligned} J_\nu = {} & -\frac{1}{2\pi i} \frac{\Gamma(n+1)\Gamma(\nu)\Gamma^2(\gamma)}{\Gamma^2(\gamma+n)k^\nu} \\ & \times \oint_{C'} (-t)^{\gamma-\nu-1}(1-t)^{\gamma+n-1} \frac{d^n}{dt^n} \left[(1-t)^{-\nu}(-t)^{\nu-\gamma}\right] dt. \end{aligned}$$

Integrating by parts n times transfers the operation $(d/dt)^n$ to $(-t)^{\gamma-\nu-1}(1-t)^{\gamma+n-1}$; expand the derivative by Leibniz's formula. The result is a sum of integrals, each reducible to the familiar Euler integral. The final expression for the integral sought is

$$J_\nu = \frac{\Gamma(\nu)n!}{k^\nu \gamma(\gamma+1)\cdots(\gamma+n-1)} \times \left\{1 + \sum_{s=0}^{n-1} \frac{n(n-1)\cdots(n-s)(\gamma-\nu-s-1)(\gamma-\nu-s)\cdots(\gamma-\nu+s)}{[(s+1)!]^2\gamma(\gamma+1)\cdots(\gamma+s)}\right\} \tag{f.7}$$

It is readily seen that the integrals J_ν satisfy the following relation (p is an integer):

$$J_{\gamma+p} = \frac{(\gamma-p-1)(\gamma-p)\cdots(\gamma+p-1)}{k^{2p+1}} J_{\gamma-1-p}. \tag{f.8}$$

In the same way one evaluates the integral

$$J = \int_0^\infty e^{-\lambda z} z^{\gamma-1} F(\alpha,\gamma,kz)F(\alpha',\gamma,k'z)\,dz. \tag{f.9}$$

Represent $F(\alpha',\gamma,k'z)$ by the integral (d.9). After integrating with respect to z by formula (f.3), with $n=0$, we obtain

$$J = -\frac{1}{2\pi i}\frac{\Gamma(1-\alpha')\Gamma^2(\gamma)}{\Gamma(\gamma-\alpha')} \oint_{C'} (-t)^{\alpha'-1}(1-t)^{\gamma-\alpha'-1} \times (\lambda - k't)^{\alpha-\gamma}(\lambda-k't-k)^{-\alpha}\,dt.$$

The substitution $t \to \lambda t/(k't+\lambda-k')$ reduces this integral to the form (e.3), giving

$$J = \Gamma(\gamma)\lambda^{\alpha+\alpha'-\gamma}(\lambda-k)^{-\alpha}(\lambda-k')^{-\alpha'} F\left(\alpha,\alpha',\gamma,\frac{kk'}{(\lambda-k)(\lambda-k')}\right). \tag{f.10}$$

If α (or α') is a negative integer, $\alpha=-n$, relation (e.7) can be used to rewrite this expression as

$$J = \frac{\Gamma^2(\gamma)\Gamma(\gamma+n-\alpha')}{\Gamma(\gamma+n)\Gamma(\gamma-\alpha')}\lambda^{-n+\alpha'-\gamma}(\lambda-k)^n(\lambda-k')^{-\alpha'} \times F\left(-n,\alpha',-n+\alpha'+1-\gamma,\frac{\lambda(\lambda-k-k')}{(\lambda-k)(\lambda-k')}\right). \tag{f.11}$$

Finally, consider integrals of the form

$$J_\gamma^{sp}(\alpha,\alpha') = \int_0^\infty \exp\left(-\frac{k+k'}{2}z\right) z^{\gamma-1+s} F(\alpha,\gamma,kz)F(\alpha',\gamma-p,k'z)\,dz. \tag{f.12}$$

The parameters are assumed to have values for which the integral converges absolutely; s,p are positive integers. By (f.10), the simplest of these integrals, $J_\gamma^{00}(\alpha,\alpha')$, is

$$J_\gamma^{00}(\alpha,\alpha') = 2^\gamma\Gamma(\gamma)(k+k')^{\alpha+\alpha'-\gamma}(k'-k)^{-\alpha}(k-k')^{-\alpha'} \times F\left(\alpha,\alpha',\gamma,-\frac{4kk'}{(k'-k)^2}\right), \tag{f.13}$$

and if α (or α') is a negative integer ($\alpha = -n$), then,
by (f.11), it may also be written as

$$J_\gamma^{00}(-n, \alpha') = 2^\gamma \frac{\Gamma(\gamma)(\gamma-\alpha')(\gamma-\alpha'+1)\cdots(\gamma-\alpha'+n-1)}{\gamma(\gamma+1)\cdots(\gamma+n-1)} \times (-1)^n (k+k')^{-n+\alpha'-\gamma}(k-k')^{n-\alpha'} \times F\left[-n, \alpha', \alpha'+1-n-\gamma, \left(\frac{k+k'}{k-k'}\right)^2\right]. \tag{f.14}$$

A general formula for $J_\gamma^{sp}(\alpha, \alpha')$ can be derived, but it is too complicated for convenient use. It is preferable to use recurrence relations that reduce $J_\gamma^{sp}(\alpha, \alpha')$ to the integral with $s = p = 0$[61]. The formula

$$J_\gamma^{sp}(\alpha, \alpha') = \frac{\gamma-1}{k}\left\{J_{\gamma-1}^{s,p-1}(\alpha, \alpha') - J_{\gamma-1}^{s,p-1}(\alpha-1, \alpha')\right\} \tag{f.15}$$

makes it possible to reduce $J_\gamma^{sp}(\alpha, \alpha')$ to an integral with $p = 0$. Thereafter the formula

$$J_\gamma^{s+1,0}(\alpha, \alpha') = \frac{4}{k^2-k'^2}\left\{\left[\frac{\gamma}{2}(k-k') - k\alpha + k'\alpha' - k's\right] J_\gamma^{s0}(\alpha, \alpha') + s(\gamma-1+s-2\alpha')J_\gamma^{s-1,0}(\alpha, \alpha') + 2\alpha' s J_\gamma^{s-1,0}(\alpha, \alpha'+1)\right\} \tag{f.16}$$

completes the reduction to the integral with $s = p = 0$.

[61] See W. Gordon // Ann. d. Phys. 1929. V. 2. P. 1031.

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